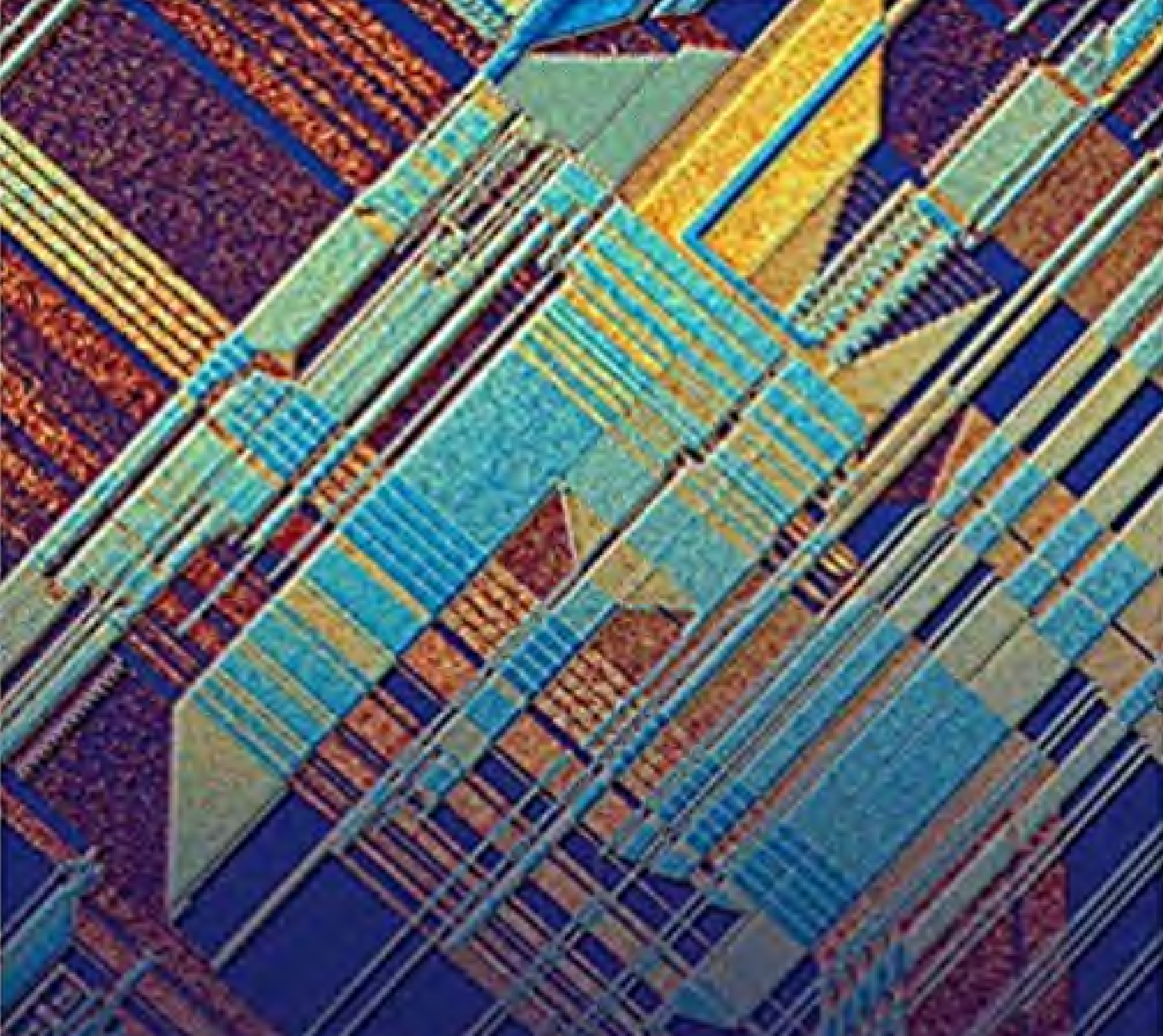

INTEGRATED MICROELECTRONIC DEVICES

Pearson

J. A. del Alamo

To Diego, Paul, and Antoine. This book grew up alongside you.

Contents

Preface

Judging by its age, now beyond 50 years old, one would think that microelectronics is a mature engineering discipline. Yet, its youthful exponential growth and its dramatic impact on human society continue unabated. For those of us that have been given the privilege of playing a role in this amazing endeavor, it has been an exhilarating experience. For years it was predicted that the relentless down scaling of transistor dimensions will hit a “fundamental limit” beyond which traditional device physics will be irreparably upset and progress will stall. Fortunately, semiconductor technologists and device engineers have been too busy to listen to the doomsayers. Innovative solutions to the challenges that continued to emerge were identified. Old taboos and preconceptions had to go by the wayside, and new materials needed to be brought to bear but progress never slowed down. The “microelectronics revolution” is perhaps the best exponent of human creativity and resourcefulness that there has ever been.

While recently, expressions of concern about the impending end of “Moore’s Law” have grown louder and the path forward with transistor scaling is not entirely apparent, it is very clear that CMOS (complementary metal–oxide semiconductor, a logic circuit family based on metal–oxide–semiconductor field-effect transistors) in whatever form it will take, will continue to be the backbone of computation, communications, power management, medical devices and many other kinds of systems for years to come. Also clear is that going forward, a richer set of technological options will need to be investigated: new materials, new structures, new geometries, and even new physics. In this context, indepth understanding of semiconductor fundamentals and device physics will be more valuable than ever.

It is in this regard that this book attempts to fill an important gap in the academic literature. While there are excellent texts on semiconductor device physics in the market, there is a need for a rigorous description that is relevant to modern nanoelectronics. In recent times, for instance, the so-called “extrinsic effects” or “parasitics” have come to play a dominant role in device operation. Moreover, device physics such as impact ionization and breakdown represent significant considerations in nanoscale device design. Another example is the role of device layout on device performance metrics that is of increasing relevance. This book additionally aims to provide a realistic technology context for the main stream devices: the metal–oxide–semiconductor field-effect transistor (MOSFET) and the bipolar junction transistor (BJT).

This book is based on my experience in teaching *6.720J/3.43J Integrated Microelectronic Devices,* a semester-long graduate student subject jointly offered in the Departments of Electrical Engineering and Computer Science (EECS) and Materials Science and Engineering (MS&E) at Massachusetts Institute of Technology (MIT). Typically, the class is composed of graduate students in EECS, Materials Science, Mechanical Engineering, Chemical Engineering and Physics plus a few seniors in the same departments. Graduate students in EECS and MS&E with interest in semiconductor materials and devices are strongly encouraged to take this subject their very first semester at MIT. While the book originated in a graduate course at MIT, it has been constructed to be productively used in an advanced undergraduate subject at the junior/senior level, as explained below.

The central goal of this book is to present the fundamentals of semiconductor device operation with relevance to modern integrated microelectronics (as opposed to, say, photonics, energy conversion devices, or power electronics). This means that no optical devices nor power devices of any kind are described. In contrast, emphasis is devoted to frequency response, layout, geometrical effects, parasitic issues and modeling in integrated microelectronics devices (transistors and diodes). In spite of this focus, the concepts learned here are highly applicable in other device contexts. This book should be a great resource for a broad range of students with a diverse set of interests.

There are two distinct parts to this book. The first five chapters introduce fundamental aspects of semiconductor physics pertaining to microelectronic devices: band structure, electron statistics, generation and recombination, drift and diffusion, and minority and majority carrier situations. Each chapter gives in its main body a general description suitable for a junior/senior-level or a first-year graduate course. These chapters also include at the end a number of advanced topics that can be selected individually to provide further depth. These can be the basis of a more advanced graduate subject.

Six device chapters follow with a similar outline. After a brief introductory section, the main body of the chapter presents a first-order, physically meaningful description of device physics and operation of an "ideal device." The ideal device is stripped down to its very essence, preserving the key physics, and is analyzed in a simple and intuitive way. The ideal device, constructed and studied this way, is therefore an excellent vehicle to learn device physics at the junior/senior level. One or more of the following sections present significant non idealities, important second-order effects and other considerations that are relevant in "real" devices. These are suitable topics for graduate courses. To some extent, teachers of graduate subjects will be able to pick and choose topics from these latter sections since they are rather independent of one another. Every chapter finishes with a set of advanced topics that contain more advanced graduate-level material also amenable to individual selection.

With this organization, the proposed text should be suitable for a one-semester junior/senior-level course by selecting the front sections of selected chapters (1 through 9 should be a popular set). At the same time, it could be used in a two-semester senior-level or a graduate-level course by taking advantage of the more advanced sections. In all cases, the book will provide plenty of reading material for personal study and future reference.

As required background for students to follow a course based on this book, in addition to freshman and sophomore physics and math, is a sophomore/junior subject on microelectronic circuits (at a level such as Sedra & Smith's *Microelectronic Circuits*). Specific essentials are basic linear and non-linear circuit understanding and an ability to work in the frequency domain. This book does not require previous knowledge of semiconductor device physics, quantum mechanics or statistical mechanics. It does however expect a certain degree of student sophistication in handling complexity, mathematical rigor, and physical intuition. This book will best serve students that have already taken a course that presents an integrated introductory view of devices and circuits, such as in Howe & Sodini's *Microelectronics: An Integrated Approach*

This book would not have been possible without the help and support of many people. It was my MIT colleague Charles Sodini who encouraged me to launch this project in 1993 (!!??) and gave me feedback and advice on its organization and on the content of many early chapters. Little did Charlie know, nor I, that this project would take the bulk of my career at MIT to complete. I inherited content and benefited from feedback and advice from several colleagues at MIT: Dimitri Antoniadis, Clifton Fonstad, Judy Hoyt, Tomás Palacios, and Stephen Senturia. The book includes specific materials generously contributed by a number of individuals: Phaedon Avouris, Douglas Buchanan, Massimo Fischetti, Matthew Fox, Paul Humphries, Sergei Krupenin, Wenjie Lu, Richard Martel, Samuel Mertens, Richard Molnar, Tassanee Payakapan, Brad Scharf, Shinichi Takagi, and Ling Xia. They are all identified at the appropriate location in the text.

My teaching assistants and students in 6.720J/3.43J supported me and encouraged me in this project over the years in numerous ways. I want to single out (apologies if I forget significant contributions!): Pablo Acosta, Tracy Adams, James Fiorenza, Iliana Fujimori, Usha Gogineni, David Greenberg, Alex Guo, Donghyun Jin, Jungwoo Joh, Dong-Seup Lee, Jianqiang Lin, Jelena Madic, Dennis Ouma, Jorg Scholvin, James Teherani, Anita Villanueva, Niamh Waldron, Chagarn Wang, Ginger Wang, Shireen Warnock, Joyce Wu, Mehmet Yanik, and Xin Zhao. I have also benefited from comments and feedback from many colleagues at MIT and beyond. Among them, I want to thank Thomas Clark, Emmanuel Crabbé, Isik Kizilyalli, Jody Lapham, Terry Orlando and Markus Zahn.

With many years in the making, this book would not have been possible without the teaching release and support of several department heads of EECS at MIT: Anantha Chandrakasan, Eric Grimson, John Guttag, Paul Penfield and Rafael Reif.

Thanks are also due to the staff at Pearson and Prentice Hall that have supported me through this project: Julie Bai, Scott Disanno, Tom Robbins, Shylaja Gattupalli, and Carole Snyder.

My passion for semiconductors was ignited by the opportunity that Prof. Antonio Luque gave me to become engaged in his research program on solar cells when I was an undergraduate student at Universidad Politécnica de Madrid in Spain. My mentor at Politécnica, Dr. Javier Eguren, patiently guided me through the very first steps of my research career. My PhD advisor at Stanford University, Prof. Richard M. Swanson, further nurtured my love for semiconductor devices through his vivid and insightful explanations of device physics. I owe an immense debt of gratitude to all three of them for their confidence in me and their encouragement over the years.

This book is dedicated to my sons, Diego, Paul, and Antoine. Writing it took a lot of time away from them. I hope they come to value the impact that this book can make on the careers of young people like themselves.

This project has been sustained by the patience and encouragement of Sophie, my wife. I am grateful for her presence in my life!

Jesús A. del Alamo

About the Author

Jesús A. del Alamo is Donner Professor and Professor of Electrical Engineering in the Department of Electrical Engineering and Computer Science at Massachusetts Institute of Technology. He is also Director of the Microsystems Technology Laboratories at MIT. He obtained a Telecommunications Engineer degree from the Universidad Politécnica de Madrid (Spain) and MS and PhD degrees in Electrical Engineering from Stanford University. From 1985 to 1988 he was with NTT LSI Laboratories in Atsugi (Japan). He joined MIT in 1988.

Over the years, Prof. del Alamo has been involved in research on transistors and other electronic devices in a variety of material systems. He has worked on Si solar cells, Si bipolar junction transistors, Si metal–oxide–semiconductor field-effect transistors (MOSFETs), SiGe heterostructure devices, GaAs pseudomorphic high electron mobility transistors (PHEMTs), InGaAs high electron mobility transistors (HEMTs) and MOSFETs, InGaSb HEMTs and MOSFETs, GaN HEMTs and MOSFETs, and more recently diamond MOSFETs. He has also investigated quantum-effect devices based on AlGaAs/GaAs heterostructures. His current research interests focus on the physics, technology, modeling and reliability of new III-V compound semiconductor field-effect transistors for future logic applications. He is also interested in fundamental reliability physics of GaN transistors for RF power amplification and power switching applications. In addition, Prof. del Alamo investigated the technology and pedagogy of online laboratories for science and engineering education (the iLab Project).

Prof. del Alamo's students have earned numerous best paper awards at national and international conferences. A paper from his group presented at the 2008 International Electron Devices Meeting (IEDM) was selected at the 60th Anniversary IEDM in 2014 as *One of the Ten Most Influential IEDM Papers of the Last Decade*. For his research on InGaAs quantum-well field-effect transistors, he was awarded the 2012 Intel Outstanding Researcher Award in Emerging Research Devices and the Semiconductor Research Corporation 2012 Technical Excellence Award.

Prof. del Alamo teaches undergraduate and graduate-level courses at MIT in electronics, electron devices and circuits, and advanced semiconductor device physics. He has received multiple teaching and achievement awards at MIT: the 1992 Baker Memorial Award for Excellence in Undergraduate Teaching, the 1993 H. E. Edgerton Junior Faculty Achievement Award, the 2001 Louis D. Smullin Award for Excellence in Teaching, and the 2002 Amar Bose Award for Excellence in Teaching. In 2003, he was named a MacVicar Faculty Fellow and in 2007, he was appointed Donner Professor. He has also been a recipient of the Class of 1960 Innovation in Education Award in three separate occasions. In 2012, Prof. del Alamo was awarded the IEEE Electron Devices Society Education Award "for pioneering contributions to the development of online laboratories for microelectronics education on a worldwide scale."

From 1991 to 1996, Prof. del Alamo was a National Science Foundation Presidential Young Investigator. He is also Corresponding Member of the Royal Spanish Academy of Engineering, Fellow of the IEEE and Fellow of the American Physical Society. In 2015, Universidad Politécnica de Madrid granted him a Doctor *Honoris Causa*.

Chapter 1

Electrons, Photons, and Phonons

Quantum mechanics, statistical mechanics, and solid-state physics constitute the fundamental underpinnings of modern semiconductor device physics. In-depth understanding of the detailed processes lurking inside microelectronic devices demands a firm foundation in these disciplines. Fortunately for microelectronics engineers, it is possible to go quite far in the analysis and design of devices armed solely with an intuitive understanding of a few basic results. The goal of this chapter is to present selected basic concepts in an intuitive and mathematically nonrigorous fashion.

In this chapter, we will concentrate our interest on the nature of the electron and its behavior in a crystalline semiconductor. Additionally, we will introduce two more "particles": the *photon* as a quantum of light and the *phonon* as a quantum of vibrational energy of the semiconductor lattice. These particles play a decisive role in semiconductor device physics.

The brief summary presented in this chapter obviously cannot do justice to the depth and breadth of these disciplines. In most educational institutions, senior or graduate-level courses are offered to provide a working-level knowledge of these topics. Students who have already studied these subjects should be able to skip the corresponding sections of this chapter.

1.1 SELECTED CONCEPTS OF QUANTUM MECHANICS

The formulation of quantum mechanics at the beginning of the 20th century revolutionized the understanding of the world that existed at that time. The "new physics" would have profound implications on science and technology to this date. Classical long-standing problems such as blackbody radiation, to select one of interest to us in this chapter, were finally solved. Many of the semiconductor pioneers had superior command of quantum mechanics as well as solid-state physics and statistical mechanics. This is splendidly illustrated in Shockley's book *Electrons and Holes in Semiconductors*, first published in 1950 (for more details on this and other suggested readings, see Section 1.5 at the end of the Chapter).

1.1.1 The dual nature of the photon

There are multiple aspects to quantum mechanics. Many are very relevant to the way semiconductor devices operate; most, unfortunately, are not very intuitive. In fact, many of them contradict outright common human perceptions of the physical world.

Light is a good example. In the late 19th century, in spite of the great progress achieved in the understanding of electromagnetic radiation, a few puzzling observations about light could not be explained. There was a great deal of interest, for example, in the spectrum of light emitted by objects. It was observed that heated objects had a prominent light emission spectrum that depended on their temperature. After correcting for the reflectivity of the surface, it was found that there was a "universal" spectrum of light emission for all objects that received the name of *blackbody radiation*.

A blackbody is an ideal object that does not reflect any light at all. It emits light, however, with a spectrum that depends only on its temperature. In fact, an ideal blackbody appears all black to the human eye unless it is so hot that it emits light in the visible portion of the spectrum. Familiar examples of blackbody radiation are hot coals in a campfire or red-hot iron in a foundry.

Figure 1.1 sketches the spectrum of an ideal blackbody at 300 K. The figure depicts the spectral power density as a function of wavelength (the units of the vertical axis are W/m^3, that is, 1 W of power falls over a 1 m^2 of surface normal to the radiation per m of wavelength). In the visible range, there is very little energy being emitted by a blackbody at room temperature. In fact, the peak of the blackbody radiation at room temperature is in the infrared.[1] At the turn of the century, the peculiar spectral distribution of the radiated energy of a blackbody seen in Fig. 1.1, with a sharp threshold at short wavelengths and a long tail at high wavelengths, defied understanding. Both the shape of the curve and its evolution with temperature could not be explained by the classical electromagnetic theory of light.

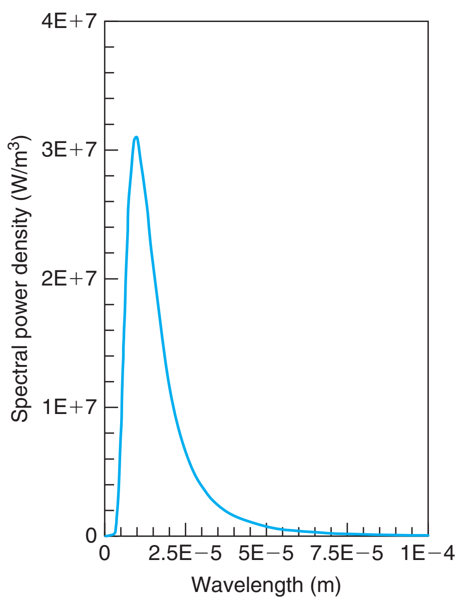

FIGURE 1.1
Blackbody light emission spectrum at 300 K.

[1]This has practical consequences. Infrared cameras create a temperature image that enables night vision. They are also used to measure temperature in integrated circuits.

A breakthrough in understanding blackbody radiation came from Planck in 1901. Planck postulated that light was emitted or absorbed by electromagnetic "oscillators" of an energy that is restricted to "quantized" values that are multiples of a fundamental energy amount:

$$E_{ph} = h\nu \tag{1.1}$$

where ν is the frequency of the light and h Planck constant (6.63×10^{-34} J · s, or 4.14×10^{-15} eV · s in units more common to microelectronics engineers; see Table I at end of the book). In Planck's hypothesis, if an oscillator lowers its energy from $nh\nu$ to $(n-1)h\nu$ (where n is an integer), it emits a quantum of electromagnetic radiation, or **photon**, of energy E_{ph} given by Eq. (1.1). Similarly, an oscillator vibrating with energy $nh\nu$ can absorb a photon of the same frequency in order to acquire a final energy $(n+1)h\nu$. Planck postulated light to be made out of tiny "clumps" that pack an amount of energy that is proportional to the frequency of the light. This revolutionary hypothesis represented a radically different way of thinking about light and became the key for the explanation of blackbody radiation. Planck's hypothesis explained, for the first time, the wavelength dependence of the blackbody radiation depicted in Fig. 1.1. The details are left to an introductory course on quantum mechanics.

EXERCISE 1.1

Find the relationship between the energy of a photon in electronvolts and its wavelength in μm.

Using Eq. (1.1), the relationship between wavelength and frequency of light, $\nu = \frac{c}{\lambda}$, and the values of the fundamental constants listed in Table I, we can easily find

$$E_{ph} = h\frac{c}{\lambda} = 4.14 \times 10^{-15}\ \text{eV} \cdot \text{s} \times \frac{3.00 \times 10^{10}\ \text{cm/s}}{\lambda\ (\text{cm})} = \frac{1.24 \times 10^{-4}}{\lambda\ (\text{cm})}\ \text{eV} = \frac{1.24 \times 10^{-4}}{\lambda\ (\mu\text{m}) \times 10^{-4}}\ \text{eV}$$
$$= \frac{1.24}{\lambda\ (\mu\text{m})}\ \text{eV} \tag{1.2}$$

This is a simple relationship that is handy to remember. For an energy of 1.1 eV (we will soon appreciate the significance of this particular number), the wavelength is 1.13 μm, which is in the infrared.

EXERCISE 1.2

Calculate the photon fluence (integrated flux) emitted by a GaAs laser that is delivering 1 mW *of light power at a wavelength of 0.85* μm.

This is another exercise where care is needed in the handling of units. The power delivered by a beam of light is equal to the photon fluence, F_{ph}, times the photon energy. Using the simple relationship derived in the previous exercise, a 0.85 μm photon has an energy of 1.46 eV. We can then calculate the fluence of the light beam in the following way:

$$F_{ph} = \frac{P}{E_{ph}} = \frac{1\ \text{mW}}{1.46\ \text{eV}} = \frac{10^{-3}\ \text{J/s}}{1.46\ \text{eV}} = \frac{10^{-3}\ \text{J/s}}{1.46\ \text{eV}} \frac{1}{1.60 \times 10^{-19}\ \text{J/eV}} = 4.28 \times 10^{15}\ \text{s}^{-1}$$

This result means that every second there are over 4,000 trillion photons with a wavelength of 0.85 μm being emitted by the laser.

Another important aspect of quantum mechanics is the *particle–wave duality*. In quantum mechanics, there is no distinction between a wave and a particle. Waves can behave like particles and vice versa. Light is again a good example.[2] Under the right conditions, electrons can be ejected from a metal by light impinging on its surface, a process called the *photoelectric* effect. When this experiment was first carried out, it was observed that the kinetic energy of the photogenerated electrons, or simply photoelectrons, was independent of the light intensity but exhibited a peculiar behavior as a function of the frequency of the incoming light. This is sketched in Fig. 1.2. There was a threshold frequency, ν_c, characteristic of each metal, below which no photoelectron was emitted. Beyond this threshold, the electron kinetic energy was found to increase with the frequency of the light.

The classical explanation was that for an electron to escape from the metal, it had to acquire enough energy to overcome the binding forces. This energy is called the *work function* of the metal, W_M. Since an electromagnetic wave has an energy density associated with it, it is reasonable to think that exposing the metal to the radiation for sufficient time would allow the metal to accumulate enough energy so as to ultimately eject an electron. In the classical picture, this should happen regardless of the frequency of the light.

An explanation based on the quantum nature of light was advanced by Einstein in 1905. Einstein postulated that the energy of an individual incoming photon is given to a single electron in a two-body collision. If the energy of the photon is lower than the metal work

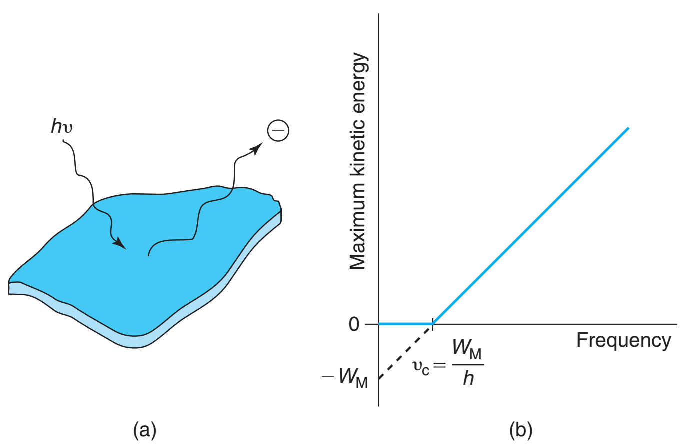

FIGURE 1.2
(a) A beam of light impinging on a metal can provoke the emission of electrons from it (the photoelectric effect). (b) Sketch of kinetic energy of electrons ejected from a metal surface as a function of the frequency of impinging light.

[2]Interestingly, at the time of Newton, a particle view of light prevailed. It was Huygens that demonstrated the wave nature of light. In some way, quantum mechanics brought together the particle and wave views of light.

function, the electron does not acquire enough energy to escape from the metal. If, on the other hand, the energy of the photon is higher than W_M, the electron is allowed to escape and the excess energy over W_M is provided to the electron as kinetic energy. Following this explanation, the threshold frequency for the light to eject an electron can be found using Eq. (1.1) to be $\nu_c = W_M/h$. The extrapolation of the electron kinetic energy to zero frequency gives W_M. In extensive experiments that followed Einstein's hypothesis, Millikan established the correctness of Einstein's predictions. As a by-product of these experiments, the work functions of many metals were measured.

At the core of Einstein's hypothesis is the exchange of energy between one electron and one photon. Since this treats the photon as a well-defined particle, it was an unprecedented concept at the time. The most direct evidence of the particle nature of electromagnetic radiation was provided by Compton, who studied the scattering of X-rays through a thin graphite foil (X-rays are simply high-energy photons). The *Compton effect*, described in detail in many quantum mechanical textbooks, could only be explained by invoking billiard-ball type interactions between photons and electrons, highlighting one more time the particle nature of light.

1.1.2 The dual nature of the electron

In 1925, inspired by the dual wave–particle nature of the photon, de Broglie postulated (in his PhD thesis) that particles also exhibit wave properties. He formulated an expression for the wavelength associated with a particle that stated:

$$\lambda_B = \frac{h}{p} \tag{1.3}$$

where p is the momentum of the particle and λ_B is known as the *de Broglie wavelength*, a cornerstone of modern quantum theory.

De Broglie hypothesized that the wave nature of a particle is only relevant when its de Broglie wavelength is comparable to the physical dimensions of its environment. If the wavelength is much smaller than the dimensions of the physical environment, classical particle behavior provides an adequate description of the relevant physics. When the wavelength is comparable or much larger than the environment, the wave nature of the particles becomes relevant and wave-like phenomena such as diffraction and interference should be observed.

In search of a confirmation of de Broglie's predictions, in 1927, Davisson and Germer studied diffraction of low-energy electrons from a crystal of Ni. They obtained diffraction patterns with a shape that could not be explained by classical physics in which electrons are considered to be particles. In fact, a peak in the diffraction pattern was observed precisely when the momentum of the incoming electrons had an associated de Broglie wavelength equal to the interatomic spacing of Ni. This constituted an experimental proof of the wave nature of the electron.

EXERCISE 1.3

Calculate the de Broglie wavelength of a free electron in vacuum with a kinetic energy of 26 meV.

A relationship between the kinetic energy of an electron and its momentum is obtained through (relativistic effect neglected):

$$E_K = \frac{1}{2}m_o v^2 = \frac{p^2}{2m_o} \tag{1.4}$$

If we solve for p and plug into Eq. (1.3), we get a relationship between the de Broglie wavelength and the kinetic energy:

$$\lambda_B = \frac{h}{\sqrt{2m_o E_K}} \tag{1.5}$$

Inserting numbers into this

$$\lambda_B = \frac{4.14 \times 10^{-15}\ \text{eV} \cdot \text{s}}{\sqrt{2 \times 5.69 \times 10^{-16}\ \text{eV} \cdot \text{s}^2/\text{cm}^2 \times 0.026\ \text{eV}}} = 7.61 \times 10^{-7}\ \text{cm} \simeq 7.6\ \text{nm}$$

The experiment of Davisson and Germer is particularly relevant for semiconductor engineers because, as we will see below, typical semiconductors are solids with interatomic distances of the order of a fraction of a nanometer. As Exercise 1.3 above showed, these dimensions are substantially smaller than the de Broglie wavelength of electrons with energies normally encountered in the operation of semiconductor devices. The quantum nature of the electron therefore plays a dominant role in many phenomena associated with semiconductors.

The wave nature of particles presents difficulties when trying to precisely pinpoint the position of any individual particle. The particle is somehow spread out in space. Quantum mechanics therefore only talks in probabilistic terms. Although in the quantum regime there is a sense of uncertainty about the precise behavior of individual particles, the properties of interest in semiconductor engineering involve the average behavior of a large population of electrons. The probabilistic language of quantum mechanics is perfectly suited to this task.

Quantum mechanics handles the wave nature of particles by defining a *wave function* whose modulus square is the probability of finding the particle at any one point in space at any one time. Knowing the wave function of an electron in a given system is the key to understanding the electronic properties of the system in question. In 1926, Schrödinger postulated an equation to describe the spatial distribution and the dynamics of the wave function. Schrödinger equation was not deduced from first principles—he hypothesized it. Seventy years of testing against experiments have amply established the validity and usefulness of this equation (for nonrelativistic physics). One of the greatest successes of the Schrödinger equation has been in its contribution to the understanding of the physics of solids. Its impact in this field has been far-reaching. Progress in semiconductor physics and technology has brought about the electronics and communications revolutions that continue on after more than five decades.

Since we will not use the Schrödinger equation in this book, it is not useful to reproduce it here. The reader can find it in all books on quantum mechanics. It is, however, necessary to appreciate a few implications of the Schrödinger equation, which are discussed in the following section. Before closing this section, it is important to note that there is still an advanced quantum mechanical concept, *tunneling*, which is of great relevance to modern devices. Tunneling describes the finite probability that particles penetrate into regions that are classically forbidden to them. Tunneling is a manifestation of quantum mechanics that is pervasive in semiconductor devices. In fact, as we discuss in this book, tunneling is at the heart of most ohmic contacts and it is also responsible for certain breakdown phenomena and parasitic currents in devices. Tunneling is also the basis of some important devices, such as EEPROM (Electrically Erasable Programmable Read-Only Memory) and Zener diodes. In spite of its importance, the eminently quantum-mechanical nature of tunneling prevents us from treating it rigorously in this book.

1.1.3 Electrons in confined environments

A certain class of electronic systems is of great interest to us. These are *bound systems* in which a field of forces would classically restrict the electron movement to a region of space. A good example of such a system is an atom (Fig. 1.3). In an atom, the Coulombic attraction between the electrons and the protons in the nucleus effectively confines the electrons to the vicinity of the nucleus, much as the Moon is restricted to rotate around the Earth due to their mutual gravitational attraction. Figure 1.3(a) sketches the Coulombic potential of an atom. This is an energy representation of the electrostatic field of forces that acts between electrons and the nucleus. As the electron moves within the sphere of influence of the nucleus, it trades back and forth potential energy E_P for kinetic energy E_K.

The wave nature of the electron imposes further restrictions to its movement. A string in a guitar, for example, can only oscillate in modes or frequencies that depend on the length, linear density, and tension of the string. In a similar way, the confinement of the electron around a nucleus results in only a discrete set of allowed "orbits" (using a particle view), each one characterized by a certain energy. An energy representation of the situation is shown in Fig. 1.3(b) for a Si atom. This figure sketches the various levels allowed for electrons when bound to it. They are obtained from a solution of the Schrödinger equation with the appropriate potential for Si.

The allowed orbits of electrons around an atom are classified using a set of characteristic *quantum numbers*. These numbers arise from solving the Schrödinger equation. They are handy

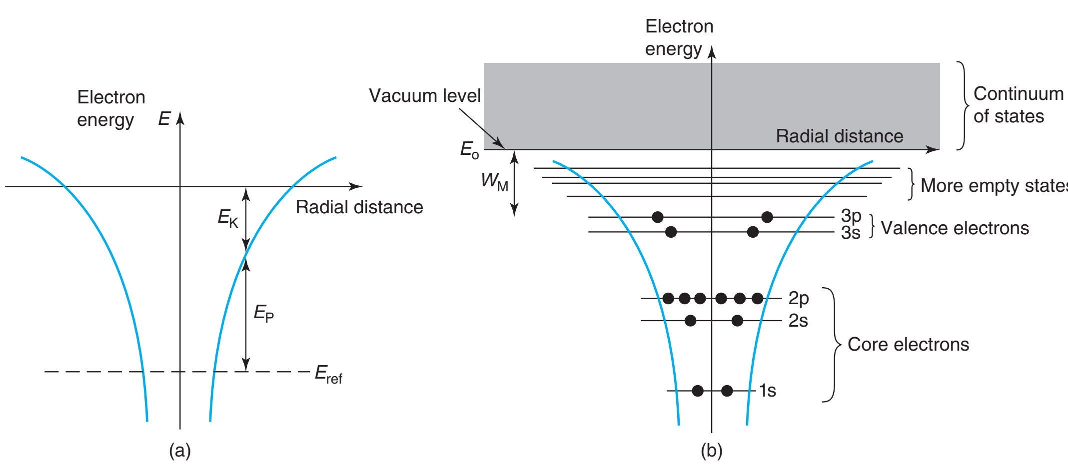

FIGURE 1.3
(a) Sketch of the Coulombic potential of an atom. As an electron wanders around the nucleus, it trades potential energy for kinetic energy. (b) Electron energy levels of a Si atom.

indices to catalog the various "shells," "subshells," "orbitals," and "spin" of the various *quantum states* that the electrons can occupy. The number of electrons of an atom and their distribution among the various quantum states determines to a great extent the electrical and chemical properties of the atom. It also greatly impacts on the electronic behavior of solids made out of these atoms.

The arrangement of the electrons in the quantum states of the atom is restricted by the *Pauli exclusion principle*, postulated by Pauli in 1925. According to this principle, no two electrons in a given quantum system can occupy the same quantum state. Without this postulate, most experimental observations, such as the shape of the blackbody spectrum, cannot be explained. In fact, without Pauli exclusion principle, most microelectronics phenomena of interest to us cannot be understood.

The Pauli exclusion principle implies that the electron distribution in a system is found by filling quantum states with one electron each starting from the lowest energy. This is shown for the 14 electrons of the Si atom in Fig. 1.3(b). The lowest energy shell, labeled 1s, contains two quantum states and therefore can hold two electrons with opposite spin. The next shell contains two subshells at different energies. The 2s subshell can hold two electrons, while the 2p subshell can accommodate six with different quantum numbers. These two subshells therefore become completely full. At this point, there are four electrons left. We proceed to the next shell. Two electrons go to the 3s subshell, which becomes full and two more to the 3p subshell. This can hold as many as six electrons so it is only partially filled. This fact has important consequences when Si atoms bond together to form a solid, as we will see. Above these subshells, there are other shells and subshells that are empty.

It is important to have a sense of how tightly bound the electrons are in a Si atom. Imagine for a moment that we try to remove one electron from a Si atom by hitting it with energetic photons, such as X-rays.[3] It takes at least 1839 eV to free up a 1s electron. Any excess energy above this value will provide extra kinetic energy to the released electron. At least 150 and 99 eV are needed to free a 2s or a 2p electron, respectively. In contrast, it takes only 8.2 eV to free an electron from the four sp^3 uppermost orbitals. Clearly, the four electrons in the outer shell are far more weakly bound to the atom than the remaining ten. These four *valence electrons*, as they are known, play a key role in the chemical reactions of Si and in the formation of bonds in a Si crystal. On the other hand, the 10 *core electrons* are much harder to dislodge and remain, in all situations that we encounter in microelectronics, tightly bound to the Si nucleus.

An electron that has just enough energy to overcome the attraction of the nucleus is represented at an energy E_o, the *vacuum level*, as shown in Fig. 1.3(b). An electron with this energy is at rest infinitely far away from the atom. Most likely, the process that frees the electron also gives it some extra kinetic energy. This electron can move around and it is represented in a "continuum" of quantum states at energies above E_o, as represented in Fig. 1.3(b). The meaning of the work function, first introduced in Section 1.1.1, now becomes clear: it is the minimum energy required to free the valence electrons, that is, the energy difference between E_o and the highest electron energy.

Before closing this section, let us note that the crucial role valence electrons play in the properties of atoms is recognized in the very organization of the periodic table. The vertical columns

[3]The technique of using X-rays to measure the electron binding energies of atoms is known as X-ray photoelectron spectroscopy, or XPS. Since the electron binding energies are peculiar to each element, XPS is widely used in the electronics industry to study the composition of materials.

IIB	IIIA	IVA	VA	VIA
	5 B	6 C	7 N	8 O
	13 Al	14 Si	15 P	16 S
30 Zn	31 Ga	32 Ge	33 As	34 Se
48 Cd	49 In	50 Sn	51 Sb	52 Te

FIGURE 1.4
Subsection of the periodic table showing the most common elements constitutive of semiconductors. The atomic number is indicated at the upper right of each chemical symbol.

of the table contain atoms with the same number of valence electrons. Figure 1.4, for example, shows a small section of the periodic table of particular interest to us as it contains the key elements constitutive of common semiconductors. The column number (in roman numbers) indicates the number of valence electrons in each atom in that column. Si belongs to column IV, with Ge and C. As one proceeds down the rows of the periodic table, the atomic number (at the upper right hand of each symbol in Fig. 1.4) increases and more shells become completed. For example, C has shell 1 completely full and shell 2 only partially full. In the case of Si, as we just saw, shells 1 and 2 are full, while shell 3 is partially full. Down the table, Ge has shells 1, 2, and 3 full, while shell 4 is partially full, and so on. We will refer back to this figure several times later in this book.

1.2 SELECTED CONCEPTS OF STATISTICAL MECHANICS

In the operation of semiconductor devices, a very large number of electrons are typically involved. In devices, we are interested in the collective behavior of the electron population rather than the details of any individual electron. Statistical mechanics is a discipline concerned with large systems of indistinguishable particles. Statistical mechanics describes the macroscopic properties of a whole system in a probabilistic fashion without inquiring about the detailed behavior of any one particle at any one time.

1.2.1 Thermal motion and thermal energy

It is a common observation that at finite temperatures, nothing in the world is really standing still. Molecules in a gas or a liquid, atoms in a solid, and electrons in a transistor are all jiggling around. This is called *Brownian motion* in honor of the botanist Robert Brown who discovered it in 1827. At a finite temperature, particles in an ensemble have a certain average kinetic energy.

Some have more, and some have less. There is a certain kinetic energy distribution around this average. While it is not possible to measure the kinetic energy of a given particle, it is not difficult to measure the average kinetic energy of an ensemble of particles. We can, for example, confine a known number of gas molecules at a given temperature in a given volume and measure the pressure that they exert on the walls of the vessel. For a gas under ideal conditions, if we do this we find that the mean value of the kinetic energy of a particle is related only to temperature and not to the nature of the particle such as its mass. This fact is used to *define* temperature itself. There is some arbitrariness in this definition and that is why there are different temperature scales. In science and engineering, the following convention for the average kinetic energy of a molecule of an ideal gas in thermal equilibrium (we define this concept below) is widely used:

$$< E_{\mathrm{K}} >= \frac{3}{2}kT \tag{1.6}$$

where k is the Boltzmann constant and T the temperature in degrees Kelvin (the temperature in degrees centigrade is equal to the value in Kelvin minus 273.15). The value of the Boltzmann constant is $k = 1.38 \times 10^{-23}$ J/K or, with energies measured in electron volts, $k = 8.62 \times 10^{-5}$ eV/K. Equation (1.6) applies to an ideal ensemble of indistinguishable classical particles with three degrees of freedom. It makes a statement about the *average* kinetic energy of a particle. There are certainly particles with higher and lower kinetic energies than this average.[4] For a classical particle at "room temperature" (defined in this book as 300 K or 27 °C), we find, using Eq. (1.6), that its average kinetic energy is 39 meV.

The value $\frac{3}{2}kT$ is an important energy reference. Processes that occur with characteristic energies smaller than $\frac{3}{2}kT$ are unlikely to be distinguishable. Processes that involve a well-defined energy exchange still will appear "blurred" in energy by an amount given by the thermal energy. This is the case, for example, for the energy spectrum of light emission in light-emitting diodes (LEDs). Since the product kT alone (without the $\frac{3}{2}$ factor) shows up in many equations in a great variety of physical processes, it is often used as an order-of-magnitude estimate of the average kinetic energy of a particle. kT is called the *thermal energy* and its value at room temperature is 26 meV, about 1/40th of an electronvolt.

1.2.2 Thermal equilibrium

Pervasive in our discussion above, but not explicitly mentioned, is the concept of *thermal equilibrium*. Let us define this concept. A system of particles is in thermal equilibrium if it fulfills two conditions: first, it does not exchange energy with the outside world, and second, it is in steady state. Let us examine these conditions in more detail.

When we picture an ideal gas confined in a vessel at a certain temperature, such as sketched in Fig. 1.5, we assume that no energy can be lost from or added to the gas ensemble through the walls of the container. This means that the collisions that the gas particles experience against the walls of the vessel are perfectly "elastic," so no energy is lost or added as a result of them. Also, external energy, such as light, cannot penetrate to the interior of the vessel. This is called a *closed system* and it is a necessary but not sufficient condition for thermal equilibrium.

[4] The factor of 3/2 in Eq. (1.6) was introduced in the theory of ideal gases as a matter of convenience.

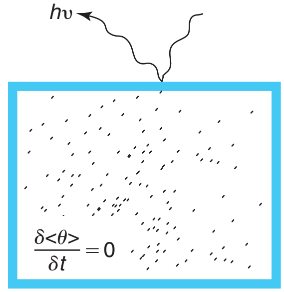

FIGURE 1.5
Illustration of a system of particles in thermal equilibrium. The system is "closed" and all time derivatives of all ensemble averages (θ in the figure) are zero.

An additional requirement for a situation to be characterized as thermal equilibrium, as Fig. 1.5 illustrates, is that all time derivatives of all ensemble averages, global and local, are zero. That is, the system is in *steady state*. It is possible for a system to be closed and yet not to be in thermal equilibrium. Suppose that at $t = 0^-$, we shine light onto a vessel containing gas, that is otherwise completely isolated from its surrounding, with an energy that is absorbed by the gas molecules. Assume furthermore that only a fraction of the vessel is exposed to the radiation. At $t = 0^+$, all energy exchanges with the outside world are stopped. In spite of this, for $t > 0^+$, the gas ensemble is not in thermal equilibrium. On average, in a region of the vessel, the gas molecules have higher kinetic energy than in the other. As time goes on, collisions between the molecules randomize the average kinetic energy. After some time, the average kinetic energy of the molecules is the same in all regions of the vessel. It is only then that we can properly refer to this situation as thermal equilibrium. The time that it takes for this condition to be established depends on the average collision time and the dimensions of the vessel.

The combination of a closed system and steady state is a very rigorous definition of thermal equilibrium.[5] When studying semiconductor devices, we will often discuss the thermal equilibrium condition in some level of detail. In strict thermal equilibrium, a device does not perform any useful function. However, this state is important because when disturbed by some outside influence, semiconductor devices react by trying to restore thermal equilibrium. If we wish to identify the bottlenecks to the reestablishment of thermal equilibrium and answer questions such as the time that it takes for equilibrium to be reached, we need to understand this state well. There are many examples in nature that illustrate this. For example, if we wish to know the frequency at which a string in a piano vibrates when struck by one of the hammers, clearly an event outside equilibrium, we must know the tension of the string. This is a parameter that applies to the string in equilibrium. A higher tension will result in a higher pitch tone.

In semiconductor devices, we will frequently encounter situations that we will denote as *quasi-equilibrium*. This is a concept that is slightly harder to understand but enormously important. Basically, in many circumstances, perturbing the device (by applying a voltage or shining light on it, for example) will not significantly modify some of the properties that we know well

[5] At times, we will refer to it simply as *equilibrium*, for short.

in equilibrium. This allows us to "stretch" results obtained in equilibrium and apply them to situations outside equilibrium.

An analogy might help to clarify the concept of quasi-equilibrium. Consider carrying out an experimental study of the physics of canoeing. An ideal place to conduct experiments is a perfectly still lake on a day without any wind. Under these conditions, one could evaluate, among other things, the movement of the water around the canoe and the energy conversion efficiency of the canoe/human/water system. Those ideal experimental conditions might be hard to get for a period of time long enough to complete the study. It is clear that one could also carry out many valid experiments on a gently flowing river or under light wind. If the water current or the wind speed is weak enough, one can surely make many useful observations about the canoe/water system. Conversely, whatever one might have learned about the canoe/water behavior in ideal conditions probably applies too to the less ideal situations of light wind or weak current. A perfectly still lake in the absence of wind can be considered a thermal equilibrium situation. A gently flowing river or a lake under a light breeze are situations outside thermal equilibrium. For many purposes, we can think of these though as thermal equilibrium situations. In contrast, if one were to study the movement of a canoe on a fast-moving river with rapids, or under strong wind that produce sizable waves, the behavior of the canoe is likely to be very different. These are clearly situations out of equilibrium that require special consideration.

The concept of quasi-equilibrium will be extensively used in this book in different contexts. Judiciously invoking quasi-equilibrium greatly simplifies the development of a wide range of analytical models for microelectronic device operation.

1.2.3 Electron statistics

In the discussion above on the thermal motion of gas molecules, no constraint was imposed on the motion of the molecules. We have already seen, however, that electrons are rarely entirely free. In any electronic system, there are only certain quantum states that electrons can occupy. Furthermore, electrons must follow the Pauli exclusion principle in occupying these states. These two restrictions must be kept in mind when dealing with the thermal properties of electrons.

Consider a confined gas of electrons in thermal equilibrium and assume that collisions among electrons are very rare, that is, the electron gas is very dilute. Now let us inquire about the energy distribution of these electrons at absolute zero temperature. The confined system in question will have a certain number of allowed quantum states. Classically, at absolute zero temperature, one would place the entire supply of electrons at the lowest energy available. Quantum mechanically, however, electrons obey Pauli exclusion principle and only one electron can occupy each quantum state. Therefore, the electron energy distribution at 0 K is obtained by filling the available quantum states with electrons starting with the lowest energy and proceeding up in energy until all electrons are placed. The energy of the top-most filled state in the distribution is called the *Fermi energy* or *Fermi level* E_F.

We can visualize the meaning of the Fermi level by looking at the example depicted in Fig. 1.6. Here we have an electronic system that consists of 21 electrons and an electron state distribution with energy in a staircase shape, as sketched in the figure. At 0 K, the lowest 21 states in energy

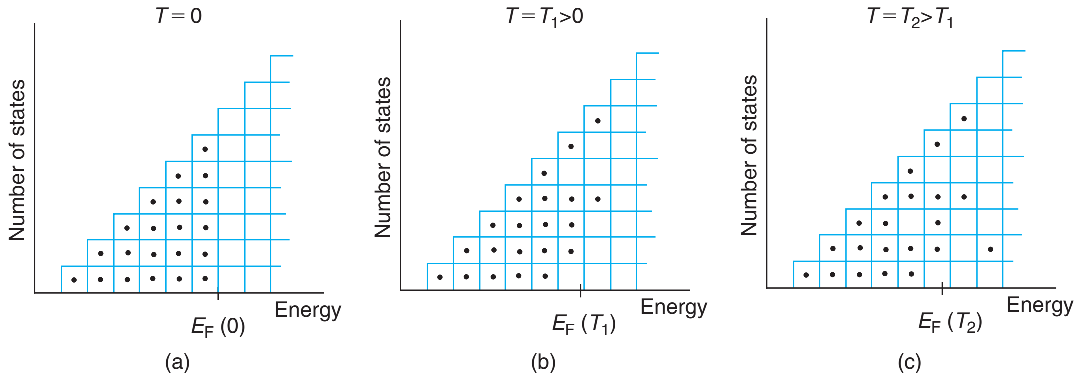

FIGURE 1.6
Example of a quantum state distribution and its electron occupation as a function of temperature. (a) Depicts the situation at $T = 0$ K, (b) and (c) at finite temperature T_1 and T_2, respectively, with $T_2 > T_1$.

get filled [see Fig. 1.6(a)]. The energy at which the state occupancy goes from 1 to 0 (or 100% to 0%) is the location of the Fermi level E_F.

At finite temperatures, the electrons possess a finite amount of kinetic energy that increases with temperature. This implies that some states above the Fermi energy, which were empty at 0 K, are now occupied, while others below E_F, which were full at 0 K, now become empty. Statistical mechanics provides us with an expression that allows us to compute the probability that a certain state is occupied in thermal equilibrium. This is called the *Fermi–Dirac distribution function*:

$$f(E) = \frac{1}{1 + \exp \dfrac{E - E_F}{kT}} \tag{1.7}$$

which is depicted in Fig. 1.7.

The Fermi–Dirac distribution function is a fundamental result of statistical mechanics. It has several interesting properties. First, for states with energies far below E_F, the probability of finding them occupied by electrons approaches unity asymptotically as the energy is reduced: deep down, all states are occupied. For states with energies well above E_F, on the other hand, the occupation probability becomes negligible: way up in energy, the states are empty. At $E = E_F$, the occupation probability is exactly 1/2. This is actually a good definition of the Fermi energy at finite temperatures.[6] A second property of the Fermi–Dirac distribution function is its symmetry. You can easily verify that $1 - f(E)$ has a mirror shape to $f(E)$ around $E = E_F$. This means that

[6]In statistical mechanics, a distinction is made between the Fermi energy, which is only defined at $T = 0$ K in the way we did above, and the *chemical potential*, defined at any temperature as the energy at which the occupation probability is 1/2. In the field of semiconductors, this distinction is not usually made and we use the term “Fermi energy” at any temperature. Furthermore, in statistical mechanics, an *electrochemical potential* is also defined in situations in which an electric field is present and a potential energy due to the field is added to every particle. This does not modify the average kinetic energy of an electron and is equivalent to a local shift of the energy distribution. Because of this, this distinction is also typically disregarded in the field of semiconductors.

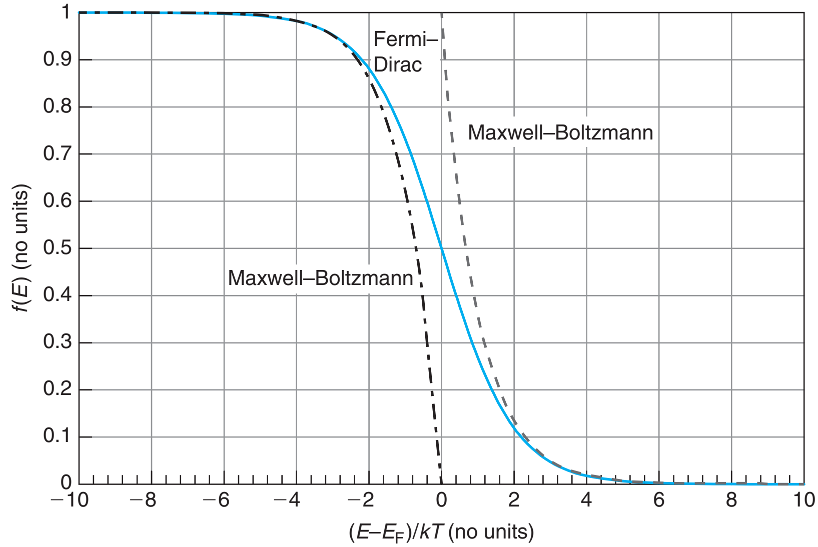

FIGURE 1.7
Fermi–Dirac distribution function. Approximate Maxwell–Boltzmann distribution functions are also shown.

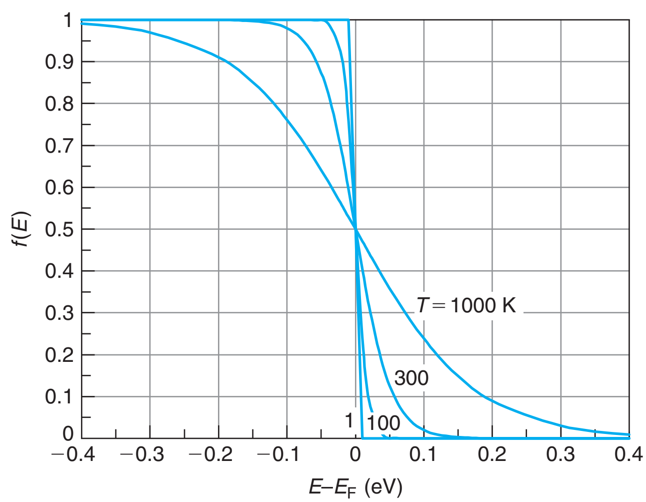

FIGURE 1.8
Fermi–Dirac distribution function for various temperatures.

the probability of finding a state at a certain energy *above* the Fermi level occupied is identical to the probability of having another state at the same energy *below* E_F empty.

A third property to remark about the Fermi–Dirac distribution function is that the transition of $f(E)$ from 1 to 0 around E_F is sharper at lower temperatures. This is better seen in Fig. 1.8

where the Fermi–Dirac distribution function is graphed for various temperatures. At $T = 0$ K, $f(E)$ abruptly jumps from 1 to 0 at E_F. As T increases above 0 K, the transition becomes softer. The width of the transition region is about $3kT$ for a 20% criterion (that is, the energy difference between $f(E) = 0.8$ and 0.2). This is a handy order of magnitude to keep in mind.

EXERCISE 1.4

Calculate the probability that a state at 0.1 eV above the Fermi level is occupied: (a) at 300 K *and (b) at 1200* K.

Use Fermi–Dirac distribution function for both parts. For part (a):

$$f(E) = \frac{1}{1 + \exp \dfrac{E - E_F}{kT}} = \frac{1}{1 + \exp \dfrac{0.1}{0.0259}} = 0.0206$$

This is a very small probability because the state is several kT's above E_F at room temperature.

To do part (b), we must first compute kT at 1200 K:

$$kT|_{1200\ \text{K}} = 4kT|_{300\ \text{K}} = 4 \times 0.0259 = 0.104\ \text{eV}$$

We now compute the probability of occupation as above:

$$f(E) = \frac{1}{1 + \exp \dfrac{0.1}{0.104}} = 0.28$$

The probability of occupation is now substantially higher because in the scale of kT, the state in question is relatively close to E_F.

The properties of the Fermi–Dirac distribution function imply that the Fermi energy in most electronic systems will be a function of temperature. We can visualize the impact of finite temperatures on the electron energy distribution by looking again at the example shown in Fig. 1.6. For a system with such few states and electrons, the statistical fluctuations among different possible configurations are significant. The sketches of Fig. 1.6(b) and (c) represent just one of the possible configurations that can be attained. Note that, at finite temperatures, in contrast with the situation at 0 K depicted in Fig. 1.6(a), a few states below the Fermi energy are empty, while others above are occupied. The Fermi energy is located where the probability of occupation is $\frac{1}{2}$. Since in this particular example the number of states increases with E, the Fermi level moves to lower energies as the temperature increases, as can be seen by comparing Fig. 1.6(a–c).

It is interesting to examine the form of the Fermi–Dirac distribution function at energies well above E_F. In this case, the exponential in the denominator is much larger than 1 and Eq. (1.7) can be approximated by

$$f(E) \simeq \exp\left(-\frac{E - E_F}{kT}\right) \qquad E - E_F \gg kT \tag{1.8}$$

This is called the *Maxwell–Boltzmann distribution function* (also graphed in Fig. 1.7). This approximation tells us that at energies much higher than the Fermi energy, the probability of finding a state occupied by an electron decreases exponentially with a characteristic energy kT. The Maxwell–Boltzmann distribution function is an excellent approximation to the Fermi–Dirac distribution function for energies a few kTs above E_F. For example, for $E - E_F = 2kT$, the error is 13%. If $E - E_F = 3kT$, the error drops to 5%.

Similarly, we can also examine the probability that a given state is occupied at energies much lower than E_F. This can be approximated by

$$f(E) \simeq 1 - \exp\frac{E - E_F}{kT} \qquad E - E_F \ll -kT \tag{1.9}$$

which we can actually rewrite as

$$1 - f(E) \simeq \exp\frac{E - E_F}{kT} \qquad E - E_F \ll -kT \tag{1.10}$$

There is a parallel between Eqs. (1.10) and (1.8). Equation (1.8) describes the probability that a state at an energy substantially above the Fermi energy is full. Equation (1.10), on the other hand, gives the probability that a state located much below the Fermi energy is *empty*. In both cases, the probability decreases exponentially at energies away from the Fermi energy. Furthermore, the characteristic energy of the exponential function is kT in both cases. The symmetry of Eqs. (1.8) and (1.10) arises from the symmetry of the Fermi function around E_F, which was mentioned above. Mathematically, it is much easier to use Eqs. (1.8) and (1.10) than the exact Eq. (1.7). Because of this, these two approximations are extensively used in the study of semiconductor devices.

EXERCISE 1.5

Solve Exercise 1.4 again under the assumption of Maxwell–Boltzmann statistics. Compute the relative error when compared with the result using Fermi–Dirac statistics.

For part (a), Maxwell–Boltzmann statistics yields

$$f(E) \simeq \exp\left(-\frac{E - E_F}{kT}\right) = \exp\left(-\frac{0.1}{0.0259}\right) = 0.0210$$

The relative error is

$$\epsilon = 100 \times \frac{0.21 - 0.206}{0.206} = 1.9\%$$

The error is small because the distance between the state in question and the Fermi level (0.1 eV) is about $4kT$.

For part (b), again using Maxwell–Boltzmann statistics results in

$$f(E) \simeq \exp\left(-\frac{0.1}{0.104}\right) = 0.38$$

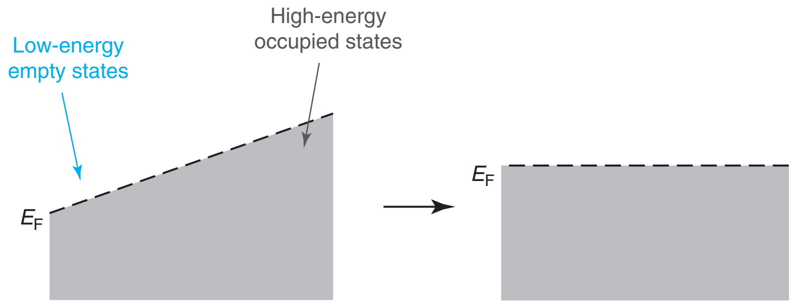

FIGURE 1.9
A system in thermal equilibrium is characterized by a Fermi energy that is constant everywhere (right). If that was not the case, as in the situation depicted on the left diagram, electrons would flow from a region where high-energy states are occupied to other regions where low-energy states are empty. In this way, the system attains an overall lower energy situation.

The relative error is now

$$\epsilon = 100 \times \frac{0.38 - 0.28}{0.28} = 36\%$$

This error is significant because the distance between the state and the Fermi level (0.1 eV) is just about equal to kT (0.105 eV).

In a final note before closing this section, the very definition of Fermi energy has an important and useful consequence. In an electronic system in thermal equilibrium, there is a unique Fermi energy that is *constant throughout the entire system*. In other words, if an electronic system is characterized by a Fermi energy that varies with location, then it is not in equilibrium. This applies regardless of the complexity of the system.

This makes intuitive sense. Consider a closed system with a Fermi energy that changes with position, as depicted on the left of Fig. 1.9. In this situation, there are regions where the Fermi energy is high and high-energy states are occupied. There are also regions where the Fermi energy is low, with empty states at low energies. The system can thus lower its overall energy by allowing electrons that occupy high-energy states to move to low-energy states that are empty. This net flow of electrons inside the system only stops when the Fermi energy becomes constant everywhere, as on the right of Fig. 1.9. This is the true equilibrium condition. The constancy of the Fermi energy in a system is a good working definition of thermal equilibrium itself.

1.3 SELECTED CONCEPTS OF SOLID-STATE PHYSICS

Most semiconductor materials used in modern microelectronics belong to the category of crystalline solids (important exceptions are amorphous semiconductors, such as amorphous silicon,

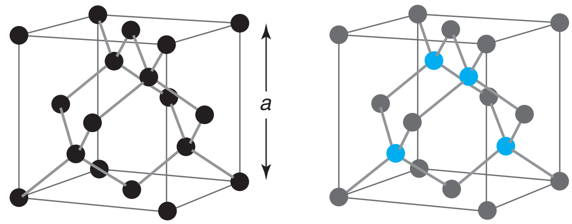

FIGURE 1.10
Sketch of unit cell of Si (left) and III–V compound semiconductor such as GaAs (right).

widely used in photovoltaics and flat-panel display applications). A perfect *crystalline solid* is a solid with an elemental atomic arrangement, or *unit cell*, that repeats itself *ad infinitum* in the three dimensions. Figure 1.10 shows a sketch of the atomic arrangement of the unit cell of the two most important semiconductors in microelectronics: Si and GaAs. For both of them, the unit cell has a cubic shape. Si is an *elemental semiconductor* since it is only made of a single kind of atom. GaAs is made out of two, Ga and As, and is thus referred to as a *compound semiconductor*. There are also compound semiconductors with more than two different kinds of atoms, such as InGaAs or InGaAsP. What is common to all these materials is a certain atomic arrangement that repeats itself in all directions in space in a perfect *lattice*. The atomic arrangement in the unit cells depicted in Fig. 1.10 appears rather complicated. In fact, a simple way to visualize it is to realize that it is made of two face-centered cubes ("fcc" or a cube with atoms at its vertices plus in the middle of all faces) with one cube displaced along the diagonal of the other a quarter of its diagonal dimension. For Si, both fcc cubes are made out of Si atoms. For GaAs, one is made out of Ga atoms and the other is made out of As atoms.

The lateral dimension of the unit cell is called the *lattice constant* (labeled *a* in Fig. 1.10). For Si and GaAs at room temperature, the lattice constant is 0.543 and 0.565 nm, respectively. Since, in general, a unit cell contains several atoms, the interatomic distance is typically smaller than the lattice constant. For Si and GaAs, the distance between neighboring atoms is 0.235 and 0.245 nm, respectively. This is an important length scale to keep in mind.

The number of atoms per unit volume, or *atomic density*, of a crystalline solid is an important reference scale for volume concentrations. For Si and GaAs, the atomic densities at room temperature are 5.0×10^{22} and 4.42×10^{22} cm^{-3}, respectively (these and other important physical parameters of Si and GaAs at room temperature are summarized in Appendix B).

The concept of a perfect crystalline solid may appear to be a pure academic fiction. In a real semiconductor crystal, there are going to be unavoidable disruptions of a perfect periodicity such as missing atoms (called *vacancies*) or atoms in wrong locations (for example, Si atoms in between lattice sites, called *interstitials*, or Ga atoms in place of As, called *Ga antisite defects*). Furthermore, foreign atoms are always present in a random way in real samples. In addition to this, at finite temperatures, the atoms in a semiconductor are not standing still but are randomly jiggling around, so that strictly speaking, we never have a perfect periodic structure. In addition, the very finite size of a device means that there are surfaces that break perfect atomic periodicity in a drastic way.

These disruptions to the ideal crystalline solid are in most cases just that, disruptions, relatively small perturbations to an otherwise very ideal structure.[7] For example, the spectacular engineering success of modern semiconductor technology is rooted in its ability of producing extremely pure and nearly perfect semiconductor crystals. Si wafers can now be fabricated with foreign atom concentrations below 1 part-per-million (ppm) or 1 every 10^6 Si atoms. Similar levels of crystalline defects can be attained. Actually, one does not need to claim such engineering prowess as a justification for the use of the "perfect" crystal concept. After all, many of the newer semiconductors have not reached the level of purity and crystalline perfection that Si enjoys. And even in Si, as Chapter 2 presents, selected foreign atoms at concentrations approaching 1 ppm to 1% are often deliberately introduced.

The ideal crystalline solid model remains a very useful view of a semiconductor because of the "effective size" of an electron in the lattice. A good estimate of this is the de Broglie wavelength that we introduced in Section 1.1.2. In vacuum and at room temperature, we estimated in Exercise 1.3 that λ_B is about 7.6 nm. This figure is not too different inside a semiconductor. Over how many lattice-sites is such an electron spread out? One can get an estimate for this by calculating the number of atoms in a sphere of 7.6 nm diameter. For typical atomic densities, this number is over 10,000 atoms. This means then that an electron in a semiconductor is really spread out over many atomic sites and it therefore "averages out" the relatively rare crystalline disruptions.

Hence, the ideal crystalline solid model is a good idealization of a modern semiconductor crystal. The unavoidable disruptions to the perfect periodic lattice are best captured in the form of "collisions" or "scattering" with the electrons, as we will see later in this chapter and in subsequent chapters.

1.3.1 Bonds and bands

A typical representation of a semiconductor lattice, such as that of Fig. 1.10, shows atoms connected by "sticks." The sticks represent bonds between atoms. In Si, GaAs, and most semiconductors, every atom is bonded to four neighbors. This number is particularly important. The top-most partially occupied subshell in the Si atom contains four empty states that can accommodate four electrons. In an elemental semiconductor such as Si, by binding with four neighbors, a Si atom shares one of its four valence electrons with each of four neighbors. On average, then, a Si atom ends up with a total of eight electrons in its valence shell. In atomic physics, this is known to be a particularly low-energy configuration. For instance, most noble gases, which are notoriously unreactive elements, have eight electrons in their outer layer. This type of bonding in which electrons are shared among neighboring atoms is called *covalent bonding*.

For a compound semiconductor such as GaAs, the situation is slightly more complex. Looking at the periodic table in Fig. 1.4, we see that Ga has three electrons in its outer layer, while As has five. In a solid in which each Ga atom is surrounded by four As atoms and each As atom is surrounded by four Ga atoms (see Fig. 1.10), each constituent atom shares on average a total of eight electrons in its outer layer. This, again, is a low-energy configuration that permits the formation of a stable crystal. In this case, however, in addition to covalent, the bonding has a small *ionic character* to it. This is because a Ga site becomes slightly negatively charged, while an

[7] A surface is not a small perturbation. Special care is needed in its treatment. This is dealt with in Chapter 8.

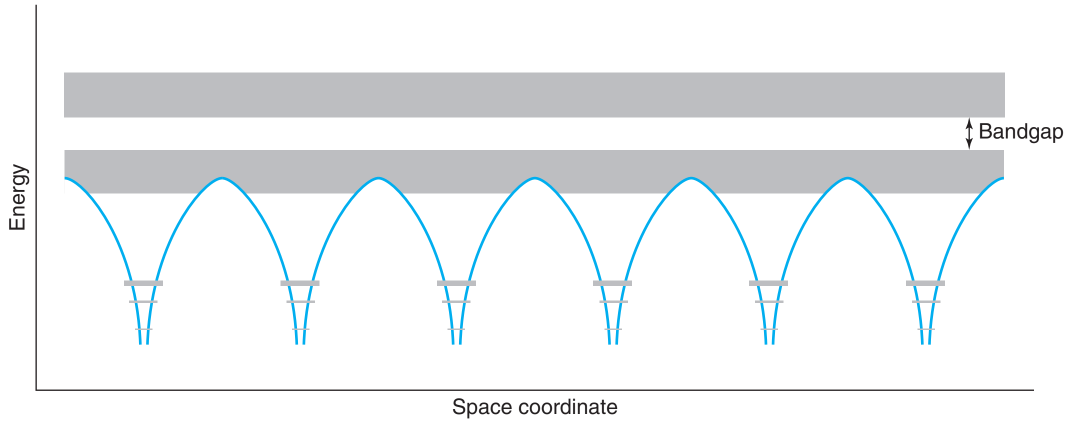

FIGURE 1.11
Sketch of bands and bandgaps in the electron state distribution of an idealized solid along a certain crystallographic direction. This energy picture can be viewed as an assembly of the Coulombic potentials of the individual atoms.

As site gets slightly positively charged as electrons are shared but protons are not. The resulting electrostatic attractions among these ions further helps to hold the crystal together.

A perfect crystalline solid is a very special electronic system. One can think of it as an entirely new system, but one can also view it as an ensemble of many smaller subsystems, the individual atoms. This mixed approach helps to understand the peculiar energy distribution of electron states in a solid that is sketched in Fig. 1.11. In a crystalline solid, the potential experienced by the electron is periodic in space. This potential largely results from the overlap of the individual potentials associated with each atom. It is also slightly affected by the electron distribution itself.

The core electrons of each atom are very tightly bound to each nucleus. Furthermore, their spatial extent is much smaller than the interatomic distance in the solid. As a result, the core electrons are for the most part unaware of the existence of the solid. The core states largely maintain their atomic character. The atomic bonding only results in a slight broadening in the energies that are allowed. For higher energies, as sketched in Fig. 1.11, the atomic level broadening becomes more significant as the atomic potentials overlap more.

For the valence electrons, the situation is very different. The spatial extent of their wavefunction is larger than the interatomic distance. When dealing with the valence electrons, it is not advantageous to think of a solid in terms of individual atoms: one should view the entire crystal instead as a whole new electronic system. In a crystalline solid, one cannot associate a valence electron with a particular pair of atoms. Electrons belong to the crystal as a whole; in a way, they are shared by all atoms. This is contrary to the core electrons that remain associated with each atom and are not shared. The quantum states for the valence electrons extend over the whole crystal, and the Pauli exclusion principle must be satisfied across all these states.

The defining characteristic of a crystalline solid is the perfect periodicity of the electrostatic potential that the electrons experience. This arises from the fact mentioned above that the unit cell repeats itself in space in three dimensions. Under these "periodic boundary conditions," as they are called, a fundamental result of solid-state physics states that the allowed electron states "cluster" in sharply defined *bands* leaving *bandgaps* of forbidden energies in between. This is shown in Fig. 1.11 by the continuous gray areas that extend over the entire crystal. Each band consists of many closely spaced but discrete electronic levels. For many purposes, they can be considered a continuum that can be described by a density of states.

1.3.2 Metals, insulators, and semiconductors

So far, we have been talking about crystalline solids in general. However, we know that there are materials with widely different electronic properties. Where do the differences among metals, semiconductors, and insulators, for example, arise from?

In order to provide a simple qualitative answer to this question, one must consider how the states in a solid get filled with the available electrons. As with an individual Si atom in Section 1.1.3, one starts by filling up the lower energy states and stops when there are no more electrons. Three different situations can arise, as sketched in Fig. 1.12.

First, let us consider a case in which a band ends up partially filled after all electrons are in place [Fig. 1.12(a)]. In this situation, there are empty states right above occupied states. As will become clearer in subsequent chapters, in these circumstances, electrons can easily move around the crystal in response to an electric field. The conductivity of materials with a partially filled band is high. These are *metals*. Metals exhibit a resistivity in the range 10^{-6} to 10^{-4} $\Omega \cdot$ cm. In a great analogy from Shockley, this situation resembles a partially filled parking lot where all cars are identical. The availability of empty slots allows any customer to move his or her car around the lot and choose the slot of his or her pleasing. In response to a stimulus, say heavy snow on the top exposed level, one expects many customers to move their cars to lower lying protected levels.

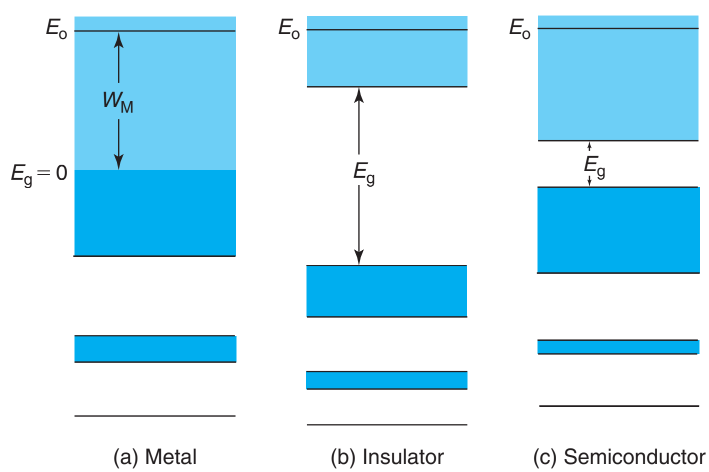

FIGURE 1.12
Sketch of band occupation for several solids: (a) metal, (b) insulator, and (c) semiconductor. Dark shading represents states occupied by electrons. Light shading represents empty states.

In a second case, all available electrons precisely fill a band. If a "wide" energy bandgap separates this full band from the next empty band, conduction is impossible because there are no available states into which electrons can easily move. This is an *insulator* and is depicted in Fig. 1.12(b). The resistivity of insulators is in the range 10^9–10^{17} $\Omega \cdot$ cm. In Shockley's parking lot analogy, a completely full lot permits cars to swap places, but since all cars are identical, one cannot observe any changes in the picture. Quantum mechanics does not allow us to tell electrons apart. If electrons merely trade locations, nothing has really changed.

We can also conceive a third situation in which one ends up with a completely full band that is separated from the next empty band by a "narrow" energy gap [Fig. 1.12(c)]. At zero absolute temperature, this situation is not different from the one just described and the material is perfectly insulating. At finite temperatures, however, the distinction between "wide" and "narrow" energy gap matters. If the bandgap is narrow enough, at finite temperatures, a few states of the upper band may actually be occupied by electrons, as implied by the Fermi–Dirac distribution discussed in Section 1.2.3. The electrons in the partially filled upper band can move around in response to a stimulus. In addition, there are empty states or *holes* (more about them are discussed in Chapter 2) in the lower band, which now allow electrons to move about. The net result is a moderate conductivity typical of a *semiconductor*. The resistivity of semiconductors is in the range 10^{-4}–10^4 $\Omega \cdot$ cm.

The parking lot analogy for semiconductors requires some elaboration. We should now consider two parking lots next to each other, one more expensive than the other because of some amenities, such as elevators, wider slots, or better lighting. If the price differential is too high, all customers flock to the cheaper one, which again gets full. With smarter pricing, some customers will be willing to pay the higher price for the up-scale lot and will park there. This leaves some slots open in the popular lot. In response to a snow storm, customers in both lots will be able to move their cars around since both have slots available.

It is clear from the above discussion that the difference between an insulator and a semiconductor is only qualitative. It really comes down to the width of the energy gap between the last filled band and the next empty band relative to the thermal energy. If the ratio between these two energies is high, very few electrons will reside in the upper band and the material will be highly insulating. If this ratio is low, substantial conductivity is expected, particularly as the temperature is raised. In fact, a pure semiconductor can behave as an insulator at low enough temperatures, while an insulator may have non-negligible conductivity at high temperatures. It is perhaps for this reason that semiconductors are named as they are. They never quite match the conductivity of metals, but their conductivity changes (and, as we will see very soon, can be engineered) over many orders of magnitude. Also in contrast with metals, the conductivity of semiconductors *increases* with temperature.

The bandgap that exists at 0 K between the last filled and the next empty band plays such a crucial role in semiconductor physics that it is often called the *fundamental energy gap* (in this book, we will refer to it simply as "the bandgap"). To calibrate our energy scale, most common semiconductors have bandgaps between 0.5 and 2.5 eV, while most common insulators have bandgaps of the order of 5–10 eV. The bandgaps of Si and GaAs at room temperature are 1.1 and 1.4 eV, respectively.

Figure 1.12 also indicates the *vacuum level* E_o. As in the case of the Si atom in Section 1.1.3, this is the energy at which electrons escape from the crystal. For metals, the energy view of Fig. 1.12 provides an intuitive picture for the photoelectric effect and the concept of work function introduced in Section 1.1.1. The work function is the minimum energy that needs to be provided to an electron in order for it to escape from the solid. In the energy view of a metal sketched in Fig. 1.12, this is clearly the energy difference between the vacuum level and the highest lying occupied state. In the photelectric effect, only photons with an energy equal or in excess of this difference would be capable of extracting electrons from the metal. For semiconductors and insulators, the concept of work function requires that we know more about the electron distribution in these materials. This is discussed in Chapter 7.

1.3.3 Density of states

The energy bands of a solid are composed of individual states that are located in energy very close to each other. At practical temperatures, the energy separation between single states is much smaller than kT and it is of no relevance to try to distinguish individual states. The total number of states present in a given window of energy is a much more important parameter. In this regard, it is useful to define a density of states, $g(E)$, such that $g(E)dE$ is the number of states in a unit volume of material that lie in a window of energy between E and $E + dE$, where dE is a small differential of energy. The units of $g(E)$ are $\mathrm{eV}^{-1} \cdot \mathrm{cm}^{-3}$.[8] In general, $g(E)$ depends on energy. A sketch of $g(E)$ versus E is shown in Fig. 1.13. This picture corresponds to the sketch in real space shown in Fig. 1.11.

In this figure, the core states are characterized by narrow bands with a delta-function density of states. States at higher energies give rise to bands with a finite density of state distribution in energy. The bandgap regions are defined by $g(E) = 0$. These regions are "forbidden" to the electrons because there are no states for them to reside in.

The concept of density of states is important. Many semiconductor physical parameters are profoundly affected by the density of states in the bands that surround the fundamental bandgap. We will see several examples of this in Chapter 2.

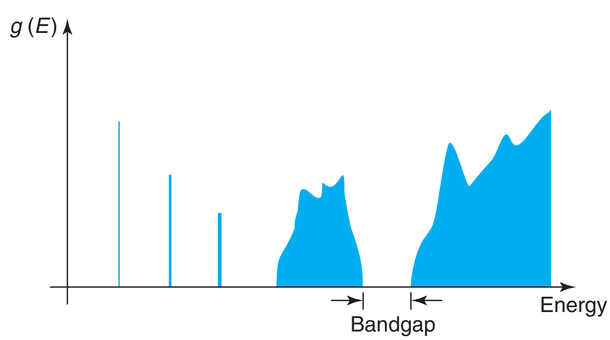

FIGURE 1.13
Sketch of density of states corresponding to electron state distribution of Fig. 1.11.

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[8]Many important properties of solids scale with the volume of the solid. Because of this, for certain variables, it is convenient to normalize the volume away and define them per unit volume. This will be done extensively in this book.

EXERCISE 1.6

Calculate the total number of states that lie at the bottom 26 meV *of a system with a density of states given by* $g(E) = 1 \times 10^{22}\sqrt{E}\ \mathrm{eV}^{-1} \cdot \mathrm{cm}^{-3}$.

The number of states that lie between energies E_1 and E_2 in a certain state distribution is simply given by the integral:

$$N(E_1, E_2) = \int_{E_1}^{E_2} g(E)dE$$

The units of $N(E_1, E_2)$ are cm^{-3}.

In the present example, if we denote as C the factor in front of $\sqrt{E}$, we must perform the integral:

$$N(0, E) = \int_0^E C\sqrt{E}\,dE = \frac{2}{3}CE^{3/2}\Big|_0^E = \frac{2}{3}CE^{3/2}$$

Putting numbers in this expression, we obtain

$$N(0, 0.026\ \mathrm{eV}) = \frac{2}{3} \times 1 \times 10^{22}\ \mathrm{eV}^{-3/2} \cdot \mathrm{cm}^{-3} \times (0.026\ \mathrm{eV})^{3/2} = 2.8 \times 10^{19}\ \mathrm{cm}^{-3}$$

1.3.4 Lattice vibrations: phonons

Atoms in a solid also possess thermal energy. At ordinary temperatures, the atoms that constitute a solid vibrate around their equilibrium positions in the lattice. In semiconductors at room temperature, the average amplitude of the vibration is of the order of a few hundredths of a nanometer. Since this is a sizable fraction of the interatomic distance, lattice vibrations affect the material properties of the semiconductor.

The lattice of a solid can be viewed as a network of atoms connected together by springs. At absolute zero temperature, the atoms will be fixed in space at the equilibrium points of these springs. At finite temperatures, they will be oscillating around these equilibrium positions. An individual atom cannot oscillate in complete ignorance of what its neighbors are doing. This is because the equilibrium position of an atom in the lattice is set by the balance between the attractive and repulsive Coulombic forces that it experiences with its neighbors. A solid is thus a large *coupled* system.

In a solid, the lattice can take many different vibrational modes. This is similar to the membrane of a drum or a string in a piano. Each mode is characterized by its wavelength and its mechanical energy. At any one time, a multiplicity of modes will be active with various energies. The resulting vibration pattern can be very complex.

The lattice can exchange energy with free electrons in the solid. Energy exchange can go two ways. An electron can give energy to the lattice or it can receive energy from it. If the electron gives some energy to the lattice, it must *excite* an available vibrational mode and provide the precise amount of energy required by that mode. When the lattice gives energy to the electron, a certain vibrational mode is extinguished and its mechanical energy is transferred to the electron.

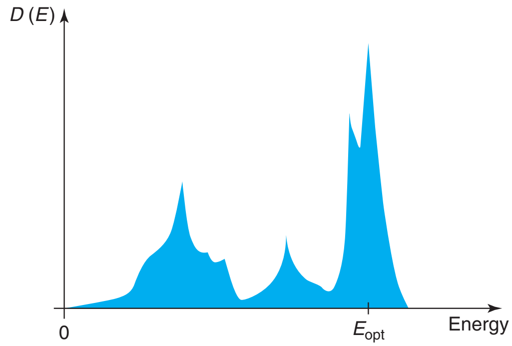


FIGURE 1.14
Sketch of the phonon spectrum of a typical semiconductor. $D(E)$ is the density of modes per unit energy per unit volume.

The quantized nature of the vibrational energy of the lattice makes it particularly convenient to visualize the vibrational modes as particles. They are called *phonons*. Energy exchanges between electrons and the lattice can be viewed as collisions between electrons and phonons. When an electron donates energy to the lattice, a phonon is “emitted.” In the reverse process, a phonon is “absorbed.”

The phonon energy spectrum, that is, the density of modes per unit energy per unit volume, plays an important role in many properties of semiconductors. Examples are their electrical and thermal resistivities. The phonon spectrum is determined by the nature of the atoms that make the solid and the bonding arrangement among them. A typical phonon spectrum for a semiconductor is sketched in Fig. 1.14. There are some aspects to it that we need to appreciate.

The most prominent feature of a semiconductor phonon spectrum is a sharp peak of phonon density beyond which no more phonons exist. These high-energy phonons are called *optical phonons* and the energy of the peak is denoted as optical phonon energy or E_{opt}. For lower energies, there is a relatively broad distribution of phonons that extends all the way down to zero energy. Sometimes this distribution has gaps in it. These low-energy phonons are called *acoustical phonons*. Comparatively, there are not many of such phonons.[9] E_{opt} is an important landmark in the energy scale of relevance to semiconductor devices. In Si, E_{opt} is 63 meV, while in GaAs, its value is 35 meV. We will explore the significance of these numbers later in this book (see, for example, Section AT4.2). At this time, let us note that E_{opt} is slightly larger than kT at room temperature but is much smaller than the bandgap of typical semiconductors.

It is interesting to compare phonons against photons and their interactions with electrons in a semiconductor. Phonons are particles with little energy and a large momentum. In contrast, photons are particles with very small momentum but relatively high energy. When an electron collides with a phonon, the electron momentum can change significantly but not its energy. In contrast, when an electron collides with a photon, the momentum of the electron is not affected but its energy can change substantially.

[9] *Optical phonons* are given such a name because their vibrational modes can be excited by the electric field of a light wave. In contrast, the vibrational modes of *acoustical phonons* cannot be excited by light.

In thermal equilibrium, there is no net energy exchange between the lattice and the electrons. On average, an equal number of phonons is emitted and is absorbed at every energy. In other words, the electron "gas" is in equilibrium with the lattice.

1.4 SUMMARY

- The electron has a dual particle–wave nature.
- The Pauli exclusion principle imposes restrictions to the dynamics of electrons. In an isolated atom, for example, certain electrons are tightly confined to the vicinity of the nucleus (core electrons), while others reside in much looser orbits (valence electrons).
- Concept of Fermi energy: at zero absolute temperature, a gas of electrons occupies all lowest energy available states up to the Fermi energy. At finite temperatures, the transition between occupied and empty states is gradual, with a softness described by a key characteristic energy, the thermal energy kT. The Fermi level is defined as the energy for which electron occupation probability is 0.5.
- A system of particles is in thermal equilibrium if it does not exchange energy with the outside world (it is closed) and if the time derivatives of all local and global ensemble averages are zero (it is in steady state). A system in thermal equilibrium has a Fermi level that is constant throughout.
- Electron occupation probability in thermal equilibrium is given by Fermi–Dirac distribution function:

$$f(E) = \frac{1}{1 + \exp \dfrac{E - E_F}{kT}}$$

  For $E \gg E_F$, or $E \ll E_F$, the Fermi–Dirac distribution function can be approximated by the Maxwell–Boltzmann distribution function:

$$f(E) \simeq \exp\left(-\frac{E - E_F}{kT}\right) \qquad E - E_F \gg kT$$

$$1 - f(E) \simeq \exp\frac{E - E_F}{kT} \qquad E - E_F \ll -kT$$

- A crystalline solid imposes periodic boundary conditions to the electrons. A consequence of this is that electron energy states cluster in bands leaving bandgaps forbidden to electrons.
- A semiconductor is a solid in which at 0 K, electrons completely fill a band leaving a bandgap between the highest energy filled state and the next available empty state. In addition, the width of this fundamental energy gap is such that at room temperature, there are a few electrons in the band just above the bandgap.
- A semiconductor lattice vibrates in certain modes. Each mode can be characterized by a phonon or quantum of mechanical energy. Electrons in a semiconductor can exchange energy with the lattice by emitting or absorbing phonons. Typically, the maximum energy of a phonon is a few tens of millielectronvolts, which is much smaller than the bandgap energy. Electrons can also exchange energy with photons or quanta of light.

1.5 FURTHER READING

The Feynman Lectures on Physics by R. P. Feynman, R. B. Leighton, and M. Sands, Addison-Wesley, 1964 (ISBN 0-201-02116-1-P, QC23.F435) have excellent reading material on elemental quantum mechanics and statistical mechanics. The style of this classic series emphasizes physical intuition. Chapters 37 and 38 of Vol. I are a great introduction to quantum mechanics, the wave-particle duality, electron behavior, and atomic structure. Chapters 39, 40, and 41 also of Volume I deal with the kinetic theory of gasses, the principles of statistical mechanics, and Brownian motion. Chapter 41, in particular, brings together results of quantum mechanics and statistical mechanics to derive the physics of blackbody radiation. Highly recommended reading.

Modern Physics and Quantum Mechanics by E. E. Anderson, Saunders, 1971 (ISBN 0-7216-1220-2, QC174.1.A527) is an excellent textbook on introductory quantum mechanics. It represents a good balance between physical intuition and mathematical rigor. Several chapters are organized with a sense of history in them, which makes them exciting and informative. The book has many examples, lots of charts and pictures, displays data from original experiments, and is in general well written and clear. Chapters 2–4 are particularly relevant to the discussion of interest to us here.

Introduction to Solid State Physics by C. Kittel, Wiley, 1976 (ISBN 0-471-49024-5, QC176.K5) is also a classic textbook. Chapters worthy of reading at this point are Chapter 1 on crystal structure, Chapter 3 on crystal binding, and Chapter 6 on electron statistics. You may also want to browse through Chapter 7 on energy bands and Chapter 4 on lattice vibrations and phonons. The book is rather rigorous mathematically, yet physically intuitive. If you have not taken a course on solid-state theory, you will have to make an effort to uncover the physics out of the mathematical clutter.

Crystal Fire—The Birth of the Information Age by M. Riordan and L. Hoddeson, Norton, 1997 (ISBN 0-393-04124-7, TK7809.R56) is an enjoyable and easy-to-read book telling the story of the invention of the transistor and the integrated circuit. Its first few chapters present an excellent description of the puzzle that physicists faced at the beginning of the 20th century and the resulting development of quantum mechanics. The book covers in simple and intuitive terms many of the topics discussed in this chapter.

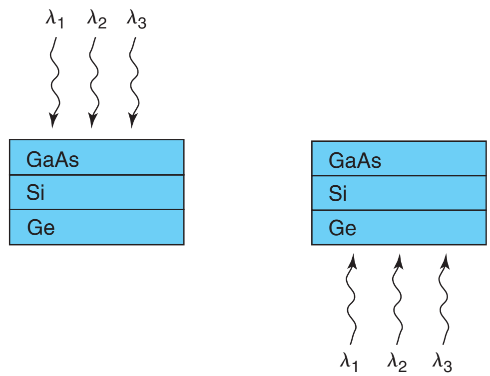
History of Semiconductor Engineering by B. Lojek, Springer, 2007 (ISBN 3-540-34257-5, TK7871.85.L65) is also a fun-to-read account of the early days of the semiconductor industry. There is a lot of detail about the beginnings of integrated circuits and the personalities involved. It includes many historical pictures and diagrams.

PROBLEMS

1.1 Compute the kinetic energy in electronvolts and the de Broglie wavelength in cm of (a) a 57-gr tennis ball served at 160 km/hr, (b) a 3000-pound car traveling at 65 miles/hr, and (c) an electron in vacuum moving at 10^9 cm/s. Comment on the suitability of the particle-type description of these objects if they are, respectively, in (a) a tennis court that is 78 feet by 36 feet, (b) a road 15 m wide, and (c) a particle accelerator with a 1-inch diameter core.

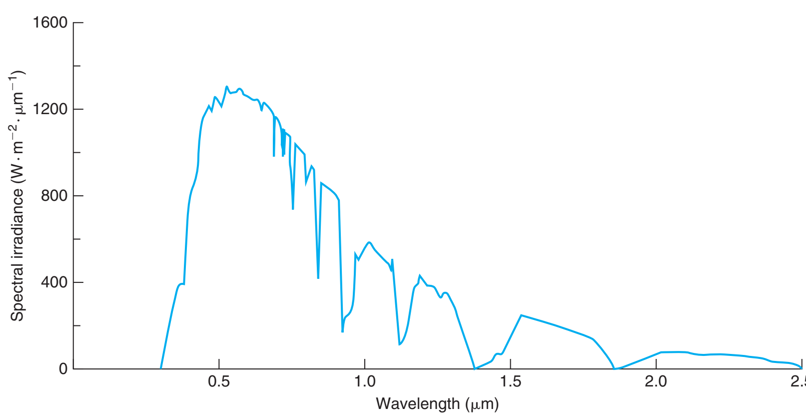
1.2 Calculate the wavelength and the frequency of a photon with the lowest possible energy required to free up a 1s core electrons from a Si atom.

1.3 Consider a stack of three wafers at room temperature. The top one is GaAs ($E_{g1} = 1.42$ eV), the middle one is Si ($E_{g2} = 1.12$ eV), and the bottom one is Ge ($E_{g3} = 0.66$ eV), as sketched below. To this stack, we shine three lasers with wavelengths $\lambda_1 = 0.85$ μm, $\lambda_2 = 1.3$ μm, and $\lambda_3 = 1.55$ μm, once from above and a second time from below. For each illuminating condition, in which wafer is each laser beam absorbed? Explain.



1.4 An **atom laser** is a coherent beam of atoms and a spectacular manifestation of matter waves. An important step toward the realization of the atom laser was the achievement of Bose–Einstein condensation. This is a cold and coherent condensate of atoms. The atom laser is realized when atoms are extracted in a coherent way from the Bose–Einstein condensate. The proof of coherence was obtained by observing an interference pattern when two Bose–Einstein condensates overlapped. Let's put some numbers to this. In its first demonstration, Na atoms were used. The diffraction pattern that was observed had a wavelength of 15 μm. *Estimate* the temperature at which the atoms were cooled down to for this to be possible. For comparison, *estimate* the de Broglie wavelength of a Na atom at room temperature. At all temperatures, consider the system of Na atoms as an ideal gas.

1.5 The spectral irradiance of the Sun under standard conditions at the Earth's surface is shown in the figure below. The peak of the radiation occurs at a wavelength of 0.6 μm where the spectral irradiance is about 1300 W · m^{-2} · μm^{-1}.



(a) Compute the spectral density of the photon flux at the surface of the Earth at a wavelength of 0.6 μm.
(b) Compute the photon flux at the Earth's surface for photons of wavelength between 0.59 and 0.61 μm wavelength.
(c) What is the maximum bandgap that a semiconductor can have and still be able to absorb 0.6 μm wavelength photons?

1.6 Starting from the atomic density of Si given in Appendix B, *estimate* the interatomic distance of Si atoms in a Si crystal in two ways. First, assume that the solid is made out of closely packed hard spherical balls that represent the Si atoms. Second, assume that the each atom can be considered a hard cube. In either case, disregard the actual crystalline structure of Si that is shown in Fig. 1.10. Compare with the actual value of the interatomic distance given in Appendix B.

1.7 This problem is about estimating the size of a hydrogen atom and the binding energy of its electron. This is an easy calculation and the result is surprisingly accurate. This is how this can be done. Consider the electron as a classical particle that is bound to the proton by the attractive electrostatic force. The electron performs a circular orbit around the proton. The quantum mechanics are introduced by assuming that the length of the orbit is equal to the de Broglie wavelength of the electron. The solution to the problem lies in computing the total energy of the electron, the sum of its kinetic energy plus its potential energy, and finding the radius that minimizes it.

Proceed as follows.

(a) Using Eqs. (1.3) and (1.4), express the kinetic energy of the electron in terms of its de Broglie wavelength. Assume that the de Broglie wavelength is equal to the circumference of the orbit and derive an expression for the kinetic energy in terms of the orbital radius.
(b) From elemental electrostatics, write an expression for the potential energy of the electron in terms of the radius of its orbit. Get the total energy of the electron. Find the radius that minimizes it. Derive a simple expression for the total energy of the electron in terms of fundamental parameters.
(c) Put numbers to these expressions. Use the SI system. Give the final result in nanometer and electronvolt.

1.8 Derive an expression for the relative error of Maxwell–Boltzmann statistics with respect to Fermi–Dirac statistics as a function of energy. Graph in a quantitative fashion.

1.9 Atoms have been guided through hollow optical fibers (see M. J. Renn et al., *Phys. Rev. Lett.* **75**, 3253, 1995). This represents a potentially convenient and flexible method for manipulating atoms. If an atom's de Broglie wavelength is comparable to the diameter of the core, the atom will propagate like a wave, as opposed to a particle. This could be used to make an atom fiber interferometer. The authors of the paper state that Rb atoms that have been cooled down to 290 nK and are launched into a fiber with a hollow core of 2 μm in diameter would travel in a single transverse atomic mode. Verify this statement by computing the de Broglie wavelength of Rb atoms in this situation. The atomic mass number of Rb is 85.5.

1.10 A certain state has a probability of 15% of being empty at 300 K. What is its location with respect to the Fermi level?

1.11 Planck's radiation law gives the frequency distribution of energy radiated by an ideal black body. It is given by

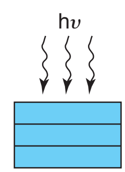
$$E(\nu) = \frac{2\pi\nu^2}{c^2} \frac{h\nu}{\exp\dfrac{h\nu}{kT} - 1}$$

In this equation, h is Planck constant, k is Boltzmann constant, c is the speed of light in vacuum, T is absolute temperature, and ν is the frequency of radiation. This equation is given in the SI system of units where E has units of $W \cdot s/m^2$.

Calculate the photon spectral density (photon flux per unit frequency) emitted by a blackbody at room temperature at an energy equal to the bandgap of Si at room temperature.

1.12 Triple junction solar cells currently hold the world record for solar cell efficiency. A typical configuration for a triple junction solar cell uses a stack of InGaP, Ge, and InGaAs layers with bandgaps of 1.86, 0.7, and 1.4 eV, respectively.

a) Determine the optimum arrangement of these three semiconductor layers that maximizes efficiency. That is, in the figure below, if the solar radiation comes from the top, what is the appropriate ordering of layers that provides maximum energy conversion efficiency? Explain your choice.



b) For the choice of layer structure that you made in the previous section and referring to the spectrum of radiation coming from the Sun at the surface of the Earth given in Problem 1.5, indicate the portion of the solar radiation spectrum that is absorbed in each of the semiconductor layers.

Chapter 2

Carrier Statistics in Equilibrium

This chapter discusses the statistics of electrons in a semiconductor in thermal equilibrium. Understanding the equilibrium situation is essential to appreciate the behavior of a semiconductor when perturbed. This is in fact a pervasive theme of this book. It is many times rather insightful to view semiconductor device operation as the process required to reestablish equilibrium following an external disturbance.

This chapter starts by discussing the distinct role of electrons in the conduction and valence bands in semiconductors. The concept of "hole" is introduced. Basically, a hole is a missing electron in the valence band. Following this, "intrinsic" and "extrinsic" semiconductors are defined and the role of a special kind of foreign atoms, "dopants," which can be introduced to engineer the equilibrium electron and hole concentrations in semiconductors, is discussed. We continue with a rather general and fairly rigorous formulation to enable the computation of the equilibrium carrier concentrations in semiconductors in a wide range of situations. This formulation is based on an energy view of the situation and exploits the concept of Fermi level.

The results obtained in this chapter are very important to semiconductor device design and will be extensively used throughout this book.

2.1 CONDUCTION AND VALENCE BANDS; BANDGAP; HOLES

We saw in Chapter 1 that a semiconductor is a crystalline solid with a band structure that at 0 K is characterized by several bands completely filled with electrons. The last full band is separated from the next empty band by a relatively small bandgap, the "fundamental" bandgap. The meaning of "relatively small" has to do with the actual temperature of operation of semiconductor devices. The proper scale of this statement is the thermal energy kT. For devices that operate near room temperature, the most widely used semiconductors have bandgaps of 0.7 to 2 eV. Si, for example, has a bandgap at room temperature of 1.12 eV, while GaAs has 1.42 eV. Wider bandgap semiconductors are currently under research for high-temperature applications, such as integrated circuits for jet engine control. A good example is SiC with a bandgap of 3 eV. Narrower bandgap semiconductors are also of interest for applications such as cooled infrared sensors. InSb with a bandgap of 0.17 eV is under investigation for this purpose.

Most of the behavior that we are likely to encounter in semiconductor devices involves an energy picture that is centered around the fundamental bandgap, or simply, "the bandgap." In semiconductor device operation, we can safely ignore the complexities of the fully occupied deeper lying bands. In fact, the two key bands around the bandgap are given special names. The band immediately below is called the *valence band*, while the band immediately above is called the *conduction band*. In an energy picture, E_v is used to denote the top edge of the valence band, E_c the location of the bottom edge of the conduction band, and E_g is used to refer to the width of the energy gap that separates these two edges. The magnitude of the bandgap depends on temperature. Advanced Topic AT2.1 discusses this.

If we zoom into the energy band structure of a semiconductor around the bandgap, the picture at the left of Fig. 2.1 emerges. It shows the top of the valence band, the bandgap, and the bottom of the conduction band. In the semiconductor literature, it has become common to extract the essence of this picture by just drawing the edges of the bands as represented on the right of Fig. 2.1. This is the first and simplest of many *energy band diagrams* that will be drawn in this book and that are widely used in the semiconductor device literature. The energy picture is a particularly intuitive and powerful way to represent many processes that take place in a semiconductor. It does not capture everything that is important about a semiconductor. For example, it does not contain enough information to understand light emission and absorption, which requires an awareness of momentum. However, this simple energy picture is very useful in understanding transport in devices such as microelectronic transistors. For this reason, it is extensively used in microelectronics engineering.

It is important not to lose sight of the physical meaning of the conduction and valence bands. To refresh our memory, we go back to the structural arrangement of atoms in a semiconductor. We discussed in the previous chapter that, in a semiconductor, atoms bond together mostly by sharing valence electrons. On average, an atom ends up with eight valence electrons around it, a particularly low-energy arrangement. This situation can be represented in a simple flat sketch as shown in Fig. 2.2(a). In this figure, each box represents an atom and each stick represents a bonding valence electron. Each atom shares its four valence electrons with its four neighbors and therefore each pair of atoms shares two electrons between them. Although Fig. 2.2 shows

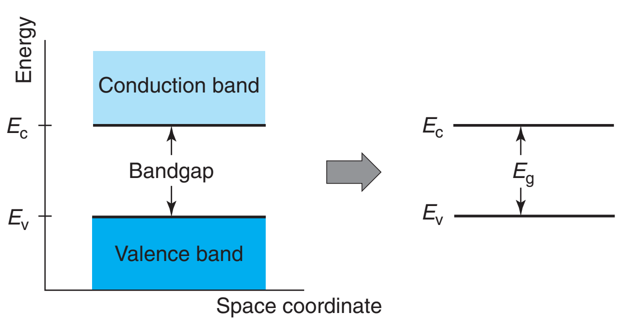


FIGURE 2.1
Left: band structure of a semiconductor around the fundamental energy gap. Dark shading represents a full band at 0 K. Light shading represents an empty band at 0 K. Right: simplified representation showing only band edges.

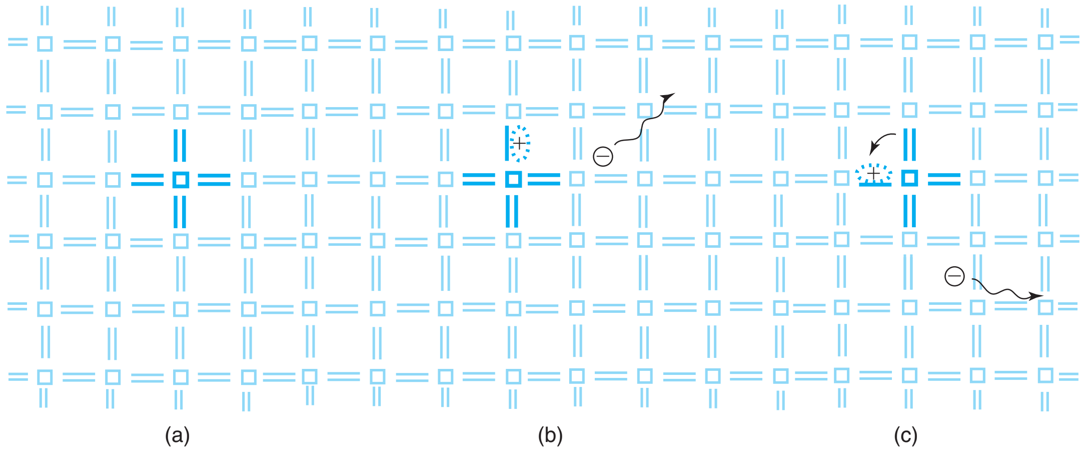


FIGURE 2.2
Flat model of the semiconductor lattice: (a) at 0 K with all valence electrons participating in bonding, (b) at finite temperature where one electron has become free leaving a hole behind, and (c) after the hole moves to a neighboring location.

a flattened array, it represents in reality a three-dimensional network of atoms. This simple flat picture does not change in any substantial way in the case of a compound semiconductor with two or more different kinds of atoms.

Figure 2.2(a) represents a 0-K situation, where all valence electrons are engaged in bonding between neighboring atoms. Figure 2.2(b) shows a picture at a finite temperature where one bonding electron has acquired enough energy from the finite thermal energy of the lattice to freely wander around the crystal. It becomes a "conduction" electron. The number of electrons that break off their ties to atoms is a function of temperature and other parameters. We will learn how to calculate this number later on in this chapter. Before that, it is important to note that in the place that was occupied by the electron that escaped, a *hole* is left. Since the electron is negatively charged with a charge of $-q$, this hole has a net positive charge of $+q$. The actual origin of this positive charge is the partially uncompensated charge associated with the protons in the two atoms involved. In spite of that, the hole can also move around. This happens when a neighboring bonding electron jumps in to complete the missing bond. This leaves two other atoms partially unbonded somewhere else. Effectively, this makes the hole move from one place to another [Fig. 2.2(c)]. Both the conduction electron and the hole are called *carriers* because as they move, they "carry" their elemental electric charge along with them.

Energy band diagrams represent the situations depicted in Fig. 2.2 in a very physical way, as shown in Fig. 2.3. In order to understand how this is done, let us first think about the 0 K condition where no covalent bonds are broken. At 0 K, the valence band is completely full of electrons while the conduction band is empty. This allows us to conclude that the bonding valence electrons are represented in the energy band diagram by levels or states inside the valence band. With a full valence band and a completely empty conduction band, no conduction is possible in an ideal semiconductor at 0 K.

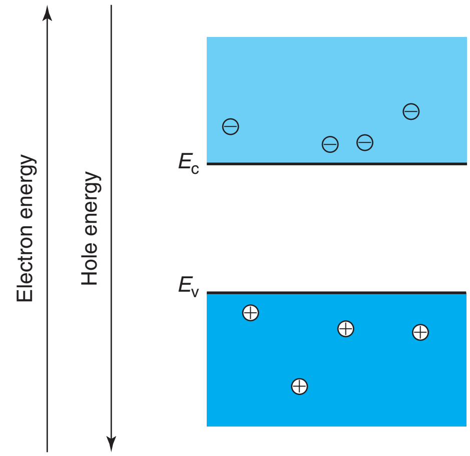


FIGURE 2.3
Sketch of free electrons and holes in energy band diagram. Note the different signs for electron and hole energy scales.

At a finite temperature, the free electron depicted in Fig. 2.2(b) and (c) has acquired enough energy to be promoted from the valence band to the conduction band where the abundance of empty states allows it to move around easily. Free electrons therefore occupy states in the conduction band. The holes left behind, however, sit inside the valence band in the energy band picture. The hole is an empty state in the valence band. Whenever there is an empty state, conduction becomes possible. Since there are many electrons in the valence band (the bonding electrons), it will not be too difficult for one of these electrons to jump to the empty position where the hole resides thereby allowing the hole to move around. The hole is therefore also free. Similar to air bubbles in water where it is easier to focus on the simple movement of the bubble rather than on the complex flow of water around it, it is much more convenient to describe the complicated movement of bonding electrons in the valence band in terms of the hole dynamics. In energy band diagrams, only holes are typically represented in the valence band. Bonding electrons can be ignored in most circumstances.

Before proceeding any further, there are two additional important observations that must be made about energy band diagrams. The first one is that at the edges of the bands, the kinetic energy of the carriers is zero. We can easily understand that with the help of Fig. 2.4(a) which depicts a photon striking a semiconductor. If the energy of this particular photon is exactly the bandgap energy of the semiconductor, then the photogenerated electron and hole cannot have any extra kinetic energy. The entire photon energy is consumed to break the bond.[1] In the energy band picture, both carriers are represented at their respective band edges.

The second observation regarding energy band diagrams is that electron energies increase upward while hole energies increase downward, as Fig. 2.3 makes explicit. If a conduction electron has higher kinetic energy, it shows up at a high energy inside the conduction band. For holes,

[1]We often refer to an electron–hole pair as resulting from the break up of a covalent bond. This term should not be interpreted to mean that the electron and hole coordinate their behavior in any way after they have been generated. Each carrier goes its own way.

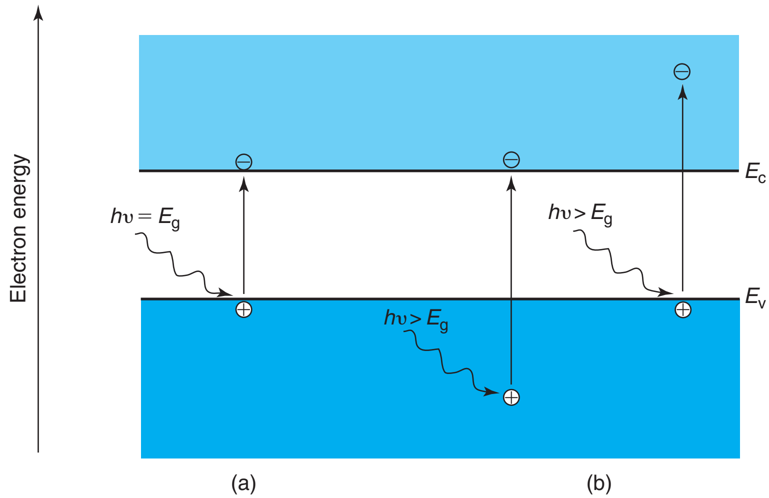


FIGURE 2.4
Energy band diagram depicting photons impinging on a semiconductor. In (a), the photon has an energy identical to the bandgap energy. As a result, the photogenerated electron and hole sit at the respective band edges. In (b), the two photons have more energy than the bandgap. In one case (right), the extra available energy beyond the bandgap is given as kinetic energy to the electron. In the other case (left), it is given to the hole.

this picture is reversed—a high kinetic energy hole is represented deep down inside the valence band. This can be better understood with the help of Fig. 2.4(b). Consider two photons of identical energy $h\nu > E_g$ impinging on different bonding electrons and producing electron–hole pairs. In one case, the electron that is promoted to the conduction band carries all the extra energy in excess of that required to create an electron–hole pair as kinetic energy. This electron is represented inside the conduction band at an energy $h\nu - E_g$ over the conduction band edge. Energy conservation leaves no extra kinetic energy for the hole, which therefore sits at E_v. In a second case, the contrary situation occurs: the electron is ejected with no extra kinetic energy so it sits at E_c and the hole carries all the extra energy as kinetic energy and it is represented at an energy $h\nu - E_g$ below the band edge. Clearly, hole energies increase downward in the energy band diagram. A high kinetic energy fast-moving hole represents a situation in which bonding electrons are jumping from bonding sites to available sites very quickly.

The flat picture of a semiconductor shown in Fig. 2.2 allows us to visualize a conduction electron and a hole in a simple and intuitive way. It is important, however, to note two significant inaccuracies that are implied in this figure. In Fig. 2.2, the electron and hole are sketched as having a "size" of the order of an interatomic distance in the semiconductor, about 0.2–0.3 nm. This is wrong by at least an order of magnitude. In fact, in Chapter 1, we estimated the effective size of an electron at room temperature to be about 7.6 nm. Since a hole is not expected to be very different from an electron (we will get to better appreciate the differences between them as we advance in this book), this is also a good estimate for the size of a hole. We also showed that a sphere of a diameter of 7.6 nm contains over 10,000 atoms. This illustrates how spread out electrons and holes really are in a semiconductor. A conduction electron is not the little lump represented in Fig. 2.2 but is spread out over thousands of atomic sites. Also, the notion that a hole jumps from a precise bonding location to a neighboring one is not very physical. A "fuzzier" probabilistic view is actually more appropriate.

The second inaccuracy implied in Fig. 2.2 comes from the view that the movement of a hole really represents the "retrograde" motion of a particular bonding electron. Not only does the large size of a hole make this view rather ambiguous, but it ignores the quantum mechanical nature of the bonding electrons in a semiconductor, which is very different from electrons in vacuum. The retrograde view of the motion of a hole can actually lead to rather unphysical conclusions. However, for the kinds of applications that we are interested on in this book, the collective dynamic behavior of the quantum-mechanical bonded electrons in a semiconductor in the presence of a broken bond is well represented by an ordinary classical particle of charge $+q$. The analogy of a bubble of air in a liquid representing the complex collective behavior of liquid molecules around it is really insightful in this regard.

As a final remark in this section, from here on in this text, we will refer to conduction electrons as simply "electrons." We will describe the behavior of bonding electrons in the valence band in terms of holes. We will totally ignore in the rest of our studies the core electrons that are tightly bound to each constituent atom of the crystal, as they play no role in semiconductor devices.

2.2 INTRINSIC SEMICONDUCTOR

In a semiconductor in thermal equilibrium at a finite temperature, some of the bonds that tie the constituent atoms are broken. In consequence, there is a certain number of free electrons in the conduction band and a number of holes left in the valence band. It is inevitable to ask the question: in this situation, how many electrons and holes are there per unit volume? How do these concentrations depend on temperature? Rigorously answering these questions is the purpose of this chapter.

Let us first say a few words about what we mean by an "ideal" semiconductor. For the time being, we will define this as a perfectly crystalline piece of a semiconductor that is 100% pure (i.e., no foreign atoms) and that is unaffected by any surface effects. An ideal semiconductor defined this way is also called an *intrinsic semiconductor*. We will soon see that we can relax the second aspect of this stringent definition to just "sufficiently" pure, that is some impurities might be present without changing the picture substantially, provided that their concentration is not too high. Also later on, we will understand the implications of the third restriction regarding surface effects. In subsequent chapters, we will learn how to deal with surfaces and to evaluate how far their influence extends into the body of a semiconductor.

So the question is again, in an intrinsic semiconductor in thermal equilibrium, how many electrons and holes are there? Answering this question requires a fairly elaborate model that is developed later in this chapter. However, a number of important dependencies can be readily identified.

It is obvious that in an intrinsic semiconductor in thermal equilibrium, the number of electrons and holes has to be identical to each other at all temperatures. This is because every bond that gets broken produces precisely one electron and one hole. Also, every bond that is formed consumes one electron and one hole. If we define n_o as the equilibrium electron concentration (number of electrons in the conduction band per unit volume) and p_o as the equilibrium hole concentration (number of holes in the valence band per unit volume), it is clear that n_o must equal p_o everywhere. This concentration is also called the *intrinsic carrier concentration*, n_i. Hence, in an intrinsic semiconductor in thermal equilibrium,

$$n_o = p_o = n_i \tag{2.1}$$

Typically, n_o, p_o, and n_i are given in units of cm^{-3}.

What are the key dependencies of n_i? Intuitively we know that n_i must have a direct dependence on temperature. The higher the temperature is the more vigorously the atoms vibrate in the semiconductor lattice and the easier it will be for a bond to break. We also expect n_i to exhibit some kind of inverse dependence on the bandgap of the semiconductor, E_g. This is because the higher the bandgap is the harder it is to break a bond and liberate an electron and a hole.

To be more specific about the dependence of n_i on T and E_g, we need a detailed model. This is presented below in Section 2.4.4. Yet, an analogy with chemical reactions can reveal the leading dependencies. Consider, for example, the well-known reaction of decomposition of water into its two constituent ions:

$$H_2O \rightleftharpoons H^+ + OH^- \tag{2.2}$$

When a water molecule decomposes, a hydrogen ion and a hydroxide ion are produced. These two ions can combine again to form a water molecule. This chemical reaction is in some way similar to the break up of a crystalline bond in a semiconductor. When a crystalline bond breaks, an electron and a hole are produced. A bond is formed when an electron and a hole come together. Bond break-up and formation in a crystalline semiconductor can be therefore thought as the following chemical reaction:

$$\text{bond} \rightleftharpoons e^- + h^+ \tag{2.3}$$

This analogy is powerful because we can exploit what we have learned in elementary chemistry to understand a great deal about the statistics of electrons and holes in semiconductors. We know, for example, that in a chemical reaction the *law of mass action* establishes a relationship between the concentrations of reactants and reaction products, through a so-called rate constant. We also know that the rate constant exhibits a peculiar dependence on temperature and on the energy required for the reaction to take place. For the water decomposition reaction, the law of mass action is written as

$$K = \frac{[\mathrm{H}^+][\mathrm{OH}^-]}{[\mathrm{H_2O}]} \sim \exp\left(-\frac{E}{kT}\right) \tag{2.4}$$

where E is the energy consumed or released in the water decomposition reaction.

Equation (2.4) exhibits a peculiar dependence on T and E that is rather common in nature. Equations such as this apply to processes that are "thermally activated," that is, processes in which the rate of reaction is limited by the need to overcome a certain energy barrier. This threshold energy is called the *activation energy*.

In analogy with Eq. (2.4), we could write a similar law of mass action for the crystalline bond dissociation reaction of (2.3), as follows:

$$K = \frac{n_o p_o}{[\text{bonds}]} \sim \exp\left(-\frac{E_g}{kT}\right) \tag{2.5}$$

where n_o and p_o have been defined above, and [bonds] refers to the concentration of unbroken bonds. For the bond dissociation reaction, the activation energy is the bandgap energy since this is the energy that is required to break a bond.

Under typical conditions, the number of broken bonds in a semiconductor is a tiny fraction of the total number of bonds (this becomes clear when we put some numbers later on). In other words, the concentration of bonds is never upset in a significant way (or the semiconductor will melt). In consequence, the concentration of unbroken bonds ([*bonds*] in Eq. (2.5)) is to first-order constant and then we can conclude that

$$n_o p_o \sim \exp\left(-\frac{E_g}{kT}\right) \tag{2.6}$$

The fact that the product of the equilibrium hole and electron concentrations has this peculiar dependence on E_g and T is a very significant result with important consequences. Although obtained here in a qualitative way, the rigorous treatment carried out later on in this chapter will lead to the same set of dependencies.

The result of Eq. (2.6), when combined with Eq. (2.1), allows us to obtain the key dependencies of n_i:

$$n_i \sim \exp\left(-\frac{E_g}{2kT}\right) \tag{2.7}$$

As expected, n_i increases with temperature and decreases with E_g. What is interesting is the specific negative exponential form of these dependencies. This is, in fact, readily observed in experiments. Figure 2.5 shows experimental measurements of n_i for Si as a function of temperature. In this figure, n_i in the y-axis is plotted in a semilog scale, while the x-axis is proportional to inverse temperature. T increases toward the left. When graphed this way, a straight line with

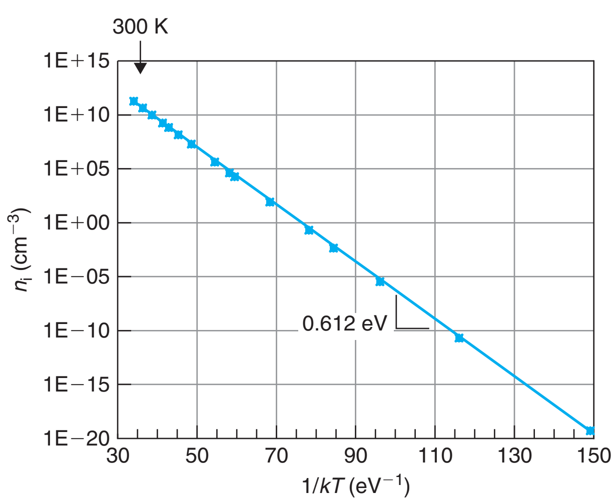


FIGURE 2.5
Arrhenius plot of n_i in Si. Measurements of Misiakos and Tsamakis [Measurements of Konstantinos Misiakos, Dimitris Tsamakis, Accurate measurements of the silicon intrinsic carrier density from 78 to 340 K, *Journal of Applied Physics*, Vol 74, Pg 3293, 1993. © J.A. del Alamo].

a negative slope is obtained. This is consistent with Eq. (2.7) and is the characteristic signature of a thermally activated process. This kind of graph is called an *Arrhenius plot* in honor of Svante Arrhenius who showed around 1889 the pervasiveness of thermally activated processes in nature. From the slope of the straight line, the bandgap of Si can be obtained. For the data of Fig. 2.5, a bandgap energy of 1.224 eV is obtained. This is close but not exactly equal to the best known value of 1.124 eV for Si at room temperature. The reason for this small discrepancy is twofold. First, the missing prefactor in Eq. (2.7) is slightly temperature dependent. This prefactor will be derived later in Section 2.4. Additionally, E_g itself also depends on temperature, as discussed in Advanced Topic AT2.1. Issues around the temperature dependence of n_i are explored in detail in Problem 2.4.

The chemical reaction analogy does not give us a complete expression for n_i. The prefactor is missing and we therefore do not know how to compute its value. To do this, we have to go through the detailed development described in Section 2.4. For reference, n_i for Si at room temperature is about 10^{10} cm^{-3}.

There is a second and hugely important result that emerges from Eq. (2.6). This equation states that the equilibrium *np* product in a semiconductor at a certain temperature is a constant that is specific to the semiconductor. Establishing a parallel again with the water dissociation reaction, this is equivalent to saying that the product of the concentration of H^+ and OH^- ions in pure water at a certain temperature is a constant. The implications of this are obvious. If we increase the concentration of H^+ in water (for example, by dissolving a small amount of acid), the concentration of OH^- will drop accordingly. This is because the equilibrium values of H^+ and OH^- are established from the balance of a dissociation reaction pointing to the right in Eq. (2.2) and a recombination reaction pointing to the left. Selectively adding H^+ to the bath enhances the rate of the left-pointing reaction bringing down the OH^- concentration.

The same arguments apply to electrons and holes in semiconductors. In the case of a perfectly pure semiconductor, the equilibrium electron and hole concentrations are identical and equal to n_i. This value arises from a balance between the rate of break up of bonds and the rate of formation of bonds. If we could find a way to selectively introduce, say, electrons so that the electron concentration increases above n_i, the rate of formation of bonds will increase and the hole concentration will drop below n_i. The product will remain a constant. Since for an intrinsic semiconductor $n_o = p_o = n_i$, in general, then,

$$n_o p_o = n_i^2 \tag{2.8}$$

This is a very important equation that will be used extensively in this and subsequent chapters. When we do our more detailed analysis in Section 2.4, we will find that this equation is of broad applicability but it is not completely general. In the language of chemical reactions again, it works in the "dilute" regime, that is, if the concentration of carriers that is added is not too high.

How carriers can be added in a selective way to a semiconductor is described in the next section.

2.3 EXTRINSIC SEMICONDUCTOR

So far, we have considered only the introduction of electrons and holes in a semiconductor via the spontaneous break up of atomic bonds. There is another way in which electrons and holes can be introduced and that is by "doping" it with relatively small amounts of selected impurities. Key to the process of doping is the fact that it is possible to selectively introduce electrons into the conduction band without producing an equal amount of holes in the valence band and vice versa. If the resulting concentration of any one of the carriers overwhelms the intrinsic carrier concentration, the semiconductor is said to be "extrinsic." When electrons are introduced in a preferential way, the semiconductor is said to be *n-type*. If holes are the majority, the semiconductor is called *p-type*.

Doping is one of the pillars over which the microelectronics industry is built. Precise placement of dopant atoms in a semiconductor allows the creation of n-type and p-type regions with nearly arbitrary shape and doping distribution. It is largely through careful engineering of doping profiles that devices are designed to deliver the required specifications for different applications. Doping can be accomplished in a variety of ways, such as solid-state diffusion or ion implantation. In diffusion, dopant atoms diffuse from a dopant-rich source into the semiconductor. In ion implantation, ionized dopant species are accelerated to high energies and "slammed" against the semiconductor penetrating to a certain depth. A thermal activation step allows the incorporation of the dopants into substitutional locations in the lattice.

The following sections describe how certain impurities are able to selectively introduce electrons or holes into a semiconductor. We will also learn to compute the equilibrium carrier concentrations in an extrinsic semiconductor.

2.3.1 Donors and acceptors

Consider what happens if, in a perfect Si lattice, a Si atom is substituted by a foreign atom, such as As or P, that comes from column V of the periodic table (see Fig. 1.4). Since these atoms are located close to Si in the periodic table, they are not very different from Si. They can take the place of a Si atom without disrupting the lattice too much. The foreign atom is said in this case to be in a substitutional position. The uniqueness of column V atoms is that they have five valence electrons. This means that after consuming four electrons to bond to the four nearest neighbors, there is a fifth electron that remains loosely attached—its binding energy is typically in the range 10–50 meV. At room temperature, this fifth electron can easily pick up enough energy from the thermal energy of the lattice to escape the attraction of the donor atom and roam freely around the crystal. This electron is indistinguishable from a thermally created electron. The substitutional column V atom has thus "donated" one of its electrons to the conduction band. The "donor" is left positively charged since there is one more proton in the nucleus than surrounding electrons. This situation is depicted in Fig. 2.6.

A parallel situation takes place if Si is doped with column III elements such as B. These atoms have three electrons in the outer layer. A substitutional B atom bonds with three Si neighbors but is unable to bond with the fourth one. There is in effect a hole in the covalent bonding structure at the location of the impurity atom. This hole can easily migrate away from the impurity site if a nearby bonding electron jumps in and satisfies the missing covalent bond in a process that typically takes less than 0.1 eV. At this point, the hole has moved to a neighboring location and is indistinguishable from a regular thermally created hole. In the process of releasing a hole to the valence band, the impurity has "accepted" a bonding electron and has become negatively charged. This is depicted in Fig. 2.7.

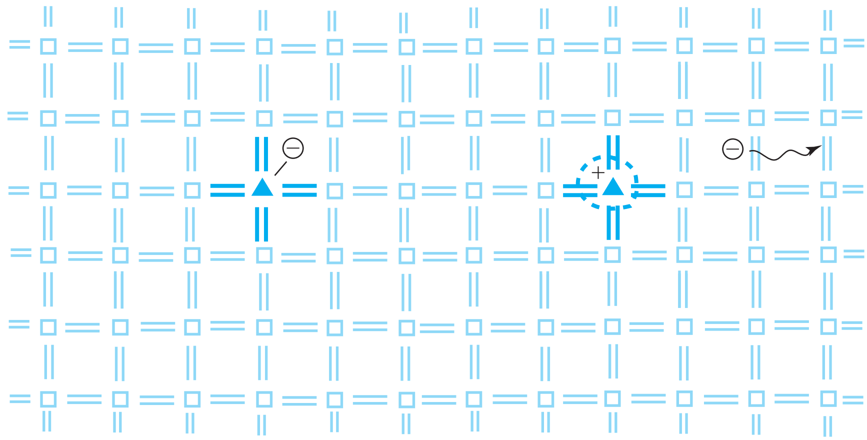


FIGURE 2.6
Sketch of a donor atom in a semiconductor lattice: (a) in a neutral state and (b) in an ionized state.

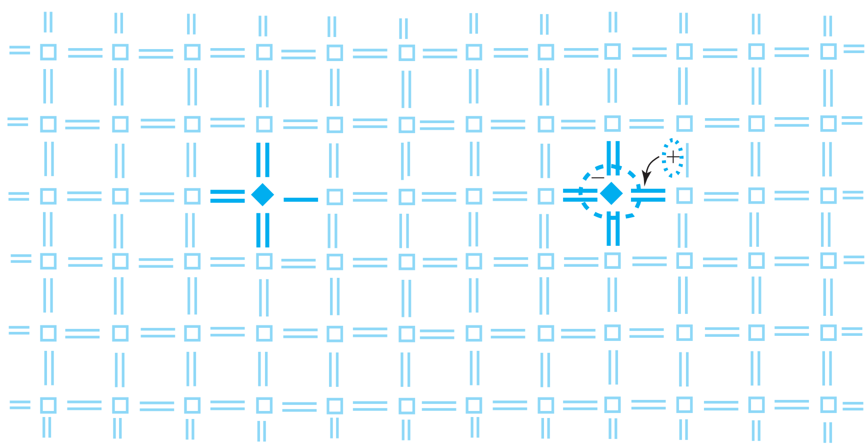


FIGURE 2.7
Sketch of an acceptor atom in a semiconductor lattice: (a) in a neutral state and (b) in an ionized state.

For GaAs and other III–V semiconductors, the situation is slightly more complicated. Elements from column VI, such as Se, are donors when placed substitutionally at an As site. Elements from column II, such as Zn or Be, behave as acceptors at a Ga site. Elements from column IV, such as Si or Ge, have an *amphoteric behavior*. This means that when placed at an As site, they behave as acceptors, but at a Ga site, they are donors. With amphoteric impurities, different

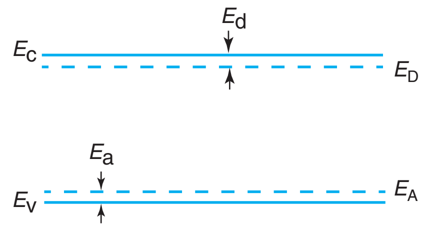


FIGURE 2.8
Representation of donor and acceptor states in the energy band diagram of a semiconductor.

doping conditions can result in either an n-type or a p-type semiconductor. In spite of this, Si is a widely used donor for III–V semiconductors. This is because it is not difficult to create conditions in which there is a high concentration of vacant Ga sites (Ga vacancies) and a small concentration of vacant As sites (As vacancies). In this way, the Si atoms preferentially find themselves in substitutional positions at Ga sites where they behave as donors.

Electron states associated with donor and acceptor atoms can be represented in the energy band diagram as shown in Fig. 2.8. Donor states are graphed inside the forbidden gap slightly below the conduction band edge at E_D. The reason is that it only takes a small amount of energy to release the loosely bound electron to the conduction band. This energy is called the *donor ionization energy* and is represented with the symbol E_d. We then have

$$E_d = E_c - E_D \tag{2.9}$$

Similarly, acceptor states are represented slightly above the valence band, at E_A, as a little energy will result in the release of a hole to the valence band. This energy is called the *acceptor ionization energy*, E_a. Then

$$E_a = E_A - E_v \tag{2.10}$$

Broken lines are used for the donor and acceptor states to indicate the fact that the impurity atoms are typically far apart from each other. Table 2.1 in Advanced Topic AT2.4 collects typical ionization energies for common Si and GaAs dopants. As is the case for Si atoms, the remaining bonding electrons of the impurity atoms are represented as full states inside the valence band.

2.3.2 Charge neutrality

A piece of a semiconductor in thermal equilibrium, as a whole, is charge neutral. This is regardless of whether it is elemental or compound, doped or undoped, at a finite temperature or at 0 K. Every atom of the semiconductor is charge neutral to begin with, that is, all atoms have matching numbers of electrons and protons. If no particles are allowed to escape from the semiconductor, overall charge neutrality must prevail.

Spatially, the picture can be more complex. First of all, the disruption to the perfect ordering of the crystal that is introduced by the surfaces can alter charge neutrality in their vicinity. This can happen even if the semiconductor is "pure," that is, without any dopant impurities. We will study the effect of surfaces later on in this book. Restricting ourselves to the bulk, far away from the surfaces, local charge neutrality might not exist either if the dopant distribution is not uniform in space. We will study this in greater detail in Chapter 4.

With these two caveats in mind, in the bulk of a uniformly doped semiconductor in thermal equilibrium, charge neutrality must prevail. Mathematically, for a general case in which the semiconductor is doped with donors and acceptors simultaneously, the balance of the negatively charged species and the positively charged species must be zero:

$$-n_o + p_o + N_D^+ - N_A^- = 0 \tag{2.11}$$

Here, N_D^+ denotes the concentration of ionized donors and N_A^- is the concentration of ionized acceptors.

The statement of charge neutrality of Eq. (2.11) is an important element in the calculation of the equilibrium carrier concentrations in doped semiconductors, as done in the next section.

2.3.3 Equilibrium carrier concentration in a doped semiconductor

In order to compute the equilibrium carrier concentrations in the bulk of a uniformly doped semiconductor, it is essential to first know the fraction of the dopant atoms that are ionized. Fortunately, the ionization energy of typical dopants is small enough that at room temperature a great majority of them are ionized. Denoting N_D as the donor concentration and N_A as the acceptor concentration, under many practical circumstances, we can write

$$N_D^+ \simeq N_D \tag{2.12}$$

$$N_A^- \simeq N_A \tag{2.13}$$

At low temperatures, this cannot be the case since there is less thermal energy available and certain dopants will be in a neutral state. When designing devices to operate at low temperatures, it is crucial to correctly account for "carrier freeze-out," as this phenomenon is known. This is described in Advanced Topic AT2.5.

It might seem counterintuitive to state that dopant atoms with an ionization energy on the order of 40–60 meV can be all ionized at room temperature when the thermal energy is only 25 meV. This apparent contradiction is resolved by realizing that in most cases there are many more states close to the band edges ready to accommodate carriers than dopant states at a slightly lower energy. As a result, the slight energetic disadvantage of the band states is more than compensated by their large numbers. In other words, for typical doping concentrations, even though it is energetically favored, it is hard for electrons in the conduction band to find an ionized donor to fall into. The same applies to acceptors and holes in the valence band. One can view this as a golfer trying to get a golf ball in the hole. From potential energy arguments, it is certainly true that the hole is a more favorable position for the ball to be in. It does not follow, however, that regardless how the golfer hits the ball, it will neatly fall in the hole! For a typical doping level of 10^{17} cm^{-3}, the ionization ratio of P in Si at room temperature is 97%, while this figure is 94% for B in Si. Advanced Topics AT2.4 and AT2.5 show how to perform these calculations.

Carrier production from dopant atoms proceeds in parallel with the thermal electron–hole pair formation from broken bonds. In a completely general case, both mechanisms must be accounted for in order to compute the equilibrium carrier concentrations. For most common semiconductors around room temperature, however, the doping concentrations overwhelm the intrinsic carrier concentration. In this instance, the equilibrium carrier concentrations are easy to compute. For an n-type extrinsic semiconductor, the equilibrium electron concentration is determined by the donor concentration:

$$n_o \simeq N_D \tag{2.14}$$

and the hole concentration can be found from Eq. (2.8):

$$p_o = \frac{n_i^2}{n_o} \simeq \frac{n_i^2}{N_D} \tag{2.15}$$

Since n_o is much higher than n_i, it follows that p_o is much smaller than n_i. In an n-type semiconductor, then, $n_o \gg p_o$. In this case, electrons are said to be *majority carriers* while holes are called *minority carriers*. Equation (2.14) could have also been obtained from Eq. (2.11) by setting $N_A = 0$ and $n_o \gg p_o$.

EXERCISE 2.1

Estimate the equilibrium electron and hole concentrations in Si uniformly doped with $N_D = 10^{16}$ cm^{-3} *at room temperature. State your assumptions.*

Assuming that all donors are ionized, Eq. (2.14) gives

$$n_o \simeq N_D = 10^{16} \text{ cm}^{-3}$$

Plugging in Eq. (2.15) we find

$$p_o \simeq \frac{n_i^2}{N_D} \simeq \frac{(10^{10} \text{ cm}^{-3})^2}{10^{16} \text{ cm}^{-3}} = 10^4 \text{ cm}^{-3}$$

where we have used an approximate value for n_i (the error in the estimation of p_o is less than 10% as a result of this approximation).

This exercise shows just how different the equilibrium electron and hole concentrations are in a typical semiconductor under fairly standard conditions.

It is important to note here that while Eq. (2.14) assumes full dopant ionization, Eq. (2.15) is more restrictive since it additionally requires the validity of Eq. (2.8). As discussed below, there are some limitations to this. Nevertheless, for many situations, Eq. (2.15) is adequate and will be used extensively in this book.

In a parallel way, for a p-type semiconductor in equilibrium, we have

$$p_o \simeq N_A \tag{2.16}$$

and

$$n_o \simeq \frac{n_i^2}{N_A} \tag{2.17}$$

In a p-type semiconductor, holes are the majority carriers and electrons are the minority carriers. Note again that Eq. (2.17) applies to the extent that Eq. (2.8) is valid.

At high temperatures, an extrinsic semiconductor eventually becomes intrinsic as the thermal break up of covalent bonds overwhelms the doping concentration. In this instance, Eqs. (2.14)–(2.17) fail and Eq. (2.1) applies. In more general terms, if a semiconductor has a concentration of dopants much smaller than n_i, the equilibrium electron and hole concentrations are basically equal to n_i. We will still denote such a semiconductor as an *intrinsic semiconductor* even though it is not perfectly pure. However, it is important to recognize that the value of n_i for Si at room temperature is so low ($\sim 10^{10}$ cm^{-3}) that it is very hard to obtain a Si wafer that is intrinsic at room temperature. The residual dopant concentration that unavoidably exists almost always overwhelms n_i. This is also the case for most common semiconductors. This situation changes at higher temperatures since n_i increases rapidly with temperature. It is in fact not difficult to have intrinsic Si at say 600 °C, as a numerical exercise in Advanced Topic AT2.1 shows.

The main results of this section are graphically summarized in the sketch of an Arrhenius plot of equilibrium electron concentration in an n-type semiconductor that is shown in Fig. 2.9. A similar picture applies for p_o in a p-type semiconductor. Figure 2.9 shows that there is a broad intermediate regime in which $n_o \simeq N_D$. In this regime, the semiconductor is extrinsic and all donors are fully ionized. This is the most common regime of device operation. At low enough temperatures (toward the right in the diagram), the semiconductor is still extrinsic but not all dopants are ionized. In this freeze-out regime, the electron concentration drops very quickly with temperature (the activation energy of n_o is shown in Advanced Topic AT2.5 to be $E_d/2$). At high enough temperatures (toward the far left in the diagram), intrinsic carrier production overwhelms the doping level and the semiconductor becomes intrinsic.

Throughout this section, we have considered semiconductors that are solely doped with donors or acceptors. It is very common in devices to have regions that contain both kinds of dopants. This happens, for example, in a PN junction fabricated by introducing n-type dopants into a p-type wafer. In this case, the n-type region is still n-type but it is partially compensated by the acceptors. It is not difficult to derive general expressions for the equilibrium carrier

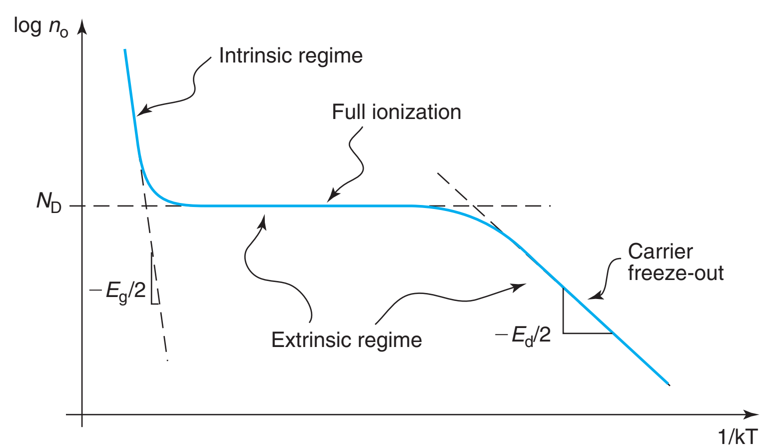


FIGURE 2.9
Arrhenius plot of n_o for an n-type semiconductor.

concentrations for a *compensated semiconductor*, as this situation is known. If there is to be compensation, sound engineering practice, that is, good process control, dictates that one type of dopant should clearly overwhelm the other, that is either $N_D \gg N_A$ or $N_A \gg N_D$. It is easy to show that in this case, all equations derived above apply if we substitute $N_d = N_D - N_A$ for N_D in Eqs. (2.14) and (2.15), and $N_a = N_A - N_D$ for N_A in Eqs. (2.16) and (2.17) (see Problem 2.9). N_d and N_a are often called *net* donor and acceptor concentrations, respectively.

2.4 CARRIER STATISTICS IN EQUILIBRIUM

The results obtained in this chapter so far, though useful and relevant, are based on a qualitative analogy between electron–hole production in an ideal semiconductor and a chemical reaction. This view has left a few important voids, such as a complete expression for n_i. It is actually not very difficult to develop a rigorous set of models for the electron and hole concentrations in thermal equilibrium in intrinsic and extrinsic semiconductors. The most physically meaningful way to do this is to adopt an energy view of semiconductors by exploiting energy band diagrams and using the concept of Fermi level introduced in Chapter 1.

In the following sections, we will develop a formulation for carrier statistics in semiconductors in thermal equilibrium in several steps. First, we will present a description of the density of states of the conduction and valence bands close to their respective band edges. Then, we will derive relationships between the equilibrium electron and hole concentrations and the relative location of the Fermi level within the band structure of the semiconductor. From this, we will be able to derive expressions for the $n_o p_o$ product and n_i. We will finish with a discussion about the location of the Fermi level in intrinsic and extrinsic semiconductors.

2.4.1 Conduction and valence band density of states

As introduced in Chapter 1, the density of states (DOS) quantitatively describes the distribution of states available for electrons in a crystal. The DOS of the valence and conduction bands in semiconductors has in general a rather complicated shape. Figure 2.10 shows, for example, calculations of the DOS for Si and GaAs as a function of energy.

As will become clear later in this book, a lot of the carrier action in a semiconductor takes place at energies close to the band edges. This implies that, in many device studies, there is no need for a detailed description of the entire conduction and valence bands. Accuracy at energies close to the edges of the bandgap will suffice.

A rather fundamental result of solid-state physics, and also an experimental observation, is that in the vicinity of the band edges, the density of states increases following a square-root dependence on energy. Consequently, the conduction band density of states of a semiconductor, $g_c(E)$, and the valence band DOS, $g_v(E)$, close to the band edges can be expressed as

$$g_c(E) = A_c\sqrt{E - E_c} \quad E \geq E_c \tag{2.18}$$

$$g_v(E) = A_v\sqrt{E_v - E} \quad E \leq E_v \tag{2.19}$$

$g_c(E)$ and $g_v(E)$ have units of $eV^{-1} \cdot cm^{-3}$. They are sketched in Fig. 2.11.

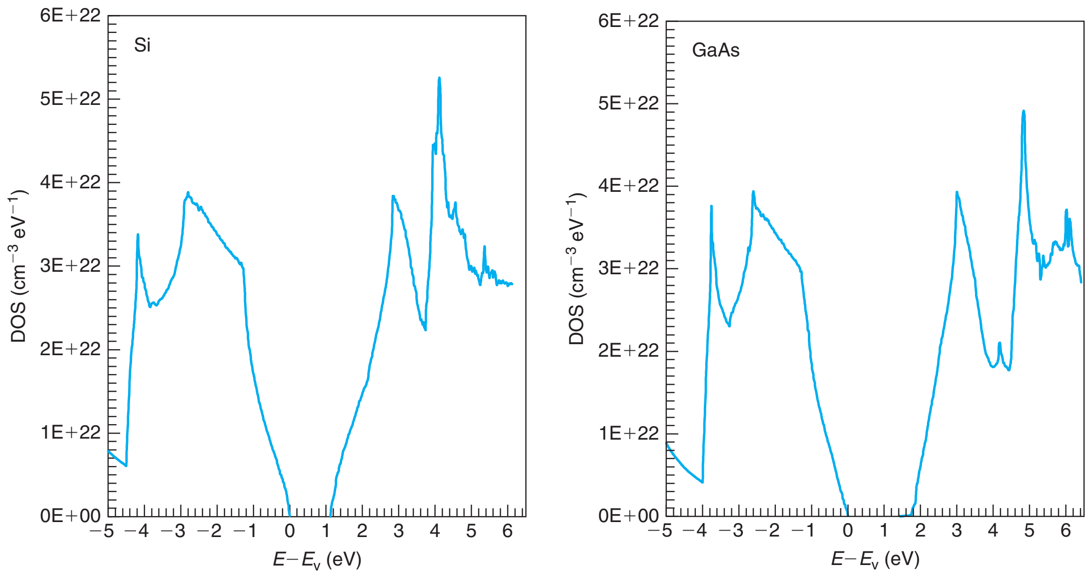


FIGURE 2.10
Density of states of the valence and conduction bands in Si (left) and GaAs (right) as a function of energy. Close to the band edges, the DOS have a square-root dependence with energy [Data Courtesy of M. Fischetti, IBM. © J.A. del Alamo].

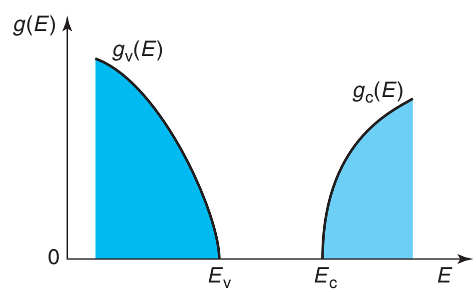


FIGURE 2.11
Sketch of conduction and valence band density of states close to the band edges.

A_c and A_v are two proportionality constants. For reasons that are best understood in a solid-state course, it is common practice to write these constants in the following way:

$$A_c = 4\pi \left(\frac{2m^*_{de}}{h^2} \right)^{3/2} \tag{2.20}$$

$$A_v = 4\pi \left(\frac{2m^*_{dh}}{h^2} \right)^{3/2} \tag{2.21}$$

m^*_{de} and m^*_{dh} defined this way have units of mass (in $\text{eV} \cdot \text{s}^2/\text{cm}^2$ in the "microelectronics" unit system). Because of this, they are called, respectively, the *density of states electron effective mass* and the *density of states hole effective mass*. Three comments should be made about these two new entities.

- First, the values of m^*_{de} and m^*_{dh}, which are semiconductor dependent, are typically given in terms of the electron rest mass m_o. For example, $m^*_{de} = 1.09\, m_o$ and $m^*_{dh} = 1.15\, m_o$ for Si at room temperature (they depend weakly on temperature).
- Second, the term "effective mass" is used with widely different meanings in the semiconductor literature. For example, you are likely to come across a conductivity effective mass, longitudinal or transverse effective masses, light-hole or heavy-hole effective masses, and others. Although closely related, these are all different physical entities and it is important not to mix them. In order to compute the DOS of a semiconductor, the *DOS effective mass* must be used.
- Finally, the DOS effective mass is definition sensitive. Some authors use slightly different expressions for Eqs. (2.20) and (2.21) (a difference of a factor of two is not uncommon). When using the DOS effective mass out of a particular source, it is important to check the expression that is being used by the author to compute the DOS and apply appropriate corrections if needed. Equations (2.20) and (2.21) are the most popular expressions.

Figure 2.11 sketches the DOS for a semiconductor such as Si in which $m^*_{de} < m^*_{dh}$. In this case, at a given distance away from the band edge, the conduction band has fewer states than the valence band.

2.4.2 Equilibrium electron concentration

We now turn our attention to the distribution of electrons in the conduction band in thermal equilibrium. In this section, we not only seek to understand the shape of the electron distribution in the conduction band but also to develop expressions that will give us the total electron concentration in the conduction band as a function of the relative position of the Fermi level with respect to the band edge.

Before we get launched in a detailed mathematical formulation of the problem, it is useful to develop some intuition as to the general dependencies that are to be expected. Assume for a moment that the Fermi level is below the conduction band edge. It is clear that the closer the Fermi level gets to the band edge is, the higher the occupation probability of the conduction band states becomes. Therefore, the total electron concentration increases. Furthermore, if the Fermi level is not too close to the conduction band edge, this relationship has to be exponential since it is the exponential tail of the Fermi–Dirac distribution function that overlaps with the conduction band.

In a situation in which the Fermi level is inside the conduction band, the electron concentration also increases the more the Fermi level penetrates into the band. However, this relationship is much weaker than in the earlier case as a much bigger portion of the Fermi–Dirac distribution function overlaps with the conduction band and not just its exponential tail. The detailed model that is developed in this section will show these dependencies.

The total electron concentration in the conduction band is computed by integrating the electron energy distribution across the entire conduction band. This is mathematically expressed as

$$n_o = \int_{E_c}^{\infty} n_o(E)\, dE \tag{2.22}$$

where $n_o(E)$ represents the concentration of electrons per unit energy (in units of $cm^{-3} \cdot eV^{-1}$) in thermal equilibrium (the subindex "o" is frequently used to denote thermal equilibrium). Note that the integral extends from the bottom of the conduction band all the way up to infinity. This is of course strictly incorrect as the conduction band does not extend that far. The error for not being more precise with the upper integration limit is negligible since significant electron concentration only exists at the bottom of the band.

The calculation of $n_o(E)$ is simple. At a given energy, the electron concentration is obtained by multiplying the conduction band density of states at that energy times the probability that a state located there is occupied by an electron:

$$n_o(E) = g_c(E)\, f(E) \tag{2.23}$$

If we use in Eq. (2.23) the Fermi–Dirac distribution function $f(E)$ [Eq. (1.7)], $g_c(E)$ from Eqs. (2.18) and (2.20), and we substitute everything into Eq. (2.22) we obtain

$$n_o = 4\pi \left(\frac{2m_{de}^*}{h^2}\right)^{3/2} \int_{E_c}^{\infty} \frac{\sqrt{E - E_c}}{1 + \exp \frac{E - E_F}{kT}} dE \tag{2.24}$$

We can rewrite this integral in a more concise form by changing variables in the following way:

$$\eta = \frac{E - E_c}{kT} \tag{2.25}$$

and

$$\eta_c = \frac{E_F - E_c}{kT} \tag{2.26}$$

where η represents the normalized energy with respect to the conduction band edge in units of kT.

After this change of variables, Eq. (2.24) becomes

$$n_o = 4\pi \left(\frac{2m_{de}^* kT}{h^2}\right)^{3/2} \int_0^{\infty} \frac{\sqrt{\eta}}{1 + e^{\eta - \eta_c}} d\eta \tag{2.27}$$

We can further define

$$N_c = 2\left(\frac{2\pi m_{de}^* kT}{h^2}\right)^{3/2} \tag{2.28}$$

N_c is called the *effective density of states of the conduction band*. Its units are cm^{-3}. The physical meaning of this parameter is explored in Problem 2.2.

EXERCISE 2.2

Compute N_c for Si at room temperature.

As in previous exercises, it is essential to be extremely careful with the units of the various fundamental constants. In computing N_c, the first step is to calculate the density of states effective mass for electrons m^*_{de}. From Appendix B, $m^*_{de}/m_o = 1.09$. Then $m^*_{de} = 1.09 \times 5.69 \times 10^{-16}$ eV $\cdot$ s^2/cm^2 $= 6.20 \times 10^{-16}$ eV $\cdot$ s^2/cm^2. We can now plug this in Eq. (2.28) to get

$$N_c = 2\left[\frac{2 \times 3.14 \times 6.20 \times 10^{-16}\ \text{eV} \cdot \text{s}^2/\text{cm}^2 \times 25.9 \times 10^{-3}\ \text{eV}}{(4.14 \times 10^{-15}\ \text{eV} \cdot \text{s})^2}\right]^{3/2} = 2.85 \times 10^{19}\ \text{cm}^{-3}$$

which is fairly close to the currently accepted value of 2.86×10^{19} cm^{-3} that is listed in Appendix B.

With the definition of N_c, Eq. (2.27) can be rewritten as

$$n_o = N_c \mathcal{F}_{1/2}(\eta_c) \tag{2.29}$$

where $\mathcal{F}_{1/2}$ is called the *Fermi–Dirac integral of order 1/2*:

$$\mathcal{F}_{1/2}(x) = \frac{2}{\sqrt{\pi}} \int_0^\infty \frac{\sqrt{\eta}}{1 + e^{\eta - x}} d\eta \tag{2.30}$$

where x is an independent variable.

The function $\mathcal{F}_{1/2}$ has some interesting properties that are summarized in Advanced Topic AT2.2 at the end of this chapter. It is also tabulated in several books for different values of x. A plot of $\mathcal{F}_{1/2}$ is shown in Fig. 2.12.[2]

There are several interesting features to the result captured in Eq. (2.29). This equation only depends on η_c, the normalized Fermi energy with respect to the conduction band edge. Since $\mathcal{F}_{1/2}(x)$ is a monotonically increasing function of x (see Fig. 2.12), the higher the Fermi level is relative to the conduction band edge, the more electrons there are in the conduction band. This is to be expected since the occupation probability of any given state in the conduction band increases the higher the Fermi level lies with respect to it. The increase in n_o with increasing $E_F - E_c$ slows down, however, when the Fermi level penetrates into the conduction band (beyond

[2]In some textbooks, the Fermi–Dirac integral of order 1/2 is defined as $F_{1/2}(x) = \frac{\sqrt{\pi}}{2}\mathcal{F}_{1/2}(x)$. To make matters more confusing, the symbols are not used in a consistent way across the literature. It is important to always pay close attention to the definition of this integral.

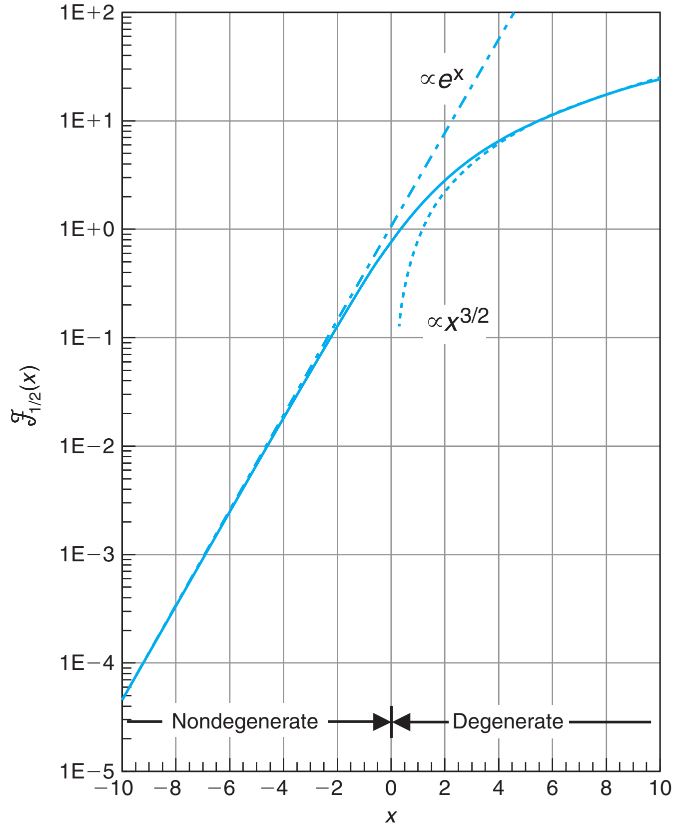


FIGURE 2.12
Fermi–Dirac integral of order 1/2. Also sketched are two simple analytical approximations.

$x = 0$ in Fig. 2.12). When this happens, the semiconductor is referred to as *degenerate*, as opposed to *nondegenerate* as it is called when E_F is below E_c.

With some restrictions in the value of x, it is possible to obtain accurate analytical approximations to $\mathcal{F}_{1/2}(x)$. Some of the most common ones are collected in Advanced Topic AT2.2 at the end of this chapter. The most useful of them is for $x \ll -1$ for which

$$\mathcal{F}_{1/2}(x) \simeq e^x \quad \text{for } x \ll -1 \tag{2.31}$$

This approximation is graphed in Fig. 2.12.

Equation (2.31) is useful because it permits us to write a simple analytical expression for n_o if the semiconductor is sufficiently nondegenerate. That is, if $\eta_c \ll -1$, n_o can be written as

$$n_o \simeq N_c \exp \frac{E_F - E_c}{kT} \tag{2.32}$$

This equation is valid when $E_c - E_F \gg kT$, or in other words, when the Fermi level is far below the conduction band edge (about $3kT$ is usually good enough). When this is the case, it also follows that $n_o \ll N_c$, a handy rule to remember.

An analytical approximation can also be obtained for a strongly degenerate semiconductor. This is also shown in Fig. 2.12. It is only valid if the Fermi level has penetrated substantially inside the conduction band. This approximation is discussed in more detail in Advanced Topic AT2.3.

EXERCISE 2.3

Estimate the location of the Fermi level with respect to the conduction band edge in a sample of Si at room temperature with (a) $n_o = 10^{17}\ cm^{-3}$, (b) $n_o = 10^{20}\ cm^{-3}$.

For case (a), $n_o = 10^{17}\ \text{cm}^{-3} \ll N_c$. We can then use Maxwell–Boltzmann statistics. Solving for $E_F - E_c$ in Eq. (2.32), we get

$$E_F - E_c = kT \ln \frac{n_o}{N_c} = 0.026 \ln \frac{10^{17}}{2.86 \times 10^{19}} = -0.15\ \text{eV}$$

Indeed E_F is below E_c by several kTs, as expected.

For case (b), $n_o > N_c$ and the semiconductor is degenerate. We must use Fermi–Dirac statistics for electrons. Solving for $\mathcal{F}_{1/2}(\eta_c)$, we get

$$\mathcal{F}_{1/2}(\eta_c) = \frac{n_o}{N_c} = \frac{10^{20}}{2.86 \times 10^{19}} = 3.5$$

To first order, we can use Fig. 2.12 to estimate $\eta_c \simeq 2.5$. Hence

$$E_F - E_c = kT\eta_c = 0.026 \times 2.5 = 0.065\ \text{eV}$$

Clearly, E_F has penetrated inside the conduction band, as expected in a degenerate n-type semiconductor.

We can more accurately estimate the location of E_F using Eqs. (2.58) and (2.59) in Advance Topic AT2.2. If we do this, we find $\eta_c = 2.44$, and $E_F - E_c = 63$ meV.

Had we estimated the location of E_F using nondegenerate statistics, as in case (a), we would have obtained $E_F - E_c = 33$ meV, an error of nearly 50%.

Qualitatively, the key difference between the nondegenerate and degenerate regimes lies in the energy distribution of electrons inside the conduction band. This is sketched in Fig. 2.13 (see also Problem 2.11). If the Fermi level is sufficiently below the conduction band edge, only the exponentially decaying high-energy tail of the Fermi distribution penetrates into the conduction band. In this case, few states are occupied and the energy distribution of conduction band electrons also decays exponentially with energy. In fact, one can directly obtain Eq. (2.32) using Maxwell–Boltzmann statistics in Eq. (2.23).

In a strongly degenerate semiconductor, the Fermi level is deep inside the conduction band and many electrons occupy the bottom of the band. As Fig. 2.13 sketches, the energy

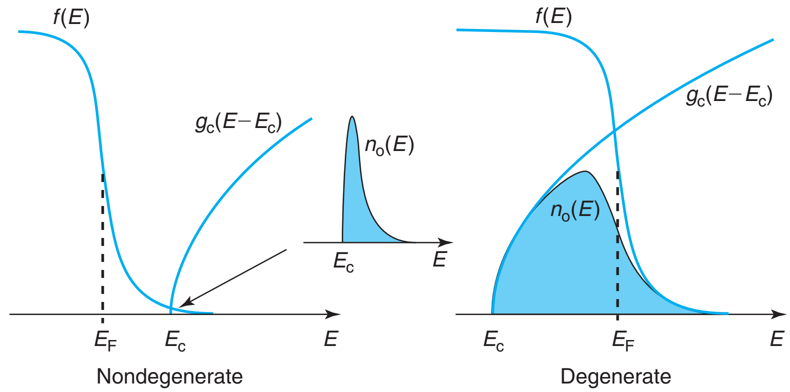


FIGURE 2.13
Sketch of electron distribution in conduction band in nondegenerate case (left) and degenerate case (right).

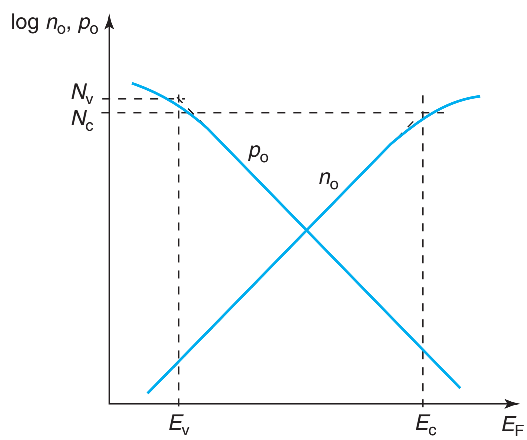


FIGURE 2.14
Sketch of electron and hole concentrations in equilibrium as a function of the relative position of the Fermi level with respect to the band structure.

distribution in this case is rather different from a nondegenerate situation. It has nearly a square-root shape at the bottom of the band with an exponentially decaying shape at higher energies.

Figure 2.14 summarizes graphically the electron concentration in equilibrium in the conduction band as a function of the relative position of the Fermi level with respect to the band edges in a semilog scale. As the Fermi level moves up inside the bandgap, n_o increases exponentially. When E_F penetrates in the conduction band and the semiconductor becomes degenerate, the dependence flattens out.

2.4.3 Equilibrium hole concentration

Similar arguments as those presented in the previous section can be applied to compute the hole concentration in the valence band in equilibrium. This time we must integrate the hole distribution in energy across the entire valence band. For an equilibrium situation,

$$p_o = \int_{-\infty}^{E_v} p_o(E)dE \tag{2.33}$$

p_o is in this case given by the product of the density of states in the valence band times the probability that a given state is *empty*:

$$p_o(E) = g_v(E)\,[1 - f(E)] \tag{2.34}$$

If we proceed as above and define

$$\eta_v = \frac{E_v - E_F}{kT} \tag{2.35}$$

and

$$N_v = 2\left(\frac{2\pi m^*_{dh} kT}{h^2}\right)^{3/2} \tag{2.36}$$

we easily obtain

$$p_o = N_v \mathcal{F}_{1/2}(\eta_v) \tag{2.37}$$

N_v is called the *effective density of states of the valence band* and η_v gives the relative position of the Fermi level with respect to the valence band edge in units of kT. The hole concentration in equilibrium follows a similar relationship to what was obtained above for electrons. In this case, however, the lower the Fermi level with respect to the valence band edge is, the higher will be the hole concentration. N_v in Si at room temperature is 3.10×10^{19} cm^{-3}.

When the Fermi level is well above the valence band edge ($\eta_v \ll -1$, or $E_F - E_v \gg kT$, or $p_o \ll N_v$), the exponential approximation to the Fermi–Dirac integral applies. This allows us to write

$$p_o \simeq N_v \exp\frac{E_v - E_F}{kT} \tag{2.38}$$

This is a case in which the hole distribution is *nondegenerate*. An analytical approximation can also be obtained for a strongly degenerate p-type semiconductor. This is shown in Advanced Topic AT2.3. The hole distribution in the valence band is sketched in Fig. 2.15 for these two cases.

Figure 2.14 illustrates the dependence of p_o on the Fermi level position with respect to the band structure. As E_F moves up inside the bandgap, p_o drops exponentially. If the E_F is close or inside the valence band, the dependence of p_o on E_F is softer.

Figure 2.16 offers a summary of the proper statistics that are to be used depending on the position of E_F with respect to the bands. If the Fermi level sits anywhere inside the forbidden gap, both electron and hole distributions are nondegenerate and Eqs. (2.32) and (2.38) are

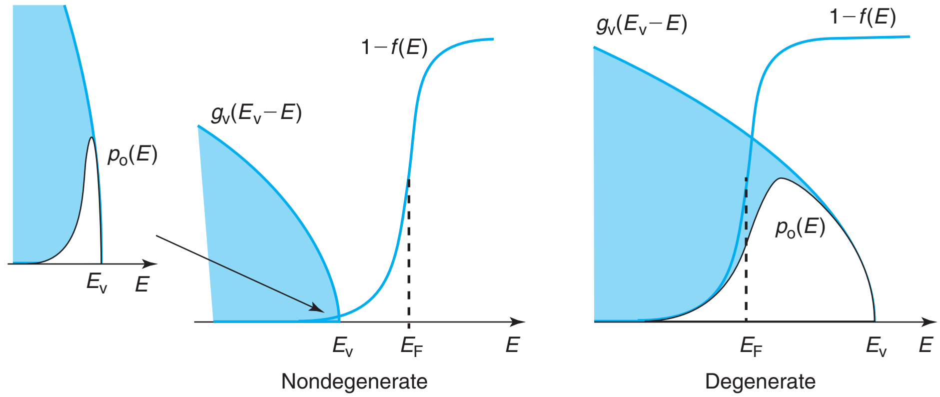


FIGURE 2.15
Sketch of hole distribution in valence band in nondegenerate (left) and degenerate case (right).

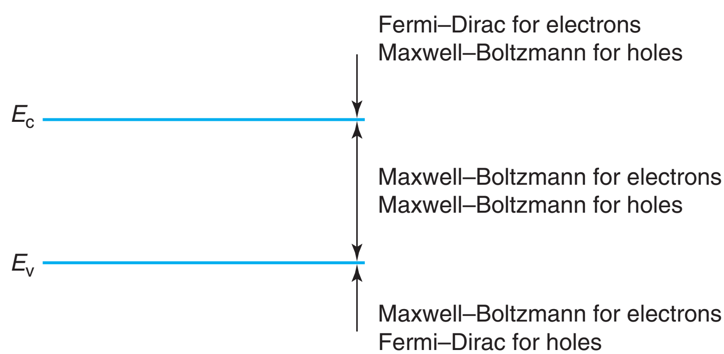


FIGURE 2.16
Summary of statistics for electrons and holes depending on the position of the Fermi level with respect to the band structure.

reasonably accurate. If the Fermi level penetrates inside anyone of the bands, then the semiconductor is degenerate. To be more specific, if the Fermi level is located inside the conduction band, then the electrons are degenerate but the holes are not. Fermi–Dirac statistics are necessary to describe the electrons but Maxwell–Boltzmann can be used for the holes. In this case, the exact expression (2.29) must be used. If the electrons are sufficiently degenerate, the strongly degenerate approximation 2.60 in Advanced Topic AT2.3 applies. If, conversely, the Fermi level has penetrated inside the valence band, the holes are degenerate but the electrons are not. A similar treatment applies when the Fermi level is in the vicinity or inside the valence band with the role of electrons and holes interchanged.

N_c and N_v represent practical benchmarks for the equilibrium carrier concentrations of electrons and holes in assessing whether the semiconductor is degenerate or not. Appendix B

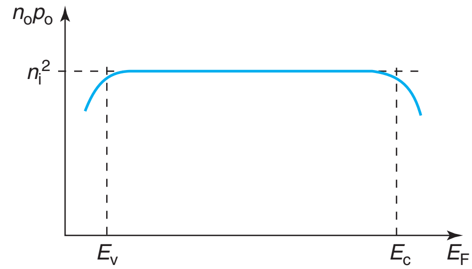


FIGURE 2.17
Sketch of $n_o p_o$ as a function of the position of the Fermi level with respect to the band structure.

collects the values for Si and GaAs at room temperature.[3] It is important to note that N_c and N_v increase with temperature in the form $\sim T^{3/2}$. The origin of this dependence is explored in Problem 2.2.

2.4.4 *np* product in equilibrium

Now that we have derived relationships between n_o and p_o and the relative location of the Fermi level with respect to the semiconductor band structure, we are in a position to derive an expression for the electron–hole product in equilibrium. Multiplying Eqs. (2.29) and (2.37), we get

$$n_o p_o = N_c N_v \mathcal{F}_{1/2}(\eta_c)\mathcal{F}_{1/2}(\eta_v) \tag{2.39}$$

This equation is sketched in Fig. 2.17 as a function of the Fermi level position with respect to the band structure. Interestingly, for a broad range of Fermi level positions inside the bandgap, $n_o p_o$ is constant. This makes sense. As Fig. 2.14 illustrates, as E_F moves through the bandgap, n_o increases at the same rate at which p_o decreases. The product of these two parameters is then constant independent of the location of the Fermi level. It is easy to find an expression for this constant value by noting that in this intermediate region, both electrons and holes are nondegenerate and Maxwell–Boltzmann statistics apply for both carriers (see Fig. 2.16). Using Eqs. (2.32) and (2.38), we get

$$n_o p_o \simeq N_c N_v \exp\left(-\frac{E_g}{kT}\right) \tag{2.40}$$

This equation is of an identical form to that of Eq. (2.6) derived using the chemical reaction arguments.

E_g, N_c, and N_v are physical constants characteristic of each semiconductor that depend only on temperature. Equation (2.40) then indeed confirms our intuitive understanding that for a given semiconductor at a given temperature, the equilibrium *np* product is a constant. Our rigorous treatment has shown, however, that this is only the case when neither the electron nor the

[3] In spite of their importance, there is still some degree of controversy about the numerical values of N_c and N_v in Si and GaAs. Different books quote different numbers. For these and other physical parameters, the values collected in this book are the most accurate in the judgment of the author.

hole distribution is degenerate. In other words, $n_o p_o$ is a constant in the "dilute" regime where the electron and hole concentrations are not too high. We will come back to this at the end of this section.

In an *intrinsic* semiconductor, Eq. (2.40) is very likely to apply under a broad temperature range since the carrier concentrations tend to be relatively low. Hence, the intrinsic carrier concentration is fairly accurately given by

$$n_i = \sqrt{N_c N_v} \exp\left(-\frac{E_g}{2kT}\right) \tag{2.41}$$

This equation is consistent with our intuitive understanding of the meaning of n_i, that is, the concentration of electrons and holes that a perfectly pure piece of semiconductor has in equilibrium at a certain temperature. In addition to the exponential dependence of n_i on E_g, already discussed in the previous section, we also see in Eq. (2.41) that n_i is proportional to N_c and N_v. This makes sense since the higher the effective density of states is, the more states there are closer to the band edges and the higher the resulting carrier concentrations at a certain temperature.

Using Eq. (2.41) is only valid when both electrons and holes can be described by Maxwell–Boltzmann statistics. This is a very good assumption for most intrinsic semiconductors. Only for semiconductors with very small bandgaps at relatively high temperatures might we need to be concerned with the accuracy of this expression. Problem 2.3 explores this issue.

EXERCISE 2.4

Compute n_i for Si at room temperature.

Appendix B lists all the needed parameters for Si at room temperature. Since E_g, k, and T all appear inside an exponential, it is important to have at least three significant digits in each of these parameters in order to have acceptable accuracy in the calculation of n_i. Using Eq. (2.41):

$$n_i = \sqrt{2.86 \times 10^{19}\ \text{cm}^{-3} \times 3.10 \times 10^{19}\ \text{cm}^{-3}} \exp\left(-\frac{1.124\ \text{eV}}{2 \times 25.9 \times 10^{-3}\ \text{eV}}\right) = 1.12 \times 10^{10}\ \text{cm}^{-3}$$

This value is fairly close to the conventionally accepted value of 1.07×10^{10} cm^{-3} listed in Appendix B. For "back-of-the-envelope" calculations, a value of 10^{10} cm^{-3} is adequate.

Coming back to Eq. (2.40), its validity does not extend to the degenerate regime (whether n- or p-type). It is reasonable that this be the case. As discussed above, when there are relatively few electrons and holes in a semiconductor, the energy required to form an extra electron–hole pair is E_g. If, say, the equilibrium electron concentration is so high that the Fermi level has penetrated inside the conduction band, the energy required to form an electron–hole pair is larger than E_g. This is because the bottom of the conduction band is full and empty states where electrons can be placed are only available at higher energies inside the band. As a result, the *np* product in equilibrium is reduced for high electron concentrations. The same happens for high hole concentrations.

An important consequence of this is that Eqs. (2.15) and (2.17) do not apply in the degenerate regime as in this case $n_o p_o < n_i^2$. In this instance, the computation of the minority carrier concentrations are a bit more elaborate since the proper Fermi–Dirac statistics must be used for the majority carriers. Exercise 2.5 illustrates how this can be done.

EXERCISE 2.5

Estimate the equilibrium hole concentration for the two cases of Exercise 2.3.

In case (a), $n_o = 10^{17}$ cm^{-3}, which is a nondegenerate situation. Then

$$p_o = \frac{n_i^2}{n_o} = \frac{1.1 \times 10^{20}}{10^{17}} = 1.1 \times 10^3 \text{ cm}^{-3}$$

In case (b), $n_o = 10^{20}$ cm^{-3}. This is a degenerate situation and we cannot use $n_o p_o = n_i^2$. In Exercise 2.3, we computed $E_F - E_c = 63$ meV for this case. This suggests that

$$E_F - E_v = E_F - E_c + E_c - E_v = E_F - E_c + E_g = 0.063 + 1.124 = 1.187 \text{ eV}$$

Since the holes are nondegenerate, we can compute p_o in the following way:

$$p_o = N_v \exp \frac{E_v - E_F}{kT} = 3.1 \times 10^{19} \exp \frac{-1.187}{0.0259} = 0.39 \text{ cm}^{-3}$$

Had we used simple nondegenerate statistics, as in case (a), the result would have been 1.1 cm^{-3} nearly a factor of three off.

In this text, we do not place a lot of emphasis on the computation of minority carrier concentrations that accounts for the need for Fermi–Dirac statistics for the majority carriers because, in reality, the situation for semiconductors with high carrier concentrations is all together more complex. In a heavily doped semiconductor, a number of special "heavy-doping effects" actually occur. They make the computation of carrier concentrations even more complicated. This is discussed in detail in Advanced Topic AT2.6. Hence, in the main body of this book, Eqs. (2.15) and (2.17) will be used regardless of the doping level. Appropriate mention will be made in cases where the errors might be significant and the special treatment of Advanced Topic AT2.6 is required.

2.4.5 Location of Fermi level

In a semiconductor in thermal equilibrium, the location of the Fermi level with respect to the band edges completely characterizes the carrier concentrations. Hence, indicating the position of the Fermi level in an energy band diagram is a simple way to represent electron and hole

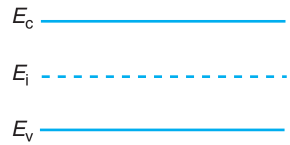


FIGURE 2.18
Sketch of the position of the Fermi level in an intrinsic semiconductor.

concentrations. We obtain this by invoking charge neutrality. Let us distinguish between intrinsic and extrinsic situations.

Intrinsic semiconductor

It is easy to find the location of the Fermi energy in an intrinsic semiconductor. Assuming that Maxwell–Boltzmann statistics apply for both electrons and holes, n_o and p_o obey the simple relationship given by Eqs. (2.32) and (2.38). Furthermore, charge neutrality in an intrinsic semiconductor implies that $n_o = p_o$. Equating them and solving for the Fermi level, we find

$$E_i = \frac{E_c + E_v}{2} + kT \ln \sqrt{\frac{N_v}{N_c}} \tag{2.42}$$

where the Fermi level in this intrinsic situation has been relabeled E_i or *intrinsic Fermi level*.

The first term in Eq. (2.42) is precisely the middle of the bandgap. The second term adds a correction to it. The magnitude of the correction depends on the degree to which N_c and N_v differ from each other. For a semiconductor in which N_c was precisely identical to N_v, then E_i would sit exactly at the middle of the forbidden gap. This makes sense as this is the way to ensure that there are identical numbers of electrons and holes in equilibrium. If N_v is bigger than N_c, as is the case for Si, then equality of electron and hole concentrations forces E_i to be above the middle of the gap to compensate for the fact that there are more states at the top of the valence band than at the bottom of the conduction band (see exercise below). The logarithm in the second term of Eq. (2.42) makes this correction in general very small and graphically, E_i is nearly always close to the middle of the gap, as sketched in Fig. 2.18.

EXERCISE 2.6

Compute E_i in Si at 300 K.

Using Eq. (2.42):

$$kT \ln \sqrt{\frac{N_v}{N_c}} = 25.9 \text{ meV} \ln \sqrt{\frac{3.1 \times 10^{19} \text{ cm}^{-3}}{2.9 \times 10^{19} \text{ cm}^{-3}}} = 1 \text{ meV}$$

E_i is just 1 meV above the middle of the gap in Si at room temperature.

Extrinsic semiconductor

For a sufficiently extrinsic n-type nondegenerate semiconductor with all dopants ionized, charge neutrality implies Eq. (2.14). Also, Maxwell–Boltzmann statistics and Eq. (2.32) apply. Equating these two relationships, we get

$$E_F - E_c = kT \ln \frac{N_D}{N_c} \tag{2.43}$$

If needed, it is straightforward to refer E_F to the valence band edge. The logarithmic relationship between $E_F - E_c$ and N_D in Eq. (2.43) is a result of the fact that in a moderately doped n-type semiconductor, the electrons are nondegenerate, and as a consequence, they are well described by Maxwell–Boltzmann statistics.

Similarly for a nondegenerate p-type semiconductor:

$$E_F - E_v = kT \ln \frac{N_v}{N_A} \tag{2.44}$$

which can easily be referred to the edge of the conduction band if required. These equations can also be used for partially compensated cases in which one type of doping overwhelms the other by substituting N_d for N_D and N_a for N_A (see Problem 2.9).

Figure 2.19 sketches the position of the Fermi level as a function of $\log N_D$. As N_D increases, E_F approaches the conduction band edge in a logarithmic manner following Eq. (2.43). For low-doping levels, the Fermi level eventually converges to E_i as the semiconductor becomes intrinsic and Eq. (2.42) has to be used. For high-doping levels, the semiconductor becomes degenerate and Eq. (2.43) also fails. To compute E_F in this case, one has to use the exact expression that accounts for Fermi–Dirac statistics, as was done in Exercise 2.3. In spite of these limitations, for n-type Si at around room temperature, Eq. (2.43) (and also Eq. (2.44) for p-type Si) is fairly accurate over 7–8 orders of magnitude of doping concentration (from $\sim 10^{11}$ to $\sim 10^{18}$ cm^{-3}).

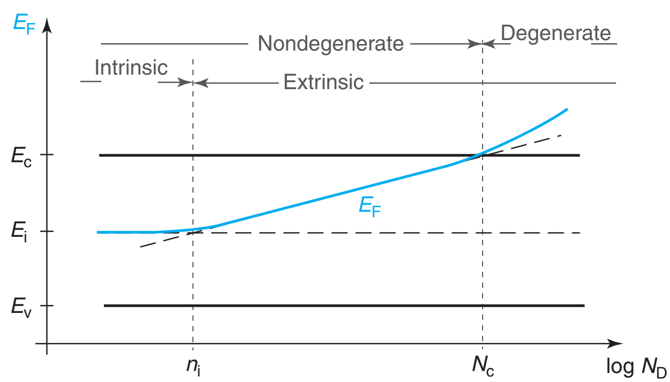


FIGURE 2.19
Sketch of the position of the Fermi level in an n-type semiconductor as a function of the doping level. Note abscissa is in a logarithmic scale.

EXERCISE 2.7

Compute the equilibrium carrier concentrations and the position of the Fermi level with respect to the conduction band edge for a uniformly doped Si region with $N_D = 1 \times 10^{16}$ cm^{-3} *and* $N_A = 1 \times 10^{17}$ cm^{-3} *at room temperature.*

This is a case of partial compensation. Since $N_A \gg N_D$, we can define a net acceptor concentration $N_a = N_A - N_D = 9 \times 10^{16}$ cm^{-3}. Using N_a instead of N_A in Eq. (2.16) we find that

$$p_o \simeq 9 \times 10^{16}\ \text{cm}^{-3}$$

Since $p_o \ll N_v$, Eq. (2.17) can be used to compute the equilibrium electron concentration:

$$n_o \simeq \frac{(1.07 \times 10^{10}\ \text{cm}^{-3})^2}{9 \times 10^{16}\ \text{cm}^{-3}} = 1.3 \times 10^3\ \text{cm}^{-3}$$

The position of the Fermi level with respect to the top of the valence band is obtained from Eq. (2.44):

$$E_F - E_v = 0.0259\ \text{eV} \ln \frac{3.1 \times 10^{19}\ \text{cm}^{-3}}{9 \times 10^{16}\ \text{cm}^{-3}} = 0.151\ \text{eV}$$

The position of E_F with respect to E_c is simply obtained by subtracting E_g:

$$E_F - E_c = E_F - E_v - E_g = 0.151 - 1.124 = -0.97\ \text{eV}$$

Since the value of n_o has already been calculated, we could have also used Eq. (2.32) directly

$$E_F - E_c = 0.0259\ \text{eV} \ln \frac{1.3 \times 10^3\ \text{cm}^{-3}}{2.9 \times 10^{19}\ \text{cm}^{-3}} = -0.97\ \text{eV}$$

The high-doping regime in which the semiconductor is degenerate or nearly degenerate is of very high practical importance. Degenerately doped n- and p-type regions are pervasive in modern Si and GaAs microelectronic devices. It is not possible, for example, to correctly predict the current gain of a bipolar transistor without adequately dealing with heavy-doping effects. In spite of the need to use Fermi–Dirac statistics for the majority carriers, modeling heavily doped regions is relatively simple at room temperature for many semiconductors. Advanced Topic AT2.6 discusses the idiosyncrasies of the high-doping regime and derives expressions for the equilibrium carrier concentrations and the position of the Fermi level in the band structure. As a simple rule, for Si at room temperature, if the carrier concentration exceeds about 10^{19} cm^{-3}, heavy-doping effects are important.

2.5 SUMMARY

- In a semiconductor, the equilibrium carrier concentrations are related to the location of the Fermi energy with respect to the band structure through:

$$n_o = N_c \mathcal{F}_{1/2}\left(\frac{E_F - E_c}{kT}\right)$$

$$p_o = N_v \mathcal{F}_{1/2}\left(\frac{E_v - E_F}{kT}\right)$$

- For the case in which the Fermi level does not get too close to either band edge (nondegenerate regime), the equilibrium carrier concentrations are related to the Fermi level location through simple exponential relations:

$$n_o \simeq N_c \exp \frac{E_F - E_c}{kT}$$

$$p_o \simeq N_v \exp \frac{E_v - E_F}{kT}$$

- For a given semiconductor, in the nondegenerate regime, the $n_o p_o$ product depends only on temperature:

$$n_o p_o \simeq N_c N_v \exp\left(-\frac{E_g}{kT}\right) = n_i^2$$

- Dopants preferentially introduce electrons or holes into a semiconductor.
- At around room temperature, under most practical circumstances, the majority carrier concentrations are set by the doping level and are independent of temperature. For an n-type semiconductor,

$$n_o \simeq N_D$$

  For a p-type semiconductor,

$$p_o \simeq N_A$$

- At low temperatures, carrier freeze-out occurs. At high temperatures, intrinsic production of electron–hole pairs eventually overwhelms the doping level.

2.6 FURTHER READING

There are several excellent books that discuss in considerable detail the statistics of electrons and holes in equilibrium.

Semiconductor Statistics by J. S. Blakemore, Dover, 1978 (ISBN 0-486-65362-5, QC611.B52). Chapters 2 and 3 of this book contain a rather comprehensive discussion of the statistics of electrons and holes in semiconductors in equilibrium in a variety of circumstances: intrinsic semiconductor, degenerate intrinsic semiconductors, as well as semiconductors with single and multiple impurity states. This text also has an appendix that describes several properties of the Fermi–Dirac integrals, presents analytical approximations, and includes tables of values for selected integrals.

Solid State and Semiconductor Physics by J. P. McKelvey, Krieger, 1986 (ISBN 0-89874-396-6, QC611.M495). Chapter 9 of this classic book contains a few sections with a treatment of the statistics of electrons and holes that is similar to the one presented here. This book is characterized by a good compromise between mathematical complexity and physical intuition.

AT2.1 TEMPERATURE DEPENDENCE OF THE BANDGAP

The bandgap of a semiconductor decreases with temperature. This is a consequence of the expansion of the lattice constant as the temperature increases. This is shown for Si and GaAs in Fig. 2.20. The dependence is accurately captured by an equation of the form:

$$E_g(T) = E_g(0) - \frac{\alpha T^2}{T + \beta} \tag{2.45}$$

where $E_g(0)$ is the bandgap at 0 K and α and β are constants. The value of these constants for Si and GaAs is given in Appendix E.

The dependence of E_g with temperature is not very strong. It is important to consider it when computing expressions that contain the bandgap inside an exponential, as the next exercise shows.

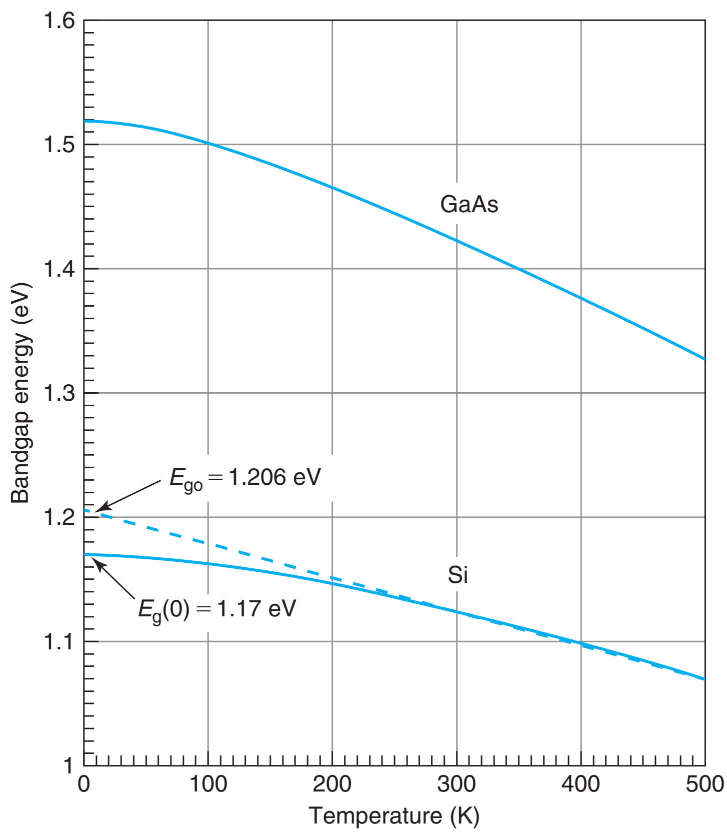


FIGURE 2.20
Bandgap energy versus temperature for Si and GaAs.

EXERCISE 2.8

Compute n_i for Si at 600 °C.

The easiest way to compute n_i at any temperature is to use its value at 300 K since this is usually relatively well known. First, let's make explicit all temperature dependencies of n_i. From Eqs. (2.41), (2.28), and (2.36), we can rewrite n_i as

$$n_i = CT^{3/2} \exp\left[-\frac{E_g(T)}{2kT}\right]$$

where C is a constant independent of temperature.

The ratio of n_i at two different temperatures is then

$$\frac{n_i(T_1)}{n_i(T_2)} = \left(\frac{T_1}{T_2}\right)^{3/2} \exp\left\{-\frac{1}{2k}\left[\frac{E_g(T_1)}{T_1} - \frac{E_g(T_2)}{T_2}\right]\right\}$$

We can now plug in numbers for $T_1 = 873$ K (600 °C) and $T_2 = 300$ K:

$$n_i(873\,\mathrm{K}) = 1.07 \times 10^{10} \times \left(\frac{873}{300}\right)^{3/2} \exp\left[-\frac{1}{2 \times 8.62 \times 10^{-5}}\left(\frac{0.950}{873} - \frac{1.124}{300}\right)\right] = 2.6 \times 10^{17}\,\mathrm{cm}^{-3}$$

This is an interesting result. It shows that a Si region doped at the 10^{17} cm^{-3} level becomes intrinsic at around 600 °C. It is easy to see that not accounting for the temperature dependence of E_g would have resulted in an error of about a factor of three.

Above 250 K, the bandgap of Si can be described to within experimental accuracy (about 1 meV) by the simpler expression:

$$E_g(T) = E_{go} - CT \tag{2.46}$$

where E_{go} is the *extrapolation* of E_g to 0 K and C is a constant (see Fig. 2.20). For Si, $E_{go} =$ 1.206 eV and $C = 2.73 \times 10^{-4}$ eV $\cdot$ K^{-1}. This is widely used around room temperature.

It is important not to confuse E_{go} with $E_g(0)$, the true bandgap at 0 K. Interestingly, E_{go} emerges frequently when carrying out experimental temperature-dependent studies of transport processes around room temperature. The reason for this is that the factor $exp(-E_g/kT)$ that is pervasive in these types of processes has an activation energy around room temperature that is precisely E_{go}. This can be easily seen by using Eq. (2.46):

$$\exp\left(-\frac{E_g}{kT}\right) \simeq \exp\left(\frac{C}{k}\right) \cdot \exp\left(-\frac{E_{go}}{kT}\right) \tag{2.47}$$

A good example of this is the data for n_i in Si of Fig. 2.5 that, in fact, shows an activation energy very close to $E_{go}/2$ (see Problem 2.4). E_{go} frequently appears in the context of the so-called *bandgap voltage reference circuits*, circuits that generate a fixed voltage regardless of temperature changes, power supply variations, or circuit loading.

AT2.2 SELECTED PROPERTIES OF THE FERMI–DIRAC INTEGRAL

The Fermi–Dirac integral of order 1/2 belongs to a family of integrals called the *Fermi–Dirac integrals* that are defined as

$$\mathcal{F}_j(x) = \frac{1}{\Gamma(j+1)} \int_0^\infty \frac{\eta^j}{1+e^{\eta-x}} d\eta \tag{2.48}$$

where Γ is the Gamma function discussed in more detail in Advanced Topic AT4.1.

A useful property of the Fermi–Dirac integrals is this

$$\frac{d}{dx}\mathcal{F}_j(x) = \mathcal{F}_{j-1}(x) \tag{2.49}$$

Other general properties of this class of integrals are described in several books (see, for example Blakemore's text cited above). Here we are almost exclusively interested in the Fermi–Dirac integral of order 1/2 for which

$$\Gamma\left(\frac{3}{2}\right) = \frac{\sqrt{\pi}}{2} \tag{2.50}$$

$\mathcal{F}_{1/2}(x)$ has been tabulated for $-4 \le x \le 10$ by Blakemore. Useful approximations are

$$\mathcal{F}_{1/2}(x) \simeq e^x \qquad \text{for } x \ll -1 \tag{2.51}$$

$$\mathcal{F}_{1/2}(x) \simeq \frac{4}{3\sqrt{\pi}} x^{3/2} \qquad \text{for } x \gg 1 \tag{2.52}$$

These are particular expressions of the more general relationships:

$$\mathcal{F}_{\mathrm{j}}(x) \simeq e^x \qquad \text{for } x \ll -1 \tag{2.53}$$

$$\mathcal{F}_{\mathrm{j}}(x) \simeq \frac{x^{j+1}}{(j+1)\Gamma(j+1)} \qquad \text{for } x \gg 1 \tag{2.54}$$

Many numerical approximations have been developed for the Fermi–Dirac integrals. For the interested reader, a review of several of them was presented by J. S. Blakemore in *Solid-State Electronics* **25**, p. 1067 (1982). The following expression, for example, has an error smaller than 0.4% over a range $-10 \le x \le +25$, which is sufficient for most applications:

$$\mathcal{F}_{1/2}(x) \simeq \frac{1}{e^{-x} + \xi(x)} \tag{2.55}$$

$$\xi(x) = \frac{3\sqrt{\pi}}{4\nu^{3/8}(x)} \tag{2.56}$$

$$\nu(x) = x^4 + 50 + 33.6\,x\,\{1 - 0.68\,\exp[-0.17(x+1)^2]\} \tag{2.57}$$

When the carrier concentration is known and the position of the Fermi energy is to be calculated, the inverse function is needed. An approximation that has an error smaller than 1% for all values of x up to $x = 20$ is

$$u = \mathcal{F}_{1/2}(x) \tag{2.58}$$

$$x \simeq \ln u + u\left[64 + 0.05524\, u(64 + \sqrt{u})\right]^{-1/4} \tag{2.59}$$

AT2.3 APPROXIMATIONS FOR STRONGLY DEGENERATE SEMICONDUCTOR

In a degenerate semiconductor, the Fermi level is located inside one of the bands. In these circumstances, the Maxwell–Boltzmann approximation to the Fermi–Dirac integral leads to a very large error (see Fig. 2.12). A better analytical approximation is given in Advanced Topic AT2.2 in Eq. (2.52) that is valid for a sufficiently degenerate semiconductor. Using this approximation, for an n-type semiconductor, we can write

$$n_o \simeq \frac{4}{3\sqrt{\pi}} N_c \left(\frac{E_F - E_c}{kT}\right)^{3/2} \tag{2.60}$$

This equation is acceptable if $\eta_c \gg 1$, $E_F - E_c \gg kT$ or $n_o \gg N_c$.

Similarly, for a sufficiently degenerate p-type semiconductor, we can write

$$p_o \simeq \frac{4}{3\sqrt{\pi}} N_v \left(\frac{E_v - E_F}{kT}\right)^{3/2} \tag{2.61}$$

valid if $\eta_v \gg 1$, $E_v - E_F \gg kT$, or $p_o \gg N_v$.

In many circumstances, the intermediate doping regime is of interest. For this, more accurate approximations are needed, such as those given in Advanced Topic AT2.2.

Notice that in Eqs. (2.60) and (2.61), the temperature dependence of N_c and N_v cancels out the $T^{-3/2}$ inside the brackets. The relationship between the majority carrier concentrations and the Fermi level is temperature independent. In fact, these equations can be easily obtained using the Fermi–Dirac distribution function in Eq. (2.23) at 0 K (see Problem (2.15) at end of the chapter).

AT2.4 STATISTICS OF DONOR AND ACCEPTOR IONIZATION

A donor atom in a semiconductor can either be ionized or neutral. The donor is ionized when it has released its electron to the conduction band. It is neutral if it still holds on to it. Defining N_D as the total donor concentration, N_D^+ as the ionized donor concentration, and N_D^o as the neutral donor concentration, mass conservation demands that

$$N_D = N_D^+ + N_D^o \tag{2.62}$$

The relative number of neutral and ionized donors is in general a function of the donor ionization energy, the location of the Fermi level and temperature. If we define $f_d(E_D)$ as the

Table 2.1
Ionization energy, impurity degeneracy factor, and Mott transition of common dopants in Si and GaAs. The Mott concentration is not very well known in all cases.

Semiconductor	Dopant	E_d (meV)	β_d	E_a (meV)	β_a	N_{Mott} (cm^{-3})
Si	As	54	2	-	-	6.4×10^{18}
	P	45	2	-	-	3.5×10^{18}
	Sb	39	2	-	-	3.0×10^{18}
	B	-	-	45	4	$\sim 4 \times 10^{18}$
GaAs	C	5.9	2	26	4	$\sim 1 \times 10^{17}$
	Si	5.8	2	35	4	$\sim 2 \times 10^{17}$
	Be	-	-	28	4	$\sim 2 \times 10^{18}$
	Zn	-	-	31	4	$\sim 3 \times 10^{18}$

probability that the donor energy level is occupied by an electron, the neutral donor concentration can be written as

$$N_{\mathrm{D}}^{o} = N_{\mathrm{D}} f_{\mathrm{d}}(E_{\mathrm{D}}) \tag{2.63}$$

One might think that the occupation probability of the donor atom is equal to the Fermi–Dirac distribution function. However, such an approximation does not account for the fact that the fifth electron can become bound to the donor atom in more than one way. This is called *impurity level degeneracy*. When this happens, statistical mechanics tells us that the probability of occupation of the donor state becomes

$$f_{\mathrm{d}}(E_{\mathrm{D}}) = \frac{1}{1 + \frac{1}{\beta_{\mathrm{d}}} \exp \dfrac{E_{\mathrm{D}} - E_{\mathrm{F}}}{kT}} \tag{2.64}$$

where β_{d} is the *donor degeneracy factor*, which is greater than or equal to unity.[4] β_{d} depends on the host semiconductor and the type of dopant (p or n), but within a given type, it is independent of the specific dopant itself. Table 2.1 summarizes donor ionization energies for typical donors in Si and GaAs.

Using Eq. (2.64) in Eq. (2.63), the neutral donor concentration is

$$N_{D}^{o} = \frac{N_{\mathrm{D}}}{1 + \frac{1}{\beta_{\mathrm{d}}} \exp \dfrac{E_{\mathrm{D}} - E_{\mathrm{F}}}{kT}} \tag{2.65}$$

The ionized donor concentration is, from Eq. (2.62):

$$N_{\mathrm{D}}^{+} = N_{\mathrm{D}} - N_{\mathrm{D}}^{o} = \frac{N_{\mathrm{D}}}{1 + \beta_{\mathrm{d}} \exp \dfrac{E_{\mathrm{F}} - E_{\mathrm{D}}}{kT}} \tag{2.66}$$

[4] The technical literature is ambiguous in the definition of impurity level degeneracy. Sometimes, β_{d} is defined so as to be smaller than or equal to one. In this case, β_{d} appears in place of the $1/\beta_{\mathrm{d}}$ factor in Eq. (2.64). This is a point where one must be careful.

Similar arguments apply for acceptors. When an acceptor captures an electron, it becomes negatively charged. Defining the neutral acceptor concentration as N_A^o and the ionized acceptor concentration as N_A^-, mass conservation again implies

$$N_A = N_A^- + N_A^o \tag{2.67}$$

In this case

$$N_A^- = N_A f_a(E_A) \tag{2.68}$$

where $f_a(E_A)$ is the occupation probability of the acceptor state. Acceptor states are also typically degenerate. As a result, $f_a(E_A)$ is given by

$$f_a(E_A) = \frac{1}{1 + \beta_a \exp \dfrac{E_A - E_F}{kT}} \tag{2.69}$$

where β_a is the *acceptor degeneracy factor* ($\beta_a \geq 1$). Values of β_a for several impurities in Si and GaAs are tabulated in Table 2.1.

In a similar way to the case of donors, using Eq. (2.69), the ionized and neutral acceptor concentrations are, respectively, given by

$$N_A^- = \frac{N_A}{1 + \beta_a \exp \dfrac{E_A - E_F}{kT}} \tag{2.70}$$

$$N_A^o = \frac{N_A}{1 + \frac{1}{\beta_a} \exp \dfrac{E_F - E_A}{kT}} \tag{2.71}$$

In order to compute the actual ionization ratio of dopants, one must know the location of the Fermi energy. This has to be calculated in a self-consistent way. Advanced Topic AT2.5 shows how to do this in a specific case.

AT2.5 CARRIER FREEZE-OUT

The energy required to ionize a dopant is small but finite. At low enough temperatures, the small thermal energy available precludes many dopants from becoming ionized. As a consequence, the carrier concentration drops with temperature. This is called *carrier freeze-out.*

The results obtained in Advanced Topic AT2.4 allow us to compute the carrier concentration as a function of temperature in the bulk of a semiconductor under charge neutrality conditions. Let us do it for an n-type semiconductor doped with a concentration N_D of donors (a similar approach can be used with a p-type semiconductor). The electron concentration in the bulk is equal to the ionized donor concentration:

$$n_o \simeq N_D^+ \tag{2.72}$$

In writing this equation, we have neglected the presence of a significant amount of electrons that arise from the natural break up of crystal bonds. This is because the energy required for this is E_g, many times larger than the typical donor ionization energy. As the temperature drops, bond break up freezes out much more quickly than donor ionization.

If we can assume Maxwell–Boltzmann statistics for the electrons in the conduction band and use the definition of E_d made in Eq. (2.9), Eq. (2.66) can be rewritten as

$$N_D^+ = \frac{N_D}{1 + \frac{\beta_d n_o}{N_c} \exp \frac{E_d}{kT}} \tag{2.73}$$

Plugging into Eq. (2.72), one obtains a quadratic equation in n_o that can easily be solved:

$$n_o = \frac{N_c}{2\beta_d} \exp\left(-\frac{E_d}{kT}\right)\left(\sqrt{1 + 4\beta_d \frac{N_D}{N_c} \exp \frac{E_d}{kT}} - 1\right) \tag{2.74}$$

In the limit of high temperature, $kT \gg E_d$, the second term inside the square root is small and a simple Taylor series expansion leads to the result $n_o \simeq N_D$ already derived above.

For low enough temperatures, $kT \ll E_d$, and the second term inside the square root overwhelms the other two inside the brackets. We then obtain

$$n_o \simeq \sqrt{\frac{N_D N_c}{\beta_d}} \exp\left(-\frac{E_d}{2kT}\right) \tag{2.75}$$

This result shows that the electron concentration at low temperatures is thermally activated with an activation energy of half of the donor ionization energy. This was plotted in Fig. 2.9 above.

In many situations, it is sufficient to compute the temperature at which freeze-out effects become significant. A good working criteria is when the two terms inside the square root in Eq. (2.74) are equal, that is

$$T_{fo} \simeq \frac{E_d}{k \ln \frac{N_c}{4\beta_d N_D}} \tag{2.76}$$

EXERCISE 2.9

Estimate the temperature at which freeze-out effects become significant for phosphorus-doped Si with $N_D = 5 \times 10^{17}$ cm^{-3}. Verify all your assumptions.

There is a small difficulty here because in order to use Eq. (2.76), we need to calculate N_c at T_{fo}, before T_{fo} is known. This situation can be resolved by writing a small computer program or using a calculator in an iterative way. In either case, N_c at T_{fo} is easily calculated starting from its room temperature value:

$$N_c(T_{fo}) = N_c(300\text{ K})\left(\frac{T_{fo}}{300}\right)^{3/2} = 2.86 \times 10^{19} \left(\frac{T_{fo}}{300}\right)^{3/2} \text{ cm}^{-3}$$

We now use Eq. (2.76):

$$T_{fo} \simeq \frac{0.045}{8.62 \times 10^{-5} \ln \dfrac{N_C(T_{fo})}{4 \times 2 \times 5 \times 10^{17}}} K$$

Solving this system of two equations, we find $T_{fo} = 280$ K, and $N_c(T_{fo}) = 2.6 \times 10^{19}$ cm^{-3}. Since at 280 K, $N_D \ll N_c$, and $n_o < N_D$, the assumption of Maxwell–Boltzmann statistics that is built into Eq. (2.76) is valid in this example. In order to verify the negligible impact of the hole concentration in Eq. (2.72), we must calculate n_o and p_o and show that $p_o \ll n_o$. This is done in the next exercise.

EXERCISE 2.10

Calculate the equilibrium electron and hole concentrations at 77 K *in a sample of P-doped Si with* $N_D = 5 \times 10^{17}$ *cm*$^{-3}$. *Verify all your assumptions.*

In the previous exercise, we estimated that freeze-out effects are significant for temperatures lower than about 280 K. We can then use Eq. (2.75) to compute n_o at 77 K. Before that, we must first compute N_c(77 K), which is found to be 3.7×10^{18} cm^{-3}, and kT(77 K), which is 6.6 meV. Then,

$$n_o = \sqrt{\frac{5 \times 10^{17} \times 3.7 \times 10^{18}}{2}} \exp\left(-\frac{0.045}{2 \times 0.0066}\right) = 3.2 \times 10^{16} \text{ cm}^{-3}.$$

In order to compute p_o, we must first calculate n_i at 77 K. Using the approach described in Advanced Topic AT2.1, we easily find $n_i(77 \text{ K}) = 2.8 \times 10^{-20}$ cm^{-3}. We then get

$$p_o = \frac{n_i^2}{n_o} = \frac{7.9 \times 10^{-40}}{3.2 \times 10^{16}} = 2.5 \times 10^{-56} \text{ cm}^{-3}$$

a truly negligible number.

The use of nondegenerate statistics for the electrons is validated since we find that $n_o \ll N_c$ at 77 K. Furthermore, not accounting for the electron contribution from the break up of bonds is validated since we found that $p_o \ll n_o$.

AT2.6 HEAVY-DOPING EFFECTS

Heavily-doped regions are pervasive in microelectronic devices. For example, the source and drain of a MOSFET have doping levels of the order of 10^{20} cm^{-3}. Analogously, the emitter of a bipolar transistor has impurity levels of similar magnitude. At high-doping levels, defined here as 10^{18} cm^{-3} and above, several special effects occur that deserve particular attention. Some of these phenomena are of relevance for important device figures of merit. For example, if heavy-doping effects are not taken into account, a calculation of the current gain of a bipolar transistor can easily yield an error exceeding a factor of 10.

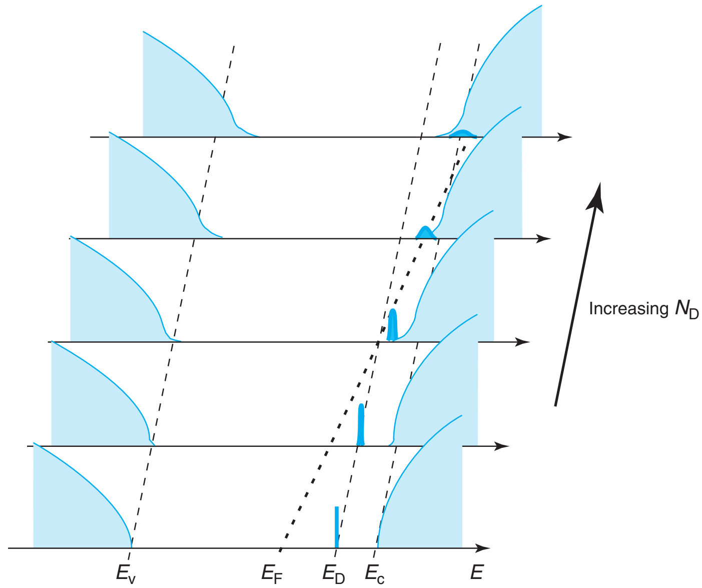


FIGURE 2.21
A sketch of heavy-doping effects in an n-type semiconductor. As N_D increases, the impurity level broadens into an impurity band and merges into the conduction band, the conduction and valence bands develop band tails, and the bandgap narrows.

Figure 2.21 illustrates the special effects that occur at high-doping levels in an n-type semiconductor:

1. The Fermi level penetrates into the majority-carrier band and the *majority carriers become degenerate*. This demands the use of Fermi–Dirac statistics for the majority carriers as discussed in the main body of this chapter and in Advanced Topic AT2.3.
2. *Bandgap narrowing* takes place, that is, the energy difference between the conduction and valence bands is reduced.
3. The impurity ionization energy drops as the doping level increases and eventually becomes zero. This is called the *Mott transition*.
4. The impurity level broadens into an *impurity band*. This is only relevant at low temperatures and will not be discussed here.
5. The density of states of the conduction and valence bands is deformed from its ideal square-root shape. *Band tailing* appears at the edges of the bands. The extent of these tails is only a few millielectronvolts and as a result, this effect is not very significant for room temperature operation. We will not consider it any further here.

With the exception of majority carrier degeneracy, whose origin we already know, these effects arise from a variety of sources. A detailed discussion is beyond the scope of this book but a few words about them are appropriate here. First of all, at high-doping levels, the dopant atoms can no longer be considered far apart from each other, in fact, their Coulombic potentials start to overlap. This reduces the impurity ionization energy. The presence of a high concentration of majority carriers shrinks the energy required to break a bond through *many-body effects*, a quantum-mechanical phenomenon. This results in bandgap narrowing. Many-body effects also contribute to a further reduction of the impurity ionization energy. Finally, at high-doping levels, *doping level fluctuations* become very important. The fact is that dopants are not uniformly dispersed inside the bulk of the semiconductor. Some regions have a higher concentration of dopants than others. At low-doping levels, these fluctuations in the doping concentration are not very significant, but at high-doping levels, coupled with other heavy-doping effects, they contribute to impurity level broadening and band tailing.

We discuss in more detail the two most important heavy-doping effects, besides majority carrier degeneracy, which are relevant for microelectronic devices, that is, the Mott transition and bandgap narrowing.

AT2.6.1 The Mott transition

As the doping level increases, it has been observed that the impurity ionization energy decreases. A simple expression that captures this phenomenon is

$$E_{\mathrm{i}} = E_{\mathrm{io}}\left[1 - \left(\frac{N_{\mathrm{i}}}{N_{Mi}}\right)^{1/3}\right] \qquad \text{for } N_{\mathrm{i}} \le N_{Mi} \tag{2.77}$$

$$E_{\mathrm{i}} = 0 \qquad \text{for } N_{\mathrm{i}} \ge N_{Mi} \tag{2.78}$$

where "i" stands for the donor or acceptor subindex, E_{io} is the low-doping value of the ionization energy given in Table 2.1, and N_{Mi} is the Mott concentration, also listed in Table 2.1 (the Mott concentration is not very well known in all cases).

Equation (2.78) shows that at a critical concentration, called the *Mott transition*, the ionization energy goes to zero. The implications of this should be immediately apparent. For doping levels higher than the Mott transition, all dopants are completely ionized at all temperatures. Carrier freeze-out does not occur at low temperatures for semiconductors doped beyond the Mott transition. As a result, the semiconductor remains conductive down to 0 K, just like a metal. For this reason, this critical concentration also receives the name of *metal-insulator transition*. This is illustrated in Fig. 2.22, which displays the ratio of the measured electron concentration at 4.2 K over 300 K in phosphorus-doped Si for several doping levels. Samples with doping levels below about 3.5×10^{18} cm^{-3} freeze out at low temperatures and $n_o(4.2\text{ K}) < n_o(300\text{ K})$. On the other hand, in samples with a doping level above 3.5×10^{18} cm^{-3}, the electron concentration is independent of temperature, that is, $n_o(4.2\text{ K}) \simeq n_o(300\text{ K})$ (the small systematic discrepancy between $n_o(4.2\text{ K})$ and $n_o(300\text{ K})$ for high-doping levels is a feature introduced in the interpretation of the data).

The implications of this phenomenon to microelectronic devices are profound. In devices designed to operate at low temperatures, regions doped below the Mott transition will significantly freeze-out (and their resistivity will substantially increase), while those regions that are doped above the Mott transition will not. The Mott transition has a noticeable influence even at room temperature. If it were not for the Mott transition, at high-doping levels, the Fermi level would cross the impurity level and significant dopant freeze-out would take place at room temperature. You can appreciate this in the next example.

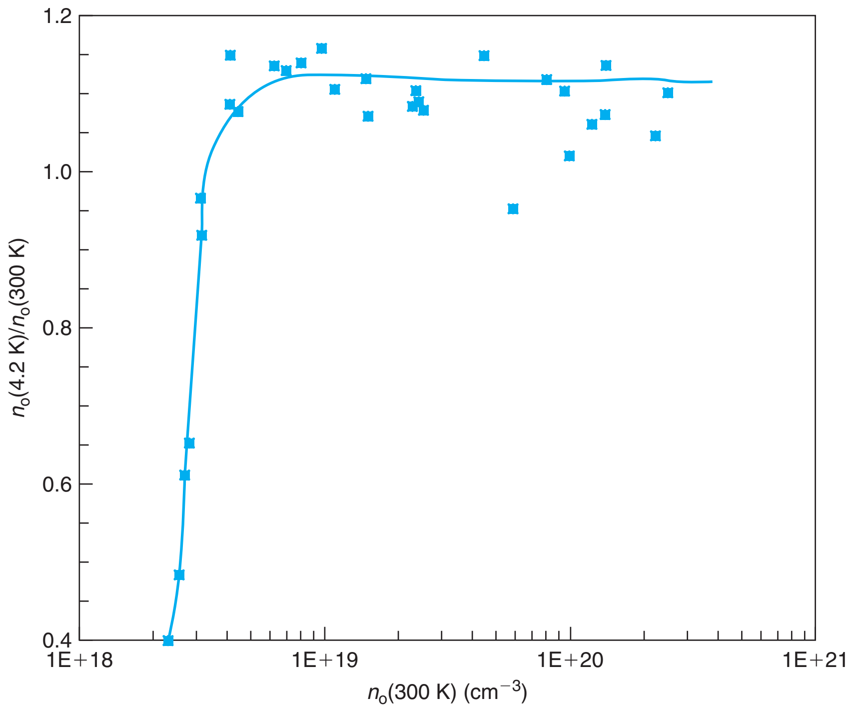


FIGURE 2.22
Ratio of the measured electron concentration at 4.2 K over 300 K in phosphorus-doped Si for several doping levels. These data clearly reveal the Mott transition for Si:P at about 3.5×10^{18} cm^{-3} (the small systematic discrepancy between n_o(4.2 K) and n_o(300 K) for high-doping levels is a feature introduced in the interpretation of the data) [Data from C. Yamanouchi et al., Electric Conduction in Phosphorus Doped Silicon at Low Temperatures, *J. Phys. Soc. Jpn.* 22, pp. 859–864 (1967). © J.A. del Alamo].

EXERCISE 2.11

Calculate the equlibrium electron concentration at 300 K *of a sample of P-doped Si with* $N_D = 2 \times 10^{18}$ cm^{-3}. *Do it with and without consideration of the Mott transition. Verify all your assumptions.*

To be expeditious, we will use directly the exact expression Eq. (2.74). This equation is valid in both cases because the only assumption built into it is the Maxwell–Boltzmann approximation for the electron statistics. This is insured in this problem, since $N_D \ll N_c$.

Without consideration of the Mott transition, E_d for P-doped Si is 0.045 eV. Plugging in Eq. (2.74), this yields

$$n_o(E_{do} = 0.045 \text{ meV}) = 1.31 \times 10^{18} \text{ cm}^{-3}$$

The dopant ionization ratio is only 66%.

The Mott transition considerably reduces E_d. Using Eq. (2.77), we find that for $N_D = 2 \times 10^{18}$ cm^{-3}, $E_d = 0.005$ eV. Plugging again into Eq. (2.74), we obtain

$$n_o(E_d = 0.005 \text{ meV}) = 1.75 \times 10^{18} \text{ cm}^{-3}$$

The dopant ionization ratio is now 87%.

The dopant concentration at which the Mott transition takes place depends on both the host semiconductor and the dopant species. Values for the Mott transition for typical dopants in Si and GaAs are listed in Table 2.1. Some of these are not very accurately known.

AT2.6.2 Bandgap narrowing

At high-doping levels, many body effects that arise from the high majority carrier concentration result in a rigid shrinkage of the bandgap. Experimentally, it is found that bandgap narrowing increases with doping level but is otherwise a property of the host semiconductor, that is, it appears to be independent of the dopant species and temperature.

Bandgap narrowing, ΔE_g, is usually defined as a *positive* quantity, that is

$$E_{g-HD} = E_{g-LD} - \Delta E_g \tag{2.79}$$

where E_{g-LD} is the low-doping value of the bandgap, and E_{g-HD} is the actual bandgap at a certain doping level. Figure 2.23 shows bandgap narrowing as a function of doping level in Si. An analytical description of this function is given in Appendix E.

Bandgap narrowing does not affect the majority carrier concentration, which is set by the doping level only. On the other hand, the minority carrier concentration is increased since the energy required to break a bond and produce an electron–hole pair is reduced. In the heavily doped regime, this is usually accounted for by defining an *effective intrinsic carrier concentration*,

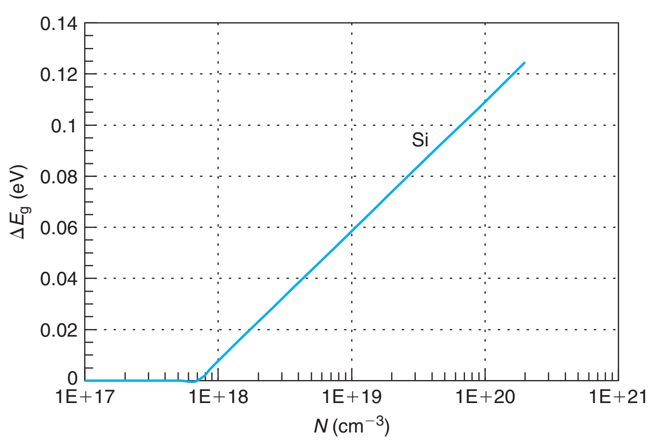


FIGURE 2.23
Bandgap narrowing versus doping level in Si.

n_{ie}, which in general is different from the low-doping value derived earlier in this chapter. Looking back at Eq. (2.41), a reduction of the bandgap results in an increase in n_i. The exponential dependence of n_i on E_g, makes small changes in the bandgap very significant.

Unfortunately, we cannot just modify Eq. (2.41) since it was deduced under the assumption that Maxwell–Boltzmann statistics are valid for both types of carriers. In the heavy-doping regime, this is clearly not the case for the majority carriers. A new expression for the $n_\mathrm{o}p_\mathrm{o}$ product that is valid in the heavy-doping regime is required. The starting point is Eq. (2.39). Multiplying and dividing by $\exp \frac{E_{g-LD}}{kT}$, we get

$$n_\mathrm{o}p_\mathrm{o} = n_\mathrm{ie}^2 = n_\mathrm{io}^2\, \mathcal{F}_{1/2}(\eta_\mathrm{c})\, \mathcal{F}_{1/2}(\eta_\mathrm{v}) \exp \frac{E_{g-LD}}{kT} \tag{2.80}$$

where, following conventional practice, we have relabeled the intrinsic carrier concentration in the low-doping regime as n_io and we have defined the $n_\mathrm{o}p_\mathrm{o}$ product in the highly doped regime as n_ie^2.

Equation (2.80) can be further simplified if we focus on a specific doping polarity. For an n-type semiconductor, for example, the holes follow Maxwell–Boltzmann statistics. This allows us to use the exponential approximation to the Fermi–Dirac integral for the term in $\mathcal{F}_{1/2}(\eta_\mathrm{v})$. After some straightforward algebra, Eq. (2.80) can be rewritten as

$$n_\mathrm{ie}^2 = n_\mathrm{io}^2 \frac{\mathcal{F}_{1/2}(\eta_\mathrm{c})}{e^{\eta_\mathrm{c}}} \exp \frac{\Delta E_\mathrm{g}}{kT} \tag{2.81}$$

where we have also used Eq. (2.79). A similar expression can be obtained for a p-type semiconductor.

Figure 2.24 graphs Eq. (2.81) for n-Si at room temperature as a function of doping level. For low-doping levels, n_ie^2 converges to n_io^2. Beyond a doping level of around $N_\mathrm{D} \simeq 10^{18}$ cm^{-3}, n_ie^2

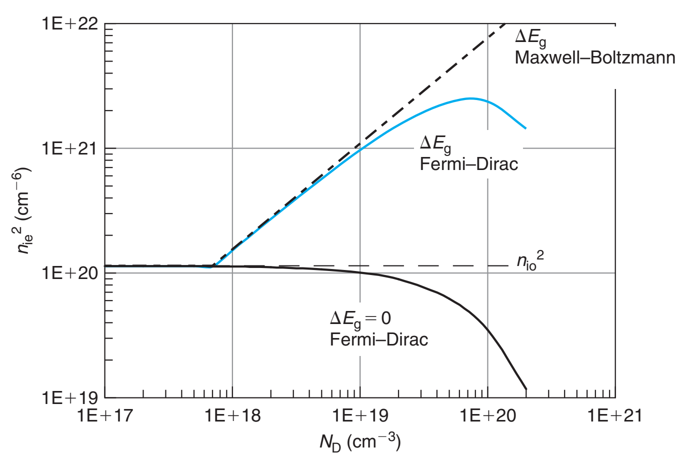


FIGURE 2.24
Effective intrinsic carrier concentration versus doping level in the heavy-doping regime for n-Si at 300 K. Also shown are the low-doped value n_io^2, and n_ie^2 separately due to bandgap narrowing and electron degeneracy.

starts to increase over the low-doped value. At about 10^{20} cm^{-3}, n_{ie}^2 peaks and starts decreasing. This peculiar behavior of n_{ie}^2 can be understood by examining separately the impact of bandgap narrowing and electron degeneracy on n_{ie}^2. These are also graphed separately in Fig. 2.24. The increase in bandgap narrowing with doping level makes n_{ie}^2 increase very quickly through the $\exp(\Delta E_g/kT)$ term in Eq. (2.81). But as the doping level increases, the Fermi level penetrates into the conduction band and the electrons become degenerate. This brings n_{ie}^2 down through the term in $\mathcal{F}_{1/2}(\eta_c)/e^{\eta_c}$, which is at most unity and decreases as η_c increases. A physical way to explain this is that as the doping level increases, the bottom of the conduction band fills up with electrons and the energy required to break a bond and produce an electron–hole pair increases. Everything else being equal, this reduces the $n_o p_o$ product.

EXERCISE 2.12

Calculate the equilibrium electron and hole concentrations at 300 K *of a sample of P-doped Si with* $N_D = 1 \times 10^{20}$ cm^{-3}*. Do it with and without consideration of bandgap narrowing. Verify all your assumptions.*

Since $N_D = 10^{20}$ cm^{-3} is beyond the Mott transition for P-doped Si,

$$n_o = N_D = 10^{20}\ \text{cm}^{-3}$$

In order to compute p_o, we must first obtain n_{ie}^2. We could simply use Fig. 2.24. But the resolution of this graph is limited, and besides, it is of value to trace the steps of the computation of n_{ie}^2.

First, we have to find the position of the Fermi level. This can be easily accomplished through the use of the approximation to the inverse Fermi–Dirac function given in Eqs. (2.58) and (2.59). This yields

$$\eta_c = 2.43$$

This means that the Fermi level has penetrated over $2kT$'s into the conduction band.

Armed with η_c we can now compute the term $\mathcal{F}_{1/2}(\eta_c)/e^{\eta_c}$ in Eq. (2.81). Note that there is no need to carry out the Fermi–Dirac integral since $\mathcal{F}_{1/2}(\eta_c) = N_D/N_c = 3.50$. We then get

$$\frac{\mathcal{F}_{1/2}(\eta_c)}{e^{\eta_c}} = 0.31$$

The degeneracy of the electron gas *reduces* $p_o n_o$ to 31% of its low-doped value.

To account for bandgap narrowing, we first obtain the value of ΔE_g that corresponds to $N_D = 10^{20}$ cm^{-3} in Fig. 2.23. We find $\Delta E_g = 109$ meV. This allows us to compute the term that enters the equation of n_{ie}^2:

$$\exp\frac{\Delta E_g}{kT} = 67.6$$

Bandgap narrowing alone increases the $p_o n_o$ product by over 67 times!

We now plug in all terms into Eq. (2.81), and get

$$n_{ie}^2 = 2.4 \times 10^{21} \text{ cm}^{-3}$$

This allows us to obtain the equilibrium hole concentration as

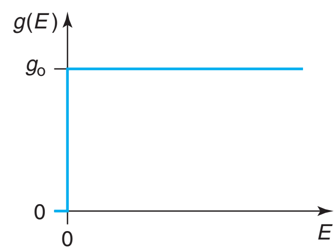
$$p_o = \frac{n_{ie}^2}{n_o} = 24 \text{ cm}^{-3}$$

Without any consideration of heavy-doping effects, we would have obtained a value of p_o of 1.1, off by more than an order of magnitude.

PROBLEMS

*Denotes a problem that requires studying an Advanced Topic

2.1 Consider a simple quantum mechanical system with an electron concentration n and a uniform density of states g_o as sketched below.



Calculate:

(a) The position of the Fermi level at 0 K.
(b) The average energy per electron at 0 K.
(c) The position of the Fermi level at a finite temperature T. Derive first an exact expression, then simplify it for the case in which $E_F \gg kT$.
(d) The *increase* in average energy per electron as the temperature is raised from 0 to T, defined in the following way:

$$<\Delta E>= \frac{1}{n}\int_0^\infty E\, g(E)[f(E,T) - f(E,0)]\, dE$$

where $f(E,T)$ and $f(E,0)$ are the Fermi–Dirac distribution function at temperature T and at zero temperature, respectively (assume $E_F \gg kT$).

Mathematical help:

$$\int_0^\infty \frac{x dx}{1+e^x} = \frac{\pi^2}{12}$$

Also, when doing the above integral, do not break it through the minus sign inside the square brackets, but rather break it in energy at E_F.

2.2 This exercise explores the physical meaning of the effective density of states of a band.

(a) Assume a fictitious semiconductor with a conduction band density of states in which there are N_c states all located at E_c. Analytically, $g_c(E) = N_c\delta(E - E_c)$, where δ is the Dirac delta function. Calculate an expression for the electron concentration as a function of the relative position of E_F with respect to E_c. Simplify for a nondegenerate situation in which E_F is far below E_c.

(b) Let us discuss now the origin of the temperature dependence $T^{3/2}$ deduced in class for N_c and N_v. Consider another fictitious semiconductor with a step-like conduction-band density of states $g_c(E) = A_c u(E - E_c)$, where A_c is a constant and u is the step function. Using Maxwell–Boltzmann statistics, calculate the electron concentration as a function of the position of E_F with respect to E_c. Why does the exponential prefactor (the effective density of states in this case) depend on T?

(c) Based on what you found above, what is the physical meaning of N_c? Can you speculate on why it is called *effective* density of states? Is N_c the total integrated density of states of the conduction band? Why does it depend on T? Does it make sense to talk about effective density of states in a degenerate case?

2.3 For high temperatures, Maxwell–Boltzmann statistics might not be suitable for an *intrinsic* semiconductor. How can this happen? In Si, which type of carrier, holes or electrons, suffers from this first as the temperature increases? *Estimate* the temperature dependence at which Maxwell–Boltzmann statistics will stop holding for intrinsic Si.

2.4 Misiakos and Tsamakis carried out measurements of the intrinsic carrier concentration in Si versus temperature (see Fig. 2.5). The data they obtained is given in the table:

T (K)	n_i (cm^{-3})
77.8	5.00×10^{-20}
100	2.00×10^{-11}
120.75	3.40×10^{-6}
137.5	4.60×10^{-3}
148.4	2.10×10^{-1}
169.7	8.40×10^{1}
195	1.78×10^{4}
199.5	4.30×10^{4}
213	4.16×10^{5}
239	1.92×10^{7}
256.5	1.45×10^{8}
270.6	6.70×10^{8}
281	1.79×10^{9}
300	9.70×10^{9}
319.5	4.51×10^{10}
340.5	1.89×10^{11}

Graph an Arrhenius plot and extract the activation energy.

Optional: do a least square fit of the data with the following equations:

(a) $n_i(T) = K_1 \exp -\dfrac{K_2}{2kT}$

(b) $n_i(T) = K_3 \left(\dfrac{T}{300}\right)^{3/2} \exp -\dfrac{K_4}{2kT}$

(c) $n_i(T) = K_5 \left(\dfrac{T}{300}\right)^{K_6} \exp -\dfrac{K_7}{2kT}$

(d) $n_i(T) = K_8 \left(\dfrac{T}{300}\right)^{K_9} \exp -\dfrac{E_g(T)}{2kT}$ with $E_g(T) = 1.206 - K_{10}T$

(e) $n_i(T) = K_{11} \left(\dfrac{T}{300}\right)^{K_{12}} \exp -\dfrac{E_g(T)}{2kT}$ with $E_g(T) = 1.17 - \dfrac{K_{13}T^2}{T + K_{14}}$

2.5 Consider a p-type Si wafer with an acceptor concentration of $N_A = 10^{17}$ cm^{-3}. Compute at room temperature under equilibrium conditions:

(a) Hole concentration.
(b) Electron concentration.
(c) Position of the Fermi level with respect to the conduction band and the valence band edges.
(d) Probability that a state at the bottom of the conduction band is occupied.
(e) Probability that a state at the top of the valence band is empty.

2.6 At which temperature will a Si-sample with 10^{17} donors become an intrinsic semiconductor? Is the Maxwell–Boltzmann approximation still valid at this temperature? If so, estimate at which temperature this approximation breaks down.

2.7 The *Fermi wavelength* is the de Broglie-wavelength of electrons located at the Fermi-energy. Estimate the Fermi wavelength of n-type Si with 3×10^{20} cm^{-3} electrons. Assume the effective mass of an electron in Si to be the same as the density of states effective mass.

2.8 InP is a semiconductor with a bandgap at room temperature of 1.35 eV, an electron density of states effective mass of 0.077 m_o, and a hole density of states effective mass of 0.64 m_o.

Calculate at room temperature:

(a) N_c, N_v, n_i, and E_i. Verify all your assumptions.
(b) In a chunk of InP that sits in thermal equilibrium at 300 K, the Fermi level is 0.2 eV below the conduction band edge. Calculate n_o and p_o (without invoking $n_o p_o = n_i^2$). Calculate the *np* product and compare with n_i^2 calculated above. Comment.
(c) In a different chunk of InP also in thermal equilibrium at 300 K, the Fermi level is now 0.1 eV below the valence band edge. Calculate n_o and p_o (without invoking $n_o p_o = n_i^2$). Calculate the *np* product and compare with n_i^2 calculated above. Comment.

2.9 Consider a compensated semiconductor uniformly doped with a donor concentration N_D and an acceptor concentration N_A, with $N_D > N_A$. Assume that conditions are such that Maxwell–Boltzmann statistics and full impurity ionization apply.

(a) Derive a general expression for the electron and hole concentrations in equilibrium.
(b) Discuss the condition that needs to be satisfied for this semiconductor to be simply considered n-type with a net doping level $N_d \simeq N_D - N_A$.

(c) If $N_D = 10^{16}$ cm^{-3} in Si at room temperature, what is the maximum compensation ratio N_A/N_D that allows the semiconductor to be simply considered n-type with $N_d = N_D - N_A$? Make explicit your error acceptance criteria.

2.10 It is possible to estimate the *donor ionization energy* in a semiconductor following a similar methodology to that followed in Problem 1.7. In very elemental terms, a donor can be considered as a hydrogen atom that is immersed in a semiconductor. The fifth valence electron of the donor atom (the one that does not participate in bonding with neighboring Si atoms) behaves in a way as the electron of the hydrogen atom. It effectively sees a positive charge of value $+q$ at the nucleus, since all the other protons are compensated by the rest of the electrons of the donor atom. The fundamental difference between a donor and a H atom is that the first is immersed in a solid. We can account for this by using the permittivity of the solid instead of that of vacuum and describe the electron by an effective mass that includes its quantum mechanical interactions with the lattice atoms.

Following the procedure of Problem 1.7, derive an expression for the radius and the binding energy of the fifth valence electron of a donor atom in Si. Use an effective mass for the electron equal to $m_e^* = 0.28m_o$. Use also the permittivity of Si that is given in Appendix B. Give the result in terms of nanometer and electronvolts. Comment. Calculate how many Si atoms are included in a sphere with a radius equal to the estimated radius of the fifth electron. Comment.

2.11 Compute and graph the *normalized* electron concentration per unit energy $n_o(E)/n_o$ in Si at room temperature for $E_F - E_c = -6kT,\ -4kT,\ -2kT,\ 0,\ 2kT,\ 4kT,$ and $6kT$. Discuss the evolution of the electron distribution as E_F penetrates into the conduction band.

2.12 Specify the range of donor concentrations in which each of the three equations below are reasonably accurate for n-type GaAs at room temperature. Assume that all donors are fully ionized and that there are no other dopants. Explain (see suitable parameters for GaAs in Appendix B).

(a) $n_o p_o = n_i^2$

(b) $E_F - E_c = kT \ln \dfrac{N_D}{N_c}$

(c) $p_o = \dfrac{n_i^2}{N_D}$

2.13 Derive relationships for n_o and p_o in terms of n_i and the relative position of E_F and E_i under the constraint of Maxwell–Boltzmann statistics.

2.14 Following a procedure similar to that of Section 2.4.2, derive an expression for the average kinetic energy of all conduction band electrons of a nondegenerate semiconductor at room temperature. Comment on the result.

2.15* Derive Eq. (2.60) directly from Eqs. (2.22) and (2.23) using the Fermi–Dirac distribution function at 0 K.

2.16 Consider an n-type Si sample with $N_D = 10^{16}$ cm^{-3} at room temperature in thermal equilibrium. Calculate the position in energy of the peak of the electron concentration inside the conduction band.

2.17 Consider an n-type semiconductor with a concentration N_D of donors. Starting from Eqs. (2.8) and (2.11), derive general relationships for n_o and p_o in terms of N_D and n_i that are valid under Maxwell–Boltzmann statistics. Discuss the condition that needs to be satisfied for Eqs. (2.14) and (2.15) to be applicable.

2.18 Derive an expression for the transition temperature between the extrinsic and intrinsic regimes in a p-type semiconductor. Discuss the dependencies that are observed. Estimate the temperature at which a p-Si wafer doped with $N_A = 10^{17}$ cm^{-3} becomes intrinsic.

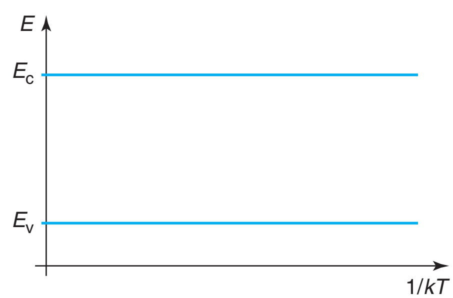
2.19 In a certain Si sample at room temperature and in thermal equilibrium, the Fermi level is 0.1 eV *below* the valence band edge.

(a) Calculate the electron concentration, n_o.
(b) Calculate the hole concentration, p_o.

2.20 Describe a semiconductor in which Maxwell–Boltzmann statistics cannot be used in thermal equilibrium for *neither electrons nor holes* at room temperature. Explain.

2.21 Other than through bandgap narrowing, does the threshold photon energy for carrier generation in a semiconductor depend on doping? Explain.

2.22 Figure 2.9 sketches the electron concentration of a piece of nondegenerate n-type semiconductor in thermal equilibrium as a function of temperature in an Arrhenius plot. In the axes provided below, sketch the evolution of the location of the Fermi level with respect to the band edges as a function of temperature for the entire temperature range. For simplicity, assume that $N_c = N_v$, and both N_c and N_v as well as E_g are all temperature independent. Explain your drawing in each of the temperature regimes.



Chapter 3

Carrier Generation and Recombination

A semiconductor in equilibrium at a finite temperature is a highly dynamic system. Electrons and holes are continuously being *generated* when covalent bonds break up and are also *recombining* reforming bonds. In thermal equilibrium, a *dynamic balance* between the rates of generation and recombination of electrons and holes is established.

A semiconductor that has been disturbed from thermal equilibrium reacts by trying to reestablish it. If the disruption is removed, the semiconductor will recover equilibrium after some time. If the disruption is maintained, the semiconductor will eventually reach a nonequilibrium steady-state situation. This behavior is at the heart of the semiconductor device operation. It is the dynamics of electrons and holes outside equilibrium that determines the behavior of devices. For example, the recombination rate of carriers in the emitter of a bipolar transistor limits its maximum current gain. Similarly, the generation rate of electrons and holes in the capacitor or parasitic junctions of a dynamic random-access memory (DRAM) sets the maximum time that the capacitor can safely store information before refreshing is required.

In Chapter 2, we calculated the electron and hole concentrations in thermal equilibrium in a variety of circumstances. In fact, we learned that the equilibrium carrier concentrations are entirely determined by the band structure of the semiconductor, the dopant concentrations, and the temperature. They have nothing to do with the particular mechanism for electron and hole generation and recombination. However, the dynamics of a semiconductor when disturbed from equilibrium are determined by the processes by which carriers are generated and recombine and the rates at which these processes take place. Understanding these mechanisms is the goal of this chapter.

Section 3.1 qualitatively describes the main generation and recombination mechanisms. Section 3.2 introduces the principle of detailed balance. This thermodynamic principle allows the formulation of the functional dependencies of the generation and recombination rates in thermal equilibrium that are presented in Section 3.3. In Section 3.4, we examine in detail two important ways in which the balance between generation and recombination can be broken by introducing excess carriers or by extracting carriers. We compute the generation and recombination rates in these two cases. Section 3.5 discusses a few typical dynamic examples that are frequently encountered in semiconductor devices. We end the chapter with a discussion on the generation and recombination mechanisms associated with a surface.

3.1 GENERATION AND RECOMBINATION MECHANISMS

In the process of recombination of an electron and a hole, energy is released. Almost without exception, the generation of an electron and a hole demands that energy be supplied (an important exception is Zener tunneling, as explained below). The various generation and recombination mechanisms in a semiconductor can be classified by the nature of the energy that is involved, whether it is thermal, optical, or other. In many cases, a generation process has associated with it a reciprocal recombination one.

We have alluded to **band-to-band thermal generation and recombination** earlier in this book. Band-to-band thermal generation of an electron–hole pair occurs when a covalent bond is ruptured by the thermal vibrations of the lattice, also referred to as phonons. A collective intervention of many phonons is required since a single one does not have enough energy to break up a bond. This process is depicted on the left of Fig. 3.1. The inverse of this process is band-to-band thermal recombination in which the energy liberated in a recombination event is released as heat to the surroundings. This process is also illustrated in Fig. 3.1.

Both of these processes are very rare in common semiconductors. The reason is the enormous difference between the largest phonon energy in a semiconductor (around 60 meV) and the bandgap energy. As a result, many phonons (about 18 for Si) need to be involved *simultaneously* in a single generation or recombination process, something that is statistically extremely unlikely. The rates of band-to-band thermal generation and recombination are much smaller than those of other processes discussed below. For this reason, we will not consider these processes any further in this book.

In **band-to-band optical generation and recombination**, the energy of the transition is supplied or released by photons. In a band-to-band optical generation process, a photon delivers the required energy to break a covalent bond. This process is also called *optical absorption*. In the inverse process, a photon is emitted in the band-to-band recombination of an electron and a hole. This mechanism is also known as *radiative recombination*. Both processes are depicted on the right of Fig. 3.1. Optical generation and recombination are at the heart of optoelectronic devices. For example, solar cells, photodetectors, and imagers rely on

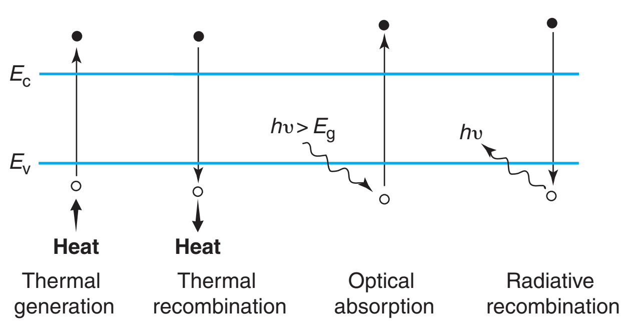


FIGURE 3.1
Schematic diagram of band-to-band generation and recombination mechanisms. From left to right, thermal generation, thermal recombination, optical generation, and optical recombination.

optical absorption; radiative recombination is behind light emission in semiconductor lasers and light-emitting diodes (LEDs).

When discussing the rates of optical generation and recombination, a crucial distinction needs to be made between two types of semiconductors: those with a *direct bandgap* and those with an *indirect bandgap*. Proper appreciation of this difference requires a more advanced study of solid-state physics that goes beyond the simplistic energy view of this textbook. Because of this, we will largely ignore this issue. Nevertheless, and just as a brief background, in our treatment of semiconductor physics, we have so far overlooked an important property of electrons and holes in a semiconductor: their momentum. It is obvious that electrons and holes in a semiconductor have a certain momentum, since they have mass and are moving around in the crystal. In a generation or recombination event, not only does energy need to be conserved, but momentum as well.

In a direct bandgap semiconductor, such as GaAs, the atomic arrangement results in a band structure in which optical generation and recombination easily satisfy the momentum conservation rule. In consequence, the rates at which these processes take place are substantial. In an indirect bandgap semiconductor, such as Si, this is not possible. A third particle needs to be involved in the process in order to supply the required change in momentum. A suitable candidate is a phonon with the appropriate momentum. The combined event, however, becomes substantially less likely. In consequence, optical generation and recombination rates are rather small in an indirect bandgap semiconductor. This fundamental difference is the reason why GaAs and other direct bandgap III–V semiconductors, such as InP, InGaAs, or InGaAsP, emit light very efficiently, while others, such as Si or Ge, do not. Semiconductor lasers and LEDs are mostly made out of direct bandgap materials.

In order to prevent undesirable carrier generation due to external light, microelectronic devices always operate in the dark. In spite of this, optical generation and recombination do actually take place. This is a consequence of the blackbody radiation that baths everything at finite temperatures. This was discussed in Chapter 1. The fact that at around room temperature the blackbody radiation is of rather low energy and intensity means that optical generation and recombination in thermal equilibrium occur at a rather reduced rate when compared with other processes.

An additional mechanism for carrier generation is **Auger generation**. In this process, the source of energy is a "hot" carrier, an electron or a hole with large kinetic energy. A few hot carriers always exist at the tail end of the distribution. A hot carrier can release its excess energy by impacting on a bond and breaking it, generating in this way an electron–hole pair. This is sketched in Fig. 3.2. The inverse process is called *Auger recombination* in which a recombination event is completed by supplying the released energy to a third carrier that in turn gets ejected with high kinetic energy. This third carrier could be an electron or a hole. Both kinds of processes are illustrated in Fig. 3.2. In all these processes, momentum is conserved. Since Auger recombination is also a three-body collision, it is relatively unlikely. An important exception is a situation in which there are many electrons or holes. This is the case, for example, of a heavily doped semiconductor. In these circumstances, carriers are close to each other and Auger recombination can become a highly efficient recombination path. This happens, for example, in the emitter of a bipolar transistor where the doping level is rather high.

Trap-assisted thermal generation and recombination are the most pervasive mechanisms in microelectronic devices. We discussed above how unlikely band-to-band thermal generation and

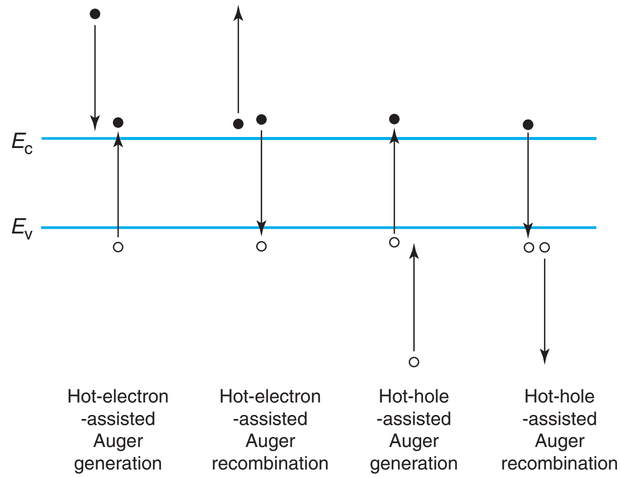


FIGURE 3.2
Auger processes. From left to right: generation by means of hot electron, recombination with ejection of hot electron, generation by means of hot hole, and recombination with ejection of hot hole.

recombination are. Thermal generation is much more probable if there are "holding places" for the electron somewhere in the middle of the bandgap. Through midgap states, instead of requiring a single "kick" with an energy larger than the bandgap, two separate events with substantially less energy are sequentially needed. This is a lot more likely to happen. Thermal recombination is also enhanced if one of the carriers can be "trapped" or immobilized at a certain site until the complementary carrier appears within range. Additionally, midgap states help to satisfy the momentum requirements of the transitions.

Electronic states inside the bandgap are not allowed in an ideal semiconductor. Impurities and crystalline imperfections (such as sites without an atom, known as *vacancies*, or atoms out of place, referred to as interstitials), however, result in electronic states in the forbidden gap. This is similar to the donor and acceptor atoms that we discussed in Chapter 2. An important difference is that dopants are foreign atoms that are fairly similar to those of the host semiconductor. As a consequence, their states are located very close to the band edges where they become effective doping centers and are thus called *shallow levels*. In contrast, thermal generation and recombination are strongly enhanced by states that are placed close to the middle of the bandgap (as discussed in Advanced Topic AT3.1, close to midgap, the probability of encountering the required number of phonons is maximized). For this reason, these states are called *deep levels* or *traps*.

Figure 3.3 schematically illustrates trap-assisted thermal generation and recombination processes. A generation event requires that a covalent bond be broken with the electron getting trapped at a deep level followed by the release of the electron to the conduction band. In a recombination event, an electron falls to an empty state inside the bandgap to later on drop into a broken bond.

Foreign atoms or crystalline defects that create traps close to the middle of the bandgap become effective generation and recombination centers. Such is the case of Au in Si that creates

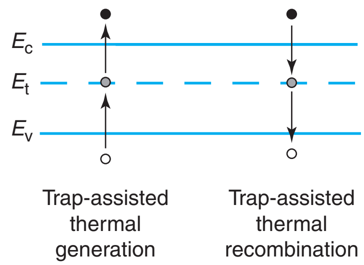


FIGURE 3.3
Trap processes. At left, generation of an electron–hole pair facilitated by a trap near the middle of the bandgap. At right, trap-assisted electron–hole pair recombination.

two levels at 0.29 and 0.54 eV above the valence band edge. Because of this, Au was used in the early days of the semiconductor industry to make very fast diodes and transistors. At the present time, the preferred techniques to introduce trap states in a controlled manner are Pt doping or electron irradiation, which can cause crystalline damage. In many circumstances, even a small concentration of foreign impurities or crystalline imperfections close to the middle of the bandgap is highly undesirable. In DRAMs, for example, they can greatly increase the leakage current, demanding in turn much more frequent refresh cycles. Of all the various possible mechanisms, trap-assisted generation and recombination are often the dominant ones in microelectronic devices. For this reason, we study them in more detail in Section 3.3.3 and in Advanced Topic AT3.1.

There are two more generation mechanisms that are important in microelectronic devices. They are both associated with high electric fields. When a large electric field is present inside a semiconductor, carriers can acquire substantial kinetic energy from the electric field and release it in a collision with a bond, generating an electron–hole pair. This is called *impact ionization*, a process that is in essence an Auger generation event, where the "hot" carrier has acquired its energy from an electric field. It is sketched on the left of Fig. 3.4. This figure will become clearer

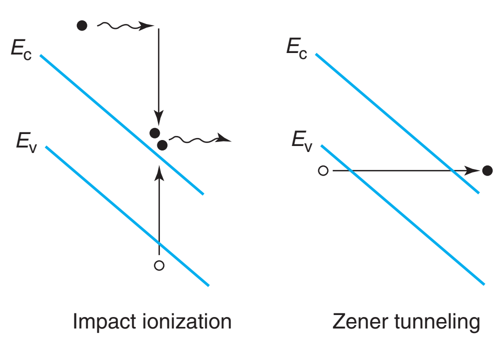


FIGURE 3.4
Schematic diagram showing two generation processes that occur in regions with a high electric field: impact ionization (left) and the Zener tunneling (right).

in Chapter 4, where we will learn that an electric field bends the bands in space. Impact ionization is very common in modern microelectronic devices because large electric fields are used to accelerate carriers to high velocities. High fields also appear in space-charge regions such as in PN junctions and metal–semiconductor junctions. Impact ionization can result in **avalanche multiplication** as the generated electrons and holes, in addition to the primary carriers that started the process, produce more impact ionization events. Unchecked, this effect can cause large current in a phenomenon known as *breakdown*, which can lead to high power dissipation and eventual device destruction. We discuss impact ionization and avalanche multiplication in more detail in Chapter 4.

An interesting generation mechanism is **field ionization**, also known as *Zener tunneling* or the *Zener effect* in which a strong electric field rips a covalent bond apart, releasing an electron and a hole. This is represented in an energy band diagram on the right of Fig. 3.4. If the band bending is very sharp, the spatial separation between occupied states in the valence band and empty states in the conduction band can become small enough for an electron to "tunnel" directly from one band to the other. A hole is left behind in the valence band. This process can become important if a very high field is present and can also lead to high current and device damage.

Energetic particles impinging onto the semiconductor can also result in the breakup of covalent bonds and the production of electron–hole pairs along their trajectory. This was in fact a very serious problem in the early days of DRAMs and SRAMs that were found to "lose" their information as a result of alpha particles emitted from the decay of radioactive contaminants in the package of the chip. A single alpha particle, typically with an energy of about 1 MeV, can generate over a million electron–hole pairs along a track that is very deep (as much as 100 μm) and narrow (a fraction of a micrometer). This sensitivity of semiconductors to energetic particles can be turned around to make detectors for high-energy physics. *Energetic electrons* incident from the outside can also produce electron–hole pairs. This has been exploited to develop sensitive electron microscope characterization techniques for semiconductors and semiconductor devices.

3.2 THERMAL EQUILIBRIUM: PRINCIPLE OF DETAILED BALANCE

Let us make two definitions. A generation process is characterized by a *generation rate* G, which is the number of electron–hole pairs created per unit volume per unit time. The units of G are $\mathrm{cm}^{-3} \cdot \mathrm{s}^{-1}$. Similarly, a recombination process is quantified by defining a recombination rate, R, which gives the number of electron–hole recombination events per unit volume per unit time. The units of R are also $\mathrm{cm}^{-3} \cdot \mathrm{s}^{-1}$.

In thermal equilibrium, a semiconductor is isolated from external sources of carrier generation. Furthermore, any past disturbances have been given ample time to die off. This was discussed in some detail in Section 1.2.2. In thermal equilibrium, a dynamic balance is established between the total generation rate and the total recombination rate. If we denote G_o and R_o as the total generation and recombination rates in equilibrium, we can write

$$R_o = \sum_i R_{oi} = G_o = \sum_i G_{oi} \tag{3.1}$$

where the subindex o indicates thermal equilibrium and the subindex i is used to denote a certain internal reciprocal generation–recombination path. The sum symbol $\sum$ is used to indicate that if there are several independent paths for recombination and generation, all the individual rates must be added.

In a thermal equilibrium situation in which there are several paths for the generation and recombination of carriers, the laws of thermodynamics impose an additional constraint. This is the *principle of detailed balance* that states that in a system in thermal equilibrium, the rate of a given process must be precisely balanced by the rate of its inverse process. For generation and recombination, this principle implies that

$$R_{\mathrm{oi}} = G_{\mathrm{oi}} \qquad \text{for all } i \tag{3.2}$$

To understand the implications of this principle, consider, for example, a semiconductor in thermal equilibrium in which a certain fraction of the electron–hole pairs are optically generated (through absorption of photons from the blackbody radiation surrounding the sample) and the remainder are generated through a thermal trap-assisted process. The principle of detailed balance not only demands that identical fractions of electron–hole recombination events occur through the reciprocal mechanisms (radiative and trap-assisted recombination, respectively) but that the spectral characteristics (that is the light intensity at different wavelengths) of the radiation involved in the radiative generation and recombination processes be identical.

It is interesting to examine what would happen if the principle of detailed balance did not apply in thermal equilibrium. Consider, for example, a hypothetical semiconductor sample in thermal equilibrium in which a thermal-generation process is completely balanced out by a radiative recombination mechanism, as sketched in Fig. 3.5. The thermal-generation process absorbs heat from the lattice, while the radiative recombination process emits light. In this fictitious situation, energy is conserved. However, if this condition is allowed to persist, as time goes on, the sample spontaneously cools down and emits more and more light along the way, a dynamic situation inconsistent with the definition of thermal equilibrium.

3.3 GENERATION AND RECOMBINATION RATES IN THERMAL EQUILIBRIUM

The rate at which the different generation and recombination processes occur *in* thermal equilibrium provides the basis for understanding the dynamics of carriers *outside* thermal equilibrium. The purpose of this section is to identify key relationships in the rates of the various internal generation and recombination processes. We will use these relationships to develop a formulation for the carrier dynamics outside equilibrium later on in this chapter.

3.3.1 Band-to-band optical generation and recombination

A semiconductor (and any other body) at a given temperature is immersed in a "bath" of electromagnetic radiation, the blackbody radiation discussed in Chapter 1. Blackbody photons with energy in excess of the bandgap energy can generate electron–hole pairs. Conversely,

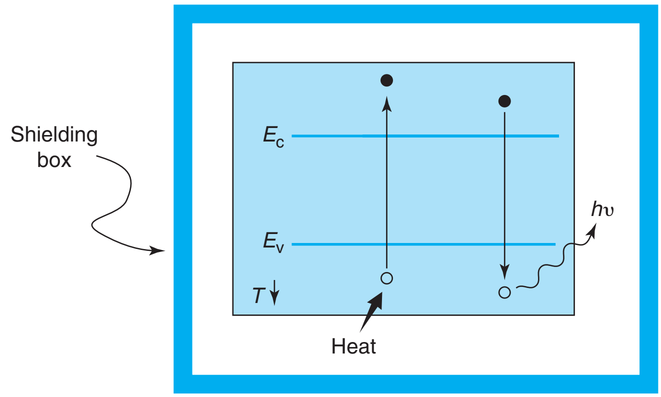


FIGURE 3.5
A semiconductor sample in thermal equilibrium in violation of the principle of detailed balance. If it were possible to balance out thermal generation and optical generation, the sample would spontaneously emit light and cool down.

recombination of electrons and holes through a band-to-band radiative process results in light emission that contributes to the blackbody radiation.

The supply of covalent bonds that can be broken is of an order of magnitude comparable to the atomic density of a semiconductor, a very large number. At around room temperature, only a tiny fraction of these bonds are broken. For all practical purposes, the supply of covalent bonds is inexhaustible. In consequence, the rate of band-to-band optical generation depends solely on temperature (since the intensity and spectrum of blackbody radiation is solely a function of temperature). We can then write

$$G_{o,rad} = g_{rad}(T) \tag{3.3}$$

The rate of recombination, however, depends not only on temperature but also on the concentration of electrons and holes that are present. The greater the number of the individual species, the greater is the opportunity for them to recombine. Since a band-to-band recombination event involves the encounter of one electron with one hole, the recombination rate depends on the product of the electron and hole concentrations. Thus

$$R_{o,rad} = k_{rad}(T)\, n_o p_o \tag{3.4}$$

where k_{rad} is a rate constant characteristic of the material and temperature.

In thermal equilibrium, the principle of detailed balance demands the equality of the band-to-band thermal generation and recombination rates, $R_{o,rad} = G_{o,rad}$. This allows us to derive a relationship between g_{rad} in Eq. (3.3) and k_{rad} in Eq. (3.4):

$$g_{rad} = k_{rad}(T) n_o p_o = k_{rad}(T) n_i^2 \tag{3.5}$$

The last equality applies if the semiconductor is nondegenerate.

Since Si is an indirect bandgap material, the constant that characterizes the rates at which these processes take place, k_{rad}, is very small and this pair of processes is negligible in front of those that we discuss below. GaAs, on the other hand, is a direct bandgap material. This strongly favors radiative band-to-band transitions and k_{rad} is significant. The values of this constant for Si and GaAs are listed in Appendix B.

EXERCISE 3.1

Consider an n-type Si sample with $N_D = 10^{17}$ cm^{-3} at room temperature. Estimate the rate at which generation and recombination events take place by means of band-to-band optical processes in thermal equilibrium.

In thermal equilibrium, the rates of generation and recombination are identical and are given by Eqs. (3.3)–(3.5). For Si with $N_D = 10^{17}$ cm^{-3} at room temperature, $n_o = 10^{17}$ cm^{-3} and $p_o \simeq 10^3$ cm^{-3}. The value of k_{rad} for Si (see Appendix B) is 2×10^{-15} cm$^3 \cdot$ s^{-1}. We then have

$$G_{o,rad} = R_{o,rad} = k_{rad}\, n_o p_o = 2 \times 10^{-15} \text{ cm}^3/\text{s} \times 10^{17} \text{ cm}^{-3} \times 10^3 \text{ cm}^{-3} = 2 \times 10^5 \text{ cm}^{-3} \cdot \text{s}^{-1}$$

3.3.2 Auger generation and recombination

In thermal equilibrium, Auger generation by means of hot electrons depends on the equilibrium electron concentration. The more electrons there are in a given semiconductor, the more likely some of them will have sufficient energy to produce electron–hole pairs. Mathematically, then

$$G_{o,eeh} = g_{eeh}(T) n_o \tag{3.6}$$

We saw in Chapter 2 that the energy distribution of electrons in the conduction band is a strong function of temperature. This implies that, for a given semiconductor, the constant g_{eeh} changes rapidly with temperature.

Auger recombination involving the ejection of a hot electron is also a three-body collision. In addition to the recombining electron–hole pair, a second electron is involved. This electron absorbs the released energy and increases its kinetic energy. The rate of this process must then be proportional to the hole concentration and the *square* of the electron concentration

$$R_{o,eeh} = k_{eeh}(T) n_o^2 p_o \tag{3.7}$$

where k_{eeh} is a rate constant characteristic of the material and temperature.

In thermal equilibrium, $G_{o,eeh} = R_{o,eeh}$. This establishes a relationship between the two rate constants as follows:

$$g_{eeh} = k_{eeh} n_o p_o \tag{3.8}$$

The same arguments and equations apply to Auger processes involving hot holes, if we replace the subindex "eeh" by "ehh" and n_o by p_o.

Auger generation and recombination rates can become very significant in situations in which there is a high concentration of one kind of carrier, such as in a heavily doped semiconductor. In Si at room temperature, as we will see later, Auger processes dominate for doping levels higher than about 10^{19} cm^{-3}.

The characteristic constants k_{eeh} and k_{ehh} receive the name of Auger coefficient for electrons and holes, respectively. Their values for Si and GaAs are listed in Appendix B.

EXERCISE 3.2

Consider an n-type Si sample with $N_D = 10^{17}\ \mathrm{cm}^{-3}$ at room temperature. Estimate the rate at which generation and recombination events take place by means of Auger processes assisted by hot electrons and hot holes in thermal equilibrium.

In thermal equilibrium, the rates of generation and recombination by means of an Auger process assisted by hot electrons are identical and are given by Eqs. (3.6)–(3.8). The value of k_{eeh} for Si is $1.8 \times 10^{-31}\ \mathrm{cm}^6/\mathrm{s}$ (see Appendix B). Hence

$$G_{o,\mathrm{eeh}} = R_{o,\mathrm{eeh}} = k_{\mathrm{eeh}}\, n_o^2 p_o = 1.8 \times 10^{-31}\ \mathrm{cm}^6/\mathrm{s} \times (10^{17}\ \mathrm{cm}^{-3})^2 \times 10^3\ \mathrm{cm}^{-3} = 1.8 \times 10^6\ \mathrm{cm}^{-3}\cdot\mathrm{s}^{-1}$$

Similarly, k_{ehh} for Si is $9.5 \times 10^{-32}\ \mathrm{cm}^6/\mathrm{s}$, and

$$G_{o,\mathrm{ehh}} = R_{o,\mathrm{ehh}} = k_{\mathrm{ehh}}\, n_o p_o^2 = 9.5 \times 10^{-32}\ \mathrm{cm}^6/\mathrm{s} \times 10^{17}\ \mathrm{cm}^{-3} \times (10^3\ \mathrm{cm}^{-3})^2 = 9.5 \times 10^{-9}\ \mathrm{cm}^{-3}\cdot\mathrm{s}^{-1}$$

Clearly, in n-type Si, electron-assisted Auger processes dominate over hole-assisted ones.

3.3.3 Trap-assisted thermal generation and recombination

We have already mentioned that traps are very effective in enhancing the thermal generation and recombination rates in a semiconductor. In Si microelectronic devices, trap-assisted processes constitute the predominant generation and recombination mechanisms. The standard treatment of trap-assisted generation and recombination is the Shockley–Read–Hall model, a rather involved derivation described in detail in Advanced Topic AT3.1. The key conclusions of the SRH model, as it is known, can be understood from a simplified version of it for the case of a trap that is located precisely at E_i, the intrinsic Fermi level. This is not a very restrictive assumption since the most effective traps are those close to the middle of the gap (see Advanced Topic AT3.1). In this study, we disregard the charge state of the trap. We can do this if the trap concentration is small relatively to the majority carrier concentration and the trap charge does not affect charge neutrality arguments.

Consider a semiconductor with traps at energy $E_t = E_i$, the intrinsic Fermi level. If the trap states could be viewed as regular electron states, the probability that the trap is occupied is given by (see Problem 3.7)

$$f(E_t) = f(E_i) = \frac{1}{1 + \exp\frac{E_i - E_F}{kT}} \simeq \frac{n_i}{n_i + p_o} \tag{3.9}$$

The approximate symbol is to reflect the fact that this expression is valid under Maxwell–Boltzmann statistics. With a trap concentration given by N_t, the concentration of traps that are occupied by electrons is then

$$n_{to} = N_t f(E_i) = N_t \frac{n_i}{n_i + p_o} \tag{3.10}$$

and the concentration of empty traps is simply

$$N_t - n_{to} = N_t - N_t \frac{n_i}{n_i + p_o} = N_t \frac{p_o}{n_i + p_o} \tag{3.11}$$

The concentration of traps that are occupied depends on where the Fermi level is and therefore on the doping type. If the Fermi level is above E_i, the case for n-type material, $p_o \ll n_i$ and $n_{to} \simeq N_t$; the traps are mainly occupied by electrons. If the Fermi level is below E_i, a p-type case, $p_o \gg n_i$ and $n_{to} \simeq N_t n_i / p_o \ll N_t$; the traps are predominantly empty. These two situations are illustrated in Fig. 3.6.

A trap is in communication with both the conduction and valence bands. Four possible processes can take place: electron capture, electron emission, hole capture, and hole emission. These are sketched in Fig. 3.7 (as usual, the term electron here refers to electrons in the conduction band). Leading dependencies for the rates at which these four processes take place can be readily obtained.

The rate of *capture of electrons* from the conduction band must be proportional to the concentration of electrons in the conduction band times the concentration of empty traps, since those are the two species involved. Thus

$$r_{o,ec} = c_e n_o (N_t - n_{to}) \tag{3.12}$$

where c_e is a proportionality constant called the *electron capture coefficient*.

The reciprocal process of *electron emission* to the conduction band should depend only on the concentration of occupied traps since the conduction band is largely empty of electrons. Then

$$r_{o,ee} = e_e n_{to} \tag{3.13}$$

where e_e is also a proportionality constant called the *electron emission coefficient.*

Similarly, the rate of *hole capture* from the valence band is proportional to the hole concentration and the concentration of occupied traps

$$r_{o,hc} = c_h p_o n_{to} \tag{3.14}$$

The rate of its reciprocal process, *hole emission* to the valence band, is proportional to the number of empty traps alone

$$r_{o,he} = e_h (N_t - n_{to}) \tag{3.15}$$

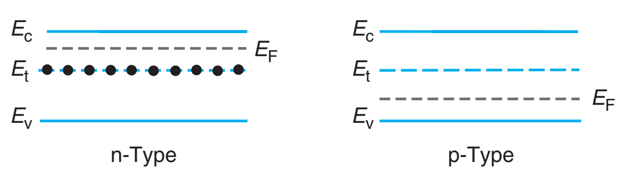


FIGURE 3.6
Illustration of occupancy of midgap traps depending on the doping level. Left: in an n-type semiconductor, the traps are mainly full. Right: in a p-type semiconductor, the traps are mainly empty.

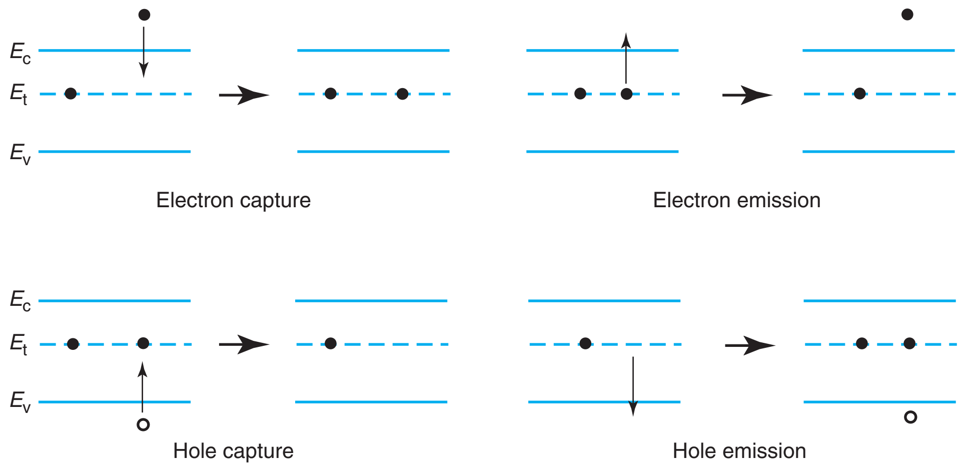


FIGURE 3.7
Trap processes. From left to right and top to bottom: electron capture, electron emission, hole capture, and hole emission.

The principle of detailed balance demands that in equilibrium all reciprocal processes take place at equal rates. Therefore, $r_{o,ec} = r_{o,ee}$ and $r_{o,hc} = r_{o,he}$ (but this does not imply $r_{o,ec} = r_{o,ee} = r_{o,hc} = r_{o,he}$). This establishes a relationship between the capture and emission coefficients for each type of carrier in thermal equilibrium:

$$e_e = c_e n_o \frac{N_t - n_{to}}{n_{to}} = c_e n_i \tag{3.16}$$

$$e_h = c_h p_o \frac{n_{to}}{N_t - n_{to}} = c_h n_i \tag{3.17}$$

where the last identity in each equation is easily established using Eq. (3.10).

The values of the capture and emission coefficients depend on the physical details of the trap involved. These constants, as well as those introduced earlier in this chapter, can be calculated from first principles. These calculations, however, are involved and contain many assumptions. Furthermore, in the case of trap-assisted generation and recombination, the identity of the trap involved in a given situation is seldom known. For these reasons, it is preferred to extract the value of these constants from actual experiments that are designed to probe the dynamics of generation and recombination.

We can now get compact expressions for the rates at which the traps communicate with the conduction and the valence bands. Plugging Eq. (3.16) into (3.13) and using Eq. (3.10), we have

$$r_{o,ec} = r_{o,ee} = c_e N_t \frac{n_i^2}{n_i + p_o} \tag{3.18}$$

Proceeding similarly with Eqs. (3.17) and (3.14) and using Eq. (3.11), we get

$$r_{o,hc} = r_{o,he} = c_h N_t \frac{n_i p_o}{n_i + p_o} \tag{3.19}$$

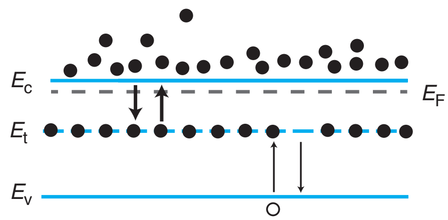


FIGURE 3.8
In an n-type semiconductor in equilibrium, a midgap trap communicates much more readily with the conduction band than with the valence band.

It is common to define:

$$\tau_{eo} = \frac{1}{N_t c_e} \tag{3.20}$$

$$\tau_{ho} = \frac{1}{N_t c_h} \tag{3.21}$$

τ_{eo} and τ_{ho} defined this way are constants that are specific to the trap in question and that scale inversely with the trap concentration. They have units of s.

With these definitions, we can rewrite Eqs. (3.18) and (3.19) in the following form:

$$r_{o,ec} = r_{o,ee} = \frac{1}{\tau_{eo}} \frac{n_i^2}{n_i + p_o} \tag{3.22}$$

$$r_{o,hc} = r_{o,he} = \frac{1}{\tau_{ho}} \frac{n_i p_o}{n_i + p_o} \tag{3.23}$$

These two equations give the rates at which the traps communicate with the conduction band and the valence band, respectively.

These equations can be further simplified for extrinsic situations. For n-type, for example

$$r_{o,ec} = r_{o,ee} \simeq \frac{n_i}{\tau_{eo}} \tag{3.24}$$

$$r_{o,hc} = r_{o,he} \simeq \frac{p_o}{\tau_{ho}} \tag{3.25}$$

For n-type material, if τ_{eo} is not very different from τ_{ho}, then $r_{o,ec} = r_{o,ee} \gg r_{o,hc} = r_{o,he}$, or in words, the rate at which the trap "talks" to the conduction band is much higher than the rate at which it talks to the valence band. This makes sense since in an n-type semiconductor, the trap is basically full of electrons and the conduction band has also lots of electrons in it. As a result, both electron capture and electron emission are very likely. On the contrary, there are few holes in the valence band and also few empty traps. Therefore, hole capture and emission are very unlikely. This situation is qualitatively illustrated in Fig. 3.8. A reverse situation occurs for a p-type semiconductor.

EXERCISE 3.3

Consider an n-type Si sample with $N_D = 10^{17}\ \mathrm{cm}^{-3}$ *at room temperature where there is a dominant trap located at* E_i *with a concentration* $N_t = 10^{16}\ \mathrm{cm}^{-3}$. *The electron and hole capture rates for this trap are* $c_e = c_h = 10^{-11}\ \mathrm{cm}^3/\mathrm{s}$. (a) *Estimate the rate at which generation and recombination events take place by means of trap-assisted processes in thermal equilibrium.* (b) *Estimate the occupation probability of the trap.*

(a) First, we compute τ_{eo} and τ_{ho}:

$$\tau_{eo} = \frac{1}{N_t c_e} = \frac{1}{10^{16} \times 10^{-11}} = 10^{-5}\ \mathrm{s}$$

$$\tau_{ho} = \frac{1}{N_t c_h} = \frac{1}{10^{16} \times 10^{-11}} = 10^{-5}\ \mathrm{s}$$

We can now estimate the rate of exchange of electrons between the trap and the conduction band in thermal equilibrium. This is given by

$$r_{o,ec} = r_{o,ee} \simeq \frac{n_i}{\tau_{eo}} = \frac{10^{10}}{10^{-5}} = 10^{15}\ \mathrm{cm}^{-3} \cdot \mathrm{s}^{-1}$$

Similarly, the rate of exchange of holes between the trap and the valence band in thermal equilibrium is given by

$$r_{o,hc} = r_{o,he} \simeq \frac{p_o}{\tau_{ho}} = \frac{10^{3}}{10^{-5}} = 10^{8}\ \mathrm{cm}^{-3} \cdot \mathrm{s}^{-1}$$

Even though the trap is located at E_i and $c_e = c_h$, the trap communicates with the conduction band at a rate seven orders of magnitude higher than with the valence band. This is because the semiconductor is n-type and most traps are filled with electrons.

(b) The trap occupation probability is given by

$$f(E_t) = f(E_i) = \frac{n_i}{n_i + p_o} \simeq \frac{10^{10}}{10^{10} + 10^{3}} \simeq 1$$

The trap occupation probability is unity because the Fermi level is well above the level of the trap.

The next section deals with the nonequilibrium situation.

3.4 GENERATION AND RECOMBINATION RATES OUTSIDE EQUILIBRIUM

When a semiconductor departs from thermal equilibrium as a result of some external influence, the rates of the reciprocal generation and recombination processes no longer necessarily balance. Consider, for example, a semiconductor in which generation and recombination in equilibrium are dominated by a band-to-band optical process. If this sample is illuminated with light of energy larger than the bandgap energy, the incoming photons will generate electrons and holes. If the intensity of the external light is not too high and the temperature of the sample does not change, the rate of the background band-to-band optical generation should not be affected since the supply of covalent bonds that can be broken is extremely large. In contrast, with more electrons and holes around, the band-to-band optical recombination rate will be enhanced since carriers are more likely to find each other. In consequence, shining light results in an increase in the band-to-band optical recombination rate over the generation rate. Interestingly, this is precisely what is required to reestablish equilibrium. If the source of radiation is removed, the preeminence of recombination over generation insures that, after a certain time, equilibrium will be recovered. The argument holds regardless of the nature of the dominant paths for generation and recombination in thermal equilibrium.

This response of semiconductors to departures from equilibrium is at the heart of semiconductor device operation. If we wish to compute the current flowing through a device under bias or its dynamic response after a certain time-dependent excitation, we must be able to calculate the electron and hole concentrations under nonequilibrium conditions. This requires us to develop an understanding of the resulting imbalance between generation and recombination, $G - R$. In thermal equilibrium, we have already learned that the total generation rate equals the total recombination rate and the carrier concentrations do not change in time. Mathematically, $n = n_o$, $p = p_o$, $G_{oi} = R_{oi}$ (for all i), and $G_o = R_o$. Outside equilibrium, that need not be the case, depending on the circumstances, the total rate of electron–hole pair generation might well be different from the total rate of electron–hole pair recombination. Then, in general, $n \neq n_o$, $p \neq p_o$, $G_i \neq R_i$ (for all i), and $G \neq R$ (notice that we reserve the subindex "o" for the thermal equilibrium situation). This is schematically illustrated in Fig. 3.9.

In these kinds of situations, it is useful to define the *net recombination rate* as the total imbalance between the rates of recombination and generation by all processes

$$U = R - G \tag{3.26}$$

In fact, under nonequilibrium conditions, all generation/recombination processes get thrown out of balance. Hence, it is also useful to define a net recombination rate for each process i, in the following way:

$$U_i = R_i - G_i \tag{3.27}$$

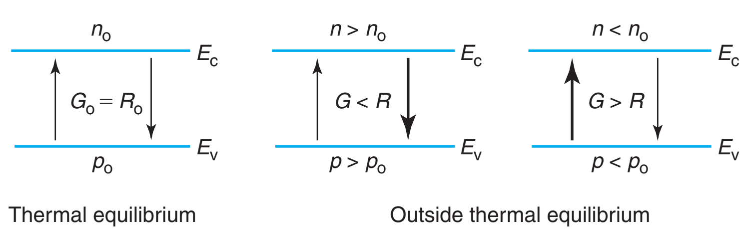


FIGURE 3.9
Schematic illustration of generation and recombination rates in and outside thermal equilibrium.

If $R_i > G_i$, then $U_i > 0$ and there is net recombination by process i taking place. Conversely, if $G_i > R_i$, then $U_i < 0$ and net generation by process i prevails.

When several generation/recombination paths act simultaneously but independently, then the overall net recombination rate is the sum of the net rate due to each process, that is

$$U = \sum_{i} U_i \tag{3.28}$$

Once a perturbation is eliminated, it is the net recombination rate that sets the dynamics of recovery of thermal equilibrium, as we will see later. In general, one needs to account for all processes when computing U. In practice, one or at most two are usually dominant. We now proceed to derive expressions for U for the three generation/recombination processes that we are studying in this chapter.

Let us consider first band-to-band optical generation and recombination. We just argued in the example above that for moderate disruptions from equilibrium, the optical generation rate is not modified from its equilibrium value. We can then use Eqs. (3.3) and (3.5) outside equilibrium

$$G_{rad} = g_{rad} = k_{rad} n_o p_o \tag{3.29}$$

The optical recombination rate, on the other hand, depends on the product of the electron and hole concentrations. If these are modified from the equilibrium values, the recombination rate will change as a result. The arguments that allowed us to write Eq. (3.4) still apply since we need one hole and one electron to complete a recombination event. We can therefore write

$$R_{rad} = k_{rad} np \tag{3.30}$$

If, as discussed above, the semiconductor is not too disturbed, the proportionality constant k_{rad} in Eqs. (3.29) and (3.30) is unmodified from its equilibrium value. If $n > n_o$ and $p > p_o$, then $R_{rad} > R_{o,rad}$. Conversely, if $n < n_o$ and $p < p_o$, then $R_{rad} < R_{o,rad}$. This is illustrated in Fig. 3.10.

The net recombination rate due to band-to-band optical process is given by

$$U_{rad} = R_{rad} - G_{rad} = k_{rad}(np - n_o p_o) \tag{3.31}$$

This equation states that the net recombination rate is proportional to the excess np product. If $np > n_o p_o$, there is net recombination. If, on the other hand, $np < n_o p_o$, generation prevails.

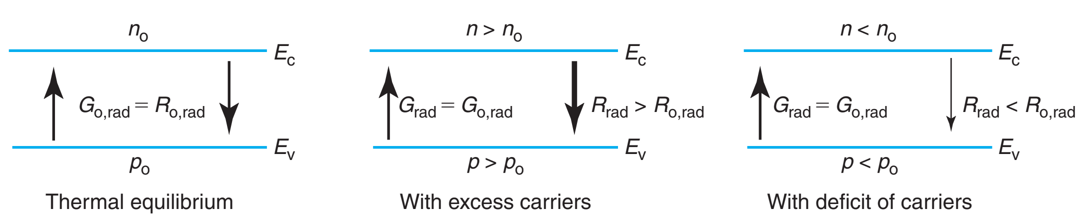


FIGURE 3.10
Schematic illustration of generation and recombination rates due to an optical band-to-band process under and outside thermal equilibrium.

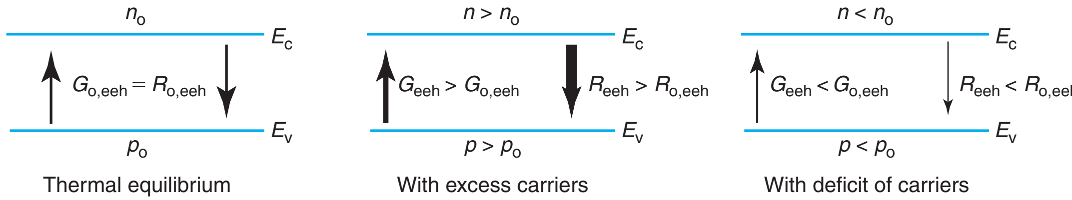


FIGURE 3.11
Schematic illustration of generation and recombination rates of a hot-electron Auger process under and outside thermal equilibrium.

The balance between Auger processes is also disturbed outside equilibrium. Let us consider a hot-electron process. In this case, the Auger generation rate [given in equilibrium by Eq. (3.6)] is modified since the number of hot electrons has increased. Outside equilibrium, we can write

$$G_{eeh} = g_{eeh}n \tag{3.32}$$

The recombination rate is also modified. Similarly to Eq. (3.7), we can write

$$R_{eeh} = k_{eeh}n^2p \tag{3.33}$$

where we have assumed that the proportionality constants g_{eeh} and k_{eeh} are unchanged from thermal equilibrium, and Eq. (3.8) still applies. This is illustrated in Fig. 3.11.

The net recombination rate is then

$$U_{eeh} = k_{eeh}n(np - n_op_o) \tag{3.34}$$

Interestingly, we also find that the net recombination rate due to a hot-electron Auger process is driven by the excess np product. A similar equation applies for a hot-hole Auger process. The total net recombination rate due to both Auger processes can be then written as

$$U_{Auger} = (k_{eeh}n + k_{ehh}p)(np - n_op_o) \tag{3.35}$$

Lastly, there are trap-assisted thermal processes. Following similar arguments to those outlined above, all four processes are perturbed from equilibrium because the electron and hole concentrations are modified from their equilibrium values, and the occupation probability of the trap is also likely to change. When we account for the relationships between the various proportionality constants given in Eqs. (3.16) and (3.17), the rates of the four processes are given by

$$r_{ec} = c_en(N_t - n_t) \tag{3.36}$$

$$r_{ee} = c_en_in_t \tag{3.37}$$

$$r_{hc} = c_hpn_t \tag{3.38}$$

$$r_{he} = c_hn_i(N_t - n_t) \tag{3.39}$$

A sketch depicting the change in these rates in the presence of excess carriers is shown in Fig. 3.12.

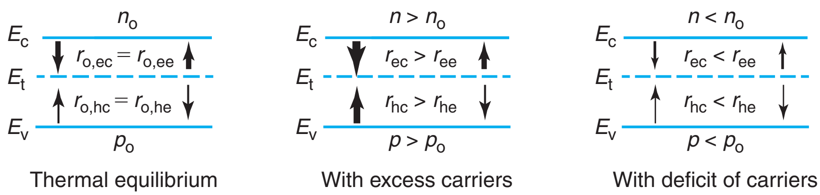


FIGURE 3.12
Schematic illustration of generation and recombination rates of a trap-assisted process under and outside thermal equilibrium.

Since the completion of a recombination event requires the capture of an electron and a hole, the net recombination rate is equal to the net rate of electron capture which in turn is equal to the net rate of hole capture. In a mathematical form

$$U_{tr} = r_{ec} - r_{ee} = r_{hc} - r_{he} \tag{3.40}$$

From the second equality and using Eqs. (3.36)–(3.39), we can derive n_t as

$$n_t = N_t \frac{c_e n + c_h n_i}{c_e(n + n_i) + c_h(p + n_i)} \tag{3.41}$$

Notice how n_t is no longer simply set by the doping level as was the case in thermal equilibrium Eq. (3.10). The trap occupation probability is in general affected when the semiconductor is disturbed from equilibrium, and this needs to be taken into account.

Substituting Eq. (3.41) into Eq. (3.40), we find after some algebra

$$U_{tr} = \frac{np - n_o p_o}{\tau_{ho}(n + n_i) + \tau_{eo}(p + n_i)} \tag{3.42}$$

Equation (3.42) shows that the net recombination rate of a trap-assisted process is also driven by the excess *np* product. However, due to the appearance of n and p in the denominator, the dependence of U_{tr} on n and p is more complicated than in the other two generation/recombination processes. More details of the Shockley–Read–Hall model are worked out in Problem 3.13.

The total net recombination rate due to all paths that we have discussed is the sum of Eqs. (3.31), (3.35), and (3.42):

$$U = U_{rad} + U_{Auger} + U_{tr} \tag{3.43}$$

For device CAD, Eqs. (3.31), (3.35), (3.42), and (3.43) are all that is needed to completely characterize the dynamic response of a semiconductor outside equilibrium. As formulated, however, these equations do not provide much physical intuition. We are going to study two specific cases for which analytical expressions can be derived. These two cases are of enormous practical importance as they appear very frequently in different device contexts.

3.4.1 Quasi-neutral low-level injection; recombination lifetime

Carrier *injection* refers to situations in which the carrier concentrations exceed their equilibrium values as a result of some outside excitation. In these circumstances, it is useful to define an *excess electron concentration* n' and an *excess hole concentration* p' as follows:

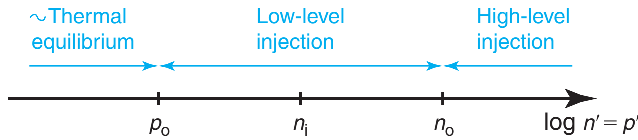


FIGURE 3.13
Schematic diagram showing notation for the different levels of carrier injection in an n-type semiconductor. The semiconductor is quasi-neutral under all conditions.

$$n = n_o + n' \tag{3.44}$$

$$p = p_o + p' \tag{3.45}$$

Since electrons and holes are generated in pairs and recombine in pairs, it is reasonable to assume that their excesses are identical, that is, $n' = p'$.[1] In a typical extrinsic semiconductor, the majority carrier concentration exceeds the minority carrier concentration by many orders of magnitude. There is then a broad intermediate carrier injection level in which the minority carrier concentration can be enhanced by several orders of magnitude, while the majority carrier concentration is negligibly modified. This is the *low-level injection regime*. For an n-type semiconductor, this condition can be mathematically expressed as

$$p_o \ll n' = p' \ll n_o \tag{3.46}$$

This is graphically illustrated in Fig. 3.13.

Correspondingly for p-type material, the low-level injection condition is written as

$$n_o \ll n' = p' \ll p_o \tag{3.47}$$

The low-level injection regime covers a very broad range of carrier injection. For example, for Si doped with $N_D = 10^{17}$ cm^{-3} at room temperature, $n_o = 10^{17}$ cm^{-3} and $p_o \sim 10^3$ cm^{-3}. The low-level injection regime applies if $n' = p'$ takes values between $\sim 10 \times 10^3 = 10^4$ and $\simeq 0.1 \times 10^{17} = 10^{16}$ cm^{-3}, about 12 orders of magnitude![2]

For a sufficiently high generation rate, the excess carrier concentration may exceed the majority carrier value. This situation is called *high-level injection* and is discussed in more detail in Advanced Topic AT3.2 at the end of the chapter.

The use of inequalities in Eq (3.46) or (3.47) greatly simplifies the expressions for the net recombination rates given above. In particular, the driving term $np - n_o p_o$ that appears in all expressions of U becomes

$$np - n_o p_o = (n_o + n')(p_o + p') - n_o p_o \simeq (n_o + p_o)p' + p'^2 \simeq (n_o + p_o)p' \tag{3.48}$$

With this simplification, all expressions of U follow an equation of the form

$$U_i \simeq \frac{p'}{\tau_i} \tag{3.49}$$

[1] We will learn in Chapter 5 that the equality of n' and p' is not generally perfect though it is a very good assumption in many cases.

[2] In this book, we will frequently use the *10% rule* as a pragmatic engineering criteria to make approximations.

where τ_i is a constant characteristic of each process. We define and discuss the physical meaning of τ_i below. The expressions for this constant for each individual process are as follows:

$$\tau_{rad} = \frac{1}{k_{rad}(n_o + p_o)} \tag{3.50}$$

$$\tau_{Auger} = \frac{1}{[k_{eeh}n_o + k_{ehh}p_o + n'(k_{eeh} + k_{ehh})](n_o + p_o)} \simeq \frac{1}{(k_{eeh}n_o + k_{ehh}p_o)(n_o + p_o)} \tag{3.51}$$

$$\tau_{tr} = \frac{\tau_{ho}n_o + \tau_{eo}p_o + n'(\tau_{eo} + \tau_{ho})}{n_o + p_o} \simeq \frac{\tau_{ho}n_o + \tau_{eo}p_o}{n_o + p_o} \tag{3.52}$$

The two approximations performed in Eqs. (3.51) and (3.52) apply if k_{eeh} and c_e are not very different from k_{ehh} and c_h, respectively (see Problem 3.8).

Equations 3.49–3.52 convey a very important message. In a semiconductor under low-level injection conditions, the net recombination rate depends linearly on the *excess* carrier concentrations through a constant that is characteristic of the material and temperature. This constant is called the *recombination lifetime*, and has a simple meaning. Equation 3.49 states that in a unit volume of semiconductor with an excess carrier concentration p', there are p'/τ_i net recombination events through process i. The mean time between recombination events is τ_i/p'. Since there are p' excess carriers, the average time that an excess carrier survives in a unit volume before recombining is τ_i, the recombination lifetime. All things being equal, the longer the lifetime, the fewer recombination events occur for a given excess carrier concentration.

Equations 3.50–3.52 define a separate lifetime for each generation/recombination path. In device analysis, however, we are generally interested in the overall lifetime that results from the combination of all recombination mechanisms. Combining Eqs. (3.43) and (3.49), we obtain

$$U \simeq \frac{p'}{\tau} \tag{3.53}$$

with

$$\frac{1}{\tau} = \sum_i \frac{1}{\tau_i} \tag{3.54}$$

This reciprocal sum of lifetimes implies that the process with the smallest lifetime dominates the generation/recombination dynamics of a semiconductor.

We can further simplify the expressions of the lifetime for an extrinsic semiconductor, the most common occurrence. For n-type, for example, $n_o \gg p_o$ and we get

$$\tau_{rad} = \frac{1}{k_{rad}n_o} \tag{3.55}$$

$$\tau_{Auger} = \frac{1}{k_{eeh}n_o^2} \tag{3.56}$$

$$\tau_{tr} = \tau_{ho} \tag{3.57}$$

Similar equations apply to a p-type semiconductor.

EXERCISE 3.4

Estimate the recombination lifetime due to each generation–recombination path as well as the overall recombination lifetime for an n-type Si sample under quasi-neutral low-level injection conditions with $N_D = 10^{17}$ cm^{-3} *at room temperature. Assume that in this sample, there is a dominant trap located at* E_i *with a concentration* $N_t = 10^{16}$ cm^{-3}. *The electron and hole capture rates for this trap are* $c_e = c_h = 10^{-11}$ cm^3/s.

The lifetime due to radiative recombination is given by Eq. (3.55):

$$\tau_{rad} = \frac{1}{k_{rad} n_o} = \frac{1}{2 \times 10^{-15}\ \text{cm}^3/\text{s} \times 10^{17}\ \text{cm}^{-3}} = 5\ \text{ms}$$

The lifetime due to Auger processes is given by Eq. (3.56):

$$\tau_{Auger} = \frac{1}{k_{eeh} n_o^2} = \frac{1}{1.8 \times 10^{-31}\ \text{cm}^6/\text{s} \times (10^{17}\ \text{cm}^{-3})^2} = 0.56\ \text{ms}$$

The lifetime due to trap-assisted processes is given by Eqs. (3.57) and (3.21):

$$\tau_{tr} = \tau_{ho} = \frac{1}{10^{16}\ \text{cm}^{-3} \times 10^{-11}\ \text{cm}^3/\text{s}} = 10\ \mu\text{s}$$

The overall lifetime is given by the inverse of the inverse sum of these three lifetimes:

$$\tau = \frac{1}{\frac{1}{\tau_{rad}} + \frac{1}{\tau_{Auger}} + \frac{1}{\tau_{tr}}} = \frac{1}{\frac{1}{5\times10^{-3}} + \frac{1}{5.6\times10^{-4}} + \frac{1}{10^{-5}}} = 9.8\ \mu\text{s}$$

Clearly, the trap-assisted process dominates this situation and sets the value of the recombination lifetime.

Figure 3.14 sketches the dependence of the three lifetimes as a function of the majority carrier concentration n_o. τ_{rad} displays a simple inverse dependence with n_o. τ_{Auger} has an inverse dependence on n_o^2. τ_{tr} is independent of n_o. It is particularly insightful to reflect on these dependencies. For a simple optical band-to-band process, the more majority carriers there are, the easier it is for a minority carrier to find one and complete a recombination event. In consequence, $\tau_{rad} \sim n_o^{-1}$. In an Auger process, two majority carriers in close proximity to the minority carrier are required, one to recombine with, the second one to absorb the energy that is released. This implies $\tau_{Auger} \sim n_o^{-2}$. For a trap-assisted process, what counts is the trap occupancy probability. In n-type material under low-level injection conditions, traps located at E_i are always full of electrons. You can easily verify this from Eq. (3.41). The bottleneck to electron–hole recombination is then the ability of a hole to find a trap (process r_{hc} is the rate-limiting step to recombination). In the timescale of hole capture, the trap always appears as if it were in equilibrium with the conduction band. This is illustrated in Fig. 3.15. The presence of more majority carriers does not change the trap occupancy probability and does not therefore make the recombination more likely. In consequence, τ_{tr} is independent of n_o.

For a given doping level, the higher the trap concentration, the smaller the trap-assisted recombination lifetime becomes. This is because in Eqs. (3.20) and (3.21), τ_{eo} and τ_{ho} depend

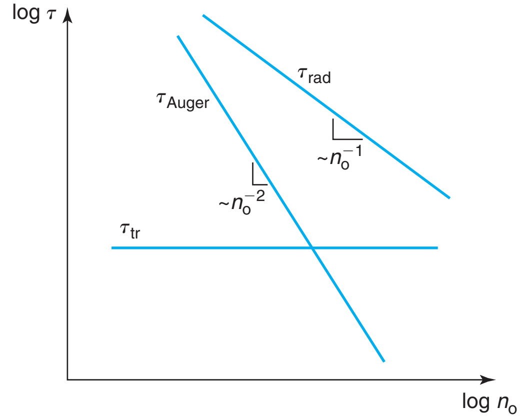


FIGURE 3.14
Schematic diagram showing the dependence of the recombination lifetime on donor concentration in an n-type semiconductor under low-level injection for all major mechanisms.

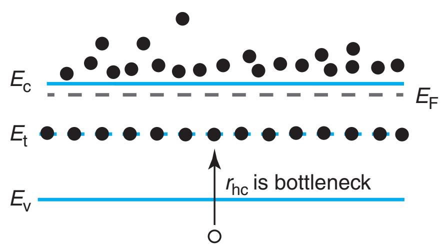


FIGURE 3.15
In an n-type semiconductor, midgap traps are all occupied by electrons. In consequence, the bottleneck for trap-assisted recombination is the capture of a hole.

inversely on N_t. This is readily seen in experiments. For example, Fig. 3.16 shows the recombination lifetime of a p-Si sample contaminated with Fe as a function of the Fe concentration. Clearly, there is an inversely proportional relationship between lifetime and Fe concentration. In order to obtain long lifetimes, it is clear that contamination needs to be minimized, particularly of those elements that introduce states close to the middle of the bandgap.

The dependence of the various lifetimes on the majority carrier concentration illustrated in Fig. 3.14 implies that in a given semiconductor, the lifetime is in general a strong function of doping level. This is illustrated in Fig. 3.17 for n- and p-type Si at room temperature. These two figures show a compilation of recently reported lifetime data in Si as a function of doping level and a simple model that captures the essence of the dependence. The analytical description of the model is given in Appendix E. These graphs indeed show that at high doping levels, the lifetime decreases with doping level, following an Auger-like inverse-square dependence. However, for low doping levels, the reported lifetime exhibits considerable scatter. This is because at low doping levels, the lifetime is controlled by traps and their nature, concentration, and location

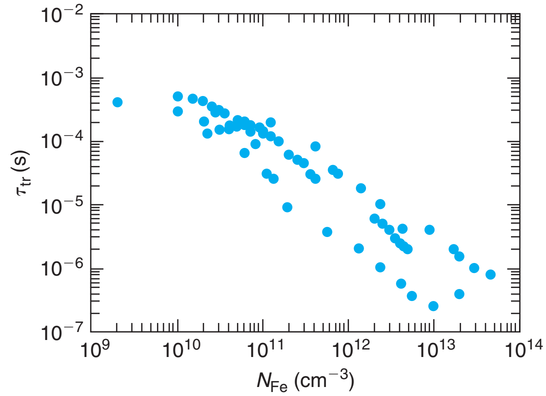


FIGURE 3.16
Recombination lifetime versus Fe concentration in p-Si ($N_A = 10^{15}$ cm^{-3}) [Republished with permission of The Institute of Electrical and Electronic Engineers, Inc. (IEEE), from Carrier Lifetimes in Silicon, *IEEE Transactions on Electron Devices*, Vol 44, Issue no. 1, Pg. 160–170, 1997; permission conveyed through Copyright Clearance Center, Inc].

inside the bandgap are all sample and process dependent. The data presented in these graphs allow us to conclude that at low doping levels, there is considerable room for "lifetime engineering." Ultrapure material subject to very clean processing, for example, can yield extraordinarily high lifetimes; judicious introduction of traps in the same material can produce very small lifetimes. At moderate and high doping levels, there is much less opportunity for this, since the lifetime becomes controlled by Auger recombination, a fundamental mechanism.

The lines in Fig. 3.17 indicate a typical doping dependence of the carrier lifetime characteristic of modern microelectronic device processing. These dependencies lie midway between the very high lifetimes needed for high-efficiency solar cells, for example, and the low lifetimes associated with suboptimal processing. The functional dependencies shown in Fig. 3.17 come in handy to carry out back-of-the-envelope estimates of carrier dynamics under many common circumstances.

3.4.2 Extraction; generation lifetime

As we deepen our understanding of semiconductor device physics, we will get to appreciate the fact that it is not only possible to inject excess carriers into a certain portion of a semiconductor, but also feasible to *extract* them from selected regions by means of electric fields. When one extracts carriers, the dynamic balance that exists between generation and recombination in thermal equilibrium is also broken. Under carrier extraction, the carrier concentrations are driven *below* their equilibrium values and all recombination rates by all possible internal mechanisms are reduced. A situation results that is characterized by *net generation* of electron–hole pairs. Once again, the semiconductor reacts so as to try to reestablish thermal equilibrium. In a case like this, if we were to eliminate the cause of carrier extraction, the semiconductor will evolve toward equilibrium over a certain time. Understanding the dynamics of this recovery process is the goal of this section.

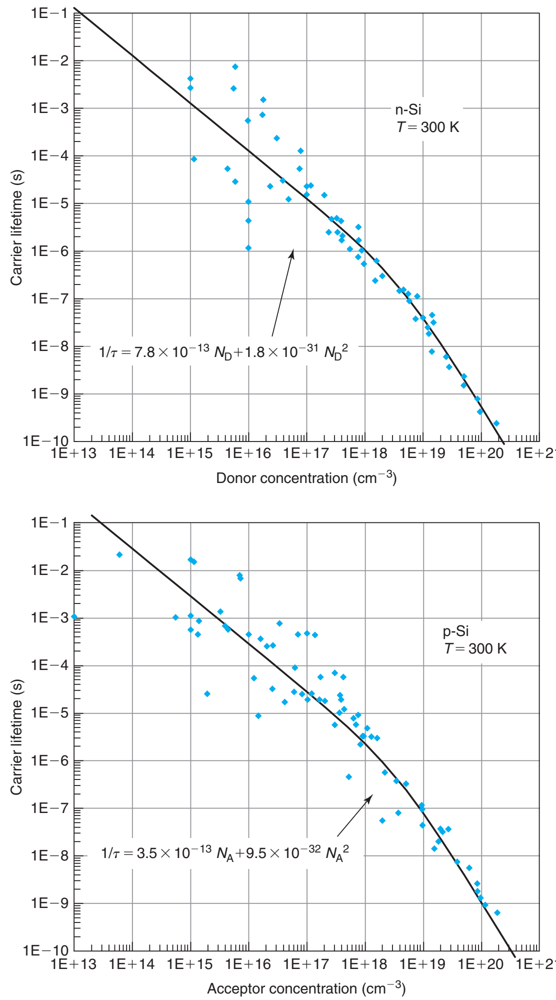


FIGURE 3.17
Recombination lifetime in n- and p-type Si as a function of doping level at 300 K. The data points represent recent experimental measurements. The lines represent typical values of recombination lifetime for microelectronic devices.

There are two different extraction cases to consider. In the first one, a few carriers are extracted without significantly upsetting the majority carrier concentration. The region under extraction remains quasi-neutral, that is, $n' \simeq p' < 0$. This is a situation qualitatively identical to the low-level injection case studied above. The only difference is that in this case we have $np < n_o p_o$ rather than $np > n_o p_o$ as in the case of low-level injection. In consequence, the equations that we derived for low-level injection apply except that U will be negative instead of positive. The proper time constant for the process is the recombination lifetime derived under low-level injection conditions.

It is possible, however, to extract significant amounts of majority carriers in addition to minority carriers. As we will see later in this book, this happens, for example, in the depletion region of a PN junction under reverse bias, or in the capacitor of a DRAM cell under certain conditions. In this case, the generation rate is substantially different.

Consider a situation in which n and p are reduced by many orders of magnitude below their equilibrium values so that $n, p \ll n_i$. The net generation rate (equal to minus the net recombination rate) due to a radiative process, from Eq. (3.31), becomes

$$-U_{rad} \simeq k_{rad} n_i^2 \tag{3.58}$$

The net generation rate due to Auger processes, from Eq. (3.35), becomes

$$-U_{Auger} = (k_{eeh} n + k_{ehh} p) n_i^2 \tag{3.59}$$

This can become truly negligible if n and p are sufficiently reduced. This makes sense since, as we saw in Section 3.3.2, Auger generation processes demand energetic carriers to provide the required energy to break the bonds. In the absence of any kind of carrier, Auger generation is suppressed.

The net generation due to a trap-assisted process, from Eq. (3.42), becomes

$$-U_{tr} \simeq \frac{n_i}{\tau_{ho} + \tau_{eo}} \tag{3.60}$$

It is easy to understand the physical meaning of this result. In the case of complete extraction, electron and hole captures are negligible. In this case, the generation rate is limited by electron emission and hole emission. These two rates must be identical, since electrons and holes are generated in pairs. Then, $U \simeq -r_{ee} = -r_{he}$ in Eqs. (3.37) and (3.39). From this argument and using also Eqs. (3.20) and (3.21), Eq. (3.60) follows in a straightforward manner.

EXERCISE 3.5

Estimate the net generation rate at room temperature in a sample of n-type Si with $N_D = 10^{17}$ cm^{-3} for which n and p are 10 orders of magnitude below their equilibrium values. Assume that in this sample, there is a dominant trap located at E_i with a concentration $N_t = 10^{16}$ cm^{-3}. The electron and hole capture rates for this trap are $c_e = c_h = 10^{-11}$ cm^3/s.

Under the indicated conditions, $n \simeq 10^7$ cm^{-3} and $p \simeq 10^{-7}$ cm^{-3}. The net generation rate by means of band-to-band radiative processes is given by Eq. (3.58):

$$-U_{rad} = k_{rad} n_i^2 = 2 \times 10^{-15} \times 10^{20} = 2 \times 10^5 \text{ cm}^{-3} \cdot \text{s}^{-1}$$

The net generation rate due to Auger processes is given by Eq. (3.59):

$$-U_{Auger} = (k_{eeh} n + k_{ehh} p) n_i^2 \simeq k_{eeh} n n_i^2 = 1.8 \times 10^{-31} \times 10^7 \times 10^{20} = 1.8 \times 10^{-4} \text{ cm}^{-3} \cdot \text{s}^{-1}$$

The net generation rate due to trap-assisted process is given by Eq. (3.60):

$$-U_{\mathrm{tr}} \simeq \frac{n_{\mathrm{i}}}{\tau_{\mathrm{ho}} + \tau_{\mathrm{eo}}} = \frac{10^{10}}{2 \times 10^{-5}} = 5 \times 10^{14}\ \mathrm{cm^{-3} \cdot s^{-1}}$$

where $\tau_{\mathrm{ho}} = \tau_{\mathrm{eo}}$ was calculated as in Exercise 3.4.

As the previous exercise illustrates, trap-assisted generation is dominant in extraction situations. Notice also that in this case, the generation rate is a constant independent of the carrier concentration. This is different from the recombination rate under low-level injection conditions given by Eq. (3.49), where U depends on the excess carrier concentration. This fact has important implications for the dynamics of recovery as we will see in the next section.

The denominator in Eq. (3.60) has units of time. Because of this, it is often given the name of *generation lifetime*, τ_g. There are two important points to be made about this concept. First, even though it is widely used, the physical significance of τ_g is unclear. One physical interpretation for it is the time required to generate n_{i} electron–hole pairs per unit volume. However, this is of doubtful usefulness since this is a condition typically far from thermal equilibrium. The second point to be made is that defined this way, the generation lifetime is typically different from the recombination lifetime, as you can see by comparing Eqs. (3.52) and (3.60). Care has to be exercised not to confuse them in a practical situation.

3.5 DYNAMICS OF EXCESS CARRIERS IN UNIFORM SITUATIONS

We are finally in a position to start examining the detailed dynamics of excess carriers in semiconductors. At this time, we are only equipped to handle some relatively simple, although fairly useful, cases. We need to restrict ourselves to an extrinsic uniformly doped semiconductor with the surfaces far away from the region of interest. Chapter 5 will address the impact of surfaces in dynamic problems. To create interesting and practical situations outside equilibrium, we irradiate the semiconductor with deep penetrating radiation that is uniformly absorbed everywhere. As a result, even outside equilibrium, the carrier concentrations are uniform in space.

In equilibrium, we saw above that the total generation rate equals the total recombination rate. Outside equilibrium, that need not be the case. If the generation and recombination rates are not identical, the carrier concentrations change with time. Since electrons and holes are generated and recombine in pairs, in general we can then write

$$\frac{dn}{dt} = \frac{dp}{dt} = G - R \tag{3.61}$$

This equation is physically intuitive. If generation prevails over recombination, the carrier concentrations increase with time. If recombination wins over generation, the carrier concentrations decrease with time.

In these situations, it is useful to distinguish between *internal* and *external* generation. Internal generation occurs through the various processes that were discussed in the previous sections: Auger, trap-assisted, and radiative. Internal generation takes place all the time, even in thermal equilibrium. In addition, external generation might occur if the semiconductor is perturbed from the outside world, such as by illumination of light of sufficient energy. This is schematically

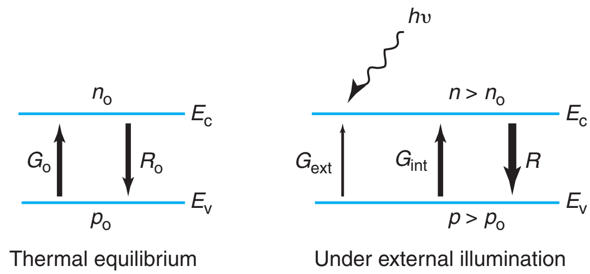


FIGURE 3.18
Schematic illustration of internal and external generation rates in a semiconductor outside equilibrium due to external illumination.

illustrated in Fig. 3.18. In general, the total generation rate is the sum of rates of generation due to internal and external mechanisms

$$G = G_{int} + G_{ext} \tag{3.62}$$

Note that the same cannot be said of the recombination rate. Recombination is all "internal."

With this definition, the imbalance between generation and recombination can be written as

$$G - R = G_{int} + G_{ext} - R = G_{ext} - U \tag{3.63}$$

where we have used the definition of net recombination rate made in Eq. (3.26), with G referring there to G_{int}.

Plugging this into Eq. (3.61), we get

$$\frac{dn}{dt} = \frac{dp}{dt} = G_{ext} - U \tag{3.64}$$

This equation also makes physical sense. If the rate of external generation exceeds the rate of net recombination in the semiconductor (by all internal processes), then the carrier concentrations increase with time. If the contrary happens, the carrier concentrations decrease with time.

Under conditions of low-level injection (and also in high-level injection, see Advanced Topic AT3.2 if the dominant recombination mechanism is through traps), we have found that U is equal to the excess carrier concentration over the recombination lifetime. Using Eqs. (3.44) and (3.45), and the quasi-neutrality statement $n' = p'$, Eq. (3.64) becomes

$$\frac{dp'}{dt} = G_{ext} - \frac{p'}{\tau} \tag{3.65}$$

In deriving this equation, we have also used the fact that n_o is time independent. This familiar first-order differential equation describes the excess carrier dynamics in a uniform situation. In the next subsections, we study some interesting examples. Before that, note that in a perfectly **static** case in which nothing changes as a function of time, Eq. (3.65) reduces to

$$p' = G_{ext}\tau \tag{3.66}$$

and the excess carrier concentration is linear in the generation rate and the carrier lifetime.

3.5.1 Example 1: Turn-on transient

Consider a semiconductor in equilibrium that for $t \geq 0$ is subject to a uniform time-independent generation rate $G_{ext} = g_l$. A particular solution to Eq. (3.65) suitable to this case is simply a constant. The homogeneous solution of the differential equation 3.65 is of the form $e^{-t/\tau}$. Applying the initial condition $p' = 0$ at $t = 0$, the solution for $t \geq 0$ is easily found to be

$$p'(t) = g_l\tau(1 - e^{-t/\tau}) \qquad \text{for } t \geq 0 \tag{3.67}$$

This is sketched in Fig. 3.19. At $t = 0$, the external generation turns on and the carrier concentration starts increasing. However, as time goes on, the rate of growth of the carrier concentration decreases and eventually saturates. Mathematically, this is captured by the $e^{-t/\tau}$ term in Eq. (3.67) that becomes very small for $t \gg \tau$. This behavior is easy to explain physically. Initially, the carrier concentration increases because the total generation rate dominates over the recombination rate. As the carrier concentration builds up, the recombination rate increases but the generation rate does not change. In due time, the recombination rate equals the generation rate and the carrier concentration does not change anymore. Utilizing Eq. (3.65) for $t \to \infty$ when the time derivative becomes negligible, we find that the excess carrier concentration needed for U to match the external generation rate g_l is $p'(t \to \infty) = g_l\tau$. This is the *steady-state* excess carrier concentration. The characteristic time constant of the exponential rise is τ, the recombination lifetime.

3.5.2 Example 2: Turn-off transient

Consider now what happens if, after a steady-state situation under constant illumination, the light is turned off at $t = 0$. Mathematically, this is also an easy problem. Since for $t > 0$, $G_{ext} = 0$, the particular solution $p'_p = 0$ is a good guess. The initial condition at $t = 0$ is $p' = g_l\tau$, since that was the steady-state excess carrier concentration before the light was turned off. The complete solution is then

$$p'(t) = g_l\tau e^{-t/\tau} \quad \text{for } t \geq 0 \tag{3.68}$$

which is plotted in Fig. 3.20.

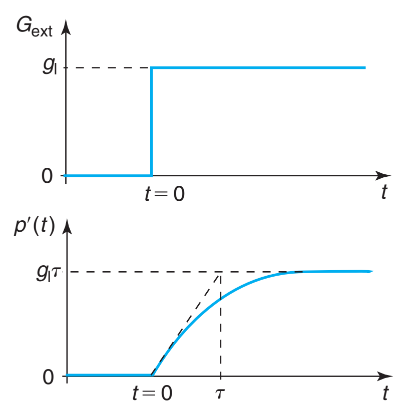


FIGURE 3.19
Turn-on transient: light excitation (above) and resulting excess carrier concentration (below) as a function of time.

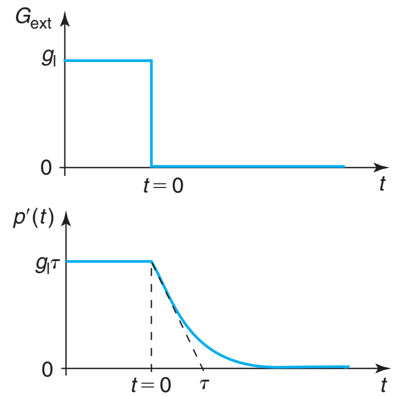


FIGURE 3.20
Turn-off transient: light excitation (above) and resulting excess carrier concentration (below) as a function of time.

This result makes good sense. After the disruption from equilibrium is eliminated, the recombination rate is not balanced anymore by the external generation rate causing the carrier concentration to decrease. This reduces the recombination rate too. Given enough time, the generation and recombination rates will match up exactly and a new steady state is obtained. Since $G_{ext} = 0$, this new steady state is precisely thermal equilibrium.

3.5.3 Example 3: A pulse of light

Consider now that at $t = 0$, a pulse of light of duration T shines over a sample that was in thermal equilibrium. We now have to consider separately the solutions for $0 \leq t \leq T$ and $t \geq T$. For $0 \leq t \leq T$, the solution is identical to the turn-on transient that we discussed above. For $t \geq T$, the solution is similar to the turn-off transient, except that the initial condition is no longer $p'(0) = g_1\tau$, but $p'(T) = g_1\tau(1 - e^{-T/\tau})$. The complete solution is then:

$$p'(t) = g_1\tau(1 - e^{-t/\tau}) \qquad \text{for } 0 \leq t \leq T \tag{3.69}$$

$$p'(t) = g_1\tau(1 - e^{-T/\tau})e^{-(t-T)/\tau} \qquad \text{for } t \geq T \tag{3.70}$$

This is graphed in Fig. 3.21.

A widely used technique to measure the recombination lifetime consists of illuminating a semiconductor sample with a short and intense light pulse. After the pulse is ended, the excess carrier concentration decays in an exponential way with a characteristic time constant that is precisely τ, as Eq. (3.70) indicates. If we had a way to monitor the time evolution of the excess carrier concentration, we would be able to determine the lifetime. In a general case, a few of the recombination events take place through a radiative process in which light is emitted. The intensity of the emitted light is directly proportional to the excess carrier concentration. Therefore, by observing the decay of the *luminescence* radiation emitted by the sample, as this process is known, we can determine the lifetime. This is illustrated in Fig. 3.22, which shows the

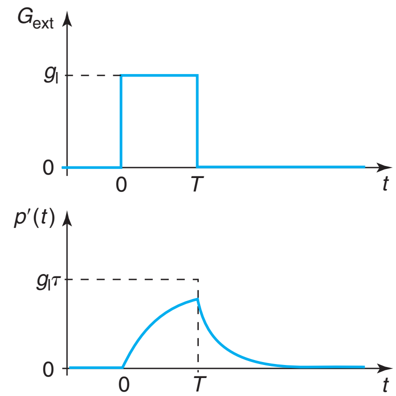


FIGURE 3.21
Pulse: light excitation (top) and time evolution of carrier concentration (bottom).

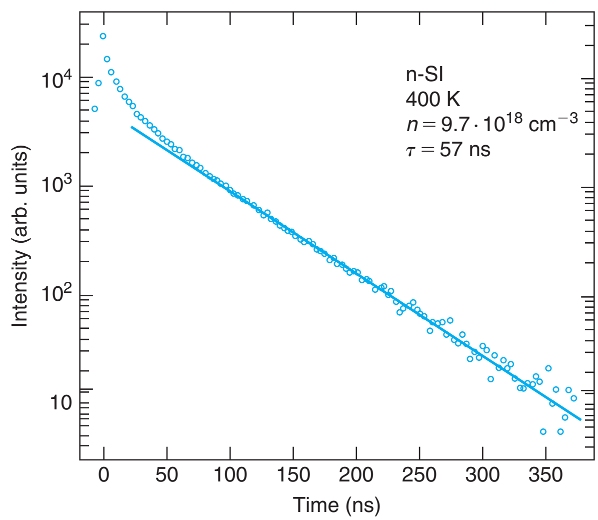


FIGURE 3.22
Decay of luminescence emitted by a Si sample after excitation by a laser pulse at $\lambda = 0.5145$ µm. The lifetime for this sample is 57 ns [Reproduced from Auger coefficients for highly doped and highly excited silicon, *Applied Physics Letters*, 31, 346, 1977 with the permission of American Institute of Physics (AIP) publishing].

luminescence emitted by a Si sample when illuminated by a laser pulse. Except for an initial faster transitory, this graph shows an almost ideal exponential decay in time.[3] The time constant of the luminescence decay is, to first order, the recombination lifetime (57 ns for the particular case shown in the figure).

[3]We will study in Section 5.8 that the initial faster decay is due to surface recombination.

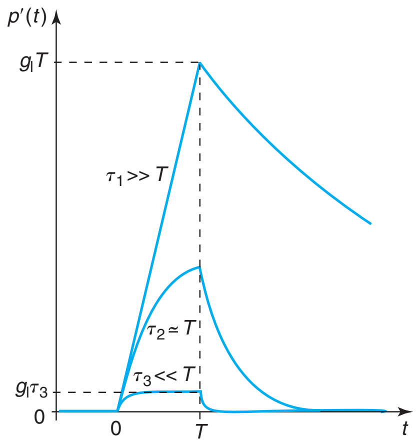


FIGURE 3.23
Excess carrier concentration in response to a pulse of light for samples with different values of the recombination lifetime. For a case in which the lifetime is much longer than the light pulse, $\tau_1 \gg T$, the carrier concentration ramps up linearly (top trace). If the lifetime, on the other hand, is much shorter than the light pulse, $\tau_3 \ll T$, the carrier concentration quickly reaches its maximum value $g_l\tau_3$. In this case, the time evolution of n' resembles $g_l(t)$. This is a quasi-static case (bottom trace). For intermediate values, $\tau_2 \simeq T$, the response has a characteristic exponential rise (middle trace).

Looking back at the general response to a pulse captured in Eqs. (3.69) and (3.70), it is interesting to reflect on two extreme cases that might arise. They are illustrated in Fig. 3.23 where a pulse of light of duration T impinges on three different samples with widely different recombination lifetimes. On all samples the light generates electron–hole pairs at a rate g_l. In a sample with recombination lifetime much longer than the pulse width, $\tau_1 \gg T$, negligible recombination takes place during the duration of the pulse. In this limit, the time evolution of the carrier concentration is approximately a ramp $p'(t) \simeq g_l t$ that reaches the value $g_l T$ by the end of the pulse (you only need to take the two leading terms of the Taylor series expansion of e^x for small x; see Appendix D for common expansions). When the light pulse is turned off, the carrier concentration drops exponentially.

In a sample with a recombination lifetime much shorter than the pulse duration, $\tau_3 \ll T$, significant recombination takes place during the pulse and the carrier concentration achieves its saturation value $g_l\tau_3$ much before the end of the pulse. In the timescale of the pulse duration, the time evolution of the carrier concentration has a nearly identical shape to the pulse itself. This is called a *quasi-static* situation.[4]

[4]It is appropriate to define a "static situation" here too. This is when nothing changes as a function of time. The situation can be out of equilibrium, such as under constant light illumination, but any transient associated with its onset has already died away.

For a case in which the recombination lifetime is comparable to the light pulse width, $\tau_2 \simeq T$, we have a typical exponential rise and decay.

It is important to understand well the quasi-static concept. The best practical definition of a quasi-static situation is one in which in a certain timescale of interest, the time transients are negligible. If we go back to Eq. (3.65) and neglect the first term of the equation, the quasi-static solution is

$$p'(t) \simeq G_{\text{ext}}(t)\tau \tag{3.71}$$

This is just like the static solution derived in Eq. (3.66). This means that a quasi-static situation is one in which there is negligible memory in the system. This happens when only small changes take place in the timescale of the characteristic time constant of the system, in this case, the recombination lifetime. For the pulse example above, this means when $T \gg \tau$. Even in a case like this, the quasi-static solution might not be acceptable if our timescale of interest is shorter than the lifetime. That might be the case, for example, if we are interested on the response to the rising edge of the pulse.

EXERCISE 3.6

A laser pulse strikes a p-Si sample with $N_A = 10^{17}$ cm^{-3} and results in a uniform generation of electron–hole pairs at a rate $g_1 = 10^{24}$ cm$^{-3} \cdot$ s^{-1} at room temperature. Estimate the maximum time duration that the pulse can have so as not to drive the sample into high-level injection.

In Fig. 3.17, we see that a p-type Si sample at room temperature has a typical carrier lifetime of about 30 μs. If the laser pulse is much longer than this time, the maximum carrier concentration is $g_1\tau \simeq 3 \times 10^{19}$ cm^{-3}. This number exceeds by over two orders of magnitude the doping level. This is high-level injection. There are two implications of this. First, the high-level injection lifetime will most surely be different from 30 μs (see AT3.2), with the consequence that the calculation that we just made is inaccurate. Second, regardless of this inaccuracy, if such a powerful laser is left shining on this sample, it will surely drive it into high-level injection.

To avoid this, the laser pulse width should be significantly shorter than the lifetime. If the pulse width is $T \ll \tau$, the carrier concentration at the end of the pulse is $g_1 T$. For the sample not to be in high-level injection, this product has to be no higher than about 10^{16} cm^{-3}. This means that $T \leq 10$ ns.

The concepts of quasi-static and **steady state** should not be confused. A steady-state situation is achieved when the *initial* transient associated with the excitation has died out. The difference between these two occurrences is best understood by considering a train of pulses that are switched on at $t = 0$, as illustrated in Fig. 3.24. If the pulse width, t_1, and spacing, t_2, are much longer than the lifetime, then a steady-state quasi-static situation is obtained right with the first pulse. If, on the other hand, the pulse width and spacing are much shorter than the lifetime, several pulses will be required to reach steady state which never becomes quasi-static (the time evolution of the carrier concentration has a very different shape from the time dependence of the light intensity). You can easily solve this problem mathematically (see Problem 3.5).

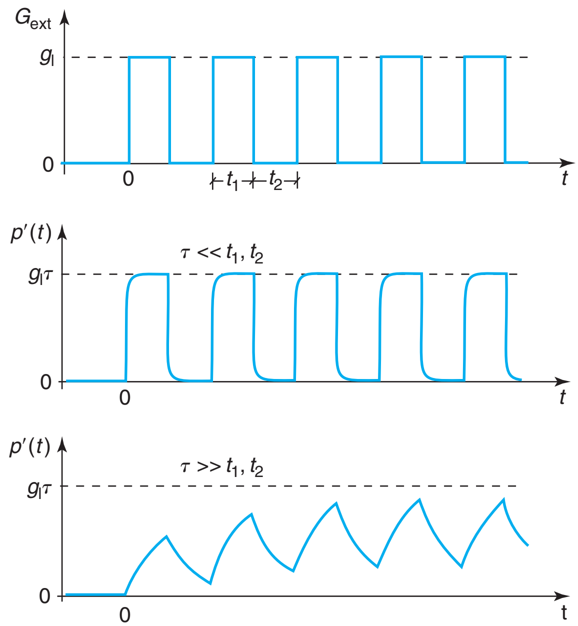


FIGURE 3.24
Response to a train of generation pulses (top). When the lifetime is much shorter than the pulse width and spacing, a steady-state quasi-static situation is achieved with the first pulse (middle). If the lifetime is much longer than the pulse width and spacing, the steady-state situation takes a few pulses to be established and a quasi-static situation is never obtained (bottom).

EXERCISE 3.7

A train of 1 ns *light pulses with a frequency of* 50 MHz *suddenly illuminates an n-Si sample with* $N_D = 10^{16}$ cm^{-3} *at room temperature. The laser intensity is such that the sample remains under low-level injection conditions at all times. How many pulses does it take for a steady-state situation to be established? Is that situation quasi-static?*

n-Si with $N_D = 10^{16}$ cm^{-3} at room temperature has a recombination lifetime of about 100 μs (see Fig. 3.17). Steady state is only established when the initial transient associated with the turning on of the pulse train has died off. This takes three lifetimes or so, or about 300 μs. At a frequency of 50 MHz, this translates into 3×10^{-4} s $\times$ 50×10^6 s^{-1} $= 15,000$ pulses. Since the pulse width is only 1 ns and the period is $T = 1/50$ MHz $= 20$ ns, both much shorter than the lifetime, the final steady state is not quasi-static.

3.6 SURFACE GENERATION AND RECOMBINATION

So far we have only considered recombination and generation that take place in the bulk of a semiconductor. At a semiconductor surface, the periodicity of the semiconductor crystal is

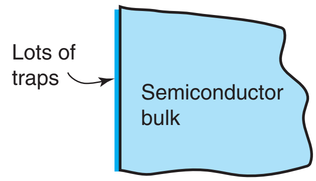


FIGURE 3.25
Sketch of semiconductor surface indicating the presence of surface generation and recombination centers.

abruptly broken. Depending on the details of the surface, whether it is covered by another material or not, many surface atoms may have "dangling bonds" that are unsatisfied, or could be bonded with a foreign material, such as O or C or a metal. We will study this in more detail in Chapter 8. As a result of these disruptions, trap states associated with the surface appear in the middle of the forbidden energy gap. These states can act as effective generation and recombination centers. This is illustrated in Fig. 3.25. As microelectronic devices become smaller, surface phenomena take a more prominent role and surface generation and recombination become increasingly important processes.

In thermal equilibrium, the principle of detailed balance demands that the generation and recombination rates at a surface precisely cancel out. Outside thermal equilibrium, this balance can be broken. For example, if there are excess carriers in the semiconductor bulk in the vicinity of the surface, there can be a net surface recombination rate. Carriers flow from the bulk toward the surface where they effectively recombine via surface traps. If the surface is very efficient at recombining, the carrier flow rate toward the surface can be high. If the surface is well "passivated," for example, there are few recombination centers, the flow rate is low.

It is not difficult to build a simple model for the net surface recombination rate via surface traps in analogy with the bulk processes discussed above. If we do that, we find, for example, that under low-level conditions, the net surface recombination rate, U_s (in units of $cm^{-2} \cdot s^{-1}$), exhibits a linear dependence on the excess carrier concentration

$$U_s \simeq Sp'(s) \tag{3.72}$$

where $p'(s)$ is the excess carrier concentration in the vicinity of the surface. Intuitively, U_s is the net electron or hole flux into the surface. Its units are $cm^{-2} \cdot s^{-1}$. More details on this are given in Chapter 5.

The proportionality constant S in Eq. (3.72) has units of cm/s and is called *surface recombination velocity*. Its physical interpretation is straightforward. S is the perpendicular component of the velocity of excess carriers "diving" into the surface to recombine. The higher S is, the more effective a surface is at recombining excess carriers. Similarly to the carrier lifetime, S depends on the density and location of traps inside the semiconductor bandgap. S is then a function of the detailed preparation technique utilized for the surface. For example, covering the surface with a thermally grown SiO_2 layer is an excellent way to "passivate" it and achieve very low values of surface recombination velocity. This is utilized, for example, in solar cells to ensure a high-energy conversion efficiency. On the contrary, if a metal is placed directly on top of a surface, the surface recombination velocity becomes very high and the surface does not tolerate any disruption from thermal equilibrium in its vicinity.

EXERCISE 3.8

Consider a p-type Si sample with $N_A = 10^{17}$ cm^{-3} *at room temperature. As a result of external light illumination, the electron carrier concentration in the vicinity of one of its surfaces is* $n(s) = 3 \times 10^{15}$ cm^{-3}. *This surface is characterized by a surface recombination velocity of* $S = 10^5$ cm/s. *Compute the net recombination rate at that surface.*

This is straightforward. In the vicinity of the surface in question, the sample is in low-level injection. Since the equilibrium electron concentration is of order $n_o \sim 10^3$ cm^{-3}, then $n'(s) \simeq n(s) = 3 \times 10^{15}$ cm^{-3}. Using Eq. (3.72), we can then compute the net recombination rate as

$$U_s = Sn'(s) = 10^5 \text{ cm/s} \times 3 \times 10^{15} \text{ cm}^{-3} = 3 \times 10^{20} \text{ cm}^{-2} \cdot \text{s}^{-1}$$

Note that the units of surface recombination or generation rate are cm$^{-2} \cdot$ s^{-1}.

If the neighborhood of a surface is driven below equilibrium by extraction, the surface traps become additional paths for the generation of carriers. In this case, the surface recombination rate is negative and follows an equation of the form:

$$U_s \simeq -S_g n_i \tag{3.73}$$

where the proportionality constant S_g is called the *surface generation velocity*. As in the case of bulk generation–recombination, the magnitude of S_g is generally different from that of S.

A study of the dynamics of the carrier concentration as a result of surface generation or recombination is complicated by the fact that carriers must flow from the bulk toward the surface or vice versa. Depending on the magnitude of S, this carrier transit time, as it is known, might limit the dynamics of the process. We will learn the mechanisms of carrier flow in Chapter 5. This will allow us to calculate the time evolution of excess carriers and their spatial dependence in the presence of surface recombination in Chapter 5.

3.7 SUMMARY

- In thermal equilibrium, there is a balance between generation and recombination. This balance is “detailed”: each generation process balances out with its reciprocal recombination process.
- If the carrier concentrations are disturbed from the thermal equilibrium values, a semiconductor reacts by trying to reestablish thermal equilibrium. For example, if excess carriers are injected into a semiconductor, net recombination results.
- In quasi-neutral low-level injection situations, the net recombination rate is proportional to the excess carrier concentration over equilibrium

$$U \simeq \frac{p'}{\tau}$$

  In these circumstances, the carrier dynamics are described by a simple first-order linear differential equation with a characteristic time constant called *recombination lifetime*

$$\frac{dn'}{dt} = G_{\text{ext}} - \frac{p'}{\tau}$$

- Recombination lifetime, τ, is the mean time that an excess carrier "survives" in a unit volume before recombining. In Si at room temperature:
  (a) for low doping levels, the recombination lifetime is dominated by trap-assisted processes and is inversely proportional to doping level, although there is strong sample to sample variation;
  (b) for high doping levels, the recombination lifetime is dominated by Auger processes and is inversely proportional to the square of the doping level.
- Time-dependent perturbations with a timescale much longer than the recombination lifetime are particularly simple to study because in this timescale, the time transients are negligible. These are called *quasi-static situations*.
- The disruption of the perfect periodicity of the lattice results in surface states through which surface generation and recombination take place.
- Under low-level injection conditions, surface recombination is characterized by the surface recombination velocity. In this case, the net surface recombination rate is given by

$$U_s \simeq Sp'(s)$$

3.8 FURTHER READING

Three books contain material that is worthwhile reading to expand on this chapter.

Semiconductor Statistics by J. S. Blakemore, Dover, 1978 (ISBN 0-486-65362-5, QC611.B52). This book has extensive discussions of carrier generation and recombination mechanisms in semiconductors. It considers in detail the dependence of the carrier lifetimes on carrier concentration and temperature. It also solves a number of nonlinear time transient problems. The presentation is always very physical and insightful. This is an excellent reference for situations that are out of the mainstream. The exhaustiveness of the presentation does not lend this book to casual reading.

Fundamentals of Solid-State Electronics by C.-T. Sah, World Scientific, 1991 (ISBN 9810206372, TK7871.85.S23). In this monumental tome, Sah presents a very organized and systematic discussion of a wealth of carrier generation and recombination mechanisms. A unifying, albeit slightly intimidating, notation allows for a comprehensive treatment. Sah emphasizes physical intuition. There are many pictorial diagrams sketching the various transitions.

Semiconductor Material and Device Characterization by D. K. Schroder, Wiley, 1990 (ISBN 0-471-51104-8, QC611.S335). This is a book that describes the most popular characterization techniques for semiconductors and semiconductor devices. Its level is accessible to students with some knowledge of the fundamentals. Chapter 8 of this book is titled *Carrier Lifetime*. It gives an introduction to the concept of recombination and generation lifetimes in the bulk and at surfaces. It also presents a variety of lifetime measuring techniques pointing out their limitations. The same author has written a number of illuminating articles on this subject. A particularly interesting and approachable one is "Carrier Lifetimes in Silicon," *IEEE Trans. Electr. Dev. 44, 160 (1997)*.

AT3.1 SHOCKLEY–READ–HALL MODEL

Since its introduction in 1952, the Shockley–Read–Hall model has become the standard treatment for trap-assisted generation and recombination. Its essence was described in the main body of this chapter for a trap located at the intrinsic Fermi level. In this Advanced Topic section, a more general derivation is presented that is valid for any location of the trap inside the semiconductor bandgap. To avoid unnecessary duplication, this derivation relies on the material presented in Sections 3.3.3 and 3.4. Unless indicated, the formulation that follows applies for n- and p-type semiconductors.

Consider a trap located at energy E_t inside the bandgap, as sketched in Fig. 3.7. In equilibrium, the occupation probability depends on the location of the Fermi level relative to the trap location

$$f(E_t) = \frac{1}{1 + \exp \frac{E_t - E_F}{kT}} \tag{3.74}$$

If the trap concentration is N_t, the concentration of traps that are occupied in thermal equilibrium is

$$n_{to} = N_t f(E_t) = \frac{N_t}{1 + \exp \frac{E_t - E_F}{kT}} \tag{3.75}$$

If the Fermi level is above the trap level, most of the traps are full; if it is below, they are predominantly empty.

The four basic mechanisms involved in the generation and recombination of carriers via traps were described in Section 3.3.3. The equations for the rates at which these mechanisms take place in thermal equilibrium, Eqs. (3.12)–(3.15), remain unchanged. The relationships between the capture coefficients and the emission coefficients are obtained by applying the principle of detailed balance. Following a similar procedure as in Section 3.3.3, we get

$$e_e = c_e n_o \frac{N_t - n_{to}}{n_{to}} = c_e n_i \exp \frac{E_t - E_i}{kT} \tag{3.76}$$

$$e_h = c_h p_o \frac{n_{to}}{N_t - n_{to}} = c_h n_i \exp \frac{E_t - E_i}{kT} \tag{3.77}$$

These expressions converge to Eqs. (3.16) and (3.17) when $E_t = E_i$, as expected.

With these relationships, the rates at which the four basic processes take place outside equilibrium are given by

$$r_{ec} = c_e n (N_t - n_t) \tag{3.78}$$

$$r_{ee} = c_e n_t n_i \exp \frac{E_t - E_i}{kT} \tag{3.79}$$

$$r_{hc} = c_h p n_t \tag{3.80}$$

$$r_{he} = c_h (N_t - n_t) n_i \exp \frac{E_i - E_t}{kT} \tag{3.81}$$

The net recombination rate is equal to the net rate of electron capture, which in turn is equal to the net rate of hole capture. This is expressed mathematically in Eq. (3.40). The second equality in this equation allows us to solve for n_t, the concentration of traps that are occupied:

$$n_t = N_t \frac{c_e n + c_h n_i \exp \dfrac{E_i - E_t}{kT}}{c_e \left(n + n_i \exp \dfrac{E_t - E_i}{kT} \right) + c_h \left(p + n_i \exp \dfrac{E_i - E_t}{kT} \right)} \tag{3.82}$$

Inserting this result back into Eq. (3.40) and using the definitions of τ_{eo} and τ_{ho} in Eqs. (3.20) and (3.21), we obtain the net recombination rate

$$U_{tr} = \frac{np - n_i^2}{\tau_{ho} \left(n + n_i \exp \dfrac{E_t - E_i}{kT} \right) + \tau_{eo} \left(p + n_i \exp \dfrac{E_i - E_t}{kT} \right)} \tag{3.83}$$

This equation reverts back to Eq. (3.42) for $E_t = E_i$.

We now distinguish between the processes of generation and recombination.

AT3.1.1 Recombination lifetime

Under excess carrier concentration conditions, there is net recombination. The concept of recombination lifetime is very general. It is the average time that an excess carrier takes to recombine. So, if we have a quasi-neutral situation with n' excess carriers per unit volume and there is net recombination given by U_{tr}, then the recombination lifetime is simply $\tau_{tr} = n'/U_{tr}$. Using Eq. (3.83), we get

$$\tau_{tr} = \frac{1}{n_o + p_o + n'} \left[\tau_{ho} \left(n_o + n' + n_i \exp \frac{E_t - E_i}{kT} \right) + \tau_{eo} \left(p_o + n' + n_i \exp \frac{E_i - E_t}{kT} \right) \right] \tag{3.84}$$

where we have also used Eqs. (3.44) and (3.45).

Equation 3.84 is the most general expression for the carrier lifetime due to trap recombination. If low-level injection conditions prevail and τ_{eo} is not too different from τ_{ho}, then the n' terms in this equation can all be neglected. The equation then simplifies to

$$\tau_{tr} \simeq \frac{1}{n_o + p_o} \left[\tau_{ho} \left(n_o + n_i \exp \frac{E_t - E_i}{kT} \right) + \tau_{eo} \left(p_o + n_i \exp \frac{E_i - E_t}{kT} \right) \right] \tag{3.85}$$

This reverts to Eq. (3.52) if $E_t = E_i$.

It is particularly interesting to discuss the effect of the trap location on the lifetime in an extrinsic semiconductor. Let us take an n-type semiconductor with $n_o \simeq N_D \gg p_o$. In this case, Eq. (3.85) simplifies to

$$\tau_{tr} \simeq \tau_{ho} \left(1 + \frac{n_i}{N_D} \exp \frac{E_t - E_i}{kT} \right) + \tau_{eo} \frac{n_i}{N_D} \exp \frac{E_i - E_t}{kT} \tag{3.86}$$

τ_{tr} is sketched in Fig. 3.26 as a function of the location of the trap inside the bandgap. This figure shows that there is a broad minimum in the lifetime that is obtained for traps located near the middle of the bandgap. For an n-type semiconductor, as sketched in Fig. 3.15, traps that are located close to the midgap are fully occupied with electrons and the bottleneck to recombination is the capture of holes. In this broad intermediate regime, the net recombination

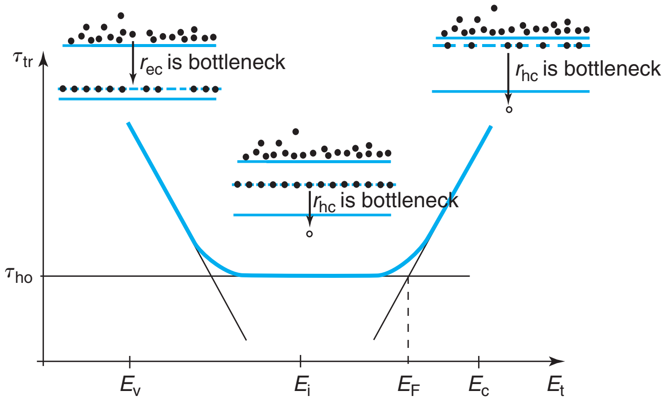


FIGURE 3.26
Sketch of recombination lifetime as a function of location of the trap inside the bandgap (n-type material, low-level injection). The insets schematically illustrate the bottleneck processes for the different situations.

rate is equal to the hole capture rate, which is given by Eq. (3.80). In this regime, $p \simeq n'$, and, from Eq. (3.82), $n_t \simeq N_t$. In consequence,

$$U_{tr} \simeq r_{hc} \simeq c_h n' N_t = \frac{n'}{\tau_{ho}} \tag{3.87}$$

This is independent of the specific location of the trap.

If the trap is close to either band edge, it loses its recombination effectiveness and the lifetime increases. The reason for this is different at both ends of the bandgap. For traps close to the conduction band, hole capture remains the bottleneck to recombination, but the electron occupation probability of the traps drops below unity. In consequence, it becomes more difficult to match an occupied trap with a hole and the lifetime increases. The expression for U_{tr} in this instance can also be easily understood. If $E_t \gg E_F$, Eq. (3.82) results in

$$n_t \simeq N_t \frac{N_D}{n_i} \exp \frac{E_i - E_t}{kT} \tag{3.88}$$

Inserting this into Eq. (3.80) we get

$$U_{tr} \simeq r_{hc} \simeq \frac{n'}{\tau_{ho}} \frac{N_D}{n_i} \exp \frac{E_i - E_t}{kT} \tag{3.89}$$

which decreases as the trap gets closer to the conduction band.

Traps close to the valence band are largely occupied by electrons. The closer it gets to the valence band, the closer the occupation probability approaches unity. Under these conditions, electron capture becomes difficult since there are no available sites for electrons to go. The net recombination rate becomes now limited by electron capture and should be given by Eq. (3.78).

The term $N_t - n_t$ can be obtained from Eq. (3.82)

$$N_t - n_t \simeq \frac{N_t n'}{n_i} \exp\frac{E_t - E_i}{kT} \tag{3.90}$$

Inserting this into Eq. (3.78) we get

$$U_{tr} \simeq r_{ec} \simeq \frac{n'}{\tau_{eo}} \frac{N_D}{n_i} \exp\frac{E_t - E_i}{kT} \tag{3.91}$$

which decreases as E_t approaches the valence band.

AT3.1.2 Generation lifetime

Under extraction conditions, with $n \ll n_i$ and $p \ll n_i$, the net recombination rate given in Eq. (3.83) becomes

$$U_{tr} = -\frac{n_i}{\tau_{ho}\exp\frac{E_t - E_i}{kT} + \tau_{eo}\exp\frac{E_i - E_t}{kT}} \tag{3.92}$$

This reverts to Eq. (3.60) if $E_t = E_i$.

The denominator of this expression has the units of time. Because of this, it is customary to denote it as *generation lifetime* and it is given by

$$\tau_g = \tau_{ho}\exp\frac{E_t - E_i}{kT} + \tau_{eo}\exp\frac{E_i - E_t}{kT} \tag{3.93}$$

The dependence of the generation lifetime on trap position is sketched in Fig. 3.27. A very sharp minimum in the generation lifetime is obtained close to (but not necessarily at) the middle of the bandgap. In order to understand the physical origin of this behavior, let us derive the trapped electron concentration from Eq. (3.82)

$$n_t = \frac{N_t}{1 + \frac{c_e}{c_h}\exp 2\frac{E_t - E_i}{kT}} \tag{3.94}$$

If the trap is located close to the conduction band, it is mainly empty. In this case, electron emission is the bottleneck to the generation process and the trapped electron concentration is, from Eq. (3.94)

$$n_t \simeq N_t \frac{c_h}{c_e}\exp 2\frac{E_i - E_t}{kT} \tag{3.95}$$

The net recombination rate is then

$$U_{tr} \simeq -r_{ee} = -\frac{n_i}{\tau_{ho}}\exp\frac{E_i - E_t}{kT} \tag{3.96}$$

which decreases exponentially as E_t approaches the conduction band.

If the trap is located close to the valence band, it is mainly full. In consequence, hole emission is the bottleneck to the generation process. In this case, from Eq. (3.94)

$$N_t - n_t \simeq N_t \frac{c_e}{c_h}\exp 2\frac{E_t - E_i}{kT} \tag{3.97}$$

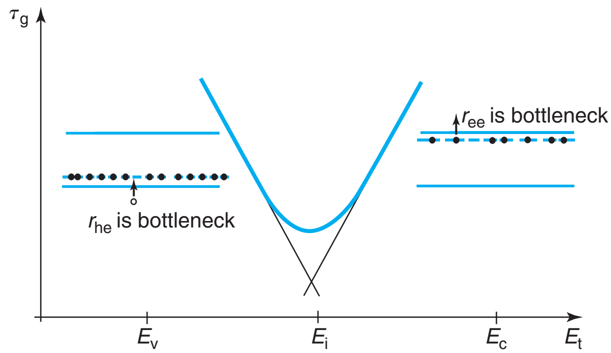


FIGURE 3.27
Sketch of generation lifetime as a function of location of the trap inside the bandgap. The insets schematically illustrate the bottleneck processes for the different situations.

The net recombination rate is given by

$$U_{tr} \simeq -r_{he} = -\frac{n_i}{\tau_{eo}} \exp \frac{E_t - E_i}{kT} \tag{3.98}$$

which decreases exponentially as the trap approaches the valence band.

An important consequence of this sharp behavior of the generation lifetime with the location of the trap is that traps close to the middle of the bandgap are particularly effective as generation centers. This contrasts with the behavior of the recombination lifetime under injection conditions in which there is a broad minimum in the lifetime in the middle of the gap. This difference opens the possibility for *lifetime engineering* of semiconductor devices in which generation and recombination lifetimes can be independently controlled for optimum performance.

AT3.2 HIGH-LEVEL INJECTION

High-level injection refers to situations where carrier levels in excess of equilibrium are high enough to overwhelm the majority carrier concentration. This happens, for example, in solar cells under concentrated sunlight, or in power devices under strong forward bias. High-level injection rarely occurs during the regular operation of microelectronic devices. It might, however, arise under special conditions. Understanding the signature and first-order physics of high-level injection is important for a microelectronics device engineer.

Mathematically, high-level injection can be stated in general as

$$n' \simeq p' \gg n_o + p_o \tag{3.99}$$

Under high-level injection, the dynamics of carrier recombination are different from the low-level injection situation we have already studied. Under high-level injection, the driving force for recombination, the excess *np* product, can be simplified to

$$np - n_o p_o \simeq n'^2 \tag{3.100}$$

This is now *quadratic* in n' as opposed to linear, as was the case of low-level injection Eq. (3.48). Because of this, the rates of recombination by the different mechanisms follow distinct dependences.

For an optical band-to-band process, Eq. (3.31) becomes

$$U_{\text{rad}} \simeq k_{\text{rad}} n'^2 \tag{3.101}$$

which exhibits a quadratic dependence on n'.

The rate of Auger recombination, from Eq. (3.35), becomes

$$U_{\text{Auger}} \simeq (k_{\text{eeh}} + k_{\text{ehh}}) n'^3 \tag{3.102}$$

now with a cubic dependence on n'.

For a trap-assisted process, on the other hand, Eq. (3.42) simplifies to

$$U_{\text{tr}} \simeq \frac{n'}{\tau_{\text{eo}} + \tau_{\text{ho}}} \tag{3.103}$$

Even in high-level injection, the net recombination rate due to a trap-assisted process is linear in the excess carrier concentration.

These results all make physical sense. For radiative recombination, the recombination rate depends on the np product [Eq. (3.30)]. With both electron and hole concentrations about equal and in very large numbers, the recombination rate must exhibit a quadratic dependence on their concentration. Similarly, the Auger recombination processes depend on n^2p or np^2, depending on whether the source of energy is an energetic electron and hole [Eq. (3.33)]. Clearly, under high-level injection conditions, Auger recombination must then follow a cubic dependence on the carrier concentration.

In trap-assisted recombination under high-level injection conditions, carrier capture vastly dominates carrier emission. In consequence, $U_{\text{tr}} \simeq r_{\text{ec}} \simeq r_{\text{hc}}$. Since both r_{ec} and r_{hc} exhibit a linear dependence on n and p, respectively, U_{tr} must also be linear. It is easy to see that using Eqs. (3.36) and (3.38) and the definitions (3.20) and (3.21), Eq. (3.103) readily follows.

Under high-level injection, we can also define a recombination lifetime for each one of these recombination processes. We use a definition identical to low-level injection, as given in Eq. (3.49). The respective high-injection lifetimes become

$$\tau_{\text{rad}}(HLI) \simeq \frac{1}{k_{\text{rad}} n'} \tag{3.104}$$

$$\tau_{\text{Auger}}(HLI) \simeq \frac{1}{(k_{\text{eeh}} + k_{\text{ehh}}) n'^2} \tag{3.105}$$

$$\tau_{\text{tr}}(HLI) \simeq \tau_{\text{eo}} + \tau_{\text{ho}} \tag{3.106}$$

Due to the nonlinear dependence of $\tau_{\text{rad}}(HLI)$ and $\tau_{\text{Auger}}(HLI)$ in high-level injection, the recombination lifetimes that characterize these processes actually depend on the injection level. $\tau_{\text{rad}}(HLI)$ goes as n'^{-1} and $\tau_{\text{Auger}}(HLI)$ goes as n'^{-2}. In contrast, the trap-assisted process is

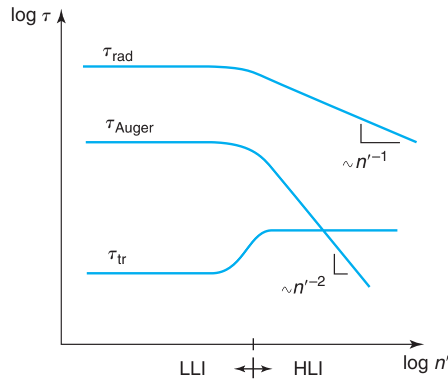


FIGURE 3.28
Schematic diagram showing the dependence of the recombination lifetime on injection level for all major mechanisms.

characterized by an injection-independent lifetime, although of different value under low-level injection conditions. This can be seen by comparing Eq. (3.106) with Eq. (3.52). In fact, for reasonably extrinsic conditions, it is easy to see that $\tau_{tr}(HLI) > \tau_{tr}(LLI)$. The injection dependence of the various lifetimes is sketched in Fig. 3.28.

Figure 3.28 indicates that as a semiconductor is driven into high-level injection, the Auger recombination lifetime decreases, while the trap-assisted recombination lifetime increases and then remains constant. Both behaviors are observed in experiments.

The peculiar behavior of the trap-assisted recombination lifetime as a function of carrier injection level derived here is not general. Equation 3.106 was obtained under the assumption that the trap is located at E_i. Advanced Topic AT3.1 discusses a more general model. From this model and following a similar approach, one can find that for traps located close to the middle of the gap, the lifetime increases as the sample goes into high-level injection. In contrast, for traps located close to the band edges, the lifetime decreases when the sample becomes highly injected.

In most high-level injection situations that are encountered in microelectronic devices, trap-assisted recombination tends to dominate. Only if the carrier concentrations reach very high values, Auger recombination may become significant.

PROBLEMS

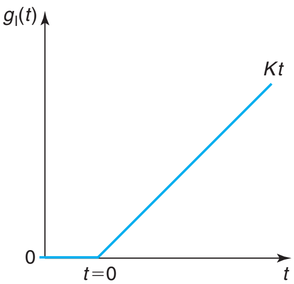
3.1 Consider a p-type Si sample with $N_A = 10^{17}$ cm^{-3} with an excess carrier concentration $n' \simeq p' = 10^{15}$ cm^{-3} at room temperature. Calculate the rates at which generation and recombination events take place by means of (i) band-to-band optical processes, (ii) Auger processes assisted by hot electrons and hot holes, and (iii) trap-assisted processes (specify the rates of emission and capture of electrons and holes). Assume that the dominant trap state in this sample is located at E_i and the concentration of traps is $N_t = 10^{16}$ cm^{-3}. The electron and hole capture rates for these traps are $c_e = c_h = 10^{-10}$ cm^3/s.

3.2 A p-type Si wafer doped with $N_A = 10^{17}\ \text{cm}^{-3}$ has been illuminated with uniformly penetrating radiation that generates $g_l = 10^{20}\ \text{cm}^{-3} \cdot \text{s}^{-1}$ for a long time. Suddenly, the radiation is turned off. Estimate the electron and hole concentrations and the net recombination rate: (a) right before the radiation is turned off, (b) 100 ns after the radiation is turned off, (c) 50 μs after the radiation is turned off, and (d) after steady state is reached.

3.3 An n-type Si wafer doped at a level of $N_D = 1 \times 10^{17}\ \text{cm}^{-3}$ is in thermal equilibrium at 300 K in the dark. At $t = 0$, light that causes uniform carrier generation throughout the wafer is turned on. The light intensity increases linearly as a function of time so that the carrier generation rate has the following time dependence:

$$g_l(t) = Kt \quad \text{for } t \geq 0$$

where K is a constant and t represents time.



(a) Derive an analytical expression for the time evolution of the minority carrier concentration in the low-level injection regime. Graph your result.
(b) Simplify result obtained in (a) above for times much shorter and much longer than the lifetime. Sketch this asymptotic behavior on the graph you obtained in (a).
(c) If the carrier lifetime of the wafer under low-level injection conditions is 15 μs and $K = 10^{22}\ \text{cm}^{-3} \cdot \text{s}^{-2}$, calculate the time at which:
- the hole concentration is doubled over the equilibrium value;
- the low-level injection solution stops being applicable.

3.4 A certain Si sample with a doping level $N_D = 10^{18}\ \text{cm}^{-3}$ exhibits a minority carrier lifetime of 1 μs at room temperature under low-level injection conditions.

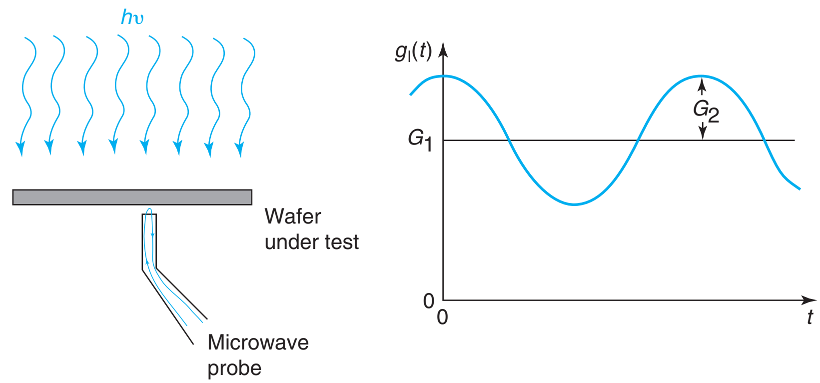
(a) Estimate the recombination rates in thermal equilibrium of as many recombination processes as you can. State your assumptions.
(b) Estimate the net recombination rate by each and every recombination process again in a situation in which $p' = n' = 10^{16}\ \text{cm}^{-3}$. State your assumptions.

3.5 Consider a general situation such as the one sketched in Fig. 3.24 in which at $t = 0$, a train of optical generation pulses is turned on over a p-type sample. The duration of each pulse is denoted as t_1, the spacing between pulses is t_2, and the period of the pulse train is $T = t_1 + t_2$. The generation rate is g_l and the minority carrier lifetime is τ.

(a) Derive an expression in terms of m for the excess electron concentration at $t = mT$.
(b) Derive an expression for the number of pulses m required for steady state to be established.

3.6 Your company wants to expand its catalog of wafer characterization tools with a contactless (and therefore clean and nondestructive) technique to measure and map minority carrier lifetimes in Si wafers. A constraint that you have been given, as lead designer, is that your company does not want to pay royalties on patents owned by competitors. This forbids you from using light pulses as the basis for the measurement.

You decide to carry out a feasibility study of a technique based on *sinusoidal* light illumination. Your idea is sketched below. A powerful lamp is modulated by a sinusoidal drive and its beam is directed to the surface of the wafer under test. The light generates electron–hole pairs that change the conductance of the wafer. This is measured by a microwave beam that bounces off the back surface of the wafer. The microwave beam can be focused to a rather small area so that a measurement of the local conductance is possible. For easiest interpretation of the results, an infrared wavelength is selected for the light so as to have uniform generation in depth throughout the entire wafer.



As a result of the sinusoidal generation on the wafer, you expect an excess carrier concentration with a sinusoidal time dependence, but, in general, phase shifted with respect to the lamp drive. You plan to measure this phase shift using an oscilloscope. You are quite sure that you can get the minority carrier lifetime from the magnitude of the phase shift. Let us examine this hypothesis first.

(a) You apply a sinusoidal drive that results in a generation function of the form

$$g_1(t) = G_1 + G_2 \cos \omega t$$

Obtain an analytical expression for the time evolution of the excess hole concentration in an n-type wafer. Assume low-level injection conditions.

(b) Discuss how the lifetime can be measured out of the phase difference, θ, between the lamp output and the carrier concentration. Make an appropriate sketch for $0 < \theta < \pi/2$.

Now that you have established the fundamentals of the technique, you proceed to examine some design issues of your tool. A popular use of a lifetime tester of this kind is to map on the wafer the spatial distribution of the lifetime. This is possible because the microwave probe samples the local conductance. By moving the wafer with respect to the beam of light and the probe, a two-dimensional map of lifetime on the wafer can be created. It is therefore important to select a modulation frequency for the light that maximizes the sensitivity of the instrumentation to changes in the lifetime.

(c) Define sensitivity $S = \frac{\partial \theta}{\partial \tau}$. Derive an expression for S. Sketch S as a function of ω. For a given lifetime, what is the optimum frequency of light modulation that one should use for maximum sensitivity? What is therefore the optimum phase shift for the measurement?

Now let us put some back-of-the-envelope numbers and examine the limitations of the technique. You find a lamp that would do the job for you. To maximize lamp reliability, you must limit its depth of modulation to 10%, that is, $G_2/G_1 = 0.1$ in the equation above. You can also span a range of G_1 between 10^{17} and 10^{22} cm^{-3}.s^{-1}. After a few experiments, you are quite confident that you can measure a phase shift of 200 mrad, but not smaller. The noise in the microwave detection system limits you to an amplitude in the excess carrier concentration sinusoidal signal equal to or larger than 10^{12} cm^{-3}.

(d) What limits the minimum lifetime that you can measure? How much is it? How high in angular frequency, ω, you need to modulate the lamp in order to measure this minimum lifetime?

(e) You must avoid that the wafer under test goes into high-level injection. Using the 10% rule (that is $p' \leq 0.1N_D$) and the charts of lifetime versus doping level shown in Fig. 3.17, what is the minimum wafer doping level that your setup will be able to handle? Estimate this for n- and p-type Si wafers. What is then the minimum value of modulation frequency that you must synthesize?

3.7 Derive Eq. (3.9). Graph $f(E_i)$ as a function of p_o in a log–log scale for Si at room temperature for values of p_o between 10 and 10^{19} cm^{-3}. Explain the dependence obtained. To do this problem, you will need to first do Problem 2.13.

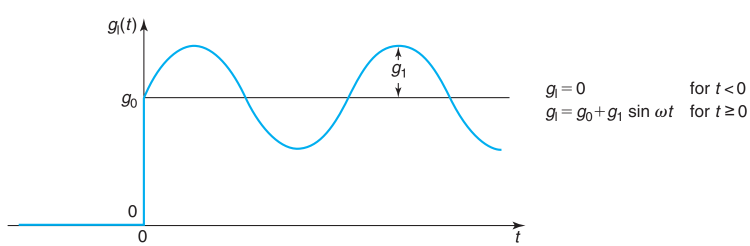
3.8 Discuss the conditions under which the approximations made in Eqs. (3.51) and (3.52) apply. Do it separately for an n- and a p-type semiconductor.

3.9 What is the highest recombination lifetime one could possibly expect in an n-type Si doped with $N_D = 10^{15}$ cm^{-3} at room temperature?

3.10 [5]A laser pulse impinges on a Si wafer with a doping level $N_D = 10^{18}$ cm^{-3}. The pulse width is 1 ns. The photon density of the beam is $N_{ph} = 10^{25}$ cm$^{-2} \cdot$ s^{-1}. The photon energy is 1.46 eV. The laser beam diameter on the wafer is 1 μm. Assume that all photons are absorbed in a band-to-band generation process.

(a) Estimate the power of the laser beam.
(b) Estimate the total energy delivered to the wafer during the duration of the pulse.
(c) Calculate the total energy delivered to the lattice during the duration of the pulse.
(d) On what timescale is all energy transferred to the lattice?

3.11 Consider a uniformly doped p-type semiconductor sample. At $t = 0$, the sample is illuminated with penetrating light such that a uniform generation rate occurs throughout the bulk. The light intensity has a sinusoidal time dependence riding on a constant value, as sketched below.



(a) Derive an expression for the excess electron concentration as a function of time for all times. Sketch the result. Comment on the various features of the result.

[5]This problems requires that you have studied Section 4.6.

(b) What condition(s) need to be fulfilled for the sample to achieve *steady state*? Derive an approximate expression for $n(t)$ in this regime.

(c) Under what condition(s) can this situation be described as *quasi-static*? Derive an approximate expression for $n(t)$ in this regime.

3.12 At room temperature and in thermal equilibrium, what is the occupation probability of a trap located at $E_t - E_v = 0.7$ eV in a p-type Si sample with $N_A = 10^{17}$ cm^{-3}?

3.13 This is an exercise to work out more details of the Shockley–Read–Hall model for trap-assisted generation and recombination of Section 3.4.

(a) Starting from Eq. (3.40), derive expressions for n_t and $N_t - n_t$ outside thermal equilibrium in terms of n and p.

(b) From these results, confirm Eq. (3.42) for U_{tr}.

(c) Simplify the expressions that you have obtained for n_t and $N_t - n_t$ for an n-type semiconductor and *moderate*-level injection ($n_o \gg n' \simeq p' \gg n_i \gg p_o$). Also assume that c_e and c_h are of the same order of magnitude.

(d) In light of these results, explain the fact that to the first order, the carrier lifetime due to trap-assisted recombination under moderate-level injection conditions is independent of n_o (see Fig. 3.14).

3.14 Consider a Si sample at room temperature with a dominant trap with a concentration $N_t = 10^{16}$ cm^{-3} located at E_i. The electron and hole capture rates for this trap are $c_e = c_h = 10^{-10}$ cm^3/s. This sample is placed under intense light illumination that causes the carrier concentrations to become $n = 2 \times 10^{17}$ cm^{-3} and $p = 10^{17}$ cm^{-3}, uniformly in space. Calculate the net recombination rate by as many generation/recombination processes as you can.

3.15 Consider a n-type Si wafer doped with $N_D = 10^{18}$ cm^{-3} that is in thermal equilibrium. Suddenly, the sample is illuminated with a pulse of uniformly penetrating radiation that generates $g_l = 10^{22}$ cm$^{-3}\cdot$ s^{-1} electron–hole pairs everywhere and lasts for $T = 1$ ns.

(a) Estimate the electron and hole concentrations and the net recombination rate right at the end of the light pulse.

(b) Estimate the electron and hole concentrations and the net recombination rate 500 ns after the light pulse is turned off.

3.16 Calculate the minimum carrier generation rate required to drive a piece of Si into high-level injection with $n = p = 10^{20}$ cm^{-3}. Assume uniform generation throughout the sample. What is the required power density in an He–Ne laser with $\lambda = 633$ nm to reach the above condition? Assume that all laser light is absorbed in a radiative process in a uniform way across the sample. The sample has a thickness of 0.1 μm.

Chapter 4

Carrier Drift and Diffusion

This chapter deals with two key processes by which carriers flow in semiconductors: drift and diffusion. Carrier drift originates from the electrostatic force that an electric field exerts over electrons and holes. Diffusion is the movement of particles in response to a concentration gradient. Carrier drift and diffusion are at the heart of device operation. Electrons, for example, drift from source to drain in an n-channel MOSFET. The magnitude of the current and the time that carriers take to traverse the device are determined by the physics of drift in the channel. In npn bipolar transistors, a combination of electron diffusion and drift in the base takes electrons from the emitter to the collector.

In order to understand the physics of carrier drift and diffusion in semiconductors, it is essential to develop an appreciation for the random motion of electrons and holes at finite temperatures. In the timescale of interest in most microelectronic devices, carriers undergo many collisions with the lattice, impurities, and each other. An important implication of this is that carriers are unlikely to maintain energy distributions that are too different from equilibrium over any meaningful periods of time. For most timescales of interest, the carriers can be considered to be in a "quasi-equilibrium" state in which, with some care, the concept of Fermi level remains useful. In these situations, we define two quasi-Fermi levels, one for electrons and the other for holes. These allow us to graph simple and intuitive energy band diagrams that effectively convey a great deal of information about device operation.

Finally, in this chapter, we also learn to compute the equilibrium carrier concentrations in situations where dopant distributions are nonuniform in space. In these instances, it is important to realize that a dynamic balance between drift and diffusion exists with no net carrier flow. Nonuniformly doped regions are pervasive and for the most part unavoidable in modern devices. Often, device characteristics are tailored through the engineering of the doping profiles. The aversion that reasonably extrinsic semiconductors have to the presence of volume charge implies that in many practical circumstances, the majority carrier concentration closely tracks the doping distribution. In other words, the semiconductor remains *quasi-neutral*. This greatly simplifies the treatment of nonuniformly doped regions.

This is a chapter rich in concepts that will be extensively used throughout the rest of the book. It is important to understand this material well before proceeding much further.

4.1 THERMAL MOTION

In this section, we look at the movement of carriers in a semiconductor in thermal equilibrium. We first look at a very idealized situation of a perfectly periodic solid. We then introduce lattice vibrations and other perturbations to this picture and consider their impact on carrier movement.

4.1.1 Thermal velocity

Let us consider a semiconductor in thermal equilibrium at a finite temperature. Let us assume a pure semiconductor with a perfect crystalline periodicity in which nothing changes in space (uniform situation) and all the surfaces are very far away. In Chapter 2, we learned how to compute the electron and hole concentrations under a variety of conditions. In this chapter, we take an interest in the movement of electrons and holes in the lattice.

If we think of a perfectly ideal semiconductor in which electrons and holes can move around without bumping into anything, at what velocity would they be traveling? We can certainly answer right away that in thermal equilibrium, by definition, the average velocity of the ensemble of electrons or holes has to be zero. This does not mean that the velocity of an individual electron or hole with a certain energy above or below their respective band edges is also zero. In fact, we have already discussed that the kinetic energy of carriers increases as they occupy states further away from the corresponding band edge. This means that their velocity increases too.

In general, the relationship between velocity (or momentum) and energy for carriers in a semiconductor is not simple. It can be computed in band calculations, and the result is very specific to each semiconductor. It is beyond the scope of this book to discuss the energy-momentum (E-k) diagrams, as these relationships are often referred to. For the purposes of this book, we can go quite far with a simple formalism that works reasonably well for most semiconductors close to the bottom of the bands. In this approach, as in classical mechanics, the carrier kinetic energy increases quadratically with velocity. This suggests that the electron velocity as a function of energy above the conduction band edge follows a simple square-root expression given by

$$v_e(E) = \sqrt{\frac{2(E - E_c)}{m^*_{ce}}} \tag{4.1}$$

A similar equation can be written for holes with the kinetic energy increasing downward from the valence band edge.

In Eq. (4.1), m^*_{ce} is known as the *electron conductivity effective mass*. This is the effective mass that connects velocity and kinetic energy of an electron. The conductivity effective mass is in general different from the free electron mass because in a semiconductor electrons are not in vacuum but are immersed in the periodic potential of the lattice. Also, m^*_{ce} is different from the density of states effective mass discussed in Chapter 2. Appendix B lists m^*_{ce}/m_o and m^*_{ch}/m_o for Si and GaAs.

Equation (4.1) gives the magnitude of the velocity of an electron with a certain energy in the conduction band. Electrons higher up in the conduction band move faster than those close to the band edge. The average magnitude of the velocity for an electron population in thermal equilibrium receives the name of *thermal velocity*. We can derive an expression for the thermal velocity by appropriately averaging the velocities of all electrons in the conduction band in the following way:

$$v_{the} = \frac{\int_{E_c}^{\infty} v_e(E) n_o(E)\, dE}{\int_{E_c}^{\infty} n_o(E)\, dE} \tag{4.2}$$

Here, $n_o(E)$ represents the electron distribution in energy in the conduction band as given by Eq. (2.23). Using this equation as well as Eq. (4.1), we can rewrite Eq. (4.2), as

$$v_{the} = \sqrt{\frac{2}{m_{ce}^*}} \frac{\displaystyle\int_{E_c}^{\infty} \frac{(E - E_c)}{1 + \exp\frac{E - E_F}{kT}}\, dE}{\displaystyle\int_{E_c}^{\infty} \frac{\sqrt{E - E_c}}{1 + \exp\frac{E - E_F}{kT}}\, dE} \tag{4.3}$$

Under Maxwell–Boltzmann statistics and redefining variables as in Section 2.4.2, we can rewrite this expression as

$$v_{the} \simeq \sqrt{\frac{2kT}{m_{ce}^*}} \frac{\int_0^{\infty} \eta e^{-\eta} d\eta}{\int_0^{\infty} \eta^{1/2} e^{-\eta} d\eta} \tag{4.4}$$

The integrals in Eq. (4.4) belong to a class called the *gamma function*. This is defined in Advanced Topic AT4.1, where some properties of this function are also given. Using the results shown there, we can rewrite Eq. (4.4) as

$$v_{the} = \sqrt{\frac{2kT}{m_{ce}^*}} \frac{\Gamma(2)}{\Gamma(\frac{3}{2})} = \sqrt{\frac{8}{\pi} \frac{kT}{m_{ce}^*}} \tag{4.5}$$

A similar equation applies to holes. The temperature dependence that emerges in this equation arises from the electron distribution in energy. The higher the temperature, the higher is the average energy of the electrons in the conduction band, and the higher is their average velocity.

Even though we have been discussing a highly idealized situation, the thermal velocity is an important benchmark for carrier transport in semiconductors that will make frequent appearances in this book. If one puts numbers in Eq. (4.3) for electrons in Si at room temperature ($m_{ce}^* = 0.28\ m_o$), one finds that the electron thermal velocity is about 2.0×10^7 cm/s. For holes ($m_{ch}^* = 0.41\ m_o$), the thermal velocity is 1.7×10^7 cm/s. Holes are somehow slower than electrons.

4.1.2 Scattering

A real semiconductor is quite different from the ideal unperturbed, perfectly periodic lattice that we discussed in the previous section. As we detail below, there are many perturbations to this idealized picture that have the effect of causing collisions, also called *scattering*, to freely moving carriers. A result of this is that carriers change directions and velocity frequently in a movement that is called *Brownian motion*. On average, just as before, carriers do not get anywhere.

Brownian motion is sketched in space in Fig. 4.1a and in energy in Fig. 4.1b. The movement of the carriers between collisions is graphed in the energy picture by horizontal (constant energy)

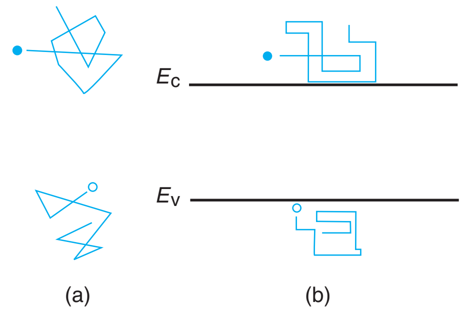


FIGURE 4.1
Schematic diagram of carrier thermal or Brownian motion in a semiconductor: (a) in real space and (b) in the energy band diagram.

lines. Some collisions change the carrier's energy. These events are represented in the energy diagram by a step change in energy. Other collisions only change the carrier's velocity but not its energy. These cannot be represented very clearly in the flattened energy view.

There are many scattering mechanisms. At finite temperatures, the semiconductor atoms vibrate about their equilibrium positions disrupting the otherwise perfect crystal potential. These perturbations behave as scattering centers for the carriers. In the phonon view of lattice vibrations introduced in Chapter 1, we can think of *lattice scattering* as a mechanical collision between the carriers and massive particles called *phonons*, hence the name *phonon scattering*. Since the phonons have fairly small energy, the energy exchanged in these collisions is relatively small. The change in the direction of the carrier velocity can however be large. In fact, phonon scattering is an effective randomizer of carrier velocity. A key feature of phonon scattering is that it increases with temperature. As the temperature is increased, the higher-energy phonon modes are proportionally more populated. Or, in simpler terms, the higher the temperature, the greater the lattice atoms vibrate about their equilibrium positions, and the more frequently carriers will scatter.

The introduction of an electrically charged impurity in a semiconductor lattice also upsets the perfect periodicity of the crystal potential. The Coulombic interaction of a carrier with a charged impurity constitutes an effective scattering mechanism. This is called *ionized impurity scattering*. Naturally, ionized impurity scattering is more important the higher the impurity concentration. Due to the Coulombic nature of the interaction, ionized impurity scattering does not change the particle's energy but it affects its velocity.

Although phonon scattering and ionized impurity scattering are the most important scattering mechanisms, there are others to be aware of. Any crystalline imperfection that results in a disruption of the perfect periodic potential becomes a potential source of collision events. Such are, for instance, interstitial atoms and lattice vacancies. Even neutral foreign atoms in a substitutional position impose slight perturbations to the crystalline periodic potential as a result of their electronic arrangement around the nucleus, which is always different from that of the host atom. This is called *neutral impurity scattering*.

Carriers can also collide with other carriers in a process known as *carrier-carrier scattering*. This is particularly important when carrier concentrations are high. In a collision event between two identical carriers, such as an electron with another electron, the carriers can exchange energy and momentum. Since this process does not impact the average behavior of the ensemble of carriers, we need not be concerned with it here. Electron–hole collisions do take place but, in most circumstances, too infrequently to concern us here.

In surface-type devices where carrier transport takes place very close to the semiconductor surface such as in the case of the MOSFET, *surface roughness scattering* also plays a key role. A surface represents a dramatic perturbation of the crystal periodicity of a semiconductor lattice. As a result, it is an effective source of scattering events for carriers in its vicinity. We will deal with it when we treat transport in an inversion layer.

The main effect of all these scattering events is to change the direction of the carrier momentum and not so much its energy. Intuitively, carriers are simply deflected but keep their energy intact. Only in phonon scattering events does a carrier change its energy by a small amount, the phonon energy. A carrier can acquire energy by absorbing a phonon or lose energy by emitting a phonon. It is through the energy exchanges associated with phonon scattering that carriers achieve thermal equilibrium with the lattice.

Scattering in a semiconductor is characterized by the *scattering time*, τ_c. This is defined as the average time between collisions. In general, the scattering time depends on the semiconductor itself, the scattering mechanism in question, and temperature. Models for scattering time associated with the various scattering mechanisms have been developed from first principles. Understanding and working with these models is beyond our purpose in this book. Fortunately, a first-order understanding of the physics of scattering can bring us very far in our quest for designing and modeling microelectronic devices. We follow this approach here.

We can get quite far by realizing that, in general, the scattering time depends on the carrier energy. Scattering becomes more likely the more states a carrier can scatter into. Therefore, the frequency of scattering should increase with the density of states. In this simplistic model, the scattering time goes as the inverse of the density of states. For electrons, using Eq. (2.18), we then have

$$\tau_{ce}(E) = \frac{M}{(E - E_c)^{1/2}} \tag{4.6}$$

where M is a constant. A similar equation applies to holes with the energy measured from the top of the valence band and increasing downward.

We can compute the average scattering time in thermal equilibrium following an approach similar to that of the thermal velocity in Eq. (4.2). In doing this, we also obtain expressions in terms of the gamma function. The result is

$$\tau_{ce} =< \tau_{ce}(E) >= \frac{\int_{E_c}^{\infty} \tau_{ce}(E) n_o(E)\, dE}{\int_{E_c}^{\infty} n_o(E)\, dE} \simeq \frac{2}{\sqrt{\pi kT}} M \tag{4.7}$$

We refer to this average value, τ_{ce}, as simply the scattering time. The temperature dependence that emerges here is due to the fact that at higher temperatures, the average energy of electrons in the conduction band increases and so does the frequency of scattering events. The value of M in a specific situation depends on many factors (nature of semiconductor, doping level, temperature, etc). We will learn below how to derive τ_{ce} from actual measurements in devices.

We complete the picture of thermal motion and scattering by introducing the notion of *mean free path*, l_c. This is the average distance traveled by a carrier between collisions. For a given carrier with a certain energy, the mean free path is the product of its thermal velocity and the corresponding scattering time. Working with electrons, since the thermal velocity goes as $(E - E_c)^{1/2}$ and the scattering time goes as $(E - E_c)^{-1/2}$, the mean free path ends up energy independent. Hence, the mean free path for an entire electron distribution in thermal equilibrium can simply be obtained by multiplying Eqs. (4.1) and (4.6):

$$l_{ce} = \sqrt{\frac{2}{m_{ce}^*}} M = \frac{\pi}{4} v_{the} \tau_{ce} \tag{4.8}$$

In this last expression, we have substituted M from Eq. (4.7) and used v_{the} in Eq. (4.3). A similar equation applies for holes. The $\pi/4$ factor in this equation emerges from the averaging in energy that is done in Eqs. (4.5) and (4.7).

As we will learn to appreciate, the mean free path is an important length scale in semiconductors. Traditionally, device dimensions have been much larger than the mean free path and carriers suffer many collisions as they travel through devices. It is under this assumption that the transport formulation introduced in this chapter is valid. Today, advanced MOSFETs are getting so small that this no longer is a good assumption. A new "ballistic" transport formulation is required in this instance.

4.2 DRIFT

Since electrons and holes are charged particles, the application of an electric field onto a semiconductor affects them. As a carrier flies in between two scattering events in the presence of an electric field, it becomes subject to a Coulombic force. As a result of this, it picks up a component of velocity in the direction of the electric field. Every collision that a carrier suffers effectively randomizes its velocity but in between collisions the carrier is under the influence of the electric field. The impact of this is a relatively slow "drift" in the direction of the electric field. This is sketched for an electron in Fig. 4.2. The average value of the *net* carrier velocity that results is called the *drift velocity*. The net current that is produced is called the *drift current*. We examine both in some detail in the following two sections.

4.2.1 Drift velocity

The drift velocity is the average net velocity that carriers acquire in an electric field. Let us develop a simple model for electrons. The electrostatic force that an electron experiences between collisions is $-q\mathcal{E}$, where $\mathcal{E}$ is the electric field and q is the absolute value of the electron charge. This force imposes a constant acceleration in the electron given by $a = -q\mathcal{E}/m_{ce}^*$. The initial velocity in the direction of the field is, on average, zero since a collision completely randomizes the electron velocity. The maximum velocity that the electron acquires in the direction of the field during a scattering time is then

$$v_e^{drift} = -\frac{q\mathcal{E}\tau_{ce}}{m_{ce}^*} \tag{4.9}$$

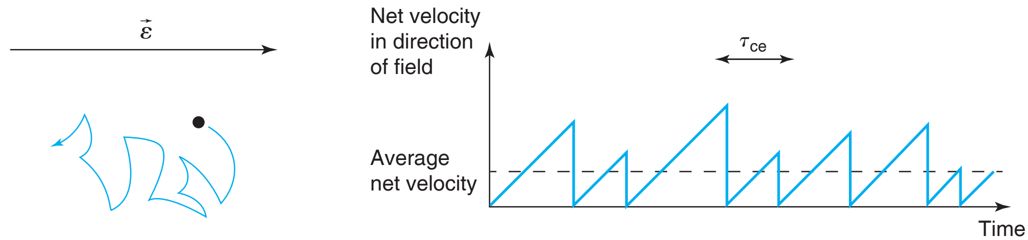


FIGURE 4.2
Left: schematic diagram of drift motion of an electron in an electric field. Right: time evolution of net electron velocity in the direction of the electric field.

Since the velocity ramps up linearly between collisions, in this simple model, the average magnitude of the velocity should actually be half of this. However, a more rigorous model that properly accounts for the detailed physics of scattering yields the result given in Eq. (4.9). A similar equation is derived for holes (with a positive sign).

The drift velocity in Eq. (4.9) is proportional to the electric field $\mathcal{E}$. The proportionality constant is called the *mobility* and is usually represented as μ, that is

$$\mu_e = \frac{q\tau_{ce}}{m_{ce}^*} \tag{4.10}$$

A similar equation applies to holes. The mobility is defined such that it is positive for both electrons and holes. In terms of mobility, the electron and hole drift velocities can thus be expressed as

$$v_e^{drift} = -\mu_e \mathcal{E} \tag{4.11}$$

$$v_h^{drift} = \mu_h \mathcal{E} \tag{4.12}$$

The mobility represents the "ease" by which a carrier drifts in an electric field. The mobility is directly proportional to the scattering time and inversely proportional to the effective mass. These dependencies make sense. The longer the time between scattering events, the faster it drifts between collisions. The heavier the carrier is, the higher its inertia and the less velocity it picks up between collisions.

The mobility is a very important physical parameter in semiconductor device engineering. The fact that the mobility is directly proportional to the scattering time implies that its value depends on the strength of the various scattering mechanisms. Figure 4.3 shows the carrier mobilities in Si at room temperature as a function of doping level. For low enough doping levels, phonon scattering dominates and the mobility is independent of doping. The value of the phonon-scattering limited mobility is about a factor of three larger for electrons than for holes in Si. This ratio is about 20 for GaAs. Analytical expressions for the dependencies shown in Fig. 4.3 are given in Appendix E.

As the doping level goes up, ionized impurity scattering becomes important and the mobility decreases very quickly. Ionized impurity scattering is not a strong function of the specific dopant atom provided that it is of the same type. That is, electrons in Si doped with the

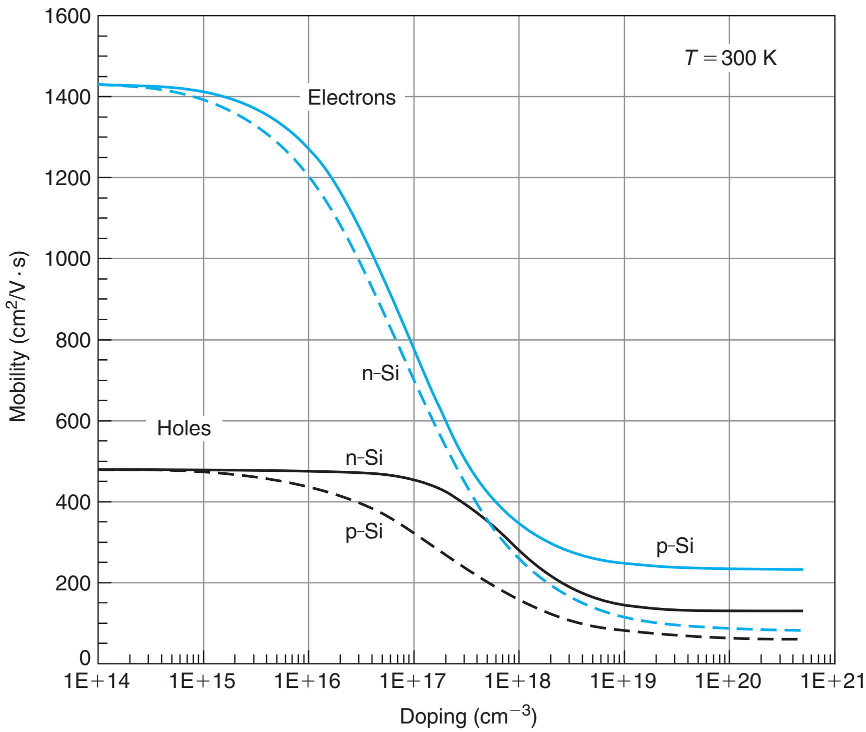


FIGURE 4.3
Minority and majority carrier electron and hole mobilities in Si at 300 K. Continuous lines refer to minority carriers and discontinuous lines refer to majority carriers.

same concentration of P, As, or Sb have nearly identical mobility. This is not the case if the type of dopant is reversed. For example, at high-doping levels, the electron mobility in an n-type Si is different from that of electrons in a p-type Si for an identical doping level. In fact, at high-doping levels and for both electrons and holes, the mobility is about a factor of two higher when the carrier is a minority carrier than when it is a majority carrier. This is because in ionized impurity scattering, *attractive scattering* is more effective than *repulsive scattering*. This arises from the peculiar shape of the potential (the so-called Yukawa or screened Coulombic potential) associated with an ionized dopant. This effect is irrelevant at low-doping levels where phonon scattering dominates. For low enough doping, the minority and majority carrier mobilities converge. Using the correct mobility for minority carriers is crucial to computing certain important parameters such as the base transit time in bipolar transistors.

As the temperature is reduced, phonon scattering becomes less important while ionized impurity scattering is enhanced. This implies that a reduction of temperature increases the mobility for low-doping levels while it decreases it for high-doping levels.

EXERCISE 4.1

Estimate the electron scattering time and the mean free path in an n-type Si sample with $N_D = 10^{17}$ cm^{-3} at room temperature.

For $N_{\mathrm{D}} = 10^{17}\ \mathrm{cm}^{-3}$, Fig. 4.3 gives $\mu_{\mathrm{e}} = 700\ \mathrm{cm}^2/\mathrm{V}\cdot\mathrm{s}$. Solving for τ_{ce} in Eq. (4.10), we get

$$\tau_{\mathrm{ce}} = \frac{\mu_{\mathrm{e}} m^*_{\mathrm{ce}}}{q} = \frac{700\ \mathrm{cm}^2/\mathrm{V}\cdot\mathrm{s} \times 0.28 \times 5.69 \times 10^{-16}\ e\mathrm{V}\cdot\mathrm{s}^2/\mathrm{cm}^2}{1\ e} = 1.1 \times 10^{-13}\ \mathrm{s} = 0.1\ \mathrm{ps}$$

The mean free path can be computed using Eq. (4.8):

$$l_{\mathrm{ce}} = \frac{\pi}{4} v_{\mathrm{the}} \tau_{\mathrm{ce}} = \frac{\pi}{4} \times 2.0 \times 10^7\ \mathrm{cm/s} \times 1.1 \times 10^{-13}\ \mathrm{s} = 1.7 \times 10^{-6}\ \mathrm{cm} = 17\ \mathrm{nm}$$

This is substantially smaller than typical device dimensions although the gate length of modern MOSFETs is approaching this value.

4.2.2 Velocity saturation

The linear relationship between the drift velocity and the electric field that we derived above, Eq. (4.9), is found experimentally to break down for high fields. This equation was obtained under the assumption of "near" thermal equilibrium or "quasi-equilibrium." That is, although thermal equilibrium is disturbed by the application of an electric field, the disruption is minor and the rates of the various scattering mechanisms remain unchanged from their thermal equilibrium values. This is a reasonable assumption if the carriers do not acquire too much energy from the electric field during their free flight between two collisions, or in other terms, when the drift velocity is much smaller than the thermal velocity, $v^{\mathrm{drift}} \ll v_{\mathrm{th}}$.

At high fields, this assumption fails. Electrons and holes can acquire substantial energy from a strong electric field. This has the effect of enhancing phonon scattering, in particular, phonon emission. For high fields, the scattering time decreases proportionally to $1/\mathcal{E}$. This leads to a saturation of the average drift velocity, as sketched in Fig. 4.4. The asymptotic value of velocity is called *saturated drift velocity* or simply saturation velocity and is represented as v_{sat}. By definition, v_{sat} is always a positive quantity. For a given semiconductor, v_{sat} only depends on temperature (weakly) and is independent of the doping level. In Si at room temperature, v_{sat} for electrons is 1×10^7 cm/s, while for holes, it is 6×10^6 cm/s.

The overall behavior of the drift velocity with electric field is relatively well represented by an equation of the form:

$$v^{\mathrm{drift}} = \mp \frac{\mu\mathcal{E}}{1 + \left|\frac{\mu\mathcal{E}}{v_{\mathrm{sat}}}\right|} \tag{4.13}$$

where the minus sign applies to electrons and the plus sign to holes. Equation (4.13) has the correct limits. For low fields, it reduces to $v^{\mathrm{drift}} = \mp\mu\mathcal{E}$. For high fields, it saturates to v_{sat}.

An estimate of the magnitude of the electric field, $\mathcal{E}_{\mathrm{c}}$, beyond which saturation effects are significant can be obtained from Eq. (4.13) by equating both terms in the denominator:

$$\mathcal{E}_{\mathrm{c}} = \frac{v_{\mathrm{sat}}}{\mu} \tag{4.14}$$

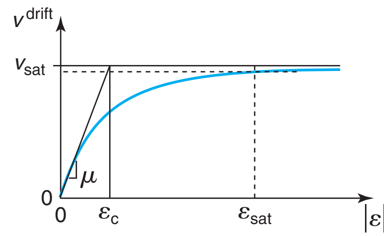


FIGURE 4.4
Sketch of velocity-field characteristics showing velocity saturation at high electric fields.

Reaching fields of the order of $\mathcal{E}_{\text{c}}$ is not difficult in modern devices. For example, in a region of a Si device with a mobility of 500 $\text{cm}^2/\text{V}\cdot\text{s}$, $\mathcal{E}_{\text{c}}$ is 2×10^4 V/cm. It only takes a voltage of 2 V applied across a length of 1 μm to reach this electric field (if it is uniformly distributed in space). The occurrence of velocity saturation, particularly for electrons, is very common in today's microelectronic devices.

At $\mathcal{E}=\mathcal{E}_{\text{c}}$, the carrier velocity is $v_{\text{sat}}/2$. For carriers to approach $v^{\text{drift}}\simeq v_{\text{sat}}$, the field has to be much higher than $\mathcal{E}_{\text{c}}$. If we denote as $\mathcal{E}_{\text{sat}}$, the field required to attain 90% of v_{sat}, for example, then we have that $\mathcal{E}_{\text{sat}}=9\mathcal{E}_{\text{c}}$.

The saturation field depends on mobility and is therefore a function of doping level. This is clearly seen in Fig. 4.5 that shows the evolution of electron drift velocity in Si at room temperature for different doping levels. As the doping level increases, the mobility decreases, and as a result, the magnitude of the field that is required to reach velocity saturation increases too. In consequence, v_{sat} effects are less likely to show up the higher the doping level. Holes display a similar behavior. Advanced Topic AT4.2 discusses in more detail velocity saturation and other "hot"-carrier effects.

4.2.3 Drift current

Carrier drift in response to an electric field produces an electric current. Once the drift velocity is known, it is easy to derive an expression for the drift current. Before doing that, let us make two important definitions that apply to any situation in which there is a net flow of particles from one point to another. *Particle flux* is defined as the number of particles that cross a unit area normal to the particle flow every unit time. In this book, flux is represented by F and has units of $\text{cm}^{-2}\cdot\text{s}^{-1}$. The flux is obviously a vector pointing in the direction of particle flow.

In semiconductors, we are particularly interested in charged particles or carriers. Carrier flux hence results in an electric current. The *current density*, J, is defined as the amount of charge that crosses a unit area placed normal to the carrier flow every unit time. The current density is also a vector and its units are $\text{C}\cdot\text{cm}^{-2}\cdot\text{s}^{-1}=\text{A/cm}^2$. There is obviously a direct relationship between flux and current density. For electrons and holes, respectively

$$J_{\text{e}}=-qF_{\text{e}} \tag{4.15}$$

$$J_{\text{h}}=qF_{\text{h}} \tag{4.16}$$

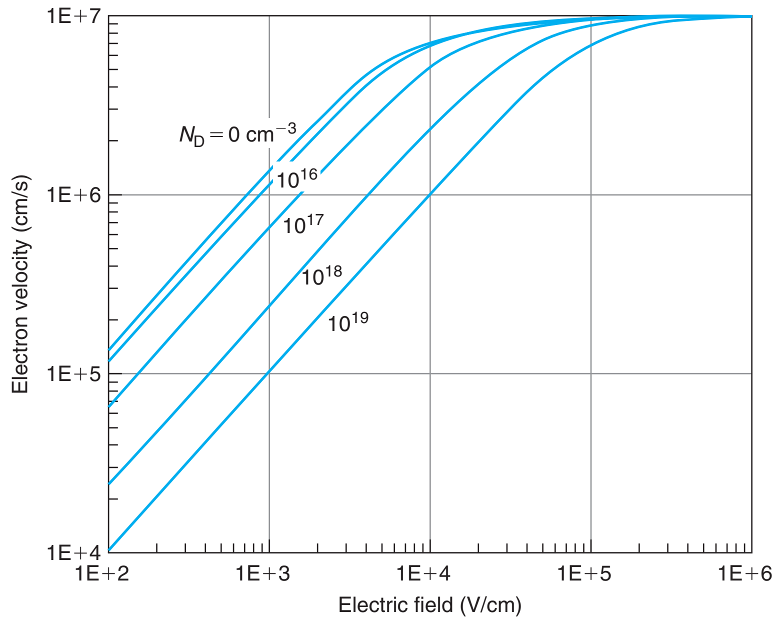


FIGURE 4.5
Velocity-field characteristics of electrons in n-type Si at 300 K for several doping levels.

The carrier flux and the current density are directly related to the net velocity of the carriers. Consider, for example, a portion of a semiconductor with n electrons per unit volume where the electrons are moving with an average net velocity v_e. Let us place our fictitious surface of unity area normal to the direction of carrier movement, as sketched in Fig. 4.6. In a time interval dt, there are $nv_e dt$ electrons crossing through this surface. The electron *flux* is then

$$F_e = nv_e \tag{4.17}$$

and the electron current density is therefore

$$J_e = -qnv_e \tag{4.18}$$

For holes with a concentration p moving at an average net velocity v_h, we analogously get

$$F_h = pv_h \tag{4.19}$$

and

$$J_h = qpv_h \tag{4.20}$$

Equations (4.18) and (4.20) are rather fundamental expressions. In their derivation, no assumptions have been made as to the detailed physical origin of the force that acts on the carriers. These equations state the simple fact that if an ensemble of carriers moves at a certain

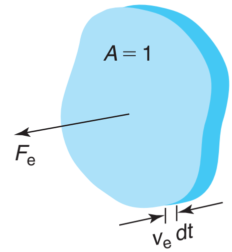


FIGURE 4.6
Sketch of unity area surface placed normal to electron flow. In a time interval dt, all particles inside the enclosed volume cross the light-shaded surface.

average velocity, by virtue of the fact that they are charged, an electric current of a certain magnitude is produced. This holds regardless of the physics of the force driving the particles.

We can now go back to the process of drift in which carriers move in response to an electric field. For low electric fields, we can substitute Eqs. (4.11) and (4.12) in Eqs. (4.18) and (4.20) to get

$$J_e = q\mu_e n\mathcal{E} \tag{4.21}$$

$$J_h = q\mu_h p\mathcal{E} \tag{4.22}$$

These are the expressions for the drift current densities for electrons and holes under low-field conditions. Since in a semiconductor, there are always electrons and holes, the total drift current density is

$$J_t = q(\mu_e n + \mu_h p)\mathcal{E} \tag{4.23}$$

This is simply *Ohm's law* for a semiconductor. It is important to remember that this equation applies only if the fields are not too high. In Ohm's law, the proportionality constant is called the *conductivity*. The units of conductivity are $\Omega^{-1} \cdot \mathrm{cm}^{-1}$.[1] One can define separately a conductivity due to electrons, another one due to holes, and a total conductivity as follows:

$$\sigma_e = q\mu_e n \tag{4.24}$$

$$\sigma_h = q\mu_h p \tag{4.25}$$

$$\sigma = q(\mu_e n + \mu_h p) \tag{4.26}$$

The inverse of the conductivity is called the *resistivity*, which is represented by the symbol ρ and has units of $\Omega \cdot \mathrm{cm}$. Both resistivity and conductivity are frequently used in a variety of contexts in semiconductor engineering. For example, the resistivity is frequently quoted by wafer vendors as a way to specify the doping level of their products. This is because it is easy to measure ρ by contactless techniques and also because it is a very strong function of doping level. In a doped semiconductor in equilibrium, the resistivity is determined by the concentration of

[1] Sometimes, Ω^{-1} is called *Mho* or *Siemens* (with symbol S). Hence, conductivity can also be given in Mho/cm or S/cm.

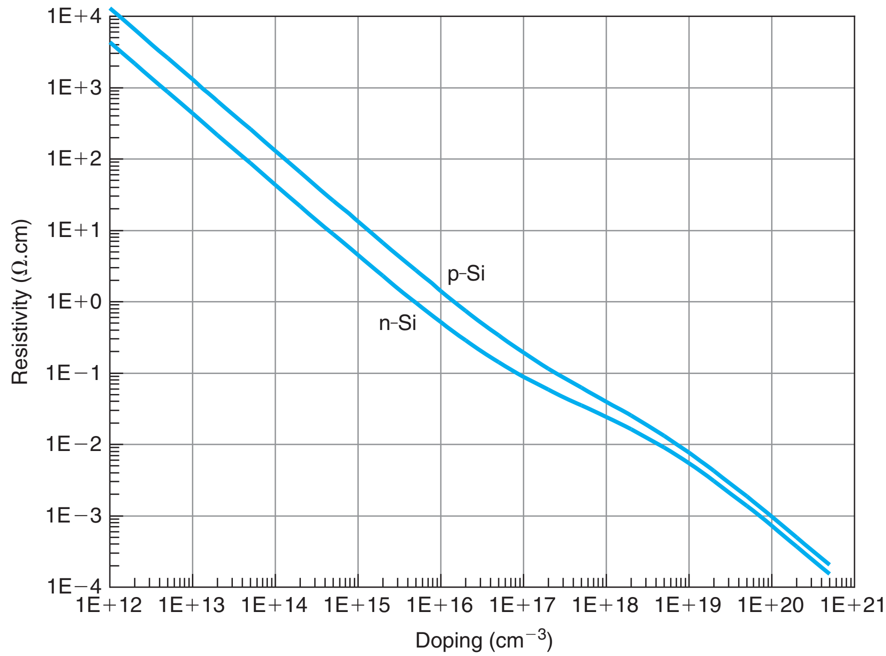


FIGURE 4.7
Resistivity of n- and p-type Si at 300 K.

majority carriers that at room temperature is nearly equal to the doping level. Expressions for the resistivity for fully ionized extrinsic n- and p-type semiconductors are, respectively,

$$\rho_n \simeq \frac{1}{q\mu_e N_D} \tag{4.27}$$

$$\rho_p \simeq \frac{1}{q\mu_h N_A} \tag{4.28}$$

Figure 4.7 shows the resistivity of Si as a function of doping level for n- and p-type Si. In both cases, the resistivity drops very quickly as the doping level increases. For an equivalent doping level, the resistivity of p-type Si is always higher than the resistivity of n-type Si. This is a consequence of the lower mobility of holes with respect to electrons that is presented in Fig. 4.3.

As the electric field increases, the linear relationships between current density and electric field, Eqs. (4.21)–(4.23), do not hold anymore. In fact, as discussed above, for high enough fields, the carrier velocity saturates. At that point, the current density reaches a maximum value. Plugging in v_{esat} and v_{hsat}, respectively, in Eqs. (4.18) and (4.20), we get

$$J_{esat} = -qnv_{esat} \tag{4.29}$$

$$J_{hsat} = qpv_{hsat} \tag{4.30}$$

These equations convey the important message that the only way to increase the drift current density in a semiconductor that is under a high electric field is to increase the carrier density. Increasing the magnitude of the field does not help at all.

EXERCISE 4.2

Consider an n-Si sample with a resistivity of $0.1\ \Omega \cdot \text{cm}$. *In a certain region of this sample, there is an electric field of* $1000\ \text{V/cm}$. *Estimate the electron and hole drift velocity and the total drift current density in that region.*

From Fig. 4.7, the doping level of the sample is $9 \times 10^{16}\ \text{cm}^{-3}$. The equilibrium hole concentration is about $1.3 \times 10^{3}\ \text{cm}^{-3}$. From Fig. 4.3, the mobility of electrons in this sample is about $800\ \text{cm}^2/\text{V} \cdot \text{s}$ and that of holes is $460\ \text{cm}^2/\text{V} \cdot \text{s}$. Let us assume that the field is small enough so that Eqs. (4.11) and (4.12) apply. Then

$$v_e^{drift} = -\mu_e \mathcal{E} = -800\ \text{cm}^2/\text{V} \cdot \text{s} \times 1000\ \text{V/cm} = -8 \times 10^5\ \text{cm/s}$$

$$v_h^{drift} = \mu_h \mathcal{E} = 460\ \text{cm}^2/\text{V} \cdot \text{s} \times 1000\ \text{V/cm} = 4.6 \times 10^5\ \text{cm/s}$$

where the minus sign of v_e^{drift} indicates that electrons are flowing against the electric field. The drift velocities are much smaller than the saturation velocity. Therefore, our initial assumption is validated. This region is operating in the mobility regime.

Since we already have the carrier drift velocities, the drift current densities are easy to calculate using Eqs. (4.18) and (4.20):

$$J_e = -qnv_e^{drift} = 1.6 \times 10^{-19}\ \text{C} \times 9 \times 10^{16}\ \text{cm}^{-3} \times 8 \times 10^5\ \text{cm/s} = 1.2 \times 10^4\ \text{A/cm}^2$$

$$J_h = qpv_h^{drift} = 1.6 \times 10^{-19}\ \text{C} \times 1.3 \times 10^{3}\ \text{cm}^{-3} \times 4.6 \times 10^5\ \text{cm/s} = 9.6 \times 10^{-11}\ \text{A/cm}^2$$

Clearly, since holes are minority carriers, the hole drift current density is negligible next to the electron drift current density. In consequence, the total drift current density is the value given by electron drift, about $1.2 \times 10^4\ \text{A/cm}^2$.

4.2.4 Energy band diagram under electric field

Under the application of an electric field, the energy band diagram of a semiconductor needs to be modified to account for the *potential energy* of the electric field. Before we learn how to do this, let us review first the familiar situation in vacuum, as sketched in Fig. 4.8.

In Fig. 4.8, an electric field points from left to right in vacuum. The electrostatic potential, ϕ, associated with this field increases toward the left. The potential energy of the field, $E_p = -q\phi$, increases toward the right. As a result, an electron inside this field moves from right to left. The

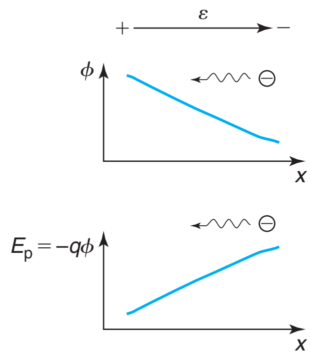


FIGURE 4.8
Sketch of electrostatic potential and potential energy associated with an electric field in vacuum.

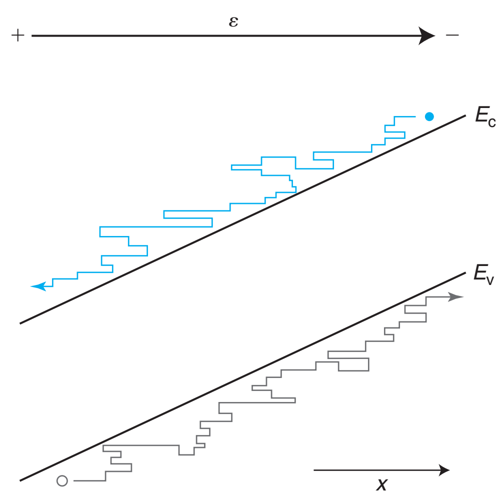


FIGURE 4.9
Sketch of an energy band diagram in the presence of an electric field. Electrons "slide" down the conduction band while holes "float" up the valence band.

electron trades potential energy for kinetic energy as it accelerates toward the left while its total energy is left unchanged.

Consider now a similar situation inside a semiconductor, as in Fig. 4.9. The potential energy due to the electric field needs to be added to the band diagram. When this is done, the bands tilt as indicated in the figure. An electron inside this semiconductor drifts toward the left. In between collisions, the electron exchanges potential energy for kinetic energy as it moves along a constant energy trajectory. As an electron gains kinetic energy from the electric field, the balance between phonon emission and phonon absorption rates is broken: it becomes more likely to emit than to absorb phonons. This is precisely the path to restoration of thermal equilibrium. So, more frequently than not, the electron emits a phonon and it loses some of its kinetic energy which is given off to the lattice in the form of heat. The end result is that the electron "slides" down the conduction band lowering its energy along the way. A hole drifting inside the same semiconductor moves from left to right "bubbling up" the valence band. This intuitive view of carrier movement in the energy domain is useful when dealing with complex energy band diagrams.

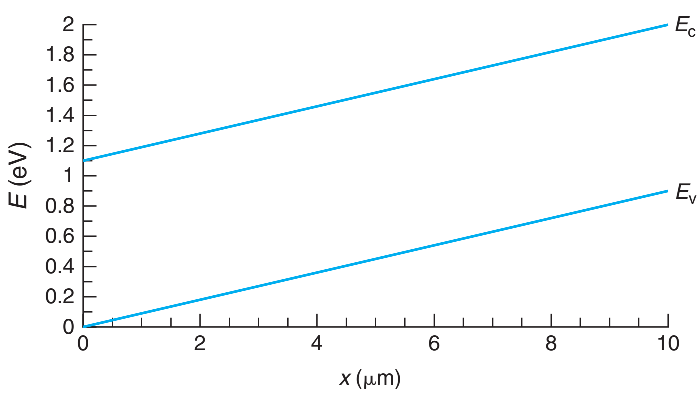
When an electric field exists inside a semiconductor, the conduction and valence bands bend to reflect the change in potential energy. This is called *band bending*. The potential energy of an electric field, as we saw above, is related to the electrostatic potential through the electron charge. Mathematically, we can then write $E_c + E_{ref} = E_p = -q\phi$, where E_{ref} is some arbitrary reference energy. A similar equation applies to the valence band. The shape of the bands therefore reflects the shape of ϕ upside down (with the proper units). In consequence,

$$\mathcal{E} = -\frac{d\phi}{dx} = \frac{1}{q}\frac{dE_c}{dx} = \frac{1}{q}\frac{dE_v}{dx} \tag{4.31}$$

This is a useful equation that permits the computation of electric fields from band diagrams and that allows the drawing of band diagrams in regions where an electric field is present.

EXERCISE 4.3

In a certain region of a semiconductor, the bands are bent as sketched in the figure below. Calculate the magnitude and direction of the electric field that exists in that region.



Equation (4.31) provides the answer. Note the special handling of the units:

$$\mathcal{E} = \frac{1}{q}\frac{dE_c}{dx} = \frac{1}{1\ e}\frac{0.9\ \text{eV}}{10 \times 10^{-4}\ \text{cm}} = 900\ \text{V/cm}$$

This field is positive since with the axis selected in the figure, the gradient of the bands is positive.

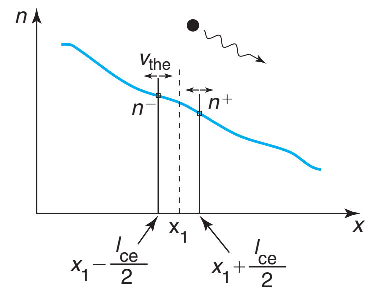


FIGURE 4.10
Sketch of electron diffusion down a concentration gradient.

4.3 DIFFUSION

Diffusion is the second physical mechanism that causes electrons and holes to flow inside semiconductors. The process of diffusion is very common in nature. It takes place whenever there is a nonuniform concentration of particles. The spread of a drop of cream in a cup of coffee or perfume from an opened bottle in a room are familiar examples of diffusion. In diffusion, the particles in question are knocked around by the medium matter. This causes them to flow from regions where their concentration is high to regions where their concentration is low. If there is nothing trying to maintain the concentration gradient, eventually the particles will be spread evenly throughout the whole volume. In the case of a drop of cream in coffee, it is the collisions of the cream molecules with those of water that disperse the cream. For perfume, it is the air molecules that spread the perfume molecules around. In a semiconductor, carrier diffusion takes place when electrons and holes get kicked around by the vibrating semiconductor atoms. Temperature is essential in diffusion. At absolute zero temperature, there is no diffusion.

4.3.1 Fick's first law

Let us look at the process of diffusion in more detail. Consider the one-dimensional situation depicted in Fig. 4.10 where an electron gradient has somehow been established in a certain region of a semiconductor. Let us attempt to compute the electron flow due to diffusion at a certain location, such as x_1. To the left of x_1, there is a higher electron concentration than to its right. If all electrons are randomly knocked around by the lattice atoms, on average, there are more electrons that flow from the left of x_1 toward the right than the other way around. In consequence, there is a net flow of electrons from left to right down the gradient.

We can easily derive a first-order expression for the electron flux at x_1. Let us assume that in the vicinity of x_1 all electrons are scattered about by the semiconductor atoms at two precise locations half a mean free path away from x_1 on either side. Let us denote the electron concentration at $x_1 - l_{ce}/2$ and $x_1 + l_{ce}/2$, as n^- and n^+, respectively. If all n^- electrons are knocked randomly at $x_1 - l_{ce}/2$, then on average $n^-/2$ electrons get scattered toward the right, and $n^-/2$ get scattered toward the left. Since $n^- > n^+$, it is clear that at x_1 there is a net flow of electrons toward the right, which is given by

$$F_e(x_1) = \frac{n^-}{2}v_{the} - \frac{n^+}{2}v_{the} = -\frac{1}{2}v_{the}(n^+ - n^-) \tag{4.32}$$

where v_{the} is the thermal velocity for the electrons. Since the mean free path is rather short in comparison with the dimensions of typical problems we are concerned with, it is advantageous to write the difference in electron concentrations $n^+ - n^-$ in terms of the electron gradient at x_1:

$$n^+ - n^- = \left.\frac{dn}{dx}\right|_{x_1} l_{\text{ce}} \tag{4.33}$$

Plugging this result into the expression for the flux, we get

$$F_{\text{e}}(x_1) = -\frac{1}{2} v_{\text{the}} l_{\text{ce}} \left.\frac{dn}{dx}\right|_{x_1} \tag{4.34}$$

There are a number of important dependences in this result. First, the flow of electrons is proportional to the gradient of their concentration. It is *the difference* in carrier concentrations that drives diffusion. This is a very fundamental result of general applicability. It applies to electrons and holes in semiconductors as well as to molecules of cream in coffee and to perfume in air. The minus sign that is obtained in Eq. (4.34) makes good physical sense. As Fig. 4.10 shows, electrons diffuse down the gradient.

The second important dependence of Eq. (4.34) is that the diffusion flux is proportional to the thermal velocity. This makes sense. The faster carriers travel between scattering events, the higher is the net flux at a given location.

The third dependence involves the *mean free path*: the longer this is, the higher the diffusion flux becomes. As Eq. (4.33) indicates, what drives diffusion is the difference of carrier concentration in the scale of the mean free path. Given a certain gradient of electron concentration, the longer the mean free path, the bigger the difference between left-going and right-going fluxes becomes.

The combined prefactor multiplying the gradient of electron concentration in Eq. (4.34) is called the *diffusion coefficient*, with symbol D, and units of cm^2/s:

$$D_{\text{e}} = \frac{1}{2} v_{\text{the}} l_{\text{ce}} \tag{4.35}$$

In terms of the diffusion coefficient, we can write the electron flux as

$$F_{\text{e}} = -D_{\text{e}} \frac{dn}{dx} \tag{4.36}$$

This is called *Fick's first law* which is derived more rigorously in many solid-state books. Similarly, for holes, we have

$$F_{\text{h}} = -D_{\text{h}} \frac{dp}{dx} \tag{4.37}$$

The diffusion coefficient gives a sense of the "ease" by which electrons diffuse in a semiconductor. For a given concentration gradient, the higher the D the higher is the diffusion flux. The value of the diffusion coefficient embodies the strength of the scattering mechanisms that carriers suffer as well as the nature of the carriers. It also depends on temperature through the temperature dependence of the thermal velocity and that of the scattering time.

4.3.2 The Einstein relation

Since the diffusion coefficient is intimately tied to scattering, it is reasonable to expect that it bears some relationship to the mobility which itself reflects carrier scattering in the presence of an electric field. This relationship is in fact very fundamental and is known as the *Einstein relation*. Using Eqs. (4.5), (4.8), (4.10), and (4.35), we can easily find

$$\frac{D_{\mathrm{e}}}{\mu_{\mathrm{e}}} = \frac{kT}{q} \tag{4.38}$$

The Einstein relation for holes can be derived in a similar way:

$$\frac{D_{\mathrm{h}}}{\mu_{\mathrm{h}}} = \frac{kT}{q} \tag{4.39}$$

The ratio kT/q is referred to as the *thermal voltage*. This is an important "tickmark" in the voltage scale that is pervasive in the analysis of semiconductor devices.[2]

The Einstein relation is quite general and is in fact valid for all systems that obey Maxwell–Boltzmann statistics. Interestingly, the Einstein relation does not depend on doping, only on temperature. The Einstein relation can also be derived for a degenerate carrier gas. In this case, a doping dependence arises (see Problem 4.9).

The Einstein relation provides a connection between mobility and diffusion coefficient in thermal equilibrium. Outside equilibrium, the Einstein relation is expected to hold if the disturbance from equilibrium is not so high that the scattering rates are modified. A good working criterion is that the electric fields must be low enough so that there is a linear proportionality between drift current and electric field.

EXERCISE 4.4

Estimate the electron and hole diffusion coefficients in $0.1\ \Omega\cdot\mathrm{cm}$ *p-Si at room temperature.*

The first step is to find the doping level of this sample. From Fig. 4.7, we find that p-type Si with a resistivity of $0.1\ \Omega\cdot\mathrm{cm}$ has a doping level of about $N_{\mathrm{A}} = 3\times10^{17}\ \mathrm{cm}^{-3}$.

The second step is to obtain the corresponding mobilities for electrons and holes. Holes are majority carriers in p-Si. Electrons are minority carriers. Care has to be exercised to select the proper curve in Fig. 4.3. For holes we find that for $N_{\mathrm{A}} = 3\times10^{17}\ \mathrm{cm}^{-3}$, the mobility is $\mu_{\mathrm{h}} \simeq 240\ \mathrm{cm}^2/\mathrm{V}\cdot\mathrm{s}$. For electrons, we find $\mu_{\mathrm{e}} \simeq 530\ \mathrm{cm}^2/\mathrm{V}\cdot\mathrm{s}$.

The diffusion coefficients are obtained using Einstein's relation. For holes

$$D_{\mathrm{h}} = \frac{kT}{q}\mu_{\mathrm{h}} = \frac{0.026\ \mathrm{eV}}{1\ e}\ 240\ \mathrm{cm}^2/\mathrm{V}\cdot\mathrm{s} = 6.2\ \mathrm{cm}^2/\mathrm{s}$$

Similarly, for electrons, we find $D_{\mathrm{e}} = 13.8\ \mathrm{cm}^2/\mathrm{s}$.

[2]In some books and articles, the thermal voltage is denoted as V_{t} or V_{th}. This notation can easily lead to confusion with the threshold voltage of a MOSFET and it will not be followed in this book.

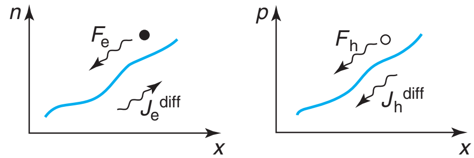


FIGURE 4.11
Sketch of relationship between fluxes and diffusion current densities for electrons and holes.

4.3.3 Diffusion current

Since electrons and holes are charged particles, their diffusion produces an electric current. Multiplying the expressions for carrier flux by the respective carrier charges, the current densities due to diffusion are obtained as follows:

$$J_e = qD_e \frac{dn}{dx} \tag{4.40}$$

$$J_h = -qD_h \frac{dp}{dx} \tag{4.41}$$

Rather than trying to remember the signs of Eqs. (4.40) and (4.41), it is much easier to obtain them in an intuitive way. The sketches of Fig. 4.11, where positive carrier gradients are illustrated, are helpful in this regard.

EXERCISE 4.5

In a certain region of a semiconductor characterized by $0 \le \mathrm{x}\ (\mu\mathrm{m}) \le 1$, *there is an electron concentration that follows a profile* $n(x) = 10^{17}(1 + 10\mathrm{x})\ \mathrm{cm}^{-3}$ *with x in* μm. *At* $x = 0$, *calculate the electron diffusion flux, the electron diffusion current density, and the electron diffusion velocity. Assume an electron diffusion coefficient of 10* cm^2/s.

Before doing any calculations, it is convenient to homogenize the units in the equation that describes the electron profile. In this equation, x needs to be given in micrometer. If instead we want to use x in cm, then we need to change x into 10^4x. Hence, the electron profile can be described as $n(x) = 10^{17}(1 + 10^5x)\ \mathrm{cm}^{-3}$ with x in cm.

At $x = 0$, the electron diffusion flux is given by Eq. (4.36)

$$F_e|_{x=0} = -D_e \left.\frac{dn}{dx}\right|_{x=0} = -10\ \mathrm{cm}^2/\mathrm{s} \times 10^{17} \times 10^5\ \mathrm{cm}^{-4} = -10^{23}\ \mathrm{cm}^{-2} \cdot \mathrm{s}^{-1}$$

The minus sign indicates that the electron flux takes place against the axis x.

The electron current density is simply obtained by multiplying the flux by the electron charge (Eq. (4.40)):

$$J_e = -qF_e = 1.6 \times 10^{-19}\ \mathrm{C} \times 10^{23}\ \mathrm{cm}^{-2} \cdot \mathrm{s}^{-1} = 1.6 \times 10^4\ \mathrm{A/cm}^2$$

The electron velocity can be obtained from the fundamental relationship that relates carrier flux and carrier velocity, Eq. (4.17):

$$v_{\mathrm{e}}^{\mathrm{diff}} = \frac{F_{\mathrm{e}}}{n} = -\frac{10^{23}\ \mathrm{cm}^{-2}\cdot \mathrm{s}^{-1}}{10^{17}\ \mathrm{cm}^{-3}} = -10^{6}\ \mathrm{cm/s}$$

The minus sign indicates that the electrons are traveling in a direction contrary to x.

4.4 TRANSIT TIME

As discussed above, if in a semiconductor region there is a concentration gradient or an electric field, there is net movement of carriers in space. It is often of interest to compute the time that it takes for a carrier, on average, to get from one point in space to another. This is called the *transit time.* Since the carrier velocity in general depends on location, the most general way of calculating the transit time in one dimension is by integrating the differential of time required to travel through a differential of space. In integral form, then, the transit time from a point at $x = 0$ to a point $x = L$ is given by

$$\tau_{\mathrm{t}} = \int_{0}^{\tau_{\mathrm{t}}} dt = \int_{0}^{L} \frac{dx}{v(x)} \tag{4.42}$$

The net carrier velocity $v(x)$ might be due to drift, diffusion, or a combination of the two. It is useful to separately examine these two cases.

For a purely diffusive case, the combination of Eqs. (4.18) and (4.40) gives an expression for the electron diffusion velocity:

$$v_{\mathrm{e}}^{\mathrm{diff}} = -D_{\mathrm{e}}\frac{1}{n}\frac{dn}{dx} \tag{4.43}$$

Inserting this in Eq. (4.42) yields

$$\tau_{\mathrm{t}} = -\frac{1}{D_{\mathrm{e}}}\int_{0}^{L} \frac{n}{\frac{dn}{dx}}\,dx \tag{4.44}$$

This is the most general expression of the transit time due to electron diffusion. It depends solely on the electron concentration profile and the electron diffusion coefficient. A simple case helps to see the leading dependencies.

Consider a linear electron concentration profile in space such as the one depicted in Fig. 4.12a. At $x = 0$, the electron concentration is n_1, and at $x = L$, the electron concentration goes to zero. This is not as unphysical a situation as it might seem. The boundary condition at $x = L$ resembles the base edge of the base–collector junction of a bipolar transistor in the forward active regime. An analytical expression for this electron profile is $n(x) = n_1(1 - \frac{x}{L})$. Plugging this profile into Eq. (4.44) yields after some simple algebra:

$$\tau_{\mathrm{t}} = \frac{L^2}{2D_{\mathrm{e}}} \tag{4.45}$$

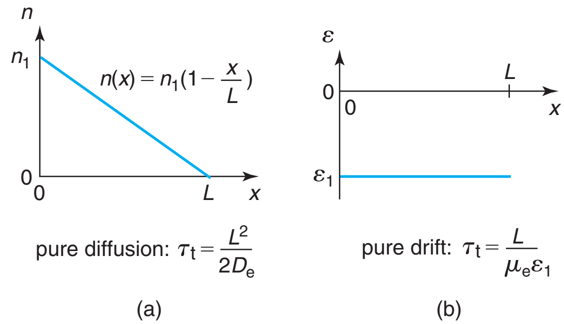


FIGURE 4.12
Two simple examples of electron flow and the resulting transit times from $x = 0$ to $x = L$: (a) diffusion down a linear gradient and (b) drift in a uniform electric field. In both cases, electrons flow from the left toward the right.

This is an interesting result. The transit time due to electron diffusion goes as the square of the length of the region through which the carriers diffuse and is inversely proportional to the diffusion coefficient. The inverse proportionality on diffusion coefficient is readily expected. The quadratic dependence on L also makes good sense. For a given value of n_1, if L increases, the electron gradient decreases proportionately and the distance that the carriers have to travel increases by the same amount. Hence, a $\tau_t \sim L^2$ dependence results.

Consider now a case of pure electron drift. Eq. (4.11) gives an expression for the electron drift velocity in terms of the electric field for small fields. Plugging this into Eq. (4.42) yields the transit time due to electron drift:

$$\tau_t = -\frac{1}{\mu_e} \int_0^L \frac{dx}{\mathcal{E}(x)} \tag{4.46}$$

This depends on electron mobility and the field distribution in space.

Let us examine the simple example depicted in Fig. 4.12b in which a uniform electric field of magnitude $\mathcal{E}_1$ is set up in a certain region in space. The electric field is negative in sign so that electrons drift from left to right. For this case, since the electric field is uniform, $\mathcal{E}_1$ can be taken out of the integral in Eq. (4.46) to yield

$$\tau_t = \frac{L}{\mu_e \mathcal{E}_1} \tag{4.47}$$

The transit time due to drift is linear in L and inversely proportional to $\mathcal{E}_1$, as expected from physical arguments.

Identical results can be obtained for the transit time for holes under similar circumstances. In situations with more complex carrier concentration or electric field profiles, the transit time expression might well be more complicated and perhaps take a nonanalytical form. The physics, however, is unchanged.

Situations that combine drift and diffusion are also straightforward to handle. In the case of electrons, for example, the total current is the sum of the drift contribution and the diffusion contribution:

$$J_e = J_e^{\text{drift}} + J_e^{\text{diff}} \tag{4.48}$$

The electron velocity in this case can be obtained using Eq. (4.18):

$$v_e = -\mu_e \mathcal{E} - D_e \frac{1}{n}\frac{dn}{dx} = v_e^{\text{drift}} + v_e^{\text{diff}} \tag{4.49}$$

In order to compute the transit time in a situation like this, the sum of the carrier drift and diffusion velocities must be used in Eq. (4.42).

4.5 NONUNIFORMLY DOPED SEMICONDUCTOR IN THERMAL EQUILIBRIUM

In thermal equilibrium, there are situations in which it is possible for an electric field to exist inside a semiconductor without applying a voltage from the outside. This is the case, for example, when the doping level inside a semiconductor region changes in space. This is quite common in devices since most doping techniques result in nonuniform impurity profiles. We study these situations here because even in thermal equilibrium, they bring into play both drift and diffusion simultaneously. Our analysis is also the basis for a future treatment of carrier behavior outside equilibrium in nonuniformly doped semiconductors.

This section introduces a number of important new concepts. The starting point is a review of one of Maxwell's equations that plays a key role in semiconductor devices: Gauss' law.

4.5.1 Gauss' law

Gauss' law states that the divergence of the electric field at a certain point in space is equal to the volume charge density at that same point divided by the permittivity of the material. In one dimension, Gauss' law can be written as

$$\frac{d\mathcal{E}}{dx} = \frac{\rho}{\epsilon} \tag{4.50}$$

where ρ is the volume charge density[3] and ϵ is the permittivity of the semiconductor.

It is important to understand Gauss' law well. If the volume charge density at a certain point is zero, it does not follow that the electric field is zero too. For example, in between the plates of a capacitor with air as dielectric and a voltage applied, the volume charge is zero but the electric field is not. The charge at the plates of the capacitor generates an electric field. If ρ is zero inside the capacitor, then $d\mathcal{E}/dx$ is zero at every point and $\mathcal{E}$ does not change from point to point. In order to have an electric field at a certain point, there is no need to have charge at that same location.

In semiconductors, there are several sources of volume charge. There is the *mobile charge* that arises from electrons and holes. There is also *fixed charge* that results from the ionized

[3] It is unfortunate that in the semiconductor literature, the symbol ρ is used for both resistivity and volume charge density. Since this is common practice, this notation will be followed in this book. The context and the units help identify which variable the symbol refers to in any particular equation.

dopants—ionized donors are positively charged and ionized acceptors are negatively charged. A general expression for the net volume charge density in a semiconductor is then

$$\rho = q(p - n + N_D^+ - N_A^-) \tag{4.51}$$

Throughout this book, we will assume that the working temperature is high enough for all dopants to be ionized. Thus $N_D^+ \simeq N_D$ and $N_A^- \simeq N_A$. Combining this with Eqs. (4.50) and (4.51), we get

$$\frac{d\mathcal{E}}{dx} = \frac{q}{\epsilon}(p - n + N_D - N_A) \tag{4.52}$$

Under thermal equilibrium conditions, in the bulk of a uniformly doped semiconductor sufficiently far away from any surface, *charge neutrality* prevails. This is because every dopant releases a carrier of the contrary charge sign. In a charge-neutral situation, Eq. (4.52) states that the electric field cannot change in space. Furthermore, since in thermal equilibrium no electric field is applied from the outside, we can conclude that $\mathcal{E}_o = 0$ everywhere (the subindex o is used to denote thermal equilibrium).

The situation is quite different in the presence of a nonuniformly doped semiconductor in thermal equilibrium. To illustrate the issues involved in a simple way, let us consider a long bar with a gradient of donors along its length as sketched in Fig. 4.13. If we can assume that nothing changes in the other two dimensions, a one-dimensional treatment should be adequate. Let us also assume that the semiconductor is sufficiently extrinsic so that we need not account for the tiny charge contribution of the minority holes. Under these conditions, Eq. (4.52) can be simplified to

$$\frac{d\mathcal{E}_o}{dx} = \frac{q}{\epsilon}(N_D - n_o) \tag{4.53}$$

where we have made explicit the thermal equilibrium situation by adding the subindex o to $\mathcal{E}$ and n.

At first sight, it may seem reasonable to expect that the electron concentration replicates exactly the donor concentration since each donor releases one electron, that is, $n_o(x) = N_D(x)$. This assures precise charge neutrality at every point of space. If that were the case, however, the electron distribution would end up being nonuniform. The electron concentration gradient would then produce a diffusion of electrons from the right where the initial concentration is high to the left where it is smaller and current would accordingly flow. This is clearly not an equilibrium situation.

We may be tempted to conclude then that as a result of carrier diffusion, electrons will flow until the electron concentration is uniform everywhere. If that were the case, on the right side of the bar, there would be a certain amount of positive charge resulting from the unbalanced concentrations of ionized donors and electrons. On the left, there would be net negative charge since the electron concentration is larger there than the donor concentration. Following Eq. (4.53), this charge distribution would result in an electric field that would pull electrons back to the right and away from the left region. This again is not an equilibrium situation.

The two extreme electron distributions that we have discussed are not possible in equilibrium. When $n_o(x) = N_D(x)$, charge neutrality at every point results in an electron diffusion current. If n_o is uniform in space, the presence of net volume charge results in a drift current. The one thing that we know must be fulfilled in thermal equilibrium is that the total electron current be zero everywhere. This implies that electron drift and diffusion must precisely balance out at every point. For a drift current to flow, we need an electric field. This can be generated if a nonuniform electron distribution is established that is somehow different from the donor concentration, that is, $n_o(x) \neq N_D(x)$. The electron gradient associated with the nonuniform electron distribution results in a diffusion current. The equilibrium carrier profile is the one for which the drift and diffusion currents cancel out at every point.

The mismatch between electron concentration and donor concentration in the nonuniformly doped bar leads to a dipole of charge, as sketched in Fig. 4.13. There is net positive charge toward the right and net negative charge toward the left. This dipole of charge produces an

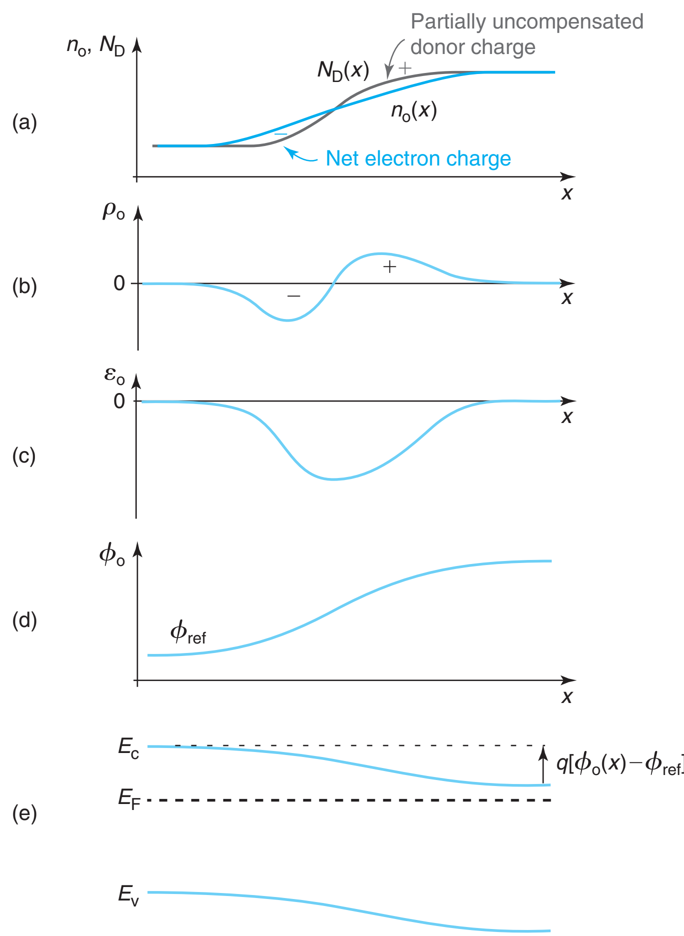


FIGURE 4.13
Nonuniformly doped semiconductor bar in thermal equilibrium: (a) doping level and resulting carrier concentration, (b) volume charge density, (c) electric field, (d) electrostatic potential, and (e) energy band diagram.

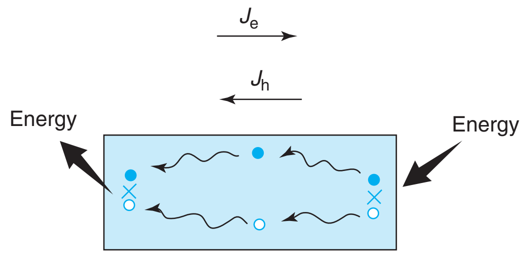


FIGURE 4.14
Sketch of impossible thermal equilibrium situation with $J_e = -J_h \neq 0$. The net carrier currents result in spontaneous absorption of energy at one end of the bar and emission of energy at the other. This cannot happen in thermal equilibrium.

electric field that points from right to left. With our choice of axis, $\mathcal{E}_o$ is negative. An equivalent way of expressing this is through an electrostatic potential ϕ_o that increases from left to right, as also sketched. The energy band diagram, in consequence, shows substantial bending. However, the Fermi level is flat throughout to reflect thermal equilibrium.

Our goal for the remainder of this section is to learn how to calculate the carrier profiles and to solve the complete electrostatic problem given an arbitrary doping distribution. Before that, we introduce a very important set of relationships connecting equilibrium carrier concentrations and electrostatic potential.

4.5.2 The Boltzmann relations

The application of the principle of detailed balance to a nonuniformly doped semiconductor in thermal equilibrium demands that the electron and hole currents be separately zero everywhere. This may appear obvious but it is instructive to consider what would happen if this were not the case (see Fig. 4.14). Assume a nonuniformly doped bar in which the electron current happens to balance out precisely the hole current so as to have net zero current. In this instance, since the two carrier types have charges of the contrary sign, electrons and holes must flow in the same direction from one side of the semiconductor to the other. In order to sustain this, electron–hole pairs must continuously be generated at one side of the bar and recombine at the other. This would imply that one side of the semiconductor is cooling down while the other is heating up. Clearly, this is not an equilibrium situation. The conclusion is that in thermal equilibrium, both carrier currents must be zero separately.

Coming back to the nonuniformly doped bar in thermal equilibrium depicted in Fig. 4.13, we just concluded that the total electron current must be zero. If the electric field resulting from the nonuniform dopant distribution is not too high, Eq. (4.21) can be used for the drift component of the current density. The total electron current density can then be written as

$$J_e = q\mu_e n_o \mathcal{E}_o + qD_e \frac{dn_o}{dx} = 0 \tag{4.54}$$

where the electron concentration has been explicitly denoted as n_o to indicate that this is an equilibrium situation. In general, n_o depends on position.

Equation (4.54) suggests that in thermal equilibrium there is a direct connection between the equilibrium carrier concentration and the electric field. The electric field drives drift, the concentration gradient drives diffusion, and they both must be in perfect balance in thermal equilibrium. Hence, n_o and $\mathcal{E}_o$ should be in some kind of relationship.

If we solve for $\mathcal{E}_o$ in Eq. (4.54), we get

$$\mathcal{E}_o = -\frac{kT}{q}\frac{1}{n_o}\frac{dn_o}{dx} \tag{4.55}$$

where we have used the Einstein relation (Eq. (4.38)).

If we apply an equivalent procedure to the minority carriers, in this case holes, we arrive at a similar equation:

$$\mathcal{E}_o = \frac{kT}{q}\frac{1}{p_o}\frac{dp_o}{dx} \tag{4.56}$$

This equation can also be derived directly from Eq. (4.55) by invoking the fact that $n_o p_o = n_i^2$.

Equations (4.55) and (4.56) are useful expressions since they allow the calculation of the electric field in a certain region of a semiconductor provided that either n_o or p_o is known. Equations (4.55) and (4.56) can also be used to derive a relationship between the electrostatic potential in thermal equilibrium ϕ_o and the equilibrium carrier concentration. From Eq. (4.55), for example, one gets the differential equation

$$\frac{d\phi_o}{dx} = \frac{kT}{q}\frac{d(\ln n_o)}{dx} \tag{4.57}$$

which after integration results in

$$\phi_o(x) - \phi_{ref} = \frac{kT}{q}\ln\frac{n_o(x)}{n_o(\text{ref})} \tag{4.58}$$

Had we started from Eq. (4.56) and proceeded in a parallel way, we would have reached the following result:

$$\phi_o(x) - \phi_{ref} = \frac{kT}{q}\ln\frac{p_o(\text{ref})}{p_o(x)} \tag{4.59}$$

These equations give a relationship between the *ratio* of the equilibrium carrier concentration at two different points and the *difference* in the electrostatic potential between the same two points. If the carrier concentration between two regions changes by a factor of 10, for example, the electrostatic potential differs by $(kT/q)\ln 10$. At room temperature, this is about 60 millivolts per decade or simply mV/dec, a handy number to remember.

ϕ_{ref} is in principle not set and, in consequence, $\phi_o(x)$ is not completely specified. This should not be a source of difficulty since in electrostatic problems the physics is always in the *potential difference* and not in its absolute value. In any one problem, one is free to choose the reference for potentials. A common choice consists on assigning the value of $\phi_{ref} = 0$ to the intrinsic carrier concentration $n_o(ref) = n_i$. With this choice, Eqs. (4.58) and (4.59), respectively, become

$$n_o = n_i \exp\frac{q\phi_o}{kT} \tag{4.60}$$

$$p_o = n_i \exp\frac{-q\phi_o}{kT} \tag{4.61}$$

Note that these equations satisfy the thermal equilibrium requirement of $n_o p_o = n_i^2$.

Both Eqs. (4.60) and (4.61) give a relationship between the equilibrium carrier concentration and the electrostatic potential in a case in which we have selected $\phi_{ref} = 0$ at the intrinsic point $n_o = p_o = n_i$. The relationship is exponential. These equations are known as the Boltzmann relations. If either one of them is known, the other one is straightforward to derive.

It cannot be stressed enough that these equations are reference sensitive. A different choice of potential reference results in expressions that look different. The physics of the situation is not changed, but its mathematical expression is. The resulting equations, in all cases, are known as the Boltzmann relations. A useful corollary is that clever choice of reference might lead to a simpler mathematical formulation. In this book, we will make different choices in different circumstances. You have to be ready to work with a menu of equivalent expressions.

EXERCISE 4.6

Derive equivalent Boltzmann relations as Eqs. (4.60) *and* (4.61) *above for a reference choice that assigns* $\phi_{ref} = 0$ *to a point where the equilibrium hole concentration is* N_A.

We can do this using Eq. (4.58) directly, or deriving an equivalent one from Eq. (4.56). Through either path, one gets the same result. If we proceed through the first route, at a point where the equilibrium hole concentration is N_A, the equilibrium electron concentration is n_i^2/N_A. At this point, we select $\phi_{ref} = 0$. Plugging this in Eq. (4.58), we get

$$n_o = \frac{n_i^2}{N_A} \exp \frac{q\phi_o}{kT} \tag{4.62}$$

Using again $n_o p_o = n_i^2$, we get

$$p_o = N_A \exp \frac{-q\phi_o}{kT} \tag{4.63}$$

As an additional algebraic exercise, show that you get the same equations starting from Eq. (4.56).

The Boltzmann relations, as described here, are nothing but expressions of a fundamental law of statistical mechanics known as Boltzmann's law. In its most generality, Boltzmann's law applies to any system of ideal particles immersed in a field of conservative forces (i.e., a field of forces that can be described by a potential) in thermal equilibrium. Boltzmann's law states that the probability of finding particles in a given spatial arrangement varies exponentially with the negative of the potential energy of that arrangement, divided by kT. For particle concentration, Boltzmann's law can be expressed as

$$n \propto e^{-E_p/kT} \tag{4.64}$$

where E_p is the potential energy.

This law applies to many other systems in nature. One particular system, the distribution of molecules in the ideal atmosphere, provides us with an intuitive and visual analogy. Imagine an ideal atmosphere that is characterized by a constant temperature (in our real atmosphere, it gets colder as we go up in altitude). All gas molecules of this ideal atmosphere are subject to the gravitational force. It is then clear that the weight of the gas column extending from a given altitude to infinity decreases as we go up in the atmosphere. The atmosphere gets "thinner" the higher we go. The concentration of molecules in height in this ideal atmosphere follows Boltzmann's law, that is, it decreases exponentially.

We can exploit this analogy to visualize the electron distribution in the conduction band of a nonuniformly doped semiconductor in thermal equilibrium. If the semiconductor is nondegenerate, electrons obey the Maxwell–Boltzmann distribution function, that is, the probability of finding an electron at a certain energy inside the conduction band goes down as the energy increases following the law $\exp[-(E - E_F)/kT]$. Since in thermal equilibrium the Fermi level is flat, conduction band states at the same energy from the Fermi level at different locations in the semiconductor have the same occupation probability. We can think of this nonuniformly doped semiconductor in thermal equilibrium as having a "stratified" distribution of electron occupation probability in the conduction band much as the gas concentration would be in an ideal atmosphere if there is a landscape in the terrain underneath. This is pictorially illustrated in Fig. 4.15. One can think of the electrostatic potential landscape as "carving out" the electron probability distribution in the conduction band. The same applies for holes in the valence band, but the picture is upside down, that is, the higher the energy, the denser the hole concentration is.

A final point to be made about the Boltzmann relations is that they have been derived under the assumption of Maxwell–Boltzmann statistics which apply to nondegenerate carrier situations. In the case of carrier degeneracy, the Einstein relation takes on a different form (see Problem 4.9) and as a consequence, the Boltzmann relations become somehow more complex. Still the basic fact remains that in thermal equilibrium, there is a one-to-one relationship between changes in electrostatic potential and changes in carrier concentration.

4.5.3 Equilibrium carrier concentration

We are finally in a position to formalize an approach to calculating the equilibrium carrier concentration in a nonuniformly doped semiconductor in thermal equilibrium. If we substitute Eq. (4.55) into Gauss' law (Eq. 4.53), we get

$$\frac{\epsilon kT}{q^2}\frac{d^2(\ln n_o)}{dx^2} = n_o - N_D \tag{4.65}$$

Given a certain doping distribution $N_D(x)$, the solution of this second-order nonlinear differential equation provides the spatial distribution of electrons in thermal equilibrium

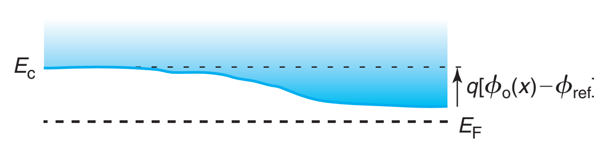


FIGURE 4.15
Pictorial view of electron occupation probability in the conduction band of nonuniformly doped semiconductor in thermal equilibrium.

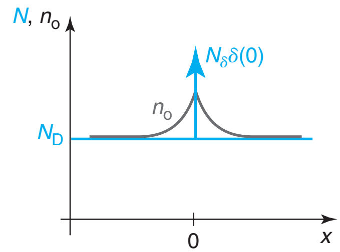


FIGURE 4.16
Schematic of delta doping. A sheet of donors with an area density N_δ is inserted in an otherwise uniformly doped n-type semiconductor. All the surfaces are very far away.

$n_o(x)$. Everything else in this equation is known. Once $n_o(x)$ is known, it is straightforward to compute $\rho_o(x)$, $\mathcal{E}_o(x)$, $\phi_o(x)$ and the energy band diagram in space.

As it turns out, Eq. (4.65) is a fairly challenging differential equation to solve. Analytical solutions are only available for a small set of simple cases. In most situations, this equation needs to be solved through numerical techniques. We study next a particularly simple example that brings out some important issues associated with solutions to this differential equation.

Example 1: delta doping

Delta doping is illustrated in Fig. 4.16. This figures sketches an extrinsic n-type semiconductor doped with a uniform doping concentration N_D. At a certain location that we label $x = 0$, there is an additional *sheet* of n-type dopants with a concentration N_δ (per unit area). The situation depicted in Fig. 4.16 is of a one-dimensional nature and all the surfaces are very far away. We are interested in calculating the equilibrium electron concentration n_o throughout this sample. Technologically, it is possible to fabricate these kinds of delta-doped layers using modern epitaxial techniques. They are widely used in GaAs high-electron mobility transistors (HEMTs).

In the absence of the delta-doped layer, charge neutrality dictates that the electron concentration everywhere be exactly equal to N_D. The delta-doped layer introduces more dopants and more electrons into the system. The dopants are positively charged and remain stuck in the lattice in a sheet-like distribution. The electrons are free to roam around. In thermal equilibrium, the balance between drift and diffusion leads to an electron distribution that is a somehow smeared version of the delta function, as indicated in Fig. 4.16. At a sufficient distance away from $x = 0$, we expect n_o to approach N_D.

The picture shown in Fig. 4.16 suggests that while there is overall charge neutrality in the system, there are regions with net charge. At $x = 0$, there is a sheet of positively charged donors with an areal density N_δ. Surrounding this, there is a region with net negative charge that decays the further away we move from the delta-doped layer. The charge imbalance creates an electric field and a potential build-up. A sketch of the complete electrostatic problem, including the energy band diagram, is shown in Fig. 4.17.

Simple as this problem seems, an analytical solution can only be obtained if the delta doping concentration is not too high with respect to the background doping level. We make this

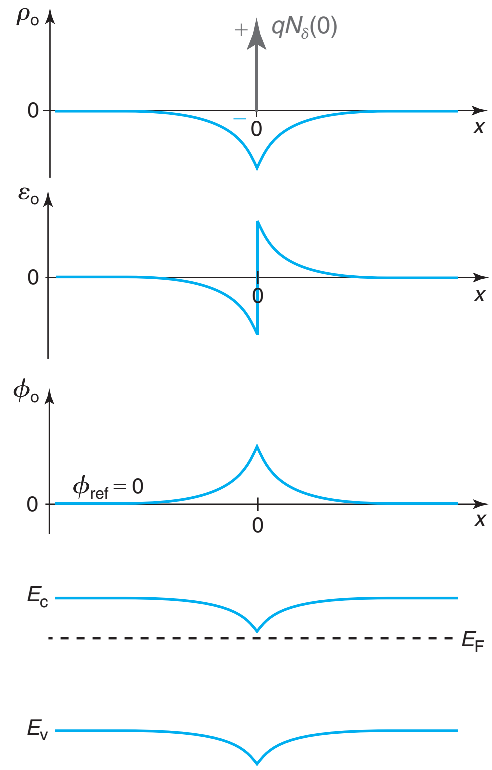


FIGURE 4.17
Sketch of solution to delta doping problem of Fig. 4.16. From top to bottom: charge density, electric field, electrostatic potential, and energy band diagram.

assumption below. After solving the problem, we study the limits imposed by this assumption on the value of N_δ.

It is easiest to work with the differential equation expressed in terms of the equilibrium electrostatic potential ϕ_o. If we choose as origin of potentials the value that corresponds to $n_o = N_D$, then following a procedure similar to that of Exercise 4.6, n_o can be written as $n_o(x) = N_D \exp(q\phi_o/kT)$. In terms of ϕ_o, we can rewrite Gauss' law in Eq. (4.53) as

$$\frac{d^2\phi_o}{dx^2} = \frac{qN_D}{\epsilon}\left(\exp\frac{q\phi_o}{kT} - 1\right) \tag{4.66}$$

This equation does not include the donor charge of the delta-doped layer and therefore applies only for $x > 0^+$ and $x < 0^-$. We treat the charge associated with the donors in the delta-doped $x = 0$ as a boundary condition to the solution of the differential equation.

This equation cannot be solved analytically in a general case. However, if the doping in the delta-doped layer is not too high, the electrostatic potential in the semiconductor will be everywhere small enough in the scale of kT/q for us to be able to expand the exponential in Eq. (4.66) around $\phi_o = 0$ and select its first two terms (see Taylor series expansion in Appendix D). When we do this, the differential equation is simplified to

$$\frac{d^2\phi_o}{dx^2} \simeq \frac{\phi_o}{L_D^2} \tag{4.67}$$

where L_D has units of length and is defined as

$$L_D = \sqrt{\frac{\epsilon kT}{q^2 N_D}} \tag{4.68}$$

We discuss the physics of L_D below.

A general solution to Eq. (4.67) can be written as

$$\phi_o(x) = Ae^{\frac{x}{L_D}} + Be^{-\frac{x}{L_D}} \tag{4.69}$$

If we focus on $x > 0^+$ (the solution for $x < 0^-$ can be easily obtained by symmetry), the coefficient A must be equal to zero since we know that ϕ_o must decay away with x. We obtain the value of B by matching boundary conditions at $x = 0$. At that location, we can create a Gaussian pill box with sides located at $x = 0^-$ and $x = 0^+$. Due to the symmetry of the problem, the field emerging through the right side of the pill box is given by

$$\mathcal{E}_o(x = 0^+) = \frac{qN_\delta}{2\epsilon} = -\left.\frac{d\phi_o}{dx}\right|_{x=0^+} = \frac{B}{L_D} \tag{4.70}$$

We solve for B here and insert the result into Eq. (4.69), to get

$$\phi_o(x) = \frac{qN_\delta L_D}{2\epsilon}\, e^{-\frac{x}{L_D}} \qquad \text{for } x > 0^+ \tag{4.71}$$

From this, the electric field can be obtained as

$$\mathcal{E}_o(x) = -\frac{d\phi_o}{dx} = \frac{qN_\delta}{2\epsilon}\, e^{-\frac{x}{L_D}} \qquad \text{for } x > 0^+ \tag{4.72}$$

The volume charge density is given by

$$\rho_o(x) = \epsilon\frac{d\mathcal{E}_o}{dx} = -\frac{qN_\delta}{2L_D}\, e^{-\frac{x}{L_D}} \qquad \text{for } x > 0^+ \tag{4.73}$$

Finally, n_o is given by

$$n_o(x) = N_D - \frac{\rho_o}{q} = N_D + \frac{N_\delta}{2L_D}\, e^{-\frac{x}{L_D}} \qquad \text{for } x > 0^+ \tag{4.74}$$

You can now verify that the integral of $n_o(x)$ from $x = 0^+$ to infinity adds up to $N_D + N_\delta/2$, as it should be. All these results are sketched in Fig. 4.17.

There is a condition that needs to be fulfilled for our solution to be acceptable. The truncated Taylor series expansion of $\exp(q\phi_o/kT)$ is reasonably valid if ϕ_o is not too high in the scale of

kT/q. The maximum value of ϕ_o occurs at $x = 0$. If we accept that the maximum tolerable value for ϕ_o is kT/q, then, from Eq. (4.71), we can conclude that $N_\delta < 2N_D L_D$. For higher values than this, our analytical solution is suspect and we need to use numerical techniques to solve this problem.

To summarize, we see that the delta-doped layer gives rise to a distribution of volume charge density that has a peculiar shape. At the location of the delta-doped layer, there is a spike of positive charge that arises from the donors. This is surrounded by a cloud of negative charge that is associated with an electron concentration that exceeds the background uniform doping level. The region with net charge is limited in space. Sufficiently far away from the delta-doped layer, charge neutrality is reestablished. The solution to the problem exhibits a characteristic length that is given by L_D. The region with substantial volume charge density is confined to a few L_D lengths around $x = 0$.

L_D emerges as the key characteristic length in many electrostatics problems in semiconductors. It is called the **Debye length**.[4] From this problem, we can see that the Debye length has to do with the ability of carriers in a semiconductor region to "screen" volume charge. More rigorously, in fact, the Debye length can be defined as

$$L_D = \sqrt{\frac{\epsilon kT}{q^2 n_o}} \tag{4.75}$$

L_D reflects the balance between drift and diffusion that must exist in a semiconductor in thermal equilibrium. If for some reason, there is net fixed charge in a semiconductor region, carriers redistribute themselves around this region in an effort to screen out the charge as much as possible. The more carriers there are, the more effective they are at this. This explains the carrier concentration dependence of L_D.

What works against perfect screening of net fixed charge is diffusion. As carriers pile up around the fixed charge, the concentration gradient that ensues drives them away through diffusion. As the temperature is increased, diffusion gets comparatively stronger (through the Einstein relation) and the Debye length increases. It is the balance between drift and diffusion that sets the value of the Debye length.

Figure 4.18 graphs the Debye length for Si at room temperature. The Debye length is the same regardless of the doping type. For typical doping levels, the Debye length is quite short, of order 10–100 nm. As the doping level increases, the Debye length gets very short. In highly doped regions, any disturbances from neutrality are confined to fairly small dimensions.

Our use of Boltzmann statistics makes the expression for Debye length in Eq. (4.68) and the plot of Fig. 4.18 not valid in the degenerate regime. New expressions can be derived that adequately deal with Fermi–Dirac statistics.

Quasi-neutral situations

The example in the previous section leaves clear how unwieldy the differential equation eq. (4.65) (or eq. 4.66) is. Analytical solutions are difficult even in relatively simple situations. There

[4] L_D as defined in Eq. (4.68) is actually called the *extrinsic* Debye length. It is also possible to define an *intrinsic* Debye length in an intrinsic semiconductor. Since we will not use this later parameter in this book, here we will refer to L_D as simply the Debye length.

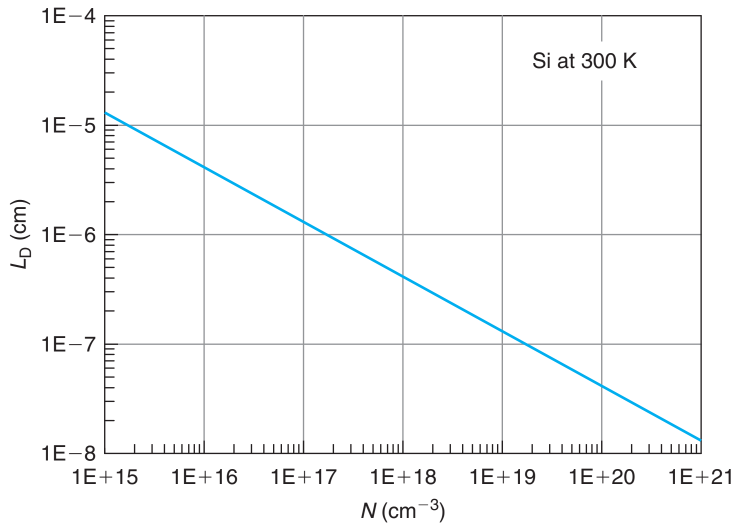


FIGURE 4.18
Debye length for Si at room temperature as a function of doping level.

is an important class of problems, however, that readily yields analytical solutions. We study them in this section.

Suppose that N_D is a function of space that changes rather slowly with x. In this case, it is reasonable to expect that n_o also displays a slowly changing spatial dependence. In fact, we saw in the delta-doped example, that n_o is a somehow "blunter" version of the doping distribution. Under these circumstances, it is possible for the left-hand side term of Eq. (4.65) to be much smaller than either term on the right-hand side. If so, it then follows that

$$n_o(x) \simeq N_D(x) \tag{4.76}$$

Expressed in a different way, if the doping profile changes slowly enough with position, the equilibrium majority carrier concentration closely tracks the doping concentration in space. In such a case, the net volume charge density is very small everywhere. It is because of this that these are called *quasi-neutral situations*.

Once we postulate $n_o(x)$ through Eq. (4.76), the electric field can be obtained from Eq. (4.55), the electrostatic potential by integration of the electric field, and the volume charge density by using Gauss' law in Eq. (4.50). Note that it would be inappropriate to conclude that Eq. (4.76) implies that $\rho_o = 0$. ρ_o is small, that is what the "quasi" in quasi-neutrality means, but not zero.

How can we assess if we are in front of a quasi-neutral situation? What really matters is the relative error in the statement in Eq. (4.76). Hence for quasi-neutrality to apply, we must demand

$$\left|\frac{n_o - N_D}{N_D}\right| \ll 1 \tag{4.77}$$

Using Eq. (4.65), we can write this as

$$\left|\frac{\epsilon kT}{q^2}\frac{d^2(\ln N_D)}{dx^2}\right| \ll N_D \tag{4.78}$$

Equation (4.78) gives a condition that one can *a priori* check in a given impurity profile. This can be expressed in a more compact way in terms of the Debye length. Using Eq. (4.68) in Eq. (4.78), we can write

$$L_D^2 \left| \frac{d^2(\ln N_D)}{dx^2} \right| \ll 1 \tag{4.79}$$

We can understand the physical meaning of this condition by inserting Eq. (4.55) in Eq. (4.77) to get

$$L_D^2 \left| \frac{d\mathcal{E}_o}{dx} \right| \ll \frac{kT}{q} \tag{4.80}$$

$L_D|d\mathcal{E}_o/dx|$ represents the change in the electric field over a Debye length. In consequence, if $\mathcal{E}_o$ is small, $L_D^2|d\mathcal{E}_o/dx|$ is roughly the change of the electrostatic potential over a Debye length. Equation (4.80) then leads to a practical rough guideline for quasi-neutrality: *For a nonuniformly doped profile to be quasi-neutral in thermal equilibrium, the change of the electrostatic potential over a Debye length must be much smaller than the thermal voltage.*

This understanding helps us to figure out when it is appropriate to consider nonuniformly doped situations as quasi-neutral. What we need is for the doping profile not to change too fast in the scale of L_D. So, the gentler the profile, the less volume charge density is going to appear and the more likely it is that the quasi-neutrality assumption applies. Also, the shorter the Debye length, the sharper the profile can be before the quasi-neutrality approximation becomes inappropriate. From looking at the reduction of L_D with doping level in Fig. 4.18, we can conclude that the higher the doping level, the more likely it is that nonuniformly doped regions can be considered quasi-neutral. This is relevant because in modern microelectronic devices, size reduction and the need to obtain low parasitic resistance has meant that doping levels have been increasing over time. In many devices, moderately and heavily doped regions can be considered quasi-neutral. This greatly facilitates their analysis.

The following example illustrates how the quasi-neutrality assumption is to be used in a nonuniformly doped situation in thermal equilibrium.

Example 2: hyperbolic doping distribution

Consider an n-type semiconductor with a doping distribution in space given by

$$N_D(x) = N_{Do} + \Delta N_D \tanh \frac{x}{L_c} \tag{4.81}$$

Depending on the relative values of N_{Do}, ΔN_D, and L_c, this profile can be quite steep or rather gradual. An example is shown in Fig. 4.19 for $N_{Do} = 10^{16}$ cm^{-3}, $\Delta N_D = 5 \times 10^{15}$ cm^{-3}, and various values of L_c spanning from 10 to 200 nm.

The figure shows also the electron concentration that results from this donor profile. This is obtained by numerically solving the differential equation eq. (4.65). As expected, the electron profile is a somehow smeared version of the donor profile. It is interesting to see the role of the characteristic length of this donor distribution, L_c, in the resulting electron concentration. A short value of L_c means that the donor concentration changes rather abruptly in space. In this case, the electron profile ends up being quite different from the donor profile. For long values of L_c, the electron concentration approaches the donor concentration.

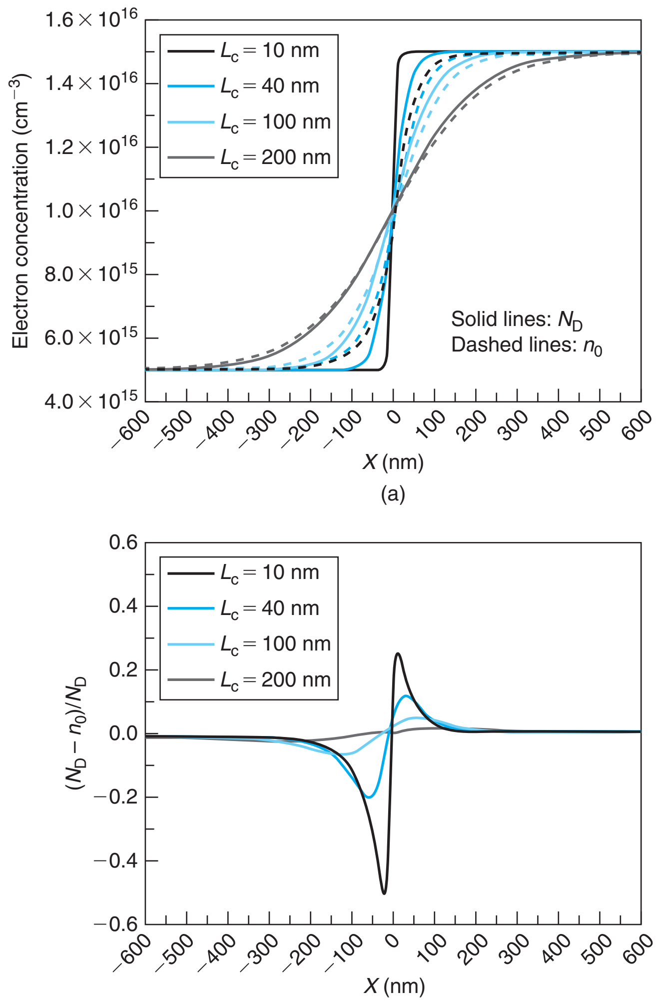


FIGURE 4.19
Example of nonuniform n-type doping situation. The doping profile is given by Eq. (4.81) **with values of** $N_D = 10^{16}\ \text{cm}^{-3}$, $\Delta N_D = 5 \times 10^{15}\ \text{cm}^{-3}$, **and various values of** L_c **spanning from 10 to 200 nm. (a) Doping profile and resulting electron concentration. (b) Relative imbalance between electron concentration and donor concentration (Calculations and Graphics by Ling Xia).**

The relative discrepancy between $n_o(x)$ and $N_D(x)$ is plotted in figure 4.19. For the shortest value of L_c, the discrepancy is found to be quite large, as high as 50%. This is clearly a situation with substantial volume charge density in space. For the longest value of L_c, the discrepancy is as small as a few percentage points. This is a quasi-neutral situation.

The Debye length corresponding to the mean doping level of this profile of $N_{Do} = 10^{16}\ \text{cm}^{-3}$ is about 40 nm. It is clear, then, that when $L_c \gg L_D$, the profile can be considered quasi-neutral. Otherwise, it cannot.

Under the assumption of quasi-neutrality, it is rather straightforward to solve this problem analytically. The electron concentration is approximately equal to the doping level everywhere:

$$n_o(x) \simeq N_D(x) = N_{Do} + \Delta N_D \tanh \frac{x}{L_c} \tag{4.82}$$

The electric field can be easily calculated using Eq. (4.55):

$$\mathcal{E}_o \simeq -\frac{kT}{q}\frac{1}{L_c}\frac{\Delta N_D}{N_{Do}}\frac{1}{\cosh^2 \dfrac{x}{L_c} + \dfrac{\Delta N_D}{2N_{Do}}\sinh \dfrac{2x}{Lc}} \tag{4.83}$$

If we select as origin of potentials $\phi(n_i) = 0$, the electrostatic potential distribution is easily obtained from Eq. (4.60):

$$\phi_o(x) \simeq \frac{kT}{q}\ln \frac{1}{n_i}(N_{Do} + \Delta N_D \tanh \frac{x}{L_c}) \tag{4.84}$$

Finally, the volume charge distribution can be obtained from Gauss' law, Eq. (4.50):

$$\rho_o \simeq \epsilon \frac{kT}{q}\frac{\Delta N_D}{L_c^2}\frac{N_{Do}\sinh \dfrac{2x}{L_c} + \Delta N_D \cosh \dfrac{2x}{L_c}}{\left(N_{Do}\cosh^2 \dfrac{x}{L_c} + \dfrac{\Delta N_D}{2}\sinh \dfrac{2x}{Lc}\right)^2} \tag{4.85}$$

For $x \geq 0$, ρ_o is positive since there are a few more donors than electrons. For $x \leq 0$, there are more electrons than donors and ρ_o is negative. All these results are graphed in Fig. 4.20.

We can now check the quasi-neutrality condition. From Eq. (4.77), quasi-neutrality prevails when

$$\left|\frac{n_o - N_D}{N_D}\right| = \left|\frac{1}{N_D}\frac{\rho_o}{q}\right| = \left|\frac{L_D^2}{L_c^2}\frac{\Delta N_D}{N_{Do}}\frac{\sinh \dfrac{2x}{L_c} + \dfrac{\Delta N_D}{N_{Do}}\cosh \dfrac{2x}{L_c}}{(\cosh^2 \dfrac{x}{L_c} + \dfrac{\Delta N_D}{2N_{Do}}\sinh \dfrac{2x}{L_c})^2}\right| \ll 1 \tag{4.86}$$

The fraction with the hyperbolic functions is at most unity. We can then rewrite the quasi-neutrality condition as

$$\left|\frac{n_o - N_D}{N_D}\right| \leq \left(\frac{L_D}{L_c}\right)^2 \frac{\Delta N_D}{N_{Do}} \ll 1 \tag{4.87}$$

In a case in which ΔN_D is of the same order as N_{Do}, quasi-neutrality is satisfied when the characteristic length L_c over which the doping profile changes is somehow larger than the Debye length L_D. This is precisely what we found in the numerical results above.

This example illustrates how in spite of assuming $n_o(x) \simeq N_D(x)$ in the quasi-neutral approximation, we can still derive a first-order expression for the electric field and charge distribution in space that result from the small imbalance that exists between $n_o(x)$ and $N_D(x)$.

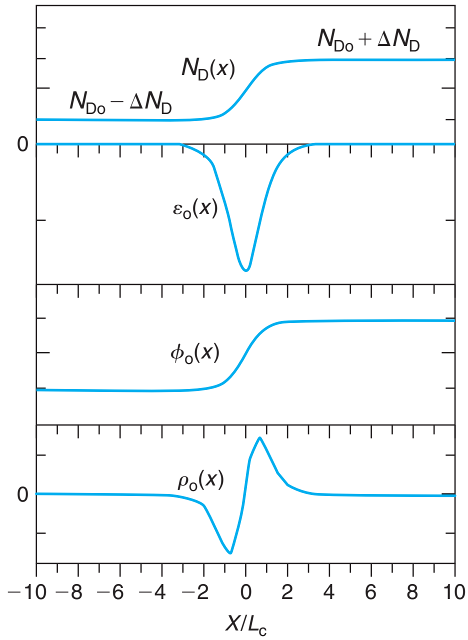


FIGURE 4.20
Example of quasi-neutral doping profile given by $N_D(x) = N_{Do} + \Delta N_D \tanh \frac{x}{L_c}$. From top to bottom: N_D, $\mathcal{E}_o$, ϕ_o, and ρ_o as a function of the normalized space coordinate x/L_c.

4.6 QUASI-FERMI LEVELS AND QUASI-EQUILIBRIUM

In Chapter 2, we learned that the Fermi level in a semiconductor uniquely characterizes the electron and hole densities in equilibrium as well as their energy distributions in their respective bands. Equations (2.29) and (2.37) related the equilibrium carrier densities to the effective density of states in the corresponding bands. A single value of E_F established this connection for both electrons and holes simultaneously.

In general, outside equilibrium the carrier densities are different from their equilibrium values. It is then not possible to find a single value of E_F that simultaneously relates n with N_c and p with N_v in the manner that Eqs. (2.29) and (2.37) do. We can however *define* two "quasi-Fermi levels," one for electrons and another one for holes such that the functional relationships between carrier densities and effective density of states given in Eqs. (2.29) and (2.37) are preserved, that is:

$$n = N_c \mathcal{F}_{1/2}\left(\frac{E_{fe} - E_c}{kT}\right) \tag{4.88}$$

$$p = N_v \mathcal{F}_{1/2}\left(\frac{E_v - E_{fh}}{kT}\right) \tag{4.89}$$

For clarity, we first simplify Eqs. (4.88) and (4.89) for nondegenerate situations:

$$n = N_c \exp \frac{E_{fe} - E_c}{kT} \tag{4.90}$$

$$p = N_v \exp \frac{E_v - E_{fh}}{kT} \tag{4.91}$$

The quasi-Fermi levels defined in this way serve as excellent visualization tools for carrier concentrations and generation and recombination in semiconductors out of equilibrium. By their very definition, carrier concentrations are directly related to energy distance of each quasi-Fermi level to its corresponding band edge through Eqs. (4.90) and (4.91). The usefulness of these two equations goes beyond this. Note that we can write the np product as

$$np = n_i^2 \exp \frac{E_{fe} - E_{fh}}{kT} \tag{4.92}$$

This implies that if E_{fe} is drawn above E_{fh}, then $np > n_i^2$. This is a situation outside equilibrium with excess carrier concentrations where net recombination is taking place. On the other hand, if $E_{fe} < E_{fh}$, then $np < n_i^2$ and the semiconductor is below equilibrium. Net generation is taking place. If the two quasi-Fermi levels coincide, then $np = n_i^2$ and the carrier concentrations are in thermal equilibrium. All these facts can be quickly verified in a band diagram by checking the relative position of the two quasi-Fermi levels.

As an illustration, Fig. 4.21 sketches four different energy band diagrams depicting different situations. The diagram on the left corresponds to an n-type sample in thermal equilibrium. In this case, there is only one Fermi level. The next diagram depicts a situation in which the same sample is under low-level injection resulting, for example, from external light illumination. Since under low-level injection $n \simeq N_D$, the quasi-Fermi level for electrons has not moved from its equilibrium position. The quasi-Fermi level for holes, on the other hand, moves toward the valence band because $p \gg p_o$. Note also that E_{fe} is above E_{fh}. This means that net recombination is taking place in this region of the semiconductor.

The third diagram is under high-level injection conditions, in which both carrier concentrations are increased from their equilibrium values. In consequence, both quasi-Fermi levels move toward their respective bands. If the sample remains quasi-neutral (as is usually the case), the

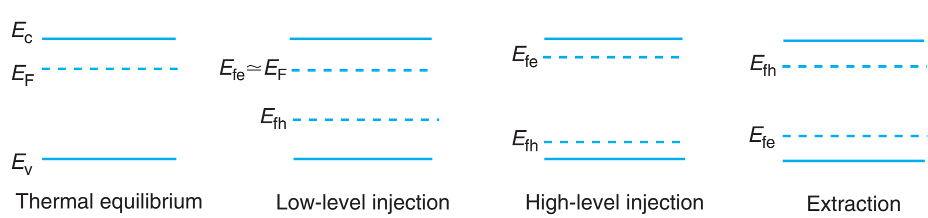


FIGURE 4.21
Illustration of the location of the Fermi level and quasi-Fermi levels in an n-type sample under four different situations: thermal equilibrium, low-level injection, high-level injection, and extraction.

distance between each quasi-Fermi level and its corresponding band is about the same (we are assuming here that the effective density of states of the conduction and valence bands are not too different from each other).

The diagram on the far right represents the same sample in extraction, with the carrier concentrations below their equilibrium values. In the energy band diagram, this is depicted by the quasi-Fermi levels becoming further away from their respective bands. In extraction, E_{fh} is above E_{fe} and net generation prevails.

EXERCISE 4.7

Calculate the relative location with respect to the band edges of the Fermi level or quasi-Fermi levels, as appropriate, for a uniformly doped n-type Si sample with $N_D = 10^{16}$ cm^{-3} at room temperature in the following situations: (a) in equilibrium; (b) in low-level injection such that there are 10^{14} cm^{-3} holes; (c) in high-level injection such that there are 10^{18} cm^{-3} electrons; and (d) in extraction where the equilibrium carrier concentrations have been reduced by 10 orders of magnitude.

(a) In thermal equilibrium, $n_o \simeq N_D = 10^{16}$ cm^{-3} and $p_o = n_i^2/n_o \simeq 10^4$ cm^{-3}. Using Eq. (2.32), the Fermi level is found to be

$$E_c - E_F = kT \ln \frac{N_c}{n_o} = 0.026 \ln \frac{2.9 \times 10^{19}}{10^{16}} = 0.21 \text{ eV}$$

(b) In low-level injection with $p = 10^{14}$ cm^{-3}, the electron concentration does not change from its equilibrium value. In consequence, E_{fe} is located with respect to E_c just where E_F was in equilibrium. Hence, $E_c - E_{fe} = 0.21$ eV. The location of E_{fh} can be obtained from Eq. (4.91):

$$E_{fh} - E_v = kT \ln \frac{N_v}{p} = 0.026 \ln \frac{3.1 \times 10^{19}}{10^{14}} = 0.33 \text{ eV}$$

(c) Under high-level injection conditions with $n = 10^{18}$ cm^{-3}, p takes the same value. In consequence,

$$E_c - E_{fe} = kT \ln \frac{N_c}{n} = 0.026 \ln \frac{2.9 \times 10^{19}}{10^{18}} = 0.088 \text{ eV}$$

$$E_{fh} - E_v = kT \ln \frac{N_v}{p} = 0.026 \ln \frac{3.1 \times 10^{19}}{10^{18}} = 0.089 \text{ eV}$$

Both quasi-Fermi levels approach their respective bands.

(d) Under extraction conditions with 10 orders of magnitude fewer carriers than in equilibrium, the location of the quasi-Fermi levels is

$$E_c - E_{fe} = kT \ln \frac{N_c}{n} = 0.026 \ln \frac{2.9 \times 10^{19}}{10^{6}} = 0.81 \text{ eV}$$

$$E_{\rm fh} - E_{\rm v} = kT \ln \frac{N_{\rm v}}{p} = 0.026 \ln \frac{3.1 \times 10^{19}}{10^{-6}} = 1.53 \text{ eV}$$

Notice that $E_{\rm fe}$ has gotten rather close to the valence band edge while $E_{\rm fh}$ has penetrated into the conduction band.

Quasi-Fermi levels not only are great tools to visualize carrier concentrations and generation and recombination but also help us picture carrier flow. To realize this, let us examine the expressions of the carrier currents outside thermal equilibrium. Let us do it first for electrons. If the electric field is not very high, from Eqs. (4.21) and (4.40), the electron current is given by

$$J_{\rm e} = q\mu_{\rm e} n\mathcal{E} + qD_{\rm e}\frac{dn}{dx} \tag{4.93}$$

This equation can be rewritten in a very interesting way. Taking the derivative with respect to x in Eq. (4.90), we find

$$\frac{dn}{dx} = \frac{n}{kT}\frac{dE_{\rm fe}}{dx} - \frac{q}{kT}n\mathcal{E} \tag{4.94}$$

Substituting this into Eq. (4.93) and using the Einstein relation Eq. (4.38), we finally obtain

$$J_{\rm e} = \mu_{\rm e} n\frac{dE_{\rm fe}}{dx} \tag{4.95}$$

Had we carried out an identical exercise for holes, we would have obtained an equivalent expression:

$$J_{\rm h} = \mu_{\rm h} p\frac{dE_{\rm fh}}{dx} \tag{4.96}$$

Equations (4.95) and (4.96) are extremely important and useful relationships. They state that the gradient of the quasi-Fermi levels can be viewed as a sort of unified driving force for carrier flow that combines drift and diffusion. The reason behind this is made obvious if we equate Eqs. (4.95) and (4.96), respectively, to the fundamental current relationship Eqs. (4.18) and (4.20). Solving for the gradient of the quasi-Fermi level, we find

$$\frac{dE_{\rm fe}}{dx} = -\frac{q}{\mu_{\rm e}}v_{\rm e} \tag{4.97}$$

$$\frac{dE_{\rm fh}}{dx} = \frac{q}{\mu_{\rm h}}v_{\rm h} \tag{4.98}$$

For both electrons and holes, we find that the gradient of the quasi-Fermi level is proportional to the carrier velocity, indeed, a very fundamental property of a carrier ensemble. It should not be surprising, then, that even though Eqs. (4.95) and (4.96) were derived under fairly restrictive conditions (low-field and Maxwell–Boltzmann statistics), they are in fact very general. The same expressions can be obtained in a degenerate semiconductor and also in the presence of high electric fields.

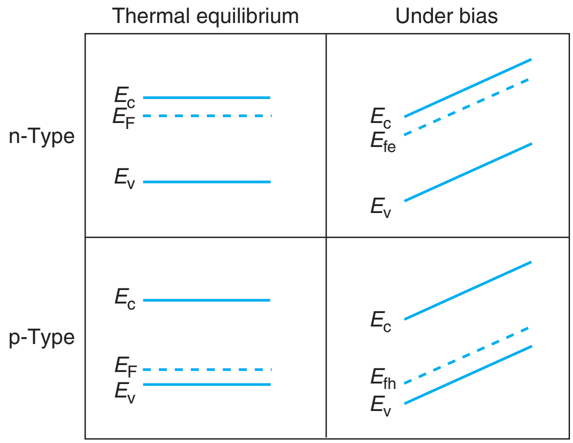


FIGURE 4.22
Illustration of current visualization in energy band diagrams through quasi-Fermi levels. Top two diagrams depict a uniformly doped n-type semiconductor in thermal equilibrium (left) and under bias (right). Bottom two diagrams show a uniformly doped p-type semiconductor in thermal equilibrium (left) and under bias (right).

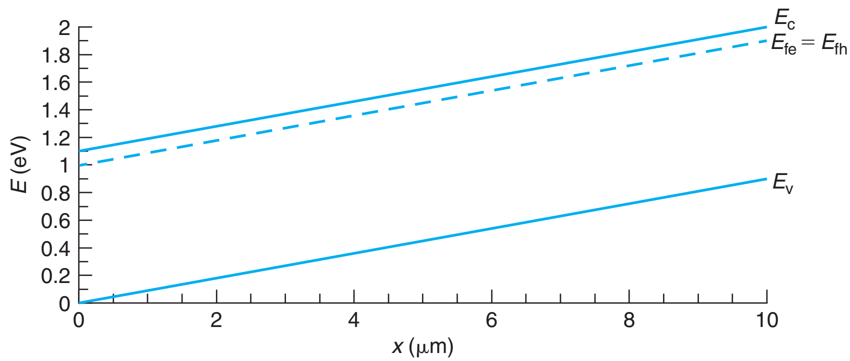
The quasi-Fermi levels provide great insight into carrier flow in a variety of situations. In the presence of a complex situation with drift and diffusion simultaneously taking place, the behavior of the quasi-Fermi levels quickly indicates things such as the net direction of carrier flow and the relative magnitude of carrier velocity. This can be understood by examining Eqs. (4.97) and (4.98) which state that the gradient of the quasi-Fermi level is proportional to the corresponding carrier velocity. This suggests that if in a certain region a quasi-Fermi level is depicted as flat, then the corresponding carrier velocity is zero or very small. If a quasi-Fermi level exhibits a slope, then carriers are flowing. The carrier velocity is proportional to the slope of the quasi-Fermi level in the band diagram.

Figure 4.22 shows several examples. The top-left diagram shows a uniformly doped n-type semiconductor in thermal equilibrium. We can judge that this is the case because the Fermi level is labeled E_F and it is sketched flat. To the right, this is the same semiconductor under bias. We understand this because the conduction and valence bands are shown inclined in space. We also see the quasi-Fermi level for electrons drawn parallel to the conduction and valence band edges. With an imaginary x-axis running from left to right in this diagram, the gradient of E_{fe} would be positive, and therefore, Eq. (4.97) would suggest that the electron velocity is negative, that is, electrons flow from right to left down the conduction band.

The two figures underneath depict similar situations for a uniformly doped p-type semiconductor under thermal equilibrium (on the left) and under bias (on the right). In this last case, hole current also flows from left to right that corresponds to a hole flow from left to right.

EXERCISE 4.8

Estimate the current density flowing through a Si sample with the energy band diagram sketched below at room temperature.



This energy band diagram reveals a sample with a uniform electric field applied to it. We could compute the current density from the drift expression (Eq. (4.23)). As we have just learned in this section, we can also use Eqs. (4.95) and (4.96). Let us proceed this way.

In the diagram, the two quasi-Fermi levels overlap. They are also fairly close to the conduction band but quite far away from the valence band. This implies that $n \gg p$ and we only need to account for electron current. The electron concentration can be estimated from the 0.1 eV distance between E_c and E_{fe}. As we learned in Chapter 2, this yields

$$n = N_c \exp \frac{E_{fe} - E_c}{kT} = 2.9 \times 10^{19} \exp \frac{-0.1}{0.0259} = 6.1 \times 10^{17} \text{ cm}^{-3}$$

This is then an n-type sample with $N_D \simeq 6.1 \times 10^{17}$ cm^{-3}. For this doping level, Fig. 4.3 shows that the electron mobility is about 300 cm^2/V.s.

The gradient of E_{fe} is

$$\frac{dE_{fe}}{dx} = \frac{0.9 \text{ eV}}{10 \times 10^{-4} \text{ cm}} = 900 \text{ eV/cm}$$

Putting it all together into Eq. (4.95), we finally get

$$J_e = 300 \text{ cm}^2/\text{V.s} \times 6.1 \times 10^{17} \text{ cm}^{-3} \times 900 \text{ eV/cm} = 1.7 \times 10^{23}\ e/\text{cm}^2 \cdot \text{s} = 2.6 \times 10^4 \text{ A/cm}^2$$

The concept of quasi-Fermi level hinges on the notion of *quasi-equilibrium*. Basically, even though we are dealing with situations outside equilibrium, the carrier distribution in energy never departs too far from thermal equilibrium. The fundamental reason for this is the fact that the scattering time, or average time between collisions, is several orders of magnitude

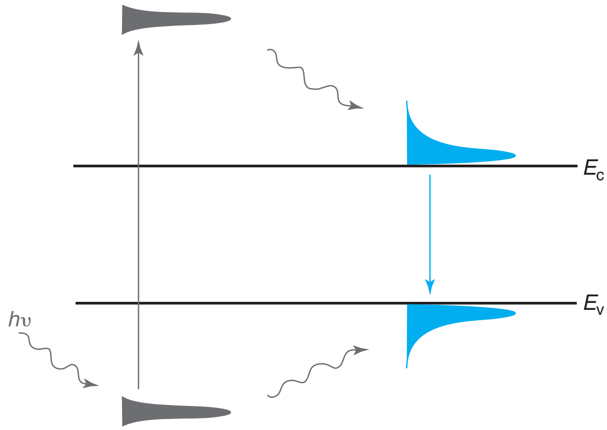


FIGURE 4.23
Sketch illustrating the applicability of the notion of quasi-equilibrium to describe the recombination process that follows a burst of hot-carrier generation by a pulse of energetic photons. Hot electrons and holes lose their extra energy on a timescale much shorter than the recombination lifetime. As a result, recombination takes place among carriers that have reached a quasi-equilibrium state with the lattice and are well described by their respective quasi-Fermi levels.

smaller than the timescales that characterize dynamic carrier behavior in semiconductor devices (depending on the particular device, this would be the carrier lifetime or the transit time due to drift or diffusion, as we will see in Chapter 5). Thus, on timescales of interest to us, carriers undergo many scattering events that keep their energy distribution in close thermal equilibrium with the lattice. It is for this reason that it is possible to define a quasi-Fermi level that plays a role equivalent to the Fermi level in strict thermal equilibrium. Often, this near equilibrium carrier distribution in energy is called a *Maxwellian distribution*.

Figure 4.23 helps to clarify the concept of quasi-equilibrium in a situation with excess carriers. The figure shows the sequence of events that follows the sudden generation of electron–hole pairs by a pulse of energetic photons. Let us assume that the energy of the photon beam is high enough so that right after the generation event hot electrons and hot holes with kinetic energies substantially higher than the thermal energy have been produced (left diagram). Right after this burst of carrier generation, two things are going to happen. First, electrons and holes will start losing excess energy by phonon emission, a process that is known as "thermalization." In addition, excess electrons and holes will start recombining with each other. For common semiconductors under typical situations, the time it takes to "thermalize" the hot carriers (a few scattering times, or < 1 ps) is much shorter than the recombination lifetimes (μs – ms). Therefore, thermalization takes place first (middle diagram) and recombination occurs after that (right diagram). In this scenario, recombination events mostly involve excess electrons and holes that have reached a quasi-equilibrium state with the lattice. The recombination dynamics can therefore be studied with the electron and hole distributions described through their respective quasi-Fermi levels.

In the situation described here, while recombination is taking place, the electrons are in near equilibrium among themselves and with the lattice (meaning, they suffer many collisions with the lattice in the timescale of the carrier lifetime). Similarly, holes are also near equilibrium among themselves and with the lattice. However, the electrons are not in equilibrium with the holes. That is what the process of recombination is trying to correct and it will take several lifetimes for equilibrium to be established. It is for this reason that we need to define two different quasi-Fermi levels that characterize the Maxwellian distributions of electrons and holes separately.

It is important to also think about the implications and constraints of quasi-equilibrium in a different situation, one in which drift is the dominant mechanism. Consider a semiconductor bar with a certain electron concentration and an electric field applied (we study how to do this in Chapter 5). In steady state, as electrons drift, they pick up kinetic energy from the electric field and release it to the lattice through phonon emission. Quasi-equilibrium means that the electron distribution does not significantly depart from the equilibrium one. We can ensure this if in the timescale in which carriers would pick up a thermal energy worth of kinetic energy from the electric field, they suffer many collisions that dissipate this energy into the lattice through phonon emission. This applies if the distance over which the potential energy changes by an amount kT is much longer than a mean free path. This is the relevant distance because if carriers suffered no energy dissipating collisions, as the potential energy associated with the electric field drops by kT, their kinetic energy increases by the same amount (Fig. 4.24).

Let's define this distance as λ_{kT}. We need that $\lambda_{kT} \gg l_{ce}$. Since, to the first order, $\lambda_{kT} = kT/\mathcal{E}$, we must then require that $\mathcal{E} \ll kT/l_{ce}$. l_{ce} depends on doping level. For the value determined in Exercise 4.1, $l_{ce} = 17$ nm, the constraint on the electric field is $\mathcal{E} \ll 1.5 \times 10^4$ V/cm. Higher electric field values are tolerable if the mean free path is shorter. The order of magnitude of this maximum electric field is quite small for electrons in Si at room temperature. This means that in many Si devices today, carriers in an electric field can get quite "hot" and quasi-equilibrium is not

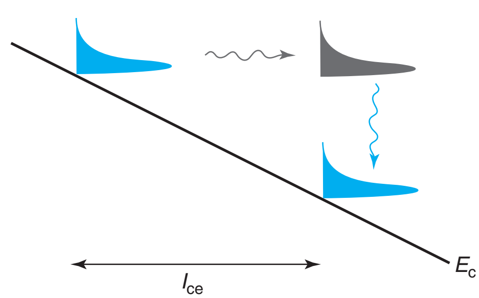


FIGURE 4.24
Sketch illustrating the applicability of the notion of quasi-equilibrium to describe electron drift in an electric field. For quasi-equilibrium to apply, electrons must not gain more than about a kT worth of kinetic energy from the electric field in a mean free path.

an appropriate concept. When this happens, the use of the quasi-Fermi level is suspect. We will still use this concept to sketch energy band diagrams in these situations mostly for convenience. It is important to realize that we are really operating beyond the usability of this concept.

4.7 SUMMARY

- Electrons and holes are charged particles that can move in a semiconductor (carriers). If they do, an electric current flows. The current density that is produced is directly proportional to the carrier concentration and the carrier velocity:

$$J_e = -qnv_e \qquad\qquad J_h = qpv_h$$

- Carriers drift in response to the application of an electric field. If the electric field is small enough, carriers drift with a velocity that is proportional to the field. The current density is then

$$J_t = q(\mu_e n + \mu_h p)\mathcal{E}$$

- For high enough fields, a saturation velocity is reached. The current density becomes

$$J_{t,sat} = q(nv_{esat} + pv_{hsat})$$

- Carriers diffuse as a result of concentration gradients. The diffusion currents for electrons and holes are, respectively:

$$J_e = qD_e\frac{dn}{dx} \qquad\qquad J_h = -qD_h\frac{dp}{dx}$$

- Detailed balance in thermal equilibrium demands that $J_e = J_h = 0$ everywhere.
- In a nonuniformly doped semiconductor in thermal equilibrium, a balance between carrier drift and diffusion results in a majority carrier profile in space that in general is different from the doping concentration distribution. This results in a spatial charge and an electric field inside the semiconductor. The electrostatic potential distribution is given by

$$\phi_o(x) - \phi_{ref} = \frac{kT}{q}\ln\frac{n_o(x)}{n_o(\text{ref})} = \frac{kT}{q}\ln\frac{p_o(\text{ref})}{p_o(x)}$$

- For a reasonably extrinsic semiconductor, if the doping distribution does not change too abruptly in space, the semiconductor remains "quasi-neutral" and the majority carrier concentration closely tracks the doping level.
- The timescale over which carriers can have energy distributions that are very different from equilibrium is much shorter than the timescales of interest in typical device operation, such as carrier lifetimes or transit times. For practical purposes, the carrier energy distributions can be considered close to equilibrium.
- Outside equilibrium, it is useful to define electron and hole quasi-Fermi levels as follows (under the Maxwell–Boltzmann approximation):

$$n = N_c \exp \frac{E_{fe} - E_c}{kT} \qquad p = N_v \exp \frac{E_v - E_{fh}}{kT}$$

- Gradients of quasi-Fermi levels can be considered as unified driving forces for carrier flow, regardless of whether the carriers move by drift, diffusion, or a combination of both processes:

$$J_e = \mu_e n \frac{dE_{fe}}{dx} \qquad J_h = \mu_h p \frac{dE_{fh}}{dx}$$

The gradient of the quasi-Fermi level is linearly proportional to the net carrier velocity.

4.8 FURTHER READING

There are several books with material relevant to this chapter.

The Feynman Lectures on Physics, Volume I by R. P. Feynman, R. B. Leighton, and M. Sands, Addison-Wesley 1963 (ISBN 0-201-02116-1-P, QC23.F435). Ch. 14 discusses forces and potential energy. Ch. 40 introduces the Boltzmann law and the velocity distribution of ideal particles. Ch. 43 contains an intuitive and easy to understand discussion about thermal motion, drift, and diffusion of gas ions and molecules. These chapters make easy but extremely illuminating reading. Highly recommended.

Introduction to Semiconductor Physics by R. B. Adler, A. C. Smith, and R. L. Longini, Wiley, 1964 (ISBN 0-471-00887-7, QC612.S4.S471 v.1). The first volume of the pioneering SEEC (Semiconductor Electronics Education Committee) series is nothing but a masterpiece. To this date, the clarity of the explanations and the physical insight that this little book provides has remained unmatched by any other textbook. In spite of its age, the selection of topics is still relevant and of high priority reading for serious students of semiconductor device physics. Material that is significant to this chapter is presented in §3.6 (nonuniformly doped situations) and 4.2 (Gauss' law and charge neutrality).

Fundamentals of Solid-State Electronics by C.-T. Sah, World Scientific, 1991 (ISBN 9810206372, TK7871.85.S23). In addition to other topics relevant to other chapters, Ch. 3 of Sah's book describes in detail drift, diffusion, and the quasi-Fermi levels. In particular, there is a fairly thorough and quantitative presentation of the various scattering mechanisms. The discussion surrounding the introduction of the quasi-Fermi levels is insightful.

Fundamentals of Semiconductor Theory and Device Physics by S. Wang, Prentice Hall, 1989 (ISBN 0-13-344409-0, QC611.W32). Always at a substantially higher level, Wang's book has a lot of material that merits reading, even if ignoring the math. The scattering mechanisms are carefully and rigorously described in §6.7. §§10.2 and 10.8 have excellent discussions on velocity saturation. §§10.3 and 10.9 describe the electron transfer effect and the peculiar velocity-field characteristics of III-V semiconductors.

Fundamentals of Carrier Transport by M. Lundstrom, 2nd Edition, Cambridge University Press, 2000 (ISBN 0-521-63134-3). This is an excellent treatment of carrier transport in semiconductors far more rigorous and at a substantially higher level than the present text. Particularly relevant here are Ch. 2 on carrier scattering, Ch. 3 on the Boltzmann transport equation, and Ch. 4 on low-field transport.

AT4.1 SELECTED PROPERTIES OF THE GAMMA FUNCTION

The gamma function is defined as

$$\Gamma(p) = \int_0^\infty \eta^{p-1} e^{-\eta} d\eta \tag{4.99}$$

In situations involving semiconductors, the gamma function typically makes an appearance for a few positive integer or fractional values of p.

For real values of p (other than zero or negative integers), the gamma function has the following property:

$$\Gamma(p+1) = p\Gamma(p) \tag{4.100}$$

For positive integer values of p, let us refer to them as n, the gamma function has the following property:

$$\Gamma(n) = (n-1)! \tag{4.101}$$

Commonly used values of the gamma function are

$$\Gamma(1) = 1 \tag{4.102}$$

$$\Gamma(\frac{1}{2}) = \sqrt{\pi} \tag{4.103}$$

Using these and the properties in Eqs. (4.100) and (4.101), other common values of the gamma function can be easily derived.

AT4.2 HOT-CARRIER EFFECTS

A charged particle immersed in an electric field drifts due to the electrostatic force that acts upon it. Between two scattering events, the carrier acquires a velocity component in the direction of the electric field. At the same time, the particle extracts kinetic energy from the potential energy of the field. If the additional kinetic energy is much smaller than the thermal energy, then the average energy of the particle is not much affected and the scattering rates are insignificantly upset from their equilibrium values. This is a quasi-equilibrium situation. If the particle acquires a kinetic energy from the field that is comparable or exceeds the thermal energy, the physics of the situation change substantially. This is known as the “hot-carrier” regime.

When carriers become hot, interesting effects occur. Earlier in this chapter we mentioned velocity saturation. In Chapter 3, we briefly described impact ionization, avalanche multiplication, and avalanche breakdown. These are all hot-carrier effects. The study of the physics of hot carriers is a rich field with a plethora of activity. As microelectronic devices shrink in size, the magnitude of the electric fields inside tends to increase and hot-carrier effects become more prominent and worrisome. Device designers ignore them at their own peril!

The aim of this Advanced Topics Section is to outline some of the underlying physics of hot-carrier effects and some of the most important consequences for microelectronic device designers. In order to do this, we must first look in more detail at the physics of phonon scattering. This leads into a discussion about the criteria for the onset of hot-carrier effects. We then study in more detail two important effects: hot-carrier transport and impact ionization. Two related physical phenomena, avalanche multiplication and avalanche breakdown are discussed in Chapter 5.

AT4.2.1 Energy relaxation versus momentum relaxation

In a solid, not all phonons are created equal. A complex lattice, such as the one of a semiconductor, allows many vibrational modes. All of these modes are very effective in exchanging *momentum* with carriers, that is, in randomizing a carrier's velocity. Some of these modes, however, are more effective than others in exchanging *energy* with carriers. In typical semiconductors, such as Si and GaAs, the most frequent collisions between phonons and carriers succeed in randomizing the velocity of the carrier but exchange little energy (a few millielectronvolts). Only collisions with special kinds of phonons, the so-called optical phonons, exchange a sizable amount of energy, the optical phonon energy, E_{opt}, a few tens of eV.

This important distinction is captured in two definitions: the *momentum relaxation time*, τ_{M}, and the *energy relaxation time*, τ_{E}. τ_{M} is the average time required to randomize the momentum of a carrier. τ_{E} is the average time required for a carrier to lose any excess energy it might have over the thermal energy. In most semiconductors $\tau_{\text{M}} < \tau_{\text{E}}$. In lowly doped Si, for example, under quasi-equilibrium conditions (for low fields), $\tau_{\text{M}} \simeq 0.2$ ps, while $\tau_{\text{E}} \simeq 0.5$ ps.

When a carrier is immersed in an electric field, this distinction between momentum and energy relaxation matters a lot. As we discussed above, between two collisions, a carrier picks up not only velocity (or momentum) in the direction of the electric field but also kinetic energy. A momentum-exchanging collision randomizes the velocity and therefore affects the net forward velocity of the carrier but only absorbs a small amount of energy. In consequence, the average energy of the carriers can build up. When this energy is high enough, optical phonon emission becomes more likely and the release of energy to the lattice is enhanced. Eventually a steady state situation is reached in which on average the carrier ensemble does not change its kinetic energy. The time that it takes for this steady-state situation to be established is on the order of the energy relaxation time. If the electric field is suddenly turned off, it will take a time of a few times τ_{E} for the carriers to thermalize by releasing their excess kinetic energy to the lattice.

The steady-state kinetic energy acquired by the carriers from an electric field is easy to estimate. Let us consider a chunk of uniformly doped n-type semiconductor of cross-sectional area A and length L (similar arguments apply for a p-type semiconductor). With a voltage V applied to it, a current I flows. The power dissipated in this piece of semiconductor is

$$W = |IV| = qnv_{\text{e}}^{\text{drift}} AL\mathcal{E} \tag{4.104}$$

where we have used Eq. (4.18) and we have assumed that the electric field is uniform along the length of the sample.

Since nAL is the total number of electrons in the sample, the power dissipated per electron is

$$w = qv_{\text{e}}^{\text{drift}}\mathcal{E} \tag{4.105}$$

For an electron, it takes a mean time τ_E to release its extra kinetic energy to the latice. On average, then, the kinetic energy picked up from the field by one electron is

$$\Delta E_K = q v_e^{\text{drift}} \mathcal{E} \tau_E \tag{4.106}$$

This result indicates that the excess kinetic energy is linearly proportional to the electric field. This is actually what is found if detailed calculations are performed, as shown in Fig. 4.25. This figure shows the average kinetic energy of electrons, E_K, in Si at room temperature in a uniform electric field, as a function of the electric field, as calculated using a Monte Carlo technique by Massimo Fischetti. Below a field of about 8 kV/cm, the average energy of the electrons is equal to the thermal energy. Above this field, the average energy increases rapidly. For a field of 200 kV/cm, for example, the average energy is about 1 eV above the thermal energy.

When the excess kinetic energy is comparable to the thermal energy, hot-electron effects become important. For n-type Si, for example, using a value of $\tau_E \simeq 0.5\,\text{ps}$, a value of $\mathcal{E} = 7.8\,\text{kV/cm}$ is required for ΔE_K to equal the thermal energy $\frac{3}{2}kT = 39\,\text{meV}$ at room temperature.

The energy relaxation time depends slightly on the average energy. Figure 4.26 shows its dependence on the magnitude of a uniform electric field in Si, also calculated by Fischetti. A consequence of the weak dependence of τ_E on $\mathcal{E}$ is that the average energy increases nearly linearly with electric field, as shown in Fig. 4.25.

AT4.2.2 Hot-electron transport

An important consequence of carriers acquiring additional kinetic energy in an electric field is that the likelihood of velocity randomizing collisions increases. This is because the higher the energy, the larger the density of states of the bands and the more states the carriers can scatter into. In consequence, the momentum relaxation time decreases as the average carrier energy increases. At high enough energy, $\tau_M \sim 1/E_K$ (this is a consequence of $g(E) \sim \sqrt{E}$). In a uniform electric field, since $E_K \sim \mathcal{E}$, then $\tau_M \sim 1/\mathcal{E}$, as seen in Fig. 4.26.

A decrease in τ_M brings with it, according with Eq. (4.10), a reduction in the mobility. Eventually, at high enough fields, since $\tau_M \sim 1/\mathcal{E}$, the mobility drops also as $\sim 1/\mathcal{E}$, and the drift velocity saturates.

A rough estimate of the saturation velocity can be obtained from the following simple argument. At high electric fields under steady-state conditions, the kinetic energy acquired from the electric field by a carrier in between two collisions must, on average, be entirely released at the next collision. If that were not the case, the average carrier energy would still be increasing and a steady-state situation would not have been reached. The maximum energy that a collision with the lattice can release, as discussed in Chapter 1, is the optical phonon energy, E_{opt}. In consequence, equating the kinetic energy acquired between collisions to E_{opt} yields (within a small numerical factor)

$$v_{\text{sat}} \simeq \sqrt{\frac{8}{3\pi}\frac{E_{\text{opt}}}{m_c^*}} \tag{4.107}$$

The factor of $8/3\pi$ takes care of the statistical distribution of the carrier velocities. This equation is derived under the assumption of an ideal parabolic band. At high energies, the bands of Si and GaAs are more complicated than this simple picture. In spite of that,

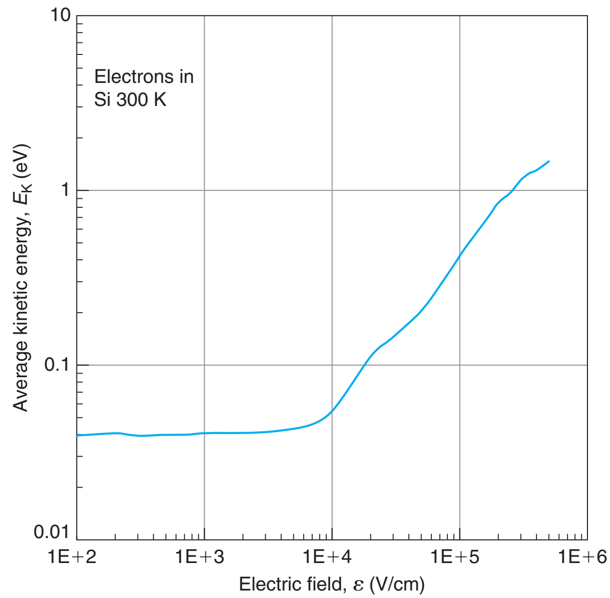


FIGURE 4.25
Average kinetic energy of electrons in uniform electric field in Si at room temperature as a function of the electric field [Data courtesy of M. Fischetti, IBM. © J.A. del Alamo].

Eq. (4.107) predicts values of v_{sat} for electrons and holes in Si and GaAs that are fairly close to the measured ones.

Equation (4.107) suggests that the saturation velocity is temperature independent. Experimentally, however, it is found that there is a weak but noticeable negative temperature dependence to it. The higher the temperature is, the lower will be the saturation velocity. The reason for this temperature dependence is not difficult to understand. The probability that an electron with energy higher than E_{opt} emits an optical phonon is not unity but it increases the higher the energy in excess of E_{opt}. At higher temperatures, the electron energy distibution is more spread out and there are proportionally more electrons at high energies. As a result, at high temperatures, the probability of phonon emission increases as E increases over E_{opt}, bringing down with it the saturation velocity.

AT4.2.3 Impact ionization

If a carrier that is sufficiently hot collides with the lattice, it can donate the kinetic energy that it has acquired from the electric field in the generation of an electron–hole pair. This is the process of impact ionization briefly discussed in Chapter 3. Impact ionization can be viewed in two ways, as a generation mechanism as in Chapter 3, or as a scattering mechanism that modifies the momentum and the energy of the impact ionizing carrier. For most purposes, we are interested in the first picture since the impact generated carriers contribute to the total

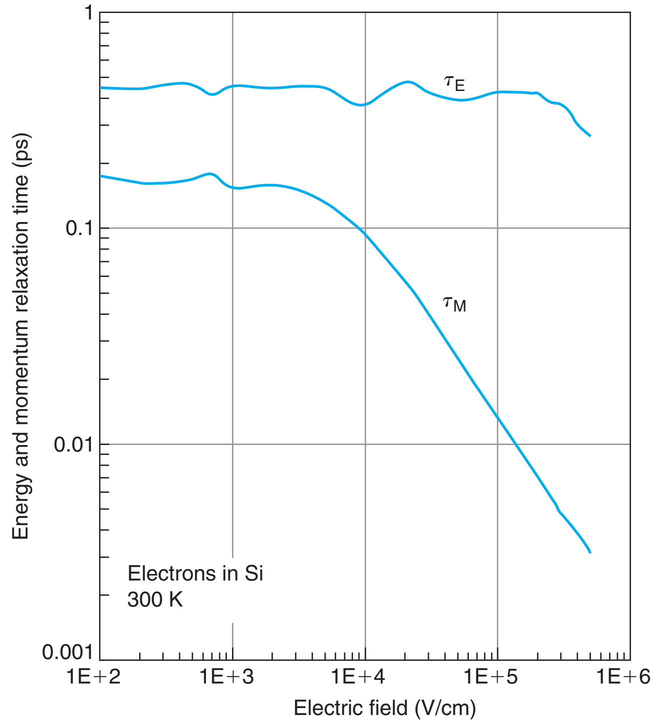


FIGURE 4.26
Energy and momentum relaxation times for electrons immersed in a uniform electric field in Si at room temperature as a function of electric field [Data courtesy of M. Fischetti, IBM. © J.A. del Alamo].

current which gets in consequence multiplied. The scattering picture is less relevant because phonon emission dominates the scattering rates of hot electrons at all energies but the highest ones.

Impact ionization is quantified by defining an *impact ionization coefficient*, α, as the average number of ionizations per unit length per carrier. It has the units of cm^{-1}. The inverse of the impact ionization coefficient can be thought of as the average distance traveled by a carrier in the field direction between two impact ionizing collisions. α is a strong function of the electric field. Figure 4.27 shows the room temperature impact ionization coefficients of electrons and holes in Si under a uniform electric field. This is a semilogarithmic plot as a function of the inverse electric field. This plot reveals that both α_e and α_h follow a law of the form:

$$\alpha \simeq A \exp\left(-\frac{B}{|\mathcal{E}|}\right) \tag{4.108}$$

with two constants A and B that depend slightly on the electric field. The values of A and B are given in Appendix E.

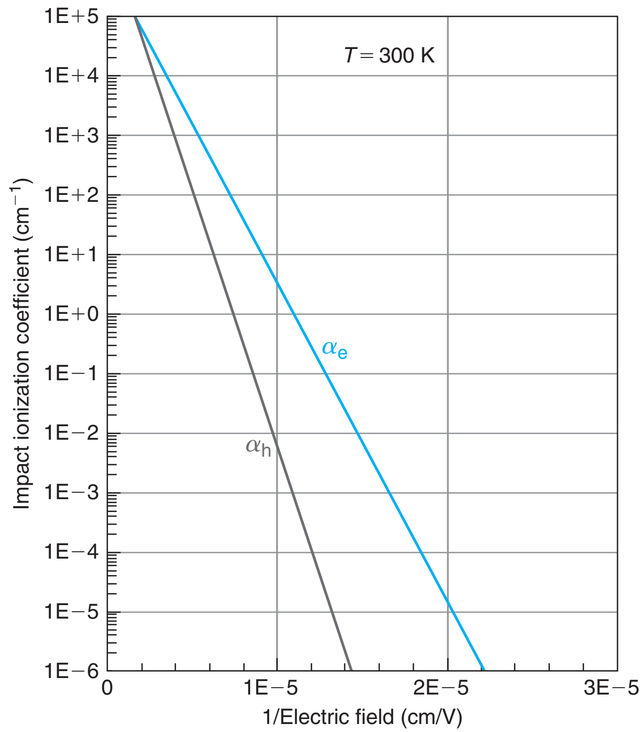


FIGURE 4.27
Impact ionization coefficient of electrons and holes as a function of electric field for Si at room temperature.

This functional relationship makes reasonable physical sense. In order for a carrier immersed in an electric field to produce an impact ionization event, it must first acquire sufficient energy from the electric field. This minimum value is known as the *impact ionization threshold energy*, E_{ii}. For a carrier to acquire this energy, it must at least travel a distance $l_{ii} = E_{ii}/q\mathcal{E}$ inside the electric field. The likelihood of a carrier traveling this distance without undergoing a single collision is $P(l_{ii}) = \exp(-l_{ii}/l_c)$, where l_c is the mean free path. In consequence, the ionization rate should go as

$$\alpha \simeq \frac{1}{l_{ii}} \exp\left(-\frac{l_{ii}}{l_c}\right) = \frac{q|\mathcal{E}|}{E_{ii}} \exp\left(-\frac{E_{ii}}{q|\mathcal{E}|l_c}\right) \tag{4.109}$$

This equation, in spite of the simplifying assumptions that were made in its derivation, captures several key dependencies that are observed in experiments. First, it reveals the inverse exponential dependence on the electric field. Second, it shows an exponential dependence, with a minus sign, on the threshold energy E_{ii}. This is consistent with the observation that semiconductors with a wide bandgap and a higher value of E_{ii} exhibit lower values of α. Finally, the dependence on the mean free path in the expression is consistent with the observation that the impact ionization coefficient decreases with temperature. The higher the temperature, the shorter the mean free path and the harder it is for a carrier to acquire the required threshold energy.

The proper value of E_{ii} and its relationship with the bandgap of the semiconductor has been a matter of discussion among researchers in the field. This is not merely an academic issue. For a long time, it was believed that $E_{ii} \sim 1.5E_g$. This suggested that using voltages below about 1.5 V or so in scaled-down Si MOSFETs would completely eliminate impact ionization which is a major reliability concern. Recently, impact ionization has been observed all the way down to sub-bandgap voltages. The reason is two-fold. On the one hand, the carrier distribution has a high-energy tail. On the other hand, it is possible for the impact ionization event to coincide with a phonon collision. In this way, E_{ii} can be as low as E_g. The way to view E_{ii} is not so much as a hard threshold below which impact ionization is impossible, but a pragmatic threshold energy above which impact ionization becomes significant. Consistent with this approach, a value of E_{ii} for Si equal to the bandgap is a sensible choice.

Impact ionization is a generation mechanism. When carriers drift in an electric field, some of the energy that they acquire from the electric field can be spent in generating electron–hole pairs. Naturally, the generation rate must depend on the carrier drift flux. In fact, it should be linear in the flux, that is, the higher the flux of carriers that drift in an electric field, the higher the generation rate due to impact ionization.

The relationship between the generation rate (in units of $\mathrm{cm^{-3} \cdot s^{-1}}$) and drift flux (with units of $\mathrm{cm^{-2} \cdot s^{-1}}$) is easy to derive. Let us do it for electrons. Looking back at Fig. 4.6, in an elemental time dt, electrons drift a distance $v_e^{drift} dt$, where v_e^{drift} is the electron drift velocity. The probability that an electron undergoes an impact ionization event in this distance is $\alpha_e v_e^{drift} dt$. If the concentration of electrons is n, the total number of generation events in time dt is $n\alpha_e v_e^{drift} dt$. Per unit time, then, the generation rate is simply $n\alpha_e v_e^{drift} = \alpha_e F_e(drift)$. Using similar arguments, it is easy to derive an expression for the generation rate due to impact ionization of holes. In both cases, the generation rate must be a positive number, regardless of the sign of the drift flux. The total generation rate due to impact ionization is then

$$G_{ii} = \alpha_e |F_e(\text{drift})| + \alpha_h |F_h(\text{drift})| \tag{4.110}$$

where the absolute symbols are used to insure that G_{ii} is always positive independently of the choice that is made for the axis.

Equation (4.110) is the basis for the treatment of carrier multiplication and avalanche breakdown that is presented in Chapter 5.

PROBLEMS

*Denotes a problem that requires studying an Advanced Topic

4.1 A device design engineer in a group in charge of developing a very aggressive Si npn bipolar process becomes concerned with the suitability of the company's device simulation tools to do the job. She argues that in an effort to push the frequency response of the transistor, the base might become so thin, that perhaps carriers could cross it without significant scattering. In this case, a device CAD tool that incorporates "ballistic" transport would be required. The group is considering using p-bases with a doping level of about 10^{18} cm^{-3}. The tentative base thickness is 50 nm.

Address the engineer's concerns by:

(a) estimating the mean free path of electrons across the base, and comparing it with the base width;
(b) estimating the ratio of the transit time for electron diffusion across the base (given by Eq. (4.45)) over the electron scattering time in the base.

4.2 Three questions regarding high-resistivity Si in thermal equilibrium at room temperature:

(a) Calculate the conductivity of intrinsic Si.
(b) Calculate the sheet resistance of a 600-μm thick intrinsic Si wafer.[5]
(c) Does intrinsic Si exhibit the lowest conductivity that Si can attain at room temperature? If not, describe the situation that leads to the lowest possible conductivity, derive an expression for it, and calculate its value.

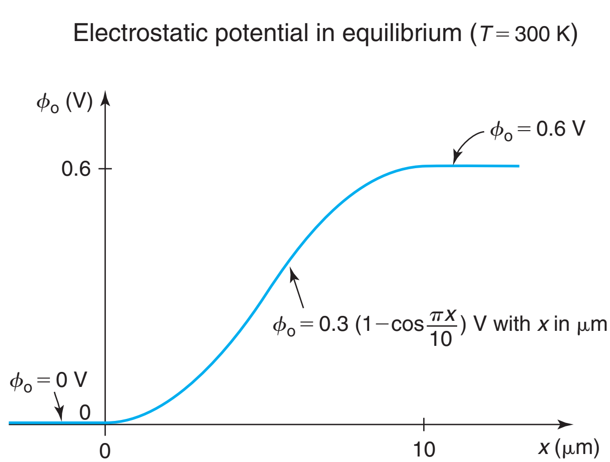
4.3 Consider a semiconductor in thermal equilibrium with a nonuniform doping level given by the functional form:

$$N_D(x) = N_{Do} + \frac{\Delta N_D}{1 + \exp \frac{x}{L_c}}$$

with $\Delta N_D < N_{Do}$.

(a) Sketch the doping distribution.
(b) Assuming quasi-neutrality, derive an expression for the electrostatic potential $\phi_o(x)$. Select as origin of potentials the point at which $N_D(x) = N_{Do}$. Sketch the potential distribution.
(c) Discuss all conditions that L_c, N_{Do}, and ΔN_D must satisfy for the given doping distribution to be considered quasi-neutral.

4.4 You are involved with an overseas team on the joint design of a Si photodiode. You often exchange information back and forth via fax. The latest fax that you got from your partners was all messed up. Only the diagram reproduced below was clearly received.



[5]Do this part after studying Section 5.5.

It is now after-hours overseas and you do not have access to your partners until tomorrow morning. You have to get as much information as you can from this diagram so as to continue working on the project. You recall that in an earlier exchange it had been decided that the left side of the device ($x \leq 0$ in the diagram above) was to have a doping level $N_A = 10^{15}$ cm^{-3}.

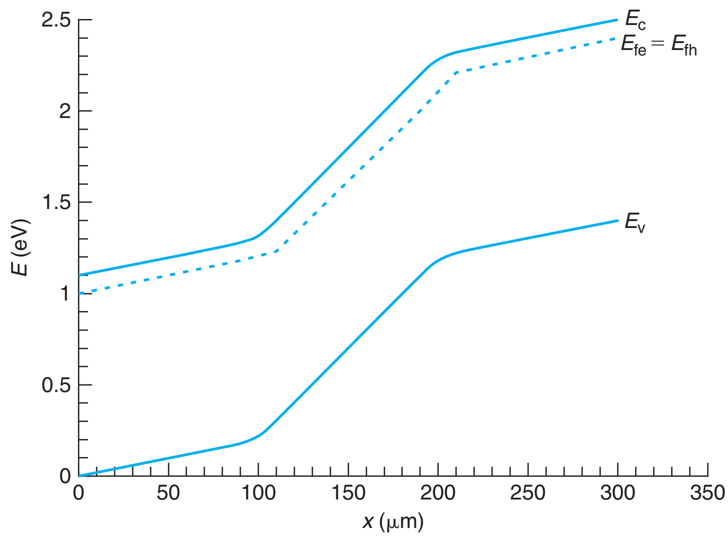
(a) Derive an algebraic expression for and *quantitatively* sketch the electric field across the structure.

(b) What can you say about the doping level and type on the right-hand side of the device (i.e., for $x \geq 10$ μm)?

(c) Is this a quasi-neutral situation? Explain.

(d) *Quantitatively* sketch the energy band diagram throughout the structure.

4.5 A Si bar at room temperature in steady state has the energy band diagram sketched below.



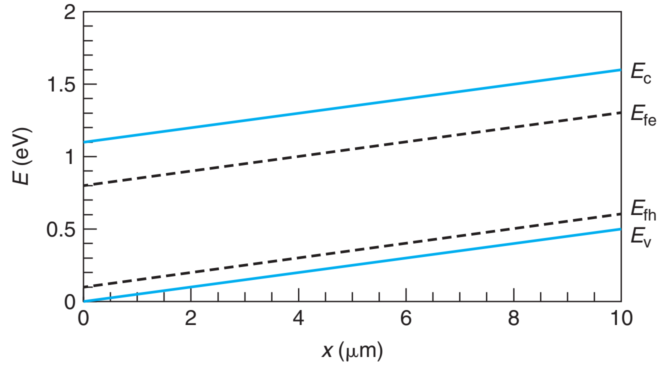
(a) Select all the terms that apply to the situation:
- thermal equilibrium/out of equilibrium
- uniformly doped/nonuniformly doped
- n-type/p-type
- degenerate/nondegenerate
- low-level injection/high-level injection/extraction
- intrinsic/extrinsic

Explain your choice.

(b) *Quantitatively* sketch the electric field, electrostatic potential, electron concentration, and hole concentration as a function of x.

(c) *Quantitatively* sketch the electron current density as a function of x.

4.6 A Si bar at room temperature in steady state has the band diagram indicated below where $E_{fh} - E_v \simeq E_F - E_v$.



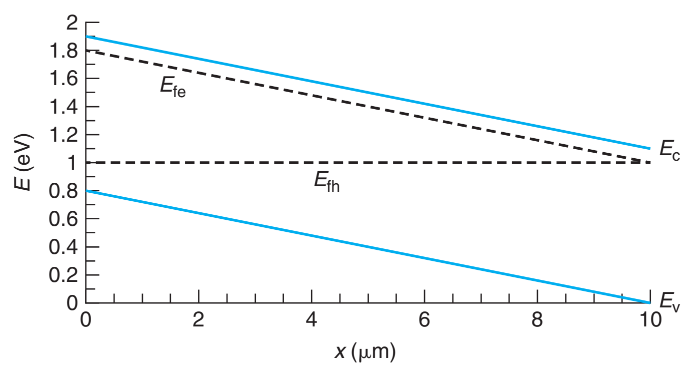
(a) Circle all the terms that apply to the situation:
   - thermal equilibrium/out of equilibrium
   - uniformly doped/nonuniformly doped
   - n-type/p-type
   - degenerate/nondegenerate
   - low-level injection/high-level injection/extraction
   - intrinsic/extrinsic

   Explain your choice.

(b) *Quantitatively* sketch the electric field, electrostatic potential, electron concentration, and hole concentration as a function of x.

(c) *Quantitatively* sketch the electron and hole current densities as a function of x.

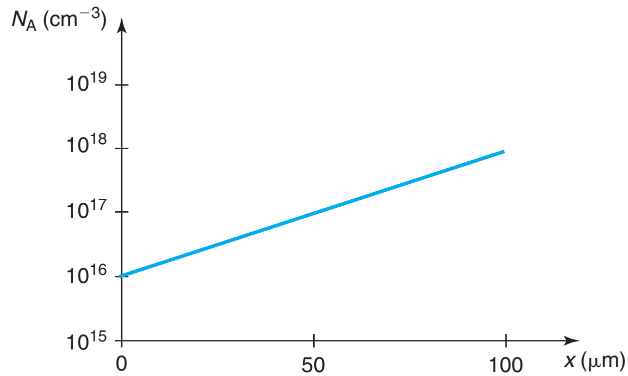
4.7 A Si bar at room temperature in steady state has the band diagram indicated below where $E_c - E_{fe} \simeq E_c - E_F$.



(a) Circle all the terms that apply to the situation:
   - thermal equilibrium/out of equilibrium
   - uniformly doped/nonuniformly doped
   - n-type/p-type
   - degenerate/nondegenerate
   - low-level injection/high-level injection/extraction
   - intrinsic/extrinsic

(b) *Quantitatively* sketch the electric field, electrostatic potential, electron concentration, and hole concentration as a function of x.
(c) *Quantitatively* sketch the net recombination rate as a function of x.

4.8 A p-type Si sample has a doping distribution in space shown in the figure below. The sample is in thermal equilibrium and at room temperature.

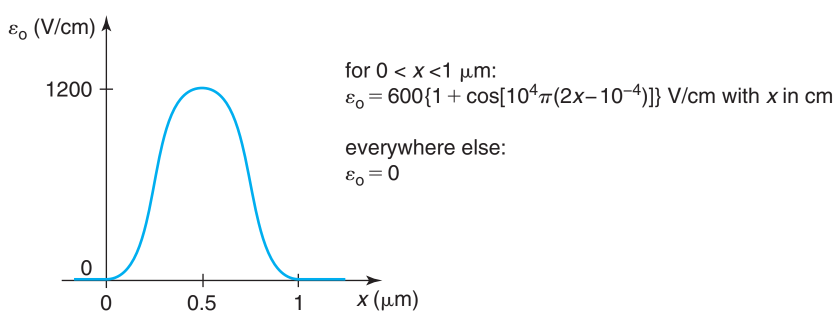


Assuming *quasi-neutrality*, derive algebraic expressions for and quantitatively sketch throughout the whole sample the spatial distributions of:
(a) the hole concentration;
(b) the electron concentration;
(c) the electric field;
(d) the electrostatic potential;
(e) the volume charge density in the portion of the sample under study; and
(f) the energy band diagram, showing the location of the Fermi level.

Calculate the Debye length and discuss whether the quasi-neutrality assumption is warranted. Disregard surface effects.

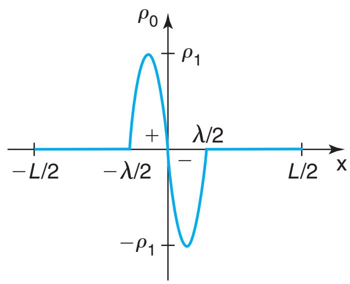
4.9* Following a similar procedure to the one outlined in Section 4.3.2, derive a general Einstein relation for electrons that is valid for all doping levels. Use the properties of the Fermi–Dirac integral of Section AT2.2. Graph your result as a function of the electron concentration. Derive analytical relations for the low- and high-electron concentration limits.

4.10 In a certain region of an n-type Si sample in thermal equilibrium at room temperature, there is an electric field distribution in space as sketched below.



(a) Derive an analytical expression and quantitatively sketch the electrostatic potential distribution in this region.
(b) Derive an analytical expression and quantitatively sketch the net volume charge density in this region.
(c) If you are told that the equilibrium electron concentration at $x = 0$ is $n_o = 10^{17}\ \text{cm}^{-3}$, compute the equilibrium electron concentration at $x = 1\ \mu\text{m}$.
(d) Can this region be considered quasi-neutral? Substantiate your answer with suitable numerical calculation.

4.11 Consider an $L = 10\ \mu\text{m}$ long bar of Si in thermal equilibrium. In the central $\lambda = 4\ \mu\text{m}$ of this bar, there is a dipole of charge with $\rho_1 = 1 \times 10^{-10}\ \text{C/cm}^3$, as sketched in the diagram below.



An analytical description of the lobule on the right is

$$\rho_o = \rho_1(x^2 - 2x) \qquad \text{for } 0 \le x \le 2$$

$$\rho_o = 0 \qquad \text{for } x \ge 2$$

with x in μm.

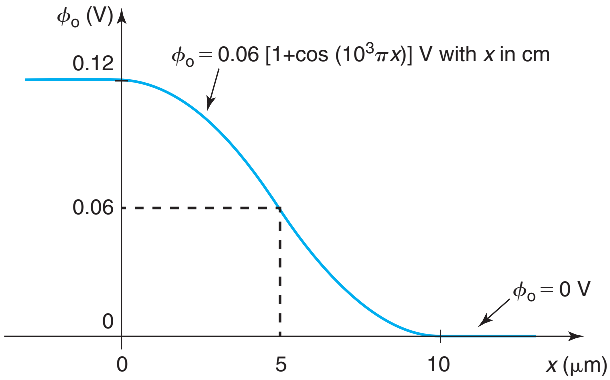
(a) Derive an expression for the electric field for $x \ge 0$. Quantitatively sketch across the entire bar.
(b) Derive an expression for the electrostatic potential for $x \ge 0$. Use as reference for potential the location at $x = 0$, that is $\phi_o(x = 0) = 0$. Quantitatively sketch across the entire bar.
(c) Calculate the ratio of the electron concentration at $x = -L/2$ and $x = L/2$.
(d) If the average electron concentration in this sample is $10^{16}\ \text{cm}^{-3}$, can this sample be considered quasi-neutral? Provide a quantitative explanation.

4.12 In a certain n-type region of a semiconductor in thermal equilibrium, there is a hole concentration with the following spatial distribution:

$$p_o(x) = 10^3(1 - 9 \times 10^3 x)\ \text{cm}^{-3} \qquad \text{for } 0 \le x \le 10^{-4} \text{ with } x \text{ in cm}$$

Assume that in this region, the electron mobility and hole mobilities are $\mu_e = 500\ \text{cm}^2/\text{V}\cdot\text{s}$ and $\mu_h = 200\ \text{cm}^2/\text{V}\cdot\text{s}$, respectively.
(a) Derive an expression for and sketch the hole diffusion current density in this region.
(b) Derive an expression for and sketch the hole diffusion velocity in this region.
(c) Derive an expression for and sketch the electric field distribution in this region.
(d) Derive an expression for and sketch the hole drift velocity in this region.
(e) Derive an expression for and sketch the electron concentration in this region.
(f) Derive an expression for and sketch the electron diffusion current density in this region.
(g) Derive an expression for and sketch the electron diffusion velocity in this region.

4.13 Consider an n-type Si region in thermal equilibrium at 300 K with an electrostatic potential distribution as sketched below. The electron concentration at $x = 0$ is $n_o(x = 0) = 10^{17}$ cm^{-3}.

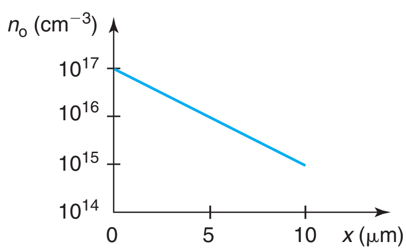


$x \le 0$	$\phi_o(x) = 0.12$ V
$0 \le x \le 10$ μm	$\phi_o(x) = 0.06[1 + \cos(10^3\pi x)]$ V with x in cm
$x > 10$ μm	$\phi_o(x) = 0$

(a) Derive an expression for the electric field, $\mathcal{E}_o$, throughout the sample. Sketch to scale.
(b) Calculate n_o and p_o at $x = 10$ μm.
(c) Calculate an expression for the net volume charge density, ρ_o, throughout the sample. Sketch to scale.
(d) Can this region be considered quasi-neutral? Substantiate your answer by identifying the worst-case point and evaluating the quasi-neutrality condition at that point.
(e) Sketch the energy band diagram across the entire region.

Note that the x-axis in the figure is in μm.

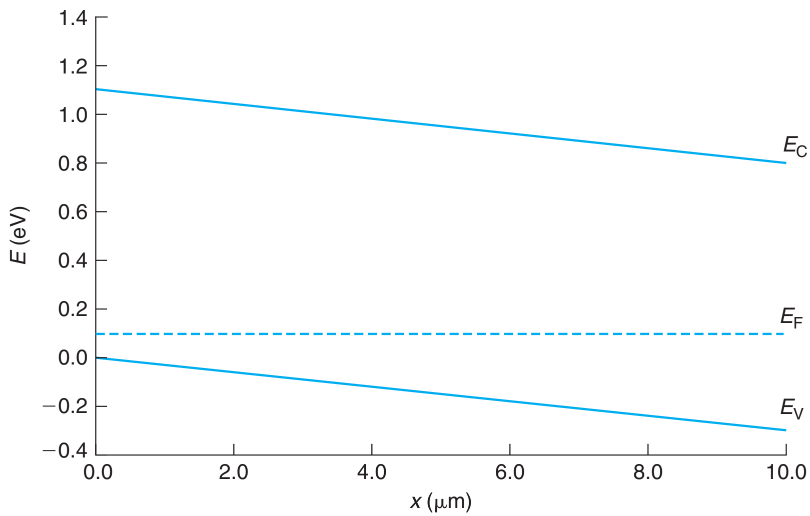
4.14 Consider a piece of n-type Si in thermal equilibrium at room temperature. In a region defined by $0 \le x$ (μm) ≤ 10, there is a spatially varying electron concentration as sketched below.



$n_o(x) = 10^{17} \times 10^{-2\times 10^3 x}$ cm^{-3} with x in cm

(a) Derive an analytical equation for the minority carrier concentration in space. Quantitatively sketch the result in a suitable diagram.
(b) Derive an analytical equation for the electrostatic potential in space. Quantitatively sketch the result in a suitable diagram.
(c) Derive an analytical equation for the electric field in space. Quantitatively sketch the result in a suitable diagram.
(d) Derive an analytical expression for the charge distribution that supports this electric field. Quantitatively sketch the result in a suitable diagram. Can this sample be considered quasi-neutral?

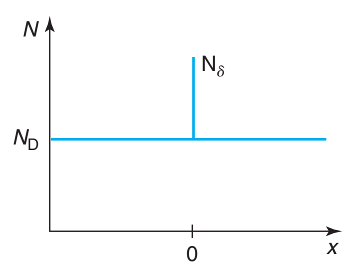
4.15 The energy band diagram below corresponds to a region of a Si bar at room temperature.



(a) Circle all the terms that apply to the situation:
- thermal equilibrium/out of equilibrium
- uniformly doped/nonuniformly doped
- n-type/p-type
- degenerate/nondegenerate
- low-level injection/high-level injection/extraction
- intrinsic/extrinsic

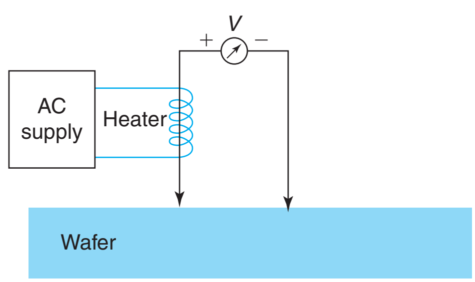
(b) Estimate the hole current density at $x = 5\ \mu\text{m}$.
(c) Estimate the electric field at $x = 5\ \mu\text{m}$.
(d) Estimate the hole diffusion flux at $x = 5\ \mu\text{m}$.

4.16 Consider an n-type semiconductor sample with a uniform doped concentration N_D (per unit volume). Inserted in this sample at $x = 0$ is a sheet of *acceptors* with a concentration N_δ (per unit area) as sketched in the figure.



(a) Qualitatively sketch the equilibrium electron concentration in space. Explain your result and state any assumptions that you make.
(b) Qualitatively sketch the volume charge density in space. Explain your result and state any assumptions that you make.
(c) Qualitatively sketch the electric field distribution in space. Explain your result and state any assumptions that you make.
(d) Qualitatively sketch the electrostatic potential distribution in space. Select as reference for potentials the potential that correspond to $n_o = N_D$. Explain your result and state any assumptions that you make.
(e) Qualitatively sketch the energy band diagram across the structure. Make sure to note the location of the Fermi level. Explain your result and state any assumptions that you make.
(f) Formulate a differential equation for the equilibrium electron concentration in space. Explain.
(g) Write a suitable boundary condition at $x = 0^+$ (just outside the delta-doped region toward the right) that you would need in order to solve this differential equation. This boundary condition must be stated in terms of $n_o(x = 0^+)$ or its derivatives at that location.

4.17 The **hot point probe** is a technique to evaluate the conductivity type of a wafer (p or n). As sketched in the diagram below, it consists of two metallic probes connected to a voltmeter. One of the probes is heated by a power supply. When both probes are brought into contact with a wafer, a voltage appears across them. The sign of the voltage indicates the polarity of the semiconductor.



With the sign convention indicated in the diagram (hot probe positive with respect to room temperature probe), what is the expected sign of the voltage when the technique is applied to an n-type sample? Explain the detailed physical origin of this voltage, that is, what is the nature of the charge that supports this voltage?

Chapter 5

Carrier Flow

In the preceding chapters, we have learned about a variety of physical mechanisms that take place in semiconductors: generation, recombination, drift, and diffusion. In a semiconductor device under operation, several of these mechanisms can be in action simultaneously. For example, in the base of a bipolar transistor, minority carriers that are injected from the emitter diffuse toward the collector. Some of them may recombine in the base. If there is an electric field, minority carrier drift can also be significant through the base. A computation of the collector current of the transistor demands a correct description of all these processes and their interactions.

This chapter starts by formulating the set of equations that describe carrier behavior in semiconductors and the various boundary conditions that can be encountered. The governing system of equations is rather complex and analytical solutions are generally not obtainable. Computer-aided design (CAD) tools have been developed to solve these equations in semiconductor devices.

For the purpose of understanding device operation and for developing simple models that can be used in design, it is essential to simplify these equations as much as possible. Fortunately, this is feasible in many practical situations. We will distinguish three broad classes of situations: majority-carrier type, minority-carrier type, and space-charge problems. In the first kind, majority carrier drift, and diffusion constitute the dominant phenomena. This is what happens, for example, in the source of a MOSFET. In the second kind, minority carrier behavior is the bottleneck. This is the situation in the base of a bipolar transistor. These two families of problems occur in quasi-neutral regions. In space-charge regions, such as in the depletion region of a p-n junction, the physics of carrier transport is somehow different and deserves special attention.

Minority-carrier type problems can be mathematically difficult, particularly in dynamic situations. This is because of the subtle interplay of carrier diffusion and drift, bulk and surface generation, or recombination. In many circumstances, device engineers do not need a complete solution to the problem. Identifying the limiting phenomena suffices to diagnose the situation and articulate a course of action. In this chapter, we solve a number of classic problems and use them to identify the key length scales and timescales for carrier behavior. Learning to do this correctly is a very valuable skill that is helpful in the design and analysis of microelectronic devices.

This chapter is, in some ways, about the management of complexity. In front of a given situation, the entire set of equations that completely describes carrier phenomena is usually a blunt instrument. It is often far more effective to analyze the physics of the problem, identify the bottlenecks, extract the essential terms from the equations, and discard all the rest. The simplified set is more likely to lead to a simple, analytical, and physically intuitive solution of great practical value. The final step in this process is to verify the assumptions that were initially made to simplify the problem. This approach is fairly common in engineering. It is crucial in effective microelectronics device engineering. It will be discussed and illustrated in detail in this chapter in a few important situations.

5.1 CONTINUITY EQUATIONS

In Chapter 3, we learned that in a certain region of a semiconductor, the carrier concentrations outside thermal equilibrium are determined by the imbalance between the rates of carrier generation and recombination in that region. Based on this understanding, we derived differential equations that allowed us to compute the time evolution of carrier concentrations in situations outside equilibrium [see Eq. (3.61)]. This picture is actually incomplete. It only applies to situations that are uniform, that is, nothing changes in space. In more general situations, in addition to generation and recombination, there is a need to account for carrier flow in and out of the region of interest. The continuity equations that we derive in this section mathematically capture this simple fact.

A continuity equation is in essence a "book-keeping relationship" that keeps track of the number of particles in a region of a semiconductor as a function of time. Consider an elemental volume of a semiconductor ΔV, such as sketched in Fig. 5.1. The number of electrons, say, in this little volume increases with time if there is generation taking place inside it. Analogously, recombination reduces the electron count. Additionally, if there is a net flow of electrons out of that volume, whether by drift or diffusion, this also reduces the number of electrons in ΔV.

We can mathematically capture this by focusing on the *rates* at which these processes take place. Specifically, the rate of increase in the number of electrons in ΔV is equal to the rate of electron generation in ΔV minus the rate of recombination in ΔV minus the net flow of electrons leaving ΔV per unit time. If ΔV is small enough, this can be written mathematically as

$$\frac{\partial(n\Delta V)}{\partial t} = G\Delta V - R\Delta V - \int \vec{F}_e \cdot d\vec{S} \tag{5.1}$$

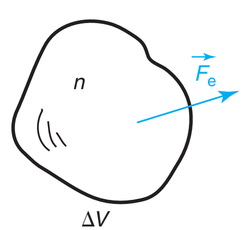


FIGURE 5.1
Sketch of small semiconductor volume ΔV with a uniform electron concentration n inside. An electron flux $\vec{F}_e$ crosses the surface of ΔV at a certain point.

where the integral on the right-hand side sums all outgoing flux through the surface of ΔV. Dividing all terms by ΔV and taking the limit for ΔV very small, we get

$$\frac{\partial n}{\partial t} = G - R - \vec{\nabla} \cdot \vec{F}_{\mathrm{e}} \tag{5.2}$$

This is the continuity equation for electrons. Following similar arguments, an identical equation is obtained for holes.

Since there is a direct relationship between carrier flux and current density [Eqs. (4.15) and (4.16)], Eq. (5.2) can easily be rewritten as

$$\frac{\partial n}{\partial t} = G - R + \frac{1}{q}\vec{\nabla} \cdot \vec{J}_{\mathrm{e}} \tag{5.3}$$

for electrons. For holes, we get

$$\frac{\partial p}{\partial t} = G - R - \frac{1}{q}\vec{\nabla} \cdot \vec{J}_{\mathrm{h}} \tag{5.4}$$

These are the most fundamental expressions of the continuity equations for electrons and holes.

It is also of interest to focus on *continuity of charge*. We saw in Chapter 4 that in a semiconductor, net volume charge density results if there is an imbalance between the total concentrations of positive species (holes and donors) and negative species (electrons and acceptors) [Eq. (4.51)]. Since the impurity concentrations cannot change in time, the rate of change of volume charge density is given by the difference of the rate of change of hole and electron concentrations multiplied by the elemental charge:

$$\frac{\partial \rho}{\partial t} = q\frac{\partial (p - n)}{\partial t} \tag{5.5}$$

This observation allows us to derive a continuity equation for charge density. We multiply Eqs. (5.3) and (5.4) by q, we subtract one from the other, and then substitute the result into Eq. (5.5) to get

$$\frac{\partial \rho}{\partial t} = -\vec{\nabla} \cdot \vec{J}_{\mathrm{t}} \tag{5.6}$$

where J_{t} is the total current (sum of electron and hole current).

This equation states that if the volume charge density is changing with time in a region of a semiconductor, it is because there is net current flowing in or out of that region. For example, if the charge density at a certain location is increasing, this means that current is flowing into that region (negative divergence).

The continuity equation for charge is very useful to analyze a variety of important situations in semiconductors. A particularly relevant one that merits being discussed here is a fully *static situation* (though not necessarily in thermal equilibrium). This situation can result, for example, if there is net carrier generation in a region of a semiconductor as a result of external illumination that is steady in time. In a case like this in which nothing changes in time, there are no sinks or sources of net charge and at any location the divergence of the total current must be zero everywhere.

The continuity equations derived here are particularly intuitive when written in integral form. This is shown in Appendix AT5.1.

5.2 SURFACE CONTINUITY EQUATIONS

A problem is not completely defined until its boundary conditions are specified. In semiconductor devices, boundary conditions often play a critical role in device operation. Many times "boundary condition engineering" is the most effective way to achieve certain device design goals. For the purpose of understanding carrier flow in semiconductor devices, we need to specify the appropriate boundary conditions that apply.

In describing a semiconductor device, the condition of all its surfaces must be specified. We are not equipped at this time to treat semiconductor surfaces rigorously. We will study the electrostatics of surfaces in detail later on in this book. Chapter 7, for example, deals with the metal–semiconductor interface and Chapter 8 discusses the insulator–semiconductor interface. For situations with excess carriers, we can make substantial progress in a number of important problems if we treat the surfaces in an empirical way. This is the approach followed in this section in which surface continuity equations are derived.

We discuss here the two simplest kinds of surfaces. First, we study a "free" surface, one in which the semiconductor is exposed or covered by a nonconductive material. Then we deal with "ohmic contacts" in which a metal is deposited on the surface. The fundamental difference between these two surfaces is that in the case of the free surface, carriers cannot escape from the semiconductor, while in the case of the ohmic contact, conduction is allowed between the metal and the semiconductor. The detailed physics of these two types of surfaces is fairly complex and a detailed analysis is left for later chapters.

5.2.1 Free surface

We denote a free surface as the boundary region of a semiconductor that is not electrically connected to anything else. As we will study in Chapter 8, at a surface, there is an energy barrier that prevents carriers from "spilling out" of the semiconductor. A consequence of this is that the current normal to a free surface must be zero:

$$J_{ts} = 0 \tag{5.7}$$

This is illustrated in Fig. 5.2.

While the total electrical current is zero at a free surface, the electron and hole currents need not be zero separately. As discussed in Chapter 4, if the surface is not properly passivated, there

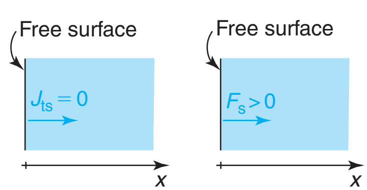


FIGURE 5.2
At a free surface, carriers cannot spill out and the net electrical current must be zero (left). This does not mean that the carrier flux is zero at the surface. If there is an imbalance between surface generation and recombination, then, net carrier flux can emerge from the surface into the semiconductor (right).

can be significant recombination and generation associated with surface states and traps. We can characterize this by defining a surface recombination rate R_s and a surface generation rate G_s.

If there is net generation or recombination at a surface, there must be a flow of carriers into or out of it that come from or go to the rest of the semiconductor. The total current at the surface is zero but the electron and hole components, separately, are not. Similar to the continuity equations derived for the bulk of a semiconductor in Section 5.2, we can write a continuity equation for a surface for each type of carrier. A fundamental difference with the bulk case is that a surface cannot store carriers since it has no volume. Particle conservation then demands that the rate of surface generation minus the rate of surface recombination, or *net surface generation rate*, equals the carrier flux out of the surface. Mathematically

$$G_s - R_s = F_s \tag{5.8}$$

The units of all terms in this equation are $\text{cm}^{-2} \cdot \text{s}^{-1}$. Note that this equation is axis sensitive. It applies for the choice of axis depicted in Fig. 5.2, but the sign of F_s is negative if the axis direction is reversed.

We can rewrite Eq. (5.8) in a slightly more useful way. First, we can express the right-hand side in terms of current density. For this, we use the results obtained in Chapter 4 in Eqs. (4.15) and (4.16). Second, the left-hand side is equal to the negative of the net recombination rate at the surface, U_s, that was introduced in Chapter 3. With these two changes, Eq. (5.8) becomes

$$U_s = \frac{1}{q} J_{es} = -\frac{1}{q} J_{hs} \tag{5.9}$$

J_{es} and J_{hs} are the current densities normal to the surface. They are identical in magnitude because electrons and holes are generated and recombine in pairs. These equations are also axis sensitive.

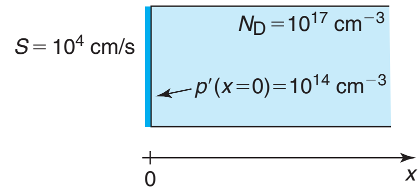
In some problems, it is useful to introduce an *external* surface generation in which electron–hole pairs are generated right at the surface. This is of course a rather unphysical occurrence. However, depending on the length scale of the problem, it can be a handy simplifying approximation for some situations. Very short wavelength light, for example, is absorbed very effectively in a semiconductor. If all length scales of the problem of interest are much longer than the absorption length of the light, for all practical purposes, the generation function can be assumed to be confined to a thin sheet at the surface. Including an external surface generation rate, Eq. (5.9) becomes

$$U_s - G_s(\text{ext}) = \frac{1}{q} J_{es} = -\frac{1}{q} J_{hs} \tag{5.10}$$

The units of $G_s(\text{ext})$ are also $\text{cm}^{-2} \cdot \text{s}^{-1}$. Once again, this equation is axis sensitive.

EXERCISE 5.1

Consider the surface of an n-type Si sample with $N_D = 10^{17}\ \text{cm}^{-3}$ *at 300* K, *as sketched below. This surface is characterized by a surface recombination velocity* $S = 10^4\ \text{cm/s}$. *At the surface, there is an excess hole concentration* $p'(x = 0) = 10^{14}\ \text{cm}^{-3}$.



Under these conditions, calculate: (a) the net recombination rate at the surface, (b) the hole flux at the surface, (c) the electron flux at the surface, (d) the hole current density at the surface, (e) the electron current density at the surface, and (f) the total current density at the surface.

(a) Since the semiconductor at the surface is under low-level injection conditions, the net recombination rate at the surface can be calculated using Eq. (3.72):

$$U_s = p'S = 10^{14} \times 10^4 = 10^{18}\ \text{cm}^{-2} \cdot \text{s}^{-1}$$

(b) The magnitude of the hole flux at the surface is equal to the net recombination rate. Since the holes are flowing into the surface to recombine (against the choice of x in the figure above), the hole flux is negative. Then

$$F_h(x=0) = -U_s = -10^{18}\ \text{cm}^{-2} \cdot \text{s}^{-1}$$

(c) The electron flux at the surface is identical to the hole flux (every hole recombines with one electron):

$$F_e(x=0) = -10^{18}\ \text{cm}^{-2} \cdot \text{s}^{-1}$$

It is also negative because the electrons are also flowing into the surface.

(d) The hole current density at the surface is simply

$$J_h(x=0) = qF_h(x=0) = -1.6 \times 10^{-19} \times 10^{18} = -0.16\ \text{A/cm}^2$$

(e) Similarly, the electron current density at the surface is

$$J_e(x=0) = -qF_e(x=0) = 1.6 \times 10^{-19} \times 10^{18} = 0.16\ \text{A/cm}^2$$

(f) The total current density at the surface is the sum of the electron and hole current densities:

$$J_t(x=0) = J_e(x=0) + J_h(x=0) = 0$$

This result makes sense since there cannot be net current into a free surface.

5.2.2 Ohmic contact

Implicit in the derivation of the surface continuity equations above is the constraint that carriers cannot "fall out" of the surface, nor be added somehow to the surface from the outside. This assumption holds for a free surface. However, in just about all semiconductor devices, there

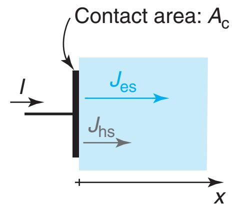


FIGURE 5.3
Sketch of ohmic contact between a metal and a semiconductor with current flowing through. The area of contact between the metal and the semiconductor is A_c.

are some surfaces that are contacted with metal so as to provide an electrical connection to the semiconductor from the outside world. These are called *ohmic contacts*.

The physics of the metal–semiconductor junction will be discussed in detail in Chapter 7, so fundamental aspects of the operation of ohmic contacts have to be left out for later on. At this time, we will have to accept some of its properties on faith until a proper justification is provided. Our goal here is to be able to write suitable continuity equations for ohmic contacts.

Consider the sketch of Fig. 5.3. It depicts a semiconductor surface covered by a metal forming an ohmic contact. The metal surface is connected to a wire through which a current I flows. The current must be continuous on the semiconductor side where in general we expect it to be supported by the flow of electrons and holes. We therefore define two current densities right at the surface on the semiconductor side one for electrons, J_{es}, and another one for holes, J_{hs}. The ohmic contact has an area A_c.

Kirchoff's law applies at an ohmic contact. Current continuity demands that the current through the wire be identical to the sum of the electron and hole currents on the semiconductor side. Mathematically,

$$I = A_c(J_{es} + J_{hs}) = qA_c(F_{hs} - F_{es}) \tag{5.11}$$

F_{es} and F_{hs} are the carrier fluxes at the semiconductor side of the metal–semiconductor interface. As before, this equation is axis sensitive. Note that there is an established sign convention for the current at a device contact. The current is considered positive if entering into the device and negative if coming out of the device. This is the convention followed in this book.

In addition to Eq. (5.11), there are a number of statements that can be made about ohmic contacts that are appropriate to list here. The reason for these will be understood when we study the metal–semiconductor junction in Chapter 7.

- First, in ohmic contacts in the absence of net generation or recombination, the metal only communicates with the *majority carriers* in the semiconductor. This means that if there is current through the wire, this current is supported entirely by *majority carrier current* in the semiconductor. This is illustrated in Fig. 5.4. Mathematically, for n-type material

$$I = A_c J_{es} = -qA_c F_{es} \tag{5.12}$$

  An equivalent equation applies for p-type material.
- If there is net generation or recombination at an ohmic contact (i.e., at the metal–semiconductor interface), this results in a *minority carrier current*. Since

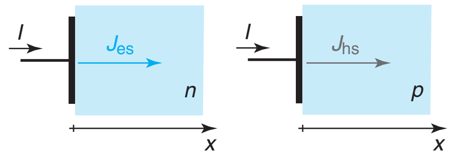


FIGURE 5.4
In the absence of net generation or recombination at an ohmic contact, terminal current is entirely supported by the majority carriers in the semiconductor (n-type on left, p-type on right).

electrons and holes are generated and recombine in pairs, this also produces an additional component to the majority carrier current of equal magnitude and contrary sign. This current component adds up to the one imposed from the outside by the wire, as mentioned above. Mathematically, for n-type material, the minority carrier current at the surface is equal to the net surface recombination rate:

$$U_s = -F_{hs} = -\frac{1}{q}J_{hs} \tag{5.13}$$

This equation applies for the choice of axis made in Fig. 5.3 or 5.4 since in the case of net recombination ($U_s > 0$), holes flow into the surface, against the x-axis. An equivalent equation applies for p-type material.

The new minority carrier surface current must be accounted for in the balance of currents. If we insert Eq. (5.13) into the general current continuity expression of Eq. (5.11), we have

$$I = A_c(J_{es} - qU_s) \tag{5.14}$$

- Finally, because of the intimate contact between the metal and the semiconductor, an ohmic contact produces a surface with *infinite surface recombination velocity*. As a result, at an ohmic contact, the minority carrier concentration is equal to its equilibrium value. The detailed reason for this will be better understood in Chapter 7. This is an important boundary condition that we will frequently encounter in this book. Mathematically,

$$n'_s = p'_s = 0 \tag{5.15}$$

To clarify the implications of these properties of ohmic contacts, we study three representative cases sketched in Fig. 5.5. In all three, there is current entering into an n-type semiconductor through the ohmic contact. For clarity in representing surface generation and recombination, in this figure, the surface has been expanded to a surface region.

The situation on the left represents an ohmic contact without any generation or recombination taking place at the interface. This is rather common in devices. It is for example the case of the contacts to the source and drain in a MOSFET and the contact to the collector of a bipolar transistor in the forward-active regime. In this instance, there is no minority carrier current, and all the wire current is supported by electrons in the semiconductor.

The case in the center represents one in which there is an ohmic contact at which net recombination prevails. This happens, for example, at the emitter contact of a BJT in the forward-active regime. As the relative size of the arrows in Fig. 5.5 attempts to indicate, this is a case in which

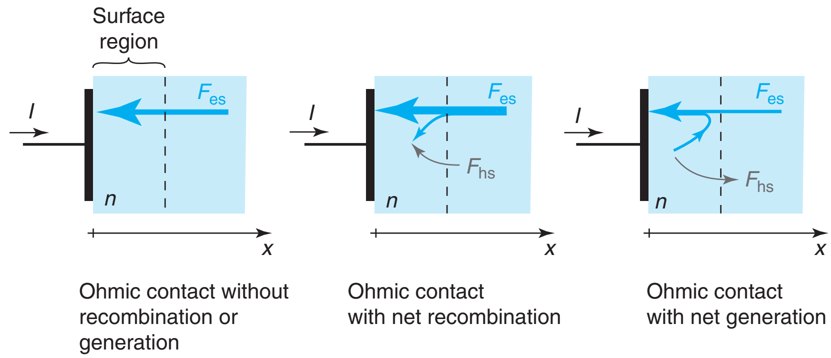


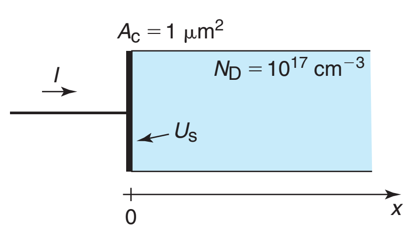
FIGURE 5.5
Three representative cases of ohmic contacts on an n-type semiconductor in which there is current coming out of the semiconductor through the metal. Depending on whether there is net generation or recombination at the metal–semiconductor interface, the carrier fluxes have different signs and magnitudes. For clarity in picturing surface generation and recombination, the surface has been expanded to a surface region.

the majority carrier flux into the surface is increased with respect to the case on the left because a few majority carriers flow in order to recombine with holes at the surface. The total current is continuous through the structure and is supported by a mixture of electron and hole flow in the semiconductor.

The situation on the right represents a contact with net generation. This happens, for example, in the emitter of a BJT in the reverse regime. In this case, the net recombination at the surface is negative, and the majority carrier flux into the surface is smaller than in the case on the left.

EXERCISE 5.2

Consider an ohmic contact to an n-type Si sample with $N_D = 10^{17}\ \text{cm}^{-3}$ *at 300* K, *as sketched below. The area of the contact is* $A_c = 1\ \mu\text{m}^2$.



Current, I, can flow through the wire into the semiconductor and there is also the possibility of net recombination with a rate U_s *at the metal–semiconductor interface. Calculate the electron* *and hole current densities at the semiconductor surface right below the contact under the following conditions:* (a) $I = 0$, $U_s = 10^{22}\ \text{cm}^{-2} \cdot \text{s}^{-1}$, (b) $I = 10\ \mu\text{A}$, $U_s = 0$, and (c) $I = 10\ \mu\text{A}$, $U_s = 10^{22}\ \text{cm}^{-2} \cdot \text{s}^{-1}$.

(a) $I = 0$ is a case equivalent to a free surface. We know in this case that the magnitude of the electron and hole fluxes at the surface are equal to the net recombination rate. However, since there is net recombination at the surface, carriers flow *toward* it, which is against the choice of axis in the figure. Therefore, both F_e and F_h are negative:

$$F_e = F_h = -U_s = -10^{22}\ \text{cm}^{-2} \cdot \text{s}^{-1}$$

The current densities are then

$$J_e = -qF_e = 1.6\ \text{kA/cm}^2$$

$$J_h = qF_h = -1.6\ \text{kA/cm}^2$$

(b) $U_s = 0$ implies that there is no minority carrier flux at the surface of the semiconductor. Hence, the hole current density is

$$J_h = 0$$

The current coming into the contact from the wire is entirely supported by majority carriers on the semiconductor side. The electron current density is then

$$J_e = \frac{I}{A_c} = \frac{10\ \mu\text{A}}{1\ \mu\text{m}^2} = 1\ \text{kA/cm}^2$$

This current is positive because it flows along x.

(c) With U_s identical to case (a) above, the hole current density is just as above:

$$J_h = -qU_s = -1.6\ \text{kA/cm}^2$$

The electron current density must be such that there is current continuity across the metal–semiconductor interface:

$$J_e = J_t - J_h = \frac{I}{A_c} - J_h = 1 + 1.6 = 2.6\ \text{kA/cm}^2$$

5.3 SHOCKLEY EQUATIONS

We now have a complete set of equations to study carrier flow in semiconductors under a great variety of circumstances. For convenience, Table 5.1 summarizes all these relationships in their most general form. They are commonly called *Shockley equations*, in honor of William Shockley, one of the inventors of the bipolar transistor.

Shockley equations are all very fundamental relations. Gauss' law states that net electrical charge of any kind produces an electric field. Next, the electron and hole current equations state

Table 5.1
Shockley equations. The continuity equation for charge is redundant with the electron and hole continuity equations. Only two of these three equations are independent.

Gauss' law:	$\vec{\nabla}\cdot\vec{\mathcal{E}} = \frac{\rho}{\epsilon} = \frac{q}{\epsilon}(p - n + N_D^+ - N_A^-)$
Electron current equation:	$\vec{J}_e = -qn\vec{v}_e^{\,drift} + qD_e\vec{\nabla}n$
Hole current equation:	$\vec{J}_h = qp\vec{v}_h^{\,drift} - qD_h\vec{\nabla}p$
Electron continuity equation:	$\frac{\partial n}{\partial t} = G_{ext} - U + \frac{1}{q}\vec{\nabla}\cdot\vec{J}_e$
or	
Hole continuity equation:	$\frac{\partial p}{\partial t} = G_{ext} - U - \frac{1}{q}\vec{\nabla}\cdot\vec{J}_h$
Charge continuity equation:	$\frac{\partial \rho}{\partial t} = -\vec{\nabla}\cdot\vec{J}_t$
Total current equation:	$\vec{J}_t = \vec{J}_e + \vec{J}_h$

that carriers in semiconductors can flow by means of drift and diffusion producing a current. Finally, the electron and hole continuity equations are particle book-keeping relationships. In Table 5.1, we have expressed the continuity equations in a slightly different way. The term $G - R$ in Eqs. (5.3) and (5.4) is rewritten in terms of an external generation rate G_{ext} and the net recombination rate U, as discussed in Chapter 3. We have also added to this set the charge continuity equation, even though it is redundant with the electron and hole continuity equations. Only two out of these three are independent. The last equation in the table states that the total current density at any point of a semiconductor is the sum of the electron and hole contributions.

This is a system of nonlinear coupled partial differential equations that generally cannot be solved in a closed form. Commercial computer programs have been developed to solve this set of equations even in three dimensions. It is of great interest to obtain, wherever possible, analytical solutions that provide physical insight and that can be used in device design. A few important cases are amenable to compact solutions. We study them in the following sections.

Before we move to study some important special cases, it is useful to reflect for a moment on Gauss' law. Gauss' law relates net volume charge density and the electric field that is generated as a result. As we saw in Chapter 4, an internal electric field in a semiconductor in thermal equilibrium can appear when the doping level is nonuniform. An electric field can also be induced by applying an external voltage. This is a situation clearly out of equilibrium. In either case, an electric field distribution can be described by its associated electrostatic potential. It is often more convenient to work with the electrostatic potential rather than the electric field because a potential difference directly relates to a voltage which is what gets applied from the outside world. However, when doing this, care has to be exercised.

Voltage is not simply equal to potential difference. We know this because in a nonuniform semiconductor in thermal equilibrium there is a certain electric field distribution and an electrostatic potential landscape yet no external voltage has been applied.

The notion of voltage is intimately associated with a nonequilibrium situation in which ϕ in a region of the semiconductor is raised with respect to another region by means of an external battery. Distinguishing between potential difference and voltage can be done if we realize that the volume charge density can, in general, be written as

$$\rho = q(p - n + N_D^+ - N_A^-) = q(p_o - n_o + N_D^+ - N_A^-) + q(p' - n') \tag{5.16}$$

The first term in brackets on the right reflects net charge density in thermal equilibrium while the second term reflects additional net charge out of equilibrium, such as when a voltage is applied.

In general, then, the electric field can be written as

$$\vec{\mathcal{E}} = \vec{\mathcal{E}}_o + \vec{\mathcal{E}}' \tag{5.17}$$

$\vec{\mathcal{E}}_o$ is the electric field in equilibrium which obeys

$$\vec{\nabla} \cdot \vec{\mathcal{E}} = \frac{q}{\epsilon}(p_o - n_o + N_D^+ - N_A^-) \tag{5.18}$$

and $\vec{\mathcal{E}}'$ is the *excess* electric field outside equilibrium which satisfies

$$\vec{\nabla} \cdot \vec{\mathcal{E}}' = \frac{q}{\epsilon}(p' - n') \tag{5.19}$$

In a given problem, $\vec{\mathcal{E}}_o$ is obtained as outlined in Chapter 4. In this chapter, we will later on show how to obtain $\vec{\mathcal{E}}'$.

Associated with $\mathcal{E}_o$ and $\mathcal{E}'$, we can define ϕ and ϕ' defined as

$$\vec{\mathcal{E}}_o = \vec{\nabla}\phi_o \tag{5.20}$$

$$\vec{\mathcal{E}}' = \vec{\nabla}\phi' \tag{5.21}$$

Voltage is defined as the difference of ϕ' at two different points. For example, the voltage between points a and b in a semiconductor (such as two contacts) is defined as

$$V_{ab} = \phi'(a) - \phi'(b) \tag{5.22}$$

In thermal equilibrium, then $\mathcal{E}' = 0$, ϕ' is constant and V from any point to any other point is zero. We will see below that V defined this way is what drives the current that flows through semiconductor structures to which an external voltage has been applied.

5.4 SIMPLIFICATIONS OF SHOCKLEY EQUATIONS TO ONE-DIMENSIONAL QUASI-NEUTRAL SITUATIONS

There are a variety of circumstances for which the Shockley equations can be substantially simplified and analytical results can be derived. Some of these situations are very important because they appear frequently even in fairly dissimilar devices. Recognizing these special cases allows understanding developed in the context of one device to port over to other devices.

The three special cases that we are going to discuss here are referred to as *minority carrier situations*, *majority carrier situations*, and *space-charge situations*. These three different regimes of carrier transport appear very frequently in the operation of microelectronic devices. Even in a

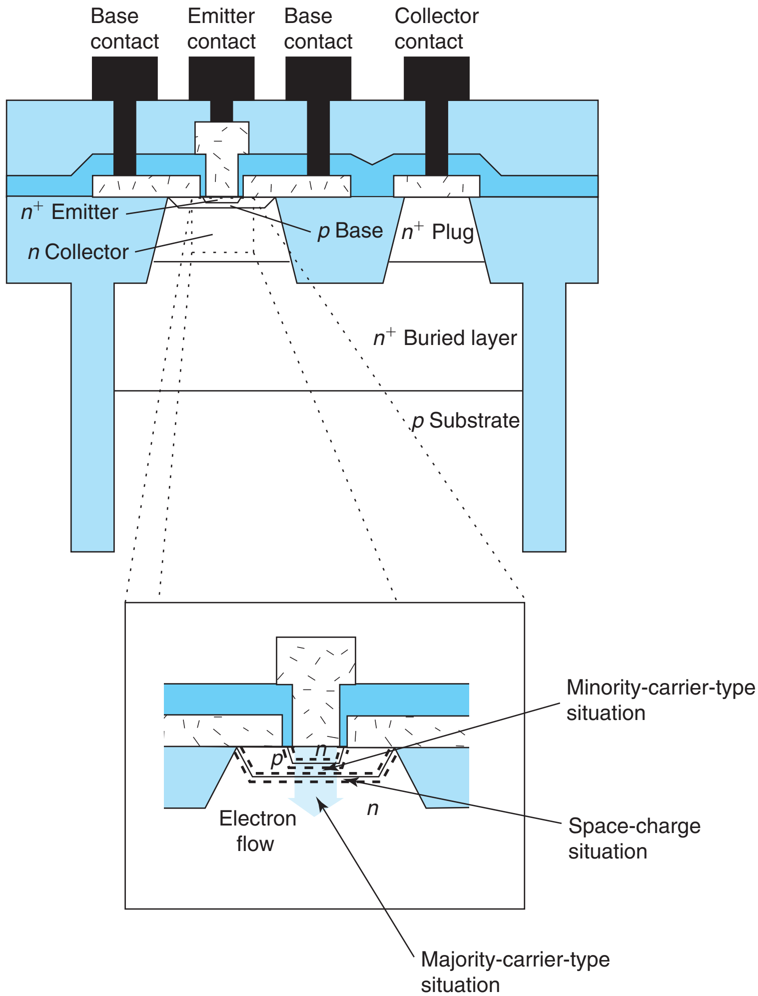


FIGURE 5.6
Cross section of modern bipolar junction transistor. The inset shows electron transport across the intrinsic portion of the device in the forward-active regime illustrating the different modes of transport that can be encountered.

given device under a specific mode of operation, these three situations can be present simultaneously. This is illustrated in Fig. 5.6 which depicts an npn bipolar transistor in the forward-active regime. If we focus on electron transport across the intrinsic portion of the device, we recognize a minority carrier situation, a space-charge region situation, and a majority carrier situation in the flow of electrons through the base, the base–collector junction, and the collector, respectively. This will become clear after studying this chapter and the basic operation of the bipolar transistor in Chapter 11.

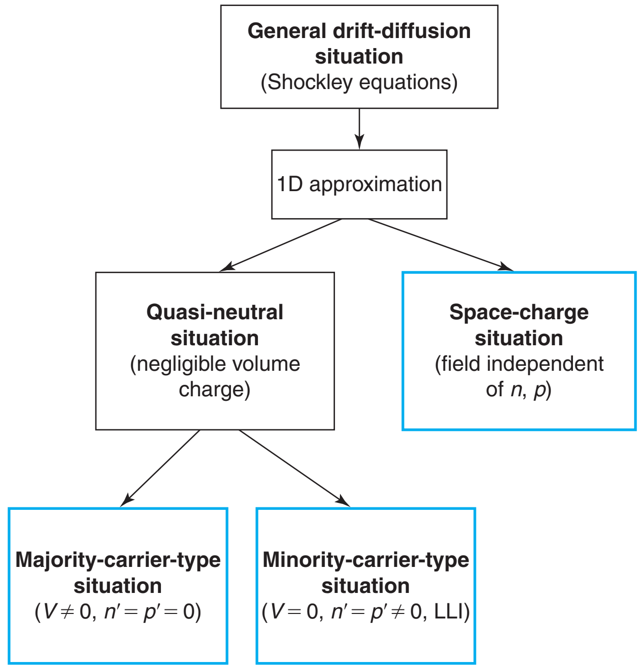


FIGURE 5.7
Chart summarizing the various approximations of the drift-diffusion formulation that are going to be performed in this chapter. V refers to the application of an external voltage to a region in a semiconductor.

The rest of the chapter is devoted to discussing in some depth these three important situations. Our goal is to develop an intuitive understanding of carrier behavior in these three cases and to build a handy tool-set with which we will analyze several microelectronic devices in later chapters.

In order to accomplish this, we will perform a number of simplifications. A judgment of when a particular simplification is viable can only be acquired with experience. There will be multiple opportunities in this book to exercise such judgment. A chart summarizing the various simplifications that we are going to perform and their interrelations is shown in Fig. 5.7. We will refer to this chart in the sections that follow.

Although microelectronic devices are three-dimensional structures, in many situations we can simplify the problem to two and even one dimension. This happens because many times the geometry is very different in the three spatial directions, there are axis or planes of symmetry, or carrier phenomena are dominated by the behavior along one particular dimension. For example, in an integrated PN diode, as sketched in Fig. 5.8, the junction is parallel to the wafer surface and the lateral dimensions are typically much larger than the vertical ones. In this case, for a reasonably well designed diode under typical operating conditions, the dominant carrier behavior under the active area takes place along the vertical dimension (x in the diagram) with the physics not changing very much for the different positions on the plane. In these types

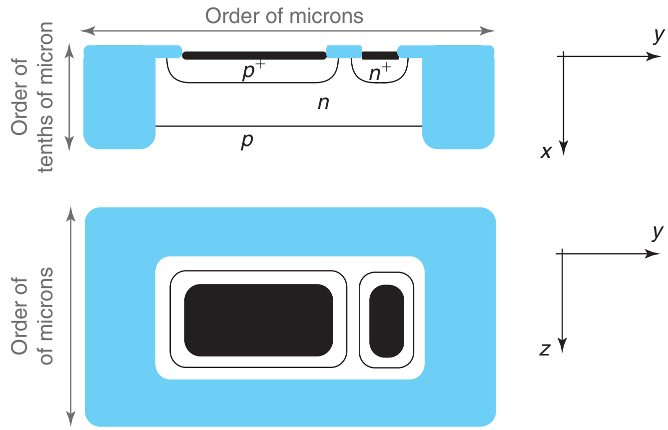


FIGURE 5.8
Cross-sectional sketch of a simple integrated PN junction diode. The PN junction lies parallel to the wafer surface. The extent of the device on the wafer is much larger than into the wafer. This justifies making one-dimensional approximation to study most aspects of device physics.

of situations, a one-dimensional treatment of carrier flow can be highly accurate under most common circumstances.[1]

Making the *one-dimensional approximation* drastically simplifies the Shockley equations. All $\vec{\nabla}$ operators become simple spatial derivatives along the selected dimension. These equations are listed in Table 5.2. This is the first important simplification to the Shockley equations that we perform, as shown in Fig. 5.7.

The second simplification addresses Gauss' law. The Shockley equation set is a difficult one to solve because of the coupling that Gauss' law introduces. Fortunately, there are two important situations for which a simplification of this equation becomes possible, drastically untying the Shockley equation set. These two are *quasi-neutral situations* and *space-charge situations*. They are both indicated in Fig. 5.7. We discuss quasi-neutral situations in the remainder of this section and space-charge situations in Section 5.9.

We introduced the concept of quasi-neutrality in Chapter 4 when we discussed nonuniform doping distributions in thermal equilibrium. At that time, we defined a quasi-neutral situation as one in which the majority carrier concentration closely tracks the doping level with the consequence that the net volume charge density is negligible. We can generalize this concept to denote as quasi-neutral any situation in which *at every location the net volume charge density that arises from a discrepancy of the concentration of positive and negative species is negligible in the scale of the charge density that is present*. Mathematically, this implies that

$$\rho \simeq 0 \tag{5.23}$$

[1]Even in a geometry like the one sketched in Fig. 5.8, certain phenomena are intrinsically of a two- or three-dimensional nature. A good example is reverse-bias breakdown that is almost always associated with the corners of the PN junction, as we will study later on in this book.

Table 5.2
Shockley equations in one dimension. The continuity equation for charge is redundant with the electron and hole continuity equations. Only two of these three equations are independent.

Gauss' law:	$\frac{\partial \mathcal{E}}{\partial x} = \frac{\rho}{\epsilon} = \frac{q}{\epsilon}(p - n + N_D - N_A)$
Electron current equation:	$J_e = -qnv_e^{drift}(\mathcal{E}) + qD_e \frac{\partial n}{\partial x}$
Hole current equation:	$J_h = qpv_h^{drift}(\mathcal{E}) - qD_h \frac{\partial p}{\partial x}$
Electron continuity equation:	$\frac{\partial n}{\partial t} = G_{ext} - U(n,p) + \frac{1}{q}\frac{\partial J_e}{\partial x}$
or	
Hole continuity equation:	$\frac{\partial p}{\partial t} = G_{ext} - U(n,p) - \frac{1}{q}\frac{\partial J_h}{\partial x}$
Charge continuity equation:	$\frac{\partial \rho}{\partial t} = -\frac{\partial J_t}{\partial x}$
Total current equation:	$J_t = J_e + J_h$

The quasi-neutral approximation, a very reasonable one in many circumstances, has its origin in the fact that electrons and holes are mobile. In response to an electric field, carriers always move in the direction of trying to erase the field. In consequence, if there are large numbers of carriers, as is often the case in semiconductors, it is simply difficult to sustain a substantial volume charge for any appreciable time duration.

The quasi-neutrality assumption drastically simplifies the solution to the Shockley equations because it breaks the coupling between $\mathcal{E}$ and n and p. In a quasi-neutral situation in thermal equilibrium $\mathcal{E}_o$, n_o, and p_o are computed as we discussed in Chapter 4. Out of equilibrium, Eq. (5.16) implies that

$$\rho' = q(p' - n') \simeq 0 \tag{5.24}$$

and therefore

$$p' \simeq n' \tag{5.25}$$

This effectively eliminates one of the carrier concentrations as an unknown.

Another important consequence of the quasi-neutrality assumption is that if the volume charge density is negligible, its change in time can also often be disregarded. This also simplifies the charge continuity equation to

$$\frac{\partial J_t}{\partial x} \simeq 0 \tag{5.26}$$

In one dimension, this is a simple statement of current continuity. If the time change in the volume charge density anywhere is negligible, then the total current density must be continuous everywhere. This will be very useful in device analysis because if we can figure out the total current density at one location in the device, we know it everywhere.

With these changes, the quasi-neutral approximation simplifies the one-dimensional Shockley equations to the set of Table 5.3. To clean up the notation, it has further been assumed that all dopants are ionized.

Table 5.3
Equation set for one-dimensional quasi-neutral situations.

$p - n + N_D - N_A \simeq 0$
$J_e = -qnv_e^{\text{drift}} + qD_e \frac{\partial n}{\partial x}$
$J_h = qpv_h^{\text{drift}} - qD_h \frac{\partial p}{\partial x}$
$\frac{\partial n}{\partial t} = G_{\text{ext}} - U + \frac{1}{q}\frac{\partial J_e}{\partial x}$ or $\frac{\partial p}{\partial t} = G_{\text{ext}} - U - \frac{1}{q}\frac{\partial J_h}{\partial x}$
$\frac{\partial J_t}{\partial x} \simeq 0$
$J_t = J_e + J_h$

In each specific situation, the conditions that need to be met for quasi-neutrality to be a reasonable assumption out of equilibrium will be different. Through several examples, we will learn how to check this condition.

We now split the discussion into two broad ways in which a semiconductor region can be brought out of equilibrium. These both are indicated in Fig. 5.7. One way is the application of a voltage from the outside. The second one is the injection or generation of excess carriers. As we will see below, the relative role played by the majority and minority carriers in these situations is very different. Because of this, we refer to the first kind, application of voltage, as "majority carrier situations," and to the second, introduction of excess carriers, as "minority carrier situations." We study them separately. These are, of course, extreme cases of a rich continuum. In real devices, mixed situations often occur. However, to deal with them effectively, good understanding of the simple cases discussed here is essential.

5.5 MAJORITY CARRIER SITUATIONS

This class of problems describes situations faced very frequently in microelectronic devices in which a voltage is applied to a semiconductor region from the outside without upsetting the carrier concentrations from their equilibrium values.

Before attacking a description of these problems, it is useful here to briefly remember what a battery does. If one connects a resistor across the two terminals of a battery, current flows. The standard notation is such that current flows out of the "long" terminal of the battery, through the resistor, into the "short" terminal of the battery, as sketched in Fig. 5.9. Current in metals and conventional resistors consists of electron flow. Since electrons are negatively charged, electrons actually flow in the contrary sense to the current, as also sketched in Fig. 5.9.

What drives the electrons to flow that way? The energy view of the situation, also sketched in Fig. 5.9, helps to understand this. The battery grabs electrons from its positive terminal and raises their energy as they cross through it. Electrons in the negative terminal have a higher energy than those in the positive terminal. If provided with a path, they will flow in an effort to lower their potential energy. The resistor is one such path (the battery itself is not, it blocks

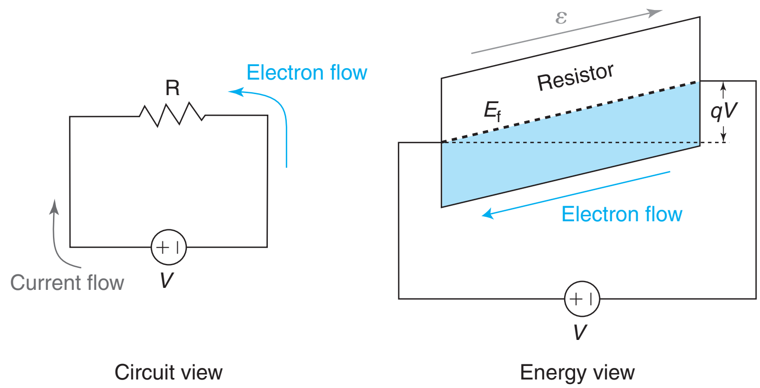


FIGURE 5.9
Left: simple circuit with resistor and battery. Right: energy view of circuit. The negative terminal of the battery raises the electron energy by qV with respect to the positive terminal. In consequence, electrons flow across the resistor from right to left.

the "backward" flow of electrons). How much energy does the battery provide to the electrons? If the battery is rated with a voltage V, the energy of an electron at the negative terminal with respect to the positive terminal is qV. If there are no ohmic losses in the wires and the contacts, the entire voltage of the battery shows up across the resistor creating an electric field and tilting the energy band structure, as shown. This presents electrons in the resistor with empty states to their left and they preferentially flow from right to left. This is what is required by the sign of the battery.

Let us now discuss majority carrier situations in which a voltage is applied to a semiconductor region. If the electric field that is introduced in the semiconductor is not too high, there is no reason for the dynamic balance that existed in thermal equilibrium between generation and recombination to be upset. This means that the carrier concentrations are not disturbed from their equilibrium values.[2] Under these circumstances, Shockley's equations simplify drastically. First of all, since the carrier concentrations are not upset from equilibrium, their time derivatives are zero. With $G_{ext} - U = 0$, this implies that $\frac{dJ_e}{dx} = \frac{dJ_h}{dx} = 0$.

We can also simplify the current equations in the following way. For electrons, for example, after substituting $n \simeq n_o$ in the equation listed in Table 5.3, we get

$$J_e = -qn_o v_e^{drift}(\mathcal{E}) + qD_e \frac{dn_o}{dx} \tag{5.27}$$

where we have made explicit the electric field dependence of the electron drift velocity.

In thermal equilibrium, that is, when $\mathcal{E} = \mathcal{E}_o$, the electron current has to be zero. Equation (5.27) becomes

$$-qn_o v_e^{drift}(\mathcal{E}_o) + qD_e \frac{dn_o}{dx} = 0 \tag{5.28}$$

[2]If the electric field gets very high, carrier generation arising from impact ionization might invalidate this assumption. This is dealt with in Section 5.10.

We can solve in this equation for the second term and insert it into Eq. (5.27) to get

$$J_e = -qn_o[v_e^{drift}(\mathcal{E}) - v_e^{drift}(\mathcal{E}_o)] \tag{5.29}$$

A similar equation can be derived for holes following a parallel procedure.

The low-field limit of this equation is of particular interest. For small fields, the drift velocity is directly proportional to the electric field. Using Eq. (5.17), we can easily rewrite Eq. (5.29) as

$$J_e = qn_o\mu_e\mathcal{E}' \tag{5.30}$$

This equation clearly shows that outside equilibrium the electron current is driven by the excess field created by the application of a voltage on the sample. That is also the case for large fields. However, the nonlinear velocity-field characteristics of carriers in semiconductors does not allow Eq. (5.29) to be written explicitly in terms of $\mathcal{E}'$.

An additional simplification that can readily be made is that, for reasonably extrinsic material, one of the carrier concentrations dominates over the other. Since, as Eq. (5.29) shows, the carrier currents are directly proportional to the carrier concentrations, in a typical majority carrier situation we can then confidently neglect the current contribution arising from the minority carriers.

This simplifies Shockley's equations to the set of Table 5.4. This equation set does not have time derivatives any more. Effectively, we are dealing with *quasi-static* situations in which the time evolution of the semiconductor is completely determined by the time dependence of the driving term (typically the applied voltage). The dynamics of majority carrier situations are discussed in more detail in Section 5.7.

In a specific majority-carrier-type situation, in order to obtain a complete solution to the problem, one needs to find a way to relate the internal electric field that is created by the application of an external applied voltage to the voltage itself. It is then best to reformulate Eq. (5.30) in terms of the electrostatic potential:

$$J_e = -qn_o\mu_e\frac{d\phi'}{dx} \tag{5.31}$$

This now connects with our definition of voltage made in Eq. (5.22). An example helps to illustrate this.

Table 5.4
Equation set for one-dimensional majority-carrier-type situations.

n-type	p-type
$n \simeq n_o \simeq N_D$	$p \simeq p_o \simeq N_A$
$J_e = -qn_o[v_e^{drift}(\mathcal{E}) - v_e^{drift}(\mathcal{E}_o)]$	$J_h = qp_o[v_h^{drift}(\mathcal{E}) - v_h^{drift}(\mathcal{E}_o)]$
$\frac{dJ_e}{dx} \simeq 0,\ \frac{dJ_h}{dx} \simeq 0,\ \frac{dJ_t}{dx} \simeq 0$	
$J_t \simeq J_e$	$J_t \simeq J_h$

5.5.1 Example 1: Semiconductor bar under voltage

Consider a uniformly doped n-type semiconductor bar of length L with contacts at both ends, as sketched in Fig. 5.10. A voltage V is applied across the contacts as indicated. This is a one-dimensional situation with all action occurring along the x-axis.

If we assume ideal contacts, the applied voltage drops entirely across the semiconductor. Since the bar is uniformly doped, this implies that the electrostatic potential builds up across the sample in a linear way, as sketched in the bottom left figure. This yields an energy band diagram as sketched on the right with straight but inclined bands. The Fermi level across the sample is labeled as E_f to indicate a situation outside equilibrium but with $E_{fe} = E_{fh} = E_f$. The difference in Fermi levels across the sample is qV, the applied voltage in units of energy.

In this picture, the electric field across the sample is uniform and is given simply by

$$\mathcal{E} = \frac{\phi(0) - \phi(L)}{L} = \frac{V}{L} \tag{5.32}$$

Note how with the choice of axis that has been made and the polarity of the battery, this electric field is positive.

In the low-field limit, we can now compute the current density using Eq. (5.31) as

$$J_t \simeq qN_D \mu_e \frac{V}{L} \tag{5.33}$$

Here we have substituted n_o by the doping level in the sample and have assumed that the hole concentration is negligible. This expression now gives the current density through the sample as a function of V. If $V = 0$, the current density is zero. In the low-field limit, the current density is linear in V.

Now, you might ask yourself, how is this electric field created? Where is the net charge that supports this electric field? Looking at the diagram on the right of Fig. 5.10, the net charge must

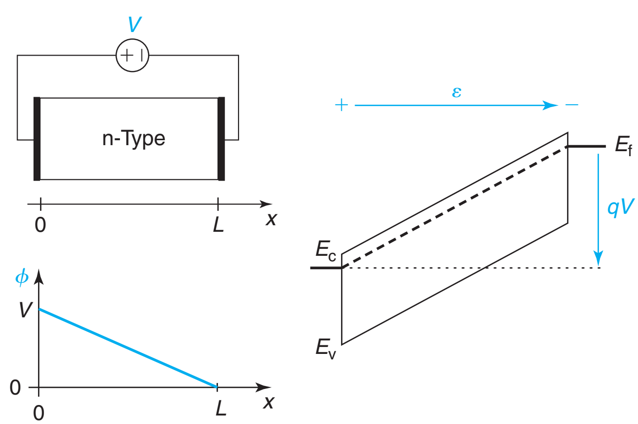


FIGURE 5.10
Top left: sketch of one-dimensional semiconductor bar under voltage. Bottom left: sketch of electrostatic potential. Right: energy band diagram.

reside close to the ends of the sample. The only way this can happen is if there is an imbalance between positive and negative charge in these regions. That means, an imbalance between donor concentration and electron concentration. Close to the terminal on the left, we expect more donors than acceptors (net positive charge). Close to the terminal on the right the situation has to be reversed. By symmetry, we expect identical charge distributions with contrary signs.

We can easily estimate the shape of the charge distribution. In general, the electron current is given by the sum of drift and diffusion currents. If the field is not too high, Eq. (5.27) can be written as

$$J_{\mathrm{e}} = qn\mu_{\mathrm{e}}\mathcal{E} + qD_{\mathrm{e}}\frac{dn}{dx} \simeq qN_{\mathrm{D}}\mu_{\mathrm{e}}\mathcal{E} + kT\mu_{\mathrm{e}}\frac{dn}{dx} \tag{5.34}$$

Here we have used the Einstein relation and assumed that the imbalance between charges is relatively small.

Under steady-state conditions

$$\frac{dJ_{\mathrm{e}}}{dx} = 0 = qN_{\mathrm{D}}\mu_{\mathrm{e}}\frac{d\mathcal{E}}{dx} + kT\mu_{\mathrm{e}}\frac{d^2n}{dx^2} \tag{5.35}$$

We now use Gauss' law for this system of donors and electrons and easily get

$$\frac{d^2(N_{\mathrm{D}} - n)}{dx^2} = \frac{N_{\mathrm{D}} - n}{L_{\mathrm{D}}^2} \tag{5.36}$$

where L_{D} is the Debye length defined in Eq. (4.68).

The solution to this differential equation is straightforward. Close to $x = 0$ (the left side of the bar), it can easily be derived as

$$N_{\mathrm{D}} - n = Ke^{-x/L_{\mathrm{D}}} \tag{5.37}$$

where K is a constant. Not surprisingly, we find a volume charge density that is confined to the edges of the sample and spreads over a length of the order of the Debye length. This is sketched in Fig. 5.11. The quasi-neutral approximation in this case applies in most of the bar if $L_{\mathrm{D}} \ll L$.

The constant K can be obtained from the application of Gauss' law. For this, we integrate the volume charge density at one end of the sample:

$$Q_{\mathrm{s}} = \int_0^\infty \rho\, dx = qK\int_0^\infty e^{-x/L_{\mathrm{D}}}dx = qKL_{\mathrm{D}} \tag{5.38}$$

Q_{s} is the sheet of charge that generates the electric field. Now using Gauss' law, the electric field that emanates from Q_{s} is

$$\mathcal{E} \simeq \frac{Q_{\mathrm{s}}}{\epsilon} = \frac{qKL_{\mathrm{D}}}{\epsilon} \tag{5.39}$$

Equating this to Eq. (5.32) yields

$$K = \frac{\epsilon V}{qL_{\mathrm{D}}L} \tag{5.40}$$

Finally,

$$N_{\mathrm{D}} - n = \frac{\epsilon V}{qL_{\mathrm{D}}L}e^{-x/L_{\mathrm{D}}} \tag{5.41}$$

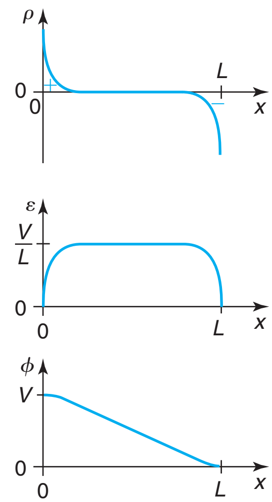


FIGURE 5.11
Electrostatics of the semiconductor bar under voltage of Fig. 5.10. From top to bottom: volume charge density, electric field, and electrostatic potential. The volume charge density at the ends of the bar extends over a region of order L_D, the Debye length.

Sketches of the volume charge density, electric field, and electrostatic potential distribution are shown in Fig. 5.11.

This result shows that the simple solutions derived in Eqs. (5.32) and (5.33) apply if the sample can be considered quasi-neutral, or, when $L_D \ll L$. First-order corrections for the impact of finite L_D can be easily obtained (see Problem 5.34).

5.5.2 Example 2: Integrated resistor

The formulation that we have developed to describe majority carrier situations is particularly useful in the analysis of the integrated resistor, a real device in its own right and an interesting case study for several important issues. A sketch of an integrated resistor is shown in Fig. 5.12. This device consists of an n-type region created inside a p-type wafer. The n-region is accessed from the outside world through two n^+ regions that are provided with ohmic contacts. Upon the application of a voltage across the contacts, current flows through the n-type region. As we will see in Chapter 6, the PN region that is formed between the n-type region and the substrate confines the current flow to the n-type region.

If we can neglect any ohmic drops across the contacts and the n^+ regions, then we are back to a situation similar to that of the previous example in which the entire applied voltage V appears across a uniformly doped n-type region. For low fields, the current density is given by Eq. (5.33) and the device terminal current is therefore obtained by multiplying this by the cross-sectional area of the device $A = Wt$:

$$I = qA\mu_e N_D \frac{V}{L} \tag{5.42}$$

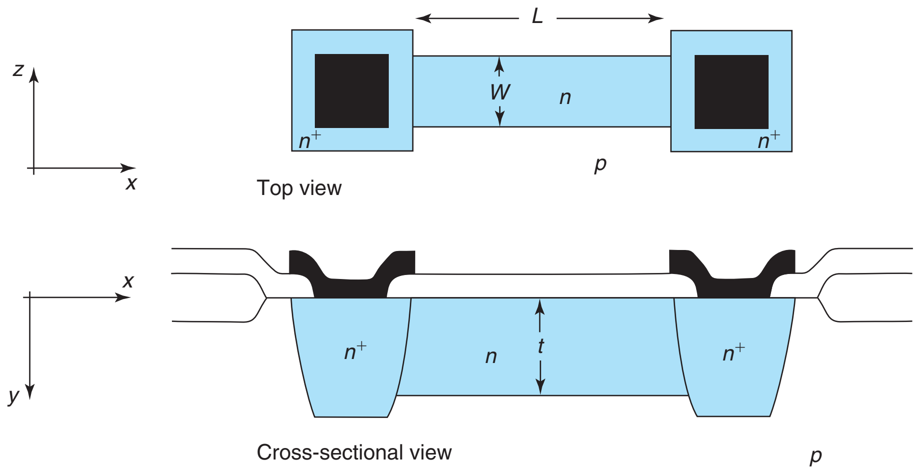


FIGURE 5.12
Sketch of integrated resistor.

The proportionality constant between current and voltage is the resistance of the sample:

$$R = \frac{L}{qA\mu_e N_D} \tag{5.43}$$

If the field is high enough, the linear relationship between electron velocity and electric field does not hold. Rather, a more complex relationship such as the one shown in Chapter 4 applies. Using Eq. (4.13) for the $v - \mathcal{E}$ relationship, solving for $\mathcal{E}$ and proceeding as above yields

$$I = \frac{qAN_D v_{sat}}{1 + \frac{v_{sat} L}{\mu_e V}} \tag{5.44}$$

For low V, this equation converges to Eq. (5.42) above. For moderate V, however, as V increases in Eq. (5.44), the current grows sublinearly with the voltage, or in other words, the resistance of the region increases. In fact, for high enough voltages, the electrons attain saturation velocity and I saturates to

$$I_{sat} = qAN_D v_{sat} \tag{5.45}$$

and the resistance increases without bounds. The I–V characteristics captured in Eq. (5.44) are sketched in Fig. 5.13. Similar results apply for p-type regions.

In ICs, it is common to fabricate integrated resistors in which the doping level is not uniform in space but rather has a peak close to the surface and decreases in depth, as sketched in Fig. 5.14. In this case, we can easily compute the resistance by viewing this resistor as the parallel of many differential resistors. The resistance is then (for small voltages)

$$R = \frac{L}{qW \int_0^t \mu_e N_D(y) dy} \tag{5.46}$$

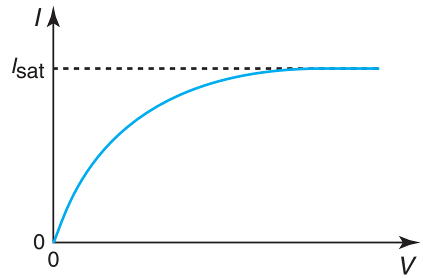


FIGURE 5.13
Sketch of nonlinear I–V characteristics of semiconductor resistor.

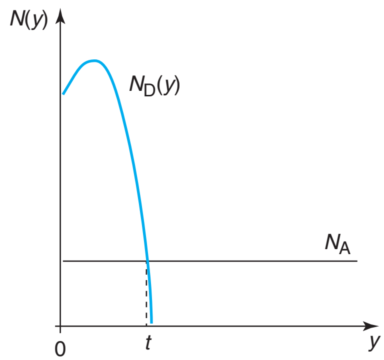


FIGURE 5.14
Typical doping profile of a semiconductor resistor. The p-type background produces a PN junction at the bottom of the resistor that effectively confines current flow to the n-type region.

This equation yields the result of Eq. (5.42) if N_D is uniform in space. In general, if the doping level changes with position, the mobility will change too. That is why it has to remain inside the integral in Eq. (5.46).

When designing integrated resistors, device and circuit engineers utilize standard doped layers that are already available in the fabrication process, such as the n^+ or p^+ source implants in a CMOS process, or the n^+-emitter or p-base in an npn bipolar process. For the purpose of designing lateral resistors, the complexity of the doping distribution in depth can be hidden under a single parameter that is called the *sheet resistance*, defined as

$$R_{sh} = \frac{1}{q \int_0^t \mu_e N_D(y) dy} \tag{5.47}$$

With this definition, the lateral resistance becomes

$$R = R_{sh} \frac{L}{W} \tag{5.48}$$

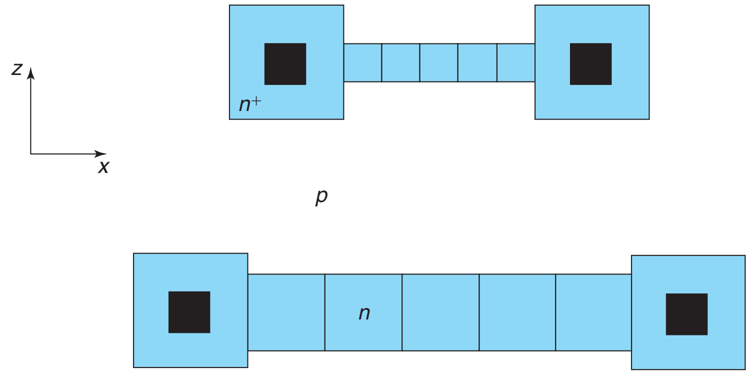


FIGURE 5.15
Two integrated resistors fabricated on the same process. The geometry is different but the resistance is the same (assuming that the parasitic resistance associated with the contacts is negligible). Coordinate x is in the direction of current flow. Coordinate y is into the semiconductor.

This is a useful equation that provides simple design rules for integrated resistors. The resistance of a planar resistor is obtained by multiplying the sheet resistance by the ratio of the length over the width of the resistor. In other words, the resistance depends only on the "number of squares" that the integrated resistor has in the direction of the current flow. This is illustrated in Fig. 5.15 where we show two resistors fabricated in the same process with identical resistance since both are made of five squares. Because of this interesting property, the units of R_{sh} are frequently given in "ohms per square," written as $\Omega/\square$.

For uniform doping distribution, the sheet resistance follows an expression:

$$R_{sh} = \frac{1}{q\mu_e N_D t} \tag{5.49}$$

The sheet resistance decreases as the doping level of the layer or its thickness increases.

EXERCISE 5.3

Consider an integrated resistor, such as the one shown in Fig. 5.12, composed of an n-type Si layer fabricated on a p-type substrate. The doping level of the body of the resistor is $N_D = 10^{19}$ cm^{-3}. *The dimensions of its active region are* $L = 5$ μm, $W = 2$ μm, *and* $t = 0.1$ μm. *At room temperature, estimate the sheet resistance of the n-type semiconductor layer and the resistance of this resistor.*

For this doping level, the electron mobility is about $\mu_e \simeq 110$ cm^2/V · s. The sheet resistance is then

$$R_{sh} = \frac{1}{q\mu_e N_D t} = \frac{1}{1.6\times10^{-19}\ \text{C}\times110\ \text{cm}^2/\text{V}\cdot\text{s}\times10^{19}\ \text{cm}^{-3}\times0.1\times10^{-4}\ \text{cm}} \simeq 570\ \Omega/\square$$

The resistance of the resistor is then obtained from:

$$R = R_{sh}\frac{L}{W} = 570\ \Omega/\square \times \frac{5\ \mu\text{m}}{2\ \mu\text{m}} \simeq 1.5\ \text{k}\Omega$$

Note how in this last expression, there is no need to convert the units of L and W to cm since they cancel out.

5.6 MINORITY CARRIER SITUATIONS

We will now deal with a completely different class of problems that also frequently appear in microelectronic devices: minority carrier situations. These are characterized by (i) quasi-neutrality, (ii) the presence of excess carriers as a result of external generation or injection from an adjacent region, and (iii) the absence of a significant external electric field. In our chart depicting the successive approximations to the Shockley equations (Fig. 5.7), these situations are shown in the box at the bottom center. Situations of this kind arise, for example, in the base of a bipolar transistor under regular bias conditions. In general, a description of these situations requires the complete set of equations of Table 5.3. Fortunately, two important approximations hold under a wide range of circumstances.

The first approximation accounts for the fact that in the absence of external electric fields, the internal electric field that might be present in equilibrium is rarely so high that velocity saturation is an issue. This allows us to assume that the drift velocity is linearly proportional to the electric field.

The second approximation is *low-level injection*. This was already discussed in Chapter 3 in the context of generation and recombination. As a reminder, low-level injection refers to a situation in which the excess carrier concentration is much higher than the equilibrium minority carrier concentration but much smaller than the equilibrium majority carrier concentration. This situation is fairly common because minority carrier devices exhibit degraded performance if the carrier concentrations reach high-injection levels. In many devices, typical operating conditions avoid high-level injection.

The restriction to low injection levels results in several simplifications:

- First, the majority carrier concentrations out of equilibrium are basically unchanged from their equilibrium values. Thus, for n-type material, $n \simeq n_o$.
- Second, the minority carrier concentrations are overwhelmed. In an n-type semiconductor, this means $p \simeq p'$.
- Third, the recombination rate is proportional to the excess carrier concentration over the carrier lifetime (see Chapter 3). For n-type, we can write $U \simeq p'/\tau$.
- Even though no external fields are applied, internal fields can be generated as a result of carrier injection. However, this is our fourth simplification, and internally generated electric fields are small enough so that minority carrier drift currents produced by them are insignificant (we will later prove this point in one of the examples). The only fields that might significantly act on the minority carriers are those that are present internally under equilibrium conditions as a result of doping gradients. We cannot extend this approximation to the majority carriers because, having so many of them around, small changes in the electric field can produce substantial changes in the current.

These simplifications allow us to rewrite the carrier current equations and the continuity equations in the following manner. For n-type material, for example, the majority carrier current can be written in the following way:

$$J_e \simeq q(n_o + n')\mu_e(\mathcal{E}_o + \mathcal{E}') + qD_e\left(\frac{\partial n_o}{\partial x} + \frac{\partial n'}{\partial x}\right) \tag{5.50}$$

where we have made explicit the excess electron concentration and the excess electric field produced outside equilibrium.

In equilibrium, that is, with $n' = 0$ and $\mathcal{E}' = 0$, the electron current has to be zero. This allows us to simplify Eq. (5.50) to

$$J_e \simeq qn_o\mu_e\mathcal{E}' + qn'\mu_e\mathcal{E}_o + qD_e\frac{\partial n'}{\partial x} \tag{5.51}$$

There are two drift terms involving electrons (as majority carriers) out of equilibrium. One is due to the excess electric field acting on the equilibrium electron concentration. The second one is the equilibrium electric field acting on the excess electron concentration. Both terms can be important. There is also one diffusion term that accounts for diffusion of excess electrons.

The expression of the minority carrier current can also be substantially simplified:

$$J_h \simeq q(p_o + p')\mu_h(\mathcal{E}_o + \mathcal{E}') - qD_h\left(\frac{\partial p_o}{\partial x} + \frac{\partial p'}{\partial x}\right) \tag{5.52}$$

Once again, we can simplify this expression by noting that in equilibrium $J_h = 0$. Equation (5.52) becomes

$$J_h \simeq qp'\mu_h\mathcal{E}_o + qp'\mu_h\mathcal{E}' - qD_h\frac{\partial p'}{\partial x} \simeq qp'\mu_h\mathcal{E}_o - qD_h\frac{\partial p'}{\partial x} \tag{5.53}$$

The term on $qp'\mu_h\mathcal{E}'$ has been neglected as discussed above. This expression of the minority carrier current now depends exclusively on the excess minority carrier concentration.

The minority carrier continuity equation is also modified to

$$\frac{\partial p'}{\partial t} = G_{ext} - \frac{p'}{\tau} - \frac{1}{q}\frac{\partial J_h}{\partial x} \tag{5.54}$$

We can further transform this equation into a particularly useful form by substituting Eq. (5.53) into Eq. (5.54). Rearranging terms, we easily get

$$D_h\frac{\partial^2 p'}{\partial x^2} - \mu_h\mathcal{E}_o\frac{\partial p'}{\partial x} - \frac{p'}{\tau} + G_{ext} = \frac{\partial p'}{\partial t} \tag{5.55}$$

where we have used the fact that $\partial\mathcal{E}_o/\partial x \simeq 0$ in a quasi-neutral region (see Section 4.5.3).

Equation (5.55) is now a differential equation with a single unknown, the excess minority carrier concentration. Once the generation function G_{ext} and the boundary conditions are specified, this equation can (in principle) be solved and p' can be obtained throughout the region of interest. From here, the complete solution quickly emerges, as we will see below.

It is interesting to see that if we follow a similar procedure with the majority carriers, that is, we introduce the majority carrier current Eq. (5.51) into its continuity equation, an equation solely on the majority carrier concentration similar to (5.55) does not emerge. There is a term in $\mathcal{E}'$ that prevents this equation from being solved all by itself. Equation (5.55) then shows that

Table 5.5
Equation set for one-dimensional minority carrier situations.

n-type	p-type
$p_o - n_o + N_D - N_A \simeq 0$	
$p' \simeq n'$	
$J_e = qn_o\mu_e\mathcal{E}' + qn'\mu_e\mathcal{E}_o + qD_e\frac{\partial n'}{\partial x}$	$J_e = qn'\mu_e\mathcal{E}_o + qD_e\frac{\partial n'}{\partial x}$
$J_h = qp'\mu_h\mathcal{E}_o - qD_h\frac{\partial p'}{\partial x}$	$J_h = qp_o\mu_h\mathcal{E}' + qp'\mu_h\mathcal{E}_o - qD_h\frac{\partial p'}{\partial x}$
$D_h\frac{\partial^2 p'}{\partial x^2} - \mu_h\mathcal{E}_o\frac{\partial p'}{\partial x} - \frac{p'}{\tau} + G_{ext} = \frac{\partial p'}{\partial t}$	$D_e\frac{\partial^2 n'}{\partial x^2} + \mu_e\mathcal{E}_o\frac{\partial n'}{\partial x} - \frac{n'}{\tau} + G_{ext} = \frac{\partial n'}{\partial t}$
$\frac{\partial J_t}{\partial x} \simeq 0$	
$J_t = J_e + J_h$	

in minority carrier situations, the dynamics of the minority carriers dominate the solution and everything else revolves around this.

To summarize, under the assumptions of low-level injection and no external field applied, the Shockley equations simplify as listed in Table 5.5. The best way to understand the nature of the approximations made in this section is to study some specific examples. We will consider two of them in some detail. Both are static. We will examine a dynamic case later on in Section 5.8.

5.6.1 Example 3: Diffusion and bulk recombination in a "long" bar

Let us consider the situation sketched in Fig. 5.16. We have a very long uniformly doped n-type semiconductor bar that is illuminated over a very narrow region with deep penetrating radiation. At this cross section, g_l electron–hole pairs per second per unit area are generated. We wish to compute the electron and hole concentrations and the respective currents throughout under static conditions. Since the doping level is uniform, the electric field inside this sample in thermal equilibrium is zero, that is, $\mathcal{E}_o = 0$.

Before attempting a complete mathematical solution to the problem, it is valuable to discuss qualitatively what is going on. Let us place the origin of the axis at the point of generation. Electron–hole pair generation at $x = 0$ causes a build up of carrier concentration above the equilibrium background. In consequence, a gradient in the carrier concentration develops which results in carrier diffusion away from the generation point. As excess carriers spill into the neighboring regions of the semiconductor, the recombination rate increases above the thermal equilibrium rate. Eventually, a steady-state situation is attained in which the total generation rate matches the total recombination rate. The same principle that we have encountered a number of times before is at play here. When thermal equilibrium is perturbed, the semiconductor reacts trying to reestablish it. In this case, as the carrier concentration builds up, the recombination rate increases in an effort to wipe out the excess carriers.

A photogenerated hole has a lifetime that is set by the material, as we discussed in Chapter 3. This means that it will only have a chance to diffuse a certain distance before it recombines. This

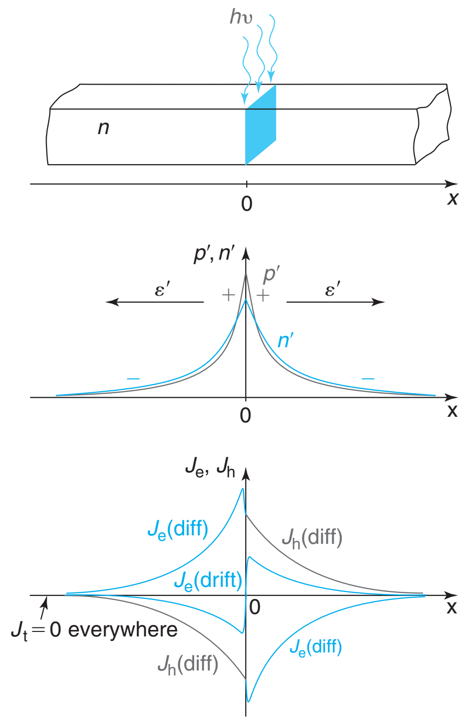


FIGURE 5.16
Top: an infinitely long bar illuminated at $x = 0$ by a sheet of light; middle: excess electron and hole concentrations; and bottom: electron and hole current components.

makes us expect a steady-state carrier profile with a shape as sketched in Fig. 5.16 in which the carrier concentration peaks at the generation point and decays to the equilibrium value at a certain distance. Far enough away from the point of generation, the semiconductor will not be aware that the region around $x = 0$ is being perturbed from equilibrium.

It is interesting to think about what happens to the photogenerated electrons. It is tempting to say that the photogenerated electrons diffuse away from the generation point, just as the photogenerated holes do. This is, however, not correct. The moment an electron is generated, it cannot be distinguished from any other one of the many majority electrons that were in the bar to begin with. The best way to think about what happens to the electrons is to realize that, by itself, the hole build up close to the generation point would drive the bar outside charge neutrality. The electric field that would result would get the electrons moving immediately in the direction of attempting to erase the drift field. This can be accomplished if the excess electron concentration precisely matches the hole electron concentration at every point. As we see next, however, this is not a sustainable situation.

If the semiconductor bar is reasonably extrinsic, we know that quasi-neutrality has to prevail. We concluded earlier in this chapter that this makes the total current constant at any cross section in the structure. Since far enough away from the generation point the semiconductor is not upset from equilibrium, the total current there is precisely zero. Hence $J_t = 0$ everywhere. Closer to $x = 0$, in the region invaded by excess carriers, the sum of the electron and hole currents has to cancel out. Summing Eqs. (5.51) and (5.53) and noting that $\mathcal{E}_o = 0$, we have

$$0 = J_t \simeq q n_o \mu_e \mathcal{E}' + q(D_e - D_h)\frac{dp'}{dx} \tag{5.56}$$

This equation suggests that if $D_e = D_h$ precisely, the electric field is exactly zero and there is no drift current of any kind. But, since in most semiconductors $D_e \neq D_h$, the diffusion currents do not cancel out. We therefore need a drift current to cancel the imbalance of the diffusion currents and provide $J_t = 0$ everywhere.

A null total current everywhere demands that p' be close but not exactly equal to n'. A small deviation from perfect charge neutrality is all that is needed to provide the necessary electric field that produces the required drift current. For $x > 0$, since $D_e > D_h$ and $\frac{dp'}{dx} < 0$, the diffusion term in Eq. (5.56) is negative. Hence, the drift term must be positive, which demands a positive electric field. This is established by having $n' < p'$ in the vicinity of $x = 0$, and $n' > p'$ far away from $x = 0$, as sketched in Fig. 5.16.

We can also conclude from this qualitative analysis that both $\mathcal{E}'$ and $|n' - p'|$ must be proportional to $D_e - D_h$. This is because it is the difference between D_e and D_h that drives the need for an electric field.

Having discussed the physics of this example in an intuitive way, let us now solve it completely. The symmetry of the situation simplifies our study to $x \geq 0$. The place to start is the minority carrier continuity equation of Table 5.5. In this uniformly doped bar, the initial electric field is zero and there is no applied field from the outside. As already argued above, the field term in the minority carrier continuity equation is negligible. Also in steady state, we can neglect the time derivative. Furthermore, with the exception of $x = 0$ (which we will treat below as a boundary condition), the bar is in the dark. The minority carrier continuity equation everywhere but at the origin becomes

$$\frac{d^2p'}{dx^2} - \frac{p'}{L_h^2} = 0 \tag{5.57}$$

where we have defined L_h as

$$L_h = \sqrt{D_h \tau} \tag{5.58}$$

L_h is called the *hole diffusion length*. Its significance will become clear very soon.

The solution of the differential Eq. (5.57) can take several mathematical forms. A suitable one in this example is the following:

$$p' = A \exp\frac{x}{L_h} + B \exp\frac{-x}{L_h} \tag{5.59}$$

This solution consists of a rising exponential and a decaying exponential in x. Since we know that eventually far away from the generation point the excess hole concentration disappears, A must be equal to zero.

We get B from considering the boundary condition at $x = 0$. A particularly easy way to think about it is to exploit the symmetry of the problem. At $x = 0$, there is a sheet of generation of holes at a rate g_l. Half of the generated holes diffuse to the right, and the other half diffuse to the left. Hence, at $x = 0^+$,

$$\frac{g_l}{2} = \frac{1}{q} J_h(0^+) = -D_h \frac{dp'}{dx}\bigg|_{x=0^+}. \tag{5.60}$$

Combining this equation with Eq. (5.59) (with $A = 0$), we easily get $B = g_l L_h/(2D_h)$ (this expression of B has the unit of cm^{-3}).

The excess hole concentration is then

$$p' = \frac{g_l L_h}{2D_h} \exp \frac{-x}{L_h} \tag{5.61}$$

This has the shape that was qualitatively predicted above. Equation (5.61) applies everywhere to the right of $x = 0$, and it also applies to $x = 0$, because p' cannot be discontinuous.

The hole current everywhere can now be easily calculated:

$$J_h = -qD_h \frac{dp'}{dx} = \frac{qg_l}{2} \exp \frac{-x}{L_h} \tag{5.62}$$

where we have assumed that $J_h(\text{drift}) \ll J_h(\text{diff})$.

The electron current can be found by first thinking of the total current J_t. As argued above, $J_t = 0$ everywhere. This means that the electron current is

$$J_e = -J_h = -\frac{qg_l}{2} \exp \frac{-x}{L_h} \tag{5.63}$$

In the electron current, we must consider both the drift and the diffusion components. The diffusion component is easy to obtain since the quasi-neutrality requirement implies that

$$n' \simeq p' = \frac{g_l L_h}{2D_h} \exp \frac{-x}{L_h} \tag{5.64}$$

and

$$J_e(\text{diff}) = qD_e \frac{dn'}{dx} = -\frac{qg_l}{2} \frac{D_e}{D_h} \exp \frac{-x}{L_h} \tag{5.65}$$

The drift component is simply the balance of the electron current minus the diffusion component:

$$J_e(\text{drift}) = J_e - J_e(\text{diff}) = \frac{qg_l}{2} \frac{D_e - D_h}{D_h} \exp \frac{-x}{L_h} \tag{5.66}$$

This is the result that was expected, that is, an electron drift current flows to the extent that D_e is different from D_h.

At this point, we have obtained expressions for the carrier and current distributions everywhere and the problem is completely solved. Still, we need to discuss the accuracy of the assumptions that were made and understand the constraints under which they work.

Let us first check the quasi-neutrality assumption. This requires obtaining the electric field distribution. From the electron drift current expression, we can easily get

$$\mathcal{E}' = \frac{J_e(\text{drift})}{q\mu_e n_o} = \frac{kT}{q} \frac{g_l}{2n_o} \frac{D_e - D_h}{D_e D_h} \exp \frac{-x}{L_h} \tag{5.67}$$

Using this expression of the electric field in Gauss' law, we can obtain the difference between n' and p':

$$p' - n' = -\frac{\epsilon kT}{q^2 n_o} \frac{g_l}{2L_h} \frac{D_e - D_h}{D_e D_h} \exp \frac{-x}{L_h} \tag{5.68}$$

which together with Eq. (5.64) can now be used to verify the quasi-neutrality condition:

$$\left| \frac{p' - n'}{p'} \right| = \left(\frac{L_D}{L_h} \right)^2 \frac{D_e - D_h}{D_e} \tag{5.69}$$

where L_D is the extrinsic Debye length defined in Chapter 4. Let us look at the order of magnitude of the various terms in this equation. The term involving the diffusion coefficients is slightly smaller than unity. The term involving the two length scales is many times smaller than one. For example, in Si with $N_D = 10^{16}$ cm^{-3} at room temperature, the hole diffusion length is 400 μm, while L_D is about 0.04 μm. Clearly, the difference between n' and p' is about eight orders of magnitude smaller than the sum of their values! The quasi-neutrality assumption is indeed extremely good.

We can now check our assumption of neglecting the drift contribution to the minority carrier current. Let's compute the relative magnitude of the hole drift current to the hole diffusion current:

$$\left| \frac{J_h(\text{drift})}{J_h(\text{diff})} \right| = \left| \frac{q\mu_h p' \mathcal{E}'}{-qD_h \frac{dp'}{dx}} \right| = \frac{1}{2} \frac{p'}{n_o} \frac{D_e - D_h}{D_e} \tag{5.70}$$

This equation says that this approximation is as good as the low-level injection condition. If the low-level injection condition is satisfied with a margin of 10 ($p' \simeq 0.1 n_o$), then the assumption that the drift contribution to the minority carrier current can be neglected is good to about 1 part in 10.

It is easy to see why these two conditions come hand in hand. As already argued, an electric field is required to the extent that D_e is different from D_h. In consequence, the resulting majority carrier drift current has an order of magnitude similar to the minority carrier diffusion current. All together, this implies that the relative magnitude of the minority carrier drift current with respect to the minority carrier diffusion current is about the same as the relative magnitude of p' with respect to n_o.

It is easy to calculate the maximum generation rate that satisfies the low-level injection condition. The worst location in the semiconductor is $x = 0$. For low-level injection to apply there, we need to demand that $p'(0) \ll n_o$. This implies that

$$g_l \ll \frac{2D_h n_o}{L_h} \tag{5.71}$$

A margin of safety of 10 in this inequality gives an error of about 10% in the computation of the current. This is sufficient for many applications.

We must also check the assumption that the drift field that is set is not so large that we need to be concerned with velocity saturation effects in the computation of the majority carrier current. The worst point is $x = 0$ where $\mathcal{E}'$ in Eq. (5.67) is maximum. At this point, using Eq. (5.67):

$$\mu_e \mathcal{E}' = \frac{g_l}{2n_o} \frac{D_e - D_h}{D_h} \ll \frac{D_e - D_h}{L_h} \tag{5.72}$$

where we have used the low-level injection condition Eq. (5.71). Typical values for the right-hand side of Eq. (5.72) are of the order of 500 cm/s, much smaller than the electron saturation velocity of 1×10^7 cm/s.

Two further comments before closing this example. First, a word about the physical meaning of the minority carrier diffusion length. The diffusion length is the average distance that a carrier travels by diffusion before recombining. In three diffusion lengths, for example, 95% of the excess carriers have recombined. The diffusion length is a very important length scale in problems in which minority carrier behavior plays a key role. More about this in Section 5.6.3.

Second is a simple calculation of the average velocity at which holes diffuse in the semiconductor bar. This is easy to obtain. From Eqs. (5.61) and (5.62), we have for $x > 0$:

$$v_h^{diff} = \frac{J_h^{diff}(x)}{qp(x)} \simeq \frac{J_h(x)}{qp'(x)} = \frac{D_h}{L_h} \tag{5.73}$$

which is independent of position. This is an important result. We will use it in the computation of the forward bias current in a PN diode.

5.6.2 Example 4: Diffusion and surface recombination in a "short" bar

Let us now consider another uniformly doped n-type bar illuminated by deep penetrating radiation over a narrow region at its center so that g_l electron–hole pairs are produced per unit second per unit area. In this case, the bar is of finite length L and it has end surfaces with infinite surface recombination velocity (Fig. 5.17). The length of the bar is much shorter than the diffusion length so that bulk recombination is very small. As in the above example, we seek to solve for the carrier concentrations and current densities everywhere in steady state.

Qualitatively we can see that this problem is similar to the previous one. The photogenerated holes diffuse away from the generation point at $x = 0$, half to each side. The important difference with the previous case is that before the holes have a chance to recombine with any electrons in the body of the bar, they reach the surfaces at $x = \pm L/2$. Since the surfaces display an infinite surface recombination velocity, all holes recombine there. In consequence, the excess hole concentration goes to zero at the surface. In other words, the surface is maintained in thermal equilibrium. The shape of the carrier concentration in this example falls again from a maximum at $x = 0$ to 0 at $x = \pm L/2$. Since there is no recombination in the bulk of the bar, the hole current is constant. This calls for a constant slope in the excess carrier concentration or a linear profile, as sketched in Fig. 5.17.

Let us now solve the problem quantitatively. The differential equation that governs hole flow in this case is simply:

$$\frac{d^2p'}{dx^2} = 0 \tag{5.74}$$

A general solution to this equation has the form:

$$p' = Ax + B \tag{5.75}$$

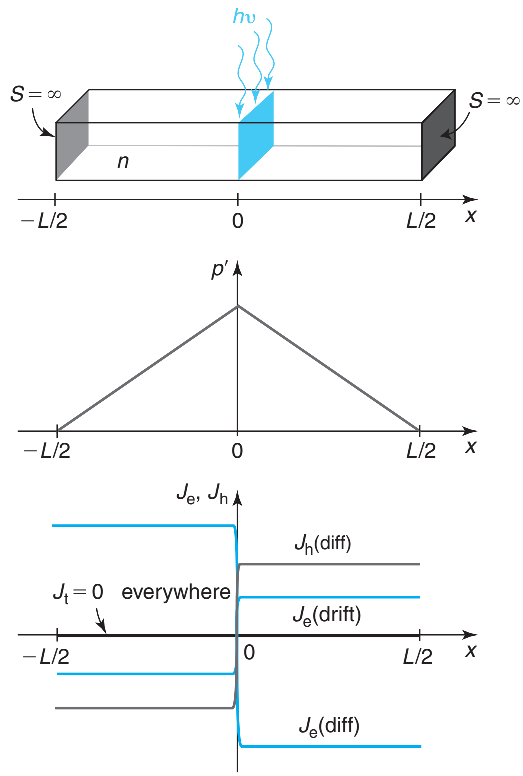


FIGURE 5.17
Top: a short bar illuminated in the middle by a sheet of generation; middle: excess hole concentration; bottom: electron and hole currents.

This problem has two boundary conditions. At $x = 0$, the boundary condition is identical to the previous case [Eq. (5.60)]. At the surfaces, the boundary condition is simply:

$$p'\left(\pm\frac{L}{2}\right) = 0 \tag{5.76}$$

For $x \geq 0$, the solution of the differential Eq. (5.74) is

$$p' = -\frac{g_1}{2D_\mathrm{h}}\left(x - \frac{L}{2}\right) \tag{5.77}$$

the result that we expected.

The hole current density is

$$J_\mathrm{h} = q\frac{g_1}{2} \tag{5.78}$$

This is a result that we could have written from the very beginning. If there is no recombination in the body of the bar, half of the generated carriers flow toward one ohmic contact and half flow to the other. The flow rate is then $g_1/2$ and the current density is $qg_1/2$.

Quasi-neutrality demands that $n' \simeq p'$, from which we can get the electron diffusion current to be

$$J_e(\text{diff}) = -q\frac{D_e}{D_h}\frac{g_l}{2} \tag{5.79}$$

In this example, the total current is also zero since the bar is not connected to anything. From this condition, we can get the electron drift current to be

$$J_e(\text{drift}) = \frac{qg_l}{2}\frac{D_e - D_h}{D_h} \tag{5.80}$$

All signs in Eqs. (5.78–5.80) are reversed for $x \le 0$.

Following a similar procedure to the previous example, the various assumptions can be verified and the maximum value of g_l allowed to maintain low-level injection conditions can be obtained.

Before closing, it is also of interest to compute the velocity at which holes diffuse through the bar. Using Eqs. (5.77) and (5.78), we get for $x > 0$:

$$v_h^{\text{diff}}(x) = \frac{J_h(x)}{qp(x)} \simeq \frac{D_h}{\frac{L}{2} - x} \tag{5.81}$$

When comparing this result with that obtained for the long bar in Eq. (5.73), we find that in the short bar, the hole diffusion velocity increases as the carriers approach the contact. This makes sense since the carrier concentration decreases but its slope is constant. A second interesting point is that the hole diffusion velocity diverges at the surface of the semiconductor. This is an artifact of our assumption that $p(x) \simeq p'(x)$. A finite velocity is obtained if we carry out a more careful analysis. In many situations, this is actually not needed and the result of Eq. (5.81) is quite adequate.

5.6.3 Length scales of minority carrier situations

The previous examples have revealed the existence of two characteristic lengths in minority-carrier type problems: the diffusion length L_{diff} (this is a generic symbol for both electrons and holes; when being specific, we use L_e for electrons and L_h for holes) and the sample length L.

L_{diff} is the average length that a carrier diffuses in a bulk semiconductor before it recombines. It was mathematically defined for holes in Eq. (5.58). A similar equation applies to electrons. L_{diff} makes a statement about the balance between diffusion and recombination. The more effective bulk recombination is (by having τ smaller), the shorter L_{diff} becomes.

Since both D and τ depend on doping, the diffusion length is a strong function of doping level. This is shown for Si at room temperature in Fig. 5.18. The lines in this figure come from combining the carrier lifetime data in Fig. 3.17 and the mobility data in Fig. 4.3, and from using the Einstein relation to obtain the diffusion coefficient from the mobility. We know that at low and moderate doping levels, the carrier lifetime is not a very tight function of doping level. In consequence, one should expect a wide range of possible values for the diffusion length in this same doping regime. The lines of Fig. 5.18 represent guidelines to reasonable values.

In a given situation, the smallest one of the two characteristic lengths, the sample length or the diffusion length, dominates. Section 5.6.1 showed an example in which the sample length is much larger than the diffusion length. The carrier profiles and the electrostatics of the problem

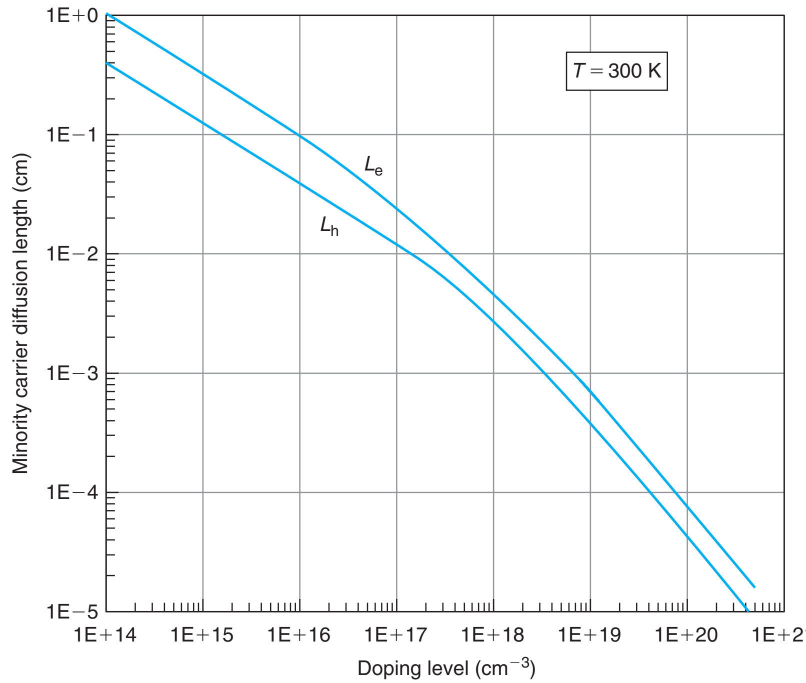


FIGURE 5.18
Minority carrier diffusion length in Si at room temperature. The lines represent typical values corresponding to the carrier lifetimes given by the lines of Fig. 3.17.

were entirely dominated by the diffusion length. In a situation like this, the semiconductor is said to be *long* from the minority carrier point of view. Section 5.6.2, on the other hand, presented an example in which the diffusion length was much longer than the sample length. In that case, the solution of the problem was entirely dominated by the sample size and bulk recombination was irrelevant. From the minority carrier point of view, the semiconductor in a situation like this is often referred to as *short*.

5.7 DYNAMICS OF MAJORITY CARRIER SITUATIONS

Time-dependent situations are particularly important in device operation. There are many cases in which we are interested in the dynamic response of a device to a time changing stimulus.

Time-dependent problems are mathematically quite difficult. Few of them result in simple analytical solutions. The goal of this section and the next is to learn to recognize in a given situation the various physical processes at play, to identify the relevant time constants, and to be able to compute order of magnitude values for these time constants.

The equation set that we derived to describe majority carrier situations in Table 5.4 does not contain any time-dependent terms. This would suggest that majority carrier situations are quasi-static, meaning that they respond instantaneously to outside stimuli without any memory of its previous state. This is a reasonable approximation in many practical situations. In an ideal integrated resistor such as the one described in Section 5.5.2, if we can disregard its parasitic capacitance, it is usually a fairly good assumption that the current follows the voltage instantaneously.

It is useful to understand where this important result derives from and what its limits are. If we look back at the simplified Shockley equations under quasi-neutrality (Table 5.3) and we assume, as we did in the simplifications leading to Table 5.4, that the carrier concentrations do not change in time, then there are no dynamics left in the description of the situation. The issue is then the quasi-neutrality approximation itself. It is in making this assumption that the time derivative of the volume charge density was dropped. The rational was that if the volume charge density is negligible, then its change in time is also negligible. But how good is this assumption?

In majority carrier situations, there are always small discrepancies to perfect charge neutrality. Net charge appears at contacts (see example in Section 5.5.1) and in the presence of dopant gradients. The quasi-neutrality assumption allows us to neglect this net charge because it is relatively small and/or it is confined to small regions in the scale of the sample size. In a dynamic situation, such as when a voltage step is applied to an integrated resistor, the net charge needs to change to satisfy the changing electrostatics. To accomplish this, some amount of charge needs to be delivered to the appropriate locations in the device. That takes a short but finite time. A derivation of the appropriate time for typical situations is performed in Advanced Topic AT5.2. There we find that the time constant for volume charge to relax is called the *dielectric relaxation time* and it is given by

$$\tau_{\mathrm{d}} = \frac{\epsilon}{\sigma} \tag{5.82}$$

where σ is the conductivity of the semiconductor region in question. The dependencies of this simple equation make sense when we realize that it is the electric field that makes charge move through the drift process.

The example in Section 5.5.1 provides us with a particularly intuitive way to derive this result. If we start with the sample at rest ($V = 0$) and at $t = 0$ we apply a voltage V, how much time does it take for the steady state charge distribution sketched at the top of Fig. 5.11 to be established? It is easy to estimate this. The total charge per unit area that needs to be delivered is obtained by plugging Eq. (5.40) into Eq. (5.38):

$$\mathrm{Q_s} = \frac{\epsilon V}{L} \tag{5.83}$$

The current density that delivers this charge is given by Eq. (5.33). Therefore, the time that it takes for this charge distribution to be built is

$$\tau = \frac{\mathrm{Q_s}}{J} = \frac{\epsilon}{qN_{\mathrm{D}}\mu_{\mathrm{e}}} \tag{5.84}$$

Since the denominator is the conductivity of this sample, this expression is identical to Eq. (5.82).

Figure 5.19 graphs τ_{d} for n- and p-type Si at room temperature. As the doping level increases, σ increases and τ_{d} decreases as a result. τ_{d} is indeed rather small for medium and highly doped regions. For example, for Si with a doping level of order 10^{16} cm^{-3}, τ_{d} is about 1 ps. In microelectronic devices, the doping levels are typically higher and they operate in timescales usually

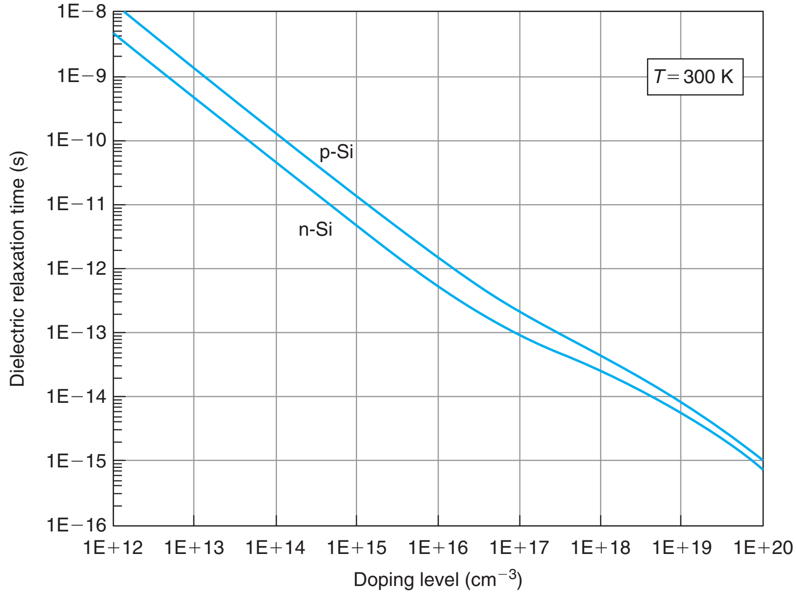


FIGURE 5.19
Dielectric relaxation time for n- and p-type Si at room temperature.

longer than this. We can then conclude that the charge redistribution that takes place in response to outside stimuli in a majority carrier situation occurs on a timescale that is much shorter than our timescale of interest. In this timescale, we can assume that majority carrier situations are indeed quasi-static.

EXERCISE 5.4

Up to what frequency can a $1\ \Omega\cdot$cm *n-type Si substrate be considered quasi-static to the application of electric fields?*

The dielectric relaxation time for this substrate is

$$\tau_d = \frac{\epsilon}{\sigma} = \epsilon\rho = 11.7 \times 8.85 \times 10^{-14} \times 1 = 1.0 \times 10^{-12}\ \text{s} = 1\ \text{ps}$$

The inverse of the dielectric relaxation time divided by 2π is a reasonable estimate of the maximum frequency at which this substrate can be considered quasi-static:

$$f_d \simeq \frac{1}{2\pi\tau_d} \simeq 150\ \text{GHz}$$

This is a frequency substantially beyond most applications of Si today (but perhaps not in the future). Hence, this substrate will respond in a perfectly quasi-static way to the application of electric fields.

Had we worked with a 100 $\Omega \cdot$ cm substrate, the frequency in question would have been 1.5 GHz. This is in the range of many applications today. The response of such a substrate to electric fields will be far from quasi-static and would have to be taken into account in equivalent circuit models of devices.

5.8 DYNAMICS OF MINORITY CARRIER SITUATIONS

In minority carrier situations, there are excess carriers that can exhibit substantial memory effects. These typically dominate the dynamics of minority carrier devices. To first order, this can be understood from the fact that carrier lifetimes are long, in the nanosecond to microsecond range. This argument is misleading, however, since often, the dynamics of minority carriers are not controlled by the recombination process but by their transport through a certain region. In this case, the proper time constant is not the lifetime but the *transit time*. Understanding this is the central goal of this section.

5.8.1 Example 5: Transient in a bar with $S = \infty$

Consider a uniformly doped n-type semiconductor bar of length L as in Fig. 5.20. The two surfaces of the bar are characterized by an infinite surface recombination velocity. For $t \leq 0$, the bar is illuminated with radiation that generates carriers uniformly everywhere at a rate g_l. At $t = 0$, the radiation is switched off. We are interested in the time evolution of the excess carrier profile everywhere in the bar. In particular, we want to identify the dominant time constant of the decay of excess carriers.

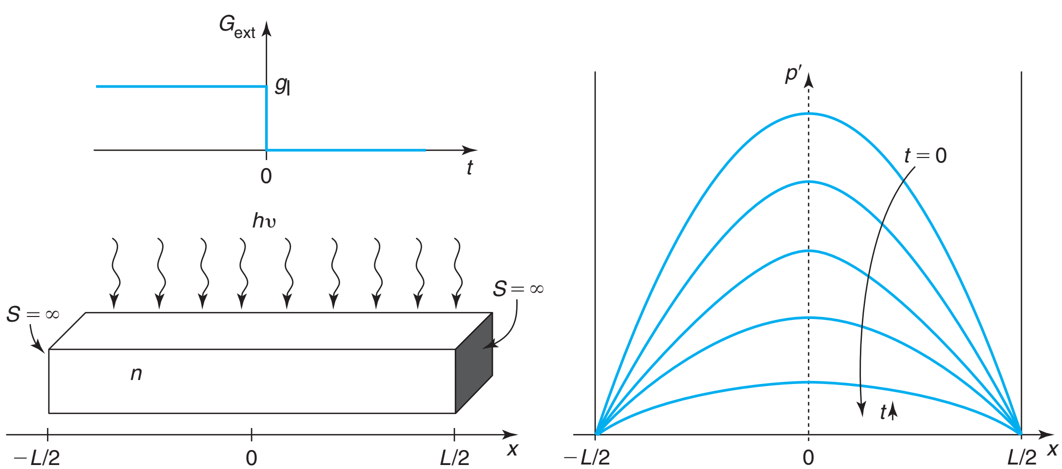


FIGURE 5.20
Sketch of time decay of excess carrier concentration in a semiconductor bar after a uniform generation function has been turned off. The surfaces are characterized by $S = \infty$.

We studied time transients in Chapter 3. The types of problems that we dealt with at that time were uniform, that is, nothing changed in space. Their characteristic time constant was the carrier lifetime τ. The present problem might appear similar at first sight. After all, the radiation is of uniform intensity in space and bulk recombination takes place everywhere in the bar. However, the presence of the surfaces with $S = \infty$ changes the situation. Excess carriers in the vicinity of these surfaces can recombine faster than in the bulk if it takes them a shorter time to diffuse to the surface. The recognition of this simple fact immediately allows us to conclude that (1) the characteristic time constant of the decay of the excess carrier concentration is *shorter* than τ and (2) the time constant associated with surface recombination is directly proportional to the length of the sample and inversely proportional to the diffusion coefficient (the longer the sample and the smaller the diffusion coefficient, the less effective surface recombination is relative to bulk recombination).

The mathematical solution of this problem is interesting. The procedure to follow is general and widely used for different kinds of transient problems. The first step is to compute the steady state excess carrier profile that exists before the pulse is turned off. Only then, the decay of carrier concentration can be studied. The problem has symmetry around $x = 0$, so only for $x \geq 0$ a solution is needed.

For $t \leq 0$, the differential equation that governs this problem is

$$D_{\mathrm{h}} \frac{d^2 p'}{dx^2} - \frac{p'}{\tau} + G_{\mathrm{ext}} = 0 \tag{5.85}$$

Substituting $G_{\mathrm{ext}} = g_{\mathrm{l}}$, and dividing all terms by D_{h}, we get

$$\frac{d^2 p'}{dx^2} - \frac{p'}{L_{\mathrm{h}}^2} + \frac{g_{\mathrm{l}}}{D_{\mathrm{h}}} = 0 \tag{5.86}$$

A solution to this differential equation is[3]

$$p' = A \cosh \frac{x}{L_{\mathrm{h}}} + B \sinh \frac{x}{L_{\mathrm{h}}} + C \tag{5.87}$$

Substitution of this expression into the differential Eq. (5.86), immediately gives $C = g_{\mathrm{l}}\tau$. Symmetry around $x = 0$ imposes the boundary condition $dp'/dx|_{x=0} = 0$. Application to Eq. (5.87) results in $B = 0$. The infinitely recombining surface at $x = L/2$ demands that $p'(L/2) = 0$. This implies that $A = -g_{\mathrm{l}}\tau/\cosh \frac{L}{2L_{\mathrm{h}}}$. All together, the steady-state solution at $t = 0$ is

$$p'(x,0) = g_{\mathrm{l}}\tau \left(1 - \frac{\cosh \frac{x}{L_{\mathrm{h}}}}{\cosh \frac{L}{2L_{\mathrm{h}}}}\right) \tag{5.88}$$

For $t \geq 0$, the governing differential equation is

$$D_{\mathrm{h}} \frac{\partial^2 p'}{\partial x^2} - \frac{p'}{\tau} = \frac{\partial p'}{\partial t} \tag{5.89}$$

[3] A solution of the form $p' = A \exp \frac{x}{L_{\mathrm{h}}} + B \exp \frac{-x}{L_{\mathrm{h}}} + C$ is also valid but results in more complex algebra.

In a standard manipulation, we first perform a change of variables. If we define

$$p' = P \exp\left(-\frac{t}{\tau}\right) \tag{5.90}$$

the differential equation that P satisfies becomes

$$D_{\mathrm{h}} \frac{\partial^2 P}{\partial x^2} = \frac{\partial P}{\partial t} \tag{5.91}$$

This equation can be attacked by the method of separation of variables. Let us postulate that the solution to Eq. (5.91) is of the form:

$$P(x,t) = X(x)T(t) \tag{5.92}$$

where X only depends on x and T only depends on t. Substitution of Eq. (5.92) in Eq. (5.91) leads to two separate differential equations:

$$\frac{1}{X}\frac{d^2X}{dx^2} = \frac{1}{D_{\mathrm{h}}}\frac{1}{T}\frac{dT}{dt} = -\frac{1}{\lambda^2} \tag{5.93}$$

The first differential equation is only in terms of the independent variable x. The second one depends only on t. The only way for them to equal each other is if they are also equal to a constant. As we will see in a moment, this separation constant must be negative. To force that, we denote it as $-1/\lambda^2$. We now proceed to look at these two equations separately.

The time-dependent equation has a solution:

$$T = K_1 \exp\left(-\frac{D_{\mathrm{h}}}{\lambda^2}t\right) \tag{5.94}$$

which is a decaying exponential. If the separation constant in Eq. (5.93) was positive, the solution given by Eq. (5.94) would be increasing with time, which is unphysical. Similarly, a separation constant of zero would imply that the only time transient is the one identified in Eq. (5.90), which we have already argued is insufficient.

The differential equation in x given by Eq. (5.93) has a solution:

$$X = K_2 \cos\frac{x}{\lambda} + K_3 \sin\frac{x}{\lambda} \tag{5.95}$$

The boundary conditions that it must satisfy are the same as for $t \leq 0$. $dX/dx|_{x=0} = 0$ implies that $K_3 = 0$. $X(L/2) = 0$ can only be satisfied if $\cos(L/2\lambda) = 0$, which in turn, imposes a restriction on the values that λ can take

$$\lambda_n = \frac{L}{(2n-1)\pi} \quad \text{for } n = 1, 2, 3, \ldots \tag{5.96}$$

We actually find that there are multiple solutions to this problem, each one characterized by a different separation constant given in Eq. (5.96). Assembling Eqs. (5.90), (5.92), (5.95), (5.94), and (5.96), the expression of one of these solutions is

$$p'_{\rm n}(x,t) = K_n \exp\left(-\frac{t}{\tau}\right) \exp\left(-\frac{D_{\rm h}}{\lambda_{\rm n}^2}t\right) \cos\frac{x}{\lambda_{\rm n}} \tag{5.97}$$

where we have absorbed all proportionality constants into one.

The most general solution of this problem is then constructed by the superposition of all these solutions:

$$p'_{\rm n}(x,t) = \exp\left(-\frac{t}{\tau}\right) \sum_{\rm n} K_{\rm n} \exp\left(-\frac{D_{\rm h}}{\lambda_{\rm n}^2}t\right) \cos\frac{x}{\lambda_{\rm n}} \quad \text{for } n = 1, 2, 3, \ldots \tag{5.98}$$

The weight given to each of these components is such that at $t = 0, p'(x, 0)$ given in Eq. (5.98) equals the solution derived above in Eq. (5.88). From Fourier analysis, it is straightforward to derive expressions for $K_{\rm n}$:

$$K_{\rm n} = \frac{1}{L}\int_{-{\rm L}/2}^{L/2} p'(x,0)\,\cos\left[\frac{(2n-1)\pi}{L}x\right] dx = \frac{4g_{\rm l}\tau \sinh\frac{L}{2L_{\rm h}}}{(2n-1)^2\pi^2\frac{L_{\rm h}}{L} + \frac{L}{L_{\rm h}}} \tag{5.99}$$

Let us analyze separately the spatial dependence and the time dependence of this result. At any time t, the solution [Eq. (5.98)] looks like a sum of cosine terms of different wavelengths given by Eq. (5.96) and different weights given by Eq. (5.99). A sketch of the first three modes ($n = 1,\ 2,\ 3$) is shown in Fig. 5.21. We think of $\lambda_{\rm n}$ as the wavelength of each of these modes.

The time evolution of the profile is not as straightforward as the case of pure bulk recombination that exhibits a simple exponential decay determined by a single time constant. Rather, the complete solution is the sum of many independent exponentially decaying functions, each characterized by a different characteristic time. The nth component exhibits a time constant:

$$\frac{1}{\tau_{\rm n}} = \frac{1}{\tau} + \frac{D_{\rm h}}{\lambda_{\rm n}^2} = \frac{1}{\tau} + D_{\rm h}\left[\frac{(2n-1)\pi}{L}\right]^2 \quad \text{for } n = 1, 2, 3, \ldots \tag{5.100}$$

All of these time constants are smaller than the carrier lifetime. The higher the order of the component, the shorter $\lambda_{\rm n}$ and the smaller is the characteristic time constant. In consequence,

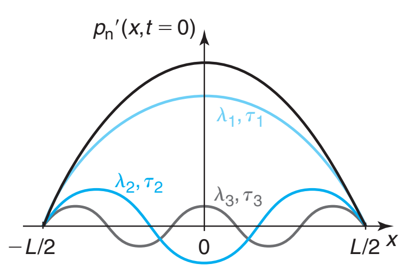


FIGURE 5.21
Spatial dependence of $p'_{\rm n}(x,t)$ and its first three modes.

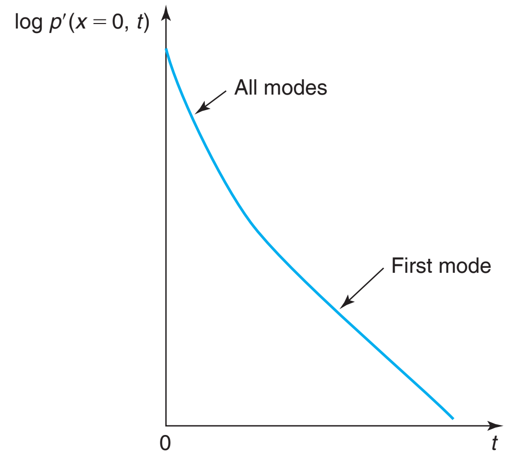


FIGURE 5.22
Sketch of time decay of excess carrier concentration at the center of a semiconductor bar after a uniform generation function has been turned off. After a fast initial decay, the dominant first mode emerges and controls the rest of the time evolution. The experimental data graphed in Fig. 3.22 shows this behavior.

after the light is switched off, there is an initial fast decay of the carrier profile that is dominated by the higher-order modes. After a short time, the higher-order terms become very small and the decay of the carrier profile from then on is dominated by the behavior of the first-order component. This is illustrated at the center of the bar in Fig. 5.22. This interesting time evolution is actually what is obtained in practice, as you can see in the experimental data of Fig. 3.22.

The time constant of the first-order mode is

$$\frac{1}{\tau_1} = \frac{1}{\tau} + D_{\mathrm{h}} \left(\frac{\pi}{L}\right)^2 \tag{5.101}$$

Since this is the longest one, this is the dominant time constant of this problem. This time constant has all the features that we expected. It consists of two terms, indicating that there are two recombination processes taking place in parallel. One of the processes is bulk recombination characterized by a time constant τ, the carrier lifetime. The other process is surface recombination characterized by a time constant $L^2/\pi^2 D_{\mathrm{h}}$. The dependencies of this surface time constant are exactly what we expected. It is directly proportional to the length of the sample and inversely proportional to the diffusion coefficient. In fact, if you work it out (see Problem 5.28), you will see that $L^2/\pi^2 D_{\mathrm{h}}$ is the *transit time* of holes through the sample to its surface. That is, the average time it takes for a photogenerated hole to diffuse from the point of generation to the sample surface where it recombines. In a more general way, we could then write

$$\frac{1}{\tau_1} = \frac{1}{\tau} + \frac{1}{\tau_{\mathrm{t}}} \tag{5.102}$$

where τ_t is the transit time. τ_t is given by

$$\tau_t = \frac{L^2}{\pi^2 D_h} \tag{5.103}$$

The concept of transit time was already presented in Section 4.4. The numerical factor in the expression of the transit time (π^2 in this case) depends on the details of the excess carrier profile ($\cos(\pi x/L)$) for this example; the linear profile discussed in Section 4.4 resulted in a numerical factor of 2). The key dependencies, however, are the same. The transit time is always proportional to the *square of the sample length* and inversely proportional to the diffusion coefficient. As a reminder, the L^2 dependence arises from: (i) the distance that carriers must diffuse is on the order of L and (ii) the concentration gradient that drives carrier diffusion is sharper if L is shorter.

Coming back to Eq. (5.101), the time constant of the first-order mode is given by the inverse of the sum of the inverse of the bulk lifetime and the transit time. This suggests that the *smallest* of these two characteristic time constants dominates the decay of the excess carrier concentration in this bar. This is understandable. In this problem, there are two recombination processes operating in parallel: bulk recombination and surface recombination. Whichever one is most effective (i.e., is characterized by the smallest time constant), dominates the overall decay. This implies, then, that in general $\tau_1 < \tau$. In the limit of slow bulk recombination, $\tau_1 \simeq \tau_t$, that is, the decay of the excess minority carrier concentration is dominated by their transit to the surface.

The key results of this section apply to other situations with identical boundary conditions. In essence, what we have done here is to examine the time decay of the Fourier components of the minority carrier profile at $t = 0$. In a situation characterized by a different spatial distribution of the generation function but identical minority carrier boundary conditions to those studied in this section, the carrier profile at $t = 0$ can be decomposed in a sum that is composed of the same Fourier terms as Eq. (5.98). The coefficients associated with each time constant are likely to be different to those derived in this section, but the time constant of each of the Fourier terms is identical. In particular, the expression of the time constant of the dominant mode should follow Eq. (5.101).

EXERCISE 5.5

Consider a situation like the one depicted in Fig. 5.20. This is an ideal one-dimensional situation involving an n-type Si bar of length $L = 100$ µm *with a doping level* $N_D = 10^{17}$ cm^{-3} *at room temperature. The surfaces located at* $-L/2$ *and* $L/2$ *are characterized by* $S = \infty$. *Estimate the value of the dominant time constant for the decay of excess minority carrier concentration after the light illumination is turned off. Assume low-level injection conditions.*

Using the fit in Fig. 3.17 for the carrier lifetime in n-type Si, for this doping level, we estimate a carrier lifetime of $\tau \simeq 13$ µs. Using the fit to the hole mobility given in Appendix E, we estimate a hole mobility in this bar of $\mu_h \simeq 455$ cm^2/V · s. This implies a hole diffusion coefficient of $D_h \simeq 12$ cm^2/s.

The time constant associated with the transit of holes to the surfaces where they recombine is then

$$\tau_t = \frac{L^2}{\pi^2 D_h} = \frac{(0.01\ \text{cm})^2}{\pi^2 \times 12\ \text{cm}^2/\text{s}} = 0.85\ \mu\text{s}$$

The dominant time constant of the excess hole decay is given by

$$\tau_1 = \frac{1}{\frac{1}{\tau} + \frac{1}{\tau_t}} = \frac{1}{\frac{1}{13\ \mu s} + \frac{1}{0.85\ \mu s}} = 0.8\ \mu s$$

In this situation, the fastest process is hole transit to the surfaces where they recombine. Therefore, the hole transit time nearly completely sets the value of the dominant time constant.

5.9 TRANSPORT IN SPACE-CHARGE AND HIGH-RESISTIVITY REGIONS

If we examine a semiconductor device from a space charge point of view, we will notice that there are regions with a very small volume charge, called *quasi-neutral regions* (QNRs), and other regions where there is substantial spatial charge called *space-charge regions* (SCRs). For example, in a PN junction, two quasi-neutral p and n regions are separated by a high-resistivity SCR located around the "metallurgical" interface where the doping level changes from p to n. In a MOSFET, an SCR also exists underneath the inversion layer.

As we have extensively discussed, QNRs are characterized by a high carrier concentration and low resistivity. SCRs, on the other hand, exhibit very low carrier concentrations and therefore have a high resistivity. Carrier transport through SCRs is of a markedly different nature than through QNRs. The dielectric relaxation time in a high-resistivity region can be very long. For example, 10 Ω · cm Si has a dielectric relaxation time of about 4 ns. This means that majority carriers can take a very long time, relative to other relevant time constants, to screen out a charge perturbation. Similarly, the Debye length can be very long in an SCR. For a carrier concentration of 10^{12} cm^{-3}, the Debye length is about 4 μm. This implies that SCRs can sustain net charge over substantial dimensions.

SCRs belong to a general class of high-resistivity regions with very low carrier concentrations. In these, transport is of a rather different nature than in QNR's. Additionally, and largely due to the low carrier concentrations, large electric fields are often present (that is the case of SCRs, in fact) or are applied from the outside.

A treatment of transport in high-resistivity regions has to be based on the Shockley equations *before* the quasi-neutral approximation was made (see Fig. 5.7). That is, we start with the equation set of Table 5.2. This brings us back to a system of equations that is highly coupled. Our strategy for uncoupling these equations again is to look at simplifying Gauss' law, although in a different way than in the QNR case. The one key feature of high-resistivity regions that we can exploit here is the fact that due to the low carrier concentrations that exist, *the electric field is independent of the carrier concentrations*. The electric field is typically set by ionized dopants (the case of SCRs) or by an external applied voltage.

A corollary of this is that *the behavior of electrons and holes in a high-resistivity region is largely uncorrelated*. The reason for this is that the coupling between the electron and hole profiles is made by the electric field. This is most clear in a quasi-neutral region in which the need to maintain $\rho \simeq 0$, tightly binds up n and p. In high-resistivity regions, on the contrary, the electric field is independent of the carrier concentrations and hence n and p are independent of each other.

Table 5.6

Equation set for space-charge and high-resistivity regions. In the entry that corresponds to Gauss' law (top line), the equation on the left applies to space-charge regions. The equation on the right applies to high-resistivity regions.

$\frac{\partial \mathcal{E}}{\partial x} \simeq \frac{q}{\epsilon}(N_D - N_A)$	or	$\mathcal{E} = \mathcal{E}'$
	$J_e = -qnv_e^{drift}(\mathcal{E}) + qD_e \frac{\partial n}{\partial x}$	
	$J_h = qpv_h^{drift}(\mathcal{E}) - qD_h \frac{\partial p}{\partial x}$	
	$\frac{\partial n}{\partial t} = G_{ext} - U(n,p) + \frac{1}{q}\frac{\partial J_e}{\partial x}$	
	$\frac{\partial p}{\partial t} = G_{ext} - U(n,p) - \frac{1}{q}\frac{\partial J_h}{\partial x}$	
	$J_t = J_e + J_h$	

A simplified set of Shockley equations for SCR and high-resistivity regions is in Table 5.6. There are two entries for Gauss' law (top line). The first one applies to space-charge regions in which the electric field is determined by the ionized impurity distributions. The second one corresponds to high-resistivity regions in which the electric field is set by the application of an external voltage.

Typically, this equation set is to be solved in the following way. The electric field is first obtained using the appropriate version of Gauss' law, as explained in the previous paragraph. Then, the electron and hole profiles are determined by solving the equation that results from plugging the current equation into the corresponding continuity equation. Finally, the current equation is used to derive the current.

The procedure is particularly easy for a steady-state situation. For electrons, for example, integrating the electron continuity equation from a point x_1 to another point x_2 yields

$$J_e(x_1) - J_e(x_2) = -q \int_{x_1}^{x_2} [G_{ext} - U(n,p)]dx \tag{5.104}$$

In situations in which there is external generation, we can often neglect the net recombination inside the region in question since n and p tend to be relatively small. In this case, we have

$$J_e(x_1) - J_e(x_2) = -q \int_{x_1}^{x_2} G_{ext}\, dx \tag{5.105}$$

from which the current can be quickly obtained.

An example is the best way to illustrate these procedures.

5.9.1 Example 6: Drift in a high-resistivity region under external electric field

This is a one-dimensional problem. Consider a high-resistivity chunk of n-type semiconductor of length L with metallic contacts applied to it. Let us assume that these contacts are ideal, that is, the bulk properties of the semiconductor extend all the way to the interface with the metal. In equilibrium, the semiconductor is charge neutral with $n_o = N_D$ and $p_o = n_i^2/n_o$.

Let us now apply a voltage V between the two contacts. This produces a uniform electric field $\mathcal{E} = V/L$ across the sample. This case appears similar to the situations that we studied in Section 5.5. In response to the electric field carriers drift and current flows through the structure. In general, the current density is given by $J = q(-n_o v_e^{drift} + p_o v_h^{drift})$, where the drift velocities v_e^{drift} and v_h^{drift} depend on the magnitude of the electric field. If N_D is small, this current can be very small. For example, if $n_o = 10^{12}$ cm^{-3}, the maximum current density in a Si sample at room temperature is obtained under velocity saturation conditions and is about 1.6 A/cm^2 (for a sense of proportion, typical microelectronic devices operate with current densities as high as 10^5 A/cm^2).

Let us now illuminate the sample with a very narrow beam of photons that produces g_l electron–hole pairs per unit area per second uniformly across the entire section of the sample at a location x_o, as sketched in Fig. 5.23. What is the resulting carrier concentration? How much current flows through the sample?

The electric field that exists inside the sample separates the photogenerated carriers. The photogenerated electrons drift to the positive terminal, while the holes drift to the negative terminal. If recombination is negligible, under steady-state circumstances, the electron current density is simply:

$$J_e = qg_l \quad \text{for } x < x_o \tag{5.106}$$

$$J_e = 0 \quad \text{for } x > x_o \tag{5.107}$$

Similarly, the hole current density is

$$J_h = 0 \quad \text{for } x < x_o \tag{5.108}$$

$$J_h = qg_l \quad \text{for } x > x_o \tag{5.109}$$

In consequence, the total current density is

$$J_t = qg_l \tag{5.110}$$

everywhere, as required by the continuity equations. This is sketched in Fig. 5.23.

The excess carrier densities are not difficult to compute. Excess electrons are only to be found to the left of the generation point. In that region, since the electron current is due to electron drift and is constant, the electron concentration is simply uniform in space. Hence

$$n' \simeq \frac{g_l}{v_e^{drift}(\mathcal{E})} \quad \text{for } 0 < x < x_o \tag{5.111}$$

$$n' \simeq 0 \quad \text{for } x_o < x < L \tag{5.112}$$

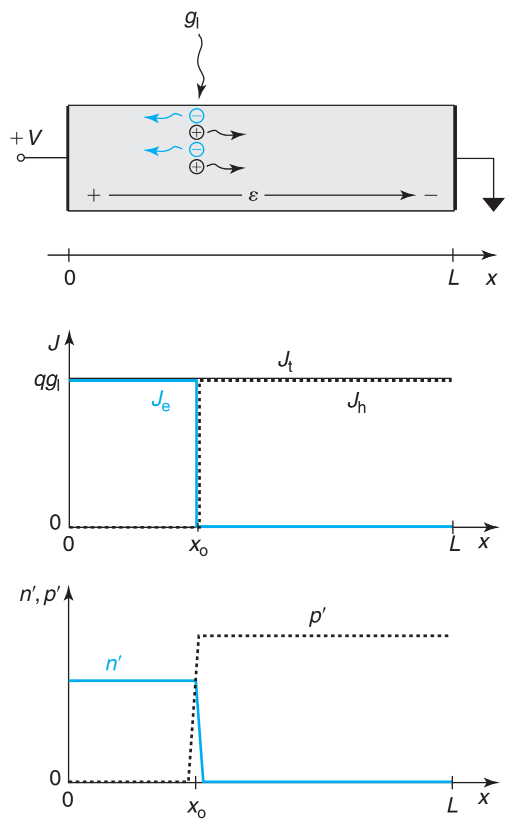


FIGURE 5.23
High-resistivity semiconductor bar under an applied electric field and a steady-state photogeneration at x_o. The middle diagram shows the carrier current densities. The bottom diagram shows the excess carrier concentrations.

Similarly for holes:

$$p' \simeq 0 \qquad \text{for } 0 < x < x_o \tag{5.113}$$

$$p' \simeq \frac{g_l}{v_h^{drift}(\mathcal{E})} \qquad \text{for } x_o < x < L \tag{5.114}$$

Since for a given electric field, in general $v_h^{drift} < v_e^{drift}$, it follows that $p' > n'$, as sketched in Fig. 5.23.

Around x_o some diffusion takes place against the electric field. In consequence, the excess carrier profile softly drops to zero. The actual shape can be computed in a similar way as in Section AT5.3.1.

The excess carrier situation depicted in Fig. 5.23 is no longer quasi-neutral. The intense electric field has separated the photogenerated carriers. A charge dipole has been established across the generation point.

In the derivation of Eqs. (5.111–5.114), we did not need to make the low-level injection approximation. In fact, since the doping level is so low, it is very likely that the carrier concentrations greatly exceed the doping level. The one assumption that we have implicitly made is that the carrier concentrations are not so large that they modify the electric field that was set up inside the bar by the application of the voltage. Without this assumption, the problem would be a lot more difficult since the field is modified by the carrier concentrations that in turn are determined by the net magnitude of the field. These kinds of situations are known as *space-charge-limited transport* and their study is beyond the objectives of this book.

5.9.2 Comparison between SCR and QNR transport

The best way to appreciate the peculiarities of SCR transport is to compare it with transport in a quasi-neutral region. Consider the example of Fig. 5.24. Shown are two uniformly doped n-type semiconductor bars with ohmic contacts at their ends. A voltage across both ohmic contacts is applied. At some location in the bar, a narrow beam of light generates electron–hole pairs at a constant rate. The bar on the left has a very low doping level and a high resistivity, as discussed in the previous subsection. The bar on the right has a low resistivity and can be considered quasi-neutral and under low-level injection. Let us assume that the bar is "short" as compared to the minority carrier diffusion length. Both are steady-state situations.

Below each bar, the excess carrier concentrations and the current densities are displayed. The case on the left has been discussed in the previous subsection in detail. The electric field separates the photogenerated carriers and a current density $J_t = qg_l$ is established along the bar and the outside circuit.

The case on the right requires a bit more thinking. This is a mixed minority-majority carrier situation. We certainly know that there is no carrier separation in this case, since quasi-neutrality imposes $n' \simeq p'$. What about the current? Since the sample has significant doping, there must certainly be a majority carrier current that results from the application of the voltage, as studied in Section 5.5. But does carrier generation also result in external current? The best way to answer this question is to go back to the general equation set for quasi-neutral carrier transport of Table 5.3. If, for simplicity, we assume that the electric field is small enough that mobility-type drift applies, summing J_e and J_h we obtain the following expression for the total current:

$$J_t = J_e + J_h = q(n\mu_e + p\mu_h)\mathcal{E} + qD_e\frac{dn}{dx} - qD_h\frac{dp}{dx} \tag{5.115}$$

Since the bar is uniformly doped, n_o and p_o do not depend on position and $\mathcal{E}_o = 0$. Also, since the bar is n-type, $n_o \gg p_o$. Equation (5.115) then simplifies to

$$J_t \simeq qn_o\mu_e\mathcal{E}' + qD_e\frac{dn'}{dx} - qD_h\frac{dp'}{dx} \tag{5.116}$$

Let us now integrate this equation along the bar, from $x = 0$ to $x = L$:

$$J_tL \simeq qn_o\mu_eV + qD_e[n'(L) - n'(0)] - qD_h[p'(L) - p'(0)] \tag{5.117}$$

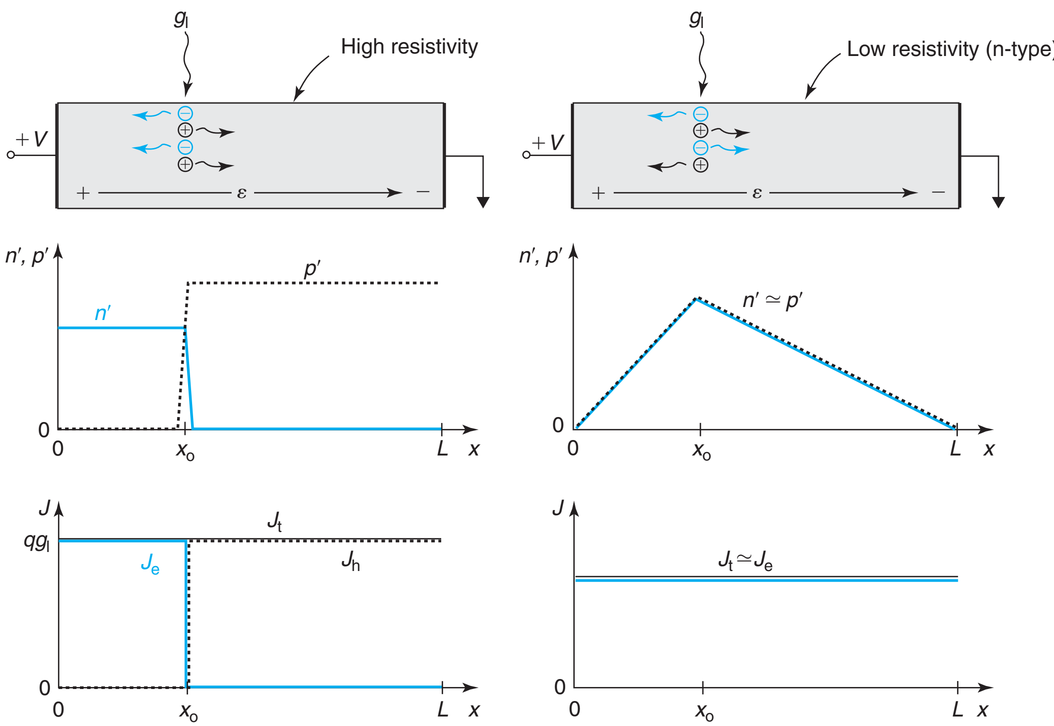


FIGURE 5.24
Comparison of SCR transport versus QNR transport. On the left is a high-resistivity semiconductor bar. On the right is a low-resistivity semiconductor bar that is shorter than the diffusion length. Both are under an applied electric field and steady-state photogeneration at x_o. The middle set of diagrams shows the excess carrier concentrations. The bottom diagram shows the carrier current densities.

Since at $x = 0$ and $x = L$ the bar has ohmic contacts where excess carrier concentrations cannot be sustained, $n'(0) = n'(L) = p'(0) = p'(L) = 0$. Hence, solving for the current, we find

$$J_t = \frac{qn_o\mu_e}{L}V \tag{5.118}$$

This is precisely the result that we obtained in Section 5.5 for an identical bar but in the absence of carrier generation. Hence, in the low-resistivity sample, carrier photogeneration does not result in external photocurrent through the circuit. Only a majority-carrier-type current flows as a result of the application of a voltage. In contrast, carrier separation in the high-resistivity sample results in a photocurrent that is directly proportional to the generation rate, as given by Eq. (5.110). In other words, in the high-resistivity sample, the photogenerated carriers are spatially separated. Because of this and also as a consequence of the low background carrier concentration, recombination cannot take place and all the photogenerated carriers get extracted out by the electric field. In contrast, in the low-resistivity sample, all the photogenerated carriers recombine at the ohmic contacts and no photogenerated current results.[4]

5.10 CARRIER MULTIPLICATION AND AVALANCHE BREAKDOWN

If a high electric field is present in a semiconductor sample, impact ionization might take place. Our analysis so far has not accounted for it. The goal of this section is to understand its implications.

As we discussed in Chapter 3, the driving force behind impact ionization is the carrier flux. When impact ionization is significant in a certain semiconductor region, the current that flows through that region can be substantially higher than expected from the analysis carried out so far in this chapter. In addition to the current contributed by the primary carriers, our focus up to this point, we must account for the secondary carriers that are generated by impact ionization. If the electric field is high enough, there might well be tertiary carriers, generated by the secondary carriers, and so on. This is sketched in Fig. 5.25. This process is called *carrier multiplication* and it is much like an atomic fission chain reaction in which neutrons, one of the products of the fission of an atom, can cause the fission of additional atoms.

The analogy between carrier multiplication and a nuclear chain reaction can be extended one step further—carrier multiplication can go "critical." If the electric field is high enough, the secondary carriers, both electrons and holes, generated by the impact ionization of the primary carriers can themselves generate more carriers. At the same time, the primary carriers continue generating more secondary carriers as they drift in the high-field region. If the electric field is

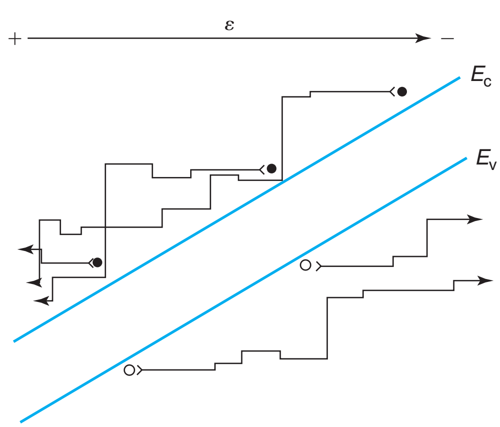


FIGURE 5.25
Sketch of carrier multiplication produced by the impact ionization in a semiconductor region with a uniform electric field.

[4]Note that there is a small *photoconductive* current in the low-resistivity sample that arises from the small modification of the conductivity of the sample as a result of carrier generation. This current is much smaller than qg_1, the maximum photogenerated current that can be produced.

high enough, a runaway situation or *carrier avalanche* can arise in which the total current grows exponentially. If left unchecked, the current density or the power dissipation can reach levels that cause device destruction. This is called *avalanche breakdown*.

Avalanche breakdown is the dominant breakdown mechanism in microelectronic devices. The importance of understanding and correctly modeling avalanche breakdown cannot be overemphasized. A miscalculation here might well mean severe device degradation or, in a worst case, device destruction. Even if the maximum current that can flow through a device is limited by some means, such as an outside circuit, so that device destruction is averted, a mode of operation in which avalanche breakdown occurs in a device is largely useless since large and uncontrolled currents flow.[5] In the design of just about any microelectronic device, the need to avoid entering, even getting close to, avalanche breakdown imposes design criteria that limit the performance of the device. For example, for a given base design, the frequency response of a bipolar transistor is essentially set by the design of the collector. This is also what controls the breakdown voltage of the device. The performance/breakdown trade-off is one of the most important design compromises that a device designer must carefully weigh.

Carrier multiplication, even if mild enough so that it does not lead to avalanche breakdown, is an important phenomenon that demands careful assessment in a device. In devices for analog applications, carrier multiplication becomes accompanied by substantial noise. In MESFETs, it results in unwanted gate current. In floating-body silicon-on-insulator (SOI) MOSFETs, it leads to the so-called kink effect.

One might think that carrier multiplication could be an excellent amplification mechanism, to implement a "semiconductor photomultiplier" for electrical and optical signals. This would be the case if impact ionization was only triggered by one type of carrier. Consider, for example, a situation where electrons are flowing through a certain semiconductor region. If the electric field increases to the point that significant electron impact ionization takes place, more electrons are produced. In consequence, the total electron current increases in a manner that is proportional to the incoming one. The trouble is with the holes generated by impact ionization. If along the way they also acquire enough energy from the electric field so that they can produce impact ionization events themselves, the tertiary electrons that are generated are not in phase with the primary electrons that started the process. This "blurs" the signal that is carried by the primary electrons. This is why carrier multiplication is very "noisy." Only if the impact ionization coefficient of one type of carrier is much higher than the other, carrier multiplication can be exploited as a gain mechanism. This can be done in specially constructed "superlattices" of two or more carefully selected semiconductors.

The general mathematical treatment of carrier multiplication needs to use the complete set of equations of Table 5.1. Simplifications are possible in steady-state situations in which the electric field is set by either the doping distribution or an externally applied voltage and is not modified by the carrier concentrations. In these circumstances, the continuity equations can be simplified if impact ionization is the dominant generation mechanism and recombination is negligible. This is often reasonable since the high electric field maintains the carrier concentrations low. In the simplified analysis here, we are also going to neglect carrier diffusion. Substituting the

[5]There are some exceptions to this. Certain voltage reference circuits are based on diodes biased at avalanche breakdown.

Table 5.7
Equation set for one-dimensional steady-state situation with imposed electric field and where carrier multiplication is the dominant generation mechanism.

$\frac{\partial \mathcal{E}}{\partial x} \simeq \frac{q}{\epsilon}(N_D - N_A)$	or	$\mathcal{E} = \mathcal{E}'$
	$J_e \simeq -qnv_e^{drift}$	
	$J_h \simeq qpv_h^{drift}$	
$-\frac{dJ_e}{dx} = \alpha_e \lvert J_e \rvert + \alpha_h \lvert J_h \rvert$	or	$\frac{dJ_h}{dx} = \alpha_e \lvert J_e \rvert + \alpha_h \lvert J_h \rvert$
	$\frac{dJ_t}{dx} = 0$	
	$J_t = J_e + J_h$	

generation function for impact ionization from Eq. (4.100) into the continuity Eqs. (5.3) and (5.4), and neglecting the time-dependent and recombination terms, we get

$$-\frac{dJ_e}{dx} = \frac{dJ_h}{dx} = \alpha_e |J_e| + \alpha_h |J_h| \tag{5.119}$$

where α_e and α_h are, respectively, the electron and hole impact ionization coefficients. The absolute signs are to ensure that the generation functions are always positive. The two continuity equations are actually redundant, since in steady state, the total current $J_t = J_e + J_h$ is independent of position. Table 5.7 summarizes the carrier equation set for steady-state carrier multiplication under a fixed electric field.

The best way to conceptually understand carrier multiplication and its implications is to study an example.

5.10.1 Example 7: Carrier multiplication in a high-resistivity region with uniform electric field

Consider a high-resistivity uniformly doped region again under an external electric field, as shown in Fig. 5.26. A narrow but penetrating beam of light impinges on the right-hand side of the sample generating g_l electron–hole pairs per cm^2 every s. This is a case similar to the example discussed in Section 5.9.1 but with $x_o = L$. Following the analysis carried out in Section 5.9.1, the holes are instantly collected at the metal contact but the electrons drift through the high-resistivity region until they reach the other end of the structure.

In the absence of significant carrier multiplication, the current through the high-field region is entirely supported by electrons and is given by $J_t = J_e = qg_l$. If carrier multiplication is taking place, as the photogenerated electrons drift down the sample, they generate secondary electrons and holes that can also, in turn, generate more carriers. In these conditions, the current profile is obtained by solving the differential Eq. (5.119). Since with our choice of axis and electric field, both J_e and J_h are positive, we have

$$-\frac{dJ_e}{dx} = \alpha_e J_e + \alpha_h J_h \tag{5.120}$$

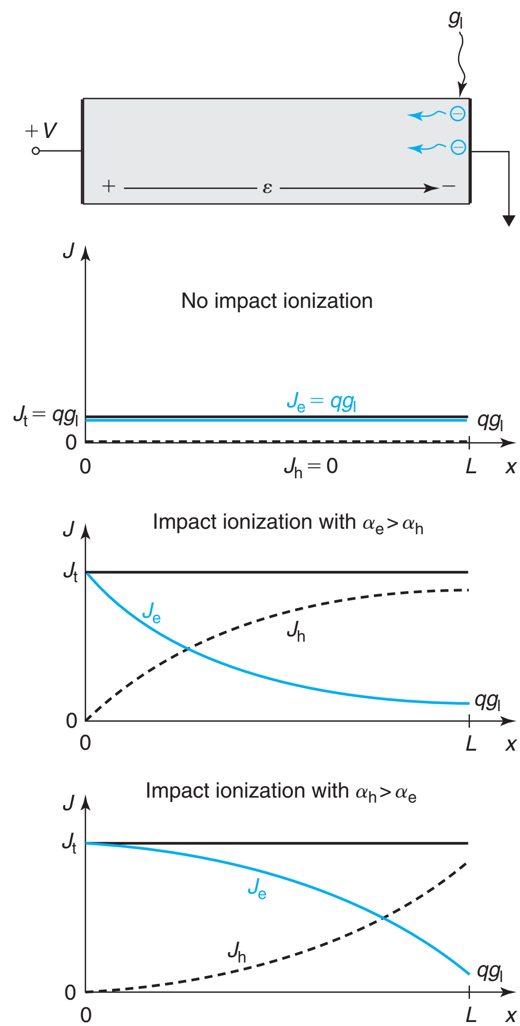


FIGURE 5.26
Top: sketch of a high-resistivity uniformly doped sample with carrier multiplication started by a sheet of electrons generated at one end. Below, the resulting electron, hole, and total currents are sketched in both cases $\alpha_e > \alpha_h$ and $\alpha_h > \alpha_e$.

It is advantageous to write this equation in terms of the total current J_t:

$$-\frac{dJ_e}{dx} = (\alpha_e - \alpha_h)J_e + \alpha_h J_t \tag{5.121}$$

This equation has to be solved subject to the boundary condition that $J_e(L) = qg_l$.

The solution to this differential equation is

$$J_e(x) = \frac{\alpha_h J_t}{\alpha_h - \alpha_e}[1 + e^{(\alpha_h - \alpha_e)x}] + Ae^{(\alpha_h - \alpha_e)x} \tag{5.122}$$

where A is a constant to be determined and J_t is unknown.

A is obtained by demanding that at $x = 0$, the total current is solely supported by electrons. This is because the sign of the electric field prevents any holes from being collected at that contact. Hence, $J_e(x = 0) = J_t$, which substituted in Eq. (5.122) results in $A = J_t(\alpha_e + \alpha_h)/(\alpha_e - \alpha_h)$.

We must also demand, as mentioned above, that $J_e(x = L) = qg_l$. This gives the expression of J_t:

$$J_t = qg_l \frac{\alpha_h - \alpha_e}{\alpha_h - \alpha_e e^{(\alpha_h - \alpha_e)L}} \tag{5.123}$$

Plugging A and J_t into Eq. (5.122) yields the final result:

$$J_e(x) = qg_l \frac{\alpha_h - \alpha_e e^{(\alpha_h - \alpha_e)x}}{\alpha_h - \alpha_e e^{(\alpha_h - \alpha_e)L}} \tag{5.124}$$

which satisfies the boundary conditions.

The hole current is simply:

$$J_h = J_t - J_e = qg_l \frac{\alpha_e[e^{(\alpha_h - \alpha_e)x} - 1]}{\alpha_h - \alpha_e e^{(\alpha_h - \alpha_e)L}} \tag{5.125}$$

All the currents are sketched in Fig. 5.26. It is interesting to discuss the shape of these currents depending on the relative magnitude of α_e and α_h. If $\alpha_e > \alpha_h$, electrons tend to be the source of most impact ionization events. In consequence, the electron current grows exponentially as the electrons travel from the generation point at $x = L$ to $x = 0$. If $\alpha_h > \alpha_e$, the contrary situation takes place. It is the hole current that grows exponentially since they are responsible for the bulk of the generation events. In both cases, however, the process is started by the electrons generated at $x = L$. In both cases too, the contribution to the total current carried out by the electrons increases from right to left as required by the sign of the electric field.

It is interesting to define a **multiplication coefficient**, M, as the ratio of the total current flowing through the structure to the current that started the multiplication process $J_e(L) = qg_l$:

$$M = \frac{\alpha_h - \alpha_e}{\alpha_h - \alpha_e e^{(\alpha_h - \alpha_e)L}} \tag{5.126}$$

There are two limits to M. If the electric field is small, both ionization coefficients are very small. This allows us to linearize the exponential in the denominator of M and obtain $M \simeq 1$. In this case, carrier multiplication is negligible.

At the other extreme, at high electric fields, since both terms in the denominator of Eq. (5.126) are positive, it is entirely possible for the denominator to go to zero and for M to diverge. When this happens, J_t grows without bounds and *avalanche breakdown* takes place. In the example considered here, this occurs when

$$(\alpha_h - \alpha_e)L = \ln \frac{\alpha_h}{\alpha_e} \tag{5.127}$$

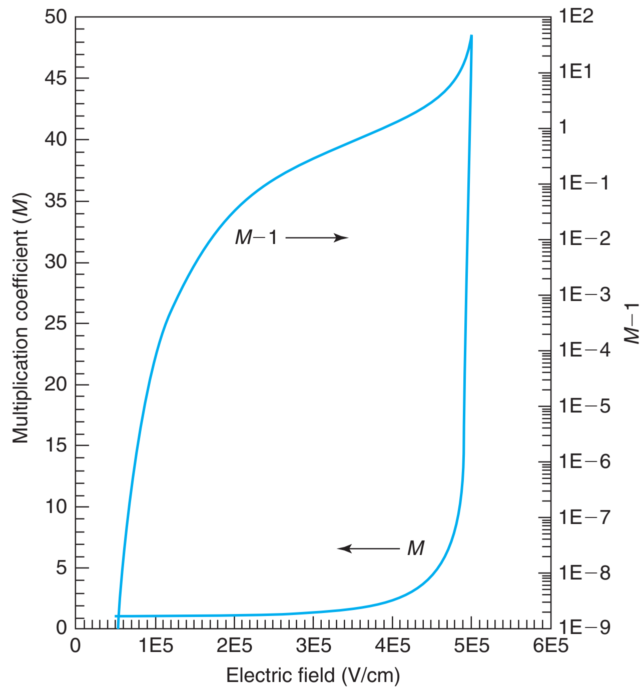


FIGURE 5.27
Multiplication coefficient M, and $M - 1$ as a function of electric field for a 0.2-μm-long Si sample with uniform electric field.

For a sample of given length L, this condition can be reached by increasing the voltage applied to the sample and hence the electric field. If $\alpha_h > \alpha_e$, for example, when the field is small, both impact ionization coefficients are small and $(\alpha_h - \alpha_e)L$ is very small. As the field increases, both coefficients increase and their difference increases too. At a certain voltage, the condition given in Eq. (5.127) is satisfied and the sample goes into breakdown. The same happens if $\alpha_e > \alpha_h$. The voltage at which this occurs is the breakdown voltage, BV.

Figure 5.27 shows an example of a calculation of the multiplication coefficient for a 0.2-μm-long Si sample under a uniform electric field (using values for the impact ionization coefficients in Appendix E). At an electric field of $\mathcal{E}_b = 5 \times 10^5$ V/cm, the sample goes into avalanche breakdown. This is called the *critical breakdown field*. The breakdown voltage of this sample is $BV = 10$ V.

The critical electric field is not a fundamental parameter of a material. Depending on the field distribution and the sample size, the critical field can be rather different. The only reliable way to compute the breakdown voltage in a given situation is to carry out a similar analysis to the one presented in this example.

It often occurs that we are interested in impact ionization in situations in which carrier multiplication is not very significant. This is the case, for example, when analyzing the substrate current in MOSFETs. In these cases, the multiplication coefficient is not a good parameter since its value is very close to unity for a wide range of electric fields. Instead, it is more useful to focus on $M - 1$, which reflects only the current produced by carriers generated through impact ionization. $M - 1$ depends very strongly on the electric field. Figure 5.27 shows $M - 1$ as a function of electric field for the example discussed in this section.

EXERCISE 5.6

Consider a 0.2-μm-*long high-resistivity region of a Si sample under an applied electric field of* $\mathcal{E} = 5 \times 10^5$ V/cm *at room temperature. (a) Calculate the multiplication coefficient. (b) If there is a triggering current of* 1 μ*A, calculate the impact ionization current that is produced and the total current that flows through the sample. (c) If we now increase the length of this high-resistivity region without changing the electric field, calculate the critical length that results in avalanche breakdown.*

(a) The first step in solving this problem is to get the appropriate values of the ionization coefficients from the analytical expressions given in Appendix E. For a field of $\mathcal{E} = 5 \times 10^5$ V/cm, the ionization coefficients are $\alpha_e = 6.0 \times 10^4$ cm^{-1} and $\alpha_h = 3.9 \times 10^4$ cm^{-1}.

We can now plug into the expression of the multiplication coefficient:

$$M = \frac{\alpha_h - \alpha_e}{\alpha_h - \alpha_e e^{(\alpha_h - \alpha_e)L}} = \frac{3.9 \times 10^4 - 6.0 \times 10^4}{3.9 \times 10^4 - 6.0 \times 10^4 e^{(3.9\times10^4 - 6.0\times10^4)0.2\times10^{-4}}} \simeq 50$$

(b) The impact ionization current is given by the triggering current times $M - 1$. Then

$$I_{ii} = (M-1)I_{trig} = (50-1) \times 1\ \mu\text{A} = 49\ \mu\text{A}$$

The total current that flows through the sample is M times the triggering current:

$$I_t = MI_{trig} = 50 \times 1\ \mu\text{A} = 50\ \mu\text{A}$$

(c) Avalanche breakdown occurs when the following condition is met:

$$(\alpha_h - \alpha_e)L_{crit} = \ln\frac{\alpha_h}{\alpha_e}$$

Solving for L_{crit}:

$$L_{crit} = \frac{1}{\alpha_h - \alpha_e}\ln\frac{\alpha_h}{\alpha_e} = \frac{1}{3.9 \times 10^4 - 6.0 \times 10^4}\ln\frac{3.9 \times 10^4}{6.0 \times 10^4} = 0.21\ \mu\text{m}$$

5.11 SUMMARY

- The systems of equations of Table 5.1 describe carrier behavior in semiconductors. These equations can be solved exactly using CAD tools in three dimensions. Appropriate simplifications for a few important families of problems can yield key insight regarding the dominant physics.
- Concept of quasi-neutrality: absence of substantial volume charge. Quasi-neutrality implies that the divergence of the total current density is zero everywhere. Quasi-neutrality is more likely the higher the mobile carrier concentrations. Quasi-neutrality is a pervasive situation in semiconductor devices.

- Majority carrier situations: characterized by the application of an external voltage without perturbing the carrier concentrations from their equilibrium values. In this family of problems, minority carriers play no role. The current through the semiconductor is determined by majority carrier drift. The relevant time constant for majority carrier situations is the dielectric relaxation time which is often much shorter than the timescale of interest.
- Minority carrier situations are those with bottlenecks dominated by minority carrier phenomena, typically diffusion, drift, recombination, and generation. Dominant processes can be identified by estimating the magnitude of the key length scales: diffusion length and the sample length. In dynamic problems, key time constants are the carrier lifetime and the transit time. The identification and estimation of the dominant length scale and timescale of minority carrier problems is a very valuable skill of great practical significance to microelectronic device engineers.
- Space-charge regions: regions with very low carrier concentrations, high resistivity, and a substantial electric field. Carrier drift is the most significant transport mechanism.
- Impact ionization in space-charge regions produces carrier multiplication and can result in avalanche breakdown. This often limits the maximum voltage that can be handled by a device (the breakdown voltage). An ability to estimate the breakdown voltage and an understanding of its leading dependencies are essential skills for a device engineer.

5.12 FURTHER READING

This chapter has covered a lot of territory. There is no single book with a treatment that is parallel to the one here. Several books, however, contain complementary material.

A First Course in Differential Equations with Applications by D. G. Zill, Prindle, Weber, and Schmidt, 1979 (ISBN 0-87150-266-6, QA372.Z54). This is a text on the theory, methods of solution, and applications of ordinary differential equations. This book is very clearly written and contains many examples from different fields of physics and engineering. The techniques taught in this book come handy to solve many different types of minority carrier problems.

Conduction of Heat in Solids by H. S. Carslaw and J. C. Jaeger, Oxford University Press, 1959 (ISBN 0-19-853303-9, QC321.C321). The mathematics of minority carrier transport share a lot with that of conduction of heat in solids. This classical text (first published in 1946) presents a very complete mathematical treatment of heat conduction in solids in several coordinate systems. This might be useful in some specific circumstances since many minority-carrier type problems have a dual heat conduction problem.

AT5.1 CONTINUITY EQUATIONS IN INTEGRAL FORM

The continuity equations derived in Section 5.1 are particularly intuitive when written in an integral form. We can easily do this by extending Eq. (5.1) to a finite volume V enclosed by a surface S:

$$\frac{\partial}{\partial t}\int_V n dV = \int_V (G - R) dV + \frac{1}{q}\int_S \vec{J}_e \cdot \vec{dS} \tag{5.128}$$

Similarly for holes:

$$\frac{\partial}{\partial t}\int_V p dV = \int_V (G-R)dV - \frac{1}{q}\int_S \vec{J}_h \cdot d\vec{S} \tag{5.129}$$

These equations clearly state that the total number of carriers inside a certain volume changes in time if there is net carrier generation or if there is net flow of carriers through its surface.

The continuity equation for charge is also very intuitive when expressed in an integral form. If we multiply both of the above equations by q and we subtract the first one from the second one and use Eq. (5.5), we get

$$\frac{\partial}{\partial t}\int_V \rho\, dV = -\int_S \vec{J}_t \cdot d\vec{S} \tag{5.130}$$

This now makes clear that if there is net charge increasing in time in a certain volume of a semiconductor it is because there is net current flowing into that volume (negative surface integral).

In steady state, the time derivatives in Eqs. (5.128–5.130) become zero. From Eq. (5.130), we can conclude that there cannot be net current into any region of the semiconductor. From Eqs. (5.128) and (5.129), if we substitute $G_{ext} - U$ for $R - G$, we can write

$$-\int_S \vec{J}_e \cdot d\vec{S} = \int_S \vec{J}_h \cdot d\vec{S} = q\int_V (G_{ext} - U)dV \tag{5.131}$$

This states that in steady state, the integral of the carrier current extended over the surface of an enclosed volume is equal (with the appropriate sign) to the net generation of carriers taking place inside the enclosed volume. This is an observation that derives directly from the continuity equations and that comes handy in a few situations.

AT5.2 DIELECTRIC RELAXATION

A uniformly doped semiconductor in thermal equilibrium, in the absence of any surface effects, is charge neutral at every point in space. Suppose that through some external influence we can somehow perturb this perfect charge neutrality by introducing a space-charge distribution. What happens if we then suddenly stop the outside influence and leave the semiconductor alone? Is charge neutrality eventually reestablished? If so, how long does it take for this to happen?

Consider, for example, an n-type semiconductor with a uniform donor concentration N_D. In thermal equilibrium, we know that $n_o = N_D$ (we focus entirely on electrons and neglect any impact of the holes). Assume now that by some external means, the electron concentration gets upset as depicted in Fig. 5.28. The electron concentration in a certain region has been increased above N_D, while in another region next to it, it has been decreased below N_D by the same amount. What happens if we stop the external influence at $t = 0$? Is this a sustainable situation?

The answers to these questions come from considering the electrostatics consequences of the situation at $t = 0$. While *global* charge neutrality is still maintained, *local* charge neutrality has been perturbed. A net volume charge density ρ appears with a dipole-like distribution, positive

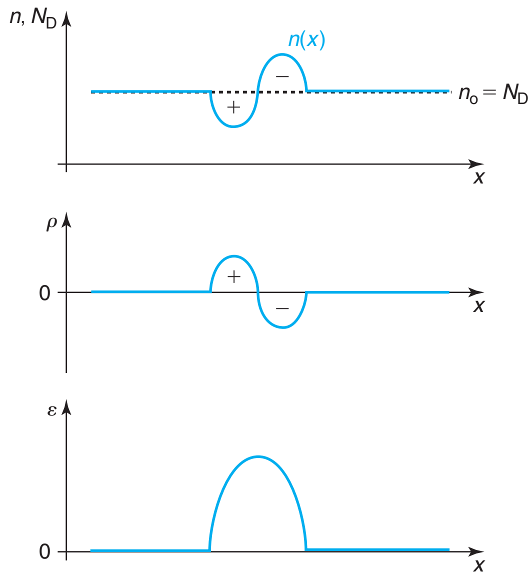


FIGURE 5.28
Electrostatics of a uniformly doped n-type semiconductor with nonuniform electron distribution that result in local deviations from charge neutrality.

on the left and negative on the right, as sketched in the figure. This charge dipole produces an electric field distribution in the same region as sketched in the bottom graph.

What is the impact of this electric field on the electrons? As we studied in Chapter 4, a positive electric field produces a negative electron flux, that is, electrons will drift from right to left. This tends to weaken the charge dipole since it redresses the imbalance between electron concentration and donor concentration. Hence, given enough time, it is clear that the charge dipole will be erased. How much time does it take for this to happen?

It is not difficult to derive an equation that describes the dynamics of charge redistribution in these kinds of situations. The starting point is the continuity equation for charge derived earlier in this chapter [Eq. (5.6)] and reproduced here for convenience:

$$\frac{\partial J_t}{\partial x} = -\frac{\partial \rho}{\partial t} \tag{5.132}$$

Neglecting carrier diffusion (which would further speed up the process) and if the electric field does not become too large, the total current in the semiconductor follows Ohm's law (see Section 4.2.3) and can be expressed as

$$J_t = \sigma \mathcal{E} \tag{5.133}$$

where σ is the conductivity of the semiconductor.

The electric field is related to the volume charge density through Gauss' law [Eq. (4.50)]. Inserting this equation into Eq. (5.133) and the result into Eq. (5.132), we get

$$\frac{\partial \rho}{\partial t} = -\frac{\sigma}{\epsilon}\rho \tag{5.134}$$

If at $t = 0$, there is a situation given by $\rho(x, t = 0)$, the solution of this first-order linear differential equation gives the evolution of the charge density as a function of time as

$$\rho(x, t) = \rho(x, t = 0)e^{-\frac{t}{\tau_d}} \tag{5.135}$$

where τ_d is called the *dielectric relaxation time* and is given by

$$\tau_d = \frac{\epsilon}{\sigma} \tag{5.136}$$

We are in front of a simple exponential-decay mechanism with a time constant that is τ_d. The inverse dependence of τ_d on σ was to be expected, since the higher the conductivity of the semiconductor, the higher a carrier current is produced in response to the presence of an electric field and the faster the charge dipole is erased. The dependence of τ_d on the dielectric constant is also reasonable since ϵ enters the relationship between the charge dipole that has been created and the electric field that is produced.

In most practical situations, the doping level and the conductivity of semiconductor regions are high enough that the dielectric relaxation time is very small. A calculation of τ_d for Si at room temperature is shown in Fig. 5.19. For any doping level above mid 10^{15} cm^{-3}, $\tau_d < 1$ ps. Since the doping level in modern devices is usually higher than this, dielectric relaxation is rarely the bottleneck to the dynamics of the semiconductor. In other words, the semiconductor can be considered quasi-neutral at all times. Perhaps an exception worth noting at this time is the use of high-resistivity substrates in high-frequency ICs. In these wafers, the dynamics of substrate effects are affected by dielectric relaxation.

AT5.3 ADVANCED TOPICS REGARDING MINORITY CARRIER SITUATIONS

This section presents advanced topics dealing with carrier flow in minority-carrier type situations. We first look at an example that illustrates the combined effect of diffusion, drift, and recombination in a short bar. This is the basis for a more general discussion on the length scales of minority carrier situations. We then examine an advanced example of a transient in a bar with finite surface recombination. This prompts a more detailed view of the timescales relevant to minority carrier situations.

AT5.3.1 Advanced Example 1: Diffusion, drift, and recombination in a short bar with internal field

This example examines the combined impact of drift and diffusion on minority carriers. Such a situation arises, for example, in the base of a bipolar transistor where a gradient of impurity concentration is created to produce an electric field that helps electron transport from the emitter to the collector.

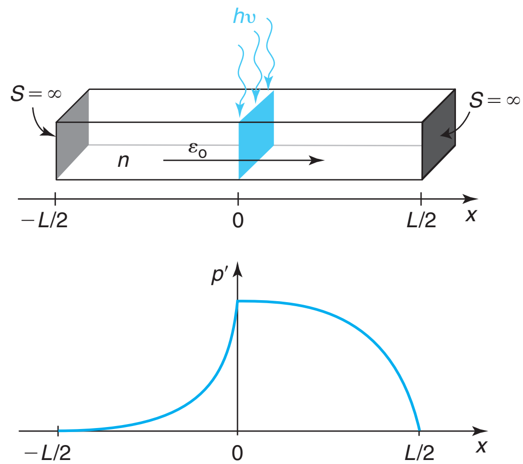


FIGURE 5.29
Top: a short bar illuminated in the middle by a sheet of generation in the presence of an electric field; bottom: excess hole concentration.

Consider a short bar that has a doping gradient inside that produces an electric field in equilibrium. The doping distribution is such that the field is constant and points as shown in Fig. 5.29. The two end surfaces are characterized by an infinite surface recombination velocity. With a narrow spike of generation at the center of the bar, we wish to solve for the carrier concentrations and current densities everywhere.

This example is similar to the one presented in Section 5.6.2, except that the symmetry of the problem has been broken. The electric field preferentially pushes photogenerated holes from the left toward the right everywhere through the bar. Hole diffusion toward the right is aided by the field, while hole diffusion toward the left is opposed by the field. As a consequence, more of the photogenerated holes will now go toward the right than toward the left. We then expect that the excess carrier concentration profiles have a shape as sketched in Fig. 5.29. Intuitively, we can see that if the field is strong enough, holes might not be able to reach the surface at the left of the bar. In this case, the solution is particularly simple. This is the one we solve here. The general case is just slightly more complicated.

An analytical solution to this problem requires that we solve the following differential equation (from Table 5.5):

$$D_h \frac{d^2 p'}{dx^2} - \mu_h \mathcal{E}_o \frac{dp'}{dx} = 0 \tag{5.137}$$

The solution is of the form:

$$p'(x) = A \exp \frac{x}{L_{\mathcal{E}}} + B \tag{5.138}$$

where we have defined

$$L_{\mathcal{E}} = \frac{kT}{q\mathcal{E}_o} \tag{5.139}$$

as the characteristic length of the problem (more about it below).

The boundary conditions at $x = \pm L/2$ are as before, $p'(\pm L/2) = 0$. For $x \geq 0$, this implies

$$p'(x) = A\left(\exp\frac{x}{L_{\mathcal{E}}} - \exp\frac{L}{2L_{\mathcal{E}}}\right) \quad \text{for } x \geq 0 \tag{5.140}$$

If the field is strong enough, as mentioned above, no carriers will be able to reach the contact at $x = -L/2$. This implies then

$$p'(x) = C\exp\frac{x}{L_{\mathcal{E}}} \quad \text{for } x \leq 0 \tag{5.141}$$

The boundary condition at $x = 0$ requires some more careful thought as the symmetry of the problem has been broken. First of all, we must demand continuity of p' at $x = 0$, otherwise, there would be an infinite diffusion current at that point. This allows us to write the coefficient C in Eq. (5.141) in terms of A in Eq. (5.140) and get

$$p'(x) = A\left(1 - \exp\frac{L}{2L_{\mathcal{E}}}\right)\exp\frac{x}{L_{\mathcal{E}}} \quad \text{for } x \leq 0 \tag{5.142}$$

Second, the flux of holes at $x = 0$ must satisfy the continuity equation, that is

$$g_l = F_h(x = 0^+) - F_h(x = 0^-) = -D_h\left(\left.\frac{dp'}{dx}\right|_{0^+} - \left.\frac{dp'}{dx}\right|_{0^-}\right) \tag{5.143}$$

Applying this boundary condition and after a few lines algebra, one gets

$$p'(x) = \frac{g_l L_{\mathcal{E}}}{D_h}\left(1 - \exp\frac{x - \frac{L}{2}}{L_{\mathcal{E}}}\right) \quad \text{for } x \geq 0 \tag{5.144}$$

$$p'(x) = \frac{g_l L_{\mathcal{E}}}{D_h}\left(1 - \exp\frac{-L}{2L_{\mathcal{E}}}\right)\exp\frac{x}{L_{\mathcal{E}}} \quad \text{for } x \leq 0 \tag{5.145}$$

These equations are graphed in Fig. 5.29. The graph shows how the holes are "windswept" by the electric field toward the right of the bar. For high electric fields, the hole concentration decays toward the left in an exponential shape with a characteristic length $L_{\mathcal{E}}$. The physical interpretation of $L_{\mathcal{E}}$ is now clear. $L_{\mathcal{E}}$ is the mean distance that a carrier can diffuse *into* an electric field. If $L_{\mathcal{E}} \ll L/2$, the hole distribution for $x \leq 0$ approaches a perfect exponential. If $L_{\mathcal{E}} \gg L/2$, the effect of the electric field is negligible and we are back to the example presented before.

It is straightforward from this point to compute current densities and to verify the adequacy and the limits of the assumptions that have been made. This is left as an exercise to the reader.

AT5.3.2 More on length scales of minority carrier situations

The previous example has revealed the existence of one more characteristic length that is relevant in minority-carrier type problems. In addition to the diffusion length L_{diff} and the sample length L, already discussed in Section 5.6.3, a field-related length $L_{\mathcal{E}}$ has appeared.

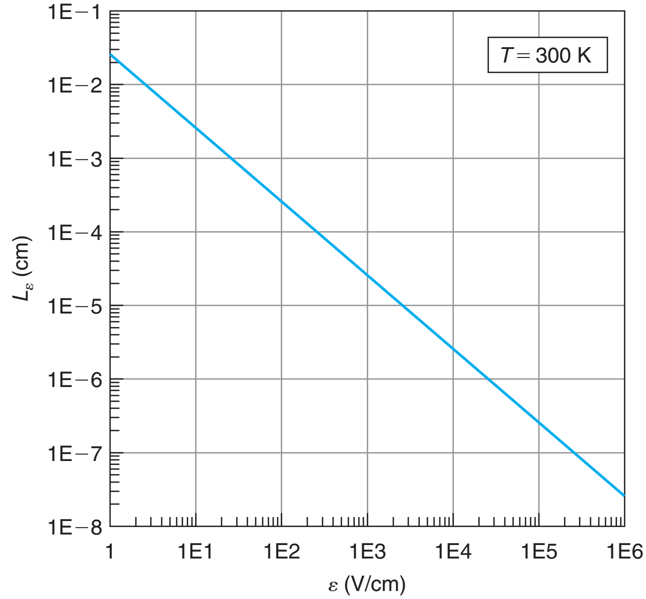


FIGURE 5.30
$L_{\mathcal{E}}$ at room temperature.

The example above reveals that $L_{\mathcal{E}}$ is the mean length that a carrier diffuses into an electric field. It gives an indication of the balance between diffusion and drift. The stronger the electric field is, the shorter $L_{\mathcal{E}}$ becomes. This is graphed in Fig. 5.30 at room temperature.

In fact, these three are not the only important length scales in minority carrier problems. There are more. A rigorous view of the situation can be obtained by writing the homogeneous differential equation that controls steady-state minority carrier behavior in the following way:

$$\frac{d^2p'}{dx^2} - \frac{1}{L_{\mathcal{E}}}\frac{dp'}{dx} - \frac{p'}{L_{\mathrm{h}}^2} = 0 \tag{5.146}$$

Solutions to this equation are of the form:

$$p'(x) = K_1 e^{x/\mathcal{L}_1} + K_2 e^{x/\mathcal{L}_2} \tag{5.147}$$

where $\mathcal{L}_1$ and $\mathcal{L}_2$ are the roots of the characteristic equation:

$$L_{\mathcal{E}}\mathcal{L}^2 + L_{\mathrm{h}}^2\mathcal{L} - L_{\mathcal{E}}L_{\mathrm{h}}^2 = 0 \tag{5.148}$$

These roots are

$$\mathcal{L}_{1,2} = \frac{1}{2L_{\mathcal{E}}}\left[-L_{\mathrm{h}}^2 \pm \sqrt{L_{\mathrm{h}}^4 + 4L_{\mathcal{E}}^2L_{\mathrm{h}}^2}\right] \tag{5.149}$$

Regardless of the sign of $\mathcal{E}_{\mathrm{o}}$, both roots are always real.

There are two characteristic scale lengths in a general problem, $\mathcal{L}_1$ and $\mathcal{L}_2$. They both exhibit a mixture of drift and diffusion. In order to understand their physical meaning, let us examine their behavior for small and large electric fields.

If $\mathcal{E}_o$ is sufficiently small, that is $\mathcal{E}_o \ll kT/qL_h$, then

$$\mathcal{L}_1 \simeq L_h = \sqrt{D\tau} \tag{5.150}$$

$$\mathcal{L}_2 \simeq -L_h = -\sqrt{D\tau} \tag{5.151}$$

On the other hand if $\mathcal{E}_o$ is large, $\mathcal{E}_o \gg kT/qL_h$, then

$$\mathcal{L}_1 \simeq L_{\mathcal{E}} = \frac{kT}{q\mathcal{E}_o} \tag{5.152}$$

$$\mathcal{L}_2 \simeq -\frac{L_h^2}{L_{\mathcal{E}}} = -\mu\mathcal{E}_o\tau \tag{5.153}$$

This behavior of $\mathcal{L}_1$ and $\mathcal{L}_2$ is graphed in Fig. 5.31. For low fields, the absolute values of $\mathcal{L}_1$ and $\mathcal{L}_2$ converge to the minority carrier diffusion length. For high electric field values, $\mathcal{L}_1$ is reduced as $\mathcal{E}_o^{-1}$, while the absolute value of $\mathcal{L}_2$ increases as $\mathcal{E}_o$. The high field value of $|\mathcal{L}_2|$ has a rich physical meaning. $\mu\mathcal{E}_o$ is the drift velocity of the minority carriers. τ is the carrier lifetime.

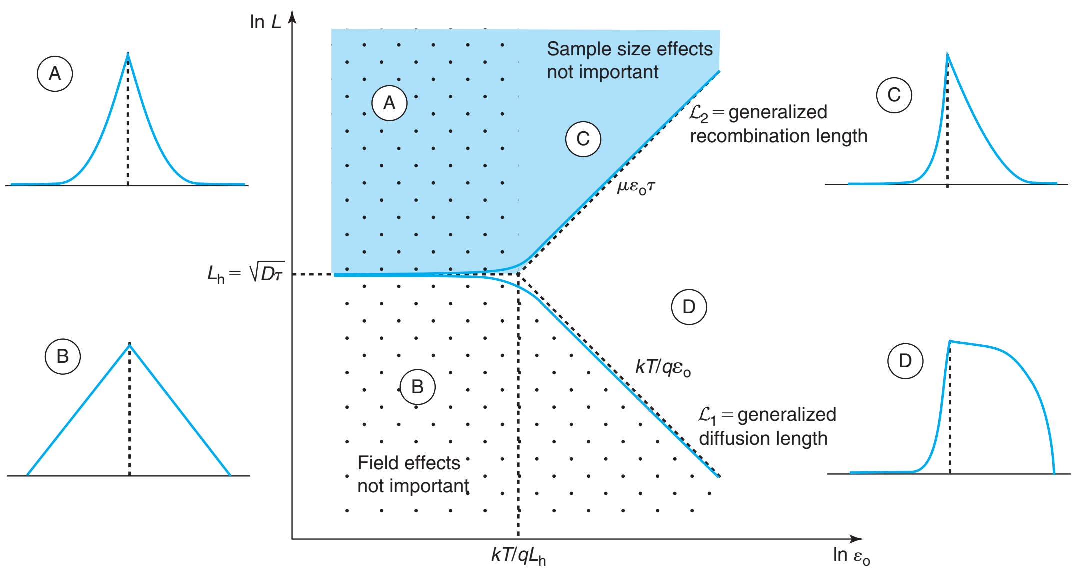


FIGURE 5.31
Sketch of the different scale lengths in minority carrier problems as function of sample size and electric field. The blue shaded area corresponds to negligible sample size effects. The polka-dotted area corresponds to negligible field effects. Sketched are also examples of carrier profiles that result from δ-generation in the various regions.

In consequence, $\mu\mathcal{E}_o\tau$ is the mean distance that a carrier travels *by drift* before recombining. Notice also that always $|\mathcal{L}_2| \geq \mathcal{L}_1$.

This discussion allows us to provide a unifying view of $\mathcal{L}_1$ and $\mathcal{L}_2$. $\mathcal{L}_1$ can be considered a *generalized diffusion length*, or mean distance that a carrier diffuses in the bulk of a semiconductor. For low values of $\mathcal{E}_o$, it coincides with the traditional diffusion length, as minority carriers diffusion is balanced against minority carrier recombination. For high values of the electric field with a sign such that it opposes diffusion, the mean distance that a carrier can diffuse is reduced, the higher the electric field, the shorter $\mathcal{L}_1$ becomes. The critical field at which drift starts affecting diffusion is kT/qL_h that has the simple physical interpretation of a thermal voltage dropping over a diffusion length.

$|\mathcal{L}_2|$ can be considered a *generalized recombination length*, or mean distance that a carrier travels in the bulk of a semiconductor before recombining. For low fields, $|\mathcal{L}_2|$ coincides with the traditional diffusion length, since diffusion is the only mechanism for carrier transport. For high fields with a sign such that it aids diffusion, the mean distance that a carrier can travel is enhanced as carriers get "dragged" by $\mathcal{E}_o$. The higher the field, the longer $|\mathcal{L}_2|$ becomes. The critical field at which drift effects become significant in $\mathcal{L}_2$ is also kT/qL_h.

In summary, $\mathcal{L}_1$ and $|\mathcal{L}_2|$ are the characteristic lengths of bulk situations, that is, those in which the surfaces of the semiconductor are farther away than the longest scale length that applies, mathematically, when $L \gg \max\{\mathcal{L}_1, |\mathcal{L}_2|\}$, where L is the sample length. This is illustrated in Fig. 5.31 with cartoons of the excess carrier profiles that result in a bar subject to δ-generation at its center, as in the examples above. Cases A and C correspond to long samples. In case A, the fields are small (example 1 above). The dominant length scales are L_h. In case C, the field is large, and in consequence, in the direction in which the field opposes diffusion, the length scale is $kT/q\mathcal{E}_o$, while if the field aids diffusion, the dominant length is $\mu\mathcal{E}_o\tau$. As the field increases and with it does $|\mathcal{L}_2|$, the sample length has to increase for surface effects to be unimportant. This is indicated in Fig. 5.31 by the blue shading.

For smaller samples, L can become the dominant scale length. We already know that for low fields, if L is much smaller than L_h, bulk recombination is negligible and the minority carrier profile is dominated by diffusion to the surface (example 2 above and cartoon B in Fig. 5.31). L becomes the dominant scale length in this case. For high fields, L remains the dominant length scale if it is much smaller than $\mathcal{L}_1$. The higher the field, the smaller the sample needs to be. For a given L, sample size effects are dominant if $\mathcal{E}_o \ll kT/qL$, a thermal voltage over the sample length, a handy rule of thumb to remember.

At high fields, there is a broad middle range of sample sizes over which the sample size is the dominant dimension in the direction in which drift aids diffusion, but $\mathcal{L}_1$ is the dominant length in the direction in which drift opposes diffusion. This was studied in Example 3 and is sketched in cartoon D in Fig. 5.31.

Table 5.8 summarizes the characteristic lengths described in this section.

AT5.3.3 Advanced Example 2: Transient in a bar with finite surface recombination

In the dynamic example with two surfaces with $S = \infty$, discussed in Section 5.8.1, the main time constant of the problem was the inverse of the inverse sum of the bulk carrier lifetime and the transit time to the surface. This was the case because once the excess carriers arrive to the

Table 5.8
Summary of dominant scale lengths in different situations.

	Dominant scale length	
Relative dimensions	Drift opposes diffusion	Drift aids diffusion
$L \gg \lvert\mathcal{L}_2\rvert \geq \mathcal{L}_1$	$\mathcal{L}_1$	$\mathcal{L}_2$
$\lvert\mathcal{L}_2\rvert \gg L \gg \mathcal{L}_1$	$\mathcal{L}_1$	L
$\lvert\mathcal{L}_2\rvert \geq \mathcal{L}_1 \gg L$	L	L

surface, they recombine instantly. If the surface recombination velocity is not infinite, clearly a new time constant is going to appear: the *surface recombination time*. We can easily estimate this time constant. The velocity at which carriers reach the surface and recombine is, by definition, S, the surface recombination velocity. In this problem, the average distance that a carrier has to travel before reaching one of the surfaces is of order $L/2$. In consequence, the average time that it takes for a carrier to reach the surface and recombine is:

$$\tau_s \approx \frac{L}{2S} \tag{5.154}$$

When this time constant is much longer than the transit time, the bottleneck to surface recombination is not the diffusion of carriers to the surface, but the actual recombination rate there. In this instance, Eq. (5.154) is the time constant that works in parallel with the bulk carrier lifetime.

It is instructive to solve this problem rigorously. From a mathematical point of view, the only difference between this problem and that in Section 5.8.1 is the boundary condition at $\pm L/2$. In the present case, for $x = L/2$, the finite recombination velocity implies at all times that

$$-D_h \frac{\partial p'}{\partial x}\Big|_{\frac{L}{2},t} = p'\left(\frac{L}{2}, t\right) S \tag{5.155}$$

For $t \leq 0$, the differential equation that governs this problem is Eq. (5.85). The solution is of the form given by Eq. (5.87). Following similar arguments as above and applying boundary condition Eq. (5.155), we obtain

$$p'(x,0) = g_l \tau \left(1 - \frac{\cosh \frac{x}{L_h}}{\cosh \frac{L}{2L_h} + \frac{D_h}{SL_h} \sinh \frac{L}{2L_h}}\right) \tag{5.156}$$

This solution is graphed in the upper trace in Fig. 5.32.

The transient problem for $x > 0$ is also governed by Eq. (5.89). In consequence, the solution is obtained in the same way as in that example. The boundary condition at $x = L/2$ now results in the following restrictions to the values that λ can take

$$\tan \frac{L}{2\lambda} = \frac{S\lambda}{D_h} \tag{5.157}$$

This implicit equation is satisfied by a number of different values of λ. That gives rise to multiple solutions that must be added to obtain a general solution that follows again the expression of

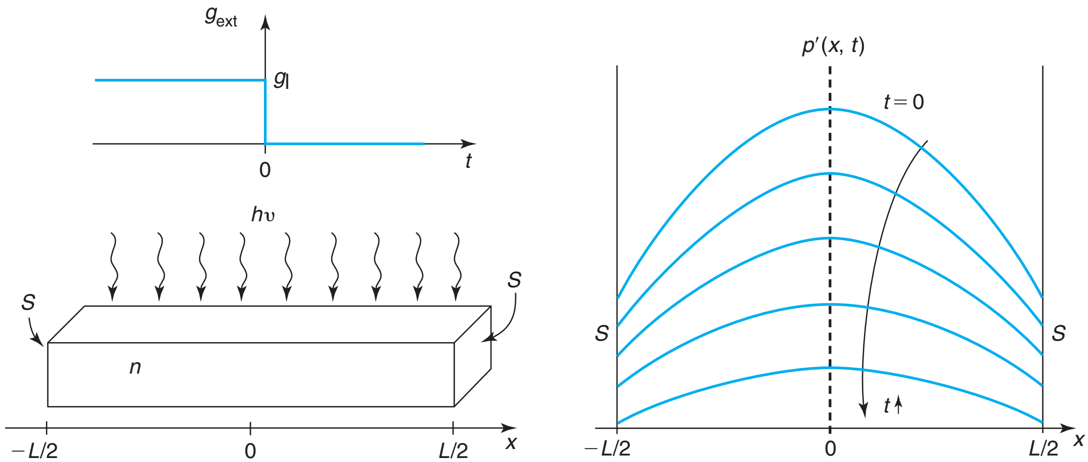


FIGURE 5.32
Sketch of time decay of excess carrier concentration in a semiconductor bar after a uniform generation function has been turned off. The surfaces are characterized by finite S.

Eq. (5.97). The relative weight of all solutions has to be obtained this time to add up to the $t = 0$ initial condition of Eq. (5.156).

We can examine the leading time constants for extreme cases of S. When S is very small, the *tan* term can be expanded around $x = 0$ to yield

$$(-1)^{n-1}\frac{L}{2\lambda_n} + (n-1)\pi = \frac{S\lambda_n}{D_h} \quad \text{for } n = 1, 2, 3, \ldots \tag{5.158}$$

The largest of these time constants corresponds to $n = 1$. The dominant time constant of the decay is

$$\frac{1}{\tau_1} = \frac{1}{\tau} + \frac{2S}{L} \quad \text{for } S \ll \frac{2D_h}{L} \tag{5.159}$$

Just as we argued above, the characteristic time constant for surface recombination in this limit of low S is the surface recombination time τ_s. Obviously, if $S = 0$, $\tau_1 \simeq \tau$, as expected.

In the other extreme, when S is very high ($S \gg \frac{2D_h}{L}$), we can expand the *tan* term in Eq. (5.157) around $\pi/2$ to obtain

$$(2n-1)\frac{\pi}{2} - \frac{D_h}{S\lambda} = \frac{L}{2\lambda} \quad \text{for } n = 1, 2, 3, \ldots \tag{5.160}$$

The largest of these time constants corresponds to $n = 1$ again. After some simple algebra, this dominant time constant is found to be identical to the one obtained in the earlier example in Eq. (5.101).

AT5.4 CARRIER MULTIPLICATION AND AVALANCHE BREAKDOWN UNDER NONUNIFORM ELECTRIC FIELD

In semiconductor devices, carrier multiplication and avalanche breakdown typically take place in space-charge regions with a nonuniform electric field. The physics of the problem do not change with respect to the uniform case studied above. Mathematically, the problem is substantially more involved.

Let us consider a case identical to the one treated in Section 5.10.1 with the exception that the electric field that is present between $x = 0$ and $x = L$ is nonuniform. In this case, the ionization coefficients α_e and α_h depend on position. It is convenient to rewrite the differential equation that governs the problem, Eq. (5.121), in the following way:

$$\frac{dJ_e}{dx} + (\alpha_e - \alpha_h)J_e = -\alpha_h J_t \tag{5.161}$$

This is a first-order linear differential equation that can be integrated by multiplying all terms by the integrating factor $\exp[\int_0^x (\alpha_e - \alpha_h)dx]$. After this is done, the solution is quickly found:

$$J_e = e^{\int_0^x (\alpha_h - \alpha_e)dx}\left[A - J_t \int_0^x \alpha_h e^{\int_0^x (\alpha_e - \alpha_h)dx}dx\right] \tag{5.162}$$

where A is an integration constant.

As in Section 5.10.1 above, A is obtained from demanding that $J_e(x = 0) = J_t$ and J_t is determined from the fact that $J_e(x = L) = qg_l$:

$$J_t = qg_l \frac{e^{\int_0^L (\alpha_e - \alpha_h)dx}}{1 - \int_0^L \alpha_h e^{\int_0^x (\alpha_e - \alpha_h)dx}dx} \tag{5.163}$$

The final result is then

$$J_e(x) = qg_l e^{\int_x^L (\alpha_e - \alpha_h)dx} \frac{1 - \int_0^x \alpha_h e^{\int_0^x (\alpha_e - \alpha_h)dx}dx}{1 - \int_0^L \alpha_h e^{\int_0^x (\alpha_e - \alpha_h)dx}dx} \tag{5.164}$$

The multiplication coefficient can be obtained by dividing both sides of Eq. (5.163) by the factor qg_l. It is convenient to transform the integral in the denominator of this equation in the following way:

$$\begin{aligned}\int_0^L \alpha_h e^{\int_0^x (\alpha_e - \alpha_h)dx}dx &= \int_0^L \alpha_e e^{\int_0^x (\alpha_e - \alpha_h)dx}dx - \int_0^L (\alpha_e - \alpha_h)e^{\int_0^x (\alpha_e - \alpha_h)dx}dx \\ &= \int_0^L \alpha_e e^{\int_0^x (\alpha_e - \alpha_h)dx}dx - e^{\int_0^L (\alpha_e - \alpha_h)dx} + 1\end{aligned}$$

In this way, the multiplication coefficient can be written in the following way:

$$M = \frac{1}{1 - \int_0^L \alpha_e e^{\int_x^L (\alpha_h - \alpha_e)dx} dx} \tag{5.165}$$

M diverges when the integral in the denominator becomes unity. This is the condition for breakdown.

Had we considered an entirely equivalent problem with the radiation impinging at $x = 0$, carrier multiplication is started by holes. In this case, the multiplication coefficient takes the expression:

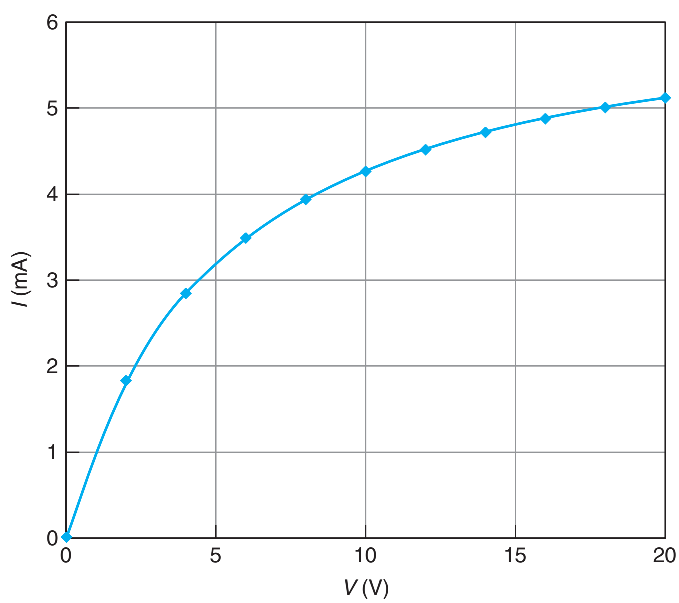
$$M = \frac{1}{1 - \int_0^L \alpha_h e^{\int_0^x (\alpha_e - \alpha_h)dx} dx} \tag{5.166}$$

PROBLEMS

* Denotes a problem that requires studying an Advanced Topic

5.1 An integrated resistor constitutes an excellent test structure to diagnose the doping level and junction depth of a doped layer. This problem illustrates this.

Consider an integrated resistor with a length $L = 3.5$ μm and a width $W = 2$ μm. The resistor is implemented on an n-layer inside a p-tub. In this resistor, you measure at room temperature the I–V characteristics that are graphed below.



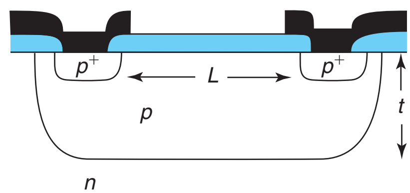
A good fit to the data is an equation of the form:

$$I = \frac{6.4}{1+\frac{5}{V}}\ \text{mA} \qquad \text{with } V \text{ in V}$$

To the first order, you can assume the resistor to have uniform doping level. Also assume that the resistor's behavior is dominated by the n-layer (there are no other significant resistances).

(a) Compute the sheet resistance of the n-layer.
(b) Extract the doping level of the n-layer.
(c) Extract the junction depth of the n-layer.

5.2 Consider an integrated resistor implemented using the p-well of a CMOS process, as sketched below. The doping level $N_A = 10^{17}$ cm^{-3} is uniform. The depth of the p-well is $t = 0.5$ μm. The source of the p-MOS transistors is used to contact the resistor. The separation between the two contacts is $L = 5$ μm. For all purposes, you can assume that the carrier flow is one dimensional and the effective length of the resistor is as indicated.



(a) Calculate the sheet resistance of the p-well.
(b) Calculate the unit width resistance of this resistor (units: $\Omega \cdot$ cm).
(c) For a voltage applied of 3 V, calculate:
- the electric field inside the resistor,
- the hole velocity inside the resistor, and
- the current per unit width flowing through the resistor.

(d) Draw to scale the energy band diagram corresponding to case (c) above. Indicate which side is positive and which side is negative.
(e) Estimate the voltage at which hole velocity saturation effects start being a concern.
(f) In the limit of very high fields, what is the maximum current per unit width that this resistor can support?

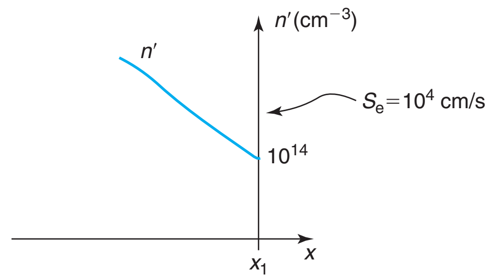
5.3 This problem is about carrying out a complete solution of the steady-state situation described in Fig. 5.20 for $t \leq 0$. In summary, we have an n-type semiconductor bar of length L with two surfaces characterized by $S = \infty$. The bar is illuminated with deep penetrating radiation that generates electron–hole pairs everywhere at a rate g_l. Under steady state,

(a) Derive an analytical expression for the excess hole concentration for $x \geq 0$. Sketch.
(b) Derive an analytical expression for the hole current density for $x \geq 0$. Sketch.
(c) Derive an analytical expression for the total current density for $x \geq 0$. Sketch.
(d) Derive an analytical expression for the total electron current density for $x \geq 0$. Sketch.
(e) Derive an analytical expression for the excess electron concentration for $x \geq 0$. Sketch.
(f) Derive an analytical expression for the electron diffusion current for $x \geq 0$. Sketch.
(g) Derive an analytical expression for the electron drift current for $x \geq 0$. Sketch.
(h) Derive an analytical expression for the net recombination rate for $x \geq 0$. Sketch.

(i) Derive an analytical expression for the electric field for $x \geq 0$. Sketch.
(j) Derive an analytical expression for the volume charge density for $x \geq 0$. Sketch.
(k) Verify the quasi-neutrality assumption. Apply order-of-magnitude numbers for $N_D = 10^{17}\ \text{cm}^{-3}$.
(l) Verify that the hole drift current density is much smaller than the hole diffusion current density.
(m) Derive an analytical expression for the highest generation rate g_l that can be tolerated for low-level injection conditions to apply.
(n) Verify the linearity between the electron drift velocity and the internal electric field. Apply order-of-magnitude numbers for $N_D = 10^{17}\ \text{cm}^{-3}$.

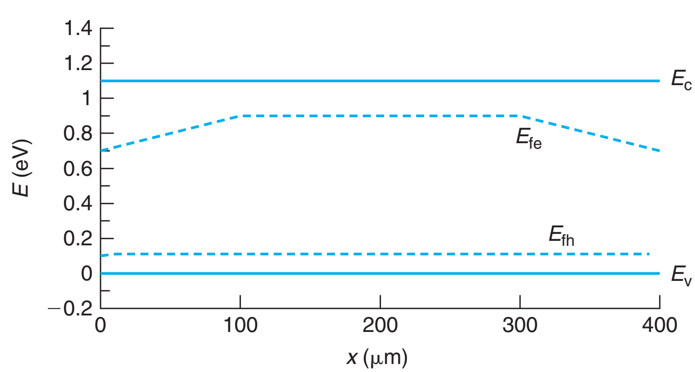
Make the usual assumptions for minority-carrier type problems described earlier in this chapter.

5.4 The figure below sketches the excess minority carrier concentration near one of the surfaces of a quasi-neutral portion of p-type Si at room temperature. The doping level is $N_A = 1 \times 10^{17}\ \text{cm}^{-3}$. The surface recombination velocity is 10^4 cm/s. The portion of semiconductor shown is in the dark.



(a) Estimate the *electron current density* at x_1.
(b) Estimate the *net recombination rate* at x_1.
(c) Estimate the *gradient* of n' at x_1.
(d) Estimate the *flux of electrons* at x_1.
(e) Estimate the *net electron velocity* at x_1.
(f) Estimate the *gradient of the electron quasi-Fermi level* at x_1.

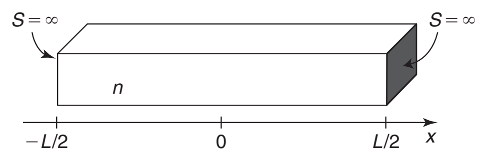
5.5 A 400-μm-long Si bar at room temperature in steady state has the energy band diagram indicated below ($E_{fh} \simeq E_F$).



(a) Circle all the terms that apply to the situation:
- thermal equilibrium/out of equilibrium
- uniformly doped/nonuniformly doped
- n-type/p-type
- low-level injection/high-level injection/extraction
- intrinsic/extrinsic

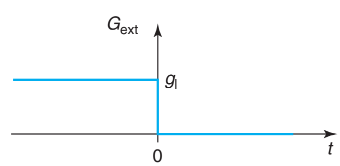
(b) *Estimate* the electron current density at $x = 0$.
(c) *Estimate* the surface recombination velocity at $x = 0$.
(d) *Estimate* the gradient of electron concentration at $x = 0$.

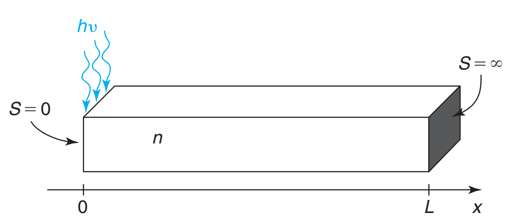
5.6 Consider a Si bar at room temperature in thermal equilibrium. The doping level is uniform and equal to $N_D = 10^{17}$ cm^{-3}. The bar is connected on two ends by ohmic contacts which from the minority carrier point of view, appear as surfaces with a surface recombination velocity $S = \infty$. All other surfaces are characterized by $S = 0$. The sample length is $L = 10$ μm.



(a) Estimate the time required for steady state to be reached after a small voltage V is suddenly applied across the ohmic contacts.
(b) Estimate the time required for steady state to be reached after deep penetrating radiation suddenly illuminates the sample. The radiation generates electron–hole pairs uniformly across the sample. The generation rate is such that the sample always remains under low-level injection conditions.

5.7 Consider a short n-type semiconductor bar of length L ($L_{\text{diff}} \gg L$) with one surface charaterized by $S = 0$, while the other surface exhibits $S = \infty$, as sketched in the diagram below. For times $t \leq 0$, the bar was illuminated for a long time with deeply penetrating radiation narrowly confined to $x = 0$, so that it can be considered a delta function in space. At $t = 0$, the radiation is turned off.

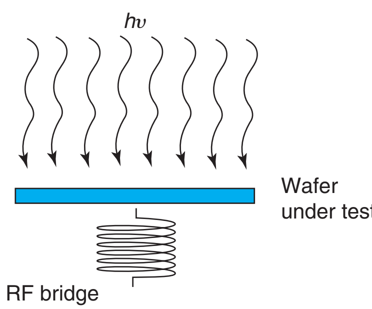




(a) Derive an analytical expression that describes the decay of excess hole concentration in time. Sketch this solution.

(b) Obtain the dominant time constant of the decay. Compare it with the result obtained in the example of Section 5.8.1. Explain differences between both sets of solutions.

5.8 Your company has just bought a lifetime tester that is based on the photoconductance decay technique. The system is sketched below. Basically, a well calibrated and powerful infrared lamp illuminates a Si wafer generating electron–hole pairs uniformly throughout its bulk. The conductance of the wafer is evaluated in a contactless mode using an inductively coupled RF bridge. This tool allows the monitoring of the decay of the conductance of the sample when the radiation is turned off. From this transient, you can extract the lifetime of the wafer. As the process control engineer in charged of wafer inspection, you are asked to verify that the tool is working properly before signing off the invoice.



(a) You select a typical product wafer, a 600-μm-thick p-type wafer with $N_A = 10^{17}$ cm^{-3}. You grow a high-quality SiO_2 on both sides in a process that you know should result in extremely low surface recombination velocity. This allows you to assume that $S = 0$ on both surfaces. You take a measurement of the conductance decay time under low-level injection conditions. You measure $\tau = 10$ μs. Now you sandblast the oxide away from both surfaces producing highly recombining surfaces with $S = \infty$ and you take another measurement of the decay time. You are very happy with the result and conclude that everything is working just as expected.

How long was the decay time obtained in the second measurement? State any assumptions you need to make.

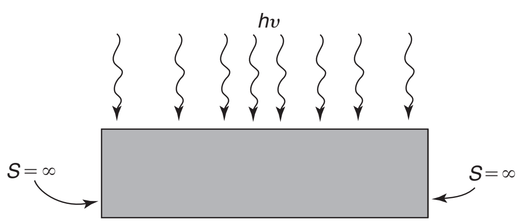
(b) You have one more idea that will allow you to double check the tool. You repeat the experiment with an identical sample covered with SiO_2 and you now take a reading of its *excess conductance* in steady state (i.e., the conductance under illumination minus the conductance in thermal equilibrium). The intensity level is such that low-level injection conditions are guaranteed. You sandblast the oxide away and you take another reading of the excess conductance of the sample *at the same illumination intensity*. From the ratio of these two measurements, you conclude that everything is working just fine. You recommend that the invoice be signed.

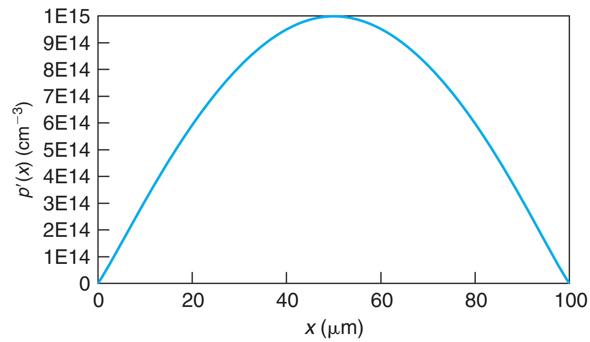
What was the ratio of the steady-state excess conductance of the sandblasted wafer to the passivated wafer that you obtained in these measurements? State any assumptions you need to make.

5.9 A piece of n-type Si with a uniform doping level $N_D = 10^{17}$ cm^{-3} is illuminated in steady state with light of nonuniform intensity, as sketched below on the left. The semiconductor region is surrounded by two surfaces characterized by a surface recombination velocity $S = \infty$. As a result of the illumination, a steady-state excess carrier profile, also graphed below, is established throughout with a functional dependence:

$$p'(x) = 10^{15} \sin\left(\frac{\pi}{100}x\right) \text{ cm}^{-3} \text{ with } x \text{ in } \mu\text{m}$$

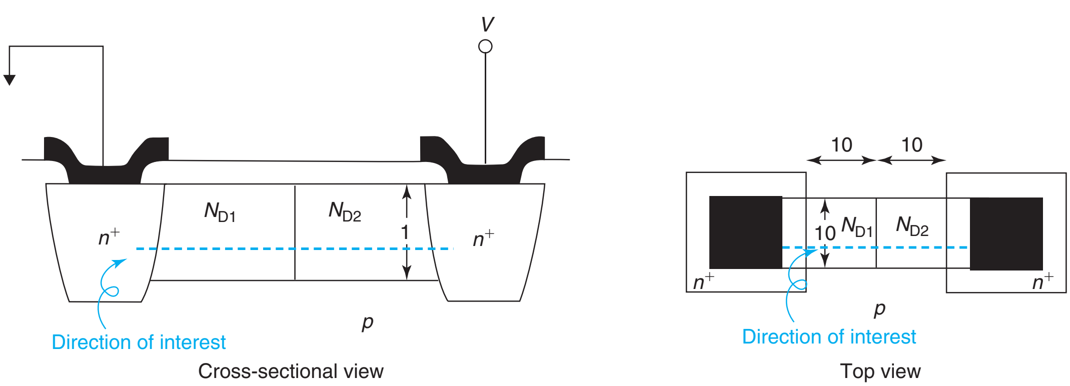
This is a one-dimensional problem.





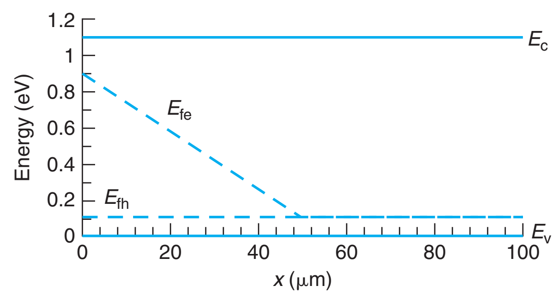
(a) Give an expression for the hole current density as a function of x everywhere. Sketch.
(b) Give an expression for the net recombination rate as a function of x everywhere. Sketch.
(c) Give an expression for the electron current density as a function of x everywhere. Sketch.
(d) Give an expression for the electric field as a function of x everywhere. Sketch.
(e) Verify the quasi-neutrality assumption.
(f) Verify that $|J_h(\text{drift})| \ll |J_h(\text{diff})|$.
(g) Synthesize the spatial dependence of the generation function that has produced the above excess carrier concentration profile. Sketch.

5.10 Consider a Si integrated resistor as sketched below. The central region that dominates the resistance features two different doping levels, as sketched. $N_{D1} = 10^{18}$ cm^{-3}, and $N_{D2} = 10^{16}$ cm^{-3}. Neglect any resistance contributions due to the contacts or the n^+ regions. The temperature is 300 K. All dimensions are in micrometers.



(a) Sketch to scale a band diagram along the indicated direction in thermal equilibrium.
(b) Sketch to scale a band diagram along the indicated direction for $V = 1$ V applied across the contacts.
(c) Estimate the current flowing through the structure with $V = 1$ V.
(d) Estimate the maximum current that can possibly flow through this structure.

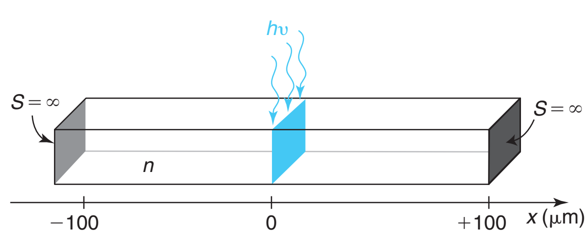
5.11 A 100-μ-m-long Si bar at room temperature in steady state has the energy band diagram indicated below ($E_{fh} \simeq E_F$). What is shown is the whole sample, no portion of it is missing.



(a) Circle all the terms that apply to the situation:
- thermal equilibrium/out of equilibrium
- uniformly doped/nonuniformly doped
- n-type/p-type
- low-level injection/high-level injection/extraction
- intrinsic/extrinsic

(b) Estimate the electron diffusion current density at $x = 0$. Make sure that sign is correct.
(c) Estimate the hole diffusion current density at $x = 0$. Make sure that sign is correct.
(d) Estimate the hole drift current density at $x = 0$. Make sure that sign is correct.
(e) Quantitatively sketch J_e^{diff}, J_e^{drift}, J_h^{diff}, J_h^{drift}, and J_T versus x.

5.12 A 200-μm-long n-type Si bar is illuminated at the center with a narrow beam of deep penetrating radiation, as sketched in the figure below. The generation rate is $g_l = 10^{22}$ cm$^{-2}\cdot$s^{-1}. The doping level of the sample is uniform and equal to $N_D = 10^{18}$ cm^{-3}. The temperature is $T = 300$ K. The two surfaces at ±100 μm are characterized by $S = \infty$. This is a one-dimensional problem: all action is occurring along x.



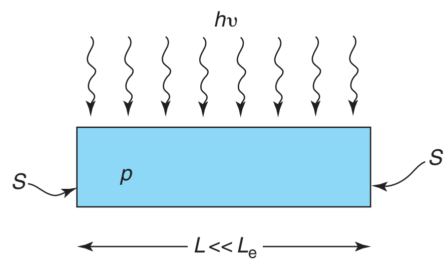
Estimate the diffusion velocity of excess holes in the x direction at $x = 0^+$.

5.13 In your company, there is a tool to measure the carrier lifetime that is based on the photoconductance decay technique studied in Problem 5.8. You need to make a measurement of the "true" bulk carrier lifetime on a certain sample and you are concerned about the impact of the surfaces on the measurement. You have heard that a way to separate the contribution of the surfaces is to take decay time measurements on two pieces of the same sample that have been thinned down to different thickness.

Your sample is 400 μm thick. You break it in half and thin down one portion to a thickness of 200 μm. You sandblast all surfaces of both pieces to insure $S = \infty$. You take a photoconductance decay measurement on each piece. On the 400-μm thick portion, you observe a dominant mode with a decay time of 6.8 μs. On the 200-μm thick portion of the sample you get a dominant mode with a decay time of 3.4 μs.

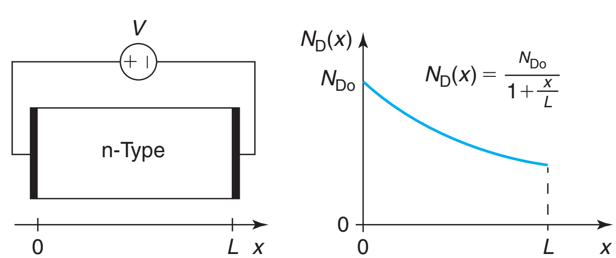
From these measurements, estimate the bulk carrier lifetime and the minority carrier mobility of the sample.

5.14 Consider an extrinsic p-type semiconductor bar under uniform light excitation throughout, as sketched below. The length of the bar, L, is much shorter than the diffusion length of excess carriers. The two surfaces are characterized by a surface recombination velocity S.



(a) Derive an expression for the excess minority carrier distribution throughout the bar. Sketch.
(b) Derive an expression for the excess minority carrier current density throughout the bar. Sketch.

5.15 Consider an extrinsic nondegenerate n-type semiconductor bar with a doping profile along its length as sketched below. The bar has ohmic contacts at both ends that allow it to be connected to a battery, as sketched in the figure. The doping distribution is such that the sample can be considered quasi-neutral everywhere. This is a one-dimensional problem: all action is occurring along x.

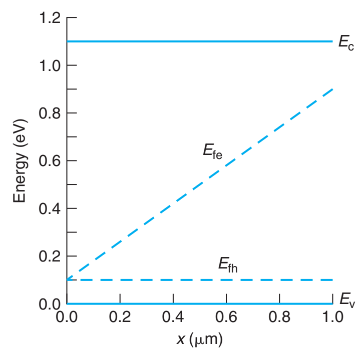


(a) In thermal equilibrium ($V = 0$), derive an expression and sketch the electric field along the length of the sample.

(b) With a voltage V applied across the ohmic contacts, as sketched in the figure, derive an expression for the current density flowing through the sample. Assume mobility-limited transport and position-independent mobility.

(c) With a voltage V applied across the ohmic contacts, as shown in the figure, derive an expression and sketch the *excess* electric field profile produced along the length of the sample.

5.16 The ohmic contact end of a Si bar at room temperature has the band diagram indicated below ($E_{fh} - E_v \simeq E_F - E_v$). The actual ohmic contact is located at $x = 0$. The ohmic contact is open-circuited, that is, there is no current coming out of the sample.



(a) Circle all the terms that apply to the situation:
- thermal equilibrium/out of equilibrium
- uniformly doped/nonuniformly doped
- n-type/p-type
- low-level injection/high-level injection/extraction
- intrinsic/extrinsic

(b) Estimate the electron current density at $x = 0$.

(c) Estimate the electron flux at $x = 0$.

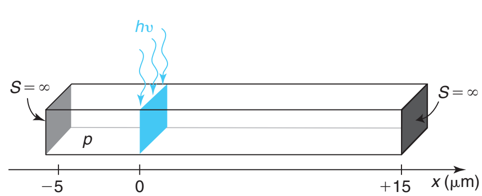
(d) Estimate the net recombination rate at $x = 0$.

(e) Estimate the net electron velocity at $x = 0$.

(f) Estimate the gradient of the electron concentration at $x = 0$.

(g) Estimate the surface recombination velocity at $x = 0$.

5.17 A $L = 20$ μm p-type Si bar is illuminated off-center with a narrow beam of deep penetrating radiation, as sketched in the figure below. The generation rate is $g_l = 10^{20}$ cm$^{-2} \cdot$ s^{-1}. The doping level of the sample is uniform and equal to $N_A = 10^{17}$ cm^{-3}. The temperature is $T = 300$ K. The two surfaces at -5 and 15 μm are characterized by $S = \infty$. This is a one-dimensional problem: all action is occurring along x.



(a) Quantitatively sketch the resulting excess carrier concentration along x.
(b) Calculate the minority carrier current density at $x = -5\ \mu\text{m}$.
(c) If we now increase the light intensity, at what generation rate the low-level injection assumption is rendered invalid?

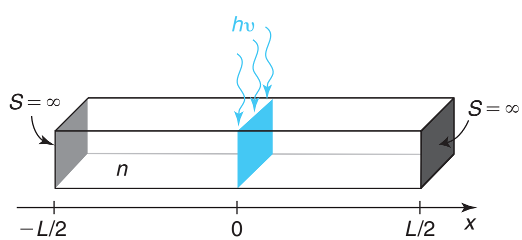
5.18 In the measurement of carrier lifetime by the photoconductive decay (PCD) technique (studied in Problem 5.8), the American Society for Testing and Materials in its ASTM:F28 specification states that the maximum bulk lifetime for a reliable measurement is 1.4 ms for a $1 \times 1 \times 5\ \text{cm}^3$ sample. They add that such a criterion for a reliable bulk lifetime measurement comes from the necessity of making the bulk recombination component large enough compared with the surface recombination component.

Estimate the error in the measurement of the lifetime using the PCD technique of a "worst-case" Si sample with a 1.4 ms bulk lifetime and with the recommended dimensions. Assume that you use deep penetrating radiation that generates electron–hole pairs uniformly throughout the entire volume of the sample. Assume low-level injection conditions. Describe the worst-case sample that you have selected.

5.19* This is a continuation of *Advanced Example 1* of Section AT5.3.1 of the notes.
(a) Derive expressions for the hole diffusion and drift currents. Sketch.
(b) Derive expressions for the electron diffusion and drift currents. Sketch.
(c) Discuss the validity of all assumptions that have been made.

Consider the field low enough for the drift velocity to be much smaller than the saturation velocity.

5.20* Consider a uniformly doped n-type semiconductor bar of length $L = 10\ \mu\text{m}$ and $S = \infty$ surfaces at two ends. All other surfaces have $S = 0$. In the middle of the bar, there is an ideal sheet of generation with a rate $g_l = 10^{21}\ \text{cm}^{-2} \cdot \text{s}^{-1}$, as sketched below. The hole mobility is $\mu_h = 400\ \text{cm}^2/\text{V} \cdot \text{s}$; the hole diffusion length L_h is much longer than the sample length L.



Address the following questions after steady state has been established.
(a) Calculate the average time that it takes for an excess hole to recombine.
(b) Quantitatively sketch the net recombination rate across the entire bar.

Now consider an identical situation except that due to some nonuniform doping distribution, a field of $\mathcal{E} = 1\ \text{kV/cm}$ is established pointing from left to right along the length of the bar. Assume that μ_h and L_h are unchanged.
(c) Calculate the average time that it takes for an excess hole to recombine.
(d) Quantitatively sketch the net recombination rate across the entire bar.

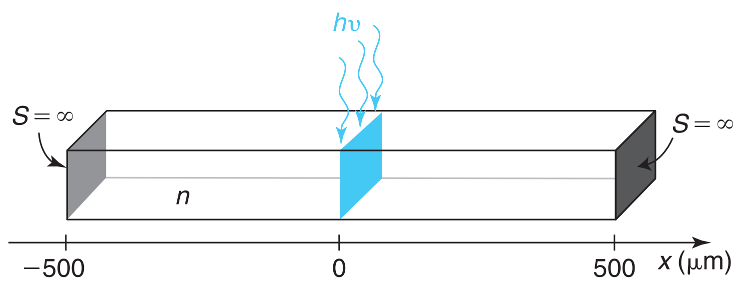
5.21 A certain doped layer in a semiconductor process has a sheet resistance $R_{sh} = 1\ \text{k}\Omega/\square$. This layer is to be used to implement integrated resistors. The minimum lateral dimension that this layer can be patterned to is 2 μm. The tolerance on any of the dimensions that can be patterned (width and length) is 0.2 μm. The tolerance of R_{sh} is 0.2% ($\frac{\Delta R_{sh}}{R_{sh}} = 0.002$).

(a) Design a 5-kΩ resistor with minimum parasitic capacitance to the substrate.

(b) Design a 5-kΩ resistor with a tolerance of 1% ($\frac{\Delta R}{R} = 0.01$). Compute overall resistor tolerance as

$$\frac{\Delta R}{R} = \sqrt{\left(\frac{\Delta R_{sh}}{R_{sh}}\right)^2 + \left(\frac{\Delta L}{L}\right)^2 + \left(\frac{\Delta W}{W}\right)^2}$$

5.22 Consider an 1-mm-long bar with a doping level $N_D = 10^{18}\ \text{cm}^{-3}$. At the two ends of the bar, there are ohmic contacts characterized by $S = \infty$. All other surfaces are perfectly passivated. This bar is subject to a sheet of deep penetrating radiation at its center that generates $g_l = 10^{20}\ \text{cm}^{-2} \cdot \text{s}^{-1}$ electron–hole pairs, as sketched below.



(a) Quantitatively sketch the resulting excess carrier concentration along x.

(b) Quantitatively sketch the energy band diagram along x. Make sure that you indicate key distances between quasi-Fermi levels and respective band edges at key points.

(c) Estimate the gradient of the quasi-Fermi level of holes at $x = 0^+$.

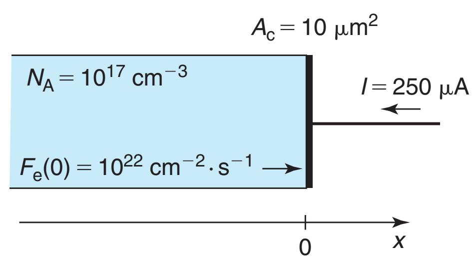
5.23 At your company, the PCD technique is used to inspect incoming wafers so as to evaluate the concentration of a certain midgap trap of great concern. This trap is characterized by $c_e = c_h = 10^{-10}\ \text{cm}^3/\text{s}$. In the evaluation of a new wafer lot, a randomly selected Si wafer is thinned down to 100 μm thickness and its surfaces are sandblasted to produce a reliable $S = \infty$ at both surfaces. In the PCD technique, deep penetrating radiation is used and the characteristic decay time of the first-order mode, τ_1, is measured at room temperature. The sample under test is under low-level injection at all times.

As a junior engineer, you have been asked to investigate a dispute between the wafer supplier and your company's wafer quality auditors. The maximum allowed trap concentration of incoming n-type wafers is $N_{tmax} = 10^{16}\ \text{cm}^{-3}$. Recently, concentrations in the $N_t \sim 10^{17}\ \text{cm}^{-3}$ range have been measured in some incoming lots. The supplier denies that the shipped wafers have such a high trap concentration level. They blame inadequate incoming wafer inspection procedures at your company.

In order to shed some light onto this dispute, you decide to investigate the sensitivity of your company's PCD technique to N_t.

(a) Estimate the lowest possible trap concentration that the PCD technique, as currently implemented at your company, can determine on n-type wafers.
(b) You realize that the recent shipments that have sparked the controversy involve wafers with a resistivity of 0.006-Ω.cm (rather than the more usual 10 Ω.cm). Estimate again the lowest possible trap concentration that the PCD technique can determine on 0.006-Ω.cm n-type Si wafers. What do you conclude from this? What is the source of the problem?
(c) While you are at it, you decide to improve the sensitivity of the technique by suggesting a change in the sample preparation. You decide that instead of sandblasting the wafer surfaces, they ought to be passivated. If this could be done and a surface recombination velocity of $S = 0$ could be attained, what would be the minimum trap concentration that could be evaluated on 10-Ω.cm n-type wafers?

5.24 Sketched below is an ohmic contact to a p-type Si sample with $N_A = 10^{17}$ cm^{-3} at 300 K. The area of the contact is $A_c = 10$ μm^2.



Flowing into this contact is a current of $I = 250$ μA. The electron flux at the surface of the semiconductor is $F_e(0) = 10^{22}$ cm$^{-2} \cdot$ s^{-1}.

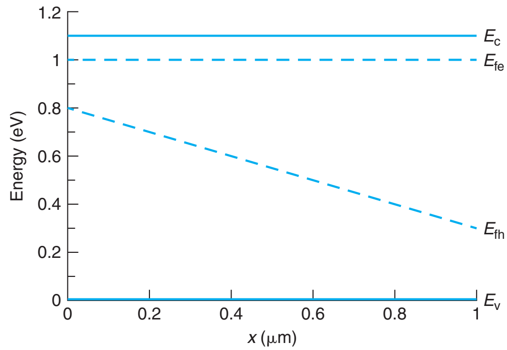
(a) Estimate the electron current density at $x = 0$, $J_e(0)$.
(b) Estimate the gradient of the electron concentration at $x = 0$.
(c) Estimate the hole current density at $x = 0$, $J_h(0)$.
(d) Estimate the hole flux at $x = 0$, $F_h(0)$.
(e) Estimate the net recombination rate at the metal–semiconductor interface, U_s.
(f) Sketch an energy band diagram of the semiconductor in the region close to the contact.

5.25 Consider a uniformly doped n-type Si sample with $N_D = 10^{17}$ cm^{-3} at room temperature. An electric field is applied from the outside world to this sample so that there is a uniform current density equal to $J_t = 1.5 \times 10^5$ A/cm^2.

(a) Estimate the electric field that is set up inside the semiconductor.
(b) Estimate the power dissipation per unit volume in this sample.
(c) If you assume that all the electrical power is dissipated in optical phonon emission, estimate the rate of optical phonon emission per unit volume in this sample.

5.26 The energy band diagram below corresponds to a region of a Si bar at room temperature next to a surface ($E_c - E_{fe} \simeq E_c - E_F$) under static conditions. The actual semiconductor surface is located at $x = 0$.

In answering the questions below, make whatever assumptions you need to make. State them and justify them clearly.



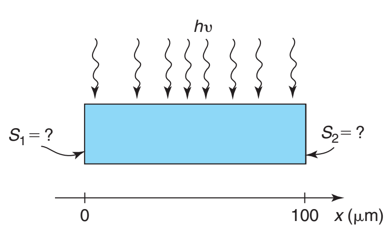
(a) Circle all the terms that apply to the situation:
- thermal equilibrium/out of equilibrium
- uniformly doped/nonuniformly doped
- n-type/p-type
- degenerate/nondegenerate
- low-level injection/high-level injection/extraction
- intrinsic/extrinsic

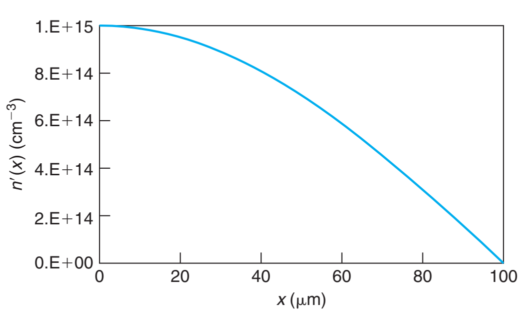
(b) Estimate the hole current density at $x = 0$.
(c) Estimate the net recombination rate at $x = 0$.
(d) Estimate the surface recombination velocity.
(e) Estimate the electron diffusion current density at $x = 0$.
(f) Estimate the electron drift current density at $x = 0$.

5.27 A piece of p-type Si with a uniform doping level $N_A = 10^{17}$ cm^{-3} is illuminated in steady state with light of nonuniform intensity, as sketched below on the left. The semiconductor region is surrounded by two surfaces characterized by different surface recombination velocities. As a result of the illumination, a steady-state excess carrier profile, also graphed below, is established throughout with a functional dependence:

$$n'(x) = 10^{15} \cos\left(\frac{\pi}{200}x\right) \text{ cm}^{-3} \qquad \text{with } x \text{ in } \mu\text{m}$$

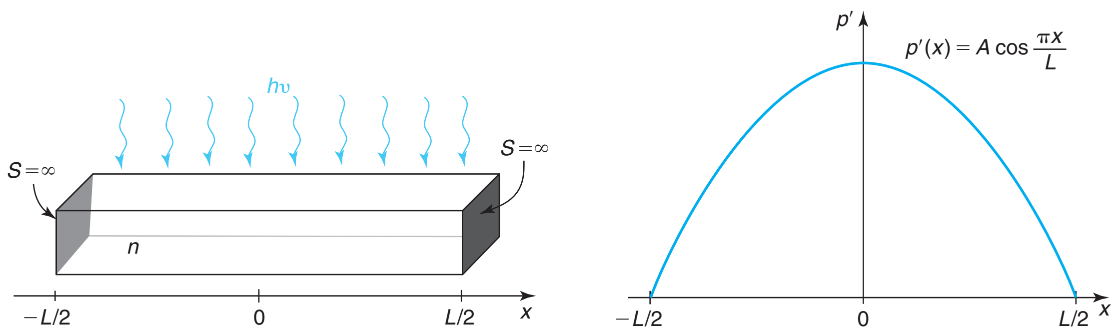
This is a one-dimensional problem.





(a) What is the value of the surface recombination velocity on the left surface? Explain your answers.
(b) What is the value of the surface recombination velocity on the right surface? Explain your answers.
(c) Give an algebraic expression for the electron current density as a function of x everywhere. Sketch.
(d) Evaluate the net recombination rate at $x = 50\ \mu\text{m}$.
(e) Evaluate the electron diffusion velocity at $x = 50\ \mu\text{m}$.
(f) Evaluate the electric field at $x = 50\ \mu\text{m}$.
(g) Evaluate the electron drift velocity at $x = 50\ \mu\text{m}$.
(h) Sketch an energy band diagram across the entire structure.

5.28 Consider a short n-type semiconductor bar of length $L \ll L_h$ with ohmic contacts at both ends illuminated by deeply penetrating radiation as indicated in the figure below. This is a low-level injection condition in one dimension.



The light intensity distribution in length is such that the resulting excess hole concentration has a shape given by

$$p'(x) = A\cos\frac{\pi x}{L}$$

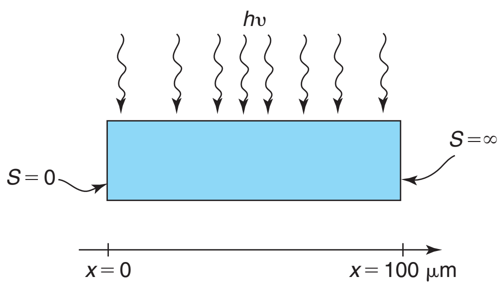
This problem is about computing the average transit time for holes in this situation. This is the dominant time constant in this minority-carrier type situation.

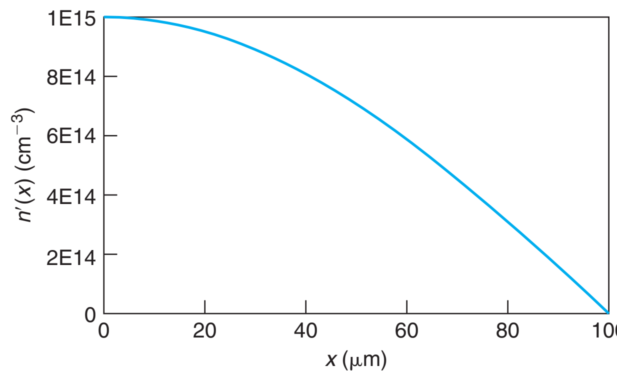
(a) Derive an algebraic expression for the transit time for holes from a location x in the bar to the nearest ohmic contact. Sketch and discuss the result that you obtain.
(b) Formulate a calculation of the average transit time for holes in this problem, that is, tell us how you would compute the average transit time.
(c) Complete the calculation of the average transit time for holes in this problem.

5.29 A piece of p-type Si with a uniform doping level of $N_A = 10^{17}\ \text{cm}^{-3}$ is illuminated in steady state with light of nonuniform intensity, as sketched below on the left. The semiconductor region is surrounded by two surfaces. The one on the left is a perfectly passivated surface and is characterized by a surface recombination velocity $S = 0$. The one on the right is an ohmic contact and is characterized by a surface recombination velocity $S = \infty$. As a result of the illumination, a steady-state excess carrier profile, also graphed below, is established throughout with a functional dependence:

$$n'(x) = 10^{15}\cos\left(\frac{\pi}{200}x\right)\ \text{cm}^{-3} \qquad \text{with } x \text{ in } \mu\text{m}$$

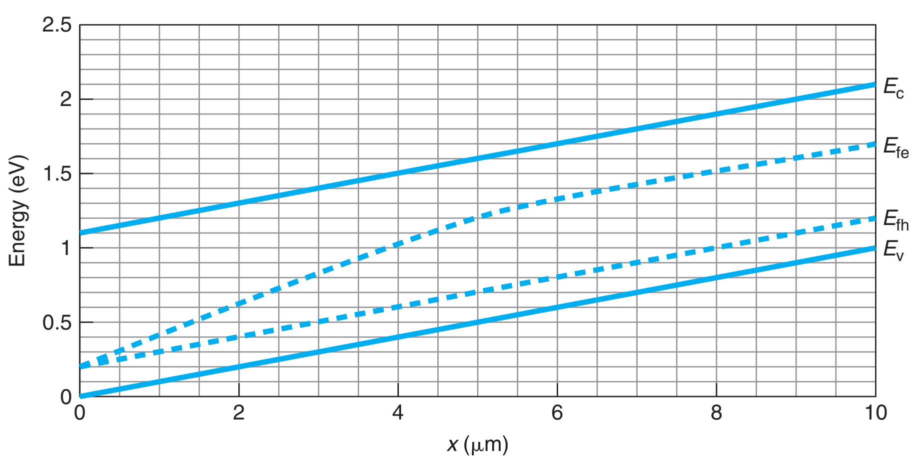
This is a one-dimensional problem. For this material, the carrier lifetime is $\tau \simeq 2 \times 10^{-5}$ s, and the electron and hole diffusion coefficients are $D_e \simeq 20\ \mathrm{cm^2/s}$ and $D_h \simeq 3\ \mathrm{cm^2/s}$, respectively.





(a) Derive an expression for the electron current density, $J_e(x)$, as a function of x. Quantitatively sketch.
(b) Derive an expression for the net recombination rate, $U(x)$, as a function of x. Quantitatively sketch.
(c) Derive an expression for the hole current density, J_h, as a function of x. Quantitatively sketch.
(d) Derive an expression for the electric field, $\mathcal{E}(x)$, as a function of x. Quantitatively sketch.
(e) Quantitatively sketch the energy band diagram across the structure.

5.30 The energy band diagram below corresponds to a region of a Si bar at room temperature under static conditions. There is an ohmic contact located at $x = 0$. The semiconductor continues beyond $x = 10\ \mu\mathrm{m}$ and there is another ohmic contact further beyond. The cross-sectional area of this piece of semiconductor is $10\ \mu\mathrm{m} \times 10\ \mu\mathrm{m}$.



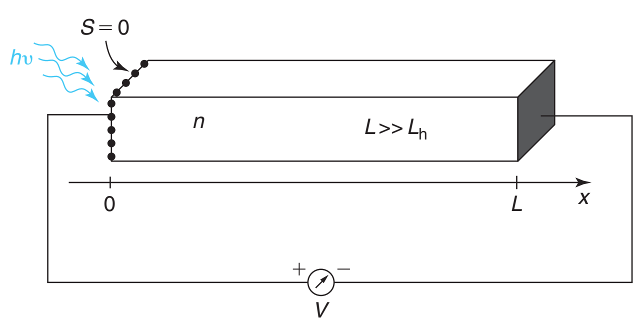
(a) Circle all the terms that apply to the situation. Briefly explain your choice.
   - thermal equilibrium/out of equilibrium
   - uniformly doped/nonuniformly doped
   - n-type/p-type
   - degenerate/nondegenerate
   - low-level injection/high-level injection/extraction
   - intrinsic/extrinsic

(b) Estimate the hole current density at $x = 8\ \mu\text{m}$.
(c) Estimate the electron current density at $x = 8\ \mu\text{m}$.
(d) Estimate the electric field at $x = 8\ \mu\text{m}$.
(e) Estimate the net recombination rate at $x = 0$.
(f) Estimate the current flowing from the wire into the contact placed at $x = 0$.

5.31 **The Dember effect** refers to the appearance of a photovoltage in a semiconductor sample that is illuminated with light such that the carrier generation rate throughout the volume is nonuniform. This problem explores the Dember effect in a simplified geometry.

Consider the one-dimensional situation sketched in the figure below. This is a long semiconductor bar that is doped n-type ($L \gg L_h$, where L is the sample length and L_h is the diffusion length for holes). The sample is illuminated on the surface at $x = 0$ with radiation that is absorbed very effectively in the semiconductor. We can simplify this by assuming that the carrier generation function is a sheet at $x = 0$ given by g_l ($\text{cm}^{-2} \cdot \text{s}^{-1}$). The generation rate is such that the sample remains under low-level injection conditions at all times. This is a static situation.

This bar is provided with a grid metal contact on the illuminated surface and another ohmic contact on the other surface so that the potential difference between these two surfaces can be measured by a voltmeter, as indicated. The metal coverage on the exposed surface is small enough that the surface recombination velocity can still be considered negligible.

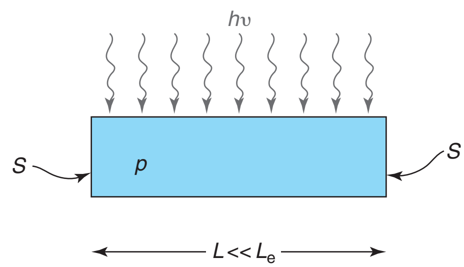


In solving this problem, you might want to recycle extensively from the results obtained in Section 5.6.1 of the class notes. Just state your assumptions.

(a) Provide an analytical equation for the excess hole concentration across the sample. Sketch your result.
(b) Derive an expression for the electric field distribution across the sample. What is the origin of this electric field? Sketch your results.
(c) Derive an expression for the Dember voltage that appears on this sample. Note the sign of the probes of the voltmeter.

(d) Estimate the Dember voltage for a sample of 1-$\Omega \cdot$ cm n-type Si when illuminated with radiation at a rate $g_l = 10^{17}$ cm$^{-2} \cdot$ s^{-1} at room temperature.

5.32 Consider an $L = 100$-μm p-type sample under steady-state illumination over its entire length, as sketched below. The sample has surfaces at $x = \pm\frac{L}{2}$ that are characterized by a surface recombination velocity S. The sample is short in the scale of the electron diffusion length.



The illumination intensity is such that the sample is under low-level injection everywhere. The profile of the excess electron concentration in this sample is given by

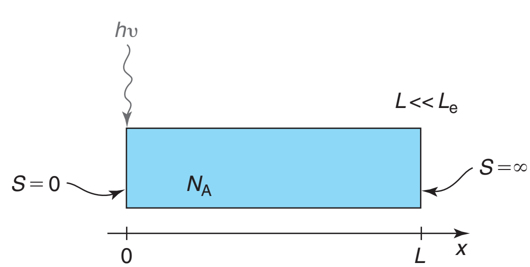
$$n'(x) = 10^{16} - 3.96 \times 10^{20} x^2 \text{ cm}^{-3}$$

with x in cm. $x = 0$ is the center of the sample. This is a one-dimensional problem where all the action is in the x direction. The doping level in this sample is $N_A = 10^{17}$ cm^{-3}.

(a) Sketch the excess electron concentration across the entire sample.
(b) Estimate the value of the surface recombination velocity at $x = L/2$.
(c) Derive an expression for the electric field, $\mathcal{E}(x)$, everywhere across this sample. Sketch.
(d) Synthesize the carrier generation function in space, $g(x)$, that gives rise to the given excess carrier profile. Sketch.

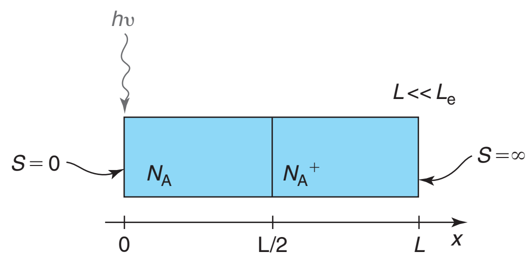
5.33 A **high-low junction** is a sudden change in doping level in a semiconductor structure. It has interesting properties that are exploited in the design of solar cells, bipolar transistors, and other semiconductor devices. This problem explores some of these properties.

Consider first a p-type semiconductor bar with uniform doping level N_A that is short in the scale of the electron diffusion length. The surface at $x = 0$ is characterized by a zero surface recombination velocity. The surface at $x = L$ is characterized by an infinite surface recombination velocity. Now consider a sheet of generation at $x = 0$, as sketched in the figure below. The generation rate g_l is such that the sample is at low-level injection everywhere. This is a steady-state situation.



(a) Sketch the resulting excess electron concentration in space. Derive an analytical expression for the excess electron concentration at $x = 0$, $n'(0)$, in terms of g_l, L, and the relevant material parameters.

Now consider a sample of identical length except that half way, the doping level abruptly increases from N_A to N_A^+, where $N_A^+ = 5N_A$. The sample is also illuminated with steady-state radiation that results on a sheet of generation at $x = 0$ with a generation rate g_l.



(b) Under low-level injection conditions, derive the boundary condition for electron concentration at $x = \frac{L}{2}$, that is, provide a relationship between $n(x = \frac{L}{2}^-)$ and $n(x = \frac{L}{2}^+)$. For this, assume that the majority carrier concentration changes from N_A to N_A^+ on a length scale much shorter than the sample length. In deriving the boundary condition at $x = \frac{L}{2}$, assume quasi-equilibrium across this junction.

(c) Sketch the resulting excess electron concentration in space across this sample. Derive an analytical expression for the excess electron concentration at $x = 0$, $n'(0)$. For this, assume that the diffusion coefficient for electrons does not change as a result of the change in doping level.

(d) Explain your result. What is the effect of the high-low junction on the minority carrier profile?

5.34 Consider an ideal one-dimensional uniformly doped n-type semiconductor bar of length L and unity cross-sectional area identical to the one illustrated in Fig. 5.10. A voltage V is applied across both ends of this bar. In Section 5.5.1, we derived an expression for the carrier distribution close to one of the ends of the bar [Eq. (5.41)]. In this problem, we study some aspects of this situation in more detail. This is a steady-state problem.

(a) Derive an expression for the electron diffusion current density close to $x = 0$. Use symmetry arguments to sketch across the entire bar.

(b) Derive an expression for the electric field close to $x = 0$. Use symmetry arguments to sketch across the entire bar.

(c) Derive an expression for the electron drift current density close to $x = 0$. Use symmetry arguments to sketch across the entire bar.

Chapter 6

PN Junction Diode

The PN junction diode (or simply "PN diode") is the first microelectronic device that we study in detail in this book. The PN diode is a two-terminal device that exploits the *rectifying characteristics* of a PN junction. What this means is that under a certain voltage polarity, current readily flows across the device, while for the reverse polarity, the device is essentially open.

The top and cross-sectional views of an integrated PN diode are shown in Fig. 6.1. It consists of a p^+ region created inside an n-type well fabricated on a p-type wafer. The contact to the n-type well is made through an n^+ region. The p region is often called the *anode* and the n region is referred to as the *cathode*.

PN diodes are very useful in many applications. They are, for example, used to provide electrostatic discharge (ESD) protection in input/output circuitry of integrated circuits. They also perform a variety of functions in analog circuits and power electronics. A number of other familiar semiconductor devices are in essence PN diodes. Examples are solar cells, light-emitting diodes, and laser diodes.

In addition, PN junctions themselves are present in virtually every microelectronic device. For example, two PN junctions back-to-back constitute the essence of the bipolar junction transistor, a very important device that we will study in detail in Chapter 11. A complementary metal-oxide-semiconductor (CMOS) transistor pair also includes several PN junctions, as can be seen in Fig. 6.2. Finally, and this is perhaps their most widespread application, reverse-biased PN junctions exhibit excellent isolation properties and are therefore employed in integrated circuits to isolate devices from each other. An example is seen in Fig. 6.1, where the n-well/p-substrate junction isolates the PN diode from the rest of the devices on the wafer.

This chapter deals with the physics of the PN junction and the PN junction diode. These are minority-carrier-type structures. In our analysis, we will heavily draw from our earlier study of minority-carrier-type transport of Chapter 5. Since minority carrier behavior is also at the heart of bipolar transistor operation, some of the materials that we develop in this chapter will reappear in Chapter 11.

This chapter is organized as follows. We start by defining the notion of the ideal PN junction diode. We follow this with an analysis of the PN diode in thermal equilibrium. We then discuss the consequences of driving the PN junction out of equilibrium through the application of a bias.

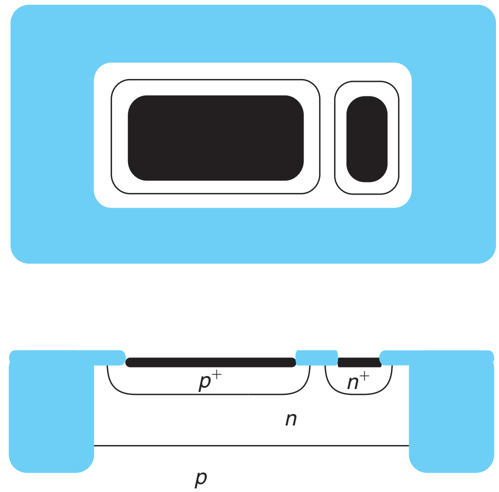


FIGURE 6.1
Top view and cross-sectional view of integrated PN junction diode.

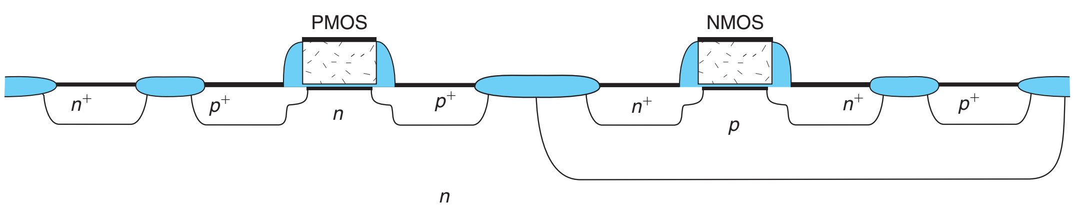


FIGURE 6.2
Cross section of a complementary metal-oxide-semiconductor (CMOS) transistor pair showing several PN junctions that naturally occur in this structure.

This culminates with the derivation of the rectifying current–voltage characteristics of the PN junction. The rest of the chapter is devoted to studying in detail the PN junction diode: equivalent circuit models, parasitics, second-order effects, and design considerations. The large-signal dynamics of the PN diode receive special attention as they are distinctly different in nature from those of the other kind of diode, the Schottky diode, that we will study in Chapter 7. We will understand the reasons behind the relatively slow dynamic response of PN diodes that makes them unsuitable for many high-frequency applications.

This is a key chapter in this book. Important concepts presented earlier are encountered again here but now in a device context. New concepts are also introduced that will reappear later on in different chapters.

6.1 THE IDEAL PN JUNCTION DIODE

It is useful to define the notion of an ideal PN junction diode. This is a concept device with a simple geometry, simplified physics, and no parasitics that captures the essence of the PN diode.

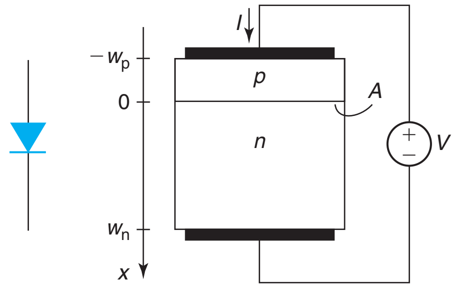


FIGURE 6.3
Left: circuit symbol for PN diode. Right: sketch of ideal PN diode.

We define this model device in this section and then analyze it in the following sections. At the end of this chapter, we will study the most significant nonidealities and the parasitics that affect real PN diodes.

Figure 6.3 shows the circuit symbol of a PN diode and a sketch of an ideal diode. This consists of a p region with doping level N_A on top of an n region with doping level N_D. The transition between p and n type regions is perfectly abrupt. Each region is electrically accessed through an ohmic contact. This device stands alone and is completely isolated from the rest of the world.

In the analysis of the ideal PN diode, we will make a number of approximations. Some of these will be "undone" later on in this chapter. These are our simplifying assumptions:

- All carrier flow is one dimensional. There are no two- or three-dimensional effects.
- Doping levels are uniform throughout (we study the impact of nonuniform doping levels in Section 6.6.5).
- We treat the p–n transition region under the so-called depletion approximation. We consider the rest of the semiconductor on both sides as quasi-neutral regions (QNRs). This is explained below.
- We assume that the QNRs are much longer than the respective minority carrier diffusion lengths (we call this a "long" diode; a "short" diode in which the QNRs are much shorter than the diffusion lengths is discussed in Section 6.6.1).
- We assume nondegenerate statistics for electrons and holes everywhere.
- Under conditions of excess carriers, low-level injection prevails at all times in all regions (some of the implications of high-level injection are examined in Section 6.6.6).
- We neglect generation and recombination of carriers in the depletion region (this is added in Section 6.6.2).
- We ignore any resistance effects from the semiconductor regions or the ohmic contacts (we study the impact of parasitic resistance in Section 6.6.3).
- We assume ideal ohmic contacts as defined in Section 5.2.2.
- We ignore any effects associated with the sidewalls of the device.

Figure 6.3 defines the axis that we will use to analyze the PN diode. Since we assume this one to be a 1D situation, there is only one axis to consider. We place its origin at the so-called metallurgical junction where the semiconductor changes from p to n type. The extent of the p region is w_p. The extent of the n region is w_n. The area of the junction is A.

The voltage and current notations are also shown in Fig. 6.3. We always define the applied voltage with the p region positive with respect to the n region. When $V > 0$, the PN diode is under forward bias. When $V < 0$, the PN diode is reverse biased. The current entering the p region is defined as positive.

6.2 IDEAL PN JUNCTION IN THERMAL EQUILIBRIUM

A good place to start the study of the ideal PN diode is with an analysis of the structure in thermal equilibrium. We can envision creating such a situation by bringing together a p and an n region in intimate contact so that charge redistribution can take place. Figure 6.4 sketches what happens. Isolated, the n side contains lots of electrons and very few holes. The p side, on the other hand, has many holes and very few electrons. A different way of looking at this is that the conduction band on the n side is populated with lots of electrons, while that on the p side is rather empty. Similarly, the valence band on the p side has a sizable amount of holes while that on the n side has very few.

Once intimate contact is established, electrons on the n side will tend to flow toward the p side since they are presented with empty states at the same energy. At the same time, holes on the p side will flow toward the n side where there are lots of available states at the same energy. We can alternatively view this intermingling of carriers as being driven by diffusion, since there are more carriers of one type on each side of the junction. While this view works well in this particular case, in general it is not a sure way to argue the sense of carrier redistribution. The energy view always provides a good understanding of the sense of carrier redistribution.

As this carrier intermingling proceeds, two things happen. First, electron–hole recombination takes place. Second and more important for us here, an electric field develops as the ionized acceptors and donors have their charge uncompensated by their respective majority carriers. The more the carrier diffusion takes place, the higher is the electric field. It is this electric field that eventually brings the diffusion process to a halt by introducing a drift force on the carriers that precisely compensates their diffusion tendency.

Once thermal equilibrium has been established and charge redistribution has stopped, a charge dipole has developed across the metallurgical junction. Sufficiently far away on either side of the junction, bulk conditions ought to prevail and the majority carrier concentrations equal the respective doping concentrations. In the vicinity of the junction, the n side has less electrons and more holes than when it was isolated. The situation is reversed on the p side. This results in net *positive charge* on the n side and net *negative charge* on the p side, as sketched in Fig. 6.5. This dipole of charge produces an electric field that results in a potential energy that must be added to the energy band diagram. The magnitude and spatial distribution of the electric field has to be such as to make the Fermi level flat everywhere, as sketched in Fig. 6.4.

The dipole of charge at the metallurgical interface of a PN junction results in a *built-in potential* ϕ_{bi}. The energy band diagram of Fig. 6.4 allows us to easily calculate ϕ_{bi}. Let us place the coordinate origin $x = 0$ at the metallurgical junction and denote $x \gg 0$ and $x \ll 0$ as the n and p regions sufficiently far away from the junction so that bulk conditions prevail. Then,

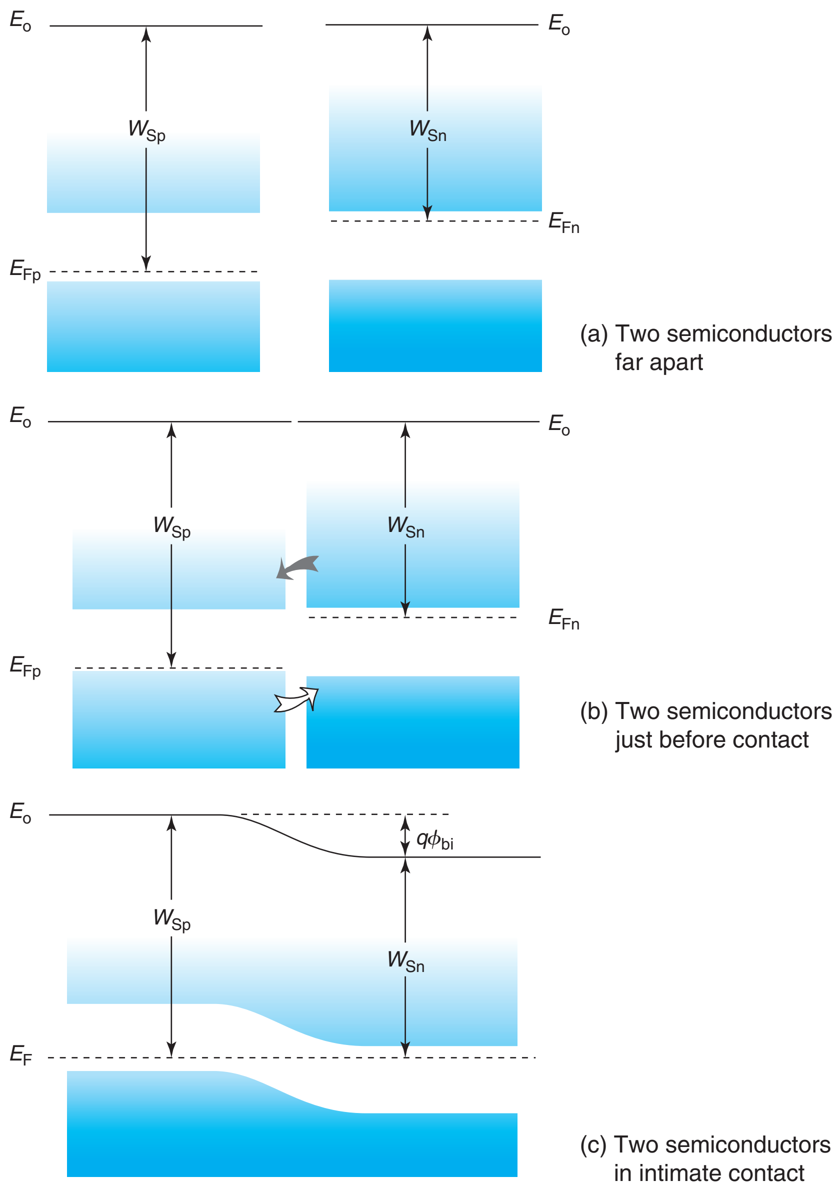


FIGURE 6.4
Formation of the energy band diagram of a PN junction in thermal equilibrium. Top: the p and n regions isolated from each other. Bottom: after contact has been established and equilibrium has been reached.

$$q\phi_{bi} = W_{Sp} - W_{Sn} = (E_C - E_F)|_{x \ll 0} - (E_C - E_F)|_{x \gg 0}$$
$$= kT \ln \frac{N_c}{n_o(x \ll 0)} - kT \ln \frac{N_c}{n_o(x \gg 0)} = kT \ln \frac{n_o(x \gg 0)}{n_o(x \ll 0)} \tag{6.1}$$

Far away on the right, where bulk conditions exist, electrons are majority carriers. Hence, $n_o(x \gg 0) = N_D$. Similarly, far away on the left, $n_o(x \ll 0) = n_i^2/N_A$. Inserting this in Eq. (6.1), we get:

$$\phi_{bi} = \frac{kT}{q} \ln \frac{N_D N_A}{n_i^2} \tag{6.2}$$

It is important to recognize the limits to the validity of this expression. Three implicit assumptions were made in its derivation. First, it was assumed that both regions are extrinsic, that is, that $N_D \gg n_i$ and $N_A \gg n_i$. This is always the case near room temperature. Second, all dopants were assumed to be ionized, also a good approximation for Si at room temperature if common dopants are utilized. Finally, in Eq. (6.1), Maxwell–Boltzmann statistics were utilized. This actually is not a good assumption in many practical PN junctions in which often one side is heavily doped. In this case, Fermi–Dirac statistics and heavy-doping effects, as described in Advanced Topic AT2.6, must be taken into account. When heavy-doping effects are important, ϕ_{bi} is slightly modified from the result given in Eq. (6.2). However, as a result of the presence of the logarithmic term, the difference is small and in general the error incurred in using Eq. (6.2) is acceptable for most applications.

Examining the volume charge density distribution in a PN junction in thermal equilibrium in Fig. 6.5, one can distinguish three different regions. Around the metallurgical junction, there is a *space-charge region* (*SCR*). On both sides of the SCR, there are two QNR, one p type and the other n type. Deep in the QNRs, $\rho \simeq 0$. As we advance toward the metallurgical junction from the n side, the electron concentration drops and the hole concentration increases. Since electrons are majority carriers, the drop of n_o is more significant from a charge point of view than the rise of p_o, and there is net positive charge due to the exposed donor atoms. Hence, $\rho \simeq qN_D$. A similar situation occurs on the p side of the junction, where $\rho \simeq -qN_A$ right next to the metallurgical junction. Because of the exponential dependence of the carrier concentration on the electrostatic potential, the transition between the QNRs and the SCR is fairly sharp.

This understanding suggests that an approximate model to the electrostatics of the PN junction in equilibrium can be obtained by performing what is called the *depletion approximation*. This approximation assumes that the two QNRs are perfectly charge neutral and that the SCR is perfectly devoid of carriers, that is, depleted. Mathematically, this can be expressed as follows:

$$\rho_o(x) \simeq 0 \qquad \text{in p-QNR: } x < -x_p \tag{6.3}$$
$$\rho_o(x) \simeq -qN_A \qquad \text{in SCR: } -x_p < x < 0 \tag{6.4}$$
$$\rho_o(x) \simeq qN_D \qquad \text{in SCR: } 0 < x < x_n \tag{6.5}$$
$$\rho_o(x) \simeq 0 \qquad \text{in n-QNR: } x_n < x \tag{6.6}$$

In these equations, the subindex "o" indicates thermal equilibrium. The location of the edges of the SCR, $-x_p$ and x_n, is determined below.

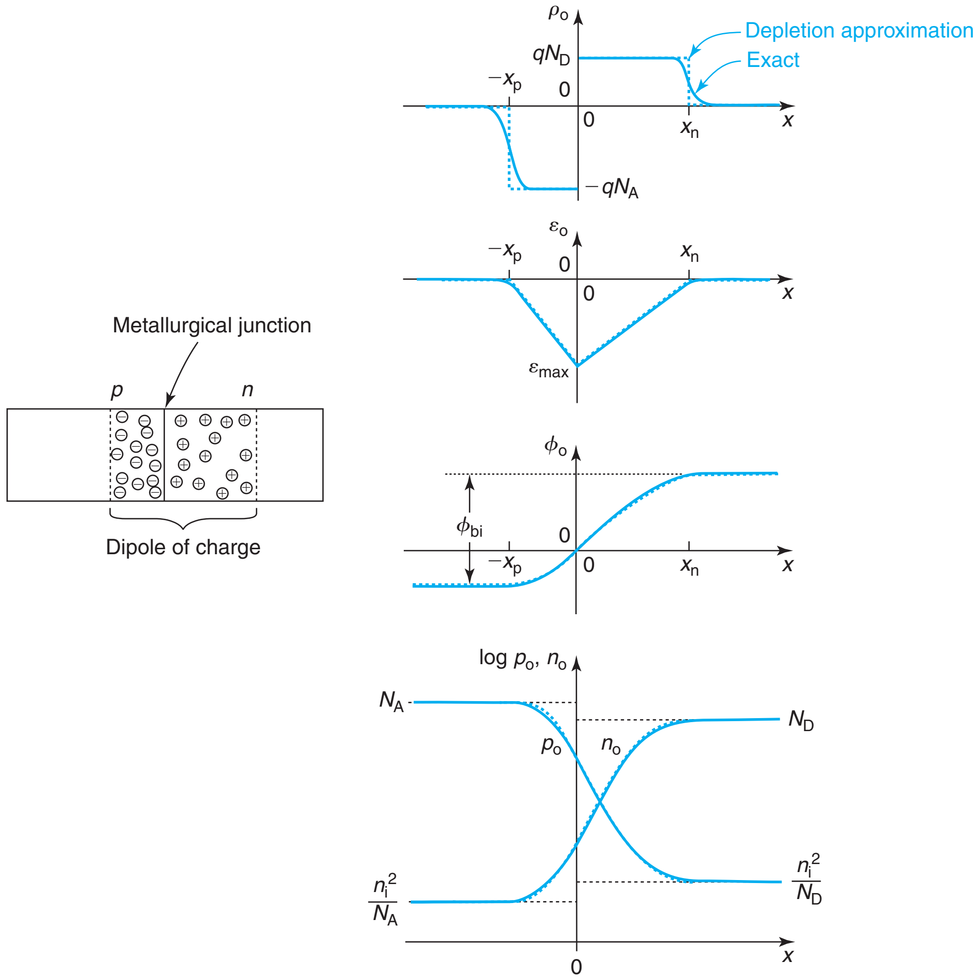


FIGURE 6.5
Electrostatics of a PN junction in thermal equilibrium. From top to bottom: volume charge density, electric field, electrostatic potential, and equilibrium carrier concentration. The continuous line is the exact solution. The discontinuous line is the calculation following the depletion approximation.

Within the depletion approximation, models for the electrostatics of the PN junction are easy to obtain. Integration of this volume charge yields the electric field profile:

$$\mathcal{E}_o(x) \simeq 0 \qquad \text{in p-QNR: } x \leq -x_p \tag{6.7}$$

$$\mathcal{E}_o(x) \simeq -\frac{qN_A}{\epsilon}(x + x_p) \qquad \text{in SCR: } -x_p \leq x \leq 0 \tag{6.8}$$

$$\mathcal{E}_o(x) \simeq \frac{qN_D}{\epsilon}(x - x_n) \qquad \text{in SCR: } 0 \leq x \leq x_n \tag{6.9}$$

$$\mathcal{E}_o(x) \simeq 0 \qquad \text{in n-QNR: } x_n \leq x \tag{6.10}$$

One more integration yields the electrostatic potential. Selecting $\phi(x = 0) = 0$, as reference:

$$\phi_o(x) \simeq -\frac{qN_A x_p^2}{2\epsilon} \qquad \text{in p-QNR: } x \leq -x_p \tag{6.11}$$

$$\phi_o(x) \simeq \frac{qN_A}{2\epsilon}(x^2 + 2x_p x) \qquad \text{in SCR: } -x_p \leq x \leq 0 \tag{6.12}$$

$$\phi_o(x) \simeq -\frac{qN_D}{2\epsilon}(x^2 - 2x_n x) \qquad \text{in SCR: } 0 \leq x \leq x_n \tag{6.13}$$

$$\phi_o(x) \simeq \frac{qN_D x_n^2}{2\epsilon} \qquad \text{in n-QNR: } x_n \leq x \tag{6.14}$$

x_n and x_p are obtained by demanding that two conditions be met. First, overall charge neutrality must be satisfied since each of the starting pieces of semiconductor were charge neutral and we have not allowed any particles to escape. This implies that:

$$qN_A x_p = qN_D x_n \tag{6.15}$$

Second, the total electrostatic potential difference across the entire structure must equal the built-in potential derived in Eq. (6.2). This results in

$$\phi_o(x_n) - \phi_o(-x_p) = \frac{qN_D x_n^2}{2\epsilon} + \frac{qN_A x_p^2}{2\epsilon} = \phi_{bi} \tag{6.16}$$

Solving in these two equations for x_n and x_p, we get:

$$x_n = \sqrt{\frac{2\epsilon N_A \phi_{bi}}{qN_D(N_D + N_A)}} \tag{6.17}$$

$$x_p = \sqrt{\frac{2\epsilon N_D \phi_{bi}}{qN_A(N_D + N_A)}} \tag{6.18}$$

The total extent of the SCR, the sum $x_n + x_p$ is

$$x_{SCR} = \sqrt{\frac{2\epsilon(N_D + N_A)\phi_{bi}}{qN_A N_D}} \tag{6.19}$$

The maximum electric field occurs at the metallurgical junction and is easily found from Eq. (6.8) or Eq. (6.9) at $x = 0$,

$$|\mathcal{E}_{o,max}| = \sqrt{\frac{2qN_A N_D \phi_{bi}}{\epsilon(N_D + N_A)}} \tag{6.20}$$

EXERCISE 6.1

Consider a PN junction like that of Fig. 6.5 with $N_A = 10^{18}$ cm^{-3} and $N_D = 10^{16}$ cm^{-3}. At room temperature, calculate (a) the built-in potential of this junction, (b) the extent of the depletion region on the p side, (c) the extent of the depletion region on the n side, and (d) the magnitude of the electric field at the metallurgical junction.

(a) To calculate the built-in potential, we use Eq. (6.2):

$$\phi_{bi} = \frac{kT}{q} \ln \frac{N_D N_A}{n_i^2} = 0.026 \text{ V} \ln \frac{10^{18} \text{ cm}^{-3} \times 10^{16} \text{ cm}^{-3}}{(1.1 \times 10^{10} \text{ cm}^{-3})^2} = 0.84 \text{ V}$$

(b) The extent of the depletion region on the p side is given by Eq. (6.18):

$$x_p = \sqrt{\frac{2\epsilon N_D \phi_{bi}}{qN_A(N_D + N_A)}} = \sqrt{\frac{2 \times 11.7 \times 8.85 \times 10^{-14} \text{ F/cm} \times 10^{16} \text{ cm}^{-3} \times 0.84 \text{ V}}{1.6 \times 10^{-19} \text{ C} \times 10^{18} \text{ cm}^{-3}(10^{16} + 10^{18}) \text{ cm}^{-3}}} = 3.3 \text{ nm}$$

(c) To obtain the extent of the depletion region on the n side, the easiest is to use the charge neutrality condition of Eq. (6.15):

$$x_n = x_p \frac{N_A}{N_D} = 3.3 \text{ nm} \frac{10^{18} \text{ cm}^{-3}}{10^{16} \text{ cm}^{-3}} = 0.33 \text{ µm}$$

(d) The electric field at the metallurgical junction can be obtained from Eq. (6.20) but is faster to use Eq. (6.8) or Eq. (6.9) at $x = 0$:

$$|\mathcal{E}_{o,max}| = \frac{qN_D}{\epsilon} x_n = \frac{1.6 \times 10^{-19} \text{ C} \times 10^{16} \text{ cm}^{-3} \times 3.3 \times 10^{-5} \text{ cm}}{11.7 \times 8.85 \times 10^{-14} \text{ F/cm}} = 5.1 \times 10^4 \text{ V/cm}$$

It is interesting to examine the electrostatics of the junction as a function of the relative doping level on both sides. If $N_A = N_D$, Eq. (6.15) implies that $x_p = x_n$. If $N_A > N_D$, then $x_p < x_n$, and vice versa. Notice that in this instance, Eqs. (6.11) and (6.14) indicate that $|\phi_o(-x_p)| < \phi_o(x_n)$. In other words, as the doping level becomes more asymmetric, the depletion region extends preferentially over the lowly doped side and most of the electrostatic potential build up occurs there.

In practical devices, it is very common to have very asymmetric junctions in which one side is much more heavily doped than the other. For example, in a P$^+$N junction, $N_A \gg N_D$. In a case like this, Eqs. (6.17–6.19) imply that

$$x_n \simeq x_{SCR} \simeq \sqrt{\frac{2\epsilon\phi_{bi}}{qN_D}} \gg x_p \tag{6.21}$$

The SCR is essentially confined to the lowly doped region *and* its thickness is basically determined by the doping level on the lowly doped side. As a consequence of this, the maximum electric field is given by

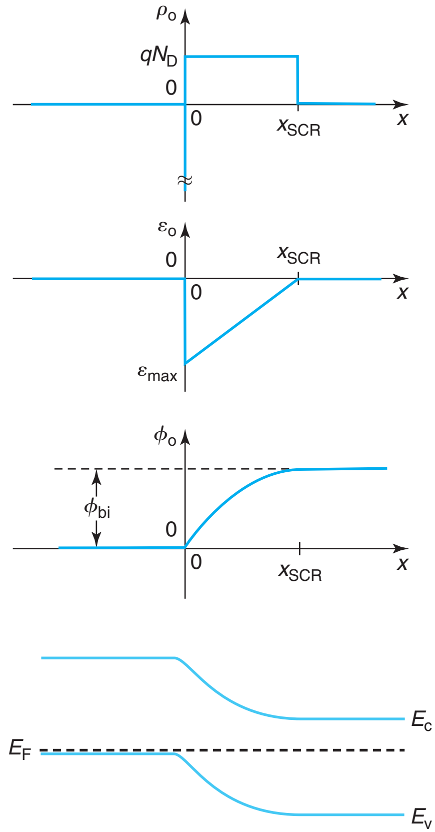


FIGURE 6.6
Electrostatics of a P$^+$N junction in thermal equilibrium. From top to bottom: volume charge density, electric field, electrostatic potential, and energy band diagram.

$$|\mathcal{E}_{o,max}| \simeq \sqrt{\frac{2qN_D\phi_{bi}}{\epsilon}} \tag{6.22}$$

and is also determined by the doping level on the lowly doped side alone. Additionally, the electrostatic potential drops mainly on the lowly doped side. From Eqs. (6.11) and (6.14), we find that

$$\phi_o(x_n) \simeq \phi_{bi} \gg |\phi_o(-x_p)| \tag{6.23}$$

This situation is depicted in Fig. 6.6. In an asymmetric junction, the lowly doped side dominates the electrostatics. As the doping level of the lowly doped side increases, $x_{SCR} \downarrow$ but $|\mathcal{E}_{o,max}| \uparrow$. There are important consequences associated with this that we will discuss later on.

A discussion on the validity of the depletion approximation is presented in Advanced Topic AT6.1.

6.3 CURRENT–VOLTAGE CHARACTERISTICS OF THE IDEAL PN DIODE

A PN diode displays rectifying behavior. With a forward bias across (p-side positive with respect to the n side), current readily flows. Under reverse bias, only a trickle current flows and the device can be considered essentially an open circuit. This section explains the origin of this unique behavior of the PN diode and builds models for its current–voltage characteristics. We start by examining the modifications to the electrostatics that follows the application of a voltage.

6.3.1 Electrostatics under bias

Applying a voltage across the two terminals of a PN junction modifies the electrostatics under equilibrium that we described in the previous section. In a first pass analysis, it is easier to understand the impact of a voltage by initially neglecting any influence of the current that flows. We can later on apply appropriate corrections to this simple analysis.

As we will see in Chapter 7, the ohmic contacts of the diode provide a way for the terminals of the battery to grab on the *majority carrier Fermi levels* on either side of the junction. This means that in forward bias, for example, the quasi-Fermi level for electrons on the n side is raised with respect to the quasi-Fermi level for holes on the p side by an energy difference qV, where V is the voltage of the battery. The contrary happens in reverse bias. Since the QNRs contain lots of majority carriers, their concentrations are hard to upset in a significant way. In consequence, as the majority quasi-Fermi levels split, the entire band diagram on each side is dragged along with them. In each QNR, the bands remain flat reflecting our assumption that if no current flows, no ohmic drop occurs in the QNRs. This is illustrated in Fig. 6.7. In this figure, only the majority quasi-Fermi levels in the QNRs are drawn. The rest of the diagram has to wait until we have had a chance to discuss the physics of the current.

As a consequence of applying a voltage, the total potential difference across the junction is upset from its equilibrium value ϕ_{bi}. In forward bias, it is reduced to $\phi_{bi} - V$. In reverse bias, it is increased to $\phi_{bi} - V$ (in reverse bias V is negative). This in turn demands that the dipole of charge that appears across the junction be modified too. In forward bias it has to be reduced, and in reverse bias it has to be increased. Since the doping levels cannot be changed, the only way to accomplish this is to change the thickness of the SCR. The changes of the electrostatics of the PN junction upon the application of a bias are summarized in Fig. 6.8.

Qualitatively, the electrostatics of the PN junction under bias do not differ from the thermal equilibrium situation. All the equations that we have derived above basically apply in forward and reverse bias if we substitute ϕ_{bi} by $\phi_{bi} - V$. In particular, the extent of the SCR becomes

$$x_{SCR}(V) = \sqrt{\frac{2\epsilon(N_D + N_A)(\phi_{bi} - V)}{qN_A N_D}} = x_{SCR}(V = 0)\sqrt{1 - \frac{V}{\phi_{bi}}} \tag{6.24}$$

The maximum electric field is

$$|\mathcal{E}_{max}(V)| = \sqrt{\frac{2qN_A N_D(\phi_{bi} - V)}{\epsilon(N_D + N_A)}} = |\mathcal{E}_{max}(V = 0)|\sqrt{1 - \frac{V}{\phi_{bi}}} \tag{6.25}$$

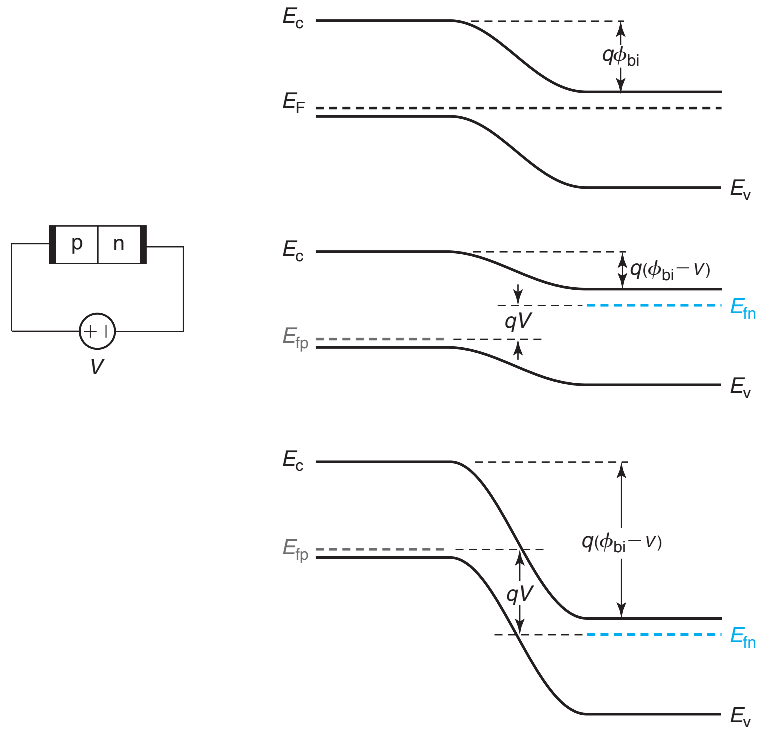


FIGURE 6.7
Energy band diagram of a PN junction in equilibrium (top), forward bias (middle), and reverse bias (bottom). Only the majority quasi-Fermi levels in the QNRs are drawn.

both valid in forward and reverse bias. With the application of a forward bias, the SCR shrinks and the maximum field decreases. A reverse bias produces the contrary effect, the SCR widens and the maximum field increases in magnitude.

If the junction is asymmetric, most of the voltage that is applied drops on the lowly doped side of the junction. The SCR widens preferentially into the lowly doped side.

A discussion on the validity of the depletion approximation under bias is presented in Advanced Topic AT6.1.

6.3.2 *I–V* characteristics: qualitative discussion

A PN junction is a *minority carrier*. It is the behavior of the minority carriers that represents the bottleneck to current flow. Let us understand why this is the case.

In thermal equilibrium, the net current everywhere along the diode is zero. Actually, the principle of detailed balance demands that separately, the hole and electron currents be zero at all locations. In general, carrier current is composed of drift and diffusion. In the QNRs, the carrier concentration is uniform and there is no field, so both drift and diffusion are zero separately.

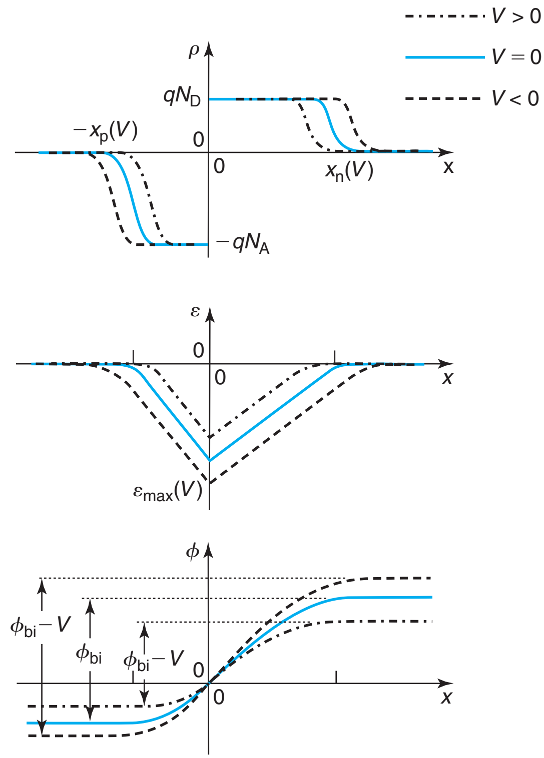


FIGURE 6.8
Volume charge density, electric field, and electrostatic potential of a PN junction as a function of bias.

Inside the SCR, there is an electric field and the gradients of carrier concentrations are very steep. Here, drift has to balance diffusion precisely at every point for thermal equilibrium to hold. That means that the diffusion tendency of carriers is precisely counterbalanced by their drift inside the electric field. It is important to notice that the sign of the electric field is the correct one for this to happen.

The application of a voltage breaks this dynamic balance inside the SCR. This can be seen in Fig. 6.9. A forward bias reduces the electric field that produces a decrease in the drift component of the carrier current. In consequence, diffusion prevails over drift and there is a net electron flow from the n side (where they are majority carriers) to the p side (where they are minority carriers). Similarly, holes flow from the p side to the n side. The term that is used to describe this process is *minority carrier injection*.

In reverse bias, the contrary happens. A negative voltage increases the electric field inside the SCR with the consequence that drift prevails over diffusion. As a result, electrons preferentially flow from the p region (where they are minority carriers) to the n region (where they are majority carriers). Similarly, holes flow from the n region to the p region. This receives the name of *minority carrier extraction*.

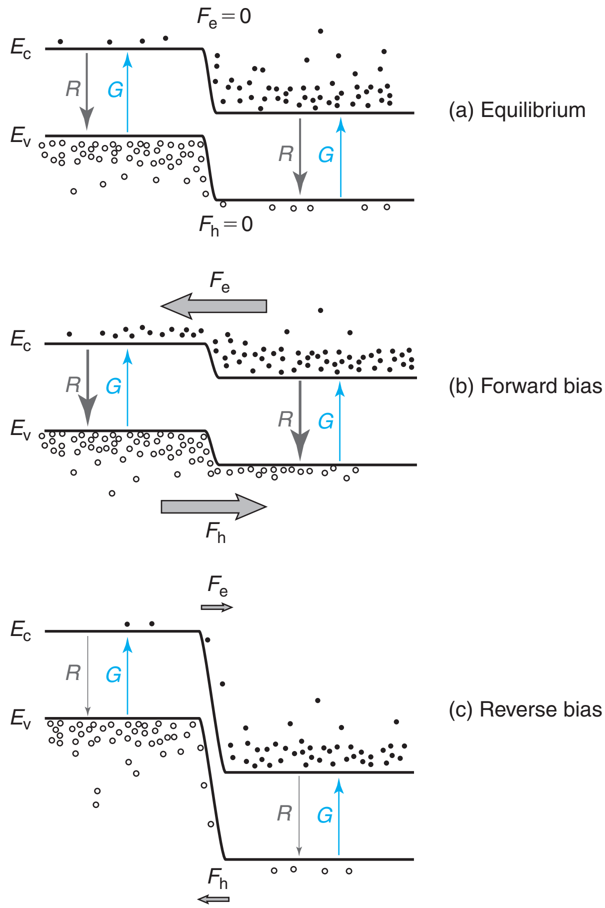


FIGURE 6.9
Current balance across SCR in thermal equilibrium (top), forward bias (middle), and reverse bias (bottom). In equilibrium, there is no net current crossing the SCR. In forward bias, minority carrier injection and recombination into the QNRs takes place. In reverse bias, minority carrier generation and extraction out of the QNRs dominates.

The energy view provides an intuitive picture of what happens under bias, as sketched in Fig. 6.9. In thermal equilibrium, electrons are in equilibrium among themselves across the whole diode. There are many more electrons on the n side than on the p side. However, there is a steep energy barrier (of a height $q\phi_{bi}$) separating these two regions. Some electrons on the n side do have enough energy to get injected into the p region. But this is precisely balanced out by electrons on the p side "falling down" or getting extracted into the n region. The net flow of electrons is precisely zero in thermal equilibrium. In forward bias, this energy barrier is reduced. In consequence, many more electrons on the n region now have enough energy to overcome the reduced barrier and therefore get injected into the p side. Electron extraction, on the other hand, is essentially unchanged since electrons can continue to easily slide down the conduction band. In contrast, in reverse bias the energy barrier at the junction is increased. Relative to equilibrium, a smaller number of electrons have now enough energy to overcome this larger barrier and get injected into the p side. Electron extraction from the p side to the n side is not modified much since the junction barrier does not prevent this flow. Then, net electron extraction prevails. A similar series of events is experienced by the holes.

But this cannot be the entire picture. In forward bias, what happens to the electrons injected into the p side? Also, what about the holes injected into the n side? In reverse bias, how are the electrons and holes that are extracted made up? We see net current flowing through the SCR, but how is this current supported through the QNRs?

Under forward bias, minority carrier injection raises the total minority carrier concentration above the equilibrium value in the QNRs. The presence of excess minority carriers also drives up the recombination rate above the value that balances the thermal generation rate. The resulting *net recombination* tends to extinguish the injected minority carriers. There is then a continuous carrier flow from the majority carrier side to the minority carrier side where they recombine with the large background of majority carriers there. Most of the recombination takes place within a few diffusion lengths of the junction where the injection takes place. Hence, at sufficient distance from the edge of the SCR, the minority carrier concentration returns to its equilibrium value. The resulting minority carrier profiles are sketched at the top of Fig. 6.10. Since the supply of majority carriers in the QNR is readily replenished by its ohmic contact, it is the dynamics of carriers as minority carriers that controls the magnitude of the current, that is, electrons in the p-QNR and holes in the n-QNR. It is because of this that the PN junction is said to be a minority carrier device.

In reverse bias, the suppression of minority carrier injection results in net minority carrier extraction. The minority carrier concentrations, consequently, drop in both QNRs below their equilibrium values. This breaks again the balance between thermal generation and recombination with *net generation* taking place to supply the minority carriers that are being extracted. A few diffusion lengths away from the edge of the SCR, the minority carrier concentrations return to the equilibrium values. The resulting carrier profiles are sketched at the bottom of Fig. 6.10. A continuous current path is established with the generated minority carriers in the QNRs being extracted by the SCR and coming out of the ohmic contact to the outside world.

Does our qualitative understanding explain the rectifying characteristics of the PN junction? Surely. Let us just focus on electrons. In forward bias, the minority carrier current is proportional to the number of electrons that have enough energy to overcome the energy barrier. Since this barrier is reduced as $\sim V$, the number of electrons with enough energy to overcome it goes as $\sim e^{qV/kT}$, and so does the current. In reverse bias, once the barrier is made large enough, electron injection from the n side to the p side is completely suppressed. The current is then entirely supported by electron generation in the p-QNR. This quickly saturates as the electron concentration at the edge of the junction approaches zero. Then, the reverse bias current also saturates.

6.3.3 *I–V* characteristics: quantitative models

Armed with the understanding produced in the previous section, we can now lay out a strategy for developing a model for the current through the PN junction. We must focus our work on the

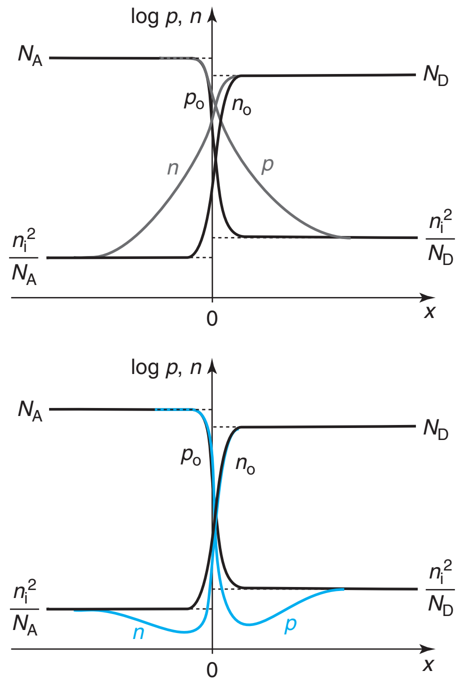


FIGURE 6.10
Sketch of carrier profiles across PN diode in forward bias (gray lines, top) and reverse bias (blue lines, bottom). In both cases the black lines represent the equilibrium carrier concentrations.

minority carriers in each QNR (in the ideal PN diode, we neglect minority carrier generation and recombination in the SCR; we will add this later in Section 6.6.2). The calculation is basically a four-step process. First, we must calculate the minority carrier concentrations at the respective edges of the QNR, $n(-x_p)$ and $p(x_n)$ (see Fig. 6.11). Second, we must derive expressions for the velocity at which minority carriers are injected into or extracted from the QNRs, $v_e(-x_p)$ and $v_h(x_n)$. Third, we compute the current that minority carriers support in their flow across the SCR, $J_e(-x_p)$ and $J_h(x_n)$. Finally, we add the contributions of electrons and holes and multiply by the area of the junction.

Boundary conditions across SCR

Let us start with the minority carrier boundary conditions across the SCR. These can be derived in various ways. An approach is to focus on the balance between drift and diffusion currents inside the SCR. In thermal equilibrium, this balance is perfect at every point in the SCR and $J_e = J_h = 0$. As argued above, out of equilibrium this balance is broken inside the SCR and $J_e \neq 0$ and $J_h \neq 0$. The difference between drift and diffusion currents results in a net current for each type of carrier. If we were to estimate how far out of balance drift and diffusion get inside the SCR for typical diode operation, we will find (see Problem 6.13) that it is actually

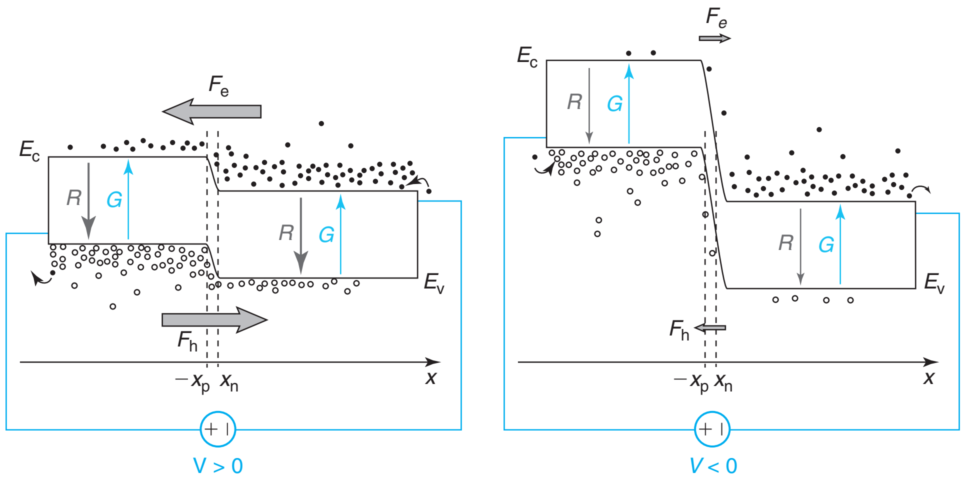


FIGURE 6.11
Sketch of carrier flux across PN junction in forward bias (left) and reverse bias (right) indicating the edges of the SCR where current calculations are performed.

very little. Relative to their magnitude, drift and diffusion never get to differ very much inside the SCR of a PN diode under bias. Only a relatively small perturbation is required to support the current that originates on the generation or recombination of minority carriers in the QNRs.

The realization of this has a very important and practical consequence that we can exploit here. In Section 4.5.2 we learned that in thermal equilibrium, there is a relationship between the *ratio* of the carrier concentrations at two points and the *difference* of the electrostatic potential between those two same points. This is the Boltzmann relation. Equation (4.58) captures this in its most general form for electrons. Across the SCR of a PN diode, the Boltzmann relation can be written, for electrons and holes, as

$$\phi_o(x_n) - \phi_o(-x_p) = \phi_{bi} = \frac{kT}{q} \ln \frac{n_o(x_n)}{n_o(-x_p)}$$

$$\phi_o(x_n) - \phi_o(-x_p) = \phi_{bi} = -\frac{kT}{q} \ln \frac{p_o(x_n)}{p_o(-x_p)}$$

Under bias, strictly speaking, these equations do not apply. However, since for each carrier type, the net carrier current is much smaller than its drift and diffusion components, then in the scale of these currents, $J_e \simeq 0$ and $J_h \simeq 0$. Hence, to first order, the Boltzmann relations should be reasonably accurate across the SCR. They can be written in the following form:

$$\phi_{bi} - V \simeq \frac{kT}{q} \ln \frac{n(x_n)}{n(-x_p)} \tag{6.26}$$

$$\phi_{bi} - V \simeq -\frac{kT}{q} \ln \frac{p(x_n)}{p(-x_p)} \tag{6.27}$$

If low-level injection conditions prevail on both sides, $n(x_n) \simeq N_D$ and $p(-x_p) \simeq N_A$ (see a discussion of the impact of high-level injection in Section 6.6.6). Using Eq. (6.2) and solving for the carrier concentrations on the minority carrier side, we get

$$n(-x_p) \simeq \frac{n_i^2}{N_A} \exp \frac{qV}{kT} \tag{6.28}$$

$$p(x_n) \simeq \frac{n_i^2}{N_D} \exp \frac{qV}{kT} \tag{6.29}$$

This result makes good physical sense. With $V = 0$, the concentrations of minority carriers at the edges of the SCR are equal to the equilibrium values. For $V > 0$, these concentrations increase by $e^{qV/kT}$ because the energy barrier across the junction has been decreased by an amount equal to qV. For $V < 0$, the minority carrier concentrations at the edges become negligible as the energy barrier is increased, carrier injection is completely suppressed, and carrier extraction dominates.

The excess carrier concentrations, as defined in Eqs. (3.44) and (3.45), are

$$n'(-x_p) \simeq \frac{n_i^2}{N_A} \left(\exp \frac{qV}{kT} - 1 \right) \tag{6.30}$$

$$p'(x_n) \simeq \frac{n_i^2}{N_D} \left(\exp \frac{qV}{kT} - 1 \right) \tag{6.31}$$

These boundary conditions have all the features that we have qualitatively discussed earlier. For electrons, for example, for $V = 0$, $n'(-x_p) = 0$. If $V \gg kT/q$, then $n'(-x_p) \gg n_i^2/N_A$ and grows exponentially with V. If, on the other hand, $V \ll -kT/q$, then $n'(-x_p)$ saturates to $\simeq -n_i^2/N_A$. With the derivation of the minority carrier boundary conditions, the first step of the diode current computation is complete.

EXERCISE 6.2

Consider a PN junction identical to that of Exercise 6.1. At room temperature and with an applied forward bias of $V = 0.6$ V, calculate: (a) the potential difference between the n side and the p side and (b) the excess minority carrier concentration at the edges of the depletion region.

(a) In Exercise 6.2, we computed the potential difference between the n side and the p side of this junction in thermal equilibrium. We found this to be $\phi_{bi} = 0.84$ V. Under forward bias, this potential difference is reduced by an amount equal to the applied voltage. Hence, we have

$$\phi(x_n) - \phi(-x_p) = \phi_{bi} - V = 0.84 - 0.6 = 0.24 \text{ V}$$

(b) The excess electron concentration at the edge of the depletion region on the p-type side can be obtained from Eq. (6.30):

$$n'(-x_p) \simeq \frac{n_i^2}{N_A} \left(\exp \frac{qV}{kT} - 1 \right) = \frac{1.1 \times 10^{20} \text{ cm}^{-6}}{10^{18} \text{ cm}^{-3}} \left(\exp \frac{0.6}{0.0259} - 1 \right) = 1.3 \times 10^{12} \text{ cm}^{-3}$$

The excess hole concentration at the edge of the depletion region on the n-type side can be obtained from the combination of Eqs. (6.30) and (6.31),

$$p'(x_n) = \frac{N_A}{N_D} n'(-x_p) = \frac{10^{18}\ \mathrm{cm}^{-3}}{10^{16}\ \mathrm{cm}^{-3}} 1.3 \times 10^{12}\ \mathrm{cm}^{-3} = 1.3 \times 10^{14}\ \mathrm{cm}^{-3}$$

In both cases, the excess minority carrier concentrations are much smaller than the corresponding doping levels. We can then conclude that this diode, under these conditions, is in low-level injection.

At this point, our presentation will treat separately forward and reverse bias, as the relevant physics are quite different.

Forward current

Let us now focus on the second step in forward bias, that is, the calculation of the velocity at which minority carriers are injected into the QNRs. Fortunately, we have already done all the work in Chapter 5. In that chapter, we studied minority carrier flow in several quasi-neutral low-level injection situations. In particular, we studied carrier flow in semiconductor bars illuminated by a thin beam of light that produced a sheet of carrier generation. Among other things, we calculated the velocity at which minority carriers diffuse away from the generation point.

Our problem here shares a lot with those already studied in Chapter 5. In a PN junction in forward bias, the reduction of potential barrier across the SCR results in minority carrier injection into the QNRs. Once inside the QNR, the behavior of the minority carriers is identical to that of the illuminated bar. In particular, if the boundary conditions are similar, the velocity at which minority carriers diffuse into the QNR should be identical to that of the photogenerated carriers.

Here, we are studying a "long" diode in which the QNRs are much longer than the respective minority carrier diffusion lengths. In Section 5.6.1, we found that the excess minority carrier profile in a long bar follows an exponential decay in space where the minority carrier diffusion length was the characteristic length of the exponential. We expect similar excess carrier profiles in the QNRs in the long diode, as sketched in Fig. 6.12. These profiles then obey the following analytical relations:

$$n'(x) = n'(-x_p) \exp \frac{x + x_p}{L_e} \quad \text{in p-QNR: } x \leq -x_p \tag{6.32}$$

$$p'(x) = p'(x_n) \exp \frac{-x + x_n}{L_h} \quad \text{in n-QNR: } x \geq x_n \tag{6.33}$$

Moreover, in Section 5.6.1, we found that the velocity at which photogenerated holes diffuse away from the generation point in a long n-type bar was given by $v_h^{diff} = D_h/L_h$ [Eq. (5.73)]. This result also applies to the PN junction. The velocity at which minority carriers diffuse into the QNRs from their respective edges is then given by

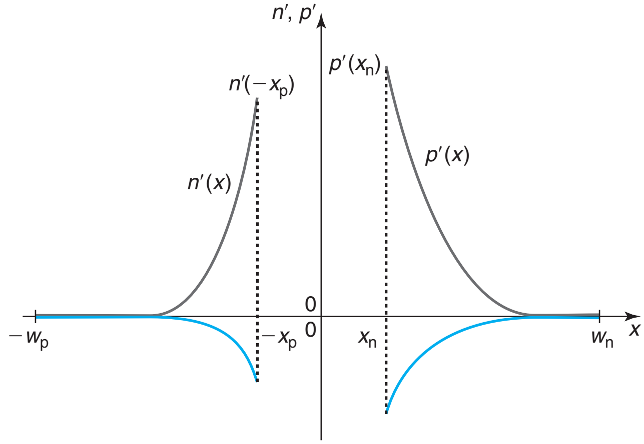


FIGURE 6.12
Excess minority carrier profiles in quasi-neutral regions of "long" diode under forward bias (gray) and reverse bias (blue).

$$v_e^{\text{diff}}(-x_p) = -\frac{D_e}{L_e} \tag{6.34}$$

$$v_h^{\text{diff}}(x_n) = \frac{D_h}{L_h} \tag{6.35}$$

where D_e and L_e pertain to minority carrier electrons in the p-type region, and D_h and L_h are associated with minority carrier holes in the n-type region. This is the result that we were looking for in step 2 of our calculation of the PN-diode current.

Step 3 consists of computing the minority carrier current injected into each QNR. Since we have both the minority carrier concentrations and the carrier velocities at the edges of the QNRs, we simply use the two pairs of results given by Eqs. (6.30) and (6.31) and Eqs. (6.34) and (6.35). We get, respectively:

$$J_e(-x_p) \simeq -qv_e^{\text{diff}}(-x_p)n'(-x_p) = q\frac{n_i^2}{N_A}\frac{D_e}{L_e}\left(\exp\frac{qV}{kT} - 1\right) = J_{es}\left(\exp\frac{qV}{kT} - 1\right) \tag{6.36}$$

$$J_h(x_n) \simeq qv_h^{\text{diff}}(x_n)p'(x_n) = q\frac{n_i^2}{N_D}\frac{D_h}{L_h}\left(\exp\frac{qV}{kT} - 1\right) = J_{hs}\left(\exp\frac{qV}{kT} - 1\right) \tag{6.37}$$

where, for convenience, we have defined J_{es} and J_{hs} as the pre-exponential factors. These equations are valid for $V > 0$.

The total current density through the junction, step 4, is the sum of these two current densities. This is consistent with our assumption that no carriers recombine inside the SCR. Adding Eqs. (6.36) and (6.37), we get

$$J = (J_{\mathrm{es}} + J_{\mathrm{hs}})\left(\exp\frac{qV}{kT} - 1\right) \tag{6.38}$$

Referring to the pre-exponential factor as the *saturation current density*, J_{s}, we can rewrite Eq. (6.38) as

$$J = J_{\mathrm{s}}\left(\exp\frac{qV}{kT} - 1\right) \tag{6.39}$$

This equation is valid, at this point, for $V > 0$, the forward-bias regime. Let us complete the model in reverse bias before we discuss this equation any further.

Reverse current

In reverse bias, the junction extracts minority carriers from the QNRs driving their concentrations below their equilibrium values. The QNRs react by generating minority carriers. The continuous process of minority carrier generation and extraction by the junction results in a reverse bias current.

The resulting excess carrier profile in the QNRs is sketched in blue in Fig. 6.12. This profile is identical in shape (exponential with characteristic length given by the diffusion length) to the forward bias profile except that the excess carrier concentration at the junction is negative instead of positive. As a result, the carrier diffusion velocities are given by the same expressions as above, except for a change of sign. We therefore have

$$v_{\mathrm{e}}^{\mathrm{diff}}(-x_{\mathrm{p}}) = \frac{D_{\mathrm{e}}}{L_{\mathrm{e}}} \tag{6.40}$$

$$v_{\mathrm{h}}^{\mathrm{diff}}(x_{\mathrm{n}}) = -\frac{D_{\mathrm{h}}}{L_{\mathrm{h}}} \tag{6.41}$$

The minority carrier currents at the respective edges of the QNRs are given by

$$J_{\mathrm{e}}(-x_{\mathrm{p}}) \simeq -qv_{\mathrm{e}}^{\mathrm{diff}}(-x_{\mathrm{p}})|n'(-x_{\mathrm{p}})| = q\frac{n_{\mathrm{i}}^2}{N_{\mathrm{A}}}\frac{D_{\mathrm{e}}}{L_{\mathrm{e}}}\left(\exp\frac{qV}{kT} - 1\right) = J_{\mathrm{es}}\left(\exp\frac{qV}{kT} - 1\right) \tag{6.42}$$

$$J_{\mathrm{h}}(x_{\mathrm{n}}) \simeq qv_{\mathrm{h}}^{\mathrm{diff}}(x_{\mathrm{n}})|p'(x_{\mathrm{n}})| = q\frac{n_{\mathrm{i}}^2}{N_{\mathrm{D}}}\frac{D_{\mathrm{h}}}{L_{\mathrm{h}}}\left(\exp\frac{qV}{kT} - 1\right) = J_{\mathrm{hs}}\left(\exp\frac{qV}{kT} - 1\right) \tag{6.43}$$

The absolute signs in $n'(-x_{\mathrm{p}})$ and $p'(x_{\mathrm{n}})$ in these equations are to correct the negative value of the excess carrier concentration that would give rise to a current in the wrong direction. These two equations are valid for $V < 0$.

Interestingly, Eqs. (6.42) and (6.43) are identical in form to Eqs. (6.36) and (6.37), respectively. Therefore, Eqs. (6.38) and (6.39) also apply for $V < 0$. We have then developed a unified model for the current density of the PN junction that applies to both forward and reverse bias. The expression that we have derived exhibits the expected classical rectifying behavior.

EXERCISE 6.3

Consider a PN diode identical to that of Exercises 6.1 and 6.2. Now consider that the p and n regions are "long" from the minority carrier point of view. At room temperature and with an applied forward bias of $V = 0.6$ V, estimate the current density flowing through this diode.

To estimate the diode current density, we need to separately compute the current density injected into each QNR and add them up. In Exercise 6.2, we already computed the excess minority carrier concentrations at the edges of the depletion region under the same conditions as given here. The best way to proceed is then to estimate the velocity at which excess carriers are injected into each QNR using Eqs. (6.34) and (6.35). For this, we need to know the respective diffusion coefficients and diffusion lengths.

To estimate the diffusion coefficients, we can use the mobility data graphed in Fig. 4.3 or the corresponding fits given in Appendix E. For electrons in the p-type region, we obtain $\mu_e = 344\ \text{cm}^2/\text{V}\cdot\text{s}$ and $D_e = 9.0\ \text{cm}^2/\text{s}$. For holes in the n-type region, we obtain $\mu_h = 476\ \text{cm}^2/\text{V}\cdot\text{s}$ and $D_h = 12.4\ \text{cm}^2/\text{s}$.

To estimate the diffusion lengths, we can either use Fig. 5.18 or we can obtain the corresponding minority carrier lifetimes and then use Eq. (5.58) to obtain the diffusion length. We proceed this way. From Appendix E, we estimate the minority carrier lifetime in the p-type region as $\tau_e = 2.3 \times 10^{-6}$ s and in the n-type region as $\tau_h = 1.3 \times 10^{-4}$ s. This yields diffusion lengths that are $L_e = 4.6 \times 10^{-3}$ cm and $L_h = 4.0 \times 10^{-2}$ cm, respectively.

Now, using Eqs. (6.34) and (6.35) we estimate the velocity at which minority carriers are injected into each QNR. For electron injection into the p-QNR, we have

$$|v_e^{\text{diff}}(-x_p)| = \frac{D_e}{L_e} = \frac{9.0\ \text{cm}^2/\text{s}}{4.6 \times 10^{-3}\ \text{cm}} = 2.9 \times 10^3\ \text{cm/s}$$

For hole injection into the n-QNR, we have

$$v_h^{\text{diff}}(x_n) = \frac{D_h}{L_h} = \frac{12.4\ \text{cm}^2/\text{s}}{4.0 \times 10^{-2}\ \text{cm}} = 3.1 \times 10^2\ \text{cm/s}$$

The current density injected into each QNR can be obtained using Eqs. (6.36) and (6.37). For the p-QNR, we have

$$\begin{aligned} J_e(-x_p) &= q\,|v_e^{\text{diff}}(-x_p)|\,n'(-x_p) = 1.6 \times 10^{-19}\ \text{C} \times 2.9 \times 10^3\ \text{cm/s} \times 1.3 \times 10^{12}\ \text{cm}^{-3} \\ &= 6.0 \times 10^{-4}\ \text{A/cm}^2 \end{aligned}$$

Similarly, for the n-QNR:

$$\begin{aligned} J_h(x_n) &= q\,v_h^{\text{diff}}(x_n)\,p'(x_n) = 1.6 \times 10^{-19}\ \text{C} \times 3.1 \times 10^2\ \text{cm/s} \times 1.3 \times 10^{14}\ \text{cm}^{-3} \\ &= 6.5 \times 10^{-3}\ \text{A/cm}^2 \end{aligned}$$

The total diode current density is the sum of these two components:

$$J = J_e(-x_p) + J_h(x_n) = 7.1 \times 10^{-3}\ \mathrm{A/cm^2}$$

If the junction has an area A, the currents injected into the p and n regions are, respectively, given by

$$I_p = AJ_e(-x_p) = AJ_{es}\left(\exp\frac{qV}{kT} - 1\right) \tag{6.44}$$

$$I_n = AJ_h(x_n) = AJ_{hs}\left(\exp\frac{qV}{kT} - 1\right) \tag{6.45}$$

and the total current is

$$I = I_p + I_n = I_s\left(\exp\frac{qV}{kT} - 1\right) \tag{6.46}$$

with $I_s = AJ_s$. I_s is known as the diode *saturation current*.

Equation (6.46) is the result that we were looking for. It gives the current through the diode as a function of the applied voltage. It is all in terms of physically meaningful parameters and fundamental constants. In forward bias, the current grows exponentially, as we expected. In reverse bias, the current saturates to $-I_S$, also as expected. This behavior is plotted in Fig. 6.13. Since I_S tends to be small, it is common to graph the I–V characteristics in a semilogarithmic scale, as shown on the right side of Fig. 6.13. In a semilog scale, the forward bias current appears as a straight line with a slope of $(q/kT)\log\ e = 0.43q/kT$, which only depends on temperature. The higher the temperature, the lower the slope. It is common to refer to this slope in units of millivolts per decade, that is, the voltage required (in millivolts) to produce an increase in current of a factor of 10. At room temperature, this figure is 60 mV/dec, a handy number to remember.

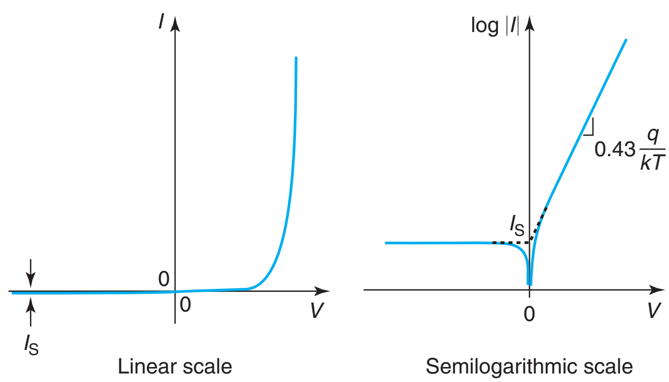


FIGURE 6.13
Sketch of *I*–*V* characteristics of ideal PN junction in linear and semilogarithmic scales.

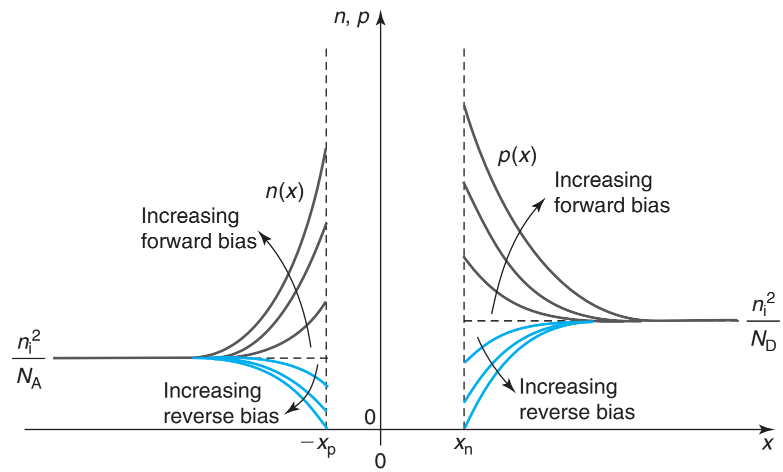


FIGURE 6.14
Evolution of minority carrier profiles in the quasi-neutral regions of a "long" diode as a function of bias.

Since the boundary conditions given by Eq. (6.30) and (6.31) capture both forward and reverse bias and the profiles Eq. (6.32) and (6.33) are valid in both forward and reverse bias, the equations that we have derived for the *I–V* characteristics, Eqs. (6.38–6.46), apply in both forward and reverse bias.

Before we proceed further, it is of value to examine the relative magnitude of the two terms of the diode current that arise from each of the diode regions. If the doping levels are similar, these two terms are of comparable order of magnitude. The diffusion velocity for electrons is higher than that of holes so, for identical doping levels, electron injection in the p-type region is somehow higher than hole injection into the n-type region. However, the current is inversely proportional to the doping level. So, in an asymmetric junction, current injection into the lowly doped region tends to dominate.

It is important to emphasize here that the rectifying behavior of the diode is entirely due to the boundary condition at the edges of the SCR. This is clearly seen in the sketch of the minority carrier profiles as a function of bias shown in Fig. 6.14. In forward bias, the higher the voltage, the higher the concentration of minority carriers that are injected and contribute to current via their recombination with majority carriers. The relationship is of a Boltzmann type. In reverse bias, our model predicts that the minority carrier concentrations at the edges of the SCR drop to negligible values when $|V| \gg kT/q$. Since the minority carrier concentrations cannot be any smaller than zero, that means that when $|V| \gg kT/q$, the generation rate everywhere in the QNRs saturates and the reverse current also saturates. The reverse current is $-I_s$ and it is usually a very small value (typically $I_s \sim 1$ pA/cm^2 in Si PN junctions at room temperature).[1]

Several key dependencies are observed in Eq. (6.46). First of all, in forward bias, the current increases exponentially with V. If plotted in a semilogarithmic scale, as sketched in Fig. 6.13, the forward characteristics appear as a straight line with a slope of $0.43q/kT$. It actually is common

[1]Obviously, the carrier concentration cannot drop to zero as it would not be possible to support a finite current this way. This is a limit of this simplified formulation. In reality, the carrier concentrations at the edges of the SCR drop to very small values that are largely bias insensitive. This produces current saturation.

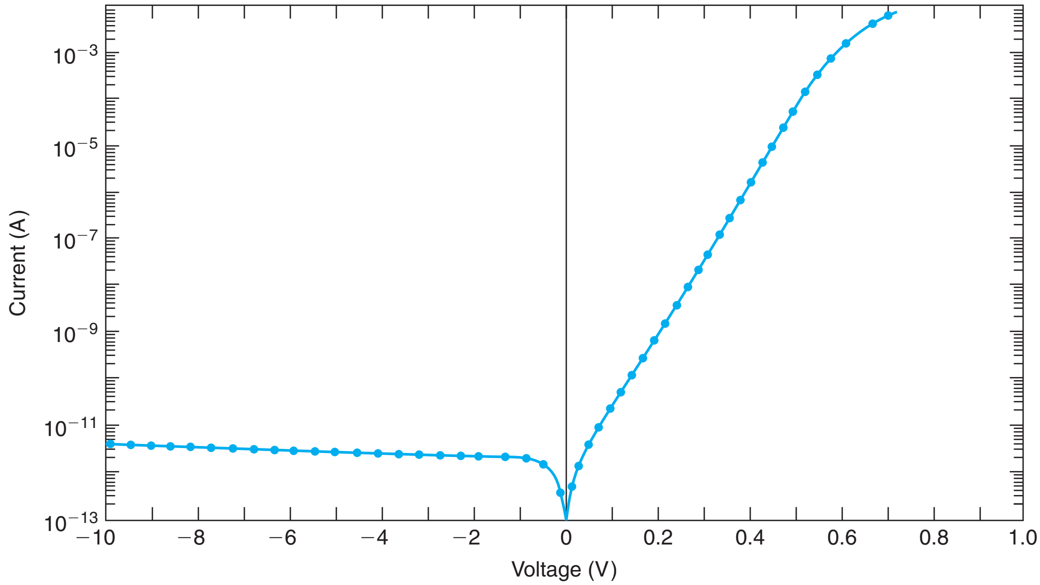


FIGURE 6.15
Experimental *I–V* characteristics of Si PN junction at room temperature.

in the literature to refer to the inverse of this slope, whose ideal value at room temperature is 60 mV/decade. The extrapolation of this straight line to $V = 0$ is I_s. This is also the magnitude of the current at a sufficient reverse bias.

These key features are actually seen in experimental PN junctions. Figure 6.15 shows a semilogarithmic plot of the I–V characteristics of a PN diode at room temperature. These characteristics display a nearly ideal behavior. In forward bias, the current increases exponentially with a slope of about 62 mV/dec. The extrapolation of this current to $V = 0$ V gives $I_s \simeq 1$ pA. In reverse bias, the current is rather independent of bias and equal to about 4 pA, which is very close to I_s. The deviation from the perfect straight line at high forward bias is due to series resistance. The slight slope of the current in reverse bias is due to generation in the SCR. These nonidealities are discussed below.

Figure 6.16 shows I/I_s versus qV/kT for a PN junction at several temperatures. For all temperatures, the data follows Eq. (6.46) fairly well. Note in particular that the extrapolations of the forward and reverse characteristics meet at $V = 0$, as predicted by simple theory.

It is instructive to think about the electron and hole current profiles across the entire PN junction. They are shown in Fig. 6.17 for forward bias. A similar picture can be constructed in reverse bias. The current profiles are similar to those of the example of Section 5.6.1. The total current is the sum of the electron-injected current to the p region plus the hole-injected current to the n region. Across the narrow SCR, we assume that there is negligible recombination and the currents do not change. In the QNRs, the minority carrier currents fall off exponentially from the edge of the SCR. Far away from the junction, the total current is entirely supported by the majority carriers. At the end of the p region, the total current is entirely due to holes, while at the end of the n region, it is completely carried by electrons. The majority carrier currents flow through their respective ohmic contacts to the outside world.

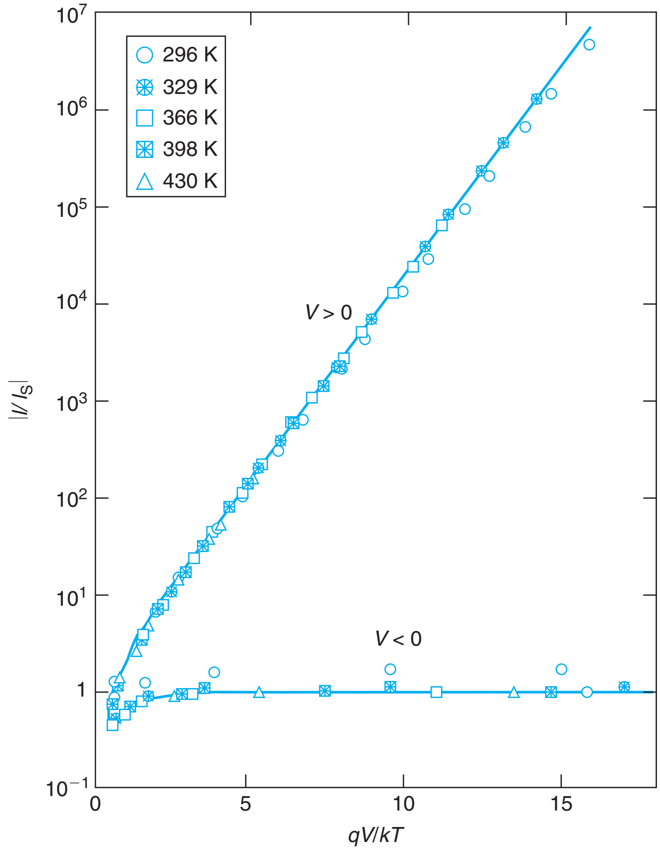


FIGURE 6.16
Normalized experimental *I–V* characteristics of Si PN junction at several temperatures [Reproduced from Current-voltage characteristics of ideal silicon diodes in the range 300–400 K, *Journal of Applied Physics*, 57, 646, 1985. with the permission of American Institute of Physics (AIP) publishing].

Now that we have computed the carrier currents at all points in a PN junction, we are in a position to complete the energy band diagrams out of equilibrium by sketching the quasi-Fermi levels everywhere. This is done in Fig. 6.18 for a junction where both QNRs are much longer than the respective minority carrier diffusion lengths. The assumption that we made inside the SCR that the carriers are in quasi-equilibrium implies that the quasi-Fermi levels are flat there. In the QNRs, the further away we move from their edge with the SCR, the closer we are to reestablishing equilibrium. A few diffusion lengths away, in fact, the carrier concentrations recover their equilibrium values. In consequence, the minority carrier quasi-Fermi levels approach the majority carrier ones and eventually merge into one as quasi-equilibrium is obtained sufficiently far away.

There is an interesting and relevant implication of quasi-equilibrium existing for electrons and holes separately and the quasi-Fermi levels being flat across the SCR. In Chapter 4, we saw

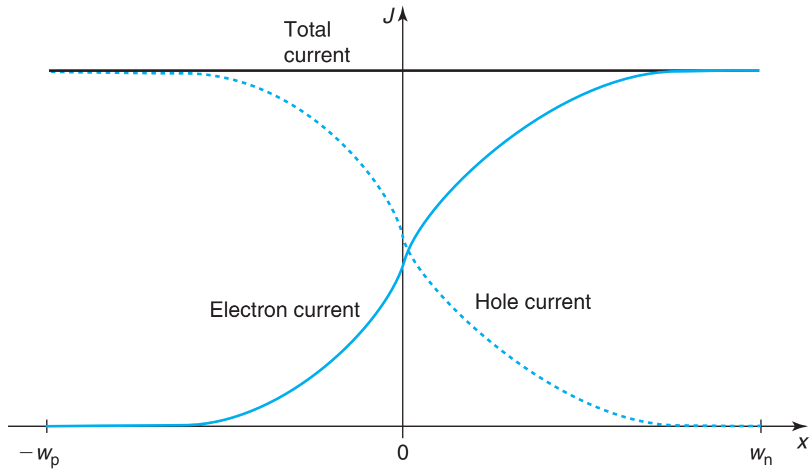


FIGURE 6.17
Sketch of current profiles across PN diode in forward bias.

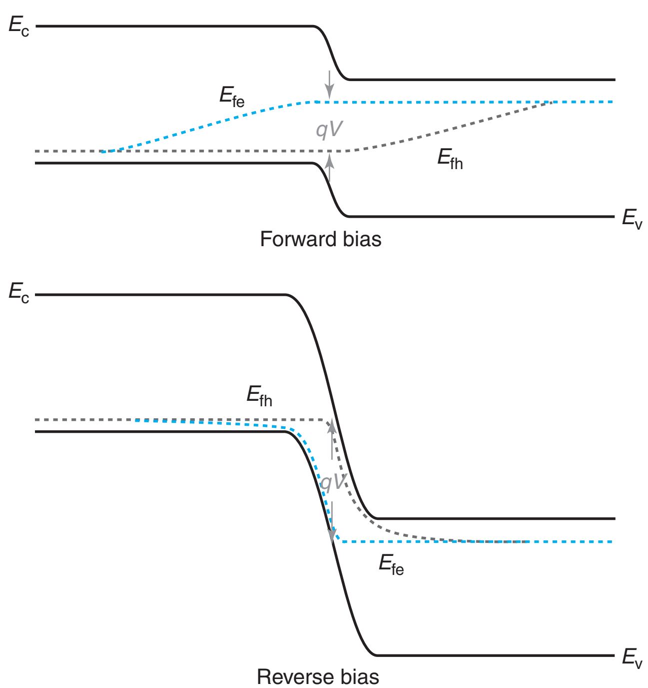


FIGURE 6.18
Schematic energy band diagram in forward and reverse bias showing the location of the quasi-Fermi levels.

that the np product was exponentially dependent on the difference of the quasi-Fermi levels [see Eq. (4.92)]. For the SCR of a PN junction, this difference is precisely qV. As a consequence, we can write

$$np = n_i^2 \exp\frac{qV}{kT} \tag{6.47}$$

From this equation, it is trivial to derive the boundary conditions Eq. (6.28) and (6.29).

6.4 CHARGE–VOLTAGE CHARACTERISTICS OF IDEAL PN DIODE

At first sight, it would seem that a formulation of the current–voltage characteristics of a diode is all we could possibly need to analyze circuits containing diodes. However, the current–voltage characteristics only provide a simple *static* description of the device. In most applications, we also care about the *dynamic* operation of the diode. In power electronics, for example, the diode voltage often changes abruptly from a large reverse bias to a forward bias and back. In many analog applications, it is common to apply to the diode a rapidly varying small-signal voltage on top of a bias. The frequency of the small signal can be quite high. To correctly describe the operation of the diode in situations like these, we need to keep track of the charge stored inside. What does this mean?

Let us think for a moment about a parallel plate capacitor, as sketched in Fig. 6.19. The static current–voltage characteristics of an ideal parallel plate capacitor are very simple: for any voltage, the current is zero (left in Fig. 6.19). However, the dynamic behavior of the capacitor is much more interesting (right in Fig. 6.19). When the voltage changes, current flows to charge the plates of the capacitor. For example, if the voltage on one plate rises with respect to the other, positive charge is deposited to that plate and the same amount of positive charge is removed from the other plate. This produces a flow of charge through the circuit, which results in current.[2] The current that flows through the circuit is given by the familiar expression $i(t) = \mathrm{d}q(t)/\mathrm{d}t$, where $q(t)$ is the charge stored in the capacitor (one plate is charged with $+q$, while the other plate is charged

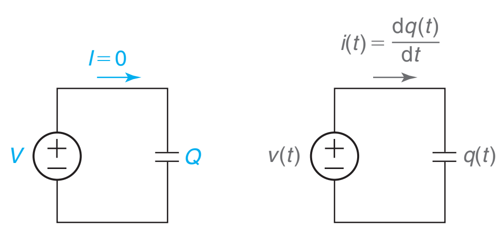


FIGURE 6.19
Behavior of capacitor to static voltage (left) and a dynamic voltage (right). In the static situation, the current is zero. In the dynamic situation, current flows as the plates of the capacitor charge or discharge.

[2]In reality, we know that current should be described by the retrograde movement of electrons. So, for the picture on the right side of Fig. 6.19, when the voltage rises on one plate with respect to the other, electrons are taken away from the first plate and the same amount of electrons are added to the second plate.

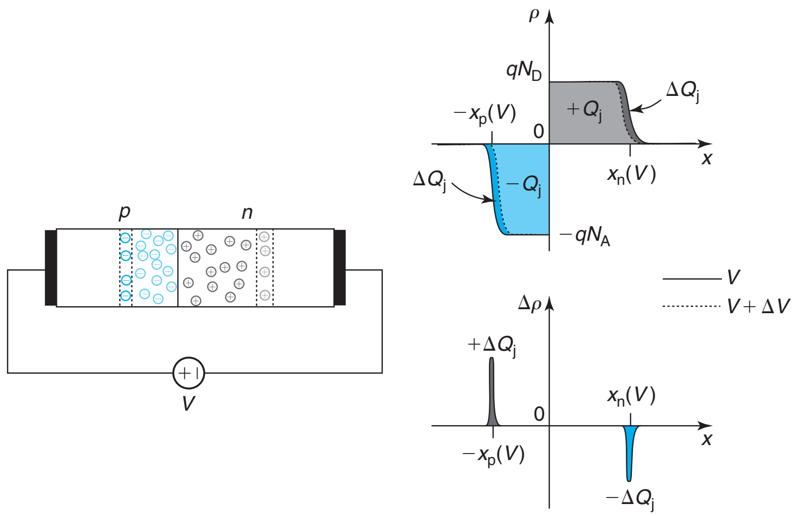


FIGURE 6.20
Sketch of volume charge density in the SCR of a PN junction at a certain bias V and its change upon the application of an additional small voltage ΔV.

with $-q$). If we were to describe the behavior of the capacitor by only its static current–voltage characteristics ($I = 0$ for any V), we would clearly miss its interesting dynamic features!

A similar situation occurs in virtually every semiconductor device. Just like in a parallel plate capacitor, in a semiconductor device there is stored charge with a magnitude that depends on the applied voltage. If the voltage changes, the stored charge has to change and that produces a current flow. This needs to be accounted for in the analysis of circuits that contain semiconductor devices. This is what the charge–voltage characteristics describe.

For the case of the PN diode, what charge is it that we need to account for? Two types, actually. One is associated with the depletion region across the metallurgical junction and the second one is associated with minority carrier injection into the QNRs. The next two sections discuss this in more detail and derive models for these two types of stored charge.

6.4.1 Depletion charge

In Section 6.3.1, we saw that the extent of the depletion region in a PN diode is modulated by the applied voltage. A reverse bias enlarges the depletion layer and a forward bias shrinks it. The depletion layer consists of two distinct regions: on the n side of the junction there is a region that is positively charged, while on the p side of the junction there is another region that is negatively charged. Overall charge neutrality demands that the absolute amount of charge in both regions be the same. This clearly resembles the two plates of a capacitor.

This situation is depicted in Fig. 6.20. This graph shows the charge distribution in the depletion region of a diode at a certain bias V and the change in this charge when the bias increases by a small amount ΔV. For simplicity, let us assume that both V and ΔV are positive, though this is not necessary. At first sight, it seems like there is a sign discrepancy in the charge with respect to

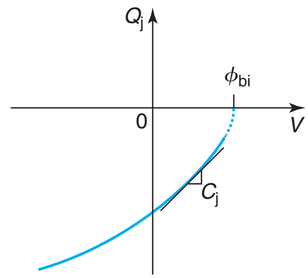


FIGURE 6.21
Sketch of the charge–voltage characteristics associated with the space-charge region of a PN diode.

the situation in a parallel plate capacitor. The positive side of the diode (the p side) has negative charge in its share of the depletion region, while the negative side of the diode (the n side) has positive charge in its portion of the SCR. This is not a problem because what matters is how this charge changes when V changes. So, if V increases by an amount ΔV, we know that the depletion region shrinks on both sides. That means that a sliver of positive charge must be deposited at the edge of the SCR on the p side and the same amount of negative charge has to be deposited at the other edge of the SCR on the n side, as sketched in the bottom figure. The *change in charge* that occurs as a result of the *change in voltage* clearly has the right sign and current flows in the circuit with the expected sign.

From the discussion on the electrostatics of the PN diode of Section 6.3.1 we can immediately write the charge on either side of the depletion region as a function of the applied voltage. For the p side, for example,

$$Q_j(V) = -AqN_A x_p = -A\sqrt{\frac{2q\epsilon N_D N_A(\phi_{bi} - V)}{N_D + N_A}} \tag{6.48}$$

The negative sign reflects the fact that the charge on the p side is due to exposed acceptors. We obtain an identical expression with a positive sign if we evaluate the charge on the n side of the SCR. This expression includes the area of the junction so the units of Q_j are Coulombs.

This expression can be rewritten in the following form:

$$Q_j(V) = Q_j(V = 0)\sqrt{1 - \frac{V}{\phi_{bi}}} \tag{6.49}$$

As forward bias increases, the depletion region shrinks and Q_j is reduced in magnitude. As the reverse bias increases, on the other hand, Q_j increases in magnitude. The voltage dependence of Q_j is square root. This is sketched in Fig. 6.21. Notice how, in spite of the negative sign in Q_j, its derivative with V (the capacitance, discussed below) is positive, as it should be.

6.4.2 Minority carrier charge

The second type of stored charge in a PN diode has to do with minority carrier injection. In a forward-biased PN junction, the two QNRs are flooded with excess minority carriers.

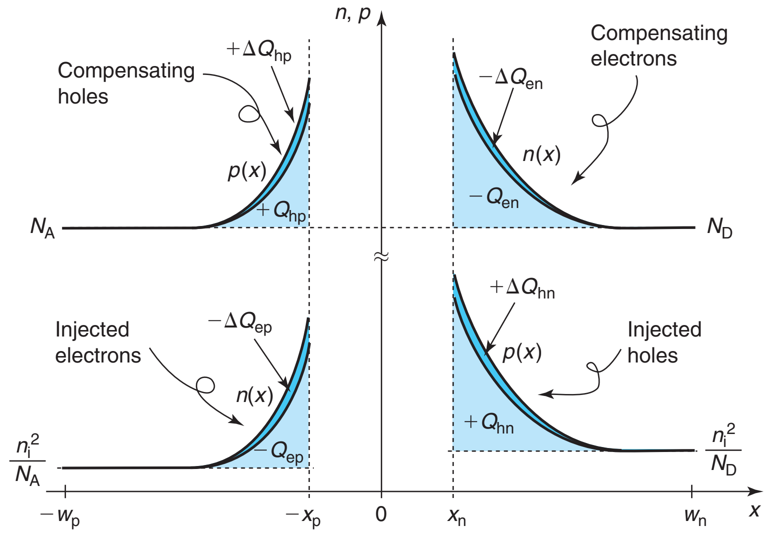


FIGURE 6.22
Sketch of minority and majority carrier profiles in a PN junction in forward bias. Both sides are long in comparison with the minority carrier diffusion lengths. In response to a small change in the voltage applied to the junction, both the minority and the majority carrier concentrations change by the same amount point by point in the QNRs.

Quasi-neutrality is preserved by piling up a nearly identical concentration of majority carriers at every point in space, as sketched in Fig. 6.22. The required majority carriers are easily delivered by the battery through the respective ohmic contact. As studied in detail in Chapter 5, quasi-neutrality is a consequence of the relatively high conductivity of extrinsic semiconductor regions. Even a very small electric field can produce a substantial majority carrier current in a moderately doped region. The current circulates until the majority carrier distribution matches the minority carrier profile. This occurs in a timescale of the dielectric relaxation time, a very short time.

The modification of the majority carrier concentrations relative to the doping level is very small if low-level injection conditions are maintained. The assumption of low-level injection demands in a p-type region, for example, that $n' \ll p_o$. Since quasi-neutrality is satisfied when $n' \simeq p'$, then $p' \ll p_o$ and $p \simeq p_o$. Therefore, for most purposes, we can still assume that the majority carrier concentration equals the doping level everywhere.

Now let us consider what happens to the carrier profiles if we increase the forward voltage by a small amount, as sketched in Fig. 6.22. A slightly higher forward voltage injects more minority carriers at the edge of the SCR. This results in an increased overall minority carrier concentration everywhere. The majority carrier concentration must match this small increase in total charge in order to preserve quasi-neutrality. We again see that the change in voltage results in a change in stored charge, just like in a capacitor.

Let us look in more detail at the n side. If the forward voltage increases by a small amount, the p side injects holes to the n-QNR. This results in an overall increase ΔQ_{hn} in the positive charge stored in the n region. In order to preserve quasi-neutrality, the ohmic contact of the n-type region delivers the same amount of electrons with a total charge $\Delta Q_{\mathrm{en}} = -\Delta Q_{\mathrm{hn}} \equiv -\Delta Q_{\mathrm{n}}$. In this way, the net charge in the n-QNR is back to zero. To maintain quasi-neutrality point by point, the spatial charge distribution of additional holes matches very closely that of extra electrons, as indicated in Fig. 6.22. The additional holes injected into the n-QNR were provided by the positive side of the battery. The electrons were provided by the negative side of the battery consistent with the sign notation for a capacitor. The moment the minority carrier profile has reached its final distribution at the new forward voltage, the flow of charge bound to get stored ceases. The normal diode current continues to flow at a slightly higher value.

A similar sequence of events takes place on the p-QNR. If the forward voltage changes, the negative terminal of the battery delivers negative charge in the form of excess electrons, ΔQ_{ep}, which is matched point by point by positive charge delivered by the positive terminal of the battery in the form of excess holes through the ohmic contact to the p-QNR, that is, $\Delta Q_{\mathrm{ep}} = -\Delta Q_{\mathrm{hp}} \equiv -\Delta Q_{\mathrm{p}}$. The p-QNR also displays the same capacitive effect as the n-QNR. Furthermore, both "capacitors" are located in parallel directly across the terminals of the PN junction.

We are now in a position to develop a model for the stored charge in the n- and p-QNRs. These can be obtained simply by integrating the respective excess minority carrier concentrations:

$$Q_{\mathrm{p}} = qA \int_{-w_{\mathrm{p}}}^{-x_{\mathrm{p}}} n'(x)\mathrm{d}x \tag{6.50}$$

$$Q_{\mathrm{n}} = qA \int_{x_{\mathrm{n}}}^{w_{\mathrm{n}}} p'(x)\mathrm{d}x \tag{6.51}$$

In these expressions, Q_{p} refers to the stored charge in the p-type QNR and Q_{n} to that of the n-type QNR.

We derived expressions for the excess carrier concentrations in the two QNRs for our ideal PN diode [Eqs. (6.32) and (6.33)]. Plugging these expressions in and integrating, we get:

$$Q_{\mathrm{p}} = qAL_{\mathrm{e}}n'(-x_{\mathrm{p}}) \tag{6.52}$$

$$Q_{\mathrm{n}} = qAL_{\mathrm{h}}p'(x_{\mathrm{n}}) \tag{6.53}$$

We can rewrite these expressions in terms of the minority carrier current density injected into the respective regions using Eqs. (6.36) and (6.37) on the one hand, and Eqs. (6.34) and (6.35) on the other hand, to obtain

$$Q_{\mathrm{p}} = A\frac{L_{\mathrm{e}}^2}{D_{\mathrm{e}}}J_{\mathrm{e}}(-x_{\mathrm{p}}) \tag{6.54}$$

$$Q_{\mathrm{n}} = A\frac{L_{\mathrm{h}}^2}{D_{\mathrm{h}}}J_{\mathrm{h}}(x_{\mathrm{n}}) \tag{6.55}$$

Finally, using the definitions made in Eq. (6.44) and noting that L^2/D is equal to the minority carrier lifetime, we obtain

$$Q_p = \tau_e I_p \tag{6.56}$$

$$Q_n = \tau_h I_n \tag{6.57}$$

where τ_e and τ_h are, respectively, the minority carrier lifetimes for electrons in the p region and holes in the n region.

The total stored charge is simply then,

$$Q_d = Q_p + Q_n = \tau_e I_p + \tau_h I_n = Q_s\left(\exp\frac{qV}{kT} - 1\right) \tag{6.58}$$

This expression and the individual components of the current given by Eqs. (6.56) and (6.57) make good physical sense. If we think about this in forward bias, when Q_d can become sizable, the current flowing through the diode is entirely due to recombination. For each region of the diode, the characteristic time constant of this process is the recombination lifetime τ. Then the amount of stored charge has to be τI. Conversely, a charge Q_d recombining in a time span of τ gives rise to a current $I = Q_d/\tau$.

An important result obtained here is that the minority carrier charge stored in a diode goes as $Q_d \propto \exp(qV/kT) - 1$. This reflects the boundary conditions for excess carrier concentrations at the edges of the SCR. In forward bias, this charge increases exponentially. In reverse bias, this charge saturates to a very small value that is in most cases negligible.

Minority carrier storage in PN diodes is an issue that affects its switching characteristics in a very significant way. It we attempt to switch a diode from a forward bias state to a reverse state, the transitory is not complete until all the minority carrier charge that exists in forward bias is disposed of. This can take substantial time. Similarly, if we try to switch a PN diode from reverse bias to forward bias, a lot of current needs to be delivered to provide the minority carrier charge that needs to be stored. This also takes time. We analyze the situation in detail in Advanced Topic AT6.4. The absence of minority carrier storage in Schottky diodes, as we will see in Chapter 7, makes these devices preferable for switching applications.

6.5 EQUIVALENT CIRCUIT MODELS OF THE IDEAL PN DIODE

An equivalent circuit model for a semiconductor device is a circuit-like description of its behavior that uses elemental circuit components such as resistors, capacitors, current sources, voltage sources, and others. One of these elemental components in the equivalent circuit models toolbox is actually a "chargeless" ideal diode. This is a two-terminal element that exhibits rectifying current–voltage characteristics as given by Eq. (6.46) but does not hold stored charge. It is then more akin to a nonlinear resistor with rectifying characteristics. A symbol for a chargeless ideal diode is shown on the left side of Fig. 6.23. A chargeless ideal diode is characterized by a single parameter, its saturation current I_s, which is often written right next to the symbol. In the chapters that follow, we will use this element when constructing equivalent circuit models for the Schottky diode, the MOSFET, and the bipolar junction transistor.

Our concept of ideal diode in this chapter does involve stored charge, as discussed in detail in the previous section. Therefore, to the chargeless ideal diode, we need to add a storage element

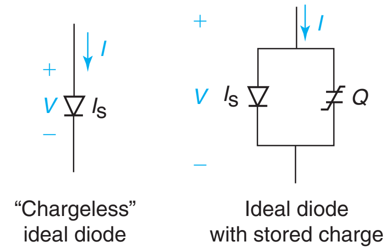


FIGURE 6.23
Left: symbol for chargeless ideal diode. I_s represents its saturation current. Right: large-signal equivalent circuit model of ideal diode with stored charge.

in parallel. The reason for the parallel configuration is the fact that the same voltage that feeds the current through the diode also drives charge storage. This combination of elements is the most elemental equivalent circuit model for an ideal diode and is shown on the right side of Fig. 6.23. The charge storage element is usually represented by a capacitor-like symbol with a characteristic broken line across, as indicated in the figure. This is to suggest that this is a nonlinear element. As we now know, in an ideal diode, charge storage exhibits a complex voltage dependence given by the sum of the depletion region capacitance, Eq. (6.48), and the minority carrier charge, Eq. (6.58).

The equivalent circuit model on the right side of Fig. 6.23 is referred to as a *large-signal equivalent circuit*. The two branches in parallel represent the direct diode current that flows upon the application of a voltage plus the current needed to charge or discharge the diode if the voltage changes. So, in a general dynamic situation, the terminal diode current is given by

$$I = I_s\left(\exp\frac{qV}{kT} - 1\right) + \frac{dQ}{dt} \tag{6.59}$$

To keep the notation simple, I, V, and Q here denote instantaneous time-dependent variables.

In many applications, it is also important to develop a good understanding of the small-signal behavior of semiconductor devices. This refers to their response to small excursions around its bias point. This is illustrated in Fig. 6.24. Devices are used this way in many analog and communications applications. The small-signal behavior of a device can be obtained by linearizing the current and charge equations.

For the diode, starting with the current–voltage characteristics, we apply a voltage across that consists of a bias V, plus a small signal v. The resulting current is

$$I + i = I_s\left[\exp\frac{q(V+v)}{kT} - 1\right] \tag{6.60}$$

If v is small in the scale of kT/q, then we can linearize the exponential by selecting the first two terms of the Taylor series expansion,

$$I + i \simeq I_s\left[\exp\frac{qV}{kT}\left(1 + \frac{qv}{kT}\right) - 1\right] = I + \frac{q(I+I_s)}{kT}v \tag{6.61}$$

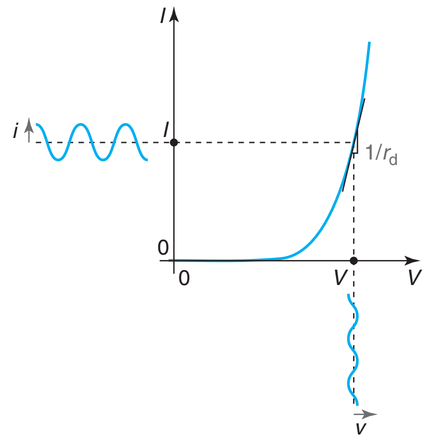


FIGURE 6.24
Illustration of small-signal behavior and dynamic resistance of a PN diode. The dynamic resistance characterizes the slope of the *I–V* characteristics of the diode at a given bias point. Its value depends on the selected bias point. For small excursions of voltage around this bias point, the dynamic resistance captures reasonably well the response of the diode.

From the small-signal point of view, the ideal diode looks like a resistor of value

$$r_\mathrm{d} = \frac{kT}{q(I + I_\mathrm{s})} \tag{6.62}$$

where r_d is referred to as the *dynamic resistance* of the diode. Its meaning is graphically illustrated in Fig. 6.24. The dynamic resistance is the inverse slope of the I–V characteristics of the diode at the selected bias point. Its value, therefore, depends on the bias that has been selected. In forward bias, if $V \gg kT/q$, then

$$r_\mathrm{d} \simeq \frac{kT}{qI} \tag{6.63}$$

This is an important result. The dynamic resistance of a diode is inversely proportional to the diode current. The dependence is completely fundamental, that is, it is completely specified by fundamental constants and the diode temperature. For a diode biased at $I = 1$ mA at room temperature, for example, $r_\mathrm{d} = 26\ \Omega$. This fundamental relationship suggests that the value of r_d is independent of the design of the diode; a modern diode and a diode from the early days of microelectronics both present an identical dynamic resistance if biased at the same current level. Of course, this is to the extent that they both exhibit I–V characteristics that approach the ideal behavior.

We turn our attention now to the charge–voltage characteristics. It is interesting to realize that, using the chain rule, the second term in Eq. (6.59) can be written as

$$\frac{\mathrm{d}Q}{\mathrm{d}t} = \frac{\mathrm{d}Q}{\mathrm{d}V}\frac{\mathrm{d}V}{\mathrm{d}t} = C\frac{\mathrm{d}V}{\mathrm{d}t} \tag{6.64}$$

where C has the units of capacitance. This suggests an alternative, but completely equivalent description of charge storage in a diode (and other semiconductor devices) in terms of a diode capacitance defined as

$$C(V) = \frac{\mathrm{d}Q}{\mathrm{d}V} \tag{6.65}$$

This equation is written in a way to emphasize the fact that a device capacitance is in general bias dependent.

Describing charge storage in terms of capacitance has the advantage that only the variables I and V are needed to completely specify the situation. Also, Eq. (6.64) suggests that the absolute value of Q does not matter but it is $\mathrm{d}Q/\mathrm{d}V$, or capacitance, what really counts for the dynamics of a device.

Since in a PN diode there are two components of Q, the diode capacitance also contains two terms. Associated with charge storage in the SCR there is a capacitance that is called *depletion capacitance* or *junction capacitance*, C_j. We can easily evaluate an expression for it. If we go back to Eq. (6.48), for example, and use the capacitance definition of Eq. (6.65), we immediately obtain

$$C_\mathrm{j}(V) = A\sqrt{\frac{\epsilon q N_\mathrm{A} N_\mathrm{D}}{2(N_\mathrm{D}+N_\mathrm{A})(\phi_\mathrm{bi}-V)}} = \frac{C_\mathrm{j}(V=0)}{\sqrt{1-\dfrac{V}{\phi_\mathrm{bi}}}} \tag{6.66}$$

This expression is sketched in Fig. 6.25. The shape of C_j with V makes good sense if we think of this as a parallel plate capacitor where the plates are separated by the thickness of the SCR, as Fig. 6.20 suggests. Starting from $V = 0$, as forward bias is applied, the SCR shrinks and the capacitance increases. If reverse bias is applied, the SCR widens and the capacitance decreases. In fact this way of thinking yields the expression of C_j very quickly if we write

$$C_\mathrm{j}(V) = A\frac{\epsilon}{x_\mathrm{SCR}(V)} \tag{6.67}$$

Substituting x_SCR from Eq. (6.24) into this expression immediately yields Eq. (6.66).

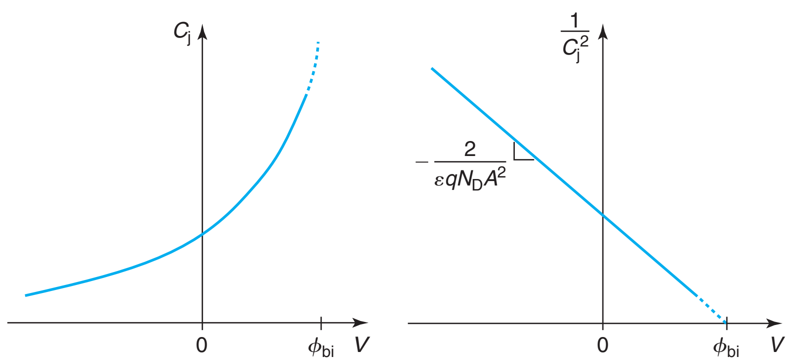


FIGURE 6.25
Sketch of C–V characteristics of a PN junction (left). Also plotted (right) is $1/C_\mathrm{j}$ versus V for an asymmetric P$^+$N junction. The doping level on the lowly doped side and the built potential can be determined from this plot.

Measuring the capacitance–voltage characteristics often constitutes a powerful way to characterize semiconductor devices. We use an asymmetric P^+N junction to illustrate this. In this case, Eq. (6.66) reduces to

$$C_j(V) = A\sqrt{\frac{\epsilon q N_D}{2(\phi_{bi} - V)}} \tag{6.68}$$

Here, we see that the capacitance is dominated by the doping on the lowly doped side of the junction. In fact, $C_j(V)$ in Eq. (6.68) only depends on N_D and ϕ_{bi}. So $C - V$ measurements can easily yield these two parameters. A particularly easy way to accomplish this is to plot $1/C_j^2$ as a function of V. From Eq. (6.68) for the p^+N junction, we find

$$\frac{1}{C_j^2} = \frac{2(\phi_{bi} - V)}{\epsilon q N_D A^2} \tag{6.69}$$

This expression shows that $1/C_j^2$ depends linearly on voltage with a negative slope that is a function of N_D, the area of the diode, and fundamental constants. Also, $1/C_j^2$ goes to zero as V goes to ϕ_{bi}. This behavior is sketched in Fig. 6.25. Figure 6.26 provides an experimental demonstration of this. It is then clear that from $C - V$ measurements, knowing the area of the diode, we can determine the doping level on the lowly doped side of the diode and the built-in potential of the junction.

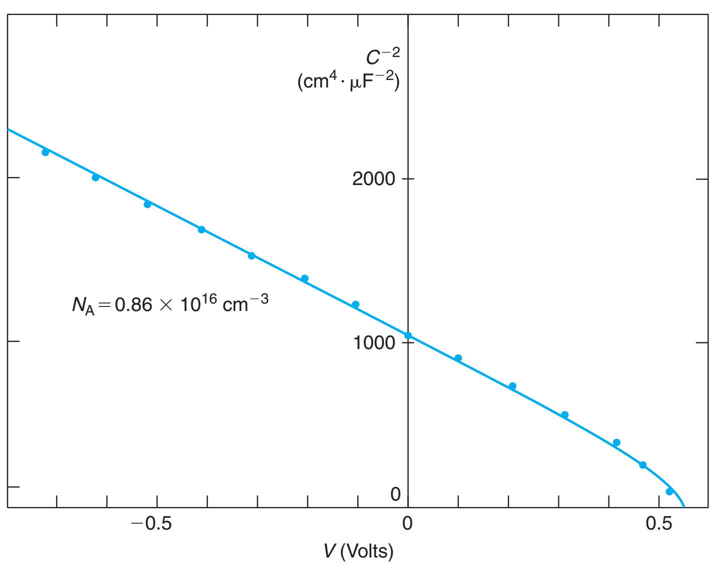


FIGURE 6.26
Experimental $1/C_j^2$ versus V characteristics of a Si PN junction. From the slope of the straight line, the doping level on the lowly doped side is determined [Republished with permission of The Institute of Electrical and Electronic Engineers, Inc. (IEEE), from Analysis and capacitive measurement of the built-in-field parameter in highly doped emitters, *IEEE Transactions on Electron Devices*, Volume: 29, Issue: 10, 1604, 1982.; permission conveyed through Copyright Clearance Center, Inc].

EXERCISE 6.4

Consider a PN diode identical to that of Exercises 6.1–6.3. The diode has a junction area of 10 μm^2*. Estimate the junction capacitance associated with this diode at room temperature and under an applied forward bias of* $V = 0.6$ V.

This can be done in different ways. The fastest way is perhaps using the result of Exercise 6.1 where we estimated the extent of the SCR in this diode under zero bias. There we found that $x_{SCR} \simeq x_n = 0.33\ \mu m$. The junction capacitance at zero bias is then

$$C_j(V=0) = A\frac{\epsilon}{x_{SCR}(V=0)} = 10\times10^{-8}\ \text{cm}^2 \times \frac{1.0\times10^{-12}\ \text{F/cm}}{0.33\times10^{-4}\ \text{cm}} = 3.1\times10^{-15}\ \text{F} = 3.1\ \text{fF}$$

The junction capacitance at a certain voltage can be obtained using Eq. (6.66),

$$C_j(V) = \frac{C_j(V=0)}{\sqrt{1-\frac{V}{\phi_{bi}}}} = \frac{3.1\ \text{fF}}{\sqrt{1-\frac{0.6}{0.84}}} = 5.8\ \text{fF}$$

The second source of stored charge in a diode is minority carrier charge. This can be described through a capacitance that is called the *diffusion capacitance*. We can obtain an expression for this using Eq. (6.58) in Eq. (6.65),

$$C_d = \frac{dQ_d}{dV} = \frac{q}{kT}Q_s \exp\frac{qV}{kT} \tag{6.70}$$

The voltage dependence of C_d is a pure exponential. C_d goes to zero for reverse bias and increases exponentially with forward bias. If the forward bias voltage exceeds kT/q, then we can approximate C_d with

$$C_d \simeq \frac{q}{kT}Q_d \tag{6.71}$$

or simply, q/kT times the total stored minority carrier charge in the diode.

The total capacitance of the diode is the sum of C_j and C_d,

$$C = C_j + C_d \tag{6.72}$$

Due to their unique voltage dependencies, in an ideal diode, C_j dominates in reverse bias and small forward bias and C_d prevails in strong forward bias. This is illustrated in Fig. 6.27.

EXERCISE 6.5

Consider a PN diode identical to that of Exercises 6.1–6.4. Note again that the p and n regions are "long" from the minority carrier point of view. At room temperature and with an applied forward bias of $V = 0.6$ V*, estimate (a) the charge density stored in the two QNRs and (b) the diffusion capacitance.*

(a) In Exercise 6.3, we obtained the magnitude of the current density injected into each QNR of this diode at this bias point. We also determined the minority carrier lifetimes. With these results, the fastest way to determine the charge stored in each QNR is using Eqs. (6.56) and (6.57). For the p-QNR, the stored charge per unit area is

$$Q_p = \tau_e A J_e(-x_p) = 2.3 \times 10^{-6}\text{ s} \times 10^{-7}\text{ cm}^2 \times 6.0 \times 10^{-4}\text{ A/cm}^2 = 1.4 \times 10^{-16}\text{ C}$$

For the n-QNR, the stored charge per unit area is

$$Q_n = \tau_h A J_h(x_n) = 1.3 \times 10^{-4}\text{ s} \times 10^{-7}\text{ cm}^2 \times 6.5 \times 10^{-3}\text{ A/cm}^2 = 8.5 \times 10^{-14}\text{ C}$$

(b) The total carrier charge is the sum of these two and it is dominated by the n-QNR. The value is $Q_d = 8.5 \times 10^{-14}$ C. The diffusion capacitance is then

$$C_d \simeq \frac{q}{kT} Q_d = \frac{8.5 \times 10^{-14}\text{ C}}{0.026\text{ V}} = 3.3 \times 10^{-12}\text{ F} = 3.3\text{ pF}$$

When compared with the result obtained in Exercise 6.4, it is clear that the diffusion capacitance dominates at this bias point.

A small-signal equivalent circuit model of a device is a handy circuit-like description of its small-signal behavior. Small-signal equivalent circuits are a very intuitive way to grasp the behavior of a device from the small-signal point of view and are extensively used by circuit designers. By definition, a small-signal equivalent circuit model is a linearized version of the large-signal equivalent circuit model.

For an ideal PN diode, the small-signal equivalent circuit model looks like a pure *RC* parallel circuit as shown in Fig. 6.28. The resistance is the dynamic resistance that comes from the linearization of the ideal chargeless diode. The capacitance is the sum of the junction and diffusion capacitances and comes from the linearization of the charge storage element.

There is an additional kind of equivalent circuit model that is worth discussing. This is the model that is used to describe a device in the computer-aided design (CAD) of integrated circuits. These models tend to be based on physics but they have to pay attention to issues of computational efficiency and numerical convergence. These models also have to allow the parametrization of the device so that they can be matched to describe the characteristics of a given device with specific dimensions. These issues are discussed in more detail in Advanced Topic AT6.3, where a simple equivalent circuit model for circuit CAD for the PN diode is presented.

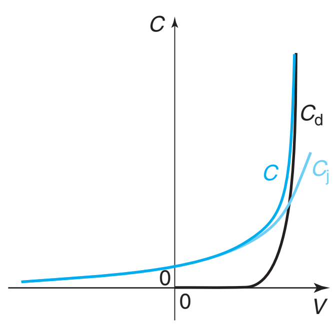


FIGURE 6.27
Sketch of diode capacitance and its components versus diode voltage. Depletion capacitance dominates in reverse and small forward bias. For strong forward bias, diffusion capacitance dominates.

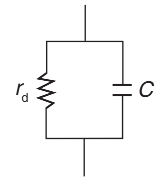


FIGURE 6.28
Small-signal equivalent circuit model of an ideal PN diode.

6.6 NONIDEAL AND SECOND-ORDER EFFECTS

In this section, we proceed to discuss the most significant nonidealities and second-order effects in PN diodes. It is in this section that we undo several of the assumptions made in Section 6.1.

6.6.1 Short diode

In our ideal PN diode we assumed that the quasi-neutral p and n regions were both much longer than their respective minority carrier diffusion lengths. Real diodes, depending on their design, can have QNRs that are long, short, or comparable to the relevant diffusion length. Also, since there are two sides to a diode, several combinations are clearly possible. In this section, we study the case of a diode in which both sides are much shorter than the respective diffusion lengths. We call this a "short diode." It is straightforward, though mathematically a bit more involved, to extend the analysis to more general situations.

The uniqueness of a short diode is that recombination and generation of minority carriers takes place at the surfaces of the QNRs (the ohmic contacts), as opposed to their bulk. The situation is similar to that of the "short" bar that we studied in Section 5.6.2, and the solutions that we obtained there can be readily used here. Other than accounting for this, the basic physics of the PN diode does not change.

If we think in terms of minority carrier injection, the excess carrier concentrations in a short diode are as depicted in Fig. 6.29. The ohmic contacts located at the surfaces of the diode (see Fig. 6.3) force the excess minority carrier concentrations there to zero. The boundary conditions at the edges of the SCR are set by the applied voltage, as determined above. In the absence of bulk recombination, the excess minority carrier profiles across the QNRs are straight lines and are therefore given by

$$n'(x) = n'(-x_p)\frac{x + w_p}{-x_p + w_p} \quad \text{in p-QNR: } x \le -x_p \tag{6.73}$$

$$p'(x) = p'(x_n)\frac{-x + w_n}{w_n - x_n} \quad \text{in n-QNR: } x \ge x_n \tag{6.74}$$

These linear minority carrier profiles lead to a minority carrier diffusion velocity at the edges of the QNRs that, in analogy with Eq. (5.81), are given by

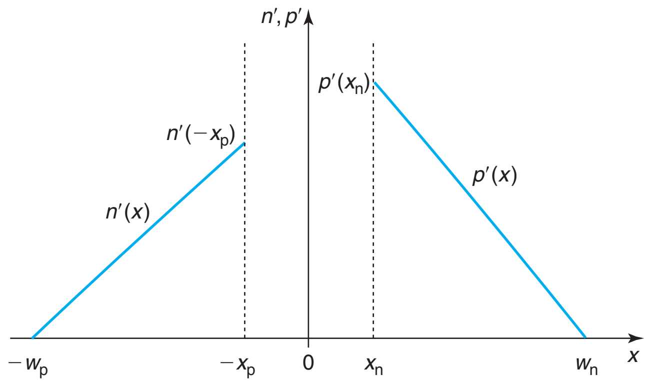


FIGURE 6.29
Schematic diagram of excess minority carrier profiles in quasi-neutral regions of a "short" diode with infinite surface recombination velocity at the ends.

$$v_e^{diff}(-x_p) = -\frac{D_e}{w_p - x_p} \tag{6.75}$$

$$v_h^{diff}(x_n) = \frac{D_h}{w_n - x_n} \tag{6.76}$$

These are the velocities that should be used in the calculation of the diode current. Following a similar path as in Section 6.3.3, we obtain the two components of the diode current as follows:

$$I_p = qA\frac{n_i^2}{N_A}\frac{D_e}{w_p - x_p}\left(\exp\frac{qV}{kT} - 1\right) \tag{6.77}$$

$$I_n = qA\frac{n_i^2}{N_D}\frac{D_h}{w_n - x_n}\left(\exp\frac{qV}{kT} - 1\right) \tag{6.78}$$

Adding up I_p and I_n yields the expression of the diode I–V characteristics. The saturation current of the short diode is given by

$$I_s \simeq qAn_i^2\left(\frac{1}{N_A}\frac{D_e}{w_p} + \frac{1}{N_D}\frac{D_h}{w_n}\right) \tag{6.79}$$

where we have further assumed that $w_p \gg x_p$ and $w_n \gg x_n$, as is commonly the case.

The ideal rectifying characteristics of the ideal PN diode are unchanged, whether the QNRs are long or short. This is because the peculiar voltage dependence of the diode emerges from the boundary conditions at the edges of the SCR and are independent of the design of the QNR.

The expression for the stored minority carrier charge in a short diode also needs to be revised. To be sure, the physics of stored charge is the same as in the long diode. The excess (or defect) of minority carrier concentrations imply a parallel modification in the majority carrier concentrations. All these charges need to be supplied by the contacts and remain stored in the diode.

With the excess minority carrier profiles being given by Eqs. (6.73) and (6.74), the minority carrier stored charge in each side of the diode is given by:

$$Q_\mathrm{p} = qA\frac{1}{2}n'(-x_\mathrm{p})(w_\mathrm{p} - x_\mathrm{p}) = \tau_\mathrm{tp} I_\mathrm{p} \tag{6.80}$$

$$Q_\mathrm{n} = qA\frac{1}{2}p'(x_\mathrm{n})(w_\mathrm{n} - x_\mathrm{n}) = \tau_\mathrm{tn} I_\mathrm{n} \tag{6.81}$$

where τ_tp and τ_tn are the transit times for minority carriers across the p- and n-QNRs, respectively. They follow the expressions:

$$\tau_\mathrm{tp} = \frac{(w_\mathrm{p} - x_\mathrm{p})^2}{2D_\mathrm{e}} \tag{6.82}$$

$$\tau_\mathrm{tn} = \frac{(w_\mathrm{n} - x_\mathrm{n})^2}{2D_\mathrm{h}} \tag{6.83}$$

This is an interesting result. It parallels what we obtained for the long diode in Eqs. (6.56) and (6.57). The charge stored in a QNR is given by the current injected into it times the relevant minority carrier time constant. For the long diode, this was the carrier lifetime. For the short diode, it obviously is the minority carrier transit time since carriers need to reach the ohmic contact before they can recombine. When the transport process across the QNR is diffusion and the profile is a straight line as in Fig. 6.29, we saw in Section 4.4 that the transit time is given by the square of the QNR length divided by two times the diffusion coefficient. This is exactly what we obtain here.

The total stored charge in a short diode is given by the sum of Eqs. (6.80) and (6.81):

$$Q_\mathrm{d} = Q_\mathrm{p} + Q_\mathrm{n} = \tau_\mathrm{tp} I_\mathrm{p} + \tau_\mathrm{tn} I_\mathrm{n} = Q_\mathrm{s}\left(\exp\frac{qV}{kT} - 1\right) \tag{6.84}$$

an expression that is functionally identical to that of the long diode.

Other than the need for using appropriate expressions for current and charge, the basic physics of the ideal diode is not modified in the short diode and the equivalent circuit models remain unchanged.

It is of interest, nevertheless, to sketch the energy band diagrams of a short diode. These are shown for forward and reverse bias in Fig. 6.30. Here, we can see that the quasi-Fermi levels are split across the entire QNRs and only merge at the ohmic contacts. This reflects minority carrier concentrations that are out of equilibrium across the entire body of the semiconductor with the exception of the surfaces where the ohmic contacts force equilibrium to be maintained.

6.6.2 Space-charge generation and recombination

In the theory of the PN diode presented so far, we have neglected any generation and recombination in the SCR around the metallurgical junction. In our model, so far, quasi-neutral generation and recombination is all we need to explain the *I–V* characteristics experimentally observed in PN diodes.

It is not uncommon, in practice, to see PN diodes that do not quite follow such an ideal behavior. Figure 6.31, for example, sketches the *I–V* characteristics of a diode in which the ideal

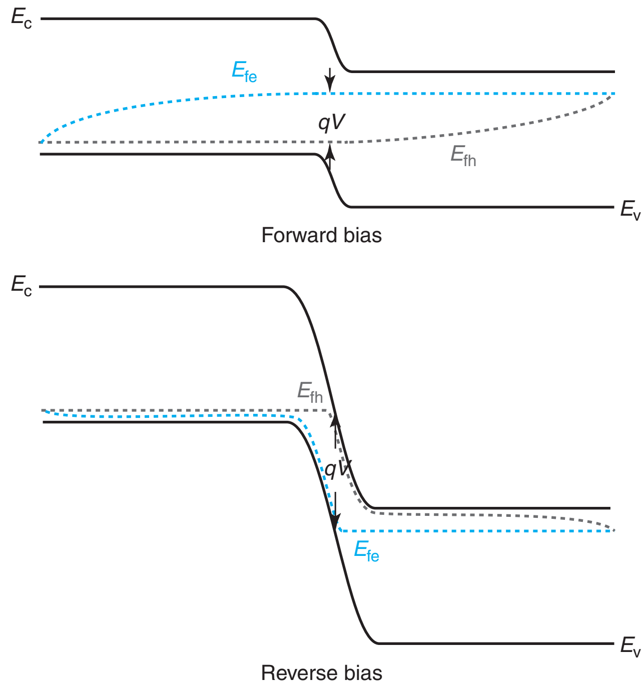


FIGURE 6.30
Schematic energy band diagram in forward and reverse bias showing the location of the quasi-Fermi levels in a short diode.

$e^{qV/kT}$ expected in forward bias does not extend all the way down to $V = 0$. In fact, for small forward bias, the current seems to follow a dependence more like $e^{qV/2kT}$. Similarly, the diode of Fig. 6.31 exhibits a reverse bias behavior in which the current is not saturated but is in excess of the ideal value extrapolated to $V = 0$ from forward bias.

These anomalies are due to recombination (in forward bias) and generation (in reverse bias) through traps located in the SCR. These mechanisms contribute additional current components to the PN diode.

A rigorous model for the diode current that includes SCR generation and recombination is mathematically involved. In most cases, it is also not useful to seek a very detailed description because the nature, concentration, and electrical characteristics of the traps are not generally well known. What is most useful is to develop a simple model that captures the essential physics and provides the key dependencies.

The model that we develop in this section assumes that SCR generation and recombination represent a small perturbation to the ideal diode picture. In particular, we will assume that the carrier concentrations across the SCR are given by the expressions obtained under the ideal diode model.

We start by going back to Eq. (3.42) that provides the most general relation for the net recombination rate due to trap-assisted processes. If we use the expression for the pn product obtained for the ideal diode in Eq. (6.47), Eq. (3.42) becomes

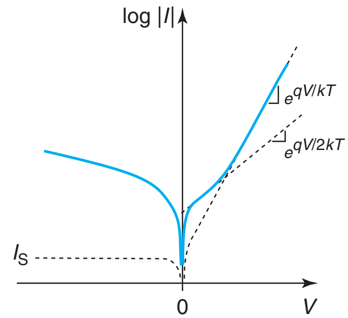


FIGURE 6.31
Sketch of *I–V* characteristics for typical Si PN junction. Notice the nonideal *I–V* characteristics for small forward and for reverse bias associated with SCR recombination and generation, respectively.

$$U_{tr}|_{SCR} = \frac{n_i^2 \left(\exp \dfrac{qV}{kT} - 1 \right)}{\tau_{ho} n + \tau_{eo} p + (\tau_{ho} + \tau_{eo}) n_i} \tag{6.85}$$

As discussed in Section 5.9, the SCR current can be computed by integrating the net recombination rate across the SCR and multiplying by the carrier charge

$$J_{SCR} = q \int_{-x_p}^{x_n} U_{tr} \mathrm{d}x \tag{6.86}$$

Inserting Eq. (6.85) into Eq. (6.86) does not lead to an analytical result. A relatively simple expression that displays all the dependencies observed experimentally can be obtained by identifying the highest value of U_{tr} anywhere in the SCR, and assuming that this value applies everywhere. This gives an upper limit to the desired result.

The denominator of Eq. (6.85) contains a term in n, and another one in p. Since the product of p and n is a constant, it is not difficult to prove that the highest value of U_{tr} is obtained when $\tau_{ho} n = \tau_{eo} p$. This yields a maximum net recombination rate given by

$$U_{tr}|_{SCR,max} = \frac{n_i \left(\exp \frac{qV}{kT} - 1 \right)}{2\sqrt{\tau_{eo}\tau_{ho}} \exp \frac{qV}{2kT} + \tau_{eo} + \tau_{ho}} \simeq \frac{n_i}{2\sqrt{\tau_{eo}\tau_{ho}}} \left(\exp \frac{qV}{2kT} - 1 \right) \tag{6.87}$$

where we have assumed $\tau_{eo} \simeq \tau_{ho}$, for simplicity.

Inserting this result into Eq. (6.86) and assuming that this maximum recombination rate applies everywhere, a worst-case scenario, results in

$$J_{SCR,max} = \frac{q n_i x_{SCR}}{2\sqrt{\tau_{eo}\tau_{ho}}} \left(\exp \frac{qV}{2kT} - 1 \right) \tag{6.88}$$

This is an expression that is valid both in forward and reverse bias.

We can also define a saturation current for SCR generation/recombination. That is the pre-exponential factor in Eq. (6.88),

$$J_s(\text{SCR}) = \frac{qn_i x_{\text{SCR}}}{2\sqrt{\tau_{eo}\tau_{ho}}} \tag{6.89}$$

There are several features to notice in the result captured in Eq. (6.88). First, the forward current grows as $e^{qV/2kT}$. This is in contrast with recombination in a QNR which goes as $e^{qV/kT}$. The coefficient in front of the kT factor in the denominator of the exponent is often called the *ideality factor*, n. $n = 1$ for QNR recombination, while $n = 2$ for SCR recombination.

The reason for this rather different voltage behavior is easy to see. Inside the SCR and at its edges, we know that $np = n_i^2 \exp(qV/kT)$. At the edges of the QNRs, the majority carrier concentration is equal to the doping level and the minority carrier concentration goes as $n_i^2 \exp(qV/kT)$. We saw earlier that in a sufficiently extrinsic QNR, it is the capture of minority carriers that limits the recombination rate. Hence, the net recombination rate must exhibit the same dependence as the minority carrier concentration and scale as $n_i^2 \exp(qV/kT)$. On the other hand, with $\tau_{eo} = \tau_{ho}$, at the point of highest recombination rate inside the SCR, $n = p = n_i \exp(qV/2kT)$. The carrier concentrations at this point increase with voltage at half the rate as they do at the edges of the SCR. Hence, the recombination rate exhibits the same dependence and goes as $n_i \exp(qV/2kT)$. An equivalent way to look at this is to realize that with $\tau_{eo} = \tau_{ho}$, all the recombination takes place precisely at the center of the SCR. As V changes, the occupation probability of states at that location changes as $\exp(qV/2kT)$, while at the edges of the SCR, it changes as $\exp(qV/kT)$. This is what drives the different dependencies of the recombination currents.

In practice, SCR recombination is characterized by an experimental ideality factor $1 < n \leq 2$, the discrepancy arising from the simplifying approximations that we have made in the derivation of Eq. (6.88). A consequence of the higher ideality factor in the SCR recombination current is that it dominates for small forward bias, while QNR recombination dominates for large forward bias, as sketched in Fig. 6.31.

Second, the SCR saturation current depends on n_i. This is in contrast with QNR current I_s that depends on n_i^2 [(see Eqs. 6.36) and (6.37)]. Because of this, the temperature dependence of these two currents is quite different. The saturation current of QNR generation and recombination is thermally activated with an activation energy of about E_g, the bandgap of the semiconductor. In contrast, the saturation current of SCR-generation/recombination current displays an activation energy of $E_g/2$.

Third, the magnitude of the SCR generation/recombination current depends on the identity and concentration of the trap or traps that are present inside the SCR. This can be seen in Eq. (6.88) in which the saturation current depends on $\sqrt{\tau_{eo}\tau_{ho}}$. In practice, this implies that the SCR generation/recombination currents are highly process and contamination sensitive. In contrast, in many diodes, recombination in the QNRs takes place at the contacts and is limited by the diffusion coefficients. This is set by doping and is independent of the bulk carrier lifetimes.

Finally, in reverse bias, Eq. (6.89) exhibits a dependence on the SCR thickness. This has the important consequence that the reverse current due to SCR generation in a PN diode is not saturated but increases as the reverse bias increases. This is also sketched in Fig. 6.31.

The nonideal currents described in this section are intimately associated with traps in the SCR. This can be the result of undesired impurities or crystallographic defects introduced by

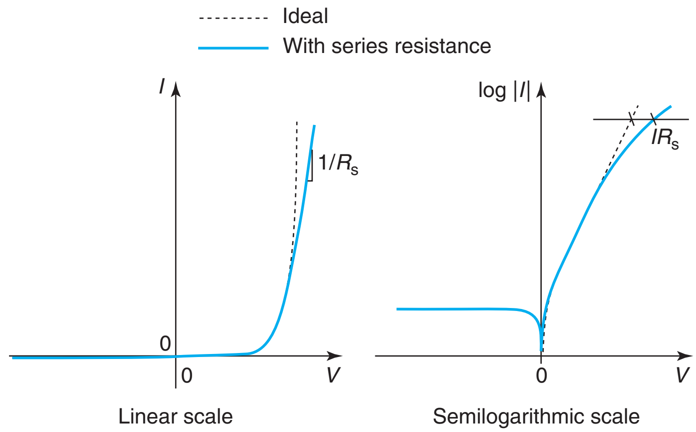


FIGURE 6.32
Sketch of I–V characteristics of an ideal PN diode and a diode with series resistance, both in a linear scale (left) and in a semilogarithmic scale (right).

the process. Since it is eminently possible to obtain very ideal diodes, as we saw in Figs. 6.15 and 6.16, the type of nonideal behavior discussed here is then a good indicator of the cleanliness and integrity of microelectronic processes and materials.

6.6.3 Series resistance

An important parasitic in a PN diode is series resistance. Series resistance arises from the finite resistance associated with the less-than-perfect ohmic contacts and the nonzero resistivity of the QNRs. If current flows through a diode, series resistance produces an ohmic drop that adds on top of the junction voltage. In an alternative view, series resistance increases the voltage that needs to be applied to a diode to obtain a desirable level of current. Hence, series resistance "stretches" the I–V characteristics sideways. The effect is more pronounced the higher the current through the diode.

The impact of series resistance in the I–V characteristics can be seen in Fig. 6.32. On the right are the characteristics in a semilog scale. We see that at low current levels, the characteristics remain ideal. However, at high current, the voltage required to supply the current increases by an amount equal to the ohmic drop through the diode, or IR_s, where R_s is the series resistance. A consequence of this is that in a linear scale, for strong forward bias, the current no longer rises exponentially but tends to increase linearly with the voltage with a slope given by $1/R_s$.

An expression for the I–V characteristics of the diode that incorporates the impact of series resistance can be readily obtained by subtracting from V in the ideal diode current expression of Eq. (6.46) the ohmic drop produced by the series resistance, IR_s,

$$I = I_s \left[\exp \frac{q(V - IR_s)}{kT} - 1 \right] \tag{6.90}$$

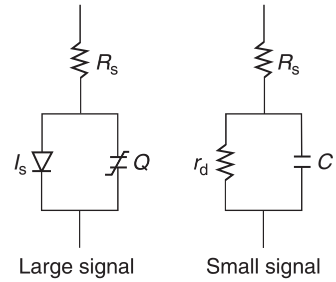


FIGURE 6.33
Equivalent circuit models of PN diode incorporating the effect of series resistance: large signal (left) and small signal (right).

It is clear in this expression that when IR_s is small in the scale of V, the impact of series resistance is negligible. However, for high values of I the impact can be very substantial. One unfortunate consequence of series resistance is that the current can no longer be solved in an explicit way in terms of the voltage (although the voltage can still be solved in terms of the current).

Series resistance can be readily incorporated into the equivalent circuit models of the diode. This only requires adding a resistance in series with the models for the ideal diode. This is shown in Fig. 6.33.

Given a certain diode structure, how does one compute its series resistance? This is not difficult for a diode with the structure shown in Fig. 6.3. There are two kinds of contributions to the series resistance. One is due to the contact resistances and the second one is due to the resistance of the QNRs. Both can be evaluated readily.

As we will discuss in Chapter 7, the resistance of an ohmic contact is characterized by its ohmic contact resistivity, ρ_c, given in $\Omega \cdot \text{cm}^2$. If a contact has an area A, the ohmic contact resistance is then ρ_c/A. This has units of Ω and scales as $1/A$, that is, the larger the contact area, the smaller its resistance. For a diode like the one shown in Fig. 6.3, where both contacts have an area equal to the junction area, the total ohmic contact resistance is given by

$$R_{cp} + R_{cn} = \frac{1}{A}(\rho_{cp} + \rho_{cn}) \tag{6.91}$$

where ρ_{cp} and ρ_{cn} refer to the contact resistivities of the ohmic contacts to the p and n regions, respectively.

The second component of the series resistance is associated with the finite resistance of the QNRs. This can be evaluated by computing their corresponding geometrical resistance as given by Eq. (5.43). In this case, we have

$$R_p + R_n = \frac{1}{A}\left(\frac{w_p - x_p}{q\mu_h N_A} + \frac{w_n - x_n}{q\mu_e N_D}\right) \tag{6.92}$$

where the first term reflects the contribution of the p-QNR and the second one that of the n-QNR.

The total series resistance of the diode is the sum of these four terms

$$R_s = R_{cp} + R_{cn} + R_p + R_n \tag{6.93}$$

In general, the series resistance of a diode is to be minimized. Equation (6.93) suggests that achieving this requires paying attention to both contact resistances and the resistance of the QNRs, which are affected by doping levels and geometrical design.

The resistance of the QNRs as given by Eq. (6.92) is simple and intuitive but it needs to be justified. It is not entirely obvious that in this complex situation in which we have excess minority carriers, a purely majority-carrier-like approach is appropriate. As it turns out, it is not difficult to show that Eq. (6.92) is an excellent approximation for this situation. Solving this problem rigorously is a great opportunity to review key concepts introduced in Chapter 5. This is done in Advanced Topic AT6.2.

6.6.4 Breakdown voltage

In reverse bias, the current in an ideal diode increases with voltage and quickly saturates to a very small value given by the saturation current. As we saw earlier, in actual diodes, the reverse bias current can be higher than expected as a result of carrier generation in the SCR. If this phenomenon is significant, the current increases with reverse voltage as the generation volume expands. Still, the reverse current is very small. However, for large enough reverse bias, a sudden increase in current is observed in all diodes. The current level can get substantial, just as high as in the forward regime, as sketched in Fig. 6.34. This condition is called *breakdown* and the voltage at which it happens is referred to as the *breakdown voltage*. The breakdown condition can be destructive since high current flows while a high voltage is being applied. The power dissipation and temperature rise that take place in the diode can create irreparable damage. The breakdown voltage is obviously a key consideration in PN diode design. In this section, we study the physics of this effect and we discuss design guidelines for the breakdown voltage.

Breakdown can occur through several effects. One is band-to-band or Zener tunneling, briefly mentioned in Section 3.1. By far the most common process is impact ionization, whose basics physics was presented in Chapters 4 and 5. This is the mechanism that we discuss here.

As the reverse bias in a PN junction is increased, the electric field inside the SCR increases too. This makes it more likely that carriers that transit through the SCR gain enough energy from the electric field to undergo impact ionizing collisions with the lattice and generate electron–hole pairs. Even though this reverse bias current is very small, for high enough fields, substantial carrier multiplication might occur. At some critical field, avalanche breakdown eventually takes place and the reverse current rises abruptly (Fig. 6.34).

It is not possible to develop an analytical model for the breakdown voltage of a PN diode due to impact ionization. The electric field is nonuniform in space and the expressions of the field-dependent impact ionization coefficient are algebraically too complicated to yield simple solutions. Through adequate simplifications, we can arrive to a model that captures the essential physics and illuminates the key dependencies.

The starting point for the model are the expressions for the multiplication factors derived in Advanced Topic AT5.4 for a general nonuniform electric field case. Let us consider a simple situation in which the ionization coefficients $\alpha_e = \alpha_h = \alpha$. Notice from the data in Fig. 4.27 that

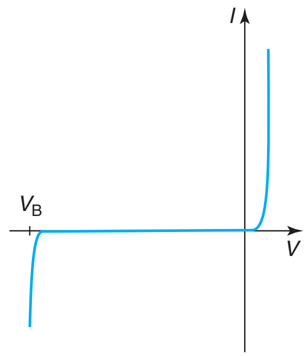


FIGURE 6.34
Sketch of reverse breakdown in a PN diode *I–V* characteristics.

this is not a bad assumption at high electric fields where breakdown occurs. In this instance, from either Eq. (5.165) or (5.166), the condition for breakdown becomes

$$\int_{-x_p}^{x_n} \alpha \mathrm{d}x = 1 \tag{6.94}$$

To further simplify the problem, we assume an asymmetric P$^+$N junction where the electric field preferentially extends through the n side of the SCR. In this case, $x_p \simeq 0$ and $x_n = x_{SCR}$ and the electric field distribution in the SCR can be written as $\mathcal{E}(x) = -|\mathcal{E}_{max}|(1 - x/x_{SCR})$. The negative sign in the electric field is a consequence of our choice of axis that is consistent with that of Section 6.2. With this and going back to Eq. (4.108), we can now write an expression for the impact ionization coefficient inside the PN junction SCR as

$$\alpha = A \exp\left[-\frac{B}{|\mathcal{E}_{max}|\left(1 - \frac{x}{x_{SCR}}\right)}\right] \tag{6.95}$$

If we use this expression, the integral in Eq. (6.94) does not yield an analytical result. However, due to the exponential dependence of α on $\mathcal{E}$, the most significant contribution to the integral in Eq. (6.94) takes place around $x \simeq 0$, where $\mathcal{E} \simeq \mathcal{E}_{max}$. Around the metallurgical junction, we can approximate

$$\frac{1}{1 - \frac{x}{x_{SCR}}} \simeq 1 + \frac{x}{x_{SCR}} \tag{6.96}$$

We can now insert this into Eq. (6.95) and the result into Eq. (6.94) to yield

$$1 = \int_0^{x_{SCR}} \alpha \mathrm{d}x \simeq A \int_0^{x_{SCR}} \exp\left[-\frac{B}{|\mathcal{E}_{max}|}\left(1 + \frac{x}{x_{SCR}}\right)\right] \mathrm{d}x \tag{6.97}$$

This can now be readily integrated to yield

$$1 \simeq \alpha(|\mathcal{E}_{\max}|)\frac{|\mathcal{E}_{\max}|x_{\mathrm{SCR}}}{B}\left[1-\frac{\alpha(|\mathcal{E}_{\max}|)}{A}\right] \tag{6.98}$$

As we will see through some numbers below, at breakdown, $\alpha(|\mathcal{E}_{\max}|) \ll A$. We also note that at breakdown, $|\mathcal{E}_{\max}|x_{\mathrm{SCR}} = 2V_{\mathrm{B}}$, where V_{B} is the breakdown voltage (defined as a positive value). We can therefore approximate the breakdown condition as

$$\alpha(|\mathcal{E}_{\max}|)2V_{\mathrm{B}} \simeq B \tag{6.99}$$

where $\mathcal{E}_{\max}$ refers to its value at breakdown. From Eq. (6.25), this can be expressed in terms of V_{B} as

$$|\mathcal{E}_{\max}(V_{\mathrm{B}})| \simeq \sqrt{\frac{2qN_{\mathrm{D}}V_{\mathrm{B}}}{\epsilon}} \tag{6.100}$$

Because of the exponential dependence of α on $\mathcal{E}$, we cannot explicitly solve for V_{B} in Eq. (6.99). However, we can solve for N_{D} in terms of everything else. Plugging Eq. (6.100) into Eq. (4.108) and this into Eq. (6.99) and then solving for N_{D} yields

$$N_{\mathrm{D}} = \frac{\epsilon B^2}{2qV_{\mathrm{B}}}\frac{1}{\ln^2\left(\dfrac{2AV_{\mathrm{B}}}{B}\right)} \tag{6.101}$$

This is now a relatively simple *implicit* model for V_{B}. The most important feature of this model is that V_{B} depends only on N_{D} and fundamental constants. The higher N_{D}, the lower V_{B}. This is understandable since the higher the N_{D}, the higher the initial electric field inside the SCR and the easier it will be to reach the breakdown condition as the reverse bias increases. Note that because of the logarithmic term in the denominator, the dependence between V_{B} and N_{D} is not quite inverse linear but somehow weaker than that, particularly for small values of V_{B}.

A chart graphing experimental measurements of V_{B} versus N_{D} for abrupt uniformly doped Si PN diodes at room temperature is shown in Fig. 6.35. As the doping level increases, the breakdown voltage decreases. The decrease slows down for high doping levels.

An interesting feature of the breakdown voltage due to impact ionization is its temperature dependence. For Si, as the temperature increases, the impact ionization coefficients decrease. This is the result of the mean free path becoming smaller due to enhanced phonon scattering. A consequence of this is that the breakdown voltage increases as the temperature increases, that is, it exhibits a *positive temperature coefficient*. This is distinct from the other major breakdown mechanism that can be observed in PN diodes, which is Zener breakdown. Zener breakdown takes place when direct tunneling occurs between the conduction band and the valence band across the SCR under strong reverse bias. As the temperature increases, the bandgap gets smaller, the tunneling barrier decreases, and the tunneling process becomes more likely. In consequence, the breakdown voltage decreases. Zener breakdown thus is characterized by a *negative temperature coefficient.*

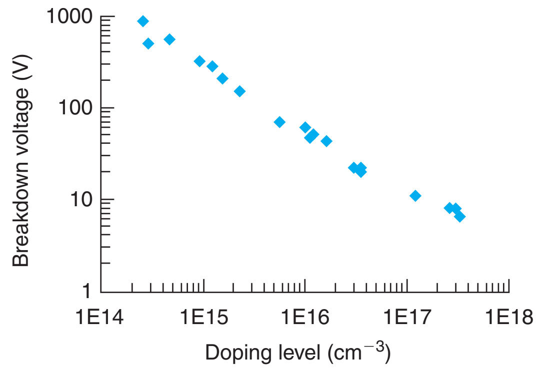


FIGURE 6.35
Experimental breakdown voltage of abrupt uniformly doped PN diodes versus doping level at 300 K.

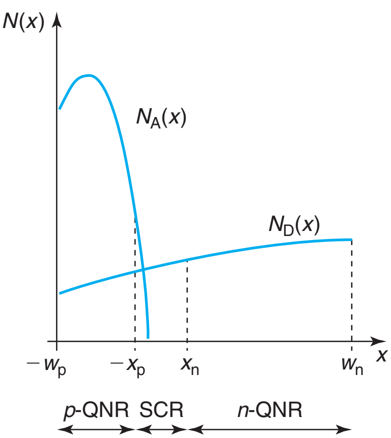


FIGURE 6.36
Sketch of a nonuniform doping level of an integrated diode. Typically, the emitter is heavily doped and highly nonuniform. The base has a lower doping level and relatively more uniform. The back of the base is not shown.

6.6.5 Nonuniform doping distributions

The fabrication processes that are used to manufacture integrated microelectronic devices often result in doping distributions that are highly nonuniform. This does not change the basic physics of operation of a PN diode. However, the actual doping profile must be taken into account when computing the various diode figures of merit or when designing a diode for a specific application. Additionally, a nonuniform doping level changes the functional dependence of some important parameters, such as the capacitance versus voltage characteristics.

A typical doping distribution of an integrated PN diode is sketched in Fig. 6.36. In this particular case, a thin and heavily doped p-type region is ion implanted or diffused into an n-type region.

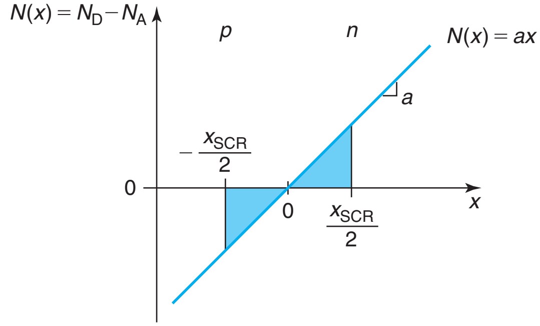


FIGURE 6.37
Doping distribution in PN junction with linearly graded impurity profile.

In analogy with bipolar transistor notation, the thin heavily doped region is often denoted as *emitter*, while the region with a lower doping level is known as *base*. As was the case in the ideal uniformly doped case studied above, from an electrical point of view, this diode can be divided into three regions: the n- and p-QNRs and the SCR in between. The basic role played by these three regions is unchanged from what we have already studied. As in the ideal diode, the *I–V* characteristics are mainly determined by recombination in the QNRs. The *C–V* characteristics are set by the electrostatics of the SCR. The dynamics are largely a function of the junction capacitance and the minority carrier storage in the QNRs. For all three, the ideal diode treatment needs to be revised. Let us examine them in turn.

Depletion capacitance

When the doping distribution in both regions of a PN junction is nonuniform, the voltage dependence of the depletion capacitance does not follow any longer the simple square root law of Eq. (6.66). Arbitrary doping profiles, in general, do not give rise to simple functional laws for the *C–V* characteristics. Device simulators can accurately account for this and provide numerical calculations of the depletion capacitance that are acceptable. In a diode model for circuit CAD, however, a simple analytical relationship is required. Fortunately, in modern devices, the doping levels are high enough and consequently, the SCR is thin enough that the doping level distribution across the SCR can often be described using a simple power law.

In order to illustrate the issues, let us solve a simple case. Consider a PN junction where the doping level changes linearly inside the SCR, that is,

$$N_D - N_A = ax \tag{6.102}$$

where a is called the *grading constant* and has units of cm^{-4}. This impurity profile is illustrated in Fig. 6.37.

We approach this problem in the context of the depletion approximation. In order to solve the electrostatics of this junction, we first make the depletion approximation. We assume that over a certain region of extent x_{SCR}, the carrier concentrations are much lower than the dopant concentrations. Outside, quasi-neutral conditions prevail. Due to the symmetry of the charge distribution, the SCR extends equal amounts into the n- and the p-type regions. Hence,

$$x_n = x_p = \frac{x_{SCR}}{2} \tag{6.103}$$

We still do not know how much x_{SCR} is. In order to derive it, we must solve completely the electrostatics of this problem.

The volume charge density is given by:

$$\rho(x) = qax \quad \text{for} - \frac{x_{SCR}}{2} \leq x \leq \frac{x_{SCR}}{2} \tag{6.104}$$

$$\rho(x) \simeq 0 \qquad \text{outside} \tag{6.105}$$

Using Gauss' law, the electric field is easily obtained:

$$\mathcal{E}(x) = \frac{qa}{2\epsilon}\left[x^2 - \left(\frac{x_{SCR}}{2}\right)^2\right] \quad \text{for} - \frac{x_{SCR}}{2} \leq x \leq \frac{x_{SCR}}{2} \tag{6.106}$$

$$\mathcal{E}(x) \simeq 0 \qquad \text{outside} \tag{6.107}$$

Strictly speaking, the electric field at the edges of the SCR is obviously not zero. If the doping level changes outside, a small field is developed. However, this field is likely to be much smaller than the field inside the SCR. For the purposes of solving the electrostatics of the SCR it can safely be neglected.

One further integration yields the electrostatic potential distribution. Selecting as reference $\phi(x = 0) = 0$,

$$\phi(x) = \frac{qa}{6\epsilon}\left[3\left(\frac{x_{SCR}}{2}\right)^2 x - x^3\right] \quad \text{for} - \frac{x_{SCR}}{2} \leq x \leq \frac{x_{SCR}}{2} \tag{6.108}$$

Outside, the potential distribution depends on the details of the impurity profile.

The electrostatic potential difference across the SCR is equal to the built-in potential minus the applied voltage, $\phi(x_{SCR}/2) - \phi(-x_{SCR}/2) = \phi_{bi} - V$. This allows us to derive an expression for x_{SCR},

$$x_{SCR} = \left[\frac{12\epsilon(\phi_{bi} - V)}{qa}\right]^{1/3} \tag{6.109}$$

ϕ_{bi} is obtained from an equation similar to Eq. (6.1),

$$\phi_{bi} = \frac{kT}{q} \ln \frac{n_o(x_{SCR}/2)}{n_o(-x_{SCR}/2)} = 2\frac{kT}{q} \ln \frac{a x_{SCR}}{2n_i} \tag{6.110}$$

Since x_{SCR} in Eq. (6.109) depends on ϕ_{bi} and ϕ_{bi} in Eq. (6.110) depends on x_{SCR}, this pair of equations needs to be solved iteratively.

The depletion capacitance of the linearly graded junction can be most easily obtained using Eq. (6.67),

$$C_j = A\left[\frac{qa\epsilon^2}{12(\phi_{bi} - V)}\right]^{1/3} = \frac{C_j(V = 0)}{\left(1 - \frac{V}{\phi_{bi}}\right)^{1/3}} \tag{6.111}$$

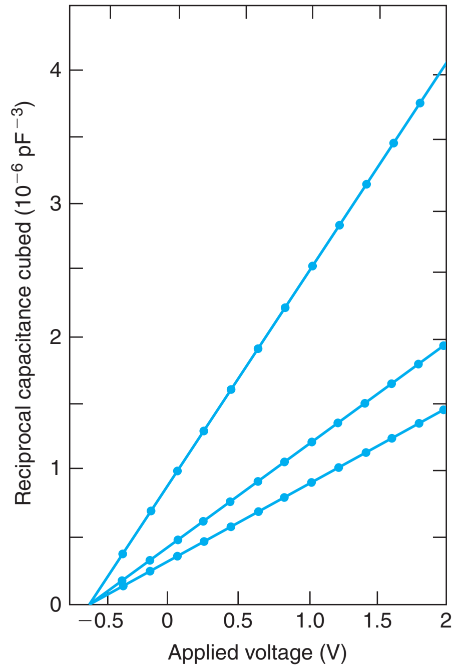


FIGURE 6.38
Experimental $1/C^3$ versus V characteristics for three different Si PN junctions. As Eq. (6.112) predicts, $1/C^3$ has a linear dependence on voltage [Reproduced by permission of the Institution of Engineering & Technology (note: the voltage convention followed by these authors is contrary to that of this book)].

This result suggests that $1/C_j^3$ has a linear dependence on V,

$$\frac{1}{C_j^3} = \frac{12(\phi_{bi} - V)}{qa\epsilon^2 A^3} \tag{6.112}$$

This is a simple relationship that can be used to extract ϕ_{bi} and the grading constant a.

Many actual PN junctions follow the linearly graded behavior studied in this section. An example is shown in Fig. 6.38. In general, it is found in practice that the depletion capacitance of most PN diodes can be approximated by the functional form,

$$C_j = \frac{C_{jo}}{\left(1 - \frac{V}{\phi_j}\right)^m} \tag{6.113}$$

with C_{jo} being the depletion capacitance that corresponds to $V = 0$, ϕ_j an effective junction built-in potential, and m a numerical exponent typically between 0.3 and 0.5.

Current–voltage characteristics

The forward I–V characteristics of the diode are dominated by recombination in the QNRs. The total current is the sum of the current injected into the n side plus the current injected into the p side. The I–V characteristics of the diode are unchanged from those of Eq. (6.46). Only the individual expressions of the two components of the saturation current I_s are affected. In a nonuniformly doped diode, the problem of computing the saturation current is complicated by the fact that in addition to minority carrier diffusion and recombination, drift takes place. This is because, as studied in Chapter 4, a nonuniform doping level produces an electric field that acts on the carriers. The equation set that governs this family of problems was given in Table 5.5. The computation of the saturation current injected into a nonuniformly doped region with an arbitrary profile is not amenable to an analytical solution and must be done by computer. For the minority carrier holes, Eq. (5.55) needs to be solved subject to the proper boundary conditions at the edge of the SCR and at the surface of the diode. A parallel approach is followed for minority carrier electrons.

In many integrated PN diodes, the depth of the doped regions is shallow enough that minority carrier recombination in the bulk of the emitter or the base is negligible. Minority carrier recombination takes place predominantly at the surfaces of the diode. This is what is called a short or *transparent region*.[3] In this case, a calculation of the saturation current is particularly easy. For the p side, in the absence of external generation, under steady state, and neglecting volume recombination, the continuity equation equivalent to Eq. (5.54) becomes simply $\mathrm{d}J_e/\mathrm{d}x = 0$, or a simple statement that the electron current does not change in space. This is obvious and not very useful in itself. Let us then use the minority carrier current equation (given in Table 5.5),

$$J_e = qn'\mu_e\mathcal{E}_o + qD_e\frac{\mathrm{d}n'}{\mathrm{d}x} \tag{6.114}$$

In a nonuniformly doped p-type region, the electric field in equilibrium is given by Eq. (4.6). In a minority carrier-type problem, we can assume that this field is negligibly modified out of equilibrium. Inserting this equation into Eq. (6.114), and using the Einstein relation for electrons, Eq. (4.38), we get

$$J_e = q\frac{D_e}{N_A}\frac{d(n'N_A)}{\mathrm{d}x} \tag{6.115}$$

After bringing the term qD_e/N_A to the left, this equation can be integrated across the nonuniformly doped region. For the emitter of the PN junction shown in Fig. 6.36, it becomes

$$\int_{-w_p}^{-x_p} J_e\frac{N_A}{qD_e}\mathrm{d}x = \int_{-w_p}^{-x_p}\frac{d(n'N_A)}{\mathrm{d}x}\mathrm{d}x = n'N_A|_{-x_p} - n'N_A|_{-w_p} \tag{6.116}$$

The boundary condition at $-x_p$ is simply as in Eq. (6.30). The boundary condition at $-w_p$ depends on the details of the surface. If the surface of the semiconductor is covered by an ohmic contact, then $n'(-w_p) = 0$. Getting J_e out of the integral in Eq. (6.116) and solving for it, we get

[3] As opposed to a long or *opaque region* in which all minority carriers recombine before reaching the surface.

$$J_e(-x_p) = \frac{qn_i^2}{\int_{-w_p}^{-x_p} \frac{N_A}{D_e} dx} \left(\exp \frac{qV}{kT} - 1 \right) \tag{6.117}$$

If the doping level is uniform, this equation simplifies to that corresponding to a short diode [Eq. (6.77)], as should be expected.

If the surface is characterized by a surface recombination S, then $n'(-w_p) = J_e(-w_p)/qS = J_e(-x_p)/qS$, and

$$J_e(-x_p) = \frac{qn_i^2}{\int_{-w_p}^{-x_p} \frac{N_A}{D_e} dx + \frac{N_A(-w_p)}{S}} \left(\exp \frac{qV}{kT} - 1 \right) \tag{6.118}$$

Equations (6.117) and (6.118) are very simple expressions. The integrals in the denominator of both the equations are easy to perform once the doping profile is known. In these integrals, the diffusion coefficient must remain inside the integral since, in general, it is a function of position if the doping level changes in space. Nevertheless, since D_e is not a strong function of N_A, sometimes D_e is taken out of the integral. In this case, the remaining integral $\int_{-w_p}^{-x_p} N_A dx$ is called the *Gummel number*, in honor of H. K. Gummel, a bipolar transistor pioneer. Gummel was first to realize that, to the first order, *the total amount of dopants* in a thin transparent region (that is when bulk recombination is negligible) determines the minority carrier injection current. Their particular spatial distribution is only of secondary importance. Interestingly, this implies that the magnitude of the minority carrier current is independent whether the electric field developed by the doping distribution aids or opposes minority carrier diffusion. This result, as you will see below, does not apply to the minority carrier storage, which is intimately related to the electric field distribution.

The current contribution arising from hole injection into the nonuniformly doped base is computed in the same way and a totally equivalent equation emerges.

Minority carrier storage

In a uniformly doped region, the dominant time constant for minority carrier behavior is the lifetime or the transit time, whichever is shortest. This does not change in a nonuniformly doped region, just the computation of these characteristic times is more complicated. In fact, in a general case, an analytical solution is not possible.

A case of importance for which an analytical solution is possible is the short or transparent case in which the carrier lifetime is much longer than the transit time. In this case, the transit time is the dominant time constant of the problem.

Let us compute the diffusion capacitance and the transit time for a p region similar to the one examined just above. The starting point is the expression of the minority carrier charge in the p region given in Eq. (6.50). For this, we need the excess electron profile across the quasi-neutral p region. This can be obtained by integrating Eq. (6.115) in a similar way as in Eq. (6.116) but from the surface at $x = -w_p$ to a point x inside,

$$\int_{-w_p}^{x} J_e \frac{N_A}{qD_e} dx = \int_{-w_p}^{x} \frac{d(n'N_A)}{dx} dx = n'N_A|_x - n'N_A|_{-w_p} \tag{6.119}$$

If the surface is covered by an ohmic contact, $n'(-w_p) = 0$, and we can then easily solve for $n'(x)$,

$$n'(x) = \frac{J_e(-x_p)}{qN_A(x)} \int_{-w_p}^{x} \frac{N_A}{D_e} dx \tag{6.120}$$

Inserting this result into Eq. (6.50) yields

$$Q_p = J_e(-x_p) \int_{-w_p}^{-x_p} \frac{1}{N_A} \left(\int_{-w_p}^{x} \frac{N_A}{D_e} dx \right) dx \tag{6.121}$$

The diffusion capacitance associated with the p region is

$$C_{dp} \simeq \frac{q}{kT} J_e(-x_p) \int_{-w_p}^{-x_p} \frac{1}{N_A} \left(\int_{-w_p}^{x} \frac{N_A}{D_e} dx \right) dx \tag{6.122}$$

From this expression, we can identify the transit time through the p region as

$$\tau_{tp} = \int_{-w_p}^{-x_p} \frac{1}{N_A} \left(\int_{-w_p}^{x} \frac{N_A}{D_e} dx \right) dx \tag{6.123}$$

You can easily verify that this expression converges to Eq. (6.82) for the uniformly doped case.

It might not be obvious just by looking at Eq. (6.123), but the presence of an electric field in a QNR can substantially affect the transit time of minority carriers. An electric field that makes minority carriers drift in the same direction as they diffuse will result in a shorter transit time. A field that opposes diffusion will increase the transit time. Problem 6.4 shows an excellent example of this.

What type of an impurity gradient does one need in order to engineer an electric field that helps diffusion and results in a fast transit time? The sketches of Fig. 6.39 for a nonuniformly doped p-type region help us to determine this. By working out the equilibrium majority carrier profile in the QNR, it is clear that a "downgoing" impurity gradient results in a field that aids electron diffusion. In contrast, an upgoing impurity gradient leads to a field that opposes electron diffusion. You can easily verify that the same happens for an n-type region.

6.6.6 High-injection effects

A basic assumption made in the derivation of the forward characteristics of the PN diode is low-level injection. Low-level injection means that the minority carrier concentration at the edges of the PN junction is always negligible in comparison with the majority carrier concentration. For high enough forward voltage, this assumption fails as the diode goes into "high-level injection." When this happens, our ideal theory does not apply any longer. You can see in

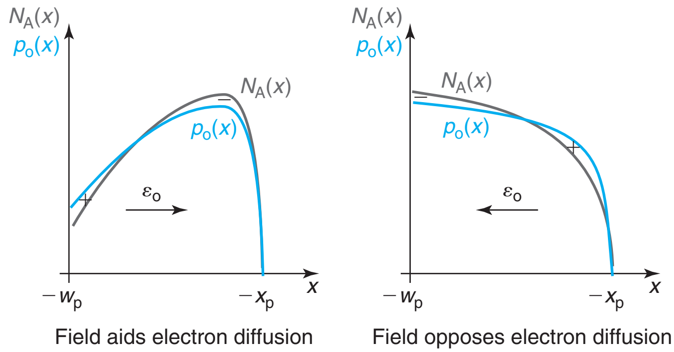


FIGURE 6.39
Sketch of majority carrier distributions and electric field in two nonuniformly doped p-type quasi-neutral regions. A "downgoing" gradient (left) results in an electric field that helps electron diffusion. An "upgoing" gradient (right) opposes electron diffusion.

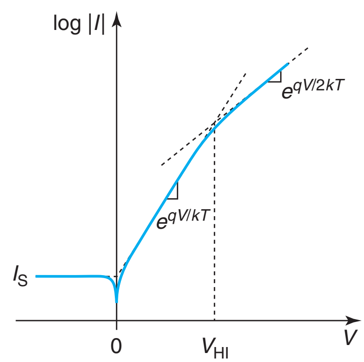


FIGURE 6.40
Sketch of forward bias *I–V* characteristics of a PN diode. For strong forward voltage, the diode can go into high-level injection and the current grows as $e^{qV/2kT}$.

Fig. 6.40 that, in fact, for strong forward voltage, the I–V characteristics deviate from their ideal $e^{qV/kT}$ behavior into what appears more to be $e^{qV/2kT}$. Let us discuss why this happens.

First, let us estimate the voltage at which high-level injection occurs. This is simple. In an asymmetric diode, high-level injection occurs first in the lowly doped side of the diode. Let us consider here a P^+N diode with a donor concentration N_D (an entirely equivalent result is obtained for a N^+P diode). Figure 6.41 shows the evolution of the minority and majority carrier concentrations (in a semilog scale) in the n region as the diode forward voltage increases. This figure clearly shows that the location that is driven first into high-level injection is the edge of the SCR. For small forward voltages, the minority carrier concentration is much less than the doping level and the majority carrier concentration is hence unaffected by minority carrier injection. For a high enough forward voltage, the minority carrier concentration at the edge of the SCR

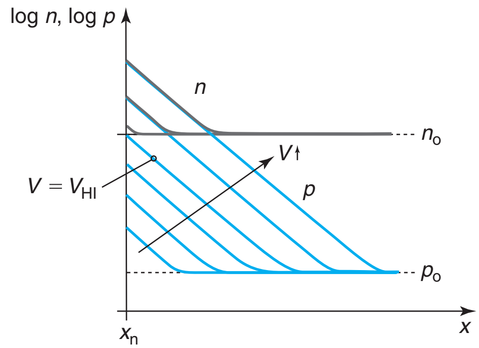


FIGURE 6.41
Minority and majority carrier concentrations in the n side of a P^+N diode as a function of forward bias. At V_{HI}, the n region goes into high-level injection.

reaches a value equal to the doping level. At this voltage, the majority carrier concentration at the edge of the SCR is twice the doping level. Clearly, it is not possible to assume any longer that the majority carrier concentration is unaffected as this would violate quasi-neutrality. The n side of the QNR has become high-level injected.

This is a reasonable definition for the onset of high-level injection. If we allow ourselves to "stretch" the low-level injection boundary condition, Eq. (6.29), up to this point, high-level injection occurs when

$$p(x_n) = \frac{n_i^2}{N_D} \exp \frac{qV_{HI}}{kT} = N_D \tag{6.124}$$

where V_{HI} refers to the voltage at the onset of high-level injection. Solving for V_{HI} in this equation, we get

$$V_{HI} = \frac{2kT}{q} \ln \frac{N_D}{n_i} \tag{6.125}$$

This result is reasonable. The higher the doping level (of the lowly doped side), the higher the voltage that it takes to drive the diode into high-level injection.

In microelectronic device operation, we are not interested in general in the high-level injection regime. This is something to be avoided. In PN junctions, for example, the I–V characteristics grow more slowly, as sketched in Fig. 6.40, something that is undesirable for many applications. In bipolar transistors, for example, high-level injection of the base brings with it a drop in gain. Unless one is working with devices that have been designed to operate in high-level injection, such as power diodes or certain kinds of solar cells, high-level injection is a syndrome to avoid all together. For this reason, we will not treat it in detail here.

It is of value, nevertheless, to understand the origin of the $e^{qV/2kT}$ dependence of the current in the high-level injection regime. This result can be easily derived from the expression of the np product inside the SCR given by Eq. (6.47). This equation remains valid in the high-level injection regime because it simply states that electrons and holes are in equilibrium among each other due to the fact that the SCR is very thin. At the edge of the SCR that goes into high-level injection first, $n \simeq p$, and

$$n \simeq p \simeq n_{\mathrm{i}} \exp \frac{qV}{2kT} \tag{6.126}$$

The nature of the diode current is still carrier recombination in the QNRs. The boundary condition of Eq. (6.126) implies that the voltage dependence of the current will be of the form $e^{qV/2kT}$.

Physically, one can understand the origin of this result by examining the evolution of the energy band diagrams around the SCR as the voltage increases. This is sketched in Fig. 6.42. The key is to examine the dependence of the energy barrier that carriers face in crossing the junction. In the low-level injection regime, as the voltage increases, the energy barrier is reduced by exactly qV. This is because the majority carrier quasi-Fermi levels are "rigidly" tied up to the respective band edges. This results from the fact that in low-level injection, the majority carrier concentrations remain unchanged.

As one of the sides of the diode goes into high-level injection, this is not the case any longer. The large minority carrier injection forces an increase in the majority carrier concentration. As a consequence, in high-level injection, the majority carrier quasi-Fermi level on the high-level injected side gets closer and closer to the majority carrier band edge (conduction band for a p^+–n diode). After this happens, the energy barrier presented by the junction is only reduced at half the rate of the quasi-Fermi level splitting (always qV). This is sketched in Fig. 6.42.

6.7 INTEGRATED PN DIODE

PN diodes are present in many integrated circuits. They are used in rectifying circuits, as detectors in communications applications, and as bias shifters and input protection devices (against electrostatic discharge or ESD). This section is dedicated to studying a few important issues involved in the design and analysis of actual integrated PN diodes.

In spite of their usefulness, PN diodes in integrated circuits are implemented without the luxury of a dedicated process. Rather, some of the existing process steps that are used to make the n- and p-MOSFETs in the case of CMOS, or the bipolar transistors in the case of a bipolar process, must be combined in creative ways to achieve diodes with the desired characteristics. This brings in some interesting issues. We discuss the most important ones in the following subsections.

6.7.1 Isolation

A first issue to think about is device isolation. In integrated electronics, multiple devices sit side by side on a common substrate. For circuits to operate properly, all these devices must be properly isolated from each other. This is particularly hard with the devices packed tightly as is desired in very dense integrated circuits. There are several techniques to achieve device isolation in ICs. One of them is using trenches dug into the substrate and filled with a dielectric. A second way is using a PN junction. This is called *junction isolation*. A combination of trenches and junction isolation can be seen in the diode of Fig. 6.43, which is built on a bipolar process (more about this in Chapter 11).

In this diode, the p-type base of the transistor is used as anode and the n-type collector is used as cathode. Isolation with other devices is provided through a dielectric trench that rings

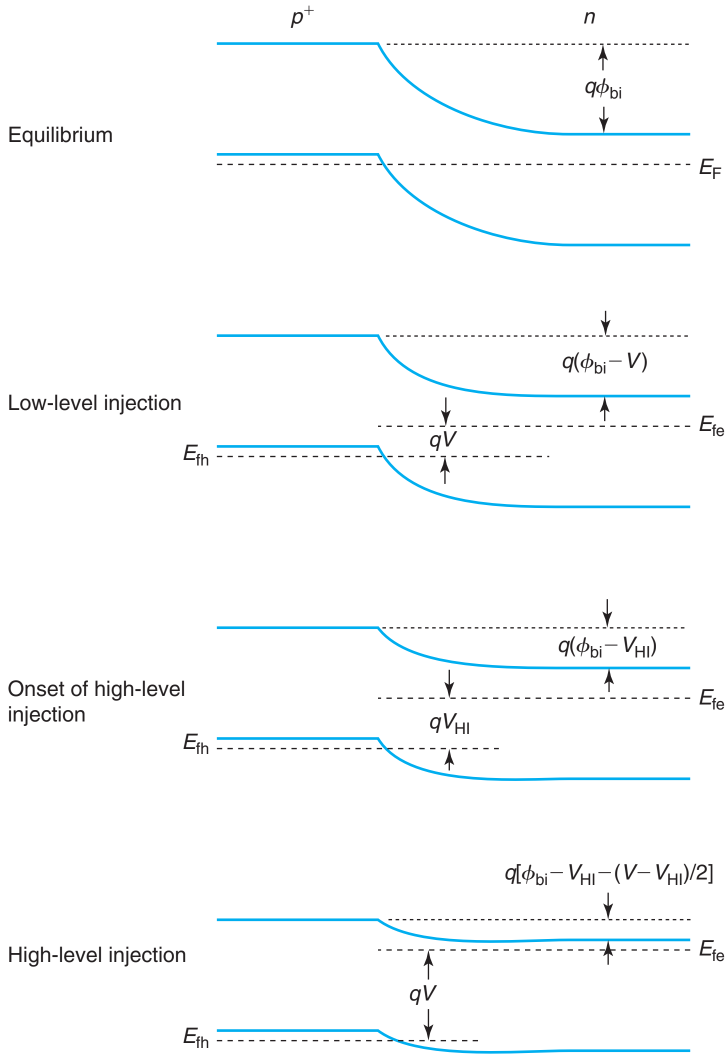


FIGURE 6.42
Energy band diagram of a PN junction as a function of injection level. In the low-level injection regime, the energy barrier across the junction is reduced in a proportional way to the applied voltage. In the high-level injection regime, the energy barrier is reduced only at half the rate of increase of the forward voltage.

around the sides and a substrate PN junction at the bottom. Using a PN junction for isolation makes sense since we know that in equilibrium or under small reverse bias, a PN junction has negligible current flowing through it. PN junction isolation is very widely used in IC design. We can see it in the integrated resistor design of Fig. 5.12 and in the CMOS transistor pair of Fig. 6.2.

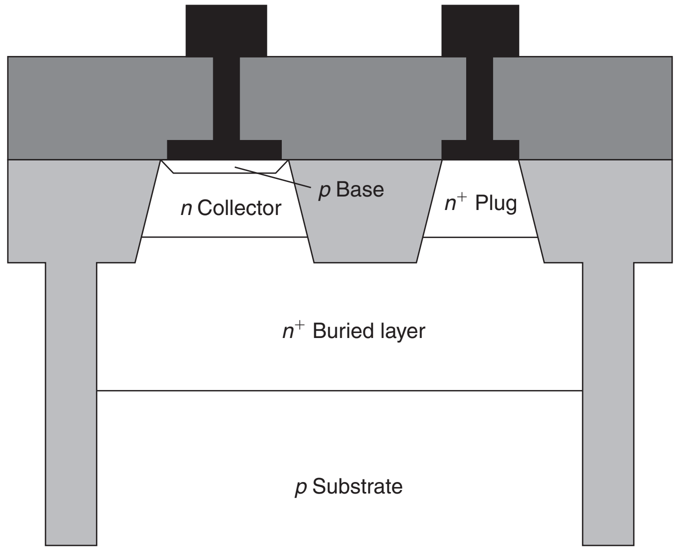


FIGURE 6.43
Sketch of integrated Si PN diode fabricated in a BJT process.

For a substrate PN junction in which the substrate is p type, reverse bias can be assured if the substrate is connected to the most negative voltage available in the circuit (if the substrate was n type, the most positive voltage would be used). In any case, the substrate junction contributes a parasitic depletion capacitance that must be accounted for when modeling this device.

The main challenge in implementing integrated PN diodes is the presence of a parasitic "bipolar" transistor that can compromise isolation. The problem is illustrated in Fig. 6.44 for a PN diode implemented on a CMOS process on a p-type substrate. In this case, the most negative voltage available in the circuit is applied to the substrate. As a consequence, the cathode/substrate PN junction is reverse biased or at zero volts. Under these conditions, a parasitic "bipolar" action can take place between the two PN junctions. The details of the problem need to be postponed until we study the bipolar transistor in Chapter 11. Basically, with the anode–cathode junction under forward bias, holes are injected downward into the cathode. If the hole diffusion length in the cathode is shorter than the cathode depth, all the holes recombine in the cathode. This is the ideal case that we have studied so far. However, if the hole diffusion length is comparable to or longer than the depth of the cathode, some of these holes will reach the back of the n region. With the cathode/substrate junction in reverse bias, the high electric field prevalent in its SCR "extracts" or "collects" holes away into the substrate where they are majority carriers. The result is that a fraction of the diode current is being diverted to the substrate. This is an undesirable current path that, if unaccounted for, can disrupt or impede proper operation of the diode in its circuit.

A solution to this problem in a CMOS process is actually very difficult. In a typical CMOS process, the wells have depths that are much shorter than the minority carrier diffusion lengths and that means that a great deal of the minority carriers injected into the well will be collected

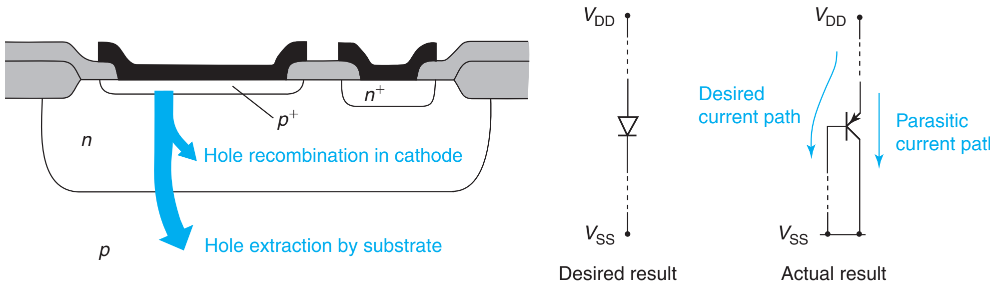


FIGURE 6.44
Sketch of isolation problem in integrated PN junction. If the hole diffusion length in the anode is longer than its depth, holes injected from the cathode can be extracted by the substrate in a *bipolar-type effect*. This is undesirable in most circuits.

by the substrate instead of recombining inside the well. Integrated diodes just simply cannot be implemented in CMOS. The only exception is the diode that has a terminal at the same voltage as the substrate. Obviously, in this case, current diversion to the substrate is not an issue.

In a bipolar transistor process, the problem is much easier to handle. In this case, as shown in Fig. 6.43, the presence of the n$^+$-buried collector effectively prevents hole transport down to the substrate. There are two reasons for it. First is the potential barrier associated with the NN$^+$ high–low junction that confines holes to the n side. Second is the small diffusion length prevalent in the buried layer as a consequence of its high doping level. Any hole that is injected into the buried layer will recombine within it before it can diffuse down toward the substrate. If properly designed, diodes such as that of Fig. 6.43 can prevent injected holes from reaching the N$^+$P substrate junction.

6.7.2 Series resistance

A second issue to consider in a practical integrated diode is its series resistance. In an actual diode, both contacts are made from the top surface. This allows the diode to be connected to other devices on the substrate. Unlike in an ideal PN diode, in a real integrated diode, current flow is two dimensional. For example, in the diode of Fig. 6.43, electron majority carrier current in the cathode enters through the top contact, goes down the n$^+$ plug, then turns sideways to flow through the buried layer, and then turns around again to eventually flow upward toward the intrinsic device. In a situation like this, how does one estimate the series resistance?

A sketch of the parasitic resistance components in the device of Fig. 6.43 is shown in Fig. 6.45. There are essentially eight resistance components. Starting from the contact to the region, we have:

- contact resistance to p region, R_{cp},
- body resistance of p region, R_p,
- body resistance of n region, R_n,
- spreading resistance of n$^+$-buried layer, R_{b1},
- lateral resistance of n$^+$-buried layer, R_{b2},

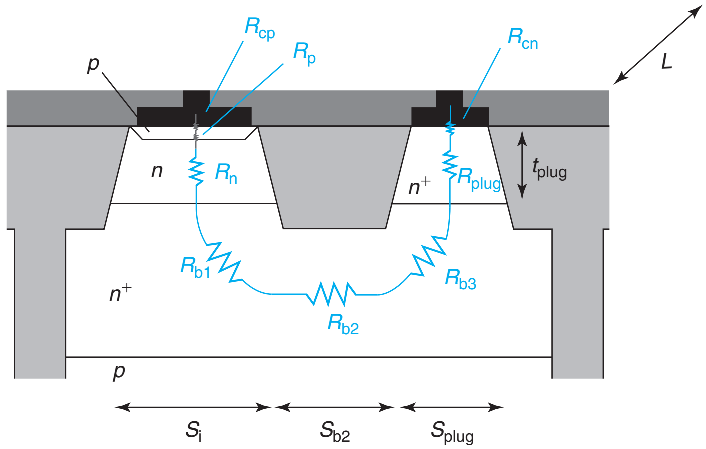


FIGURE 6.45
Parasitic resistances associated with integrated Si PN diode fabricated in a BJT process.

- spreading resistance of n$^+$-buried layer, R_{b3},
- n$^+$-plug region resistance, R_{plug}, and
- contact resistance to n region, R_{cn}.

Of these, R_{cp}, R_{cn}, R_p, and R_n are computed as indicated earlier in this chapter for the ideal diode. New resistances are those associated with the n$^+$-plug region, R_{plug}, and the n$^+$-buried layer, $R_{b1} + R_{b2} + R_{b3}$.

R_{plug} can be easily computed simply by knowing the geometry of the plug and its resistivity, ρ_{plug} (we assume the doping level to be uniform). If we denote the depth of the plug as t_{plug}, its width as S_{plug}, and its length (which is also the length of the entire diode) as L, then we have

$$R_{plug} \simeq \rho_{plug} \frac{t_{plug}}{S_{plug} L} \tag{6.127}$$

The resistance associated with the buried layer is more complex because the current flow is eminently two dimensional. We can simplify the problem if we consider that the lateral dimensions of a typical diode are much larger than the vertical dimensions. In consequence, we can neglect the component of resistance associated with vertical current flow in the buried layer and just focus on the lateral resistance. In a problem like this, it suffices to characterize the buried layer through its sheet resistance R_{shb}.

Of the three components of the buried layer resistance, R_{b2} is easy to compute. In this region, electron flow is essentially one dimensional. We can then write

$$R_{b2} \simeq R_{shb} \frac{S_{b2}}{L} \tag{6.128}$$

We could think of using a similar approach to compute the resistance associated with lateral carrier flow through the two other regions of the buried layer, but doing this would overestimate it. This is because the total diode current does not flow through the entire lateral extent of the regions $S_{\rm i}$ and $S_{\rm plug}$. For example, in the intrinsic device region, we can assume that the diode current flows downward uniformly across the entire dimension $S_{\rm i}$. When it hits the buried layer, it turns around and it starts flowing sideways from left to right in the figure. As a result, as we advance from left to right along the buried layer, the current through it is increasing in a linear way until it reaches its maximum value (the total diode current) right at the edge of the intrinsic region on the right-hand side. Similarly, the lateral current through the buried layer under the plug region drops linearly as we advance from its left edge toward the right as the current is deviated upward. Obviously, the resistance associated with these current paths is *less* than the geometrical resistance of the regions.

Solving this problem is not difficult but it requires some careful work. We postpone a solution until we treat the base resistance in a bipolar transistor which is a similar problem. This is done in Section 11.5.6. The result is actually quite simple. For a problem like this, the lateral resistance is one-third of the geometrical resistance. Then,

$$R_{\rm b1} \simeq \frac{1}{3} R_{\rm shb} \frac{S_{\rm i}}{L} \tag{6.129}$$

$$R_{\rm b3} \simeq \frac{1}{3} R_{\rm shb} \frac{S_{\rm plug}}{L} \tag{6.130}$$

With this, we now have a complete first-order model for the series resistance of an integrated diode.

6.7.3 High–low junction

An additional important issue that emerges in an integrated PN diode implemented in a bipolar process derives from the presence of the $\rm n^+$-buried layer below the n-type region and the boundary condition that this represents for holes injected into this region. In a diode of this kind, there is a sudden increase in doping level inside the n-type region. This is known as a *high–low junction*. Its presence can have a substantial impact on the minority carrier distribution in the n side of the diode under forward bias and therefore on the diode current.

The relevant case to study is that of a "short" diode in which the diffusion length of holes in the n-type region is much longer than the extent of this region. It is in this case that injected holes diffuse downward and reach the high–low junction.

Qualitatively, it is easy to appreciate the impact of a high–low junction on minority carrier flow by drawing an energy band diagram of the structure in equilibrium. This is shown in Fig. 6.46. The sudden increase of n-type doping level across the high–low junction brings the conduction band closer to the Fermi level. This creates an energy barrier for holes in the valence band that acts to prevent hole injection into the $\rm n^+$ region. Therefore, with the diode in forward bias, when holes are injected into the n-type region, they diffuse away from the junction and reach the high–low junction where they pile up.

Intuitively, we can then understand that a high–low junction exerts a blocking action for the minority carriers. It confines them to the lightly doped side of the junction where carrier lifetimes tend to be long. In consequence, minority carrier current tends to be suppressed.

Let us now develop a model for the situation. Let us assume the simple situation depicted in Fig. 6.47. This figure shows the n side of a PN diode with a doping level distribution that abruptly

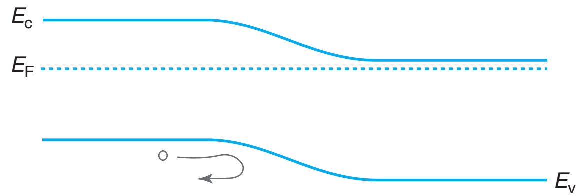


FIGURE 6.46
Energy band diagram of an NN$^+$ high–low junction in equilibrium. For holes in the n region, the high low-junction appears as an energy barrier that blocks their flow into the n$^+$ region.

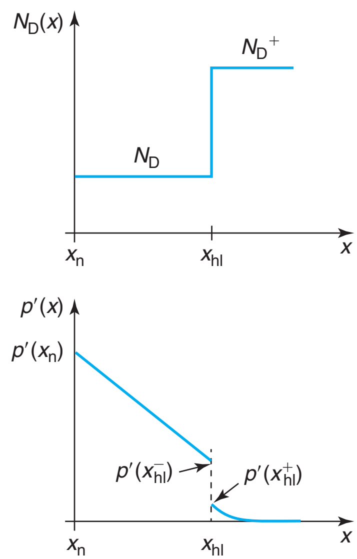


FIGURE 6.47
Doping distribution and excess hole concentration across high–low junction.

changes from N_D to N_D^+ (with $N_D^+ > N_D$) at $x = x_{hl}$. The lightly doped region is short in the scale of the relevant minority carrier diffusion length. The highly doped side, on the other hand, is long. Let us label the diffusion length there as L_h^+.

Under forward bias, holes are injected into the n side of the diode and they diffuse in. The presence of the high–low junction restricts the number of holes that can overcome the energy barrier there to a small number. These diffuse and recombine in the highly doped region and the excess hole concentration drops with a characteristic exponential shape with L_h^+ as the decay length. In the lightly doped side of the junction, the profile is a straight line since this region is short in the scale of the diffusion length.

In order to solve this problem, we need to first think about the boundary conditions at the high–low junction. The first one is about the excess hole concentration. It is clear that there is a sudden drop as we cross the junction. We can think of this in two ways. At the high–low junction, there is a potential barrier of magnitude,

$$\Delta E_{\mathrm{hl}} = kT \ln \frac{N_{\mathrm{D}}^{+}}{N_{\mathrm{D}}} \tag{6.131}$$

In consequence, the ratio of the hole concentration across the junction is

$$p(x_{\mathrm{hl}}^{+}) = p(x_{\mathrm{hl}}^{-}) \exp\left(-\frac{\Delta E_{\mathrm{hl}}}{kT}\right) = p(x_{\mathrm{hl}}^{-}) \frac{N_{\mathrm{D}}}{N_{\mathrm{D}}^{+}} \tag{6.132}$$

A second way to think about this is to assume quasi-equilibrium across the high–low junction. What do we mean by this? This means that a single hole quasi-Fermi level completely describes the hole distribution in the vicinity of the high–low junction. In this case, we can assume that the *pn* product does not change across the junction. Since the n-type doping changes from N_{D} to N_{D}^{+}, the hole concentration must drop by the same ratio. That is exactly what Eq. (6.132) says.

The second boundary condition at $x = x_{\mathrm{hl}}$ is that of continuity of hole current. Over the very short distance from $x = x_{\mathrm{hl}}^{-}$ to $x = x_{\mathrm{hl}}^{+}$, hole recombination is negligible. Then, $J_{\mathrm{h}}(x_{\mathrm{hl}}^{-}) = J_{\mathrm{h}}(x_{\mathrm{hl}}^{+})$. This implies that

$$D_{\mathrm{h}} \left.\frac{\mathrm{d}p'}{\mathrm{d}x}\right|_{x_{\mathrm{hl}}^{-}} = D_{\mathrm{h}}^{+} \left.\frac{\mathrm{d}p'}{\mathrm{d}x}\right|_{x_{\mathrm{hl}}^{+}} \tag{6.133}$$

where D_{h} and D_{h}^{+} represent the diffusion coefficients for holes in the n and n^{+} regions, respectively. In general, these are not equal since the doping level is not the same.

High-low junctions are particularly useful to reduce recombination at an ohmic contact. As we know, excess minority carrier recombination at an ohmic contact is very efficient. Inserting a high–low junction in the path of the minority carriers blocks their flow toward the ohmic contact and reduces the overall recombination current. This is instrumental in minimizing recombination, for example, in solar cells where high–low junctions are widely used.

With these two boundary conditions at x_{hl}, the problem can be solved quickly. The hole current density injected into the n region is continuous all the way to $x = x_{\mathrm{hl}}^{+}$. Then, at this point, we have

$$J_{\mathrm{h}} = q \frac{D_{\mathrm{h}}^{+}}{L_{\mathrm{h}}^{+}} p'(x_{\mathrm{hl}}^{+}) = q \frac{D_{\mathrm{h}}^{+}}{L_{\mathrm{h}}^{+}} \frac{N_{\mathrm{D}}}{N_{\mathrm{D}}^{+}} p'(x_{\mathrm{hl}}^{-}) \tag{6.134}$$

where we have used Eq. (6.132). In the n-type region, we can write

$$J_{\mathrm{h}} = q D_{\mathrm{h}} \frac{p'(x_{\mathrm{n}}) - p'(x_{\mathrm{hl}}^{-})}{x_{\mathrm{hl}} - x_{\mathrm{n}}} \tag{6.135}$$

Equating these two expressions of J_{h} allows us to solve for $p'(x_{\mathrm{hl}}^{-})$. Inserting this result into either one of them, finally yields the following expression for the hole current:

$$J_{\mathrm{h}} = q \frac{n_{\mathrm{i}}^{2}}{N_{\mathrm{D}}} \frac{D_{\mathrm{h}}}{x_{\mathrm{hl}} - x_{\mathrm{n}}} \frac{1}{1 + \frac{D_{\mathrm{h}}}{D_{\mathrm{h}}^{+}} \frac{L_{\mathrm{h}}^{+}}{x_{\mathrm{hl}} - x_{\mathrm{n}}} \frac{N_{\mathrm{D}}^{+}}{N_{\mathrm{D}}}} \left(\exp \frac{qV}{kT} - 1\right) \tag{6.136}$$

If we compare this result with that of the short diode with an ohmic contact placed at x_{hl}, obtained in Section 6.6.1, we can see that the hole injection current has been suppressed by the factor in the denominator of the third fraction. If $N_D^+ \gg N_D$, as often in practice, this factor is large and the injection current is suppressed in a very significant way. This is important in solar cells and other applications where high–low junctions are widely used.

6.8 SUMMARY

- In a PN junction, a space-charged region forms around the metallurgical interface with a volume charge density that is set by the ionized dopant concentration. The SCR is surrounded by two QNRs.
- In thermal equilibrium, there is an energy barrier that appears for holes and electrons across the metallurgical junction. The height of this barrier is modulated when voltage is applied. This results in excess minority carrier concentrations at the edges of the SCR that go as $e^{qV/kT} - 1$. This is the origin of the rectifying behavior of a PN diode.
- In forward bias, the current in an ideal diode is dominated by minority carrier injection and recombination in the QNRs. In reverse bias, the current is dominated by minority carrier extraction and generation in QNRs.
- The width of the SCR is also modulated by the applied voltage. This yields a charge storage effect that is described by the depletion capacitance.
- In a forward biased PN junction, carrier storage takes place in the QNRs. This also introduces a capacitive effect described by the diffusion capacitance.
- For strong forward bias, diffusion capacitance dominates. For moderate forward bias or reverse bias, depletion capacitance dominates.
- SCR generation and recombination currents are frequently observed, respectively, in reverse and small forward bias.
- In most PN diodes, breakdown at high reverse bias occurs due to an avalanche process. The breakdown voltage is set by the doping level.
- A parasitic bipolar effect compromises the isolation of integrated PN diodes and makes it difficult to synthesize PN diodes in a CMOS process.

6.9 FURTHER READING

The treatment of the PN diode is fairly standard in most textbooks. The approach followed here mirrors this well-established path. It is difficult to find any books that offer something substantially different or that go beyond the treatment presented here.

Semiconductor Device Fundamentals by R. F. Pierret, Addison-Wesley, Dover, 1996 (ISBN 0-201-54393-1, TK7871.85.P484). The description of the PN diode in this book is similar to that of the present text. The presentation is clear and emphasizes physical intuition. There are also a number of practical problems that do a good job in placing the PN junction and PN diodes in the real world. Pierret goes further than the present text in a number of areas such as the admittance of the PN diode as a function of frequency and a more detailed study of the turn-on and turn-off transitories.

AT6.1 VALIDITY OF THE DEPLETION APPROXIMATION

We discuss here the validity of the depletion approximation that lies at the heart of the analysis of the PN junction. We examine a general case with $N_A \geq N_D$, as depicted in Fig. 6.5, under applied voltage V. Similar conclusions apply for $N_D \geq N_A$. As seen in that figure, the depletion approximation demands that $n(0), p(0) \ll N_A, N_D$. Of these, $p(0) \ll N_D$ is the hardest condition to fulfill. We examine this in more detail.

For convenience, we denote as ϕ_{bn} and ϕ_{bp} the potential build up across the n- and p-type portions of the SCR, respectively. Based on the results obtained in Sections 6.2 and 6.3.1, it is easy to show that

$$N_D \phi_{bn} = N_A \phi_{bp} \tag{6.137}$$

In general, $\phi_{bn} + \phi_{bp} = \phi_{bi} - V$. We then have

$$\phi_{bp} = \frac{N_D}{N_A + N_D}(\phi_{bi} - V) \tag{6.138}$$

$p(0)$ can then be written as

$$p(0) = N_A \exp \frac{-q\phi_{bp}}{kT} \tag{6.139}$$

where we have assumed the quasi-equilibrium approximation.

Plugging Eq. (6.138) into Eq. (6.139) and solving for $\phi_{bi} - V$, for $p(0) \ll N_D$ to be fulfilled, we must demand

$$\phi_{bi} - V \gg \frac{kT}{q}\left(1 + \frac{N_A}{N_D}\right) \ln \frac{N_A}{N_D} \tag{6.140}$$

In the case of a perfectly symmetric junction, the depletion approximation is always valid (V never gets too close to ϕ_{bi} due to external ohmic drops). For asymmetric junctions, the higher the doping asymmetry, the larger a potential build up across the junction is required for Eq. (6.140) to be fulfilled. For example, for $N_A/N_D > 10$, $\phi_{bi} - V \gg 0.66$ V. This means that for strong forward bias, this condition will fail. For a larger asymmetry, say $N_A/N_D = 100$, $\phi_{bi} - V \gg 12$ V. This implies that only under substantial reverse bias the depletion approximation is fulfilled.

This simple analysis reveals that the depletion approximation does not apply in many situations. Yet, it is widely used to analyze the PN junction. The reason for this will become clearer in Chapter 8. There, we will study the electrostatics of the metal-oxide-semiconductor structure and we will learn about the inversion regime. The situation in an asymmetric PN junction is quite similar to this. If $N_A \gg N_D$, on the n side of the metallurgical junction, an "inversion region" of holes is formed with a concentration that exceeds the background doping level. Technically, the depletion approximation fails. However, as it turns out, many aspects of its predictions remain rather valid. The reason is, as we will learn in Chapter 8, that the inversion layer is very thin and that the thickness of the depletion region on the n side of the junction is not much affected by its presence. Recall that it is the physics of the lowly doped side that controls most aspects of the operation of the PN junction.

There is an additional aspect of the depletion approximation that we need to verify. It assumes that the charge distribution is "boxy," that is, its edges are rather sharp. This is a good assumption if the softness of the edges is small in the scale of the depletion region thickness itself. It will become clear in Chapter 8, and it is easy to understand based on what we know at this point, that the charge density at the edges of the depletion region decays exponentially with a characteristic length given by the Debye length. So, essentially, we need to verify that $x_n \gg L_{Dn}$ and $x_p \gg L_{Dp}$. As before, the lowly doped side is the most stringent. For an asymmetric PN junction with $N_A > N_D$, going back to Eq. (6.17) (but using $\phi_{bi} - V$ instead of ϕ_{bi} to make the result more general) and Eq. (4.68), $x_n \gg L_{Dn}$ implies that

$$\phi_{bi} - V \gg \frac{kT}{q}\left(1 + \frac{N_A}{N_D}\right) \tag{6.141}$$

In most practical circumstances, this condition is easier to fulfill than Eq. (6.140).

This discussion, however, suggests that the “boxy” description of the charge distribution of the PN junction is not appropriate under strong forward bias when the depletion region thickness is very small. This impacts the evolution of the junction capacitance. We saw in Section 6.5 that for strong forward bias, as the depletion region shrinks, the junction capacitance diverges to infinity [Eq. (6.66)]. This equation derived under the depletion approximation that we have just seen is not valid under strong forward bias.

A more appropriate description of the electrostatics of the PN junction can be obtained following an approach similar to the Poisson–Boltzmann formulation of the MOS electrostatics (Advanced Topic AT8.3). In analogy with the depletion regime of the MOS structure, the PN junction capacitance for strong forward bias should approach an identical value to that of the semiconductor capacitance associated with the MOS structure at flat band [Eq. (8.43) and Fig. 8.22],

$$C_j(V \simeq \phi_{bi}) \simeq A\frac{\epsilon}{L_D} \tag{6.142}$$

This resolves the unphysical diverging result that was obtained in Eq. (6.66). In any case, there is not much practical significance to this since, as we saw in Section 6.5, under strong forward bias, the capacitance of a PN junction is dominated by the diffusion capacitance.

AT6.2 QUASI-NEUTRAL REGION RESISTANCE IN IDEAL DIODE

In Section 6.6.3, we wrote expressions for the resistance associated with the QNRs of a long diode. Deriving them is the goal of this section. The derivation represents a great review of key materials from Chapter 5.

We do the analysis for the n-type QNR that extends from $x = x_n$, the edge of the SCR, to $x = w_n$, the ohmic contact. We follow an approach similar to that of Section 5.6.1. As we do this, we will think of carrier injection with $V > 0$, but our development also includes carrier extraction for $V < 0$.

In essence, what we need to accomplish is to derive an expression for the excess electric field, $\mathcal{E}'$, that appears across the n-QNR and then integrate this to obtain the voltage drop. The origin of $\mathcal{E}'$ in Section 5.6.1 was to balance out currents in a situation in which $D_e \neq D_h$, as is often the case. Here, we will also find that $\mathcal{E}'$ is also required to support the diode current through majority carrier drift.

The starting point is the excess minority carrier profile (holes) across the n-QNR. This was derived in Eq. (6.33). The hole current density associated with them is

$$J_{\mathrm{h}} \simeq J_{\mathrm{h,diff}} = q\frac{D_{\mathrm{h}}}{L_{\mathrm{h}}}p'(x_{\mathrm{n}})\exp\frac{-x+x_{\mathrm{n}}}{L_{\mathrm{h}}} \tag{6.143}$$

The excess majority carrier profile (electrons) is to the first order identical to the excess minority carrier profile. This leads to an electron diffusion current given by

$$J_{\mathrm{e,diff}} = -q\frac{D_{\mathrm{e}}}{L_{\mathrm{h}}}p'(x_{\mathrm{n}})\exp\frac{-x+x_{\mathrm{n}}}{L_{\mathrm{h}}} \tag{6.144}$$

The total current density at any point is I/A. Then, the electron drift current is given by

$$J_{\mathrm{e,drift}} \simeq q\mu_{\mathrm{e}}N_{\mathrm{D}}\mathcal{E}' = \frac{I}{A} - J_{\mathrm{e,diff}} - J_{\mathrm{h}} = \frac{I}{A} + q\frac{D_{\mathrm{e}}-D_{\mathrm{h}}}{L_{\mathrm{h}}}p'(x_{\mathrm{n}})\exp\frac{-x+x_{\mathrm{n}}}{L_{\mathrm{h}}} \tag{6.145}$$

where we have assumed that $n \simeq N_{\mathrm{D}}$ consistent with our low-level injection assumption.

We can now solve for $\mathcal{E}'$ and integrate across the n-QNR to obtain the ohmic drop there,

$$V_{\mathrm{n}} = -\int_{x_{\mathrm{n}}}^{w_{\mathrm{n}}}\mathcal{E}'\mathrm{d}x = -\frac{I(w_{\mathrm{n}}-x_{\mathrm{n}})}{Aq\mu_{\mathrm{e}}N_{\mathrm{D}}} - \frac{D_{\mathrm{e}}-D_{\mathrm{h}}}{\mu_{\mathrm{e}}N_{\mathrm{D}}}p'(x_{\mathrm{n}}) \tag{6.146}$$

From Eqs. (6.37) and (6.45), we can write

$$p'(x_{\mathrm{n}}) = \frac{I_{\mathrm{n}}L_{\mathrm{h}}}{qAD_{\mathrm{h}}} \tag{6.147}$$

Substituting this into Eq. (6.146), we obtain

$$V_{\mathrm{n}} = -\frac{I(w_{\mathrm{n}}-x_{\mathrm{n}})}{Aq\mu_{\mathrm{e}}N_{\mathrm{D}}} - \frac{I_{\mathrm{n}}L_{\mathrm{h}}}{Aq\mu_{\mathrm{e}}N_{\mathrm{D}}}\frac{D_{\mathrm{e}}-D_{\mathrm{h}}}{D_{\mathrm{h}}} \tag{6.148}$$

To the extent that I_{n} is a fraction of I and L_{h} is much smaller than $w_{\mathrm{n}} - x_{\mathrm{n}}$, the second term in Eq. (6.148) is much smaller than the first term and the ohmic drop that is obtained for the n-QNR is consistent with the expression for its resistance that we wrote in Eq. (6.92). Once again, if D_{e} was precisely equal to D_{h}, then this expression would be rigorously correct. To the extent that $D_{\mathrm{e}} \neq D_{\mathrm{h}}$, a small electric field needs to be set up to keep the semiconductor quasi-neutral. By the way, the negative sign in Eq. (6.148) simply is a reflection of the fact that in our notation, the p region, located on the left, is positive with respect to the n region located on the right. The axis is pointing in the direction of decreasing voltages.

AT6.3 EQUIVALENT CIRCUIT MODEL FOR CIRCUIT DESIGN

Circuit computer-aided design (CAD) tools, such as SPICE, capture the behavior of a PN diode in an equivalent circuit model that is described by a system of equations. These models tend to be proprietary and differ somehow from company to company. In a given CAD system, a device equivalent circuit is "coded in," that is, its topology cannot be changed. Typically, a CAD system offers different topologies suitable for different technological implementations. The value of the elements of the equivalent circuit model can be adjusted by means of *model parameters* (in **bold** below) selected to best reproduce the I–V and C–V characteristics and any other measurements that are taken on the diodes. A successful CAD model for a device reproduces the experimental characteristics accurately, is computationally efficient, and the parameters that describe the model can be easily extracted from experiments. Effective CAD models tend to be strongly based on physical models.

CAD models can get quite complex as engineers attempt to describe subtle but important features of the device, strive for their models to predict the device characteristics over a broad range of temperatures, and allow them to appropriately scale for different device layouts. In this section, we describe a typical, though relatively simple, CAD model for a PN diode.

Typically, the I–V characteristics are captured by

$$I = I_S \left(\exp \frac{qV_j}{\mathbf{N}kT} - 1 \right) \tag{6.149}$$

where V_j is the internal voltage applied across the ideal diode. V_j is computed by the CAD program in the course of the simulations after accounting for the series resistance modeled using model parameter **RS**. V_j is computed from the external voltage V by means of

$$V_j = V - I\,\mathbf{RS} \tag{6.150}$$

Equation (6.149) contains several parameters. **N** is a model parameter called the *ideality factor* or *emission coefficient* and it should be unity for an ideal PN diode. Its value can be adjusted, however, to accommodate nonidealities. The values of the physical constants q and k are programmed in the simulator. The temperature T is specified by the user.

I_S in Eq. (6.149) plays the role of the diode saturation current. It is specified at any temperature by three model parameters through the following expression:

$$I_S = \mathbf{IS} \left(\frac{T}{T_M} \right)^{\mathbf{XTI}} \exp \left[\frac{-q\,\mathbf{EG}}{k} \left(\frac{1}{T} - \frac{1}{T_M} \right) \right] \tag{6.151}$$

IS is a model parameter that represents the saturation current measured at a specified temperature T_M. The parameters **EG** and **XTI** account for the temperature dependence of the prefactor of I_S in the PN diode current. This arises mainly from the temperature dependence of n_i^2 which we know goes as

$$I_S \propto n_i^2 \propto T^3 \exp - \frac{E_g}{kT} \tag{6.152}$$

Hence, the energy bandgap parameter **EG** should to first order have a value equal to the bandgap of the specific semiconductor (hence the name of this parameter) and the exponent **XTI** should be 3. To the second order, however, **EG** is a bit higher than the semiconductor bandgap due to the temperature dependence of E_g (see Section AT2.1) and **XTI** is not exactly 3 due to the temperature dependence of the rest of the parameters in I_S, such as mobility and carrier lifetime.

The junction capacitance in a CAD model of the PN diode is typically represented as

$$C_{\mathrm{j}} = \frac{\mathbf{CJO}}{\left(1 - \frac{V_{\mathrm{j}}}{\mathbf{VJ}}\right)^{\mathbf{M}}} \tag{6.153}$$

CJO is the capacitance of the junction at zero bias. **VJ** is the junction *built-in potential*.[4] **M** is the so-called grading coefficient. For the uniformly doped case studied in this book, $\mathbf{M} = \mathbf{0.5}$. This value can be adjusted if the doping profile is not flat. A separate perimeter capacitance term is frequently available with different voltage dependence. This comes handy to accurately model PN diodes with different area to perimeter ratios fabricated on the same process. Note also that Eq. (6.153) blows up if V_{j} exceeds **VJ**. This often results in convergence problems in the course of circuit simulation. Different simulators have different ways of dealing with this problem.

In addition to the junction capacitance, there is the diffusion capacitance that accounts for minority carrier storage in the PN diode. This is modeled in the following way:

$$C_{\mathrm{d}} = \frac{q}{kT}\,\mathbf{TT}\,I_{\mathrm{S}} \exp\frac{qV_{\mathrm{j}}}{kT} \qquad \text{for } V_{\mathrm{j}} \geq 0 \tag{6.154}$$

where V_{j} is the internal voltage across the PN junction. The parameter **TT** is called the *transit time*. By comparing Eq. (6.154) with Eqs. (6.58) and (6.70), we can determine the relationship between the CAD parameter **TT** and the physics of the device,

$$\mathbf{TT} = \frac{\tau_{\mathrm{e}} I_{\mathrm{ps}} + \tau_{\mathrm{h}} I_{\mathrm{ns}}}{I_{\mathrm{s}}} \tag{6.155}$$

where I_{ps} and I_{ns} represent the contributions to the saturation current originating from the p and the n regions, respectively, and τ_{e} and τ_{h} represent the carrier lifetimes in these regions.

If these regions are short in comparison with the diffusion length, then the dominant time constant is the transit time. The form of Eq. (6.154) does not need to change to accommodate this or other more complex situations.

The total capacitance of the PN diode is given by the sum of the diffusion and the depletion capacitances.

Table 6.1 summarizes the most important PN diode parameters for circuit CAD. The table also gives the ideal value of these parameters according to the models presented in this chapter.

AT6.4 SWITCHING CHARACTERISTICS OF PN DIODE

Diodes are attractive devices for switching applications. These are situations, such as in electrical power management, in which the diode is supposed to change from a highly conducting ON state to a highly blocking OFF state. The transition often needs to be very fast. In these kinds of applications, a fundamental weakness of the PN diode emerges. Typical PN diodes are quite slow, that is, it takes a significant length of time for the transitories associated with the switching

[4] **VJ**, the circuit model parameter for the junction built-in potential, should not be confused with V_{j}, the internal voltage across the edges of the PN diode SCR.

Table 6.1
Parameters for PN-diode model for circuit CAD.

Name	Parameter Description	Units	Ideal Value
IS	Saturation current	A	–
N	Ideality factor	–	1
EG	Energy bandgap	V	$E_g(300\text{ K})$
RS	Series resistance	Ω	–
CJO	Zero bias depletion capacitance	F	–
VJ	Built-in potential	V	–
M	Grading coefficient	–	0.5
XTI	Saturation current temperature exponent	–	–
TT	Transit time	s	–
BV	Breakdown voltage	V	–

events to be extinguished. In contrast, as we will see in the following chapter, Schottky diodes offer a very significant advantage and are the device of choice for many of these applications.

The main reason for the slow switching of the PN diode is that in forward bias, there is a substantial amount of minority carrier charge stored in the device. This is negligible in reverse bias. In order for the PN diode to switch from a forward state to a reverse state, the minority carrier charge must be disposed off. Similarly, in switching from a reverse state to a forward state, minority carrier charge needs to be provided. Depending on the details of the diode and the outside circuit, it can take significant time to achieve steady state after a switching event.

In order to appreciate the significance of this, consider the case illustrated in Fig. 6.48 in which a PN diode is switched back and forth between a forward voltage V_f and a reverse voltage V_r (note that, defined this way, V_r is negative). Under DC conditions, the forward state is characterized by a large forward current flow, $I_f(V_f)$, while the reverse state is characterized by small reverse current, $I_r(V_r)$.

Let us further assume that this is an abrupt diode where one side (the lowly doped one) dominates its minority carrier behavior and that this side is "long" in the scale of the relevant minority carrier diffusion length. The evolution of the excess minority carrier concentration in time during the switch-on and switch-off transients is shown in Fig. 6.49.

Let us consider first the switch-off transient depicted on the left side of Fig. 6.49 and, in a circuit form, in Fig. 6.50. At $t = 0^-$, a steady-state forward current I_f flows. In consequence, if we denote as τ the minority carrier lifetime in this region, the total stored minority carrier charge is $Q = \tau I_f$. At $t = 0^+$, the external voltage across the diode abruptly switches from V_f to $V_r < 0$. The internal voltage, however, cannot change immediately since for that to happen, the excess minority carrier concentration at the edge of the QNR must change too. This can only be accomplished by one of the two ways. One is to wait until they recombine. A second one is to make them flow out the diode. The fastest of these mechanisms controls the minority carrier discharge of the diode.

Minority carrier recombination is characterized by the carrier lifetime τ. Minority carrier discharge through the outside circuit is limited by the series resistance R_s of the diode. As the circuit schematic in Fig. 6.50 suggests, right after switching, at $t = 0^+$, the current through the diode suddenly reverses sign and becomes

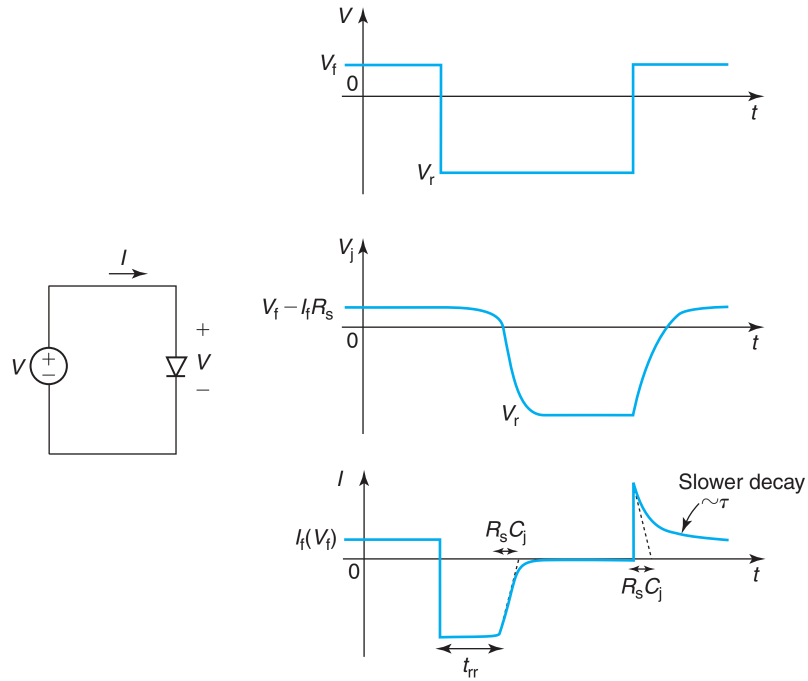


FIGURE 6.48
Switching transients in a PN diode. In addition to the $R_s C_j$ time constants, the recovery of the PN diode is delayed by the need to extract (in the switch-off transient) or supply (in the switch-on transient) minority carrier charge to the quasi-neutral regions.

$$I(t = 0^+) = \frac{V_r - (V_f - I_f R_s)}{R_s} \simeq \frac{V_r}{R_s} \tag{6.156}$$

Here, we have assumed that $V_f \ll |V_r|$. Note that $I(t = 0^+)$ is negative.

For $t > 0^+$, the minority carrier charge discharges through the resistor. Because of the exponential dependence of minority carrier charge and junction voltage, as the discharge proceeds, V_j initially changes very slowly. As a result, as Fig. 6.48 indicates, $I(t > 0^+)$ essentially remains unchanged from the value given in Eq. (6.156). Only when the minority carriers have bled off, the junction voltage will reach zero voltage and eventually start evolving toward V_r.

This understanding allows us to estimate, to the first order, the time that it takes for the minority carrier charge to bleed off. This is given by the ratio of the stored charge to the discharge current,

$$t_{rr} \simeq \frac{Q}{|I(t = 0^+)|} \simeq \tau \frac{I_f(V_f) R_s}{|V_r|} \tag{6.157}$$

This is called the *reverse recovery time*. Note that in a well-designed diode, the ohmic drop across the series resistance in forward bias, $I_f R_s$, is much smaller than the forward voltage V_f. Since we are assuming that $V_f \ll |V_r|$, it then follows that $I_f R_s \ll |V_r|$. As a consequence, $t_{rr} \ll \tau$.

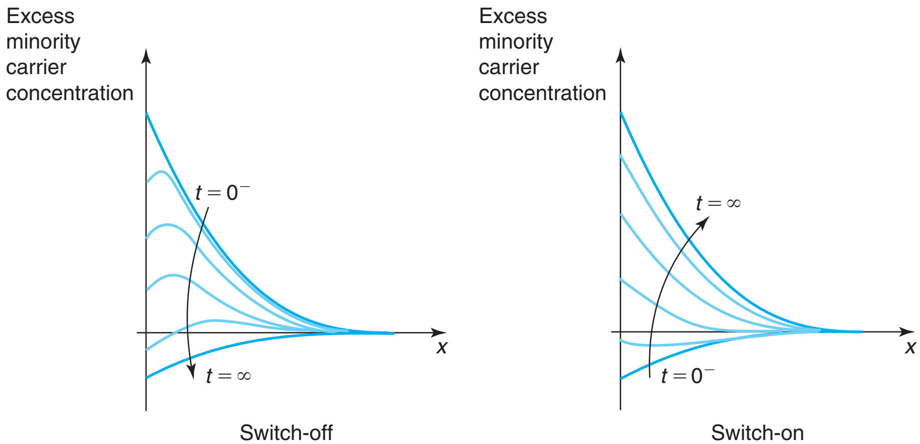


FIGURE 6.49
Evolution of the excess minority carrier concentrations in the quasi-neutral region of a PN diode during a switch-off transient (left) and a switch-on transient (right).

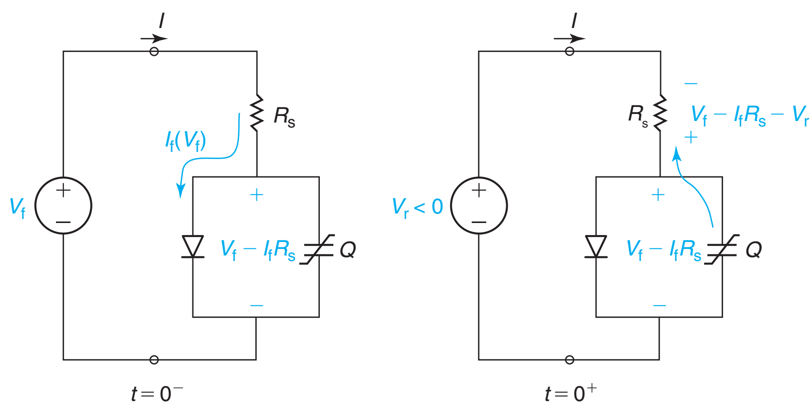


FIGURE 6.50
Internal and external currents in a PN diode right before and after a switch-off event.

Getting the minority carriers out through the external circuit is faster than waiting for them to recombine.

When the internal voltage of the diode goes through zero, the excess minority carrier discharge is extinguished. From there on, the discharge of the junction capacitance is what limits the transient characteristics of the diode. This is a simple exponential process characterized by a time constant $R_s C_j$.

All together, the switch-off transient evolves as sketched in Fig. 6.48. For a period of time equal to t_{rr}, a fairly constant reverse current flows, after which an exponential decay to the final steady-state value follows.

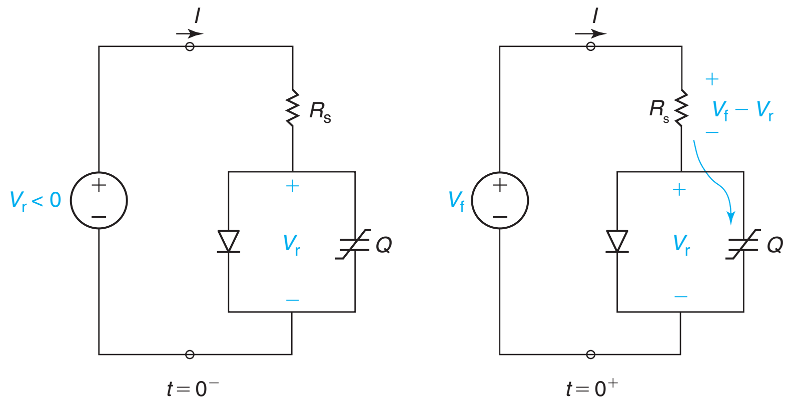


FIGURE 6.51
Internal and external currents in a PN diode right before and after a switch-on event.

The dependencies observed in Eq. (6.157) make sense. The higher τ or $I_f(V_f)$, the higher the amount of minority carrier charge that is stored and the longer it takes to bleed it off. The higher R_s or the lower $|V_r|$, the lower the magnitude of the discharge current during the recovery period $|I(t = 0^+)|$ and the longer it takes for the discharge to be completed.

Let us now shift our attention to the turn-on transient (Fig. 6.51). When the external voltage abruptly switches from V_r to V_f, the junction capacitance starts charging through R_s. This produces a peak forward current of magnitude,

$$I(t = 0^+) = \frac{V_f - V_r}{R_s} \simeq \frac{|V_r|}{R_s} \tag{6.158}$$

As the junction capacitor charges up and the internal voltage rises toward V_f, the forward current drops in an exponential way. The characteristic time constant of this process is of the order of $R_s C_j$, where C_j is again the junction capacitance. When the diode becomes forward biased, two things happen that delay the attainment of steady state. First, the diode starts to conduct. This robs current that could be used to charge up the capacitor. Second, the battery must supply extra current to store the required minority carrier charge to the QNRs. This process has a characteristic time that is again on the order of the dominant time constant for minority-carrier-type behavior, τ. It then takes a few τs to attain the final steady state. The complete transient is sketched in Fig. 6.48. The temporal evolution of the excess minority carrier concentration is sketched in Fig. 6.49.

EXERCISE 6.6

Simulate a switching transient of a PN diode in SPICE, as sketched in Fig. 6.48. The forward voltage is +0.89 V. The reverse voltage is −5 V. The SPICE parameters of the diode are: **IS** $= 6e - 17$, **N** $= 1$, **EG** $= 1.12$, **RS** $= 8$, **CJO** $= 1.2e - 13$, **VJ** $= 0.6$, **M** $= 0.41$, **XTI** $= 3.814$, *and* **TT** $= 4e - 9$. *Extract the characteristic time constants of the transients and compare them with simple estimates.*

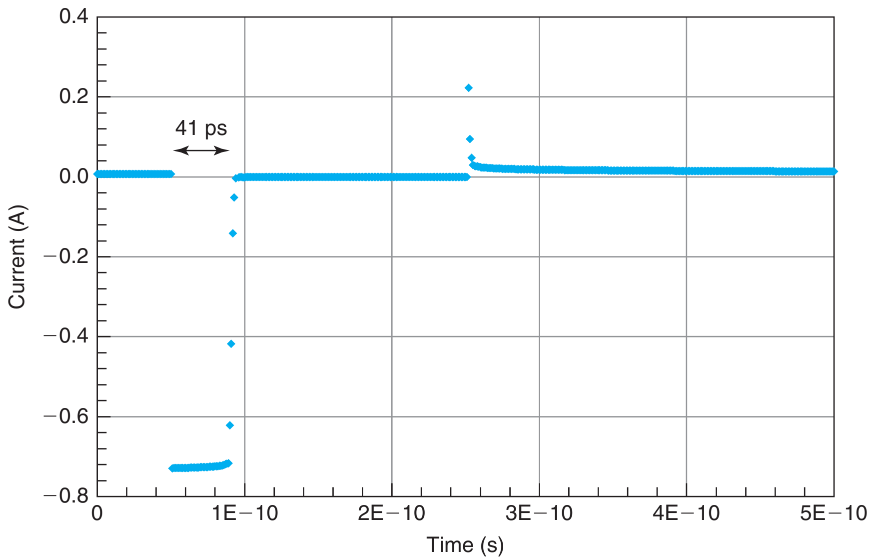


FIGURE 6.52
Switching transients in a PN diode as simulated by HSPICE (see exercise 6.6). For $0 \le t \le 50$ ps, $I = I_f = 7.2$ mA.

Figure 6.52 shows the output obtained from HSPICE for a case in which there is a delay time of 50 ps, the reverse voltage is applied during 200 ps, and the forward voltage is applied during 250 ps.

The turn-off characteristics show pronounced minority carrier storage. The reverse recovery time that is extracted from the simulations is about 41 ps. This compares very well with the value that can be calculated using Eq. (6.157), which is 46 ps (the DC forward current that corresponds to $V_f = 0.89$ V is 7.2 mA; in order to compute this number, you need to account for the series resistance of the diode). Following the reverse recovery time, the current through the diode is seen to drop very fast to the value that corresponds to the reverse bias of −5 V. The expected value of the $R_s C$ time constant is about 1 ps.

The turn-on transient displays an initial fast decay associated with the charging of the junction capacitance followed by a much slower tail toward the final $I_f(V_f)$ value. If you blow up this region and examine the time constant of this tail, you will find that it is about 2.6 ns. This compares rather well with the dominant time constant of the diode which is 4 ns.

The reverse recovery current obtained in SPICE is about 730 mA. This compares well with the expected value of 736 mA. The peak current following the turn-on transient should also have this magnitude. However, the initial decay is so fast that with the selected time discretization in SPICE this peak is not completely resolved.

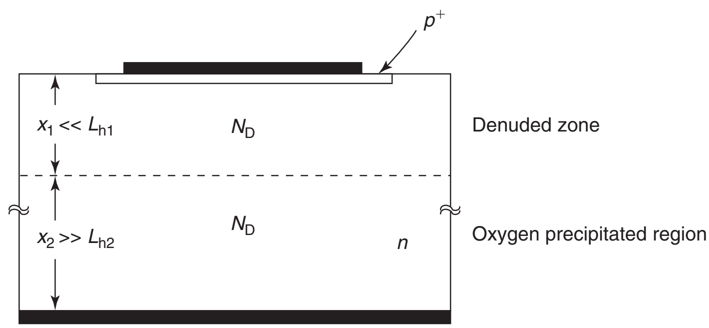
This example dramatically illustrates how drastically the switching of a PN diode is slowed down by minority carrier storage.

PROBLEMS

* Denotes a problem that requires studying an Advanced Topic.

6.1 Modern IC processing utilizes a technique called *oxygen precipitation* to getter metallic contaminants out of a thin layer at the top surface of the wafer. It is in this so-called denuded zone, where the devices are fabricated. Typically, the denuded zone is thin and is characterized by a fairly long lifetime, while the rest of the wafer where oxygen precipitation has taken place exhibits much lower lifetimes. The denuded zone results in device reverse-bias leakage currents that are much reduced.

The change in lifetime in the wafer presents an interesting problem for the computation of the current injected downward in a bulk p^+–n diode, as sketched in the figure below.

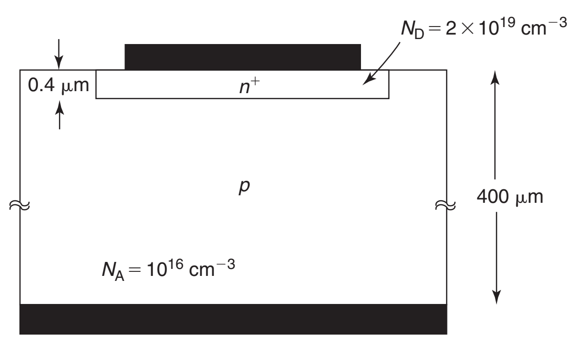


Make the following assumptions. The doping level is the same in both regions of the n substrate. The junction depth of the p^+ region and its associated depletion region are much thinner than x_1, the denuded zone thickness. x_1 is much shorter than the local diffusion length L_{h1}. The thickness of the oxygen precipitated region x_2 is much longer than the local diffusion length L_{h2}.

(a) Sketch the shape of the excess hole concentration in the n region in forward bias.

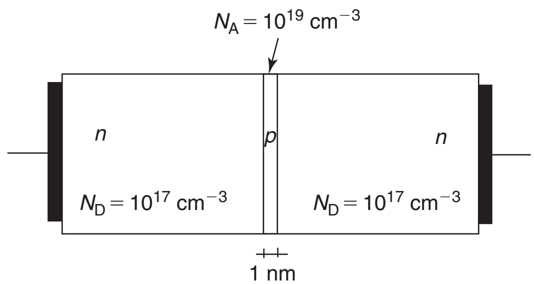
(b) Derive an expression for the hole current density injected into the n-type substrate in forward bias in terms of the hole diffusion length in the oxygen precipitated region, L_{h2}, the denuded zone thickness, x_1, and the usual parameters.

6.2 *In the list of SPICE parameters that has been given to you for the abrupt PN diode sketched below, you read **TT** = 24 ns. This seems to you so way off that you think it is a typo.



Estimate **TT** for this diode.

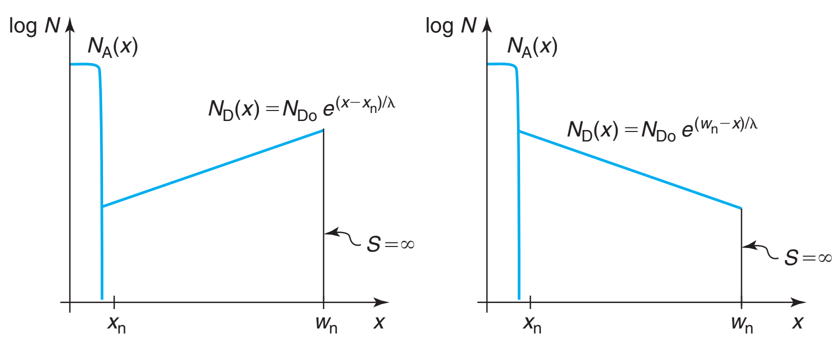
6.3 Consider a Si device with a structure as sketched in the following diagram:



This device basically consists of a thin p-type region inserted in the middle of an n-type body. Two ohmic contacts are provided to the ends of the n regions. The device sits at room temperature.

(a) Quantitatively sketch the volume charge density throughout the structure in thermal equilibrium. Justify any assumptions you make.
(b) Quantitatively sketch the electric field throughout the structure in thermal equilibrium.
(c) Quantitatively sketch the electrostatic potential throughout the structure in thermal equilibrium.
(d) Quantitatively sketch the energy band diagram throughout the structure in thermal equilibrium.
(e) Now apply a voltage of 0.1 V to the right contact with respect to the left contact. Explain what happens. Qualitatively sketch the electrostatics of the device (volume charge density, electric field, and electrostatic potential).
(f) Quantitatively sketch the energy band diagram for the situation described in (e).

6.4 Consider two P^+N junctions with nonuniform donor distributions as sketched below. Notice that the doping profiles are basically identical but flipped over in x. $x = x_n$ is the edge of the depletion region on the n side and $x = w_n$ is the ohmic contact that is characterized by $S = \infty$. The extent of the QNR on the n side, w_n, is much shorter than the diffusion length for holes, that is, this is a "short" region for the minority carriers. The doping levels are such that $x_n \ll w_n - x_n$.

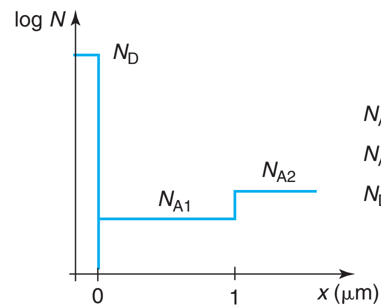


Assuming that D_h is independent of position,

(a) Calculate expressions for the Gummel number and the hole saturation current density, $J_{hs}(x_n)$, injected into both n regions;
(b) For both profiles, sketch $J_{hs}(x_n)$ versus $\lambda/(w_n - x_n)$ and discuss limits when $\lambda \to \infty$;
(c) Calculate expressions for the hole transit time, τ_{th}, across both n regions;
(d) For both profiles, sketch τ_{th} versus $\lambda/(w_n - x_n)$ and discuss limits when $\lambda \to \infty$;
(e) For both profiles, sketch the excess minority carrier profiles and comment on the differences that you found between them.

It will be helpful to use the Taylor series expansions of Appendix E.

6.5 A PN junction diode has a heavily doped emitter and a lightly doped base. The base doping profile changes in space as sketched below. Quantitatively graph the $1/C^2$ versus V characteristics of this PN diode in reverse bias between $V = 0$ and $V = -10$ V.

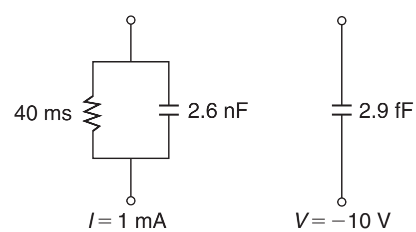


$N_{A1} = 10^{16}\ \text{cm}^{-3}$

$N_{A2} = 4 \times 10^{16}\ \text{cm}^{-3}$

$N_D = 10^{20}\ \text{cm}^{-3}$

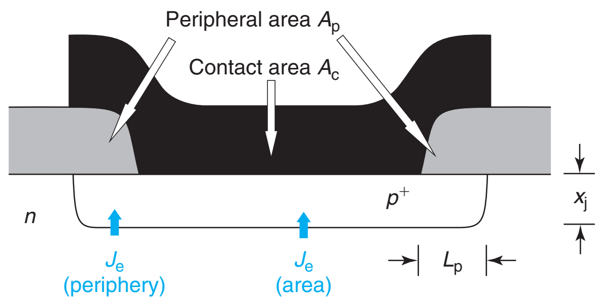
6.6 The small-signal equivalent circuit of a P$^+$N diode that you are trying to reverse engineer is measured at room temperature for two different bias points, one forward ($I = 1$ mA) and another one reverse ($V = -10$ V). The results are sketched below:



Based on your understanding of the technology that has been used, you are pretty sure that you can assume that the diode is very abrupt (that is $N_A \gg N_D$), that the doping level is uniform on the n side, and that, from the point of view of the minority carriers, the n side can be considered short with an ohmic contact at the end. Furthermore, assume that all minority carrier behavior is dominated by the lowly doped side. On the microscope, you measure the area of the diode and you find it to be 10 μm^2.

(a) Estimate the doping level on the n side, N_D.
(b) Estimate the thickness of the n region, w_n.
(c) Verify whatever assumptions you need to make.

6.7 Consider the emitter of a PN junction diode as sketched below. The central portion of the emitter is covered by a metal contact where $S = \infty$. This region has an area A_c. The peripheral portion of the diode is covered by oxide and is characterized by $S = 0$. The area of this region is A_p. The length of the oxide covered area in the periphery, L_p, is much longer than the junction depth, x_j.



Let us do some relevant calculations for a case in which $N_A = 10^{19}$ cm^{-3}, $x_j = 0.1$ μm, $A_c = 50$ μm^2, and $A_p = 5$ μm^2.

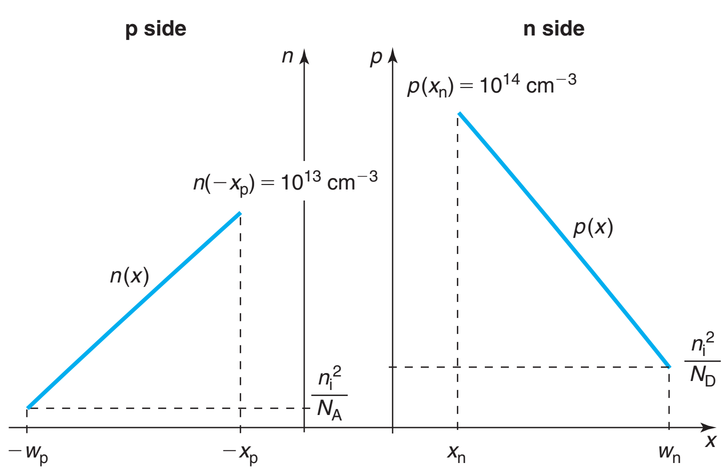
(a) Calculate the ratio of the forward bias current injected to the peripheral portion of the emitter over that injected to the contacted portion.

(b) Calculate the ratio of the minority carrier charge stored in the peripheral portion of the emitter over that stored under the contact area.

(c) Identify and evaluate the dominant minority carrier time constant for the region underneath the contact.

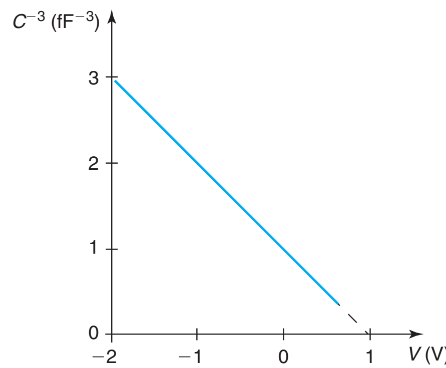
(d) Identify and evaluate the dominant minority carrier time constant for the peripheral region of the emitter.

6.8 Below is a sketch *not to scale* of the minority carrier distribution across the QNRs of a forward-biased PN diode. For this diode, $W_p - x_p = 4$ μm, $W_n - x_n = 3$ μm, $D_e = 25$ cm^2/s, and $D_h = 10$ cm^2/s. The area of the junction is 10 μm^2.



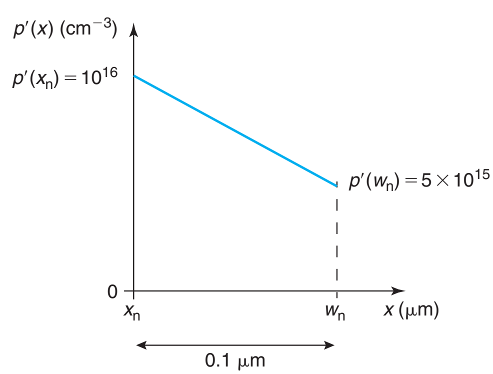
(a) Calculate the hole current injected into the n side of the diode.
(b) Calculate the electron current injected into the p side of the diode.
(c) Calculate the diffusion capacitance associated with carrier storage on the n side of the diode.
(d) Calculate the diffusion capacitance associated with carrier storage on the p side of the diode.
(e) How much should the voltage across the junction increase if we wish to double the total current through the diode?
(f) If we increase the voltage in the manner suggested in the previous question, what happens to the total diffusion capacitance of the diode?
(g) What is the ratio of the doping levels across the junction: N_A/N_D?
(h) In what direction should N_A/N_D change if we wish to redesign the diode so as to get less diffusion capacitance *at the same current level*? (Assume that in redesigning the diode D_e, D_h, $W_n - x_n$, and $W_p - x_p$ do not change.) Choose one: N_A/N_D *must increase*. N_A/N_D *must decrease*. Explain.

6.9 A PN diode with a junction area $A = 1\ \mu\text{m}^2$ has the following C–V characteristics:



Estimate the extent of the depletion region at $V = 0$ V.

6.10 Consider an abrupt asymmetric P^+N junction diode. All the action in this device is dominated by the lowly doped n-type region. At a certain bias point, the excess minority concentration in the quasi-neutral n-type region has a distribution as sketched below:

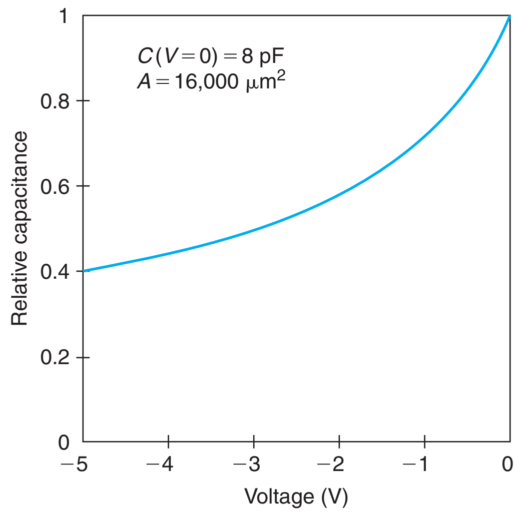


x_n is the edge of the SCR. w_n is the surface. In this region, $D_h = 5\ \text{cm}^2/\text{s}$. Assume low-level injection.

Answer the following questions for the bias point that corresponds to the diagram:

(a) Calculate the diode current density.
(b) Calculate the total amount of excess minority carrier charge.
(c) Calculate the diffusion capacitance.
(d) Calculate the dominant time constant.

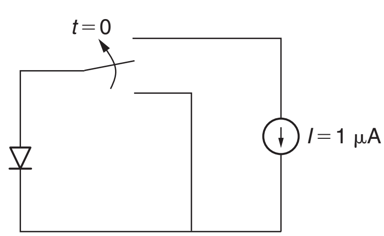
6.11 In a paper on Si PN junction varactors, you see the following graph with the capacitance–voltage characteristics of the diode at room temperature:



Assuming that the diode is highly asymmetrically doped, reverse engineer the diode.

(a) Estimate the built-in potential of the junction.
(b) Estimate the doping level of the lowly doped side, N_L.
(c) Estimate the doping level of the highly doped side, N_H.

6.12 Consider an ideal P^+N Si diode with a doping level in the lowly doped n-type region of $10^{15}\ \text{cm}^{-3}$. The saturation current is 1 pA and the diode area is $10^{-4}\ \text{cm}^2$. Initially, the diode is in thermal equilibrium with its terminal shorted. At $t = 0$, a current source bias of 1 μA is applied in the reverse conduction mode as indicated in the figure below.



(a) After a while, what happens? Explain.
(b) How much time does it take for *it* to happen? State any assumptions you might need to make.

6.13 This problem is about testing the quality of the quasi-equilibrium approximation across the SCR of an ideal PN junction under bias. This is to be done by comparing the magnitude of the minority carrier current, which is equal to the imbalance between drift and diffusion currents inside the SCR, with either one of these two current components.

Consider an N^+P junction with "long" QNRs. Focus on electron current. Assume negligible SCR generation and recombination.

(a) Derive a first-order expression for the electron drift current at the metallurgical junction, $J_e^{drift}(x=0)$. Assume linearity between electron drift velocity and electric field (small electric field). Make other suitable approximations.
(b) Derive an expression for the net electron current across the SCR, J_e.
(c) Express the ratio $J_e/J_e^{drift}(x=0)$ in terms of $\phi_{bi} - V$, the electron diffusion length, and the Debye length in the p-QNR.
(d) For typical doping levels, discuss the conditions that must be met for the quasi-equilibrium assumption to hold.

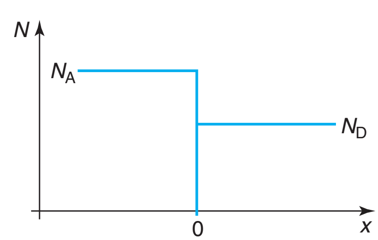
6.14 *Consider an asymmetric P^+N diode at room temperature. This diode is designed in a way that its lightly doped n-type side dominates all minority carrier behavior. From a minority carrier point of view, this n-type side can be considered *long*.

This diode has been designed for a switching application from a certain forward voltage V_f to a certain reverse voltage V_r. Upon testing, it has been found that the *reverse recovery time*, t_{rr}, is too long. This problem is about considering options to reduce t_{rr}.

Indicate below the impact in t_{rr} of the specified change. Circle one: t_{rr} increases $= t_{rr}\uparrow$, t_{rr} decreases $= t_{rr}\downarrow$, or no effect in t_{rr}. For each item, give the reason for your choice.

Increase doping level, $N_D \uparrow$:	$t_{rr}\uparrow$	$t_{rr}\downarrow$	no effect in t_{rr}
Increase operational temperature, $T\uparrow$:	$t_{rr}\uparrow$	$t_{rr}\downarrow$	no effect in t_{rr}
Increase n-type QNR width, $W_n\uparrow$:	$t_{rr}\uparrow$	$t_{rr}\downarrow$	no effect in t_{rr}
Irradiate diode so that carrier lifetime is reduced, $\tau\downarrow$:	$t_{rr}\uparrow$	$t_{rr}\downarrow$	no effect in t_{rr}
Increase area of diode, $A\uparrow$:	$t_{rr}\uparrow$	$t_{rr}\downarrow$	no effect in t_{rr}

6.15 Consider an abrupt PN junction with $N_A = 10^{17}\ \text{cm}^{-3}$ and $N_D = 10^{16}\ \text{cm}^{-3}$, as sketched below.



(a) In thermal equilibrium, compute the value of the electrostatic potential difference between $x = 0$ and deep inside the p-type QNR.
(b) Compute n_o and p_o at $x = 0$ in thermal equilibrium.
(c) Compute the value of x for which $n_o = p_o = n_i$ in thermal equilibrium.
(d) Compute the total amount of charge per unit area on the p side of the junction when a reverse bias voltage of 5 V is applied to the diode.

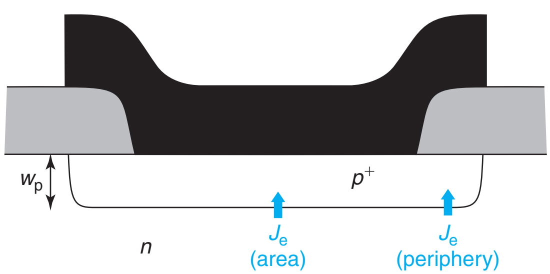
6.16 A PN junction diode in forward bias can be used as a thermometer. This is best done by biasing the diode at *constant forward current* and then measuring the voltage across as the temperature changes. This problem is about deriving a relationship for the forward bias voltage versus temperature for an ideal PN diode.

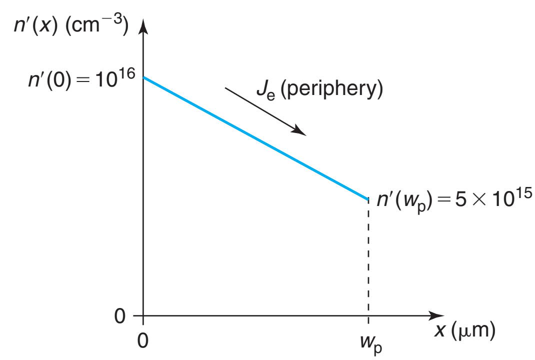
Make the following assumptions: no series resistance, sufficient forward voltage ($V \gg \frac{kT}{q}$), and neglect power law-type temperature dependences (i.e., $\sim T^{\alpha}$) next to Boltzmann-type temperature dependences (i.e., $\sim \exp \frac{-E}{nkT}$).

(a) Consider an ideal PN diode at a temperature T_o, biased at a forward current I_f with a voltage across $V(I_f, T_o)$. Derive an expression for V versus T in terms of $V(I_f, T_o)$, T, T_o, and fundamental constants. Sketch.

(b) Calculate the temperature sensitivity, in units of mV/K, of an ideal PN diode with $I_s = 10$ pA biased at a current of $I_f = 1$ mA at $T_o = 300$ K.

6.17 Consider a planar PN junction as sketched below on the left. This problem is about quantifying recombination at the peripheral surface of the p$^+$-type region, right next to the ohmic contact. In this region, at a certain forward bias point, an excess electron concentration profile is set as indicated in the figure below on the right. The depth of the p$^+$ region is $w_p = 0.1$ μm. You can assume that the extent of the depletion region into the p$^+$ region is a negligible portion of this. You can also assume that the region is short to the electrons ($L_e >> w_p$) and that the electron mobility in this region is 250 cm^2/V · s.



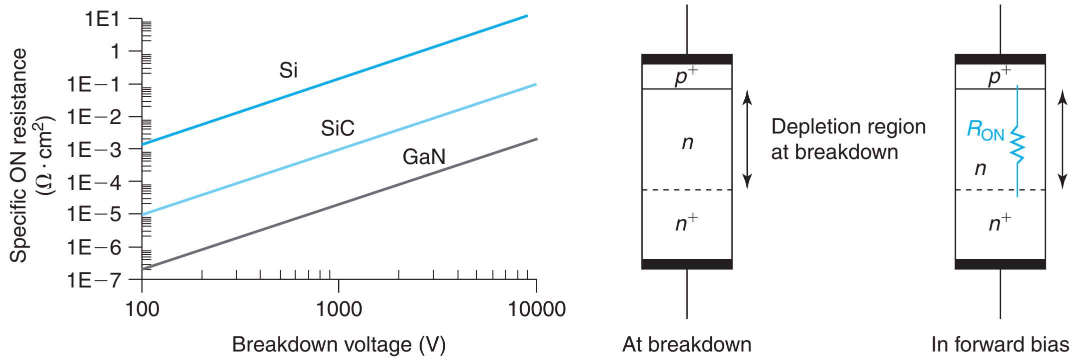


(a) Estimate the electron current density in the peripheral portion of the p$^+$ region.

(b) Estimate the net recombination rate at the surface of the peripheral portion of the p$^+$ region.

(c) Estimate the surface recombination velocity at the surface of the peripheral portion of the p$^+$ region.

6.18 When comparing the suitability of different semiconductors for power electronics applications, a common graph that is used plots the specific ON resistance (in Ω · cm^{-2}) against the breakdown voltage of an ideal PN diode. The specific ON resistance is the minimum resistance in the forward regime you could possibly get out of a PN diode with unit cross-sectional area that has been designed to deliver a certain reverse bias breakdown voltage. This graph is shown below. Lines for Si, SiC, and GaN are sketched. For the same breakdown voltage, GaN offers far less ON resistance than Si, which is why there is so much excitement these days about GaN power electronics. Your task in this problem is to understand the meaning of this graph by calculating these metrics for an ideal Si PN diode.

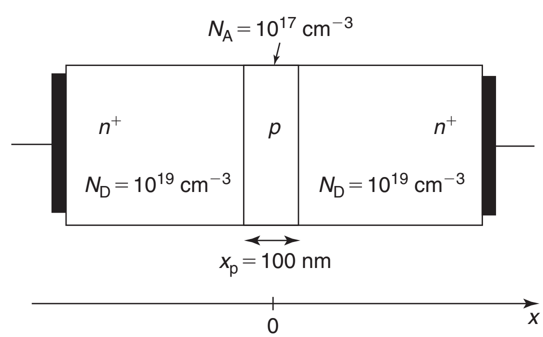


Consider an ideal strongly asymmetric p^+–n–n^+ diode with unit cross-sectional area at room temperature. The n^+ region is introduced to minimize the resistance of the quasi-neutral n region. The breakdown voltage and the resistance characteristics of this diode are dominated by the lowly doped n-type region. The two key design parameters of this diode from the point of view of power electronics are the doping level of the n region and its thickness. A higher reverse breakdown voltage requires a thicker layer with a lower doping level. Hence, it offers higher resistance in the forward regime. The optimum thickness of the n region is that required to accommodate the depletion region at breakdown.

Let us do some numbers. Consider an ideal p^+–n–n^+ diode with a doping level in the lightly doped region of $N_D = 10^{15}$ cm^{-3}.

(a) Estimate the breakdown voltage of this device.
(b) Estimate the extent of the depletion region at breakdown.
(c) Estimate the specific ON resistance contributed by the lowly doped n-type region of a thickness equal to the one required to support the breakdown voltage of part (a).
(d) Plot the point corresponding to the design in the graph on the left and comment.

6.19 Consider the Si structure sketched below at room temperature.



(a) Evaluate the built-in potential of this structure, from n^+ region on the left to n^+ region on the right.
(b) Estimate the hole concentration at the center of the p-type region.

(c) Carefully sketch the energy band diagram across the structure, from n^+ region to n^+ region.
(d) Calculate the capacitance of this structure at a bias voltage of 0 V.

6.20 In a certain PN junction diode at room temperature at a particular forward bias voltage, the current supported by hole injection in the n side of the diode is 100 μA. The width of the quasi-neutral n-type region is 1 μm. This is much shorter than the corresponding hole diffusion length. At the surface of the n-type region there is an ohmic contact. The hole diffusion coefficient is 10 cm^2/s. The PN junction area is 10 μm^2.

In answering the questions that follow, make and state suitable approximations.

(a) Estimate the excess hole concentration at the SCR edge of the n quasi-neutral region.
(b) Estimate the velocity at which holes are injected at the edge of the n-QNR.
(c) Estimate the hole flux arriving at the ohmic contact of the n-type QNR.
(d) Estimate the diffusion capacitance associated with hole storage in the n-QNR.

6.21 *A certain integrated PN diode is characterized at room temperature by the following set of SPICE parameters: **IS** $= 10^{-16}$, **N** $= 1$, **RS** $= 10$, **CJO** $= 10^{-13}$, **VJ** $= 0.9$, **M** $= 0.5$, **XTI** $= 3.9$, and **TT** $= 10^{-8}$.

(a) Estimate the current flowing through the diode at $V = 0.8$ V at room temperature.
(b) Estimate the reverse recovery time when this diode switches from $V = 0.8$ V to $V = -5$ V at room temperature.
(c) Estimate the total excess minority carrier stored charge at $V = 0.8$ V at room temperature.
(d) Estimate the capacitance at $V = 0.8$ V at room temperature.

Chapter 7

Schottky Diode and Ohmic Contact

This chapter treats the metal–semiconductor junction and two important applications: Schottky barrier diodes (or simply Schottky diodes) and ohmic contacts. Metal–semiconductor junctions are present in virtually all microelectronic devices. Their most pervasive form is ohmic contacts that are used to provide electrical access to devices from the outside world. However, under the right conditions, metal–semiconductor junctions can display rectifying behavior, just like PN junctions. This is exploited to make Schottky diodes. An example is shown in Fig. 7.1. The uniqueness of the Schottky diode, in comparison with the PN diode, is its fast dynamic response. This arises from the fact that the Schottky diode is largely a majority-carrier-type device. Because of this, Schottky diodes have found use in many analog, digital, power, and communications applications.

A top view and a cross-sectional view of an integrated Schottky diode is shown in Fig. 7.2. It consists of an n-type well on a p-type substrate. A metal layer is in direct contact with the n-well. If appropriately designed, this junction exhibits rectifying characteristics. The contact to the body is made through an n^+region with a metal contact applied to it. The design here is optimized to make an ohmic contact that exhibits minimum contact resistance. One of the goals of this chapter is to understand when a metal–semiconductor junction shows rectifying characteristics and when it exhibits ohmic behavior. For the diode shown in this figure, the metal is referred to as the *anode* and the n-type semiconductor is the *cathode*. It is also possible to make metal/p-type Schottky diodes. In this case, the metal is the cathode and the p-type semiconductor is the anode.

This chapter studies in detail the physics of the metal–semiconductor junction and its use in Schottky diodes and ohmic contacts. It is organized as follows. We start by defining the notion of an "ideal Schottky diode." This is a hypothetical device of simplified geometry and physics that helps us focus on the most important issues. The metal–semiconductor junction in thermal equilibrium is discussed next. The treatment relies heavily on the energy band diagram view of semiconductors and metals. The consequences of applying a voltage to a Schottky diode are discussed next, first qualitatively and then quantitatively. We study both the current–voltage and the charge–voltage characteristics of Schottky diodes. We then discuss equivalent circuit models for Schottky diodes and some of the most significant nonideal and second-order effects.

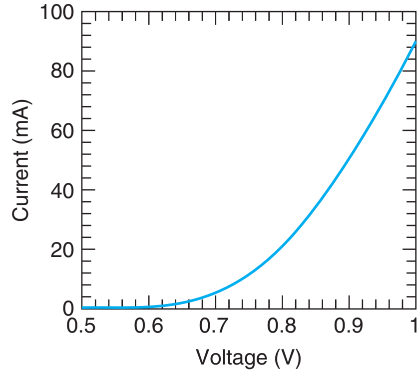


FIGURE 7.1
Rectifying current–voltage characteristics of a 1N5711 Schottky diode [Reprinted with permission from Matt Fox].

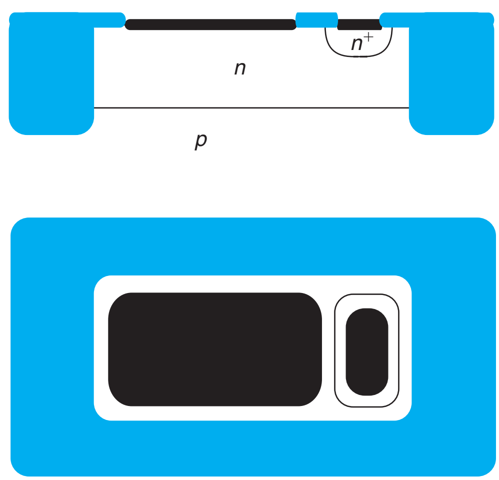


FIGURE 7.2
Cross section and top view of integrated Schottky diode.

We devote a few pages to a few important technology and design issues in practical Schottky diodes. The chapter finishes with a detailed discussion of ohmic contacts, their physics, and key design issues. A number of Advanced Topics at the end of the chapter allow the student to deepen their understanding in several areas.

7.1 THE IDEAL SCHOTTKY DIODE

We start this chapter by defining the concept of an ideal Schottky diode. This is a device with simplified geometry and physics and no parasitics. The ideal Schottky diode captures the essence

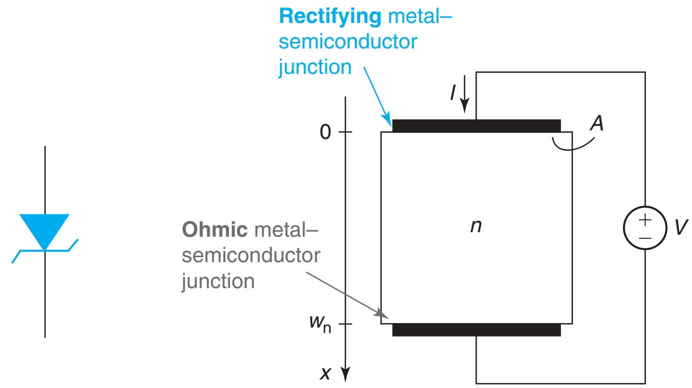


FIGURE 7.3
Left: circuit symbol of Schottky diode. Right: sketch of ideal Schottky diode.

of the rectifying metal–semiconductor junction and hides some of its complexities. Later on in this chapter, we will relax some of the assumptions that we make here and we will also study the most significant nonidealities.

The circuit symbol and a sketch of an ideal Schottky diode are shown in Fig. 7.3. In this book, the ideal Schottky diode consists of an n-type semiconductor region with a Schottky metal on top (we use this term to refer to the metal of a metal–semiconductor junction that exhibits rectifying characteristics). The semiconductor is contacted by means of an ideal ohmic contact at the bottom. As a short hand, when we simply talk about "the metal" in the context of a Schottky diode, we refer to the Schottky metal that yields the rectifying characteristics. The metal that makes the ohmic contact is usually referred to as *ohmic metal* or simply *contact*.

In the analysis of the ideal Schottky diode, we are going to make the following assumptions:

- All carrier flow is one dimensional. There are no 2D or 3D effects.
- The metal–semiconductor interface is smooth with ideal bonding for the semiconductor atoms preserved.
- The doping level in the semiconductor is uniform throughout.
- We assume that nondegenerate carrier statistics apply in all situations.
- We assume that the Schottky barrier height is given by the difference between the work function of the metal and the electron affinity of the semiconductor (for n-type semiconductor, this is explained below). This assumption is discussed in Section AT7.1.
- We assume that the alignment between metal and semiconductor work functions is such that in equilibrium, there is a depletion region on the semiconductor side.
- We treat the metal–semiconductor junction under the depletion approximation. We consider the rest of the semiconductor as quasi-neutral.
- We disregard the minority carriers and we therefore neglect generation and recombination.
- We ignore any resistance effects associated with the Schottky metal, the semiconductor, or the ohmic contact (we study the impact of parasitic resistance in Section 7.6.1).
- We assume ideal ohmic contact as defined in Section 5.2.2.
- We ignore any edge effects or other effects associated with the sidewalls of the device.

Figure 7.3 defines the axis that we will use in our analysis of the Schottky diode. We place its origin at the metallurgical interface between the metal and the semiconductor. The extent of the semiconductor is w_n. The area of the device is A.

For a metal/n-semiconductor junction, such as the one depicted in Fig. 7.3, the voltage polarity is usually defined as in the figure. With this polarity, when $V > 0$, the Schottky diode is in forward bias. When $V < 0$, the Schottky diode is in reverse bias. The metal is the anode and the semiconductor is the cathode. Everything is reversed for a metal/p-semiconductor junction.

7.2 IDEAL SCHOTTKY DIODE IN THERMAL EQUILIBRIUM

A metal–semiconductor junction is an artificial structure made out of two rather dissimilar materials. Whenever two materials with different properties are brought together in intimate contact, charge redistribution takes place. This has often fascinating consequences. The PN junction that was studied in Chapter 6 was an example of this. The metal–semiconductor junction is unique in a number of ways. Before studying in detail the metal–semiconductor junction, a number of important concepts can be better grasped in a simpler metal–metal junction.

7.2.1 A simpler system: a metal–metal junction

Figure 7.4 illustrates the evolution of the electronic structure of two different metals as they are brought in contact to form a metal–metal junction. Figure 7.4(a) shows the energy band diagram of each metal when placed far apart from each other. Each metal has its own peculiar energy distribution of bands and bandgaps. The work function (the minimum energy required to free up an electron) is also different in the two materials.

As the two metals are brought together, some electrons in the low work function metal, metal 2 on the right of Fig. 7.4, are presented with empty states at lower energy in the high work function metal 1 [Fig. 7.4(b)]. As soon as contact is established, electrons from metal 2 rush to metal 1 to lower their energy. This makes metal 1 negatively charged and metal 2 positively charged. As time goes on and electron flow continues, an electric field builds up across the interface that in due time stops further electron migration. Equilibrium is eventually reached.

This is a dynamic equilibrium situation. The initial tendency of electrons to flow from metal 2 to metal 1 in order to lower their energy is precisely counterbalanced by the electric field that gets set up at the interface. This electric field originates on the *dipole of charge* that is formed as a consequence of electron migration. This is graphed in Fig. 7.5. The electric field has a sign such that it opposes further electron transfer from metal 2 to metal 1. In Fig. 7.5, $\mathcal{E}$ is negative as a consequence of the choice of axis. The electric field in turn results in a potential difference between the two materials that is called the *built-in potential*, ϕ_{bi}. The potential energy associated with the electric field must be added to the energy band diagram. As a consequence, all the bands bend with a shape that is identical to the electrostatic potential with a minus sign [Fig. 7.4(c)].

When the two metals are sufficiently far apart, they can be considered as two different electronic systems with their own separate Fermi levels. The moment the two materials are brought in intimate contact and electrons are allowed to freely flow from one to the other, they constitute

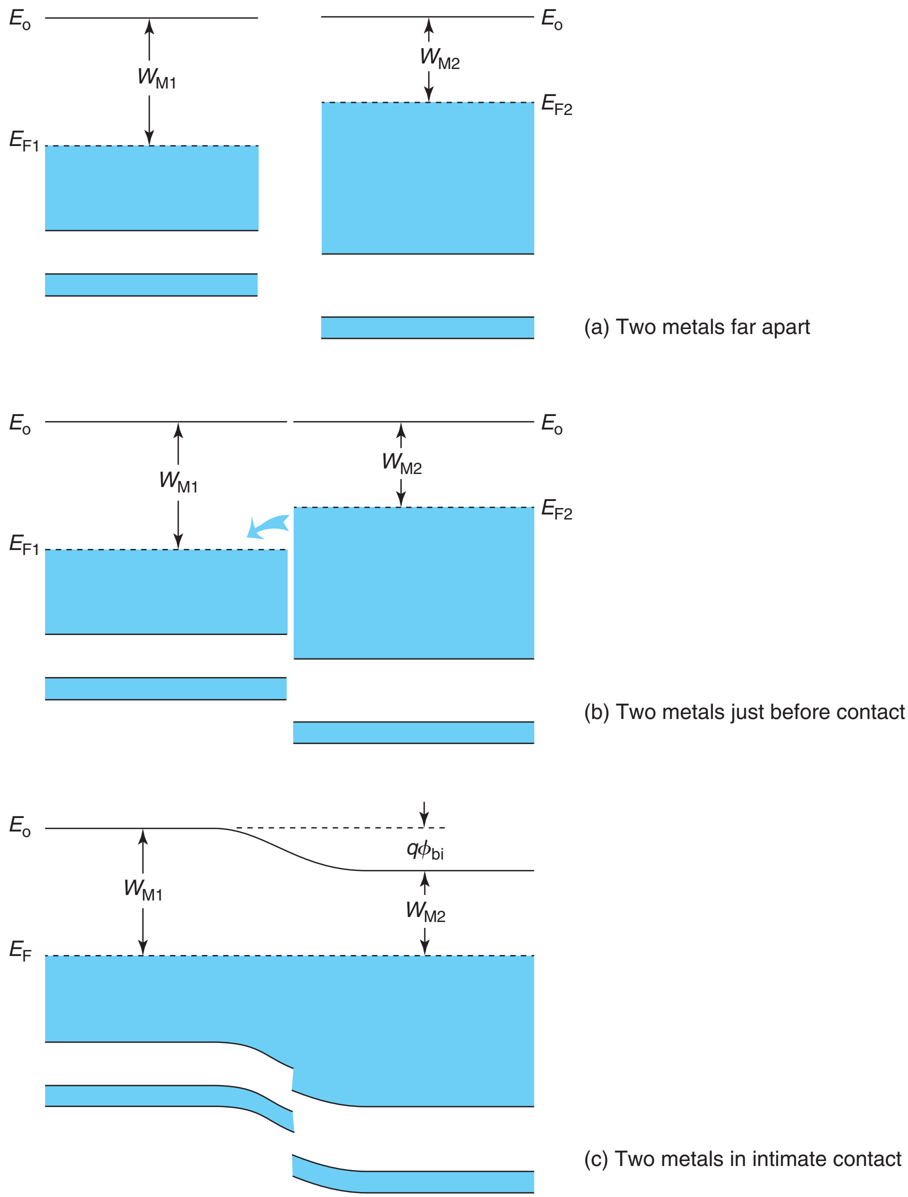


FIGURE 7.4
Energy band diagrams illustrating two different metals with different work functions: (a) far apart from each other, (b) just before contact is established, and (c) in intimate contact.

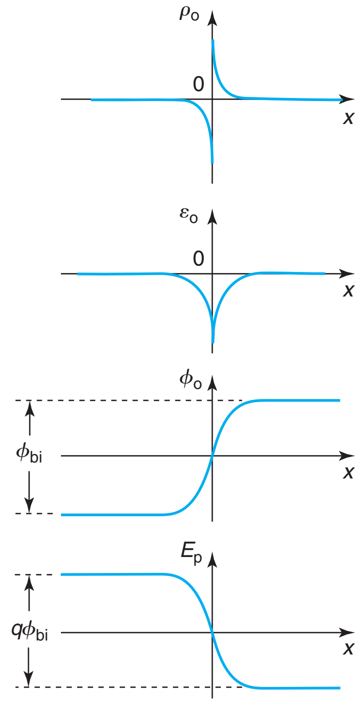


FIGURE 7.5
Volume charge density, electric field, electrostatic potential, and potential energy corresponding to the metal–metal junction of Fig. 7.4 in equilibrium.

a new unified electronic system. In consequence, when thermal equilibrium is established, the Fermi level must be flat everywhere. This is the situation depicted in Fig. 7.4(c).

At this point, it is legitimate to ask: what happens to the work function? How much energy does it take to extract an electron from the new combined electronic system? The best way to answer these questions is to carry out a fictitious photoelectric effect experiment, as sketched in Fig. 7.6. Suppose we have a tunable light source that can emit a spatially narrow beam of photons of any arbitrary energy. Suppose also that we have a way to detect when electrons are extracted from any one of the metals. If the beam of light has a narrow spatial spread, we can measure the *local work function* by shining light at a spot of the surface of the metal and observing the light energy at which electrons start escaping from that location.

When the two metals are apart, the energy required to free up an electron is well defined, that is, W_{M1} for metal 1 and W_{M2} for metal 2. When the metals are in contact, the situation is not so clear. Sufficiently far away from the interface, each metal should not be aware of the presence of the other. Since the nature of each one has not been modified by bringing them into contact, it is then reasonable that the threshold light energy that frees up an electron remains unchanged. Far away from the interface, then, the work function of metal 1 is still W_{M1} and for metal 2 it is W_{M2}. As we approach the interface, electrons have migrated from metal 2 to metal 1. In consequence,

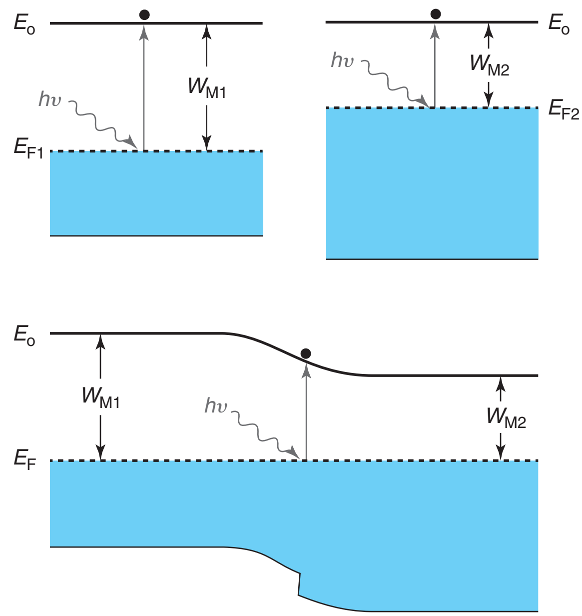


FIGURE 7.6
The photoelectric effect can be used to define a local work function in a metal–metal junction. Top: with the metals isolated, the threshold energy for extracting an electron is equal to the regular work functions W_{M1} and W_{M2}. Bottom: in a metal–metal junction, the local work function changes smoothly across the interface from W_{M1} to W_{M2}.

the threshold photon energy of metal 2 slowly *increases* as we approach metal 1, that is, we have to "dig deeper" into the electronic structure of metal 2 to find electrons that can be freed up. When we cross the interface into metal 1, previously empty electronic states are now full. As a result, the energy required to free up some of these electrons will be smaller than W_{M1}. As we go further and further away from the interface, the local work function approaches W_{M1}. This is sketched in Fig. 7.6.

We can then define, without any ambiguity, a *local work function*. We find that this parameter changes smoothly from one metal to the other and sufficiently far away from the interface it recovers the value that the metal exhibits when in isolation. It is clear that this local work function cannot change abruptly in space. If it did, electrons at the location where the discontinuity exists could move a small distance and easily lower their energy. This would not be an equilibrium situation.

Through the local work function, we can also define a *local vacuum energy*, E_o as in Fig. 7.4. At any point in space, E_o is located at an energy above the Fermi level equal to the local work function. The shape of the local vacuum energy is identical to the potential energy of Fig. 7.5 and the total difference in energy of the local vacuum level across the structure is $W_{M1} - W_{M2}$. This allows us to conclude that the built-in potential of this structure is $\phi_{bi} = (W_{M1} - W_{M2})/q$.

A rigorous solution of the electrostatics of a metal–metal junction in thermal equilibrium, although not being very difficult to carry out, is of no great use to us. The one aspect of it that is important is the length scale of the space-charge regions at the interface between the two metals. Because of the high electron density of a metal (about $\sim 10^{22}$ cm^{-3}), a metal does not tolerate net charge in its bulk. If there is no overall charge neutrality in a metal, its net charge is confined to a thin sheet at the surface. In our case, this is the metal/metal interface. The space-charge region at the interface is only a fraction of a nanometer thick, much smaller than all scale lengths of interest in microelectronic devices. For all purposes, it is then safe to consider this as a delta function and that is what we will do in this book.

We can now turn to the metal–semiconductor junction, the main topic of this chapter.

7.2.2 Energy band lineup of metal–semiconductor junction

A metal–semiconductor junction shares a lot of similarities with the metal–metal junction. While a metal is characterized by a partially full band, the most prominent feature of the energy band structure of a semiconductor is a complete band separated by a bandgap from the next empty band. Depending on the doping concentration and type of the semiconductor, the Fermi level can be located just about anywhere in the bandgap; it can even penetrate to some extent into the conduction or valence bands at high doping levels. In consequence, the detailed charge redistribution that takes place between a metal and a semiconductor when they are brought in intimate contact depends on the doping type and doping level of the semiconductor. So will the electrical properties of the resulting metal–semiconductor junctions.

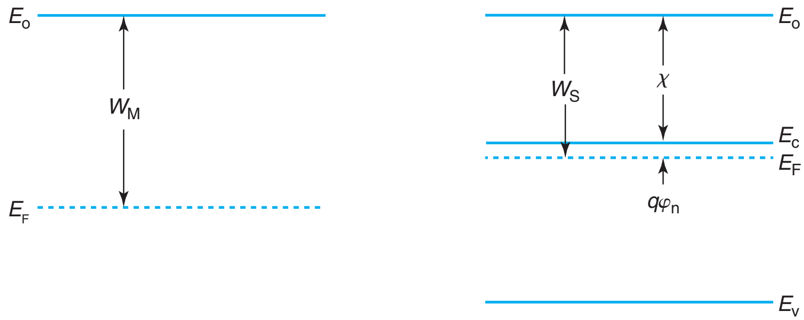
Figure 7.7 shows the energy band diagram of a typical metal and a nondegenerate n-type semiconductor. As in a metal, a semiconductor is characterized by a work function W_S. In a semiconductor, the work function is defined as the energy difference between the vacuum level and the Fermi level in equilibrium. Because of this, W_S is a function of the doping level. In order to completely specify the relationship between the work function and the doping level in a semiconductor, an additional parameter is required. This is the *electron affinity*, χ, which measures the energy difference between the vacuum level and the edge of the conduction band. χ is a property of the host semiconductor. For Si, for example, $\chi = 4.04$ eV at room temperature. The relationship between the electron affinity and the semiconductor work function is given by

$$W_S = \chi + q\varphi_n \tag{7.1}$$

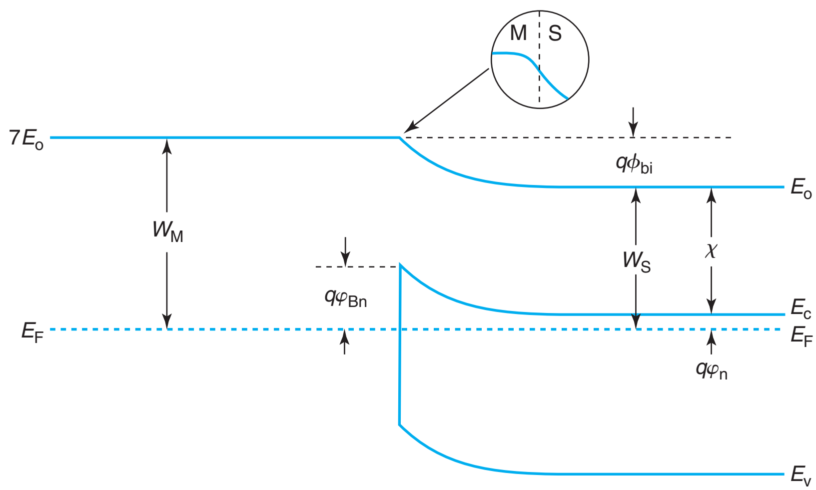
where $q\varphi_n$ is the energy difference between E_c and E_F and is calculated from the doping level as discussed in Chapter 2.

Figure 7.7 illustrates a case in which the work function of the metal is higher than the work function of the semiconductor. This is the most typical situation for Si. In this instance, when the metal and the semiconductor are brought together, the difference in work functions results in electron redistribution in which electrons preferentially flow from the semiconductor to the metal. As in the metal–metal junction, this creates a charge dipole at the interface of the two materials. The semiconductor side becomes positively charged as a consequence of the exposed ionized donors, while the metal gets flooded by additional electrons and becomes negatively charged. The resulting energy band diagram in equilibrium is shown at the bottom of Fig. 7.7.

This energy band diagram is qualitatively identical to the one drawn in Fig. 7.4 for the metal–metal junction. Fundamental parameters of the semiconductor, such as χ and E_g, are



(a) Metal and semiconductor far apart



(b) Metal and semiconductor in intimate contact

FIGURE 7.7
Energy band diagrams illustrating a metal and an n-type semiconductor: (a) far apart from each other and (b) in intimate contact.

not affected by the presence of the metal. Additionally, sufficiently far away from the interface, the semiconductor and the metal are not upset by the presence of the junction. In consequence, their properties must remain the same as when they were isolated. For the semiconductor, this means that far away on the right, E_F is located at a distance W_S below the vacuum level and the energy difference between E_F and E_c is unchanged from the isolated case. In consequence, the built-in potential of this junction is given by

$$\phi_{bi} = \frac{1}{q}(W_M - W_S) \tag{7.2}$$

What is substantially different in the case of the metal–semiconductor junction with respect to the metal–metal junction is the relative band bending in the metal and the semiconductor. Since a metal has a carrier concentration several orders of magnitude higher than a semiconductor, the potential distribution across a metal–semiconductor junction is extremely asymmetric, with the semiconductor absorbing the great majority of it. The metal energy bands bend a negligible fraction of $q\phi_{bi}$ over a very small distance from the interface. In contrast, the bands in the semiconductor bend up substantially over a much longer length scale. The closer we get to the interface, the further away the conduction band is from the Fermi level and the fewer electrons there are. The detailed shape of the band bending in the semiconductor will be calculated in the following section. We will see that it is nearly parabolic.

At the metal–semiconductor interface, an electron with an energy equal to the Fermi level has an abrupt barrier to overcome in order to get from the metal to the semiconductor. This energy barrier plays a crucial role in the operation of metal–semiconductor junctions out of equilibrium and is called the *Schottky barrier height*. From Fig. 7.7, it is easy to see that

$$q\varphi_{Bn} = W_M - \chi \tag{7.3}$$

This equation says that the Schottky barrier height is a property of the metal–semiconductor pair and it is not a function of the doping level of the semiconductor. This is often referred to as the *Schottky–Mott relation*.

Combining Eqs. (7.1–7.3), we obtain an expression for the built-in potential of a Schottky diode in terms of the Schottky barrier height:

$$q\phi_{bi} = q\varphi_{Bn} - q\varphi_n \tag{7.4}$$

Figure 7.8 shows the energy band diagram for a typical metal/p-semiconductor junction. With the two materials far apart, the Fermi level in the metal is higher than the Fermi level in the semiconductor. In this instance, as we bring the materials together, electrons flow from the metal to the semiconductor. On the metal side, this leads to a surface with a deficiency of electrons, that is, positively charged. Since the semiconductor is p-type, metal electrons rush to the semiconductor and preferentially occupy the holes at the top of the valence band. In consequence, there are fewer holes there when equilibrium is established and the semiconductor becomes negatively charged.

In thermal equilibrium, the bands in the semiconductor bend down reflecting the reduction of holes on the semiconductor side as the interface is approached. There is also an energy barrier that appears at the metal–semiconductor interface. It is customary to refer to it as the Schottky barrier height for holes $q\varphi_{Bp}$ (always a positive quantity). From the figure, we can see that $q\varphi_{Bp}$ is given by

$$q\varphi_{Bp} = \chi + E_g - W_M \tag{7.5}$$

This is also independent of doping. In terms of $q\varphi_{Bp}$, ϕ_{bi} is

$$q\phi_{bi} = q\varphi_{Bp} - q\varphi_p \tag{7.6}$$

where $q\varphi_p$ is the energy distance between E_F and E_v.

An interesting property of a metal–semiconductor pair is obtained by adding Eqs. (7.3) and (7.5):

$$q\varphi_{Bn} + q\varphi_{Bp} = E_g \tag{7.7}$$

The Schottky barrier heights of a metal on a semiconductor when p-type doped and n-type doped add up to the bandgap of the semiconductor. This is a handy relationship

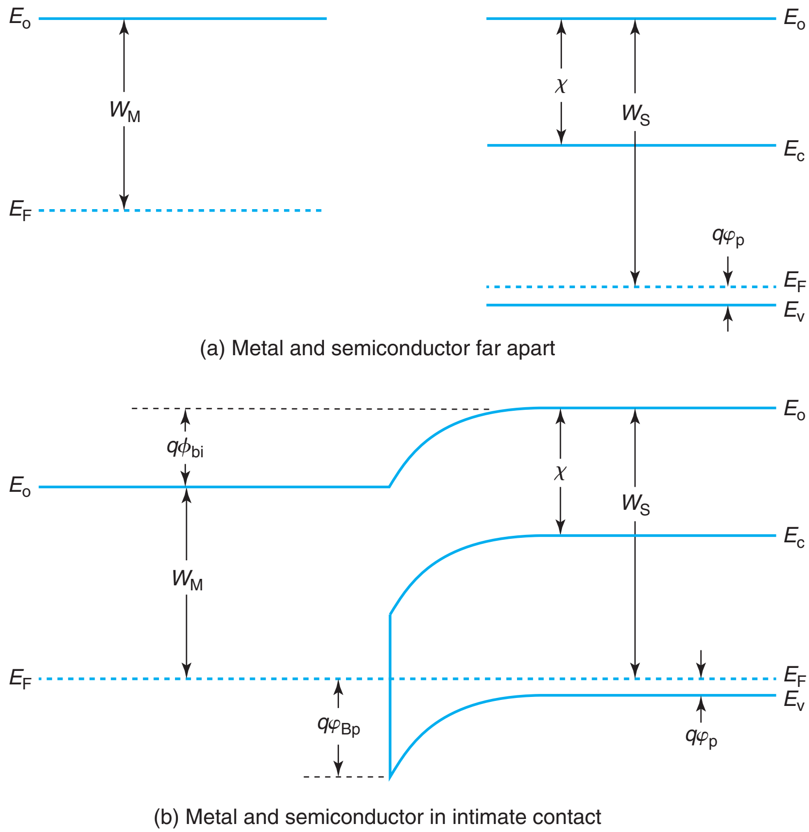


FIGURE 7.8
Energy band diagrams illustrating a metal and a p-type semiconductor: (a) far apart from each other and (b) in intimate contact.

that can be exploited in practice if we are missing the Schottky barrier height for a certain metal–semiconductor pair but we have its value for the same system with the contrary polarity of the semiconductor.

Before we move on, it is important to note that Eqs. (7.3) and (7.5) should not be used to estimate the Schottky barrier height of a metal–semiconductor pair. These equation suggest that the Schottky barrier height is a fundamental property of the bulk materials involved. In practice, it is found that for a given metal–semiconductor pair, $q\varphi_{Bn}$ and $q\varphi_{Bp}$ depend on the details of the fabrication process and the crystalline orientation of the semiconductor. This strongly suggests that the Schottky barrier height is affected to some extent by the interfacial chemistry and is not just a property of the bulk materials. For many metals, the predictions of Eq. (7.3) can be significantly off. More details of this are given in Advance Topic AT7.1. The practical approach is not to rely on Eq. (7.3) or (7.5) but rather design experiments to measure the actual value of the Schottky barrier height of interest. Equations (7.4) and (7.6) remain correct and can be safely used to relate the built-in potential, the Schottky barrier height, and the relative location of the

Fermi level in the semiconductor. As is shown in Advanced Topic AT7.1, Eq. (7.7) also remains correct.

In the next section, we study the electrostatics of the metal–semiconductor junction in thermal equilibrium in a more quantitative way.

7.2.3 Electrostatics of metal–semiconductor junction in equilibrium

The goal of this section is to develop a first-order model for the electrostatics of a metal/semiconductor junction in thermal equilibrium, that is, to calculate the volume charge density, electric field, electrostatic potential, and carrier concentration distributions in space across the structure. This understanding is essential to deal with situations out of equilibrium later on in this chapter. This section treats the metal/n-semiconductor junction. The procedure is similar to the metal/p-type semiconductor junction and leads to very similar equations.

The problem to be solved is a typical one in semiconductor devices and it resembles the PN diode. In most metal–semiconductor junctions, the net volume charge density in the semiconductor in the vicinity of the metal is high enough that it cannot be considered quasi-neutral any longer. Because of this, the region close to the metal–semiconductor interface is called the *space-charge region*, or SCR. Unlike the relatively gradual nonuniformly doped distributions that we studied in Chapter 4, a space-charge region many times appears in situations in which abrupt transitions occur. It is the case, for example, in a PN junction and also that of a metal-oxide-semiconductor structure, as we will describe later on in this book. Sufficiently far away from the metal–semiconductor interface, in the bulk of the semiconductor, charge neutrality should prevail. This region is called the *quasi-neutral region* or QNR. The boundary between the SCR and the QNR is fairly sharp.

The first step in thinking about the electrostatics of this problem is to consider the volume charge density, ρ_o. In general, for a metal/n-semiconductor junction, and in the absence of a substantial concentration of compensating acceptors, ρ_o is given by

$$\rho_o = q(p_o - n_o + N_D) \tag{7.8}$$

Deep in the quasi-neutral bulk of the semiconductor $\rho_o \simeq 0$. As we advance toward the metal–semiconductor interface, n_o drops and p_o consequently rises (the $n_o p_o$ product must remain constant in equilibrium). This makes ρ_o positive. Since p_o starts from an extremely small value in the QNR, the drop in n_o is most significant. If the built-in potential of the junction is not too small, as we get closer to the interface, we will reach a point in which n_o has become negligible in front of N_D. Similarly, if ϕ_{bi} is not too high either, p_o might never become significant next to N_D. When all this happens, $\rho_o \simeq qN_D$. Because of the exponential dependence of n_o on ϕ_o, the transition between $\rho_o \simeq 0$ and $\rho_o \simeq qN_D$ is actually quite sharp. This suggests a simple approximation to the charge distribution in the semiconductor:

$$\rho_o(x) \simeq qN_D \qquad \text{in SCR: } 0 \le x < x_d \tag{7.9}$$

$$\rho_o(x) \simeq 0 \qquad \text{in QNR: } x_d < x \tag{7.10}$$

This assumption of a box-like shape for the charge distribution is the *depletion approximation*, something that we have already become familiar within the analysis of the PN diode.

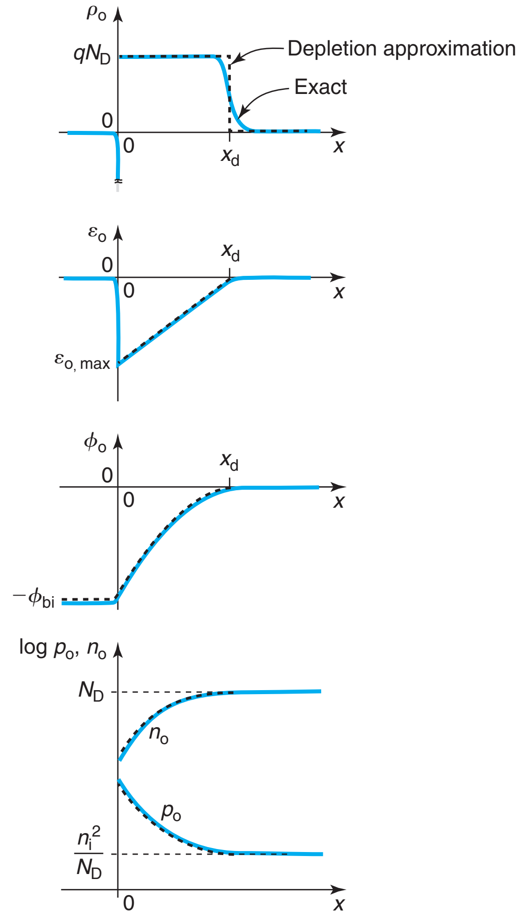


FIGURE 7.9
Volume charge density, electric field, electrostatic potential, and equilibrium carrier concentrations across metal/n-semiconductor junction.

Establishing the location of the boundary between the SCR and the QNR, x_d in Eqs. (7.9) and (7.10), is one of the goals of the calculations that follow. This charge distribution is shown in Fig. 7.9.

Integration of this volume space-charge density distribution yields the electric field profile:

$$\mathcal{E}_o(x) \simeq \frac{qN_D}{\epsilon}(x - x_d) \quad \text{in SCR: } 0 \le x \le x_d \tag{7.11}$$

$$\mathcal{E}_o(x) \simeq 0 \quad \text{in QNR: } x_d \le x \tag{7.12}$$

This field is negative because it points from the semiconductor into the metal, contrary to our choice of axis, as shown in Fig. 7.9.

One more integration yields the electrostatic potential:

$$\phi_o(x) = -\frac{qN_D}{2\epsilon}(x^2 - 2x_d x + x_d^2) \quad \text{in SCR: } 0 \le x \le x_d \tag{7.13}$$

$$\phi_o(x) = 0 \quad \text{in QNR: } x_d \le x \tag{7.14}$$

where we have selected $\phi_o(bulk) = 0$ as the reference for potentials. The potential is also graphed in Fig. 7.9.

The extension of the depletion region, x_d, is obtained by demanding that the total potential difference across the semiconductor be ϕ_{bi} which we know from energy arguments [Eq. (7.4)]. With our selected potential reference, it implies that $\phi_o(0) = -\phi_{bi}$. Solving for x_d in Eq. (7.13), we get

$$x_d = \sqrt{\frac{2\epsilon\phi_{bi}}{qN_D}} \tag{7.15}$$

The maximum electric field occurs at the interface and it is given by

$$|\mathcal{E}_{o,max}| = \frac{qN_D x_d}{\epsilon} = \sqrt{\frac{2qN_D\phi_{bi}}{\epsilon}} \tag{7.16}$$

These two equations state that the higher the doping level, the thinner the depletion region is and the higher the electric field becomes at the metal/semiconductor interface. This is a consequence of the electrostatics of a charge dipole. To attain a certain potential buildup, a more spatially compact charge dipole requires a higher charge and results in a bigger field inside the dipole. This is what happens in the depletion region of a metal–semiconductor junction when the doping level in the semiconductor increases.

It is interesting to realize that Eqs. (7.15) and (7.16) are identical to that of a highly asymmetric P$^+$N junction [Eqs. (6.21) and (6.22), respectively]. This makes sense as in that case we learned that the electrostatic potential drops entirely in the lowly doped n-type semiconductor. That is also the case in the metal–semiconductor junction studied here.

EXERCISE 7.1

Calculate the built-in potential, the depletion region thickness and the maximum electric field at 300 K *of an Al–Si junction with* $q\varphi_{Bn} = 0.68$ eV *in thermal equilibrium. The doping level of the Si is* $N_D = 10^{17}$ cm^{-3}.

For $N_D = 10^{17}$ cm^{-3} Si at 300 K, the energy difference between the the conduction band and the Fermi level is $q\varphi_n = 0.15$ eV. Using Eq. (7.4), ϕ_{bi} is

$$\phi_{bi} = \varphi_{Bn} - \varphi_n = 0.53 \text{ V}$$

The depletion region thickness can be obtained from Eq. (7.15):

$$x_d = \sqrt{\frac{2\epsilon\phi_{bi}}{qN_D}} = \sqrt{\frac{2 \times 1.04 \times 10^{-12} \text{ F/cm} \times 0.53 \text{ V}}{1.6 \times 10^{-19} \text{ C} \times 10^{17} \text{ cm}^{-3}}} = 8.3 \times 10^{-6} \text{ cm} = 83 \text{ nm}$$

The maximum electric field occurs at the metal–semiconductor interface. From Eq. (7.16), it is given by

$$|\mathcal{E}_{o,max}| = \frac{qN_D x_d}{\epsilon} = \frac{1.6\times10^{-19}\ \mathrm{C}\times10^{17}\ \mathrm{cm}^{-3}\times8.3\times10^{-6}\ \mathrm{cm}}{1.04\times10^{-12}\ \mathrm{F/cm}} = 1.4\times10^{5}\ \mathrm{V/cm}$$

With the electrostatic potential now completely determined, we are in a position to calculate the equilibrium carrier concentrations and verify our initial assumptions. For this, we use the Boltzmann relations, which for our choice of potential reference are written as

$$n_o(x) = N_D \exp\frac{q\phi_o(x)}{kT} \tag{7.17}$$

$$p_o(x) = \frac{n_i^2}{N_D}\exp\frac{-q\phi_o(x)}{kT} \tag{7.18}$$

with $\phi_o(x)$ given by Eqs. (7.13) and (7.14). $n_o(x)$ and $p_o(x)$ are also sketched in Fig. 7.9.

We can now verify the assumptions that were made in formulating the depletion approximation. First, in going from Equations (7.8) to (7.9), we neglected the equilibrium hole concentration. The location where p_o is highest, as can be seen in Fig. 7.9, is at the metal–semiconductor interface. At that point,

$$p_o(0) = \frac{n_i^2}{N_D}\exp\frac{q\phi_{bi}}{kT} = N_v\exp\frac{-(E_g - q\varphi_{Bn})}{kT} \tag{7.19}$$

where we have also used Eqs. (7.4) and (2.41.) This equation implies that for $p_o(0)$ to be negligible next to N_D, $q\phi_{Bn}$ must not get too close to E_g. As the exercise below shows, for typical doping levels and common metals, this condition is readily satisfied.

We also neglected n_o next to N_D everywhere in the depletion region. We must make sure that at the interface, n_o is sufficiently small for this assumption to hold everywhere else. At $x = 0$, we have

$$n_o(0) = N_D\exp\frac{-q\phi_{bi}}{kT} = N_c\exp\frac{-q\varphi_{Bn}}{kT} \tag{7.20}$$

where we have also used Eqs. (7.4) and (2.41.) This condition is satisfied if ϕ_{Bn} is at least several kT/q's.

EXERCISE 7.2

Specify the range of values of Schottky barrier height that at 300 K *allow the use of the depletion approximation in equilibrium for metal–semiconductor junctions built on* $N_D = 10^{17}\ \mathrm{cm}^{-3}$ Si.

The upper limit of $q\varphi_{Bn}$ is set by the maximum tolerable p_o at the metal–semiconductor interface. This is given by Eq. (7.19):

$$p_o(0) = N_v\exp\frac{-(E_g - q\varphi_{Bn})}{kT} \ll N_D$$

Solving for $q\varphi_{\mathrm{Bn}}$, we get

$$q\varphi_{\mathrm{Bn}} \ll E_{\mathrm{g}} - kT \ln \frac{N_{\mathrm{v}}}{N_{\mathrm{D}}}$$

For $N_{\mathrm{D}} = 10^{17}\ \mathrm{cm}^{-3}$ at 300 K, this demands that $q\varphi_{\mathrm{Bn}}$ be a few kT's smaller than 0.98 eV.

The lower limit of $q\varphi_{\mathrm{Bn}}$ is set by the maximum tolerable n_{o} at $x = 0$ given by Eq. (7.20):

$$n_{\mathrm{o}}(0) = N_{\mathrm{c}} \exp \frac{-q\varphi_{\mathrm{Bn}}}{kT} \ll N_{\mathrm{D}}$$

For $N_{\mathrm{D}} = 10^{17}\ \mathrm{cm}^{-3}$ at 300 K, this demands that $q\varphi_{\mathrm{Bn}}$ be at least several kT's over 0.15 eV.

Since most metals on Si fall within these two limits, the depletion approximation is of very wide applicability.

7.3 CURRENT–VOLTAGE CHARACTERISTICS OF IDEAL SCHOTTKY DIODE

When a voltage is applied to the two terminals of a metal–semiconductor junction, current flows. As in the case of the PN diode, the current through a Schottky diode exhibits rectifying behavior. Before we can understand the origin of this, we must study the modifications that the electrostatics of the junction undergo upon the application of a voltage. This is the topic of the next section.

7.3.1 Electrostatics under bias

Let us connect a battery with voltage V across a Schottky diode. Through the contact to the Schottky metal, the positive side of the battery "grabs" on the Fermi level in the metal. Through the ohmic contact, as we will see in Section 7.8, the negative side of the battery grabs on the majority carrier quasi-Fermi level of the semiconductor. The application of a voltage V between the Schottky metal and the ohmic contact therefore results in a split between the Fermi level of the metal and the quasi-Fermi level for electrons in the semiconductor of a magnitude qV, as shown in Fig. 7.10 for forward bias and Fig. 7.11 for reverse bias. What is the resulting energy band diagram? Do the electrostatics change across the structure? If so, how do they change?

A complete answer to these questions must be deferred until we have a way to calculate the current that flows when a voltage is applied. This is a chicken and egg problem. We cannot calculate the current without knowing the potential distribution through the structure, which we cannot compute without knowing the currents. To break this circle, let us first assume that the current has a negligible impact on the electrostatics. We will then use this to compute the currents. At that point, we can estimate the order of magnitude of the required corrections and perform them if they are not too large or ignore them if they are very small.

In the case of a resistor, when a voltage was applied across its terminals, we made the implicit assumption that the voltage dropped in a uniform way along its length. That is why the energy band diagram is tilted with a constant slope everywhere. This is reasonable if the resistor has uniform properties (resistivity and cross section) along its length. In the case of the Schottky diode, the situation is rather different. This structure has basically five rather different regions

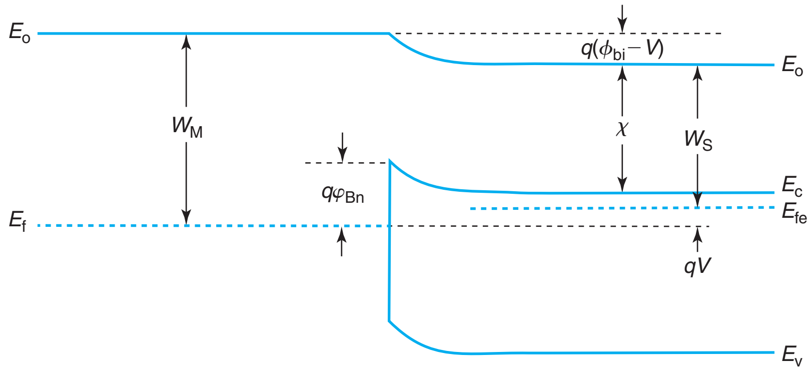


FIGURE 7.10
Energy band diagram of a metal/n-semiconductor junction in forward bias ($V > 0$).

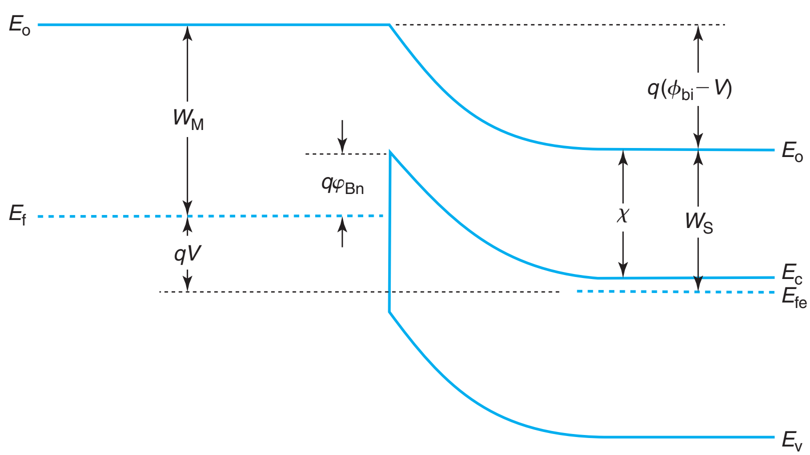


FIGURE 7.11
Energy band diagram of a metal/n-semiconductor junction in reverse bias ($V < 0$).

where the voltage can drop: the Schottky metal bulk, the interface of the Schottky metal with the semiconductor, the space-charge region in the semiconductor, the quasi-neutral region or bulk of the semiconductor and the ohmic contact to the semiconductor. Since the Schottky metal is very conducting in comparison with the semiconductor, the voltage drop there is negligible. At its interface with the semiconductor, the Schottky metal can sustain some electrical charge. This is a very thin region and cannot absorb any significant voltage. Moving into the semiconductor, if we compare the QNR with the SCR, we can see that the QNR has many more carriers than the SCR; the SCR is a lot more "resistive" than the QNR and it is reasonable to assume that most of the voltage drops there. In an ideal Schottky diode, we also neglect the ohmic drop in the ohmic contact. This all leads to assuming in a first pass that the voltage that is applied to a Schottky diode drops entirely across the SCR. We will review this after we discuss the *I–V* characteristics of the junction.

Figure 7.10 shows the energy band diagram in forward bias. In this case, the quasi-Fermi level for electrons in the QNR of the semiconductor is raised over the Fermi level of the metal by an amount qV. Since we assume that there is no ohmic drop across the QNR in the semiconductor, E_{fe} remains flat throughout. The difference in the local vacuum level across the structure is reduced from a value of $q\phi_{bi}$ in equilibrium to a value $q(\phi_{bi} - V)$ in forward bias. The total built-in potential has accordingly shrunk from ϕ_{bi} to $\phi_{bi} - V$. How can this be accommodated? By reducing the magnitude of the dipole of charge that is set on both sides of the Schottky interface. Since the volume charge density, qN_D, cannot be changed, the only way to accomplish this is by shrinking the extent of the space-charge region.

In reverse bias, the situation is similar. In this case, the battery lowers the Fermi level in the semiconductor with respect to the Fermi level in the metal. This *increases* the total band bending across the structure to $q(\phi_{bi} - V)$ (V is negative). In order to accommodate this, the space-charge region must widen. The energy band diagram is shown in Fig. 7.11. In both cases, forward and reverse, continuity of the local vacuum potential prevents any changes on the Schottky barrier height as a result of the bias application.

Figure 7.12 shows the distribution of volume charge density, electric field, and electrostatic potential under forward and reverse bias. The electrostatics are essentially identical to

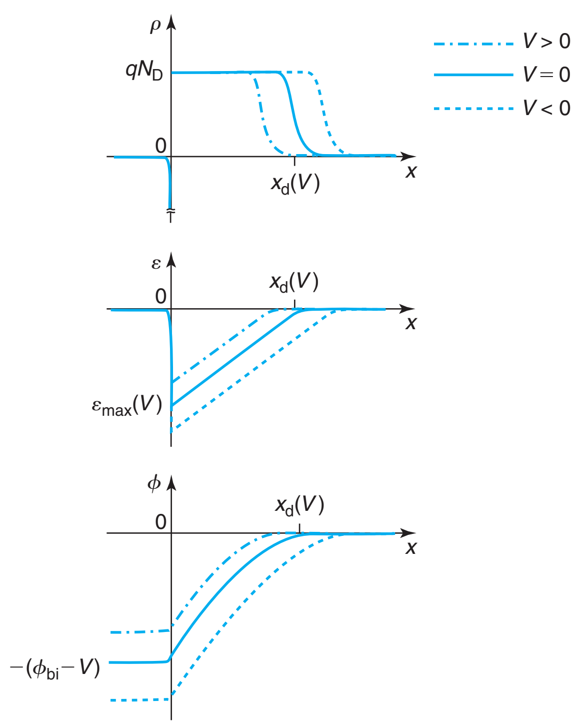


FIGURE 7.12
Sketches of volume charge density, electric field, and electrostatic potential in equilibrium and under bias in a metal/n-semiconductor junction.

equilibrium, except that the potential difference across the structure is not ϕ_{bi} but $\phi_{bi} - V$. This allows us to reuse the solutions to the electrostatics of the equilibrium situation derived above, except where we wrote ϕ_{bi}, we must now write $\phi_{bi} - V$. The depletion region thickness is, for example

$$x_d(V) = \sqrt{\frac{2\epsilon(\phi_{bi} - V)}{qN_D}} = x_d(V=0)\sqrt{1 - \frac{V}{\phi_{bi}}} \tag{7.21}$$

The maximum electric field under bias is

$$|\mathcal{E}_{max}(V)| = \frac{qN_D x_d(V)}{\epsilon} = \sqrt{\frac{2qN_D(\phi_{bi} - V)}{\epsilon}} = |\mathcal{E}_{max}(V=0)|\sqrt{1 - \frac{V}{\phi_{bi}}} \tag{7.22}$$

These expressions are valid provided that the depletion approximation continues to apply. This is the case if the forward voltage does not get too close to ϕ_{bi}.

Sketching the carrier concentration distribution and the location of the Fermi level across the structure requires solving the current–voltage characteristics. This is postponed until Section 7.3.3.

EXERCISE 7.3

Calculate the extent of the depletion region and the peak electric field in an Al/n-Si Schottky diode with $N_D = 10^{17}$ cm^{-3} *at a forward voltage of* 0.5 V *at* 300 K.

We use the results of Exercise 7.1. The built-in potential of this structure is $\phi_{bi} = 0.53$ V. Then,

$$\sqrt{1 - \frac{V}{\phi_{bi}}} = \sqrt{1 - \frac{0.5\text{ V}}{0.53\text{ V}}} = 0.24$$

The extent of the depletion region is

$$x_d(0.5\text{ V}) = x_d(V=0)\sqrt{1 - \frac{V}{\phi_{bi}}} = 8.3 \times 10^{-6}\text{ cm} \times 0.24 = 20\text{ nm}$$

The peak electric field is

$$|\mathcal{E}_{max}(0.5\text{ V})| = |\mathcal{E}_{max}(V=0)|\sqrt{1 - \frac{V}{\phi_{bi}}} = 1.4 \times 10^5\text{ V/cm} \times 0.24 = 3.4 \times 10^4\text{ V/cm}$$

7.3.2 *I–V* characteristics: qualitative discussion

A Schottky diode is essentially a *majority carrier device*. Minority carriers only play a secondary role in its behavior. This is not hard to understand since in equilibrium there is not a significant amount of minority carriers anywhere in the semiconductor that can flow and recombine.

It is not uncommon in semiconductor devices under bias to find that a specific region constitutes the bottleneck to current flow while the rest of the device adapts as needed. In these kinds of situations, the strategy to derive a model for the current–voltage characteristics is to focus on this bottleneck and construct a model for transport there. In a Schottky diode, the lowest concentration of carriers is found in the space-charge region. This is where we will center our attention. Figure 7.13 qualitatively helps us to consider what happens there.

Figure 7.13 shows three sketches of energy band diagrams for our Schottky diode in three different situations. In equilibrium, Fig. 7.13(a), electrons face an energy barrier of height $q\varphi_{\mathrm{Bn}}$ as they attempt to flow from the metal to the semiconductor or vice versa. The electron flow from the metal to the semiconductor is balanced out by the flow from the semiconductor to the metal. The net flow of electrons across the barrier is zero and the net current is zero.

In forward bias, Fig. 7.13(b), the Fermi level on the semiconductor is raised by an amount qV with respect to the Fermi level in the metal. Looking from the point of view of the electrons, the energy barrier facing metal electrons attempting to enter into the semiconductor is $q\varphi_{\mathrm{Bn}}$, just as in equilibrium. In contrast, the energy barrier preventing the flow of electrons from the semiconductor to the metal has *decreased* to $q(\varphi_{\mathrm{Bn}} - V)$. Clearly, this results in a net flow of electrons from the semiconductor to the metal producing a current in the contrary sense.

In reverse bias, Fig. 7.13(c), the Fermi level on the semiconductor is lowered by an amount qV with respect to the Fermi level in the metal. The energy barrier facing the metal electrons is still unchanged from equilibrium, while the energy barrier in front of semiconductor electrons has now *increased* to $q(\varphi_{\mathrm{Bn}} - V)$ (V is negative in reverse bias). In consequence, there is a net flow of electrons from the metal to the semiconductor and a current in the contrary sense.

The sketches of Fig. 7.13 also allow us to recognize the functional dependence of the current on the voltage applied across the junction. In forward bias, as V is increased, the number of electrons in the semiconductor that have enough energy to overcome the energy barrier at the Schottky interface increases exponentially with V. This is a consequence of the exponentially decaying distribution of electrons in the conduction band of the semiconductor. The electron flux from the semiconductor to the metal should therefore increase as $e^{qV/kT}$.

In reverse bias, on the other hand, the number of electrons in the semiconductor with enough energy to overcome the interfacial barrier is also exponentially suppressed as the reverse bias $|V|$ increases. When $|V|$ exceeds a few kT/q's, the injection of electrons from the semiconductor to the metal is completely suppressed and the current saturates to the value given by the injection of electrons from the metal to the semiconductor which is unchanged from equilibrium.

Our qualitative arguments have brought us quite far. We have identified the fact that overcoming the energy barrier at the Schottky interface is the bottleneck to electron flow. Thinking in terms of the electron population at that bottleneck qualitatively explains the rectifying behavior of the Schottky diode. We are just a step away from developing a simple model for the current–voltage characteristics. What we are missing is a way to figure out the *rate* at which electrons flow through the SCR.

Let us start by thinking about the situation in thermal equilibrium. With $V = 0$, the net current anywhere is zero. In the SCR, where an electric field is present, this means that there is a drift current that must be perfectly balanced everywhere by a diffusion current. This originates in the gradient of electron concentration that exists from the bulk, where it is highest, to the metal–semiconductor interface where it is lowest.

Under forward bias, the field distribution in the SCR shrinks in magnitude and spatial extent. This breaks the balance of drift and diffusion with diffusion prevailing over drift. This yields a net electron flow from the semiconductor toward the metal. As these electrons enter the metal,

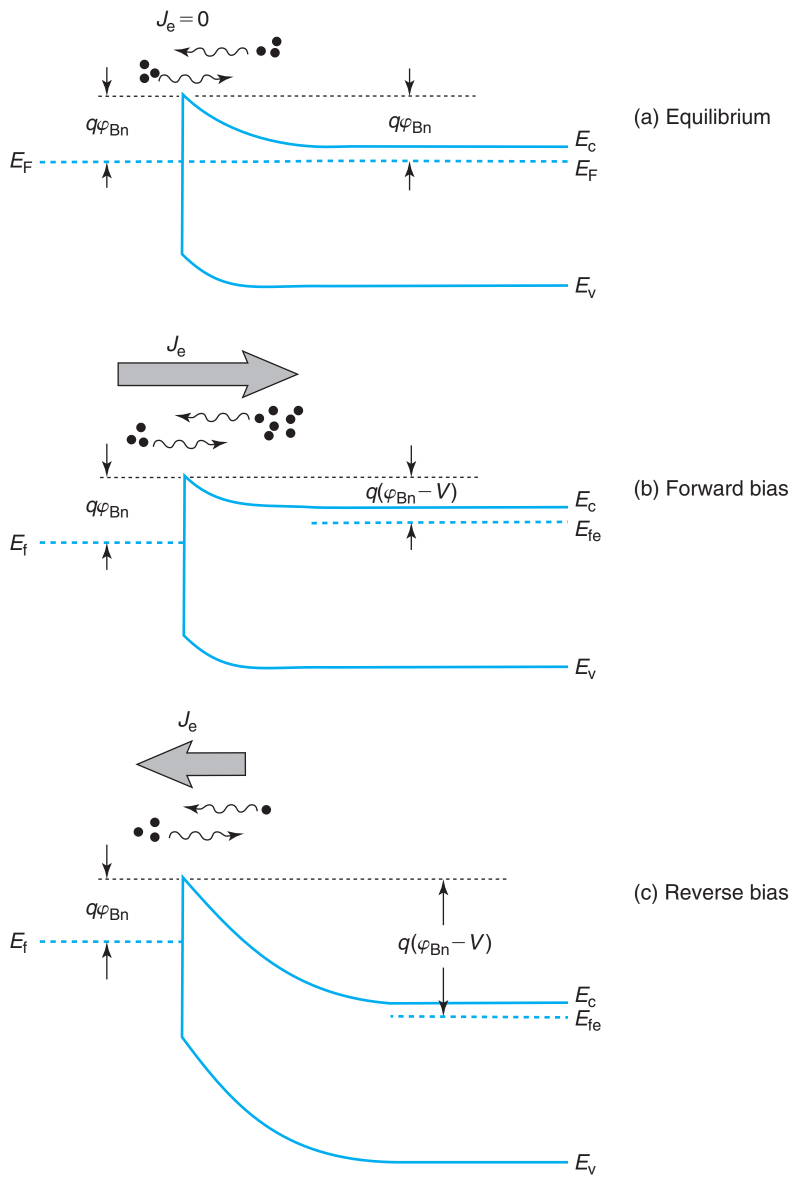


FIGURE 7.13
Sketch of electron flow across metal–semiconductor junction in (a) equilibrium, (b) forward bias, and (c) reverse bias. In equilibrium, there is no net flow of electrons. In forward bias, there is a net flow of electrons from the semiconductor to the metal. In reverse bias, there is net electron flow from the metal to the semiconductor.

they quickly lose their energy thermalizing with the many electrons there. Somehow, what limits the forward bias current is the ability of electrons to reach the metal–semiconductor interface. Once there, they are immediately "sucked" by the metal.

We can contemplate two extreme situations here. In one, the rate-limiting step to the current is the process of electron diffusion against the strong electric field in the SCR. In this limit, the electron concentration drops as we advance from the edge of the SCR toward the Schottky interface. At the metal–semiconductor interface, the electron concentration approaches the equilibrium value as these electrons are in a quasi-equilibrium state with those in the metal. A model that describes the current in this limit is known as the *drift-diffusion model*. This situation is more likely to arise in semiconductors with small mobilities and for low forward bias when the opposing electric field is relatively high.

At the other limit, if the mobility is high and the field is small as a result of strong forward bias, electrons readily reach the metal–semiconductor interface. In this case, it is the process of "emission" into the metal that limits the current. What is exactly this process? We focus at the tip of the energy barrier on the semiconductor side of the the metal–semiconductor interface because that is where the electron concentration is the lowest anywhere in the semiconductor. At that location, the electron concentration is finite and the velocity at which they flow into the metal is also finite. This limits the current to some amount. So, in the limit in which electrons can easily arrive to the Schottky interface from the bulk, it is this maximum "emission current" at the tip of the barrier that sets the diode current. This is known as the *thermionic emission model*.

As it turns out, for Si, GaAs, and many common semiconductors under most conditions, the thermionic emission current is found to be the rate-limiting step. This is because the mobilities are high enough and carriers can readily reach the Schottky interface. The next section formulates the thermionic emission model. Advanced Topic AT7.2 describes the drift-diffusion model.

7.3.3 *I–V* characteristics: thermionic emission model

As we have discussed in the previous section, in the thermionic emission model, the current is limited by the electron emission process over the tip of the energy barrier at the Schottky interface. We then focus at the semiconductor side of the Schottky interface located at $x = 0^+$ and we write the diode current as the net electron current at that location. This is given by the balance between current due to electron flow from the semiconductor to the metal minus the reverse electron flow from the metal to the semiconductor:

$$J_t \simeq J_e(0^+) = J_{e,SM}(0^+) - J_{e,MS}(0^+) \tag{7.23}$$

In forward bias, $J_{e,MS}(0^+)$ is very small next to $J_{e,SM}(0^+)$. We then neglect this component for the time being and focus on this first term that accounts for electrons emitted from the semiconductor into the metal. This current can be approximately expressed as

$$J_{e,SM}(0^+) \simeq -qn(0^+)v_e(0^+) \tag{7.24}$$

Here, the minus sign comes from the fact that with our definition of axis, the velocity of these electrons is against x or negative ($J_{e,SM}(0^+)$ ends up positive). Note also that we assume that all electrons at $x = 0^+$ contribute to this current component. The reverse current is very small and, for the time being, we can ignore the few electrons that support it.

To proceed, we need to exploit the assumption that electrons have no difficulty reaching the Schottky interface from the bulk. This is in essence, the *quasi-equilibrium approximation* in which electrons across the SCR are all in equilibrium with each other and also with electrons in the bulk of the semiconductor. Without further thinking, the Boltzmann relation would allow us to express $n(0^+)$ in terms of the electron concentration in the bulk, N_D, and the difference of potentials between the surface and the bulk, $\phi_{bi} - V$. We would be then tempted to write

$$n(0^+) = N_D \exp \frac{-q(\phi_{bi} - V)}{kT} \tag{7.25}$$

But there is a problem here. At $x = 0^+$, the electron distribution cannot be close to equilibrium. In fact, it has to be very different from an equilibrium Maxwellian distribution. A carrier distribution can only be considered in thermal equilibrium when electrons undergo many collisions with the lattice that randomize their velocity. Right at the metal–semiconductor interface, any electron that has velocity components pointing toward the metal will be "sucked" by the metal but no electron in the metal can make it into the semiconductor because there is a large energy barrier at the interface that prevents it. So, right at the tip of the barrier at $x = 0^+$, there are very few electrons with velocity components pointing toward the semiconductor (this, to the extent that we can neglect $J_{e,MS}(0^+)$ next to $J_{e,SM}(0^+)$). In effect, at $x = 0^+$, we have an *hemi–Maxwellian distribution*. This is sketched in Fig. 7.14.

This figure sketches the conduction band diagram of a Schottky diode in forward bias. It also sketches the distribution of the x-component of the electron velocity at three different locations. In the bulk of the semiconductor, there is a large electron concentration that can be considered to be in near thermal equilibrium with the lattice. The distribution of electron concentration in energy and the X-component of velocity are very close to perfect equilibrium (dashed lines). Only a small difference exists between the distribution of electron velocities pointing toward the metal and away from the metal that gives rise to a net electron flow toward the Schottky interface. Under the quasi-equilibrium approximation, this situation persists across nearly the entire SCR.

On average, a mean free path away from the metal–semiconductor interface, at $x = l_{ce}$, the electrons that ultimately are emitted into the metal suffer their last collision. From there on, those that have enough energy to overcome the energy barrier at the interface and that after their last collision have a component of velocity pointing toward the metal succeed in getting emitted into the metal and contribute to diode current. Those that have the right velocity component but do not have enough energy eventually are reflected by the energy barrier and do not contribute to current. All those that scatter back into the semiconductor also do not contribute to current. The end result is that the distribution of electron velocities right at the Schottky interface at $x = 0^+$ is very far from thermal equilibrium as only the negative velocity half of the Maxwellian distribution is present.

This understanding now allows to correctly estimate $n(0^+)$. We can easily express $n(0^+)$ in terms of $n(l_{ce})$, the electron concentration a mean free path away from the interface. On average, only half of the electrons at $x = l_{ce}$ have a velocity that points toward the interface. Of those, only a fraction have enough kinetic energy to make it over the potential barrier that exists between $x = 0$ and $x = l_{ce}$. The height of this potential barrier is given by the electrostatic

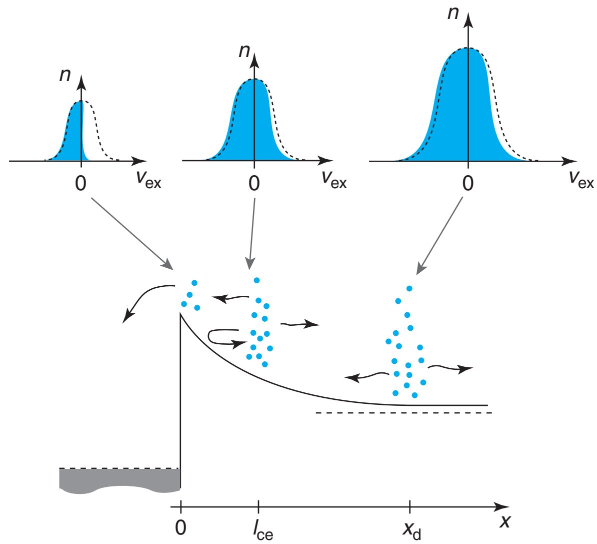


FIGURE 7.14
Sketch of conduction band across SCR in Schottky diode. Also shown are the distributions of x-component of electron velocity at three locations. Across most of the SCR, electrons are in quasi-equilibrium with the bulk and the electron distribution is Maxwellian. On the semiconductor side of the Schottky interface, the velocity distribution is hemi-Maxwellian as only electrons with velocity components pointing toward the metal are present.

potential difference between $x = 0^+$ and $x = l_{ce}$, that is $q[\phi(l_{ce}) - \phi(0^+)]$. Then, the electron concentration at $x = 0$ can be written as

$$n(0^+) = \frac{n(l_{ce})}{2} \exp \frac{q[\phi(0^+) - \phi(l_{ce})]}{kT} \tag{7.26}$$

The electron concentration at $x = l_{ce}$ can be computed by solving the drift-diffusion problem across the rest of the SCR. As we have mentioned before, in semiconductors with relatively high mobilities, such as Si and GaAs, a quasi-equilibrium situation prevails over the SCR from the bulk up to $x = l_{ce}$. We can then use the Boltzmann relation to relate $n(l_{ce})$ and $n(x_d) \simeq N_D$:

$$n(l_{ce}) \simeq N_D \exp \frac{q\phi(l_{ce})}{kT} \tag{7.27}$$

We can now substitute Eq. (7.27) into (7.26) and use the fact that $\phi(0^+) = -(\phi_{bi} - V)$ obtained in Section 7.3 to get

$$n(0^+) = \frac{N_D}{2} \exp \frac{-q(\phi_{bi} - V)}{kT} = \frac{N_c}{2} \exp \frac{-q(\varphi_{Bn} - V)}{kT} \tag{7.28}$$

Equation (7.28) has a simple and intuitive physical interpretation. The electron concentration at $x = 0^+$ is exactly half of what one would have if it was in thermal equilibrium with the bulk

[Eq. (7.25)]. The factor of 1/2 originates from the lack of scattering to the left of $x = 0$ to bring any electrons back. All electrons at $x = 0^+$ are injected into the metal. Notice that, as expected, $n(0^+) \sim e^{qV/kT}$.

The second step in the construction of a model for current is the computation of the electron velocity at the interface pointing toward the metal. In Chapter 4, we introduced the notion of thermal velocity of electrons, v_{the}. This is the average instantaneous velocity of electrons over distances shorter than a mean free path. At the metal–semiconductor interface, however, most electrons point into the metal with velocities at an angle, therefore contributing a smaller net forward velocity. When the statistics of the electron velocity distribution are properly taken into account, the velocity at which electrons are emitted from the semiconductor into the metal (and vice versa in reverse bias) is actually $v_{\text{the}}/2$. Using Eq. (4.5), we can then write

$$v_e(0^+) = -\frac{v_{\text{the}}}{2} = -\sqrt{\frac{2kT}{\pi m^*_{\text{ce}}}} \tag{7.29}$$

where the minus sign simply indicates that electrons are flowing in the contrary sense to our choice of axis. For Si at room temperature, $v_e(0^+)$ is just about 10^7 cm/s.

Plugging Eqs. (7.28) and (7.29) into (7.24), we get

$$J_{\text{e,SM}}(0^+) = qN_D\sqrt{\frac{kT}{2\pi m^*_{\text{ce}}}}\exp\frac{-q(\phi_{\text{bi}} - V)}{kT} = qN_c\sqrt{\frac{kT}{2\pi m^*_{\text{ce}}}}\exp\frac{-q\varphi_{\text{Bn}}}{kT}\exp\frac{qV}{kT} \tag{7.30}$$

where we have used Eq. (7.4) and the relationship given in Chapter 4 among φ_n, N_D, and N_c.

Using now the expression for N_c given in Eq. (2.28), we can write

$$J_{\text{e,SM}}(0^+) = A^*T^2\exp\frac{-q\varphi_{\text{Bn}}}{kT}\exp\frac{qV}{kT} \tag{7.31}$$

where A^* is

$$A^* = \frac{4\pi qk^2 m_o}{h^3}\sqrt{\frac{(m^*_{\text{de}}/m_o)^3}{(m^*_{\text{ce}}/m_o)}} \tag{7.32}$$

A^* is called *Richardson constant* and is a characteristic of the material and the type of carrier. The quantity *in front of* the square root has a value of $A_o = 120\ \text{A}\cdot\text{cm}^{-2}\cdot\text{K}^{-2}$.

We have solved half of the problem. We still need to get $J_{\text{e,MS}}(0^+)$ to complete the calculation. However, we know that the current due to electron flow from the metal to the semiconductor is unaffected by the application of a voltage. This is again due to the fact that the energy barrier that is presented to electrons in the metal does not change with bias. Therefore, we can get an expression for $J_{\text{e,MS}}(0^+)$ by equating it to $J_{\text{e,SM}}(0^+)$ in Eq. (7.31) at zero volts when the net current should be zero. This yields

$$J_{\text{e,MS}}(0^+) = A^*T^2\exp\frac{-q\varphi_{\text{Bn}}}{kT} \tag{7.33}$$

The final net diode current is then

$$J_t = A^*T^2\exp\frac{-q\varphi_{\text{Bn}}}{kT}\left(\exp\frac{qV}{kT} - 1\right) \tag{7.34}$$

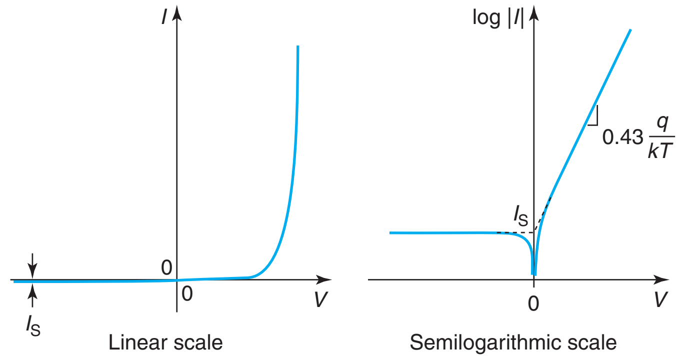


FIGURE 7.15
Sketch of *I–V* characteristics of metal–semiconductor junction, in a linear scale on the left, and in a semilogarithmic scale on the right.

This equation is valid in forward as well as in reverse bias since in reverse bias, the current saturates to Eq. (7.33) (with a minus sign) as the forward emission of electrons from the semiconductor to the metal is completely suppressed.

In a diode that has a junction area A_j, the current flowing through the diode is then

$$I = I_s\left(\exp\frac{qV}{kT} - 1\right) \tag{7.35}$$

where I_s is called the *saturation current*. This is identical to a PN diode and is plotted again in Fig. 7.15 in linear and semilog scales. In a semilog scale, the forward bias current appears as a straight line with a slope of $(q/kT)\log\ e = 0.43q/kT$ or 60 mV/dec at room temperature.

Figure 7.16 shows experimental *I–V* characteristics of a PtSi/n-Si junction. The experimental data follows the shape predicted by Eq. (7.35) very accurately.

The saturation current I_s has a very peculiar set of dependencies in itself:

$$I_s = A_j A^* T^2 \exp\frac{-q\varphi_{Bn}}{kT} \tag{7.36}$$

This bears some similitude to the corresponding expression for the PN junction but there are also some differences. The pre-exponential factor in I_s depends on the square of the absolute temperature. For the PN junction, there is a dependence of T^{4+a} where a reflects the temperature dependence of the relevant transport parameters.

I_s also depends exponentially with a minus sign on the Schottky barrier height (in the PN junction, there is a similar exponential dependence on the bandgap of the semiconductor). This dependence is reasonable since the higher the barrier, the lower (exponentially) is the flow of electrons over it. Since A^* does not depend on temperature, I_s/T^2 is thermally activated with an activation energy equal to $q\varphi_{Bn}$. This suggests an experimental procedure to determine $q\varphi_{Bn}$. First, *I–V* characteristics are measured as a function of temperature and I_s is extracted at all temperatures. Then I_s/T^2 is graphed in an Arrhenius plot. The slope of the resulting straight line gives $q\varphi_{Bn}$. This is experimentally illustrated in Fig. 7.17 for the same diode of Fig. 7.16.

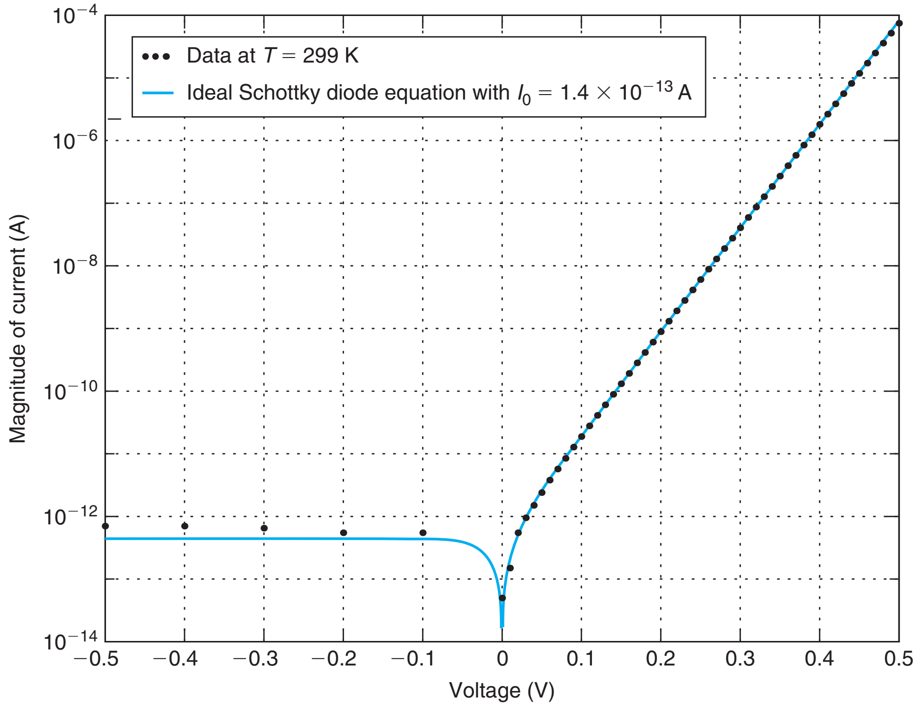


FIGURE 7.16
Experimental *I–V* characteristics of PtSi/n-Si Schottky diodes at room temperature [Reprinted with permission from Sergei Krupenin].

A word about Richardson constant A^*. As seen in Eq. (7.32), the thermionic emission theory predicts it to depend only on the semiconductor and to be independent of the metal. For n-Si, if one plugs in the appropriate values of effective masses, one obtains $A^* \simeq 258\ \mathrm{A \cdot cm^{-2} \cdot K^{-2}}$. A more refined theory predicts that A^* depends somehow on the crystalline orientation of the material. It is not the same to make a Schotky diode on (100) Si than on (111) Si. In practice, one can experimentally extract A^* from actual measurements on Schottky diodes. What is found is that A^* depends a lot on the nature of the metal itself, its thickness, and the details of the process: surface preparation prior to metal deposition, the deposition parameters and any postdeposition treatments. The reasons for these dependencies are various: surface roughness, local strain, the formation of metal–semiconductor compounds, interfacial oxides, and so on. The bottom line is that for a given metal–semiconductor system, we cannot simply take the theoretically predicted value of A^* given by the thermionic emission theory and expect to accurately predict I_s in Schottky diodes. The proper action is to build some test diodes and measure A^* and then use this in the model.

Let us now discuss the key assumption that we have made in the development of our theory. In deriving the previous equations, we assumed that thermionic emission of electrons over the Schottky barrier is the rate-limiting mechanism for electron transport and that drift and diffusion through the SCR are in near perfect balance from $x = l_{ce}$ to the end of the depletion region

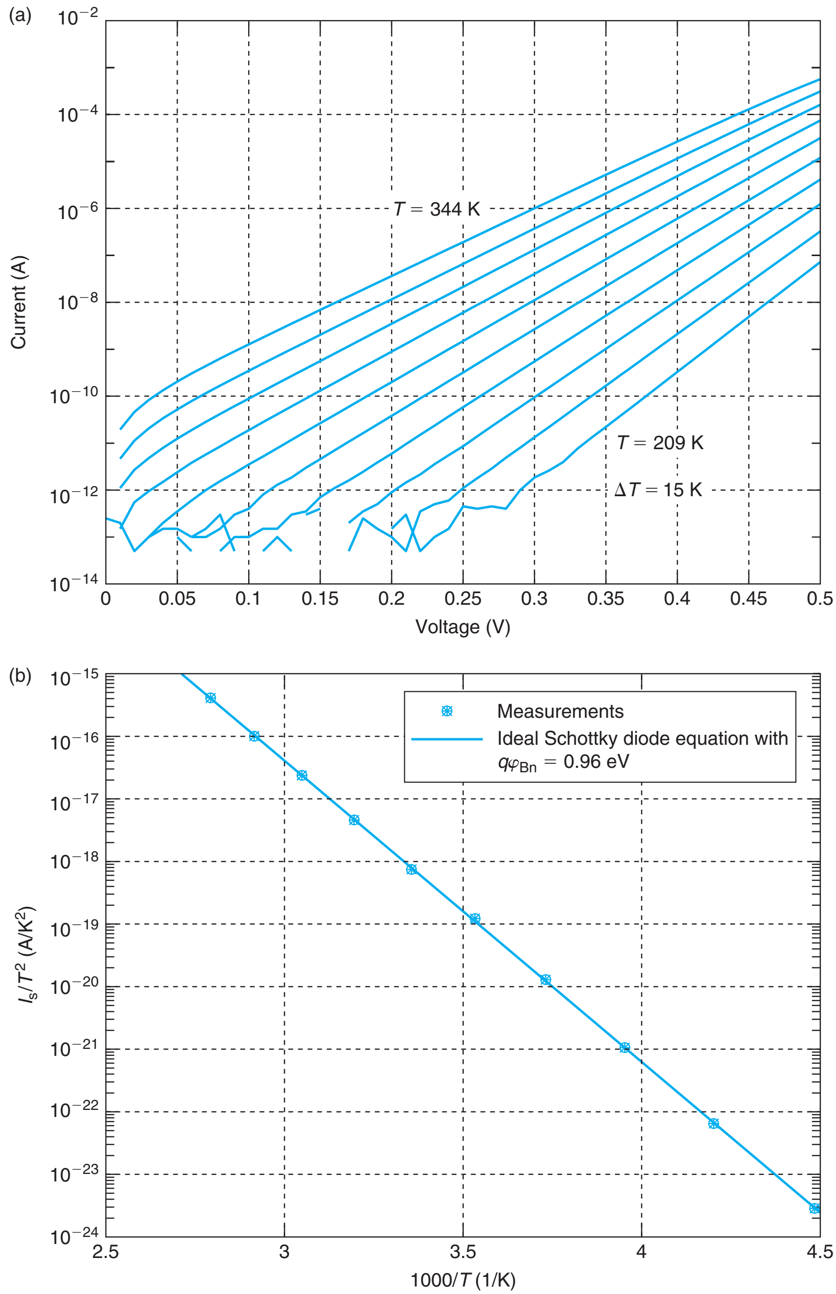


FIGURE 7.17
(a) Forward *I–V* characteristics of PtSi/n-Si Schottky diode for different temperatures. (b) Arrhenius plot of I_s/T^2 for the same diode. The slope of the straight line gives $q\varphi_{Bn}$ [Reprinted with permission from Sergei Krupenin].

toward the body. This assumption allowed us to focus on the emission process over the tip of the energy barrier at the metal–semiconductor interface in order to compute the current. Elsewhere in the SCR, we used the Boltzmann relation to calculate the electron concentration.

For the quasi-equilibrium assumption to apply in this manner, the net current that flows through the junction must be much smaller than either the drift or the diffusion currents. In this way, only a small imbalance between these two currents needs to exist. It is particularly intuitive to focus on the drift current. Quasi-equilibrium is guaranteed up to $x = l_{ce}$ if the thermionic emission current is much smaller than the drift current at that location, that is,

$$|J_{e,SM}(0^+)| \ll |J_{e,drift}(l_{ce})| \tag{7.37}$$

Using Eqs. (7.24), (7.26), and (7.29) on the left-hand side, and the usual expression for the drift current on the right-hand side ($J_{e,drift} = q\mu_e n\mathcal{E}$), we can rewrite this as

$$\frac{1}{4}v_{the} \exp\left(-\frac{q\Delta\phi_{lce}}{kT}\right) \ll \mu_e|\mathcal{E}(l_{ce})| \tag{7.38}$$

where we have defined $\Delta\phi_{lce}$ as the potential build-up over the first mean free path of the SCR starting from the metal–semiconductor interface.

Equations (4.5), (4.8), and (4.10) allow us to express the mobility in terms of the velocity and the mean free path as $\mu_e = 4q\ell_{ce}/(\pi m^*_{ce} v_{the})$. Plugging this on the right-hand side of Eq. (7.38), and noting that $l_{ce}|\mathcal{E}(l_{ce})| < \Delta\phi_{lce}$, we can rewrite this expression as

$$\frac{q\Delta\phi_{lce}}{kT} \exp\left(\frac{q\Delta\phi_{lce}}{kT}\right) \gg \frac{1}{2} \tag{7.39}$$

This is readily satisfied if

$$\Delta\phi_{lce} > 1.4\frac{kT}{q} \tag{7.40}$$

or, in other words, if the potential drop in the first mean free path from the interface exceeds $1.4kT/q$. This is called the *Bethe condition*. In Si, this condition is easy to satisfy at all practical voltages because the mean free paths are rather long. An example is given in Exercise 7.4.

EXERCISE 7.4

Evaluate the validity of the Bethe condition for an Al/n-Si Schottky diode with $N_D = 10^{17}$ cm^{-3} *at a forward voltage of* 0.5 V *at 300* K.

For this calculation, we need the mean free path in the SCR. If we assume that the electron mobility in an SCR is identical to that of a QNR with an identical doping level, we can use the result of Exercise 4.1 where we obtained for this same doping level $l_{ce} \simeq 17$ nm.

From Exercise 7.3, we know that for this Schottky diode at this forward voltage, the peak field at the Schottky interface is $|\mathcal{E}_{max}| \simeq 3.4 \times 10^4$ V/cm. Then, the potential drop in the first mean free path away from the Schottky interface is

$$\Delta\phi_{lce} \simeq l_{ce}|\mathcal{E}_{max}| \simeq 17 \times 10^{-7} \text{ cm} \times 3.4 \times 10^4 \text{ V/cm} = 0.058 \text{ V}$$

This is comfortably bigger than 1.4 $kT/q \simeq$ 36 meV and the Bethe condition is fulfilled. For this Schottky diode, even for such a high forward voltage, the thermionic emission model is the appropriate one.

We are finally in a good position to draw the electron quasi-Fermi level, E_{fe}, across the device. The assumption of quasi-equilibrium across the entire device with the exception of the very first mean free path implies that the quasi-Fermi level for electrons is flat in this entire region. Its location is set by the doping level in the QNR. This is sketched in Fig. 7.18.

Within the very first mean free path away from the metal–semiconductor interface, the concept of quasi-Fermi level is not appropriate as the electron distribution is very far from Maxwellian. We indicate this in Fig. 7.18 by means of a dotted line that suggests that the electron distribution comes to equilibrium with the electrons in the metal at the metal–semiconductor interface.

Figure 7.18 also draws the hole quasi-Fermi level across the structure. We can think of this in a manner similar to a PN junction. At the metal–semiconductor interface, we have a certain equilibrium hole concentration that is determined by the Schottky barrier height. These holes are in excellent communication with the metal since there is no energy barrier for them (or valence band electrons) at the interface. When the energy barrier is reduced under forward bias, the interface holes are injected into the semiconductor. These holes diffuse and eventually recombine. The quasi-Fermi level distribution is then similar to that of a forward-biased PN junction. Within a few diffusion lengths away from the interface, the quasi-Fermi level for holes merges with the quasi-Fermi level for electrons, as indicated in the diagram. In reverse bias, we also have a similar situation to that of a reverse-biased PN junction. Holes are generated in the semiconductor and swept into the metal–semiconductor SCR where they are extracted by the metal.

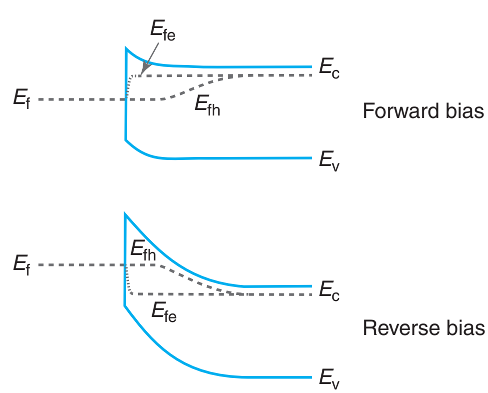


FIGURE 7.18
Energy band diagrams in metal–semiconductor junction in forward and reverse bias showing the location of the quasi-Fermi level for electrons.

This discussion reveals that there is an additional minority carrier component to the Schottky diode current. Under most situations, this component is very small and can be safely neglected. Nevertheless, under special conditions (high Schottky barrier height, strong forward voltage), it can play a significant role and might need to be considered.

7.4 CHARGE–VOLTAGE CHARACTERISTICS OF IDEAL SCHOTTKY DIODE

As we learned in the PN diode, in order to describe the dynamic behavior of a Schottky diode in a circuit, we need a model for the stored charge. If the voltage across the diode changes, the stored charge must change too and the outside circuit must deliver it. This is a real current that needs to be accounted for.

The ideal Schottky diode is a majority carrier device. The current that flows upon the application of a voltage arises from the flow of majority carriers over the energy barrier at the metal–semiconductor interface. In an ideal Schottky diode, unlike a PN diode, the minority carrier concentration is unperturbed from its equilibrium value. There is then no minority carrier storage to contend with. In an ideal Schottky diode, the only stored charge to keep track of is that of the space-charge region which thickness is modulated with the applied voltage as we discussed in Section 7.3.1.

The situation is similar to that of a PN diode. Figure 7.19 sketches the volume charge density in the space-charge region of a Schottky diode with a certain voltage V applied. Whether the voltage is forward or reverse, there is a dipole of charge of a magnitude Q_j across the metal–semiconductor interface. For the example depicted in Fig. 7.19, there is a charge $+Q_j$ in the SCR of the semiconductor that is imaged in an equal amount of charge $-Q_j$ due to a pile up of electrons at the metal surface.

If the voltage that is applied across the diode is increased (made more positive) by a small amount, ΔV, the SCR shrinks a small amount and the pile up of electrons at the metal surface is

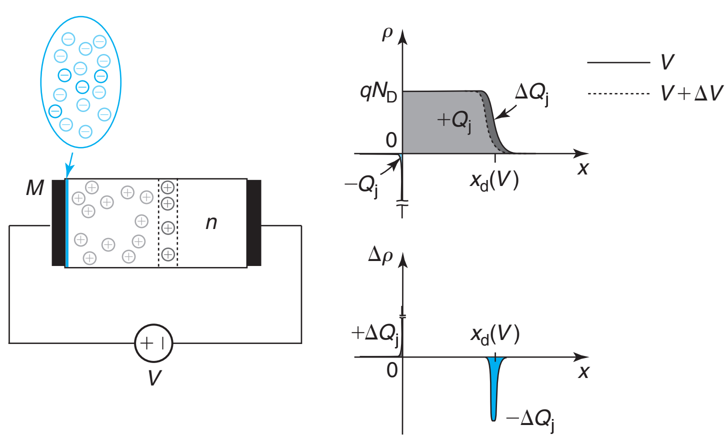


FIGURE 7.19
Change in the volume charge density in a metal/n-semiconductor junction as a result of increasing the voltage across.

also reduced somehow. This means that a small quantity of positive charge, $+\Delta Q_j$, is delivered to the metal (the positive plate) and the same amount of negative charge, $-\Delta Q_j$, is delivered to the semiconductor (the negative plate). The overall charge dipole shrinks, as required.

The stored charge in the SCR can easily be obtained within the depletion approximation. From Eq. (7.21), we have

$$Q_j(V) = -AqN_D x_d(V) = -A\sqrt{2qN_D\epsilon(\phi_{bi} - V)} \tag{7.41}$$

As for the PN diode, this equation can be rewritten as

$$Q_j(V) = Q_j(V = 0)\sqrt{1 - \frac{V}{\phi_{bi}}} \tag{7.42}$$

It should not be surprising that we obtain an identical $Q_j - V$ relationship for the Schottky diode and the PN diode. Essentially, the electrostatics of the SCR for both devices is identical. The graph of Q_j versus V for the PN diode of Fig. 6.21 applies here.

7.5 EQUIVALENT CIRCUIT MODELS FOR THE IDEAL SCHOTTKY DIODE

A device equivalent circuit model is a circuit-like description of its electrical behavior. With the physics of the Schottky diode and the PN diode related in so many ways, their equivalent circuit models are also bound to be very close. In fact, topologically, the equivalent circuit models for these two devices are identical.

We studied the equivalent circuit models for the ideal PN diode in Section 6.5. We saw that the large-signal ideal PN diode model contains a "charge-less" ideal diode element that embodies its rectifying I–V characteristics plus a charge storage element in parallel that represents the SCR and the minority carrier charge. The equivalent circuit model for the Schottky diode should look exactly the same except for the absence of the minority carrier charge in the charge storage element. For convenience, we graph it again on the left of Fig. 7.20. In this figure, I_s represents the saturation current of the Schottky diode [Eq. (7.36)] and $Q = Q_j$ represents the SCR charge given in Eq. (7.42).

When we discussed the PN diode equivalent circuit models, we also noted the interest in developing a small-signal model that represents in a circuit form the behavior of the device to

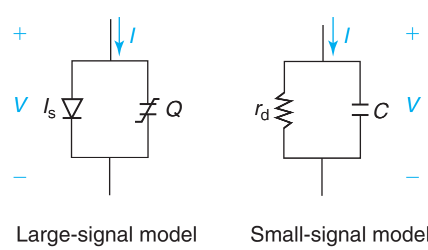


FIGURE 7.20
Large-signal and small-signal equivalent circuit models for the Schottky diode.

small excursions around a bias point. For the PN diode, the ideal diode element is linearized into a resistor and the charge storage element turns into a capacitor. The small-signal equivalent circuit model for an ideal Schottky diode is identical. As graphed on the right of Fig. 7.20, it contains a resistor in parallel with a capacitor. The small-signal resistor is given by an identical expression to that of the PN diode, reproduced here for convenience:

$$r_{\mathrm{d}} = \frac{kT}{q(I + I_{\mathrm{s}})} \simeq \frac{kT}{qI} \tag{7.43}$$

where the approximation applies for moderate and strong forward bias. This is the dynamic resistance of the ideal Schottky diode.

The capacitor is associated with SCR charge storage. As in the case of the PN diode, we can obtain it by differentiating Eq. (7.41) or (7.42) with respect to voltage:

$$C(V) = A\sqrt{\frac{\epsilon q N_{\mathrm{D}}}{2(\phi_{\mathrm{bi}} - V)}} = \frac{C(V=0)}{\sqrt{1 - \frac{V}{\phi_{\mathrm{bi}}}}} \tag{7.44}$$

This is an identical functional expression to that of a PN diode (sketched in Fig. 6.25). As in that case, this expression for the SCR capacitance can also be obtained in analogy with a parallel plate capacitor using $C(V) = A\epsilon/x_{\mathrm{d}}(V)$ and then using Eq. (7.21).

Also in a close parallelism with a strongly asymmetric PN diode, the inverse square of $C(V)$ in Eq. (7.44) is linearly proportional to N_{D}, the doping level in the semiconductor. This is a standard way to measure this plus the Schottky barrier height in Schottky diodes, as shown in the example of Fig. 7.21.

7.6 NONIDEAL AND SECOND-ORDER EFFECTS

In this section, we chip away at the long list of assumptions that was made in the definition of the ideal Schottky diode and discuss some important nonidealities and second-order effects.

7.6.1 Series resistance

In our analysis of the ideal Schottky diode of Fig. 7.3, we neglected the presence of any series resistance. When we derived the expressions for the I–V characteristics, we assumed that the entire voltage applied to the terminals of the device appears across the SCR. But we know that there are ohmic drops at the bottom contact of the diode and in the quasi-neutral body of the device. Just like in the PN diode, these ohmic drops can affect the I–V characteristics of the device at high currents in a significant way.

Since the I–V characteristics of the Schottky diode are functionally identical to those of the PN diode, the role of series resistance is also the same in both devices. We refer to Eq. (6.90) for a modified diode equation that incorporates series resistance and to Fig. 6.32 for a manifestation of the series resistance in its I–V characteristics. Series resistance is also captured in an equivalent circuit model of the Schottky diode in the same way as for the PN diode (see Fig. 6.33).

The simple Eq. (6.90) does a remarkably good job in describing the I–V characteristics of a Schottky diode. An example is shown in Fig. 7.22 that shows the I–V characteristics of a 1N5711

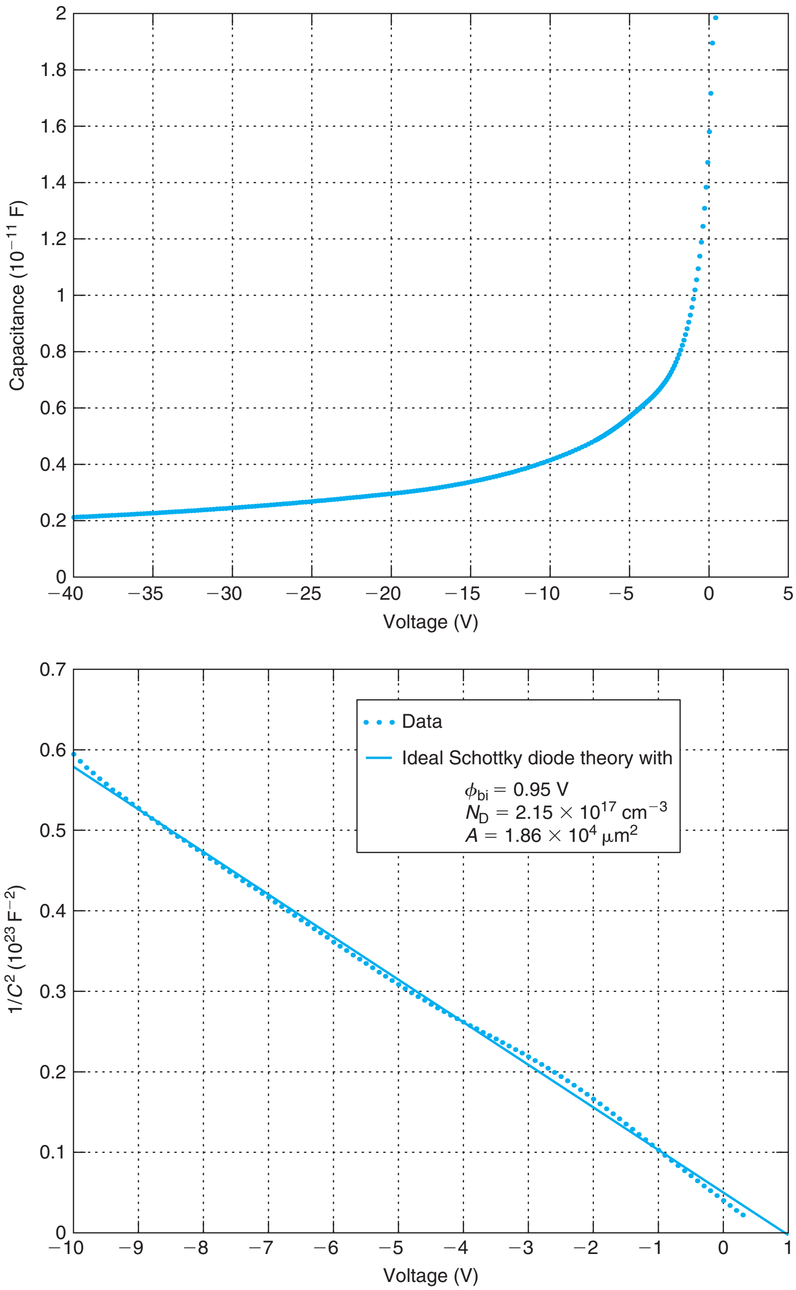


FIGURE 7.21
C versus V and $1/C^2$ versus V for a GaN Schottky diode [Reprinted with permission from Sergei Krupenin].

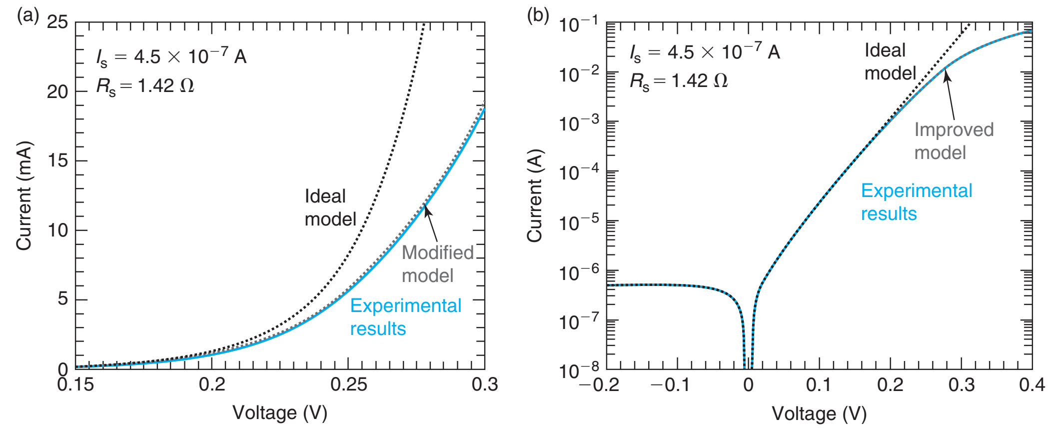


FIGURE 7.22
Experimental and modeled *I–V* characteristics of a 1N5711 Schottky diode. (a) In linear scale; and (b) in semilog scale. The ideal diode model refers to Eq. (7.35). The improved model refers to Eq. (6.90) which also applies to the Schottky diode [Reprinted with permission from Matt Fox].

Schottky diode and the predictions of the model given by the ideal diode equation as well as Eq. (6.90) with suitable values for I_s and R_s. As you can see, the modified diode equation incorporating series resistance describes very well the I–V characteristics of the diode.

There are two components to the series resistance of the Schottky diode: the contact resistance and the body resistance. Between the bottom metal and the semiconductor, there is a contact resistance. We study the physics of ohmic contacts in detail later on in this chapter. As we discussed in the context of the PN diode, a metal–semiconductor contact resistance is characterized through its contact resistivity, ρ_c. The contact resistance is given by the product of ρ_c and the area of contact. Therefore, for the Schottky diode of Fig. 7.3, we have

$$R_c = \frac{\rho_c}{A} \tag{7.45}$$

Then there is the resistance of the body. This is associated with the QNR as majority carrier electrons drift through this region to support the diode current. The resistance is simply the geometrical resistance of this region given by

$$R_n = \frac{1}{A}\frac{w_n - x_d}{q\mu_e N_D} \tag{7.46}$$

The voltage dependence of R_n through that of x_d is generally negligible. The total resistance of the diode is given by the sum of R_c and R_n.

Schottky diodes are frequently utilized in applications in which it is important to minimize any parasitic ohmic drop as this adds to power dissipation. In high-frequency applications, R_s degrades the dynamic response of the diodes, as we will discuss in Section AT7.4. For these reasons, a key consideration in Schottky diode design is to minimize series resistance.

7.6.2 Breakdown voltage

In a Shottky diode in reverse bias, a small trickle current flows due to electrons in the metal that have enough energy to overcome the Schottky barrier height at the metal–semiconductor interface and get injected into the semiconductor. As the reverse voltage increases, the magnitude of the electric field in the depletion region increases too. For sufficiently high reverse bias, the flowing electrons might gain enough energy from the electric field and suffer impact ionizing collisions as they travel through the depletion region. The additional carriers that are generated contribute to the total current which in this way increases over the ideal value given by Eq. (7.36) above. For high enough reverse bias, avalanche breakdown might take place and the current increases abruptly to very high values. The manifestation of breakdown in the *I–V* characteristics of a Schottky diode is similar to that of a PN diode (Fig. 6.34). The voltage at which avalanche breakdown occurs is the *breakdown voltage* of the device. This is a very important parameter since it limits the highest reverse bias that the device can support.

As in so many other aspects, the physics of avalanche breakdown in Schottky diodes and PN diodes are closely related. So, in principle, we could use the theory presented in Section 6.6.4 to model the breakdown voltage of a Schottky diode. From our study of PN diode breakdown, we would then conclude that the breakdown voltage in a Schottky diode is mostly set by the doping level of the semiconductor and should be rather independent of the metal that is used. We can also expect that the higher the doping in the semiconductor, the lower the breakdown voltage.

In practice what is found is that for the same doping level in the semiconductor, the breakdown voltage of a Schottky diode is significantly smaller than what is typically obtained in a PN diode. The reason for this has to do with the presence of a high electric field at the sharp edges of the metal that leads to premature breakdown. The situation is illustrated in Fig. 7.23. This problem does not exist in PN diodes because the high-field region is inside the semiconductor and away from the metal contacts.

Several solutions to this problem are sketched in Fig. 7.23. One of them relies on the *metal overlap* over an SiO_2 window that defines the active area of the diode. The edges of the metal in this way are not in direct contact with the semiconductor. A *moat* structure sometimes is feasible in certain processes. In this approach, as shown, the semiconductor is slightly etched in the active area of the diode. The curvature of the periphery of the moat smoothes out the electric field from the edge of the metal.

A PN junction *guard ring* is also effective. In a design on an n-type well, a p-region is inserted right underneath the periphery of the diode, as sketched in Fig. 7.23. At the edges of the metal, this produces a PN junction in parallel with the Schottky junction. This combination can support a higher reverse voltage. This approach has the drawback of the extra space needed in the lateral dimension to accommodate the p-region. This makes the diode bigger and in consequence less attractive economically. There is also the danger in this design that the PN junction might turn on in forward bias. If that happens, the minority carrier charge associated with the PN diode considerably slows down the combined device. This can occur, in particular, if a metal with a high Schottky barrier height is being used. Careful design of the p-doping level is required to avoid this problem. This is typically done using a device CAD tool since the problem is intrinsically two dimensional.

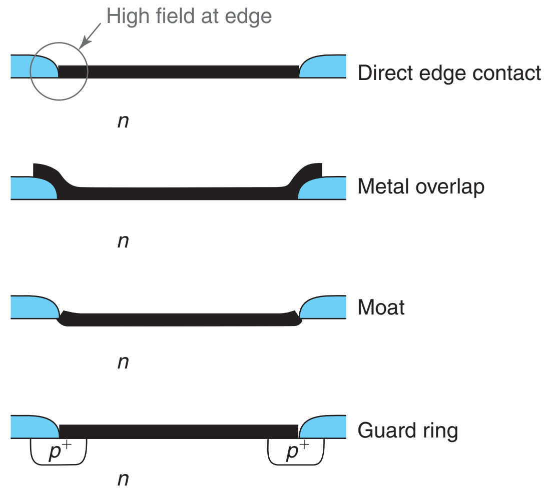


FIGURE 7.23
A direct overlap of the metal edge and the semiconductor results in high electric fields at the corner that lead to premature reverse breakdown (top). This can be mitigated through the use of a metal/oxide overlap, a moat, or a PN junction guard ring.

7.7 INTEGRATED SCHOTTKY DIODE

Schottky diodes have been in use in a variety of applications for many years. For a long time, they were used as “clamps” in the bipolar transistor TTL digital logic family. When placed in parallel with the base–collector junction of the BJT, the forward branch of the *I–V* characteristics of the Schottky diode limits the maximum forward voltage in the base–collector junction and prevents the bipolar transistor from entering saturation. In modern analog circuits, integrated Schottky diodes are very frequently used in such diverse applications as track and hold circuits in analog-to-digital converters and in the pin driver circuits of IC test equipment. The Schottky diode is the preferred choice for such applications because of its small forward voltage and its fast on–off switching speed. Schottky diodes are particularly useful as detectors and mixers in communications and radar applications at the very high frequency end of the spectrum where active devices (that is, devices with gain, such as transistors) have poor performance or are not available altogether. Their low intrinsic capacitance allows them to handle signals at frequencies beyond most other devices.

In spite of their great usefulness in modern microelectronics, cost considerations dictate that Schottky diodes be engineered by taking advantage of process modules that have been developed for other circuit components. Rarely are new process modules introduced for the sole purpose of fabricating Schottky diodes. For example, the metal used for the Schottky barrier is often the ohmic contact metal. Device engineers must show resourcefulness and imagination to design Schottky diodes with low parasitics that meet the required specifications based on existing process modules.

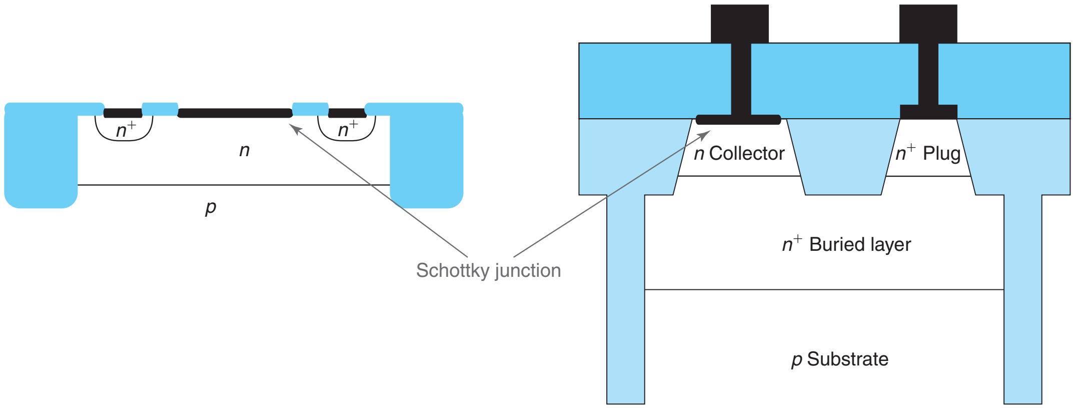


FIGURE 7.24
Schematic cross sections of two integrated Schottky diodes fabricated on a CMOS process (left) and on a bipolar process (right).

Figure 7.24 shows the cross section of two typical integrated Schottky diodes. One is fabricated in a CMOS process (left) and the other one is made on a bipolar process (right). In the CMOS process example, the n-well of the p-MOSFET is used as cathode and a metal placed in direct contact to the well serves as anode (the Schottky junction itself). The n-well is contacted using the body contacts of the p-MOSFET. In the bipolar design, the collector of a Si npn bipolar transistor serves as cathode and a metal placed in direct contact to it is the anode. The n^+-subcollector or buried layer and the n^+-collector plug provide a low resistance path for the current from the cathode to the surface of the semiconductor.

An integrated Schottky diode does not have the isolation problems of an integrated PN diode that were discussed in Section 6.7.1. Because minority carrier injection does not take place, there is no "hidden" parasitic bipolar transistor to worry about. Still, the device needs to be properly isolated. In both examples in Fig. 7.24, the Schottky diode incorporates a parasitic substrate PN diode. Recognizing this is important for two reasons. To insure proper circuit operation, the n-well should never become forward biased with respect to the p-type substrate. Under DC, this can be insured by connecting the substrate to the most negative power supply that is available. Under transient conditions, making sure of this is harder and requires appropriate circuit simulations. The second reason to be aware of regarding the parasitic substrate PN diode is its associated junction capacitance. Depending on the circuit configuration, this can seriously affect the dynamics of the diode and needs to be part of the design process.

An additional consideration in an integrated Schottky diode is its series resistance. R_s affects the performance of a Schottky diode in just about any circuit application. All components of the series resistance must be well understood and modeled. In Section 6.7.2, we showed how to compute the series resistance of a PN diode in a typical case. The procedure is very much geometry dependent but there is no fundamental difference between the Schottky diode and the PN diode. Refer to Sections 6.7.2 and 7.6.1 for more details.

Given that in a typical IC process, there is some freedom to choose the body and the metal of an integrated Schottky diode, it is important to critically think through this and be mindful

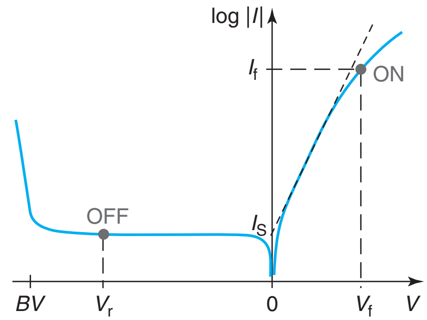


FIGURE 7.25
***I–V* characteristics of Schottky diode showing the two typical operating points in a switching application. For small-signal applications, the device is usually biased in the ON state.**

of the trade-offs involved. To discuss this, consider the I–V characteristics of a Schottky diode in Fig. 7.25. For the purpose of this discussion, let us assume that what is desired is a diode that switches from an ON state where it must support a certain amount of forward bias current I_f while dropping a very low forward voltage V_f, and an OFF state where the diode must be capable of blocking a sufficient reverse voltage V_r while conducting a very small amount of reverse bias current.

The choice of Schottky metal is of prime importance. The higher the Schottky barrier height of the metal, the more forward voltage in the ON state is needed to supply a given current, but the smaller the reverse leakage current in the OFF state. Also, a high Schottky barrier height results in increased temperature sensitivity of the I–V characteristics. The doping level in the bulk is also a key design parameter. A high doping level minimizes series resistance and switching speed but results in more capacitance and a smaller breakdown voltage.

The vertical extension of the bulk is also important. In designs with a buried layer, such as the one shown on the right of Fig. 7.24, the vertical thickness of the bulk of the diode should be enough to entirely accommodate the depletion region at breakdown. Otherwise, premature breakdown will occur. Excessive body thickness, on the other hand, adds series resistance.

Left in the design process is selecting the area of the diode. This is the prerogative of the circuit designer who does this taking into consideration the specific requirements of each diode in its circuit context. The smaller the diode area, the lower the capacitance, and the lower the reverse leakage current. A smaller diode footprint is also more economical. However, in small diodes, the series resistance and the forward voltage that is required to deliver a certain current are larger.

7.8 OHMIC CONTACTS

Electrical current flows in and out of microelectronic devices through its ohmic contacts. A perfect ohmic contact does not present any resistance to current. Real ohmic contacts, on the other hand, drop a small voltage when current flows and therefore present a parasitic resistance. If excessive, this can degrade device performance.

Ohmic contacts are in essence metal/semiconductor junctions with a very large saturation current density. They can support substantial current in forward or reverse bias with little voltage across. When $V \ll kT/q$ in a metal–semiconductor junction, the exponential in the $J-V$ characteristics can be linearized ($e^{qV/kT} \simeq 1 + qV/kT$), to obtain

$$J \simeq A^* T^2 \exp \frac{-q\varphi_{\mathrm{Bn}}}{kT} \frac{qV}{kT} = \frac{V}{\rho_{\mathrm{c}}} \tag{7.47}$$

In the limit of small V, the current through the ohmic contact is *linear* in V for forward and reverse bias. This allows us to define a figure of merit for an ohmic contact called the *ohmic contact resistivity*, ρ_{c}, as in Eq. (7.47). The units of ρ_{c} are $\Omega \cdot \mathrm{cm}^2$. A good ohmic contact is characterized by a very small value of ρ_{c}. For Si, good ohmic contacts display $\rho_{\mathrm{c}} \leq 10^{-7}\ \Omega \cdot \mathrm{cm}^2$ (or $10\ \Omega \cdot \mu\mathrm{m}^2$, an easy number to remember). This means that when a current density of $10^5\ \mathrm{A/cm}^2$ flows through them, the ohmic drop across the contact is only 10 mV.

How does a metal–semiconductor junction become a good ohmic contact? Equation (7.47) seems to indicate that the only degree of freedom provided to the engineer is the selection of a metal with a very small Schottky barrier height. While that certainly helps, something else can also be done that is far more effective, i.e. increasing the doping level right below the metal. To understand the reasons for this, we need to look beyond thermionic emission limited transport.

As the doping level increases in a metal–semiconductor junction, the extent of the depletion region decreases. This can be seen in equilibrium in Eq. (7.15). If you put numbers to this expression, you will realize that for a typical ϕ_{bi} of 0.5 V, at a doping level of about $N_{\mathrm{D}} \simeq 10^{19}\ \mathrm{cm}^{-3}$, the depletion region thickness in thermal equilibrium is of a magnitude comparable to the de Broglie wavelength at room temperature (which is about 7.6 nm, see Section 1.1.2). At these dimensions, electron tunneling suddenly becomes rather likely.

Tunneling is a quantum mechanical phenomenon that enables carriers with insufficient energy to "bore" through energy barriers. While classically forbidden, the quantum mechanical nature of the electron allows tunneling to happen with a finite probability. The energy band diagrams of Fig. 7.26 schematically illustrate tunneling in n-type and p-type ohmic contacts. In this figure, the Fermi level has been drawn inside the conduction band for n-type semiconductor and the valence band for the p-type semiconductor to indicate the fact that tunneling becomes prevalent when the doping levels are degenerate. Examining an n-type ohmic contact first (left of Fig. 7.26), the application of a voltage across the contact misaligns the Fermi level on the metal with respect to the semiconductor. For both polarities of the voltage, this presents electrons on the semiconductor or the metal with empty states on the other side of the metal–semiconductor barrier. By tunneling through the barrier, electrons can then lower their energy. In this manner current flows. The thinner the barrier, the higher the tunneling probability, and the higher the current that will flow through the contact for a given voltage.

For a p-type ohmic contact (right of Fig. 7.26), we need to examine this picture in somehow more detail since conduction in the metal takes place through electrons, but conduction through the semiconductor involves holes. When the quasi-Fermi level for holes on the semiconductor is slightly above the Fermi level on the metal, electrons in the valence band of the semiconductor can tunnel to empty states into the metal. This leaves holes behind on the semiconductor that can now support the current away from the contact into the body of the semiconductor. For

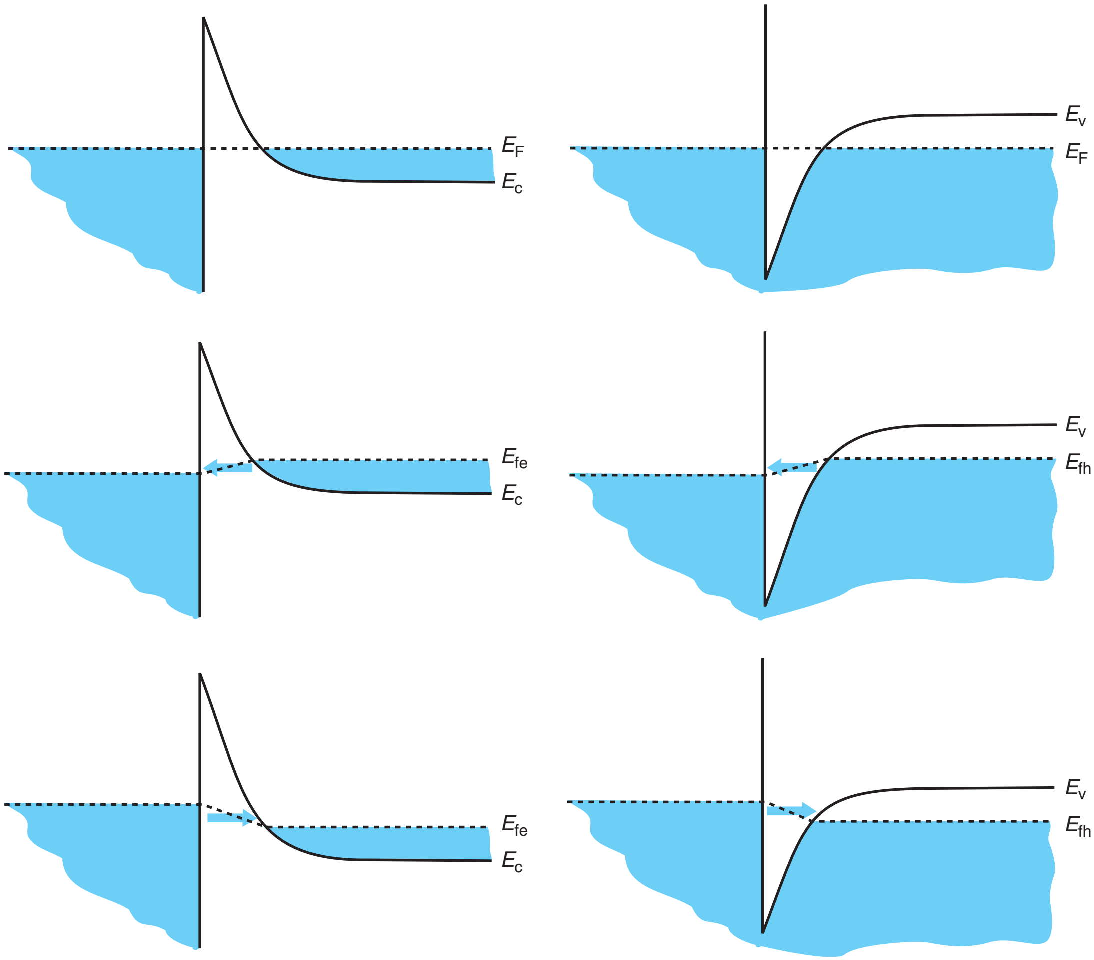


FIGURE 7.26
Energy band diagrams illustrating electron tunneling in n-type (left) and p-type (right) ohmic contacts. Top row represents thermal equilibrium. Second row and third row represent situations for $V > 0$ and $V < 0$, respectively. The arrow represents direction of electron tunneling.

the reverse polarity, electrons in the metal tunnel into the valence band of the semiconductor consuming holes and terminating a hole current that flows from the body of the semiconductor into the contact.

A property of tunneling is that the tunneling current depends exponentially on the inverse thickness of the barrier. Once tunneling becomes significant, the tunneling probability increases very quickly with further enhancements of the doping level. This is then by far the most effective and practical way of engineering low-resistivity ohmic contacts. Using metals with small

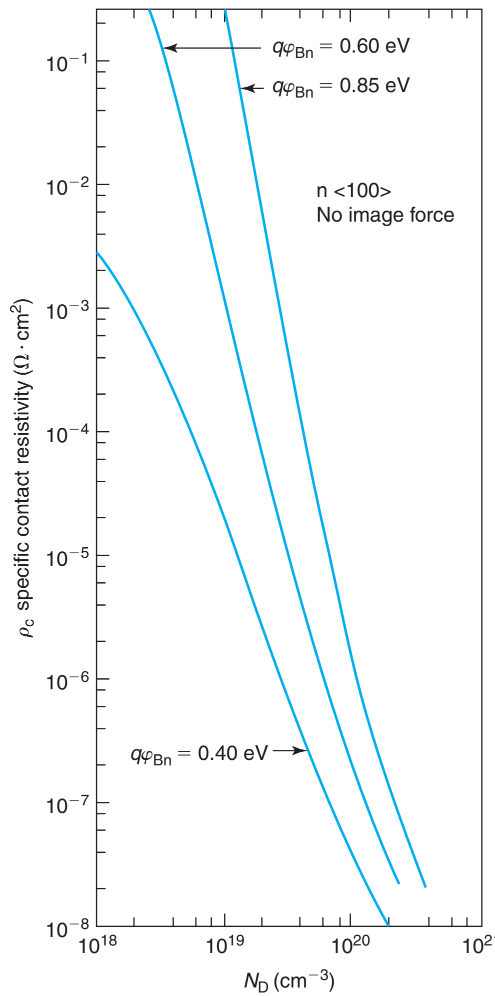


FIGURE 7.27
Calculation of ohmic contact resistivity for ohmic contacts on n-Si as a function of doping level and Schottky barrier height [Adapted from S. E. Swirhun, PhD thesis, Electrical Engineering at Stanford University, 1987].

Schottky barrier heights also helps, but it is not required. Figure 7.27 shows ohmic contact resistance as a function of doping level and Schottky barrier height. As seen, ρ_c decreases steeply at high doping levels and reaches values of the order of $10^{-8}\ \Omega \cdot cm^2$ at about a doping level in the mid $10^{20}\ cm^{-3}$ regime.

In a well-designed device, wherever an ohmic contact is needed, a heavily doped region is introduced. The contact resistance of an ohmic contact of area A_c is to the first order:

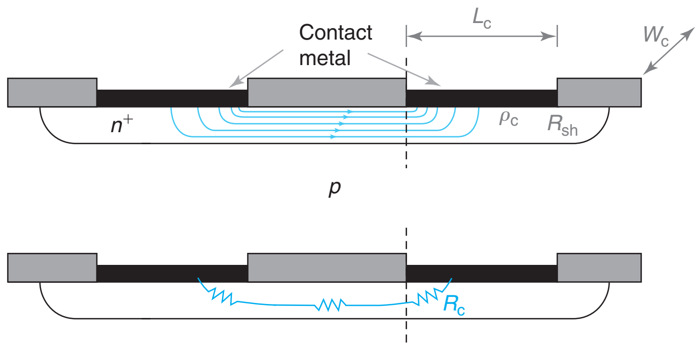


FIGURE 7.28
Top: schematic of current flow in lateral ohmic contact structure (the blue lines represent electric current lines). Bottom: equivalent resistance model defining the contact resistance.

$$R_c = \frac{\rho_c}{A_c} \tag{7.48}$$

The bigger the contact area, the smaller the contact resistance.

Equation (7.48) applies only if the current flow is normal to the metal–semiconductor interface across the entire contact area. This equation, however, is not suitable for lateral contacts, that is, contacts in which the current in the semiconductor ends up flowing parallel to the semiconductor surface. These are very common contacts in semiconductor devices and require special treatment. This is presented next.

7.8.1 Lateral ohmic contact: transmission-line model

Consider the situation of Fig. 7.28. It depicts two ohmic contacts applied to a thin n^+ layer on a p-type substrate. When a voltage is applied to one ohmic contact with respect to the other, current flows down one contact, then laterally through the n^+ layer and then up the other contact. The current is confined to the n^+ layer due to the presence of the PN junction. This kind of lateral contact structure is widely used in devices. Of this type are, for example, the source and the drain contacts of a MOSFET and the contacts to the base of a bipolar transistor.

An equivalent circuit model of this structure is shown in the bottom diagram of Fig. 7.28. We can think of this as three resistances in series. Two are associated with the contacts and one with the lateral n^+ region in between. The lateral contact resistance, R_c, lumps the resistance all the way from the ohmic metal to the leading edge of the contact (dashed line). We wish to derive an expression for R_c in terms of the relevant parameters. The ohmic contacts are characterized by a contact resistivity ρ_c. The length of the contacts is L_c and their width is W_c (into the paper). The n^+ layer has a sheet resistance R_{sh}.

It is clear that R_c is not given by an expression like Eq. (7.48). That expression implicitly assumes that the current density is uniform across the metal/semiconductor interface. Looking at Fig. 7.28, the situation here is quite different. Because of the finite resistance of the semiconductor layer, the current will tend to bunch on the leading edge of the contact. The current density will be higher there and drop as we proceed further away from that edge. If the contact

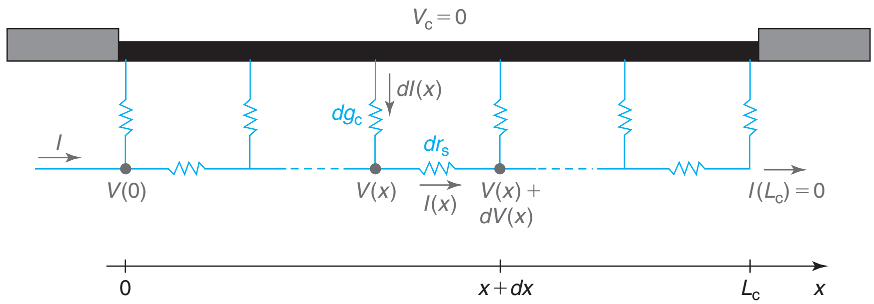


FIGURE 7.29
Transmission-line equivalent circuit model for ohmic contact on right (from dashed line) of Fig. 7.28.

is long enough, the current density will become negligible sufficiently far away from the leading edge of the contact. Clearly, the resistance of this contact (as defined above) is higher than given by Eq. (7.28) on two counts. First, current only flows through a fraction of the contact area. Second, there is a contribution to the contact resistance that comes from the semiconductor.

This is an interesting problem that is best solved through what is called a *transmission-line model*. This is an effective way to analyze distributed problems such as this one. A transmission-line equivalent circuit of the contact structure on the right (from the dashed line on toward the right) is shown in Fig. 7.29.

This electrical problem can be described by a distributed resistance network as shown in Fig. 7.29. A pair of resistors is associated with each elemental length of the contact dx. In each segment, we can define a differential contact *conductance* as $dg_c = (W_c/\rho_c)dx$ (the conductance of the elemental contact vanishes as dx goes to zero) and a differential lateral semiconductor resistance as $dr_s = (R_{sh}/W_c)dx$. We assume that a current I flows into this contact structure from the left. $x = 0$ is the leading edge of the contact and we place its voltage at $V(0)$. The contact has a length L_c. At the other end of the contact, no current can emerge and $I(L_c) = 0$. With these definitions, the contact resistance is $R_c = V(0)/I$. In order to get R_c, we need to solve for $V(x)$ and $I(x)$.

At location x, Kirchoff's voltage and current laws yield, respectively

$$dV(x) = -I(x)dr_s = -I(x)\frac{R_{sh}}{W_c}dx \tag{7.49}$$

$$dI(x) = -V(x)dg_c = -V(x)\frac{W_c}{\rho_c}dx \tag{7.50}$$

From this, we can write

$$\frac{dV(x)}{dx} = -I(x)\frac{R_{sh}}{W_c} \tag{7.51}$$

$$\frac{dI(x)}{dx} = -V(x)\frac{W_c}{\rho_c} \tag{7.52}$$

Taking a derivative on both sides of Eq. (7.52) and substituting Eq. (7.51) yields

$$\frac{d^2I(x)}{dx^2} - \frac{I(x)}{L_t^2} = 0 \tag{7.53}$$

Here, we have defined the *transfer length* as

$$L_t = \sqrt{\frac{\rho_c}{R_{sh}}} \tag{7.54}$$

It is best to write the solutions to this second-order linear differential equation as a sum of hyperbolic functions:

$$I(x) = I_1 \sinh\left(\frac{x}{L_t}\right) + I_2 \cosh\left(\frac{x}{L_t}\right) \tag{7.55}$$

We obtain coefficients I_1 and I_2 by matching boundary conditions $I(0) = I$ and $I(L_c) = 0$ to finally obtain

$$I(x) = I\left[\cosh\left(\frac{x}{L_t}\right) - \coth\left(\frac{L_c}{L_t}\right)\sinh\left(\frac{x}{L_t}\right)\right] \tag{7.56}$$

Inserting this result into Eq. (7.52) yields the voltage distribution along the semiconductor:

$$V(x) = \frac{\sqrt{\rho_c R_{sh}}}{W_c} I\left[\coth\left(\frac{L_c}{L_t}\right)\cosh\left(\frac{x}{L_t}\right) - \sinh\left(\frac{x}{L_t}\right)\right] \tag{7.57}$$

At $x = 0$, this equation yields

$$V(0) = \frac{\sqrt{\rho_c R_{sh}}}{W_c} I \coth\left(\frac{L_c}{L_t}\right) \tag{7.58}$$

The contact resistance of this contact, defined as $V(0)/I$ is

$$R_c = \frac{\sqrt{\rho_c R_{sh}}}{W_c} \coth\left(\frac{L_c}{L_t}\right) \tag{7.59}$$

We find that, in general, the contact resistance of a lateral contact depends on the contact resistivity itself, the sheet resistance of the semiconductor layer underneath and the length and width of the contact. Note how $R_c \sim 1/W_c$, as it should be.

Figure 7.30 sketches the dependence of R_c on L_c, the length of the contact. There are two interesting limits to this expression. If the contact length is much shorter than the transfer length, $L_c \ll L_t$, we can consider this a *short contact* and R_c becomes (see Taylor series expansion of $\coth x$ around $x = 0$ in Appendix D):

$$R_c \simeq \frac{\rho_c}{W_c L_c} \tag{7.60}$$

In this limit, the resistivity of the contact dominates the contact resistance and the contribution to this from the resistance of the semiconductor is negligible because the contact is very short. Notice how in this case, the current density is uniform across the contact.

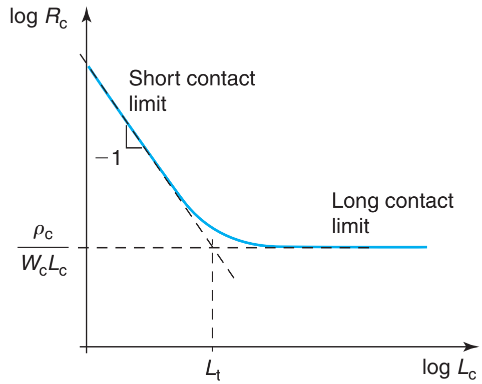


FIGURE 7.30
Evolution of contact resistance with contact length in a lateral ohmic contact.

The other limit is that of the *long contact*. In this case, $L_c \gg L_t$. Since for $x \gg 1$, $\coth x \simeq 1$, we have

$$R_c \simeq \frac{\rho_c}{W_c L_t} \tag{7.61}$$

In this case, the contact resistance scales with the transfer length L_t and is independent of the actual contact length. In this limit, the transfer length becomes the effective contact length. For long contacts, the current and the voltage drop exponentially away from the leading edge of the contact. After a few transfer lengths, there is no more current left and the actual length of the contact is irrelevant.

Understanding the scalability of lateral ohmic contacts is important in device design. It is obvious that one does not want to design a contact that is longer than two or three transfer lengths as this does not reduce the contact resistance but it increases the area that the contact consumes.

7.8.2 Boundary conditions imposed by ohmic contacts

We are now ready to justify two assumptions made elsewhere in this book about ohmic contacts. The first one refers to our statement that through an ohmic contact, we can "grab" on the majority carrier Fermi level at the semiconductor surface. Figure 7.26 justifies this assertion. It is clear in this figure that if the contact resistance is very small, only a small split of the Fermi level between the metal and the semiconductor is required to support a sizable current. In consequence, if the Fermi level of the metal is pulled up or down by a battery, the Fermi level of the semiconductor is brought along with it.

The second assumption refers to the excess minority carrier concentration at an ohmic contact. We assumed in Chapter 5 that it is always zero, or in other words, that the minority carrier concentration is in equilibrium at an ohmic contact. This is now easy to understand. Figure 7.31 shows the energy band diagram of an ohmic contact on an n-type semiconductor in a situation where there are excess holes in the neighborhood. The electric field associated with the

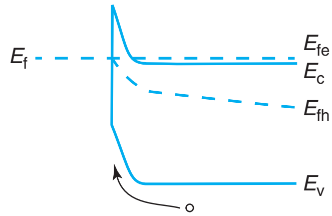


FIGURE 7.31
Energy band diagram of an ohmic contact to an n-type semiconductor with excess hole concentration in its vicinity. The electric field associated with the metal–semiconductor interface effectively "pulls" holes toward it where they recombine.

metal–semiconductor junction pulls holes from the semiconductor toward its interface with the metal where they recombine due to the high density of defects and dangling bonds. In consequence, in the vicinity of the ohmic contact, the hole concentration is never too far away from its equilibrium value. The same applies for electrons in a p-type ohmic contact.

7.9 SUMMARY

- At the interface of a typical metal–semiconductor junction, charge redistribution takes place. A consequence of this is that a depletion region is created on the semiconductor side. A second consequence is that an energy barrier to majority carrier flow appears between the metal and the semiconductor.
- Under bias, the barrier that blocks carrier flow from the semiconductor to the metal is modulated by the applied voltage, while the barrier in the reverse direction is unchanged. This results in the rectifying behavior of the Schottky diode.
- In most circumstances, current in a Schottky diode in forward bias is dominated by the process of emission of carriers into the metal.
- Under bias, the extension of the depletion region in the semiconductor of a Schottky diode is changed by the applied voltage. This produces a capacitive effect.
- Under most circumstances, a Schottky diode under bias does not store minority carriers. This is the key reason for their fast response to dynamic signals. The dominant time constant of a Schottky diode is the RC time constant of the depletion capacitance and the total series resistance.
- Practical design of Schottky diodes attempts to achieve a balance among a small time constant, sufficient forward conduction, small reverse current, high maximum reverse voltage, and small area. The challenge to the microelectronic device designer is to accomplish this constrained by the use of established processes. This is because dedicated processes are rarely available to the fabrication of Schottky diodes.
- Good ohmic contacts present a negligible barrier to current flow in and out of the semiconductor. The key to accomplish this is to heavily dope the semiconductor directly underneath the metal. The higher the doping level, the lower the contact resistivity, the most important figure of merit of an ohmic contact.

- In lateral ohmic contacts, the sheet resistance of the semiconductor layer underneath the contact plays a role. There is a transfer length beyond which longer contacts do not result in smaller contact resistance.

7.10 FURTHER READING

Metal-Semiconductor Contacts and Devices by S. S. Cohen and G. Sh. Gildenblat, VLSI Electronics Microstructure Science, Vol. 13, Academic Press 1986 (ISBN 0-12-234113-9, TK7874.V56 vol. 13). This high-level book is entirely dedicated to Schottky structures. Several chapters are of interest to us. Chapter 2 describes a generalized theory of transport in the metal–semiconductor junction including drift and diffusion, tunneling, and minority carrier injection. Chapter 3 presents experimental techniques to determine the Schottky barrier height. Chapters 4–6 deal with ohmic contact characterization and technology. Although the technology portions are quite dated, the book is still a useful reference for fundamental concepts.

AT7.1 NONIDEAL SCHOTTKY BARRIER HEIGHT OF METAL–SEMICONDUCTOR JUNCTIONS

When discussing the band line up of a metal/n-type semiconductor junction in thermal equilibrium in Section 7.2.2, we concluded that the Schottky barrier height is given by the difference in the work function of the metal and the electron affinity of the semiconductor [Eq. (7.3)]. This is very relevant from an engineering point of view as it suggests that by appropriate selection of the metal, a wide range of Schottky barrier heights should be available to the device designers.

The reality is somehow different. This can be seen in Fig. 7.32 that graphs the experimental Schottky barrier height of metal–semiconductor junctions prepared with different metals on n-type and p-type Silicon. This figure shows that $q\varphi_{Bn}$ tends to increase as W_M increases but the dependence is significantly weaker than expected from simple theory (line). A similar weak dependence of $q\varphi_{Bp}$ with W_M is observed. A wide distribution of Schottky barrier heights is available by using different metals but the range is much narrower than expected from the Schottky–Mott relation. Interestingly, for a given metal, the sum $q\varphi_{Bn} + q\varphi_{Bp}$, within experimental error, adds up to the bandgap of Si, just as stated in Eq. (7.7).

Understanding at a fundamental level the situation depicted in Fig. 7.32 is an active topic of research in modern solid-state physics. In this book, we cannot do justice to the various theories that are currently under consideration. Nevertheless, it is important for device designers to develop an appreciation of some of the critical issues involved and how to think about energy band line ups that are not ideal (as defined in the main body of this chapter). This is the purpose of this Advanced Topic.

When one stops to think about it, it is not surprising that the simple approach to predicting Schottky barrier heights described in Section 7.2.2 fails. Through Eq. (7.3), the Schottky barrier height, which is a purely interfacial property of a metal and a semiconductor, is calculated using information from the bulk characteristics of its constituents. This approach ignores the very presence of an interface between these two materials. It is right at that interface where everything happens.

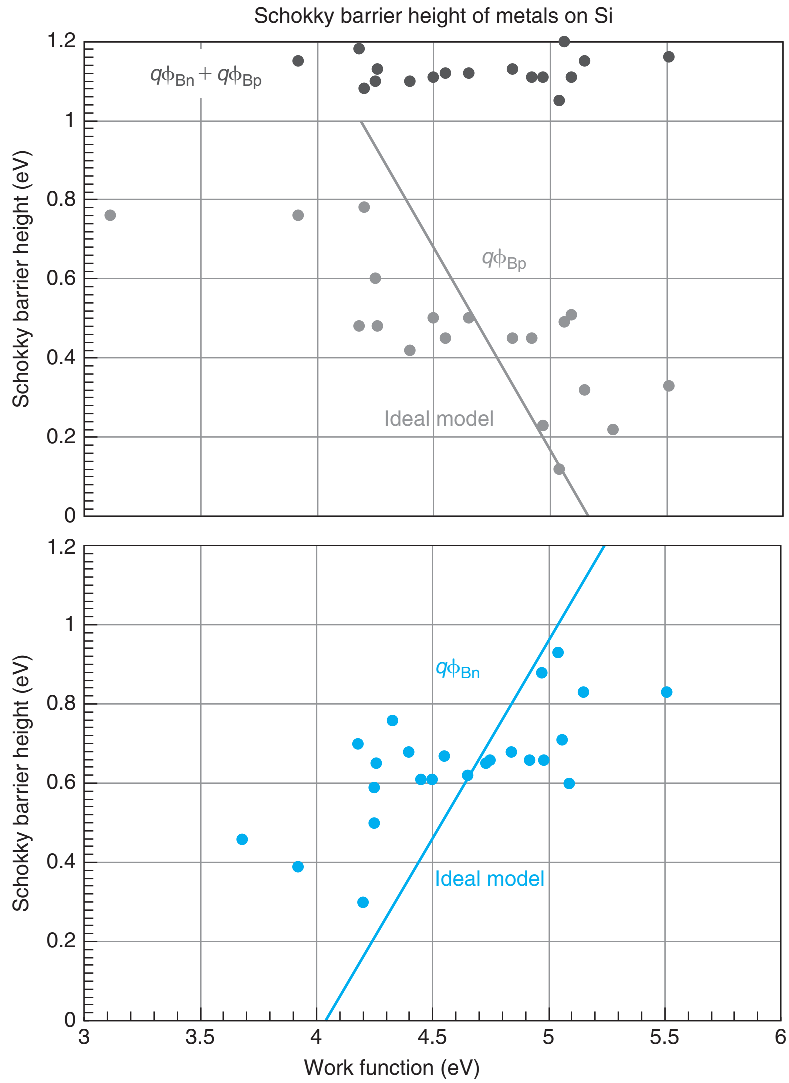


FIGURE 7.32
Schottky barrier height of several metals on n-type and p-type Si, as well as their sum, as a function of metal work function. [© J.A. del Alamo].

Just about every metal under almost any circumstance reacts in some form when deposited over a semiconductor. In careful experiments it has been observed that under the right conditions, the Schottky barrier height of a given metal–semiconductor junction depends on the cleaning process, the deposition and postdeposition annealing techniques that are used as well as the surface orientation of the semiconductor and the thickness of the metal. This strongly suggests that the details of the atomic bonding and interfacial structure of the

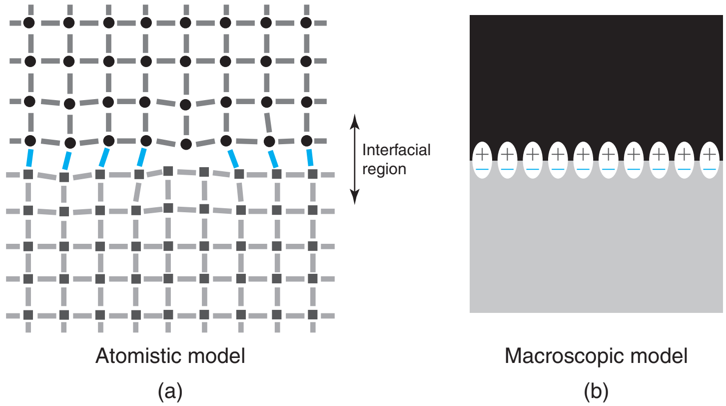


FIGURE 7.33
Models for metal–semiconductor interface. (a) Atomistic model showing interfacial bonding, dangling bonds on either side and lattice deformation that can extend a few monolayers into each material. (b) Macroscopic model showing a fixed dipole of charge at the interface.

metal–semiconductor interface should affect the Schottky barrier height. Atomistic calculations of metal–semiconductor interfaces actually confirm this. How can we understand this?

The interface between a metal and a semiconductor is actually a messy affair. A conceptual sketch is shown on the left of Fig. 7.33. This figure illustrates how, in general, the semiconductor and the metal lattices are different (in fact, metals are often polycrystalline) and as a result, atoms do not precisely line up across the interface. Right at the boundary, some metal–semiconductor atomic bonds are formed leaving some dangling bonds on either side. Also, the crystal lattices on both sides get deformed to a few monolayers in depth into each material. It is then reasonable to expect that the interfacial region of a metal semiconductor junction ends up with different electronic properties than the bulk of either material.

From an electrical point of view, this relatively disorganized interface region when averaged over enough area is very likely to produce a charge dipole across the interface (right of Fig. 7.33). This does not mean that there is an overall loss of charge neutrality but that there is a slight separation of positive and negative charge at the interface (electrons and core ions) with a few more electrons ending up on one side than on the other. The sign and strength of this dipole depends on the atomic details of the interface.

A dipole of charge at the interface results in a sharp electric field confined to a small region and a small but finite electrostatic potential buildup that affects the overall band line up. From simple electrostatics, a charge dipole with a density N_δ and an average charge separation d gives rise to a potential buildup given by

$$\Delta\phi_\delta = \frac{qN_\delta d}{\epsilon} \tag{7.62}$$

We can put some numbers to this expression. If we have a charge dipole density of 10^{13} cm^{-2} with an average charge separation of 1 nm and we use the permittivity of Si, the potential build up at the interface ends up being 0.16 V. This is well in line with the discrepancies that

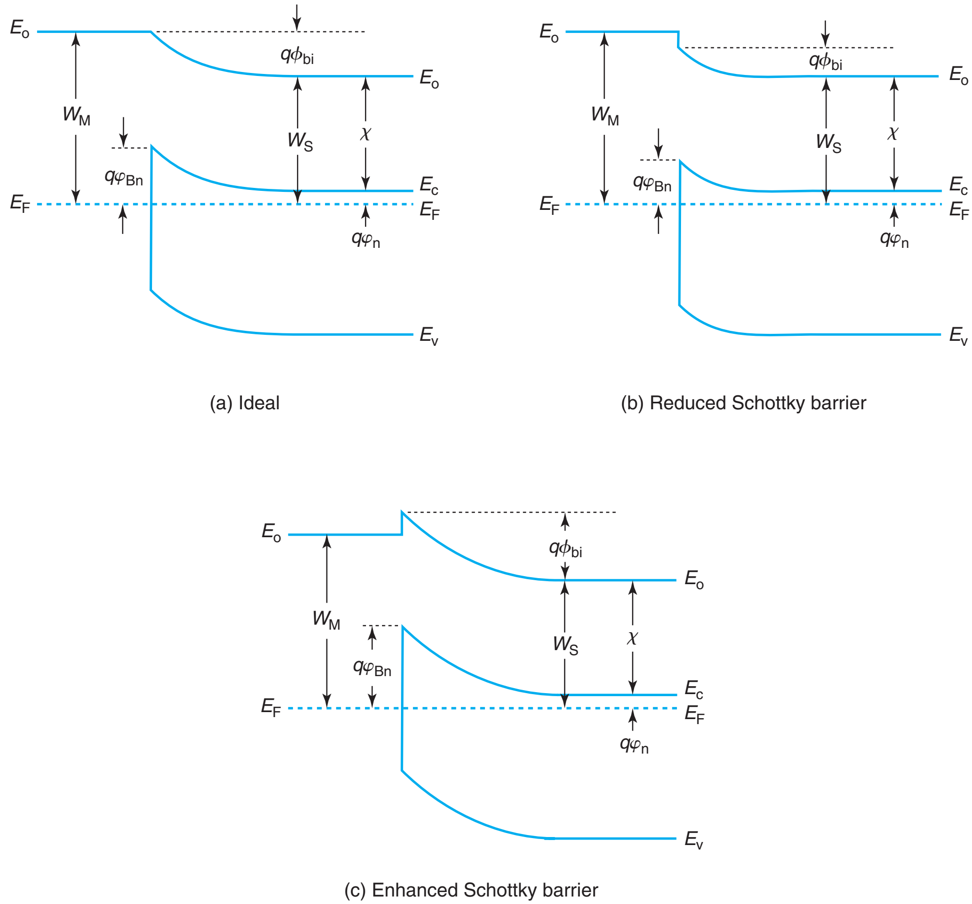


FIGURE 7.34
Energy band diagrams of metal–semiconductor interfaces in the presence of interfacial charge dipole. (a) Ideal case; (b) charge dipole reduces Schottky barrier height; and (c) charge dipole enhances Schottky barrier height.

we see between the simple theory and the experiments in Fig. 7.32. Note that a dipole density of 10^{13} cm^{-2} is not that large when compared with the atomic density of a typical semiconductor surface. For example, in (100) Si, the atomic density is 6.7×10^{14} cm^{-2}.

How is the band diagram modified if there is a charge dipole at the metal–semiconductor interface? This is shown in Fig. 7.34. Depending on the sign of the interfacial dipole, the Schottky barrier height can be reduced or enhanced with respect to the ideal value. As we have studied before, in the absence of a charge dipole, the difference between vacuum levels in the metal and the semiconductor of a metal–semiconductor junction in thermal equilibrium is given by the difference in work functions. The Schottky barrier height is then the difference between the metal work function and the electron affinity (left figure). Depending on the sign of the interfacial charge dipole, the potential build up across the semiconductor will have to adapt so that the overall vacuum level difference ends up unchanged. This can then reduce (middle figure) or enhance (right picture) the Schottky barrier height with respect to the ideal value.

The key lesson from this section is that the Schottky barrier height of a metal–semiconductor system is strongly affected by the details of the interfacial region and should be measured in a practical situation. This understanding also opens the possibility for interface engineering as a way to tune the Schottky barrier height of a metal–semiconductor pair.

A final observation. Remember how for a given metal–semiconductor pair, we concluded in Section 7.2.2 that the Schottky barrier height on the n-type semiconductor and on the p-type semiconductor add up to the bandgap [Eq. (7.7)]. The presence of a dipole of charge at the metal–semiconductor interface is evidently a property of the metal and the semiconductor and the details of fabrication and has nothing to do with the doping type of the semiconductor. As a result, the shifts that the interfacial charge dipole introduces are identical for both semiconductor types and they cancel out when the Schottky barrier heights are added. Equation (7.7) remains valid.

AT7.2 DRIFT-DIFFUSION MODEL FOR *I–V* CHARACTERISTICS

This Advanced Topic presents the drift-diffusion model for the *I–V* characteristics of the Schottky diode. Here we think in terms of a metal/n-type semiconductor Schottky diode that is in forward bias. The drift-diffusion model assumes that the rate-limiting step to the current in the device is the process of electron transport across the SCR and not the process of electron emission into the metal.

Under forward bias, as discussed in the main body of this chapter, diffusion prevails over drift in the SCR and there is a net flow of electrons from the body of the device, across the SCR toward the metal/semiconductor interface. If we continue to neglect the role of minority carrier holes, the diode current is all supported by electrons and, in general, can be written as the sum of its drift and diffusion components:

$$J_t \simeq J_e = q\mu_e n\mathcal{E} + qD_e\frac{dn}{dx} = qD_e\left(-\frac{qn}{kT}\frac{d\phi}{dx} + \frac{dn}{dx}\right) \tag{7.63}$$

We can multiply both sides of this expression by $\exp\left(-\frac{q\phi}{kT}\right)$, to get

$$\begin{aligned} J_t \exp\left(-\frac{q\phi}{kT}\right) &= qD_e\left[-\frac{qn}{kT}\frac{d\phi}{dx}\exp\left(-\frac{q\phi}{kT}\right) + \frac{dn}{dx}\exp\left(-\frac{q\phi}{kT}\right)\right] \\ &= qD_e\frac{d}{dx}\left[n\exp\left(-\frac{q\phi}{kT}\right)\right] \end{aligned} \tag{7.64}$$

We now integrate across the SCR:

$$J_t\int_{0^+}^{x_d}\exp\left(-\frac{q\phi}{kT}\right)dx = qD_e n\exp\left(-\frac{q\phi}{kT}\right)\Bigg|_0^{x_d} \tag{7.65}$$

where we have brought J_t out of the integral because in steady state, the total current is constant in space.

To perform the integral on the left-hand side, we need an expression for $\phi(x)$. For a general case under bias, starting from Eq. (7.13), we can easily write

$$\phi(x) = -(\phi_{bi} - V)\left(\frac{x^2}{x_d^2} - \frac{2x}{x_d} + 1\right) \qquad \text{for } 0 \le x \le x_d \tag{7.66}$$

Integrating this does not yield an analytical solution. However, since the biggest contribution to the integral is around $x = 0$, we can expand $\phi(x)$ in that region:

$$\phi(x) \simeq -(\phi_{bi} - V)\left(1 - \frac{2x}{x_d}\right) \qquad \text{around } x = 0 \tag{7.67}$$

The integral can now be easily performed:

$$\int_{0^+}^{x_d} \exp\left(-\frac{q\phi}{kT}\right) dx \simeq \frac{kTx_d}{2q(\phi_{bi} - V)} \exp\frac{q(\phi_{bi} - V)}{kT} \tag{7.68}$$

For the right-hand side, we note that $x = x_d$ is the edge of the depletion region where $\phi(x_d) = 0$ and $n(x_d) = N_D$. $x = 0$ is the metal–semiconductor interface where $\phi(0^+) = -(\phi_{bi} - V)$. Under the drift-diffusion approximation, the electron concentration at $x = 0^+$ is unperturbed from equilibrium. This is consistent with the notion that the rate-limiting step is electrons reaching the interface. Electron emission into the metal can happen at much faster rate. We can then use the result from thermal equilibrium that was given in Eq. (7.20). All together, we have

$$qD_e n \exp\left(-\frac{q\phi}{kT}\right)\Bigg|_0^{x_d} = qD_e N_D\left(1 - \exp\frac{-qV}{kT}\right) \tag{7.69}$$

Assembling now Eqs. (7.65), (7.68), and (7.69) and solving for J_t, we get:

$$J_t = q\mu_e\sqrt{\frac{2q(\phi_{bi} - V)N_D}{\epsilon}}N_c \exp\frac{-q\varphi_{Bn}}{kT}\left(\exp\frac{qV}{kT} - 1\right) \tag{7.70}$$

where we have again used Eq. (7.19) as well as Eq. (7.21).

Multiplying by the area A_j allows us to formulate again the *I–V* characteristics in the classical rectifying form of Eq. (7.35).

The key dependencies of our result in Eq. (7.70) are very much similar to those of the thermionic emission model given by Eq. (7.34). It has the same rectifying voltage dependence and the same exponential dependence on the Schottky barrier height. Actually, if we look at the reverse bias current predicted by this model the result is very intuitive. For large enough reverse bias, Eq. (7.70) simplifies to

$$J_t \simeq -q\mu_e\sqrt{\frac{2q(\phi_{bi} - V)N_D}{\epsilon}}N_c \exp\frac{-q\varphi_{Bn}}{kT} = -q\mu_e|\mathcal{E}_{max}|n(0) \tag{7.71}$$

The reverse bias current is entirely supported by drift at the metal–semiconductor interface and it then takes the classic drift current expression with the electron concentration and the electric field computed at $x = 0$.

It is interesting to look at the ratio of the current predicted by the drift-diffusion model and the thermionic emission model. Using Eqs. (7.70) and (7.34), after some simple algebra, we can find

$$\frac{J_t(DD)}{J_t(TE)} = 4\frac{\mu_e|\mathcal{E}_{max}|}{v_{the}} \tag{7.72}$$

This is really interesting and also makes good sense. The ratio of the currents can be phrased as a ratio of velocities. In the case of the drift-diffusion model, the velocity that matters is the drift velocity at the metal-semiconductor interface. For the thermionic emission model, it is a quarter of the thermal velocity of electrons that matters. Whichever one of these velocities is the smallest limits current transport. Since the peak electric field in a Schottky diode can be quite high even under strong forward bias, in materials in which μ_e is relatively high, it is often the case that thermionic emission is the most restrictive process. The thermionic emission process is therefore the appropriate model to use to describe the *I–V* characteristics of Schottky diodes.

EXERCISE 7.5

Evaluate the ratio of the current obtained in an Al/n-Si Schottky diode with $N_D = 10^{17}$ cm^{-3} *at a forward voltage of* 0.5 V *at 300* K *by the drift-diffusion and thermionic emission models.*

We assume that the electron mobility in the SCR is identical to that of a QNR of the same doping level. Therefore, for this doping level, $\mu_e \simeq 700$ cm^2/V · s.

From Exercise 7.3, we know that for this Schottky diode at this forward voltage, the peak electric field at the Schottky interface is $|\mathcal{E}_{max}| \simeq 3.4 \times 10^4$ V/cm.

We also know that the thermal velocity for electrons at 300 K is about $v_{the} \simeq 2.0 \times 10^7$ cm/s.

All together, then

$$\frac{J_t(DD)}{J_t(TE)} = 4\frac{\mu_e \mathcal{E}_{max}}{v_{the}} = 4 \times \frac{700 \text{ cm}^2/\text{V} \cdot \text{s} \times 3.4 \times 10^4 \text{ V/cm}}{2.0 \times 10^7 \text{ cm/s}} = 4.8$$

Under these conditions, the drift-diffusion model predicts a current that is about 4.8 times higher than the thermionic emission model. Therefore, the process of emission from electrons into the metal is the rate-limiting step and the thermionic emission model is the correct one to use here. This finding is consistent with the verification of the validity of the Bethe condition that was carried out in Exercise 7.4.

AT7.3 EQUIVALENT CIRCUIT MODEL OF SCHOTTKY DIODE FOR CIRCUIT DESIGN

In the modeling of Schottky diodes for circuit CAD, the equivalent circuit model of a PN diode is often used. This works well because the behavior of these two devices is very similar. Referring to Fig. 6.33, in its simplest form, this circuit consists of an ideal diode, a capacitor in parallel with it, and a series resistance.

Several of the model parameters in the diode CAD model have a similar meaning when describing the PN diode and the Schottky diode. Referring to Section AT6.3 that describes the equivalent circuit model for a PN diode for circuit CAD, this is the case for model parameters **N**, **RS**, **IS**, **CJO**, **VJ**, **M**, and **BV**.

Other model parameters take on a different meaning due to the subtle difference in physics between both diodes. For example, looking at the expression of **IS** of Eq. (6.151), and comparing it with I_s for the Schottky diode derived in Eq. (7.36), it is clear that model parameter **EG** plays the role here of the Schottky barrier height. Its value should therefore be smaller than in a PN diode where it plays the role of the bandgap. Also, model parameter **XTI** ideally should have a value of 2, which tends to be smaller than the values needed in PN diodes.

The biggest difference between the Schottky diode model and the PN diode model is in parameter **TT** (transit time). This captures minority carrier storage in the PN diode which does not exist in an ideal Schottky diode. Therefore, **TT** should be set to zero. Actual Schottky diodes in strong forward bias do show some minority carrier effects. Describing this accurately will require the introduction of a finite value to **TT**.

Table 7.1 summarizes the basic set of model parameters used to describe Schottky diodes in a circuit CAD environment. When appropriate, their ideal values are also given.

Table 7.1
Summary of simplest set of Schottky diode model parameters for circuit CAD. Ideal value refers to the simple Schottky diode described in these pages.

Name	Parameter Description	Units	Ideal Value
IS	Saturation current	A	-
N	Ideality factor	-	1
EG	Schottky barrier height	V	-
RS	Series resistance	Ω	-
CJO	Zero bias depletion capacitance	F	-
VJ	Built-in potential	V	-
M	Grading coefficient	-	0.5
XTI	Saturation current temperature exponent	-	2
TT	Transit time	s	0
BV	Breakdown voltage	V	-

As in the case of the PN diode, additional model parameters are found in more advanced models to describe nonideal leakage currents in reverse bias, temperature dependence, parasitic diode isolation, and the noise characteristics. Also, scaling parameters are introduced to scale the diode geometry from the "nominal" device that was experimentally characterized to extract the model, to any other device size.

AT7.4 SWITCHING CHARACTERISTICS OF SCHOTTKY DIODE

A real uniqueness of Schottky diodes is their fast switching characteristics. In many large-signal applications, Schottky diodes must quickly switch from a strong forward bias with substantial current flow to a high reverse bias characterized by negligible current and vice versa. How quickly a Schottky diode is able to switch between these two states is a key figure of merit for many large-signal applications.

We studied these kinds of transients in the PN diode in Section AT6.4. We learned that they can be quite slow due to minority carrier storage. Since this is absent in a Schottky diode, these devices can switch much faster. This makes them preferable in many applications such as power management and ICs for automatic test equipment.

An appreciation for the limiting mechanisms can be obtained by considering the simple switching example depicted in Fig. 7.35. In this exercise, we have a Schottky diode switching from a forward voltage V_f to a reverse voltage V_r and back (note that the way the voltage is defined in Fig. 7.35, V_r is a negative number). The current through the diode behaves as shown in the figure. Right after the switch-off transient (from V_f to V_r), the current through the diode shows a prominent negative spike that decays away in an exponential way. Similarly, right after the swith-on transient (from V_r to V_f), the diode current also exhibits a prominent positive spike

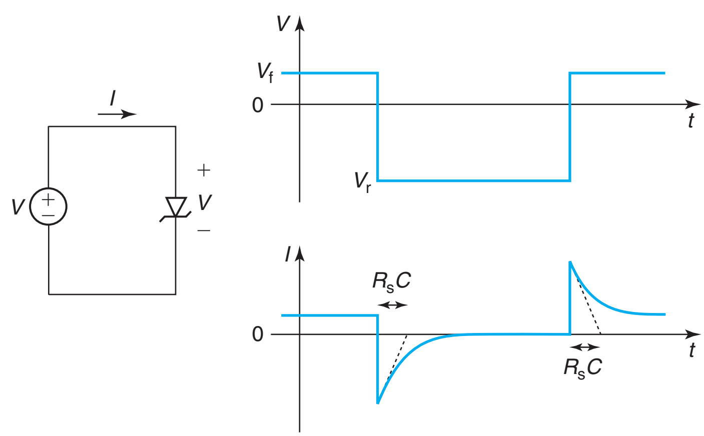


FIGURE 7.35
Switching transients in a Schottky diode. In response to a sudden switch in the voltage applied to the diode, the diode current shows spikes that decay with a characteristic time constant R_sC.

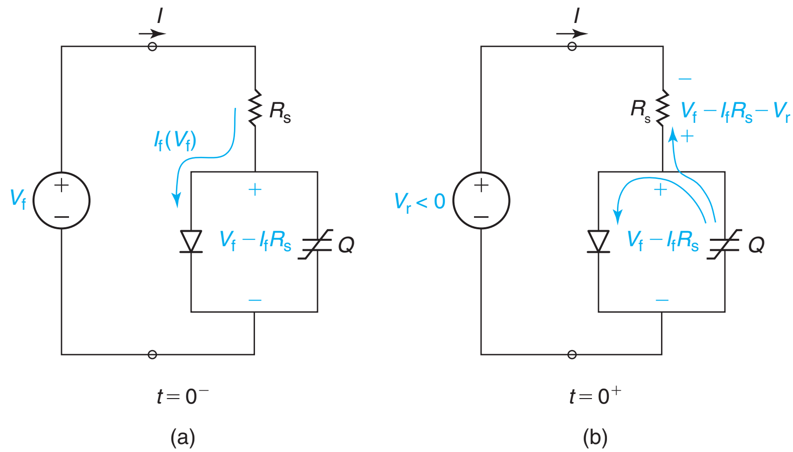


FIGURE 7.36
Switch-off transient in a Schottky diode. (a) At $t = 0^-$. (b) At $t = 0^+$.

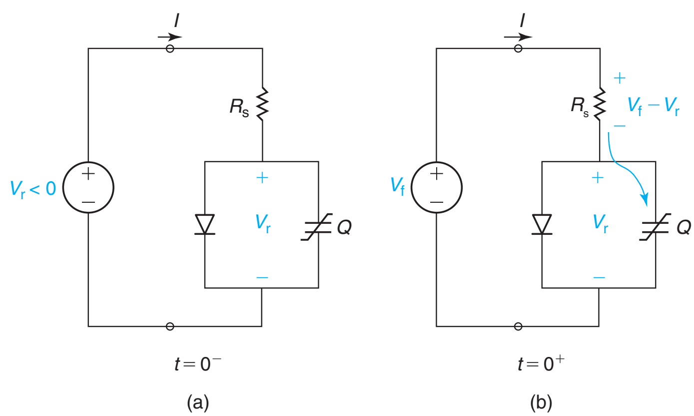


FIGURE 7.37
Switch-on transient in a Schottky diode. (a) At $t = 0^-$. (b) At $t = 0^+$.

that also decays away in an exponential fashion. In order to calculate the magnitude of the current spikes and, more importantly, the time constants of the exponential decays, we need to examine in detail the switching transients. This is best done by substituting the diode for its equivalent circuit model, as shown in Figs. 7.36 and 7.37 for the switch-off and switch-on transients, respectively.

Figure 7.36 shows the details of the current paths during the switch-off transient. At $t = 0^-$, the diode is in forward bias with a voltage V_f across. There is then a forward bias current flowing through the diode of magnitude I_f. This current produces an ohmic drop in the series resistance. As a result, the junction voltage is $V_f - I_f R_s$. As V switches from V_f to V_r, the junction voltage cannot abruptly change and remains with a value $V_f - I_f R_s$. Hence, at $t = 0^+$, the resistor has a voltage $V_f - I_f R_s - V_r$ across. The polarity of this voltage is reversed with respect to its sign at $t = 0^-$. Hence, the current flowing through the diode terminals at $t = 0^+$ is

$$I(t = 0^+) \simeq \frac{V_r - V_f}{R_s} + I_f \tag{7.73}$$

Notice that this current is negative, that is, it is a reverse current.

As $t > 0^+$, the charge storage element associated with the junction starts discharging through R_s as well as through the diode. However, due to the exponential I–V characteristics of the diode, the diode current path gets shut off very quickly. After that, the junction charge element discharges only through the series resistance R_s until it attains its final value V_r.

A rigorous analytical solution of this switching event is complicated because of two reasons. The first one is the initial fast discharge through the diode. Additionally, as we studied above, the junction charge in a Schottky diode depends on voltage. If Q was voltage independent, past a short initial transitory, the time evolution of the diode current would be exponential with a characteristic time constant of a value $R_s C$, where C is the associated junction capacitance. With Q changing with voltage, the decay is not a pure exponential. Still, the characteristic time of the current decay is of the order of R_s times the average C.

In the switch-on transient (Fig. 7.37), V switches from $V_r < 0$ to V_f. At $t = 0^-$, the capacitor associated with the junction has a voltage V_r across and the current going through the diode is the saturation current $-I_s$, a very small value. At $t = 0^+$, the voltage across the diode changes abruptly to V_f, but the junction voltage cannot change. Hence, a voltage $V_f - V_r$ appears across the series resistance of the diode with the polarity indicated in the figure. This produces a current of magnitude:

$$I(0^+) \simeq \frac{V_f - V_r}{R_s} \tag{7.74}$$

Since the junction remains reverse biased, this current spike flows into the junction capacitance that as a result starts charging up. As it does, the voltage across the junction becomes less negative, the voltage across the resistor is accordingly reduced and the charging current is reduced as well. This is a typical RC-dominated transient characterized by an exponential evolution with a time constant equal to $R_s C$. Again, the fact that C depends on the junction voltage results in a transient that is not a pure exponential.

The result of this detailed analysis is that the Schottky diode can respond to large-signal voltage switching in the scale of $R_s C$. Fast switching demands minimization of both C and R_s. This contrasts with the switching times of the PN diode, which in addition to this RC time constant, are delayed by the need to provide or eliminate the stored minority carriers in the quasi-neutral regions.

EXERCISE 7.6

Simulate a switching transient of a Schottky diode in SPICE, as sketched in Fig. 7.35. The forward voltage is +0.45 V. *The reverse voltage is* −3 V. *The SPICE parameters of the diode are* **IS** = 5.5*e* − 13, **N** = 1.03, **EG** = 0.89, **RS** = 11, **CJO** = 3.24*e* − 13, **VJ** = 0.5, **M** = 0.339, **XTI** = 2, *and* **TT** = 0 *(units in Table 7.1). Extract the characteristic time constants of the transients and compare them with simple estimates.*

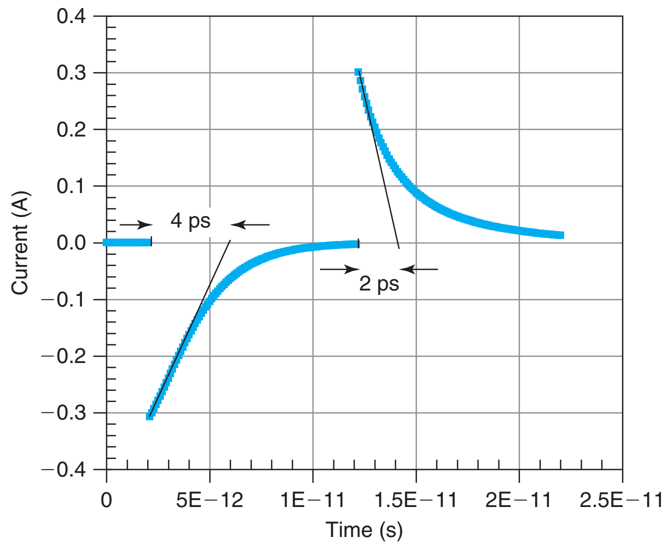


FIGURE 7.38
Switching transients in a Schottky diode as simulated by HSPICE (see Exercise 7.6).

Figure 7.38 shows the output obtained from HSPICE for a case in which there is a delay time of 2 ps, the reverse voltage is applied for 10 ps and the forward voltage is applied for 10 ps. Both turn-on and turn-off transients exhibit a near exponential dependence. For the turn-off transient, the extracted characteristic time constant is about 4 ps. The R_sC time constant expected from the junction capacitance at $V = 0.45$ V is about twice that, 7.8 ps. This suggests that the diode itself is effective in contributing to the discharge of the capacitor at the beginning of the transient.

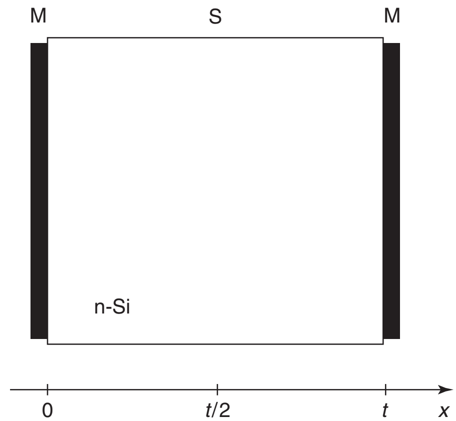
The turn-on transient is characterized by a time constant of about 2 ps. The expected value of the R_sC product corresponding to $V = -3$ V is 1.9 ps. The excellent agreement reflects the fact that in reverse bias, the diode does not prevent the charge of the capacitor.

Right after the transients, the terminal currents are about 300 mA. This agrees with the expected value of $3.45/11 = 0.31$ A.

PROBLEMS

* Denotes a problem that requires studying an Advanced Topic.

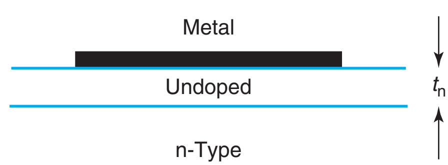
7.1 Consider a thin Si sample with two metallic contacts on each side, as shown in the figure. The structure is at room temperature in thermal equilibrium and the doping is $N_D = 10^{15}\ \text{cm}^{-3}$ everywhere. The Schottky barrier height of both metal–semiconductor junctions is $q\varphi_{Bn} = 0.7$ eV.



For the following three situations, (*i*) calculate the electron and hole concentrations and the electric field at the center of the sample $x = t/2$, (*ii*) sketch the corresponding energy band diagram throughout the structure, and (*iii*) compute the built-in potential of each metal–semiconductor junction.

(a) For a sample thickness of $t = 5$ μm.
(b) For a sample thickness of $t = 1$ μm.
(c) In the limit of a very thin sample.

7.2 The **Mott diode** consists of a metal–semiconductor junction in which the semiconductor layer immediately adjacent to the interface is undoped, as indicated in the figure. Consider a case in which $W_M > W_s$, where W_s refers to the work function of the n-type bulk.

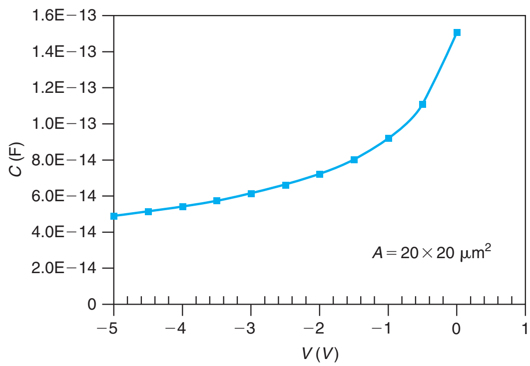


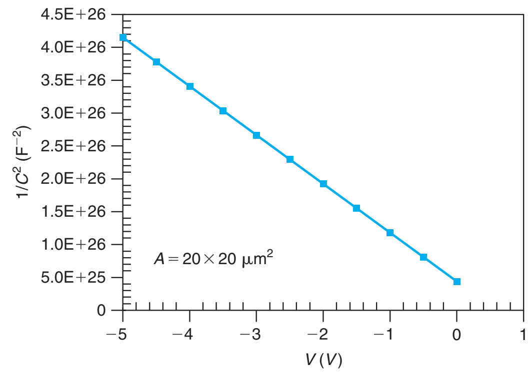
(a) Sketch the space charge, ρ_o, electric field, $\mathcal{E}_o$, electrostatic potential, ϕ_o, and energy band diagram in thermal equilibrium.
(b) Under the depletion approximation, calculate expressions for ρ_o, $\mathcal{E}_o$, and ϕ_o as a function of x in the semiconductor. Leave everything in terms of x_d, the depletion region extension into the n-type region.
(c) Compute an expression for x_d in terms of material parameters.
(d) Modify the expressions for ρ, $\mathcal{E}$, ϕ, and x_d for forward and reverse bias.
(e) Derive an expression for the *C–V* characteristics. Sketch.

7.3 You have received a Si Schottky diode from your research supervisor with the instructions of measuring the Schottky barrier height. This is a rather novel device fabricated by some collaborators of your supervisor using an exotic metal. You are given a single sample.

You have a lot of experience in this procedure. Even though it was not asked of you, you start by carrying out *C–V* measurements at room temperature. Your plan is to proceed with temperature-dependent *I–V* characteristics from which you will extract the Schottky barrier height.

Tragically, as the C–V trace comes to its end, the device blows up. You panic. Another student that is working in your lab suggests that you can extract the Schottky barrier height from the C–V characteristics that you have just obtained. Through a microscope you measure the area of the metal as $20 \times 20\ \mu m^2$. You forgot to ask whether the substrate is n-type or p-type. Below are your measurements. V represents the voltage of the metal with respect to the semiconductor.





Extract the Schottky barrier height from the available information.

7.4 Consider an ideal metal/n-type semiconductor junction. If without changing anything else, you change metals so that the Schottky barrier height $q\varphi_{Bn}$ goes up, how do the following device parameters change? Provide reasons for all your answers.

Current at a certain forward voltage, $I(V_f)$:	$\uparrow$	$\downarrow$	no effect	it depends
Capacitance at a certain forward voltage, $C(V_f)$:	$\uparrow$	$\downarrow$	no effect	it depends
Activation energy of saturation current, $E_a(I_s)$:	$\uparrow$	$\downarrow$	no effect	it depends
Charge in depletion region in equilibrium, $Q_d(V = 0)$:	$\uparrow$	$\downarrow$	no effect	it depends
Dynamic resistance at a certain forward current, $r_d(I_f)$:	$\uparrow$	$\downarrow$	no effect	it depends

7.5* A certain n-Si Schottky diode is characterized by the following set of SPICE parameters: **IS** $= 9 \times 10^{-14}$ A, **N** $= 1.00$, **EG** $= 0.95$, **RS** $= 20\ \Omega$, **CJO** $= 3 \times 10^{-13}$ F, **VJ** $= 0.8$ V, **M** $= 0.5$, **XTI** $= 2$, **TT** $= 0$, **BV** $= 20$ V, all determined at 300 K.

(a) Carry out SPICE simulations of the I–V characteristics of the device at room temperature and plot $|I|$ versus V in a semilog scale from $V = -5$ V to $V = 0.5$ V.

(b) Carry out I–V simulations at different temperatures ($T = 300, 325, 350, 375,$ and 400 K). For each temperature, extract I_s by extrapolating the forward-bias characteristics to $V = 0$. Make an Arrhenius plot of I_s/T^2 versus $1000/T$. Extract the Schottky barrier height from this plot. Compare with **EG**.

(c) Perform SPICE simulations of the C–V characteristics at room temperature. Plot C versus V in a linear scale from $V = -5$ V to $V = 0.5$ V.

(d) Perform SPICE simulations of a switching event as in Fig. 7.35 in the notes from $V = 0.45$ V to $V = -3$ V and back. Extract the characteristic time constants of both switching transitions and compare them with simple estimations carried out as in the notes. Also extract the peak currents at $t = 0^+$ and compare with simple calculations.

7.6* Consider an integrated Al/n-Si Schottky diode implemented in a bipolar process, as shown on the right of Fig. 7.24. The area of the metal/semiconductor interface is 40 μm^2. The total area of the n-tub is 800 μm^2. The doping level in the n-region is 1×10^{17} cm^{-3}. The doping level in the substrate is 1×10^{16} cm^{-3}. $q\varphi_{Bn}$ for Al on n-Si is 0.68 eV.

(a) Derive values for standard model parameters for this diode (except **RS**). Include parameters to characterize the capacitance to the substrate. Model this as an asymmetric PN diode with $\phi_{bi} = 0.9$ V, as discussed in Chapter 6. Refer to the new parameters as **CJOS**, **VJS**, and **MS**.

(b) Compute and sketch the I–V and C–V characteristics of the diode for a wide range of voltages. Ignore parasitic effects. What is the maximum positive voltage that can be applied to the diode? What about the maximum negative voltage?

7.7[1] Consider an ideal metal/n-type semiconductor junction at room temperature. The metal has a work function $W_M = 4.05$ eV, the semiconductor is Si and is doped with $N_D = 10^{17}$ cm^{-3}.

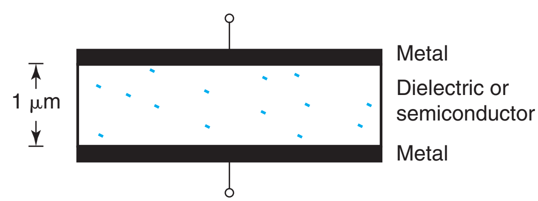
(a) Sketch to scale the energy band diagram in equilibrium.

(b) Compute the built-in potential.

(c) Compute the electron and hole concentrations right at the metal/semiconductor interface.

(d) Estimate the spatial extent of the space-charge region in the semiconductor.

7.8 Consider a parallel-plate capacitor with Al electrodes as sketched below. The separation between the plates is 1 μm.



Calculate the capacitance per unit area of this device when the following materials are utilized as dielectric:

(*i*) SiO_2
(*ii*) intrinsic Si
(*iii*) n-type Si with $N_D = 1 \times 10^{15}$ cm^{-3}
(*iv*) n-type Si with $N_D = 1 \times 10^{17}$ cm^{-3}

In all cases, the contact between the Al plate and the dielectric material is intimate.

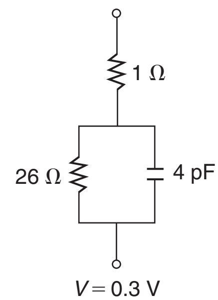
7.9 A Schottky diode biased at a forward voltage of $V = 0.3$ V has the small-signal equivalent circuit indicated below at room temperature. The Schottky barrier height of this diode is $\varphi_B = 0.9$ V.

(a) Estimate the current through the diode for $V = -1$ V. State any assumptions you need to make.

(b) Estimate the forward voltage across the diode for $I = 100$ mA. State any assumptions you need to make.

(c) Estimate the capacitance of the diode for $I = 100$ mA. State any assumptions you need to make.

[1]You will need to study the contents of Chapter 8 before attempting to solve part (d) of this problem.



7.10 Consider a Schottky barrier diode on an n-type Si substrate. Suppose we *increase* the doping level in the semiconductor N_D. Nothing else is changed. Indicate the impact that this would have on the parameters and figures of merit listed below.

Circle one: *increase* =↑, *decrease* =↓, or *no effect*. For each item, give the reason for your choice.

Built-in potential, ϕ_{bi}:	↑ ↓ *no effect*
Current at a certain forward voltage, $I(V_f)$:	↑ ↓ *no effect*
Capacitance at a certain forward voltage, $C(V_f)$:	↑ ↓ *no effect*
Switching time constant, τ:	↑ ↓ *no effect*
Equilibrium electron concentration at the metal–semiconductor interface, $n_o(x = 0)$:	↑ ↓ *no effect*

7.11 Consider a Schottky diode with a 10 μm^2 active metal–semiconductor junction area and a Schottky barrier height of 0.85 eV. This diode is forward biased with $V = 0.6$ V at room temperature. Answer the following questions in the context of the thermionic emission theory of transport in the metal–semiconductor junction.

(a) Calculate the electron concentration at the semiconductor side of the metal–semiconductor interface.

(b) Calculate the net electron velocity at the semiconductor side of the metal–semiconductor interface.

(c) Calculate the diode current.

7.12 *A certain IC foundry offers a process that includes a "nominal" Schottky diode characterized by the following set of SPICE parameters at 300 K: **IS** $= 1e - 13$, **N** $= 1.0$, **EG** $= 0.9$, **RS** $= 10$, **CJO** $= 1e - 12$, **VJ** $= 0.7$, **M** $= 0.5$, **XTI** $= 2$, **TT** $= 0$, and **BV** $= 10$. This "nominal" Schottky diode has a junction area of 10 μm^2. Estimate the 3 dB bandwidth, $f_{3dB} = \frac{\omega_{3dB}}{2\pi}$, of this "nominal" Schottky diode at a forward current of 1 mA and at 300 K.

7.13 Can the built-in potential of a metal–semiconductor junction in thermal equilibrium exceed the Schottky barrier height? Explain your answer. Draw relevant energy band diagrams as needed.

7.14 This problem is about making some early design decisions for a process for a Schottky diode varactor (variable capacitor). The output of this exercise is a first-order sense of the Schottky barrier height of the metal and the doping level of the semiconductor.

The room-temperature specifications of this varactor are (i) a capacitance per unit area at 0 V of $C_o = 1\ \text{fF}/\mu\text{m}^2$ and (ii) the capacitance must change by a factor of 2 between 0 and 2 V (an 100% tuning range). To minimize power consumption, the Schottky diode must operate in reverse bias.

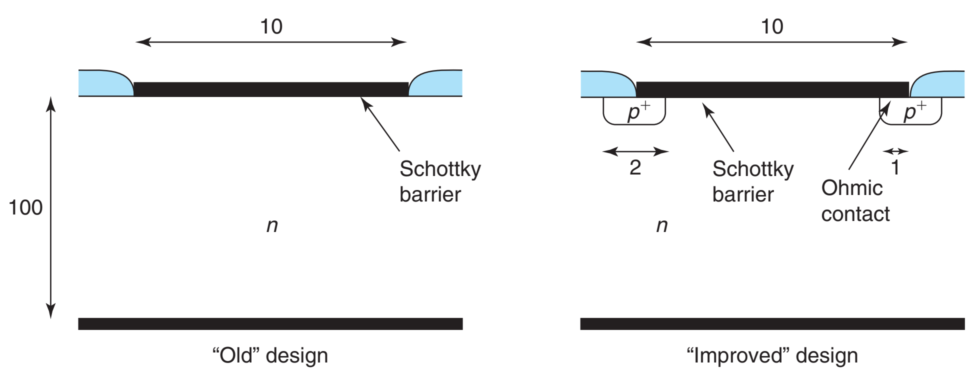
Assume a metal/n-Si structure. Provide values of N_D and $q\varphi_{Bn}$ that meet the design specs.

7.15 [2]In a metal–semiconductor junction that uses a metal with a large Schottky barrier height, an inversion layer of minority carriers might get formed at the metal–semiconductor interface in thermal equilibrium. Consider such a situation on n-type Si with $N_D = 10^{17}\ \text{cm}^{-3}$.

(a) Estimate the minimum Schottky barrier height for which this becomes a concern.

(b) Sketch to scale the energy band diagram of such a metal–semiconductor junction in thermal equilibrium, that is, label all key energy differences and horizontal dimensions.

7.16 In order to improve the breakdown voltage of a Schottky barrier diode, a designer decides to introduce a p$^+$ guard ring, as sketched below. This problem is about evaluating the penalty in capacitance associated with this change.



In both designs, the Schottky barrier height is 0.8 eV, the doping of the semiconductor is $N_D = 10^{16}\ \text{cm}^{-3}$, and the wafer thickness is 100 μm. The ohmic contact to the body of the semiconductor is placed on the bottom surface of the wafer.

In the old design, the Schottky metal contacts the semiconductor over a $10 \times 10\ \mu\text{m}^2$ square window. In the new design, the Schottky metal is also $10 \times 10\ \mu\text{m}^2$. The p$^+$ guard ring is 2 μm wide and is centered around the edge of the metal. The p$^+$ region has a doping level of $N_A = 10^{19}\ \text{cm}^{-3}$.

Assume that any minority carrier physics in the PN diode are dominated by the n-type substrate. Answer all the following questions at room temperature.

(a) Estimate the capacitance of the old design at $V = 0$ V.

(b) Estimate the capacitance of the new design at $V = 0$ V.

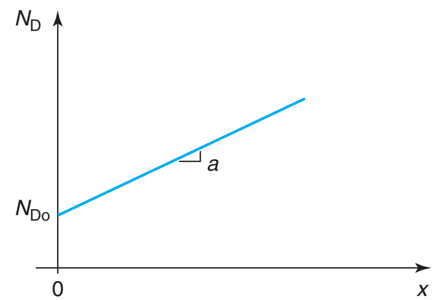
(c) Estimate the capacitance of the old design at $V = 0.5$ V.

(d) Estimate the capacitance of the new design at $V = 0.5$ V.

7.17 Consider a metal–semiconductor junction fabricated on a linearly graded n-type doped semiconductor. The doping distribution in the semiconductor is sketched in the figure below. The grading

[2]Attempt this problem after studying Chapter 8.

coefficient is a. $x = 0$ denotes the metal–semiconductor interface. The Schottky barrier of this structure is $q\varphi_B$.



This problem is about formulating the electrostatics of this problem under the depletion approximation.

(a) Derive an algebraic expression for the spatial dependence of the volume charge density across the structure. Leave expression in terms of the depletion region depth. Sketch.
(b) Derive an algebraic expression for the spatial dependence of the electric field across the structure. Leave expression in terms of the depletion region depth. Sketch.
(c) Derive an algebraic expression for the spatial dependence of the electrostatic potential across the structure. Explicitly state your choice of potential reference. Leave expression in terms of the depletion region depth. Sketch.
(d) Derive an expression that allows for the solution of the depletion region width in terms of known parameters.

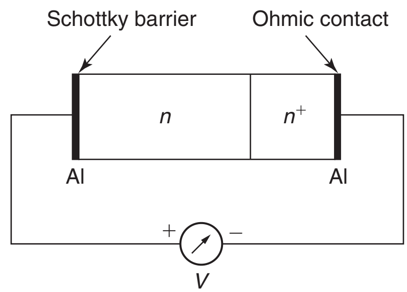
7.18 Consider a metal–semiconductor junction made out of Al on n-type Si with $N_D = 10^{19}$ cm^{-3} at room temperature. For this system, the Schottky barrier height is $q\varphi_{Bn} = 0.68$ eV.

(a) Compute the built-in potential for this structure. Sketch the energy band diagram in thermal equilibrium indicating all key energies. Neglect carrier degeneracy.

Now consider inserting a thin p layer between the metal and the n-type semiconductor. The doping level is $N_A = 10^{17}$ cm^{-3} and the thickness is $x_j = 0.1$ μm.

(b) Qualitatively sketch the volume charge density of this structure in thermal equilibrium. Exploit the fact that $N_D \gg N_A$. Clearly label all instances of net electrical charge. Explain.
(c) Qualitatively sketch the electric field across the structure in thermal equilibrium. Explain.
(d) Qualitatively sketch the electrostatic potential across the structure in thermal equilibrium. Select as reference $\phi = 0$ in the bulk of the n-type region. Explain.
(e) Qualitatively sketch the energy band diagram of this structure in thermal equilibrium. Explain.
(f) Based on what you see in the energy band diagram, at a given forward voltage (metal positive with respect to n-type substrate), would this structure have more or less current than the structure in part (a)? At a given forward voltage, would it have more or less capacitance than the structure in part (a)? Explain.
(g) Calculate the electric field at the metal–semiconductor interface in thermal equilibrium.

7.19 Consider a Schottky diode built on n-type Silicon, as sketched in the figure. The Schottky metal is Al that has a Schottky barrier height on n-type Si of $q\varphi_{Bn} = 0.68$ eV. The doping level in the n region is $N_D = 10^{16}$ cm^{-3}. In order to provide a good ohmic contact to this diode, there is an n$^+$ region with a doping level of $N_D^+ = 10^{19}$ cm^{-3}. Al is also used as the ohmic metal.



Consider this structure in thermal equilibrium at room temperature. Use Boltzmann statistics to treat the n$^+$ region.

(a) Sketch a complete energy band diagram across the entire structure (from metal to metal). Indicate the location of the Fermi level. Assume that both semiconductor regions are wide enough to fully accommodate the depletion regions associated with the metal–semiconductor junctions. Be neat in your sketch.
(b) Compute the built-in potential of the metal–semiconductor junction on the left.
(c) Compute the built-in potential of the n–n$^+$ junction.
(d) Compute the built-in potential of the metal–semiconductor junction on the right.
(e) Is there a voltage difference across the terminals of this device?

7.20 The Bethe condition establishes the applicability of the thermionic emission model to the current–voltage characteristics of a Schottky diode. This problem explores the voltage range of applicability of the Bethe condition. Consider an WSi_2/n-Si Schottky diode at room temperature. The Schottky barrier height for this system is $q\varphi_{Bn} = 0.64$ V. The doping level of the semiconductor is $N_D = 4 \times 10^{17}$ cm^{-3}.

Follow this approach:

(a) In what bias regime, forward or/and reverse, does the Bethe condition become problematic? Explain.
(b) Estimate the mean free path of electrons in the semiconductor.
(c) Estimate the maximum forward and/or reverse voltage for applicability of the Bethe condition in this diode.
(d) At this (these) voltage(s), estimate the current density flowing through the diode.

Chapter 8

The Si Surface and the Metal–Oxide–Semiconductor Structure

This chapter deals with what is perhaps the most important material system in modern microelectronics: the metal–oxide–semiconductor structure or MOS. The MOS structure is at the heart of integrated microelectronics. A basic CMOS inverter is made out of two metal–oxide–semiconductor field-effect transistors (MOSFETs) in which carrier transport is controlled by the field-effect action of a MOS structure. Hundreds of millions of them are now integrated together on a single chip. MOS capacitors are also the storage element of dynamic random-access memories (DRAM). Billions of them are routinely integrated on a memory chip. MOS structures also appear when interconnect metal lines run over an oxide-covered semiconductor surface. In a typical modern microprocessor, there are tens of meters worth of them.

In spite of its seemingly restrictive name, the MOS structure is to be viewed here as a generic sandwich of a highly conducting material on top of an insulator, all on top of a semiconductor (a better term might be MIS, for metal–insulator–semiconductor). The treatment of the MOS structure presented in this chapter can be easily extended to heavily doped polysilicon gates that are the preferred choice in modern CMOS. Similarly, insulators other than silicon dioxide are widely used in modern microelectronics. A prominent example is silicon nitride (Si_3N_4). With some precautions, the MOS theory can be used to treat MIS structures based on semiconductors other than Si and in which the "insulator" is actually a wide bandgap semiconductor. A prominent example is the metal/AlGaAs/InGaAs heterostructure at the heart of modern high-electron mobility transistors (HEMTs).

The goal of this chapter is to understand the changes brought to the electrostatics of the MOS structure upon the application of a voltage to the metal with respect to the semiconductor. Even though current cannot flow through the insulator, the charge distribution in the semiconductor close to the insulator/semiconductor interface is affected in a major way. One can induce a depletion region, accumulate majority carriers, or all together "invert" the semiconductor type and create a thin layer of minority carriers. This "inversion layer" constitutes the conducting channel in most MOSFETs.

An additional goal of this chapter is to continue building up the tool kit for studying semiconductor devices by introducing the Poisson–Boltzmann formulation. This is a powerful formalism to rigorously analyze the electrostatics of the MOS structure. The usefulness of the

Poisson–Boltzmann formulation transcends this structure as it can be applied to other junctions that appear in semiconductor devices.

This chapter places special emphasis in the understanding of the capacitance–voltage characteristics of the MOS structure. As in the metal–semiconductor and PN junctions, the *C*–*V* characteristics summarize in a compact way the electrostatics of the structure. *C*–*V* characteristics form also the basis for a variety of techniques to characterize and diagnose semiconductor surfaces, interfaces, dielectrics, and devices.

This is one of the most important chapters in this book. Solid command of the physics of the MOS structure is essential to understand MOSFETs and CMOS.

8.1 THE SEMICONDUCTOR SURFACE

At a semiconductor surface, the perfect crystalline periodicity of the solid comes to an abrupt end. This affects the transport properties of electrons and holes in its vicinity in a major way. Providing a simple and intuitive picture of what happens at a surface is the goal of this section. Before we do that, let us introduce the concept of *ideal semiconductor surface.*

An "ideal surface" can be thought of as one in which the semiconductor lattice comes to an end but the bulk electronic properties are unmodified by this disruption. Schematically, we can visualize an ideal surface if we introduce an imaginary plane inside the bulk of a semiconductor that carriers cannot traverse. This is illustrated on Fig. 8.1. On one side is the real crystal, on the other side is a "virtual" crystal. Along the ideal surface plane, the semiconductor bonding

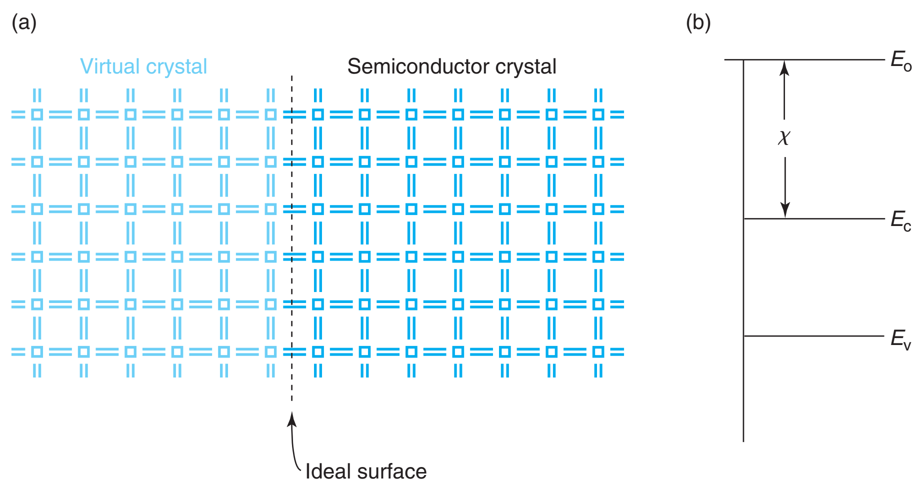


FIGURE 8.1
(a) Schematic diagram of an ideal surface with unperturbed semiconductor bonds. This is as if the real crystal continued into a "virtual crystal" where carriers are forbidden. (b) Energy band diagram in the vicinity of an ideal surface. The bands are "bulk-like" all the way to the surface.

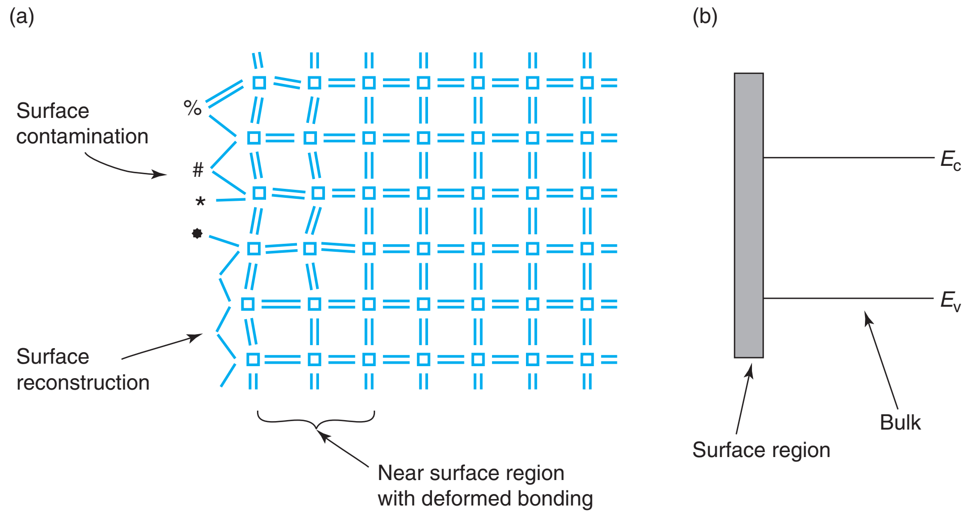


FIGURE 8.2
(a) Schematic illustration of a "real" surface with surface contaminants, surface reconstruction, and a subsurface region with deformed bonding. (b) Schematic energy band diagram in the vicinity of a "real" surface showing a different electron energy distribution near the surface.

arrangement remains undisturbed. Carriers can come up to that plane without feeling its presence in any way. They just simply cannot cross it because there are no allowed states on the other side of the interface for the carriers to go to. Another view of this situation is to consider that a large energy barrier exists right at the surface, as sketched in Fig. 8.1(b). The barrier height for electrons in the conduction band is equal to the electron affinity of the semiconductor. It is even higher for the holes. As a consequence, the hole and electron current components normal to that plane are both zero. This concept of an ideal surface, while naive, is useful. We already encountered it in our discussion on the metal–semiconductor junction. We will invoke it many more times in this book.

In a "real" surface, the four-fold coordination of the semiconductor atoms cannot be preserved. Even if nothing else happens, unsatisfied bonds will be present. These give rise to *surface states* or allowed electronic states at the surface of the semiconductor with an energy distribution that is quite different from that of the bulk. Often, surface states have energies inside the bandgap of the semiconductor. The implications of this are discussed in more detail in Advanced Topic AT8.1. Unsatisfied bonds at the semiconductor surface also represent a rather unstable situation as the atoms are eager to lower their energy by attempting to emulate four-fold coordination. As a consequence, a surface with unsatisfied bonds is very reactive. The surface atoms can find various ways to lower their total energy, as schematically illustrated in Fig. 8.2. One of them is to bond with foreign atoms or molecules, such as O or C or organic contaminants, that might be present in the atmosphere surrounding the surface. A semiconductor surface, if exposed to air, oxidizes and gets contaminated very easily. A second avenue for surface energy lowering is *surface reconstruction* in which the surface atoms bond among themselves in an

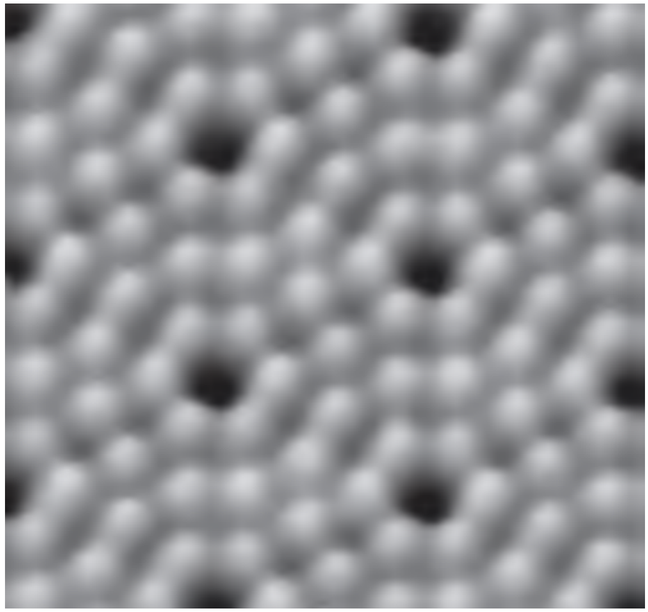


FIGURE 8.3
Scanning tunneling microscope picture of the 7×7 reconstruction of the (111) Si surface. Each round gray feature is associated with one distinct Si atom. The peculiar atomic arrangement pattern that is observed arises from minimization of the surface energy [Courtesy of R. Martel and Ph. Avouris, IBM].

arrangement that is rather different from the one in the bulk. A (111) Si surface, for example, reconstructs itself into a spectacular diamond-shape 7×7 arrangement, pictured in Fig. 8.3.

When considering the impact that a "real" surface has on semiconductor device behavior, we might be tempted to believe that whatever it is, it should be confined to the immediate vicinity of the surface, that is the region where the semiconductor bonding is upset from its bulk arrangement. This deformed region is actually very thin, perhaps two or three monolayers. In consequence, we might be lead to believe that the influence of the surface does not extend more than a few nanometers below the surface. From a crystalline point of view, this argument is true, but from an electrostatic and transport one, it is not. We will learn in this chapter that a surface can modify the electrostatics of the bulk underneath to distances that can be as high as a few microns. Furthermore, we already saw in Chapter 5 that in minority carrier-type situations the influence of surface recombination reaches several diffusion lengths into the bulk. Depending on doping level and processing details, this can be as far as hundreds of microns.

The high reactivity of an exposed semiconductor surface makes it unsuitable for semiconductor device applications. The microelectronics revolution would have not taken place without the discovery of *surface passivation*. This consists of the deposition of a dielectric material or the formation of a thin film on top of a semiconductor surface such that bulk-like bonding at the interface is preserved. This is schematically illustrated in Fig. 8.4.

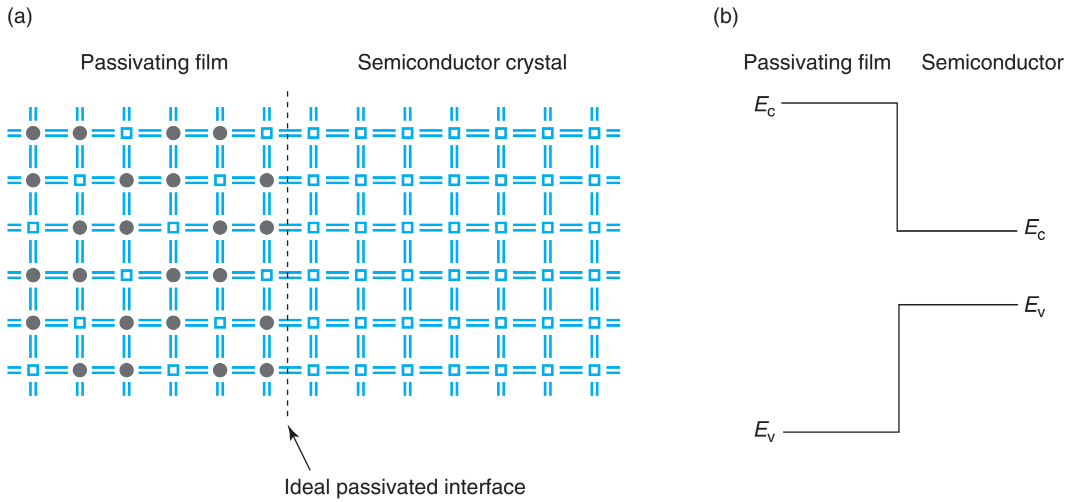


FIGURE 8.4
(a) Schematic illustration of surface passivation. A dielectric film has been placed on top of the semiconductor surface so that bulk-like bonding exists at the dielectric/semiconductor interface. (b) Energy band diagram in the vicinity of a passivated interface. The bands in the semiconductor are "bulk-like" all the way to the interface.

The most important passivating film is SiO_2. When Si is oxidized in an ultraclean environment, Si reacts with O to form SiO_2, which grows on the Si surface. SiO_2 formed this way is an amorphous dielectric material with remarkable properties. One of them is that at the Si/SiO_2 interface, nearly all Si bonds are satisfied. If the gray circles are taken as the O atoms, Fig. 8.4 can be interpreted as an ideal SiO_2 film on a Si surface. Notice that there are twice as many O atoms as Si atoms (the open squares) in the dielectric film. Note also that all atoms in the dielectric as well as at the dielectric/semiconductor interface share a total of eight valence electrons, the ideal bonding configuration.

This simple picture is however misleading in at least two important ways. First, SiO_2 films used in microelectronics are amorphous, that is, they do not have the long-range crystalline order that is suggested in Fig. 8.4. There is some short-range order, to be sure, with a bonding configuration similar to the one implied in this figure, but the film is really made out of multiple tiny crystallites with all kinds of orientations. A second problem is that Fig. 8.4 suggests that the Si/SiO_2 interface is perfectly flat. This is also not the case. Even carefully prepared interfaces have surface roughness on the scale of one to two monolayers. This is seen in the transmission-electron microscope picture of Fig. 8.5 that shows a cross section of a MOS structure used in modern CMOS applications. The MOS structure of the picture has a polysilicon gate on top of a 2.7-nm-thick SiO_2 layer grown on a (100) Si substrate. If one looks in detail, one can see an Si/SiO_2 interface that is not flat but exhibits a roughness of about two monolayers. Surface roughness has very important implications for carrier transport along the Si/SiO_2 interface, as it happens in a MOSFET. This will be explored in Chapter 10. Interestingly, the extraordinary resolution of

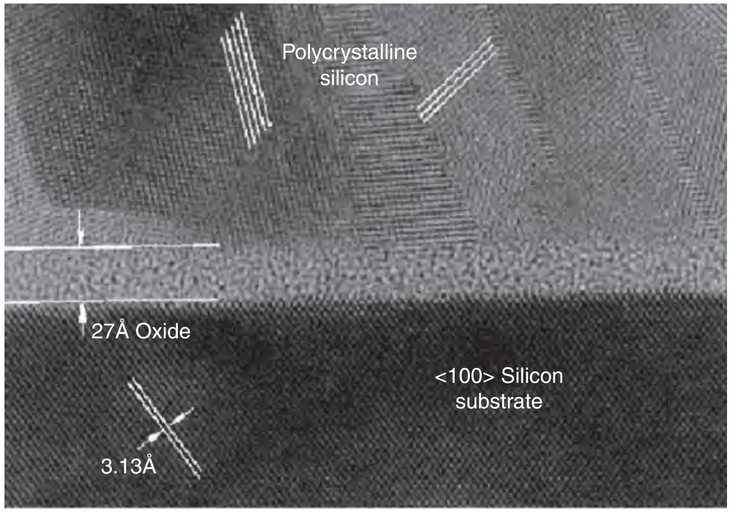


FIGURE 8.5
Transmission electron micrograph of polysilicon MOS structure showing 1-2 monolayer surface roughness at the Si/SiO_2 interface. The gate, made out of heavily doped polysilicon, shows several grains with different orientations. The extraordinary resolution of this picture permits the direct observation of the lattice planes that provides an absolute length calibration [Courtesy of D. Buchanan, IBM].

the picture in Fig. 8.5 allows the direct observation of the lattice planes in the Si substrate. This provides an absolute length calibration for the picture. Also, the polycrystalline nature of the gate is apparent.

In an ideal surface, the energy band diagram is bulk-like all the way to the surface, as sketched in Fig. 8.1(b). The electrons do not “fall off” the semiconductor at the surface because they face an energy barrier equal to the electron affinity of the semiconductor. Electrons need to have an energy equal to the vacuum energy to escape from the semiconductor.

The energy band diagram of a perfectly passivated semiconductor surface is similar to an ideal surface, that is, the semiconductor bands remain bulk-like all the way to the interface. As we discussed in Chapter 1, the band diagram of an insulator is qualitatively identical to that of a semiconductor, except that the bandgap is much wider. The bandgap of SiO_2, for example, is about 9 eV. In consequence, at the insulator/semiconductor interface, there is a large discontinuity in the conduction band and the valence bands that prevents carriers from escaping. This is shown on Fig. 8.4 (b).

The energy band description of the bulk semiconductor is inappropriate in the vicinity of an unpassivated surface. The nonideal bonding arrangement of the surface, the presence of broken bonds and contamination results in an electron state configuration that is rather different from the one that prevails in the bulk. More importantly, the existence of a bandgap devoid of electron states can no longer be guaranteed, and in fact, a general continuum of states is more appropriate. This is pictured on Fig. 8.2(b). A more detail discussion of an unpassivated surface is presented in Advanced Topic AT8.1.

8.2 THE IDEAL METAL–OXIDE–SEMICONDUCTOR STRUCTURE

As in the case of the PN diode and the Schottky diode, it is useful to define the concept of an "ideal MOS structure." This captures the essence of the MOS physics while hiding some of the second-order effects. Later in this chapter, we will relax some of the assumptions made in this section.

A sketch of an ideal MOS structure is shown in Fig. 8.6. It consists of a p-type semiconductor (one can equally define it on an n-type substrate), an oxide of thickness x_{ox}, and a metal film. the metal is referred to as the "gate." The semiconductor is contacted at the bottom by an ideal ohmic contact.

The following assumptions characterize the ideal MOS structure:

- This is a one-dimensional situation. There are no 2D or 3D effects.
- We assume an oxide free of volume charge. The impact of oxide charge is discussed in Advanced Topic AT8.2.1.
- We assume an ideal semiconductor surface at the oxide-semiconductor interface, that is, smooth and defect free. The impact of interface states is discussed in Advanced Topic AT8.2.2.
- The doping level in the semiconductor is uniform throughout.
- We assume that the metal work function and the doping level in the semiconductor are such that under zero bias, there is a depletion region in the semiconductor.
- In situations in which there is a depletion region in the semiconductor, we treat it under the depletion approximation.
- Under the application of a DC voltage, we assume that no DC current flows.
- The semiconductor is contacted through an ideal ohmic contact, as defined earlier in this book.

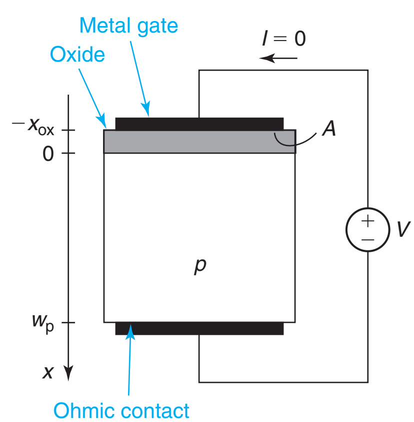


FIGURE 8.6
Sketch of ideal MOS structure on *p*-type substrate.

- We neglect any sidewall effects associated with the edges of the device.
- Maxwell–Boltzmann statistics apply to both types of carriers under all conditions.

Figure 8.6 also defines the axis that we will use in our analysis. The origin is placed at the oxide/semiconductor interface. The area of the MOS structure is A. The voltage convention that we will follow is also shown in the figure. For a MOS structure on a p-type substrate, V is defined as the potential difference between the gate and the semiconductor.

8.3 THE IDEAL METAL–OXIDE–SEMICONDUCTOR STRUCTURE AT ZERO BIAS

We start by analyzing the MOS structure at zero bias. The first question that one might ask is why do we use this notation, "zero bias," as opposed to "thermal equilibrium" used earlier in this book. Is it the same?

In an ideal MOS structure with a perfectly insulating oxide, no electron flow can take place between the metal and the semiconductor and the system cannot really attain thermal equilibrium. We can deal with this problem by shorting together the metal and the semiconductor through an external wire so that electron redistribution can take place through it. This is equivalent to applying a zero bias across the structure.

Consider a metal/oxide/p-type semiconductor structure with the three materials far apart, as sketched in an energy band diagram in Fig. 8.7(a) (dual situations occur with an n-type semiconductor substrate, as we will summarize later). The common reference energy for the three band diagrams is the vacuum level. The electronic structure of the metal is characterized by its work function. The semiconductor and the insulator are characterized by their electron affinity and their bandgaps, as well as the location of the Fermi level. In the case of the semiconductor, the Fermi level is set by doping. For the insulator, in principle, one could establish the location of the Fermi level as if it was an intrinsic semiconductor. In practice, the bandgap of the insulator is so large, that the Fermi level can swing widely through most of it without changing in any significant way the occupation of the bands. For all practical purposes, the valence band is always completely full of electrons and the conduction band is perfectly empty. Because of this, the Fermi level is not usually drawn inside an insulator.

The situation shown in Fig. 8.7(a) is one in which the Fermi level in the semiconductor is lower than in the metal. In consequence, if these three materials are brought in intimate contact through an external wire, electrons will tend to flow from the metal to the semiconductor. When this happens, the metal becomes positively charged and the semiconductor acquires negative charge. This sets up an electric field that eventually brings carrier exchange to a halt establishing thermal equilibrium. The bottom of Fig. 8.7 shows the resulting energy band diagram at zero bias.

There are several interesting features in this energy band diagram. First of all, far away from their interfaces with the insulator, bulk thermal equilibrium conditions prevail in the metal and the semiconductor. In the vicinity of the oxide/semiconductor interface, the charge exchange that has taken place between the metal and the semiconductor implies that the band diagram of the semiconductor bends *down*. In addition to getting more electrons close to the insulator, thermal equilibrium demands that fewer holes are present there. The semiconductor has become

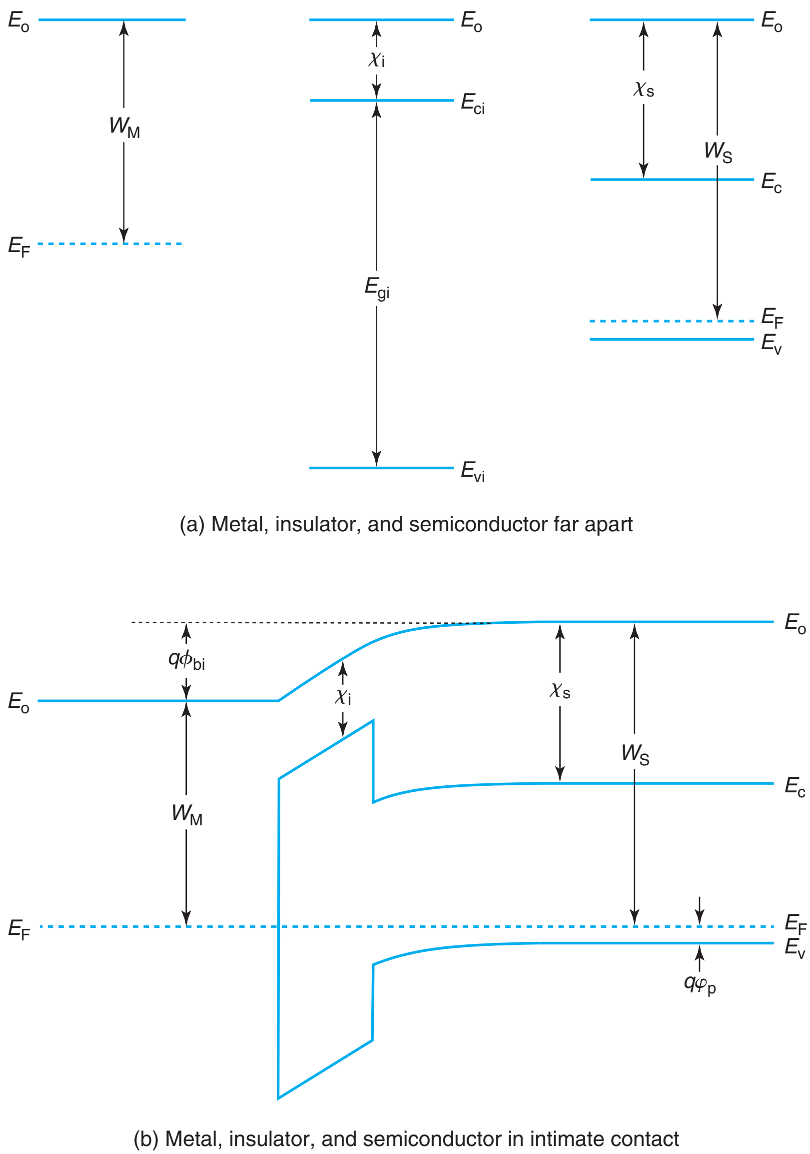


FIGURE 8.7
Energy band diagrams (a) for isolated metal, insulator, and semiconductor; and (b) after bringing them in intimate contact at zero bias.

negatively charged. In fact, from a charge point of view, since holes are the majority carriers, the reduction of the hole concentration is far more significant than the increase in the electron concentration. The net effect, in any case, is that the semiconductor is negatively charged in the neighborhood of the interface with the insulator. This negative charge is imaged at the metal-insulator interface where there is a deficit of electrons. In the length scale of interest, this does not bend the band diagram of the metal in a significant way and it is not shown (refer to a related discussion in Section 7.2).

The dipole of charge that has appeared across the MOS structure results in two features in the band diagram. First, it bends the bands in the insulator. The fact that there is not bulk charge inside an ideal insulator implies that the bands are inclined but straight. Second, there is a built-in potential difference across the entire structure. As in previous junctions of dissimilar materials, the built-in potential is entirely set by the work functions of the end materials. In this case, it is given by

$$q\phi_{\text{bi}} = W_{\text{S}} - W_{\text{M}} \tag{8.1}$$

A final observation to be made about the energy band diagram of Fig. 8.7 is the appearance of large conduction band and valence band discontinuities at the semiconductor/insulator interface. They constitute huge energy barriers to carrier flow and ideally prevent the injection of electrons and holes from the semiconductor to the insulator.

Before we proceed to analyze in detail the electrostatics of this situation, let us first derive some general relationships for the electrostatics of the MOS structure. We will do this at zero bias, but with suitable modification, they can also be used under bias.

8.3.1 General relations for the electrostatics of the ideal MOS structure

Figure 8.8 shows a general MOS structure at zero bias. The coordinate system is selected to point into the wafer, with the origin placed at the semiconductor/insulator interface, as sketched on the left of Fig. 8.8. The right of the figure shows the volume charge density, the electric field, and the electrostatic potential along the space coordinate x.

The top of Fig. 8.8 shows the charge distribution in space. In the semiconductor, we depict a generic charge distribution that extends to some depth from the Si/SiO_2 interface and that integrates to a total charge areal density Q_s (in units of C/cm^2). The insulator is assumed to be ideal and devoid of any charge inside. The metal also rejects volume charge but allows it at its surfaces. We can think of this charge as a "sheet" located at the metal–oxide interface with an areal density Q_g (in units of C/cm^2). Overall charge neutrality demands that

$$Q_g = -Q_s \tag{8.2}$$

The field lines produced by this dipole of charge originate in the sheet charge at the metal/insulator interface and terminate in the semiconductor. As a result, the field inside the metal as well as sufficiently deep inside the semiconductor are zero. Application of Gauss law to this situation is straightforward. If we construct a "pill box" that includes all Q_s and penetrates

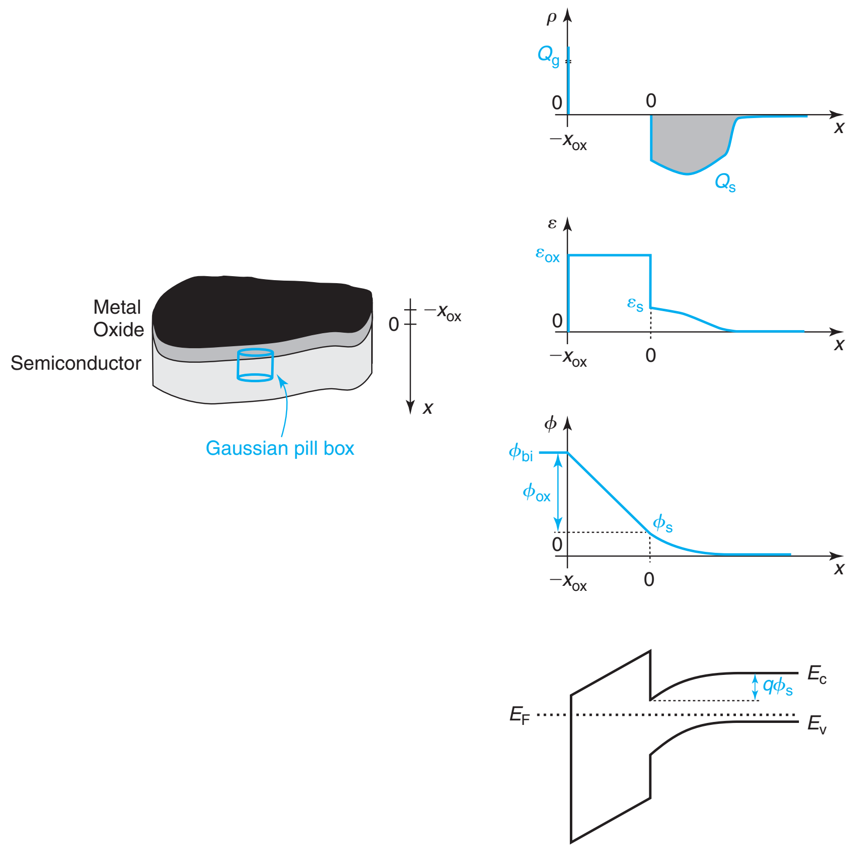


FIGURE 8.8
Schematic of MOS structure defining the coordinate axis and a Gaussian pill box (left). Generic sketches of, from top to bottom, volume charge density, electric field, electrostatic potential, and energy band diagram for the MOS structure (right).

partially into the oxide, as shown on the left of Fig. 8.8, we find the electric field in the oxide to be:

$$\mathcal{E}_{ox} = -\frac{Q_s}{\epsilon_{ox}} \tag{8.3}$$

where ϵ_{ox} is the permittivity of the oxide[1]. $\mathcal{E}_{ox}$ is uniform in space inside the oxide since there are no charges there.

The permittivities of the insulator and the semiconductor are generally different. This implies that at the Si/insulator interface, the electric field is discontinuous. The relationship between

[1]It is important to keep in mind the distinction between *permittivity* (ϵ, in F/cm^2) and *dielectric constant* (κ, no units). They are related through $\epsilon = \kappa\epsilon_o$ where ϵ_o is the permittivity of vacuum (value given in Appendix A).

the normal components of the electric field on both sides of the interface is obtained from the continuity of the electric displacement vector, that is

$$\epsilon_s \mathcal{E}_s = \epsilon_{ox} \mathcal{E}_{ox} \tag{8.4}$$

The permittivity of SiO_2 is 3.9. This is exactly three times smaller than the permittivity of Si (see Appendix B). Therefore, the field inside the oxide is three times the field at the semiconductor surface.

Substituting Eq. (8.4) into (8.3) provides a relationship between the electric field at the semiconductor/insulator interface on the semiconductor side and Q_s:

$$\mathcal{E}_s = -\frac{Q_s}{\epsilon_s} \tag{8.5}$$

We could have obtained this result by selecting a different pillbox that includes all of Q_s and has one of its surfaces placed exactly at $x = 0$.

The electric field distribution results in an electrostatic potential profile that is sketched in Fig. 8.8. In this graph, the origin of potentials is selected deep in the bulk of the semiconductor. The total potential difference across the structure at zero bias is ϕ_{bi}. The total potential drop across the structure is equal to the sum of the drop in the semiconductor, ϕ_s, plus the drop in the insulator, ϕ_{ox}, that is

$$\phi_{bi} = \phi_s + \phi_{ox} \tag{8.6}$$

ϕ_s is called the *surface potential*. It is related to the detailed charge distribution in space inside the semiconductor. ϕ_{ox} is the potential build-up across the insulator. Since the field in the oxide is uniform, ϕ_{ox} is given by

$$\phi_{ox} = x_{ox} \mathcal{E}_{ox} \tag{8.7}$$

Equation (8.6) is the basis for a general relationship between ϕ_s and Q_s. Toward deriving this, let us first define the capacitance per unit area of the insulator:

$$C_{ox} = \frac{\epsilon_{ox}}{x_{ox}} \tag{8.8}$$

If we now plug Eq. (8.7) into Eq. (8.6) and use Eqs. (8.4) and (8.5), we can write

$$\phi_{bi} = \phi_s - \frac{Q_s}{C_{ox}} \tag{8.9}$$

The relationships just derived apply to any ideal MOS structure under any circumstances. They have been obtained from fundamental statements about the electrostatics of the structure. In particular, they all apply to situations under bias if we substitute ϕ_{bi} by the appropriate expression for the total potential build-up across the structure.

The bottom sketch in Fig. 8.8 is the energy band diagram. A significant point to be made is that with the electrostatic potential reference that was selected above, the total band bending inside the semiconductor is given by $q\phi_s$.

We can now proceed to study in some detail the electrostatics of the zero-bias situation outlined above.

EXERCISE 8.1

A certain MOS structure with $x_{ox} = 4.5$ nm exhibits $\phi_{bi} = 1$ V and $\phi_{ox} = 0.42$ V at zero bias. Calculate ϕ_s, Q_s, $\mathcal{E}_s$, and $\mathcal{E}_{ox}$.

There are multiple ways to go about solving this problem. We can use Eq. (8.7) to get $\mathcal{E}_{ox}$:

$$\mathcal{E}_{ox} = \frac{\phi_{ox}}{x_{ox}} = \frac{0.42\ \text{V}}{4.5 \times 10^{-7}\ \text{cm}} = 9.3 \times 10^5\ \text{V/cm}$$

From here, we can calculate $\mathcal{E}_s$ using Eq. (8.4):

$$\mathcal{E}_s = \frac{\epsilon_{ox}}{\epsilon_s}\mathcal{E}_{ox} = 0.33 \times 9.3 \times 10^5\ \text{V/cm} = 3.1 \times 10^5\ \text{V/cm}$$

We can get Q_s from either Eq. (8.3) or (8.5). Using the later one:

$$Q_s = \epsilon_s\mathcal{E}_s = 11.7 \times 8.85 \times 10^{-14}\ \text{F/cm} \times 3.1 \times 10^5\ \text{V/cm} = 3.2 \times 10^{-7}\ \text{C/cm}^2$$

Finally, ϕ_s can be obtained from Eq. (8.6):

$$\phi_s = \phi_{bi} - \phi_{ox} = 0.58\ \text{V}$$

8.3.2 Electrostatic of the MOS structure under zero bias

For a uniformly doped semiconductor, deriving an analytical formulation of the electrostatics of the MOS structure under zero bias is relatively straightforward under the *depletion approximation*. If one focuses on the semiconductor in Fig. 8.7, one sees that close to the interface with the insulator there is a region with a space charge region, while far away from it, the bulk is quasi-neutral. This is similar to the situations that we observed in the PN diode and the Schottky diode. Consistent with the treatment that was followed there, we can perform the depletion approximation and assume that up to a certain depth, x_d, the semiconductor is devoid of majority carriers (holes in this case) and beyond x_d, the semiconductor is perfectly neutral. Under this approximation, the volume charge density, electric field, electrostatic potential, and equilibrium carrier concentrations and energy band diagram of Fig. 8.9 result.

Within the depletion approximation, the volume charge density inside the depletion region is uniform and equal to $\rho = -qN_A$. Hence, the integrated semiconductor charge is

$$Q_s \simeq -qN_A x_d \tag{8.10}$$

The electric field distribution in the semiconductor has a triangular shape with a maximum at the semiconductor interface with the oxide. Plugging Eq. (8.10) into Eq. (8.5), the electric field at the semiconductor surface is given by

$$\mathcal{E}_s \simeq \frac{qN_A x_d}{\epsilon_s} \tag{8.11}$$

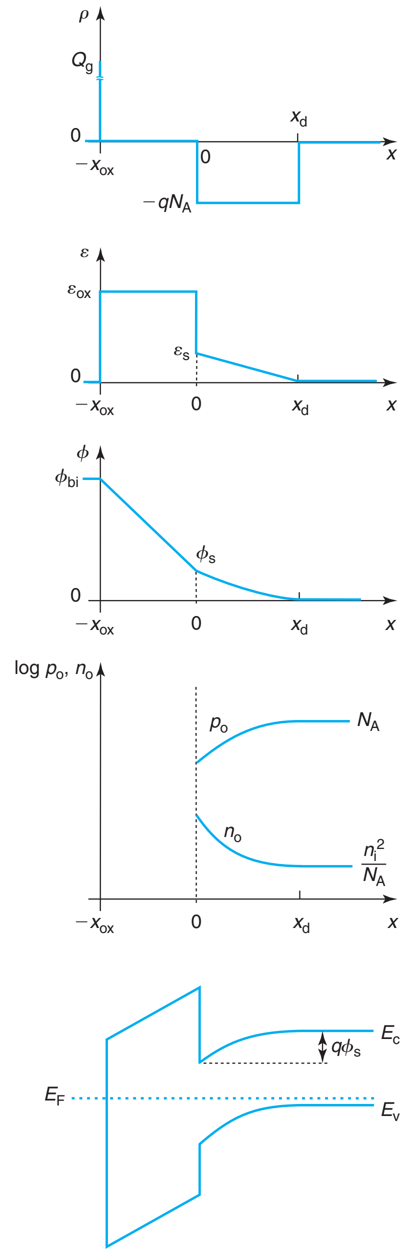


FIGURE 8.9
From top to bottom: sketches of volume charge density, electric field, electrostatic potential, equilibrium carrier concentration, and energy band diagram in a MOS structure under zero bias under the depletion approximation (p-substrate).

Equation (8.3) gives an expression that allows the calculation of the electric field inside the oxide. We can then write

$$\mathcal{E}_{ox} \simeq \frac{qN_A x_d}{\epsilon_{ox}} \tag{8.12}$$

The electrostatic potential in the semiconductor has a parabolic shape. The surface potential is obtained by integrating the triangularly shaped electric field. This yields

$$\phi_s \simeq \frac{qN_A x_d^2}{2\epsilon_s} \tag{8.13}$$

The only unknown at this time is the depth of the depletion region x_d. This can be obtained by plugging Eqs. (8.13) and (8.10) into the fundamental electrostatics relation (8.9) to yield

$$\phi_{bi} \simeq \frac{qN_A x_d^2}{2\epsilon_s} + \frac{qN_A x_d}{C_{ox}} \tag{8.14}$$

Solving this quadratic equation we obtain

$$x_d \simeq \frac{\epsilon_s}{C_{ox}}\left(\sqrt{1+\frac{4\phi_{bi}}{\gamma^2}}-1\right) \tag{8.15}$$

where γ is the so-called *body factor*:

$$\gamma = \frac{1}{C_{ox}}\sqrt{2\epsilon_s q N_A} \tag{8.16}$$

which has units of $V^{1/2}$. Equation (8.15) states that the larger the difference in work functions between the metal and the semiconductor, the larger x_d is. This makes sense since the higher ϕ_{bi}, the larger the charge exchange that needs to take place.

It is also of interest to focus on the total charge in the depletion layer. This is given by

$$Q_d \simeq -qN_A x_d \simeq -\sqrt{2\epsilon_s q N_A \phi_s} \tag{8.17}$$

where we have used Eq. (8.13). As in a PN junction and a Schottky junction, there is a square-root relationship between depletion charge and potential build up across a depletion region.

γ is an important parameter of a MOS structure. As Eq. (8.16) indicates, γ depends on the doping level of the substrate and the capacitance per unit area of the insulator. As it will be better understood later on in this chapter, γ makes a statement about the relative magnitude of the depletion region capacitance to the oxide capacitance. The higher the doping level, the thinner the depletion region in the semiconductor, and the higher the depletion capacitance, hence γ goes up. Similarly, the thinner the oxide, the higher the oxide capacitance, and the lower γ gets. Both dependencies are seen in Fig. 8.10 which graphs γ versus N_A and the SiO_2 thickness.

In a well-designed MOSFET, a proper balance between the oxide and depletion region thicknesses is required. Hence, γ is typically on the order of 0.1–1 $V^{1/2}$. In time, γ has tended to decrease as MOSFET size has shrunk.

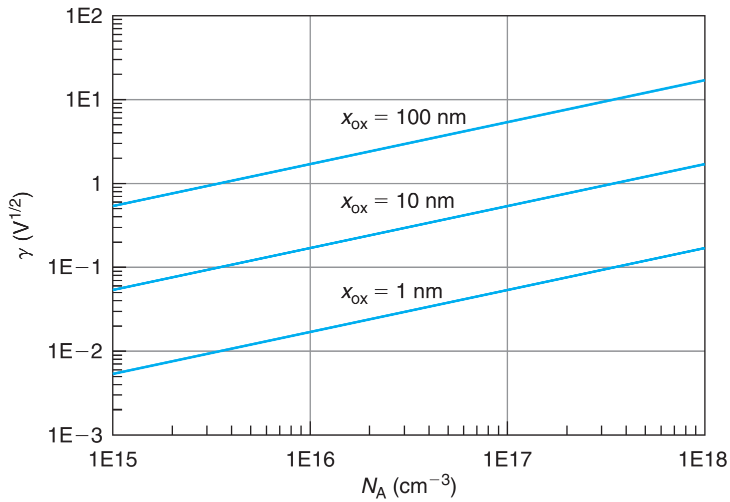


FIGURE 8.10
Body factor, γ, versus substrate doping level for different values of the SiO_2 thickness.

Once the electrostatic potential in the semiconductor is known, it is straightforward to obtain first-order expressions for the zero bias carrier densities in the semiconductor. The procedure is identical to that one followed in the case of the metal–semiconductor junction in Section 7.2.3.

EXERCISE 8.2

Consider a MOS structure made out of a W gate ($W_M = 4.54$ eV), a 4.5 nm SiO_2 insulator, and a p-type doped Si substrate with $N_A = 6 \times 10^{17}$ cm^{-3}. Estimate the electric field in the oxide and the surface potential at zero bias.

Since W_M is about $\chi + E_g/2$, this structure is likely to be in depletion. In order to compute $\mathcal{E}_{ox}$, we first need to calculate x_d. Before that we need to compute ϕ_{bi}, C_{ox}, and γ, which can be obtained, respectively, from Eqs. (8.1, 8.8), and (8.16). In order to get ϕ_{bi}, we need to first compute W_S:

$$W_S = \chi + E_g + kT \ln \frac{N_A}{N_v} = 4.04 + 1.12 + 0.026 \times \ln \frac{6 \times 10^{17}}{3.1 \times 10^{19}} = 5.06 \text{ eV}$$

Now

$$\phi_{bi} = \frac{1}{q}(W_S - W_M) = 5.06 - 4.54 = 0.52 \text{ V}$$

$$C_{ox} = \frac{\epsilon_{ox}}{x_{ox}} = \frac{3.45 \times 10^{-13} \text{ F/cm}}{4.5 \times 10^{-7} \text{ cm}} = 7.7 \times 10^{-7} \text{ F/cm}^2$$

$$\gamma = \frac{1}{C_{ox}}\sqrt{2\epsilon_s q N_A} = \frac{\sqrt{2 \times 1.04 \times 10^{-12}\ \mathrm{F/cm} \times 1.6 \times 10^{-19}\ \mathrm{C} \times 6 \times 10^{17}\ \mathrm{cm^{-3}}}}{7.7 \times 10^{-7}\ \mathrm{F/cm^2}} = 0.58\ \mathrm{V^{1/2}}$$

We can now use Eq. (8.15) to obtain x_d:

$$x_d = \frac{\epsilon_s}{C_{ox}}\left(\sqrt{1 + \frac{4\phi_{bi}}{\gamma^2}} - 1\right) = \frac{1.04 \times 10^{-12}\ \mathrm{F/cm}}{7.7 \times 10^{-7}\ \mathrm{F/cm^2}}\left(\sqrt{1 + \frac{4 \times 0.52\ \mathrm{V}}{0.58^2\ \mathrm{V}}} - 1\right)$$
$$= 2.3 \times 10^{-6}\ \mathrm{cm} = 23\ \mathrm{nm}$$

$\mathcal{E}_{ox}$ can now be obtained using Eq. (8.12):

$$\mathcal{E}_{ox} = \frac{qN_A x_d}{\epsilon_{ox}} = \frac{1.6 \times 10^{-19}\ \mathrm{C} \times 6 \times 10^{17}\ \mathrm{cm^{-3}} \times 2.3 \times 10^{-6}\ \mathrm{cm}}{3.45 \times 10^{-13}\ \mathrm{F/cm}} = 6.4 \times 10^5\ \mathrm{V/cm}$$

The surface potential can be obtained from Eq. (8.13):

$$\phi_s = \frac{qN_A x_d^2}{2\epsilon_s} = \frac{1.6 \times 10^{-19}\ \mathrm{C} \times 6 \times 10^{17}\ \mathrm{cm^{-3}} \times (2.3 \times 10^{-6}\ \mathrm{cm})^2}{2 \times 1.04 \times 10^{-12}\ \mathrm{F/cm}} = 0.24\ \mathrm{V}$$

In the next section, we will learn how to confirm that indeed this structure is in depletion.

8.4 THE IDEAL METAL–OXIDE–SEMICONDUCTOR STRUCTURE UNDER BIAS

The application of a voltage to the gate with respect to the body of a MOS structure affects the potential and charge distributions throughout. For the case of a p-type substrate, if we define as positive the voltage in the metal with respect to the semiconductor, the total potential difference across the structure goes from a zero bias value of ϕ_{bi} to a new value equal to $\phi_{bi} + V$, where V is the applied voltage. The charge dipole across the MOS structure has to adapt itself so as to produce such a potential build up. In what way is it modified and what are the consequences of this? These are the topics that we study in this section.

Before discussing the details, we need to recognize that the presence of the oxide in the MOS structure blocks the flow of current even when a voltage is applied. Current continuity then implies that no current flows across the semiconductor. So, even though a bias has been applied, the semiconductor remains in a quasi-equilibrium condition. We can therefore conclude that similar to the PN diode or the Schottky diode, the electrostatics of a MOS structure with a built-in potential ϕ_{bi} under a voltage V are identical to the electrostatics of a MOS structure with a built-in potential equivalent to $\phi_{bi} + V$ under zero bias.[2] As a consequence, the general

[2]In the Schottky junction and the PN junction, we assumed that the electrostatics under bias were identical to equilibrium even though current was flowing by. In most cases, this assumption gives reasonable results. In the case of the MOS structure, this assumption is particularly well justified since no current flows.

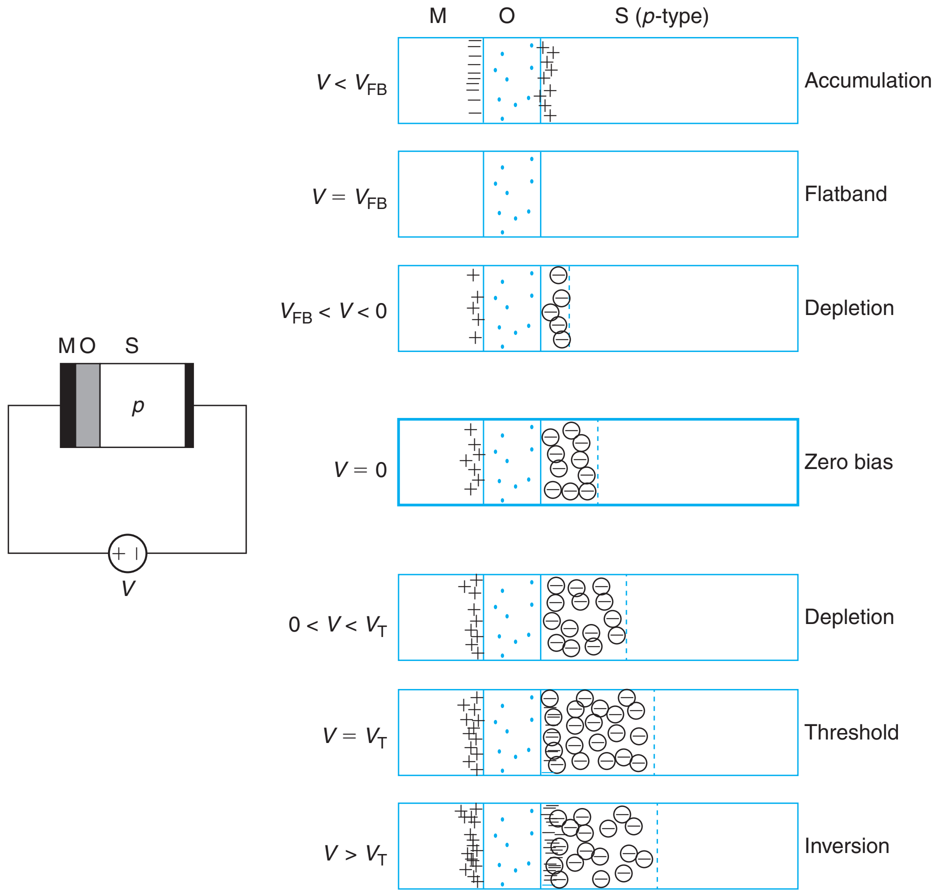


FIGURE 8.11
Impact of the application of a voltage to the charge distribution of a MOS structure. For negative voltages of the metal with respect to the semiconductor, the depletion region shrinks. For a large enough negative voltage, the structure is driven into accumulation. For positive voltages, the depletion region widens. For large enough voltage, the structure is driven into inversion.

relation for the electrostatics of the MOS structure derived in Eq. (8.9) applies here for a total potential difference that is given by $\phi_{bi} + V$:

$$\phi_{bi} + V = \phi_s(V) - \frac{Q_s(V)}{C_{ox}} \tag{8.18}$$

where, in general, we note that ϕ_s and Q_s now depend on V.

Let us now qualitatively discuss the changes that take place in the electrostatics of a MOS structure as a result of the application of a voltage. A pictorial view is sketched in Fig. 8.11.

The center diagram shows the structure at zero bias in which a depletion region exists in the semiconductor. An identical amount of positive charge resides at the metal–oxide interface. If a negative voltage is applied, the potential difference across the structure is reduced. In consequence, the magnitude of the charge dipole must decrease. This implies that the depletion layer shrinks. This voltage regime in which the depletion region thickness is being modulated by the applied bias is referred to as *depletion*.

For sufficient negative bias, the potential difference across the structure is wiped out. This calls for the charge dipole to disappear and the depletion region to completely collapse. This condition is called *flatband*. If a more negative voltage than this is applied, the potential difference across the structure reverses sign. This now demands the appearance of positive charge in the semiconductor and a negative sheet of charge at the metal–oxide interface. This can only be accomplished by accumulating holes against the oxide–semiconductor interface, a condition that is referred to as *accumulation*. This is shown in the top diagram of Fig. 8.11.

Moving downward from $V = 0$ now, if a positive voltage is applied to the gate with respect to the semiconductor, the total potential difference across the structure increases and the depletion region has to widen to satisfy the electrostatics. If the voltage is made sufficiently large, eventually electrons start piling up at the insulator/semiconductor interface. The onset of this phenomenon is called *threshold*. For voltages higher than this, the electron concentration at the oxide–semiconductor interface becomes significant, a condition that is known as *inversion*. This is the regime that is exploited in MOSFETs.

We proceed to analyze these different situations in more detail in the following subsections.

8.4.1 Depletion

A detailed view of the MOS electrostatics in the depletion regime is shown in Fig. 8.12. Starting with the potential distribution, the application of a voltage to the gate with respect to the body changes the potential difference across the MOS structure. This requires a modulation of the charge distribution. For $V > 0$, the potential difference increases from the zero bias value of ϕ_{bi} to $\phi_{bi} + V$. In consequence, the depletion region widens and the electric field everywhere increases. For $V < 0$, the contrary happens and the potential difference is reduced, the depletion region shrinks and the electric field is lowered throughout.

The situation is well reflected in the evolution of the energy band diagram, which is also shown in Fig. 8.12. For $V > 0$, the Fermi level in the metal is pulled down with respect to the Fermi level in the semiconductor. As a result, the band bending across the oxide and the semiconductor increases. The contrary happens for $V < 0$. Note how the Fermi level in the semiconductor is drawn flat and that the hole and electron quasi-Fermi levels overlap. This reflects the absence of current through the structure and the consideration that the semiconductor remains in a quasi-equilibrium state.

It is interesting to reflect on what happens to the carrier concentrations. This is sketched in the top right of Fig. 8.12. For $V > 0$, the hole concentration drops close to the semiconductor/oxide interface as the depletion region widens. The contrary happens when $V < 0$. Since the *pn* product has to remain constant (consistent with zero current), for $V > 0$, the electron concentration close to the oxide/semiconductor interface rises while it drops for $V < 0$.

In the depletion regime, the change in the potential build up across the structure modulates the depletion region thickness. Nevertheless, the essential character of the electrostatics remains

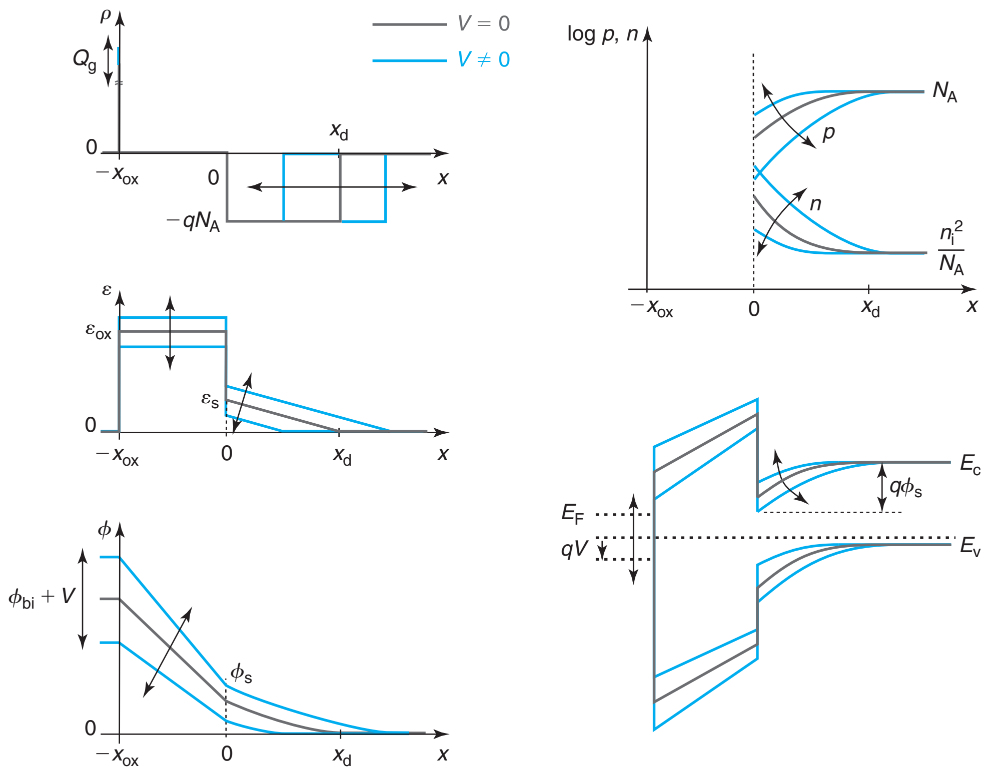


FIGURE 8.12
From top to bottom and left to right: sketches of volume charge density, electric field, electrostatic potential, carrier concentration, and energy band diagram of a MOS structure under zero bias and for bias negative and positive but close to zero (depletion regime).

unchanged from the situation that we discussed under zero bias. In consequence, an analytical description of the depletion regime can be obtained simply by going to our zero bias results and substituting ϕ_{bi} by $\phi_{bi} + V$.

In particular, going back to the results obtained in Section 8.3.2, the depletion region thickness as a function of V is given by

$$x_d(V) \simeq \frac{\epsilon_s}{C_{ox}} \left(\sqrt{1 + 4\frac{\phi_{bi} + V}{\gamma^2}} - 1 \right) \tag{8.19}$$

The total charge in the depletion region, from Eq. (8.10), can be written as:

$$Q_s(V) = Q_d(V) \simeq -\frac{1}{2}\gamma^2 C_{ox} \left(\sqrt{1 + 4\frac{\phi_{bi} + V}{\gamma^2}} - 1 \right) \tag{8.20}$$

where we have used the definition of γ in Eq. (8.16).

The evolution of the surface potential with V can be easily obtained from Eq. (8.13):

$$\phi_s(V) \simeq \frac{1}{4}\gamma^2\left(\sqrt{1+4\frac{\phi_{bi}+V}{\gamma^2}}-1\right)^2 \tag{8.21}$$

Similar relations can be obtained for any other variable of interest by starting from the zero bias equations derived in Section 8.3.2.

EXERCISE 8.3

Consider a MOS structure identical to that of Exercise 8.2 except for the use of a poly-Si gate ($W_M = 4.04$ eV). *Calculate* Q_s *and its extent into the semiconductor for* $V = -0.5$ V.

Let us assume that for $V = -0.5$ V this structure is in depletion (we will verify this in Exercise 8.4). The first step is to compute ϕ_{bi} as in Exercise 8.2. This yields $\phi_{bi} = 1.0$ V. Q_s can be obtained from Eq. (8.20) (the values of $C_{ox} = 7.7 \times 10^{-7}$ F/cm^2, and $\gamma = 0.58$ V$^{1/2}$ were also obtained in Exercise 8.2):

$$Q_s = -\frac{1}{2}\gamma^2 C_{ox}\left(\sqrt{1+4\frac{\phi_{bi}+V}{\gamma^2}}-1\right)$$

$$= -\frac{1}{2}(0.58)^2\ \text{V} \times 7.7\times 10^{-7}\ \text{F/cm}^2\left[\sqrt{1+4\frac{(1.0-0.5)\ \text{V}}{(0.58\ \text{V})^2}}-1\right] = -2.1\times 10^{-7}\ \text{C/cm}^2$$

The extent of the depletion region is simply:

$$x_d = \frac{Q_s}{-qN_A} = \frac{-2.1\times 10^{-6}\ \text{C/cm}^2}{-1.6\times 10^{-19}\ C\times 6\times 10^{17}\ \text{cm}^{-3}} = 22\ \text{nm}$$

8.4.2 Flatband

For large enough negative voltage applied to the gate with respect to the body of the MOS structure, the depletion region is wiped out. This is referred to as *flatband*. In this instance, the potential build up across the structure is precisely zero. As a result, there is no charge dipole, and there is no electric field anywhere. As sketched in Fig. 8.13, the energy bands are perfectly flat, hence the name, and the carrier concentrations are constant throughout the semiconductor, which is then perfectly charge neutral.

The voltage required to drive the structure to flatband is referred to as the *flatband voltage*. Since there is no charge in the semiconductor, $Q_s = 0$. As a result, the band bending is zero and the surface potential $\phi_s = 0$. Then, from the general relation for electrostatics in Eq. (8.18), the flatband voltage is given by

$$V_{FB} = -\phi_{bi} \tag{8.22}$$

Looking back at the expression of ϕ_{bi} given in Eq. (8.1), it is clear that the flatband voltage is entirely set by the bulk properties of the metal and the semiconductor.

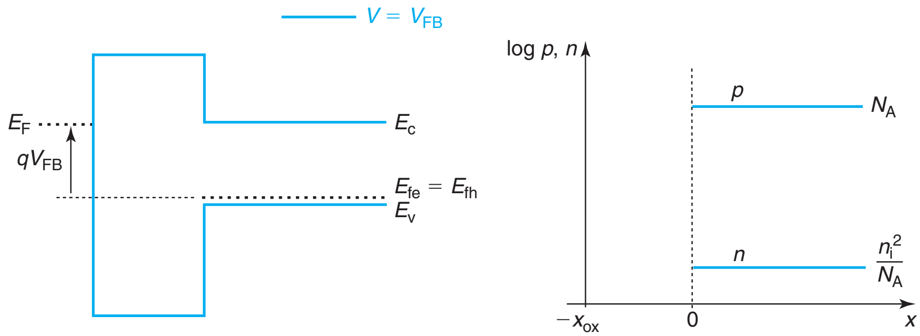


FIGURE 8.13
Energy band diagram and carrier concentration distribution in the MOS structure at flatband.

The flatband voltage is an important reference voltage in the operation of MOS devices. We will extensively use it in the treatment of the MOS structure and in the analysis of MOSFETs.

8.4.3 Accumulation

Nothing prevents us from applying a voltage across the MOS structure that is more negative than V_{FB}. When this is done, the gate is placed at a more negative potential than the body of the semiconductor. This forces a charge dipole with positive charge in the semiconductor and an identical amount of negative charge at the metal–oxide interface. Positive charge on the semiconductor can be obtained if the hole concentration is higher than the acceptor concentration. Since holes are mobile, they will tend to pile up, or accumulate, at the oxide–semiconductor interface. This is the accumulation regime.

A detailed view of the electrostatics in accumulation is shown in Fig. 8.14. The thin hole accumulation layer is imaged at the gate–oxide interface where there is a negative sheet of charge. This yields electric field and electrostatic potential distributions as sketched in the figure and a band structure that now bends upwards toward the metal.

It is relatively straightforward to derive first-order expressions for the hole accumulation charge and other electrostatic parameters under the *sheet-charge approximation*. This assumes that the accumulation layer is much thinner than any other dimensions in the structure. The hole distribution in the semiconductor reflects a balance between drift and diffusion. Hence, its extent is expected to be of the order of the Debye length, which can be quite small.

Under the sheet charge approximation, the surface potential in accumulation is

$$\phi_s \simeq 0 \tag{8.23}$$

This is because a sheet of charge produces a sharp change in the electric field over zero distance and this does not accrue any potential buildup.

If we denote as Q_a the sheet hole charge density in the accumulation layer, then, using Eq. (8.18), we can easily obtain

$$Q_a \simeq C_{ox}(V_{FB} - V) \tag{8.24}$$

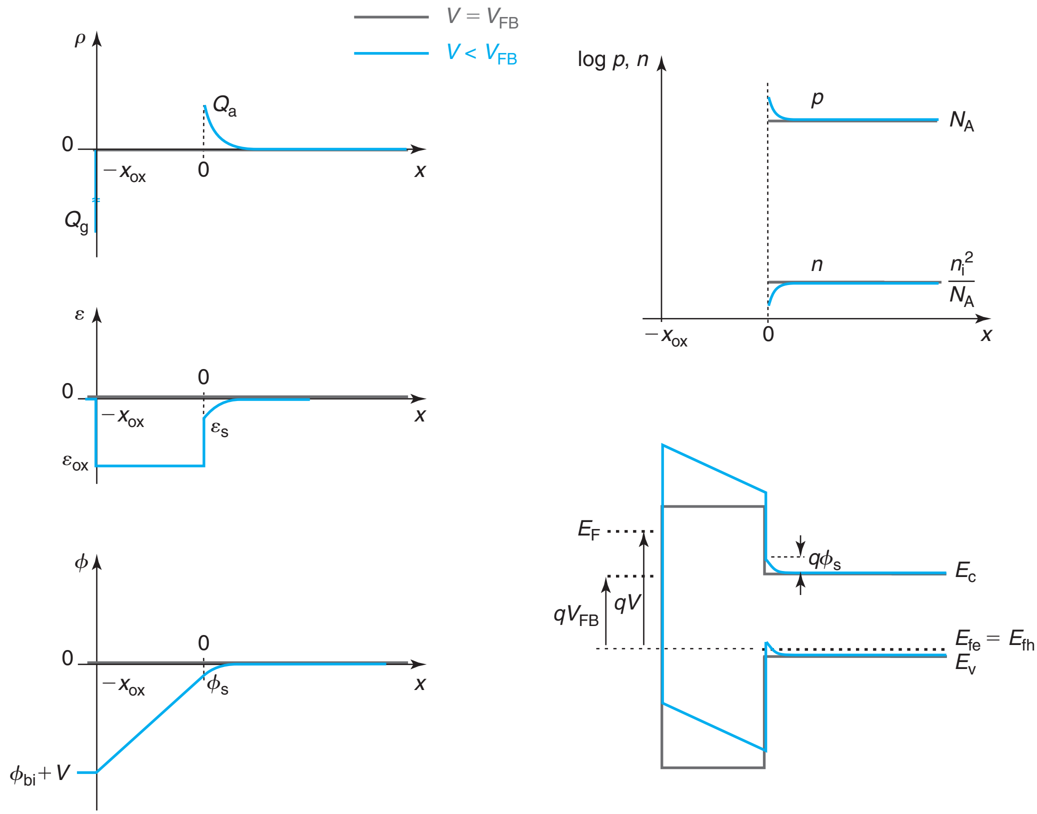


FIGURE 8.14
From top to bottom and left to right: sketches of volume charge density, electric field, electrostatic potential, carrier concentration, and energy band diagram of a MOS structure at flatband and in accumulation.

where we have also used Eq. (8.22). This is an important result. Once in accumulation, the hole accumulation charge increases linearly with the voltage that is applied beyond the flatband voltage.

8.4.4 Threshold

We now come back to applying positive voltages to the gate with respect to the body. As we discussed above, for $V > 0$, the depletion region in the semiconductor widens. This is necessary to accommodate the additional negative charge required to satisfy the higher external potential difference between the gate and the body (see Fig. 8.12). A consequence of this can also be seen in this figure and that is that the electron concentration at the surface of the semiconductor increases. This is also reflected in the energy band diagram in which we see that as a result of the increased band bending associated with the enlarged depletion region, the conduction band edge at the surface of the semiconductor gets closer to the Fermi level.

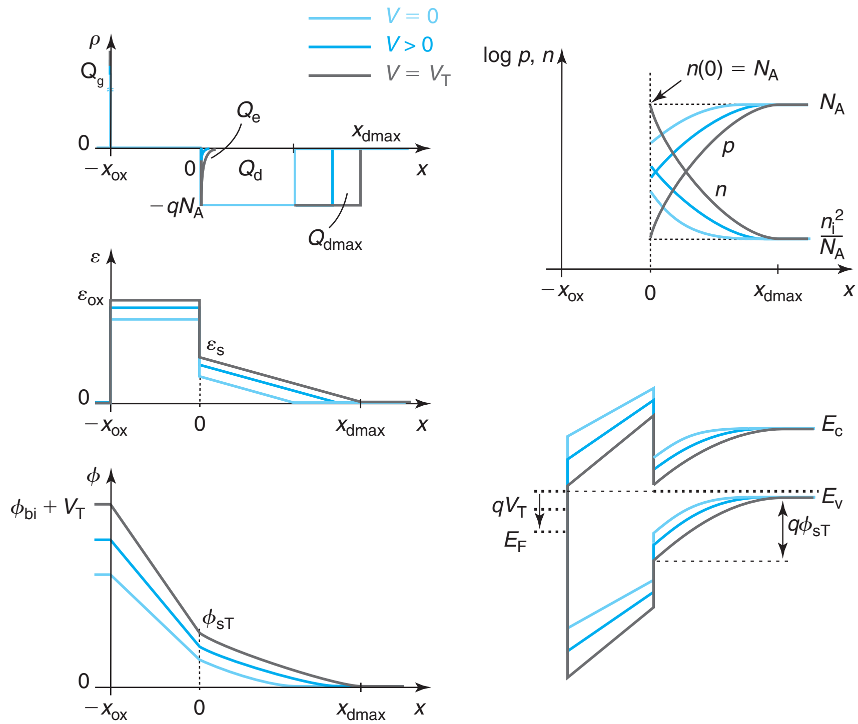


FIGURE 8.15
From top to bottom and left to right: sketches of volume charge density, electric field, electrostatic potential, carrier concentration, and energy band diagram of a MOS structure at zero volts, in depletion regime and at threshold.

It is clear that there is a voltage for which the electron concentration at the surface becomes equal to the hole concentration in the body of the semiconductor (see Fig. 8.15). This condition is called *threshold* and it marks the onset of the inversion regime. "Threshold" is a very appropriate name as at this voltage, the semiconductor type changes at the surface from p-type to n-type. For gate voltages more positive than the threshold voltage, the electrostatics of the MOS structure drastically change in character.

It is relatively straightforward to derive an expression for the threshold voltage of a MOS structure. We start by finding an expression for the surface potential at threshold, ϕ_{sT}. For this, we can use the Boltzmann relation and relate the difference in electrostatic potential between the surface and the bulk, precisely ϕ_{sT}, to the ratio of electron concentration at these two locations. This yields

$$\phi_{sT} = \frac{kT}{q} \ln \frac{n(x=0)|_T}{\frac{n_i^2}{N_A}} = 2\frac{kT}{q} \ln \frac{N_A}{n_i} = 2\phi_f \tag{8.25}$$

where we have taken $n(x=0)|_T = N_A$. We discuss the meaning of ϕ_f below.

We next need an expression for the charge in the semiconductor at threshold. If we can neglect the contribution from the electrons at the surface (not a bad assumption since the depletion region is much thicker than this thin electron layer), then the semiconductor charge at threshold is associated with the depletion region. For reasons that will become clear soon, we denote the thickness of the depletion region at threshold as x_{dmax} and the depletion charge as Q_{dmax}. With a potential build up across the depletion layer of ϕ_{sT}, using Eq. (8.17), x_{dmax} and Q_{dmax} are, respectively, given by

$$x_{dmax} \simeq \sqrt{\frac{2\epsilon_s \phi_{sT}}{qN_A}} \tag{8.26}$$

and

$$Q_{dmax} \simeq -\sqrt{2\epsilon_s q N_A \phi_{sT}} \tag{8.27}$$

We can now use the general relation for electrostatics of the MOS structure [Eq. (8.18)] at threshold to obtain

$$V_T = -\phi_{bi} + \phi_{sT} - \frac{Q_{dmax}}{C_{ox}} = V_{FB} + \phi_{sT} + \gamma\sqrt{\phi_{sT}} \tag{8.28}$$

where we have also substituted Eqs. (8.16), (8.22), and (8.27).

V_T is the threshold voltage of the MOS structure and it plays a key role in MOSFET operation. V_T depends on all the key parameters of the MOS structure: semiconductor doping level, oxide thickness and gate work function. Figure 8.16 shows V_T as a function of N_A and x_{ox} for a n^+-polySi gate and SiO_2 dielectric. As N_A or x_{ox} increase, so does V_T. Notice that the dependence on doping is much weaker for thin oxides. This is because the doping dependence of V_T is mainly through the square root doping dependence of γ [Eq. (8.16)]. For a thin oxide, C_{ox} is high, and as Eq. (8.16) shows, the absolute magnitude of γ is very small.

In Eq. (8.25), the surface potential at threshold was defined in terms of ϕ_f. Figure 8.17 illustrates this definition. $q\phi_f$ is the energy difference between the intrinsic Fermi level and the Fermi level in the body of the semiconductor. When the band bending in the semiconductor is precisely equal to $2q\phi_f$, then the electron concentration at the surface is identical to the hole concentration in the bulk.

8.4.5 Inversion

Threshold marks the onset of the creation of an inversion layer of electrons at the oxide-semiconductor interface of a MOS structure. This does not occur abruptly at the treshold voltage but for $V > V_T$ its existence can no longer be ignored. In fact, for $V > V_T$, the electrostatics of the MOS structure are profoundly affected by the presence of this inversion layer.

Applying a gate voltage that exceeds V_T demands an increase in the magnitude of the charge dipole across the MOS structure. To understand what this implies on the semiconductor side, we need to note the dependencies of the inversion layer charge, Q_i, and depletion region charge, Q_d, on the surface potential. While Q_i is expected to increase exponentially with ϕ_s (more on this later), Q_d evolves as the square root of ϕ_s [Eq. (8.17)]. Hence, any increase in ϕ_s brought about by an increase in voltage will mostly grown the inversion layer while it will only produce a minor increase in the depletion region thickness beyond its value at threshold. It is because of

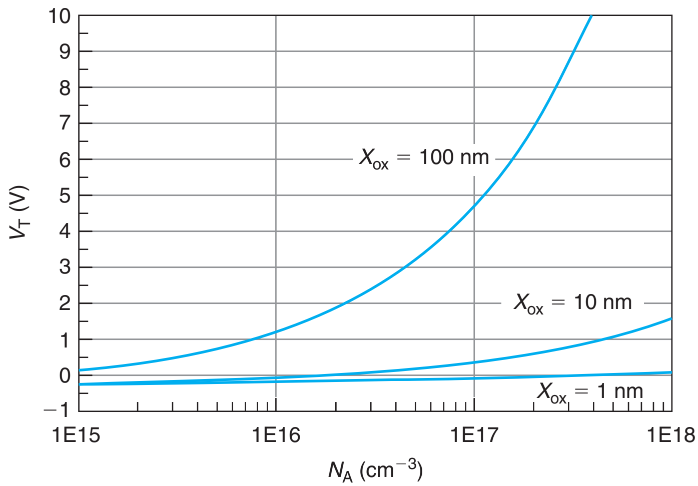


FIGURE 8.16
Threshold voltage, V_T, versus substrate doping level for different values of the oxide thickness. The gate is made out of n$^+$-polySi ($W_M = 4.04$ eV) and the oxide is SiO_2.

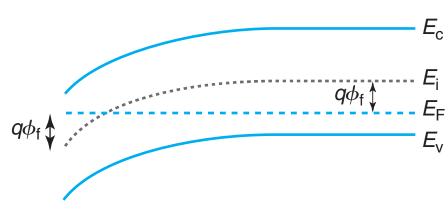


FIGURE 8.17
Energy band diagram of semiconductor at the onset of inversion (p-substrate). The definition of ϕ_f is indicated.

this that we denote the thickness and charge associated with the depletion region at threshold as x_{dmax} and Q_{dmax}, respectively.

Figure 8.18 illustrates the electrostatics of the MOS structure at threshold and in inversion. Starting from the top-right sketch, in inversion, the electron concentration at the surface of the semiconductor increases in a prominent way. This is more clearly seen in the top-left sketch that shows the charge distribution across the structure. In inversion, the electron concentration at the surface exceeds the background acceptor concentration. As a result of the appearance of the inversion layer, the electric field exhibits a rather abrupt jump at the semiconductor surface. However, since the inversion layer is expected to be very thin (the electrons are "jammed" against the oxide), the electrostatic potential does not change very much. Due to the exponential dependence on Q_i on ϕ_s, not much of a change in ϕ_s is needed to induce the required

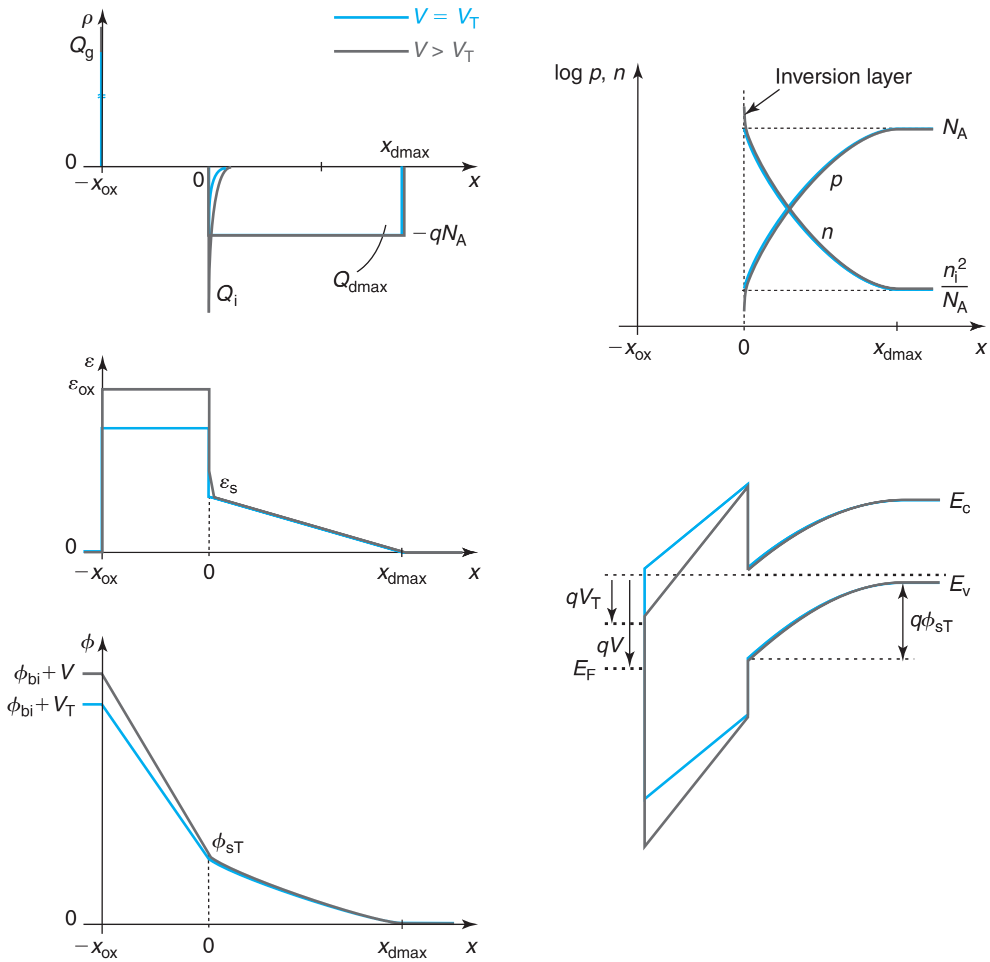


FIGURE 8.18
From top to bottom and left to right: sketches of volume charge density, electric field, electrostatic potential, carrier concentration and energy band diagram of a MOS structure at threshold and in inversion.

amount of Q_i to satisfy the electrostatics. This is also reflected in the band diagram on the bottom right that shows a band bending across the semiconductor that has not changed much beyond threshold with most of the extra applied gate voltage used to bend the bands across the oxide.

It is relatively straightforward to obtain a first-order expression for the evolution of the inversion charge with the gate voltage. This can easily be done under the context of the *sheet-charge approximation*. As in accumulation, we assume that the inversion layer charge is very thin in the scale of the other dimensions of the problem (x_{ox} and x_{dmax}). Under this approximation, the surface potential in inversion is approximately equal to the value that it has at threshold:

$$\phi_s \simeq \phi_{sT} \tag{8.29}$$

We can now use the general relation for the electrostatics of the MOS structure [Eq. (8.18)] in inversion:

$$\phi_{bi} + V = \phi_{sT} + \frac{Q_{dmax} + Q_i}{C_{ox}} \tag{8.30}$$

We now solve for Q_i and substitute the expressions of V_{FB} [Eq. (8.22)] and V_T [Eq. (8.28)] to get

$$Q_i = -C_{ox}(V - V_T) \qquad \text{for } V \geq V_T \tag{8.31}$$

This is a very important equation. It is often called the *charge–control relation of the inversion layer*. It states that once reached threshold, the application of additional voltage across the MOS structure grows the inversion layer charge in a linear manner.

While we have obtained important and useful results using a simple view of the electrostatics of the MOS structure, a more rigorous analysis is often desirable. Advanced Topic AT8.3 at the end of this chapter presents the so-called Poisson–Boltzmann formulation of the MOS electrostatics. This approach yields more accurate analytical models for many important aspects of the MOS electrostatics.

EXERCISE 8.4

Consider a MOS structure identical to the one studied in Exercise 8.3. Compute V_T and Q_i for $V = 2$ V.

We use results obtained for ϕ_{bi}, C_{ox}, and γ from Exercise 8.3. The first step is to compute ϕ_{sT} using Eq. (8.25):

$$\phi_{sT} = 2\frac{kT}{q}\ln\frac{N_A}{n_i} = 2 \times 0.026 \times \ln\frac{6 \times 10^{17}}{1.02 \times 10^{10}} = 0.93 \text{ V}$$

We then use Eq. (8.28) to obtain V_T:

$$V_T = V_{FB} + \phi_{sT} + \gamma\sqrt{\phi_{sT}} = -1.0 \text{ V} + 0.93 \text{ V} + 0.58 \text{ V}^{1/2} \times \sqrt{0.93 \text{ V}} = 0.49 \text{ V}$$

The calculation of Q_i is now straightforward through Eq. (8.31):

$$Q_i = -C_{ox}(V - V_T) = -7.7 \times 10^{-7} \text{ F/cm}^2 \times (2 - 0.49) \text{ V} = -1.2 \times 10^{-6} \text{ C/cm}^2$$

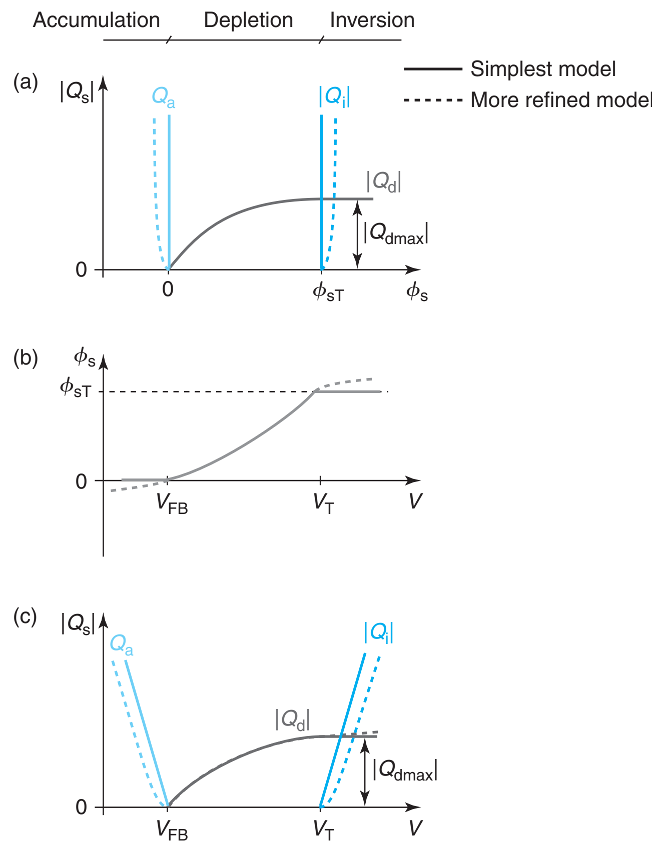


FIGURE 8.19
Summary of charge–voltage characteristics of MOS structure. (a) Semiconductor charge versus surface potential. (b) Surface potential versus voltage. (c) Semiconductor charge versus voltage. Dashed lines represent improved dependencies obtained from the Poisson–Boltzmann model of Appendix AT8.3.

8.4.6 Summary of charge–voltage characteristics

Figure 8.19 summarizes the charge–voltage characteristics of the MOS structure discussed in this section. On the top, the absolute of the semiconductor charge is graphed against ϕ_s. In the accumulation regime, $Q_s = Q_a > 0$. Under the sheet-charge approximation, in accumulation $\phi_s \simeq 0$. In depletion, $Q_s = Q_d < 0$ and $|Q_d|$ evolves in a square-root form with ϕ_s. In inversion, $Q_s = Q_{dmax} + Q_i < 0$. Under the sheet-charge approximation, $\phi_s \simeq \phi_{sT}$.

The graph in the middle shows ϕ_s versus V. For $V < V_{FB}$, we have the accumulation regime with $\phi_s \simeq 0$. In depletion, ϕ_s evolves with V according to Eq. (8.21). In inversion, for $V > V_T$, $\phi_s \simeq \phi_{sT}$.

At the bottom of Fig. 8.19, the absolute of the semiconductor charge is graphed against the voltage. For $V < V_{FB}$, the structure is in accumulation and Q_s is linear on $V_{FB} - V$. Between V_{FB} and V_T, the structure is in depletion and the dependence of Q_s with V follows the square-root form given by Eq. (8.20). For $V > V_T$, the structure is in inversion and Q_i is linear on $V - V_T$.

In the simple model developed in this section, the use of the sheet-charge approximation "pins" the surface potential in accumulation and inversion at 0 and ϕ_{sT}, respectively. In reality, carriers at the semiconductor surface are distributed in depth to some extent. This implies that ϕ_s becomes slightly negative in accumulation and it also increases somehow past ϕ_{sT} in inversion. This is indicated by the dashed lines in Fig. 8.19.

A more rigorous model that accounts for this is developed in Appendix AT8.3. This is referred to as the *Poisson–Boltzmann formulation*. This model does not make any assumptions about the spatial distribution of carriers at the semiconductor surface. It therefore yields a more accurate description of the electrostatics that returns, among other results, the evolution of carrier charge as a function of surface potential, and the surface potential versus the applied voltage. These key results are summarized here.

In accumulation, the accumulation charge is found to depend on ϕ_s according to the following expression [from Eq. (8.124)]:

$$Q_a \simeq \sqrt{2\epsilon_s kTN_A}\sqrt{\exp\frac{-q\phi_s}{kT} + \frac{q\phi_s}{kT} - 1} \qquad \text{for } \phi_s < 0 \tag{8.32}$$

Notice how Q_a goes to zero, as ϕ_s goes to zero. The leading term of Q_a for strong accumulation ($|\phi_s| \gg kT/q$) depends exponentially on $-q\phi_s/2kT$.

Another important result from the Poisson–Boltzmann formulation is the relationship between ϕ_s and V. In accumulation, this is approximately [from Eq. (8.126)]:

$$\phi_s \simeq -\frac{kT}{q}\ln\left\{1 + \left[\frac{C_{ox}(V_{FB} - V)}{\sqrt{2\epsilon_s kTN_A}}\right]^2\right\} \qquad \text{for } V < V_{FB} \tag{8.33}$$

At flatband, $\phi_s = 0$, but as $V < V_{FB}$, ϕ_s becomes negative but it does so quite slowly.

In inversion, the dependence of Q_i on ϕ_s is found to be [from Eq. (8.137)]

$$Q_i \simeq -\sqrt{2\epsilon_s kTN_A}\left[\sqrt{\frac{q2\phi_f}{kT} + \exp\frac{q(\phi_s - 2\phi_f)}{kT} - 1} - \sqrt{\frac{q2\phi_f}{kT}}\right] \qquad \text{for } \phi_s > 2\phi_f \tag{8.34}$$

In this expression, Q_i goes to zero as ϕ_s approaches $2\phi_f$. In strong inversion, the leading term of Q_i goes as $\sim \exp(q\phi_s/2kT)$.

The dependence of ϕ_s on V is given by [from Eq. (8.140)]

$$\phi_s \simeq 2\phi_f + \frac{kT}{q}\ln\left\{\left[\frac{C_{ox}(V - V_T)}{\sqrt{2\epsilon_s kTN_A}} + \sqrt{\frac{q2\phi_f}{kT}}\right]^2 - \frac{q2\phi_f}{kT} + 1\right\} \qquad \text{for } V > V_T \tag{8.35}$$

At $V = V_T$, $\phi_s = 2\phi_f$, consistent with our definition of threshold. In inversion, ϕ_s does increase somehow above $2\phi_f$ but it does so in a logarithmic way.

Other interesting results for the MOS electrostatics under the Poisson–Boltzmann formulation are derived in Advanced Topic AT8.3.

8.5 DYNAMICS OF THE MOS STRUCTURE

A MOS structure looks in many ways like a capacitor. On one side, there is a metal plate; on the other side, there is the semiconductor which is in effect another conducting "plate." In the middle, there is a dielectric, the oxide. Depending on the voltage that is applied to the MOS structure, net electrical charge exists at the metal/dielectric interface and in the semiconductor. The total charge at these two plates is of equal magnitude but opposite sign. If the voltage is changed, the charge on both plates is modified by the same amount in absolute terms. At all times, the magnitude of the charge on each plate is identical. The MOS structure behaves as a true capacitor. A simple equivalent model description of the MOS structure consists then of a capacitor, capturing the MOS electrostatics, in series with a resistor that accounts for the finite conductance of the substrate. This is sketched on Fig. 8.20(a). This simple equivalent circuit provides a reasonable description of the dynamics of the MOS structure.

For the simple *RC* circuit of Fig. 8.20(a) to be useful, all the dependencies of the MOS capacitance need to be properly described. There are two critical dependencies to understand and represent. The first one is the impact of voltage. The second one, perhaps less obvious, is time or the dynamic behavior of the electrostatics themselves. We study these two factors separately in the next two sections.

8.5.1 Quasi-static *C–V* characteristics

The capacitance–voltage characteristics of a MOS structure have tremendous practical importance as a diagnostic tool. From the *C–V* characteristics, one can extract the flat-band voltage, the threshold voltage, the oxide thickness and the semiconductor doping level. Also, a detailed study of the *C–V* characteristics can reveal nonidealities in the structure such as mobile oxide

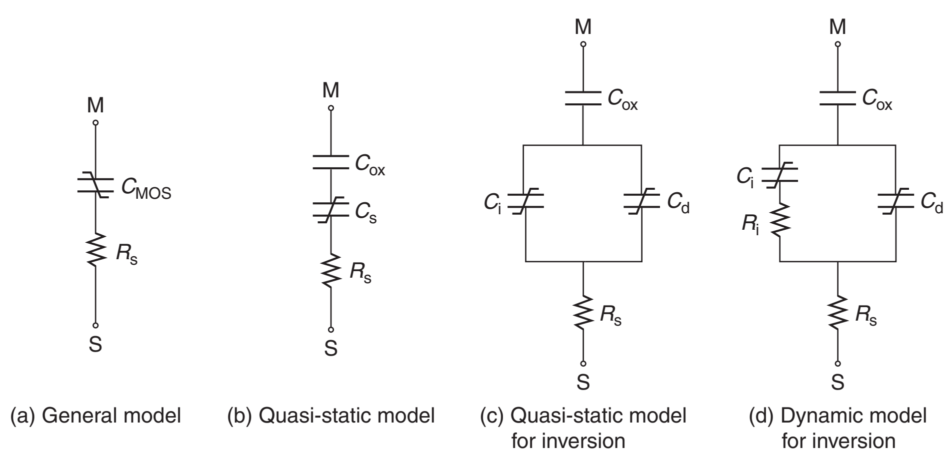


FIGURE 8.20
Equivalent circuit models for the MOS structure: (a) general model; (b) quasi-static model; (c) quasi-static model for inversion; and (d) dynamic model for inversion.

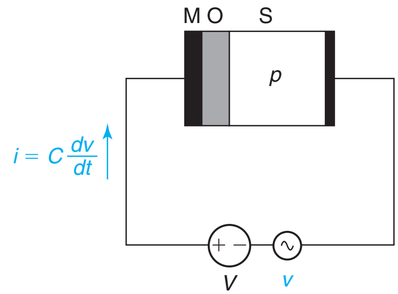


FIGURE 8.21
Circuit for the measurement of the *C–V* characteristics of a MOS structure. On top of a bias V, a small-signal v is applied. The current that flows is proportional to the time derivative of the small-signal v and the capacitance of the MOS structure.

charge (see Advanced Topic AT8.2.1), interface states (see Advanced Topic AT8.2.2), or gate doping depletion (in the case of a polySi gate, see Section 10.3.2). The *C–V* characteristics of a MOS structure are also of capital importance in just about any of its applications. The *C–V* characteristics summarize the complex behavior of the MOS structure that was described in detail in the preceding sections. Studying in detail the *C–V* characteristics is an opportunity to review and solidify our understanding of the physics of the MOS structure.

As we have encountered before in this book, a quasi-static situation is one without significant memory effects. Effectively, at all times, the MOS structure behaves as if it was in DC. The term quasi-static *C–V* characteristics refers to the capacitance that is measured at a certain bias voltage in a small-signal configuration when the frequency of the driving signal is low enough for the electrostatics to be independent of frequency (how low the frequency needs to be will become clear later). A circuit schematic is shown in Fig. 8.21. The formalism of the electrostatics of the MOS structure described so far in this chapter completely applies to a quasi-static situation and is therefore the basis for a derivation of the quasi-static *C–V* characteristics.

A general definition of capacitance of the MOS structure is the change of charge on the gate as a result of a small change in voltage across the structure:

$$C = \frac{dQ_g}{dV} \tag{8.36}$$

Since $Q_g = -Q_s$, using the chain rule for derivatives, the capacitance can be expressed as

$$C = -\frac{dQ_s}{dV} = -\frac{dQ_s}{d\phi_s}\frac{1}{\frac{dV}{d\phi_s}} \tag{8.37}$$

From the general relationship given by (8.18), we have

$$\frac{dV}{d\phi_s} = 1 - \frac{1}{C_{ox}}\frac{dQ_s}{d\phi_s} \tag{8.38}$$

Let us define the semiconductor capacitance, C_s, as

$$C_s = -\frac{dQ_s}{d\phi_s} \tag{8.39}$$

As we will see below, C_s defined this way is always positive. Armed with this definition, Eq. (8.38) can be rewritten as

$$\frac{dV}{d\phi_s} = 1 + \frac{C_s}{C_{ox}} \tag{8.40}$$

Substituting this in Eq. (8.37), we get

$$C = \frac{1}{\frac{1}{C_{ox}} + \frac{1}{C_s}} \tag{8.41}$$

There is a simple physical interpretation to the result embodied in Eqs. (8.39) and (8.41). The capacitance of the MOS structure can be thought of as two capacitances in series, one associated with the oxide, C_{ox}, and another one associated with the semiconductor C_s. The definition of C_s of Eq. (8.39) is consistent with this interpretation since it represents the change of the charge in the semiconductor as a result of a change in the electrostatic potential build-up across the semiconductor. This simple picture of two capacitors in series is schematically illustrated at the center of Fig. 8.20.

We now derive first-order expressions for C_s in the various regimes. In accumulation, $Q_s = Q_a$. Equation (8.32) gives Q_a versus ϕ_s. From this, we can easily derive C_s in accumulation. For strong accumulation ($|\phi_s| \gg kT/q$), we find

$$C_s = -\frac{dQ_a}{d\phi_s} \simeq \frac{q}{2kT} C_{ox}(V_{FB} - V) \qquad \text{for } V < V_{FB} \tag{8.42}$$

where we have used Eq. (8.24). In strong accumulation, C_s increases linearly with the gate voltage below flatband.

Close to flatband, when $\phi_s \simeq 0$, the semiconductor capacitance is

$$C_s = -\frac{dQ_a}{d\phi_s} \simeq \frac{\epsilon}{L_D} \qquad \text{for } V \simeq V_{FB} \tag{8.43}$$

where L_D is the Debye length. This is a particularly intuitive result since we know that the Debye length gives a sense of the distance over which small perturbations to charge neutrality in an extrinsic semiconductor are screened out.

In depletion, $Q_s = Q_d$. The dependence of the depletion charge with ϕ_s is given by Eq. (8.17). Using also the dependence of ϕ_s on V given by Eq. (8.21) and the definition of γ in Eq. (8.16), we can obtain

$$C_s = -\frac{dQ_d}{d\phi_s} \simeq \frac{C_{ox}}{\sqrt{1 + 4\frac{V - V_{FB}}{\gamma^2}} - 1} \qquad \text{for } V_{FB} < V < V_T \tag{8.44}$$

This expression for the semiconductor capacitance in depletion can also be obtained if we realize that

$$C_s \simeq \frac{\epsilon_s}{x_d} \qquad \text{for } V_{FB} < V < V_T \tag{8.45}$$

and we use the expression of $x_d(V)$ given in Eq. (8.19). In the depletion regime, there is a depletion region underneath the oxide. When the surface potential changes, the edge of the depletion region moves. This is at a distance x_d away from the semiconductor/oxide interface.

Looking at the Poisson–Boltzmann formulation in Appendix AT8.3, Eq. (8.44) is only valid for $\phi_s \gg kT/q$ or with V not close to V_{FB}. As V goes to zero, this equation clearly diverges. Using the more rigorous model of Appendix AT8.3, we can easily find that in the depletion regime for V close to zero, the semiconductor capacitance approaches the value given by Eq. (8.43).

In inversion, $Q_s = Q_i + Q_d$. This implies

$$C_s = -\frac{dQ_i}{d\phi_s} - \frac{dQ_d}{d\phi_s} = C_i + C_d \tag{8.46}$$

C_i is the inversion layer capacitance and C_d is the depletion capacitance. An equivalent circuit model description of this is two capacitors in parallel, as illustrated in Fig. 8.20.

To the first order, $Q_d \simeq Q_{dmax}$ and therefore $C_d \simeq 0$. Then $C_s \simeq C_i$. Equation (8.34) gives the dependence of Q_i on ϕ_s. In strong inversion, we also use Eq. (8.31) to get

$$C_s = -\frac{dQ_i}{d\phi_s} \simeq \frac{q}{2kT}Q_i = \frac{q}{2kT}C_{ox}(V - V_T) \qquad \text{for } V > V_T \tag{8.47}$$

The overall evolution of the semiconductor capacitance with V is shown at the top of Fig. 8.22. In accumulation and inversion, C_s increases in a linear fashion with V. In depletion, it evolves in a square-root form. The minimum capacitance is obtained around threshold.

The overall MOS capacitance is obtained by inserting these expressions for C_s into Eq. (8.41). In accumulation, we have

$$C \simeq C_{ox}\frac{1}{1 + \frac{2kT}{q(V_{FB}-V)}} \qquad \text{for } V < V_{FB} \tag{8.48}$$

Around flatband, we get

$$C \simeq C_{ox}\frac{1}{1 + C_{ox}\frac{L_D}{\epsilon_s}} \qquad \text{for } V \simeq V_{FB} \tag{8.49}$$

In depletion, we obtain

$$C \simeq \frac{C_{ox}}{\sqrt{1 + 4\frac{V-V_{FB}}{\gamma^2}}} \qquad \text{for } V_{FB} < V < V_T \tag{8.50}$$

In inversion, we get

$$C \simeq C_{ox}\frac{1}{1 + \frac{2kT}{q(V-V_T)}} \qquad \text{for } V > V_T \tag{8.51}$$

The bottom of Fig. 8.22 sketches C versus V. Since we are looking at two capacitances in series, the smaller one tends to dominate. Hence, under strong accumulation and inversion, the overall MOS capacitance approaches C_{ox}. In depletion, C_s decreases as V increases beyond V_{FB}, and the overall capacitance can become quite small.

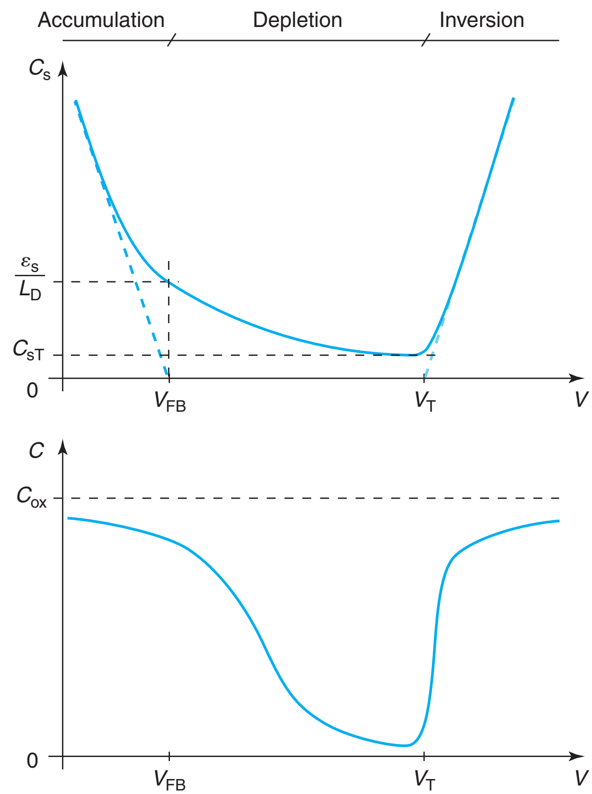


FIGURE 8.22
Top: Sketch of semiconductor capacitance of MOS structures in different regimes. Bottom: Overall MOS capacitance versus voltage.

The capacitance of the semiconductor in the depletion regime at threshold is of special significance as it enters in models for the subthreshold behavior of the MOS structure which we study below. An expression for this can be obtained from Eq. (8.44) by setting $V = V_T$. After some algebra, this can be written as

$$C_{sT} \simeq C_{ox} \frac{\gamma}{2\sqrt{2\phi_f}} \tag{8.52}$$

This result can also be obtained if we realize that $C_{sT} = \epsilon_s / x_{dmax}$ and use the expression for x_{dmax} in Eq. (8.26).

8.5.2 High-frequency *C–V* characteristics

The quasi-static *C–V* characteristics described above are obtained for a driving signal with low enough frequency. If a high frequency is employed, the physics change. This is described in this section. Let us examine the three voltage regimes separately.

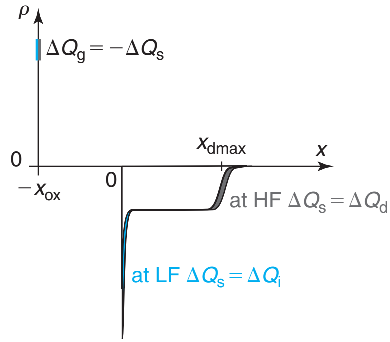


FIGURE 8.23
Illustration of change in charge in semiconductor in response to an AC voltage signal at low frequencies and high frequencies.

The application of an AC voltage modulates the charge in the semiconductor. In the accumulation regime, this charge is made out of holes in the accumulation layer. Since holes are majority carriers in the semiconductor, they are in good communication with the outside world through the ohmic contact. The proper time constant to get holes in and out of the accumulation layer is the dielectric relaxation time of the substrate or the *RC* time constant of the whole structure, whichever is largest. Either case, it is a rather fast phenomenon. In consequence, the *C*–*V* characteristics in accumulation derived in the previous section apply up to fairly high frequencies. The situation is identical in depletion, where the hole concentration is modulated at the edge of the depletion region in response to the application of an AC voltage. The *C*–*V* characteristics in depletion obtained above also apply up to rather high frequencies.

The situation is quite different in inversion. As we discussed above, in strong inversion, the electron concentration in the inversion layer is modulated in response to the AC voltage. The thickness of the depletion region is basically unchanged. The extra electrons required to satisfy the electrostatics have to be generated in the semiconductor. The proper time constant for this is the generation lifetime, typically in the microsecond to millisecond range. In consequence, the *C*–*V* characteristics derived above only apply if the AC signal changes slowly in the scale of the inverse of τ_g, this typically means frequencies in the *kHz* range or below. For higher frequencies, the inversion capacitance picture ought to look quite different from what has been discussed so far.

In fact, if the frequency is high enough, the electron concentration in the inversion layer is unable to respond to the AC voltage signal. Since the voltage is nevertheless changing, charge modulation must take place somewhere to satisfy the electrostatics. The only other place where charge can be modulated is at the edge of the depletion region with the substrate. This can be accomplished by getting holes in and out of the substrate through the ohmic contact, a process that as we discussed above, is limited by the *RC* time constant of the structure. This is sketched in Fig. 8.23.

This understanding allows us to conclude that the high-frequency inversion semiconductor capacitance is, to the first order, equal to the depletion capacitance at threshold [derived in Eq. (8.52)]:

$$C_{s,HF} \simeq C_{sT} \tag{8.53}$$

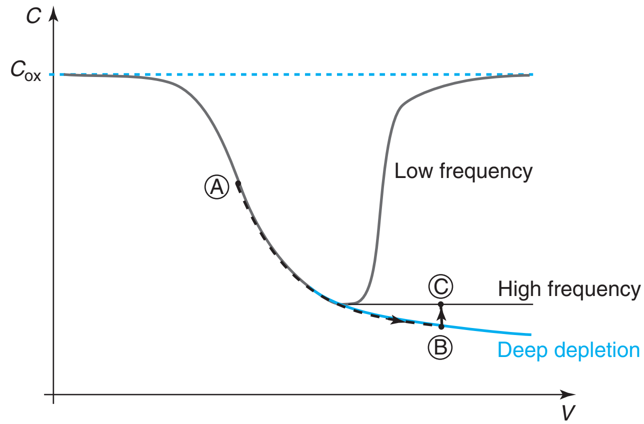


FIGURE 8.24
C–V characteristics of typical MOS structure at low frequency (quasi-static), high frequency, and under a fast ramp or after a step (deep depletion). The marked points refer to the transient described in Fig. 8.26.

This is because in inversion $\phi_s \simeq \phi_{sT}$ and $x_d \simeq x_{dmax}$. C_{sT} can be a substantially lower value than the quasi-static capacitance of the structure for the same bias point [given by Eq. (8.47)].

A comparison of the theoretical *C–V* characteristics of a typical MOS structure at low frequency and high frequency is shown in Fig. 8.24. In inversion, *C* remains at the lowest value attained at threshold. Experimental results showing the evolution of the *C–V* characteristics of a typical MOS with frequency are depicted in Fig. 8.25.

An equivalent circuit description of the high-frequency behavior of the MOS structure is shown on Fig. 8.20(d). In this model, in series with the inversion layer capacitance, we have placed a resistor R_i. This resistor captures the difficulty of providing electrons to the inversion layer. The $R_i - C_i$ branch behaves as a low-pass filter. Its time constant, R_iC_i, is the inverse of the generation lifetime τ_g that controls electron generation in the depletion region. For low frequencies, the impedance of this branch is dominated by C_i. Since it is in parallel but it is much larger than C_d, the capacitance of the MOS structure is approximately C_i. At high frequencies, on the other hand, the impedance is dominated by R_i. In consequence, the capacitance of the MOS structure is approximately $C_d \simeq C_{sT}$.

8.5.3 Deep depletion

The previous section has revealed the rich dynamics of the MOS structure in the inversion regime that originate from the "sluggishness" of response of the inversion layer to changes in applied voltage. The condition just studied was one in which a small high-frequency signal was applied on top of a DC bias. Another dynamic situation of great interest is one that results when a MOS structure is suddenly switched into inversion by the application of a step or a fast voltage ramp. This condition is called *deep depletion*.

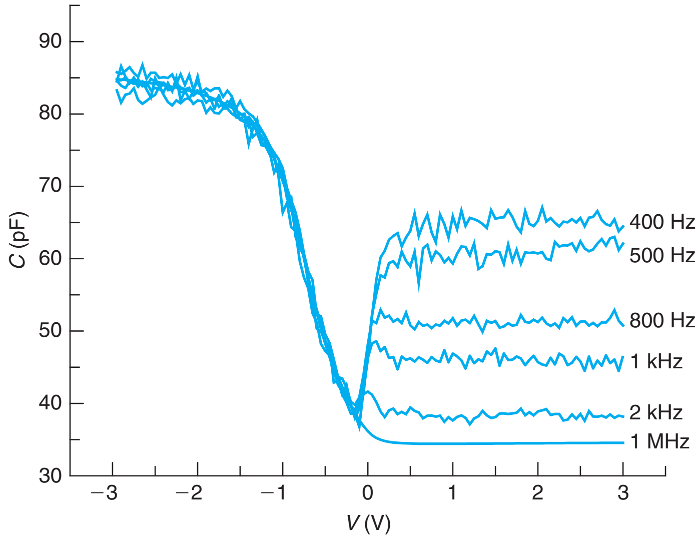


FIGURE 8.25
Experimental *C–V* characteristics of a MOS structure as a function of frequency. At low frequency, the measurement becomes noisy and in general it is not possible to obtain the ideal quasi-static behaviour sketched in Fig. 8.22 [Reprinted with permission from Tassanee Payakapan].

Consider a MOS structure at zero bias. Let us assume that at $V = 0$ it is in depletion. Let us consider what happens if at $t = 0$, we apply a voltage step, as sketched in Fig. 8.26. If the step is negative and of a magnitude that exceeds the flat-band voltage, the structure will be driven into accumulation. After an *RC* time constant, a relatively short time, the depletion region will be wiped out and holes will accumulate at the semiconductor/oxide interface. If the step is negative but of a magnitude smaller than V_{FB} or positive but smaller than V_T, the depletion region thickness will change. This also happens in a timescale of the corresponding *RC* time constant.

If the voltage step is positive and exceeds the threshold voltage, the situation gets complicated. A short time after the application of the step (following the inevitable *RC* delay), the only way to satisfy the electrostatics of the structure is for the depletion region to widen beyond x_{dmax}. This is because even though the voltage is such that an inversion layer should be formed, there has not been enough time for the electrons to be generated. For this to happen, we must wait a time of order of the generation lifetime, τ_g, a fairly long time in the scale of many useful electronic functions. The MOS structure has gone into what is called *deep depletion* because for a while the depletion region thickness widens beyond the value that corresponds to the inversion regime. If the step has enough duration, eventually the required electrons will be generated, the inversion layer will be formed and the depletion region will collapse down to x_{dmax}. This is sketched in Fig. 8.26.

The dynamics of the deep depletion regime can be observed experimentally by measuring the capacitance of the MOS structure during the transient. This can be accomplished by overlapping a high-frequency sinusoidal signal on top of the voltage step. The result is sketched in

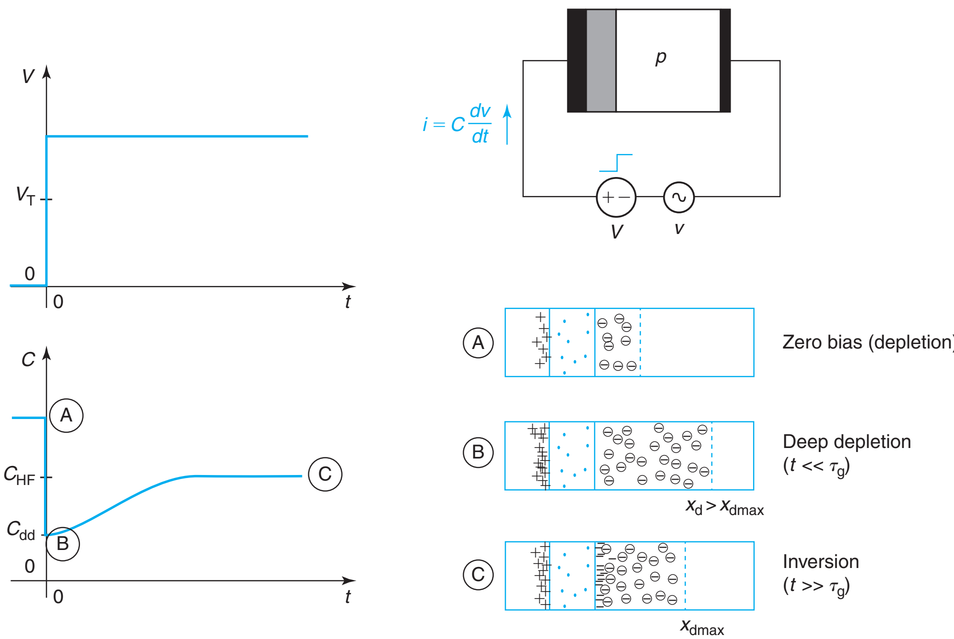


FIGURE 8.26
Evolution of the electrostatics of the MOS structure after the application of a voltage step exceeding V_T. The marked points correspond to the locations indicated in Fig. 8.24. Immediately following the application of the step, the inversion layer does not have enough time to form and the depletion region widens beyond x_{dmax}. This is called *deep depletion*. Given enough time, electrons are generated, the inversion layer fully forms and the depletion region collapses to x_{dmax}. If the capacitance is being measured as a function of time by means of a high-frequency AC voltage overlapping the step, we will observe that immediately upon the application of the step, the capacitance drops below its inversion value. As time goes on, the capacitance relaxes to the value corresponding to inversion.

Fig. 8.26. After the *RC* transitory has died off but before a significant amount of electrons has been generated, the capacitance in deep depletion will be lower than the high-frequency capacitance in the inversion regime. This is because $x_d > x_{dmax}$. Given enough time, the inversion layer will eventually grow, the depletion region will collapse back to x_{dmax} and the capacitance will revert to the high-frequency inversion value discussed in the previous section.

Deriving a model for the electrostatics of the MOS structure in deep depletion is quite simple because it is a fairly straightforward extension of the model for the electrostatics of the depletion regime for $V > V_T$ that we have derived earlier in this chapter. At $t = 0^+$, right after the voltage step is applied, the depletion region widens to a thickness that we can denote as $x_{d,dd}$. In analogy with Eq. (8.19), $x_{d,dd}$ is given by

$$x_{d,dd} = \frac{\epsilon_s}{C_{ox}}\left(\sqrt{1 + 4\frac{V - V_{FB}}{\gamma^2}} - 1\right) \tag{8.54}$$

Since at $t = 0^+$ the charge in the semiconductor is entirely given by the depletion charge, the semiconductor capacitance at this point in time, is simply given by

$$C_{s,dd} \simeq \frac{\epsilon_s}{x_{d,dd}} \tag{8.55}$$

The overall capacitance of the MOS structure is then

$$C_{dd} = \frac{1}{\frac{1}{C_{ox}} + \frac{1}{C_{s,dd}}} \simeq \frac{C_{ox}}{\sqrt{1 + 4\frac{V - V_{FB}}{\gamma^2}}} \tag{8.56}$$

For $V > V_T$, it follows that, $x_{d,dd} > x_{dmax}$, $C_{s,dd} < C_{sT}$, and $C_{dd} < C_{HF}$. In deep depletion, the higher the voltage the structure is stepped into, the more the depletion region widens and the lower the capacitance. This is often sketched in the C–V characteristics as shown in Fig. 8.24.

To further the understanding of the deep depletion regime, Fig. 8.27 illustrates the evolution of the energy band diagrams of the MOS structure at key points in the transient of Fig. 8.26. The band diagrams at points A and C correspond to depletion and inversion and have already been studied before. The interesting new band diagram is the one corresponding to point B or deep depletion. The extended depletion region produces a large band bending in the semiconductor. In spite of this, there is no inversion layer because there has not been enough time for the electrons to be generated. How could these two facts be reconciled in the energy band diagram? Since the electron concentration is given by the energy difference between the quasi-Fermi level for electrons and the edge of the conduction band, this situation requires that E_{fe} be substantially below E_c at the surface of the semiconductor. At the same time, the hole quasi-Fermi level must remain flat across the structure as these are majority carriers and they respond quickly to the change of voltage across the structure. Interestingly, satisfying these constrains implies that the electron quasi-Fermi level ends up below the hole quasi-Fermi level across the depletion region, as indicated in Fig. 8.27. This is consistent with the occurrence of electron–hole pair generation which we have argued needs to take place for the structure to eventually reach the steady-state situation of point C.

In deep depletion, the semiconductor is out of equilibrium, as in a reverse bias PN junction. The resulting electron generation contributes to growing the inversion layer and the hole generation shrinks the depletion layer. If the body of the semiconductor extends to enough depth, in deep depletion, E_{fe} remains below E_{fh} over a length in the scale of the electron diffusion length.

For deep depletion to occur, there must not be a supply of electrons readily available to form the inversion layer. That is the situation studied here where the electrons need to be generated. As we will see later in this chapter, in a three-terminal MOS structure in which there is an n^+-region right next to the channel that can supply the required electrons, deep depletion cannot be produced.

The dynamic behavior of the MOS structure under pulsed conditions that has been discussed in this section is at the heart of charge-coupled devices (CCDs) and CCD-based image sensors (see Problem 8.28). It is also exploited as a characterization technique. The recovery time from

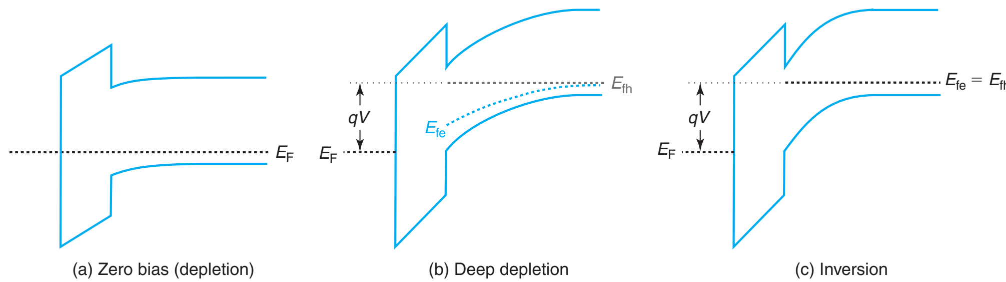


FIGURE 8.27
Evolution of the energy band diagrams in the MOS structure of Fig. 8.26 at key points in time.

deep depletion in a MOS structure is related to the generation lifetime that itself depends on the purity and defect concentration in the vicinity of the oxide–semiconductor interface. A large concentration of defects leads to a fast recovery while the contrary happens if the MOS structure is of high quality.

EXERCISE 8.5

Consider a MOS structure identical to that of Exercises 8.3 and 8.4. Calculate the quasi-static capacitance and the high-frequency capacitance at a bias $V = 2$ V. Calculate also the capacitance immediately after pulsing the structure from $V = 0$ to $V = 2$ V.

Since for this structure $V = 2$ V corresponds to inversion, to the first order, the quasi-static capacitance is simply the oxide capacitance:

$$C \simeq C_{ox} = 7.7 \times 10^{-7}\ \mathrm{F/cm^2}$$

The high-frequency capacitance is the parallel of C_{sT} and C_{ox}. C_{sT} is given by Eq. (8.52):

$$C_{sT} = C_{ox}\frac{\gamma}{2\sqrt{2\phi_f}} = \frac{0.58\ \mathrm{V}^{1/2}}{2\sqrt{0.93\ \mathrm{V}}}7.7 \times 10^{-7}\ \mathrm{F/cm^2} = 2.3 \times 10^{-7}\ \mathrm{F/cm^2}$$

Then

$$C_{HF} = \frac{1}{\frac{1}{C_{ox}} + \frac{1}{C_s}} = \frac{1}{\frac{1}{7.7\times10^{-7}} + \frac{1}{2.3\times10^{-7}}} = 1.8 \times 10^{-7}\ \mathrm{F/cm^2}$$

When the structure is pulsed from $V = 0$ to $V = 2$ V, it goes into deep depletion. In order to compute the capacitance, we must first calculate the extent of the depletion region right after the step. This is given by Eq. (8.54):

$$x_{d,dd} = \frac{\epsilon_s}{C_{ox}}\left(\sqrt{1 + 4\frac{V - V_{FB}}{\gamma^2}} - 1\right) = \frac{1.04 \times 10^{-12}\ \mathrm{F/cm}}{7.7 \times 10^{-7}\ \mathrm{F/cm^2}}\left(\sqrt{1 + 4\frac{2+1}{0.58^2}} - 1\right) = 6.9 \times 10^{-6}\ \mathrm{cm}$$

$C_{s,dd}$ is, then, from Eq. (8.55):

$$C_{s,dd} = \frac{\epsilon_s}{x_{d,dd}} = \frac{1.04 \times 10^{-12}\ \text{F/cm}}{6.9 \times 10^{-6}\ \text{cm}} = 1.5 \times 10^{-7}\ \text{F/cm}^2$$

When put in parallel with C_{ox}, we obtain the capacitance of the whole structure in deep depletion:

$$C_{dd} = \frac{1}{\frac{1}{C_{ox}} + \frac{1}{C_{s,dd}}} = \frac{1}{\frac{1}{7.7\times10^{-7}} + \frac{1}{1.5\times10^{-7}}} = 1.3 \times 10^{-7}\ \text{F/cm}^2$$

8.6 WEAK INVERSION AND THE SUBTHRESHOLD REGIME

The subthreshold regime, also sometimes called *weak inversion*, refers to the regime of operation of a MOS structure for gate voltages just below threshold. In this range of voltages, the electron concentration at the oxide–semiconductor interface is lower than the acceptor doping level in the bulk. From an electrostatic point of view, the dominant charge originates in the uncompensated acceptors in the depletion region and that is why up to this point, we have labeled this mode of operation the depletion regime. However, immediately below threshold, the electron concentration is of some significance and in a MOSFET, it can result in non-negligible current that is known as the *subthreshold current*. An example of subthreshold current in a MOSFET can be seen in Fig. 9.44.

In modern microelectronics, there is strong emphasis on low-power operation. One of its implications is the need to tightly control the trickle current that inevitably flows through a MOSFET when it is supposed to be OFF. This is often called the *OFF current*, I_{off}, and is a key consideration in device design. In addition, in order to predict power dissipation in a circuit, it is important to model this current correctly. Furthermore, in dynamic circuits, the time that a certain amount of charge can get stored in a dynamic node depends inversely on the leakage current that flows to neighboring nodes. In MOSFET circuits, this is often set by the subthreshold current of the transistors. In order to design devices with minimum OFF current, good physical understanding of the subthreshold regime is required. In this section, we develop a first-order model for the total electron concentration in the semiconductor in the subthreshold regime. This will be the basis for the derivation of a model for the subthreshold current of a MOSFET in Chapter 9.

To start, we need to revisit our discussion of the depletion regime in Section 8.4.1. Our interest here is in the total electron charge in the semiconductor. This can be obtained using the Boltzmann relationship for electrons as follows [3]:

$$Q_e = -q\int_0^\infty n(x)dx = -q\frac{n_i^2}{N_A}\int_0^\infty \exp\frac{q\phi}{kT}dx = -q\frac{n_i^2}{N_A}\int_{\phi_s}^0 \exp\frac{q\phi}{kT}\left(\frac{dx}{d\phi}\right)d\phi \tag{8.57}$$

[3] We use here Q_e rather than Q_i to emphasize the important point that the MOS structure is not in inversion.

In the last step, we have changed variables from x to ϕ. The fact that the upper limit of the integral goes to $x = \infty$ or $\phi = 0$ is not a problem since its most significant contributions are right next to the oxide–semiconductor interface.

$d\phi/dx$ is the minus electric field in the depletion region. In the depletion regime, this can be written in terms of ϕ as

$$\frac{d\phi}{dx} = -\mathcal{E} \simeq -\sqrt{\frac{2qN_A\phi}{\epsilon_s}} \tag{8.58}$$

We plug this into Eq. (8.57) to obtain

$$Q_e \simeq -q\frac{n_i^2}{N_A}\sqrt{\frac{\epsilon_s}{2qN_A}}\int_0^{\phi_s}\frac{1}{\sqrt{\phi}}\exp\frac{q\phi}{kT}d\phi \tag{8.59}$$

As stated above, the major contributions to this integral come around $\phi = \phi_s$, so one should not be concerned with the divergence at $\phi = 0$. Therefore, we can approximately write

$$Q_e \simeq -q\frac{n_i^2}{N_A}\sqrt{\frac{\epsilon_s}{2qN_A\phi_s}}\int_0^{\phi_s}\exp\frac{q\phi}{kT}d\phi \simeq -\frac{kT}{q}\sqrt{\frac{q\epsilon_s N_A}{2\phi_s}}\exp\frac{q(\phi_s - 2\phi_f)}{kT} \tag{8.60}$$

where we have assumed that $\phi_s \gg kT/q$ and we have used the definition of ϕ_f in Eq. (8.25).

We obtain here the important result that in the subthreshold regime, $|Q_e| \sim \exp\frac{q\phi_s}{kT}$, which makes good sense. The rigorous numerical calculation presented in Appendix AT8.3 confirms the validity of this result.

We now need to relate ϕ_s to V. We could use Eq. (8.21) which is valid in the depletion regime. However, since we are interested in a relatively small range of voltages immediately below threshold, we can obtain a simpler expression by linearizing the voltage dependence of ϕ_s around threshold in the following way:

$$\phi_s - \phi_{sT} \simeq \phi_s - 2\phi_f \simeq \left.\frac{d\phi_s}{dV}\right|_T (V - V_T) \tag{8.61}$$

The derivative $d\phi_s/dV$ around threshold can be obtained from Eq. (8.40):

$$\left.\frac{d\phi_s}{dV}\right|_T = \frac{1}{\left.\frac{dV}{d\phi_s}\right|_T} = \frac{1}{1 + \frac{C_{sT}}{C_{ox}}} \tag{8.62}$$

Substituting this into Eq. (8.60), approximating $\phi_s \simeq 2\phi_f$ inside the square root and using the definition of C_{sT} in Eq. (8.52), we finally get

$$Q_e \simeq -\frac{kT}{q}C_{sT}\exp\frac{q(V - V_T)}{\left(1 + \frac{C_{sT}}{C_{ox}}\right)kT} \qquad \text{for } V \le V_T \tag{8.63}$$

Equation (8.63) is a very important result. Below threshold, $|Q_e|$ drops exponentially with V. The sharpness of this drop is determined by the ratio of C_{sT} to C_{ox}. The lower this ratio, the sharper the drop. We can understand this result if we realize that, differentially, we are in front of a capacitive divider with C_{ox} and C_{sT} in series, and ϕ_s being the potential at the common

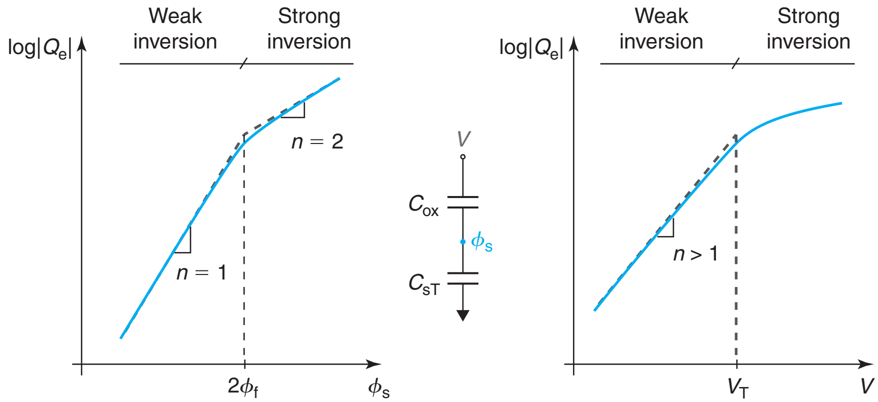


FIGURE 8.28
Sketches of logarithm of electron charge at MOS oxide–semiconductor interface from weak inversion to strong inversion. Left: versus surface potential. Right: versus voltage. Indicated is the ideality factor of the exponential regions. Inset at center: equivalent circuit model around threshold.

node (see inset in Fig. 8.28). The higher C_{ox} is with respect to C_{sT}, the more closely ϕ_s follows changes in V. Or, in other words, the tighter the electrostatic control of the gate over the surface potential, the sharper the inversion layer can be turned on and off. The factor multiplying kT in the denominator is often called the *ideality factor* of the subthreshold regime. It is clear that it is always larger than unity.

Figure 8.28 sketches the log of $|Q_e|$ versus ϕ_s on the left and versus V on the right from weak inversion to strong inversion. In weak inversion, the integrated electron charge increases as $\sim \exp(q\phi_s/kT)$. In contrast, in strong inversion, the dependence is $\sim \exp(q\phi_s/2kT)$. In weak inversion, $|Q_e|$ increases exponentially with V with an ideality factor that is greater than 1, while in strong inversion, $|Q_e|$ increases linearly with $V - V_T$.

The sharpness of the drop of $|Q_e|$ with V below threshold is characterized by the so-called *inverse subthreshold slope* or more commonly referred to as the *subthreshold swing*, S. This is defined as the voltage required to increase the magnitude of Q_e by a decade. From Eq. (8.63), we have

$$S = n\frac{kT}{q}\ln 10 = \left(1 + \frac{C_{sT}}{C_{ox}}\right)\frac{kT}{q}\ln 10 \tag{8.64}$$

S is usually given in mV/dec. At best, if $n = 1$, $S = 60$ mV/dec at room temperature. Typically, S is higher than that, meaning, it is less sharp than the ideal value.

EXERCISE 8.6

Consider a MOS structure such as the one of Exercises 8.3–8.5. Calculate the subthreshold swing S at room temperature.

First, we compute the ideality factor n:

$$n = 1 + \frac{C_{sT}}{C_{ox}} = 1 + \frac{\gamma}{2\sqrt{2\phi_f}} = 1 + \frac{0.58}{2\sqrt{0.93}} = 1.3$$

The subthreshold swing is then obtained from Eq. (8.64). At room temperature:

$$S = n\frac{kT}{q}\ln 10 = 1.3 \times 0.026\ \text{V} \times \ln 10 = 0.078\ \text{V} = 78\ \text{mV/dec}$$

8.7 THREE-TERMINAL MOS STRUCTURE

The great usefulness of the MOS structure in modern electronics comes from the possibility of contacting the inversion layer *independently* from the rest of the structure. This allows its use as a conducting "channel" with a conductivity that can be controlled through the gate voltage. This is the essence of the MOSFET. Making a contact to the inversion layer is not difficult. One only has to place an n^+-doped region right next to the edge of the gate, as sketched in Fig. 8.29. This n^+-region communicates with the outside world through an ohmic contact. It is important that there be a small overlap between the n^+-region and the actual MOS structure so that a reliable, low-resistance connection to the inversion layer is obtained. A feature of this three-terminal MOS structure is that the depletion regions of the N^+P region and the MOS structure overlap.

A contact to the inversion layer allows the application of a voltage between the inversion layer and the semiconductor. This enables the manipulation of the surface potential in the semiconductor and the inversion layer charge independently from the gate. Before we examine the consequences of doing this, let us label the voltages. The n^+-region that provides access to the

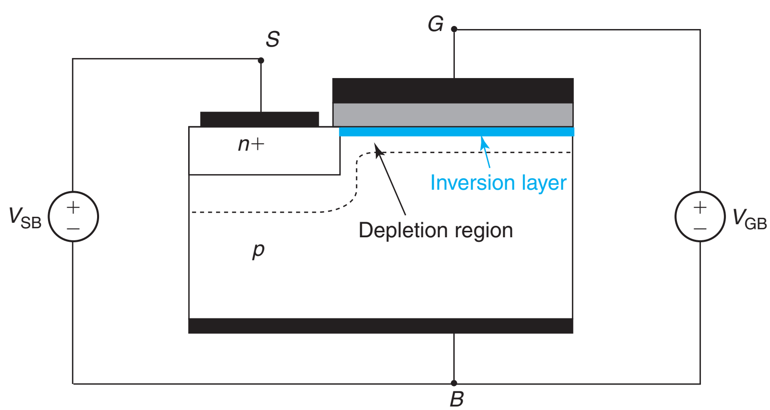


FIGURE 8.29
Schematic diagram of a MOS structure with a contact to the inversion layer.

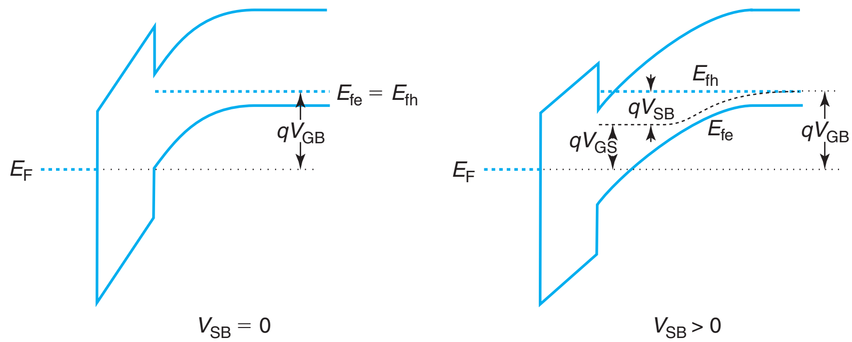


FIGURE 8.30
Energy band diagram of three-terminal MOS structure with $V_{SB} = 0$ (left) and $V_{SB} > 0$ (right).

inversion region is called the *source*, since it will play the role of source of electrons for the MOSFET. The semiconductor substrate is often referred to also as the *body*. The metal continues to be denoted as *gate*. Therefore, the source to body voltage is denoted as V_{SB} and the gate-to-body voltage is labeled as V_{GB}.

The range of voltages that in practice we might be interested in applying to the inversion layer is restricted by the presence of the N^+P source-body junction diode. Under forward bias, substantial forward diode current can flow, a largely undesirable condition in most applications. Hence, we will discuss the physics of the situation for reverse bias applied to the N^+P junction, that is, for $V_{SB} \geq 0$. Our models, however, also apply for $V_{SB} \leq 0$. In this discussion, we will keep V_{GB} constant and at a level such that an invesion layer is present under all conditions.

The best way to discuss the effect of $V_{SB} \geq 0$ is to sketch appropriate energy band diagrams, as done in Fig. 8.30. The left of this figure shows a diagram that corresponds to a MOS structure in inversion with $V_{SB} = 0$. The right diagram shows the same structure under an identical value of V_{GB} but with $V_{SB} > 0$. On the left, $E_{fe} = E_{fh}$ throughout the semiconductor body which is in a quasi-equilibrium estate. On the right, this is not the case. Through the n^+-region, the ohmic contact "grabs" on the majority carrier quasi-Fermi level of the inversion layer, that is, E_{fe}. Since the substrate contact holds onto the hole quasi-Fermi level in the body, the Fermi level is split across the semiconductor by qV_{SB} in the manner that is sketched in the diagram. It is clear that a consequence of this is that the surface potential, the total potential difference across the semiconductor, increases by about the same amount, that is

$$\phi_{sT}(V_{SB}) = \phi_{sT}(V_{SB} = 0) + V_{SB} \tag{8.65}$$

The most interesting question to try to answer is what happens to the absolute magnitude of the inversion layer charge, that is, does it change, and if so, does it increase or decrease? It is clear from Fig. 8.30 that the charge in the depletion layer has to increase in absolute terms in order to accrue a higher surface potential. This happens through a widening of the depletion region. It is less evident how the inversion charge evolves. In order to answer this, it is convenient to look in some detail at the changes that occur in the electrostatics. These are sketched in Fig. 8.31.

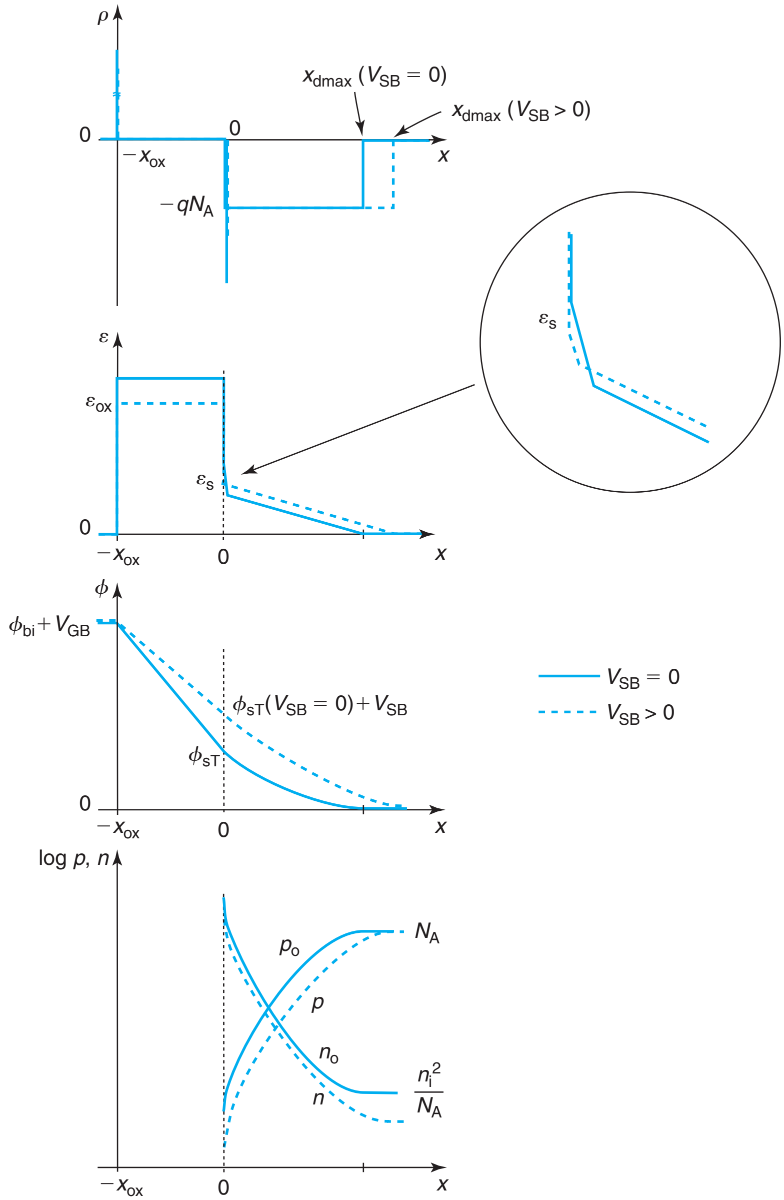


FIGURE 8.31
Evolution of the electrostatics of the three-terminal MOS structure upon the application of $V_{SB} > 0$.

It is easiest to start with the third diagram from the top in Fig. 8.31 that sketches the potential distribution across the structure. Since V_{GB} is unchanged, the total potential build up from gate to substrate contact remains unchanged and equal to $\phi_{bi} + V_{GB}$. The fact that ϕ_s has increased upon the application of $V_{SB} > 0$ implies that ϕ_{ox} must have decreased. This in turn means a reduction of $\mathcal{E}_{ox}$ and as a consequence of $\mathcal{E}_s$. Gauss's law requires that the magnitude of the total charge in the semiconductor, $|Q_s|$, must also decrease. Since $Q_s = Q_i + Q_d$, and we argued above that $|Q_d|$ was increased, the only possible way to reconcile the picture is for $|Q_i|$ to be reduced. In summary, the application of $V_{SB} > 0$ results in a reduction of the electron concentration in the inversion layer.

An important consequence of the application of a voltage between the inversion layer and the body is the split of quasi-Fermi levels in the semiconductor. In particular, for $V_{SB} > 0$, $E_{fe} < E_{fh}$. This is a situation in which the carrier concentrations are driven *below* the thermal equilibrium values, as sketched in the bottom diagram of Fig. 8.31. As we learned in Chapter 4, the semiconductor reacts by trying to reestablish equilibrium, that is, by generating electron–hole pairs. A consequence of this is the presence of a small leakage current flowing through the source and body terminals. This adds up to the reverse current of the N^+P source-body junction. The magnitude of this current depends on the nature and concentration of traps inside the depletion region. As in a reverse-biased PN junction, the minority carrier concentration gets reduced in the body below equilibrium over a distance of the order of the diffusion length.

The impact of V_{SB} on the electrostatics of the MOS structure can be viewed also as producing a shift in the threshold voltage. Since the application of $V_{SB} > 0$ results in a reduction of the magnitude of the inversion charge that corresponds to a certain V_{GB}, this can be captured by an *increase* in the threshold voltage. Through Eq. (8.31), we see that this produces the desired result. It is straightforward to derive a new expression for V_T that accounts for the impact of V_{SB}. Since the application of V_{SB} increases ϕ_s by V_{SB}, at threshold to the first order, the surface potential goes from ϕ_{sT} to $\phi_{sT} + V_{SB}$. Hence, the expression of the threshold voltage of Eq. (8.28) becomes

$$V_T^{GB} = V_{FB} + \phi_{sT} + V_{SB} + \gamma\sqrt{\phi_{sT} + V_{SB}} \tag{8.66}$$

This can also be written as

$$V_T^{GB}(V_{SB}) = V_{To} + V_{SB} + \gamma\left(\sqrt{\phi_{sT} + V_{SB}} - \sqrt{\phi_{sT}}\right) \tag{8.67}$$

where we have defined $V_{To} = V_T^{GB}(V_{SB} = 0)$.

Since we are now in front of a device with three terminals, it is important to specify the two terminals that the threshold voltage refers to. This is reflected in the new notation used in the equations above. For MOSFET operation, we are not as much interested in the gate-to-body threshold voltage, V_T^{GB}, as we have been considering until now, but in the gate-to-source threshold voltage V_T^{GS}. Since $V_{GB} = V_{GS} + V_{SB}$, the two threshold voltages are related by

$$V_T^{GS}(V_{SB}) = V_T^{GB}(V_{SB}) - V_{SB} = V_{FB} + \phi_{sT} + \gamma\sqrt{\phi_{sT} + V_{SB}} \tag{8.68}$$

This can be easily rewritten as

$$V_T^{GS}(V_{SB}) = V_{To} + \gamma(\sqrt{\phi_{sT} + V_{SB}} - \sqrt{\phi_{sT}}) \tag{8.69}$$

which increases as V_{SB} increases. Note that $V_{To} = V_T^{GB}(V_{SB} = 0) = V_T^{GS}(V_{SB} = 0)$.

As V_T shifts with the application of V_{SB}, the charge in the inversion layer is affected. The charge control relation, Eq. (8.31), must be modified to account for this. It can be written in two ways, in terms of the gate-to-body voltage:

$$Q_i = -C_{ox}[V_{GB} - V_T^{GB}(V_{SB})] \tag{8.70}$$

or, as more commonly done, in terms of the gate-to-source voltage:

$$Q_i = -C_{ox}[V_{GS} - V_T^{GS}(V_{SB})] \tag{8.71}$$

Both formulations give, of course, the same result.

EXERCISE 8.7

Consider a MOS structure such as the one of Exercises 8.3–8.6 with an additional contact to the source that allows the application of V_{SB}. Calculate ϕ_{sT}, V_T^{GB}, V_T^{GS}, and C_{sT} for $V_{SB} = 3$ V.

Let us first calculate ϕ_{sT}. From Exercise 8.4, we know that for $V_{SB} = 0$, $\phi_{sT} = 0.93$ V. With $V_{SB} = 3$ V, Eq. (8.65) implies

$$\phi_{sT}(V_{SB} = 3\text{ V}) = \phi_{sT}(V_{SB} = 0) + V_{SB} = 0.93 + 3 = 3.93\text{ V}$$

V_T^{GB} is obtained from Eq. (8.67):

$$\begin{aligned} V_T^{GB}(V_{SB} = 3\text{ V}) &= V_{To} + V_{SB} + \gamma(\sqrt{\phi_{sT} + V_{SB}} - \sqrt{\phi_{sT}}) \\ &= 0.5 + 3 + 0.58(\sqrt{0.93 + 3} - \sqrt{0.93}) = 4.1\text{ V} \end{aligned}$$

V_T^{GS} is obtained simply from

$$V_T^{GS}(V_{SB}) = V_T^{GB}(V_{SB}) - V_{SB} = 4.1 - 3 = 1.1\text{ V}$$

C_{sT} is lowered by the application of V_{SB} because the depletion region under the inversion layer widens. It can be obtained by adapting Eq. (8.52) in the following way:

$$C_{sT}(V_{SB}) = \frac{\gamma C_{ox}}{2\sqrt{2\phi_f + V_{SB}}} = \frac{0.58\text{ V}^{1/2} \times 7.7 \times 10^{-7}\text{ F/cm}^2}{2\sqrt{0.93 + 3\text{ V}}} = 1.1 \times 10^{-7}\text{ F/cm}^2$$

The phenomenon that we have just discussed, the *back-bias effect* as it is sometimes referred to, has important implications for MOSFET and CMOS circuit operation and is a major concern to device and circuit designers. Three examples follow.

- The optimum MOSFET technology to be used for a given application depends on the desired circuit topologies and their sensitivity to the back-bias effect. This is illustrated in Fig. 8.32 that shows a typical CMOS NAND logic gate[4]. Let us focus on the n-MOS

[4]In logic circuits, the term "gate" is used to refer to elemental circuit configurations that perform simple logic functions. The use of the word "gate" here should not be confused with the metal gate of a transistor.

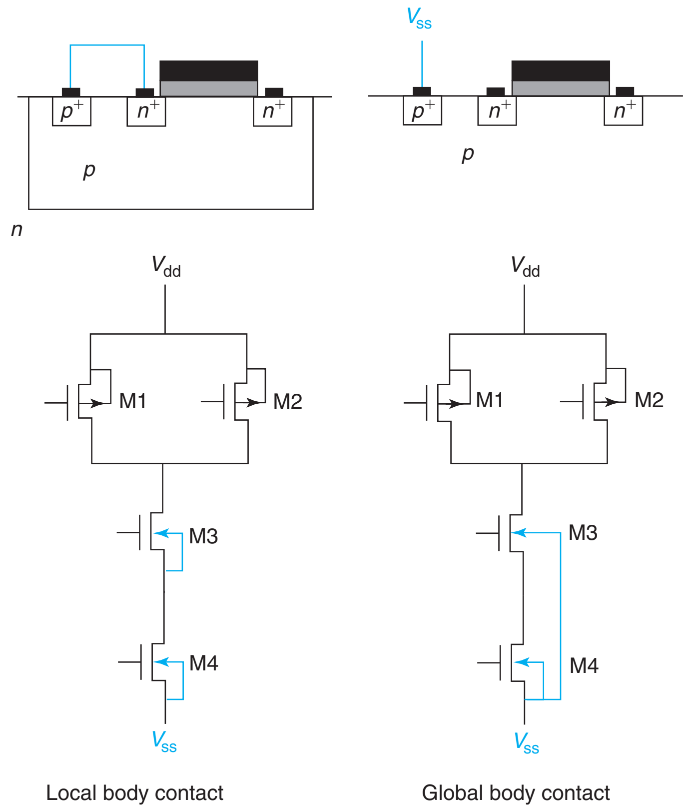


FIGURE 8.32
Illustration of two approaches to contact bodies of transistors. The left diagrams depict a local body contact where the body of every n-MOSFET is tied up to its own source. On the right is depicted a global body contact with all bodies of n-MOSFETs tied up to V_{ss}. The bottom diagrams show the schematic of a NAND gate making explicit the body connections.

devices at the bottom. The technology of the left of the figure features a p-well process on an n-type substrate. In consequence, a local body contact is available for every transistor (the p^+-region). The technology on the right has a p-type substrate that is shared among all n-MOSFETs. In this technology, only a global contact is available. Ideal circuit operation calls for each transistor to have its body shorted to its own source. This can be accomplished in the technology on the left, as is made explicit in the diagram, but for the technology on the right, this is not possible. In this case, transistor M4 has its body connected to its source but M3 does not because the source sits at a higher voltage than the bottom rail. There are two deleterious consequences of this. First, the logic gate on the left switches faster than the logic gate on the right. This is because as the source voltage of transistor M3 on the right circuit rises, its current driving capability is degraded. Second, the operation of the logic gate on the right highly depends on the relative timing of the signals arriving to the gates of transistors M3 and M4 and hence displays a lot more "jitter" or uncertainty in its switching.

- The *body effect* refers to the shift in the threshold voltage along the inversion layer in a MOSFET biased in a current flowing state. This originates in the lateral ohmic drop that occurs in the inversion layer with respect to the body that is produced by the current. The details of this effect will be studied in Chapter 9. Suffice it to say here that it is a deleterious effect that results in lower transistor current that one would otherwise obtain.
- The shift in V_T that results from the application of a back-bias can be exploited to turn off a transistor, a logic gate, or entire circuits in a system. In spite of the extra circuit complexity required to implement this concept, this feature is becoming increasingly useful in the development of low-power systems in which certain functions can be disabled in a selective way when they are not needed.

There are several additional consequences of applying a bias to the inversion layer with respect to the source in a MOS structure. For $V_{SB} > 0$, the depletion region widens. This implies that the capacitance associated with the substrate for a given V_{GB} is reduced. In particular, at threshold, C_{sT} is smaller the higher V_{SB} is. An interesting consequence of this is the reduction of the subthreshold swing, that is, the device exhibits sharper subthreshold characteristics.

One final point about the energy band diagram on the right of Fig. 8.30. In this diagram, it is clear that the quasi-Fermi level for electrons is below the quasi-Fermi level for holes in the top region of the semiconductor. This will extend an electron diffusion length into the substrate. A consequence of this is that the carrier concentrations are below the thermal equilibrium values and the semiconductor responds by generating electron–hole pairs. The generated holes are swept toward the body and the generated electrons are extracted by the inversion layer. Hence, in a three-terminal MOS structure biased in this way, there is a small current that enters the source and flows out of the body. This adds to the current associated with the reverse-biased source-body PN junction.

8.8 SUMMARY

- The Si/SiO_2 interface is nearly ideal. Most Si bonds are satisfied at the interface. However, the interface exhibits a roughness of the order of two monolayers.
- At the Si/SiO_2 interface, the semiconductor can swing all the way from accumulation to inversion by selecting a proper metal or through the application of a voltage to the metal with respect to the semiconductor.
- Beyond threshold, the absolute inversion charge increases linearly with voltage.
- To the first order, the surface potential in inversion and accumulation does not change with V.
- In the subthreshold regime, the magnitude of the minority carrier charge at the oxide-semiconductor interface drops exponentially with voltage below threshold.
- The inversion capacitance depends on the frequency of the measurement due to the relatively slow generation lifetime.
- If the MOS structure is driven quickly from below threshold to above threshold, a *deep depletion* condition is established until the carriers that form the inversion layer have had enough time to be generated.

- Application of a body bias with respect to the inversion layer increases the threshold voltage.
- The Poisson–Boltzmann formulation is a powerful tool to examine electrostatics of complex structures in a rigorous way.

8.9 FURTHER READING

Operation and Modeling of the MOS Transistor (Third Edition) by Y. P. Tsividis and C. McAndrew, Oxford, 2011 (ISBN 978-0-19-517015-3, TK7871.99.M44T77) is a classic textbook on MOSFET physics and modeling that is now in its third edition. The level of treatment is similar to the one adopted here though it goes in greater depth in some areas. A drawback of earlier editions in the absence of energy band diagrams has now been corrected. Chapters relevant here are Chapter 2 that deals with the two-terminal MOS structure and Chapter 3 that addresses the three-terminal structure. In these chapters, there is a detailed treatment of threshold. It also discusses in some depth the various regimes of inversion. This is an important reference book.

AT8.1 SURFACE STATES

A surface represents a severe disruption to the periodic crystalline structure of a semiconductor. Even utilizing the best passivation techniques, the bonding arrangement of the semiconductor atoms at the surface is inevitably disturbed from its bulk configuration. One particularly insidious consequence of this is dangling bonds or unsatisfied semiconductor bonds at the surface. These result in energy levels with a broad energy distribution that are known as *surface states*. Surface states can trap electrons or holes and become charged. In this way, they can affect the electrostatics inside the semiconductor down to some depth. In this section, we study some of the most important issues associated with surface states.

Surface states generally come in two kinds: *donor states* and *acceptor states*. Donor states are positively charged when empty and they become neutral when they trap an electron. Acceptor states are neutral when empty and become negatively charged when they trap an electron. Surface states can have energies all across the band structure of the semiconductor at the surface. Their greatest impact is when these states appear inside the bandgap. Generally, it is found that surface states located toward the bottom of the bandgap are of a donor character while states that are placed toward the top of the bandgap tend to be of an acceptor nature. This has interesting implications.

A surface in isolation, that is, if electron exchange with the bulk is not possible, must be charge neutral. The location of the Fermi level at the surface in this instance is called the *surface neutrality level*. The location of E_N depends on the details of the surface state distribution. Figure 8.33 shows a typical example. Shown here is a surface with a broad distribution of donor states toward the bottom of the bandgap and another distribution of acceptor states toward the top. The axis pointing to the left sketches the surface state density, D_{ss}, given in units of $eV^{-1}.cm^{-2}$. For reference, the location of the conduction and valence bands in the bulk are also indicated. As always, states below the Fermi level are full while states above the Fermi level are

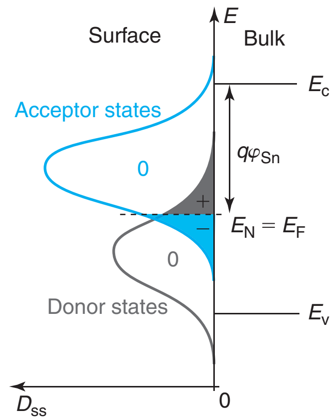


FIGURE 8.33
Example of energy distribution of surface states in a semiconductor. With the surface in isolation from the bulk, the Fermi level sits at the surface neutrality level, as indicated.

empty. Occupied donor states are charge neutral but empty donor states are positively charged. Conversely, occupied acceptor states are negatively charged while empty ones are neutral. This means that for the distribution of Fig. 8.33, the Fermi level will sit at a location where the positive charge due to empty donor states precisely balances the negative charge of occupied acceptor states, as indicated in the figure. An interesting implication of this is that in a distribution like the one indicated in the figure with a large peak of acceptor states right above another peak of donor states, the surface neutrality level will tend to be placed somewhere between the two peaks, as sketched.

How does this picture change if the surface can exchange carriers with the rest of the semiconductor, as is generally the case? The Fermi level in the bulk is set by the doping level and will generally not line up with the surface neutrality level. This implies that upon intimate contact, some charge redistribution between the bulk and the surface takes place. If the semiconductor is n-type, then it is quite likely that E_F in the bulk is higher than E_N at the surface. Upon contact, electrons will tend to flow from the semiconductor toward the surface states. In equilibrium, then, the surface gains electrons and becomes negatively charged. The bulk loses electrons and becomes positively charged. Since electrons and holes must be in equilibrium in the bulk, we know that this implies that a depletion region is formed below the semiconductor. The situation is sketched in Fig. 8.34. If the semiconductor is p-type, electrons flow from the surface toward the semiconductor leaving the surface positively charged and the semiconductor negatively charged. Again, a depletion region appears right below the surface.

In the limiting case of a very high surface state density, after equilibrium is established between the surface and the bulk, the Fermi level at the surface does not get to move much from its neutral level. This is a condition known as *Fermi level pinning*. In these types of surfaces, the Fermi level at the surface is essentially stuck in place regardless of the doping level and type of the bulk. This is a condition that affects poorly passivated surfaces, especially in compound semiconductors such as GaAs.

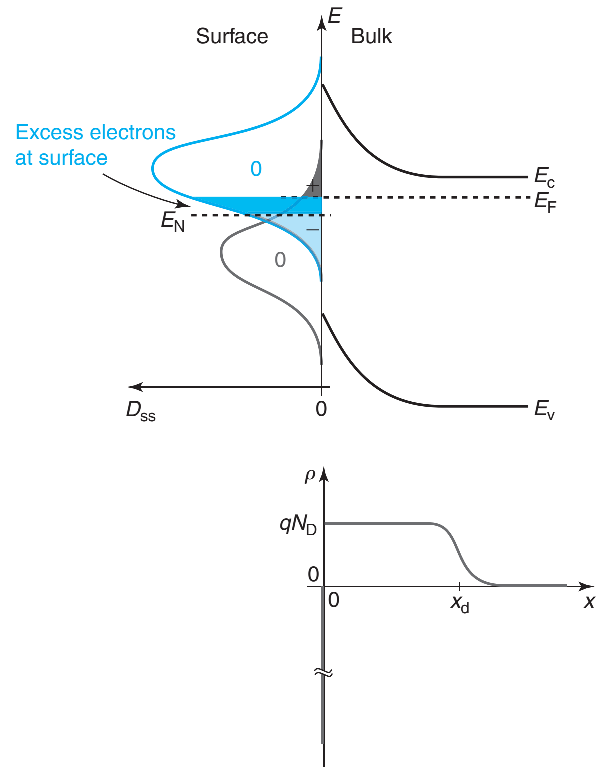


FIGURE 8.34
Band structure in thermal equilibrium in the neighborhood of the surface of an n-type semiconductor in the presence of surface states. The charge redistribution that takes place between the surface and the semiconductor results in a depletion region right below the surface.

In the limit of Fermi level pinning, it is easy to model the electrostatics of the problem. First, note that this situation is virtually identical to that of a metal–semiconductor junction on an n-type semiconductor studied in Section 7.2. In the case of a surface, the role of the Schottky barrier height is played by the difference between the conduction band edge and the charge neutrality level at the surface. This is a property of the defect distribution at the surface of the semiconductor, as discussed above. If we denote this energy difference as $q\varphi_{Sn}$ (Fig. 8.33), then the total band bending in the semiconductor, $q\phi_{bis}$, is equal to $q\varphi_{Sn}$ minus the difference between the conduction band and the Fermi level in the quasi-neutral bulk which is set by the doping level. We then have

$$q\phi_{bis} = q\varphi_{Sn} - kT \log \frac{N_c}{N_D} \tag{8.72}$$

In analogy with Eq. (7.15), the depletion region thickness that appears right below the surface can be found from

$$x_{\mathrm{ds}} = \sqrt{\frac{2\epsilon\phi_{\mathrm{bis}}}{qN_{\mathrm{D}}}} \tag{8.73}$$

If we know the density of surface states at the charge neutrality level, we can estimate the degree to which our assumption of Fermi level pinning applies. If the total (acceptor plus donor) surface state density is denoted as $D_{\mathrm{ss}}(E_{\mathrm{N}})$, then the electron charge at the surface is approximately given by $qD_{\mathrm{ss}}(E_{\mathrm{N}})(E_{\mathrm{F}} - E_{\mathrm{N}})$. This must equal the charge in the semiconductor given by $qN_{\mathrm{D}}x_{\mathrm{ds}}$. From this, and using Eq. (8.73), we can solve for $E_{\mathrm{F}} - E_{\mathrm{N}}$:

$$E_{\mathrm{F}} - E_{\mathrm{N}} \simeq \frac{1}{D_{\mathrm{ss}}(E_{\mathrm{N}})}\sqrt{\frac{2\epsilon\phi_{\mathrm{bis}}N_{\mathrm{D}}}{q}} \tag{8.74}$$

It is clear that if $D_{\mathrm{ss}}(E_{\mathrm{N}})$ is large, then the difference between E_{F} and E_{N} is small and surface Fermi level pinning is a good assumption. Conversely, in a passivated surface with small $D_{\mathrm{ss}}(E_{\mathrm{N}})$, then the Fermi level will move to a location at the surface that depends on the doping level in the body of the semiconductor. For a precise calculation, the details of the surface state distribution are required.

AT8.2 NONIDEAL EFFECTS IN MOS STRUCTURE

Highly perfect MOS structures have been fabricated in the SiO_2/Si system with characteristics that approach the ideal behavior presented in the main body of this chapter. In many practical cases, however, a number of nonidealities are observed. It is of some importance to understand their main physical origin and characteristic signature. Achieving this is the purpose of this advanced topic. The main vehicle for the analysis of nonidealities in MOS structures is *C–V* characterization. As we have seen in previous chapters, this is a powerful and widely used technique that is rather sensitive to many deviations from the ideal behavior.

We analyze two broad types of nonidealities: oxide charge and interface states. Oxide charge refers to the presence of charged centers inside the bulk of the oxide. This obviously affects the electrostatics of the MOS structure. Oxide charge can come from several sources that we discuss below. A complication in the practical analysis of oxide charge is that in some cases, the species responsible can move in response to the application of an electric field across the oxide. This can lead to device instabilities that upset its normal behavior.

The second type of nonideality is interface states. These are electronic states that exist inside the bandgap of the semiconductor at the oxide–semiconductor interface as a result of the disrupted atomic bonding there. When charged, interface states can also affect the MOS electrostatics in various significant ways. What is unique about interface states is that they can communicate with the semiconductor exchanging charged carriers. This again introduces new dynamic behavior that is visible in many situations.

AT8.2.1 Oxide charge

In our treatment of the MOS structure, we have considered the oxide as completely devoid of any type of charge, whether mobile or fixed. In practice, it is not uncommon for the oxide to

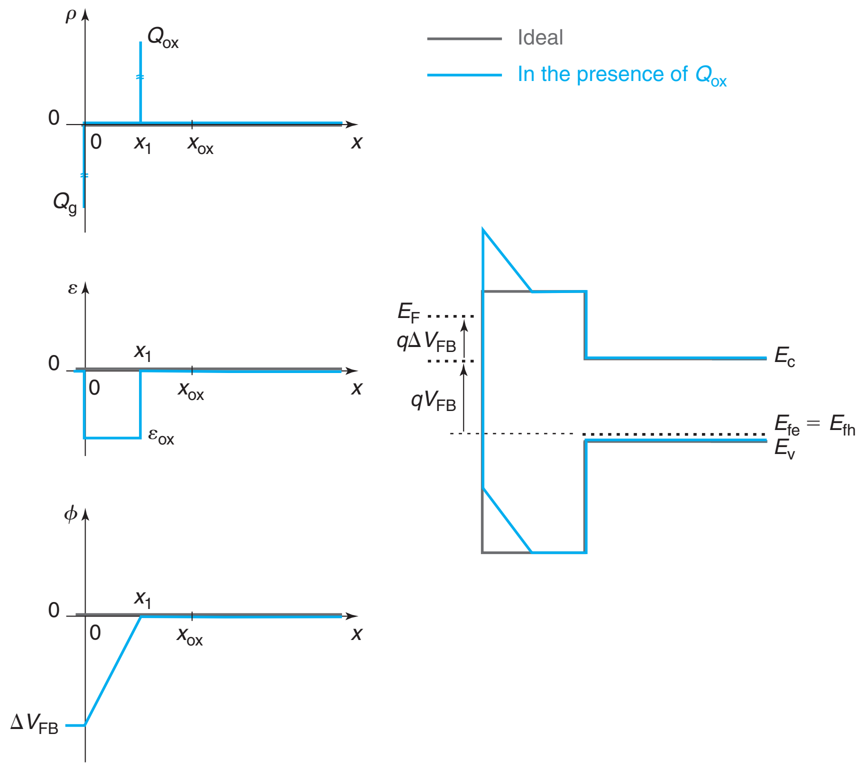


FIGURE 8.35
From top to bottom and left to right: sketches of volume charge density, electric field, electrostatic potential, and energy band diagram of a MOS structure at flatband with and without a sheet of positive charge in the oxide.

contain trapped charge such as ions of various kinds (including Si), charged defects, and trapped carriers. When this happens, the electrostatics of the MOS structure are upset. This manifests itself in a variety of ways depending on the nature of the charge and its location within the oxide.

Let us first analyze a simple situation in which there is a sheet of positive charge at some location in the oxide, Q_{ox}. The electrostatics of this problem at flatband are shown in Fig. 8.35. Notice that with respect to the main body of this chapter, we have moved the origin of the x-axis to the metal–oxide interface to simplify the mathematics of the problem. With the semiconductor at flatband, the sheet of oxide charge is imaged at the metal.

The presence of a sheet at positive charge in the oxide creates an electric field that emanates from it and terminates at the metal gate. This creates a potential difference that, as shown in the figure, results in a *negative flatband voltage shift*. It is easy to derive an expression for this.

Using Gauss theorem, the electric field that emanates from the oxide charge is directed to the left and is given by

$$\mathcal{E}_{ox} = -\frac{Q_{ox}}{\epsilon_{ox}} \tag{8.75}$$

If the sheet of charge is placed at a generic location x_1, then the potential build up from 0 to x_1 is precisely the shift in the flat-band voltage with a minus sign:

$$\Delta V_{FB} = -\frac{Q_{ox}}{\epsilon_{ox}} x_1 \tag{8.76}$$

There is already an important conclusion here. The location where oxide charge has the most impact on the MOS electrostatics is right at the interface with the semiconductor. From there on, its impact decreases linearly the closer it gets to the metal–oxide interface. At that location, $x = 0$, the impact of oxide charge vanishes.

It is easy to generalize this result to an arbitrary charge density ρ_{ox} distributed across the oxide. The electrostatics of this case are shown in Fig. 8.36. At any location x, the electric field in the oxide is given by

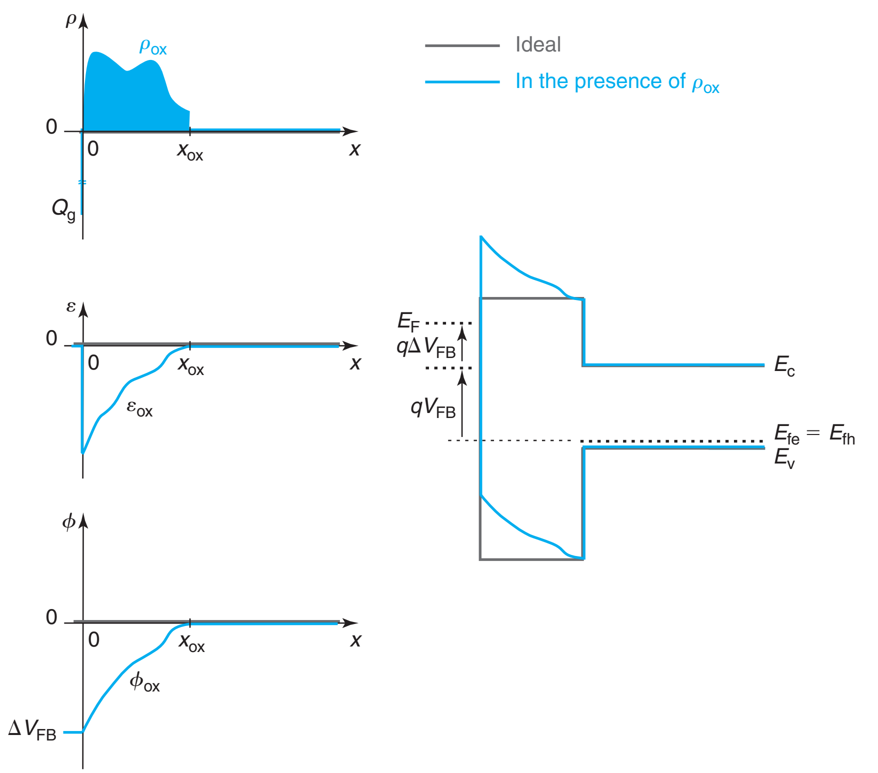


FIGURE 8.36
From top to bottom and left to right: sketches of volume charge density, electric field, electrostatic potential, and energy band diagram of a MOS structure at flatband with and without a distribution of charge within the oxide.

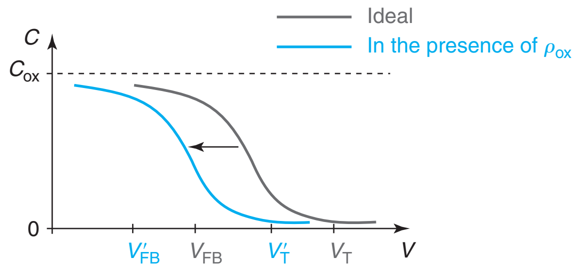


FIGURE 8.37
Impact of oxide charge on the high-frequency *C–V* characteristics of a MOS structure. There is a rigid sideways shift of the characteristics by an amount that can be calculated through Eq. (8.79). The indicated shift corresponds to positive oxide charge. In the case of negative charge, the characteristics shift to the right.

$$\mathcal{E}_{ox}(x) = -\frac{1}{\epsilon_{ox}} \int_{x}^{x_{ox}} \rho_{ox}(x)dx \tag{8.77}$$

This electric field distribution results in a potential build up across the oxide which is equal to the minus flat-band voltage shift and is given by

$$\Delta V_{FB} = -[\phi(x_{ox}) - \phi(0)] = -\frac{1}{\epsilon_{ox}} \int_{0}^{x_{ox}} \int_{x}^{x_{ox}} \rho_{ox}(x')dx'dx \tag{8.78}$$

We can perform this integral by parts to easily obtain

$$\Delta V_{FB} = -\frac{1}{\epsilon_{ox}} \int_{0}^{x_{ox}} x\rho_{ox}(x)dx \tag{8.79}$$

Again we find that the impact of the oxide charge on the electrostatics increases the further away the charge is located from the gate.

For simplicity, we have performed our derivation at the flat-band condition. We could have equally carried it out at any other bias point and we would have obtained an identical gate voltage shift. The oxide charge results in a rigid shift of the characteristics of the MOS structures by an amount given by Eq. (8.79). The clearest manifestation of the presence of oxide charge is therefore a sideways shift of the *C–V* characteristics with respect to where we expect them based on the work function of the metal, the oxide thickness, and the doping level in the semiconductor, as sketched in Fig. 8.37. The characteristics shift to the left if the weighted oxide charge is positive and to the right if negative.

An interesting situation arises when the charge in the oxide is mobile. That is the case, for example, of light alkali ions, such as Na^+, Li^+, or K^+ which are quite abundant. If present in the oxide, these ions will produce a negative shift of the *C–V* characteristics, as described above. At

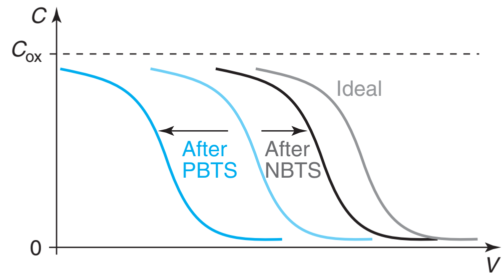


FIGURE 8.38
Impact of positive-bias temperature stress (PBST) and negative-bias temperature stress (NBST) on high-frequency *C–V* characteristics of a MOS structure that contains mobile positive ionic charge. The movement of the ions in response to the applied electric field across the oxide affects the *C–V* characteristics.

moderate temperatures, however, these ions can readily move. The direction in which they move depends on the bias that is applied to the MOS structure. With the gate positive with respect to the body (known as *positive-bias temperature stress* or PBST), the alkaly ions will move away from the metal yielding an increase in the negative shift of the *C–V* characteristics. With the gate negative with respect to the body ("negative-bias temperature stress" or NBST), the ions move toward the metal and the *C–V* characteristics shift again to approach their ideal distribution. This is illustrated in Fig. 8.38.

In the early days of MOSFETs, device instability was a serious problem. The threshold voltage changed in seemingly random ways during device operation. This was eventually traced to alkaly ion contamination introduced by improper handling. The problem was solved by introducing strict cleaning protocols that prevented contact between the operators and the wafers during processing.

AT8.2.2 Interface states

When we introduced the ideal MOS structure at the beginning of this chapter, we assumed that the semiconductor atomic bonding at the oxide–semiconductor interface is unperturbed from the bulk four-fold coordination. Even in a high-quality MOS gate stack, this is not quite the case as there are always a few broken bonds at the interface that create surface states. In the case of a MOS structure, these are referred to as *interface states.* If present in excessive amounts, interface states become an important consideration in MOS electrostatics, inversion layer transport, and reliability. We discuss some of the key issues in this section.

The first order of business is to describe how interface states affect the electrostatics of the MOS structure. Our study of the role of fixed charge in the previous section and of surface states in Section AT8.1 allows us to readily understand this. Let us assume that we have a positive sheet of charge Q_{it} at the oxide–semiconductor interface from charged interface states. At any bias point of the otherwise ideal MOS structure, this sheet of charge results in a change in the electric field across the oxide given by

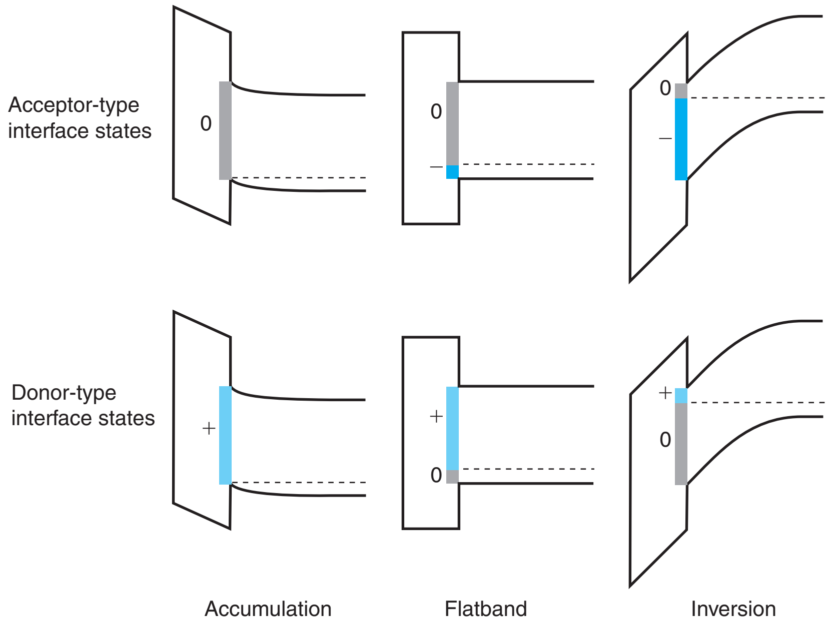


FIGURE 8.39
Evolution of charge at interface states of MOS structure as a function of bias point. Top three images correspond to a uniform density of acceptor-type interface states. The bottom three images correspond to donor-type interface states.

$$\Delta\mathcal{E}_{ox} = -\frac{Q_{it}}{\epsilon_{ox}} \tag{8.80}$$

This in turn yields a *reduction* in the gate to body voltage by an amount:

$$\Delta V = -\frac{Q_{it}}{C_{ox}} \tag{8.81}$$

All else constant, then, adding positive charge at the oxide–semiconductor interface results in a negative shift of the gate voltage.

From this result, we can visualize the impact of the presence of interface states on the C–V characteristics of a MOS structure. To illustrate the issues, it is best to work with two simple cases involving uniform densities of acceptor-like or donor-like interface states across the bandgap. These are both depicted in Fig. 8.39 for a MOS structure on a p-type substrate.

On the top of Fig. 8.39, we illustrate the situation of acceptor-type interface states. In accumulation, the Fermi level is right at the bottom of the interface state distribution which is then completely empty. There is then no charge at the interface. As the device is driven into flatband and eventually into inversion, the interface gets negatively charged. Conversely, with donor-type interface states, the interface is positively charged in accumulation and as the gate voltage increases and the Fermi level slides through the interface, the positive charge is reduced.

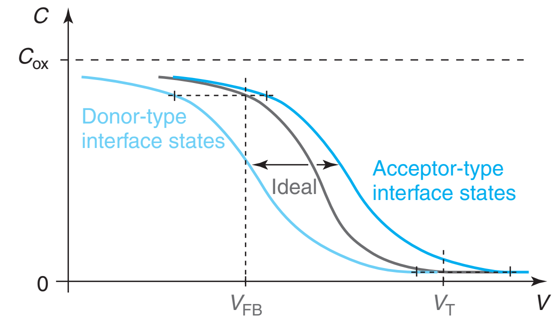


FIGURE 8.40
Impact of interface state distributions of Fig. 8.39 on high-frequency *C–V* characteristics. The presence of interface states deforms the *C–V* characteristics. Acceptor-type states shift the characteristics to the right while donor-type states shift the characteristics to the left.

The impact of this on the C–V characteristics is illustrated in Fig. 8.40. In general, interface states deform the C–V characteristics by stretching them along the V-axis. For an interface with acceptor-type interface states, the C–V characteristics are unaffected in accumulation (the interface is neutral) but are increasingly stretched toward positive voltages as V moves toward inversion and the interface charges up negatively. For an interface with donor-type interface states, the C–V characteristics are stretched toward negative voltages in accumulation when the interface is positively charged. As the voltage increases toward inversion, the positive interfacial charge shrinks and the C–V characteristics increasingly approach the ideal case. In a practical case, from a comparison between the ideal C–V characteristics expected from the design of the MOS structure and the actual ones, one can determine the interface state distribution across the bandgap. This assumes that there are no other nonidealities in the system.

The changes in the C–V characteristics just discussed suggest that the equivalent circuit model for the MOS structure that was presented in Section 8.5 needs to be modified. In general, the charge at the MOS gate now images not only charge in the semiconductor but also charge at the oxide–semiconductor interface. In analogy with Eqs. (8.36) and (8.37), we can then write

$$C = \frac{dQ_g}{dV} = -\frac{d(Q_s + Q_{it})}{dV} = \left(-\frac{dQ_s}{d\phi_s} - \frac{dQ_{it}}{d\phi_s}\right)\frac{1}{\frac{dV}{d\phi_s}} \tag{8.82}$$

We can associate a capacitance with the interface state charge Q_{it}. Let us denote this as C_{it} defined as

$$C_{it} = -\frac{dQ_{it}}{d\phi_s} \tag{8.83}$$

Notice that this capacitance is positive since as ϕ_s increases, the Fermi level at the surface moves up in energy with respect to the band structure and the interface charges up negatively (that is, the net interface charge decreases if positive or increases if negative).

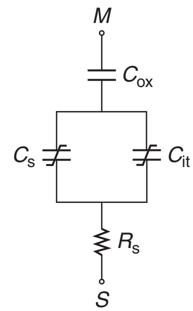


FIGURE 8.41
Quasi-static equivalent circuit model for the MOS structure that includes the effect of interface states.

The general relationship for the electrostatics of the MOS structure given in Eq. (8.18) also needs to be modified to include the interface charge. It becomes

$$\phi_{bi} + V = \phi_s - \frac{Q_s + Q_{it}}{C_{ox}} \tag{8.84}$$

From this, we obtain

$$\frac{dV}{d\phi_s} = 1 + \frac{C_s + C_{it}}{C_{ox}} \tag{8.85}$$

where we have also used the definition of C_s made in Eq. (8.39). We can now plug Eq. (8.85) into Eq. (8.82) and use again the definitions of C_s and C_{it} to get

$$C = \frac{1}{\frac{1}{C_{ox}} + \frac{1}{C_s + C_{it}}} \tag{8.86}$$

When compared with the corresponding equation for the ideal MOSFET [Eq. (8.41)], the impact of interface states is to introduce a new capacitance C_{it} in parallel with the semiconductor capacitance C_s. A quasi-static equivalent circuit model for this situation is as sketched in Fig. 8.41.

One of the most noticeable and important consequences of the presence of interface states is in the subthreshold characteristics of the MOS structure. Just as the C–V characteristics, they get smeared in the presence of interface states. This is described by an increase in the subthreshold swing. A simple model for this can be easily developed. In our treatment of the ideal subthreshold characteristics in Section 8.6, we derived an expression for the subthreshold swing in Eq. (8.64). With interface states, a capacitance C_{it} must be introduced in parallel with C_{sT}, the semiconductor capacitance at threshold. The subthreshold swing is then given by

$$S = \left(1 + \frac{C_{sT} + C_{it}}{C_{ox}}\right) \frac{kT}{q} \ln 10 \tag{8.87}$$

If C_{it} is comparable to C_{sT}, then interface states will have a noticeable degrading impact on the subthreshold characteristics. Note also that if the interface state density is a function of

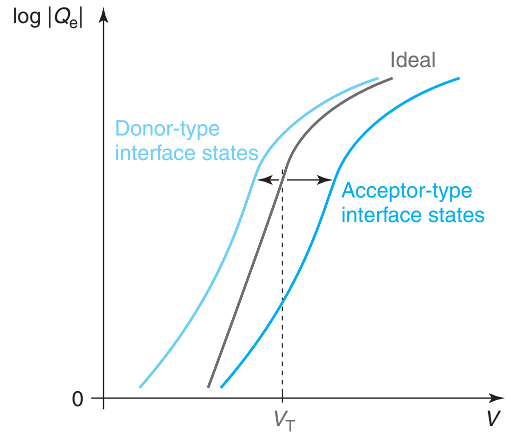


FIGURE 8.42
Impact of interface state distributions of Fig. 8.39 on subthreshold characteristics of MOS structure. The presence of interface states deforms the subthreshold characteristics. Acceptor-type states shift the characteristics to the right while donor-type states shift the characteristics to the left. Regardless of the type of interface state, the subthreshold swing increases as a result of the presence of interface states.

energy across the bandgap, then the subthreshold swing will depend on bias and we no longer have a relatively straight subthreshold characteristics in a semilog scale.

Sketches of the subthreshold characteristics that correspond to the two interface state distributions depicted in Fig. 8.39 are shown in Fig. 8.42. Note that interface states have two distinct effects. The first is that they shift the characteristics to the right in the presence of acceptor-type interface states or to the left if the states are donor-type. In addition, the characteristics generally soften resulting in a higher subthreshold swing.

The subthreshold swing provides us with a direct way to estimate the density of interface states at the Fermi level. We refer to this as D_{it} with units of $eV^{-1}.cm^{-2}$. For a given position of the Fermi level across the bandgap at the surface, the subthreshold swing depends only on the total interface state density at that point, regardless of whether the states are donor or acceptor type. This is because if the Fermi level moves up in the bandgap by an amount ΔE, then the surface potential changes by $\Delta\phi_s = \Delta E/1e$ and the charge at the interface increases by an amount $\Delta Q_{it} = -qD_{it}\Delta E$. In consequence, using Eq. (8.83), C_{it} can then be written as

$$C_{it} = qD_{it} \tag{8.88}$$

Technically, to make the units work, the term on the right should be multiplied by $1e$ where e refers to the electron charge. This is to cancel out the e in the units of D_{it}. Because there is no numeric consequence to C_{it}, this $1e$ term is not usually written in Eq. (8.88).

In general, D_{it} in Eq. (8.88) refers to the sum of the density of states of acceptor-type and donor-type states. In both cases, as the Fermi level sweeps up through the distribution, the interface becomes more negatively charged. The subthreshold swing cannot distinguish the nature of the interface states. On the other hand, the shift along the voltage axis does reflect the character of the states.

AT8.3 POISSON–BOLTZMANN FORMULATION OF MOS ELECTROSTATICS

This section presents a general formulation for the electrostatics of the MOS structure under a variety of conditions. The Poisson–Boltzmann formulation, as it is known, allows us to get detailed descriptions of ϕ, $\mathcal{E}$, and ρ, in the oxide and the semiconductor. It also forms the basis for a detailed discussion of the C–V characteristics. A similar formulation can be constructed to treat the electrostatics of the PN junction and metal–semiconductor junction.

We assume here a MOS structure with a geometry identical to that studied in the main body of this chapter. We also assume that the semiconductor is in a quasi-equilibrium condition. We work with Boltzmann statistics. The starting point to solve the electrostatics of this situation is the expression of the volume charge density. For an uncompensated uniformly doped p-type semiconductor, this can be written as

$$\rho = q(p - n - N_A) \tag{8.89}$$

This allows to write Poisson's equation in the form:

$$\frac{d^2\phi}{dx^2} = -\frac{q}{\epsilon_s}(p - n - N_A) \tag{8.90}$$

With the semiconductor under quasi-equilibrium, we can use the Boltzmann relations to express the carrier concentrations in terms of the electrostatic potential. Selecting as origin of potentials deep in the bulk, $\phi(bulk) = 0$, we have

$$n = n_{oB} \exp\frac{q\phi}{kT} \tag{8.91}$$

$$p = p_{oB} \exp\frac{-q\phi}{kT} \tag{8.92}$$

where n_{oB} and p_{oB} are, respectively, the equilibrium electron and hole concentrations in the bulk.

In the bulk, charge neutrality prevails. Mathematically

$$p_{oB} - n_{oB} - N_A = 0 \tag{8.93}$$

Plugging Eqs. (8.91)–(8.93) into Eq. (8.90), we get

$$\frac{d^2\phi}{dx^2} = -\frac{qN_A}{\epsilon_s}\left[\left(\exp\frac{-q\phi}{kT} - 1\right) - \frac{n_i^2}{N_A^2}\left(\exp\frac{q\phi}{kT} - 1\right)\right] \tag{8.94}$$

where we have also assumed that $p_{oB} \simeq N_A$ and $n_{oB} \simeq n_i^2/N_A$.

Equation (8.94) is called the *Poisson–Boltzmann equation*. It contains a single unknown, the electrostatic potential ϕ. A double integration of this equation entirely solves the problem.

The first integration is performed using the following mathematical equality:

$$\frac{d}{dx}\left(\frac{d\phi}{dx}\right)^2 = 2\frac{d\phi}{dx}\frac{d^2\phi}{dx^2} \tag{8.95}$$

Multiplying both sides of the differential equation (8.94) by $2\frac{d\phi}{dx}$ and using the relationship just presented, we get

$$\frac{d}{dx}\left(\frac{d\phi}{dx}\right)^2 = -\frac{2qN_A}{\epsilon_s}\left[\left(\exp\frac{-q\phi}{kT}-1\right)-\frac{n_i^2}{N_A^2}\left(\exp\frac{q\phi}{kT}-1\right)\right]\frac{d\phi}{dx} \tag{8.96}$$

We can now integrate this equation across the semiconductor. We start from the bulk, $x = \infty$ and integrate toward the surface. On the left-hand side

$$\int_{\infty}^{x}\frac{d}{dx}\left(\frac{d\phi}{dx}\right)^2 dx = \left(\frac{d\phi}{dx}\right)^2\Bigg|_x - \left(\frac{d\phi}{dx}\right)^2\Bigg|_\infty = \left(\frac{d\phi}{dx}\right)^2 \tag{8.97}$$

$\frac{d\phi}{dx}\big|_\infty = 0$ because sufficiently far away from the surface in the bulk of the semiconductor, the electric field is zero.

The right-hand side of Eq. (8.96) can also be easily integrated:

$$-\frac{2qN_A}{\epsilon_s}\int_{\infty}^{x}\left[\left(\exp\frac{-q\phi}{kT}-1\right)-\frac{n_i^2}{N_A^2}\left(\exp\frac{q\phi}{kT}-1\right)\right]\frac{d\phi}{dx}dx =$$

$$-\frac{2qN_A}{\epsilon_s}\int_{0}^{\phi}\left[\left(\exp\frac{-q\phi}{kT}-1\right)-\frac{n_i^2}{N_A^2}\left(\exp\frac{q\phi}{kT}-1\right)\right]d\phi =$$

$$\frac{2kTN_A}{\epsilon_s}\left[\left(\exp\frac{-q\phi}{kT}+\frac{q\phi}{kT}-1\right)+\frac{n_i^2}{N_A^2}\left(\exp\frac{q\phi}{kT}-\frac{q\phi}{kT}-1\right)\right] \tag{8.98}$$

Assembling now Eqs. (8.97) and (8.98), we get

$$\frac{d\phi}{dx} = \pm\sqrt{\frac{2kTN_A}{\epsilon_s}}\left[\left(\exp\frac{-q\phi}{kT}+\frac{q\phi}{kT}-1\right)+\frac{n_i^2}{N_A^2}\left(\exp\frac{q\phi}{kT}-\frac{q\phi}{kT}-1\right)\right]^{1/2} \tag{8.99}$$

We now have to select the proper sign of this equation. This is best done with the help of Fig. 8.43. When ϕ is positive in the semiconductor, as in depletion and inversion, then the sketch of Fig. 8.43 indicates that $\frac{d\phi}{dx} < 0$. When ϕ is negative, as in accumulation, $\frac{d\phi}{dx} > 0$. In general, we can then write

$$sign\left(\frac{d\phi}{dx}\right) = -sign(\phi) = -\frac{\phi}{|\phi|} \tag{8.100}$$

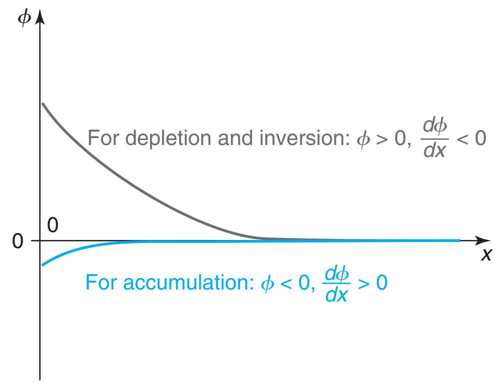


FIGURE 8.43
Sketch of electrostatic potential in the semiconductor. When ϕ is positive, $d\phi/dx$ is negative; when ϕ is negative, $d\phi/dx$ is positive.

We can define the function $F(\phi)$:

$$F(\phi) = \frac{\phi}{|\phi|}\left[\left(\exp\frac{-q\phi}{kT} + \frac{q\phi}{kT} - 1\right) + \frac{n_\mathrm{i}^2}{N_\mathrm{A}^2}\left(\exp\frac{q\phi}{kT} - \frac{q\phi}{kT} - 1\right)\right]^{1/2} \tag{8.101}$$

Using Eqs. (8.100) and (8.101), we can rewrite Eq. (8.99) in the following way:

$$\frac{d\phi}{dx} = -\sqrt{\frac{2kTN_\mathrm{A}}{\epsilon_\mathrm{s}}}F(\phi) \tag{8.102}$$

A second integration of Eq. (8.102) gives ϕ versus x. This constitutes a complete solution to the electrostatics problem:

$$\int_{\phi_\mathrm{s}}^{\phi} \frac{d\phi}{F(\phi)} = -\sqrt{\frac{2kTN_\mathrm{A}}{\epsilon_\mathrm{s}}}x \tag{8.103}$$

Unfortunately, an analytical integration of this equation is not possible in general. If some simplifications can be made, analytical formulations for $\phi(x)$ can be derived. This is carried out in the three sections below.

Even before the additional integration, Eq. (8.102) leads to several useful results. The electric field in the semiconductor is the negative derivative of ϕ in space, hence:

$$\mathcal{E} = \sqrt{\frac{2kTN_\mathrm{A}}{\epsilon_\mathrm{s}}}F(\phi) \tag{8.104}$$

In particular, at the insulator/semiconductor interface:

$$\mathcal{E}_\mathrm{s} = \sqrt{\frac{2kTN_\mathrm{A}}{\epsilon_\mathrm{s}}}F(\phi_\mathrm{s}) \tag{8.105}$$

Using this result in Eq. (8.5) gives us a general expression for the total charge in the semiconductor:

$$Q_s = -\sqrt{2\epsilon_s kTN_A}\,F(\phi_s) \tag{8.106}$$

We can also break the semiconductor charge into its two components, that is, those due to the electron and hole concentrations being different from their respective bulk values:

$$Q_e = -q\int_0^\infty (n - n_{oB})dx \simeq -q\frac{n_i^2}{N_A}\int_0^\infty \left(\exp\frac{q\phi}{kT} - 1\right)dx \tag{8.107}$$

$$Q_h = q\int_0^\infty (p - p_{oB})dx \simeq qN_A\int_0^\infty \left(\exp\frac{-q\phi}{kT} - 1\right)dx \tag{8.108}$$

Changing variables from x to ϕ in these two integrals and using Eq. (8.102), one gets

$$Q_e = q\frac{n_i^2}{N_A}\sqrt{\frac{\epsilon_s}{2kTN_A}}\int_{\phi_s}^{0}\frac{\exp\frac{q\phi}{kT} - 1}{F(\phi)}d\phi \tag{8.109}$$

$$Q_h = -\sqrt{\frac{\epsilon_s q^2 N_A}{2kT}}\int_{\phi_s}^{0}\frac{\exp\frac{-q\phi}{kT} - 1}{F(\phi)}d\phi \tag{8.110}$$

Finally, we can also plug Eq. (8.106) into Eq. (8.18) and get a relationship between ϕ_s and V:

$$V = -\phi_{bi} + \phi_s + \frac{\sqrt{2\epsilon_s kTN_A}}{C_{ox}}F(\phi_s) = V_{FB} + \phi_s + \gamma\sqrt{\frac{kT}{q}}F(\phi_s) \tag{8.111}$$

If V is given, this equation can be easily solved in an iterative fashion to obtain ϕ_s. Once ϕ_s is known, Eq. (8.105) allows the calculation of the electric field at the oxide/semiconductor interface. Additionally, Eq. (8.106) can be used to get Q_s as a function of V. We can break Q_s on its two components Q_e and Q_h by numerically integrating Eqs. (8.109) and (8.110). A complete description of ϕ versus x can only be obtained by integrating Eq. (8.103). This also yields n and p versus x from Eqs. (8.91) and (8.92), and ρ versus x from Eq. (8.89).

In order to better appreciate how the physics of the MOS structure evolves with applied voltage, it is worthwhile to first examine results obtained from the Poisson–Boltzmann formulation. To this end, Figs. 8.44–8.50 display a complete set of results for a typical case with $x_{ox} = 4.5$ nm, $N_A = 6 \times 10^{17}$ cm^{-3}, $W_M = 4.04$ eV, and $T = 300$ K as a function of applied voltage. For this case, $V_T = 0.5$ V, and $V_{FB} = -1$ V, $\phi_{sT} = 0.93$ V.

Figure 8.44 graphs ϕ_s versus V. As V increases, so does ϕ_s. The most prominent feature of this figure is its "S"-shape. For sufficiently negative values of V ($V < -1$ V), the surface potential ϕ_s tends to saturate to a value that is not too different from zero. This is the accumulation regime. For intermediate values of V ($-1 < V < 0.5$ V), ϕ_s increases rapidly with V. This is the depletion

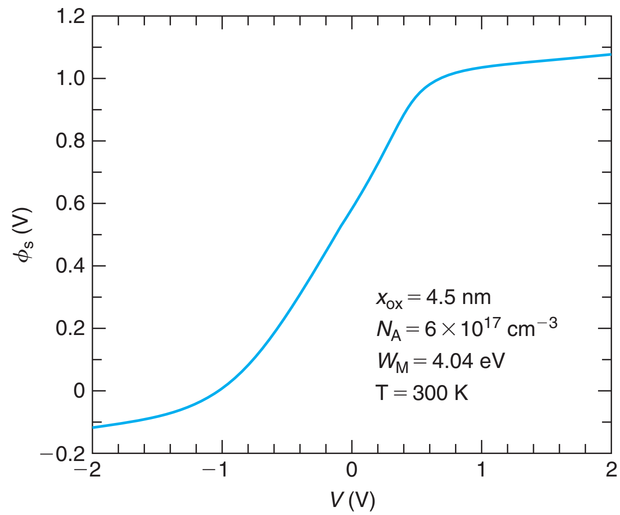


FIGURE 8.44
Surface potential versus applied voltage for a typical MOS structure [Reprinted with permission from Wenjie Lu, MIT].

regime. For sufficiently positive values of V ($V > 0.5$ V), ϕ_s tends to saturate again. This is the inversion regime.

Figure 8.45(a) graphs Q_s, Q_e, and Q_h versus V. Three regimes are apparent again. For negative and slightly positive voltages, $Q_s \simeq Q_h$. These are accumulation and depletion, respectively. For sufficiently positive voltages, Q_h saturates and Q_e starts rising in absolute terms, as we argued it should happen in inversion. Note that Q_e is always negative, while Q_h is both positive (in accumulation) and negative (in depletion and inversion).

Figure 8.45(b) shows $\mathcal{E}_s$ versus V. Again the three regimes are apparent. For negative V, $\mathcal{E}_s$ is negative and depends linearly on V (accumulation). For positive but small V, $\mathcal{E}_s$ becomes positive and displays a sublinear dependence on V (depletion). For positive enough V, $\mathcal{E}_s$ increases linearly again with V (inversion).

The physics of the various regimes is illuminated by examining the evolution of the relevant variables as a function of spatial location in the semiconductor. Figure 8.46(a) displays the volume charge density in the semiconductor. For $V < 0$, there is a prominent positive peak at the interface. This is the hole accumulation region. As V is made positive, a negatively charged region grows from the interface into the semiconductor. This is the depletion region. Beyond $V = 0.5$ V, threshold, the depletion region stops growing. Instead, a prominent negative peak due to the electron inversion layer appears at the oxide/semiconductor interface.

Consistent with this, the electric field in the semiconductor, Fig. 8.46(b), displays a prominent peak at the interface for negative V, is zero everywhere for $V = -1.0$ V, but increases and penetrates deeper into the semiconductor as V is made more positive. However, beyond $V = 0.5$ V, the field in the bulk of the semiconductor does not change much but increases substantially at the surface. This is consistent with the identification of the various regimes.

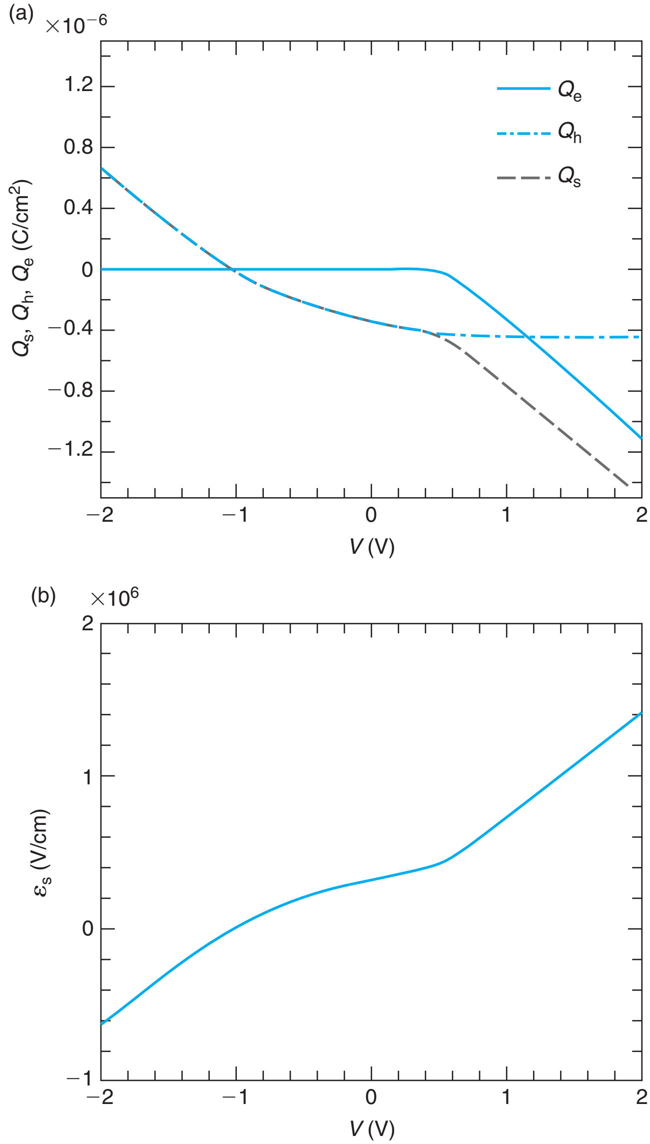


FIGURE 8.45
(a) Semiconductor charge versus applied voltage for the MOS structure of Fig. 8.44; graphed also are the two components of Q_s: Q_e, and Q_h. Electric field at the insulator/semiconductor interface versus applied voltage for the same MOS structure [Reprinted with permission from Wenjie Lu, MIT].

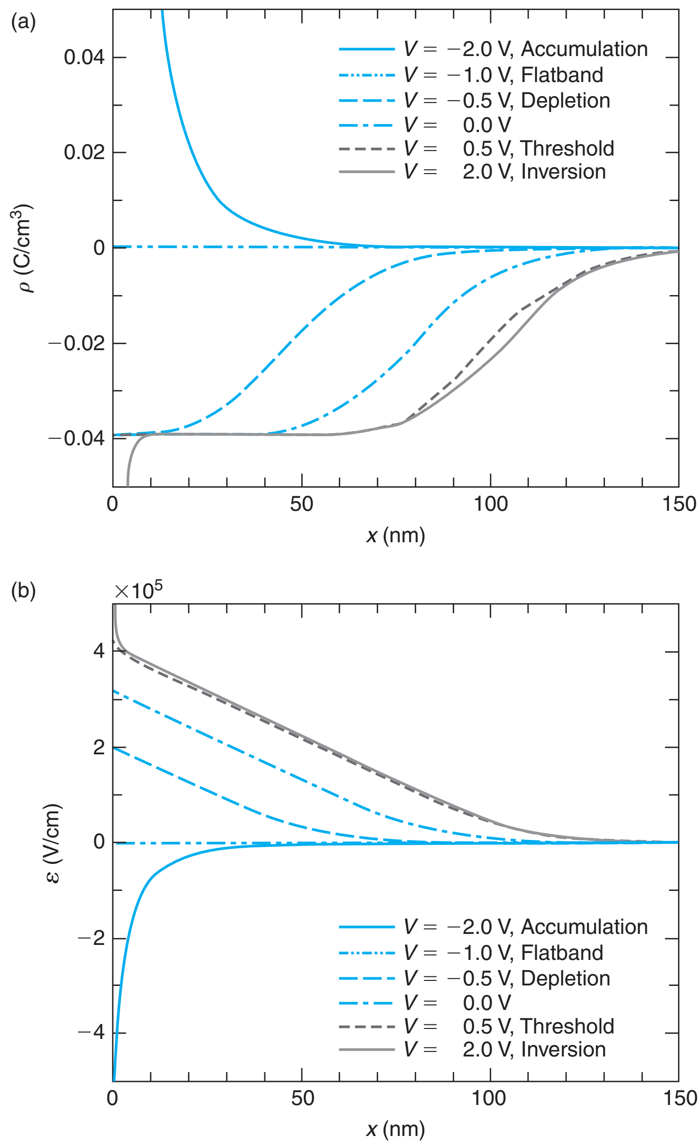


FIGURE 8.46
Volume charge density (a) and electric field (b) versus space coordinate for several voltages for the MOS structure of Fig. 8.44 [Reprinted with permission from Wenjie Lu, MIT].

The spatial dependence of ϕ is shown on Fig. 8.47(a) for six selected voltages. For $V = -2$ V, ϕ is relatively small and negative and confined to the vicinity of the interface. This is accumulation. For $V = -1.0$ V, $\phi = 0$ everywhere. This is flatband. As V becomes positive, ϕ increases (depletion) but beyond $V = 0.5$ V (threshold) it does not change much.

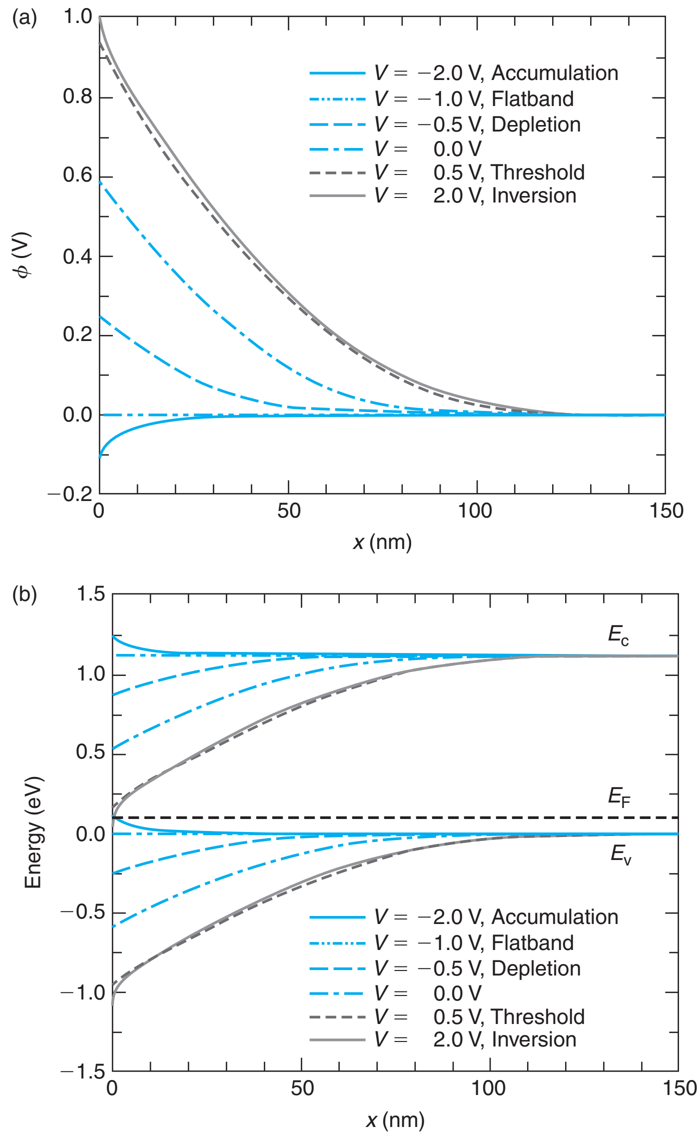


FIGURE 8.47
Electrostatic potential (a) and energy band diagrams (b) versus space coordinate for several voltages for the MOS structure of Fig. 8.44 [Reprinted with permission from Wenjie Lu, MIT].

Figure 8.47(b) shows the energy band diagrams in the semiconductor for several bias points. At flatband, the bands are flat. In accumulation, the bands bend up right at the interface. In depletion, the bands bend down. In inversion, the conduction band closely approaches the Fermi level at the interface, indicative of the presence of an inversion layer.

Figure 8.48 shows the electron (a) and hole (b) profiles across the semiconductor for the same voltage values. Deep into the semiconductor, p and n attain constant values that are unaffected by the applied voltage. These are the bulk values, p_{oB} and n_{oB}, respectively, imposed by the doping level. In accumulation, for negative V, the electron concentration dips at the interface below the bulk value, but the hole concentration increases substantially there. As V is made more positive, n increases at the interface but also deeper into the semiconductor and p decreases accordingly. At $V = 0.5$ V, the electron concentration at the interface equals the hole concentration in the bulk. This is threshold. For values of V larger than this threshold value, the electron concentration at the interface rises quickly in inversion. Note that at all points, $np = n_i^2$.

Figure 8.45 (a) showing the evolution of charge with voltage is drawn on a linear scale. Using this scale, it is not possible to appreciate the evolution of the electron charge below threshold. The subthreshold regime is more clearly seen when we graph $|Q_e|$ versus V in a semilog scale as done on Fig. 8.49(a). This figure makes clear that below threshold ($V_T \simeq 0.5$ V in this case), Q_e drops in a nearly exponential fashion, as we discussed in Section 8.6.

The evolution of $|Q_e|$ with ϕ_s in the subthreshold regime is shown on Fig. 8.49(b). Below threshold, we observe a $|Q_e| \sim \exp \frac{q\phi_s}{kT}$ dependence, in agreement with Eq. (8.60). Above threshold, the dependence of $|Q_e|$ on ϕ_s softens and turns into a $\exp \frac{q\phi_s}{2kT}$ dependence, as predicted by Eq. (8.34).

The Poisson–Boltzmann formulation also yields a rigorous model for the C–V characteristics of the MOS structure. Looking back at the model constructed in Section 8.5.1 and using the expression of Q_s in Eq. (8.106), the semiconductor capacitance can be expressed as

$$C_s = \sqrt{2\epsilon_s kTN_A}\,\frac{dF(\phi_s)}{d\phi_s} \tag{8.112}$$

This is always positive, as a capacitance should be.

Using Eq. (8.101), we can calculate the derivative of the $F(\phi)$ function:

$$\frac{dF(\phi)}{d\phi} = \frac{q}{kT}\frac{1}{2F(\phi)}\left[\left(-\exp\frac{-q\phi}{kT}+1\right)+\frac{n_i^2}{N_A^2}\left(\exp\frac{q\phi}{kT}-1\right)\right] \tag{8.113}$$

Inserting this into Eq. (8.112), yields

$$C_s = \frac{\epsilon_s}{\sqrt{2}L_D}\frac{1}{F(\phi_s)}\left[\left(-\exp\frac{-q\phi_s}{kT}+1\right)+\frac{n_i^2}{N_A^2}\left(\exp\frac{q\phi_s}{kT}-1\right)\right] \tag{8.114}$$

where L_D is the extrinsic Debye length.

An exact calculation of C_s versus V using Eq. (8.114) for the same MOS structure as in Figs. 8.44–8.48 is shown on Fig. 8.50(a). The overall capacitance, obtained by the series of C_s and C_{ox} as in Eq. (8.41), is shown on Fig. 8.50(b).

These two figures display the three regimes of operation of the MOS structure that were discussed in Section 8.5.1. On Fig. 8.50(a), we see that in accumulation, for $V < -1$ V, C_s increases in a roughly linear fashion as V_{GS} drops below the flatband voltage. This is what is predicted by Eq. (8.42). In depletion, C_s evolves in a rough square-root way, as given by Eq. (8.44). In inversion, C_s increases beyond threshold in an approximately linear way with a slope similar to that of the accumulation regime. This is consistent with the predictions of Eq. (8.47).

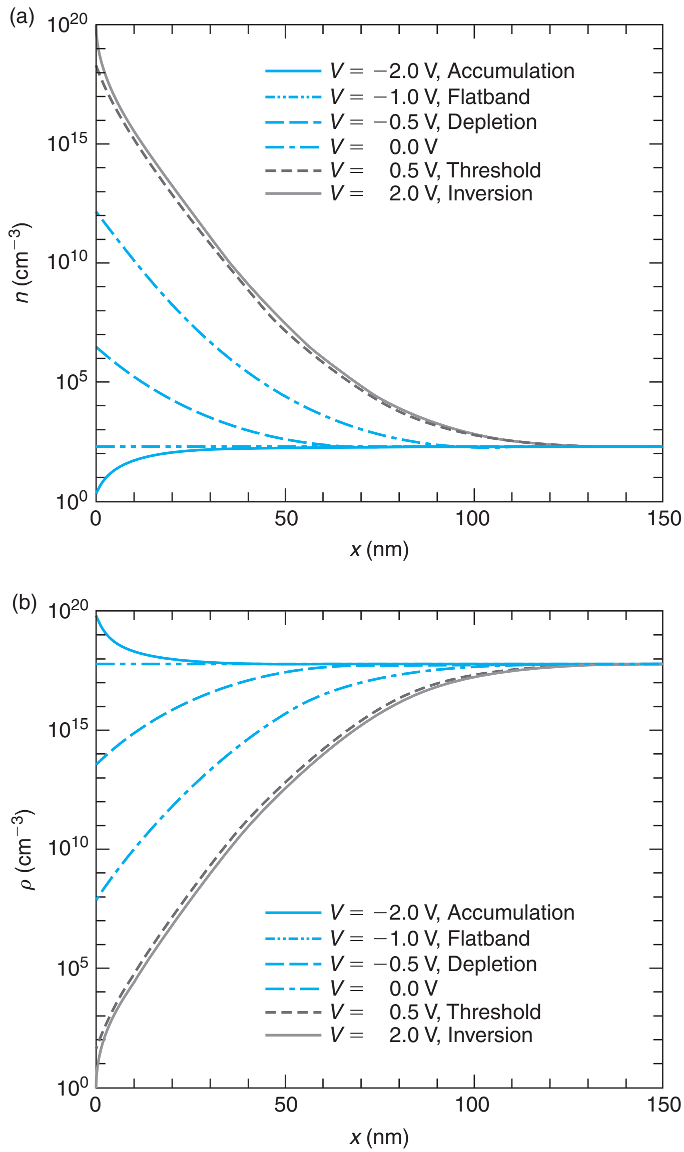


FIGURE 8.48
Electron (a) and hole (b) concentrations versus space coordinate for several voltages for the MOS structure of Fig. 8.44 [Reprinted with permission from Wenjie Lu, MIT].

The overall MOS capacitance is shown on Fig. 8.50(b). In accumulation and inversion, C approaches C_{ox} the further we bias the structure below V_{FB} or above V_T, respectively. In depletion, the overall MOS capacitance evolves in a square root way consistent with Eq. (8.50).

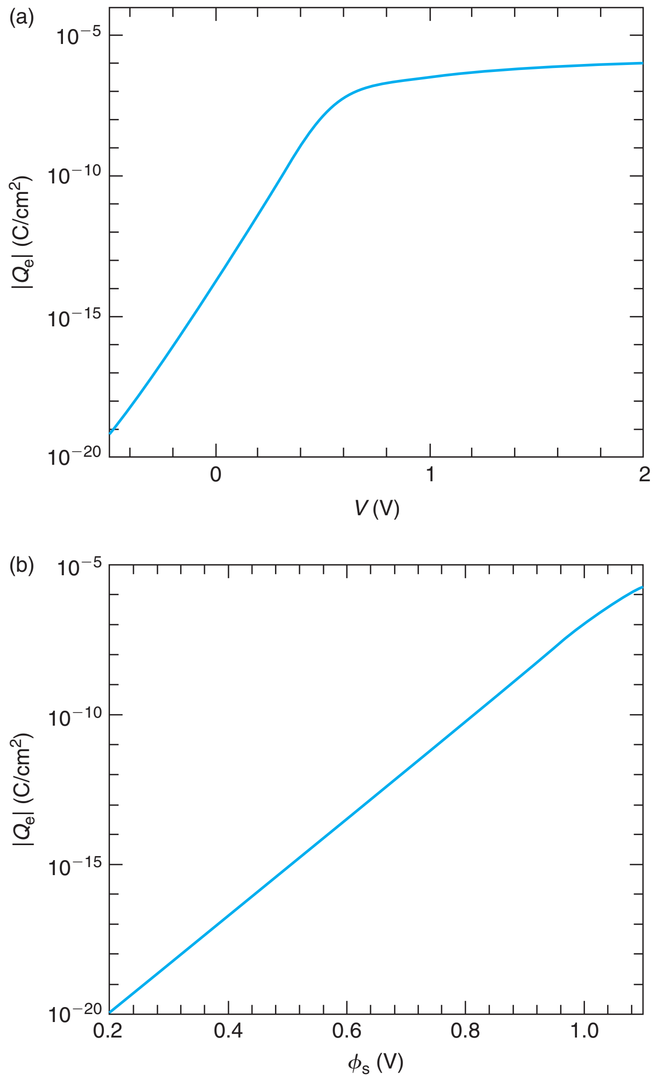


FIGURE 8.49
Q_e versus V (a) and ϕ_s (b) in a semilogarithmic scale for the MOS structure of Fig. 8.44. For small V, Q_e increases exponentially with V [Reprinted with permission from Wenjie Lu, MIT].

The physical understanding gained in this section is essential to devise useful mathematical simplifications of the exact formalism. We study the three main regimes separately below.

To accomplish this, it is helpful to first examine the key dependences of the function $F(\phi)$ defined in Eq. (8.101). A sketch of $|F(\phi)|$ in a semilog scale is shown in Fig. 8.51 for different values of n_i^2/N_A^2. This graph illustrates several features of the function $F(\phi)$:

- For $\phi = 0$, $F(0) = 0$.
- For $\phi = 0$, $\frac{d^2F}{d\phi^2}\big|_0 = 0$.

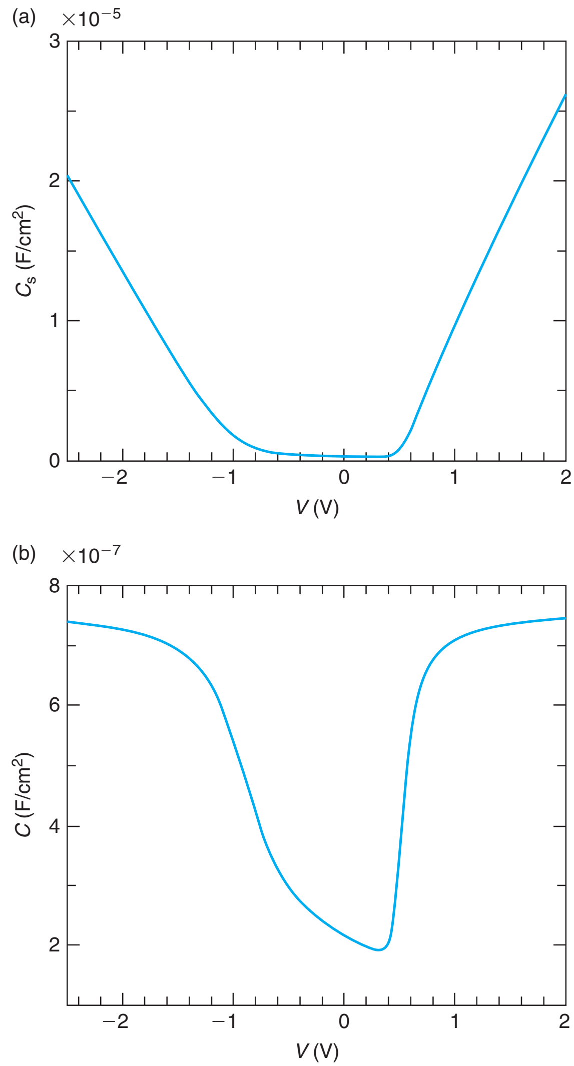


FIGURE 8.50
Semiconductor capacitance (a) and total capacitance of MOS structure (b) versus applied voltage for the MOS structure of Fig. 8.44 under quasi-static conditions [Reprinted with permission from Wenjie Lu, MIT].

- For $\phi < 0$, $F(\phi) < 0$. For $\phi > 0$, $F(\phi) > 0$.
- For all ϕ, $\frac{dF}{d\phi} > 0$.
- The F function in Eq. (8.101) contains six terms inside its square brackets. The last three are multiplied by a prefactor that represents the ratio of the minority carrier concentration to the majority carrier concentration in the bulk, a very small number (of order $\sim 10^{-10} - 10^{-14}$).

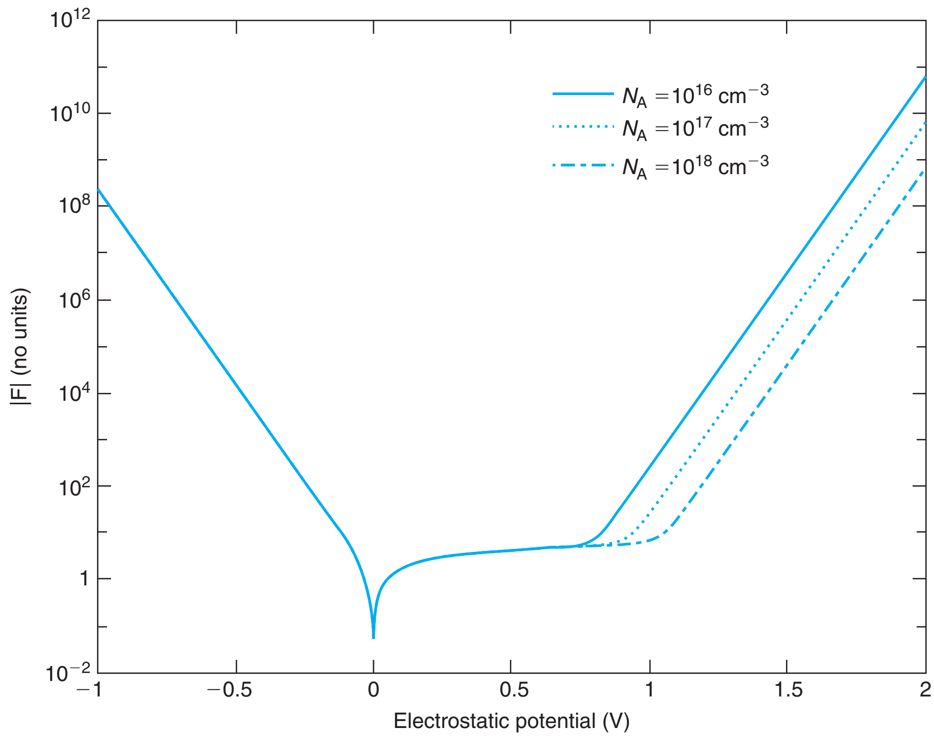


FIGURE 8.51
Graph of $|F|$ from Eq. (8.101) for several values of the acceptor doping level N_A, as a function of the electrostatic potential ϕ.

- For $\phi < 0$ and $|\phi|$ more than a few $\frac{kT}{q}$, the first term in Eq. (8.101) dominates and F displays an exponential behavior of the form:

$$F(\phi) \simeq -\exp\frac{-q\phi}{2kT} \tag{8.115}$$

- For $\phi > 0$ and ϕ more than a few $\frac{kT}{q}$, but not too large, the second term of Eq. (8.101) dominates and F displays a behavior of the form:

$$F(\phi) \simeq \sqrt{\frac{q\phi}{kT}} \tag{8.116}$$

- When $\phi > 0$ and sufficiently large, the fourth term eventually dominates and F follows a behavior:

$$F(\phi) \simeq \frac{n_i}{N_A}\exp\frac{q\phi}{2kT} \tag{8.117}$$

As the prefactor n_i^2/N_A^2 becomes larger for smaller values of N_A, the onset of this last exponential behavior occurs earlier, as Fig. 8.51 shows.

We are now in a position to develop appropriate simplifications for the three key regimes in the MOS structure.

AT8.3.1 Approximations for depletion

In the depletion regime, the semiconductor charge is dominated by fixed mobile ions, the substrate dopants. In this regime, Fig. 8.44 shows that ϕ_s changes quickly with V, and Fig. 8.45 shows that Q_s and $\mathcal{E}_s$ both have a square root dependence in V. It is easy to see why this should be the case.

As we saw in Eq. (8.116), in the depletion regime, the second term in $F(\phi)$ [Eq. (8.101)] dominates. Inserting it into Eq. (8.103), we get

$$\int_{\phi_s}^{\phi} \frac{d\phi}{\sqrt{\phi}} = -\sqrt{\frac{2qN_A}{\epsilon_s}}x \tag{8.118}$$

which can be easily integrated to yield

$$\sqrt{\phi} = \sqrt{\phi_s} - \sqrt{\frac{qN_A}{2\epsilon_s}}x \tag{8.119}$$

This equation is only valid up to the point at which ϕ goes to zero, the potential in the bulk. This point is the end of the depletion region, x_d. Solving from Eq. (8.119), we find

$$x_d = \sqrt{\frac{2\epsilon_s\phi_s}{qN_A}} \tag{8.120}$$

This is identical to Eq. (8.13). Inserting this expression back in Eq. (8.119), we get

$$\phi = \phi_s\left(1 - \frac{x}{x_d}\right)^2 \qquad \text{for } x \leq x_d \tag{8.121}$$

which is the expected result.

An expression for ϕ_s as a function of V can be obtained from Eq. (8.111). Inserting Eq. (8.116) into it, we get

$$V = V_{FB} + \phi_s + \gamma\sqrt{\phi_s} \tag{8.122}$$

The solution of this quadratic equation is

$$\phi_s = \frac{\gamma^2}{4}\left(\sqrt{1 + 4\frac{V - V_{FB}}{\gamma^2}} - 1\right)^2 \tag{8.123}$$

This is as obtained in Eq. (8.21). For small values of V in excess of V_{FB}, this equation is quadratic in V. For large values of V, it is linear. This is precisely the behavior observed in Fig. 8.44.

The rest of the results obtained in Section 8.4.1 follow easily.

AT8.3.2 Approximations for Accumulation

For sufficiently negative bias, the depletion layer disappears and in its place a thin layer of holes is induced at the semiconductor interface. This is the accumulation regime. In this regime, Fig. 8.44 shows that ϕ_s saturates with V. Additionally, Fig. 8.45 shows that Q_h and $\mathcal{E}_s$ increase in magnitude linearly with V. Let us understand where these dependencies come from.

In accumulation, the first three terms of the $F(\phi)$ function in Eq. (8.101) are relevant. Plugging this into Eq. (8.106) gives us the hole accumulation sheet charge density as a function of ϕ_s:

$$Q_a = Q_s \simeq \sqrt{2\epsilon_s kTN_A}\sqrt{\exp\frac{-q\phi_s}{kT} + \frac{q\phi_s}{kT} - 1} \tag{8.124}$$

In strong accumulation, Q_a increases exponentially as $\exp\frac{-q\phi_s}{2kT}$ (remember that in accumulation ϕ_s is negative).

The relationship between the applied voltage and the surface potential can be obtained from Eq. (8.111):

$$V = V_{FB} + \phi_s - \frac{\sqrt{2\epsilon_s kTN_A}}{C_{ox}}\sqrt{\exp\frac{-q\phi_s}{kT} + \frac{q\phi_s}{kT} - 1} \tag{8.125}$$

This equation is not amenable to an explicit solution since there are several terms that depend on ϕ_s. Of them, however, the exponential term is the dominant one. We can then obtain a first-order solution by setting the linear terms to zero since $\phi_s = 0$ marks the onset of the accumulation regime. After performing this substitution, we can solve for ϕ_s in the exponential term of Eq. (8.125) to get

$$\phi_s \simeq -\frac{kT}{q}\ln\left\{1 + \left[\frac{C_{ox}(V_{FB} - V)}{\sqrt{2\epsilon_s kTN_A}}\right]^2\right\} \tag{8.126}$$

This equation exhibits the right dependence. For $V = V_{FB}$, $\phi_s = 0$. As V increases in absolute terms below flatband, ϕ_s increases in magnitude, although rather weakly.

An approximate expression for Q_a versus V can be obtained from the fundamental charge control relationship of Eq. (8.18) using Eq. (8.126) for ϕ_s. This yields

$$Q_a \simeq C_{ox}\left(V_{FB} - V - \frac{kT}{q}\ln\left\{1 + \left[\frac{C_{ox}(V_{FB} - V)}{\sqrt{2\epsilon_s kTN_A}}\right]^2\right\}\right) \tag{8.127}$$

In strong accumulation, when $|V| \gg |V_{FB}|$, the logarithmic term becomes negligible next to the linear term and we have the classic result:

$$Q_a \simeq C_{ox}(V_{FB} - V) \tag{8.128}$$

Equation (8.128) explains the observation made in Fig. 8.45 that the accumulation charge in strong accumulation depends linearly in V. In fact, the slope of the dependence is precisely C_{ox} the capacitance per unit area of the oxide. Equation (8.128) in only valid for $V < V_{FB}$. Only the voltage in excess of the flat-band voltage is used to grow the accumulation layer.

We can get a sense of the approximate distribution for the electrostatic potential in space and the thickness of the accumulation layer in strong accumulation using the dominant term of $F(\phi)$ (the exponential term) and inserting this into Eq. (8.103). This yields

$$\int_{\phi_s}^{\phi} \exp \frac{q\phi}{2kT} d\phi = \sqrt{\frac{2kTN_A}{\epsilon_s}}x \tag{8.129}$$

Integration of this equation leads to

$$x = \sqrt{2}L_D \left(\exp \frac{q\phi}{2kT} - \exp \frac{q\phi_s}{2kT} \right) \tag{8.130}$$

This allows us to solve for the thickness of the accumulation layer that corresponds to $\phi = 0$:

$$x_{acc} \simeq \sqrt{2}L_D \left(1 - \exp \frac{q\phi_s}{2kT} \right) \simeq \sqrt{2}L_D \tag{8.131}$$

The simplification that has been made in Eq. (8.131) is possible because in accumulation ϕ_s is negative. Equation (8.131) tells us that the accumulation layer is very thin, of the order of the Debye length. Also, interestingly, the thickness of the accumulation layer only depends on the doping level of the semiconductor and temperature and is independent of other parameters such as oxide thickness or the magnitude of ϕ_s. For $N_A = 6 \times 10^{17}$ cm^{-3}, for example, $x_{acc} \simeq 7.5$ nm.

EXERCISE 8.8

Consider a MOS structure such as the one of Exercises 8.3–8.7. Calculate Q_s and ϕ_s for $V = -2$ V.

In Exercise 8.3, we obtained for this structure $\phi_{bi} = 1.0$ V. V_{FB} is then -1 V. At $V = -2$ V, the structure is then in accumulation. Hence, $Q_s = Q_h$ which in a first pass can be computed in a straightforward way using Eq. (8.24):

$$Q_s = Q_h = C_{ox}(V_{FB} - V) \simeq 7.7 \times 10^{-7} \text{ F/cm}^2(-1+2) \text{ V} = 7.7 \times 10^{-7} \text{ C/cm}^2$$

ϕ_s can be obtained from Eq. (8.126):

$$\begin{aligned}\phi_s &\simeq -\frac{kT}{q} \ln \left\{ 1 + \left[\frac{C_{ox}(V_{FB} - V)}{\sqrt{2\epsilon_s kTN_A}} \right]^2 \right\} \\ &= -0.026 \text{ V} \ln \left\{ 1 + \left[\frac{7.7 \times 10^{-7} \text{ F/cm}^2(-1+2) \text{ V}}{\sqrt{2 \times 1.04 \times 10^{-12} \text{ F/cm} \times 0.026 \text{ eV} \times 1.6 \times 10^{-19} \text{ C} \times 6 \times 10^{17} \text{ cm}^{-3}}} \right]^2 \right\} \\ &= -0.12 \text{ V}\end{aligned}$$

As expected, this is much smaller than V.

With this estimate for ϕ_s, we can now reevaluate Q_h more precisely using Eq. (8.127):

$$Q_s = Q_h \simeq C_{ox}(V_{FB} - V + \phi_s) = 7.7 \times 10^{-7}\ \text{F/cm}^2(-1 + 2 - 0.12)\ \text{V} = 6.8 \times 10^{-7}\ \text{C/cm}^2$$

This represents about a 10% correction over the result obtained above.

AT8.3.3 Approximations for Inversion

In the inversion regime, a thin sheet of electrons appears at the semiconductor surface. In this regime, as Fig. 8.44 shows, ϕ_s tends to saturate with V. Additionally, Fig. 8.45 shows that Q_h saturates while Q_e and $\mathcal{E}_s$ increase linearly with V. Furthermore, the extent of the depletion region into the substrate stops growing with V. Let us understand more rigorously where these dependencies come from.

The leading term in the F function in inversion is indicated in Eq. (8.117). However, to smoothly connect with the depletion regime, we have to keep two additional terms, as in

$$F(\phi_s) \simeq \sqrt{\frac{q\phi_s}{kT} - 1 + \frac{n_i^2}{N_A^2}\exp\frac{q\phi_s}{kT}} = \sqrt{\frac{q\phi_s}{kT} - 1 + \exp\frac{q(\phi_s - 2\phi_f)}{kT}} \tag{8.132}$$

where we have used the definition of ϕ_f in Eq. (8.25). This function smoothly connects with the F function in depletion when $\phi_s = 2\phi_f$.

The charge in the semiconductor in inversion is obtained from Eq. (8.106):

$$Q_s \simeq -\sqrt{2\epsilon_s kTN_A}\sqrt{\frac{q\phi_s}{kT} - 1 + \exp\frac{q(\phi_s - 2\phi_f)}{kT}} \tag{8.133}$$

Q_s is the sum of the inversion charge plus the depletion charge. To separate the two terms, we need to consider the spatial distribution of ϕ.

In strong inversion, ϕ increases from zero in the bulk to a $\phi_s > 2\phi_f$ at the surface. While ϕ is not too large, the second term of $F(\phi)$ in Eq. (8.101) dominates. Close to the surface, ϕ becomes large enough that the fourth term of $F(\phi)$ becomes dominant. A suitable way to deal with this transition is through a piecewise approximation for $F(\phi)$. To the first order, we can select $2\phi_f$ as the value of ϕ at the boundary between these two regions. Hence, in the inversion regime:

$$F(\phi) \simeq \sqrt{\frac{q\phi}{kT}} \qquad \text{for } \phi < 2\phi_f \tag{8.134}$$

$$F(\phi) \simeq \sqrt{\frac{q\phi}{kT} - 1 + \exp\frac{q(\phi - 2\phi_f)}{kT}} \qquad \text{for } \phi > 2\phi_f \tag{8.135}$$

Equation (8.135) applies to the electron inversion layer, while Eq. (8.134) applies to the depletion layer underneath. This piecewise description of $F(\phi)$ is reasonably simple and continuous and represents a good basis for obtaining a number of useful first-order results.

The electron charge in the inversion layer can be obtained by inserting Eq. (8.135) into Eq. (8.109) and integrating through the inversion layer:

$$Q_i = Q_e \simeq q\frac{n_i^2}{N_A}\sqrt{\frac{\epsilon_s}{2kTN_A}}\int_{\phi_s}^{2\phi_f}\frac{\exp\frac{q\phi}{kT}}{\sqrt{\frac{q2\phi_f}{kT}-1+\exp\frac{q(\phi-2\phi_f)}{kT}}}d\phi \tag{8.136}$$

In the linear term inside the square root in the denominator, ϕ has been substituted by a constant value of $2\phi_f$ which is a reasonable approximation when compared with the exponential term on ϕ next to it.

It is not difficult to perform this integral and obtain

$$Q_i \simeq -\sqrt{2\epsilon_s kTN_A}\left[\sqrt{\frac{q2\phi_f}{kT}+\exp\frac{q(\phi_s-2\phi_f)}{kT}-1}-\sqrt{\frac{q2\phi_f}{kT}}\right] \tag{8.137}$$

This result shows that in strong inversion, the inversion layer charge grows *exponentially* with the magnitude of ϕ_s in excess of $2\phi_f$. Also, the leading dependence is $\sim\exp(q\phi_s/2kT)$.

The depletion charge can be computed from Eq. (8.110) in a similar way. Alternatively, it can also be derived from the difference between Q_s in Eq. (8.133) and Q_i in Eq. (8.137), to obtain

$$Q_d \simeq -\sqrt{2\epsilon_s qN_A 2\phi_f} = -C_{ox}\gamma\sqrt{2\phi_f} \tag{8.138}$$

This is a constant value, independent of ϕ_s or V, just as observed in Fig. 8.45.

The relationship between ϕ_s and V can be obtained by inserting Eq. (8.133) into Eq. (8.111):

$$V = V_{FB}+\phi_s+\frac{\sqrt{2\epsilon_s kTN_A}}{C_{ox}}\sqrt{\frac{q\phi_s}{kT}-1+\exp\frac{q(\phi_s-2\phi_f)}{kT}} \tag{8.139}$$

The ϕ_s dependence on the right-hand side is complex and an explicit solution is not possible. However, the exponential term exhibits a much stronger dependence on ϕ_s than the other two terms. We can therefore obtain an approximate solution by equating $\phi_s \simeq 2\phi_f$ in the two other terms and solving for ϕ_s in the exponential. We obtain

$$\phi_s \simeq 2\phi_f+\frac{kT}{q}\ln\left\{\left[\frac{C_{ox}(V-V_T)}{\sqrt{2\epsilon_s kTN_A}}+\sqrt{\frac{q2\phi_f}{kT}}\right]^2-\frac{q2\phi_f}{kT}+1\right\} \tag{8.140}$$

This equation exhibits the right limit. For $V = V_T$, with V_T as defined in Eq. (8.28), $\phi_s = 2\phi_f$. Also, as V increases above V_T, ϕ_s increases too but it does so in a logarithmic way. This is consistent with what we observed in Fig. 8.44.

In the inversion regime, the dependence of the inversion layer charge on voltage is of great importance. This can be obtained from the fundamental charge control relationship in Eq. (8.18). Again, noting that $Q_s = Q_i + Q_d$ and using the expressions for Q_d in Eq. (8.138) and for V_T in Eq. (8.28), we can write

$$\begin{aligned}Q_i &\simeq -C_{ox}(V-V_T-\phi_s+\phi_{sT})\\ &= -C_{ox}\left(V-V_T-\frac{kT}{q}\ln\left\{\left[\frac{C_{ox}(V-V_T)}{\sqrt{2\epsilon_s kTN_A}}+\sqrt{\frac{q2\phi_f}{kT}}\right]^2-\frac{q2\phi_f}{kT}+1\right\}\right)\end{aligned} \tag{8.141}$$

where we have assumed $\phi_{sT} = 2\phi_f$.

In strong inversion, the logarithmic term becomes less relevant and

$$Q_{\mathrm{i}} \simeq -C_{\mathrm{ox}}(V - V_{\mathrm{T}}) \tag{8.142}$$

This is the charge control relation for the inversion layer that was derived in Eq. (8.31). In the main body of this chapter, this equation was obtained under two assumptions: that the inversion layer thickness is much thinner than any other vertical dimension in the problem (the *sheet-charge approximation*) and that beyond threshold, the depth of the depletion region stops increasing. Both assumptions have been mathematically justified in this section and their limits of applicability can be gauged from the more accurate formulation that is presented here.

The choice of threshold surface potential $\phi_{\mathrm{sT}} \simeq 2\phi_{\mathrm{f}}$ is a reasonable one and is widely used in the literature. You can see, however, that in the more exact charge control relationship of Eq. (8.141), there is an additional term with a negative sign that is increasing logarithmically with $V - V_{\mathrm{T}}$. If we want to preserve the simple relationship of Eq. (8.142), then it would be advisable to define V_{T} at a slightly higher surface potential than $2\phi_{\mathrm{f}}$. Often times, ϕ_{sT} is defined as

$$\phi_{\mathrm{sT}} \simeq 2\phi_{\mathrm{f}} + m\frac{kT}{q} \tag{8.143}$$

with values for m between 1 and 6 often used.

The potential distribution in space can be obtained from integrating Eq. (8.103). For the inversion layer, we have to use Eq. (8.135). This is a rather messy integral. An acceptable description of the situation is obtained for ϕ_{s} sufficiently larger than $2\phi_{\mathrm{f}}$, by just keeping the exponential term in Eq. (8.135). In this case, Eq. (8.103) becomes

$$\int_{\phi_{\mathrm{s}}}^{\phi} \exp\frac{-q(\phi - 2\phi_{\mathrm{f}})}{2kT}\, d\phi = -\sqrt{\frac{2kTN_{\mathrm{A}}}{\epsilon_{\mathrm{s}}}}\, x \qquad \text{with } \phi > 2\phi_{\mathrm{f}} \tag{8.144}$$

Integrating, we obtain

$$\exp\frac{-q(\phi - 2\phi_{\mathrm{f}})}{2kT} - \exp\frac{-q(\phi_{\mathrm{s}} - 2\phi_{\mathrm{f}})}{2kT} = \frac{x}{\sqrt{2}L_{\mathrm{D}}} \qquad \text{with } \phi > 2\phi_{\mathrm{f}} \tag{8.145}$$

where we have used the definition of Debye length made in Eq. (4.68).

This equation allows us to estimate the inversion layer thickness. To be consistent with our piecewise model above, this is defined as the location at which $\phi = 2\phi_{\mathrm{f}}$. If ϕ_{s} is larger than $2\phi_{\mathrm{f}}$ by a few kT/q's at least, then Eq. (8.145) gives

$$x_{\mathrm{inv}} \simeq \sqrt{2}L_{\mathrm{D}}\left[1 - \exp\frac{-q(\phi_{\mathrm{s}} - 2\phi_{\mathrm{f}})}{2kT}\right] \simeq \sqrt{2}L_{\mathrm{D}} \tag{8.146}$$

The last approximation applies for the case of strong inversion.

This is a simple and easy to remember result. The inversion layer thickness is of the order of the Debye length. For $N_{\mathrm{A}} = 6 \times 10^{17}\ \mathrm{cm}^{-3}$, for example, x_{inv} is about 7.5 nm.

EXERCISE 8.9

Consider a MOS structure such as the one of Exercises 8.3–8.8. Estimate Q_s and ϕ_s for $V = 2$ V.

Since for this structure $V_T = 0.5$ V, at $V = 2$ V, it operates in inversion. Hence $Q_s = Q_i + Q_d$. Q_i, the inversion charge, can be evaluated using Eq. (8.142):

$$Q_i \simeq -C_{ox}(V - V_T) = -7.7 \times 10^{-7}\ \text{F/cm}^2(2 - 0.5)\ \text{V} = -1.2 \times 10^{-6}\ \text{C/cm}^2$$

Q_d is the depletion charge. It is obtained from Eq. (8.138):

$$Q_d \simeq -C_{ox}\gamma\sqrt{2\phi_f} = -7.7 \times 10^{-7}\ \text{F/cm}^2 \times 0.58\ \text{V}^{1/2}\sqrt{0.93} = -4.3 \times 10^{-7}\ \text{Q/cm}^2$$

Q_s is the sum of these two components:

$$Q_s = Q_i + Q_d = -1.2 \times 10^{-6}\ \text{C/cm}^2 - 4.3 \times 10^{-7}\ \text{Q/cm}^2 = -1.6 \times 10^{-6}\ \text{Q/cm}^2$$

ϕ_s is evaluated using Eq. (8.140):

$$\phi_s \simeq 2\phi_f + \frac{kT}{q}\ln\left\{\left[\frac{C_{ox}(V - V_T)}{\sqrt{2\epsilon_s kTN_A}} - \sqrt{\frac{q2\phi_f}{kT}}\right]^2 - \frac{q2\phi_f}{kT} - 1\right\}$$

$$= 0.93\ \text{V} + 0.026\ \text{V}\ln\left\{\left[\frac{1.2 \times 10^{-6}\ \text{C/cm}^2}{7.2 \times 10^{-8}\ \text{C/cm}^2} - \sqrt{\frac{0.93\ \text{V}}{0.026\ \text{V}}}\right]^2 - \frac{0.93\ \text{V}}{0.026\ \text{V}} - 1\right\} = 1.04\ \text{V}$$

This is about 4 kT's over $2\phi_f$. In this calculation, we have used our evaluation of the square root in the denominator that was performed in Exercise 8.8.

Using this value of ϕ_s, we can estimate Q_i more precisely using Eq. (8.141):

$$Q_i \simeq -C_{ox}(V - V_T - \phi_s + \phi_{sT}) = 7.7 \times 10^{-7}\ \text{F/cm}^2(2 - 0.5 - 0.11)\ \text{V} = -1.1 \times 10^{-6}\ \text{C/cm}^2$$

This is less than 10% different from our rough estimate above.

PROBLEMS

8.1 Estimate the conduction band and valence band discontinuities at the Si/SiO_2 interface.

8.2 Select the work function of the gate material of a MOS structure that is needed to obtain $V_{FB} = 0$ V on an n-type substrate with $N_D = 10^{17}$ cm^{-3} and $x_{ox} = 10$ nm. Sketch the energy band diagram of this structure at zero bias.

8.3 Consider an ideal MOS structure that consists of a n$^+$-poly-Si gate, a 9-nm SiO_2 insulator, on a p-Si substrate with a doping level $N_A = 3 \times 10^{17}$ cm^{-3}. At room temperature and for $V_G = -3, 0, 0.3$, and 3 V, estimate numerical values for

(a) the surface potential;
(b) the total charge per unit area in the semiconductor and its breakdown into electron charge and hole charge;
(c) the electric field in the oxide;
(d) the extension of the depletion region in the semiconductor;
(e) the low-frequency capacitance; and
(f) the high-frequency capacitance.

Do not solve the problem numerically. Use the analytical first-order models described in this chapter. Treat the n^+-polySi gate as a metal in which the Fermi level sits at the conduction band edge irrespective of bias.

Additionally, calculate:

(g) the threshold voltage;
(h) the accumulation layer charge per unit area at the oxide breakdown condition; and
(i) the inversion layer charge per unit area at the oxide breakdown condition.

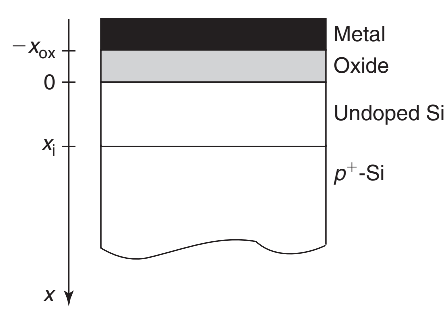
The oxide breakdown field is 4 MV/cm.

8.4 Consider a MOS structure using a p-Si substrate. At zero bias, this structure is in the depletion regime. Suppose we can now change gate materials and use a new one with a *higher* work function, W_M. Nothing else is changed. Indicate the impact that this would have on the parameters and figures of merit listed below.

Circle one: *increase* =↑, *decrease* =↓, or *no effect*. For each item, give the reason for your choice.

flatband voltage, V_{FB}:	↑ ↓ *no effect*
surface potential at zero bias, $\phi_s(V = 0)$:	↑ ↓ *no effect*
surface potential at threshold, ϕ_{sT}:	↑ ↓ *no effect*
high-frequency capacitance in inversion, $C_{HF}(inv)$:	↑ ↓ *no effect*
how does N_A have to change to maintain V_T unaffected?	↑ ↓ *no effect*

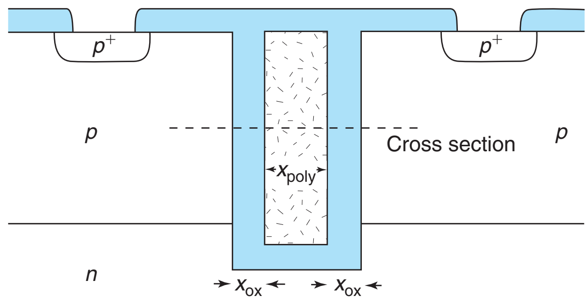
8.5 Consider a metal–oxide–semiconductor structure as sketched below. In this structure, directly underneath the oxide over a depth x_i, the doping level in the semiconductor is negligible. Below x_i the semiconductor is heavily doped p-type. The work function of the metal is smaller than the work function of the p^+ semiconductor, but the difference in work functions is smaller than the bandgap of the semiconductor.



(a) Sketch the volume charge density distribution along x at zero bias.
(b) Sketch the electric field distribution along x at zero bias.
(c) Sketch the electrostatic potential distribution along x at zero bias.

(d) Sketch the energy band diagram along x at zero bias.
(e) Derive an analytical expression for the *charge in the semiconductor* at zero bias as a function of relevant material-related parameters.
(f) Derive an analytical expression for the *threshold voltage* of this MOS structure in terms of relevant material-related parameters. Define threshold as the condition that brings the conduction band edge in contact with the Fermi level at $x = 0$.

8.6 Trench isolation is a compact way to isolate devices. Typically, the trench has a lining of SiO_2 and is filled with poly. There is no contact to the polySi filling. This is illustrated in the schematic below.



An important concern with trench isolation is the capacitance of the structure. This depends on the thickness of the SiO_2 lining, but also on the doping level and type of the polySi filling. Let's consider here the two limiting possibilities.

I. First assume that the trench is filled with p^+-polySi.

(a) Sketch the energy band diagram at zero bias along the indicated cross section.
(b) Calculate the low-frequency capacitance per unit area of trench that is seen between the ohmic contacts for $x_{ox} = 80$ nm, $x_{polySi} = 100$ nm, and $N_A = 2 \times 10^{16}$ cm^{-3}. No DC bias is applied.
(c) *Estimate* the electric field across the oxide lining.

II. Now consider the trench filled up with n^+-polySi.

(d) Sketch the energy band diagram at zero bias along the indicated cross section.
(e) Calculate the low-frequency capacitance per unit area of trench that is seen at the ohmic contacts for $x_{ox} = 80$ nm, $x_{polySi} = 100$ nm, and $N_A = 2 \times 10^{17}$ cm^{-3}. No DC bias is applied.
(f) Calculate the electric field across the oxide lining.

Regarding the polySi filling, assume that the Fermi level sits at E_c for the n^+ case and at E_v for the p^+ case. For all calculations, assume that no bias is applied between the two p^+ contacts.

8.7 Consider a MOS structure with a n^+-Si gate on a p-type Si substrate characterized by the following parameters: $W_M = \chi_s = 4.04$ eV, $x_{ox} = 9$ nm, $N_A = 1.2 \times 10^{17}$ cm^{-3}, $T = 300$ K.

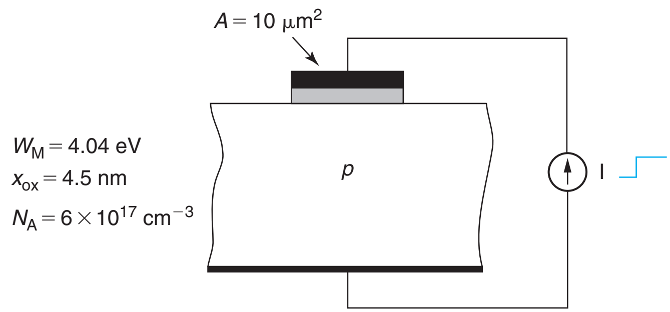
For $t < 0$, a voltage $V_1 = -2$ V has been applied from the gate to the bulk of the semiconductor for a long time. At $t = 0$, the applied voltage abruptly changes to $V_2 = 3$ V.

(a) Calculate the semiconductor charge and the electric field in the oxide at $t = 0^-$. What regime is the MOS structure in?

(b) Calculate the semiconductor charge and the electric field in the oxide at $t = 0^+$. Consider any RC delays negligible. What regime is the MOS structure in?

(c) Calculate the semiconductor charge and the electric field in the oxide for $t \to \infty$. What regime is the MOS structure in?

8.8 Consider a MOS structure as sketched in the figure below. For $t < 0$, the structure is at zero bias. In this condition, there is a depletion region in the semiconductor with a charge density of $Q_d = -3.4 \times 10^{-7}\ \mathrm{C/cm^2}$.



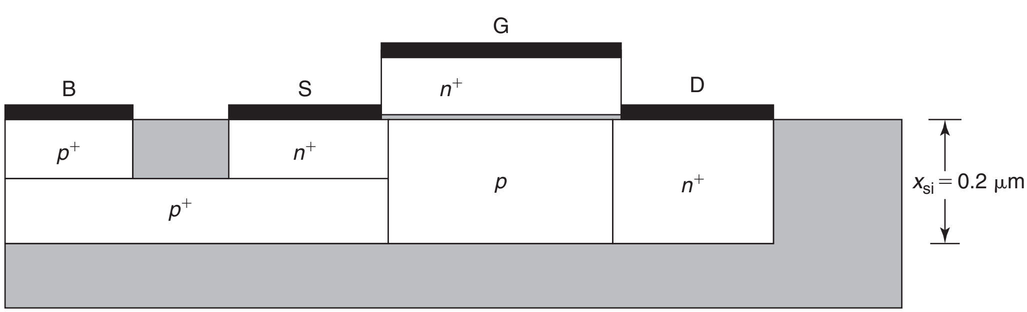
At $t = 0$, a current source is switched on with the sign indicated in the figure. *Estimate* the time that it takes for the MOS structure to be brought to the onset of inversion for the three cases below. The generation lifetime in the depletion region is $\tau_g = 1\ \mu s$.

(a) If $I = -100$ nA.

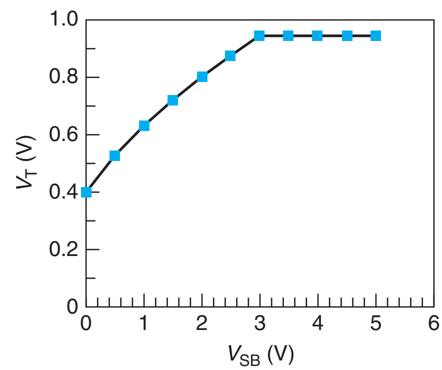
(b) If $I = 100$ nA.

(c) If $I = 1$ nA.

8.9 One of your competitors is working on a new body-contacted SOI (silicon-on-insulator) MOSFET technology. In this approach, a MOSFET is fabricated on an SOI substrate with a body contact on the side, as sketched below.

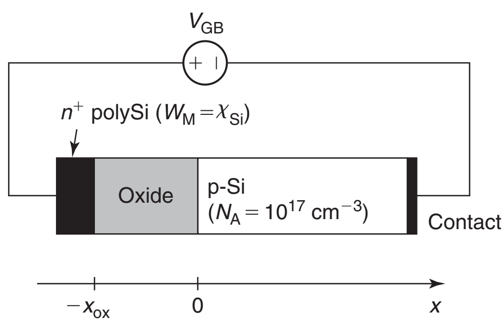


You have very little information about your competitor's design. You know that they use an n$^+$-polySi gate and a standard Si film thickness of $x_{si} = 0.2\ \mu$m. That is all you know. At a recent conference, your competitor presented a graph showing the dependence of the threshold voltage of the MOSFET as a function of the source to body voltage, V_{SB} (see below). Based on this new information, you can learn a bit more about their design.



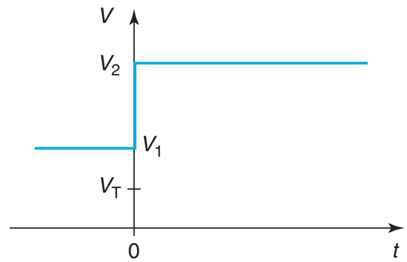
(a) Explain the origin of the peculiar dependence of V_T on V_{SB}.
(b) From the available information, extract the doping level, N_A, of the Si film.
(c) From the available information, extract the gate oxide thickness, x_{ox}, of the MOSFET.

8.10 Consider the following MOS structure:



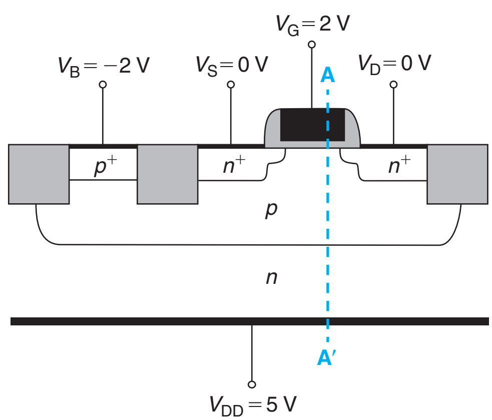
(a) Calculate the flatband voltage.
(b) Calculate the extent of the depletion region in the semiconductor at threshold.
(c) Calculate the electric field in the oxide at threshold.
(d) Calculate the inversion layer sheet charge when the electric field in the oxide is $\mathcal{E}_{ox} = 10^6$ V/cm.
(e) Calculate the accumulation layer sheet charge when the electric field in the oxide is $\mathcal{E}_{ox} = -10^6$ V/cm.

8.11 Consider a MOS capacitor on a p-type substrate. The gate material is n$^+$-polySilicon ($W_M = \chi_{Si} =$ 4.04 eV), the oxide thickness is 10 nm, the doping level of the substrate is $N_A = 1 \times 10^{17}$ cm^{-3}. This structure has a $V_T = 0.33$ V. For $t < 0$, the device is biased in inversion with $V_1 = 2$ V. At $t = 0$, V is suddenly increased to $V_2 = 5$ V and remains at this value for a long time, as sketched below.



(a) Sketch the evolution of the high-frequency capacitance of the MOS structure as a function of time. Explain the physics of the transitory.
(b) Estimate the high-frequency capacitance of the MOS structure at $t = 0^-$.
(c) Estimate the high-frequency capacitance of the MOS structure at $t = 0^+$.
(d) Estimate the high-frequency capacitance of the MOS structure at $t \to \infty$.

8.12 Consider an ideal n-channel MOSFET created on a p-type tub on an n-type substrate, as shown in the figure below. The structure is biased as indicated. This is at room temperature.



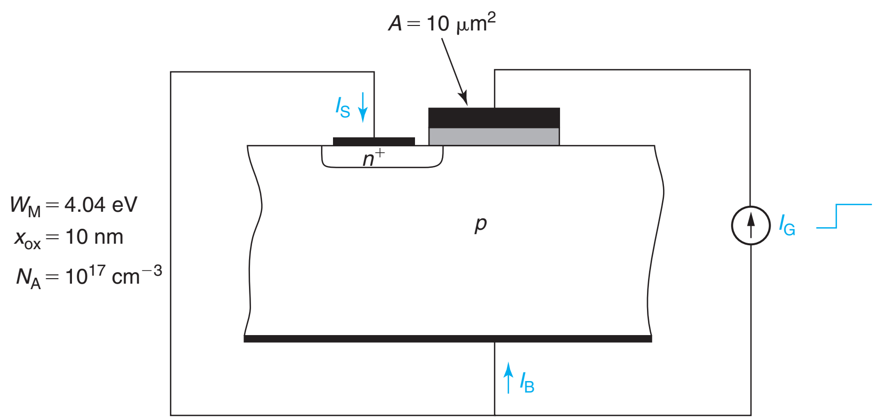
The MOS structure is identical to that of Exercises 8.3–8.8 in the notes, that is, n$^+$-poly Si gate ($W_M = 4.04$ eV), $x_{ox} = 4.5$ nm, and $N_A = 6 \times 10^{17}$ cm^{-3}. The depth of the p-type tub is $w_p = 1$ μm. The doping of the substrate is $N_D = 6 \times 10^{17}$ cm^{-3} and is 250 μm thick.

Under the indicated conditions, answer the questions below. To save time, you can recycle any results obtained in Exercises 8.3–8.8.

(a) In what regime is the MOS structure biased?
(b) Estimate the surface potential of the MOS structure.
(c) Estimate the charge density at the metal/oxide interface of the gate.
(d) Carefully sketch a qualitative energy band diagram across line A-A′ indicated above. Treat the n$^+$ gate as a simple metal. Indicate the location of the quasi-Fermi levels for electrons and holes across the entire semiconductor.

8.13 Consider a MOS structure with an n$^+$-Si gate on a p-type Si substrate characterized by the following parameters: $W_M = \chi_s = 4.04$ eV, $x_{ox} = 10$ nm, $N_A = 10^{17}$ cm^{-3}. Calculate the electron and hole concentrations (in cm^{-3}) at the oxide–semiconductor interface at zero bias at $T = 300$ K.

8.14 Consider a three-terminal MOS structure at $T = 300$ K as sketched in the figure below. Notice that this structure is characterized by the same physical parameters as that of the previous problem.



For $t < 0$, the structure is at zero bias. At $t = 0$, a current source of a magnitude $I_G = 10$ nA is switched on with the sign indicated in the figure.

Quantitatively sketch the time evolution of I_B and I_S with t for $t \geq 0$. Explain.

8.15 Consider an n$^+$-poly Si gate ($W_M = \chi_S = 4.04$ eV) MOS structure with a gate oxide thickness of $x_{ox} = 15$ nm and a semiconductor doping level of $N_A = 10^{17}$ cm^{-3}.

Answer the following questions at 300 K. For each question, state in what regime the MOS structure is operating. V refers to the voltage of the gate with respect to the body of the semiconductor.

To save you time, for this structure $C_{ox} = 2.3 \times 10^{-7}$ F/cm^2 and $\gamma = 0.8$ V$^{1/2}$.

(a) Compute the electric field across the oxide for $V = -2$ V.

(b) Compute the hole concentration at the oxide/semiconductor interface when there is a depletion region in the semiconductor of $x_d = 50$ nm.

(c) Compute the electric field on the semiconductor side of the oxide–semiconductor interface when, in steady state, there is a inversion layer charge of $|Q_i| = 5 \times 10^{-7}$ C/cm^2.

(d) Estimate the surface potential right after a pulse of $V = 4$ V is applied at $t = 0$ to the MOS structure in equilibrium.

(e) Estimate the charge at the gate/oxide interface under the same conditions as (d).

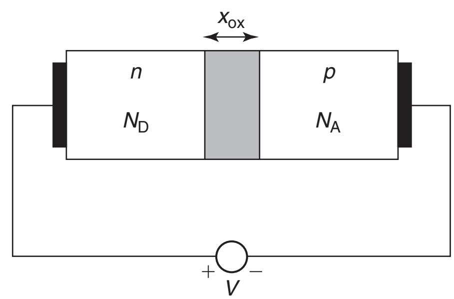
(f) Estimate the high-frequency capacitance of the MOS structure under the same conditions as (d).

8.16 You have learned that there is a simple technique for extracting the flat-band voltage from the high-frequency C–V characteristics of a MOS structure. The only thing you know about this technique is that it is somehow based on graphing $1/C_{HF}^2$ vs. V.

In order to figure out how this technique might work, derive a complete analytical description of $1/C_{HF}^2$ versus V for a MOS structure with a p-type body from accumulation to inversion. Sketch this relationship in a linear scale.

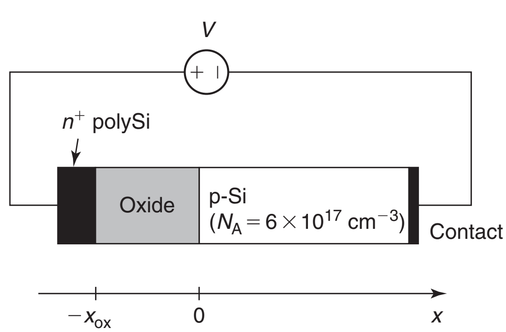
Discuss how one could extract V_{FB} from this graph.

8.17 Consider a n-Si/oxide/p-Si (semiconductor–oxide–semiconductor or SOS) structure as sketched below. The doping level in both regions is identical: $N_A = N_D$. The oxide thickness is x_{ox}.



(a) Sketch the charge distribution across the SOS structure at $V = 0$.
(b) Sketch the energy band diagram at $V = 0$.
(c) Derive an expression for the built-in potential of this structure at $V = 0$.
(d) Now apply a negative voltage, V, across the structure. At a certain voltage, a characteristic situation occurs. How would you describe it? Derive an expression for the applied voltage required to produce this situation. Sketch the energy band diagram across the SOS structure at this bias.
(e) Now apply a positive voltage across the structure. At a certain voltage, a characteristic situation occurs. How would you describe it? Derive an expression for the applied voltage required to produce this situation. Sketch the energy band diagram across the SOS structure at this bias.
(f) Sketch the C–V characteristics of this structure at low frequency. Place suitable tickmarks in the C and V axes.

8.18 Consider the following MOS structure at room temperature:



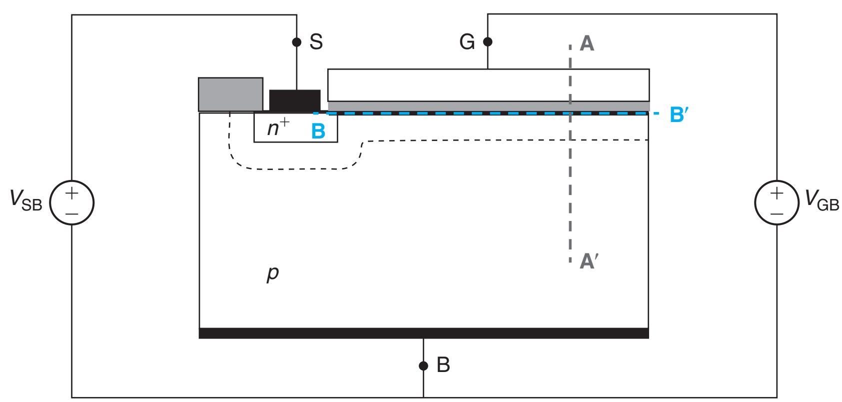
The oxide thickness is $x_{ox} = 5$ nm. The gate is made out of n^+-polySi with $W_M = \chi_s = 4.04$ eV. To save you time, for this structure $C_{ox} = 6.9 \times 10^{-7}$ F/cm^2 and $\gamma = 0.65$ V$^{1/2}$.

(a) What is the magnitude of $\mathcal{E}_{ox}$ (electric field across the oxide) for a condition in which $Q_G = -2 \times 10^{-7}$ C/cm^2?
(b) What is the capacitance of the MOS structure at a bias point for which the total charge in the semiconductor is equal to -2×10^{-7} C/cm^2?

8.19 Consider a two-terminal MOS structure identical to that of Exercises 8.3–8.9 in the notes. The key parameters of this structure are $N_A = 6 \times 10^{17}$ cm^{-3}, $x_{ox} = 4.5$ nm, and $W_M = 4.04$ eV. For this problem, you can reuse as many results as you want from any of these exercises.

(a) Estimate the surface potential at the onset of electron degeneracy, that is, at a bias point beyond which Fermi–Dirac statistics ought to be used to describe the statistics of electrons in the inversion layer. Sketch an energy band diagram for the semiconductor at this bias point.
(b) At this same bias point, estimate the hole concentration at the surface of the semiconductor.
(c) At this same bias point, estimate the inversion layer sheet-carrier density.
(d) Estimate the gate voltage that leads to this condition.

8.20 Consider the three-terminal MOS structure sketched below. The n$^+$ region has a doping level $N_D = 10^{19}$ cm^{-3}. The MOS structure itself is identical to that of Exercises 8.3–8.9 in the notes. The key parameters of this structure are $N_A = 6 \times 10^{17}$ cm^{-3}, $x_{ox} = 4.5$ nm, and $W_M = 4.04$ eV. For this problem, you can reuse as many results as you want from any of these exercises.

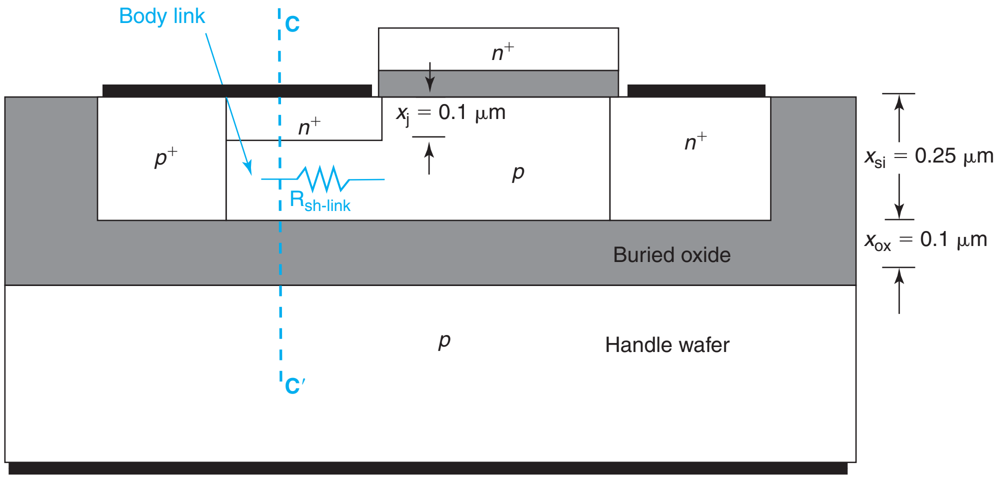


This problem is about studying the electrostatics of this structure across sections AA′ and BB′, indicated in the diagram, under several conditions. Section AA′ cuts across the MOS structure vertically at a location far away from the n$^+$ region. $x = 0$ is located at the oxide–semiconductor interface. Section BB′ runs along the oxide–semiconductor interface (on the semiconductor side). It starts just inside the n$^+$ region and it runs until far away from it. $y = 0$ is the location of the N$^+$P metallurgical junction.

In each part that follows, *quantitatively* sketch the electrostatic potential, ϕ, along sections AA′ and BB′. The sketches should include the values of the potential at key points and the spatial coordinate of the relevant point. Select $\phi = 0$ in the quasi-neutral p-type bulk of the semiconductor.

(a) $V_{GB} = 0$ V, $V_{SB} = 0$ V.
(b) $V_{GB} = 5$ V, $V_{SB} = 0$ V.
(c) $V_{GB} = 5$ V, $V_{SB} = 3$ V.

8.21 Consider an n-channel silicon-on-insulator MOSFET, as sketched below.



In this device design, there is a concern about the body resistance. This is because there is a relatively thin link between the body and its body contact underneath the source. This problem is about estimating the sheet resistance of this link and the body resistance associated with it.

The doping levels in the various regions are as follows:

- source-drain regions: $N_D^+ = 10^{19}$ cm^{-3}
- body and body link region: $N_A = 10^{17}$ cm^{-3}
- body contact: $N_A^+ = 10^{19}$ cm^{-3}
- handle wafer: $N_A = 10^{17}$ cm^{-3}

Note that the body contact, the handle wafer contact and the source are all shorted out.

(a) Qualitatively sketch an energy band diagram along cross section CC′ indicated in the figure. What is the regime of operation of the semiconductor–oxide–semiconductor (SOS) structure associated with the body link/buried oxide/handle wafer?

(b) Estimate the sheet resistance of the p-type link.

(c) Estimate the resistance of the link for a device that has a link length of 2 μm and a gate width of $W = 100$ μm.

(d) Suppose that the value of the link resistance that you obtain is too high. You wish to reduce it by applying a voltage to the handle wafer with respect to the body. What sign of the voltage should you apply: $V_{HB} > 0$? $V_{HB} < 0$? Explain and sketch an energy band diagram along CC′ of the resulting situation.

8.22 Consider a two-terminal MOS structure identical to that of Exercises 8.3–8.9 in the notes. The key parameters of this structure are $N_A = 6 \times 10^{17}$ cm^{-3}, $x_{ox} = 4.5$ nm, and $W_M = 4.04$ eV. For this problem, you can reuse as many results as you want from any of these exercises.

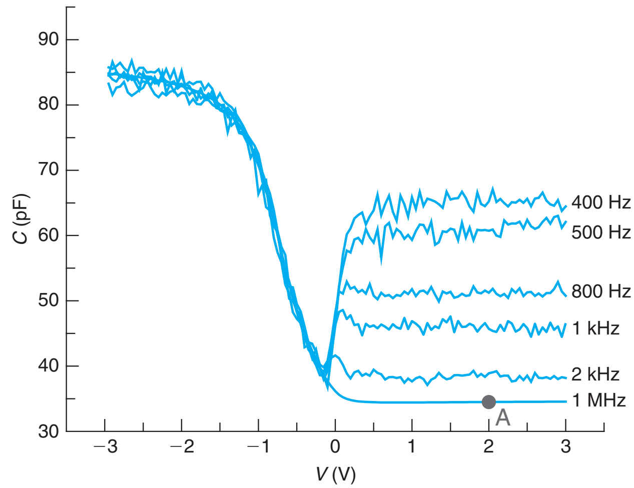
(a) Estimate the surface potential at the bias that yields $n(x = 0) = p(x = 0)$ at the surface of the semiconductor. In what regime is the MOS structure operating?

(b) At this same bias point, estimate the charge in the semiconductor.

(c) At this same bias point, estimate the electric field across the oxide.

(d) Estimate the gate voltage that leads to this condition.

8.23 The figure below graphs the experimental C–V characteristics obtained on a Si/SiO$_2$ MOS structure at room temperature at different frequencies. The metal is unknown. The area of the structure is $100 \times 100\ \mu\text{m}^2$.

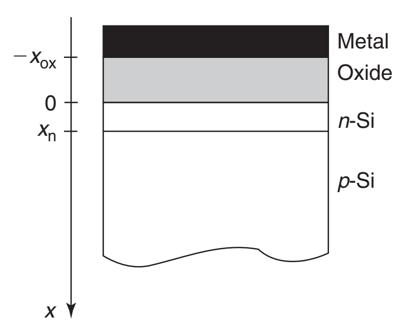


Using the data shown in this graph and making reasonable assumptions, answer the following questions:

(a) Estimate the thickness of the oxide.
(b) Estimate the extent of the depletion region in the semiconductor at threshold.
(c) Estimate the doping level in the substrate.
(d) Estimate the inversion layer sheet charge density when the structure is operated at point **A** in the figure.

8.24 In some applications, MOSFETs with different threshold voltages are desired. A way to accomplish this is to introduce a threshold control implant. How this works is explored in this problem.

Consider a regular MOS structure on a p-type Si substrate. Assume that the structure is designed in such a way as to yield a positive threshold voltage. Now consider introducing a very thin n-type doped layer at the surface of the semiconductor, as sketched in the figure below. This is easily accomplished by implanting donor atoms under the right conditions. For simplicity, we will assume that the ion implantation results in a uniform doping distribution with a doping level of N_D and a thickness x_n.



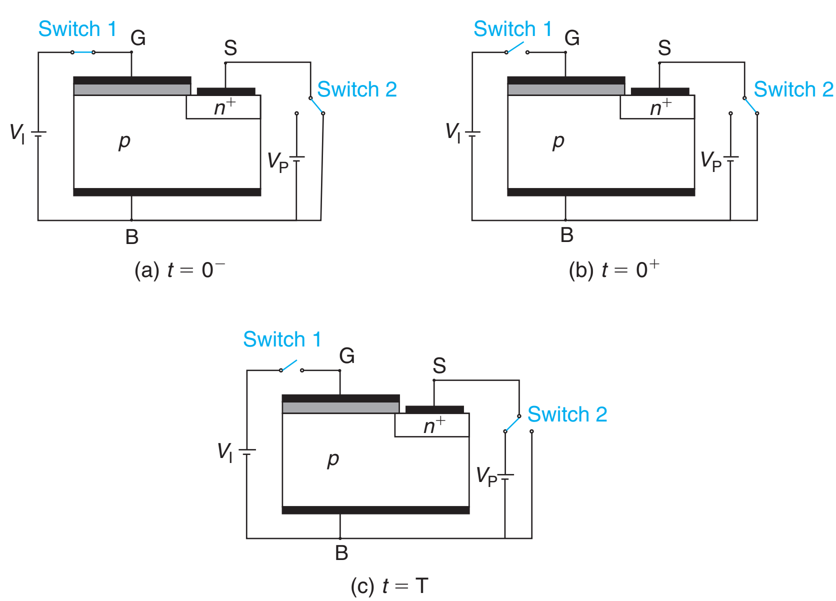
Your assignment is to derive an expression for the shift in the threshold voltage that the introduction of the n-type layer causes. The n-type threshold control layer has been designed so that it is fully depleted at threshold. You can assume that $x_n \ll x_{dmax}$ (but not that $N_D x_n \ll N_A x_{dmax}$). Follow the steps below:

(a) Qualitatively sketch the *volume charge density* in space at threshold.
(b) Qualitatively sketch the *electric field* in space at threshold.
(c) Qualitatively sketch the *electrostatic potential* distribution in space at threshold.
(d) Derive an expression for the electric field at the surface of the semiconductor, $\mathcal{E}_s$ in terms of parameters that characterize the semiconductor, such as N_A, N_D, x_n, x_{dmax}, etc.
(e) Is the surface potential at threshold, ϕ_{sT}, affected by the introduction of the n layer? If so, derive an expression for the new surface potential.
(f) Is the extent of the depletion region in the p region at threshold, x_{dmax}, affected by the introduction of the n layer? If so, derive an expression for the new x_{dmax}.
(g) Derive an expression for the threshold voltage of this structure. By comparing it with the threshold voltage of the conventional structure, give an expression for the shift in threshold voltage that the introduction of the n region produces.

8.25 An ideal MOS structure on a p-type Si substrate is characterized by a threshold voltage $V_T = 0.6$ V, a flat-band voltage $V_{FB} = -1$ V, and a subthreshold swing $S = 86$ mV/dec. All these values are given at room temperature. From this, you can reverse engineer the physical parameters of the structure. Answer the questions below. State any approximations that you need to make.

(a) Estimate the doping level in the semiconductor.
(b) Estimate the gate oxide thickness.
(c) Estimate the work function of the gate metal.

8.26 The three-terminal MOS structure can be used to achieve what is called *parametric operation*. Consider the sequence of events graphed below:

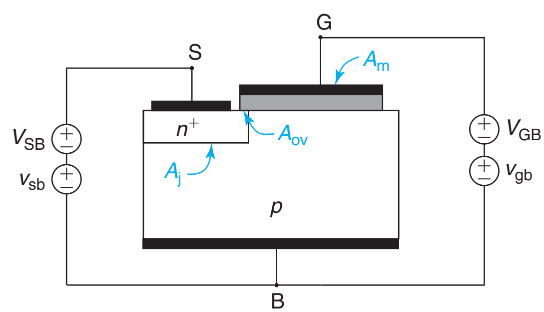


In (a), the MOS structure has been biased for a long time with $V_{GB} = V_I > V_T$ and the source is shorted to the body. At $t = 0$, the gate is open and left to float. This leads to situation (b). Some time later in (c), the source is applied a voltage $V_{SB} = V_P$ that is large enough so that the inversion layer is just turned off.

The reference for all voltages is the body voltage. V_T refers to the threshold voltage of the MOS structure measured from the gate to the body. Answer the questions below in terms of the usual MOS parameters: N_A, V_{FB}, V_T, x_{ox}, ϕ_{sT}, C_{ox}, and so on.

(a) At $t = 0^-$, provide expressions for the sheet charge density in the gate, the inversion layer, and the depletion region of the MOS structure.
(b) At $t = 0^+$, what is the voltage at the gate V_{GB}?
(c) At $t = 0^+$, provide expressions for the sheet charge density in the gate, the inversion layer, and the depletion region of the MOS structure.
(d) Derive an expression for the voltage V_P that is required to *just* turn the inversion layer off at $t = T$.
(e) Derive an expression for the voltage at the gate at $t = T$. Is this higher or lower than the gate-body voltage obtained in (b)? By how much?

8.27 Consider the three-terminal MOS structure sketched below. The MOS structure is characterized by $V_{FB} = -1$ V, $V_T = 1$ V, $\gamma = 0.7$ V$^{1/2}$, $C_{ox} = 5 \times 10^{-8}$ F/cm^2 and a metal area $A_m = 10$ μm^2. The PN junction is characterized by $C_{jo} = 10^{-7}$ F/cm^2, $\phi_{bi} = 1$ V and an area $A_j = 5$ μm^2. There is a small region where the MOS gate overlaps the n$^+$ region of area $A_{ov} = 1$ μm^2.



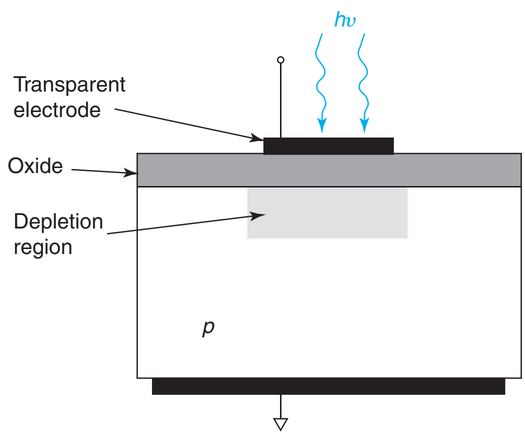
This problem is about the capacitance measured across different pairs of terminal under different conditions. The capacitance is measured by superimposing a high-frequency small signal on top of a bias source, as indicated in the figure. Assume ideal MOS and PN junction as defined in this text. State any other assumptions you need to make.

(a) For $V_{SB} = 0$, $v_{sb} = 0$, and $V_{GB} = 0$, estimate the high-frequency capacitance measured across the *gate-body* terminals.
(b) For $V_{SB} = 0$, $v_{sb} = 0$, and $V_{GB} = -2$ V, estimate the high-frequency capacitance measured across the *gate-body* terminals.
(c) For $V_{SB} = 2$ V, $V_{GB} = 0$, and $v_{gb} = 0$, estimate the high-frequency capacitance measured across the *source-body* terminals.
(d) For $V_{SB} = 0$, $v_{sb} = 0$, and $V_{GB} = 2$ V, estimate the high-frequency capacitance measured across the *gate-body* terminals.
(e) For $V_{SB} = 0$, $V_{GB} = 2$ V, and $v_{gb} = 0$, estimate the high-frequency capacitance measured across the *source-body* terminals.

8.28 The invention of the charge-coupled device (CCD) image sensor was honored in 2009 by the awarding of Nobel Price in physics to Willard Boyle and George Smith. They did their seminal work in the 70s at Bell Labs. CCD imagers are today widely used in all kinds of cameras and imaging systems.

The CCD imager relies on a MOS capacitor pulsed into deep depletion. If illuminated, the photogenerated electrons will collect at the oxide–semiconductor interface. The amount of electrons that is collected is proportional to the light intensity. There is a narrow window of time for this charge to be collected and to be sensed and that is the generation lifetime of the structure. This problem explores some of the issues involved.

Consider a pixel in a CCD imager that consists of a MOS capacitor with a transparent gate electrode, as pictured below:



The MOS structure is characterized by the following parameters: transparent metal gate with $W_M = \chi_s = 4.04$ eV, $x_{ox} = 15$ nm, uniform $N_A = 10^{17}$ cm^{-3}. To save you time, for this structure $C_{ox} = 2.3 \times 10^{-7}$ F/cm^2, $\gamma = 0.8$ V$^{1/2}$, $\phi_{sT} = 0.84$ V, $V_{FB} = -1$ V, $V_T = 0.6$ V. The active area of the pixel is 10×10 μm^2.

(a) Consider first this structure in the dark. For $t = 0^-$, the MOS structure is biased at $V = 0$ for a long time. At $t = 0$, a voltage pulse of $V_G = 5$ V is suddenly applied to the gate with respect to the body. Calculate the thickness of the depletion region at $t = 0^+$.

(b) Now, holding $V_G = 5$ V, a very short time after $t = 0$ the shutter opens and light illuminates the structure. The light intensity and wavelength are such that photogeneration takes place right at the surface of the semiconductor at a rate of $g_l = 10^{15}$ cm$^{-2} \cdot$ s^{-1}. How many photogenerated electrons are collected at the oxide–semiconductor interface after 1 ms of exposure? Assume that this span of time is much shorter than the generation lifetime for this structure.

(c) After 1 ms of illumination, the shutter is closed while V_G remains at 5 V. What is the thickness of the depletion region right after the shutter closes?

8.29 An alternative threshold definition in a MOSFET is the condition that leads to the oxide capacitance being equal to the semiconductor capacitance, that is, $C_{ox} = C_s$.

(a) Derive a simple expression for this alternative definition of the threshold voltage, V_T', in terms of the threshold voltage defined in class, V_T. State any assumptions that you need to make.

(b) Derive an expression for the inversion layer charge at this new threshold voltage.

Chapter 9

The "Long" Metal–Oxide–Semiconductor Field-Effect Transistor

The Metal–Oxide–Semiconductor Field-Effect Transistor or MOSFET is arguably the most important semiconductor device ever invented. Integrated circuits based on MOSFETs are ubiquitous in our modern society. There are many that we readily see, such as microprocessors in computers or digital-signal processors in wireless phones, but there are many more that we do not see, such as chips embedded in home appliances, industrial tools, toys, cars, and so on. As a result of the explosion of logic ICs, MOSFET technology has become a commodity: It is widely standardized and it is readily available to IC designers around the world. This has fueled, in turn, the widespread use of MOSFETs in many other applications that go beyond logic, such as analog and communications systems. Current examples are wireless LANs, cell phones, and optical fiber transceiver modules, among many others.

There have been many contributors to the conception and development of the MOSFET. Ross postulated in 1955 that an inversion layer could be induced electrostatically by an electrode placed close to the surface of a semiconductor. The first reduction to practice of the MOSFET had to wait until SiO_2 with sufficient quality was developed. The first MOSFET was demonstrated in 1959 by Dawon Kahng and Martin Atalla at Bell Labs. The integrated MOSFET and MOSFET ICs quickly followed. While these represented huge advances in microelectronics, a truly revolutionary event was the invention of CMOS in 1963 by Frank Wanlass at Fairchild. CMOS refers to complementary MOS, or the pairing of an n-channel MOSFET (based on electrons) and a p-channel MOSFET (based on holes) to form a logic gate. The distinctive feature of CMOS, in contrast with other logic families, is that it performs logic while consuming no DC power. This unique characteristic, perhaps more than anything else, underpins the microelectronics revolution and has led to ultra-large-scale integrated circuits. More details about the evolution of MOSFET design and key milestones along the way are given in Section 10.5.6.

A schematic cross section of a modern integrated planar *n-channel* MOSFET is shown in Fig. 9.1. At its heart, the MOSFET consists of a metal–oxide–semiconductor structure with an n^+ region on each side. The metal of the MOS structure is referred to as *gate*. The two n^+ regions are referred to as *source* and *drain*. The p-type well surrounding the device is referred to as *body*. A MOSFET is a four-terminal device. The gate, source, drain, and body regions are all contacted separately. For a long time, the gate of a MOSFET was made out of n^+-polySi. For reasons that will become clear in Chapter 10, in modern devices, the gate is made out of metal. The surfaces of the source and drain, as well as the body contact, are all silicided for the same reason. To facilitate

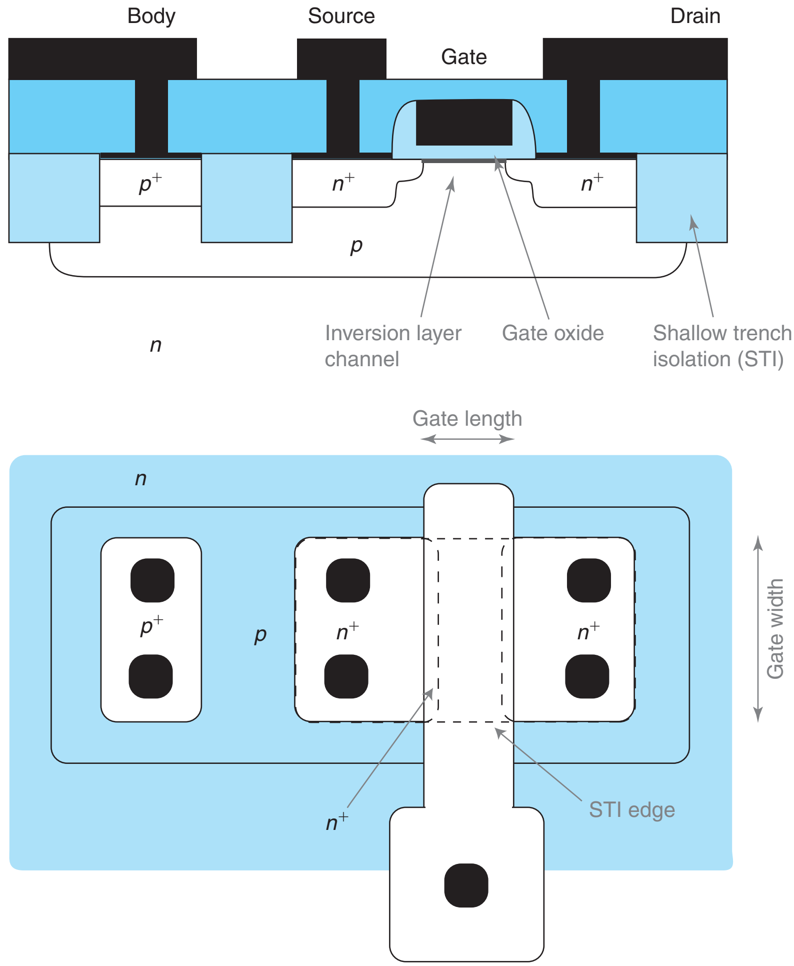


FIGURE 9.1
Simplified schematic diagram of a modern MOSFET: cross section (top) and layout at the wafer surface (bottom).

electrical contact, a p^+ region is implanted at the surface of the body contact. The device is isolated through shallow trenches that cut below the heavily doped regions of the source, drain, and body. In the planar view shown at the bottom of Fig. 9.1, two key dimensions are defined. *Gate length* and *gate width* refer to the length and width of the intrinsic portion of the device defined by the overlap of the polySi gate and the area enclosed by the shallow trench isolation. These are critical dimensions with a major impact on most electrical figures of merit of the MOSFET.[1]

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[1]Notice the odd choice of notation: the width of the gate tends to be larger than its length. This is for historical reasons.

The complementary p-channel MOSFET looks similar with p and n regions interchanged. Its operation is similar to that of the n-channel MOSFET but the signs of voltages and currents are reversed. Since it is based on hole transport, the p-channel MOSFET has lower performance.

The heart of the MOSFET is the intrinsic region, also referred to as *channel* region. This is the MOS portion of the device directly underneath the gate between source and drain. The rest is often referred to as the extrinsic portions of the MOSFET. When a gate voltage in excess of the threshold voltage is applied to the gate of the MOS structure, an inversion layer forms in the semiconductor right below the oxide/semiconductor interface. By virtue of the existence of a finite overlap of the gate above the n^+ source and drain regions (this is very important to insure low parasitic resistance), the inversion layer creates a conducting channel that connects the source and the drain. If a positive voltage is applied to the drain with respect to the source, electrons will flow from the source to the drain through the inversion layer. When the gate voltage is brought below the threshold voltage, there is no inversion layer in the MOS structure and the conducting path between source and drain is broken. In this manner, the MOSFET behaves as a switch. Electron flow from source to drain is controlled by the gate voltage; it is enabled when the gate voltage is above threshold and it is forbidden in the opposite case.

The MOSFET can also behave as an amplifier. With the gate voltage above threshold and an inversion layer connecting source and drain, the higher the gate voltage, the more electrons are induced in the inversion layer, and the higher the current that flows between source and drain. As we will study in this chapter, under the right conditions, the current between the source and drain can be made independent of the drain voltage. This "isolation between input and output" is an essential property of a good amplifying device.

A word on notation. In most cases, a MOSFET features a rather symmetrical design. The source and drain are interchangeable. What makes the source and the drain be identified as such is the voltage that is applied to them. The drain is always biased positive with respect to the source. In this way, electrons flow out of the source and into the drain. If the voltage difference between the source and drain was to change its sign, we would change the names too.

In this book, the MOSFET is presented in two chapters. The present chapter deals with the "long" MOSFET, while the next one is concerned with the "short" MOSFET. The terms "short" and "long" refer to the length of the gate of the transistor (marked in Fig. 9.1). As we will learn, these are somehow relative terms to other key dimensions of the device. Modern MOSFETs are very small and they are getting smaller all the time. Because of this, understanding the short MOSFET is of great importance for microelectronics engineers. This is the topic of Chapter 10. Before the short MOSFET can be productively attacked, one has to develop a clear picture of the operation of the "long" MOSFET. This is the purpose of this chapter.

The chapter starts with a definition of the concept of the ideal MOSFET. This is a useful simplification that captures the essence of the MOSFET and allows us to quickly understand its basic operation. The ideal MOSFET is amenable to developing first-order models that provide substantial physical insight. Section 9.2 qualitatively describes the operation of the ideal MOSFET using a simple water analogy. The basic physics of electron transport in the inversion layer is discussed in Section 9.3. Section 9.4 proceeds to describe the current–voltage characteristics of the ideal MOSFET and develops a simple analytical model for it. Section 9.5 presents the charge–voltage characteristics of the ideal MOSFET. After that, Section 9.6 discusses its small-signal behavior in the saturation regime that is the most useful regime of operation. The chapter ends with a section on the most important nonideal effects of the long MOSFET.

9.1 THE IDEAL MOSFET

As in the other chapters in this book, it is of great usefulness to define the concept of the ideal MOSFET. This is a model device that captures the essence of the MOSFET but hides away the complicating details and the second-order effects. The ideal MOSFET is a useful framework on top of which to construct a solid understanding and a useful set of models for the MOSFET. This will be of great value in this chapter about the long MOSFET, and also in Chapter 10 when studying the short MOSFET.

A sketch of the ideal MOSFET is shown in Fig. 9.2. This is a perfectly symmetrical device. It consists of uniformly doped source, drain, gate, and body regions. The edges of the source and drain n^+ regions are perfectly lined up with the edges of the MOS structure. The body contact is made at the bottom of the device.

The ideal MOSFET contains a number of simplifying assumptions:

- The MOS structure is ideal as defined in Chapter 8.
- All carrier flow is one dimensional.
- Doping levels are uniform throughout.
- We assume Maxwell–Boltzmann statistics (nondegenerate).
- Electron transport along the inversion layer takes place only by drift (i.e., we neglect diffusion). This assumption is verified in Problem 9.15.
- Electrons drift along the inversion layer in the mobility regime, that is, the electron velocity is proportional to the lateral electric field along the inversion layer. We undo this assumption in Chapter 10.
- We neglect the *body effect*, that is, the dependence of the threshold voltage on the space coordinate along the channel. This is fixed in Section 9.7.1.
- We ignore any resistance effects associated with the quasi-neutral regions or the Ohmic contacts to these regions. The impact of parasitic source and drain resistance is discussed in Section 9.7.5.

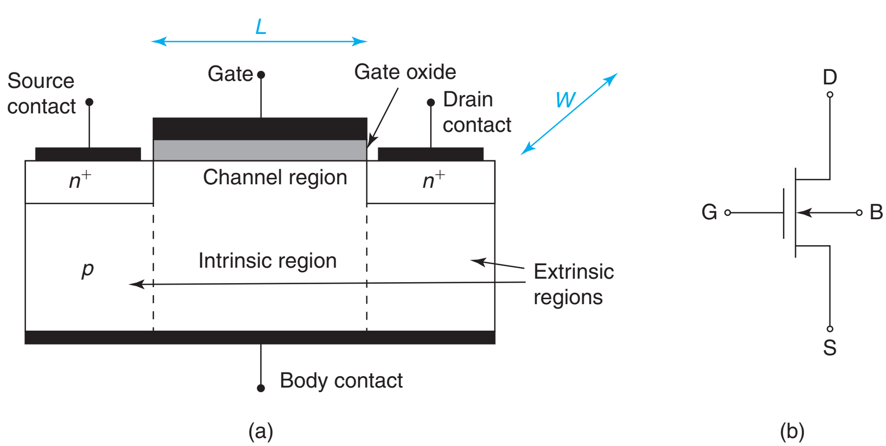


FIGURE 9.2
(a) Sketch of the cross section of an ideal MOSFET. This conceptual device is a good basis for developing first-order understanding and models for MOSFET operation. (b) Circuit symbol for a MOSFET.

- We ignore any effects associated with the sidewall of the source-body and drain-body junctions. This implies that the extent of the depletion region underneath the MOS structure is assumed to be unaffected by the presence of the source and drain regions. The impact of this on the electrostatics of the channel is discussed in Section 10.2.
- We ignore the saturation current of the reverse-biased source/body and drain/body PN junctions.
- There are no three-dimensional effects, that is, the device scales perfectly with its width.
- We ignore any effects associated with the substrate that surrounds the transistor (not shown in Fig. 9.2), including the presence of a PN junction with the MOSFET body with rectifying *I–V* characteristics.
- We neglect impact ionization anywhere in the device. Its role is discussed in Section 10.4.2.

Figure 9.3 shows a cross section of the ideal MOSFET indicating the coordinate axis convention used here. The x-axis is selected into the wafer with its origin at the semiconductor surface. The y-axis is selected along the channel from the source to the drain with its origin at the source edge. Indicated is also the inversion layer and the depletion region that under the right conditions exist underneath the source, channel, and the drain regions of the device. This figure suggests that the depletion regions of the source and drain merge with that of the channel at the edges of the channel. This is obviously the case. However, in our ideal MOSFET we assume that the depletion region underneath the channel is of uniform thickness throughout. That is, we neglect any effect of the source and drain sidewalls.

Figure 9.3 also shows the voltage and current notations that are used in this book. All voltages are referred to the source. As usual, entering terminal currents are positive. In our ideal

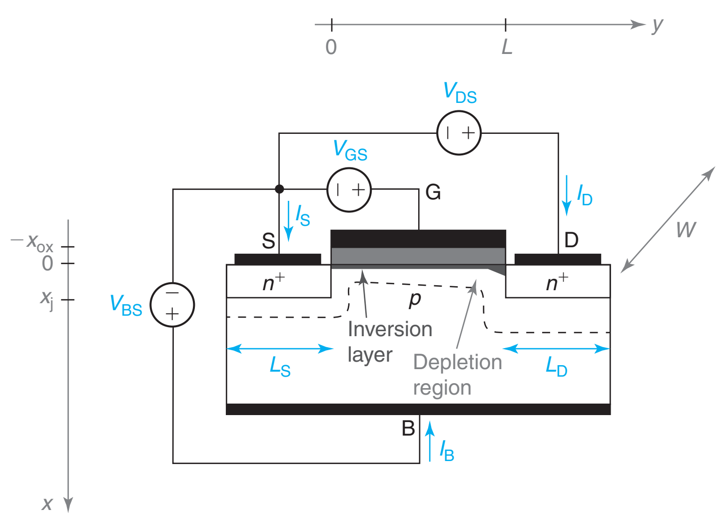


FIGURE 9.3
Sketch of the cross section of an ideal MOSFET defining spatial coordinates as well as voltage and current notation. Shown are also sketches of the inversion layer and the depletion region that under appropriate conditions exist underneath the source, gate, and the drain.

MOSFET, the gate and the body currents are assumed to be zero. Hence, the source current is equal to minus the drain current.

9.2 QUALITATIVE OPERATION OF THE IDEAL MOSFET

The operation of the MOSFET can be qualitatively understood by establishing an analogy with a water reservoir system, as sketched in Fig. 9.4. In this analogy, the source and drain can be thought of as water reservoirs with a relative height that can be adjusted. The gate plays the role of barrier separating the two water reservoirs. The relative height of the two reservoirs of water corresponds to the drain-to-source voltage. The higher V_{DS}, the *lower* the level of the water on the drain is with respect to that of the source. The relative height of the barrier above the water level of the source corresponds to the gate to source voltage in excess of threshold. The higher V_{GS}, the lower the gate barrier. At a value of $V_{GS} = V_T$, the top of the barrier is leveled with the water surface. In this analogy, we ignore the effect of the MOSFET body.

The water analogy of the MOSFET reveals three possible regimes of operation. These are sketched in Fig. 9.5. When the barrier level is above the water level on the source, water cannot flow between the source and drain. This is shown in Fig. 5(a). This occurs regardless of the relative levels of the source and drain water surfaces. In the MOSFET, this corresponds to the *cut-off* regime with $V_{GS} < V_T$. In the cut-off regime, there is no inversion layer under the gate and $I_D = 0$ regardless of the value of V_{DS}.

A second regime of operation corresponds to the case depicted in Fig. 9.5(b). When the barrier is below the water levels of the source *and* drain, water flows above the barrier from one reservoir to the other. It is clear that in this case, the higher the water level difference between the source and drain, or the lower the barrier level with respect to the water surfaces, the higher the current too. In the MOSFET, this situation corresponds to the *linear* regime, also called the *triode* regime. In this mode of operation, $V_{GS} > V_T$ and $V_{GD} > V_T$. As a consequence, an inversion layer is formed below the gate that constitutes a path that connects the source and drain. With $V_{DS} > 0$, an electric field is created along the inversion layer and electrons drift from

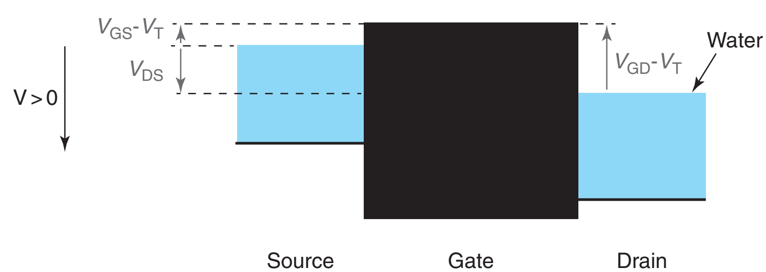


FIGURE 9.4
Water analogy for ideal MOSFET. The source and drain can be thought of as pools of water separated by a barrier. The drain-to-source voltage is equivalent to the relative heights of the water levels in the two pools of water. The gate to source voltage above threshold is equivalent to the relative height of the barrier above the water level on the source.

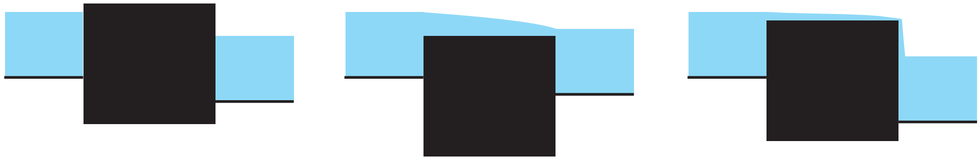


FIGURE 9.5
Regimes of operation of the water analogy of the MOSFET. The drain water level is always below the source water level. In the cut-off regime (a), the barrier is above the water level on the source and drain and water does not flow. In the triode or linear regime (b), the barrier is below the water level of both the source and drain. Water flows from source to drain. In the saturation regime (c), the barrier is below the water level of the source but above that of the drain. Water flows from source to drain, but the water flow rate is independent of the relative levels of the source and drain.

source to drain. As in the water analogy, the current flowing through the channel depends both on V_{DS} and V_{GS}. If V_{DS} increases, the lateral electric field along the inversion layer increases and the current increases. If V_{GS} increases, the amount of electrons induced in the inversion layer increases and the current also increases.

Finally, the third regime of operation of the MOSFET is depicted in Fig. 9.5(c). In this regime, the top of the barrier is below the water level on the source but above the water level on the drain. Water still can flow over the barrier, but there is a key difference with the regime in the middle of the figure. The water flow is independent of the relative height of the water level on the drain side with respect to the source. As the water flows over the barrier, when it reaches the end, it simply falls down freely. Lowering the water level at the drain does not increase the flow of water.

In the MOSFET, this regime of operation corresponds to a situation in which $V_{GS} > V_T$ but $V_{GD} < V_T$. As we will see, this creates an inversion layer at the surface of the semiconductor that thins down along the channel. As in the water analogy, the electron velocity increases along the channel from source to drain. At the drain side of the channel, it is said that the inversion layer is *pinched off*. The electrons simply "drop" into the drain. The remarkable aspect of this regime is that once V_{DS} is high enough for this regime to be established, the drain voltage has no further impact on the electric field that is set up along the channel. Beyond this value of V_{DS}, referred to as V_{DSsat}, the electric field in the channel is only affected by V_{GS}. In consequence, the channel current is independent of V_{DS}. This is called the *saturation* regime and it is the most important regime of operation of the MOSFET. It is in the saturation regime where a MOSFET is biased as an amplifier.

Armed with this qualitative understanding, we are now in a position to start looking in a more rigorous manner at the physics of the MOSFET in its various regimes of operation and start developing models for it. The next section discusses the physics of electron transport along the inversion layer that is at the heart of the MOSFET operation.

9.3 INVERSION LAYER TRANSPORT IN THE IDEAL MOSFET

The first step in the construction of a model for the ideal MOSFET is to develop a suitable formulation for electron transport in the inversion layer. While this is fairly straightforward, there are some subtleties to it. A detailed study is presented in Advanced Topic AT9.1. This section captures the most important results. Relevant variables are shown in Fig. 9.6.

In the inversion layer of a MOSFET under normal operation, electrons are trapped in a potential barrier that prevents them from escaping to the gate or the body. In consequence, electron transport only takes place laterally along the surface of the semiconductor from source to drain, right under the gate oxide. In this situation, there is no compelling interest in describing the behavior of electrons as a function of the *normal* direction into the semiconductor (the x-direction). The vertical dimension can be abstracted away by integrating the electron concentration across the inversion layer. So, instead of thinking of $n(x, y)$, the electron concentration as a function of location in x and y at any one location in the inversion layer, it is much more productive to think of the *sheet carrier concentration* in the inversion layer, $n_s(y)$, at a particular location along the channel, where

$$n_s(y) = \int_0^{\infty} n(x, y)dx \tag{9.1}$$

n_s has units of cm^{-2}. This integrates all electrons in the inversion layer at a location y and ignores their detailed distribution in x. The upper integration limit in this integral can be extended to ∞ without introducing any problems because the electron concentration peaks at the surface and drops very fast with x.

With the MOSFET biased in one of its conducting regimes, the electron current along the inversion layer can be easily expressed in terms of n_s. In the ideal MOSFET, we assume that electron flow takes place by drift and we neglect any electron diffusion (see Advanced Topic AT9.1). Under this assumption, the electron current is simply given by the product of the electron

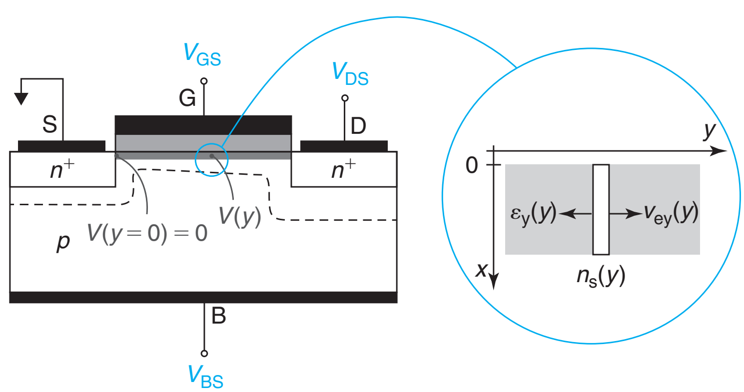


FIGURE 9.6
Schematic diagram of a MOSFET illustrating local gate voltage overdrive.

charge, the electron velocity, and the sheet electron concentration, times the width of the device

$$I_e = -qWv_{ey}(y)n_s(y) \tag{9.2}$$

where v_{ey} is the average velocity for the electrons in the inversion layer in the y-direction from source to drain. Note that while v_{ey} and n_s depend on y, I_e does not. This is a consequence of the fact that electrons cannot escape from the channel and must flow from source to drain.

Equation (9.2) can also be written in terms of the *sheet-charge density* of the inversion layer, Q_i. This is related to n_s through

$$Q_i(y) = -qn_s(y) \tag{9.3}$$

In terms of Q_i, Eq. (9.2) becomes

$$I_e = Wv_{ey}(y)Q_i(y) \tag{9.4}$$

In more rigorous terms, the key assumption that led us to Eqs. (9.2) and (9.4) is called the *sheet-charge approximation* (SCA). This approximation makes it physically meaningful to define an *average* lateral velocity for all the electrons in the inversion layer. Intuitively, this is possible whenever the distribution of lateral velocities in depth does not change too rapidly in the scale of the changes that are taking place in the electron concentration. It essentially requires that the inversion layer be very thin in the scale of other normal dimensions of the MOS structure (the oxide thickness and the thickness of the depletion region underneath the inversion layer). This is discussed in detail in Advanced Topic AT9.1.

According to our definition of the ideal MOSFET, we assume that the lateral electric field along the inversion layer is small enough for electrons to drift in the mobility regime. In this regime, the lateral electron velocity is proportional to the lateral electric field. As it turns out, this assumption is quite poor in MOSFETs of any gate length. Nevertheless, it is important and useful to proceed with this assumption for a while as the resulting models are an excellent starting point for a more detailed study of the MOSFET. We will undo this assumption and study the impact of velocity saturation in Chapter 10. In the mobility regime, we then have

$$v_{ey}(y) \simeq -\mu_e \mathcal{E}_y(y) \tag{9.5}$$

where $\mathcal{E}_y(y)$ is the average longitudinal electric field in the inversion layer. In this regime, Eq. (9.4) becomes

$$I_e \simeq -W\mu_e \mathcal{E}_y(y)Q_i(y) \tag{9.6}$$

An alternate way to express Eq. (9.6) is in terms of the voltage $V(y)$ of the inversion layer (also referred to as *channel voltage*). Since the source is our reference for all voltages, we define V as the surface potential at a location y with respect to the surface potential at the source end of the channel,

$$V(y) = \phi_s(y) - \phi_s(y = 0) \tag{9.7}$$

The lateral electric field in the inversion layer can now be expressed in terms of V,

$$\mathcal{E}_y(y) = -\frac{d\phi_s(y)}{dy} = -\frac{dV(y)}{dy} \tag{9.8}$$

Inserting Eq. (9.8) into (9.6) yields

$$I_e = W\mu_e Q_i(y)\frac{dV(y)}{dy} \tag{9.9}$$

To make further progress we need to find a way to relate $Q_i(y)$ with $V(y)$. The *gradual-channel approximation* (GCA) allows us to do that. Before we introduce this approximation, it is necessary to remember the fundamental charge-control relationship of the inversion layer in inversion. This was studied in Chapter 8. In a two-terminal MOS structure under inversion, the charge in the inversion layer is to the first order determined by the voltage applied to the gate with respect to the source in excess of the threshold voltage,

$$Q_i = -C_{ox}(V_{GS} - V_T) \tag{9.10}$$

This expression was derived for a MOS structure with the source as voltage reference and with the body tied up to the source ($V_S = V_B = 0$). In this case, the inversion layer voltage is zero everywhere and as a consequence, the inversion layer charge is uniform across the structure. In a MOSFET, unlike the simpler two-terminal MOS structure, this expression only applies at the source end of the channel, where $V = 0$. Anywhere else down the channel, V is different from zero and this expression needs to be reformulated.

The GCA allows us to recycle Eq. (9.10) to situations in which V changes in space but it does so relatively slowly. The key assumption is that at any one location, the inversion layer charge is only set by the local *vertical* electrostatics with the *lateral* electrostatics having a negligible effect on it. This means that we can reuse Eq. (9.10) provided that we subtract the *local* voltage of the inversion layer, $V(y)$, which in general is not zero. This is sketched in Fig. 9.7. Here, we isolate a sliver dy of channel at location y. Electrical contact to this sliver of channel is provided from the left, as in the three-terminal MOS structure that we studied in Chapter 8. When there is no

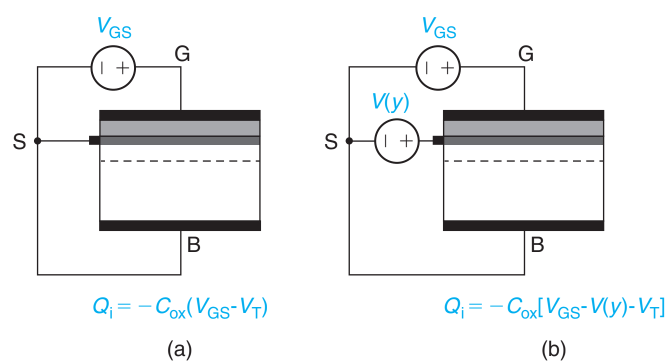


FIGURE 9.7
Analogy between a sliver of channel dy located at position y and the three-terminal MOS structure studied in Section 8.7. In the absence of a voltage drop along the channel (a), $V_{GS} - V_T$ determines the inversion charge at location y. However, when there is a voltage drop along the channel and location y is at a voltage $V(y)$ (b), then $V_{GS} - V_T - V(y)$ is what sets the inversion layer charge at y.

voltage drop along the channel, the voltage at location y is zero and the charge density at that location is given by Eq. (9.10). This is the situation depicted in Fig. 9.7(a). Figure 9.7(b) shows a situation in which there is a voltage drop along the channel and the sliver of channel at location y sits at a voltage $V(y)$ with respect to the source. The voltage difference between the gate and the inversion layer at location y is now $V_{GS} - V(y)$, instead of simply V_{GS}, and the inversion layer charge at that location is then given by

$$Q_i(y) \simeq -C_{ox}[V_{GS} - V(y) - V_T] \tag{9.11}$$

Notice that in the situation depicted on in Fig. 9.7(b), the appearance of a voltage difference between the local inversion layer and the body also produces a shift in the local threshold voltage. In this first-pass analysis we neglect this. The consequence of a location-dependent V_T shift is the so-called *body effect* that is discussed later on in this chapter.

The GCA effectively breaks up the two-dimensional electrostatics of the MOSFET into two simpler one-dimensional problems: the vertical electrostatics control the charge in the inversion layer, while the lateral electrostatics control its lateral flow along the channel. In the GCA, the lateral electrostatics represent a relatively "small" perturbation of the vertical electrostatics. As a consequence, the inversion layer charge at any one location can be computed using the one-dimensional MOS theory developed in Chapter 8, with the simple precaution of using the *local* inversion layer voltage, $V(y)$, instead of $V = 0$.

Advanced Topic AT9.1 discusses the GCA and the SCA in more detail and explores the limits of applicability of these two approximations. It is shown that for the GCA to apply, the lateral electric field must be smaller than an "effective" vertical electric field. In turn, this demands that the aspect ratio of the device be large enough, that is, that the channel length be sufficiently longer than a vertical characteristic length. In AT9.1, it is also shown that if the GCA is fulfilled, the SCA also applies.

Equation (9.11) allows us to formulate the MOSFET current equation in terms of a single variable, $V(y)$. Plugging Eq. (9.11) into (9.9), we get

$$I_e = -W\mu_e C_{ox}[V_{GS} - V_T - V(y)]\frac{dV(y)}{dy} \tag{9.12}$$

This is a first-order differential equation in terms of $V(y)$. Its solution with appropriate boundary conditions yields the channel current in the ideal MOSFET under a variety of conditions.

9.4 CURRENT–VOLTAGE CHARACTERISTICS OF THE IDEAL MOSFET

In the previous section, we developed a quasi-1D formulation for inversion layer transport in a MOSFET. This is the basis for a first-order formulation of the current–voltage characteristics of the MOSFET under the cut-off, linear, and saturation regimes. This is carried out in the following three subsections.

9.4.1 The cut-off regime

The cut-off regime is defined for values of $V_{GS} < V_T$ and $V_{DS} \geq 0$. In a simple first-order formulation, since V_{GS} is below threshold, there is no inversion layer and the current is zero, regardless of the value of V_{DS}. In the cut-off regime, the transistor is *off*. This is one of the two logic states in which a MOSFET is biased when operating as a switch.

To the second order, we know that below threshold, there are still electrons at the surface of the semiconductor. We have referred to it as the weak inversion regime of the MOS structure or the subthreshold regime. As a consequence, even for $V_{GS} < V_T$, there will be a small amount of current flowing through the MOSFET. This is the *subthreshold regime* of operation of the MOSFET and it will be discussed in Section 9.7.4.

9.4.2 The linear regime

The linear regime is defined for values of $V_{GS} > V_T$ and $V_{GD} > V_T$. This situation is sketched in Fig. 9.8. Under these conditions, an inversion layer extends all the way under the gate from the source to drain, a lateral field is set up along the inversion layer, and current flows.

With our current understanding, we should expect the drain current in the linear regime to increase with V_{GS} and V_{DS}, but for different reasons. All things being equal, if V_{GS} increases, the inversion layer charge increases in absolute magnitude and the channel current must increase along with it. On the other hand, if V_{DS} increases, the electric field along the channel is also enhanced as a result, and more channel current ought to flow. This behavior resembles that of the triode vacuum tube and for this reason, the linear regime is also referred to as the *triode* regime.

Deriving an expression for the current–voltage characteristics of the MOSFET in the linear regime is fairly straightforward. We start where we left off in the previous section, with Eq. (9.12). Solving this differential equation in this case is rather easy. By bringing dy to the left-hand side, variable separation is achieved,

$$I_e dy = -W\mu_e C_{ox}(V_{GS} - V_T - V)dV \tag{9.13}$$

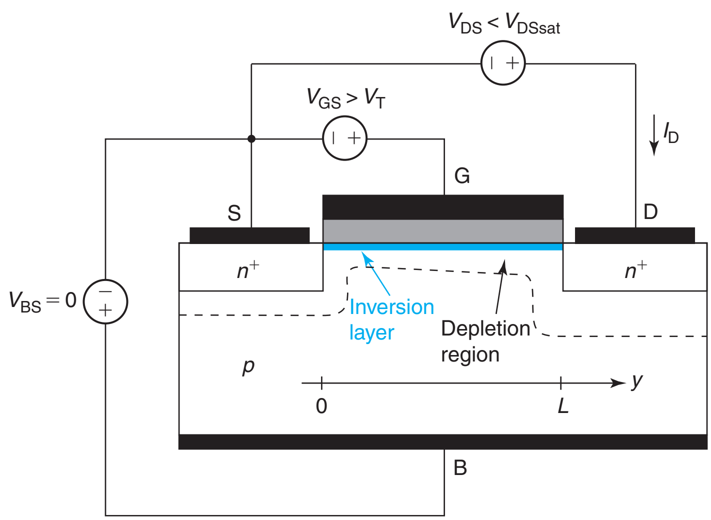


FIGURE 9.8
Sketch of MOSFET in linear regime and with no back-bias.

We can then integrate the left-hand side along the channel from $y = 0$ to $y = L$, and the right-hand side from $V = 0$ to $V = V_{DS}$, to get

$$I_e \int_0^L dy = -W\mu_e C_{ox} \int_0^{V_{DS}} (V_{GS} - V_T - V)dV \tag{9.14}$$

or, after integration,

$$I_e = -\frac{W}{L}\mu_e C_{ox}(V_{GS} - V_T - \frac{1}{2}V_{DS})V_{DS} \tag{9.15}$$

Going from channel current to drain terminal current is simple. Since electrons flow through the inversion layer from source to drain, this must necessarily result in an *entering* drain terminal current which, under standard sign convention, is positive. Since in our choice of y-axis, I_e is negative, we therefore need to apply a negative sign. The drain current is then

$$I_D = \frac{W}{L}\mu_e C_{ox}(V_{GS} - V_T - \frac{1}{2}V_{DS})V_{DS} \tag{9.16}$$

This equation gives the current–voltage characteristics of the ideal MOSFET in the linear regime. Before graphing it for different values of V_{DS} and V_{GS}, we must understand its limits of applicability. Equation (9.16) was derived under the assumption that the inversion layer extends underneath the entire channel. However, the charge distribution in the inversion layer is not uniform along the channel: it drops in absolute magnitude from source to drain. This is because V increases along the inversion layer from 0 to V_{DS} and, as a result, Q_i is reduced in absolute magnitude [see Eq. (9.11)]. This is discussed in detail later on in this section. The lowest inversion layer charge occurs at the drain end of the device where it is given by

$$Q_i(y = L) = -C_{ox}(V_{GS} - V_T - V_{DS}) \tag{9.17}$$

For $|Q_i(y = L)| > 0$, $V_{DS} < V_{GS} - V_T$ must be satisfied. This is the restriction on Eq. (9.16.)

Under this constraint, Eq. (9.16) is graphed in Fig. 9.9 that sketches I_D as a function of V_{DS} for different values of V_{GS}. The linear regime is the area of this graph comprised by the vertical axis and the line labeled "$V_{DS} = V_{GS} - V_T$." There are several features in this graph that are worth noting. First, for a given value of V_{GS}, I_D increases with V_{DS}. Initially, the increase is relatively fast but then it appears to saturate. The reason for the increase was mentioned before: a higher V_{DS} results in a larger lateral field and a larger current. The reason for the saturation will be discussed below. Second, for a given value of V_{DS}, I_D increases with V_{GS}. This was also expected. A higher V_{GS} induces a higher concentration of electrons in the inversion layer that results in increased current. Finally, for $V_{GS} = V_T$, $I_D = 0$, as it is to be expected since the inversion layer disappears. Notice that for $V_{GS} = V_T$, the model only applies for $V_{DS} = 0$ for which $I_D = 0$ anyway.

In order to better understand the physics of the MOSFET in the linear regime, in particular, its saturating behavior as V_{DS} increases, Fig. 9.10 sketches $|Q_i(y)|$, $|\mathcal{E}_y(y)|$, $V(y)$, and $V_{GS} - V(y)$ along the channel of the MOSFET from source to drain. The mathematical expressions for these parameters as a function of y are easy to derive (see Problem 9.2). As Fig. 9.10 indicates, Q_i drops in absolute magnitude along the channel from source to drain. This is because V increases along

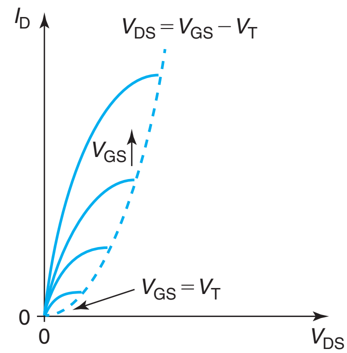


FIGURE 9.9
Sketch of *I–V* characteristics of a MOSFET in the linear regime. Each line corresponds to a different value of V_{GS}.

the channel from a value of zero at the source, to V_{DS} at the drain. This “debiases” the inversion layer as we proceed from source to drain. In order to maintain a constant current, $\mathcal{E}_y(y)$ must increase in magnitude from source to drain so that there is an increasing electron velocity. This yields a voltage distribution along the channel that is superlinear in y. The amount of local “gate overdrive,” $V_{GS} - V(y) - V_T$, is reduced along the channel from source to drain.

It is illuminating to consider how the profiles of Fig. 9.10 change as V_{DS} increases. This is shown in Fig. 9.11. For $V_{DS} = 0$, the lateral field across the inversion layer is zero and the electron concentration is uniform. For low values of V_{DS}, a nearly uniform electric field is set along the channel and the voltage distribution is nearly linear. As V_{DS} increases, the lateral electric field increases but the drain end of the device starts getting debiased, that is, the electron concentration toward the drain end of the channel drops as y approaches L. For large V_{DS}, channel debiasing becomes severe. The electron concentration on the drain side of the channel drops very low and the majority of the voltage is absorbed there.

To understand how channel debiasing results in current saturation, one must focus on its effect on the situation at the source end of the channel. Toward understanding this, it is useful to go back to the very fundamental Eq. (9.4) that expressed the channel current at any one location as the product of the channel charge times the electron velocity at that particular location. At the source end of the channel ($y = 0$), this expression becomes

$$I_e \simeq W v_{ey}(0) Q_i(0) \tag{9.18}$$

We can write an equation like this for any location of the channel. What is significant about the source end is that the inversion layer charge at that location is set only by V_{GS} through

$$Q_i(0) = -C_{ox}(V_{GS} - V_T) \tag{9.19}$$

since $V(0) = 0$. In particular, $Q_i(0)$ is independent of what happens down the channel.

The V_{DS} dependence enters Eq. (9.18) through the electron velocity, $v_{ey}(0)$. This is set by the local lateral electric field. With transport in the mobility regime,

$$v_{ey}(0) \simeq -\mu_e \mathcal{E}_y(0) \tag{9.20}$$

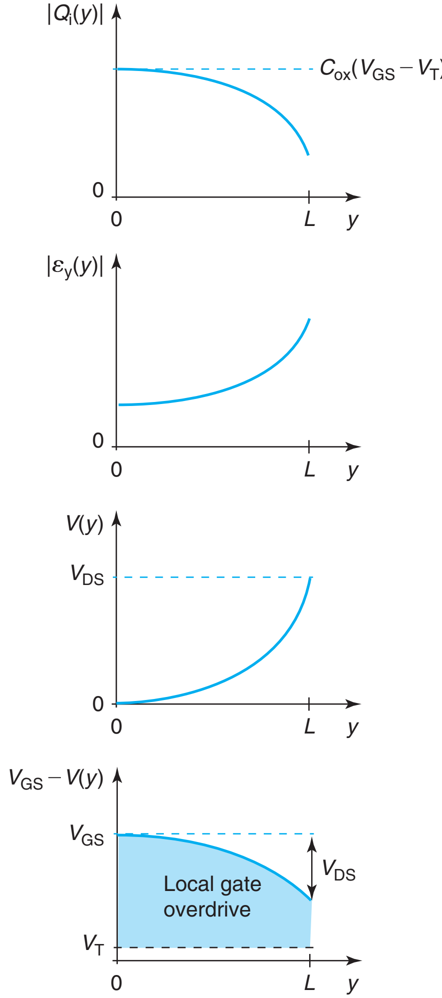


FIGURE 9.10
Inversion layer charge, lateral electric field, channel voltage, and local gate overdrive as a function of location along the channel for a MOSFET biased in the linear regime.

The key to understanding the saturating behavior of I_D with V_{DS} is to focus on the V_{DS} dependence of $\mathcal{E}_y(0)$. As Fig. 9.11 shows, as V_{DS} increases, so does the electric field at the source. However, when channel debiasing becomes prominent at high V_{DS}, the rise of $\mathcal{E}_y(y)$ slows down as higher values of V_{DS} are applied and eventually it does not increase any more. In consequence, at high V_{DS}, the velocity at which electrons are extracted from the source saturates and so does the current.

This line of argumentation is productive in one more and very general way. In the MOSFET, when dealing with transport phenomena, it is often most useful to focus on what is happening on

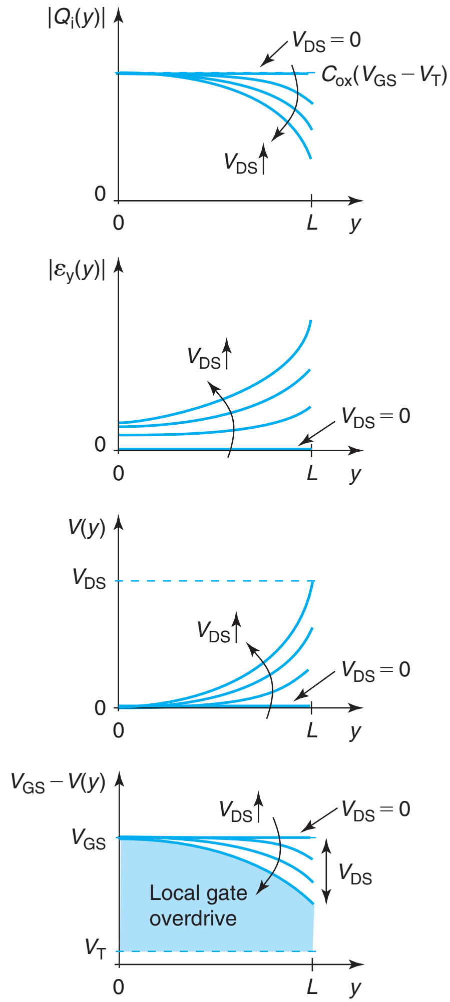


FIGURE 9.11
Inversion layer charge, lateral electric field, channel voltage, and local gate overdrive as a function of location along the channel for a MOSFET biased in the linear regime for increasing values of V_{DS}. For high V_{DS}, channel debiasing at the drain end of the channel is apparent.

the source side, as opposed to the drain side of the channel. It is at the source end of the channel that we can think in terms of the number of electrons that are extracted from the source and the velocity at which they come out. This is a simple and powerful picture that often provides great physical insight.

It is also interesting to reflect on what is going on at the drain end of the channel as V_{DS} approaches $V_{GS} - V_T$. This is the point with the lowest concentration of electrons in the inversion layer. As V_{DS} approaches $V_{GS} - V_T$, Eq. (9.17) indicates that Q_i goes to zero at $y = L$. In this limiting case, the location at the drain end of the channel is known as the *pinch-off* point. Complete depletion of electrons is of course not possible, otherwise, the current could not flow. What is happening and how could our formalism deal with it?

There are several problems with our formalism close to $y = L$ for values of V_{DS} that approach $V_{GS} - V_T$. First of all, at $y \simeq L$ for $V_{DS} \simeq V_{GS} - V_T$, the local gate overdrive goes to zero. As a consequence, the fundamental charge-control relationship stops applying and the GCA fails. This should make the use of Eq. (9.17) suspect. An additional problem with our formalism around the pinch-off condition is that the assumption of linearity between electric field and electron velocity is unlikely to apply as the lateral field gets high enough for velocity saturation to occur. For these reasons, the current–voltage characteristics given by Eq. (9.16) do not apply when V_{DS} approaches $V_{GS} - V_T$.

Developing a detailed model of current transport at the drain end of the channel when V_{DS} approaches and exceeds $V_{GS} - V_T$ is a major challenge. Fortunately, for a first-order analysis, we do not need to do it. The reason is that the model embodied in Eq. (9.16) is fairly accurate over nearly the entire range of drain current. This is understood by examining Fig. 9.9 in detail. For small values of V_{DS}, increasing V_{DS} brings about a significant increase in I_D. For higher values of V_{DS}, however, a further increase in V_{DS} produces a proportionally smaller increase in I_D. For V_{DS} values close to $V_{GS} - V_T$, I_D is essentially saturated. The origin of this diminishing return in I_D for high values of V_{DS} is the "channel debiasing" discussed above. Even before we reach pinch-off and the simple model developed above becomes invalid, the drain current has essentially stopped increasing.

It is not difficult to quantify how close V_{DS} can get to $V_{GS} - V_T$ before the simple model developed above fails. As Problem 9.3 shows, for typical MOSFET designs, V_{DS} can get up to about 80% of $V_{GS} - V_T$ before the GCA fails. This means that I_D can build up to about 96% of the maximum value predicted by Eq. (9.16) and still be fairly accurate.

EXERCISE 9.1

Consider a long n-channel MOSFET characterized by the following parameters: n^+*-polysilicon gate* ($W_M = \chi_S = 4.04$ eV), $x_{ox} = 15$ nm, *uniform* $N_A = 10^{17}$ cm^{-3}, $L = 1$ µm, $W = 10$ µm *at a bias given by* $V_{GS} = 2.5$ V, $V_{DS} = 1$ V, *and* $V_{BS} = 0$. *Estimate the sheet charge density in the inversion layer at the source and drain ends of the channel. Estimate the current flowing through the drain of this transistor. Use* $\mu_e = 500$ cm^2/V · s.

Proceeding as in Chapter 8, for this MOS structure we can easily find: $C_{ox} = 2.3 \times 10^{-7}$ F/cm^2, $\gamma = 0.79$ V$^{1/2}$, $\phi_{sT} = 0.84$ V, $V_{FB} = -1$ V, and $V_T = 0.56$ V.

To obtain the sheet charge density at the source end of the channel at the indicated bias, we can use Eq. (9.19),

$$Q_i(0) = -C_{ox}(V_{GS} - V_T) = -2.3 \times 10^{-7}\ \text{F/cm}^2\ (2.5 - 0.56)\ \text{V} = -4.5 \times 10^{-7}\ \text{C/cm}^2$$

Before we can obtain the corresponding value at the drain end of the channel, we need to assert the regime of operation of the transistor. For the indicated bias, $V_{GS} - V_T = 1.94$ V $> V_{DS}$ $= 1$ V. Hence, this MOSFET is biased in the linear regime. We can then use Eq. (9.17) to obtain

$$Q_i(L) = -C_{ox}(V_{GS} - V_T - V_{DS}) = -2.3 \times 10^{-7}\ \text{F/cm}^2\ (2.5 - 0.56 - 1)\ \text{V} = -2.2 \times 10^{-7}\ \text{C/cm}^2$$

The drain current can be obtained from Eq. (9.16),

$$\begin{aligned} I_D &= \frac{W}{L}\mu_e C_{ox}(V_{GS} - V_T - \frac{1}{2}V_{DS})V_{DS} \\ &= \frac{10\ \mu\text{m}}{1\ \mu\text{m}} 500\ \text{cm}^2/\text{V}\cdot\text{s} \times 2.3 \times 10^{-7}\ \text{F/cm}^2 (2.5 - 0.56 - \frac{1}{2}1\ \text{V}) \times 1\ \text{V} = 1.7\ \text{mA} \end{aligned}$$

9.4.3 The saturation regime

In the previous section we have developed a simple model for the linear regime of operation of the MOSFET. We found that I_D depends on V_{DS} and V_{GS} and we derived an expression for I_D that applies up to essentially $V_{DS} = V_{GS} - V_T$. What happens if V_{DS} reaches or exceeds this value? This is the saturation regime.

It is clear from our discussion above that around $V_{DS} = V_{GS} - V_T$, the electron concentration at the drain end of the channel drops substantially. In fact, it can become rather small in comparison with the doping level of the body. If this happens, we can think about this as a small depletion region appearing right at the drain end of the channel. This is sketched in Fig. 9.12. This depletion region does not represent in any way a barrier to electron flow. On the contrary, the lateral electric field in this region "pulls" electrons into the drain.[2]

What is significant about the appearance of this depletion region is that if V_{DS} is increased beyond $V_{GS} - V_T$, the extra applied voltage drops entirely in this pinch-off region by widening its lateral extension. This is also sketched in Fig. 9.12. What happens here is entirely analogous to the electrostatics of a reverse-biased PN junction that widens as the reverse bias increases. If the pinch-off region widening is small in the scale of the channel length, the electrostatics of the

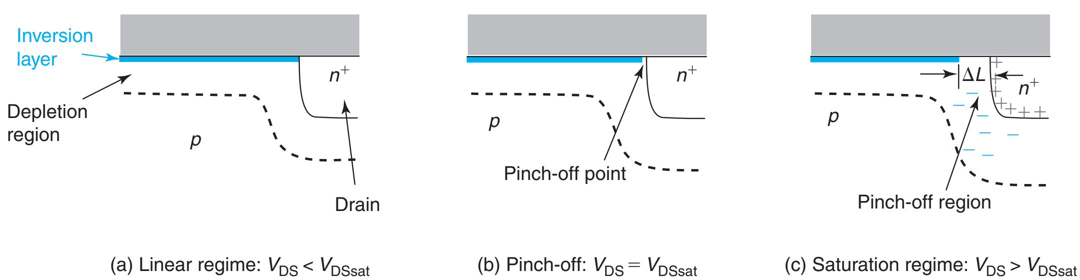


FIGURE 9.12
Sketch of electrostatics in MOSFET for different values of V_{DS}. (a) Linear regime: $V_{DS} < V_{DSsat}$; (b) pinch-off $V_{DS} = V_{DSsat}$; and (c) saturation regime $V_{DS} > V_{DSsat}$. At pinch-off and in saturation the electric field at the drain end of the channel pulls electrons into the drain.

[2]Transport through the pinch-off point in a MOSFET is just like through the base-collector depletion region of a bipolar transistor biased in the forward active regime, as will become clear in Chapter 11.

channel are not significantly affected by an increase in V_{DS} past $V_{GS} - V_T$. Hence, the channel current does not change from the value that it has at $V_{DS} = V_{GS} - V_T$. In other words, the drain current has *saturated* and the MOSFET is said to be in the *saturation regime*.

To further clarify the physics of the MOSFET around pinch-off, Fig. 9.13 sketches the inversion layer charge, the lateral electric field, the voltage, and the local gate overdrive along the channel in the linear regime, at pinch-off, and in saturation. The top diagram shows that at pinch-off the inversion charge reaches a minimum at the drain end of the channel. For higher values of V_{DS}, the pinch-off point widens and extends slightly into the channel. The third diagram from the top shows that all the extra voltage applied past pinch-off drops in the pinch-off region and that the voltage distribution in the rest of the channel is negligibly affected. This results in a negative gate overdrive in the pinch-off region at the drain end of the channel, as seen in the bottom diagram. The second diagram from the top shows a similar picture for the electric field. This figure shows that as V_{DS} initially increases beyond 0, the electric field at the source increases and the MOSFET current should increase. The rate of increase diminishes as more voltage is consumed on the drain end of the channel. When V_{DS} increases beyond V_{DSsat}, the electric field at the source does not increase anymore. This is responsible for drain current saturation. The electric field at the source is sketched on the right-hand side of Fig. 9.13.

Before we write an expression for the drain current in the saturation regime, we need to discuss several aspects of the pinch-off model that we have just introduced. Current saturation in the pinch-off model has been predicated upon the assumption that a depletion region forms at the drain end of the channel, that is, that the electron concentration drops below the acceptor concentration in the body. This assumption might be satisfied at low current levels, but it is certain to be violated if the current level is high enough. It turns out that the appearance of a depletion region is not an essential condition for current saturation. Electron velocity saturation at the drain end of the channel also causes it. Velocity saturation of electrons on the drain side of the channel is likely to occur as the electron concentration drops but current continuity must be maintained. We will understand in the next chapter how velocity saturation causes current saturation.

A second ingredient of current saturation is that the extent of the depletion region (or the velocity saturated region) is small in the scale of the channel length. This, of course, depends on the design of the MOSFET. For well-designed devices, this is always the case, but proving it requires a detailed model of the two-dimensional electrostatics of the drain end of the channel.

We can now derive a first-order expression for I_{Dsat}, the drain current in the saturation regime. Since we argued above that Eq. (9.16) predicts I_D with good accuracy nearly all the way to $V_{DS} = V_{GS} - V_T$ and I_D does not increase beyond this value of V_{DS}, then I_{Dsat} can be estimated by substituting $V_{DS} = V_{GS} - V_T$ in Eq. (9.16) to yield

$$I_{Dsat} \simeq \frac{W}{2L}\mu_e C_{ox}(V_{GS} - V_T)^2 \tag{9.21}$$

Figure 9.14 shows the so-called output characteristics of the MOSFET. In this graph, the drain current is graphed unchanged with V_{DS} for $V_{DS} \geq V_{GS} - V_T$. This is the manifestation of the saturation regime.

A key result that emerges in Eq. (9.21) is that in the saturation regime, the drain current increases with *the square* of the gate overdrive (the excess gate-to-source voltage above threshold). This is sketched in Fig. 9.15. The reason for the square-law dependence deserves some

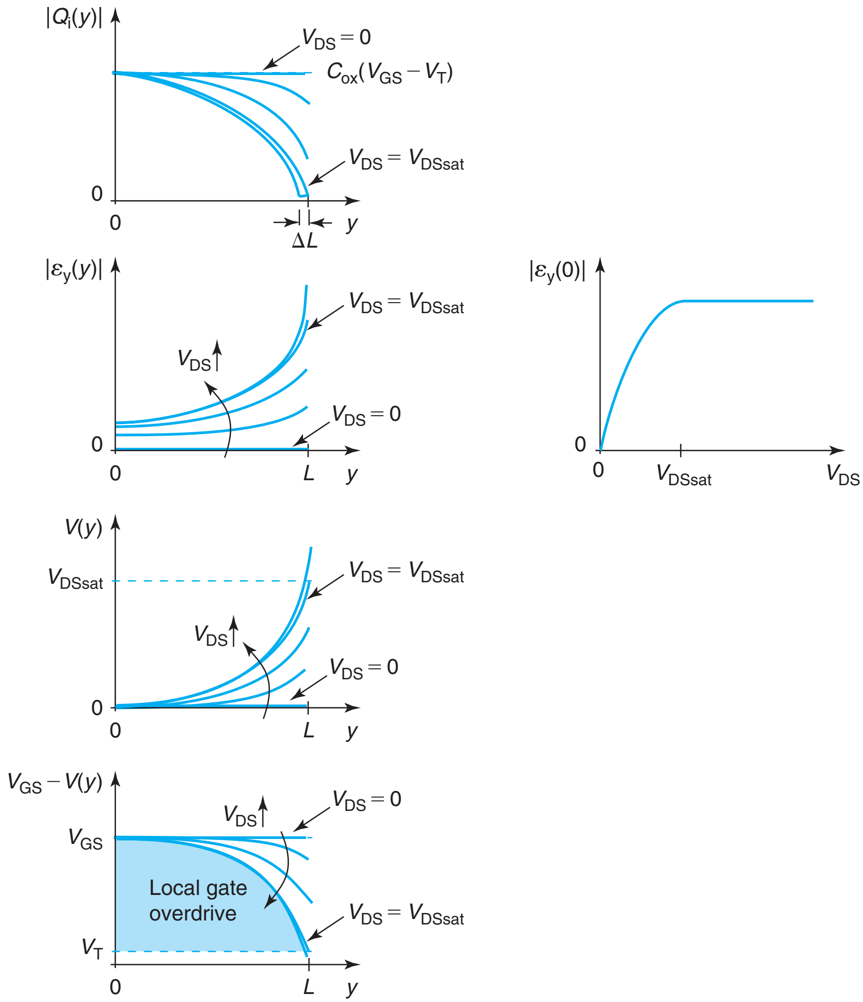


FIGURE 9.13
From top to bottom on the left: inversion layer charge, lateral electric field, channel voltage, and local gate overdrive as a function of location along the channel for a MOSFET for increasing values of V_{DS}. For $V_{DS} > V_{DSsat}$, the electrostatics of the channel are not significantly affected by the increase in V_{DS} and the channel current is saturated. On the right, electric field at the source as a function of V_{DS}. When $V_{DS} \geq V_{DSsat}$, the electric field at the source stops increasing. This is responsible for drain current saturation.

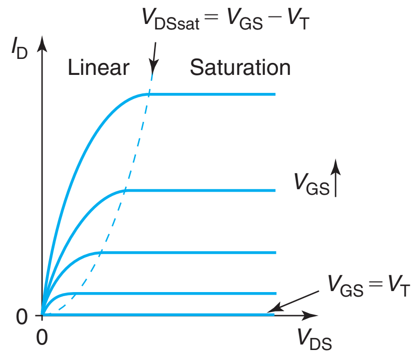


FIGURE 9.14
Sketch of output *I–V* characteristics of a MOSFET in the linear and saturation regimes.

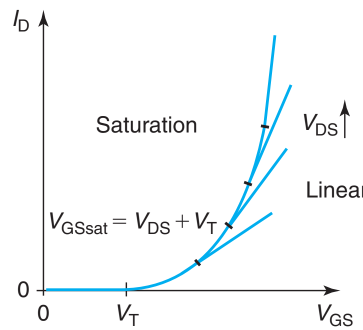


FIGURE 9.15
Sketch of transfer characteristics of a MOSFET , that is, I_D versus V_{GS} for different values of V_{DS}. In the saturation regime, I_D increases quadratically with gate overdrive. When V_{GS} exceeds $V_{DS} + V_T$, the transistor enters the linear regime and I_D rises linearly with gate overdrive.

thought. The higher V_{GS}, the higher the inversion charge in the channel that is available for transport. This is a linear dependence. Additionally, the higher V_{GS}, the higher the value of V_{DS} that can be applied before the channel is "pinched-off." As a consequence, when the MOSFET is saturated, the field inside the channel is also higher and the current is also larger. This is also a linear relationship. The combination of both linear dependences results in a square-law dependence for I_D on $V_{GS} - V_T$.

The value of V_{DS} required to pinch-off the device is commonly referred to as V_{DSsat}. In the simple formulation derived here, V_{DSsat} is given by

$$V_{DSsat} = V_{GS} - V_T \tag{9.22}$$

The locus of V_{DSsat} in the output characteristics is indicated in Fig. 9.14. As a result of the square-law dependence of I_{Dsat} on $V_{GS} - V_T$, this locus has a parabolic shape.

The I–V characteristics of a MOSFET when graphed as in Fig. 9.15 are known as the *transfer characteristics*. For a given value of V_{DS}, as V_{GS} increases above V_T, the transistor turns on directly into the saturation regime and the current rises quadratically with gate overdrive. As V_{GS} increases, eventually the transistor is pushed into the linear regime. This happens when $V_{GSsat} = V_{DS} + V_T$. Beyond this point, I_D rises linearly with gate overdrive.

Figure 9.16 shows the experimental output and transfer characteristics of a 2N7000 MOSFET. The behavior of this transistor is very much as discussed in this section.

EXERCISE 9.2

Consider the same long n-channel MOSFET as in Exercise 9.1 but now biased with $V_{GS} = 2.5$ V, $V_{DS} = 4$ V, *and* $V_{BS} = 0$. *Estimate the sheet charge density in the inversion layer at the source and drain ends of the channel. Estimate the current flowing through the drain of this transistor. Use* $\mu_e = 500\ \text{cm}^2/\text{V}\cdot\text{s}$.

This transistor is now biased in the saturation regime. This is because $V_{DS} = 4\ \text{V} > V_{DSsat} = V_{GS} - V_T = 1.94$ V.

The expression for the sheet charge density at the source end of the channel only depends on V_{GS}. Since the value of V_{GS} here is identical to that in Exercise 9.1, $Q_i(0)$ is also the same and given by $4.5 \times 10^{-7}\ \text{C/cm}^2$.

With the MOSFET biased in saturation, at the drain end of the channel there is a pinch-off point. Hence, the sheet charge density of electrons there is quite small and at this moment, we do not know how to estimate it. In Chapter 10, we will learn how to do this.

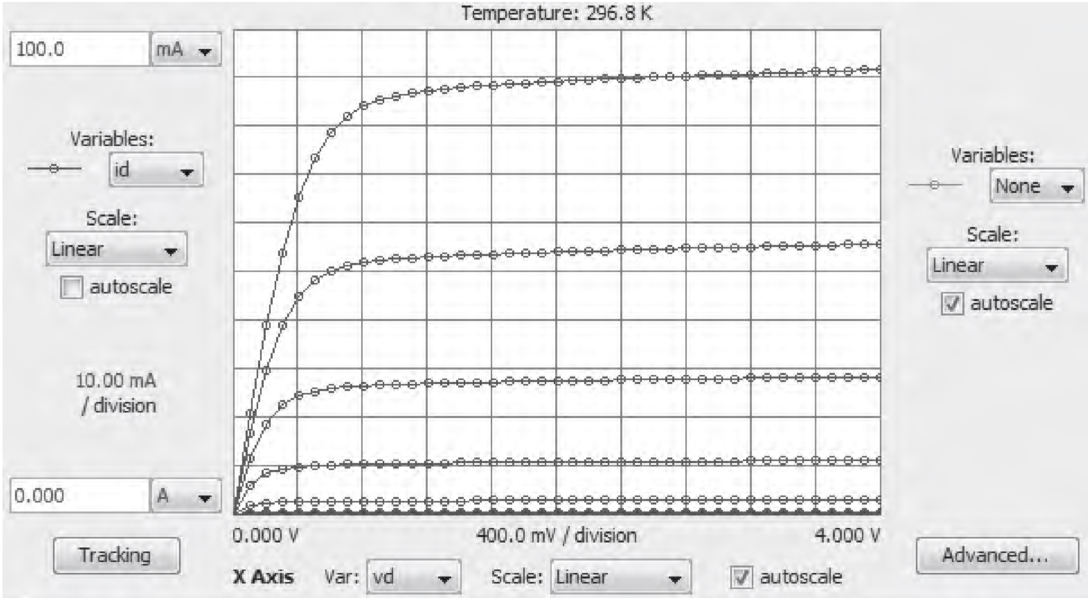
The drain current can be obtained from Eq. (9.21),

$$I_D = \frac{W}{2L}\mu_e C_{ox}(V_{GS} - V_T)^2 = \frac{10\ \mu\text{m}}{2 \times 1\ \mu\text{m}} 500\ \text{cm}^2/\text{V}\cdot\text{s} \times 2.3 \times 10^{-7}\ \text{F/cm}^2 (2.5 - 0.56)\text{V}^2$$
$$= 2.2\ \text{mA}$$

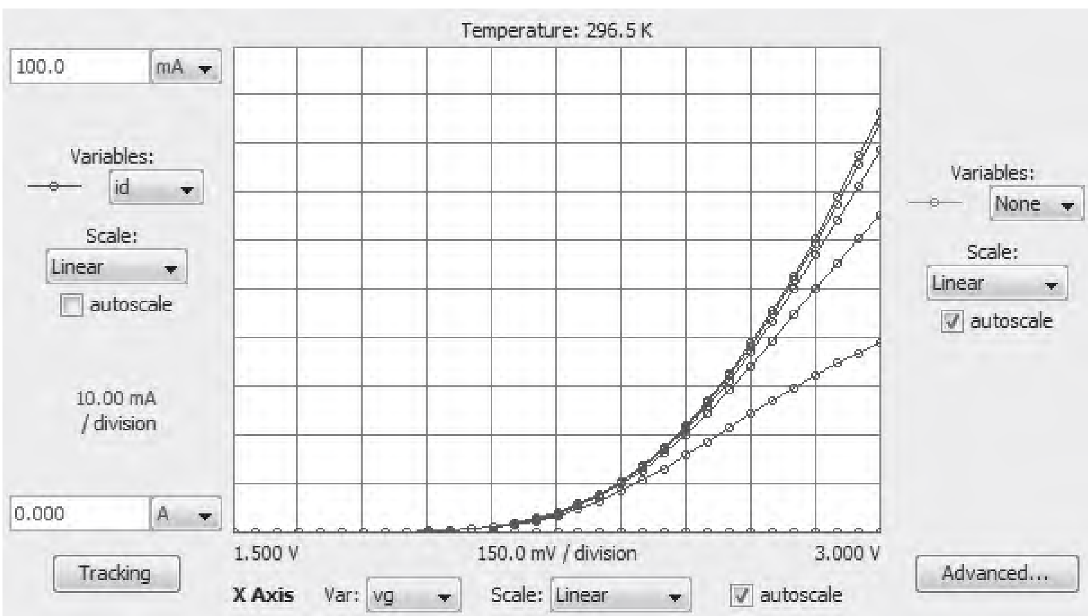
9.4.4 DC large-signal equivalent-circuit model of ideal MOSFET

An equivalent circuit model that captures the DC I–V characteristics of the ideal MOSFET is fairly straightforward. In the ideal MOSFET, the drain current also flows through the source and it is controlled by V_{DS} and V_{GS}. This behavior can be captured by means of a controlled current source. The dependence of the drain current on V_{DS} and V_{GS} is given by Eqs. (9.16) and (9.21) in the linear and saturation regimes, respectively.

In the ideal MOSFET, the gate does not draw any DC current. It is therefore an open circuit. Also, in the ideal MOSFET, we ignore the presence of the source/body and drain/body PN junctions. Hence, the body contact is also an open. The entire DC large-signal equivalent circuit model of the ideal MOSFET is graphed in Fig. 9.17. The diamond-shaped current source indicates that this is a dependent source that is controlled by V_{GS} and V_{DS}.



(a)



(b)

FIGURE 9.16
Measured *I*–*V* characteristics of a 2N7000 MOSFET. (a): Output characteristics with V_{GS} stepped from 1 V to 3 V in steps of 0.2 V. (b): Transfer characteristics with V_{DS} stepped from 0 to 1 V in steps of 0.2 V (Reprinted with permission from MIT Microelectronics iLab).

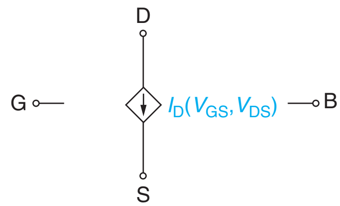


FIGURE 9.17
DC large-signal equivalent circuit model of ideal MOSFET.

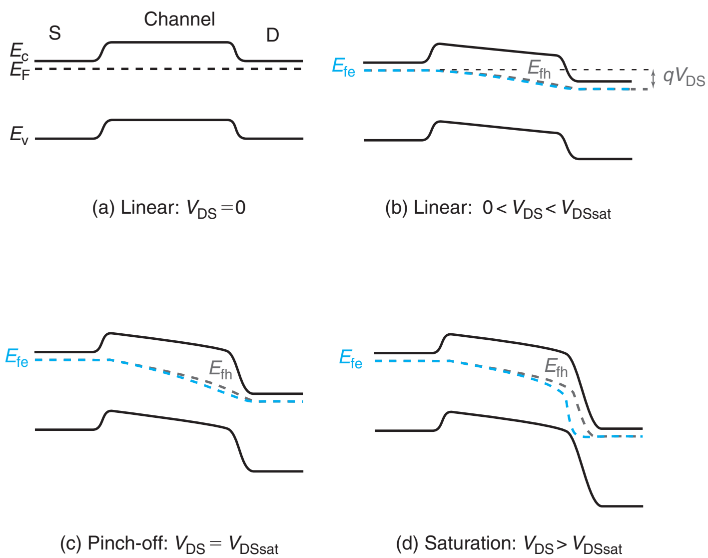


FIGURE 9.18
Energy band diagram along semiconductor surface from source to drain for four different values of V_{DS} (in all cases $V_{GS} > V_T$).

9.4.5 Energy band diagrams

Our understanding of the I–V characteristics of the MOSFET would not be complete without energy band diagrams. Figure 9.18 sketches energy band diagrams along the semiconductor/insulator interface for a value of V_{GS} in strong inversion and for several values of V_{DS}.

In the linear regime with $V_{DS} = 0$, the Fermi level is flat along the channel from the source to drain. The conduction band is close to the Fermi level in the source and drain but a bit further away in the channel. The bands are flat in the channel.

For a value of $V_{DS} > 0$ in the linear regime, the electron Fermi level is lowered at the drain with respect to the source by an amount qV_{DS}. Since the drain is heavily doped, this brings down with it the entire band structure on the drain. A lateral electric field appears in the channel and in consequence, the band structure bends along the channel. Current flows. This also means that the quasi-Fermi level for electrons also tilts along the channel.

At pinch-off, for $V_{DS} = V_{DSsat}$, the bending of the bands and the electron quasi-Fermi level along the channel is exacerbated. As one proceeds from source to drain along the channel, the bending increases, although for different reasons. E_c and E_v bend because the electric field increases toward the drain end of the channel. E_{fe} bends because the electron concentration drops toward the drain but the current is constant along the channel. In consequence, the gradient of E_{fe} must increase as we proceed from source to drain.

In saturation, for $V_{DS} > V_{DSsat}$, the band structure and electron quasi-Fermi level along the channel do not change with respect to the situation observed at $V_{DS} = V_{DSsat}$. The applied voltage beyond V_{DSsat}, that is, $V_{DS} - V_{DSsat}$, drops entirely at the pinch-off point where the bands and the quasi-Fermi level show a steep dive. It is clear from this diagram that the pinch-off point does not represent at all a bottleneck to the drain current. On the contrary, at the pinch-off point, the electrons are in a "free fall" down the channel to the drain.

In saturation, the bottleneck to the drain current is the energy barrier that electrons at the source have to overcome before they can enter the channel and the electric field at that same point. The energy barrier at the source end of the channel is controlled by V_{GS}. The higher V_{GS}, the lower the energy barrier and the higher the concentration of electrons that have enough energy to get injected into the channel. In particular, the height of this energy barrier is independent of what happens down the channel and hence the electron charge at $y = 0$ is independent of V_{DS}. The tilt of the bands at the source end of the channel determines the electric field there and the velocity at which electrons are extracted from the source. This is set by the potential distribution along the channel. This is unchanged for $V_{DS} > V_{DSsat}$, since all the extra voltage applied beyond V_{DSsat} drops at the pinch-off point.

Figure 9.18 also shows the quasi-Fermi level for holes along the channel at the different bias conditions. With $V_{DS} = 0$, both quasi-Fermi levels coincide. As V_{DS} increases, the quasi-Fermi levels split with $E_{fh} > E_{fe}$ toward the end of the channel. This is similar to the situation depicted on Fig. 8.30(b) in which a positive bias is applied between the inversion layer and the body, as will become clear later on in this chapter.

9.5 CHARGE–VOLTAGE CHARACTERISTICS OF THE IDEAL MOSFET

The I–V characteristics of a MOSFET, as is the case in most semiconductor devices, represent only a partial view of the behavior that is generally relevant in circuit applications. In particular, to capture the dynamics of a MOSFET, we need to complete the picture with the charge–voltage or capacitance–voltage characteristics. These describe the charge that is stored in the device and that must be provided to it or removed from it every time any terminal voltage changes.

In a MOSFET there are two types of stored charge. There is first the depletion charge in the depletion regions associated with the source-body and drain-body PN junctions as well as the depletion charge in the body of the semiconductor that is associated with the MOS structure itself. In addition to this, there is the electron charge stored in the inversion layer. If $V_{DS} > 0$, this charge is in transit from source to drain and results in current. Nevertheless, this is the charge that must have been provided once and must be removed if the transistor is to be switched off. We construct first-order models for these two types of stored charge in the following two subsections.

9.5.1 Depletion charge

In an ideal MOSFET, as sketched in Fig. 9.3, there are three distinct depletion regions. First, there is the depletion region associated with the source-body PN junction. Second, there is the depletion region associated with the drain-body PN junction. Finally, there is the depletion region associated with the MOS structure itself. The charge–voltage characteristics of these three depletion regions have been studied earlier in the previous chapters.

For the source-body and drain-body junctions, we can recycle results from Section 6.4. The charge associated with the source-body depletion region is given by

$$Q_{\mathrm{jS}} = L_{\mathrm{S}} W \sqrt{2qN_{\mathrm{A}}\epsilon_{\mathrm{s}}(\phi_{\mathrm{bij}} - V_{\mathrm{BS}})} = Q_{\mathrm{jSo}}\sqrt{1 - \frac{V_{\mathrm{BS}}}{\phi_{\mathrm{bij}}}} \tag{9.23}$$

where L_{S} is the extent of the source in the y-dimension and ϕ_{bij} the built-in potential of the source-body junction. Q_{jSo} is the depletion charge in thermal equilibrium,

$$Q_{\mathrm{jSo}} = L_{\mathrm{S}} W \sqrt{2qN_{\mathrm{A}}\epsilon_{\mathrm{s}}\phi_{\mathrm{bij}}} \tag{9.24}$$

In writing Eqs. (9.23) and (9.24), we have assumed that this is an asymmetric junction, or that the doping level of the source is much higher than that of the body. This is a very good assumption in practice.

Similarly, the depletion charge associated with the drain-body junction is given by

$$Q_{\mathrm{jD}} = L_{\mathrm{D}} W \sqrt{2qN_{\mathrm{A}}\epsilon_{\mathrm{s}}(\phi_{\mathrm{bij}} - V_{\mathrm{BD}})} = Q_{\mathrm{jDo}}\sqrt{1 - \frac{V_{\mathrm{BS}} - V_{\mathrm{DS}}}{\phi_{\mathrm{bij}}}} \tag{9.25}$$

with Q_{jDo} given by

$$Q_{\mathrm{jDo}} = L_{\mathrm{D}} W \sqrt{2qN_{\mathrm{A}}\epsilon_{\mathrm{s}}\phi_{\mathrm{bij}}} \tag{9.26}$$

This is equal to Q_{jSo} if $L_{\mathrm{S}} = L_{\mathrm{D}}$ as is commonly the case. We have assumed that the doping level on both junctions is the same and that the built-in potential is then identical.

The depletion region charge associated with the MOS structure depends on the regime of operation of the MOSFET. For $V_{\mathrm{GS}} < V_{\mathrm{T}}$, the MOSFET is in cut-off and there is no inversion layer anywhere in the channel. In this case, the charge in this depletion region was calculated in Eq. (8.20) and is given by

$$Q_{\mathrm{jB}} = LW\frac{1}{2}\gamma^2 C_{\mathrm{ox}}\left[\sqrt{1 + 4\frac{V_{\mathrm{GB}} - V_{\mathrm{FB}}}{\gamma^2}} - 1\right] = LW\frac{1}{2}\gamma^2 C_{\mathrm{ox}}\left[\sqrt{1 + 4\frac{V_{\mathrm{GS}} - V_{\mathrm{BS}} - V_{\mathrm{FB}}}{\gamma^2}} - 1\right] \tag{9.27}$$

where V_{FB} is the flatband voltage of the MOS structure. The depletion region charge increases from zero at $V_{\mathrm{GS}} = V_{\mathrm{FB}}$ to its maximum value when $V_{\mathrm{GS}} = V_{\mathrm{T}}$.

For gate voltages in excess of V_T, an inversion layer is formed under the gate and the depletion region thickness does not increase any more. Hence, for the MOSFET in the linear and saturation regimes, the charge associated with the depletion region in the channel is given by

$$Q_{jBmax} = LW\frac{1}{2}\gamma^2 C_{ox}\left[\sqrt{1+4\frac{V_T - V_{BS} - V_{FB}}{\gamma^2}} - 1\right] \tag{9.28}$$

When the voltage applied across any of the terminals of a MOSFET changes, the stored depletion charge also changes. This gives rise to capacitive effects. For the source-body and drain-body junctions, the junction capacitances is given, respectively, by:

$$C_{jS} = \frac{C_{jSo}}{\sqrt{1 - \frac{V_{BS}}{\phi_{bij}}}} \tag{9.29}$$

$$C_{jD} = \frac{C_{jDo}}{\sqrt{1 - \frac{V_{BS}-V_{DS}}{\phi_{bij}}}} \tag{9.30}$$

with

$$C_{jSo} = L_S W\sqrt{\frac{\epsilon_s q N_A}{2\phi_{bij}}} \tag{9.31}$$

$$C_{jDo} = L_D W\sqrt{\frac{\epsilon_s q N_A}{2\phi_{bij}}} \tag{9.32}$$

In the cut-off regime, the depletion region charge underneath the gate is modulated by V_{GB}. Hence, there is also a capacitive effect associated with this. This capacitance is given by

$$C_{jB} = \frac{LWC_{ox}}{\sqrt{1 + 4\frac{V_{GB}-V_{FB}}{\gamma^2}}} \tag{9.33}$$

This is a result already obtained in Eq. (8.50). It can also be easily obtained by differentiating Eq. (9.27) with respect to V_{GB}. This result applies when $V_{GS} < V_T$.

When $V_{GS} > V_T$, the MOSFET is in the linear or saturation regime and Q_{jB} becomes independent of V_{GB}. The associated capacitance is zero.

EXERCISE 9.3

Consider once again the long n-channel MOSFET of Exercises 9.1 and 9.2. The source and drain regions have a doping level $N_S = N_D = 10^{19}$ cm^{-3} *and a length* $L_S = L_D = 5$ μm. *Estimate the values of* C_{jS}, C_{jD}, *and* C_{jB} *at the following two bias points: (i)* $V_{GS} = 0$ V, $V_{DS} = 1$ V, *and* $V_{BS} = 0$ *and (ii)* $V_{GS} = 2.5$ V, $V_{DS} = 1$ V, *and* $V_{BS} = 0$.

At the first bias point, the transistor is in cut-off. We start by computing the capacitance per unit area of the source and drain junctions in equilibrium. These are identical and given by Eqs. (9.31) and (9.32). Proceeding as in Chapter 6, we start by calculating the built-in potential of these junctions which we find to be $\phi_{bij} = 0.94$. Then,

$$C_{\mathrm{jSo}} = C_{\mathrm{jDo}} = L_{\mathrm{S}} W \sqrt{\frac{\epsilon_{\mathrm{s}} q N_{\mathrm{A}}}{2\phi_{\mathrm{bij}}}} = L_{\mathrm{D}} W \sqrt{\frac{\epsilon_{\mathrm{s}} q N_{\mathrm{A}}}{2\phi_{\mathrm{bij}}}}$$

$$= 5 \times 10^{-4}\ \mathrm{cm} \times 10 \times 10^{-4}\ \mathrm{cm} \sqrt{\frac{1.04 \times 10^{-12}\ \mathrm{F/cm} \times 1.6 \times 10^{-19}\ \mathrm{C} \times 10^{17}\ \mathrm{cm}^{-3}}{2 \times 0.94\ \mathrm{V}}}$$

$$= 4.7 \times 10^{-14}\ \mathrm{F}$$

With $V_{\mathrm{BS}} = 0$, the capacitance of the source-body junction (Eq. 9.29) is given by the result just obtained,

$$C_{\mathrm{jS}} = C_{\mathrm{jSo}} = 4.7 \times 10^{-14}\ \mathrm{F}$$

There is a bias applied between the drain and the body. C_{jD} is then given by [Eq. (9.30)]

$$C_{\mathrm{jD}} = \frac{C_{\mathrm{jDo}}}{\sqrt{1 - \frac{V_{\mathrm{BS}} - V_{\mathrm{DS}}}{\phi_{\mathrm{bij}}}}} = \frac{4.7 \times 10^{-14}\ \mathrm{F}}{\sqrt{1 - \frac{0-1\ \mathrm{V}}{0.94\ \mathrm{V}}}} = 3.3 \times 10^{-14}\ \mathrm{F}$$

For the first bias point, the capacitance associated with the depletion region of the MOS structure is given by Eq. (9.33),

$$C_{\mathrm{jB}} = \frac{LWC_{\mathrm{ox}}}{\sqrt{1 + 4\frac{V_{\mathrm{GB}} - V_{\mathrm{FB}}}{\gamma^2}}} = \frac{1 \times 10^{-4}\ \mathrm{cm} \times 10 \times 10^{-4}\ \mathrm{cm} \times 2.3 \times 10^{-7}\ \mathrm{F/cm^2}}{\sqrt{1 + 4\dfrac{0 + 1\ \mathrm{V}}{0.79^2\ \mathrm{V}}}} = 1.8 \times 10^{-14}\ \mathrm{F}$$

At the second bias point, V_{GS} has increased to 2.5 V, which is above threshold. Hence, the MOSFET is now in the linear regime and an inversion layer forms. This breaks the electrostatic coupling between the gate and the body and $C_{\mathrm{jB}} = 0$. Additionally, V_{BS} and V_{BD} are unchanged from the first bias point, so C_{jS} and C_{jD} have the same values that were obtained at the first bias point.

9.5.2 Inversion charge

In addition to the depletion region charge, in a MOSFET in the linear or saturation regimes of operation, there is a certain amount of electron charge that is stored in the inversion layer. When turning on the MOSFET, this charge must be provided, and when turning it off, it has to be removed. The presence of inversion layer charge clearly impacts the dynamics of the MOSFET. In this section, we compute the integrated inversion layer charge for the ideal MOSFET. We also capture its dynamic behavior by means of two intrinsic capacitances.

We seek an expression for the integral of charge in the inversion layer of an ideal MOSFET. This can be expressed mathematically as the integral of Q_{i} along the entire channel,

$$Q_{\mathrm{I}} = W \int_0^L Q_{\mathrm{i}}(y)\,dy \tag{9.34}$$

Since we have derived an expression for Q_i in terms of V, rather than y, it is best to change variables in this integral and rewrite it as

$$Q_I = W \int_0^{V_{DS}} Q_i(V) \frac{dy}{dV} dV \tag{9.35}$$

The term in dy/dV can be obtained from Eq. (9.9),

$$\frac{dy}{dV} = -\frac{W\mu_e}{I_D} Q_i(V) \tag{9.36}$$

Combining this with Eq. (9.35), we get

$$Q_I = -\frac{W^2\mu_e}{I_D} \int_0^{V_{DS}} Q_i^2(V) dV \tag{9.37}$$

For the ideal MOSFET, $Q_i(V)$ was given in Eq. (9.11). Inserting this in Eq. (9.37) yields

$$Q_I = -\frac{W^2\mu_e C_{ox}^2}{I_D} \int_0^{V_{DS}} (V_{GS} - V - V_T)^2 dV = \frac{1}{3} W^2 \mu_e C_{ox}^2 \frac{(V_{GS} - V_{DS} - V_T)^3 - (V_{GS} - V_T)^3}{\frac{W}{L}\mu_e C_{ox}(V_{GS} - V_T - \frac{1}{2}V_{DS})V_{DS}} \tag{9.38}$$

After a relatively simple mathematical manipulation,[3] we can express Q_I as

$$Q_I = -\frac{2}{3} WLC_{ox} \frac{(V_{GS} - V_T)^2 + (V_{GS} - V_T)(V_{GD} - V_T) + (V_{GD} - V_T)^2}{(V_{GS} - V_T) + (V_{GD} - V_T)} \tag{9.39}$$

Note that since we have used Eq. (9.11), this expression is only valid in the linear regime. As was the case of the MOSFET current, Q_I in the saturation regime can be obtained from this by setting $V_{DS} = V_{DSsat} = V_{GS} - V_T$, or $V_{GD} = V_T$. This yields

$$Q_I = -\frac{2}{3} WLC_{ox}(V_{GS} - V_T) \tag{9.40}$$

This interesting and simple result can be best understood by obtaining Q_I also in the limit of small V_{DS}. From Eq. (9.39), we get

$$Q_I \simeq -WLC_{ox}(V_{GS} - V_T) \tag{9.41}$$

This result makes good sense. In the limit of small V_{DS}, at any one location in the inversion layer, the charge density is given by $-C_{ox}(V_{GS} - V_T)$. Hence, the total charge in the inversion layer can be obtained by multiplying this by the geometrical area of the gate.

---

[3]Note that

$$(V_{GS} - V_T - \frac{1}{2}V_{DS})V_{DS} = \frac{1}{2}[(V_{GS} - V_T)^2 - (V_{GD} - V_T)^2]$$

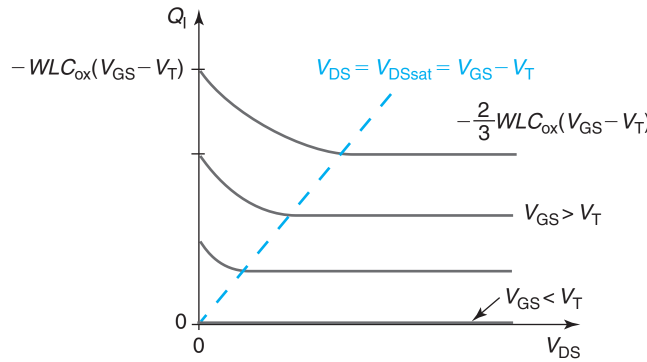


FIGURE 9.19
Inversion layer charge versus V_{DS} and V_{GS} for an ideal MOSFET in the linear or saturation regimes.

The $\frac{2}{3}$ factor that is present in the expression of the inversion layer charge in the saturation regime in Eq. (9.40) deserves some thinking. If we sketch the inversion charge versus V_{DS} for a given value of $V_{GS} > V_T$, as in Fig. 9.19, we see that as V_{DS} increases, Q_I monotonically decreases from the value given by Eq. (9.41) toward that given by Eq. (9.40). This is the consequence of the channel debiasing that was mentioned earlier. For values of $V_{GS} > V_T$, as V_{DS} increases, the inversion layer charge at any location in the channel, except at the source, gets reduced. This is most prominent toward the end of the channel. This was sketched in the top diagram of Fig. 9.11. Beyond $V_{DS} = V_{DSsat}$, the inversion layer charge profile along the channel does not change anymore and Q_I saturates. This is seen in the top diagram of Fig. 9.13. The factor of $\frac{2}{3}$ comes from the shape of the evolution of Q_i with y along the channel.

As we attempt to draw a large-signal equivalent circuit model for the MOSFET that accounts for the charge storage discussed in this section, we encounter an interesting problem. It is clear that the depletion charge storage associated with the source-body junction depletion region, Q_{jS}, hangs from the source and body terminals. Similarly, Q_{jD} hangs from the drain and body terminals. Additionally, the depletion charge associated with the MOS depletion region, Q_{jB}, hangs from the gate and body terminals. The interesting question is what to do about Q_I?

One of the terminals of Q_I is clear, it is the gate. The other terminal is less obvious, since the inversion layer is in contact with both the source and the drain. It then seems that an appropriate way to proceed is to split Q_I into two components: one that hangs between gate and source, and a second one that hangs between gate and drain. We call these Q_{IS} and Q_{ID}, respectively. It is clear that $Q_I = Q_{IS} + Q_{ID}$. It is not obvious how to compute Q_{IS} and Q_{ID}, individually. In fact, a detailed calculation of Q_{IS} and Q_{ID} is a bit involved. Fortunately, as it will become evident below, in many applications we do not need detailed models for Q_{IS} and Q_{ID}. It suffices to know the fact that they depend, in general, on both V_{GS} and V_{DS}.

The topology of the large-signal equivalent circuit model of the MOSFET incorporating all the storage elements discussed here is shown in Fig. 9.20.

An alternate way to describe the charge–voltage characteristics associated with the inversion layer is through the capacitance. In the case of a microelectronics device with more than two

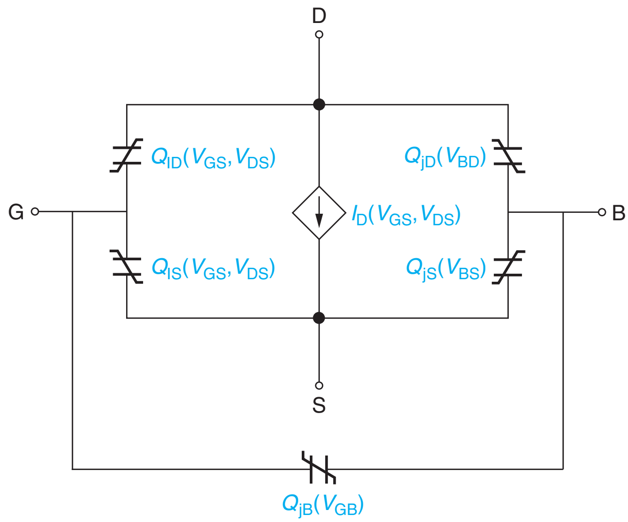


FIGURE 9.20
Large-signal equivalent circuit model for ideal MOSFET including depletion charge and inversion charge storage.

terminals, one has to exercise some caution when defining a capacitance and selecting the terminals across which it is to be placed. The best way to do this is to keep physical insight at all times and to look for the terminals that deliver the charge in both lobes of the charge dipole.

The inversion charge in an ideal MOSFET is supplied by the source and drain. The inversion layer charge is imaged at the gate. The gate charge is delivered by the gate contact. We should therefore expect two capacitors, one placed between the gate and the source and a second one between the gate and the drain. We will refer to these capacitors as C_{gsi} and C_{gdi}, where the i refers to the intrinsic device. C_{gsi} and C_{gdi} can be easily obtained by computing suitable derivatives of Q_{I} in Eq. (9.39) with respect to the appropriate terminal voltages, holding all other voltages constant,

$$C_{\mathrm{gsi}} = -\left.\frac{\partial Q_{\mathrm{I}}}{\partial V_{\mathrm{GS}}}\right|_{V_{\mathrm{GD}}} = \frac{1}{2}WLC_{\mathrm{ox}}(V_{\mathrm{GS}} - V_{\mathrm{T}})\frac{V_{\mathrm{GS}} - V_{\mathrm{T}} - \frac{2}{3}V_{\mathrm{DS}}}{(V_{\mathrm{GS}} - V_{\mathrm{T}} - \frac{1}{2}V_{\mathrm{DS}})^2} \tag{9.42}$$

$$C_{\mathrm{gdi}} = -\left.\frac{\partial Q_{\mathrm{I}}}{\partial V_{\mathrm{GD}}}\right|_{V_{\mathrm{GS}}} = \frac{1}{2}WLC_{\mathrm{ox}}(V_{\mathrm{GS}} - V_{\mathrm{DS}} - V_{\mathrm{T}})\frac{V_{\mathrm{GS}} - V_{\mathrm{T}} - \frac{1}{3}V_{\mathrm{DS}}}{(V_{\mathrm{GS}} - V_{\mathrm{T}} - \frac{1}{2}V_{\mathrm{DS}})^2} \tag{9.43}$$

As was the case of Eq. (9.39), these expressions only hold for the linear regime. For $V_{\mathrm{DS}} = V_{\mathrm{DSsat}}$, they become:

$$C_{\mathrm{gsi}} = \frac{2}{3}WLC_{\mathrm{ox}} \tag{9.44}$$

$$C_{\mathrm{gdi}} = 0 \tag{9.45}$$

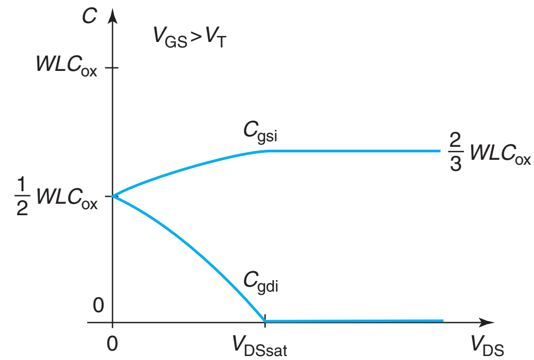


FIGURE 9.21

C_{gsi} **and** C_{gdi} **versus** V_{DS} **for an ideal MOSFET in the linear and saturation regimes for a given value of** $V_{GS} > V_T$**.**

which are the appropriate values for the saturation regime ($V_{DS} > V_{DSsat}$). For very small V_{DS}, it is interesting to see that

$$C_{gsi} \simeq C_{gdi} \simeq \frac{1}{2}WLC_{ox} \tag{9.46}$$

The evolution of C_{gsi} and C_{gdi} with V_{DS} is shown in Fig. 9.21. This behavior makes good physical sense. For small V_{DS}, we have a very symmetric situation where the charge in the inversion layer splits even between the source and the drain. As V_{DS} increases, the electrostatic influence of V_{GD} wanes, while that of V_{GS} strengthens. At $V_{DS} = V_{DSsat}$, the drain loses control of the inversion layer charge and $C_{gdi} = 0$. The channel has been pinched off. For higher values of V_{DS}, the picture does not change.

EXERCISE 9.4

Consider once again the long n-channel MOSFET of Exercises 9.1–9.3. Estimate the values of C_{gsi} *and* C_{gdi} *at the following three bias points: (i)* $V_{GS} = 2.5$ V, $V_{DS} = 0$ V, *and* $V_{BS} = 0$; *(ii)* $V_{GS} = 2.5$ V, $V_{DS} = 1$ V, *and* $V_{BS} = 0$; *and (iii)* $V_{GS} = 2.5$ V, $V_{DS} = 4$ V, *and* $V_{BS} = 0$.

We start by computing the geometrical capacitance of the gate. This is given by

$$C_g = LWC_{ox} = 1 \times 10^{-4}\ \text{cm} \times 10 \times 10^{-4}\ \text{cm} \times 2.3 \times 10^{-7}\ \text{F/cm}^2 = 2.3 \times 10^{-14}\ \text{F}$$

At the first bias point, the transistor is in the linear regime with $V_{DS} = 0$. Hence, the geometrical capacitance of the gate is split equally between the source and drain. Then,

$$C_{gsi} = C_{gdi} = \frac{1}{2}LWC_{ox} = 1.2 \times 10^{-14}\ \text{F}$$

At the second bias point, the MOSFET is in the linear regime. Hence, we have to use equations (9.42) and (9.43). For C_{gsi}, we have

$$
\begin{aligned}
C_{gsi} &= \frac{1}{2}WLC_{ox}(V_{GS}-V_T)\frac{V_{GS}-V_T-\frac{2}{3}V_{DS}}{(V_{GS}-V_T-\frac{1}{2}V_{DS})^2} \\
&= 1.2\times 10^{-14}\ \mathrm{F}(2.5-0.56\ \mathrm{V})\frac{2.5-0.56-\frac{2}{3}\times 1\ \mathrm{V}}{(2.5-0.56-\frac{1}{2}\times 1\ \mathrm{V})^2} = 1.4\times 10^{-14}\ \mathrm{F}
\end{aligned}
$$

Proceeding in a similar way with Eq. (9.43), we obtain $C_{gdi} = 8.8\times 10^{-15}$ F.

At the third bias point, the MOSFET is in saturation. Hence, according to Eq. (9.44), C_{gsi} is given by

$$
C_{gsi} = \frac{2}{3}WLC_{ox} = \frac{2}{3}2.3\times 10^{-14}\ \mathrm{F} = 1.5\times 10^{-14}\ \mathrm{F}
$$

And $C_{gdi} = 0$ [Eq. (9.45)].

9.6 SMALL-SIGNAL BEHAVIOR OF IDEAL MOSFET

In analog and mixed-signal applications, MOSFETs are often used in the small-signal mode. In these situations, the device is biased by means of DC sources in a certain regime of operation, typically saturation, and then a small signal is applied to one or more terminals. There is interest in evaluating the response of the device to the small signals alone. In a general way, we can consider these kinds of situations in Fig. 9.22(a). This figure shows a MOSFET biased through three DC voltage sources, V_{GS}, V_{DS}, and V_{BS}, referred to the source. In addition, three small-signal sources: $v_{gs}(t)$, $v_{ds}(t)$, and $v_{bs}(t)$, are also applied as shown. In general, we are interested in the time-dependent drain current, $i_D(t)$, that results.

The equivalent circuit model that we developed in the previous section can certainly be used to solve this problem. This is illustrated in Fig. 9.22. In this approach, the transistor is substituted by its equivalent circuit model and standard circuit solving techniques are used to determine $i_D(t)$. However, since the model elements are nonlinear, this approach can become quite tedious. Fortunately, many circuit simulation tools, such as SPICE, are available to solve the problem this way. In this approach, actual waveforms are introduced to a device model and the output waveforms are computed. A problem with this approach is that it is hard to develop an intuitive understanding for the connection among the figures of merit of interest, such as amplitude and phase of the drain current waveform, and the bias point or the device parameters. Fortunately, for a class of problems referred to as "small-signal problems," there is a more expeditious and intuitive approach.

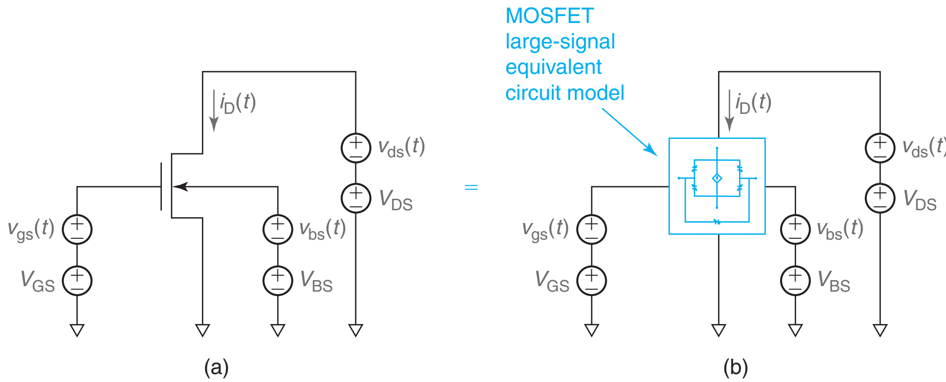


FIGURE 9.22
Illustration showing how a mixed-signal situation can be studied using the large-signal equivalent circuit model developed in the previous section.

9.6.1 Small-signal equivalent circuit model of ideal MOSFET

If the amplitude of the small signals is small enough, superposition suggests that the time-dependent response of the device, $i_D(t)$, should be the sum of the DC bias current I_D that is imposed by the DC voltage sources plus a small time-dependent component $i_d(t)$. This is sketched in Fig. 9.23. Mathematically, this can be expressed as

$$i_D(V_{GS}, V_{DS}, V_{BS}; v_{gs}, v_{ds}, v_{bs}) \simeq I_D(V_{GS}, V_{DS}, V_{BS}) + i_d(V_{GS}, V_{DS}, V_{BS}; v_{gs}, v_{ds}, v_{bs}) \qquad (9.47)$$

Note how the small-signal response in general depends on the bias point that is selected.

The small-signal equivalent circuit model of the MOSFET is a circuit representation of the set of dependencies of the small-signal current $i_d(t)$ on the small-signal voltages v_{gs}, v_{ds}, and v_{bs}. This is the blue box figure in Fig. 9.23(c). Since the small-signal sources are of small magnitude, the nonlinear behavior of the MOSFET gets linearized. Linearity then allows us to express $i_d(t)$ as the sum of three independent terms in each of the small-signal voltage sources,

$$\begin{aligned} &i_d(V_{GS}, V_{DS}, V_{BS}; v_{gs}, v_{ds}, v_{bs}) \\ &\quad = i_d(V_{GS}, V_{DS}, V_{BS}; v_{gs}) + i_d(V_{GS}, V_{DS}, V_{BS}; v_{ds}) + i_d(V_{GS}, V_{DS}, V_{BS}; v_{bs}) \end{aligned} \qquad (9.48)$$

Depending on the complexity of the small-signal equivalent circuit model, this approach to solving the problem can be fairly fast and yield very meaningful analytical results. So the key question is: what is a suitable small-signal equivalent circuit model for the MOSFET? Since we already have a large-signal equivalent circuit model (Fig. 9.20), this requires only that we linearize it. This is fairly straightforward. In the first step, the charge storage elements become simple capacitors. Their expressions were already derived in the previous section. We are then left to deal with the nonlinear voltage-controlled current source.

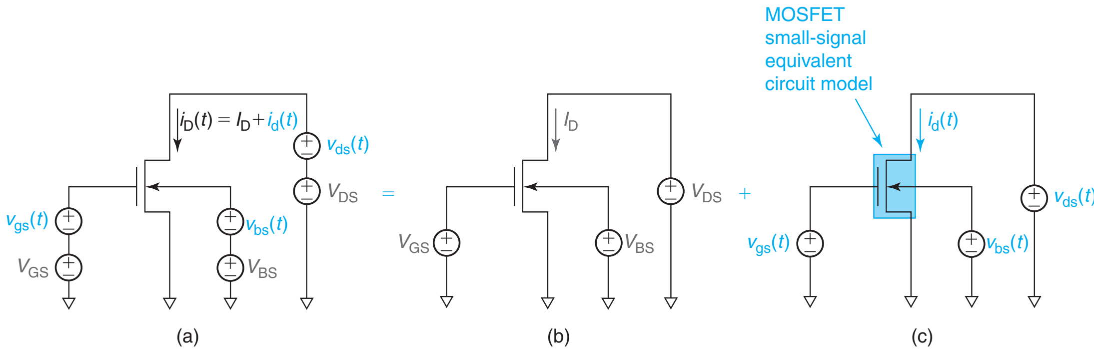


FIGURE 9.23
Illustration of superposition as a way to break a mixed-signal situation (a) into a DC bias situation (b) plus a small-signal situation (c).

In a most general way, linearizing I_D demands that we take the linear terms of its Taylor series expansion,

$$\begin{aligned} &i_D(V_{GS}, V_{DS}, V_{BS}; v_{gs}, v_{ds}, v_{bs}) \\ &\simeq I_D(V_{GS}, V_{DS}, V_{BS}) + \left(\frac{\partial I_D}{\partial V_{GS}}\bigg|_{V_{DS},V_{BS}} \right) v_{gs} + \left(\frac{\partial I_D}{\partial V_{DS}}\bigg|_{V_{GS},V_{BS}} \right) v_{ds} + \left(\frac{\partial I_D}{\partial V_{BS}}\bigg|_{V_{GS},V_{DS}} \right) v_{bs} \\ &= I_D + g_m v_{gs} + g_o v_{ds} + g_{mb} v_{bs} \end{aligned} \tag{9.49}$$

The coefficients of v_{gs}, v_{ds}, and v_{bs} receive the names of *transconductance*, *output conductance* (also called *drain conductance*), and *back transconductance*, respectively. In the ideal MOSFET, I_D does not depend on V_{BS}. Hence, g_{mb} is zero.[4] There are therefore two terms left in i_d. How do they get captured in the small-signal equivalent circuit model? The first term gives us a sense of remote control. It states that in response to a small wiggle in the gate voltage, the drain current changes a bit. The current flowing between the drain and the source is then controlled by the gate-to-source voltage. This is, of course, the transistor effect. To represent this, we need a voltage-controlled current source. In the second term, the drain-to-source current is modulated by v_{ds}, which is the voltage across the same two terminals. In the ideal MOSFET, this is the case in the linear regime, though this does not happen in the saturation regime. This effect, nevertheless, can be captured by means of a simple resistor of value $1/g_o$, or of a conductance equal to g_o. All together, the small-signal equivalent circuit model for the MOSFET is shown in Fig. 9.24.

We derived the above expressions for all the capacitors in Fig. 9.24. We now derive expressions for g_m and g_o for the MOSFET operating in the various regimes.

By definition, the transconductance g_m is

$$g_m = \frac{\partial I_D}{\partial V_{GS}}\bigg|_{V_{DS},V_{BS}} \tag{9.50}$$

[4]In Section 9.7.2, we study the impact of applying a voltage to the MOSFET substrate and we will derive a first-order expression for g_{mb}.

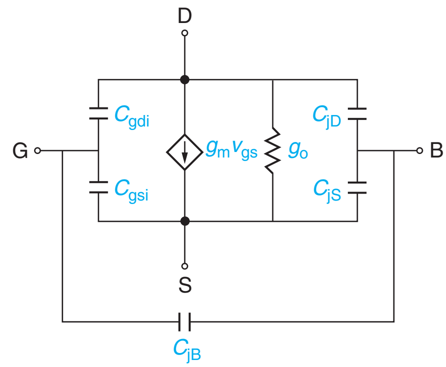


FIGURE 9.24
Small-signal equivalent circuit model of an ideal MOSFET.

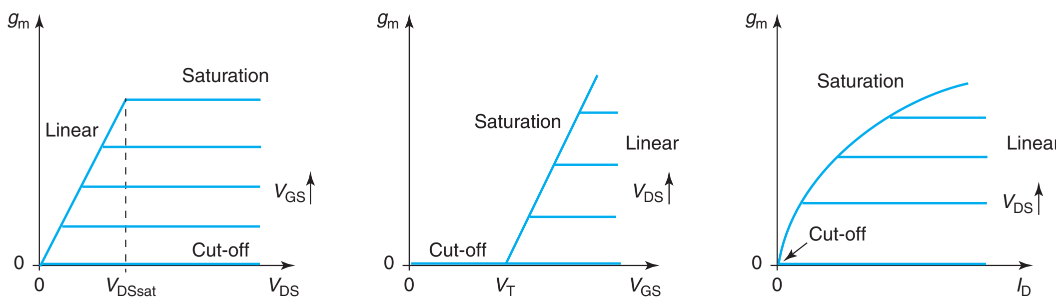


FIGURE 9.25
Evolution of g_m with V_{GS} and V_{DS} in ideal MOSFET plotted in various ways.

In cut-off, $I_D = 0$ and $g_m = 0$. In the linear regime, we go back to Eq. (9.16) and get

$$g_m = \frac{W}{L}\mu_e C_{ox} V_{DS} \tag{9.51}$$

In the saturation regime, we go back to Eq. (9.21), and get

$$g_m = \frac{W}{L}\mu_e C_{ox}(V_{GS} - V_T) = \sqrt{2\frac{W}{L}\mu_e C_{ox} I_D} \tag{9.52}$$

Figure 9.25 shows the evolution of g_m with the bias point graphed in various ways. In the linear regime, g_m is independent of V_{GS} and linearly dependent on V_{DS}. For a fixed V_{DS}, g_m in the linear regime is independent of I_D. In the saturation regime, g_m is independent of V_{DS} and is linearly proportional to V_{GS}. Hence, for a fixed V_{DS}, g_m follows a square-root evolution with I_D.

Now we turn our attention to g_o. The output conductance is defined as

$$g_o = \left.\frac{\partial I_D}{\partial V_{DS}}\right|_{V_{GS}, V_{BS}} \tag{9.53}$$

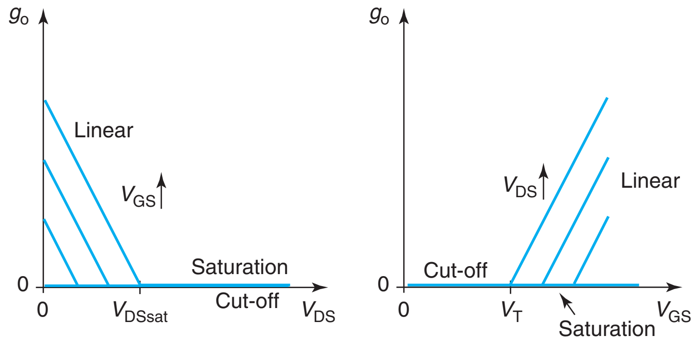


FIGURE 9.26
Evolution of g_o with V_{GS} and V_{DS} in ideal MOSFET.

In cut-off, $I_D = 0$ and $g_o = 0$. In the linear regime, we go back to Eq. (9.16) again and get

$$g_o = \frac{W}{L}\mu_e C_{ox}(V_{GS} - V_T - V_{DS}) \tag{9.54}$$

In the saturation regime, the device is pinched off and I_D is independent of V_{DS}. Then,

$$g_o = 0 \tag{9.55}$$

The evolution of g_o with V_{DS} and V_{GS} is shown in Fig. 9.26. In the ideal MOSFET, g_o is finite in the linear regime where it increases linearly with V_{GS} and decreases linearly with V_{DS} with an identical slope.

9.6.2 Short-circuit current-gain cut-off frequency, f_T, of ideal MOSFET in saturation

The short-circuit current-gain cut-off frequency, f_T, of a MOSFET in saturation is a figure of merit that is widely used to assess the suitability of MOSFETs for high-frequency mixed-signal applications. f_T is a widely quoted figure of merit that is indicative of electron transport through the intrinsic device. In this section, we present a definition of f_T, we derive an expression for it for the ideal MOSFET, and we provide a physically intuitive picture for the meaning of f_T.

f_T is defined with the MOSFET biased as shown in Fig. 9.27. Voltage sources on the gate and the drain are selected to bias the device in the saturation regime. The body is shorted out to the source (though this is not strictly necessary). A small-signal voltage source is applied on top of the gate bias. We are interested in evaluating the small-signal *current gain* of the MOSFET, i_d/i_g, in these conditions. This is often referred to as h_{21}.

In DC, an ideal MOSFET does not draw gate current. As a result, its DC current gain is infinite. If we now wiggle the gate voltage up and down a little bit, the drain current will follow but, in addition, a displacement current flows into the gate to charge and discharge all the capacitors that hang from the gate terminal. This now results in finite current gain. The magnitude of the gate current increases as the frequency of the input signal increases, and in consequence, the current gain decreases. At some frequency, the current gain goes to unity. This is f_T. f_T therefore

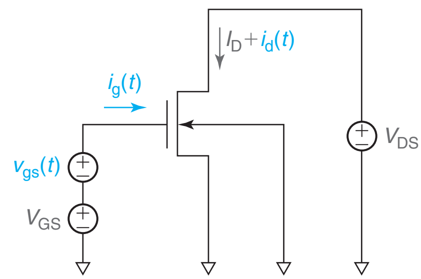


FIGURE 9.27
Circuit configuration for the evaluation of f_T in a MOSFET.

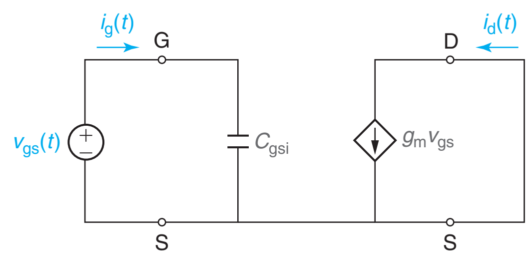


FIGURE 9.28
Small-signal equivalent circuit model for the computation of f_T in an ideal MOSFET.

provides an assessment of the high-frequency suitability of MOSFET and, in general, transistors. It is a widely watched and quoted figure of merit. Mathematically, then we have

$$|h_{21}(f_T)| = \left|\frac{i_d}{i_g}\right|_{v_{ds}=0} = 1 \tag{9.56}$$

Deriving an expression for f_T for the ideal MOSFET is fairly straightforward. The first step is to draw a small-signal equivalent circuit model for the circuit configuration. This is shown in Fig. 9.28. In this circuit, the MOSFET is in saturation. Hence, $C_{jB} = C_{gdi} = 0$. Also, $g_o = 0$. Since the body is shorted to the source, C_{jS} is shorted out. Also, from a small-signal point of view, C_{jD} is also shorted out since it is connected to a DC voltage source.

With $v_{gs}(t)$ applied at the input of this circuit, the gate and drain currents are given by:

$$i_g = j\omega C_{gsi} v_{gs} \tag{9.57}$$

$$i_d = g_m v_{gs} \tag{9.58}$$

h_{21} is then given by

$$h_{21} = \frac{g_m}{j\omega C_{gsi}} \tag{9.59}$$

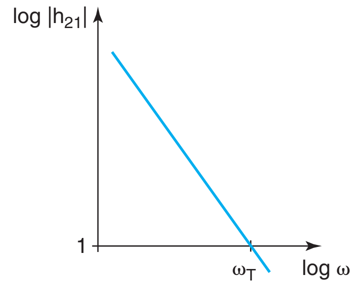


FIGURE 9.29
Bode plot of magnitude of $|h_{21}|$ versus frequency indicating ω_T.

The magnitude of h_{21} is then

$$|h_{21}| = \frac{g_m}{\omega C_{gsi}} \tag{9.60}$$

The evolution of $|h_{21}|$ with frequency is sketched in a Bode plot in Fig. 9.29. As expected, $|h_{21}|$ decreases with frequency. At a certain frequency, $|h_{21}| = 1$. The frequency at which this happens is given by

$$f_T = \frac{g_m}{2\pi C_{gsi}} \tag{9.61}$$

This equation makes good physical sense. The higher g_m, the higher the drain current that results in response to a wiggle in v_{gs} at the input and hence the higher the frequency at which the transistor exhibits current gain. On the other hand, the higher C_{gsi}, all things being equal, the higher the gate current that the device draws, and the lower the frequency up to which the transistor offers current gain.

Let us discuss now the bias dependence of f_T. With an ideal MOSFET in saturation, we know that C_{gsi} is completely independent of the bias point. On the other hand, g_m increases linearly with V_{GS} and is independent of V_{DS}. Hence, f_T increases linearly with V_{GS} above threshold. f_T is also independent of V_{DS}. This can be seen if we substitute Eqs. (9.44) and (9.52) into Eq. (9.61) to get

$$f_T = \frac{1}{2\pi}\frac{3}{2}\frac{\mu_e(V_{GS} - V_T)}{L^2} \tag{9.62}$$

This is sketched in Fig. 9.30.

Equation (9.62) also gives a sense of the materials and structural parameters of the MOSFET that affect f_T. We note that f_T is linearly dependent on the electron mobility in the channel. This makes good sense as, all things being equal, a higher value of μ_e results in more drain current. f_T depends inversely on the *square* of the gate length. This is also reasonable since the shorter the gate length, the more drain current the device produces and the smaller the input capacitance.

It is also of interest to reflect on the dependences that are *not* seen in Eq. (9.62). First, f_T is independent of W. This is because both C_{gsi} and g_m are linearly proportional to the gate width of the MOSFET. Note also that f_T is independent of C_{ox} or x_{ox}. This is again because both C_{gsi} and g_m have the same linear dependence on the capacitance of the MOS structure per unit area.

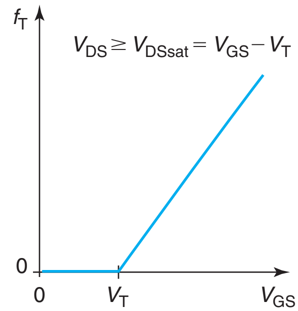


FIGURE 9.30
In a MOSFET in saturation, f_T is linearly dependent on V_{GS} and is independent of V_{DS}.

An intuitive picture of the physical meaning of f_T can be drawn from the realization that f_T has the units of inverse time. We can then define

$$\tau_d = \frac{1}{2\pi f_T} = \frac{2}{3}\frac{L^2}{\mu_e(V_{GS} - V_T)} \tag{9.63}$$

τ_d is called the *delay time*. What is the physical meaning of τ_d?

We can understand this by asking the following question. How much time does it take for a perturbation applied to the gate of a MOSFET to propagate into the drain current?

Consider an ideal MOSFET biased in saturation with a certain gate voltage V_{GS}. Now consider that at a certain instant, $t = 0$, the bias of the gate is suddenly increased by a small amount ΔV_{GS}. The drain current at $t = 0^-$ is $I_D(0^-) = \frac{W}{2L}\mu_e C_{ox}(V_{GS} - V_T)^2$. We know that, given enough time, the drain current will increase by an amount $\Delta I_D = \frac{W}{L}\mu_e C_{ox}(V_{GS} - V_T)\Delta V_{GS}$. The question is how much time does it take for this to happen?

At $t = 0^+$, the source current suddenly increases by an amount $\Delta I_S = -\frac{W}{L}\mu_e C_{ox}(V_{GS} - V_T)\Delta V_{GS}$. This change is instantaneous. This is because the energy barrier that controls the entrance of electrons from the source into the channel gets instantaneously reduced by the change in V_{GS}. However, before this pulse of current shows up at the drain terminal, the inversion layer needs to be "charged up." At $t = 0^-$, we have $Q_I(0^-) = -\frac{2}{3}WLC_{ox}(V_{GS} - V_T)$. By the time the step in current reaches the drain, the inversion layer charge must have increased by an amount $\Delta Q_I = -\frac{2}{3}WLC_{ox}\Delta V_{GS}$. Since it is the source that is providing these electrons, the time that it takes for this charge to be delivered is

$$\tau_d = \frac{\Delta Q_I}{\Delta I_S} = \frac{2}{3}\frac{L^2}{\mu_e(V_{GS} - V_T)} \tag{9.64}$$

It is only after this time has passed and the inversion layer has been fully charged up that the drain current increases. Note how this equation is identical to Eq. (9.63).

The delay time is then the time that it takes to charge up the inversion layer after a perturbation has been applied to the gate of a MOSFET. This delay sets the maximum frequency at which the MOSFET exhibits current gain. Notice that the delay time is different from the *transit* *time*. We can define this as the average time that it takes for an electron to cross from source to drain. The transit time in a MOSFET can be calculated as

$$\tau_t = \int_0^L \frac{dy}{v_e(y)} = \frac{4}{3}\frac{L^2}{\mu_e(V_{GS} - V_T)} \tag{9.65}$$

This result can be easily derived (see Problem 9.4) once an expression for $v_e(y)$ along the channel in saturation is obtained. Notice how $\tau_t > \tau_d$. The longer transit time is due to the fact that in charging up the channel, not all electrons have to travel all the way from source to drain. Most of them end up stored at intermediate positions in the channel.

EXERCISE 9.5

Consider once again the long n-channel MOSFET of Exercises 9.1–9.4. Estimate g_m and f_T at a bias point in saturation for which $I_D = 2$ mA.

For g_m, we can directly use Eq. (9.52),

$$g_m = \sqrt{2\frac{W}{L}\mu_e C_{ox} I_D} = \sqrt{2\frac{10\ \mu\text{m}}{1\ \mu\text{m}} 500\ \text{cm}^2/\text{V}\cdot\text{s} \times 2.3\times10^{-7}\ \text{F/cm}^2 \times 2\times10^{-3}\ \text{A}} = 2.1\ \text{mS}$$

For f_T, we could use Eq. (9.61),

$$f_T = \frac{g_m}{2\pi C_{gsi}} = \frac{1}{2\pi}\frac{2.1\times10^{-3}\ \text{A/V}}{1.5\times10^{-14}\ \text{F}} = 22\ \text{GHz}$$

In this, we have used our estimate of C_{gsi} from Exercise 9.4.

9.7 NONIDEAL EFFECTS IN MOSFET

We have completed a first pass through the current–voltage and charge–voltage characteristics of the ideal MOSFET. We have obtained sound physical understanding about its operation and we have developed simple models for its behavior. We are now in an excellent position to evaluate some of the simplifying assumptions that were made in the definition of the "ideal" MOSFET, understand the physics behind a few important "nonideal" effects, and introduce corrections to the models to account for them.

9.7.1 Body effect

The body effect and the impact of back bias were briefly mentioned in Chapter 8. The underlying physics of both of these effects is identical and was explained in Section 8.7. In essence, in a MOS structure when a voltage is applied to the inversion layer with respect to the body underneath, the threshold voltage of the structure is modified. For an electron inversion layer on a p-type substrate, the application of a positive voltage to the inversion layer with respect to the body results in an increase of the threshold voltage. The body effect and the role of back bias are both manifestations of this piece of physics in a MOSFET.

Let us consider first the *body effect*. Consider a MOSFET biased in the linear regime, as depicted in Fig. 9.8, with the body tied up to the source. As we discussed, there is a voltage drop along the channel from source to drain that we labeled $V(y)$. This was illustrated in Figs. 9.11 and 9.13. A consequence of the appearance of $V(y)$ is that there is a channel-to-body voltage difference. Since the body is tied up to the source, this voltage difference is precisely $V(y)$.

On the source side of the channel, the channel-to-body voltage remains always at zero. However, as we advance toward the drain, it increases and it becomes V_{DS} at the drain end of the channel. This rise in channel voltage is what produces the lateral electric field that makes the electrons drift from source to drain. A consequence of the nonzero channel to body voltage is that the *local* threshold voltage in the inversion layer actually depends on channel location. Since V rises along the inversion layer from source to drain, the local V_T shifts positive along the way.

There are obvious consequences to this. At any one location other than the source end of the channel, the inversion layer charge will be smaller in absolute magnitude than in the ideal MOSFET. This is sketched in Fig. 9.31 that depicts the voltage drop along the channel of a MOSFET in the linear regime as well as the gate overdrive with and without the body effect. It is clear from this figure that the positive V_T shift that takes place along the channel reduces the gate overdrive throughout the device. The main consequence of this is a reduction in the current drivability of the MOSFET. Refining the ideal MOSFET model to account for the body effect is rather straightforward.

The starting point is Eq. (9.9) that describes the drift current in the channel of a MOSFET. The inversion layer charge at any one location in the channel is given by expression (9.11). This expression should explicitly acknowledge the fact that V_T actually depends on position. We rewrite it here:

$$Q_i(y) \simeq -C_{ox}[V_{GS} - V(y) - V_T(y)] \tag{9.66}$$

Now, we need to find a way to describe the spatial dependence of V_T. In Chapter 8, we studied the effect of applying a voltage to the inversion layer with respect to the body in a MOS structure. We argued there that this produced a change in the threshold voltage (referred from gate to source) that was given by Eq. (8.69). This is precisely the same situation that we have here. At a location y, there is a potential difference between the inversion layer in a MOSFET and the body. Hence, the local value of V_T will also follow Eq. (8.69). Adapted to the notation used in the present chapter, this equation can be rewritten as

$$V_T(V) = V_{To} + \gamma(\sqrt{\phi_{sT} + V} - \sqrt{\phi_{sT}}) \tag{9.67}$$

where V_{To} refers to the threshold voltage of the two-terminal MOS structure for $V_{SB} = 0$.

Inserting this equation into Eq. (9.66) and the result into Eq. (9.9), we get

$$I_e = W\mu_e C_{ox}\left[V_{GS} - V - V_{To} - \gamma(\sqrt{\phi_{sT} + V} - \sqrt{\phi_{sT}})\right]\frac{dV}{dy} \tag{9.68}$$

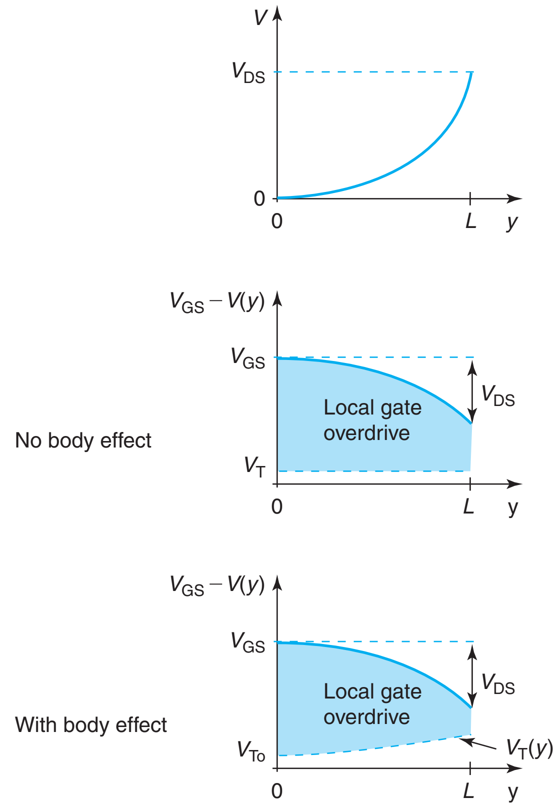


FIGURE 9.31
From top to botton, voltage, gate overdrive in the absence of body effect, and gate overdrive with body effect for a MOSFET in the linear regime.

This differential equation can be integrated in an identical way as Eq. (9.12). The resulting expression for the drain current is

$$I_D = \frac{W}{L}\mu_e C_{ox}\left\{\left(V_{GS} - V_{To} + \gamma\sqrt{\phi_{sT}} - \frac{1}{2}V_{DS}\right)V_{DS} - \frac{2}{3}\gamma[(\phi_{sT} + V_{DS})^{3/2} - (\phi_{sT})^{3/2}]\right\} \tag{9.69}$$

If we compare this expression for the drain current with the one derived earlier in Eq. (9.16), we see three new terms that are multiplied by the body parameter γ. Clearly, if γ goes to zero, we revert back to Eq. (9.16). This is to be expected from our presentation in Chapter 8 where we saw that in the limit of $\gamma = 0$, the impact of the back bias in the electrostatics of the MOS structure becomes negligible.

It might not be clear just by looking at Eq. (9.69) that the aggregate effect of all these new terms is to *reduce* the drain current from the ideal value given in Eq. (9.16), but that is what happens. The magnitude of the reduction gets worse as γ increases. In spite of this, the basic behavior of the drain current with V_{DS} and V_{GS} is not seriously affected by the body effect. In particular, channel pinch-off and drain current saturation take place just as before except that the final expression of the saturated drain current and the value of V_{DSsat} are modified.

To get the new expression for V_{DSsat}, we follow an identical procedure as for the ideal MOSFET. We need to look at the expression for the inversion layer charge at $y = L$, and let that go to zero. From Eqs. (9.17) and (9.67), we have

$$Q_i(y = L) = -C_{ox}\left[V_{GS} - V_{DSsat} - V_{To} - \gamma\left(\sqrt{\phi_{sT} + V_{DSsat}} - \sqrt{\phi_{sT}}\right)\right] = 0 \tag{9.70}$$

Solving for V_{DSsat}, we find

$$V_{DSsat} = V_{GS} - V_{To} + \gamma\sqrt{\phi_{sT}} - \frac{\gamma^2}{2}\left[\sqrt{1 + \frac{4}{\gamma^2}(V_{GS} - V_{FB})} - 1\right] \tag{9.71}$$

where we have also used Eq. (8.28). As shown below, the body effect reduces the value of V_{DSsat}. In the limit of $\gamma = 0$, V_{DSsat} reverts to $V_{GS} - V_{To}$, as one should expect.

An expression for the saturated drain current can be obtained by substituting V_{DSsat} for V_{DS} in Eq. (9.69),

$$I_{Dsat} = \frac{W}{L}\mu_e C_{ox}\left\{\left(V_{GS} - V_{To} + \gamma\sqrt{\phi_{sT}} - \frac{1}{2}V_{DSsat}\right)V_{DSsat} - \frac{2}{3}\gamma\left[(\phi_{sT} + V_{DSsat})^{3/2} - (\phi_{sT})^{3/2}\right]\right\} \tag{9.72}$$

Figure 9.32 shows the impact of the body effect on the output I–V characteristics of a typical MOSFET. On this figure, the solid lines represent the I–V characteristics calculated using the first-order model [Eqs. (9.16) and (9.21)]. The dashed lines are computed using the model that

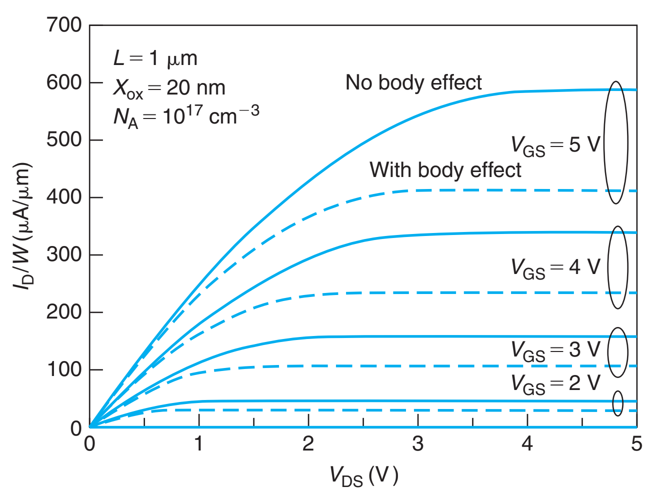


FIGURE 9.32
Calculated output *I*–*V* characteristics of a typical MOSFET using the first-order model without the body effect (solid lines) and including the body effect (dashed lines).

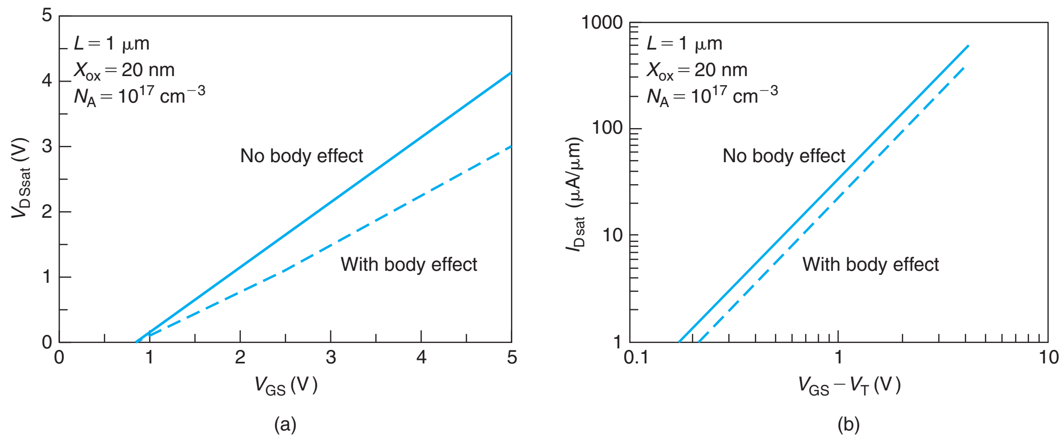


FIGURE 9.33
Impact of the body effect on V_{DSsat} (a) and I_{Dsat} (b) for a typical MOSFET. The device is identical to that of Fig. 9.32. Calculations using the first-order model without the body effect are graphed on solid lines, and including the body effect are graphed in dashed lines.

includes the body effect [Eqs. (9.69) and (9.72)]. Three features are noticeable in this figure. First, for all values of V_{GS} and V_{DS}, including the body effect results in a reduction of the drain current. This was expected. Second, the deleterious effect of the body effect goes away as the device is turned off. This is easily understandable since in this instance, the potential drop along the channel becomes negligible. Third, at a given V_{GS}, the body effect reduces the value of the drain-to-source voltage required to saturate the transistor.

The impact of the body effect on V_{DSsat} is more clearly seen in Fig. 9.33(a). This figure makes evident that V_{DSsat} is reduced by the body effect. Interestingly, the dependence of V_{DSsat} on V_{GS} remains rather linear. This suggests that the dependence of I_{Dsat} on $V_{\mathrm{GS}} - V_{\mathrm{To}}$ might remain quadratic in the presence of the body effect. In fact, the log–log plot of I_{Dsat} as a function of $V_{\mathrm{GS}} - V_{\mathrm{To}}$ in Fig. 9.33(b) shows just this. The body effect reduces I_{Dsat}, but does not significantly change the nature of its dependence on gate overdrive.

These observations suggest that simpler but still fairly accurate expressions of V_{DSsat}, I_{D}, and I_{Dsat} exist. These can be obtained by going back to Eq. (9.67) and linearizing the square root in the limit in which V is small with respect to ϕ_{sT}. This yields

$$V_{\mathrm{T}}(V) \simeq V_{\mathrm{To}} + \frac{\gamma}{2\sqrt{\phi_{\mathrm{sT}}}}V \tag{9.73}$$

This Taylor series expansion works well for small values of V but it should get progressively inaccurate as V increases. Fortunately, V in the channel never gets too high due to pinch-off.[5]

[5]Empirically, it has been found that a factor of 3, instead of 2, in the denominator of the second term in the right-hand side of Eq. (9.73) works better.

We now retrace our steps one more time and insert Eq. (9.73) into Eq. (9.66) and the result into Eq. (9.9) to get a simple differential equation that when solved yields

$$I_D \simeq \frac{W}{L}\mu_e C_{ox}\left(V_{GS} - V_{To} - \frac{m}{2}V_{DS}\right)V_{DS} \tag{9.74}$$

where m is given by

$$m = 1 + \frac{\gamma}{2\sqrt{\phi_{sT}}} \tag{9.75}$$

The expression of V_{DSsat}, obtained in the same way as before, becomes

$$V_{DSsat} \simeq \frac{1}{m}(V_{GS} - V_{To}) \tag{9.76}$$

Finally, the expression for the saturated drain current becomes

$$I_{Dsat} \simeq \frac{W}{2mL}\mu_e C_{ox}(V_{GS} - V_{To})^2 \tag{9.77}$$

Equations (9.74), (9.76), and (9.77) are very similar to the expressions we derived for an ideal MOSFET earlier in this chapter, Eqs. (9.16), (9.22), and (9.21), respectively. The only difference is the appearance of the parameter m, the so-called body-effect coefficient. As seen in Eq. (9.75), m is always positive and larger than unity. As a result, all things being equal, I_D, I_{Dsat}, and V_{DSsat} get reduced by the incorporation of the body effect. These new simplified equations are fairly accurate. This can be seen in Fig. 9.34. They are also easier to remember than the exact equations derived earlier in this section.

Interestingly, m has an identical expression to the ideality factor of the subthreshold regime discussed in Section 8.6. This should not be very surprising. In both the subthreshold regime and the body effect, there is a competition between the electrostatics of the gate and the electrostatics of the body for control of the inversion layer charge. If the controlling action of the gate dominates over that of the body, then the subthreshold regime is sharp and the body effect is small. The contrary happens if the controlling action of the body becomes significant.

It is a common practice when dealing with the long MOSFET to treat the parameter m as an adjustable one. It is found that values of m of the order of 1.1–1.4 work quite well.

9.7.2 Effect of back bias

A MOSFET is a *four-terminal* device. In addition to the source, drain, and gate, there is also the region right underneath the transistor, also called the back or the *body*. In an integrated device, the body is contacted at the surface of the wafer. So far, we have not considered the impact of applying a body voltage on the *I–V* characteristics of the transistor. For the most part, any voltage that would be applied to the body with respect to the source will be negative or slightly positive. If a relatively large positive bias is applied, the body-source PN diode turns on adding an undesirable current path to the MOSFET. Hence, we should mostly be concerned with applying negative values of V_{BS}.

We learned in Section 8.7 that the application of a negative voltage to the body with respect to the inversion layer results in a positive shift of the threshold voltage. All things being equal, then, if we have a MOSFET that is *on*, the application of $V_{BS} < 0$ should result in a decrease

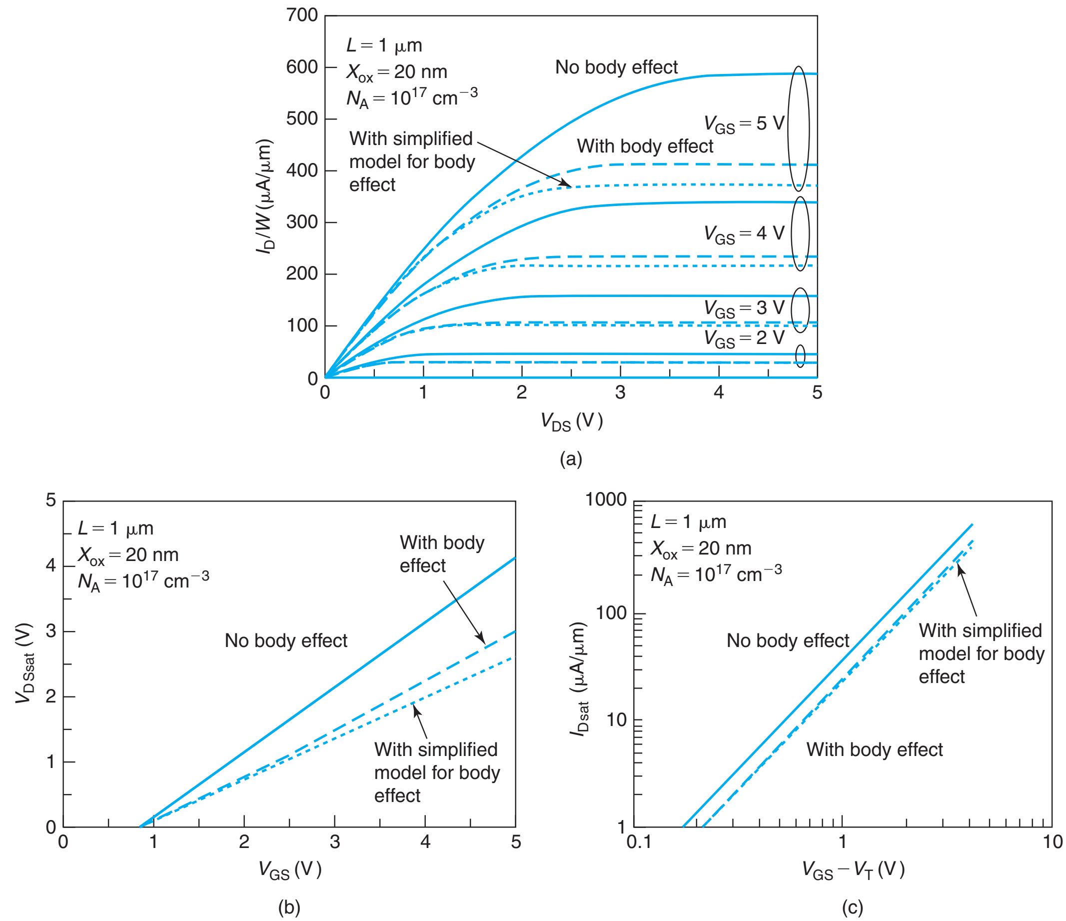


FIGURE 9.34
Comparison of behavior of simplified equations for output characteristics (a), V_{DSsat} (b), and I_{Dsat} (c) describing the body effect (short dash) against the exact equations (long dash). For reference, calculations without the body effect are also shown (solid). The device is identical to that of Fig. 9.32.

in the current flowing through the transistor. This is because as V_{T} shifts positive, the overdrive on the gate of the MOSFET is reduced. The evolution of the transfer characteristics of an ideal MOSFET as a function of back bias is illustrated in Fig. 9.35.

Modifying the simple first-order model developed above to account for this is straightforward. The shift in the threshold voltage as a result of applying a back bias is given by Eq. (8.69). Using the notation of this chapter, we can rewrite it as

$$V_{\text{T}}(V_{\text{BS}}) = V_{\text{To}} + \gamma\left(\sqrt{\phi_{\text{sT}} - V_{\text{BS}}} - \sqrt{\phi_{\text{sT}}}\right) \tag{9.78}$$

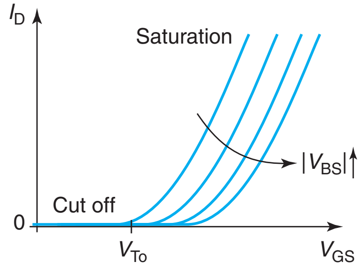


FIGURE 9.35
Transfer characteristics of ideal MOSFET as a function of backbias with $V_{BS} < 0$.

The ideal I–V characteristics of the MOSFET captured in Eqs. (9.16) and (9.21) can then be made to account for the back bias effect provided that Eq. (9.78) is used for the threshold voltage. It is clear in this formulation that as V_{BS} becomes more negative, V_T in Eq. (9.78) goes up along with it and I_D, given by Eqs. (9.16) and (9.21), goes down.

In addition to the straightforward impact of the application of a back bias on the threshold voltage just described, the characteristics of the MOSFET are also affected through the term that accounts for the body effect. It is easy to see why this should be the case. Both the body effect and the back bias effect arise from the electrostatic influence of the body on inversion layer charge. The application of a back bias with $V_{BS} < 0$ results in a widening of the depletion region underneath the inversion layer. This brings the "back gate" further away from the inversion layer and weakens the impact of the body effect on the transistor. Let us see how the model that accounts for the body effect is to be modified to include the impact of back bias also.

We start from Eq. (9.67) that gives the expression of the threshold voltage along the inversion layer accounting for the potential drop that occurs along the channel. In the presence of a back bias, this equation becomes

$$V_T(V_{BS}, V) = V_{To} + \gamma\left(\sqrt{\phi_{sT} - V_{BS} + V} - \sqrt{\phi_{sT}}\right) \tag{9.79}$$

To keep the model simple, we can expand this equation around $V = 0$, to get

$$V_T(V_{BS}, V) \simeq V_T(V_{BS}) + \frac{\gamma}{2\sqrt{\phi_{sT} - V_{BS}}}V \tag{9.80}$$

where $V_T(V_{BS})$ is given by Eq. (9.78).

Equation (9.80) is nearly identical to Eq. (9.73) except for the addition of V_{BS} to ϕ_{sT} inside the square root. Following the development carried out in the previous subsection, an identical set of equations for I_D emerges if we use a body-effect coefficient that depends on the back bias, as given by

$$m(V_{BS}) = 1 + \frac{\gamma}{2\sqrt{\phi_{sT} - V_{BS}}} \tag{9.81}$$

It is then clear that the application of $V_{BS} < 0$ increases V_T and reduces m. The dominant effect is a reduction in I_D and I_{Dsat} given by Eqs. (9.74) and (9.77), respectively, as expected. This can be seen in the output characteristics of a MOSFET for a back bias of 0 V and 5 V in Fig. 9.36.

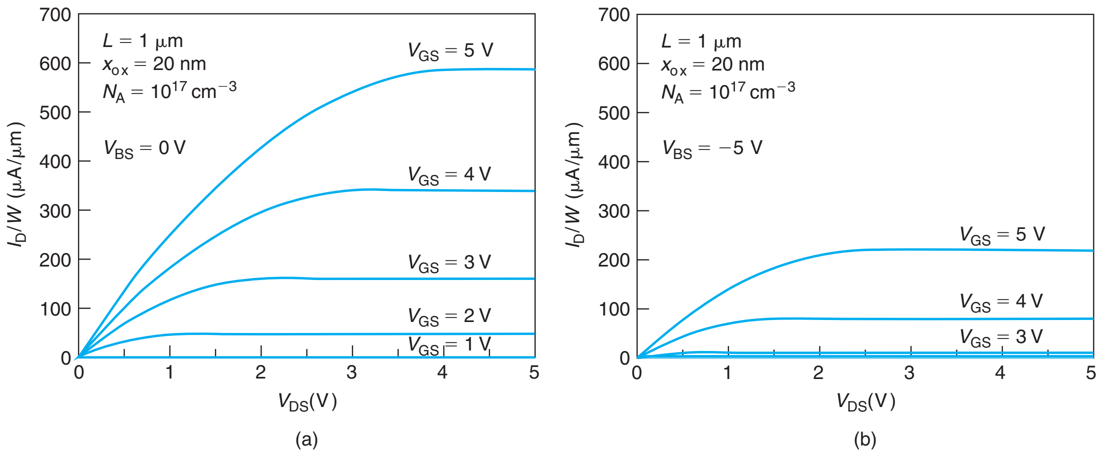


FIGURE 9.36
Effect of back bias on output characteristics of long MOSFET: $V_{BS} = 0$ V (a) and $V_{BS} = -5$ V (b). The device is identical to that of Fig. 9.32.

The existence of a dependence of the drain current on the back bias implies that if a small-signal is applied to the substrate terminal with respect to the source, this signal will propagate to the drain current. We actually recognized this when in Section 9.6.1 we developed the small-signal model for the MOSFET and defined a *back transconductance*, g_{mb} [Eq. (9.49)]. Now we are in a position to derive an expression for g_{mb} for the ideal MOSFET. This is straightforward because a small-signal change in V_{BS} impacts the drain current through a small change in V_T. We can then write

$$g_{mb} = \left.\frac{\partial I_D}{\partial V_{BS}}\right|_{V_{DS},V_{GS}} = \left.\frac{\partial I_D}{\partial V_T}\right|_{V_{DS},V_{GS}} \times \left.\frac{\partial V_T}{\partial V_{BS}}\right|_{V_{DS},V_{GS}} \tag{9.82}$$

In an ideal MOSFET in saturation, the first derivative in this equation is precisely $-g_m$. This is because in saturation the drain current depends on $V_{GS} - V_T$, so the dependence of I_D on V_T and $-V_{GS}$ are identical. The second derivative can be easily performed from Eq. (9.78) and expressed in terms of m defined in Eq. (9.81). All together we obtain

$$g_{mb} = (m - 1)g_m \tag{9.83}$$

It is not surprising that the body-effect coefficient, m, makes an appearance here. As discussed above, m reflects the competition between the substrate and the gate for control of the channel surface potential. In general $m > 1$ and g_{mb} is finite. The closer m gets to 1, the more the gate prevails over the body. In the limit of $m = 1$, the body exerts no control over the channel and therefore g_{mb} goes to zero.

Equivalent circuit models incorporating the back bias effect and the associated back transconductance are sketched in Fig. 9.37. In the large-signal model (a), an additional dependence of I_D on V_{BS} is explicitly noted. In the small-signal equivalent circuit model (b), a new current source is added between the drain and the source that is controlled by v_{bs}.

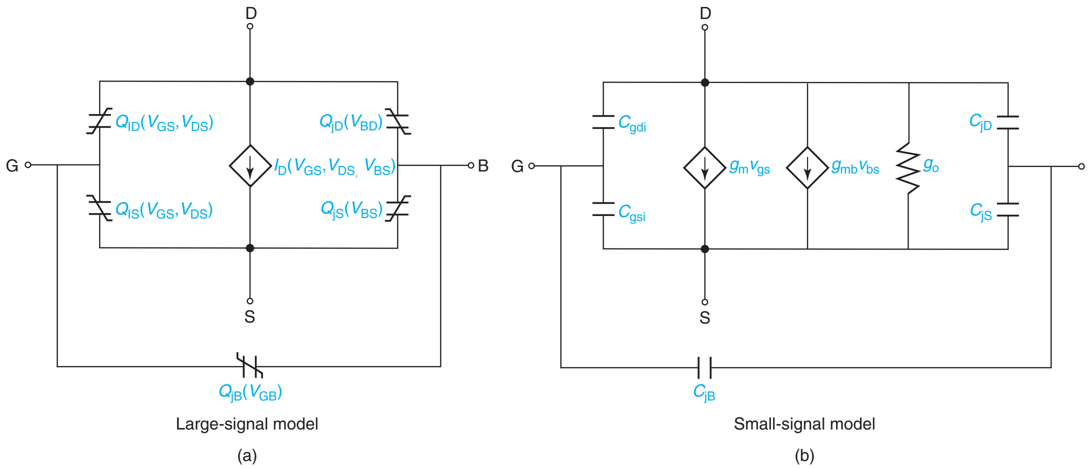


FIGURE 9.37
MOSFET equivalent circuit models incorporating the impact of the back contact. (a): Large-signal model with a V_{BS} dependence made explicit. (b): Small-signal model incorporating a new current source driven by v_{bs}.

9.7.3 Channel-length modulation

In the ideal MOSFET model developed earlier in this chapter, the current–voltage characteristics are perfectly saturated for $V_{DS} > V_{DSsat}$. This is the saturation regime of operation. However, if we look at a real MOSFET, such as the 2N7000 depicted in Fig. 9.16, we see a small but distinct slope in the I_D–V_{DS} characteristics. The slope also increases with V_{GS}. This imperfectly saturated drain current greatly impacts many circuit applications of MOSFETs, such as the voltage gain in MOSFET amplifiers and the noise margins of CMOS logic gates, just to cite two examples. This section explains the physical origin of this effect and presents a simple model that accounts for the key dependencies.

The finite output conductance of a MOSFET has its origin in the electrostatics of the pinch-off point. An appropriate treatment is beyond our reach at the moment and has to wait until the short MOSFET is discussed in the next chapter. Nevertheless, reasonable understanding and a first-order model can be derived by borrowing a few results from Chapter 10.

In Section 9.4.3, we learned that with the MOSFET in the linear regime, the inversion layer electron velocity increases and the electron concentration drops as we proceed from source to drain. We also learned that the drop in carrier concentration is more pronounced the higher V_{DS} is. We called this *channel debiasing*. At pinch-off, our very simplistic model predicted that the electron concentration at the drain end of the channel becomes zero and the electron velocity becomes infinite. Obviously, this is not possible. Electrons cannot travel at velocities higher than the saturation velocity and since the current is finite, the electron concentration cannot drop to zero.

In fact, as we will study in detail in Chapter 10, current saturation in a MOSFET actually occurs when the electron velocity at the drain end of the channel reaches the saturation velocity. At this condition, the electron concentration at the pinch-off point is small but finite and the current that flows is also finite. Ideally, a further increase in V_{DS} beyond the bias that yields this condition does not directly increase the electron velocity any further and the current therefore should not change.

An important word in the previous phrase is "directly." To the first order, an increase in the drain voltage beyond V_{DSsat} should not translate into an increased potential build-up along the channel of a MOSFET. Instead, as we will see in Chapter 10, it results in an enlargement of the pinch-off region where the carrier velocity is saturated. This effectively shortens the electrical channel length. Since in this instance the same potential build-up (V_{DSsat}) takes place over a shorter length, this implies that the electric field along the channel increases a little bit and so does the drain current. It is because of this effective shortening of the channel length that this effect is given the name of "channel-length modulation."

Building a model for channel-length modulation essentially means solving this complex two-dimensional electrostatics problem. A simplified treatment is presented in the next chapter. We qualitatively discuss the key findings here.

Figure 9.38 zooms into the drain end of the channel of a MOSFET in saturation with $V_{DS} > V_{DSsat}$. As discussed, at the drain end of the channel, a velocity saturation region of length l_{sat} appears. Figure 9.39 sketches the volume charge density, electron velocity, lateral electric field, and lateral potential along the semiconductor surface with the MOSFET in saturation for $V_{DS} = V_{DSsat}$ and $V_{DS} > V_{DSsat}$. This is an exaggerated picture to highlight the key issues.

At the onset of saturation, for $V_{DS} = V_{DSsat}$, we see that the electron concentration drops along the channel and reaches a minimum value right at the drain end. At that same point, the electron velocity peaks at its saturation velocity value. The electric field increases along the channel and peaks also at the drain end at a value that saturates the electron velocity. We label this as $\mathcal{E}_{sat}$. The channel voltage builds up along the channel up to a value of V_{DSsat} at its drain end.

When $V_{DS} > V_{DSsat}$, the additional voltage that is applied is used to extend the region over which the electron velocity is saturated to a length l_{sat}. The effective length of the channel over which the voltage builds up to V_{DSsat} is therefore shortened from L to L'. This results in a small increase of the electric field everywhere including at the source. Since the charge is constant at the source, this implies that the drain current increases a little bit. This is what is responsible for the output conductance of the device.

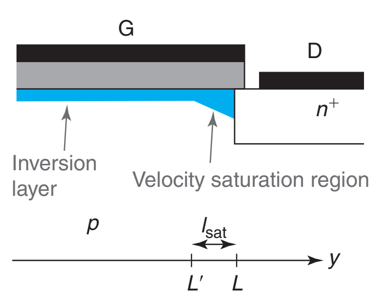


FIGURE 9.38
Geometry of the pinch-off region in a MOSFET for $V_{DS} > V_{DSsat}$.

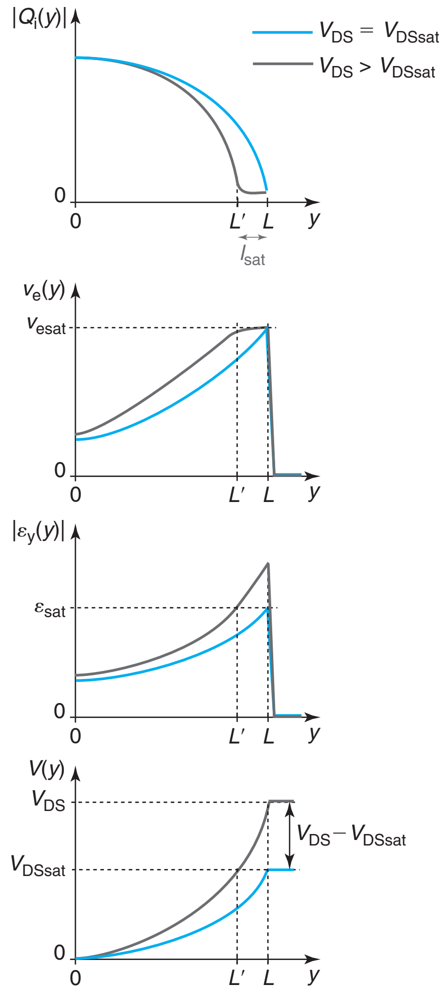


FIGURE 9.39
Electrostatics along the surface of the semiconductor for the pinch-off region of a MOSFET in saturation. From top to bottom: inversion layer charge density, electron velocity, lateral electric field, and channel voltage.

Modeling this effect requires deriving an expression for l_{sat} as a function of V_{DS}. This is what is done in Chapter 10. To the first order, if we assume that velocity saturation occurs at a constant lateral field $\mathcal{E}_{sat}$, as indicated in Fig. 9.39, and that the effect of V_{DS} is relatively small, we should expect, approximately [Eq. (10.79)]

$$l_{sat} \simeq \frac{V_{DS} - V_{DSsat}}{\mathcal{E}_{sat}} \tag{9.84}$$

We can now reuse the drain current model developed earlier in this chapter except that we substitute L for $L - l_{\mathrm{sat}}$. Therefore, we have

$$I_{\mathrm{Dsat}} \propto \frac{1}{L - l_{\mathrm{sat}}} = \frac{1}{L\left(1 - \frac{l_{\mathrm{sat}}}{L}\right)} \simeq \frac{1}{L}\left(1 + \frac{l_{\mathrm{sat}}}{L}\right) = \frac{1}{L}\left(1 + \frac{V_{\mathrm{DS}} - V_{\mathrm{DSsat}}}{\mathcal{E}_{\mathrm{sat}} L}\right) \tag{9.85}$$

where we have assumed that in a well-designed device, $l_{\mathrm{sat}} \ll L$.

The second term inside the parentheses represents the correction term that is introduced in the MOSFET saturation current model as a result of channel-length modulation. It is proportional to $V_{\mathrm{DS}} - V_{\mathrm{DSsat}}$. The proportionality constant is often written as λ, defined as the *channel-length modulation parameter*, with units of inverse volt,

$$\lambda = \frac{1}{\mathcal{E}_{\mathrm{sat}} L} \tag{9.86}$$

Using λ, we can write the complete expression of the saturation current of a MOSFET in the presence of channel-length modulation as[6]

$$I_{\mathrm{Dsat}} = \frac{W}{2L}\mu_{\mathrm{e}} C_{\mathrm{ox}} (V_{\mathrm{GS}} - V_{\mathrm{T}})^2 [1 + \lambda(V_{\mathrm{DS}} - V_{\mathrm{DSsat}})] \tag{9.87}$$

The correction term on the saturation current depends only on $V_{\mathrm{DS}} - V_{\mathrm{DSsat}}$. However, this term is multiplied by the ideal saturation current. Hence, it becomes more visible in the output characteristics the higher the value of V_{GS}. This is shown in the sketch of the output characteristics of Fig. 9.40 and agrees with what is observed in practice.

Channel-length modulation impacts the small-signal equivalent circuit model of the MOSFET. In saturation, the output conductance of the ideal MOSFET model is zero [Eq. (9.55)].

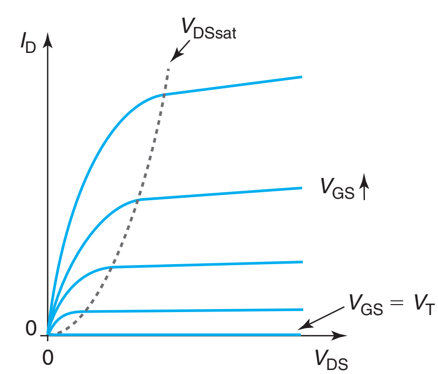


FIGURE 9.40
Sketch of the output characteristics of the MOSFET in the presence of channel-length modulation.

[6]In analogy with the Early effect in the bipolar transistor that also results in a finite output conductance, sometimes $1/\lambda$ in a MOSFET is referred to as the *Early voltage*.

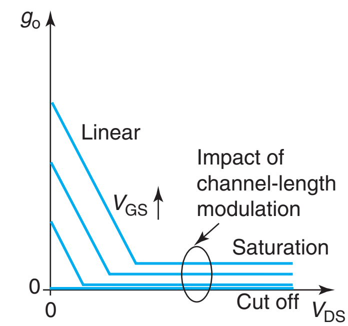


FIGURE 9.41
Evolution of g_o with V_{GS} and V_{DS} in MOSFET showing the impact of channel-length modulation.

Obviously, channel-length modulation introduces a finite amount of output conductance. This can be easily obtained by applying the definition of Eq. (9.53),

$$g_o = \left.\frac{\partial I_D}{\partial V_{DS}}\right|_{V_{GS},V_{BS}} \simeq \lambda I_{Dsat} \tag{9.88}$$

The important result here is that in a given device, g_o increases with I_{Dsat}. This matters a lot in analog circuit design, where g_o is often a crucial consideration. In a given transistor, the only way to improve g_o (i.e., make it smaller) is to operate it at a lower current level. Alternatively, in applications where low g_o is critical, analog circuit designers often choose devices with longer gate lengths. As seen in Eq. (9.86), this results in a lower value of λ and a reduced output conductance at a certain current level.

Figure 9.41 sketches the evolution of g_o in a MOSFET now accounting for the impact of channel-length modulation. Figure 9.42 shows the output conductance of the 2N7000 MOSFET whose output characteristics are shown in Fig. 9.16.

9.7.4 The subthreshold regime

In the ideal MOSFET model, we assumed that the drain current is zero for $V_{GS} < V_T$. We referred to this as the cut-off regime. In a real MOSFET, the current decreases as V_{GS} drops below V_T, but it never quite becomes zero. For example, Fig. 9.43 shows the output characteristics of the 2N7000 MOSFET plotted in a semilog scale. Graphing the characteristics in this way allows us to examine in more detail the behavior of the MOSFET over a broader range of currents. It is clear from this figure that below a gate-source voltage of about 2 V (which seems a reasonable estimate of the threshold voltage according to Fig. 9.16), the current drops very fast as V_{GS} is reduced. However, down to $V_{GS} = 1$ V, it is still fairly sizable.

There are two interesting features that are noticeable in the data of Fig. 9.43. First, the current drops in what appears to be an exponential way. For values of V_{GS} between 1 and 2 V, the current lines appear relatively evenly spaced. Since V_{GS} is being stepped in a linear way, this suggest that I_D is evolving in an exponential way with V_{GS}. This can be better seen in the data of Fig. 9.44

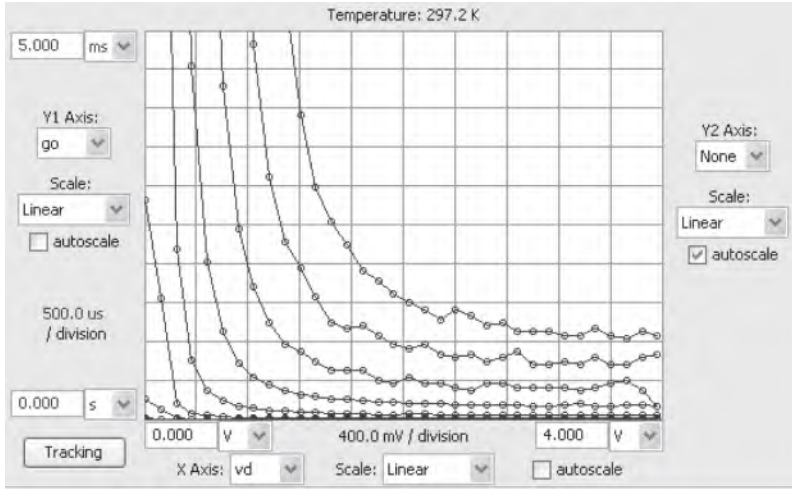


FIGURE 9.42
Measured output conductance of a 2N7000 MOSFET corresponding to the output characteristics shown in Fig. 9.16. V_{GS} is stepped from 1 to 3 V in steps of 0.2 V (Reprinted with permission from MIT Microelectronics iLab).

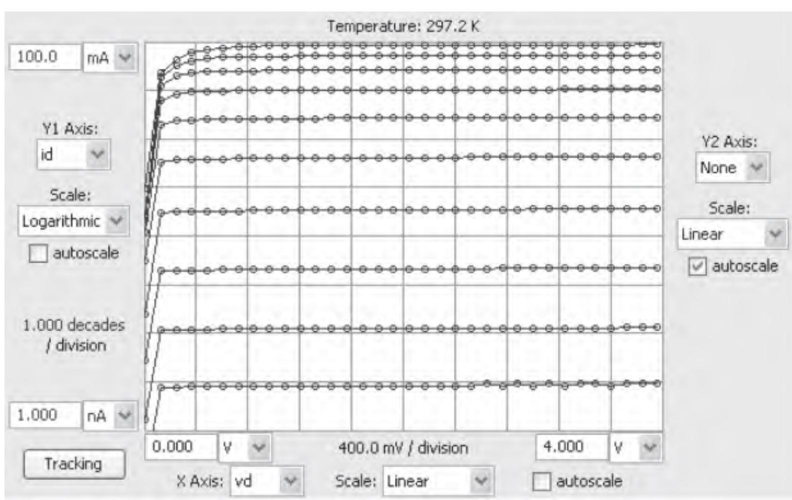


FIGURE 9.43
Measured output *I–V* characteristics of a 2N7000 MOSFET. These characteristics are identical to those of Fig. 9.16 except that I_D is shown in a semilogarithmic scale. V_{GS} is stepped from 1 to 3 V in steps of 0.2 V (Reprinted with permission from MIT Microelectronics iLab).

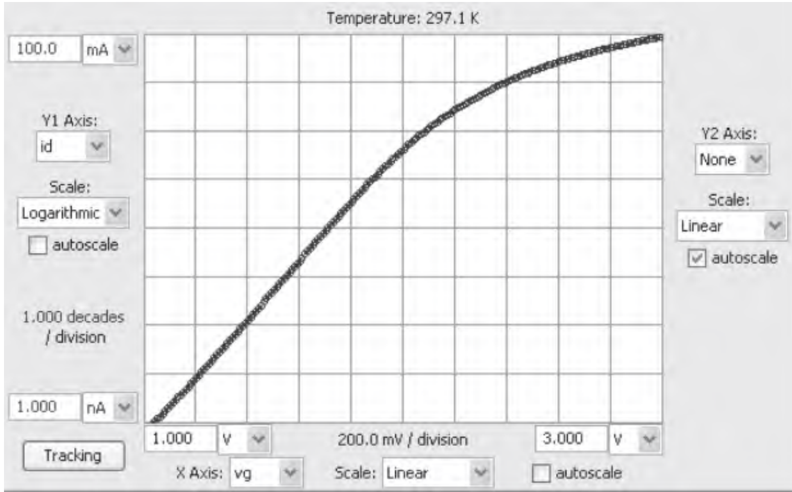


FIGURE 9.44
Measured drain current versus gate–source voltage for a 2N7000 MOSFET in a semilog scale. $V_{DS} = 2$ V (Reprinted with permission from MIT Microelectronics iLab).

that shows the drain current as a function of V_{GS} in a semilog scale. Indeed, below a value of $V_{GS} \simeq 2$ V, the current–voltage characteristics follow a straight line.

The second interesting feature that can be seen in Fig. 9.43 is that the current is perfectly saturated with V_{DS} down to the lowest value of V_{DS} that can be resolved, which in this measurement is 120 mV. In contrast, for $V_{GS} > V_T$, it takes quite a bit of drain-to-source voltage to saturate the drain current.

We are in front of what is termed the subthreshold regime. This is a regime of operation of great significance for analog and logic applications. In logic circuits, for example, we look at the MOSFET as a switch. This means that for $V_{GS} < V_T$, we really want the switch to be open, that is, the drain current to be zero. However, as we can see in Figs. 9.43 and 9.44, the drain current never quite becomes zero, the switch "leaks" a little bit. In consequence, there is some power consumption associated with the imperfectly open switch. Since the current is small, this power can be rather small. However, in a logic IC that integrates billions of transistors, this can add up to a sizable amount of power that needs to be delivered and dissipated.

In this section, we wish to understand the physics of the subthreshold current and develop a simple model that captures the leading dependencies.

At the heart of the subthreshold regime of a MOSFET is the *weak inversion* regime of the MOS structure that we discussed in Section 8.6. Below threshold, we found that electron charge at the surface of the semiconductor (it is not proper to denote this as inversion layer) did not quite go to zero but dropped in an exponential way with V, the gate to body voltage [see Eq. (8.63)]. This suggests that the exponential dependence of the subthreshold current in the MOSFET must arise from the charge-control relationship in weak inversion. The interesting question is how does this charge flow from source to drain to produce drain current?

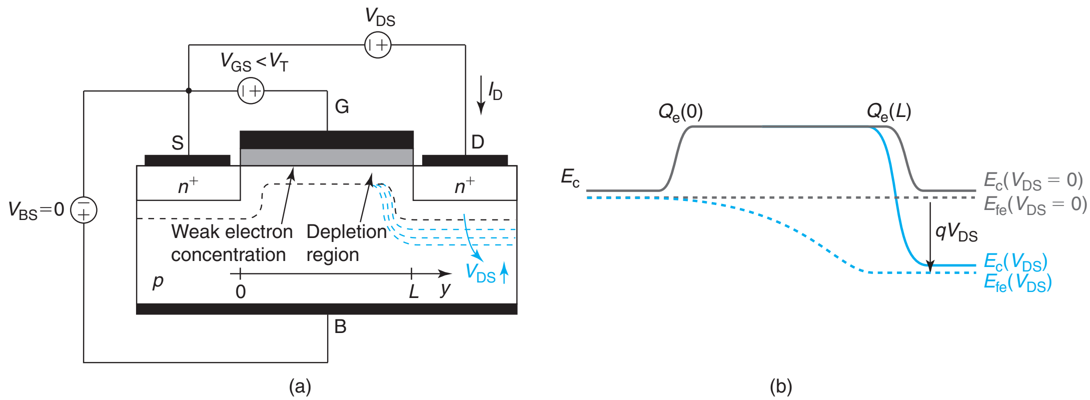


FIGURE 9.45
(a): Sketch of MOSFET in subthreshold regime showing the evolution of the depletion region under the source, gate, and drain for different values of V_{DS}. (b): Sketch of the conduction band edge along the surface of the semiconductor for different values of V_{DS}.

In a MOSFET below threshold, the dominant electrostatic feature underneath the gate is the presence of a depletion region. To be sure, there are a few electrons inside this depletion region. These are those that are responsible for transport in the subthreshold regime. However, their concentration is much smaller than the acceptor concentration. The electrostatics are then dominated by the volume charge density that arises from the uncompensated acceptors. All this charge is imaged at the gate. In this situation, the drain has no way to manipulate the surface potential under the gate. This is unlike in the linear regime in which the drain electrically "grabs" on the inversion layer at the end of the channel and can therefore affect the surface potential along the entire length of the channel. There is no electrical channel to grab on in the subthreshold regime. How, then, can current flow?

We can understand what is happening by looking at Fig. 9.45 that sketches the evolution of the depletion region (a) in a MOSFET biased in subthreshold, and the conduction band edge (b) along the channel under the source, gate, and drain for two values of V_{DS}.

With $V_{DS} = 0$, the depletion region under the source and drain are identical. The electron quasi-Fermi level is completely flat along the surface of the semiconductor and the electron concentration is uniform everywhere under the channel and given by Eq. (8.63), where $V = V_{GS} = V_{GD}$.

Increasing V_{DS} widens the depletion region under the drain and slightly sideways under the channel. From an energy point of view, increasing V_{DS} brings the conduction band edge and the quasi-Fermi level for electrons in the drain down with respect to the source by an amount equal to qV_{DS}. Note, however, that the conduction band edge under the gate remains flat. This is because the application of a voltage to the drain with respect to the source has not introduced a lateral electric field on the semiconductor underneath the gate. The electric field in the semiconductor under the gate points vertical.

The interesting question now is what happens to the electron concentration underneath the gate. It is clearly not going to be uniform since the distance between the conduction band edge and the quasi-Fermi level for electrons changes along the surface of the semiconductor from source to drain.

At the source end, the energy band diagram is not modified at all with respect to the situation that we had at $V_{DS} = 0$. Hence, from Eq. (8.63), we have

$$Q_e(y = 0) = -\frac{kT}{q} C_{sT} \exp \frac{q(V_{GS} - V_T)}{nkT} \tag{9.89}$$

where C_{sT} is the capacitance associated with the depletion region in the semiconductor at threshold and was given by Eq. (8.52), and n is the ideality factor given by

$$n = 1 + \frac{C_{sT}}{C_{ox}} \tag{9.90}$$

Under the drain, the situation is quite different. The application of V_{DS} brings down the conduction band edge and the quasi-Fermi level of electrons inside the drain with respect to the situation at $V_{DS} = 0$. Since the conduction band diagram is flat under the gate, then right under the drain edge of the gate, the quasi-Fermi level for electrons drops below the conduction band edge by an amount equal to qV_{DS}. This means that the electron concentration must have dropped too. In fact, comparing energy barriers for electrons in the channel on the source side and the drain side, the barrier on the drain side is qV_{DS} higher than on the source side. Assuming quasi-equilibrium across the thin depletion regions on both sides of the channel, we can then conclude that the ratio of the electron charge at the drain end of the channel is related to that at the source end of the channel through a Boltzmann factor. That is,

$$Q_e(y = L) = Q_e(y = 0) \exp \frac{-qV_{DS}}{kT} \tag{9.91}$$

We have an interesting situation here. The electron concentration at the semiconductor surface under the drain end of the gate is much smaller than under the source end. This is going to result in *electron diffusion* from source to drain and a current flow! This is the subthreshold current.

Obtaining an expression for the subthreshold current is now straightforward. It must be given by Fick’s second law which states that electron diffusion current is directly proportional to the gradient of electron concentration. In this situation, this implies that

$$I_D = WD_e \frac{dQ_e(y)}{dy} \tag{9.92}$$

Since electrons have nowhere to go but from source to drain (they cannot escape to the substrate because there is a potential barrier due to the band bending associated with the depletion region), the electron current must be constant along the surface of the semiconductor. This means that the gradient of electrons must also be a constant or that the electron profile is a straight line. From Eq. (9.92), it then follows that:

$$I_D = WD_e \frac{Q_e(y = L) - Q_e(y = 0)}{L} = \frac{W}{L} D_e Q_e(y = 0) \left[\frac{Q_e(y = L)}{Q_e(y = 0)} - 1 \right] \tag{9.93}$$

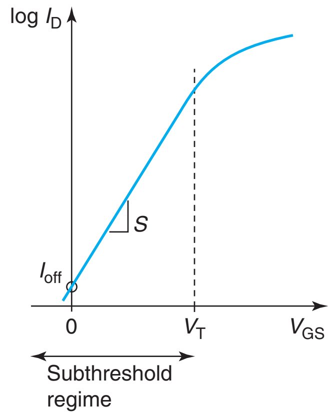


FIGURE 9.46
Sketch of subthreshold characteristics of a MOSFET.

Plugging in the results obtained in Eqs. (9.89) and (9.91), we get

$$I_D \simeq \frac{W}{L} D_e \frac{kT}{q} C_{sT} \exp \frac{q(V_{GS} - V_T)}{nkT} \left(1 - \exp \frac{-qV_{DS}}{kT}\right) \tag{9.94}$$

This is the result that we were looking for and is graphed in Fig. 9.46.

Let us examine some of the features of this equation. First, it is clear that I_D evolves with V_{GS} following an exponential law. The ideality factor of the exponential dependence is given by Eq. (9.90). Second, for $V_{DS} = 0$, I_D is zero as it should be. Third, as V_{DS} increases, its impact on the subthreshold current quickly vanishes. This is because the $\exp \frac{-qV_{DS}}{kT}$ rapidly becomes negligible as V_{DS} increases beyond a few thermal voltages. This is clearly seen in the experimental data of Fig. 9.43.

The slope of the subthreshold characteristics is characterized through the *subthreshold swing* (confusingly, also referred to as "subthreshold slope" or "inverse subthreshold slope") which is defined as

$$S = \frac{nkT}{q} \ln 10 = \left(1 + \frac{C_{sT}}{C_{ox}}\right) \frac{kT}{q} \ln 10 \tag{9.95}$$

An ideality factor of 1 corresponds to a subthreshold swing of 60 mV/dec at room temperature. This is what the PN diode exhibits. In comparison with this, since in a MOSFET $n > 1$, the subthreshold swing is higher and the subthreshold current rises more gently with V_{GS}.

There are two key dependencies in the subthreshold swing. One is temperature. The higher the temperature, the higher the subthreshold swing. This is just like in the PN diode. The second one is in the ratio C_{sT}/C_{ox}. Anything that minimizes this ratio will sharpen the subthreshold swing of the MOSFET. What we essentially want is $C_{ox} \gg C_{sT}$, or that the gate exercises a much tighter electrostatic control over the surface potential than the substrate. This suggests the use of thinner gate oxides and lower body doping levels.

One final point to make about the subthreshold swing is that the body voltage, in addition to shifting the subthreshold characteristics by acting on the threshold voltage, also impacts the subthreshold swing. The more negative the body voltage, the deeper the depletion layer becomes underneath the gate and the lower C_{sT} ends up being. In consequence, the subthreshold swing becomes sharper.

The subthreshold swing is important because it sets the value of the off-state current, I_{off}, a key figure of merit in a MOSFET that is used in CMOS logic applications. I_{off} is defined as $I_D(V_{GS}=0, V_{DS}=V_{DD})$, where V_{DD} is the maximum voltage available in the circuit. I_{off} is marked in Fig. 9.46.

As expression for I_{off} can be easily obtained from Eq. (9.94),

$$I_{off} = I_D(V_{GS}=0, V_{DS}=V_{DD}) \simeq \frac{W}{L} D_e \frac{kT}{q} C_{sT} \exp\frac{-qV_T}{nkT} \tag{9.96}$$

With the subthreshold swing defined as in Eq. (9.95), this can also be written as

$$I_{off} \propto 10^{-\frac{V_T}{S}} \tag{9.97}$$

Writing I_{off} this way brings in evidence the two key factors that affect it: the threshold voltage and the subthreshold swing. Increasing V_T decreases I_{off} exponentially. The key trade-off here is that the gate overdrive, $V_{DD} - V_T$, is reduced and the drain current in saturation drops. One can also obtain low I_{off} by lowering the subthreshold swing. The relevant dependencies of S have been studied above.

9.7.5 Source and drain resistance

Our model for the ideal MOSFET so far does not include any resistance effects. In real MOSFETs, there are two resistances associated with the source and drain that are connected in series with the channel. They arise from the contact resistance of the metal contact to the n^+ regions, the finite lateral resistance of these regions and their linkage to the inversion layer under the gate edge. These series resistances represent significant parasitics with often very visible and deleterious impact on the device's electrical characteristics.

The most important consequence of the presence of the source and drain resistances in a MOSFET is a reduction in the current driving ability of the device. This can be understood through the simple model depicted in Fig. 9.47. This model explicitly places the source and drain resistances in series with the ideal device. When current flows through the MOSFET, the ohmic drops in the source and drain resistances reduce the voltage drive that is applied across the internal nodes of the transistor. This results in a reduction of the drain current below that of the ideal device. We can quantify this through a simple model.

If we denote V_{DS} and V_{GS} as the external terminal voltages, and V_{DSi} and V_{GSi} as the internal node voltages (see Fig. 9.47), we can write:

$$V_{GSi} = V_{GS} - I_D R_S \tag{9.98}$$

$$V_{DSi} = V_{DS} - I_D (R_S + R_D) \tag{9.99}$$

In the saturation regime, the drain current of the MOSFET is set by the internal gate overdrive $V_{GSi} - V_T$. Then, using Eq. (9.21) we have

$$I_{Dsat} = \frac{W}{2L} \mu_e C_{ox} (V_{GSi} - V_T)^2 = \frac{W}{2L} \mu_e C_{ox} (V_{GS} - I_D R_S - V_T)^2 \tag{9.100}$$

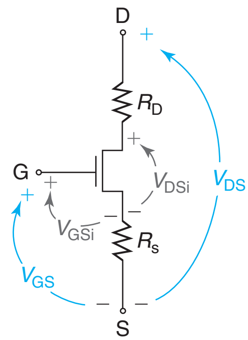


FIGURE 9.47
Schematic diagram showing parasitic source and drain resistance in MOSFET. The impact of the source and drain resistances is to reduce the internal voltages that are applied to the MOSFET.

For a well-designed transistor in which resistance effects are not too large, when we expand the quadratic term in this equation, it is reasonable to assume that $I_D R_S \ll V_{GS} - V_T$. In other words, the ohmic drop on the source resistance is much smaller than the gate overdrive. In this instance, we can easily solve for I_{Dsat},

$$I_{Dsat} \simeq \frac{\frac{W}{2L}\mu_e C_{ox}(V_{GS} - V_T)^2}{1 + \frac{W}{L}\mu_e C_{ox}(V_{GS} - V_T)R_S} \tag{9.101}$$

This equation has the right limit as R_S goes to zero for which we recover the current–voltage characteristics of the ideal device. For nonnegligible values of R_S, the term in the denominator reduces the drain current from its ideal value. This term becomes more significant the higher the gate overdrive. A consequence of this is that at high gate overdrive, the transfer characteristics of the transistor are no longer quadratic in $V_{GS} - V_T$ but they exhibit a weaker dependence. The absence of any role for R_D in I_{Dsat} is because the transistor is saturated and the drain current is insensitive to the value of V_{DS}.

In the linear regime, both R_S and R_D play a role. We can derive a first-order expression for the linear drain current starting from the ideal Eq. (9.16) and then plugging in Eqs. (9.98) and (9.99). After assuming a well-behaved transistor with moderate values of parasitic resistances, we obtain

$$I_D = \frac{W}{L}\mu_e C_{ox}\left(V_{GSi} - V_T - \frac{1}{2}V_{DSi}\right)V_{DSi} \simeq \frac{\frac{W}{L}\mu_e C_{ox}(V_{GS} - V_T - \frac{1}{2}V_{DS})V_{DS}}{1 + \frac{W}{L}\mu_e C_{ox}(V_{GS} - V_T - \frac{1}{2}V_{DS})(R_S + R_D)} \tag{9.102}$$

Once again, this equation converges to the ideal expression if $R_S + R_D = 0$. As the parasitic resistances become significant, the linear drain current drops below its ideal value. The drop becomes more prominent the harder we drive the transistor.

Figure 9.48(a) illustrates the changes to the output characteristics of an ideal MOSFET as a result of the presence of the source and drain resistance. For a given V_{GS}, the drain current is

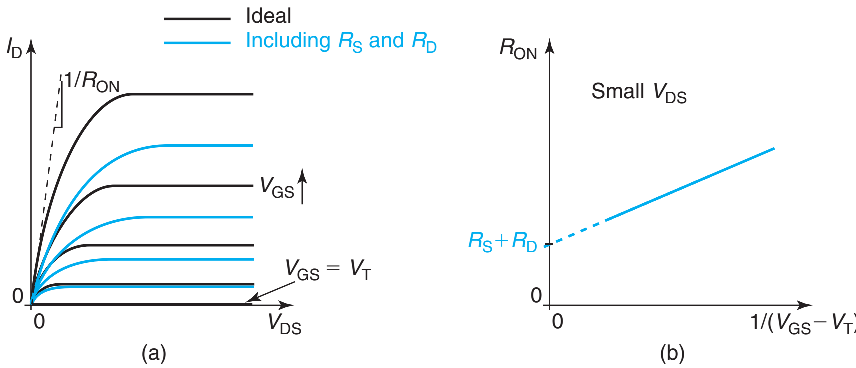


FIGURE 9.48
(a) Impact of R_S and R_D on the output characteristics of a MOSFET. The definition of R_{ON} is indicated. (b) Illustration of technique to extract $R_S + R_D$ in a MOSFET from measurements of R_{ON} as a function of gate overdrive in the linear regime.

reduced. The higher the gate overdrive, the bigger the effect. The value of V_{DS} that saturates the transistor is also increased. Finally, for very small V_{DS}, the slope of the I–V characteristics is reduced.

The change in the linear characteristics of the transistor as a result of the source and drain resistance is of particular importance in certain applications such as power management. It is quantified through the so-called ON resistance, R_{ON}. This is defined as the ratio of V_{DS} to I_D in the limit of small V_{DS}. From Eq. (9.102), we can easily get

$$R_{ON} = \left.\frac{V_{DS}}{I_D}\right|_{\text{small}\,V_{DS}} \simeq R_S + R_D + \frac{1}{\frac{W}{L}\mu_e C_{ox}(V_{GS} - V_T)} \tag{9.103}$$

This result has a simple physical interpretation. The total resistance between the source and drain terminals of a MOSFET for small V_{DS} is given by the sum of the source and drain parasitic resistances plus the channel resistance. Paths to minimize R_{ON} include shrinking R_S and R_D and mitigating the channel resistance by selecting the geometry of the transistor (large W/L) and applying a large gate overdrive.

Equation (9.103) also suggests a method for extracting $R_S + R_D$ in a MOSFET. If we plot R_{ON} as a function of $1/(V_{GS} - V_T)$, a straight line should result that intersects the y-axis at $R_S + R_D$. This is illustrated in Fig. 9.48(b).

An important consequence of the presence of the source and drain resistances is the degradation of the transconductance of the device. It is clear from looking at Fig. 9.48 that g_m is reduced from the ideal value. In saturation, an expression for g_m that accounts for R_S (R_D has no impact) can be easily obtained by differentiating Eq. (9.98), dividing throughout by dI_D and then solving for g_m, to yield

$$g_m = \frac{g_{mi}}{1 + g_{mi}R_S} \tag{9.104}$$

In this equation, g_{mi} is the *intrinsic transconductance*, or the transconductance of the ideal device,

$$g_{mi} = \left.\frac{\partial I_D}{\partial V_{GSi}}\right|_{V_{DSi}} \tag{9.105}$$

and g_m is the *extrinsic transconductance*, defined as

$$g_m = \left.\frac{\partial I_D}{\partial V_{GS}}\right|_{V_{DS}} \tag{9.106}$$

Notice that in these derivatives holding constant V_{DS} in one case and V_{DSi} in the other is not an issue since adding R_S and R_D to the ideal MOSFET does not perturb its perfect saturation.[7]

It is clear from the result of Eq. (9.104) that the transconductance of a MOSFET is reduced as a result of adding source resistance and that the effect is the greatest, the higher the intrinsic transconductance of the device.

R_S and R_D should be added to the small-signal equivalent circuit model of the MOSFET. When we do this, we need to be clear that the transconductance and output conductance elements in it represent the intrinsic values characteristic of the ideal MOSFET. A small-signal equivalent circuit model for a MOSFET that includes R_S and R_D is shown in Fig. 9.49. Notice that in this model, C_{jS} and C_{jD} have been placed on the *inside* of R_S and R_D. A better model

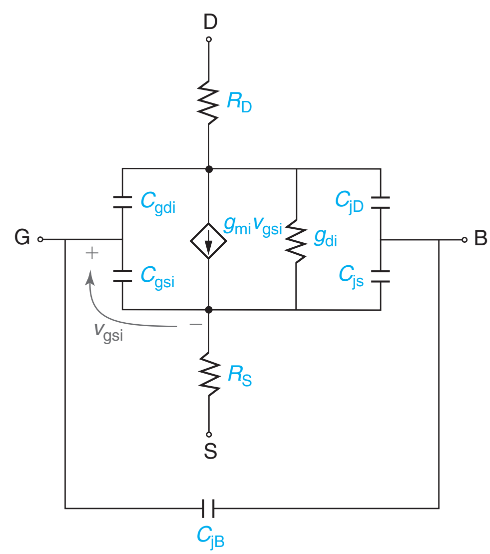


FIGURE 9.49
Small-signal equivalent circuit model of a MOSFET including R_S and R_D. The transconductance and output conductance generators are those of the ideal MOSFET.

[7]If g_d is included, one has to be more careful. See S. Y. Chou and D. A. Antoniadis, *IEEE Trans. Electron Dev.* ED-32, p. 448 (1987).

would break R_S and R_D into a contact resistance component and a lateral component and place the junction capacitances at the common node.

9.8 SUMMARY

- The SCA states that the inversion layer in a MOSFET is very thin in the scale of the vertical dimensions of the device. It allows us to formulate the MOSFET current in terms of the sheet-charge density.
- The GCA breaks the 2D electrostatic problem into two 1D simpler ones. The vertical electrostatics control the inversion layer charge and the lateral electrostatics control the flow of inversion charge from source to drain.
- In the linear regime, the drain current of an ideal MOSFET increases with V_{GS} and V_{DS}. In the saturation regime, the drain current depends only on V_{GS}. The dependence is quadratic. In the cut-off regime, the drain current through an ideal MOSFET is zero.
- The transconductance of an ideal MOSFET in saturation increases with the square root of the drain current.
- The frequency response of a MOSFET is quantified through the short-circuit current-gain cut-off frequency or f_T. Physically, this is the time it takes for a perturbation applied to the gate of a MOSFET to propagate into the drain current.
- The "body effect" arises from an increase in the local V_T along the channel. It results in premature drain current saturation and reduces the current drive of the transistor.
- The application of a back bias to a MOSFET shifts V_T and affects the drain current.
- In a refined MOSFET model, drain current saturation and channel pinch-off occur when electrons reach velocity saturation at the drain end of the channel. For V_{DS} values higher than V_{DSsat}, the pinch-off point broadens into a region with finite length where the electron velocity is saturated. This effectively reduces the channel length and increases the drain current of the device. Channel-length modulation, as this phenomenon is called, is responsible for the finite output conductance of a MOSFET in the saturation regime.
- For V_{GS} values below V_T, the drain current of a MOSFET drops exponentially. This is known as the subthreshold regime. The sharpness of the subthreshold regime is characterized by the subthreshold swing. This reflects a competition between the gate and the body of the transistor for electrostatic control over the surface potential. The sharpness of the subthreshold regime is enhanced the tighter the electrostatic influence of the gate of the transistor.

9.9 FURTHER READING

Operation and Modeling of the MOS Transistor (Third Edition) by Y. P. Tsividis and C. McAndrew, Oxford, 2011 (ISBN 978-0-19-517015-3, TK7871.99.M44T77) is a classic textbook on MOSFET physics and modeling. The treatment of the MOSFET is rather comprehensive and in places goes beyond that of the present book. Chapters relevant here are Chapter 4 that mostly deals with the long-channel MOSFET structure, Chapter 6 describing large-signal dynamic operation, and Chapters 7 and 8 addressing small-signal modeling, including noise. This is an important reference book.

AT9.1 A MORE DETAILED STUDY OF INVERSION LAYER TRANSPORT

In general, the inversion layer current has two terms: drift and diffusion. The application of a potential difference across the ends of the inversion layer imposes an electric field that results in a drift current. As we saw in the main body of this chapter, there is also a lateral gradient of electron concentration that is produced. This causes a diffusion current. In general, then, the electron current in a differential of volume inside the inversion layer (see Fig. 9.50) is given by

$$J_{ey}(x,y) = -qn(x,y)v_{ey}^{\text{drift}}(x,y) + qD_e \frac{\partial n(x,y)}{\partial y} \tag{9.107}$$

Understanding the notation used in this equation is important. J_{ey} refers to the electron current along the y-direction. n is the electron concentration at a certain location in the inversion layer. v_{ey}^{drift} is the electron drift velocity along the y-direction. All of them, in general, depend on x and y.

To keep things simple, let us proceed under the assumption that the electric field is small enough so that there is a linear relationship between drift velocity and electric field. In this case, Eq. (9.107) can be written as

$$J_{ey}(x,y) = qn(x,y)\mu_e \mathcal{E}_y(x,y) + qD_e \frac{\partial n(x,y)}{\partial y} \tag{9.108}$$

In this equation, $\mathcal{E}_y$ refers to the lateral electric field set up in the inversion layer, which in general, depends also on x and y.

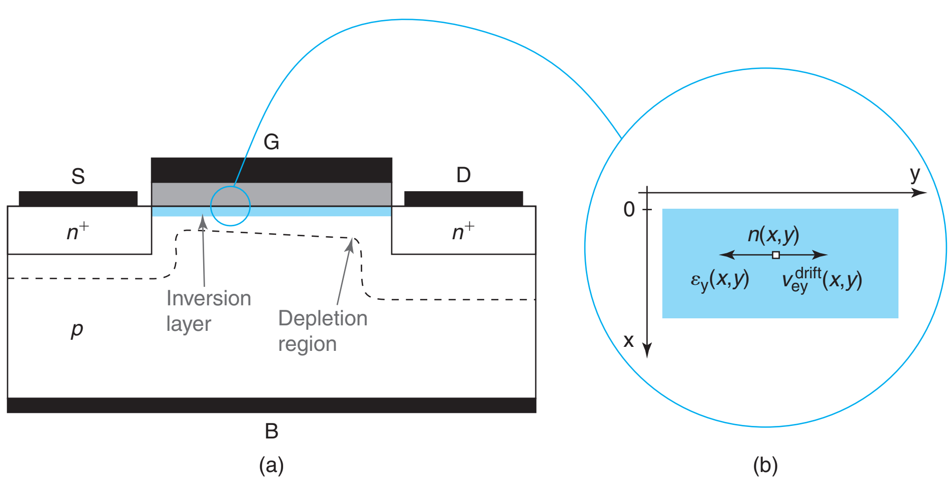


FIGURE 9.50
(a): Sketch of a four-terminal MOS structure biased in the linear regime.
(b): Detail of the electrostatics of a small element of the inversion layer.

We are rarely interested in the details of the electron current distribution vertically in the inversion layer, but on the total longitudinal electron current supported by the inversion layer. In consequence, we can eliminate unnecessary detail by integrating Eq. (9.108) in depth from the oxide–semiconductor interface to the bottom of the inversion layer,

$$J_e = \int_0^\infty J_{ey}(x,y)dx = q\mu_e \int_0^\infty n(x,y)\mathcal{E}_y(x,y)dx + qD_e \int_0^\infty \frac{\partial n(x,y)}{\partial y}dx \tag{9.109}$$

The integral starts at $x = 0$, the oxide–semiconductor interface, and should extend across the entire inversion layer. Since the electrons are all piled up against the interface, we make an insignificant mistake by extending the integral all the way to infinity. Also, since there are no vertical currents in the structure, we have dropped the y subindex in the current expression.

Three comments are proper here. First, the units of J_e in this equation are ampere per centimeter or current per unit width of the MOS structure. The *wider* the structure, the more total current it supports. In contrast, J_{ey} in Eqs. (9.107) and (9.108) is in ampere per square centimeter. Second, since there are no sources or sinks of electrons inside the inversion layer, J_e does not depend on y. And third, an assumption that has been made also here is that μ_e and D_e are independent of x. We can also view them as suitable average values for the entire electron distribution in depth. μ_e and D_e satisfy Einstein's relation.

AT9.1.1 The sheet-charge approximation

A significant simplification of this formulation is possible if the inversion layer can be considered very thin. As in Chapter 8, we call this the SCA. For a thin inversion layer, $\mathcal{E}_y$ in the drift term of Eq. (9.109) does not depend very much on x. We can then drop its x coordinate dependence and bring it out of the integral as $\mathcal{E}_y(y)$. The limits of validity of this assumption are explored later on in this section. Additionally, the $\partial/\partial y$ term in the diffusion term can be brought out of the second integral since its limits are independent of y (Leibniz rule). All this results in

$$J_e = q\mu_e n_s(y)\mathcal{E}_y(y) + qD_e \frac{dn_s(y)}{dy} \tag{9.110}$$

where n_s is the *sheet electron concentration* of the inversion layer (in units of cm^{-2}), defined in Eq. (9.1). In general, n_s depends on the lateral coordinate y.

A completely equivalent description can be given in terms of the inversion layer sheet charge density, Q_i, given by Eq. (9.3),

$$J_e = -\mu_e Q_i(y)\mathcal{E}_y(y) - D_e \frac{dQ_i(y)}{dy} \tag{9.111}$$

The SCA has allowed us to express the inversion layer current in terms of the inversion layer charge and the lateral electric field. The next step is to find a way to determine $Q_i(y)$.

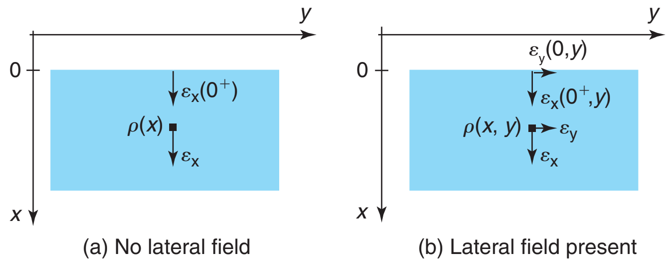


FIGURE 9.51
Electrostatics of the inversion layer. (a) In the absence of a lateral field, the electrostatics problem is simply one dimensional. (b) With a lateral field applied, the electrostatics problem becomes two dimensional.

AT9.1.2 The gradual-channel approximation

Before attempting to find an expression for $Q_i(y)$, it is useful to review the electrostatics of a MOS structure in the case in which no lateral field is applied to the inversion layer. This was studied in great detail in Chapter 8. As sketched in Fig. 9.51(a), this is strictly a one-dimensional electrostatic problem along the x-direction. Gauss' law for this problem can be written as

$$\frac{d\mathcal{E}_x}{dx} = \frac{\rho(x)}{\epsilon_s} \tag{9.112}$$

where $\mathcal{E}_x$ is the vertical field in the semiconductor. In this situation, $\mathcal{E}_x$ is independent of y. From here, we derived in Chapter 8 a simple relationship between Q_i and the gate voltage V_G that was valid for strong inversion and that was given by $Q_i = -C_{ox}(V_G - V_T)$. Note that this equation uses the inversion layer and the substrate as voltage reference ($V_S = V_B = 0$).

With a lateral field applied across the inversion layer, the situation gets more complicated. The electrostatic problem is now of a two-dimensional nature. Gauss' law for this situation is

$$\frac{\partial\mathcal{E}_x}{\partial x} + \frac{\partial\mathcal{E}_y}{\partial y} = \frac{\rho(x,y)}{\epsilon_s} \tag{9.113}$$

where, in general, $\mathcal{E}_x$ and $\mathcal{E}_y$ depend both on x and y.

In strong inversion and when current flows, $\mathcal{E}_y \neq 0$. In general, then, $\frac{\partial\mathcal{E}_y}{\partial y}$ does not vanish. The two-dimensional problem embodied in Eq. (9.113) is substantially harder than the one-dimensional problem solved in Chapter 8.

A simple approximate solution can be obtained under what is called the *GCA*. If the change of $\mathcal{E}_y$ along the y-direction is much slower than the change of $\mathcal{E}_x$ along the x-direction, that is, $\frac{\partial\mathcal{E}_x}{\partial x} \gg \frac{\partial\mathcal{E}_y}{\partial y}$, then Eq. (9.113) can be simplified to

$$\frac{\partial\mathcal{E}_x}{\partial x} \simeq \frac{\rho(x,y)}{\epsilon_s} \tag{9.114}$$

This equation has a similar solution to that of Eq. (9.112). Under strong inversion, the inversion-layer charge at location y is determined by the voltage overdrive applied to the gate with respect to the inversion layer at the same location, that is,

$$Q_i(y) \simeq -C_{ox}[V_G - V(y) - V_T] \tag{9.115}$$

Q_i now depends on lateral location y through the local inversion layer voltage $V(y)$ that was defined in Eq. (9.7).

The GCA allows the break up of the two-dimensional electrostatics problem into two simpler one-dimensional problems: the vertical electrostatics control the charge in the inversion layer, while the lateral electrostatics control the lateral flow of inversion layer charge. In the GCA, the lateral electrostatics represent a small perturbation of the vertical electrostatics. As a consequence, the inversion layer charge at any one location can be computed using the one-dimensional MOS theory developed in Chapter 8, with the simple precaution of using the *local* inversion layer voltage. At the end of this section, we will discuss the constraints on the applicability of the gradual-channel approximation.

We can now proceed with the derivation of a first-order model for the current. Taking the derivative of $Q_i(y)$ with respect to y in Eq. (9.115), while assuming that V_T does not change in space (negligible body effect), we can write

$$\frac{dQ_i(y)}{dy} \simeq C_{ox}\frac{dV(y)}{dy} \tag{9.116}$$

We can plug this into Eq. (9.111), apply Einstein’s relation, and also use Eq. (9.8) to get

$$J_e = \mu_e\left[Q_i(y) - \frac{kT}{q}C_{ox}\right]\frac{dV(y)}{dy} \tag{9.117}$$

Notice that since Q_i is negative, diffusion actually helps drift in electron transport along the inversion layer. This makes sense since we know that n_s drops along the channel from source to drain.

We can now discuss the relative magnitude of the drift and diffusion terms in Eq. (9.117). Since Q_i is given by Eq. (9.115), it is clear that drift prevails over diffusion if the amount of gate overdrive is much larger than the thermal voltage, that is, if $V_G - V(y) - V_T \gg \frac{kT}{q}$. This occurs when the MOS structure is in strong inversion. In this case, the expression of the inversion current simplifies to

$$J_e \simeq \mu_e Q_i(y)\frac{dV(y)}{dy} \tag{9.118}$$

This is the equation that we used to derive the I–V characteristics of the MOSFET in the linear and saturation regimes in the main body of this chapter.

Close to threshold, diffusion becomes important. In fact, below threshold, as we saw in Sec. 9.7.4, current conduction is dominated by electron diffusion. This is the so-called subthreshold current [beware, in this regime, Eq. (9.115) does not apply since it was derived for strong inversion; as a consequence, Eq. (9.117) does not apply either].

AT9.1.3 Validity of approximations

We can now discuss the limits of the GCA. As stated above, the GCA is valid when

$$\left|\frac{\partial \mathcal{E}_x}{\partial x}\right| \gg \left|\frac{\partial \mathcal{E}_y}{\partial y}\right| \tag{9.119}$$

Let us evaluate this inequality at $x = 0^+$, the semiconductor side of the oxide–semiconductor interface. The first term can be estimated by taking the difference in electric field at the surface and at the bottom of the inversion layer,

$$\left.\frac{\partial \mathcal{E}_x}{\partial x}\right|_{x=0^+} \simeq \frac{\mathcal{E}_x(x = x_{inv}) - \mathcal{E}_x(x = 0^+)}{x_{inv}} = \frac{Q_i}{\epsilon_s x_{inv}} \tag{9.120}$$

The second term can be estimated from Eq. (9.8),

$$\left.\frac{\partial \mathcal{E}_y}{\partial y}\right|_{x=0^+} = -\frac{d^2V}{dy^2} \tag{9.121}$$

The second derivative of V can be obtained from Eq. (9.117) by taking the derivative with respect to y and realizing that $dJ_e/dy = 0$. Hence,

$$\frac{d^2V}{dy^2} \simeq -\frac{1}{Q_i}\frac{dQ_i}{dy}\frac{dV}{dy} = -\frac{C_{ox}}{Q_i}\left(\frac{dV}{dy}\right)^2 = -\frac{C_{ox}}{Q_i}[\mathcal{E}_y(y)]^2 \tag{9.122}$$

where Eqs. (9.116) and (9.8) have been used [the term on $\frac{kT}{q}C_{ox}$ in Eq. (9.117) was neglected, see justification below].

Combining Eqs. (9.119), (9.120), (9.121), and (9.122), the condition for the GCA can be rewritten as

$$|\mathcal{E}_y(y)| \ll \frac{|Q_i|}{\sqrt{\epsilon_s x_{inv} C_{ox}}} \tag{9.123}$$

And using the expression of Q_i given by Eq. (9.115), it can also be expressed as

$$|\mathcal{E}_y(y)| \ll \frac{V_G - V(y) - V_T}{\sqrt{\frac{\epsilon_s}{\epsilon_{ox}} x_{inv} x_{ox}}} \tag{9.124}$$

In essence, for the GCA to be fulfilled, the lateral electric field must be much smaller than an effective vertical field given by the voltage overdrive on the gate divided by an effective vertical characteristic length [the denominator in Eq. (9.124)]. This vertical characteristic length is a function of the oxide thickness, the inversion layer thickness, and the permittivities of both materials. The smaller this characteristic length, the easier it is for the GCA to be fulfilled.

Let us put some typical numbers to examine the validity of the GCA. For a MOSFET structure with an oxide thickness of $x_{ox} = 4.5$ nm and a p-type substrate of $N_A = 6 \times 10^{17}$ cm^{-3}, $x_{inv} \simeq 7.5$ nm, and the vertical characteristic length turns out to be about 10 nm. For a location in which $V_G - V(y) - V_T = 1$ V, the lateral field must be much smaller than 10^6 V/cm, which is an extraordinarily high value. Hence, except right around threshold when $V_G - V(y) - V_T \simeq 0$, the gradual-channel approximation is indeed an excellent one.

We now turn our attention to verifying the SCA that was made at the beginning of this section when $\mathcal{E}_y(x, y)$ was taken out of the integral in Eq. (9.109). This can be done with minimum error if $n(x, y)$ changes with x much faster than $\mathcal{E}_y(x, y)$. Mathematically, this condition can be expressed as

$$\left|\frac{1}{\mathcal{E}_y}\frac{\partial \mathcal{E}_y}{\partial x}\right| \ll \left|\frac{1}{n}\frac{\partial n}{\partial x}\right| \tag{9.125}$$

Let us examine this inequality at $x = 0^+$. It is handy to rewrite the left-hand side in the following way. Since $\vec{\nabla} \times \vec{\mathcal{E}} = 0$, it follows that

$$\frac{\partial \mathcal{E}_y}{\partial x} = \frac{\partial \mathcal{E}_x}{\partial y} \tag{9.126}$$

inside the inversion layer.

In the context of the GCA, at $x = 0^+$,

$$\mathcal{E}_x|_{x=0^+} \simeq -\frac{Q_i + Q_d}{\epsilon_s} \tag{9.127}$$

Hence,

$$\left.\frac{\partial \mathcal{E}_x}{\partial y}\right|_{x=0^+} \simeq -\frac{1}{\epsilon_s}\frac{dQ_i}{dy} = \frac{C_{ox}}{\epsilon_s}\mathcal{E}_y(y) \tag{9.128}$$

where we have used Eqs. (9.116) and (9.8).

Combining Eq. (9.128) and (9.126), we can write

$$\left|\frac{1}{\mathcal{E}_y}\frac{\partial \mathcal{E}_y}{\partial x}\right|_{x=0^+} = \left|\frac{1}{\mathcal{E}_y}\frac{\partial \mathcal{E}_x}{\partial y}\right|_{x=0^+} \simeq \frac{C_{ox}}{\epsilon_s} \tag{9.129}$$

Now we work on the right-hand side of Eq. (9.125). From Chapter 8, we know that

$$n(x) = N_A \exp\frac{q[\phi(x) - 2\phi_f]}{kT} \tag{9.130}$$

Therefore,

$$\frac{1}{n}\frac{\partial n}{\partial x} = \frac{q}{kT}\frac{\partial \phi}{\partial x} = -\frac{q}{kT}\mathcal{E}_x \tag{9.131}$$

$\mathcal{E}_x$ is given by Eq. (9.127). Hence,

$$\left|\frac{1}{n}\frac{\partial n}{\partial x}\right|_{x=0^+} \simeq \left|\frac{q}{kT}\frac{Q_i + Q_d}{\epsilon_s}\right| \tag{9.132}$$

Plugging in Eqs. (9.129) and (9.132), the condition in Eq. (9.125) at $x = 0^+$ can be written as

$$\frac{C_{ox}}{\epsilon_s} \ll \left|\frac{q}{kT}\frac{Q_i + Q_d}{\epsilon_s}\right| \tag{9.133}$$

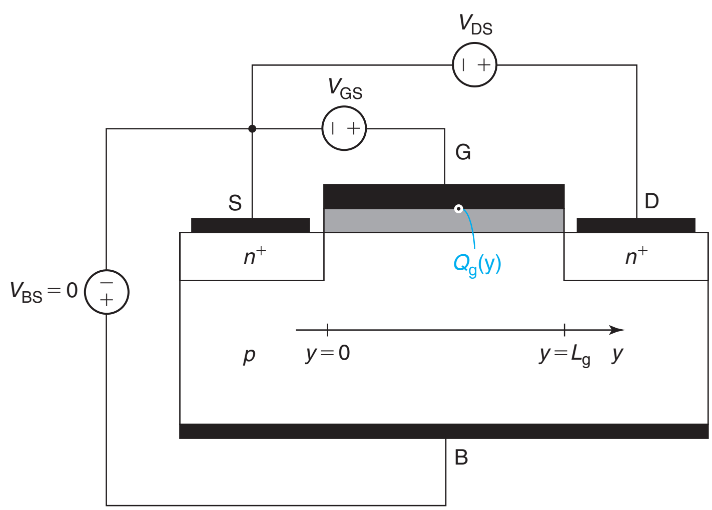
Since Q_i and Q_d have the same sign, this condition is satisfied if

$$|Q_i| \gg \frac{kT}{q}C_{ox} \tag{9.134}$$

This is exactly the mathematical expression derived above for the GCA. Hence, under conditions in which the gradual approximation applies, $\mathcal{E}_y$ changes sufficiently slowly inside the inversion layer that it can safely be taken out of the integral in Eq. (9.109) and the SCA also holds.

PROBLEMS

9.1 Consider a long n-channel MOSFET biased in the linear regime. This problem is about answering a simple question: does the sheet charge density imaged on the *gate*, $Q_g(y)$, increase, decrease, or remains constant as one goes from source to drain ($y = 0$ to $y = L_g$)?



You are to answer this question in a quantitative way by calculating the sheet charge density imaged on the gate of a long MOSFET characterized by the following parameters: n^+-polysilicon gate ($W_M = \chi_S = 4.04$ eV), $x_{ox} = 15$ nm, uniform $N_A = 10^{17}$ cm^{-3}, at a bias given by $V_{GS} = 2.5$ V, $V_{DS} = 1$ V, and $V_{BS} = 0$.

To solve this problem, you need to account for the impact of the *body effect*. Otherwise, consider this MOSFET to be ideal.

To save you time, for this structure:

$$C_{ox} = 2.3 \times 10^{-7}\ \text{F/cm}^2,\ \gamma = 0.8\ \text{V}^{1/2},\ \phi_{sT} = 0.84\ \text{V},\ V_{FB} = -1\ \text{V},\ V_{To} = 0.56\ \text{V}$$

(a) Compute the sheet charge density of electrons in the inversion layer at the *source* end of the channel, $Q_i(y = 0)$.
(b) Compute the charge in the depletion region under the channel at the *source* end of the channel, $Q_d(y = 0)$.
(c) Compute the sheet charge density of electrons in the inversion layer at the *drain* end of the channel, $Q_i(y = L_g)$.

(d) Compute the charge in the depletion region under the channel at the *drain* end of the channel, $Q_d(y = L_g)$.

(e) Compute the sheet charge density imaged at the gate/oxide interface at the source and drain ends of the channel, $Q_g(y = 0)$ and $Q_g(y = L_g)$. Which one is highest?

9.2 Consider a long-channel n-MOSFET in the linear regime with $V_{BS} = 0$. Neglecting the body effect, derive analytical expressions for $V(y)$, $\mathcal{E}_{ox}(y)$, $\mathcal{E}_y(x = 0, y)$, $v_e(y)$, and $Q_i(y)$ from source to drain.

9.3 Derive an analytical expression for the condition that gives how close V_{DS} can get to $V_{GS} - V_T$ in a MOSFET biased in the linear regime before the GCA fails. Express your result in terms of $\Delta V_{DS} = V_{GS} - V_T - V_{DS}$. Do not include the body effect. $V_{SB} = 0$. You will need to solve Problem 9.2 first. Compute ΔV_{DS} for a MOSFET characterized by $L = 0.1$ µm, $x_{ox} = 4.5$ nm, and $N_A = 6 \times 10^{17}$ cm^{-3}. What fraction of the maximum current predicted by Eq. (9.16) is attained at this maximum tolerable value of V_{DS}?

9.4 Consider an n-channel MOSFET in the saturation regime.

(a) Derive expressions and sketch the spatial dependence of $V(y)$, $\mathcal{E}_{ox}(y)$, $\mathcal{E}_y(y)$, $v_e(y)$, and $Q_i(y)$ from source to drain. Explain your results.

(b) Derive also an expression for the electron velocity due to diffusion. Compare the diffusion velocity with the drift velocity obtained in part (a). Under what conditions is electron diffusion safely ignored?

(c) Obtain an expression for the transit time of electrons through the channel from source to drain by computing

$$\tau_t = \int_0^L dt = \int_0^L \frac{dy}{v_e(y)}$$

(d) Compute the transit time again using the following expression:

$$\tau_t = \frac{|Q_I|}{I_D} = \frac{W_g |\int_0^L Q_i(y)dy|}{I_D}$$

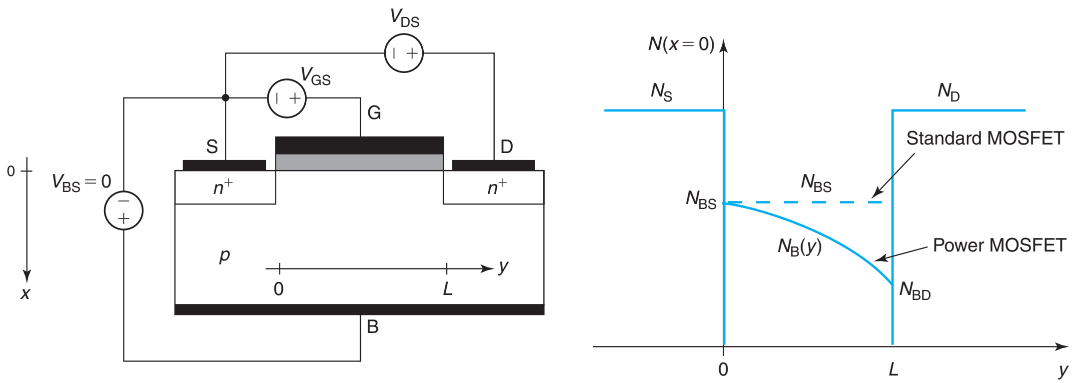
Compare this result with the one obtained in part (c).

For all parts, neglect the body effect, that is, assume that the current of the transistor is given by Eq. (9.21).

9.5 In power MOSFETs, the body doping is often non-uniform. This problem examines some of the consequences of this.

Consider a power MOSFET as sketched in the diagram below. In this device, the doping level in the body drops along the channel from source to drain, as sketched in the diagram on the right. This diagram shows the doping distribution right along the surface of the device. N_S and N_D refer to the n-type doping level of the source and drain. N_A is the p-type doping level in the body that changes underneath the channel along the y-dimension, but not along the x-dimension. N_{BS} refers to the body doping at the source end of the channel ($y = 0$). N_{BD} refers to the body doping at the drain end of the channel ($y = L$).

In this problem, we will compare the operation of the power MOSFET with that of a regular MOSFET with a uniform doping level in the body equal to N_{BS}, as indicated by the dashed line in the diagram. In this way, these two devices have an identical threshold voltage.



This is a long-channel MOSFET problem. Other than a nonuniform body doping, consider this device, as well as the reference standard MOSFET, as an ideal long-channel MOSFET as defined in this chapter.

(a) Due to the fact that the body doping changes along the channel, the *local* threshold voltage of the inversion layer is also a function of position. How does $V_T(y)$ evolve along the channel? Sketch a diagram of $V_T(y)$ versus y for the power MOSFET and the regular MOSFET. Explain your graph.

(b) Now consider these two devices biased in the linear regime with an identical $V_{GS} > V_T$ and an identical and small V_{DS}. Sketch the absolute of the inversion layer charge $|Q_i(y)|$ along the channel from source to drain for both devices. Explain your graph.

(c) Now consider these devices biased in saturation at an identical $V_{GS} > V_T$ and an identical and large $V_{DS} > V_{DSsat}$. Sketch a diagram of the voltage along the channel $V(y)$ versus y for the power MOSFET and the regular MOSFET. Explain your graph.

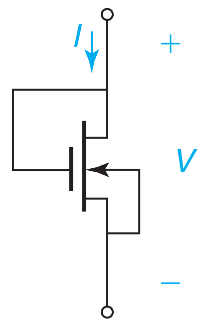
(d) Under the bias condition of part (c), which device has the highest drain current? Why?

(e) For an identical value of $V_{GS} > V_T$, sketch a diagram of I_D versus V_{DS} for the power MOSFET and the regular MOSFET from $V_{DS} = 0$ to $V_{DS} > V_{DSsat}$. Explain your graph.

(f) Under the bias condition of part (c), which device has the highest transconductance? Why?

9.6 You have been given the specs of a foundry's digital CMOS technology that you are considering for one of your company's designs. Some of the values that characterize the n-MOSFET at room temperature are: $I_{off} = 1$ nA/μm, $V_T = 0.5$ V, and $S = 70$ mV/dec. For your design, it is crucial that you know I_{off} at 85 °C, which is not given in the spec sheet. You consult some books and you find that V_T changes typically about -4 mV/°C. Also, the inversion layer mobility goes approximately as T^{-2}. With all this, *estimate* I_{off} at 85 °C. State any assumptions that you need to make.

9.7 Analog designers often need to shift DC bias levels in circuits. This is best accomplished using PN diodes. In standard CMOS, floating PN diodes (that is, diodes not directly connected to the power rails) are not readily available. A common way to "synthesize" a diode is to tie up an enhancement-mode MOSFET ($V_T > 0$) with the gate and drain shorted as sketched below. For this, a long MOSFET is typically used.



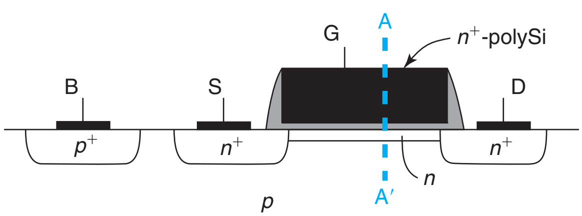
In responding to the questions below, use "long-MOSFET" theory without body effect nor back bias effect.

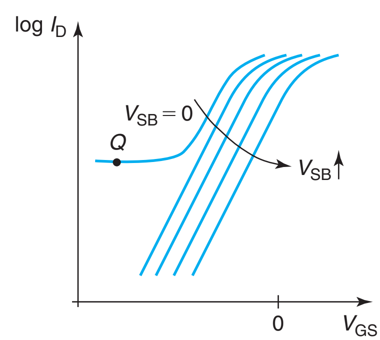
(a) Derive suitable equations for the I–V characteristics of the "diode," I vs. V for $V < V_T$ and $V > V_T$. Sketch the I–V characteristics in a linear scale.

(b) Sketch a high-frequency small-signal equivalent circuit model for this "diode" for situations in which $V > V_T$. Derive expressions for all important small-signal elements as a function of bias.

(c) Derive and sketch the I–V characteristics of the "diode" if implemented with a depletion-mode device ($V_T < 0$).

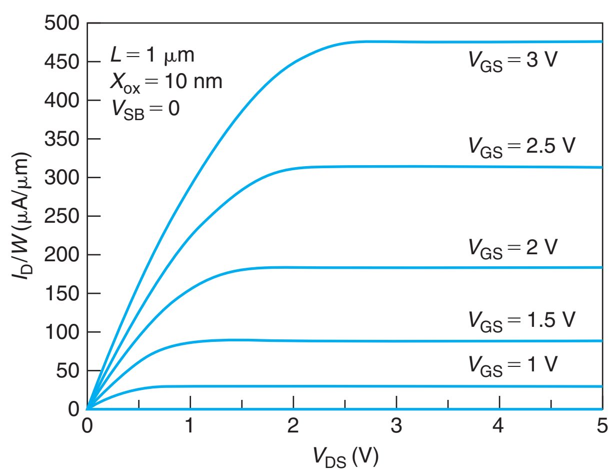
9.8 A depletion-mode MOSFET is an FET that is *on* at $V_{GS} = 0$ V. For an n-MOSFET, this is usually accomplished by purposely introducing an n-type doped region underneath the gate, as sketched below. This is called a *buried-channel MOSFET*.





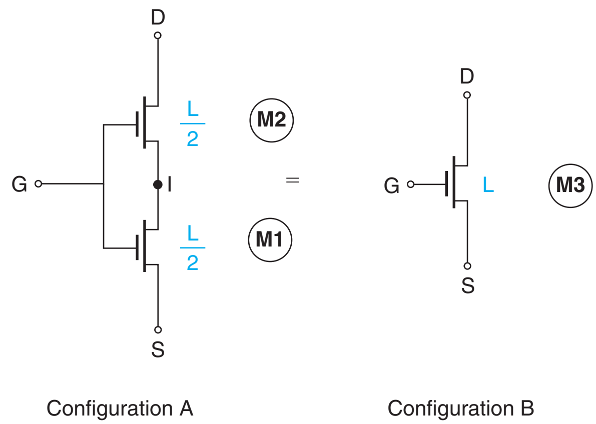
(a) Sketch an energy band diagram for $V_{GS} = V_{DS} = V_{BS} = 0$ through the cross section A–A′ as shown in the figure on page 604. Indicate the presence of depletion, accumulation, or inversion regions.
(b) Sketch an energy band diagram with the MOS structure *at flat-band* across the A–A′ section. Provide an expression for the flat-band voltage V_{FB} in terms of appropriate parameters. V_{FB} should be referred to as V_{GS}.
(c) The subthreshold characteristics of a poorly designed buried-channel n-MOSFET are also sketched above as a function of the back bias. It is found that for V_{SB} sufficiently positive, the transistor turns off nicely. However, for $V_{SB} = 0$, the transistor fails to turn off completely. What is going on? Sketch the energy band diagram across the A–A′ section at bias point **Q**.

9.9 Consider the output I–V characteristics of a long-channel n-MOSFET for $V_{SB} = 0$ below. The device has a gate length $L_g = 1$ μm and a gate oxide thickness $x_{ox} = 10$ nm. The output characteristics have been normalized for a unity width device.



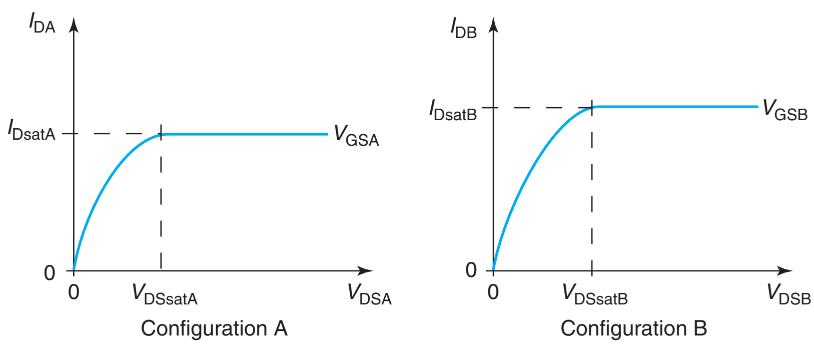
(a) Assuming that this device is an ideal long-channel MOSFET, estimate its threshold voltage.
(b) Assuming that this device is an ideal long-channel MOSFET, estimate the mobility of the electron inversion layer.
(c) Assuming that this device is an ideal long-channel MOSFET, estimate the transconductance g_m at a bias of $V_{GS} = 3$ V and $V_{DS} = 3$ V for $W_g = 10$ μm.
(d) At a bias of $V_{GS} = 3$ V and $V_{DS} = 3$ V, estimate the gate-source capacitance C_{gs} of a $W_g =$ 10 μm device.
(e) Now you are given the additional information that the subthreshold swing of this transistor technology at room temperature is $S = 78$ mV/dec. At a bias of $V_{GS} = 3$ V and $V_{DS} = 3$ V, estimate the back-gate transconductance g_{mb} of a $W_g = 10$ μm device.
(f) At a bias of $V_{GS} = 3$ V and $V_{DS} = 3$ V, how much less current do you expect to have as a result of the body effect? (Consider the additional subthreshold current information, if needed.)

9.10 One of my colleagues asserts that:



This basically means that a configuration of two MOSFETs in series with a common gate voltage is equivalent to a single MOSFET with a channel length that is twice as long; all other aspects of the device are unchanged. This problem is about evaluating this assertion for an ideal MOSFET (no body effect, no back bias, no channel-length modulation, and no short-channel effects).

Consider the following sketches of the I–V characteristics for these two configurations for the same value of $V_{GS} > V_T$:



Label the internal node in configuration A as "I", that is, refer to its voltage with respect to the source as V_{IS}.

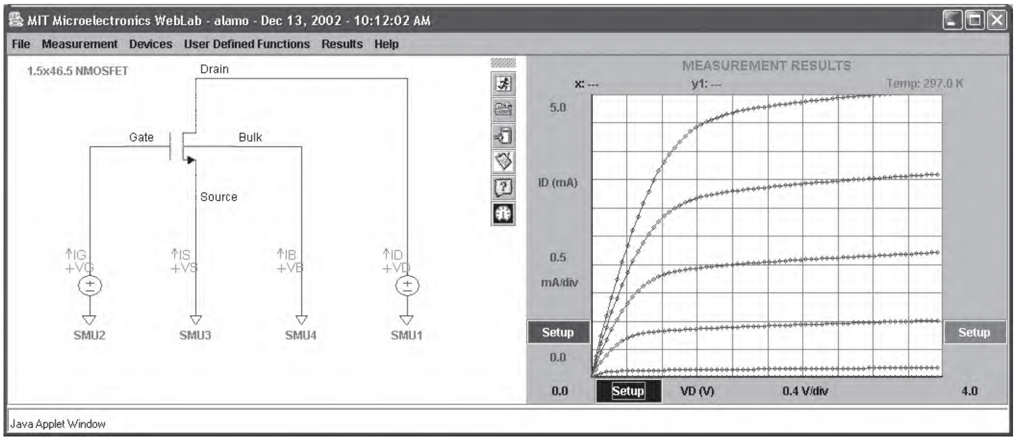
(a) For $V_{DS} > V_{DSsat}$, in what regime is each of the transistors in configuration A biased?

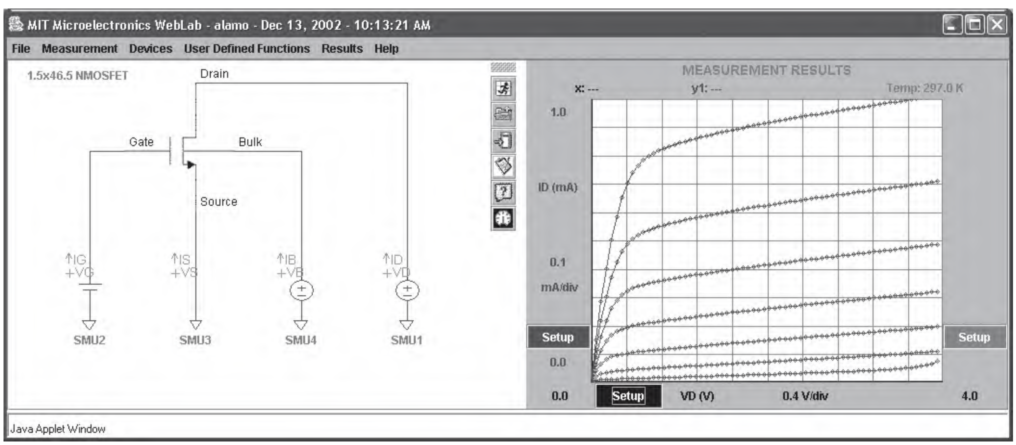
(b) For configuration A and for $V_{DS} > V_{DSsat}$, derive equations for the drain current through each transistor as a function of V_{GS}, V_{DS}, V_T, and V_{IS}. From this, derive an expression for V_{IS}.

(c) For configuration A, derive an expression for I_{Dsat} as a function of V_{GS}, V_{DS}, V_T, and structural parameters. How does this compare with I_{Dsat} of configuration B for the same values of V_{GS} and V_{DS}?

(d) For configuration A, derive an expression for V_{DSsat}. How does it compare with that of configuration B for the same V_{GS}?

9.11 Below are two screen shots of measurements taken on a 1.5×46.5 μm NMOSFET. The first one shows the output characteristics obtained for $V_{SB} = 0$ V and V_{GS} from 0 to 3 V, in steps of 0.5 V. The second one shows the output characteristics obtained with the roles of the gate and body reversed. In these, $V_{GS} = 1.5$ V and V_{SB} is stepped from 0 to 3 V, in steps of 0.5 V.





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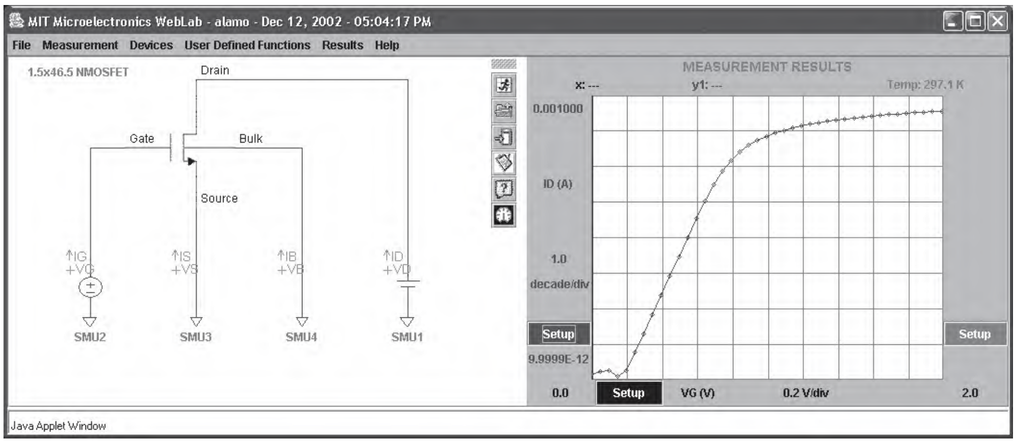
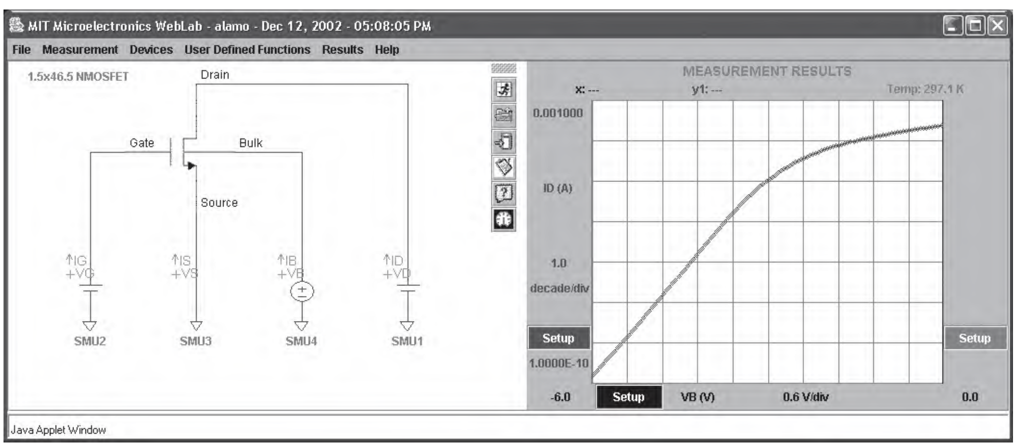
As we discussed in class, the body of the MOSFET behaves as a gate and reasonably looking output characteristics are obtained.

(a) One of the most striking differences between the two sets of output characteristics is V_{DSsat}. The characteristics obtained with the body operating as the main gate seem to display remarkably smaller values of V_{DSsat} than in the normal mode. This question is about understanding the origin of this.

Derive a simple expression for V_{DSsat} as a function of V_{SB} for an ideal MOSFET. Do not include the “body effect” (but obviously include the “back bias” effect). Is this expression consistent with the data above? Explain.

Below are the subthreshold characteristics obtained for this MOSFET for both modes of operation. In the normal mode (top), $V_{SB} = 0$. In the mode in which the body is used as gate (bottom), $V_{GS} = 1.5$ V. In both cases, $V_{DS} = 0.1$ V.





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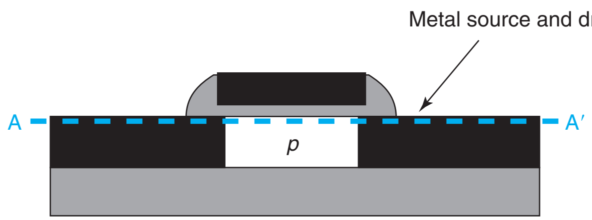
A remarkable difference is also observed here: the subthreshold swing of the device in the normal mode is significantly sharper than in the mode in which the body is used as the main gate.

(b) Derive an expression for the relationship between the subthreshold swing of a MOSFET in the normal mode and in the gate-reversed mode. Explain your result.

(c) From the experimental characteristics of the previous page, extract the subthreshold swing of this device for both modes of operation. Compare their relationship with that predicted with the equation derived above. Discuss.

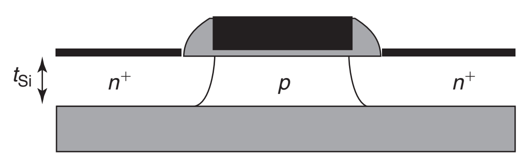
9.12 A Schottky barrier source and drain MOSFET has been under study as an approach to reduce the source and drain resistance and eliminate latch up. A schematic of this device is shown below. This problem is about drawing energy band diagrams in order to understand its basic operation.



Consider a device with a threshold voltage $V_T = 0.5$ V, body doping level $N_A = 10^{17}$ cm^{-3}, and source and drain Schottky barrier height $q\varphi_{Bp} = 0.5$ eV. The gate is made up of n$^+$ polySi.

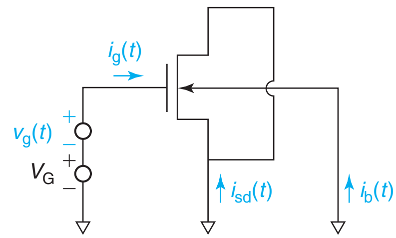
(a) Sketch an energy band diagram right along the surface of the device (cross section A–A′ above) for $V_{GS} = 1$ V and $V_{DS} = 0$. Give as many values of key energy features as you can.
(b) Sketch an energy band diagram right along the surface of the device (cross section A–A′ above) for $V_{GS} = V_{DS} = 1$ V. Give as many values of key energy features as you can.
(c) How does this device operate? Can you see any drawbacks?

9.13 The below figure depicts a fully depleted silicon-on-insulator (SOI) n-channel MOSFET. This is a transistor in which the body is fully depleted in the regular region of operation. For this transistor, $L = 1$ μm, $W = 5$ μm, $x_{ox} = 20$ nm, and $N_A = 10^{17}$ cm^{-3}. The Si thickness is $t_{Si} = 10$ nm. The gate is made up of n$^+$ polySi.



(a) With $V_{DS} = 0$ V, calculate the value of V_{GS} that fully depletes the body of the transistor. Sketch the energy band diagram vertically through the middle of the device at this bias point.
(b) Calculate the threshold voltage of this device. Sketch the energy band diagram vertically through the middle of the device at this bias point.
(c) Estimate the subthreshold swing of the device at room temperature.

9.14 The "split C–V technique" is widely used to characterize MOSFETs. This problem is about understanding some fundamental aspects of this technique. The configuration for the split C–V technique is shown in the figure below:

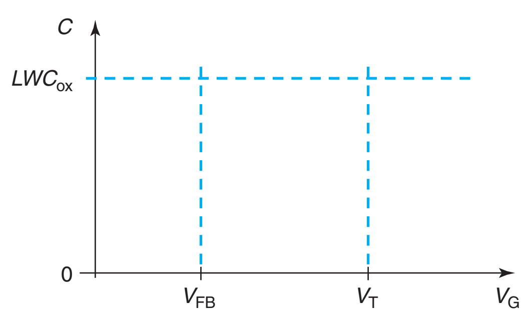


In essence, the source and drain are shorted together and grounded. The body is also grounded. The gate bias is set to V_G through a DC source. A small-signal AC signal v_g is applied on top of V_G. The current through all the terminals is measured as indicated in the figure. From these currents, the gate-to-channel capacitance, C_{gc}, and the gate-to-body capacitance C_{gb} are obtained. These capacitances are defined as

$$C_{gc} = -\frac{i_{sd}}{\frac{dv_g}{dt}} \qquad\qquad C_{gb} = -\frac{i_b}{\frac{dv_g}{dt}}$$

The physical dimensions of the gate of this transistor are L and W.

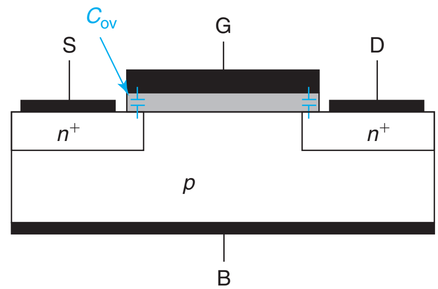
(a) In the axes shown below, sketch the evolution of C_{gc} with V_G across the entire V_G range for a low-frequency AC signal. Explain the key features of your drawing.



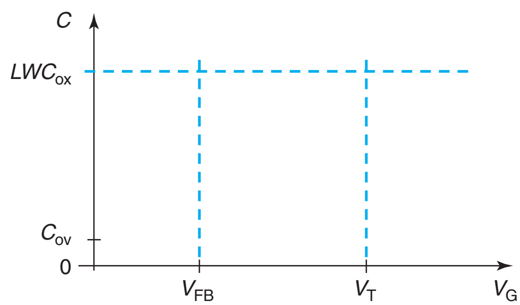
(b) In the same axes as in part (a), sketch the evolution of C_{gb} with V_G across the entire V_G range for a low-frequency AC signal. Explain the key features of your drawing.

(c) In the same axes as in part (a), sketch the evolution of C_{gc} and C_{gb} with V_G across the entire V_G range for a *high-frequency* AC signal.

(d) Now consider the impact of the overlap capacitances between the gate and the source and drain extensions of the transistor.



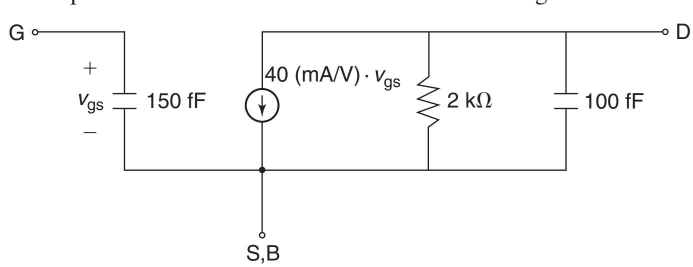
In the axes below, sketch again the evolution of C_{gc} and C_{gb} with V_G at low frequency now accounting for the presence of C_{ov}. The relative value of C_{ov} is indicated in the figure below.



(e) The split C–V technique when combined with DC measurements of the drain current of the MOSFET in the linear regime can be used to experimentally obtain the mobility dependence on $\mathcal{E}_{eff}$. Explain how this would work. Do this ignoring C_{ov}.

9.15 In the calculation of the drain current in an ideal MOSFET we neglected the contribution of diffusion. Justify this assumption by calculating an expression for the ratio of the diffusion velocity over the drift velocity as a function of position in the channel of a long n-channel MOSFET biased in the saturation regime. Discuss your result. In this problem, neglect the body effect. You might want to use results that you obtained from solving Problem 9.2.

9.16 Below is the small-signal equivalent circuit model of an ideal MOSFET biased in saturation at a certain V_{DS} and V_{GS}. The body is shorted to the source ($V_{BS} = 0$). The model does include a nonideal finite output conductance that arises from channel-length modulation.



In each of the sections that follow, calculations are to be performed of the new values of the elements of the small-signal equivalent circuit model of the transistor if one parameter changes with respect to the situation shown above. If there is not enough information for a numerical answer to be obtained, explain and give the formula that would allow you to calculate the value. In every case, only one parameter is changed at a time.

(a) V_{GS} increases such that the drain current is doubled.
(b) V_{DS} is doubled.
(c) The device width W is doubled (V_{GS} and V_{DS} do not change).
(d) The channel length L is doubled (V_{GS} and V_{DS} do not change).

Chapter 10

The “Short” Metal–Oxide–Semiconductor Field-Effect Transistor

The MOSFET is the work horse of the IC industry. 50 years after the invention of the integrated circuit, MOSFET ICs can be found in digital systems, solid-state memories, analog products, power electronics, imagers, and radio-frequency circuits, among many other applications. What accounts for the success of the MOSFET? From our study of the long-channel MOSFET in Chapter 9, we understand that one of its most interesting and relevant attributes can be summarized in *“smaller is better.”* What is meant by this is that in many applications, most notably logic, smaller MOSFETs have more desirable properties. We learned in Chapter 9 that shortening the gate length and other dimensions of the MOSFET not only allows more of them to be packed together but it also increases the drain current density, transconductance, and frequency response. The improvements in these key performance metrics also enable their partial trade-off against reductions in the operating voltage, something very valuable in many applications as it results in lower power consumption.

The success of the MOSFET is also due to the unique characteristics of CMOS, a simple circuit configuration that pairs an n-channel and a p-channel transistor. CMOS allows the synthesis of logic circuits with negligible standby power consumption, a key for the integration of complex ICs, such as microprocessors, with millions of transistors. CMOS has also been instrumental in obtaining high-performance and low-cost analog and communications circuits. CMOS and MOSFET scaling have kept the IC industry youthful by regularly deploying new technology with ever-improving performance, power efficiency, transistor density, and at dramatically falling costs.

MOSFET size reduction presents some interesting theoretical and practical problems. In the models of Chapter 9, for example, we assumed mobility-limited transport. This is a suitable approximation if the electric field along the channel does not get too high. But, size scaling means that electric fields will tend to increase. At some point, velocity saturation effects become relevant. If neglected, transistor performance will be overestimated. This is just one example of the so-called short-channel effects or phenomena that must be taken into account to correctly describe short MOSFETs. This chapter is dedicated to describing unique issues of relevance to scaled MOSFETs and building simple models that can be used to guide short-channel MOSFET design.

This chapter starts with three sections dedicated to short-channel effects. The first section deals with transport issues where topics such as mobility degradation and velocity saturation are discussed. The second section is dedicated to electrostatics. This refers to understanding the dependence of the threshold and subthreshold behavior of the MOSFET as a function of geometry and voltage. The third section deals with gate stack scaling, the evolution of gate capacitance, and gate leakage current as the transistor is scaled in the vertical dimension.

MOSFET scaling has enabled voltage scaling. However, as we will see in this chapter, voltage has not scaled as fast as MOSFET dimensions. This means that electric fields have gradually been going up. In addition to velocity saturation and gate leakage current, there are other high-field effects that have become important and need to be studied. A separate section deals with some of the most significant ones.

The final section of this chapter is dedicated to describing how MOSFET scaling has actually taken place. Helpful in this description are two simple scaling scenarios: constant-field scaling in which scaling takes place keeping the lateral and vertical electric fields constant, and constant-voltage scaling, in which the operating voltage is kept constant. For many years, MOSFET scaling took place, in essence, as successive waves of these two modes of scaling. In more recent times, scaling has become more complex and the notion of "generalized scaling" was introduced. The rest of the section is dedicated to providing a historical perspective of scaling and the evolution of MOSFET design.

This chapter is not the end of the MOSFET in this book. There are a few important bipolar issues in MOSFETs that cannot be described until the bipolar transistor is introduced in the next chapter. A few selected topics are discussed in an appendix to Chapter 11.

10.1 MOSFET SHORT-CHANNEL EFFECTS: TRANSPORT

This section discusses short-channel effects that affect the transport of electrons in the channel of a MOSFET. The simple formulation of the ideal MOSFET that was presented in Chapter 9 fails in two significant ways. First, it has been experimentally observed that the inversion layer mobility decreases as the carrier concentration increases in the inversion layer. This means that at high gate overdrive, the electron current will be smaller than otherwise expected.

In addition to this, there is the impact of velocity saturation. An assumption in the ideal MOSFET formulation of Chapter 9 is that the electron velocity in the channel is linearly proportional to the electric field along the channel. We know that due to velocity saturation, this is not the case. At high enough lateral fields, the velocity saturates to a maximum value given by the saturation velocity. This obviously limits the current that flows through the MOSFET.

The combined impact of these two phenomena in the current–voltage characteristics of a MOSFET is very substantial. For submicron devices, the discrepancy between the experimentally measured current and the theoretically expected value from the ideal MOSFET formulation can easily be larger than one order of magnitude.

The next two sections discuss these two short-channel effects.

10.1.1 Mobility degradation

The mobility of carriers in the inversion layer of a MOSFET is significantly lower than in bulk silicon. The abrupt disruption to the crystalline periodicity of the semiconductor lattice at its

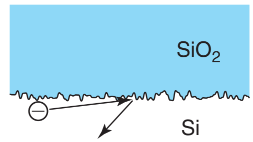


FIGURE 10.1
Sketch of electron scattering due to interface roughness in an inversion layer at the Si/SiO$_2$ interface.

interface with the oxide aggravates carrier scattering. A new scattering mechanism known as *interface roughness scattering* appears. The origin of this is the fact that the Si/SiO$_2$ interface is not perfectly specular but has some degree of roughness, as was discussed in the context of Figure 8.5. As carriers drift along the surface of the semiconductor from source to drain, a rough interface makes some of them bounce back, thus reducing the overall forward flow down the inversion layer. This is sketched in Figure 10.1.

In addition to the introduction of interface roughness scattering, phonon scattering is also enhanced at an inversion layer. This stems from the spatial confinement of the carriers in the vertical dimension that increases their ability to couple with vibrational modes of the lattice.

Both phonon scattering and interface roughness scattering become more severe the more the inversion layer carriers are jammed against the oxide–semiconductor interface. This is exacerbated for high gate overdrive when the inversion layer well is deepest. As a consequence, the higher the inversion layer sheet carrier concentration, the lower the mobility. This can be seen in Figure 10.2 where the experimental inversion layer mobilities for electrons and

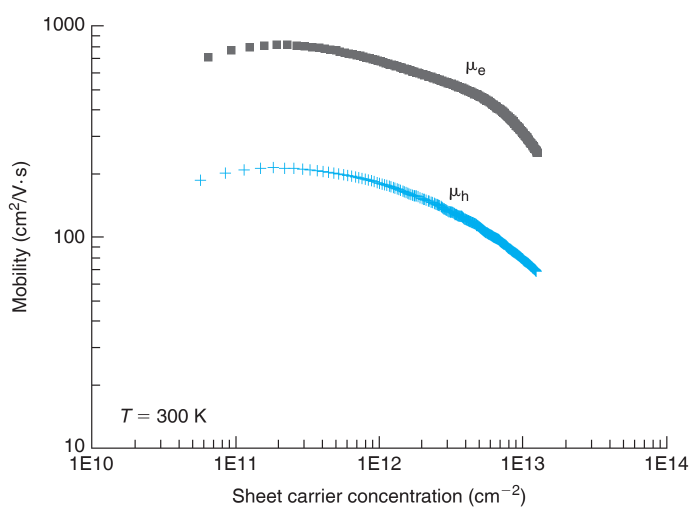


FIGURE 10.2
Electron and hole inversion layer mobility as a function of their respective sheet carrier concentrations for (100) Si at 300 K [Data courtesy of Shinichi Takagi (S. Takagi et al., IEEE Trans. *Electron Dev*. Vol. 41, p. 2357 (1994))].

holes are graphed as a function of their respective sheet carrier concentrations in the channel. In the limit of very weak confinement and very low sheet carrier concentration, the carrier mobilities approach values typical of bulk material. However, as the sheet carrier concentration increases and electrons become more tightly squeezed against the Si/SiO_2 interface, the carrier mobilities are reduced dramatically. This is what is known as *mobility degradation*.

The degradation in mobility with sheet carrier concentration is very significant. For example, from the lowest sheet electron concentration to the highest value plotted in Figure 10.2, the electron mobility drops from above 900 to below 300 $cm^2/V \cdot s$. Needless to say, this has a very sizable impact on the current drive that the inversion layer can support in a MOSFET.

The cleanest experimental manifestation of mobility degradation in the *I–V* characteristics of a MOSFET is observed in the linear regime. Figure 10.3 shows the drain current and the transconductance of an otherwise ideal MOSFET for a small value of V_{DS} such that the device is in the linear regime. As we discussed in Chapter 9, for low values of V_{DS}, soon after the device turns on, it enters the linear regime and the drain current should rise linearly with V_{GS}. In this regime, we ideally expect the transconductance to be constant with V_{GS}. As can be seen in the sketches in Figure 10.3, in reality, it is observed that the current rises more slowly than linearly and that the transconductance decreases as V_{GS} increases.

How mobility degradation is responsible for this can be easily understood by looking at the expression of the transconductance in the linear regime in Eq. (9.51) in which we see that the transconductance is directly proportional to the electron mobility.[1] Mobility degradation results in a loss of current drive not only in the linear regime [Figure 10.3(a)] but also in the saturation regime.

A striking feature of mobility degradation is what is referred to as its *universal* behavior. It has been found that when measurements of the inversion layer mobility obtained under a variety of conditions in different MOS structures are plotted against an "effective vertical field" (defined below), a nearly universal relationship emerges that applies to many MOS structures

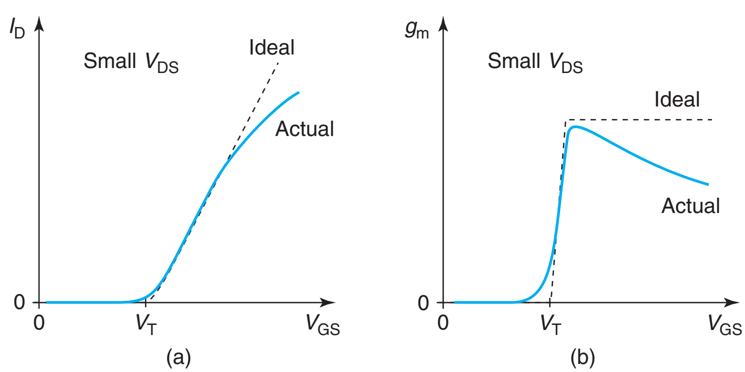


FIGURE 10.3
Sketch of drain current and transconductance in a MOSFET in the linear regime. The discontinuous line represents the predictions of the ideal MOSFET. The continuous lines are experimental observations.

[1]One should be mindful here that Eq. (9.51) was derived under the assumption that μ_e is constant which is strictly not the case in the presence of mobility degradation.

regardless of their physical parameters and the details of the applied bias. This is shown in Figure 10.4 for electrons and holes and is known as the *universal mobility curve*.

The two graphs in Figure 10.4 plot the electron and hole mobilities against an effective electric field for several substrate doping levels. For each doping level value, there is a characteristic shape to the evolution of mobility with effective field. As $\mathcal{E}_{eff}$ increases, the mobility initially increases. For high enough values of $\mathcal{E}_{eff}$, the mobility starts decreasing. It is this decreasing

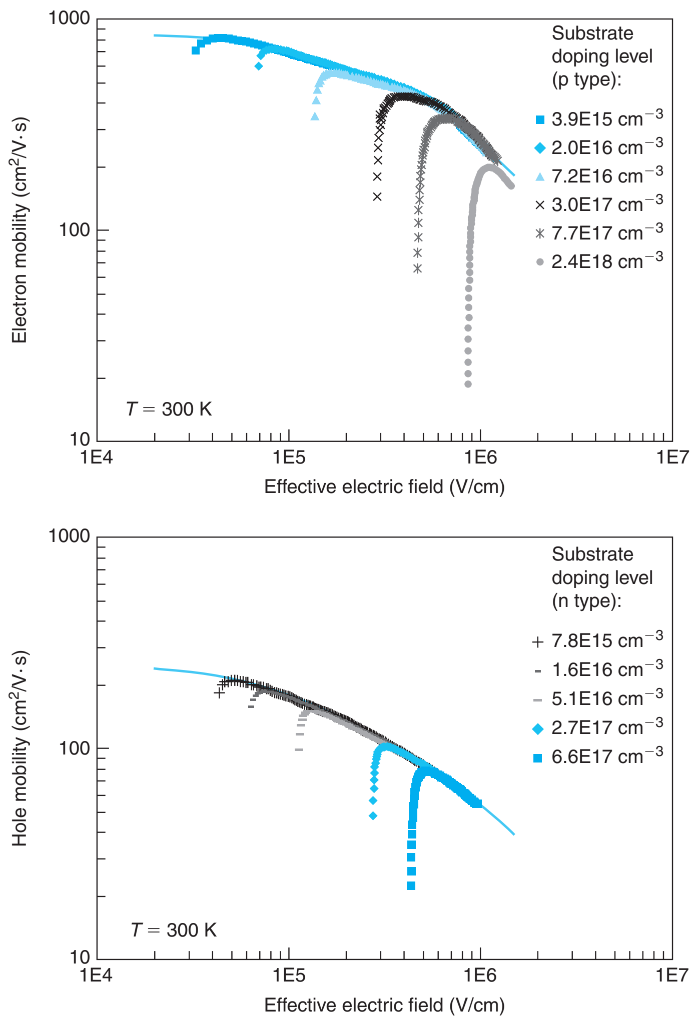


FIGURE 10.4
Electron and hole inversion layer mobility versus effective electric field for (100) Si at 300 K. The lines represent simple model that provide a reasonable description to the data [Data courtesy of Shinichi Takagi (S. Takagi et al., IEEE Trans. *Electron Dev*. Vol. 41, p. 2357 (1994))].

branch that exhibits the so-called universal behavior. For low values of $\mathcal{E}_{\text{eff}}$, interface roughness scattering and phonon scattering are mitigated and the dominant scattering mechanism is ionized impurity scattering. That is why the mobility is a very strong function of the substrate doping level. However, as $\mathcal{E}_{\text{eff}}$ increases, roughness and phonon scattering become dominant and in this regime, the mobility depends exclusively on the value of $\mathcal{E}_{\text{eff}}$, hence the name of "universal."

The effective vertical field in the abscissa of Figure 10.4 is given by

$$\mathcal{E}_{\text{eff}} = \frac{Q_{\text{dmax}} + \eta Q_{\text{inv}}}{\epsilon_{\text{s}}} \tag{10.1}$$

Q_{dmax} and Q_{inv} are the charge in the depletion region and inversion layer, respectively. A universal mobility relationship is obtained only for precise values of the parameter η. For electrons, the value is $\eta = 1/2$ and for holes, $\eta = 1/3$.[2] As V_{GS} increases beyond threshold, Q_{dmax} does not change but Q_{inv} increases. As a result, $\mathcal{E}_{\text{eff}}$ increases too. As we see in Figure 10.4, this brings down the mobility.

Figure 10.4 also shows lines that provide a reasonable fit to the universal inversion layer electron and hole mobilities as a function of $\mathcal{E}_{\text{eff}}$ that can be used for problems. The algebraic expressions of these fits can be found in Appendix E.

An interesting observation is that for $\eta = 1/2$, $\mathcal{E}_{\text{eff}}$ given by Eq. (10.1) is the average vertical field of the inversion layer. This is a simple interpretation of $\mathcal{E}_{\text{eff}}$ that helps us to intuitively understand the impact of mobility degradation on device operation.

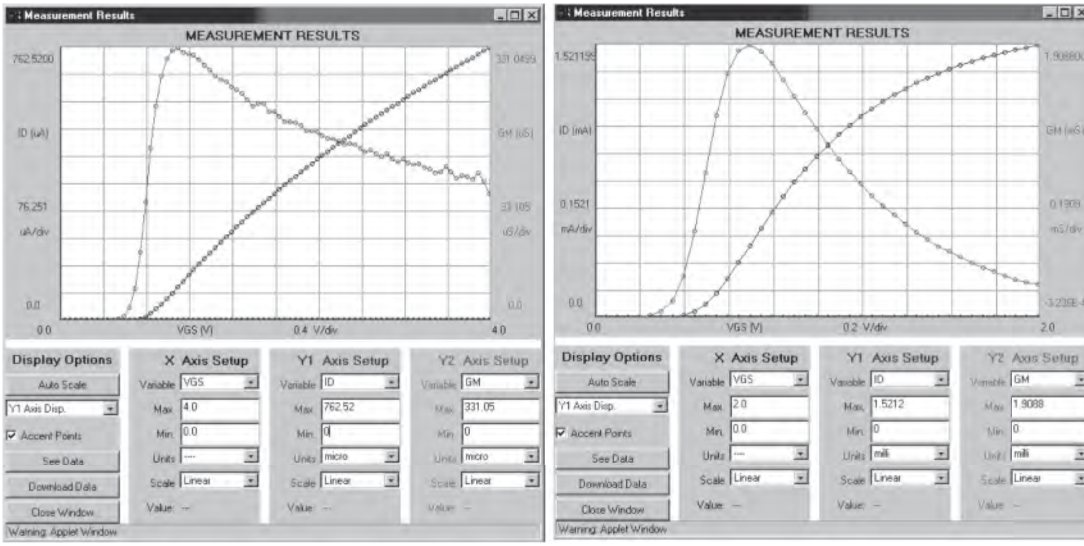
Actually, it is easy to derive a first-order expression for $\mathcal{E}_{\text{eff}}$ that allows for a quick estimate of inversion mobility in the case of a polysilicon-gate n-channel device. The charge in the depletion layer, Q_{dmax}, can be expressed in terms of V_{T} as in Eq. (8.28). A simplification is possible for an n$^+$-polysilicon gate since V_{FB} and $\phi_{\text{sT}} \simeq 2\phi_{\text{f}}$ almost precisely cancel out (see Exercise 8.4). Q_{dmax} can therefore be approximated as $Q_{\text{dmax}} \simeq -C_{\text{ox}}V_{\text{T}}$. Combined with the fundamental charge–control relationship for the MOSFET [Eq. (8.31)], we have

$$|\mathcal{E}_{\text{eff}}| \simeq \left| \frac{Q_{\text{dmax}} + \frac{1}{2}Q_{\text{inv}}}{\epsilon_{\text{s}}} \right| \simeq \frac{C_{\text{ox}}V_{\text{T}} + \frac{1}{2}C_{\text{ox}}(V_{\text{GS}} - V_{\text{T}})}{\epsilon_{\text{s}}} = \frac{\epsilon_{\text{ox}}}{\epsilon_{\text{s}}} \frac{V_{\text{GS}} + V_{\text{T}}}{2x_{\text{ox}}} \tag{10.2}$$

This equation is valuable because it intuitively tells us that increasing V_{GS} or V_{T}, or reducing x_{ox} increases $\mathcal{E}_{\text{eff}}$ and degrades the channel mobility. But note again, Eq. (10.2) is only appropriate for an n$^+$-polysilicon gate.

It is nothing but remarkable that the inversion layer mobility exhibits this "universal" dependence on $\mathcal{E}_{\text{eff}}$. If one plots mobility against surface potential or inversion layer sheet carrier concentration (as in Figure 10.2), a universal behavior is not obtained. Only when mobility is plotted as a function of $\mathcal{E}_{\text{eff}}$, such universal dependence is obtained. $\mathcal{E}_{\text{eff}}$ itself is a function of a number of physical parameters, such as the oxide thickness, substrate doping level, and gate metal work function, as well as the gate voltage and the substrate voltage in the case of a three- or four-terminal MOSFET structure. Regardless of the individual values of the physical parameters

[2] The data in Figure 10.4 and the values of η of 1/2 and 1/3 for electrons and holes (respectively) are valid for a Si surface with (100) orientation. The appropriate values of η are different for other surface orientations.



(a) (b)

FIGURE 10.5
Drain current and transconductance in the linear regime ($V_{DS} = 0.1$ V) for an $L_g = 1.5$ μm MOSFET (a) and an $L_g = 0.18$ μm MOSFET (b). Clearly, the short-channel device suffers much more from mobility degradation at high values of V_{GS}. (MIT Microelectronics iLab).

or applied voltages, once $\mathcal{E}_{eff}$ is large enough, what matters for the inversion layer mobility is the value of the $\mathcal{E}_{eff}$ "package."

There are very fundamental reasons for this universal behavior in the dependencies of phonon scattering and interface roughness scattering of inversion layer carriers. Understanding the details is beyond the scope of this book. It can be shown from fairly fundamental principles that both scattering mechanisms depend on $\mathcal{E}_{eff}$ as given in Eq. (10.1) with an appropriate value of η that embodies the peculiarities of the atomic arrangement of the semiconductor at its surface.

An interesting consequence of the universal mobility dependence on effective vertical field is that mobility degradation generally gets worse as the vertical dimensions of the device scale down. This is because, all things being equal, for a thinner oxide, the effective vertical field increases. Since lateral scaling and vertical scaling have to proceed in a harmonious way, MOSFET scaling brings along a degradation in inversion layer mobility. It is because of this that mobility degradation is considered a "short-channel effect" even though it really has nothing to do with the lateral dimensions of the device.

As sketched in Fig. 10.3(b), the cleanest manifestation of mobility degradation is observed in the transconductance characteristics in the linear regime. Figure 10.5 shows two examples for a long device ($L_g = 1.5$ V) and a short device ($L_g = 0.18$ μm). Both show evidence of mobility degradation in the drop of the linear transconductance for high V_{GS}. Clearly, the short device suffers much more from this effect.

EXERCISE 10.1

Estimate the value of the universal inversion layer mobility of (i) the long-channel MOSFET studied in Exercise 9.1 at $V_{GS} = 2.5$ V and (ii) a short-channel MOSFET characteristic of the 65 nm node with $x_{ox} = 1.3$ nm and $V_T = 0.35$ V at $V_{GS} = 1$ V.

We start with the long-channel MOSFET. We retrieve appropriate data from Exercise 9.1. The first step is to compute the effective field at the indicated conditions. For this, we use Eq. (10.2):

$$|\mathcal{E}_{eff}| \simeq \frac{\epsilon_{ox}}{\epsilon_s}\frac{V_{GS}+V_T}{2x_{ox}} = 0.33\frac{2.5+0.56\ \text{V}}{2\times 15\times 10^{-7}\ \text{cm}} = 1.5\times 10^5\ \text{V/cm}$$

We then use the fit to the universal electron inversion layer mobility that is provided in Appendix E to get

$$\mu_e = \frac{\mu_o}{1+\left(\frac{|\mathcal{E}_{eff}|}{\mathcal{E}_o}\right)^{\theta}} = \frac{880}{1+\left(\frac{1.5\times10^5\ \text{V/cm}}{4.2\times10^5\ \text{V/cm}}\right)^{1.05}} = 657\ \text{cm}^2/\text{V}\cdot\text{s}$$

If we proceed in a similar way for the short-channel MOSFET, we obtain $|\mathcal{E}_{eff}| \simeq 1.7\times 10^6$ V/cm and $\mu_e \simeq 165$ cm^2/V $\cdot$ s. This dramatic difference in μ_e illustrates the large degradation in inversion layer mobility that takes place in short-channel MOSFETs.

10.1.2 Velocity saturation

Besides mobility degradation, an important additional reason for the reduced current of MOSFETs with respect to what is expected from the long MOSFET theory is carrier velocity saturation. Mobility degradation is caused by high *vertical* electric fields. In contrast, velocity saturation effects arise from high *longitudinal* electric fields in the channel. As discussed in Chapter 4, at high electric field, the drift velocity of carriers no longer increases linearly with the electric field but tends to saturate. The reason for velocity saturation is optical phonon emission, a scattering process in which hot carriers collide with the lattice at increasing rates as they become more energetic and they lose energy and forward velocity along the way.

Velocity saturation has a huge impact on the performance of MOSFETs. For example, a 1 V voltage drop on a 0.1-μm-long channel roughly translates into an electric field of order 10^5 V/cm. If we assume an electron mobility of 500 cm^2/V $\cdot$ s, in the mobility regime, electrons should drift at a velocity of 5×10^7 cm/s. Obviously, this is not possible since it exceeds the electron saturation velocity of 10^7 cm/s. If we use Eq. (4.13) that includes velocity saturation effects, we find that the electron velocity for this electric field is actually 8.3×10^6 cm/s, which is six times less than expected from simple theory. This translates into a current level in a MOSFET that is significantly lower than predicted by the models developed in Chapter 9.

It is not difficult to develop an analytical model for the *I–V* characteristics of a MOSFET that incorporates velocity saturation effects. The procedure is very much the same as that followed in Section 9.4 for the ideal MOSFET. It starts by obtaining an expression for the drain current in the linear regime. Then, the value of V_{DS} beyond which the current cannot increase anymore is computed. This allows the calculation of the current that flows through the transistor in the saturation regime.

The starting point is then Eq. (9.4) which is the most general expression for the channel current in a MOSFET under the sheet-charge approximation, which we continue to assume. For convenience, we reproduce this expression here in terms of the drain terminal current:

$$I_D \simeq -W v_{ey}(y) Q_i(y) \tag{10.3}$$

In this expression, $v_{ey}(y)$ is the electron velocity in the channel, which, using Eqs. (4.13) and (4.14) can be expressed as

$$v_{ey}(y) = -\frac{\mu_e \mathcal{E}_y(y)}{1 + \left|\frac{\mu_e \mathcal{E}_y(y)}{v_{sat}}\right|} = -\frac{\mu_e \mathcal{E}_y(y)}{1 - \frac{\mathcal{E}_y(y)}{\mathcal{E}_c}} \tag{10.4}$$

Here, we have accounted for the fact that in our definition of axis, $\mathcal{E}_y$ is negative.

Under the gradual-channel approximation, the charge in the channel is given by the fundamental charge control relationship [Eq. (9.11)]:

$$Q_i(y) \simeq -C_{ox}[V_{GS} - V(y) - V_T] \tag{10.5}$$

Putting it all together, we have

$$I_D = -W \frac{\mu_e \mathcal{E}_y(y)}{1 - \frac{\mathcal{E}_y(y)}{\mathcal{E}_c}} C_{ox}[V_{GS} - V(y) - V_T] \tag{10.6}$$

If we aggregate all terms in $\mathcal{E}_y(y)$ to the right and then we use Eq. (9.9) to express $\mathcal{E}_y(y)$ in terms of $V(y)$, we obtain

$$I_D = \left\{ W \mu_e C_{ox}[V_{GS} - V(y) - V_T] - \frac{I_D}{\mathcal{E}_c} \right\} \frac{dV(y)}{dy} \tag{10.7}$$

This is a differential equation similar to Eq. (9.12) but in terms of I_D. As in Chapter 9, this equation can be integrated after variable separation. Integrating from $y = 0$ to $y = L$ which corresponds to $V = 0$ and $V = V_{DS}$, respectively, yields

$$I_D = \frac{W}{L} \frac{\mu_e C_{ox}}{1 + \frac{V_{DS}}{\mathcal{E}_c L}} \left(V_{GS} - V_T - \frac{1}{2} V_{DS} \right) V_{DS} \tag{10.8}$$

This is an interesting equation. It looks very much like the expression of the drain current in the linear regime in the ideal MOSFET [Eq. (9.16)] except that the mobility appears divided by a factor $1 + \frac{V_{DS}}{\mathcal{E}_c L}$. All things being equal, this new term results in reduced drain current with respect to the ideal MOSFET. As is to be expected, for very low V_{DS}, when the lateral field in the channel is small and velocity saturation effects are unimportant, this equation converges to the ideal MOSFET equation [Eq. (9.16)]. However, as V_{DS} increases, the new factor in the denominator becomes increasingly more important and I_D is reduced with respect to its ideal value.

Equation (10.8) has a behavior similar to that obtained for the ideal MOSFET. For a given V_{GS}, I_D increases with V_{DS} until a maximum current is obtained. Beyond that, the equation predicts that I_D decreases with V_{DS}. This is not physically possible so the current saturates at this maximum value. The value of V_{DS} at which current saturation occurs can be easily obtained by solving $dI_D/dV_{DS} = 0$. This leads to

$$V_{DSsat} = \mathcal{E}_c L \left(\sqrt{1 + 2\frac{V_{GS} - V_T}{\mathcal{E}_c L}} - 1 \right) \tag{10.9}$$

The value of the saturated drain current is then

$$I_{Dsat} = \frac{W}{L} \frac{\mu_e C_{ox}}{1 + \frac{V_{DSsat}}{\mathcal{E}_c L}} \left(V_{GS} - V_T - \frac{1}{2} V_{DSsat} \right) V_{DSsat} \tag{10.10}$$

It is interesting to see that in the presence of velocity saturation, current saturation in a MOSFET occurs at a lower value of V_{DS} than in an ideal MOSFET. This can be seen in Fig. 10.6 that sketches Eq. (10.9) and the ideal MOSFET behavior [Eq. (9.22)]. This *premature* current saturation occurs before channel pinch off which one expects at $V_{DS} = V_{GS} - V_T$. This is one of the consequences of velocity saturation. The drain current stops increasing with V_{DS} the moment the electrons at the drain-end of the channel hit velocity saturation.

This recognition allows us to write the drain current in a particularly compact way. If we use Eq. (10.3) at the drain end of the channel at $V_{DS} = V_{DSsat}$ with electrons traveling at velocity saturation, we obtain

$$I_{Dsat} = W v_{sat} C_{ox} (V_{GS} - V_T - V_{DSsat}) \tag{10.11}$$

It is easy to prove that Eq. (10.11) and (10.10) are identical if one uses the expression of V_{DSsat} given by Eq. (10.9).

Due to velocity saturation, current saturation occurs at a lower value of V_{DS} and therefore the drain current saturates to a lower value than in the ideal MOSFET. Physically, this can be understood by examining Figs. 10.7 and 10.8. These sketch the evolution of inversion layer charge, channel potential, electric field, and velocity along the channel in the linear regime and at pinch-off for both the ideal MOSFET (operating in the mobility regime) and a MOSFET with velocity saturation effects. In Fig. 10.7, both sets of curves are sketched in the linear regime at the same value of V_{DS} and for the same gate overdrive. It is clear that the main effect of carrier velocity saturation is to produce a sharper electric field distribution when compared with

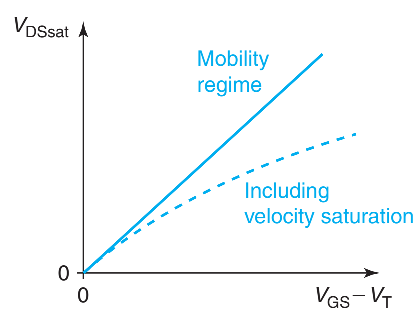


FIGURE 10.6
Sketch of evolution of V_{DSsat} with V_{GS} in the mobility regime and the velocity saturation regime.

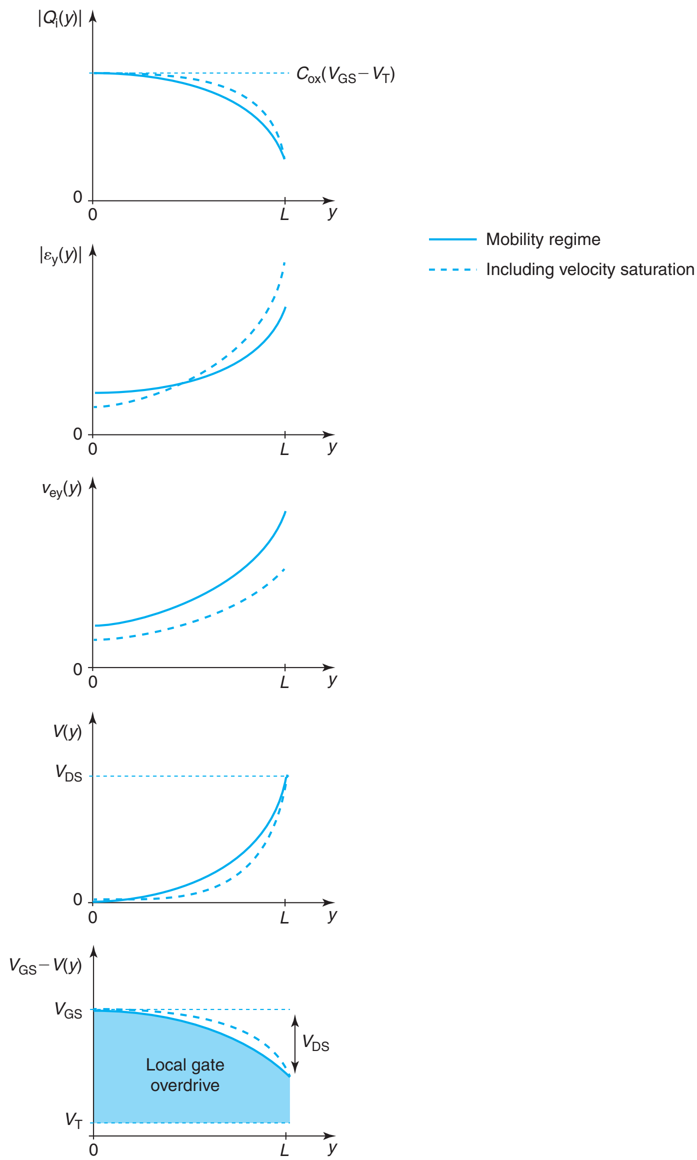


FIGURE 10.7
Sketch of inversion layer charge, lateral electric field, electron velocity, channel voltage, and $V_{GS} - V(y)$ as a function of position along the channel for a MOSFET in the linear regime with and without velocity saturation effects. $V_{GS} - V_T$ and V_{DS} are identical in both situations.

the mobility regime. This is because channel debiasing is more pronounced in the presence of velocity saturation effects. That is, in order to maintain the current constant along the channel, there is a need for a higher field toward the drain side of the channel to compensate for the lower velocity. A consequence of this is that the field at the source is lower and therefore the overall channel current is lower.

Figure 10.8 shows the lateral electrostatics at V_{DSsat} for both situations with the same gate overdrive. In an ideal MOSFET, drain current saturation occurs due to pinch off at the drain end of the channel where the charge goes to a very small value and the field peaks at a very high value. With velocity saturation effects, current saturation occurs at a lower value of V_{DSsat} for which the sheet carrier concentration at the drain end of the channel can still be sizeable but the velocity hits v_{sat}.

Since the velocity-field characteristics that we are using in the development of this model contain the correct low-field limit, the expressions that we have derived above should have the correct limits for very low fields or for a very high value of the saturation velocity. This is easy to verify. In the mobility regime, when v_{sat} is made very large, or $\mathcal{E}_c L \gg (V_{GS} - V_T,\ V_{DSsat})$, Eqs. (10.10) and (10.11) for I_{Dsat} converge toward the ideal expression (9.21). Also, Eq. (10.9) for V_{DSsat} reduces to the ideal expression given by Eq. (9.22).

Of interest is also the limit of very short MOSFETs with strong velocity saturation effects for which $\mathcal{E}_c L \ll (V_{GS} - V_T,\ V_{DSsat})$. In this high-field limit, Eqs. (10.9) and (10.10) become, respectively

$$V_{DSsat} \simeq \sqrt{2\mathcal{E}_c L(V_{GS} - V_T)} \tag{10.12}$$

and

$$I_{Dsat} \simeq W v_{sat} C_{ox}(V_{GS} - V_T) \tag{10.13}$$

The interpretation of this last equation is straightforward. In the limit of very short MOSFET and very high fields, electrons travel at saturation velocity through the entire channel. In particular, electrons emerge out of the source already at saturation velocity. Using Eq. (10.3) at the source end of the channel immediately yields Eq. (10.13). In this limit, the sheet charge concentration along the channel is constant and equal to $C_{ox}(V_{GS} - V_T)$.

The model that we have just developed for the *I–V* characteristics of a MOSFET in the presence of velocity saturation allows us to make a number of contrasting observations to the behavior of the ideal MOSFET. We have already mentioned premature current saturation as a result of velocity saturation at the drain end of the channel. We also find, as sketched in Fig. 10.6, that the dependence of V_{DSsat} with V_{GS} is no longer linear but square root. This is readily observed in experiments.

From the point of view of the current, we also mentioned the substantially reduced drain current. This can be appreciated in Fig. 10.9 where we show the calculated output characteristics for a 1-μm MOSFET with typical parameters as expected from the ideal MOSFET model and after adding velocity saturation effects. As it is easy to see, even at the 1-μm-gate length level, the loss of current is very substantial indeed!

In addition to the loss of current, it is evident in Fig. 10.9 that the V_{GS} dependence of the drain current in saturation is rather different. In the ideal model, the relationship is quadratic. However, with velocity saturation, the dependence is weaker than quadratic and for high V_{GS} it becomes almost linear. This is also seen in Fig. 10.9(b) that shows calculations of I_{Dsat} versus gate overdrive for a typical 1-μm MOSFET.

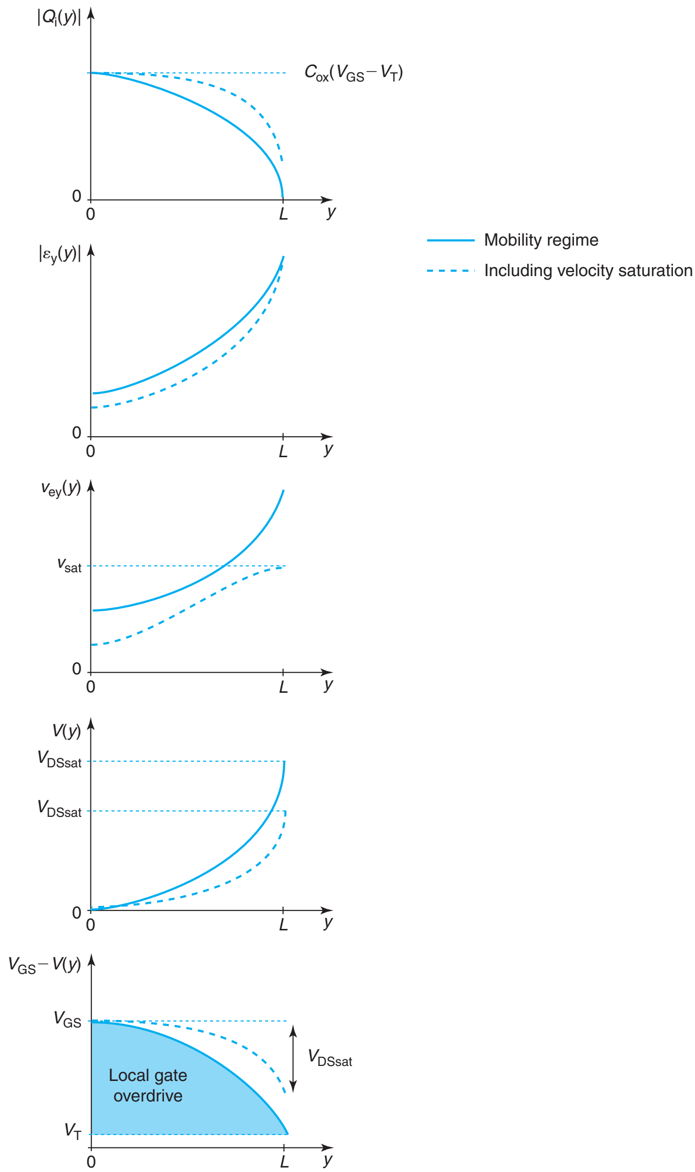


FIGURE 10.8
Sketch of inversion layer charge, lateral electric field, electron velocity, channel voltage, and $V_{GS} - V(y)$ as a function of position along the channel for a MOSFET biased at V_{DSsat} with and without velocity saturation effects. The gate overdrive, $V_{GS} - V_T$, is identical in both situations.

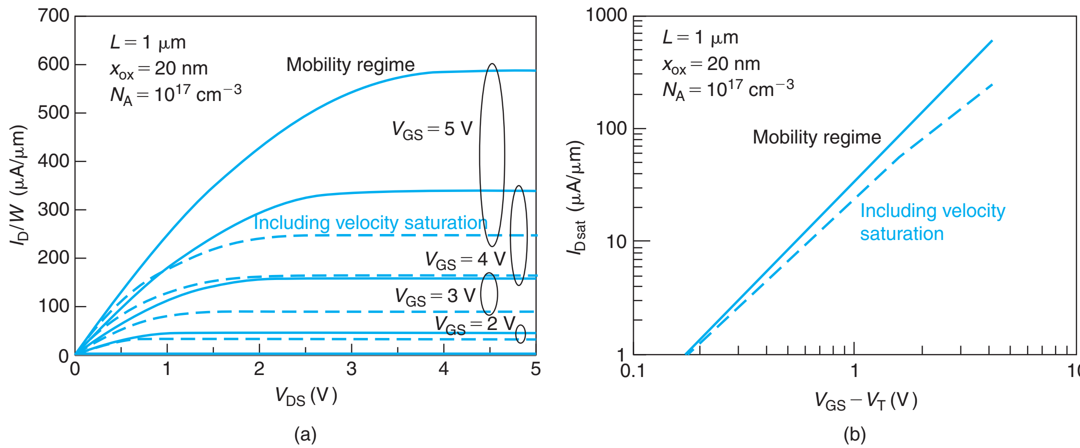


FIGURE 10.9
Calculated output characteristics (a) and saturated drain current versus gate voltage overdrive (b) for a 1-μm MOSFET under the mobility and velocity saturation regimes.

The loss of drain current and the change in its V_{GS} dependence greatly affects the transconductance of a short MOSFET, a key parameter in many applications. An expression for the transconductance in the saturation regime can be easily obtained by differentiating Eq. (10.11):

$$g_m = \frac{dI_{Dsat}}{dV_{GS}} = W v_{sat} C_{ox}\left(1 - \frac{dV_{DSsat}}{dV_{GS}}\right) \tag{10.14}$$

The last derivative can be obtained from Eq. (10.9) to yield

$$g_m = W v_{sat} C_{ox}\left(1 - \frac{1}{\sqrt{1 + 2\frac{V_{GS}-V_T}{\mathcal{E}_c L}}}\right) \tag{10.15}$$

In the limit of a very long MOSFET or high saturation velocity, this expression reverts to the transconductance expression that we obtained for the ideal MOSFET in Eq. (9.52). For a very short MOSFET at high gate overdrive, this expression becomes

$$g_m = W v_{sat} C_{ox} \tag{10.16}$$

In this limit, the transconductance becomes a constant regardless of bias and its magnitude is entirely set by the value of the saturation velocity and the gate oxide thickness.

EXERCISE 10.2

Consider the long-channel MOSFET studied in Exercise 9.2. At the same bias point as in that exercise, $V_{GS} = 2.5$ V, $V_{DS} = 4$ V, and $V_{BS} = 0$, estimate the drain current and the electron *velocity and sheet electron concentration at the drain end of the channel. For consistency with the calculations of Exercise 9.2, use* $\mu_e = 500\ \text{cm}^2/\text{V}\cdot\text{s}$.

We start by estimating V_{DSsat} using Eq. (10.9). For this, we first need to first calculate $\mathcal{E}_c$ using Eq. (4.14):

$$\mathcal{E}_c = \frac{v_{sat}}{\mu_e} = \frac{1\times10^7\ \text{cm/s}}{500\ \text{cm}^2/\text{V}\cdot\text{s}} = 2\times10^4\ \text{V/cm}$$

For this transistor, then, $\mathcal{E}_c L = 2\times10^4\ \text{V/cm}\times10^{-4}\ \text{cm} = 2\ \text{V}$. We can now proceed to estimate V_{DSsat}:

$$V_{DSsat} = \mathcal{E}_c L\left(\sqrt{1+2\frac{V_{GS}-V_T}{\mathcal{E}_c L}}-1\right) = 2\ \text{V}\left(\sqrt{1+2\frac{2.5-0.56\ \text{V}}{2\ \text{V}}}-1\right) = 1.4\ \text{V}$$

This is significantly smaller than the long-channel value $V_{GS}-V_T = 1.9$ V suggesting that velocity saturation effects are of some significance even in this long-channel device.

We can now proceed to estimate the drain current using Eq. (10.11):

$$\begin{aligned} I_{Dsat} &= W v_{sat} C_{ox}(V_{GS}-V_T-V_{DSsat}) \\ &= 10^{-3}\ \text{cm}\times10^7\ \text{cm/s}\times2.3\ \times10^{-7}\ \text{F/cm}^2(2.5-0.56-1.4\ \text{V}) = 1.2\ \text{mA} \end{aligned}$$

Even though this is a long-channel device, this value of I_D is significantly smaller than what was estimated under mobility-limited transport in Exercise 9.2.

At the drain end of the channel, the electrons are traveling at velocity saturation:

$$v_e(L) = v_{sat} = 10^7\ \text{cm/s}$$

The sheet electron concentration at the drain end of the channel is then obtained from

$$n_s(L) = \frac{Q_i(L)}{q} = \frac{I_D}{qWv_e(L)} = \frac{1.2\times10^{-3}\ \text{A}}{1.6\times10^{-19}\ \text{C}\ \times10^{-3}\ \text{cm}\times10^7\ \text{cm/s}} = 7.5\times10^{11}\ \text{cm}^{-2}$$

10.2 MOSFET SHORT-CHANNEL EFFECTS: ELECTROSTATICS

This section discusses short-channel effects that affect the electrostatics of MOSFETs. This term usually refers to the relative importance of the lateral field versus the vertical field in controlling charge and charge transport under the gate. In an ideal MOSFET biased above threshold, we argued that the vertical electric field imposed by the gate sets the charge in the channel, while the lateral electric field established by the drain–source voltage controls the velocity at which electrons drift. Below threshold, with the channel depleted, the depletion charge under the gate is entirely determined by the gate voltage and the drain voltage has no impact on this whatsoever. A consequence of this is that figures of merit such as the threshold voltage and the subthreshold swing are entirely set by the doping level of the body and the details of the gate stack (oxide thickness and metal work function).

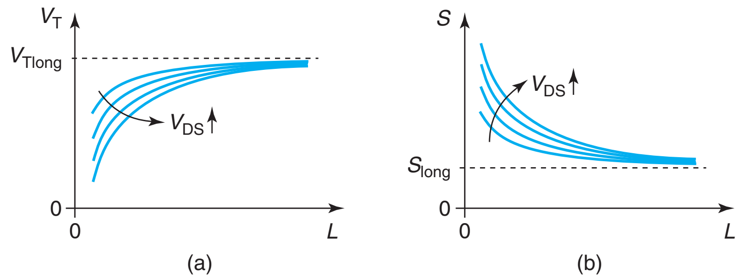


FIGURE 10.10
Sketch of evolution of V_T (a) and S (b) in a given MOSFET technology as a function of gate length and V_{DS}. For long devices, V_T and S are independent of L. Short devices exhibit a negative V_T shift that becomes bigger the shorter the gate length and the higher the value of V_{DS}. Similarly, in short MOSFETs, as L is reduced or V_{DS} is increased, S increases.

In short MOSFETs, it is an experimental observation that the threshold voltage and the subthreshold swing depend on gate length and the value of V_{DS}. Otherwise identical devices, except for having different gate lengths, have different values of V_T and S. The shorter devices have a more negative V_T and exhibit a larger value of S. In addition, in short MOSFETs, increasing V_{DS} produces a shift in V_T in the negative direction and increases S. Simple sketches are shown in Fig. 10.10.

These effects are undesirable. Small variations in gate length or supply voltage are unavoidable in complex circuits. Yet, V_T needs to be tightly controlled for accurate circuit operation, while S needs to be kept in check since it impacts the off-state current which is critical in power consumption. To deal with ambiguity in the values of V_T and S, circuit designers must design for worst-case conditions which often times comes at the cost of performance, area, and energy efficiency.

This section studies these important short-channel effects. We first study the V_T dependence on gate length and V_{DS}, and we then discuss the impact of these two parameters on the subthreshold swing. At its heart, the physics of all these effects is the same and that is the nonideal electrostatics of a MOSFET when its *aspect ratio*, the gate length to gate-oxide thickness ratio (we introduce a more rigorous definition below), becomes relatively small.

10.2.1 Threshold voltage dependence on gate length: V_T rolloff

In the real world, not all MOSFETs have the exact same gate length. In fact, there is a natural variation in the gate length of nominally identical transistors that arises from inevitable tolerances in the fabrication process. The gate length changes from transistor to transistor inside a die, from die to die in a wafer, across wafers in a run, and from run to run. Even in an ideal MOSFET, gate length variations translate into performance variations since the gate length enters into many important transistor figures of merit. This makes the operation of a given device in a circuit somehow uncertain.

In a short MOSFET, the situation becomes more problematic because V_T is found to depend on gate length. Shorter devices have more negative values of V_T, as shown in Figure 10.10. In long devices, V_T is independent of L, however, as L decreases, V_T shifts negative. This is often referred to as *threshold voltage rolloff*. If the V_T shift is excessive, this can become a serious problem as different devices will have quite different values of V_T. In this section, we study the physical origin of this effect and we develop a first-order model that captures its key dependencies. This model provides practical device design guidelines.

The reason for the V_T dependence on L comes from the penetration under the gate of the depletion region associated with the source and drain junctions, as illustrated in Fig. 10.11. This figure sketches the shape of the depletion region in a MOSFET biased at threshold. The presence of the source and drain junctions creates a slightly deeper depletion region under the gate at both ends of the channel. In a long-channel device, this effect is comparatively small and under the center of the gate, the band structure is unaffected by this. In a long MOSFET, the highest point of the energy barrier located at the center of the gate remains firmly controlled by the gate and V_T does not depend on L.

In a short-channel MOSFET, however, the source and drain depletion regions penetrate under the gate and merge to some extent. In this case, for a value of V_{GS} that leads to the threshold condition in the long-channel device, the conduction band in the short-channel device is closer to the Fermi level. This implies a higher electron concentration at the surface of the semiconductor than in the long-channel device. Since we can recover the carrier concentration at the threshold definition in the long-channel device by biasing V_{GS} a little bit more negative, in effect, this represents a negative shift in the threshold voltage. The shorter the gate length is, the

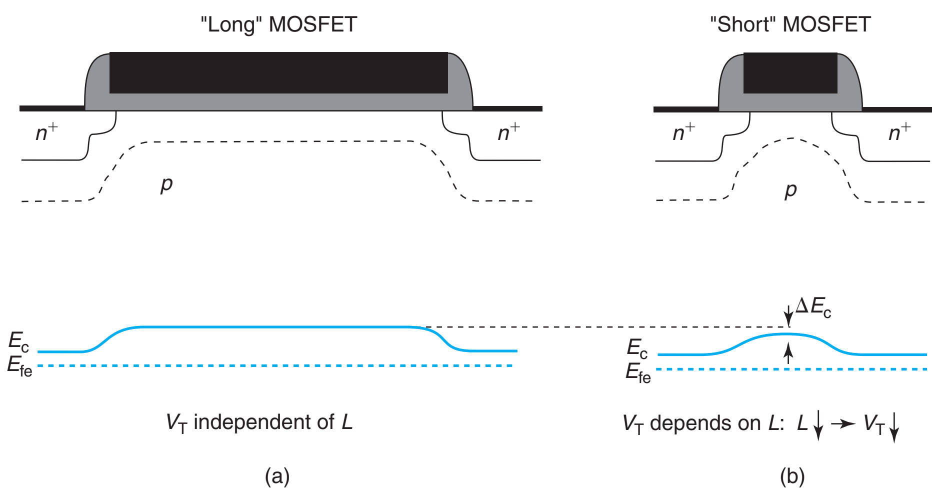


FIGURE 10.11
Sketch of depletion region from the source to the drain for a long-channel MOSFET (a) and a short-channel MOSFET (b). Both devices have the same bias that corresponds to the threshold condition for the long-channel device with $V_{DS} = 0$. Below are the respective shapes of the conduction band from the source to the drain.

closer the source and the drain are to each other and the more the energy barrier underneath the center of the gate is depressed. V_T becomes dependent on L.

The key to mitigating V_T rolloff is to minimize the penetration under the gate of the depletion regions associated with the source and the drain. That calls for a transistor design with a high *electrostatic integrity*. This means that in the region under the gate the vertical electrostatics must dominate over the lateral electrostatics or, in other words, that most of the field lines that emanate from the depletion region under the gate terminate at the gate metal and not at the source and the drain. Intuitively we can see that this is more likely to be the case when the depletion region under the gate and under the source and drain are very thin and when the gate oxide is also thin.

This is now an eminently two-dimensional electrostatics problem. CAD tools can readily analyze this situation. As always, it is of great value to derive first-order analytical relationships that capture the essence of the problem and that we can use to guide design. For example, it is not altogether clear what the shape of the dependence of V_T on L is. Also, it is useful to get an appreciation for the value of L at which the shift in V_T starts getting significant and how this value depends on x_{ox} and N_A. For this, we need a simplified analytical model that captures the essential physics. In the next few pages, we present a relatively simple model that has been shown to apply to well-designed MOSFETs.

The geometry of the problem is the same as in Figure 9.3. The MOSFET is biased with $V_{DS} = 0$ and the gate bias is such that the MOS structure is in depletion. For simplicity, in the following treatment, we assume that the body is shorted to the source. In our analysis, we can neglect the inversion charge at the semiconductor surface.

Figure 10.12 illustrates the electrostatic potential distribution in this situation. Figure 10.12(a) shows ϕ as a function of depth at a location y along the channel. Selecting the body of the transistor as reference for potentials, the gate potential is equal to $V_{GS} - V_{FB}$, where V_{FB} is the flat-band voltage which is equal to the minus built-in potential of the MOS structure. The potential at the semiconductor surface is labeled as ϕ_s and this value controls the electron concentration at the surface.

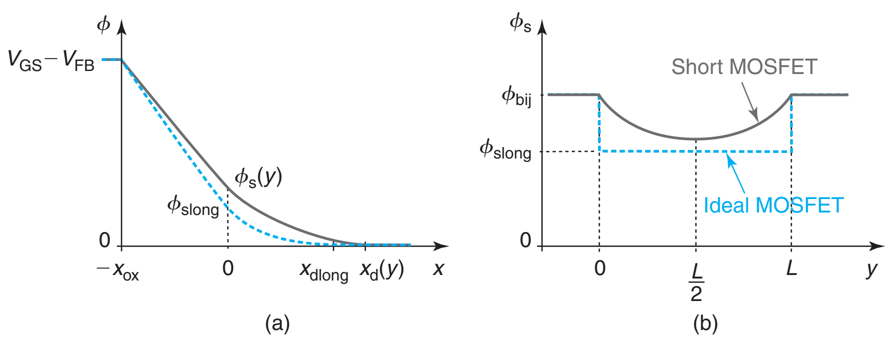


FIGURE 10.12
Electrostatic potential distribution in depth at location y in the channel (a), and along semiconductor surface from the source to the drain (b) corresponding to long- and short-channel MOSFETs biased around threshold with $V_{DS} = 0$.

Due to the two-dimensional nature of this situation, both ϕ_s and the depletion region width, x_d, depend on y. Figure 10.12(b) sketches the surface potential along the channel. With the semiconductor body as the reference for all potentials and $V_{DS} = 0$, the potential inside the source and drain regions is equal to ϕ_{bij}, the built-in potential of the PN junctions. The potential distribution along the channel exhibits a minimum at the center of the gate. This corresponds to the highest point in the conduction band in the channel. With $V_{DS} = 0$, the center of the channel is the location that controls V_T. In this diagram, we also show the surface potential in an ideal MOSFET at the same bias point. In the absence of short-channel effects and all things being equal, the surface potential in the ideal MOSFET drops abruptly from ϕ_{bij} at the source and drain regions to a value that we denote as ϕ_{slong} in the channel. For the same value of V_{GS}, ϕ_s in the short device is higher than ϕ_{slong}. This results in a higher sheet electron concentration at the surface in the short-channel device. To bring this value down to that of the ideal MOSFET, one has to reduce V_{GS}. Effectively, this means that V_T has shifted negative.

An analytical solution to $\phi_s(y)$ is not possible without some approximations. As worked out in detail in Appendix AT10.1, a suitable expression is

$$\phi_s(y) \simeq \phi_{slong} + (\phi_{bij} - \phi_{slong}) \frac{\sinh \frac{y}{\lambda} + \sinh \frac{L-y}{\lambda}}{\sinh \frac{L}{\lambda}} \tag{10.17}$$

where λ is the characteristic length of this electrostatic problem which is given by

$$\lambda = \sqrt{\frac{\epsilon_s}{\epsilon_{ox}} x_{ox} x_{dlong}} \tag{10.18}$$

In this expression, x_{dlong} is the thickness of the depletion region under the specified conditions for the ideal MOSFET.

λ can be rewritten as

$$\frac{\lambda}{\epsilon_s} = \sqrt{\frac{x_{ox}}{\epsilon_{ox}} \frac{x_{dlong}}{\epsilon_s}} \tag{10.19}$$

This suggests that λ can be thought of as the geometric mean of the oxide thickness and the depletion region width of the MOS structure at threshold weighted by their respective dielectric constants.

The solution of Eq. (10.17) is symmetric around the center of the channel at $y = L/2$. Since the dominant term in the *sinh* function is an exponential, to the first order, the shape of the surface potential contains decaying exponentials placed at the edges of the channel with a characteristic constant equal to λ. Equation 10.17 thus has the right limits. When $L \gg \lambda$, ϕ_s drops to ϕ_{slong} a few λ away from the ends of the channel. In this case, the threshold voltage of the device has its ideal value. On the other hand, when L is comparable to λ, the two decaying *sinh* functions overlap in the middle and the minimum surface potential is higher than ϕ_{slong}. This produces a negative shift in V_T.

In order to compute ΔV_T, we focus on the potential at the center of the gate. From Fig. 10.12, this is

$$\phi_s\left(\frac{L}{2}\right) = \phi_{slong} + (\phi_{bij} - \phi_{slong}) \frac{1}{\cosh \frac{L}{2\lambda}} \tag{10.20}$$

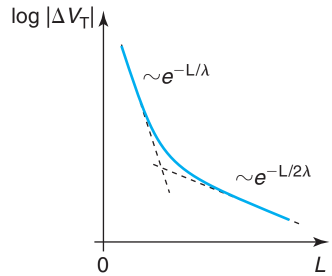


FIGURE 10.13
Sketch of shift in threshold voltage (in absolute value) with gate length.

As in the ideal MOSFET, we define threshold as the condition that yields $\phi_s\left(\frac{L}{2}\right) = \phi_{sT}$. At this point, solving for ϕ_{slong} in Eq. (10.20), we obtain

$$\phi_{slong} = \frac{\phi_{sT}\cosh\frac{L}{2\lambda} - \phi_{bij}}{\cosh\frac{L}{2\lambda} - 1} \tag{10.21}$$

The correct limits also emerge here. For $L \gg \lambda$, $\phi_{slong} = \phi_{sT}$. For values of L comparable to λ, $\phi_{slong} < \phi_{sT}$. The difference is precisely the threshold voltage shift (see Appendix AT10.1):

$$\Delta V_T = \phi_{slong} - \phi_{sT} = -\frac{\phi_{bij} - \phi_{sT}}{\cosh\frac{L}{2\lambda} - 1} \tag{10.22}$$

This result makes good sense. When $L \gg \lambda$, ΔV_T goes to zero and we are in the long-channel limit. When L approaches zero, ΔV_T goes to $-\infty$, that is, the transistor cannot be turned off.

For reasonably well-designed devices, $L \gg \lambda$, and the *cosh* term in Eq. (10.22) can be expanded to yield

$$\Delta V_T \simeq -2(\phi_{bij} - \phi_{sT})\left(e^{-L/2\lambda} + 2e^{-L/\lambda}\right) \tag{10.23}$$

This result also has the correct limit as for very long L, ΔV_T goes to zero. For relatively long devices, the dominant exponential is the first term in $e^{-L/2\lambda}$, so as L becomes shorter, $|\Delta V_T|$ increases exponentially. For much shorter values of L, the second exponential also becomes important and $|\Delta V_T|$ grows much faster. This behavior of ΔV_T with L is illustrated in Fig. 10.13.

The formulation presented here reveals the key role that the characteristic length λ plays in V_T rolloff. Mitigation of this problem calls for an enhancement of the channel aspect ratio by using a thin gate oxide or/and a thin depletion region in the body. This, in turn, demands a higher substrate doping level.

10.2.2 Threshold voltage dependence on V_{DS}: drain-induced barrier lowering (DIBL)

In a short MOSFET, V_T not only depends on L but also on V_{DS}. Different values of V_{GS} are required to turn on a MOSFET when biased with different values of V_{DS}. The V_T dependence on V_{DS} is a short-channel effect as it becomes more prominent the shorter the gate length of the device is. This was sketched in Fig. 10.10. In short devices, the higher the value of V_{DS}, the lower V_T. This introduces the notion of *linear* V_T, to denote the value of V_T that corresponds to small values of V_{DS}, and *saturated* V_T, to refer to the value of V_T that corresponds to a high V_{DS} characteristic of saturated transistor operation.

The dependence of V_T on V_{DS} is undesirable. During operation of a MOSFET, V_{DS} is not always held tight. It can change depending on the operation of other transistors. It can also be affected by voltage supply spikes or droop, ground bounce, and noise coupling. If V_T depends on V_{DS}, it is not possible to precisely predict the current through a MOSFET in a given circuit at a certain instant. Circuit designers have a way to compensate for this but always at some cost in device size, performance, or power consumption. Device designers strive to minimize the dependence of V_T on V_{DS}.

It should not be surprising that the physics behind this effect are essentially identical to those that underpin the dependence of V_T on L studied in the previous section. Figure 10.14 shows sketches of depletion region thickness and conduction band profile for long and short MOSFETs biased in depletion for different values of V_{DS}. In a long MOSFET, increasing V_{DS} widens the depletion region associated with the drain-body junction. This enhances its penetration under the drain end of the channel. Since the channel is very long, the conduction band profile around the center of the gate which is what sets the threshold voltage is unaffected and V_T does not change.

In the short MOSFET, however, as V_{DS} increases, the penetration of the drain-body junction depletion region affects the conduction band profile under most of the gate in a significant way.

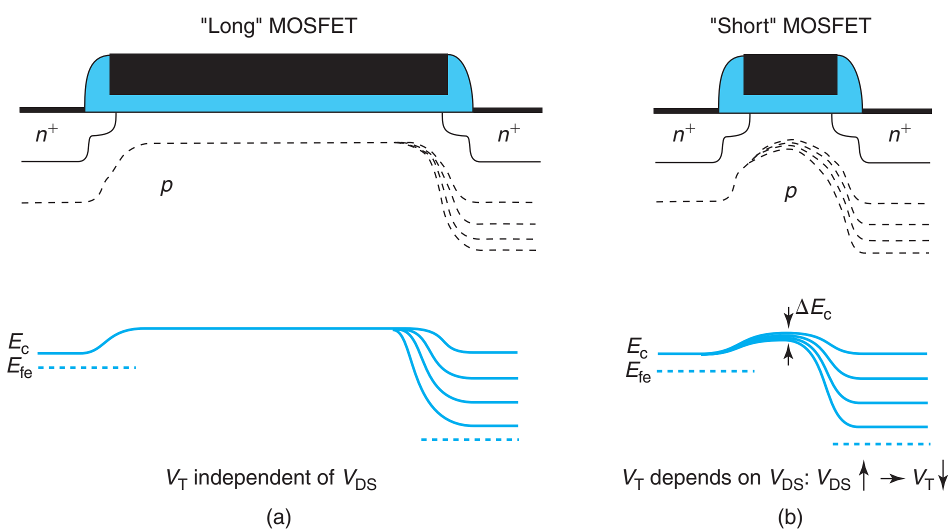


FIGURE 10.14
Sketch of depletion region from the source to the drain for a long-channel MOSFET (a) and a short-channel MOSFET (b). Both devices have the same bias that corresponds to the threshold condition for the long-channel device with $V_{DS} = 0$. Below are the respective shapes of the conduction band from the source to the drain.

In fact, the peak of the barrier in the conduction band, which is what sets the magnitude of the current, drops and its location moves toward the source. All else being equal, there is more current than under ideal circumstances which corresponds to a negative shift in V_T. Figure 10.10 also suggests that the shorter the gate length, the bigger the V_T shift.

Because at the heart of the V_T shift with V_{DS} is a reduction of the conduction band barrier, this phenomenon is known as *drain-induced barrier lowering*, or DIBL. Once again, we see here a competition between lateral and vertical electrostatics. The stronger the vertical electrostatics with respect to the lateral electrostatics, the smaller the impact of V_{DS} on V_T. Minimizing DIBL calls for enhancing the aspect ratio of the device.

An approximate model for the shift of V_T with V_{DS} can be derived from the same formulation that was used in the previous section for dealing with the change in V_T due to L. Appendix AT10.1 derives the differential equation that governs the 2D electrostatic problem. The general solution contains both DIBL and V_T rolloff. In this more general case, the surface potential can be approximated by

$$\phi_s(y) \simeq \phi_{slong} + (\phi_{bij} - \phi_{slong} + V_{DS})\frac{\sinh\frac{y}{\lambda}}{\sinh\frac{L}{\lambda}} + (\phi_{bij} - \phi_{slong})\frac{\sinh\frac{L-y}{\lambda}}{\sinh\frac{L}{\lambda}} \tag{10.24}$$

Notice that this equation reverts to Eq. (10.17) for $V_{DS} = 0$. The shape of ϕ_s versus channel location is sketched in Fig. 10.15.

To derive an expression for V_T, we first need to find the point where the conduction band is the highest. This is no longer the center of the channel but it shifts toward the source as V_{DS} increases. Let us denote this location as y_o. We can compute y_o by noting that at this point, $d\phi_s/dy = 0$. If we take the leading terms in the hyperbolic functions, y_o is approximately:

$$y_o \simeq \frac{L}{2} - \frac{\lambda}{2}\ln\frac{\phi_{bij} - \phi_{slong} + V_{DS}}{\phi_{bij} - \phi_{slong}} \tag{10.25}$$

As expected, y_o starts in the middle of the channel for $V_{DS} = 0$ and moves toward the source as V_{DS} increases.

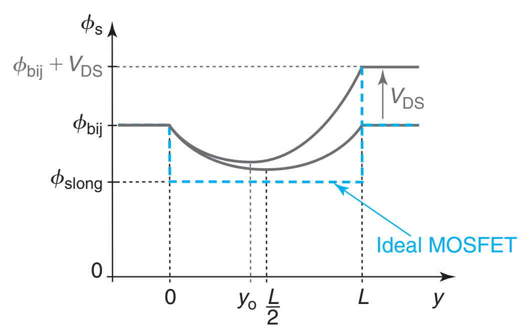


FIGURE 10.15
Sketch of surface potential vs. channel location for a short and a long-channel MOSFET biased as in Fig. 10.14 for $V_{DS} = 0$ and $V_{DS} > 0$.

At y_o, the surface potential is approximately (again, taking the leading terms in the hyperbolic functions):

$$\phi_s(y_o) \simeq \phi_{slong} + 2\sqrt{(\phi_{bij} - \phi_{slong})(\phi_{bij} - \phi_{slong} + V_{DS})}e^{-L/2\lambda} \tag{10.26}$$

For $V_{DS} = 0$, it is easy to verify that this equation is identical to the leading term in Eq. (10.20).

The threshold condition occurs when $\phi_s(y_o) = \phi_{sT}$. Equating Eq. (10.26) and solving for ϕ_{slong}, after some tedious algebra, one obtains

$$\begin{aligned}\phi_{slong} \simeq \phi_{sT} &- 2[2(\phi_{bij} - \phi_{sT}) + V_{DS}]e^{-L/\lambda} \\ &- 2\sqrt{(\phi_{bij} - \phi_{sT})(\phi_{bij} - \phi_{sT} + V_{DS})}e^{-L/2\lambda}\end{aligned} \tag{10.27}$$

ΔV_T is the difference between ϕ_{slong} and ϕ_{sT}:

$$\Delta V_T \simeq -2[2(\phi_{bij} - \phi_{sT}) + V_{DS}]e^{-L/\lambda} - 2\sqrt{(\phi_{bij} - \phi_{sT})(\phi_{bij} - \phi_{sT} + V_{DS})}e^{-L/2\lambda} \tag{10.28}$$

In the limit of $V_{DS} = 0$, this equation reverts to Eq. (10.23). This is very convenient because Eq. (10.28) contains both linear V_T rolloff and DIBL at the same time.

The behavior of ΔV_T with V_{DS} for different values of L is illustrated in Fig. 10.16. Long-channel devices suffer little from DIBL and the dependence is roughly square root. Short-channel devices exhibit much stronger V_T shifts as V_{DS} increases and the dependence becomes more linear.

The cleanest manifestation of DIBL is observed by graphing the subthreshold characteristics of a device for different values of V_{DS}. A typical case is shown in Fig. 10.17 for a 0.18-μm-gate length MOSFET. We see here a parallel negative shift in the subthreshold characteristics of the device as V_{DS} increases. It is clear in the closeup on the right that the subthreshold characteristics are roughly equally spaced along the V_{GS} axis. This reflects the nearly linear shift in V_T with V_{DS} that is characteristic of short-channel devices.

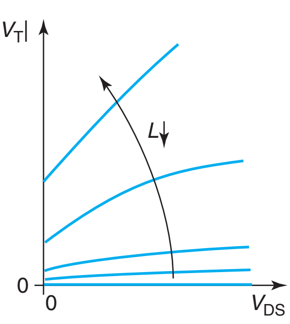


FIGURE 10.16
Sketch of shift in threshold voltage (in absolute value) with V_{DS} for different values of gate length.

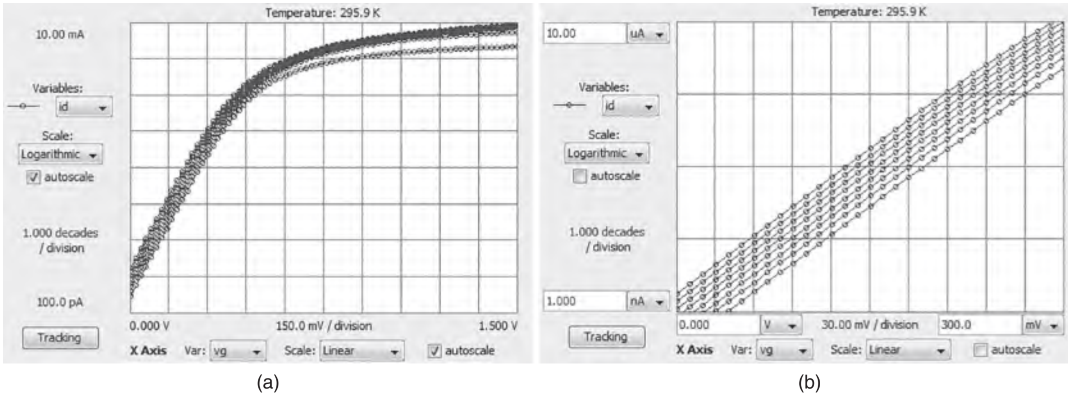


FIGURE 10.17
(a) Subthreshold characteristics of a 0.18-μm MOSFET for different values of V_{DS} from 0.1 to 1.5 V in steps of 0.2 V. (b) close up of the subthreshold characteristics indicating a nearly linear shift in V_T with V_{DS} (MIT Microelectronics iLab).

As a result of the roughly linear dependence of V_T on V_{DS} for short devices, a figure of merit that is broadly used to quantify DIBL is defined in the following way:

$$DIBL = \left| \frac{V_T(V_{DS} = V_{DD}) - V_T(V_{DS} = V_{DSsmall})}{V_{DD} - V_{DSsmall}} \right| \tag{10.29}$$

where V_{DD} is the operating voltage and $V_{DSsmall}$ is a small value of V_{DS} such as 50 or 100 mV. Although DIBL as defined above does not have units, it is usually quoted in mV/V. Typical values for state-of-the-art transistors are below 100 mV/V, which means that V_T shifts by less than 100 mV from its linear value to its saturated value at the operating bias of 1 V.

10.2.3 Subthreshold swing dependence on gate length and V_{DS}

The third short-channel effect related to electrostatics that we discuss here is the dependence of the subthreshold swing on gate length and V_{DS} that is observed in short-channel devices [Fig. 10.10(b)]. To recall, in the ideal MOSFET, the drain current drops exponentially as V_{GS} is reduced below V_T. The sharpness of the drop of the subthreshold current is characterized through the subthreshold swing, S, which was defined in Eq. (9.95). The smaller the value of S, the sharper the current drop. In the ideal MOSFET, the subthreshold swing depends solely on temperature and the design of the gate stack, that is, oxide thickness, gate metal work function, and the body doping level. Nowhere in our model of the subthreshold swing do we see a dependence on gate length or V_{DS}. Yet, this is what is seen in short devices, where as the gate length decreases or V_{DS} increases, S increases.

The dependence of S on V_{DS} and L is important because S sets the off-state current and the static power consumption. All else equal, if S increases, I_{off} increases too. This relationship is exponential. In ICs that contain hundreds of millions of transistors, excessive I_{off} quickly adds up to intolerable power dissipation. Tight control of the subthreshold swing is essential to prevent this.

The physical origin behind the V_{DS} or L dependence of the subthreshold swing is in essence the same as we saw in action in the previous two sections. It reflects the competition by the gate and the drain for electrostatic control of the energy barrier that is present in the channel of a MOSFET biased below threshold. This energy barrier sets the subthreshold current and therefore the threshold voltage *and* the subthreshold swing. As we discussed earlier in this chapter, in short devices, the lateral field that emanates from the source and the drain raises the potential along the channel with respect to a long MOSFET. This weakens the control that the gate exerts over the channel. This manifests itself as a negative shift in V_T and a slower change in the interface electron charge with V_{GS}. The subthreshold swing is a measure of this.

Since the physics are essentially the same, all the key results required to derive a first-order model for the change of S with L and V_{DS} in a short MOSFET are already in place. First, let us recall the definition of subthreshold swing. From Eqs. (9.95) and (8.40), we can write

$$S = \frac{nkT}{q}\ln 10 = \frac{\frac{kT}{q}\ln 10}{\left.\frac{d\phi_s}{dV_{GS}}\right|_T} \tag{10.30}$$

The derivative in the denominator is the change in surface potential that occurs as a result of changing V_{GS} for values around threshold.

It is particularly straightforward to derive an expression for S for low V_{DS}. For this, we work with Eq. (10.20) which gives the surface potential at the center of the channel around threshold. Using this expression, we can obtain

$$\left.\frac{d\phi_s(L/2)}{dV_{GS}}\right|_T = \left.\frac{d\phi_{slong}}{dV_{GS}}\right|_T\left(1 - \frac{1}{\cosh\frac{L}{2\lambda}}\right) \tag{10.31}$$

The derivative on the right-hand side is related to the subthreshold swing of the long-channel device. Hence, substituting Eq. (10.31) into (10.30), we have

$$S = S_{long}\frac{\cosh\frac{L}{2\lambda}}{\cosh\frac{L}{2\lambda} - 1} \tag{10.32}$$

where S_{long} refers to the subthreshold swing of the ideal long-channel device. This expression has the correct limits. For $L \gg 2\lambda$, $S \simeq S_{long}$. For $L \ll 2\lambda$, S blows up and the device cannot be shut off.

For reasonably well-designed devices that are not too short, we can take the leading term in the *cosh* function to obtain

$$S \simeq S_{long}\left(1 + 2e^{-L/2\lambda}\right) \tag{10.33}$$

This expression also has the correct limit for long devices. It shows that as L becomes comparable to about 2λ, S starts to increase above its ideal value.

Starting from Eq. (10.26) and following a similar procedure (though somehow more tedious), we can obtain a first-order expression that also includes the impact of V_{DS}. The result is

$$S \simeq S_{long}\left[1 + \frac{2(\phi_{bij} - \phi_{sT}) + V_{DS}}{\sqrt{(\phi_{bij} - \phi_{sT})(\phi_{bij} - \phi_{sT} + V_{DS})}}e^{-L/2\lambda}\right] \tag{10.34}$$

This result reverts to Eq. (10.33) for $V_{DS} = 0$. For $V_{DS} > 0$, the value of S predicted by Eq. (10.34) is higher than that of Eq. (10.33) which is consistent with experimental observations.

The right of Eq. (10.10) sketches the evolution of subthreshold swing with gate length for different values of V_{DS}. For long gate lengths, the subthreshold swing is insensitive to V_{DS} and converges to its long gate length value. For short gate lengths, S increases and so does its V_{DS} sensitivity. Note From Eq. (10.34) that this follows a roughly square-root behavior.

It is clear from this study that preventing the degradation of the subthreshold swing as a result of short-channel effects requires the design of devices with short values of λ relative to the gate length or a high channel aspect ratio.

10.3 MOSFET SHORT-CHANNEL EFFECTS: GATE STACK SCALING

In the previous section on the electrostatics of the short-channel MOSFET, we have learned about the important role of the characteristic electrostatic length, λ, and the aspect ratio of the device, L/λ. These determine what we term the electrostatic integrity of the device and several important figures of merit, such as V_T rolloff, DIBL, and subthreshold swing. The characteristic length λ itself depends on the gate oxide thickness. The need to preserve electrostatic integrity as device size scales down demands that the gate oxide is also thinned down. As we will see in the next section, gate oxide thickness has indeed been scaling in a harmonious way with gate length throughout the 40 years of the IC industry.

There are a number of complications associated with very thin gate oxides that require special consideration. This section discusses some of the most important. Among the most significant is scaling of the gate capacitance. An assumption so far in our treatment of the MOSFET is that the capacitance between the gate and the inversion layer is inversely proportional to the gate oxide thickness. There are a number of reasons why this is not entirely the case. For thick gate oxides, these do not matter very much, but for thin oxides, they become increasingly important. It is in this way that this can be considered a “short-channel” issue.

Second is the assumption that the gate oxide constitutes a perfect insulator and that the gate current is zero. As it turns out, a thin oxide does allow a small but non-negligible amount of current to flow through. This current increases exponentially as the oxide thickness decreases. This “gate leakage” current adds to the *Off* current of the device and results in static power dissipation, a key concern in modern technology.

The next sections discuss these issues in some detail.

10.3.1 Gate capacitance

The starting point in our treatment of the MOSFET current–voltage characteristics was the charge control relationship for the inversion layer. This states that the inversion layer charge increases linearly with the gate overdrive above threshold. For convenience, we reproduce this expression here

$$Q_i = -C_{ox}(V_{GS} - V_T) \tag{10.35}$$

This charge control relationship has been central to our development of the models for the long- and short-channel MOSFET and for our treatment of short-channel effects.

Underlying this expression was the assumption that in strong inversion, the gate capacitance is approximately equal to the oxide capacitance, $C_g \simeq C_{ox}$. That is why C_{ox} appears in Eq. (10.35). This is a reasonable assumption for thick gate oxides when C_{ox} is relatively small but is an increasingly bad assumption as the oxide gets thinner, the case of short-channel MOSFETs.

The assumption $C_g \simeq C_{ox}$ ignores two significant effects associated with the inversion layer and the gate electrode, respectively. Under the sheet-charge approximation, the inversion layer is assumed to have zero thickness and therefore infinite capacitance. We know from Section 8.5.1 that this is not the case. The inversion layer has a thin but nonzero thickness and therefore a finite capacitance which must be considered. Also important is the fact that the gate electrode is not a metal but heavily doped polysilicon. In strong inversion, a thin depletion layer appears at its interface with the gate oxide. This effectively removes the electrical gate slightly away from the gate/oxide interface. This phenomenon is known as *polysilicon depletion* and can be described by an additional series capacitor. These two effects are illustrated in Fig. 10.18.

In consideration of these two new issues, a more appropriate model for the gate capacitance of a MOS structure contains three capacitors in series, C_{ox} for the gate oxide, C_i for the inversion layer, and C_p for the polysilicon gate. The overall gate capacitance is then

$$C_g = \frac{1}{\frac{1}{C_{ox}} + \frac{1}{C_i} + \frac{1}{C_p}} \tag{10.36}$$

In thick gate oxide devices, the thickness of the inversion layer and the depletion layer at the gate/oxide interface are comparatively negligible and $C_g \simeq C_{ox}$ represents a reasonable approximation. For deeply scaled MOSFETs with thin gate oxides, C_{ox} is high and neglecting the role of C_i and C_p can significantly *overestimate* C_g and the drain current. In the rest of this section, we address these two effects and introduce appropriate corrective terms.

Starting with the inversion layer capacitance, when deriving Eq. (10.35), we performed the sheet-charge approximation which assumes that the inversion layer is an infinitely thin sheet located at the semiconductor surface. A consequence of this is that the surface potential is pinned at the value that corresponds to threshold, ϕ_{sT}. In reality, we know that the inversion layer has a finite thickness and that ϕ_s increases beyond ϕ_{sT}. This is shown, for example, in Figs. 8.46 and 8.44, respectively.

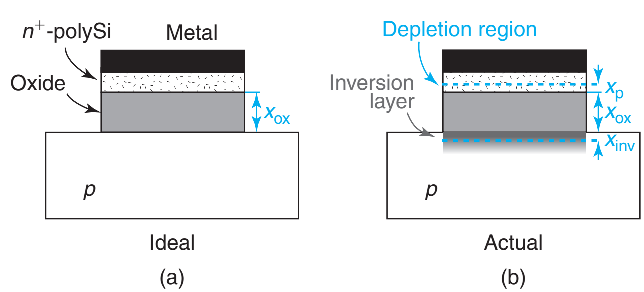


FIGURE 10.18
Schematic of ideal MOS polysilicon gate stack (a) and more realistic gate stack (b) illustrating polysilicon depletion and finite thickness of inversion layer.

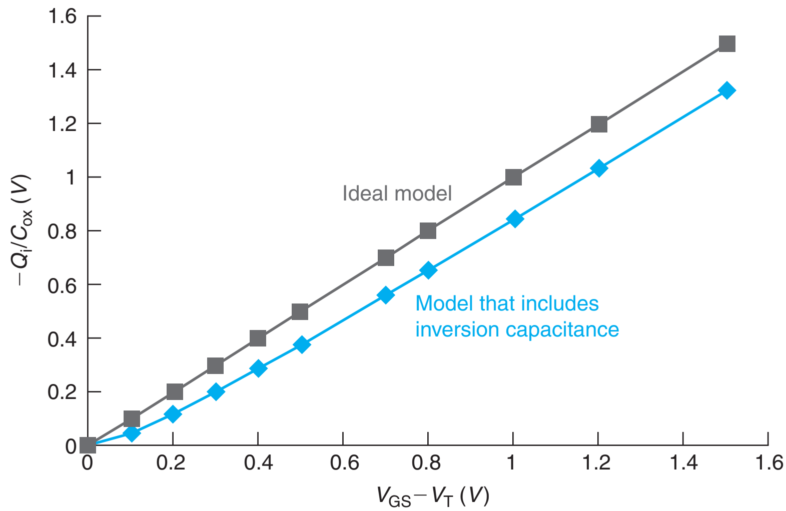


FIGURE 10.19
$-Q_i/C_{ox}$ for the ideal charge control model and a revised model that includes the effect of inversion capacitance [Eq. (10.37)].

In Section 8.5.1, we developed a model for the MOS gate capacitance. We derived a simple expression for the gate capacitance in inversion [Eq. (8.51)]. We can use this expression to obtain Q_i by simply multiplying it by $-dV_{GS}$ and integrating from V_T to V_{GS}.[3] This is a straightforward integral that yields

$$Q_i \simeq -C_{ox}\left\{V_{GS} - V_T - \frac{2kT}{q}\ln\left[1 + \frac{q}{2kT}(V_{GS} - V_T)\right]\right\} \tag{10.37}$$

This is an interesting expression that is graphed in Figure 10.19 together with the ideal charge control relationship [Eq. (10.35)]. The first two terms inside the curly brackets represent the traditional charge control relationship. These are followed by a new negative term that goes to zero at $V_{GS} = V_T$ but it grows as the inversion layer is turned on. So, for any amount of gate overdrive, this new expression yields a lower value of $|Q_i|$ than the traditional expression. The correction increases as the overdrive increases but only in a logarithmic form. The lower $|Q_i|$ translates into lower current than predicted by the simple model. The relative discrepancy is larger for low gate overdrive which is the relevant case for deeply scaled transistors.

The second correction to the gate capacitance model deals with the polysilicon gate. Strictly speaking, when the MOS structure is in strong inversion, the polysilicon gate is in depletion and a thin depletion region appears at its interface with the gate oxide. This is illustrated in Fig. 10.20. Since the doping level is very high ($> 10^{20}$ cm^{-3}, typically), this depletion region is very thin and in many cases can be neglected. However, as the gate oxide thickness scales to extremely small dimensions, this effect becomes increasingly important and cannot be ignored. The existence of this depletion region in the polysilicon gate electrode effectively separates it from the electrical

[3] Here we follow Y. Taur and T. H. Ning, *Fundamentals of Modern VLSI Devices*, Second Edition, 2009, pp. 173–174.

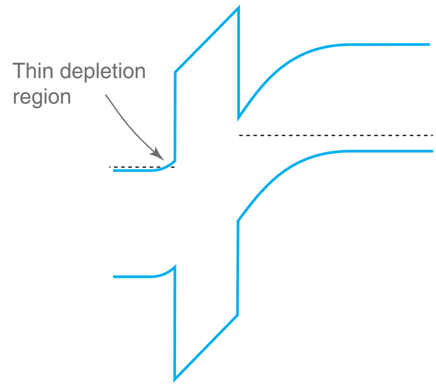


FIGURE 10.20
Energy band diagram with a MOS structure with an n^+ polysilicon gate in strong inversion. The existence of a thin depletion region at the polysilicon/oxide interface is indicated.

point of view further away from the inversion layer. For a given gate overdrive, this reduces the inversion layer charge.

Polysilicon depletion can be modeled by adding a capacitance in series with C_{ox} and C_i as done in Eq. (10.36). To the first order, this capacitance can be modeled as

$$C_p \simeq \frac{\epsilon_s}{x_p} \tag{10.38}$$

where x_p represents the thickness of the polysilicon depletion region.

We can easily estimate x_p. With the MOS structure biased in strong inversion, the charge in the polysilicon depletion region images all the charge in the semiconductor which is the sum of the charge in the inversion layer plus that in the depletion layer underneath. The charge in the inversion layer is to the first order given by Eq. (10.35). The charge in the depletion layer is Q_{dmax} which for a polysilicon-gate MOSFET, we saw in Section 10.1.1, can be approximated by $Q_{dmax} \simeq -C_{ox}V_T$. All together, then, the charge in the depletion region of the polysilicon gate is approximately given by

$$Q_g \simeq C_{ox}V_{GS} \tag{10.39}$$

With a doping level in the polysilicon gate of N_p (n-type for an n-MOSFET), the extent of the depletion region can be obtained from $qN_px_p = Q_g$ to yield

$$x_p \simeq \frac{Q_g}{qN_p} = \frac{C_{ox}V_{GS}}{qN_p} \tag{10.40}$$

The capacitance associated with this depletion region is then

$$C_p \simeq \frac{qN_p\epsilon_s}{C_{ox}V_{GS}} \tag{10.41}$$

Ignoring the role of C_i, the gate capacitance is

$$C_g = C_{ox} \frac{1}{1 + \frac{C_{ox}}{C_p}} = C_{ox} \frac{1}{1 + \frac{C_{ox}^2 V_{GS}}{q N_p \epsilon_s}} \tag{10.42}$$

To compute Q_i, we proceed as before. We multiply $-C_g$ by dV_{GS} and integrate from V_T to V_{GS} to yield

$$\begin{aligned} Q_i &= -C_{ox} \left(\frac{q N_p \epsilon_s}{C_{ox}^2} \right) \left[\ln(1 + \frac{C_{ox}^2 V_{GS}}{q N_p \epsilon_s}) - \ln\left(1 + \frac{C_{ox}^2 V_T}{q N_p \epsilon_s}\right) \right] \\ &\simeq -C_{ox}(V_{GS} - V_T) \left[1 - \frac{C_{ox}^2}{2 q N_p \epsilon_s} (V_{GS} + V_T) \right] \end{aligned} \tag{10.43}$$

In deriving Eq. (10.43), we have further assumed that the correction imposed by polysilicon depletion is relatively small. This allows us to Taylor series expand the logarithm expressions and select the two leading terms.

We find that polysilicon depletion introduces a new term in the charge control relationship of the MOS structure. The new term is less than one and multiplies C_{ox}. This is quite physically intuitive as the capacitance associated with the polysilicon gate is in series with C_{ox}. The presence of this new term clearly detracts from the inversion charge at any gate overdrive. Interestingly, the correction becomes more important as the gate overdrive increases. This is to be expected since the depletion region in the polysilicon increases with overall band bending. In addition, the correction becomes more important as the gate oxide thickness is reduced and as the polysilicon doping level decreases, as expected from our physical discussion above.

Polysilicon depletion is illustrated in Fig. 10.21 that sketches the *C–V* characteristics of a MOS structure with and without polysilicon depletion. In accumulation and in depletion, the *C–V* characteristics are unchanged. In inversion, the gate capacitance with polysilicon depletion is lower than for the ideal case. The discrepancy increases as V_{GS} increases. This anomaly in the *C–V* characteristics in strong inversion is the clearest manifestation of polysilicon depletion.

Polysilicon depletion and the finite inversion capacitance introduce significant corrections to the gate capacitance of short MOSFETs with important consequences to their drive current. As discussed, both effects become more significant the thinner the gate oxide. A way to account for these corrections in simple terms is the introduction of the notion of *Capacitance Equivalent Thickness*, or CET.[4] This is the gate oxide thickness that one would have to use in the ideal charge control relationship [Eq. (10.35)] in order to get the same inversion charge obtained from the correct calculations that include the two effects discussed here. Obviously, looking at Eqs. (10.37) and (10.43), defining CET this way makes it dependent on the value of V_{GS}. Typically, the value of V_{GS} that corresponds to the *on* current is selected.

[4] *Capacitance equivalent thickness* (CET) should not be confused with *equivalent oxide thickness* (EOT). They both represent an equivalent oxide thickness but the definitions are different. EOT is defined in Section 10.5.6.

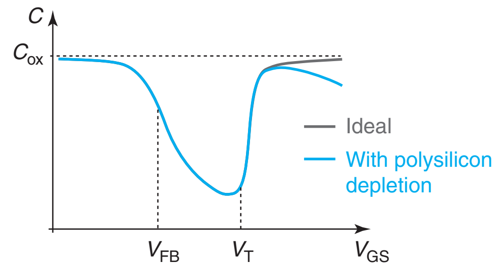


FIGURE 10.21
Illustration of polysilicon depletion. The graph sketches the *C–V* characteristics of an ideal MOS structure and one suffering from polysilicon depletion. The characteristics differ in strong inversion where under polysilicon depletion, the gate capacitance actually drops as V_{GS} increases.

We can easily derive an expression for CET by equating the general expression of the gate capacitance [Eq. (10.36)] to an effective oxide capacitance that is entirely determined by an oxide thickness equal to CET. Solving for CET yields

$$CET = x_{ox}\left(1 + \frac{C_{ox}}{C_i} + \frac{C_{ox}}{C_p}\right) = x_{ox} + \frac{\epsilon_{ox}}{\epsilon_s}(x_i + x_p) \tag{10.44}$$

where x_{ox} refers to the physical thickness of the gate oxide and x_i is an effective thickness for the inversion layer. Obviously $CET > x_{ox}$.

EXERCISE 10.3

Consider a MOS gate stack consisting of 2 nm *SiO*$_2$ *dielectric and a poly-Silicon gate with a doping level of* $N_p = 10^{20}\ \mathrm{cm}^{-3}$ *yielding a* $V_T = 0.3$ V. *Estimate the CET of this gate stack at* $V_{GS} = 1.2$ V.

For this gate oxide, $C_{ox} = \epsilon_{ox}/x_{ox} = 1.7 \times 10^{-6}\ \mathrm{F/cm^2}$. We start by estimating C_{ox}/C_i. For this, we can use Eq. (8.47):

$$\frac{C_{ox}}{C_i} \simeq \frac{2kT}{q(V_{GS} - V_T)} = \frac{2 \times 0.026\ \mathrm{V}}{1.2 - 0.3\ \mathrm{V}} = 5.8 \times 10^{-2}$$

To estimate C_{ox}/C_p we use Eq. (10.41):

$$\frac{C_{ox}}{C_p} \simeq \frac{C_{ox}^2 V_{GS}}{qN_p\epsilon_s} = \frac{(1.7 \times 10^{-6}\ \mathrm{F/cm^2})^2 \times 1.2\ \mathrm{V}}{1.6 \times 10^{-19}\ C \times 10^{20}\ \mathrm{cm}^{-3} \times 1.04 \times 10^{-12}\ \mathrm{F/cm}} = 0.21$$

Then, using Eq. (10.44), we have

$$CET = x_{ox}(1 + \frac{C_{ox}}{C_i} + \frac{C_{ox}}{C_p}) = 2\ \mathrm{nm}(1 + 0.06 + 0.21) = 2.5\ \mathrm{nm}$$

This is a non-negligible increase over the physical oxide thickness ($x_{ox} = 2$ nm) that should noticeably affect current drive and short-channel effects.

10.3.2 Gate leakage current

The gate current of an ideal MOSFET was assumed to be zero. We can argue this from the fact that dielectrics are in general excellent electrical insulators but, more rigorously, as a result of the large conduction and valence band discontinuities that are present at the Si/oxide interfaces (a few electron volt). These represent an effective block to the flow of carriers across the oxide (see, for example, Figure 8.7). This picture should hold regardless of the oxide thickness. However, it is experimentally observed that when the gate oxide becomes very thin, as is the case of short MOSFETs, the gate actually "leaks" and a finite amount of gate current flows. This gate leakage current is seen to increase exponentially as the oxide thickness is reduced. It also increases exponentially with voltage. In deeply scaled devices, the current can become so large that the power consumption associated with it comes to represent a significant fraction of the standby power budget. In recent times, this has emerged as such a severe problem that addressing it has profoundly shaped the evolution of MOSFET design, as we discuss later on in this chapter.

The physics of gate leakage current in very thin gate oxides is a quantum mechanical phenomenon that derives from the wave nature of the electron. Classically, electrons cannot flow through the gate oxide if their energy is not sufficient to overcome the conduction band discontinuity with the semiconductor. Quantum mechanics, however, allows for a finite chance for the electron to punch through the oxide (see Fig. 10.22). The probability is usually very

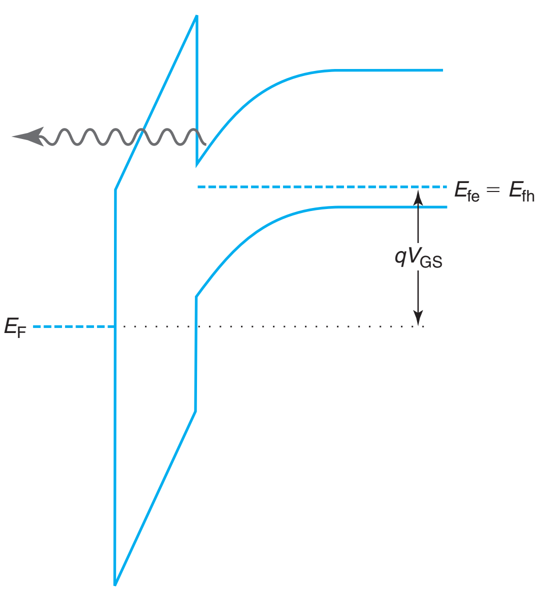


FIGURE 10.22
Illustration of electron tunneling through gate oxide.

small but it becomes sizeable when the wavelength of the electron (the de Broglie wavelength, λ_B, introduced in Chapter 1) becomes comparable to the gate oxide thickness. With λ_B in the few nanometer range, oxide tunneling has become a very real issue in the last few years.

The development of a detailed model for gate leakage current is beyond the scope of this text as it requires dealing with its quantum mechanical nature. However, it is appropriate to summarize here the result of a simple model that captures the key dependencies.

From quantum mechanics, the probability that an electron penetrates through a classically forbidden barrier is exponentially dependent on the inverse of the barrier thickness and the square root of its height. In a MOSFET, this leads to a gate current that exhibits the following dependencies:

$$J_g = A\mathcal{E}_{ox}^2 \exp\left(-\frac{B}{\mathcal{E}_{ox}}\right) \tag{10.45}$$

where $\mathcal{E}_{ox}$ is the electric field through the oxide and A and B are constants that contain relevant physical and material parameters.

The most significant dependence of the gate current in Eq. (10.45) is its inverse exponential dependence on the electric field across the gate oxide. This implies that, for a constant gate voltage, as the gate oxide thickness is reduced, the gate current increases very fast. This is sketched in Fig. 10.23. A rule of thumb used by device designers is that the tolerable limit for gate leakage is a current density of 1 A/cm^2 at 1 V at room temperature. Experiments indicate that this level of current is reached for an SiO_2 thickness of about 1.7–1.8 nm. This point was attained in the mid-late 2000s. Managing these potentially large gate leakage currents while continuing to scale has required the introduction of new gate dielectrics.

As it has become clear, scaling gate oxide thickness is essential to maintaining good short-channel effects and continue to enhance performance as devices get smaller. Gate leakage current brings oxide thickness scaling to a halt and therefore threatens overall transistor footprint scaling. A way around this limit is to change gate dielectrics, that is, to

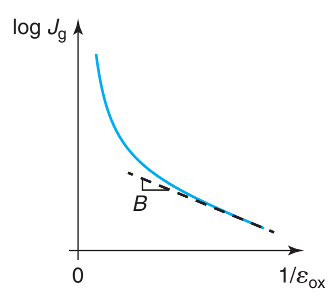


FIGURE 10.23
Sketch of gate current due to tunneling as a function of inverse of electric field across the gate oxide [Eq. (10.45)].

move away from SiO_2 and instead use a gate dielectric that exhibits a higher permittivity. This is often referred to as a *high-K dielectric*.[5] The reason for this approach is clear. For a given layer thickness, all other things being equal, a higher permittivity gate dielectric yields more capacitance, more current drive and better short-channel effects than SiO_2. This improved performance can be partially traded away against gate leakage current by using a thicker dielectric layer. As Eq. (10.45) indicates, this reduces $\mathcal{E}_{ox}$ in a linear manner and thefore the gate leakage current in an exponentially way.

When discussing alternative gate dielectrics, it is common to refer to a figure of merit called *equivalent SiO_2 thickness* (or simply EOT, for "equivalent oxide thickness"). EOT of a given dielectric layer is the thickness of an SiO_2 layer that yields the same capacitance. EOT is determined by its physical thickness and the ratio of its permittivity to that of SiO_2:

$$EOT_{\text{high-K}} = \frac{\epsilon_{SiO_2}}{\epsilon_{\text{high-K}}} x_{\text{high-K}} = \frac{\kappa_{SiO_2}}{\kappa_{\text{high-K}}} x_{\text{high-K}} \tag{10.46}$$

The oxide capacitance associated with the dielectric in the gate stack, C_{ox}, as used throughout this book, is related to EOT in the following way:

$$C_{ox} = \frac{\epsilon_{\text{high-K}}}{x_{\text{high-K}}} = \frac{\epsilon_{SiO_2}}{EOT_{\text{high-K}}} \tag{10.47}$$

This equation leaves clear that higher value of C_{ox} require the use of thinner EOT gate dielectrics. The use of high-K dielectrics allows EOT scaling without physical oxide thickness scaling, therefore limiting gate leakage current.

EXERCISE 10.4

Consider the MOS gate stack of Exercise 10.3. Estimate the EOT of this gate stack. Estimate also the CET and EOT of an alternative gate stack that uses a metal gate and a 4-nm-*thick HfO_2 gate oxide* ($\kappa = 25$) *with an identical* $V_T = 0.3$ V *at* $V_{GS} = 1.2$ V.

The EOT of the gate stack of Exercise 10.3 is simply 2 nm. This is because the gate oxide is SiO_2. With a 4-nm-thick HfO_2 gate dielectric, the EOT is, from Eq. (10.46):

$$EOT_{\text{high-K}} = \frac{\kappa_{SiO_2}}{\kappa_{\text{high-K}}} x_{\text{high-K}} = \frac{3.9}{25} 4 \text{ nm} = 0.62 \text{ nm}$$

This is a very significant improvement over the SiO_2 gate stack.

For CET, we have to go back to Eq. (10.44). By introducing a metal gate, the poly-Silicon depletion problem disappears and the term on C_{ox}/C_p goes to zero (effectively $C_p = \infty$). The term on C_{ox}/C_i is unchanged from what we calculated in Exercise 10.3 since it is entirely set by $V_{GS} - V_T$. Hence, we have

$$CET = EOT\left(1 + \frac{C_{ox}}{C_i}\right) = 0.62 \text{ nm}(1 + 0.06) = 0.66 \text{ nm}$$

This is a remarkable improvement over the CET that was obtained in Exercise 10.3.

[5]This is pronounced as the English letter "K" even though this name comes from the Greek letter κ (kappa) for permittivity.

10.4 MOSFET HIGH-FIELD EFFECTS

The electric field distribution inside a MOSFET is under many conditions highly nonuniform. In a MOSFET in saturation, in particular, the field sharply peaks at the drain end of the channel. For high values of V_{DS}, this field can have a sizable magnitude. This brings a number of concerns.

Electrons traveling down the channel in a MOSFET in saturation can gain substantial kinetic energy from the high electric field and in this way get very "hot." Hot electrons can in turn generate electrons and holes through *impact ionization* (studied in Section AT4.2). The extra electrons contribute additional drain current and the holes are extracted by the substrate and contribute to *substrate current*. With high enough voltage, the electric field in the device can get so high that *avalanche breakdown* occurs and the device current grows out of control. This can damage or even destroy the transistor, therefore setting a limit to the maximum voltage that the MOSFET can handle. Under the right conditions, hot electrons can also overcome the energy barrier at the Si/SiO_2 interface and escape through the gate. This creates *gate current*.

Hot electrons not only result in additional terminal currents in the MOSFET but also affect device reliability. Under certain conditions, hot electrons can induce *device degradation* through the creation of *interface traps* at the Si/SiO_2 interface under the gate. These are dangling bonds that can trap carriers shifting the local threshold voltage and degrading the channel mobility in that region. This potentially affects many electrical figures of merit of the device and compromises circuit operation.

Managing the electric field distribution particularly under high-voltage conditions is a crucial aspect of MOSFET design. As we will see below, MOSFET scaling generally results in increasing electric fields inside the device. This makes high-field effects much more important in very small devices.

In this book, we cannot do proper justice to all the interesting and important high-field effects that take place in MOSFETs. Our goal here is to gain an appreciation and develop first-order models for some of the most important phenomena. This should be a good basis for further self-study. We start by discussing the electric field distribution at the drain end of the channel with the MOSFET biased in saturation. This is the condition that leads to some of the most prominent high-field effects that we discuss in the following sections.

10.4.1 Electrostatics of velocity saturation region

The most problematic situation for high electric field effects in a MOSFET takes place when the device is biased in saturation with a high V_{DS}. We know that under these conditions, the electric field distribution along the channel is very nonuniform with a high peak appearing at its drain end. Many of the physical phenomena that we care about are quite nonlinear in the electric field. As a result, estimating the value of this peak electric field and understanding its key dependencies is of great importance. This forces us to deal with something that we have avoided so far and that is the electrostatics of the pinch-off region.

In a MOSFET in saturation, the drain current is weakly dependent on V_{DS}. In Chapter 9, we attributed this to the appearance of a pinch-off point in the channel beyond which the electron concentration at the surface of the semiconductor drops below the background acceptor concentration. This meant that there is in fact a depletion region at the drain end of the channel and that the extra voltage applied to the drain beyond V_{DSsat} is used to increase the extent of this depletion region. This only affects in a minor way the electric field along the rest of the channel all the way back to the source with the consequence of an almost perfectly saturated drain current.

Earlier in this chapter, we learned that the picture presented in Chapter 9 was too simplistic. As we studied the impact of velocity saturation in the current–voltage characteristics of the MOSFET, we learned that current saturation can also occur due to carrier velocity saturation and not just pinch-off. Even in relatively long MOSFETs operating at fairly benign bias points, the electric field at the drain end of the channel can get high enough to saturate the channel carrier velocity. Increasing V_{DS} beyond the value that leads to this condition, V_{DSsat}, does not increase the carrier velocity and the current saturates.

In Section 10.1.2, we did not deal with the high-field region at all. In fact, if one looks in detail (see Problem 10.2), the formulation that we developed predicts that for $V_{\mathrm{DS}} = V_{\mathrm{DSsat}}$, the electric field at the drain end of the channel goes to infinity. This is obviously not possible and it suggests that something is wrong with this formulation. What is wrong is that the gradual channel approximation fails at the drain end of the channel as V_{DS} approaches V_{DSsat} (see Problem 10.4). Hence, the formalism of Section 10.1.2 is only valid for values of V_{DS} close to V_{DSsat} but not higher.

Detailed computer simulations have provided insight into the relevant physics for V_{DS} around and beyond V_{DSsat}. This has allowed the development of a relatively simple model that does a reasonable job in capturing the essential physics. This is described next.

From simulations we know that as V_{DS} gets close to V_{DSsat} and the electron velocity at the drain end of the channel approaches the saturation velocity, the lateral electric field in the channel gets very large. With a very high and fast increasing lateral electric field, the gradual channel approximation no longer applies. In fact, Gauss' law requires that the vertical electric field shrink which, in turn, weakens the confinement of the electrons at the surface of the semiconductor. A consequence of this is that at the drain end of the channel, the electron distribution spreads out to some depth into the substrate as electrons proceed toward the drain. In this regime and in this region, the sheet-charge approximation fails.

If V_{DS} is increased beyond V_{DSsat}, the extra voltage has to be accommodated by a larger lateral electric field. This necessarily moves the point at which the GCA fails further up channel toward the source and broadens the region in which velocity saturation effects are prominent and the GCA does not apply. It is this “velocity saturation region” at the drain end of the channel that we need to model in order to understand high-field effects.

The electrostatics of the velocity saturation region are rather complex. However, it is possible to develop a simple model that captures the essence of the situation. A schematic diagram of the high-field region is shown in Fig. 10.24. The details of the model are given in Appendix AT10.2.

For a transistor biased with $V_{\mathrm{DS}} > V_{\mathrm{DSsat}}$, the velocity saturation region extends from a point in the channel at a coordinate L' to the drain end of the channel at $y = L$. We select L' as the point at which $V(L') = V_{\mathrm{DSsat}}$. At this point, we assume that the lateral electric field is equal to $|\mathcal{E}_{\mathrm{y}}(L')| = \mathcal{E}_{\mathrm{sat}}$, the magnitude of the field beyond which prominent velocity saturation effects occur (see Fig. 4.4). The bottom boundary of the velocity saturation region is placed at $x = x_{\mathrm{j}}$, the junction depth. In this region, we assume that the electron concentration is uniform in depth.

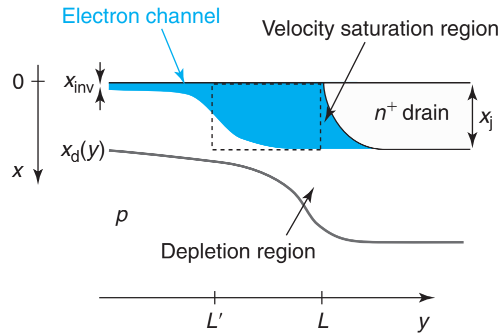


FIGURE 10.24
Velocity saturation region at drain end of channel of MOSFET biased in the saturation regime with $V_{DS} > V_{DSsat}$.

This is not unreasonable as the electrons ultimately flow toward the drain with a depth of x_j. Solving this problem for a given value of V_{DS} means figuring out the location of L' and obtaining an approximate distribution for the lateral electric field $\mathcal{E}_y(y)$ in the velocity saturation region.

Appendix AT10.2 presents the details of the problem. The electrostatic problem is formulated in terms of $V(y)$, the channel voltage. The differential equation that governs the evolution of $V(y)$ in the velocity saturation region is found to be

$$\frac{d^2V}{dy^2} - \frac{V}{l^2} = -\frac{V_{DSsat}}{l^2} \tag{10.48}$$

where l is a characteristic length for this electrostatic problem given by

$$l = \sqrt{\frac{\epsilon_s}{\epsilon_{ox}} x_{ox} x_j} \tag{10.49}$$

A solution of this differential equation has the form:

$$V(y) = V_{DSsat} + A \sinh \frac{y - L'}{l} + B \cosh \frac{L - y + L'}{l} \tag{10.50}$$

A and B can be obtained from two boundary conditions of this problem:

$$V(y = L') = V_{DSsat} \tag{10.51}$$

$$V(y = L) = V_{DS} \tag{10.52}$$

This leads to

$$V(y) = V_{DSsat} + (V_{DS} - V_{DSsat}) \frac{\sinh \frac{y-L'}{l}}{\sinh \frac{L-L'}{l}} \tag{10.53}$$

At this point, we still do not know the value of L'. We can get this by deriving an expression for the lateral electric field distribution and assigning a value of $-\mathcal{E}_{sat}$ at $y = L'$. A general expression for the lateral electric field can be obtained by differentiating Eq. (10.53):

$$\mathcal{E}_y(y) = -\frac{dV}{dy} = -\frac{V_{DS} - V_{DSsat}}{l}\frac{\cosh\frac{y-L'}{l}}{\sinh\frac{L-L'}{l}} \tag{10.54}$$

At $y = L'$, the lateral field is $-\mathcal{E}_{sat}$:

$$\mathcal{E}_y(y = L') = -\mathcal{E}_{sat} = -\frac{V_{DS} - V_{DSsat}}{l}\frac{1}{\sinh\frac{L-L'}{l}} \tag{10.55}$$

At $y = L$, the peak field occurs. In absolute magnitude, we denote this as $\mathcal{E}_{max}$:

$$\mathcal{E}_y(y = L) = -\mathcal{E}_{max} = -\frac{V_{DS} - V_{DSsat}}{l}\frac{1}{\tanh\frac{L-L'}{l}} \tag{10.56}$$

Squaring these last two relationships and taking the difference allows us to express $\mathcal{E}_{max}$ in terms of $\mathcal{E}_{sat}$:

$$\mathcal{E}_{max} = \sqrt{\mathcal{E}_{sat}^2 + \left(\frac{V_{DS} - V_{DSsat}}{l}\right)^2} \tag{10.57}$$

Plugging this into Eq. (10.56) yields the length of the high-field region:

$$L - L' = l \ln\left[\frac{V_{DS} - V_{DSsat}}{l\mathcal{E}_{sat}} + \sqrt{1 + \left(\frac{V_{DS} - V_{DSsat}}{l\mathcal{E}_{sat}}\right)^2}\right] \tag{10.58}$$

All these expressions have the right limits. For $V_{DS} = V_{DSsat}$, the length of the high-field region goes to zero and $\mathcal{E}_{max} = \mathcal{E}_{sat}$. As V_{DS} increases beyond V_{DSsat}, the high-field region grows and $\mathcal{E}_{max}$ increases beyond $\mathcal{E}_{sat}$.

To finish, the expression of $\mathcal{E}_{sat}$ obtained in Eq. (10.55) allows us to simplify the expressions of $V(y)$ and $\mathcal{E}_y(y)$ in the following way:

$$V(y) = V_{DSsat} + l\mathcal{E}_{sat}\sinh\frac{y - L'}{l} \tag{10.59}$$

and

$$\mathcal{E}_y(y) = -\mathcal{E}_{sat}\cosh\frac{y - L'}{l} \tag{10.60}$$

Figure 10.25 sketches the voltage and field distributions along the channel for $V_{DS} > V_{DSsat}$. In the velocity saturation region, the channel charge is independent of y and the electron velocity is saturated to v_{sat}.

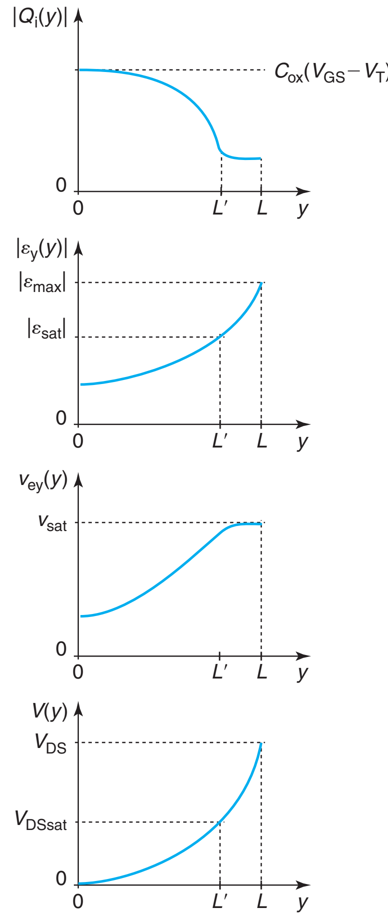


FIGURE 10.25
Sketch of channel charge, lateral electric field, electron velocity, and channel voltage as a function of position along the channel for a MOSFET in the saturation regime with $V_{DS} > V_{DSsat}$. The high-field region is located between $y = L'$ and L. In this region, the electron velocity is saturated, and the integrated electron charge is constant.

The most important result of this section is Eq. (10.57) which gives a first-order expression for the peak electric field in the high-field region. $\mathcal{E}_{max}$ increases as V_{DS} increases past V_{DSsat}. For high enough voltage, $\mathcal{E}_{max}$ evolves in a roughly linear way with V_{DS}.

10.4.2 Impact ionization and substrate current

As we have just seen, in a MOSFET in saturation, the electric field at the drain end of the channel can get quite high. Electrons traveling down the channel are therefore capable of gaining enough kinetic energy to have a finite probability of generating electron–hole pairs through

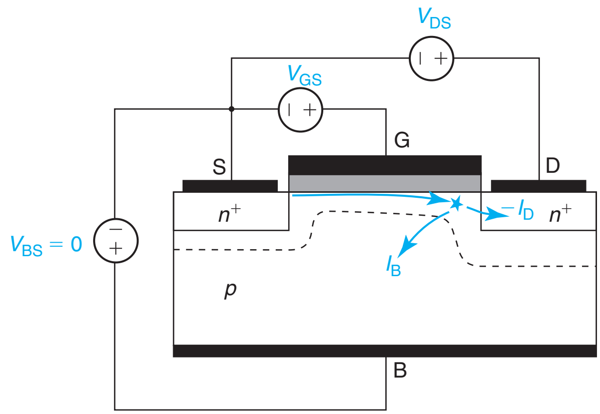


FIGURE 10.26
Sketch of impact ionization and substrate current in a MOSFET biased in the saturation regime.

impact ionization. The electrons that are produced represent an additional contribution to the drain current. The holes are extracted by the substrate and flow out of the device through the substrate contact contributing in this way to negative substrate current. Figure 10.26 shows a schematic diagram of the situation.

In a MOSFET biased in saturation with $V_{BS} = 0$, in the absence of impact ionization, the substrate current is simply the reverse bias current through the drain-body PN junction. This is usually very small and fairly independent of V_{DS}. When V_{DS} is increased to high enough values, impact ionization takes place and negative substrate current starts flowing. Because of the exponential dependence of impact ionization and electric field [see Eq. (4.108)], the substrate current increases roughly exponentially with V_{DS}. This is clearly seen in Fig. 10.27 that graphs the experimental output characteristics of a MOSFET with a 3-μm-gate length (left axis) and the absolute of the substrate current (right axis) in a semilogarithmic scale. For V_{DS} below 2.8 V, the substrate current is negligible. However, when V_{DS} increases beyond this value, I_B increases very fast.

Impact ionization not only produces substrate current but also affects the drain current that now contains an additional component due to the generated electrons that flow toward the drain. In the experimental characteristics of Fig. 10.27, this component is relatively small and its effect on the overall drain current is imperceptible. However, since impact ionization grows exponentially with V_{DS}, it can easily become significant and must be accounted for. On the other hand, to the first order, the source current is unaffected by the presence of impact ionization and remains well described by the current models developed in earlier sections.[6] A consequence of impact ionization is then that the source and drain currents are no longer equal. Their ratio is precisely the *multiplication coefficient* that we introduced in Chapter 5 (Section 5.10).

[6]We will see in Section AT11.1 that a “hidden” bipolar transistor effect can enter the picture and modify the source current too.

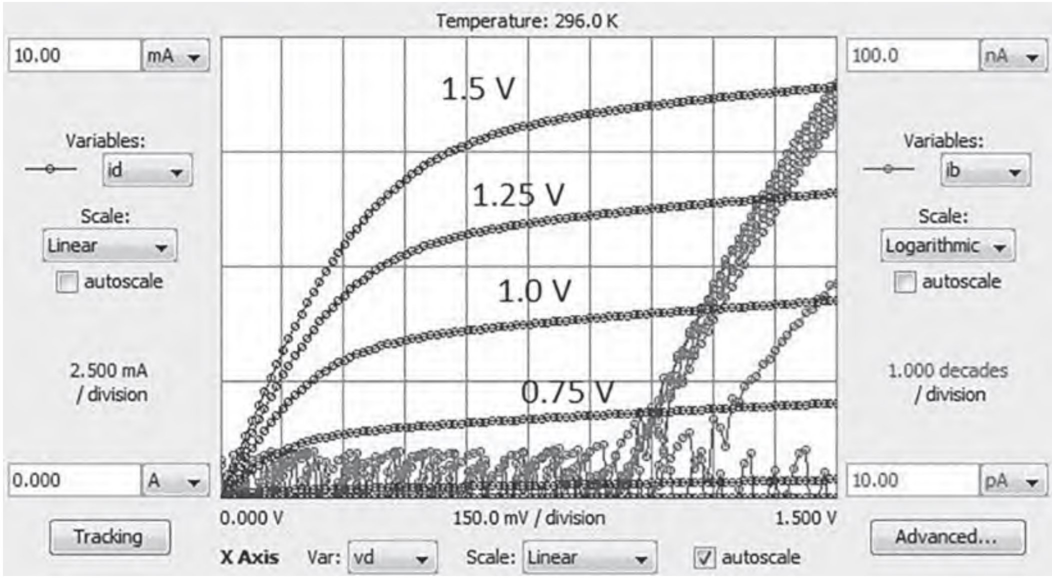


FIGURE 10.27
Output characteristics and substrate current in a MOSFET with a 0.18-μm-gate length for different values of V_{GS} (Data obtained through the MIT Microelectronics iLab).

Developing a simple model for the MOSFET currents that includes impact ionization is not difficult. We start by recognizing that, as indicated above, the ratio of the drain to the source current is the multiplication coefficient. Then

$$I_D = M|I_S| \tag{10.61}$$

The substrate current is the difference between these two:

$$I_B = -I_D - I_S = -(M-1)|I_S| \tag{10.62}$$

Writing the substrate current this way emphasizes the fact that it is negative since it is carried by holes coming out of the body contact.

The calculation of the multiplication coefficient must account for the fact that the electric field is highly nonuniform inside the transistor. We dealt with this situation in Advanced Topic **??**. For a case in which impact ionization is started by electrons, as relevant here, Eq. (5.165) applies. This is a complicated expression particularly considering the fact that the impact ionization coefficients depend exponentially on the electric field [see Eq. (4.108)].

Two simplifications yield a reasonably simple and insightful analytical solution. First, we are often interested in the weak impact ionization regime, that is, the regime in which the electric field is not too large and $M \ll 1$. In this case, the impact ionization coefficients α_e and α_h are very small. This is, for example, the situation depicted in Fig. 10.27 in which the substrate current is several orders of magnitude smaller than the drain current. Second, in this regime, as indicated in Figure 4.27, $\alpha_e \gg \alpha_h$. These two realizations simplify Eq. (5.165) to

$$M \simeq \frac{1}{1 - \int_0^L \alpha_e dy} \simeq 1 + \int_0^L \alpha_e dy \tag{10.63}$$

$M - 1$ is then

$$M - 1 \simeq \int_0^L \alpha_e(\mathcal{E}) dy \tag{10.64}$$

In Section AT4.2.3, we discussed the basic physics of impact ionization and argued that to the first order, the impact ionization rate evolves as

$$\alpha \simeq A \exp\left(-\frac{B}{|\mathcal{E}|}\right) \tag{10.65}$$

where A and B are constants that are characteristic of the material and temperature. Plugging this into Eq. (10.64) yields

$$M - 1 \simeq \int_0^L A \exp\left(-\frac{B}{|\mathcal{E}|}\right) dy = \int_0^L A \exp\left(\frac{B}{\mathcal{E}}\right) dy \tag{10.66}$$

where we have noted the fact that with the choice of axis that we have made in the analysis of the MOSFET, $\mathcal{E}$ is negative, and $|\mathcal{E}| = -\mathcal{E}$.

Before we attempt to perform this integral, it is best to change variables from y to $\mathcal{E}$:

$$M - 1 = A \int_{-\mathcal{E}_{sat}}^{-\mathcal{E}_{max}} \exp\left(\frac{B}{\mathcal{E}}\right) \frac{dy}{d\mathcal{E}} d\mathcal{E} \tag{10.67}$$

where the minus signs in the limits of the integral come because we define both $\mathcal{E}_{sat}$ and $\mathcal{E}_{max}$ as positive quantities.

We now need to use the expression of the electric field derived in the previous section. In general, what results is not amenable to an exact analytical solution. A reasonable approximation is obtained if we note that due to the exponential dependence of the integrand on the electric field, the greatest contribution to this integral comes from the drain end of the channel where the electric field is the highest in absolute value.

Around $y = L$, the electric field [as given by Eq. (10.60)], can be approximated by

$$\mathcal{E}(y) \simeq -\frac{1}{2}\mathcal{E}_{sat} \exp\frac{y - L'}{l} \tag{10.68}$$

and, then

$$\frac{d\mathcal{E}}{dy} \simeq -\frac{1}{2}\frac{\mathcal{E}_{sat}}{l} \exp\frac{y - L'}{l} \simeq \frac{\mathcal{E}}{l} \tag{10.69}$$

We plug this result in Eq. (10.67) and perform another variable transformation:

$$M - 1 = Al \int_{-\mathcal{E}_{\text{sat}}}^{-\mathcal{E}_{\text{max}}} \frac{1}{\mathcal{E}} \exp\left(\frac{B}{\mathcal{E}}\right) d\mathcal{E} = -Al \int_{-1/\mathcal{E}_{\text{sat}}}^{-1/\mathcal{E}_{\text{max}}} \mathcal{E} \exp\left(\frac{B}{\mathcal{E}}\right) d\left(\frac{1}{\mathcal{E}}\right) \tag{10.70}$$

Since the greatest contribution to this integral comes in the region where $-\mathcal{E} \simeq \mathcal{E}_{\text{max}}$, we can bring the linear term out of the integral as a constant equal to $-\mathcal{E}_{\text{max}}$ and then perform the integral:

$$M - 1 \simeq Al\mathcal{E}_{\text{max}} \int_{-1/\mathcal{E}_{\text{sat}}}^{-1/\mathcal{E}_{\text{max}}} \exp\left(\frac{B}{\mathcal{E}}\right) d\left(\frac{1}{\mathcal{E}}\right) = \frac{A}{B} l\mathcal{E}_{\text{max}} \left[\exp\left(-\frac{B}{\mathcal{E}_{\text{max}}}\right) - \exp\left(-\frac{B}{\mathcal{E}_{\text{sat}}}\right)\right] \tag{10.71}$$

For sufficiently large V_{DS} in excess of V_{DSsat},

$$\mathcal{E}_{\text{max}} \simeq \frac{V_{\text{DS}} - V_{\text{DSsat}}}{l} \gg \mathcal{E}_{\text{sat}} \tag{10.72}$$

and

$$M - 1 \simeq \frac{A}{B}(V_{\text{DS}} - V_{\text{DSsat}}) \exp\left(-\frac{Bl}{V_{\text{DS}} - V_{\text{DSsat}}}\right) \tag{10.73}$$

Plugging this now into the expression of the substrate current [Eq. (10.62)], we obtain

$$I_{\text{B}} = -(M-1)|I_{\text{S}}| \simeq -\frac{A}{B}|I_{\text{S}}|(V_{\text{DS}} - V_{\text{DSsat}}) \exp\left(-\frac{Bl}{V_{\text{DS}} - V_{\text{DSsat}}}\right) \tag{10.74}$$

and for a situation in which the drain current is negligibly upset:

$$I_{\text{B}} \simeq -\frac{A}{B} I_{\text{D}}(V_{\text{DS}} - V_{\text{DSsat}}) \exp\left(-\frac{Bl}{V_{\text{DS}} - V_{\text{DSsat}}}\right) \tag{10.75}$$

This is the classic expression for the MOSFET substrate current that applies when $|I_{\text{B}}| \ll I_{\text{D}}$.

There are two key dependencies here. First and most important, the substrate current is, within a relatively small correction, exponentially dependent on $-1/(V_{\text{DS}} - V_{\text{DSsat}})$. As V_{DS} increases, $V_{\text{DS}} - V_{\text{DSsat}}$ increases, $1/(V_{\text{DS}} - V_{\text{DSsat}})$ decreases, and the exponential therefore increases. This is exactly what is seen in the experimental data of Fig. 10.27. This dependence on V_{DS} makes sense since it mimics the dependence of the impact ionization rate on the maximum electric field and this is roughly linear on $V_{\text{DS}} - V_{\text{DSsat}}$. The second key dependence is that I_{B} is linear on I_{D}. This is also physically sound since the more carriers drift down the channel, the higher the total number of impact ionization events that take place and the bigger the total hole generation rate.

The combined dependence of the substrate current on $V_{\text{DS}} - V_{\text{DSsat}}$ and I_{D} yields an interesting overall dependence of I_{B} on V_{GS}. For high V_{DS}, as V_{GS} increases above threshold, the substrate current increases right along with I_{D}. However, this rise soon starts to slow down as V_{DSsat} also increases and $V_{\text{DS}} - V_{\text{DSsat}}$ decreases. Since the dependence of the substrate current on this is exponential, eventually the substrate current peaks and starts decreasing. This is seen in the experimental data of Fig. 10.28.

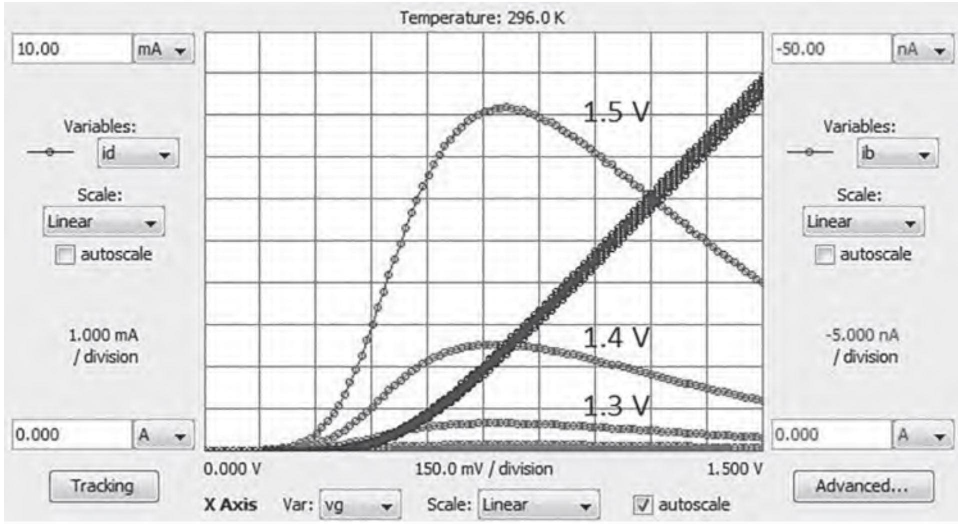


FIGURE 10.28
Substrate current versus V_{GS} for the MOSFET of Fig. 10.27 for different values of V_{DS} (indicated in the label). (MIT Microelectronics iLab).

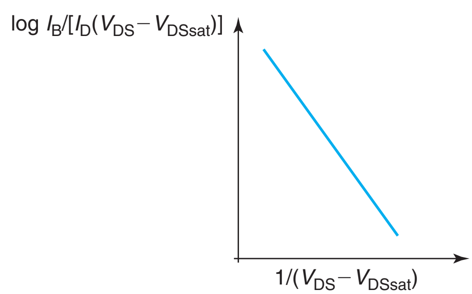


FIGURE 10.29
Ideal behavior of substrate current normalized by $I_D(V_{DS} - V_{DSsat})$ as a function of $1/(V_{DS} - V_{DSsat})$ in a semilog scale. Equation (10.75) predicts that when experimental data is graphed in this way, a straight line is obtained.

Equation (10.75) has another interesting implication that can be often observed in experiments. The substrate current normalized by $I_D(V_{DS} - V_{DSsat})$ depends only on $\exp(-B/(V_{DS} - V_{DSsat})$ for any value of V_{GS}. When graphed in a semilog plot as a function of $1/(V_{DS} - V_{DSsat})$, this dependency is a straight line. This is sketched in Fig. 10.29. This is a classic plot. It demonstrates the impact ionization origin of the substrate current.

So far, our discussion has taken place in the "weak" impact ionization regime in which V_{DS} is high enough for significant substrate current to flow but not so high that the drain current is visibly affected. However, for high enough V_{DS}, the multiplication coefficient can become quite large and the drain current can increase over its low V_{DS} value. If V_{DS} is increased further, eventually *avalanche breakdown* can take place and the drain current grows in an uncontrolled manner. Indeed, at high V_{DS}, it is experimentally observed that the drain current in MOSFETs suddenly increases. If nothing limits this current to a safe value, the device can self-destruct due to the high power that is dissipated. This is the breakdown condition and the value of V_{DS} at which this condition is reached is referred to as the *breakdown voltage*.

When examining in detail the breakdown condition in a MOSFET, it is often found that it is reached at a voltage that is *smaller* than what would be expected from the simple theory presented in this section. In other words, the transistor undergoes *premature breakdown*. The reason for this is the existence of a "hidden" bipolar transistor inside the MOSFET. This parasitic bipolar transistor is made out by the n-source, the p-body, and the n-drain. We will study bipolar transistor breakdown in Chapter 11.

10.4.3 Output conductance

An important aspect of the MOSFET that we have yet to treat is the output conductance. This refers to the slope of the output characteristics at constant V_{GS}. The ideal MOSFET when biased in saturation exhibits a drain current that is completely independent of V_{DS}. This can be characterized by a zero output conductance or, equivalently, by an infinite output resistance. In a real MOSFET, as shown in Fig. 10.30, the slope is finite. The output conductance is of great importance in many applications, especially in analog circuits. It is then essential to develop an understanding about the physics behind the output conductance and its dependence on bias point and device design.

It is only at this point in this chapter that we can do some degree of justice to this important topic. This is because there are two different pieces of physics that affect the output conductance of a MOSFET in saturation and both require the advanced treatment of this chapter. One is *drain induced barrier lowering* (DIBL) and the second one is *channel length modulation* (CLM). Let us discuss these first separately.

DIBL, introduced earlier in this chapter, reflects the change in the threshold voltage with V_{DS}. DIBL originates from the electrostatic influence of the drain on the channel surface potential. If DIBL is prominent, its effect is visible in the output characteristics of the device. The reason for this is that in a transistor in saturation, if we increase V_{DS}, V_T decreases due to DIBL. As a consequence, the gate overdrive increases and the output current increases. An important consequence of this understanding is that the better an electrostatic integrity a device exhibits, the lower its output conductance due to DIBL will be.

A second source of output conductance in a MOSFET in saturation is CLM. We learned in the previous section that in a MOSFET biased in saturation, a region appears at the drain end of the channel where electrons travel at saturation velocity. We also learned that the length of this region increases as V_{DS} increases beyond V_{DSsat}, the value that saturates the transistor. A consequence of this is that the effective electrical length of the channel, the distance from the source to the beginning of the velocity saturation region, shortens as V_{DS} increases. This results in an increase in the drain current and therefore a finite slope in the output characteristics.

Developing a first-order model for the output conductance that includes both of these effects is not difficult. The starting point is the drain current equation with the MOSFET biased in saturation for $V_{DS} > V_{DSsat}$. For this, we use an earlier result in this chapter that includes velocity

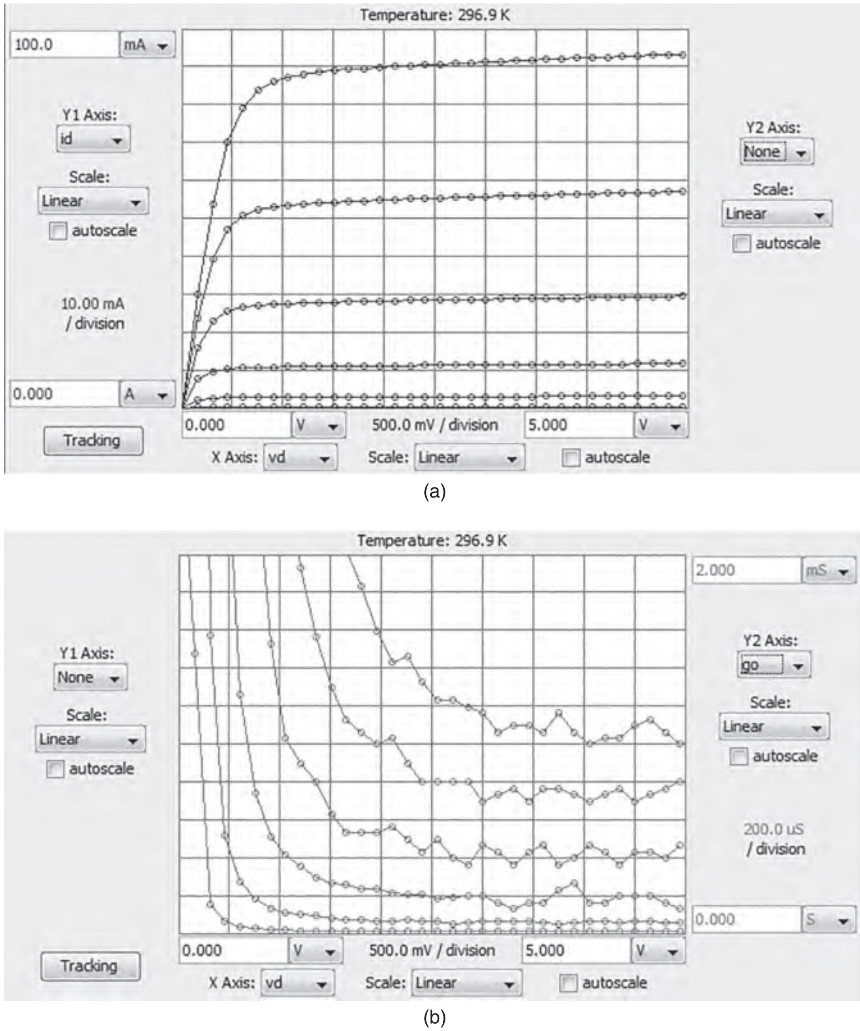


FIGURE 10.30
Output characteristics (a) and conductance (b) of 2N7000 MOSFET. (MIT Microelectronics iLab).

saturation [Eq. (10.10)]. To include CLM, in that expression, we substitute the gate length of the device L by the effective gate length after the length of the velocity saturation region has been subtracted, L'. L' is a function of V_{DS}. The drain current equation in the saturation regime is then

$$I_D = \frac{W}{L'} \frac{\mu_e C_{ox}}{1 + \dfrac{V_{DSsat}}{\mathcal{E}_c L'}} \left(V_{GS} - V_T - \frac{1}{2} V_{DSsat} \right) V_{DSsat} = I_{Dsat} \frac{\mathcal{E}_c L + V_{DSsat}}{\mathcal{E}_c L' + V_{DSsat}} \quad \text{for} \quad V_{DS} \geq V_{DSsat} \tag{10.76}$$

where I_{Dsat} refers to the saturated drain current as given by the velocity saturation model [Eq. (10.10) or Eq. (10.11)], which are equivalent]. In this new expression, V_{DSsat} is still given by Eq. (10.9) since at pinch off, $L' = L$.

The output conductance is defined as the change in I_{D} with V_{DS}, while holding V_{GS} constant. Looking at Eq. (10.76), we see that I_{D} in the saturation regime can change with V_{DS} if L' changes or if V_{T} changes. Using the chain rule, we then have

$$g_{\mathrm{o}} = \left.\frac{\partial I_{\mathrm{D}}}{\partial V_{\mathrm{DS}}}\right|_{\mathrm{V_{GS}}} = \left.\frac{\partial I_{\mathrm{D}}}{\partial L'}\right|_{\mathrm{V_{GS}}} \left.\frac{\partial L'}{\partial V_{\mathrm{DS}}}\right|_{\mathrm{V_{GS}}} + \left.\frac{\partial I_{\mathrm{D}}}{\partial V_{\mathrm{T}}}\right|_{\mathrm{V_{GS}}} \left.\frac{\partial V_{\mathrm{T}}}{\partial V_{\mathrm{DS}}}\right|_{\mathrm{V_{GS}}} = g_{\mathrm{o,CLM}} + g_{\mathrm{o,DIBL}} \tag{10.77}$$

The first term on the right-hand side captures CLM and the second one accounts for DIBL.

Starting with CLM, the derivative of the current with L' can be readily obtained from Eq. (10.76):

$$\left.\frac{\partial I_{\mathrm{D}}}{\partial L'}\right|_{\mathrm{V_{GS}}} = I_{\mathrm{Dsat}} \frac{-(\mathcal{E}_{\mathrm{c}} L + V_{\mathrm{DSsat}})\mathcal{E}_{\mathrm{c}}}{(\mathcal{E}_{\mathrm{c}} L' + V_{\mathrm{DSsat}})^2} \simeq -I_{\mathrm{Dsat}} \frac{\mathcal{E}_{\mathrm{c}}}{\mathcal{E}_{\mathrm{c}} L + V_{\mathrm{DSsat}}} \tag{10.78}$$

where we have assumed that in a well-designed device, L and L' are not too different, that is, that the length of the velocity saturation region is a small fraction of the channel length. The derivative in Eq. (10.78) is negative because as L' decreases, I_{D} increases.

The derivative of L' with V_{DS} can be readily obtained from Eq. (10.58). To keep things relatively simple, we can take the leading terms of that expression in the limit in which V_{DS} exceeds V_{DSsat} by a relatively small amount, as is often the case. Then, L' is approximately:

$$L' \simeq L - l \ln\left(1 + \frac{V_{\mathrm{DS}} - V_{\mathrm{DSsat}}}{l\mathcal{E}_{\mathrm{sat}}}\right) \tag{10.79}$$

Its derivative with V_{DS} is then

$$\left.\frac{\partial L'}{\partial V_{\mathrm{DS}}}\right|_{\mathrm{V_{GS}}} = -\frac{1}{\mathcal{E}_{\mathrm{sat}}} \frac{1}{1 + \frac{V_{\mathrm{DS}} - V_{\mathrm{DSsat}}}{l\mathcal{E}_{\mathrm{sat}}}} \tag{10.80}$$

The minus sign in this expression indicates that L' shortens when V_{DS} increases. The leading term in this expression is $1/\mathcal{E}_{\mathrm{sat}}$ which makes sense since that is the electric field at L'. The additional term gives a small correction as V_{DS} increases past V_{DSsat}.

Before assembling the expression of the output conductance due to CLM, we note that Eq. (10.9) allows us to rewrite the term $\mathcal{E}_{\mathrm{c}} L + V_{\mathrm{DSsat}}$ in Eq. (10.78) as $\mathcal{E}_{\mathrm{c}} L\sqrt{1 + 2\frac{V_{\mathrm{GS}} - V_{\mathrm{T}}}{\mathcal{E}_{\mathrm{c}} L}}$. With this, g_{o} due to CLM can be written as

$$g_{\mathrm{o,CLM}} \simeq I_{\mathrm{Dsat}} \frac{1}{L\mathcal{E}_{\mathrm{sat}}\sqrt{1 + 2\frac{V_{\mathrm{GS}} - V_{\mathrm{T}}}{\mathcal{E}_{\mathrm{c}} L}}} \frac{1}{1 + \frac{V_{\mathrm{DS}} - V_{\mathrm{DSsat}}}{l\mathcal{E}_{\mathrm{sat}}}} \tag{10.81}$$

We discuss the key dependencies of this expression below.

We now focus on the output conductance due to DIBL. The derivative of I_{D} with V_{T} can be easily obtained by noting that in the expression of I_{D}, V_{T} and V_{GS} always appear in the form $V_{\mathrm{GS}} - V_{\mathrm{T}}$ (this also includes the expression of V_{DSsat}). We can therefore write

$$\left.\frac{\partial I_{\mathrm{D}}}{\partial V_{\mathrm{T}}}\right|_{\mathrm{V_{GS}}} = -\left.\frac{\partial I_{\mathrm{D}}}{\partial V_{\mathrm{GS}}}\right|_{\mathrm{V_{T}}} = -g_{\mathrm{m}} \tag{10.82}$$

The minus sign makes sense because as V_T increases while holding V_{GS} constant, I_D decreases.

The derivative of V_T with V_{DS} is very close to what we defined as DIBL in Eq. (10.29):

$$\left.\frac{\partial V_T}{\partial V_{DS}}\right|_{V_{GS}} \simeq -DIBL \tag{10.83}$$

The negative sign is also appropriate because as V_{DS} increases, V_T shifts negative. The parameter $DIBL$ is always defined as positive.

Assembling everything, we get

$$g_{o,DIBL} \simeq g_m\, DIBL = W v_{sat} C_{ox} DIBL \left(1 - \frac{1}{\sqrt{1 + 2\frac{V_{GS}-V_T}{\mathcal{E}_c L}}}\right) \tag{10.84}$$

With our analytical first-order models in place, we can now discuss the leading dependencies. First, it is important to realize that $g_{o,DIBL}$ is linearly proportional to DIBL itself. Since long-channel devices suffer from little DIBL, this implies that the output conductance of long-channel MOSFETs is dominated by CLM. In short-channel devices, DIBL becomes important and it often dominates the behavior of g_o.

The most important dependence of $g_{o,CLM}$ is its nearly linear dependence on I_{Dsat}. The dependence is slightly sublinear because of the $V_{GS} - V_T$ term inside the square root in the denominator. Still, as I_{Dsat} increases, $g_{o,CLM}$ increases too. This is easily visible in the characteristics of a long-channel MOSFET, as seen in Fig. 10.30. In the saturation regime, for higher values of V_{GS}, I_D increases and g_o increases too. However, the rate of increase of g_o with V_{GS} is not as fast as that of I_D, reflecting the sublinear behavior. Equation (10.81) further indicates that $g_{o,CLM}$ also depends on V_{DS}. Due to the term in the denominator, as V_{DS} increases, $g_{o,CLM}$ decreases. This dependence is not very strong but it is noticeable in the experimental characteristics. In Fig. 10.30, for values of V_{DS} for which the transistor is fairly well saturated g_o continues to decrease.

In short devices for which DIBL is of some importance, $g_{o,DIBL}$ can become dominant. The voltage dependencies are quite different. Looking at Eq. (10.84), $g_{o,DIBL}$ is independent of V_{DS} and increases with V_{GS} for low gate overdrive. For high gate overdrive, $g_{o,DIBL}$ tends to saturate to

$$g_{o,DIBL} \simeq W v_{sat} C_{ox} DIBL \tag{10.85}$$

which is independent of V_{GS} or I_D.

This behavior of the output conductance is readily seen in short-channel devices as shown on Fig. 10.31 for the 0.18-μm-gate length MOSFET of Fig. 10.27. This chart plots g_o vs. I_D in the saturation regime. Initially, g_o increases linearly with I_D, consistent with CLM-limited output conductance. For high enough current, the output conductance approaches a constant value, as expected from a DIBL limited mechanism. The graph also plots g_m for the same value of V_{DS}. For high gate overdrive, the transconductance tends to saturate consistent with velocity saturation.

From a structural point of view, the most significant dependence of g_o is on the gate length. In this regard, in the long gate length regime when $g_{o,CLM}$ dominates, there is an inverse dependence of g_o on L. In this regime, at the same bias point, shorter transistors display a higher

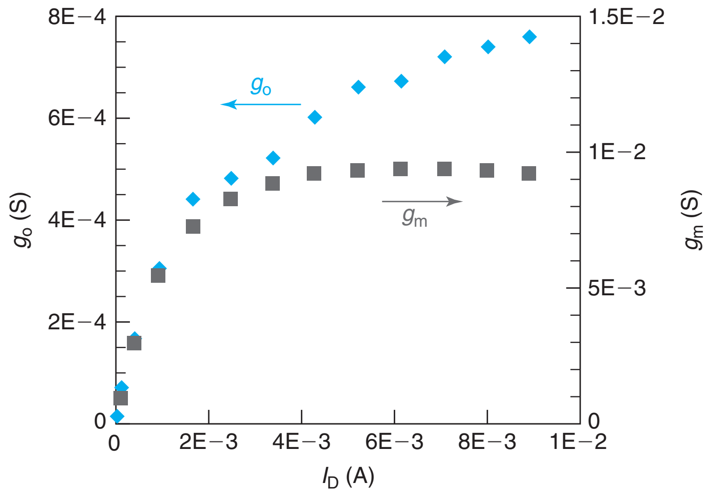


FIGURE 10.31
Output conductance and transconductance as a function of drain current of 0.18-μm MOSFET of Fig. 10.27 at $V_{DS} = 1.5$ V. A saturating behavior of g_o as V_{GS} increases is characteristic of DIBL.

output conductance. With very short devices when DIBL becomes dominant, g_o increases as DIBL increases. For these two reasons, when a very low output conductance is needed, long gate length devices with very high electrostatic integrity are favored. Also, a bias point close to threshold is selected.

10.4.4 Gate-induced drain leakage

The subthreshold behavior of MOSFETs is of great importance in logic applications. We have already discussed in some detail the physics of the subthreshold current and the key dependencies of the subthreshold swing. A commonly observed behavior in short MOSFETs is that below threshold, the drain current initially drops exponentially as V_{GS} decreases (this is the common subthreshold regime), but at some point, it reaches a minimum and starts increasing again. This is sketched in Fig. 10.32. For sufficiently low V_{GS} below V_T, a new drain current component emerges that increases with decreasing V_{GS} and increasing V_{DS}. This new current component is of some concern since it can contribute to the off-current of the MOSFET, as is the case in Fig. 10.32.

This new drain current component arises from a mechanism known as *gate-induced drain leakage* or GIDL. The essence of this current is electron tunneling across the bandgap at the high field region that exists at the overlap of the gate and the drain extension at high drain-gate bias. This is schematically illustrated in Fig. 10.33.

Figure 10.33(a) depicts the drain end of the channel of a MOSFET biased with $V_{GS} \ll V_T$ and large V_{DG}. It also assumes a body voltage around zero. Under these conditions, there is a thin depletion region right at the surface of the n^+ drain in the overlap region with the gate. If

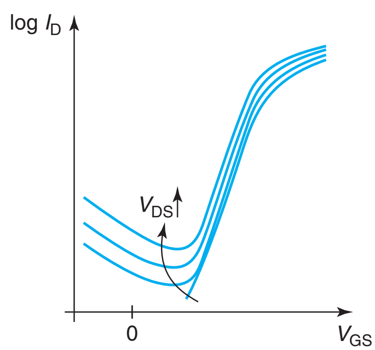


FIGURE 10.32
Sketch of subthreshold characteristics of a short MOSFET showing the characteristic signature of GIDL. This is an increase in the drain current for values of V_{GS} much below threshold.

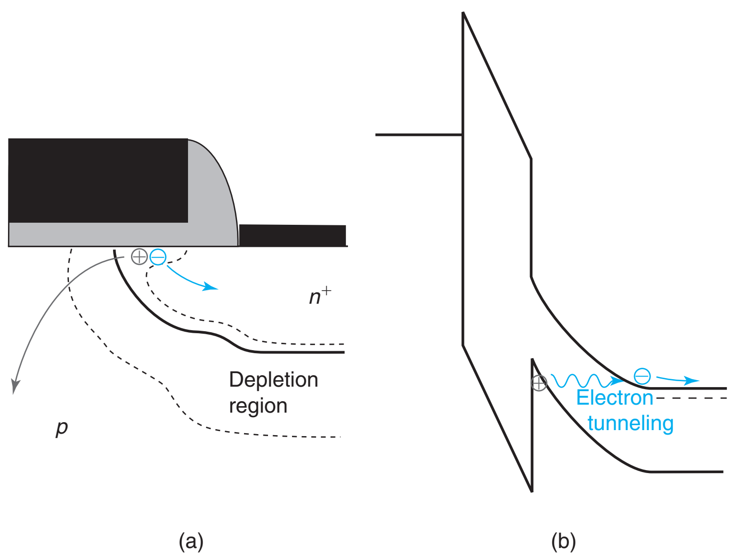


FIGURE 10.33
Illustration of gate-induced drain leakage (GIDL) at the gate–drain overlap region of a MOSFET biased in the subthreshold regime with a large drain-to-gate voltage. (a) In the overlap region, there is a depletion region at the semiconductor surface with a large vertical electric field. Band-to-band tunneling of electrons from the substrate to the drain contributes to drain current. The generated holes flow to the substrate. (b) Energy band diagram across the vertical dimension in the gate–drain overlap region showing band-to-band tunneling of electrons.

V_{DG} is large enough, due to the high doping in the semiconductor in this overlap region, the electric field across this depletion region can be quite high.

In Figure 10.33(b), an energy band diagram is depicted across the vertical dimension in this overlap region. It shows how the large voltage difference between the drain and gate results in a large field across the oxide and also through the depletion region at the surface of the drain extension. If the field is such that the top of the valence band right at the semiconductor surface comes above the bottom of the conduction band deeper into the quasi-neutral drain extension, then occupied states in the valence band become energetically aligned with empty states in the conduction band. With a high enough drain doping level, the spatial separation between the top of the valence band and the bottom of the conduction band can be quite small. Also, the energy barrier in the middle has a height equal to the semiconductor bandgap, about 1.1 eV for Si, which is not so large. So, under appropriate conditions, band-to-band tunneling of electrons can take place, as indicated in Fig. 10.33(b). The generated electrons become majority carriers in the drain and they pop out the drain contact contributing to drain current. The holes are collected by the substrate where they also are majority carriers and contribute to substrate current.

The dependencies of the GIDL current observed in Fig. 10.32 are now easily understood. As V_{DG} increases, whether by reducing V_{GS} or increasing V_{DG}, the vertical electric field is enhanced and the GIDL current increases too.

An interesting feature is that the GIDL current is independent of gate length. This is because tunneling only takes place in the drain–gate overlap region and the rest of the gate plays no role. Also, the GIDL current is rather temperature independent, something characteristic of pure tunneling.

Since we are in front of a tunneling current, a purely quantum-mechanical phenomenon, developing a simple model for the GIDL current is beyond the scope of this book. Nevertheless, the functional dependencies of the current are simple and worth mentioning. As in the case of the gate current through the gate dielectric of a MOSFET discussed in Section 10.3.2, we are looking at tunneling through a triangular barrier. The functional dependence is then similar to the one we described for gate tunneling and is given by

$$I_{D,GIDL} = A\mathcal{E}_s \exp\left(-\frac{B}{\mathcal{E}_s}\right) \tag{10.86}$$

where $\mathcal{E}_s$ is the electric field at the surface of the semiconductor where tunneling takes place. A and B are two constants that characterize the process.

A first-order expression for $\mathcal{E}_s$ as a function of V_{DG} can be easily obtained by noting that at the onset of *band-to-band tunneling* (BTBT), the energy build up across the depletion region in the semiconductor is just about equal to the bandgap of the semiconductor, as indicated in Fig. 10.34. For simplicity, let us work with a case in which the work function of the metal is such that the flatband voltage of the MOS structure is zero, such as the case of an n^+-polySi gate. If we assume that as a result of the heavy doping, the Fermi level inside the drain sits close to the conduction band edge, then, the field across the oxide in this situation is, approximately:

$$\mathcal{E}_{ox} \simeq \frac{V_{DG} - \frac{E_g}{q}}{x_{ox}} \tag{10.87}$$

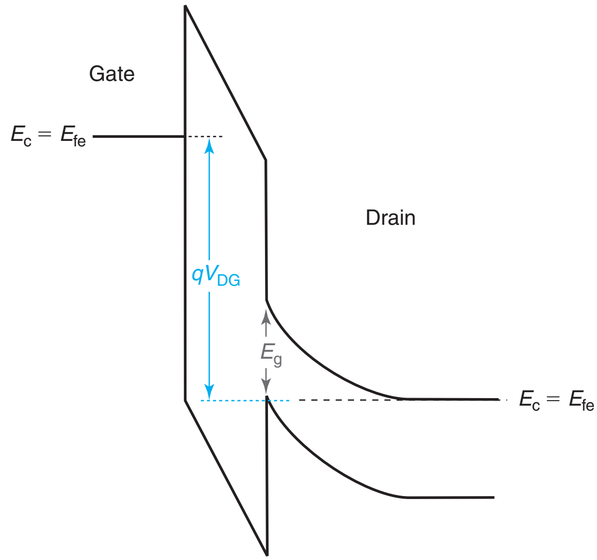


FIGURE 10.34
Simplified vertical energy band diagram across gate/oxide/drain in the overlap region at a voltage at which GIDL current starts.

The field at the surface of the semiconductor is then

$$\mathcal{E}_s \simeq \frac{\epsilon_{ox}}{\epsilon_s}\frac{V_{DG} - \frac{E_g}{q}}{x_{ox}} \tag{10.88}$$

Plugging this result into Eq. (10.86), we get a dependence of the form:

$$I_{D,GIDL} = A'\left(V_{DG} - \frac{E_g}{q}\right)\exp\left(-\frac{B'}{V_{DG} - \frac{E_g}{q}}\right) \tag{10.89}$$

where A' and B' are suitably modified constants from A and B in Eq. (10.86). Note that $A' \propto 1/x_{ox}$ and $B' \propto x_{ox}$. This peculiar behavior of the GIDL current is readily seen in experiments.[7]

GIDL is an excess drain current mechanism that needs to be kept under tight control as it contributes to the MOSFET off-current. The most effective way to do this is to limit the operating voltage. It is clear that if the operating voltage does not exceed the bandgap energy (in the appropriate units), GIDL will be irrelevant. In recent times with the operating voltage of CMOS being around 1 V, GIDL has become less of a concern.

[7]See, for example, T. Y. Chan et al., *Int. Electron Dev. Meet.*, p. 718, 1987.

10.5 MOSFET SCALING

There is something really magic about MOSFETs as it comes to their use in digital circuits and that is *"smaller is better."* When discussing MOSFET scaling, focusing on logic applications is appropriate as they have largely driven the evolution of integrated MOSFET technology over its 40-year span. In logic applications, in which a MOSFET behaves as a switch, there are huge benefits that derive from making transistors smaller. Perhaps the most important one is that the number of transistors that can be put together in an integrated circuit in an economical manner increases. This is often known as *Moore's Law*, to credit Gordon Moore who in 1965 first discussed the benefits of integration and predicted its rapid progress. Moore's Law is illustrated in Fig. 10.35, which plots the number of transistors in main stream microprocessors for the last nearly 50 years. The progress since the introduction of the first microprocessor, Intel's 4004 in 1971, has been staggering: the transistor count has increased by over six orders of magnitude. This represents a growth rate of about a factor of 2 every 2 years. Transistor count in an IC is a good measure of progress: the more transistors one manages to pack together, the more complex the logic operations that can be performed and the more useful the products that can be created.

A corollary of increasing transistor count is that the cost per transistor drops at a similar rate. This is shown in Fig. 10.36 for mainstream microprocessors as a function of year of introduction. In the nearly 50 years that this graph charts, transistor cost has shrunk by nearly six orders of magnitude. The cost per transistor drops as its footprint is reduced. Along with this, the cost of implementing a given function also tumbles. This translates into ever more affordable electronics that continue to enable new applications and create new markets.

A second significant benefit of MOSFET size scaling is performance. Performance in the context of logic circuits usually refers to the speed at which logic functions are executed. A traditional measure of performance is the clock frequency at which integrated circuits run. Figure 10.37 shows the clock frequency of mainstream microprocessors as a function of their year of introduction. As you can see, for a long time, clock frequency was rising in an exponential

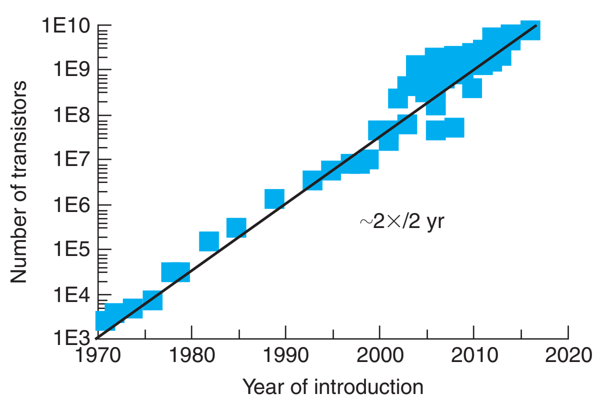


FIGURE 10.35
Moore's Law–transistor count in mainstream microprocessors as a function of year of introduction.

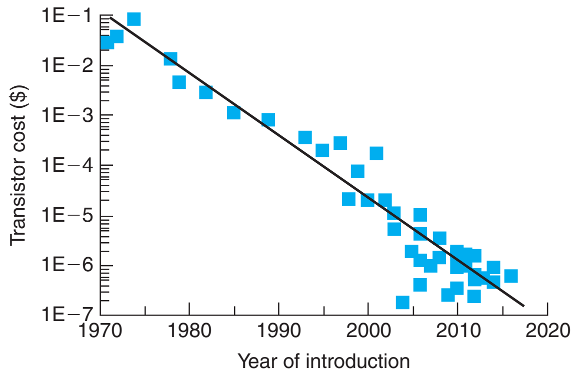


FIGURE 10.36
Average transistor cost in mainstream microprocessors as a function of year of introduction.

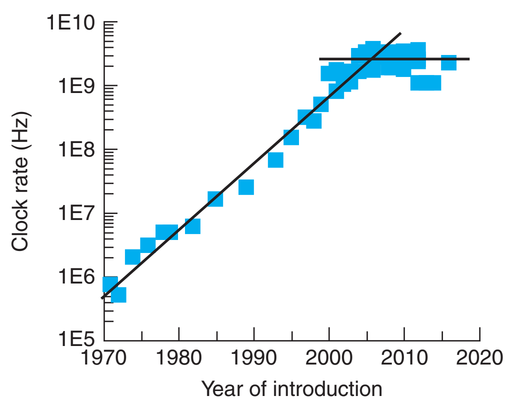


FIGURE 10.37
Evolution of clock frequency in mainstream microprocessors as a function of year of introduction.

manner (this is a semilog plot), but in recent times, it has hit a limit of about 4 GHz. Later in this section, we will discuss why this has happened.

In spite of the saturation of clock frequency that is evident in Fig. 10.37, the logic performance of microelectronics technologies has continued to increase at a brisk pace. This has been possible through architectural innovations that have created *multicores* or separate processors that are integrated together on the same chip. These processors operate in parallel so that a given task can be partitioned among several of them and therefore be executed much faster. With this new architecture, the appropriate figure of merit for logic performance is no longer the clock rate at which the individual processors run, but the aggregated throughput that they are capable of when working together on a given problem. This is measured through various performance benchmarks that assess the execution of portfolios of common tasks. When performance scaling

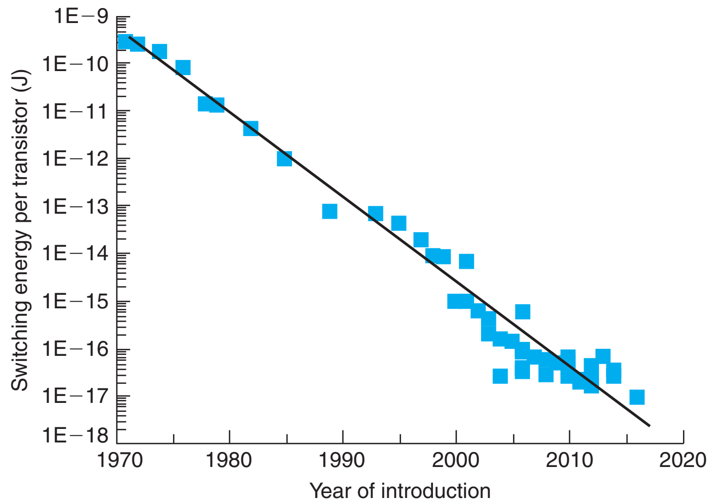


FIGURE 10.38
Evolution of energy consumption per transistor in mainstream microprocessors as a function of year of introduction.

is examined this way, continued exponential advances continue to accrue as more cores are integrated together on a chip.

The third major benefit of MOSFET scaling is energy consumption, specifically, the amount of energy required to switch the state of the transistor. As transistors have shrunk in size, their energy needs have also dropped in a very significant way. This is illustrated in Fig. 10.38 which shows chip power divided by the clock frequency and the number of transistors in mainstream microprocessors. This is a reasonable estimation of the average energy consumed in switching the state of a transistor from *off* to *on* and back. As you can see, this measure of transistor energy consumption has dropped by over seven orders of magnitude in the last nearly 50 years. This phenomenal progress underlies the explosion in recent years of many kinds of extremely power efficient portable systems.

The benefits of MOSFET size scaling are evident. But where do these come from? How exactly has scaling been taking place? How is MOSFET design likely to evolve in the future? These are the questions that we wish to answer in the following sections. We start by briefly discussing the device-level figures of merit that one needs to focus on in order to understand MOSFET scaling.

10.5.1 The MOSFET as a switch

Key to understanding MOSFET scaling is identifying the right figures of merit to focus on. What follows is a simplified analysis that provides a great deal of valuable intuition.

MOSFET scaling has been mainly driven by logic applications in which the transistor behaves as a switch—its operating point changes back and forth between two well-defined states. A rigorous analysis requires that we embed the transistor in its proper circuit environment. This is

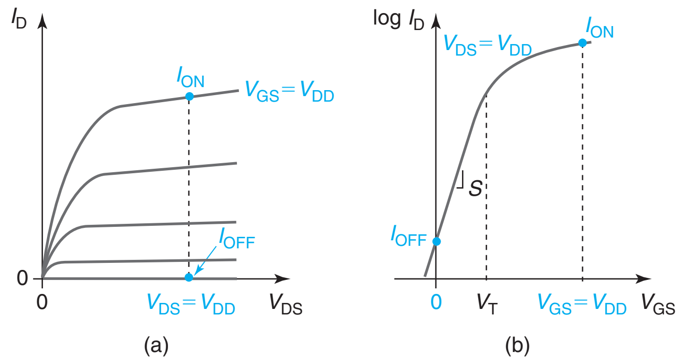


FIGURE 10.39
Illustration of the *on* and *off* states in a logic MOSFET: (a) output characteristics and (b) subthreshold characteristics.

beyond the scope of this book. For our purpose, it suffices to consider the MOSFET as switching between a high conducting state, the *on* state, and a low conducting state, the *off* state. These two states are illustrated in Fig. 10.39 in the output and subthreshold characteristics of a typical MOSFET. The goal is to be able to switch as fast as possible between these two states while consuming a minimum amount of energy.

In the *on* state, the transistor is biased in the saturation regime with a drain current equal to I_{ON}:

$$I_{ON} = I_D(V_{GS} = V_{DD}, V_{DS} = V_{DD}) \tag{10.90}$$

V_{DD} is the supply voltage used in the circuit.

In the *off* state, the transistor is biased in the cutoff regime with $V_{GS} = 0$. Ideally, there is zero drain current. However, due to subthreshold conduction, a small amount of leakage current flows through the channel. This is the *off* current or I_{OFF}:

$$I_{OFF} = I_D(V_{GS} = 0, V_{DS} = V_{DD}) \tag{10.91}$$

I_{ON} and I_{OFF} are obviously important figures of merit for logic. Key to setting the values of these two currents is the threshold voltage V_T.

Switching between the *on* and *off* states requires moving a certain amount of charge. In the *off* state, there is negligible channel charge but a depletion region exists in the semiconductor under the gate. In the *on* state, the channel is fully formed and underneath it, we also have a depletion region in the semiconductor that extends to a slightly greater depth when compared with the *off* state. If we neglect the somehow different amounts of charge stored in the depletion region under the gate in these two states, to the first order, the difference between the charge stored in the transistor in the *on* and *off* states is $\sim C_G V_{DD}$, where C_G is the total gate capacitance. If this charge is delivered at a rate I_{ON}, it will then take a certain amount of time for the transient to be finished. This is called the *switching delay*, which is approximately:

$$\tau_d \simeq \frac{C_G V_{DD}}{I_{ON}} \tag{10.92}$$

Switching from the *off* state to the *on* state and back consumes energy. This is roughly the electrostatic energy associated with the gate charge, which is delivered by the power supply in the *off–on* transition and then dumped in the *on–off* transition. Since the amount of charge that is moved is $\sim C_G V_{DD}$ and the supply voltage that delivers this charge is V_{DD}, then the *switching energy* can be roughly estimated as

$$E_d \simeq C_G V_{DD}^2 \tag{10.93}$$

This is also often called the *power-delay product*.

This switching energy is drawn from the battery every clock cycle, at the end of which it is completely dissipated. If the clock rate is f, then the power dissipation (rate of energy transfer) in the transistor is roughly:

$$P_d = fE_d = fC_G V_{DD}^2 \tag{10.94}$$

This is also called the *dynamic power* since it is associated with the charging and discharging of the transistor every switching cycle. If switching occurs at the maximum possible rate, then $f = 1/\tau_d$ and:

$$P_d = \frac{E_d}{\tau_d} \simeq I_{ON} V_{DD} \tag{10.95}$$

In addition to dynamic power, there is also *static power*. This is associated with the OFF current that flows when the transistor is biased in the OFF state:

$$P_s = I_{OFF} V_{DD} \tag{10.96}$$

In an integrated circuit at any one time, on average, half of the transistors are in the OFF state. For large circuits, static power can therefore add up to an amount just as significant as dynamic power.

The total power density dissipated in the transistor determines the temperature rise. This is proportional to $P_d + P_s$ divided by the footprint of the transistor.

Understanding MOSFET scaling is about appreciating the evolution of these key figures of merit as transistors are scaled down in size. Clearly, reducing the lateral dimensions of the device alone runs into serious trouble very soon. As the gate length approaches the electrostatic characteristic length, severe short-channel effects ensue that eventually will prevent the transistor from turning off. MOSFET scaling is about reducing transistor footprint and accruing the expected improvements in performance and energy consumption while maintaining acceptable short-channel effects. Over time, this has been accomplished in a spectacular way, as Figs. 10.35 through 10.38 testify. Understanding the details of this requires us to study a number of rather idealized scaling approaches. These are analyzed in the next sections.

10.5.2 Constant field scaling of the ideal MOSFET

Constant field scaling was proposed by Dennard in 1974.[8] Constant field scaling consists of scaling the device while maintaining the vertical and lateral electric fields constant. This is accomplished by decreasing all transistor dimensions in a proportional way and by decreasing the supply voltage by the same factor. Paying close attention to internal fields is appropriate because they play a crucial role in device reliability. By insuring a constant field as the technology evolves, no new reliability issues should emerge.

To analyze constant field scaling, we define a scaling factor $\kappa > 1$ with which all device dimensions and V_{DD} scale as specified at the top of Table 10.1.[9] In this scenario, the lateral field along the channel, roughly V_{DD}/L, and the vertical field across the oxide, roughly V_{DD}/x_{ox}, remain unchanged, hence the name of this approach. If all the lateral device dimensions scale down in a proportional way, the device footprint scales as $1/\kappa^2$, which is one of the central goals of scaling.

In the next few lines, we carry out an analysis of the changes in the major figures of merit of the MOSFET. In our notation, the primed variables represent the scaled values and the unprimed ones are the original values.

The capacitance of the gate per unit area scales up linearly as the device shrinks:

$$C'_{ox} = \frac{\epsilon_{ox}}{x'_{ox}} = \frac{\epsilon_{ox}}{\frac{x_{ox}}{\kappa}} = \kappa C_{ox} \tag{10.97}$$

Table 10.1
Evolution of ideal MOSFET characteristics under constant field scaling

Parameter	**Symbol**	**Scaling Factor ($\kappa > 1$)**
Device dimensions	L, W, x_{ox}, x_j	$1/\kappa$
Supply voltage	V_{DD}	$1/\kappa$
Body doping level	N_A	1
Device area	A	$1/\kappa^2$
Gate capacitance	C_G	$1/\kappa$
Threshold voltage	V_T	$1/\kappa$
ON current	I_{ON}	$1/\kappa$
ON current density	I_{ON}/W	1
Gate delay	τ_d	$1/\kappa$
Switching energy	E_d	$1/\kappa^3$
Dynamic power	P_d	$1/\kappa^2$
Dynamic power density	P_d/A	1
Electrostatic characteristic length	λ	$1/\sqrt{\kappa}$

[8] R. H. Dennard et al., *J. Solid St. Cir.* SC-9, p. 256 (1974).

[9] The presentation followed in this book differs slightly from the original one of Dennard. Here, the doping level does not change as the device scales. In Dennard's original proposal, the doping level scaled as κ. The difference lies in the fact that at the time that Dennard performed his study, a body voltage was applied to fine-tune V_T. Dennard scaled this voltage so as to obtain a V_T that scaled along with V_{DD}. This approach ran its course early in MOSFET scaling. For a long time since, the standard body voltage has been zero. In order to have V_T scale appropriately, the body doping is held constant.

As a consequence, the gate capacitance of the MOSFET (ignoring parasitics) scales down linearly:

$$C'_G = W'L'C'_{ox} = \frac{W}{\kappa}\frac{L}{\kappa}\kappa C_{ox} = \frac{C_G}{\kappa} \tag{10.98}$$

The threshold voltage is a bit more complicated. The original expression [see Eq. (8.28)] contains a term that scales and two others that do not. However, as we saw in Exercise 8.4 for the technologically important case of an n^+-polysilicon gate, the two nonscaling terms nearly cancel each other out. Therefore

$$V'_T = V'_{FB} + \phi'_{sT} + \gamma\sqrt{\phi'_{sT}} \simeq \frac{1}{C'_{ox}}\sqrt{2\epsilon_s q N'_A \phi'_{sT}} \simeq \frac{1}{\kappa C_{ox}}\sqrt{2\epsilon_s q N_A \phi_{sT}} \simeq \frac{V_T}{\kappa} \tag{10.99}$$

where we have neglected the logarithmic dependence of ϕ_{sT} on N_A.

The *on* current evolves in the following way:

$$I'_{ON} = \frac{W'}{2L'}\mu_e C'_{ox}(V'_{DD} - V'_T)^2 = \frac{\frac{W}{\kappa}}{2\frac{L}{\kappa}}\mu_e \kappa C_{ox}\left(\frac{V_{DD}}{\kappa} - \frac{V_T}{\kappa}\right)^2 = \frac{I_{ON}}{\kappa} \tag{10.100}$$

Since the device width W has also scaled by κ, the *on* current density remains constant.

Following Eq. (10.92) and again ignoring parasitics, the gate delay evolves as

$$\tau'_d = \frac{C'_G V'_{DD}}{I'_{ON}} = \frac{\frac{C_G}{\kappa}\frac{V_{DD}}{\kappa}}{\frac{I_{ON}}{\kappa}} = \frac{\tau_d}{\kappa} \tag{10.101}$$

This represents a big win in performance which is one of the key goals of scaling.

The switching energy, using Eq. (10.93), scales as

$$E'_d = C'_G {V'_{DD}}^2 = \frac{C_G}{\kappa}\left(\frac{V_{DD}}{\kappa}\right)^2 = \frac{E_d}{\kappa^3} \tag{10.102}$$

The energy required to switch a transistor drops with the cube of the scaling factor! This arises from the simultaneous reduction of the capacitance load and the operating voltage. The dramatic drop in switching energy represents a huge benefit of constant field scaling.

The dynamic power [Eq. (10.95)] goes as

$$P'_d = I'_{ON}V'_{DD} = \frac{I_{ON}}{\kappa}\frac{V_{DD}}{\kappa} = \frac{P_d}{\kappa^2} \tag{10.103}$$

This is also quite significant. However, the power density, which is what matters for thermal management, evolves as

$$\frac{P'_d}{A'} = \frac{\frac{P_d}{\kappa^2}}{\frac{A}{\kappa^2}} = \frac{P_d}{A} \tag{10.104}$$

which indicates that the chip temperature will remain constant as the technology scales.

Table 10.1 summarizes the changes in key device logic figures of merit in constant field scaling.

Our analysis so far has ignored the changes in I_{OFF} and static power dissipation. Historically, constant field scaling was formulated at a time when these were not primary concerns. Nevertheless, the subthreshold swing scales in the following way:

$$S' = \left(1 + \frac{C'_{sT}}{C'_{ox}}\right)\frac{kT}{q}\ln 10 = \left(1 + \frac{C_{sT}}{\kappa\, C_{ox}}\right)\frac{kT}{q}\ln 10 \tag{10.105}$$

To the first order, C_{sT}, the capacitance associated with the depletion region under the gate at threshold, does not scale as the doping level is kept constant.

Equation (10.105) indicates that constant field scaling sharpens the subthreshold swing. However, I_{OFF} depends on the ratio of V_T and S [the higher this ratio, the lower I_{OFF} is, see Eq. (9.97)]. This scales as

$$\frac{V'_T}{S'} = \frac{\frac{V_T}{\kappa}}{(1 + \frac{C_{sT}}{\kappa C_{ox}})\frac{kT}{q}\ln 10} \tag{10.106}$$

As a result of the presence of the nonscaling factor 1 inside the brackets in the denominator, the threshold voltage scales faster than the subthreshold swing. V'_T/S' drops and, in consequence, I_{OFF} and the static power dissipation increase in a very significant way.

Also worth noting is the change in short channel effects. Since the doping level of the body does not change, the depletion region thickness under the gate at threshold, x_{dmax}, does not scale. In consequence, the scaling length for electrostatics, λ [Eq. (10.18)] scales as $1/\sqrt{\kappa}$, which is not as fast as the gate length. Hence, short-channel effects get worse as scaling proceeds.

To summarize, constant field scaling brings about very significant improvements in performance and switching energy. The downside is static power dissipation which prominently increases. In addition, there is a need to continue to change the operating voltage. Throughout most of the history of MOSFET scaling, this was deemed an undesirable requirement as it made it difficult to mix and match different chips when building a system. Furthermore, constant field scaling only goes so far as the electrostatic scaling length that controls short-channel effects does not scale as fast as the gate length.

The analysis that has been performed here has been carried out for the ideal MOSFET using the formulation of Chapter 9. The basic findings of this analysis have been validated in short-channel devices using two-dimensional simulations that include the various short-channel effects discussed earlier in this chapter.[10] The reason for this is understood if we examine the scaling of I_{ON} in a short transistor [Eq. (10.11)]. It is straightforward to see that under constant field scaling, this expression also scales as $1/\kappa$ just as the long-channel expression.

10.5.3 Constant voltage scaling of the ideal MOSFET

A drawback of constant field scaling is the need to change the operating voltage from generation to generation. In response to this, constant voltage scaling emerged. As its name suggests, in constant voltage scaling, MOSFET scaling proceeds without changing the operating voltage. A consequence of this is that, unless something is done, the internal fields in the device increase with a potential negative impact on reliability. We will study later on in this chapter some of the techniques used to mitigate internal electric fields in MOSFETs.

[10] See P. Vande Voorde, *HP Journal*, Article 12, August 1997.

Table 10.2
Evolution of ideal MOSFET characteristics under constant voltage scaling

Parameter	**Symbol**	**Scaling Factor ($\kappa > 1$)**
Device dimensions	L, W, x_{ox}, x_j	$1/\kappa$
Supply voltage	V_{DD}	1
Body doping level	N_A	κ^2
Device area	A	$1/\kappa^2$
Gate capacitance	C_G	$1/\kappa$
Threshold voltage	V_T	1
ON current	I_{ON}	κ
ON current density	I_{ON}/W	κ^2
Gate delay	τ_d	$1/\kappa^2$
Switching energy	E_d	$1/\kappa$
Dynamic power	P_d	κ
Dynamic power density	P_d/A	κ^3
Electrostatic characteristic length	λ	$1/\kappa$

A summary of the scaling trajectory of the various device parameters under constant voltage scaling is shown in Table 10.2. In this approach, the device dimensions scale by $1/\kappa$, the doping level scales up by κ^2, and the voltage remains unchanged. If we carry out a similar analysis as in the previous section, we find that the device area and the gate capacitance scale as in constant field scaling, V_T remains unchanged, and the on current scales up as κ. This means that the current density increases as κ^2. This has important implications not only for performance but also for power dissipation.

With V_{DD} unchanged, the gate delay scales as $1/\kappa^2$, which represents a huge boost in performance. The switching energy also decreases as $1/\kappa$. However, the dynamic power increases as κ and the dynamic power density goes up as κ^3. This has serious implications for power dissipation and chip temperature.

Looking into I_{OFF}, since the subthreshold swing does not change and V_T is also unmodified, the exponential term in I_{OFF} does not change. However, the pre-exponential term scales as κ. Hence, the static power dissipation scales in the same way.

The scaling length for electrostatics, λ, scales as $1/\kappa$. That means that short-channel effects do not degrade as the technology scales.

To summarize, constant voltage scaling has the advantage that the operating voltage does not need to change from technology generation to technology generation. It also results in fast increasing performance as scaling proceeds. However, the internal electric fields increase with a negative impact on reliability. Even worse, in constant voltage scaling, the power dissipation explodes. This increases packaging and cooling costs and limits the use of the technology in many applications.

10.5.4 Generalized scaling of short MOSFETs

Both constant field scaling and constant voltage scaling have severe drawbacks, although of a different nature. A compromise approach that attempts to balance between the two schemes is often referred to as *generalized scaling*.

Generalized scaling accepts the notion that V_{DD} needs to scale in order to manage power dissipation. Accruing performance gains requires that V_T scales right along. However, in order to mitigate the degradation of short-channel effects, V_{DD} and V_T scale more slowly than the device geometry. This implies that power dissipation and the internal fields increase as scaling proceeds. New device structures were introduced to deal with these shortcomings.

Table 10.3 summarizes the situation. In this table, the geometry of the transistor scales with the scaling factor $\kappa > 1$. However, the supply voltage scales with scaling factor $\eta < \kappa$. For V_T to scale right along with V_{DD}, the body doping level has to scale as $\kappa^2/\eta^2 > 1$.

This table has separate entries for the long MOSFET and the short MOSFET. The difference is that the short MOSFET uses the model for the drain current that includes saturation velocity [Eq. (10.11)]. The scaling of the long MOSFET in the generalized model reverts to constant field scaling if $\eta = \kappa$ and to constant voltage scaling if $\eta = 1$.

Looking now at the short MOSFET entries, we see that under generalized scaling with $\kappa > \eta > 1$, the gate delay scales as $1/\kappa$, which means that there is a performance improvement but it is not as large as under constant voltage scaling. The switching energy and the dynamic power also scale but the dynamic power density increases. However, the increase in power density and temperature is not as severe as under constant voltage scaling. The short-channel effects also get worse under generalized scaling since the electrostatic characteristic length does not decrease as fast as the gate length. However, the problem is mitigated with respect to the situation under constant field scaling.

With V_T scaling, I_{OFF} is bound to increase exponentially though not as fast as under constant field scaling. This results in an exponential increase in static power dissipation.

Table 10.3
Evolution of ideal MOSFET characteristics under generalized scaling

Parameter	Symbol	Scaling factor ($\kappa > \eta > 1$) Long MOSFET	Short MOSFET
Device dimensions	L, W, x_{ox}, x_j	$1/\kappa$	
Supply voltage	V_{DD}	$1/\eta$	
Body doping level	N_A	κ^2/η^2	
Device area	A	$1/\kappa^2$	
Gate capacitance	C_G	$1/\kappa$	
Threshold voltage	V_T	$1/\eta$	
ON current	I_{ON}	κ/η^2	$1/\eta$
ON current density	I_{ON}/W	κ^2/η^2	κ/η
Gate delay	τ_d	η/κ^2	$1/\kappa$
Switching energy	E_d	$1/\left(\kappa\eta^2\right)$	
Dynamic power	P_d	κ/η^3	$1/\eta^2$
Dynamic power density	P_d/A	κ^3/η^3	κ^2/η^2
Electrostatic characteristic length	λ	$\sqrt{\eta}/\kappa$	

To summarize, generalized scaling in which the voltage scales at a slower rate than the device geometry allows a more balanced approach. Performance improves at the cost of worse short-channel effects and higher fields. The big drawback is increased static and dynamic power density which, though ameliorated somehow, continue to increase.

10.5.5 MOSFET scaling: a historical perspective

What route has MOSFET scaling actually taken over its 40 years of existence? In discussing this, it is useful to first define the concept of *technology generation* or *technology node*. Historically, CMOS technology development has occurred in spurts, as opposed to through a continuous improvement process. New CMOS technologies are introduced to the marketplace with a distinct cadence, roughly every 2 years. Each one of these technologies is referred to as a *generation* or a *node*.

The life cycle of a CMOS technology has three distinct phases. There is an early *research phase* in which new device and process design options are developed and evaluated. In this phase, oftentimes, new materials and structures are investigated. At the end of this phase, a selection of features is made and prototype devices and circuits are put together. A *development phase* follows in which a manufacturing process is developed for the selected design. A key requirement here is that the process must be amenable to cost-effective volume manufacturing. Typically, this phase culminates with the demonstration of a set of fairly complex circuits. At this point, the process is ready for the final *manufacturing ramp-up phase* which is focused on attaining high yields and achieving low manufacturing costs. When this is achieved, the technology is frozen and formally released to volume manufacturing. As volume expands, the technology is replicated at other fabs. From there on, this process is used to fabricate many kinds of ICs over several years.

At any one time, several CMOS technology nodes coexist at different stages in their life cycle. While several are in volume manufacturing, including the latest release, the following node in the pipeline is in production ramp-up, the next one is under development and the next one after that is in research. These successive generations of technology are typically phased so that a new one is deployed every 2 years. Each technology enables a higher transistor density (historically about 2) and delivers improved performance (historically 35%, more recently about 20%, less than this in the last few years).

Rather unique to the IC industry is that, with some variations, different companies deploy largely similar technologies at about the same time and therefore have a pipeline of technology development that is also quite similar. The reason for this coordinated action in an otherwise extremely competitive industry is the cost of the manufacturing tools involved. Each new MOSFET technology presents fresh manufacturing challenges. Those are compounded by the need to fabricate increasingly complex integrated circuits with extraordinarily high yields. In essence, each new MOSFET technology reaches to smaller dimensions and higher sophistication and requires a new manufacturing tool set. These new tools represent a large fraction of the cost of a new fab. New tool development is in itself very expensive and tool manufacturers must amortize their own development costs among many customers. The economics only work out if enough semiconductor IC companies are willing to buy the same set of tools at about the same time. So this forces the industry to work in rough lock step and deploy a new technology node at about the same time. Hence the notion of technology generation. For the same reasons, this

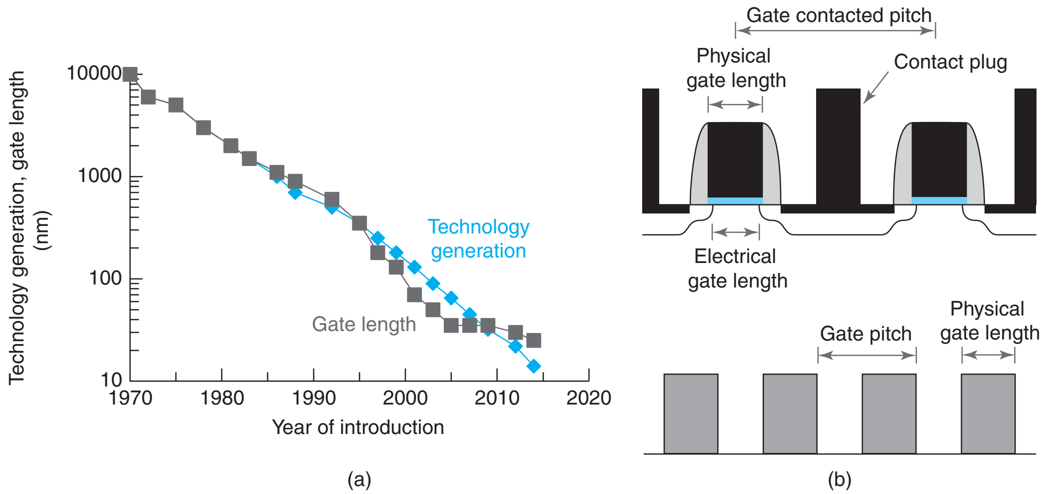


FIGURE 10.40
(a) Evolution of minimum half pitch (also known as *technology generation*) and gate length of CMOS as a function of year of introduction into volume manufacturing. (b) Illustration of definitions of physical gate length, electrical gate length, gate pitch, and gate contacted pitch.

coordinated development also applies to Si wafer diameter which has increased over the years in a discrete closely staged way.

Each technology generation is characterized by its *minimum half gate pitch*. This refers to half of the distance between the most closely spaced gates that can be fabricated on the wafer [Fig. 10.40(b)]. We talk, for example, about the 0.25 μm node, introduced into volume manufacturing in 1997, or the 32 nm node, introduced in 2009. The choice of the half gate pitch as key metric reflects the critical importance of the gate length in setting the electrical performance of the technology as well as the difficulty in fabricating it in a reproducible way. Nevertheless, the gate length of the transistor is not exactly equal to the half gate pitch but it is usually close to it, sometimes slightly bigger, sometimes slightly smaller. Both are shown in Fig. 10.40(a) as a function of the year of introduction of the technology. The most advanced CMOS technology generation at the time of this writing has gate lengths that are on the order of 25 nm.

The chart in Figure 10.40 shows that CMOS scaling has been taking place at a steady pace for now well over 40 years. It is this relentless shrinking in transistor size that underlies the exponential growth in transistor count that is known as *Moore's law* (Fig. 10.35).

As the gate length has gotten smaller, so have the other two key geometrical dimensions of the device: the gate oxide thickness and the junction depth of the source and drain extensions right next to the gate. These are both shown in Fig. 10.41 as a function of gate length in a log–log plot. Also indicated in this graph is a unity slope line. This figure indicates that these three geometrical dimensions of the device have scaled in a harmonious way for the last nearly 50 years. Only for very short gate lengths, we see a sudden jump in gate oxide thickness. This will be explained below.

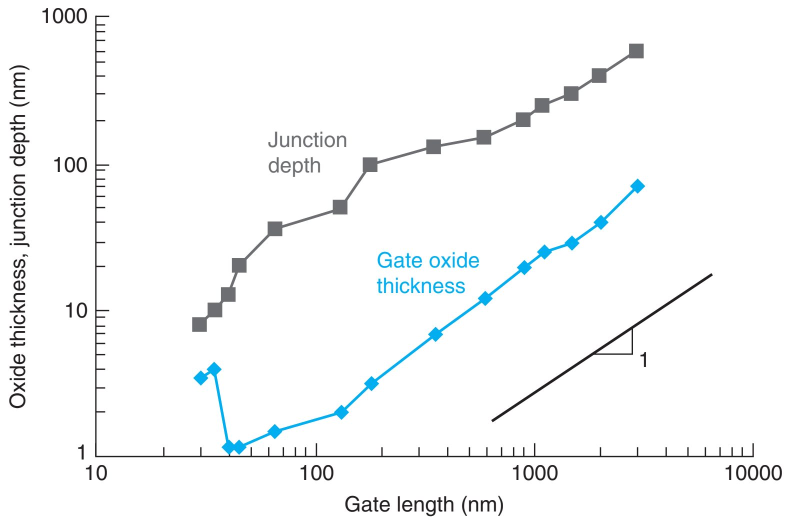


FIGURE 10.41
Junction depth and gate oxide thickness of CMOS n-channel MOSFETs as a function of gate length. Only planar transistors are graphed.

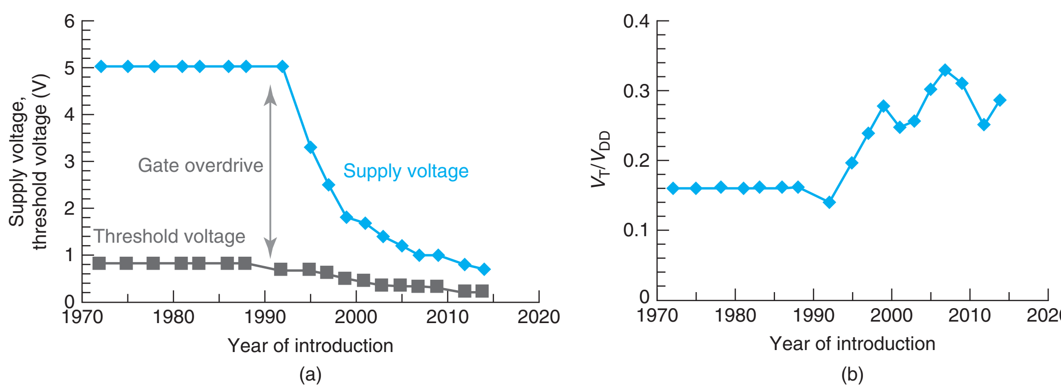


FIGURE 10.42
(a) Operating voltage and threshold voltage of CMOS technologies as a function of year of introduction. (b) Ratio of threshold voltage to supply voltage versus year of introduction.

From Figs. 10.40 and 10.41 alone, it is not possible to appreciate the actual trajectory that MOSFET scaling has followed. To comprehend this, one needs to map the evolution of the supply voltage as a function of time. This is shown in Fig. 10.42 along with the threshold voltage. Up to about 1992, transistor evolution proceeded along a *constant voltage scaling* path in which the supply voltage was 5 V. In this long early phase, a number of innovations were introduced to mitigate the inevitable increase in electric fields. By about 1992, these had become too high

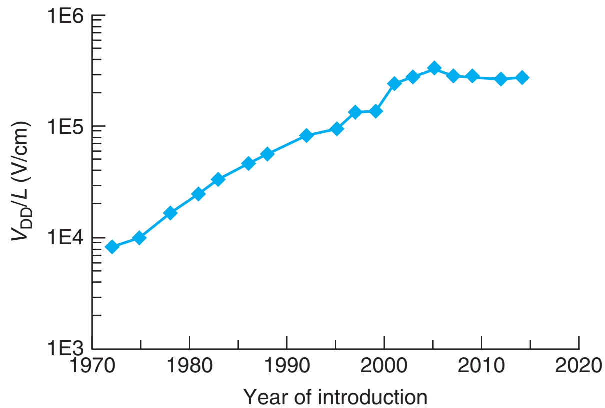


FIGURE 10.43
Ratio of supply voltage to gate length of CMOS technologies as a function of year of introduction.

and the supply voltage started scaling down. The first stop on the way down was 3.3 V but it continued from there with virtually every generation of technology. In more recent years, the supply voltage reduction has slowed down and it currently stands at around 1 V. We discuss the reason for this slowdown below. The figure also shows the evolution of the threshold voltage. While the supply voltage was constant at 5 V, the threshold voltage was fixed at around 0.8 V. As V_{DD} started scaling down, so did V_T. Current values of V_T are around 0.2–0.3 V.

A rough sense of the evolution of the internal electric field can be gauged by plotting V_{DD}/L as a function of time. This is shown in Fig. 10.43. Throughout the first 20 years of constant voltage scaling, the electric field increased by more than one order of magnitude. Around 1992, the fields got too high and the voltage started scaling down. This somehow slowed down the increase in the internal electric field. V_T also shrank but initially not as fast, so that the ratio of V_T to V_{DD} increased. This is seen in Fig. 10.42(b). This resembles *generalized scaling* but with V_T shrinking slower than V_{DD}.

Beyond about 2000, the electric field inside the device seems to have reached a maximum value and from there on, it has remained rather constant (Fig. 10.43). This is akin to a *constant field scaling* situation. In this new phase, V_{DD} and V_T have been scaling at roughly similar rates and their ratio has remained rather constant, as we see in Fig. 10.42(b). Interestingly, the rate of scaling of V_{DD}, V_T, and gate length (Fig. 10.41) have all significantly slowed down in recent years. Nevertheless, transistor footprint scaling and density (Fig. 10.35) have remained unabated. This is a tribute to the ingenuity of transistor and process designers that have continued to increase the efficiency of the use of the Si real state.

It is of interest to track the evolution of performance throughout all this time. A common way to do this is to map the progress in *on* current density. This is a good predictor of performance because the gate delay is inversely proportional to it. The current density of the n-MOSFET as a function of year of introduction is shown in Fig. 10.44. There are three distinct phases in this plot. In the first 20 years of CMOS evolution, the *on* current density was increasing at a brisk average rate of 37% per CMOS generation. This was thanks to constant voltage scaling. Beyond

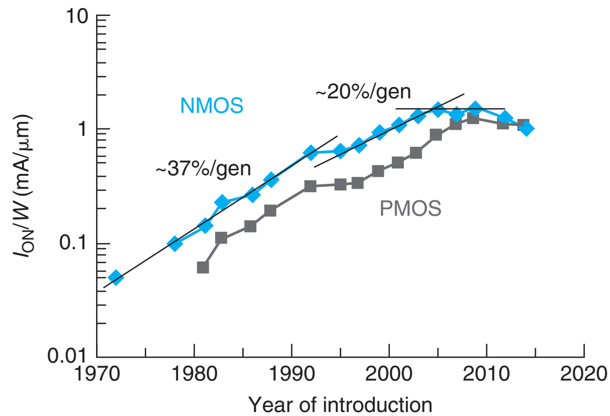


FIGURE 10.44
ON current density for n-channel and p-channel MOSFETs as a function of year of CMOS technology introduction.

1990 or so, generalized scaling brought along continued progress but at a reduced rate of about 20% per CMOS generation. In the last few years, the *on* current density seems to have reached a plateau. This is in fact what we expect from constant field scaling, as studied in Section 10.5.2 and it illustrates the increase difficulty in obtaining increased performance as the operating voltage continues to drop.

Figure 10.44 also shows the evolution of the *on* current density of the p-channel device. Up to around the year 2000, both transistors scaled in a similar way and the performance evolved following parallel paths. Beginning with the 90 nm generation, introduced in 2003, mechanical strain was incorporated into the transistors. This was done to enhance transport, as described in the next section. As it turns out, for comparable levels of strain, hole transport in the p-channel MOSFET is enhanced significantly more than electron transport in the n-channel device. As a result, as mechanical strain has increased with subsequent generations of technology in search of improved performance, the p-channel device has almost completely closed the performance gap with the n-channel device.

Greater insight into this most recent scaling phase can be obtained by examining the evolution of power density in main stream microprocessors as a function of year of introduction. This is shown in Figure 10.45. For several years, power density was rising exponentially. This is consistent with our analysis of constant voltage scaling and generalized scaling from earlier in this chapter. However, since the mid-2000s, microprocessor power density has hit a plateau of around 100 W/cm^2. This practical limit is set by the difficulty and cost of dissipating increasing amounts of power and the packaging required to prevent the chip temperature from getting excessively high. For most commercial applications, these costs become prohibitive beyond about 100 W/cm^2. It is really the need to limit power density that has brought about our current era of constant field scaling.

It is interesting to examine the evolution of the two main components of power. Both active and passive power (also known as *standby* or *static power*) have been scaling exponentially as transistor size has shrunk. For a long time, passive power was negligible in comparison with

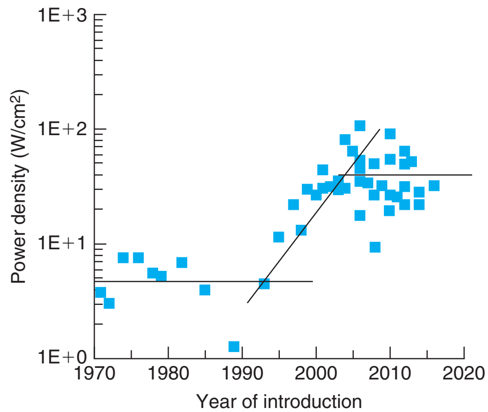


FIGURE 10.45
Power density of mainstream microprocessors as function of year of introduction.

dynamic power. However, in recent times, they have become comparable. We know that one of the most significant drawbacks of reducing the threshold voltage is the exponential increase in off current and static power dissipation. This has added to the power dissipation problems in recent years.

Our current scaling era could indeed be referred to as *power constrained scaling.* Its key attribute is that transistor size scaling proceeds unabated but power density is constrained to remain constant. Looking back at Table 10.1, we see that a constant dynamic power density is indeed a feature of constant field scaling. But we need to make room in our power budget for an increasing amount of static power density as V_T scales down. How is this to be accomplished?

Looking back at the expression for dynamic power in Eq. (10.94), we see that the dynamic power density can be written as

$$\frac{P_d}{A} = f\frac{C_G}{A}V_{DD}^2 \tag{10.107}$$

Under constant field scaling as studied earlier in this chapter, dynamic power density remains constant as a result of the competing scaling trends of f, C_G/A, and V_{DD}. The switching frequency increases linearly with the scaling factor as the gate delay becomes smaller. The gate capacitance per unit area also increases linearly as the oxide thickness shrinks. The evolution of these two terms is precisely compensated by a reduction in V_{DD} which scales down linearly with the scaling factor.

With an exploding static power density brought about by a reduction in V_T, the dynamic power density cannot remain constant, it must actually be reduced so that the total power density remains unchanged. The way to accomplish this is not to run the clock at the maximum possible rate that the technology can support. As we see in Fig. 10.37, the historical exponential rise in the clock rate of microprocessors has recently stopped. Current products operate in the 2–4 GHz range.

Constant field scaling with an awareness to the increasing role of static power dissipation provides a reasonable description of CMOS scaling over the last few years. However, the simple model behind constant field scaling does not comprehend the fact that as transistor size scaling advances, the various parasitics (resistance and capacitance) play an increasingly important role. This means that in the last few years, there has been a need to improve the intrinsic performance of the transistor. This has been accomplished by enhancing the carrier velocity through the introduction of mechanical strain, as discussed in more detail in the next section. Mechanical strain was first introduced with the 90-nm generation and every technology generation ever since has incorporated increasing levels of strain in the channel. The increase in saturation velocity has helped maintain the *on* current density roughly constant in the last few nodes, as we can see in Fig. 10.44.

10.5.6 Evolution of MOSFET design

To complement the birds eye view of MOSFET scaling presented in the previous section, this section now provides a broad description of the evolution of MOSFET device design in the last 50 years.[11]

We place our starting point not with the demonstration of the first MOSFET (credited to Dawon Kahng and Martin Atalla at Bell Labs in 1959) nor the availability of the first commercial MOSFETs (RCA, 1964) but with the first demonstration of the MOSFET integrated circuit. This is credited to Steven Hofstein and Thomas Stanley of RCA in 1962. Commercialization of the first MOSFET ICs by Fairchild, RCA and General Microelectronics followed in 1964. A sketch of these early integrated MOSFETs is shown in Fig. 10.46.

The first integrated MOSFETs were p-type, that is, based on holes and not electrons. At the time, it was recognized that n-type MOSFETs offered higher performance but their threshold voltage was hard to control. This was much less of a problem in p-type MOSFETs and, as a result, they were chosen for the first ICs. The operating voltage of these devices was 15 V and they had gate lengths around 20 μm. Other key parameters are listed in Fig. 10.46. P-type MOSFETs ruled MOSFET ICs for several years. In fact, the first microprocessor, Intel's 4004 introduced in 1971, was based on p-MOSFETs.

By the late 1960s, the origin of the threshold voltage instability of n-channel MOSFETs had been identified and the problem solved. The source was found to be Na contamination from the

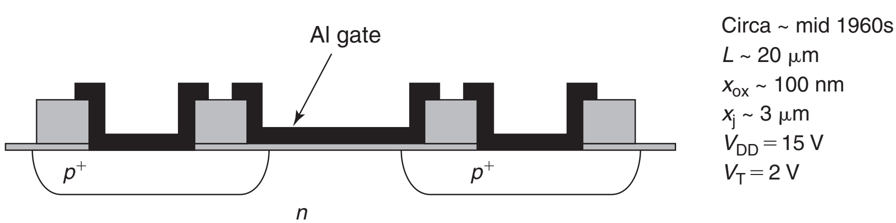


FIGURE 10.46
Sketch of an early PMOS used in the first MOSFET integrated circuits. Notable design aspects are a p-type channel and Al gate and ohmic contacts that are deposited at the same time.

[11] A source for historical details given in this section is "The Silicon Engine: A Timeline of Semiconductors in Computers" available at www.computerhistory.org/semiconductor/.

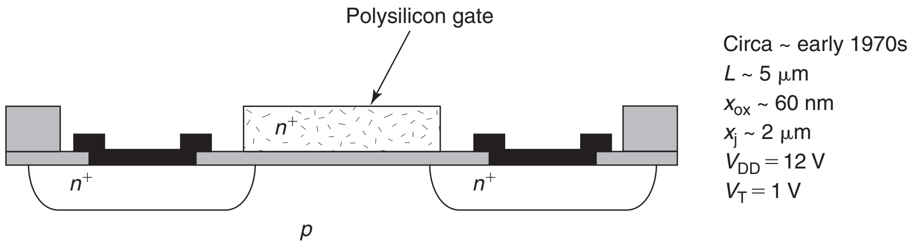


FIGURE 10.47
Sketch of an early integrated n-channel MOSFET. The most notable design aspect is a heavily doped polysilicon gate. This enables source and drain implants that are self-aligned to the gate.

fab operators. Inside the SiO_2 gate dielectric, Na is a positively charged ion that moves quite fast. Depending on the electrical history of the transistor, its threshold voltage would change in seemingly uncontrolled ways. This problem affected p-MOSFETs much less. By preventing contamination from humans, n-channel MOSFETs with reproducible V_T's became possible and eventually n-MOSFET ICs took hold.

The first commercial n-channel MOSFETs featured a significant innovation: a polysilicon gate. A sketch of such a device is shown in Fig. 10.47. A problem with the early metal gate transistors is that because Al cannot withstand the high temperature needed to diffuse the source and drain dopants, gate definition comes very late in the process. As shown in Fig. 10.46, this means that the alignment between the gate and the source and drain regions must be done "by hand" by an operator. In MOSFETs, adequate overlap between the gate and the source and drain regions is essential for the channel to connect properly to the source and drain regions. To allow for misalignment tolerances in the early Al gate designs, a large overlap between the gate and the source and drain regions must be designed which results in large parasitic capacitance.

A solution to this problem was found by using a heavily doped polysilicon gate instead of a metal gate. From an electrical point of view, if sufficiently doped, a polysilicon gate behaves very much like a metal gate. The key motivation for the change of gate materials was that a polysilicon gate can withstand high temperatures and allows the drain/source diffusion to be performed *after* the gate formation (rather than before as in the metal gate process). In this manner, the gate can be used as a mask during the source and drain diffusion. By controlling the conditions of the diffusion step, the overlap between the gate and the source and drain regions can be precisely controlled. This is a so-called self-aligned design since no alignment of the gate and source and drain regions needs to be performed by the operator. Self-aligned processes such as this one have become a main-stay of semiconductor technology.

The compact nature of the self-aligned polysilicon gate MOSFET architecture, first with the pMOS and then with the nMOS, enabled progress in transistor scaling and integration. In 1965, right at the dawn of scaling, Gordon Moore formulated his famous "law." With it, transistor integration had emerged as a path for the cost-effective manufacturing of complex, high-performance and reliable integrated circuits and systems.

In microelectronics, hand in hand with technology innovation, there has always been circuit innovation. In the early 1960s, there was an invention that would have huge implications to the future of the industry. That is CMOS, or complementary-metal–oxide–semiconductor. Early logic circuits utilized two types of FETs of the same polarity (both either p-type or n-type) with different threshold voltages: an enhancement-mode device that is normally *off* and a depletion-mode FET that is normally *on*. Complex logic functions could be synthesized using E/D logic, as it is referred to. In 1963, as reproducible NMOS transistors became more broadly available, CMOS was invented by Frank Wanlass at Fairchild. CMOS pairs enhancement-mode n-channel MOSFETs and enhancement-mode p-channel MOSFETs to synthesize highly complex logic functions in a power efficient manner. The big advantage of CMOS was the absence of stand-by power consumption while still enabling fast switching. This was the key to be able to integrate complex circuits on the same chip and still manage the power dissipation. In due time, CMOS would enable billions of transistors to be integrated together in the same die.

By the mid 1970s, transistor scaling was well established as the key driver of the microelectronics revolution. In 1974, Robert Dennard introduced his scaling laws that charted the path of scaling, mapped the performance gains and identified the challenges. Among these, appropriate scaling of parasitic resistance was essential while mitigating short-channel effects. As scaling proceeded through the 1970s, by now operating at a voltage of 5 V, managing short-channel effects became harder.

A key step forward was the introduction of gate "sidewall spacers" and "source and drain extensions." A sketch of a typical device is shown in Fig. 10.48. In this device architecture, dielectric spacers are placed on both sides of the polysilicon gate. This creates shallow lightly doped source and drain regions that are self-aligned to the polysilicon gate (these are known as *source and drain extensions*). Self-aligned with the spacers and further away from the channel are more heavily doped and deeper regions that are used for the source and drain contacts. This design allows for a balance between low resistance access to the intrinsic device and acceptable short-channel effects. Pulling the deeper junctions away from the gate edge mitigates short-channel effects at the cost of a higher access resistance. This structure is sometimes referred to as *LDD* or *lightly doped drain*. The lighter doping on the drain side also helps mitigate high-electric field effects.

The next major development in MOSFET design was the introduction of a self-aligned silicide (or "salicide") to reduce device footprint and source and drain resistance. A sketch of a typical design is shown in Fig. 10.49. A problem with the design of Fig. 10.48 is that while the

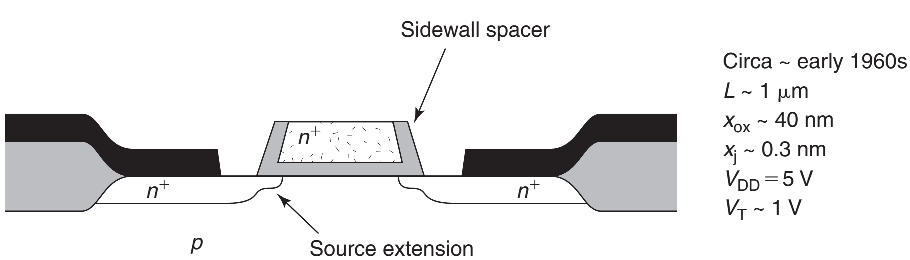


FIGURE 10.48
Sketch of an n-channel MOSFET with gate sidewall spacers and the source and drain extensions. This device architecture enabled further geometrical scaling and minimization of parasitics.

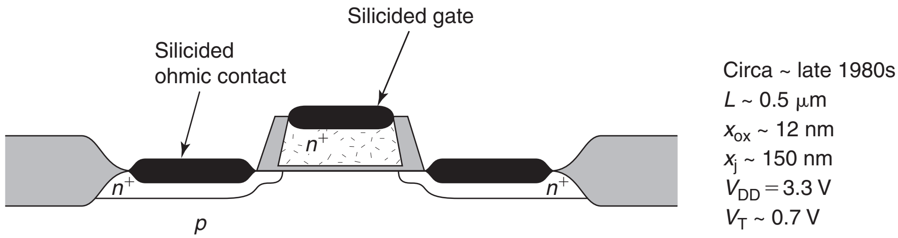


FIGURE 10.49
Sketch of a self-aligned silicided n-channel MOSFET. This device design reduces the source and drain resistance and allows further device scaling.

source and drain diffussed regions are self-aligned with the gate, the source and drain metal contacts are not. This means that the source and drain regions have to be made large enough to accommodate misalignments of the positioning of the metal contacts. This limits the ability to scale the footprint of the device and results in high source and drain resistance.

Silicides are metallic compounds that are formed when certain metals are deposited on Si and annealed. Silicides have become widely used in Si technology because they exhibit much lower resistivity than Si, they can make good ohmic contacts to Si and can be fabricated in a self-aligned fashion. Several silicides have become widely used in the Si industry: NiSi, PtSi, TiSi, WSi, etc.

The fact that silicides only form when the right metal is brought into intimate contact with Si and annealed under appropriate conditions is what enables the fabrication of self-aligned contacts. This can be understood by picturing a MOSFET structure such as the one shown in Fig. 10.48 but without the metal layers. If on top of such a structure one deposits a suitable metal, such as Ni or Ti, the metal only comes into intimate contact with Si at the exposed source, drain and gate regions. The sidewalls of the gate and the MOSFET isolation regions are covered with dielectric. During a subsequent high-temperature step, the silicide is only formed on top of the source, drain and gate regions but the metal remains unreacted on the other regions. The residual metal can now be removed by a selective etchant that dissolves the metal but not the silicide. After this, what is left is what is seen in Fig. 10.49 where the source, drain, and gate regions have highly conducting contact regions perfectly aligned with the sidewall spacers and the isolation regions.

The introduction of the salicide MOSFET architecture allowed the further shrinking of device footprint and a reduction in parasitic resistances which enhanced performance. Self-aligned silicides have remained to this day a key feature of advanced MOSFETs.

Continued MOSFET scaling over the 1990s made control of short-channel effects increasingly important. Several innovations were introduced to mitigate short-channel effects. Two of the most significant ones are shown in Fig. 10.50.

Scaling theory suggests that short-channel effects can be improved by raising the body doping. The problem with this is that the mobility in the channel is degraded due to ionized impurity scattering. This detracts from performance gains. The introduction of p^+ *pockets* represents a clever way to deal with this. In this approach, illustrated in Fig. 10.50(a), pockets of high p^+ doping are created at the source and drain ends of the channel. This is where the doping matters

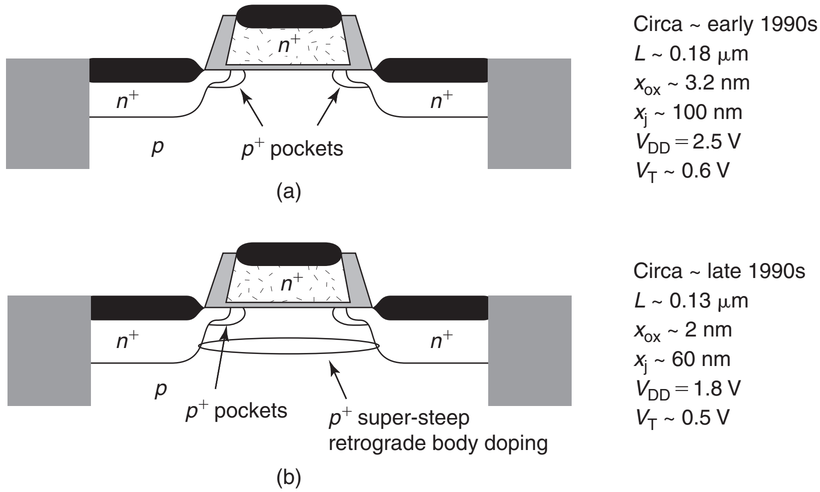


FIGURE 10.50
Sketch of n-channel MOSFETs with features to minimize short-channel effects: (a) p^+ pockets, (b) p^+ pockets plus super-steep retrograde body doping.

most from an electrostatics point of view, since high doping here helps to screen the electric field lines that emanate from the drain at high V_{DS}. These p^+ pockets can be fabricated in a self-aligned way by using angled ion implantation of the p-type species once the gate is formed. By implanting at an angle, some of the p-type dopants can laterally reach quite deep underneath the gate in a controlled manner.

A more advanced approach to control short-channel effects was introduced a few years later and is illustrated in Fig. 10.50(b). This is the so-called *super-steep retrograde (SSR) body doping*. The idea is to form a relatively highly doped buried layer under the gate while preserving a low doping level right at the surface of the wafer where the channel is induced. In this way, the channel mobility is prevented from degrading due to ionized impurity scattering but the high doping level of the buried layers mitigates short-channel effects. This balancing act can be accomplished if the p^+ buried layer has a very steep profile at the top, hence its name. This buried layer can be formed under appropriate conditions by ion implantation performed in a blanket way on the wafer prior to the formation of the gate. The p^+ SSR body doping layer is compensated under the source and the drain due to the even higher n-type doping in these regions but it remains active under the gate where it is effective in electrostatically shielding the channel from the drain. With the introduction of p^+-pockets and the SSR body doping layer, short-channel effects were effectively managed and the MOSFET scaled into the sub-0.1-μm-gate length regime.

Continued MOSFET size scaling and the increasing effort dedicated to controlling short-channel effects made improvements in performance that much harder to obtain. By around the turn of the century, the *on* current (Fig. 10.44) was only increasing at about 20% per generation, much lower than the historical 37% per generation. The bulk MOSFET was at a cross roads. Further scaling threatened to exact a heavy performance cost.

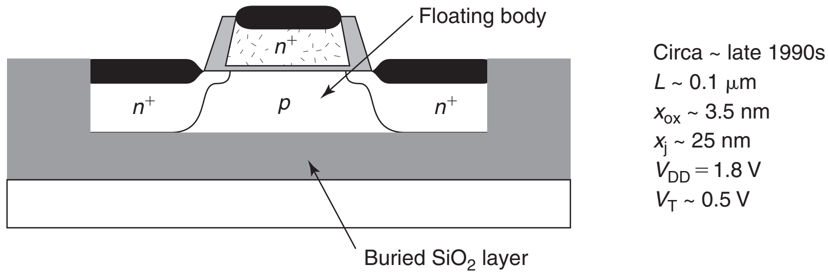


FIGURE 10.51
Sketch of a silicon-on-insulator (SOI) MOSFET. This device design features a buried SiO_2 layer that reduces junction capacitance. Even though not indicated, this device features the body doping techniques of Fig. 10.50 in order to help manage short-channel effects.

In the mid 1990s, an approach emerged that promised a one-time increase in performance that could be sustained to future generations. This came through the use of a silicon-on-insulator (SOI) substrate. An SOI–MOSFET is illustrated in Fig. 10.51.

An SOI substrate consists of a thin “device layer” created on top of a buried SiO_2 layer on a Si wafer. Because of the presence of the buried oxide, a MOSFET created on an SOI wafer enjoys reduced junction capacitance which translates into improved switching speed. Another reason for the improved performance of SOI comes from the absence of a contact to the body of the transistor which as a result floats. A consequence of this is that under normal operating conditions, the body gets forward biased somehow due to holes produced through impact ionization events. The holes cannot escape and pile up in the body. This reduces the threshold voltage of the device and enhances the gate overdrive leading to more drain current. This is an interesting bipolar effect that takes place in a MOSFET. This will be studied in more detail in Chapter 11. How much improvement the use of an SOI substrate offers has been a matter of great debate. At the upper end of the estimate is about a generation’s worth of performance advantage arising from the use of SOI.

There are some complications to the use of SOI that also arise from its floating body. A key issue is that the threshold voltage of the MOSFET is somehow unpredictable as it depends on the details of the transistor operation and its history. A consequence of this is that evaluating the operation of digital circuits in SOI is more complex and circuit designs that have been developed for bulk CMOS do not port over well to SOI CMOS. Because of these and other issues, SOI has remained to date a niche technology used for very high-performance applications, such as processors for video game consoles or servers. The extra performance of SOI can also be traded away for lower power by reducing the operating voltage. This has made this technology attractive in some low power systems.

As CMOS size scaling has proceeded, the battle to manage short-channel effects has taken an increasingly heavy toll in performance. This is clear in Fig. 10.44 where performance improvements, as measured by I_{ON}, have slowed down in the last decade and a half. To counteract this, mechanical strain was introduced in the channel in the early 2000s. The idea is simple, even though its implementation is far from straightforward. Applying stress to Si modifies its lattice constant.[12] This changes the band structure of silicon in various ways that depend on strain orientation, whether it is uniaxial (along one axis) or biaxial (along two axis), and whether it is tensile (larger lattice constant) or compressive (shorter lattice constant). The changes in the band structure propagate into changes in effective mass, mobility and saturation velocity, and the transport characteristics of short gate length MOSFETs that incorporate strain. With strain, under the appropriate conditions, significant gains in electron velocity and ON current are possible.

Sketches of strained n-channel and p-channel MOSFETs are shown in Fig. 10.52. For electrons, the most favorable strain from the point of view of transport is uniaxial tensile strain along the channel direction. This is commonly introduced by adding a "stressor film" on top

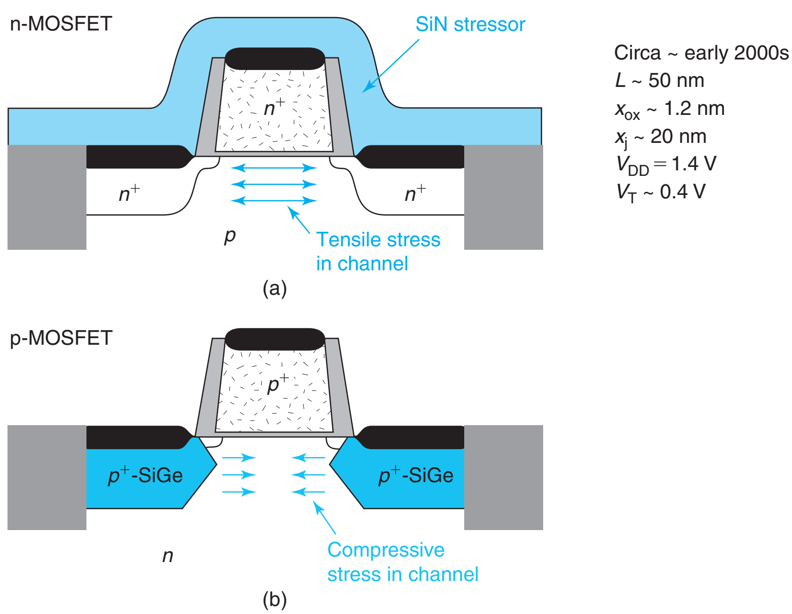


FIGURE 10.52
Sketch of strained n-channel MOSFET (a) and p-channel MOSFET (b). For the n-MOSFET, an SiN stressor is deposited on top of the device to introduce uniaxial tensile stress in the channel along the channel direction. For the p-MOSFET, uniaxial compressive stress in the channel is created by selectively growing SiGe source and drain regions. SiGe has a larger lattice constant than Si. When grown in registry with the Si substrate, it creates strong compressive stress as indicated. Even though not drawn, this device features the body doping techniques of Fig. 10.50 in order to help manage short-channel effects.

[12] Stress refers to the force that is applied to the semiconductor. In MKS, the units of stress are Newton/m^2 or Pascal. Strain refers to the relative deformation that is produced in the lattice as a result of applying stress. Strain is unit-less and is usually given in %.

of the device, as shown in Fig. 10.52(a). For holes, compressive strain in the channel direction yields the best results. This can be introduced by creating stressed isolation regions or by etching and regrowing the p^+ source and drain regions with SiGe, instead of pure Si. This is sketched in Fig. 10.52(b). SiGe has a larger lattice constant than Si and if used for the source and drain, it compresses the channel. Strained Si made its appearance with the 90-nm CMOS generation. Further improvements in performance were obtained in the 65-nm generation by adding more strain.

By the time, the 65-nm CMOS generation was reached in the mid-2000s, gate oxide thickness had been reduced to about 1.2 nm. Further transistor size scaling made imperative continued scaling of the gate oxide, but there was little room room left. Gate leakage current had become so large that it represented by then a sizable share of off-state power consumption. This was added on top of very large dynamic power that came with the increased clock frequency. By the mid-2000s, the maximum power density that could be economically dissipated had been reached and further scaling would have to take place under the constraint of constant power dissipation. This marked the beginning of a new area of power constrained scaling.

Under power-constrained scaling, transistor footprint scaling has to proceed without increasing leakage current. Yet, scaling the gate oxide thickness is essential to maintaining good short-channel effects and continuing to enhance performance as devices get smaller. The only way around this conundrum was to change gate dielectrics: to move away from SiO_2 and use a gate dielectric with a higher permittivity. This is often referred to as a *high-K dielectric*, as introduced in Section 10.3.2. The reason for this is clear. For a given layer thickness, all other things being equal, a higher permittivity dielectric yields more capacitance, more current drive, and better short-channel effects than using SiO_2. This improved performance can be partially traded away against gate leakage current by using a thicker dielectric layer. As we saw in Section 10.3.2, gate leakage current depends exponentially on dielectric thickness.

This is illustrated in Fig. 10.53 which sketches gate leakage current density at a certain value of V_{GS} for various MOS gate stacks with different dielectrics as a function of EOT [defined in Eq. (10.46)]. For a given dielectric, as EOT (and physical thickness) is reduced, the gate current increases roughly exponentially. For a given EOT, higher permittivity dielectrics show less leakage current. This is because their physical thickness is greater.

An alternative way to look at Fig. 10.53 consists of examining the EOT of different dielectrics that yields a maximum tolerable gate leakage current. Let us denote this as $J_{G,max}$ (a typical value if 1 A/cm^2 at $V_{GS} = 1$ V at room temperature). It is clear that a higher permittivity gate dielectric allows EOT to be much reduced. This, in turn, enables further gate length scaling before short-channel effects become problematic.

High-K gate dielectrics arrived on the CMOS scene with the 45-nm generation in 2007. For the first time, instead of SiO_2, HfO_2 was used as gate dielectric. A cross section of a high-K dielectric transistor is shown in Fig. 10.54. It is not an overstatement to assert that the introduction of high-K dielectrics into CMOS constituted the most disruptive technological innovation since the development of the first CMOS IC. To start with, one should not underestimate the challenge involved in reengineering what was always considered the "holiest" element of a traditional MOSFET, the Si/SiO_2 interface. For a long time, the high quality of this interface in terms of interface states, roughness, and reliability was thought not to be possible with any other dielectric material. But this was not all. In order to get the required MOS threshold voltage, the incorporation of a high-K dielectric demanded a change in the material of the gate electrode.

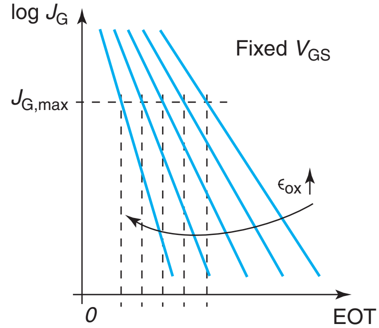


FIGURE 10.53
Sketch of gate current density versus equivalent oxide thickness for different dielectrics at a certain forward voltage. For a given maximum tolerable gate current, gate dielectrics with a higher permittivity yield lower EOT and greater scaling potential.

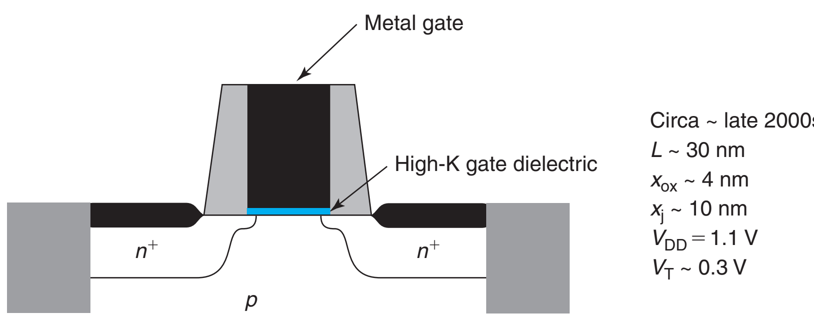


FIGURE 10.54
Sketch of n-channel MOSFET with high-K gate dielectric and metal gate.

So, heavily doped polysilicon was out and a new metal gate was in. Actually, *two* different gate metals had to be introduced in order to get both the n-channel and p-channel devices to have the correct threshold voltages. An additional advantage of this change in gate materials was the elimination of poly depletion that we discussed in Section 10.3.1.

High-K/metal gate constitutes a great leap forward in the quest for ever-increasing transistor density. This new gate technology has enabled two, perhaps three, more generations of planar CMOS. But, new technology does not completely displace earlier concepts. In fact, CMOS technology is largely cumulative. Successful innovations stay in place for multiple generations. The first and second high-K/metal gate CMOS generations also include self-aligned silicides, gate spacers, source and drain extensions, pocket implants, super-steep-retrograde body doping, and strained silicon. Significant innovations, to be cost effective, need to have the potential of being used in more than one generation of technology. The high-K/metal gate design will also endure.

Looking ahead, planar CMOS faces very serious challenges in the quest for continued scaling. Squeezing a transistor into an ever-shrinking footprint while maintaining acceptable short-channel effects and reasonable performance is becoming very hard. The planar transistor

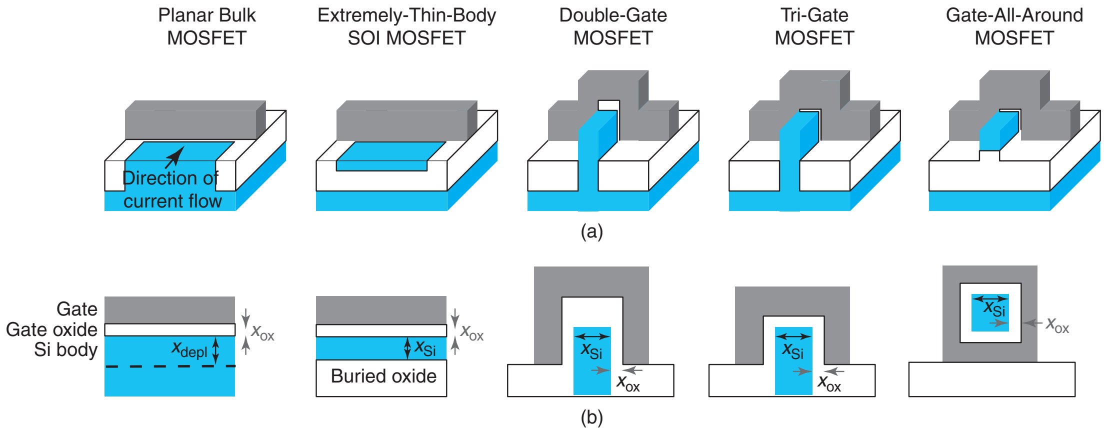


FIGURE 10.55
(a) Three-dimensional sketches of different planar and 3D MOSFET structures with current flowing into the paper, as indicated.
(b) Transversal cross-sectional geometries of the intrinsic region of the same devices.

design is limited by its characteristic electrostatic length which was derived in Eq. (10.18). A high-K dielectric allows us to scale the effective oxide thickness but the depletion length under the channel does not scale fast enough. This puts a limit to the scaling potential of the planar MOSFET. A path forward exists if we can devise new device structures with characteristic electrostatic lengths that are intrinsically *shorter* than the planar bulk MOSFET. Four alternative designs are sketched in Fig. 10.55.

The first alternative is the *Extremely-Thin-Body SOI* or ET–SOI MOSFET. As its name indicates, this is in essence an SOI MOSFET with a very thin body. For a sufficiently thin body, the channel gets completely depleted and the characteristic electrostatic length becomes

$$\lambda_{\text{ET-SOI}} = \sqrt{\frac{\epsilon_s}{\epsilon_{ox}} x_{ox} x_{Si}} \tag{10.108}$$

Here, the thickness of the Si body plays the role of the depletion region depth in Eq. (10.18).

The ET–SOI MOSFET offers a scaling path by thinning down the body thickness. The major concern about this approach is the degradation in performance that comes about when the body is made very thin. This is due to an increase in the source and drain resistance as well as a reduction in mobility that is the result of increased scattering. To reach gate lengths in the 20–25 nm range, a body thickness below 10 nm is required for which these problems become quite severe.

A second approach is the so-called *FinFET* which comes in two flavors as, sketched in Figure 10.55. In a FinFET, the channel no longer lies flat on the wafer surface but it is placed perpendicular to it resembling the fin of a fish sticking out of the water, hence its name. In one configuration, current flows between the source and the drain along the sidewalls of the fin. The key advantage of this device architecture is that with a sufficiently narrow fin, x_{Si} in Fig. 10.55, the Si body is fully depleted and for $V_{GS} > V_T$, there are not two channels on each sidewall of the fin but a single one that is controlled by both gates. This is what is referred to as a *double-gate* MOSFET since two gates control one conducting channel. Effectively, this halves the body thickness and the characteristic electrostatic length becomes

$$\lambda_{\text{DG-FET}} = \sqrt{\frac{\epsilon_s}{\epsilon_{ox}} x_{ox} \frac{x_{Si}}{2}} \tag{10.109}$$

Improved electrostatics can result if the gate is allowed to further exert control over the channel on the top surface. This is called a *Trigate MOSFET* as sketched in Fig. 10.55. In a simple design with a square body cross section, the electrostatic length now becomes

$$\lambda_{\text{TriFET}} = \sqrt{\frac{\epsilon_s}{\epsilon_{ox}} x_{ox} \frac{x_{Si}}{3}} \tag{10.110}$$

In the Trigate FET, the channel potential is modulated by three gates and hence, the control is much tighter. This is reflected in the shorter electrostatic length.

The process can be carried one step forward into what is known as *Gate-All-Around MOSFET* (GAA–MOSFET), also sketched in Fig. 10.55. With a square cross section, this design allows for electrostatic control from all four sides of the channel. This further shortens the electrostatic length which becomes

$$\lambda_{\text{GAA-FET}} = \sqrt{\frac{\epsilon_s}{\epsilon_{ox}} x_{ox} \frac{x_{Si}}{4}} \tag{10.111}$$

The Double-Gate FET, TriGate, and Gate-All-Around transistors all belong to a class of devices known as *three-dimensional MOSFETs* because the channel is no longer planar on the wafer surface but sticks out of it. The major concern with these designs is the complexity of their fabrication and the ability to meet the parasitic capacitance and resistance goals. An additional concern is width and current scaling which is less flexible than in conventional planar MOSFETs. In all of these 3D designs, the current carrying capacity of the structure is completely defined by its geometry. Transistors with different levels of drain current are obtained by placing multiple channels in parallel. Since the geometry of the fin is fixed, the current scales in a discrete way as opposed to the continuous way of planar transistors. This takes valuable flexibility away from circuit designers.

Among the four alternative designs to the planar bulk MOSFET of Fig. 10.55, the Trigate FET has been the first to reach the marketplace. The 22-nm CMOS node was deployed by Intel for volume manufacturing in 2011 with a Trigate structure. The 14 nm node with a scaled down version followed in 2014.

In the quest for an ever smaller footprint transistor with sufficient performance that can be realized in a cost-effective way, three-dimensional structures offer the promise for Moore's law to continue to march forward for a few more generations. Along this path, staying with a silicon channel, even if strained, is becoming very challenging. The problem is the simultaneous need to scale transistor dimensions and operating voltage to meet the constant power density constraint. The problem with Si is that as the operating voltage is reduced, performance drops precipitously. A way out of this is offered by substituting the Si channel by one made out of a new semiconductor with improved transport properties.

Ge is a column IV elemental semiconductor that offers both improved electron and hole mobilities with respect to Si. The hole mobility is particularly remarkable with values reaching

2000 $cm^2/V \cdot s$ for inversion layers having been demonstrated at room temperature. Ge is also a Si compatible material with relatively well-established technology since it has been incorporated in SiGe heterojunction bipolar transistors (HBTs) for now many years. Ge or SiGe alloys appear promising as channel materials for future CMOS generations. The main difficulty is obtaining a high-quality Ge/dielectric interface for the gate stack.

Compound semiconductors such as InGaAs, InAs, or InSb have electron mobilities and velocities many times over that of silicon. The mobility of electrons in InAs at room temperature, for example, is anywhere between 13,000 and 25,000 $cm^2/V \cdot s$, depending on channel configuration. These extraordinary mobilities translate into velocities that can be several times that of Silicon at significantly reduced voltages. For p-channel devices, other than Ge, antimonide semiconductors, such as InSb, GaSb, and their ternary alloy InGaSb, also offer significant improvements over Si. As with Ge, with all these materials, obtaining a high-quality and reliable dielectric/semiconductor interface in the gate stack is the main challenge.

The use of non-Si channel materials would represent a disruption even greater than the substitution of SiO_2 with a high-K dielectric in the gate stack. To be sure, a future CMOS technology involving non-Si channels will have to strongly leverage the established Si manufacturing base, as this is the only way to achieve the economies of scale required for success. Hence, integration of structures based on new channel materials onto Si wafers is a significant challenge, particularly if the n- and p-channel devices are based on different materials.

Whatever the future brings, whether using Si in the channel or not, it is reasonable to expect that field-effect transistors will remain at the center of the action for still several generations of technology. Society's hunger for ever more inexpensive, higher performing, and more power efficient electronics will continue to stimulate engineers to find new ways to leverage the unique properties of FETs and CMOS. The physics of short MOSFETs studied in this chapter will remain a valuable foundation to understand the detailed physics of future logic transistors for generations to come.

10.6 SUMMARY

- High vertical electric fields degrade the carrier mobility in the inversion layer. This makes the drain current be lower than predicted by the ideal MOSFET model. Over a broad range of parameters, the mobility follows a universal relation with an effective vertical electric field across the inversion layer. Mobility degradation becomes more important in scaled devices.
- Carrier velocity saturation in the channel due to large lateral electric fields reduces the drain current below the predictions of the ideal MOSFET model. Carrier velocity saturation also results in premature drain current saturation.
- In short MOSFETs, the threshold voltage becomes dependent on gate length and V_{DS}. Threshold voltage rolloff and DIBL, as these phenomena are respectively known, are both produced by the electrostatic impact of the drain on the channel. They are both controlled by an electrostatic characteristic length given by

$$\lambda = \sqrt{\frac{\epsilon_s}{\epsilon_{ox}} x_{ox} x_{dlong}}$$

These undesirable phenomena are minimized by making the gate length sufficiently larger than λ.

- In short MOSFETs, the subthreshold swing also becomes gate length and V_{DS} dependent. S is degraded as the gate length gets sorter and V_{DS} increases. This is also a manifestation of the electrostatic influence of the drain on the channel charge and is controlled by the same characteristic length as V_T rolloff and DIBL.
- In a regular polysilicon-gate MOSFET, the gate capacitance is smaller than the value determined by the thickness of the gate oxide. The discrepancy arises from two factors. One is polysilicon depletion, or the appearance of a thin depletion layer at the polysilicon–oxide interface that removes the electrical gate further away from the channel. The second one is the inversion layer capacitance which stems from the finite thickness of the inversion layer. The inversion capacitance is large but not infinite. In scaled MOSFETs characterized by very thin gate oxides, these two effects become important and detract from the current drive of the transistor.
- Scaled MOSFETs also suffer from gate leakage current. This is due to tunneling of carriers across the gate oxide which becomes important when the gate oxide thickness approaches the wavelength of the electron. Gate leakage current is exponentially dependent on the field across the oxide.
- In a MOSFET biased in saturation, there is a large lateral electric field at the drain end of the channel. The magnitude of this field is governed by a characteristic length given by

$$l = \sqrt{\frac{\epsilon_s}{\epsilon_{ox}} x_{ox} x_j}$$

MOSFET scaling decreases l which makes the peak electric field higher.

- A consequence of high lateral electric field at the drain end of the channel in a MOSFET in saturation is the appearance of impact ionization, the generation of holes through impact ionization processes. This results in substrate current. The impact ionization rate and the substrate current are exponentially dependent on the peak lateral electric field.
- The finite output conductance in a MOSFET in saturation emerges from two different effects. CLM originates from the increased extension of the high-field region at the drain end of the channel as V_{DS} increases. This shortens the effective electrical length of the channel and increases the drain current. The second effect is connected to DIBL, the reduction in threshold voltage as V_{DS} increases. This increases gate overdrive and the drain current. The key signature of CLM is a linearly increasing g_o with I_D. g_o due to DIBL is much less dependent on I_D. CLM tends to dominate in long-channel devices and DIBL tends to be more important in short-channel devices.
- Another potential problem associated with high fields in short MOSFETs is gate-induced drain leakage. This contributes to the *off* current of the MOSFET. GIDL originates on band-to-band tunneling between the body and the drain in the overlap region between the gate and the drain. GIDL is gate length independent.
- Constant field scaling refers to an approach to MOSFET scaling in which the dimensions of the MOSFET and the operating voltage scale so that the electric fields are maintained roughly constant. Constant voltage scaling brings about significant improvements in MOSFET performance and switching energy. The main problems are an increase in power dissipation and continuous changes in the operating voltage.

- Constant voltage scaling is a scaling approach in which the dimensions of the device scale in a harmonious way but the operating voltage does not. This has the advantage of maintaining the operating voltage stable from generation to generation and it also results in strong improvements in performance. However, power dissipation explodes and internal electric fields increase.
- Generalized scaling refers to a scaling approach in which the device operating voltage and V_T scale but they do so more slowly than the geometry of the device. This mitigates the growth of off current and static power dissipation while maintaining performance gains.
- Power constrained scaling is a form of generalized scaling in which total power density remains constant from generation to generation. This is required in modern scaling since power dissipation has reached levels that are economically unmanageable for many applications.
- In the last 40 years of MOSFET scaling, there has been a major early phase of constant voltage scaling followed by a short period of constant field scaling, then a phase of generalized scaling, and more recently we have entered an era of power constrained scaling. Along the way, exponential improvements in transistor density (popularly known as *Moore’s Law*), transistor switching energy and system performance have taken place.

10.7 FURTHER READING

Fundamentals of Modern VLSI Devices by Y. Taur and T. H. Ning, Cambridge University Press, 2nd Edition, 2009 (ISBN 978-0-521-83294-6, TK7871.99.M44T38) provides a fairly detailed description of the short MOSFET and many relevant issues by two major contributors to the field. In Chapters 3 and 4 there are excellent descriptions of short-channel effects, MOSFET scaling, and other practical matters such as MOSFET design and gate length measurement techniques. Chapter 5 is entirely dedicated to CMOS performance factors where the impact of the various device design parameters on the switching speed of basic CMOS circuits is analyzed.

Operation and Modeling of the MOS Transistor (Third Edition) by Y. P. Tsividis and C. McAndrew, Oxford, 2011, ISBN 978-0-19-517015-3, TK7871.99.M44T77) describes basic aspects of the long and short-channel MOSFET in a very rigorous way. Particularly valuable are the descriptions of the dynamics of the MOSFET and dynamic equivalent circuit models which reach far beyond what is offered in the present book.

Modern Semiconductor Devices for Integrated Circuits by C. C. Hu, Prentice Hall, 2010 (ISBN 0-13-608525-3, TK7871.85.H746) presents a fast description of a few key issues of relevance in short MOSFETs that does not attempt to be mathematically rigorous. The description is at all times very intuitive from a physics point of view. Prof. Hu has been a key contributor to MOSFET physics and technology and this book is rich in interesting comments to real world issues and references to relevant literature.

AT10.1 ELECTROSTATICS OF SHORT MOSFET AROUND THRESHOLD

In an ideal long MOSFET biased in depletion, the semiconductor charge under the gate is entirely imaged at the gate. This is a purely one-dimensional electrostatics problem. The

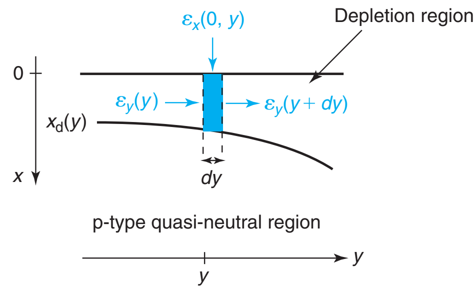


FIGURE 10.56
Sketch of Gaussian pill box in the depletion region of a short MOSFET biased around threshold.

surface potential is independent of lateral location and the threshold voltage depends only on the parameters of the gate stack: substrate doping level, gate oxide thickness, and metal work function.

In a short MOSFET, due to their relative proximity, some of the semiconductor charge is imaged at the source or drain regions. The electrostatics are now of a two-dimensional nature. In this case, the surface potential becomes spatially dependent and the threshold voltage is affected by the lateral electrostatics.

In this advanced section, we present a simplified formulation of the electrostatics of the short MOSFET around threshold, in which the MOS structure is in depletion. This is the basis for the derivation of simple expressions for the shift in V_T that results from short-channel effects.

Fig. 10.56 zooms on a small region under the gate of a short MOSFET. The top boundary represents the oxide–semiconductor interface. The bottom boundary is the edge of the depletion region, which, in general, depends on location. The blue region represents a thin Gaussian pill box placed at location y that extends in depth across the entire depletion region. The length of the pill box along y is dy.

We wish to apply Gauss' law to this pill box. $\mathcal{E}_x(0, y)$ represents the electric field emanating from the gate that penetrates into the pill box through the oxide–semiconductor interface (this field is on the semiconductor side of that interface). $\mathcal{E}_y(y)$ is the average field in depth entering along the left surface of the pill box. Similarly, $\mathcal{E}_y(y + dy)$ is the average field emanating from the pill box through its right surface. Since the bottom of the pill box is placed at the depletion region edge, there is no field crossing this surface. Application of Gauss' law yields

$$-\mathcal{E}_x(0, y)dy - \mathcal{E}_y(y)x_d + \mathcal{E}_y(y + dy)x_d = \frac{qN_A x_d dy}{\epsilon_s} \tag{10.112}$$

In an ideal MOSFET, the charge in the semiconductor is entirely imaged at the gate and $\mathcal{E}_y = 0$. In a short MOSFET, some of the charge in the semiconductor is imaged at the source or drain region, and as a result, there is a component of electric field along the channel. This field changes with location and along with it, so does the depletion region width.

If we divide all terms in Eq. (10.112) by $x_d dy$ and take the limit for short dy, we get

$$\frac{d\mathcal{E}_y(y)}{dy} - \frac{1}{x_d}\mathcal{E}_x(0, y) = \frac{qN_A}{\epsilon_s} \tag{10.113}$$

We can now rewrite this differential equation in terms of ϕ_s. $\mathcal{E}_y(y)$ can be written as

$$\mathcal{E}_y(y) = -\frac{d\phi_s}{dy} \tag{10.114}$$

Looking at the left side of Fig. 10.12, $\mathcal{E}_x(0, y)$ can be expressed as

$$\mathcal{E}_x(0, y) = \frac{\epsilon_{ox}}{\epsilon_s}\frac{V_{GS} - V_{FB} - \phi_s(y)}{x_{ox}} \tag{10.115}$$

Inserting Eqs. (10.114) and (10.115) into 10.113, we get

$$\frac{d^2\phi_s}{dy^2} + \frac{\epsilon_{ox}}{\epsilon_s x_d x_{ox}}[V_{GS} - V_{FB} - \phi_s(y)] = \frac{qN_A}{\epsilon_s} \tag{10.116}$$

Strictly speaking, in this equation, x_d depends on location. However, since we are interested in well-designed MOSFETs for which the change in ϕ_s with location should be quite slow, it is reasonable to assume that x_d changes slowly with y and we can therefore take it to the first order as a constant. A reasonable value for it is $x_d \simeq x_{dlong}$, its value in the ideal MOSFET. We know that in general the depletion width is a little bit deeper than this in a short MOSFET, but the error is expected to be relatively small.

This approximation allows us to further use the fact that in the ideal MOSFET, the fundamental charge control relationship for the MOS electrostatics [see Eq. (8.9)] applied to the present circumstances yields

$$V_{GS} - V_{FB} = \phi_{slong} + \frac{x_{ox}}{\epsilon_{ox}} qN_A x_{dlong} \tag{10.117}$$

Plugging Eq. (10.117) into Eq. (10.116) results in:

$$\frac{d^2\phi_s}{dy^2} - \frac{\phi_s}{\lambda^2} = -\frac{\phi_{slong}}{\lambda^2} \tag{10.118}$$

where we have used the definition of λ made in Eq. (10.18).

A general solution to this differential equation can be written as

$$\phi_s(y) = \phi_{slong} + A \sinh\frac{y}{\lambda} + B \sinh\frac{L-y}{\lambda} \tag{10.119}$$

A and B can be obtained by matching boundary conditions. In general, at $y = 0$ (source end of the channel) and $y = L$ (drain end of the channel), we have, respectively

$$\phi_s(y = 0) = \phi_{bij}$$

$$\phi_s(y = L) = \phi_{bij} + V_{DS}$$

The general solution becomes

$$\phi_s(y) \simeq \phi_{slong} + (\phi_{bij} - \phi_{slong} + V_{DS})\frac{\sinh\frac{y}{\lambda}}{\sinh\frac{L}{\lambda}} + (\phi_{bij} - \phi_{slong})\frac{\sinh\frac{L-y}{\lambda}}{\sinh\frac{L}{\lambda}} \tag{10.120}$$

For $V_{DS} = 0$, this equation becomes Eq. (10.17).

AT10.2 ELECTROSTATICS OF THE VELOCITY SATURATION REGION

In this appendix, we formulate the electrostatics of the velocity saturation region that appears at the drain end of the channel in a MOSFET biased in saturation.

The geometry of the problem is shown in Fig. 10.57. The velocity saturation region extends from lateral location L' in which $V(L') = V_{DSsat}$ and $\mathcal{E}_y(0, L') = -\mathcal{E}_{sat}$, the field at which prominent velocity saturation effects take place, to the drain end of the channel located at $y = L$, where $V(y = L) = V_{DS}$ and $\mathcal{E}$ peaks in absolute magnitude at $-\mathcal{E}_{max}$. In depth, the velocity saturation region extends from the surface of the semiconductor down to x_j, the drain junction depth. We assume that in the velocity saturation region, the electron distribution is uniform along the length coordinate, as required by electron velocity saturation. In addition, we assume that the electron distribution is also uniform in depth. This is a consequence of the weak vertical electric field emanating from the gate and the fact that the electrons eventually flow into the drain but not deeper.

Our goal in this appendix is to formulate a differential equation that allows to solve for the channel voltage and lateral electric field distribution in the velocity saturation region.

We construct a Gaussian pill box as indicated in Fig. 10.57. We assume that there is negligible electric field that emanates from the bottom of this pill box. This is not unreasonable as electrons toward the bottom of the pill box are flowing almost perfectly laterally toward the drain. This assumption implies that the charge in the depletion region underneath this pill box is imaged in the n^+ drain region.

Applying Gauss' law to this pill box, we get

$$-\mathcal{E}_x(0, y)dy - \mathcal{E}_y(y)x_j + \mathcal{E}_y(y + dy)x_j = \frac{q(N_A + n)x_j}{\epsilon_s}dy \qquad (10.121)$$

Here we have assumed that $\mathcal{E}_y$ does not change too much in depth so that we can work with an average vale. n represents the electron concentration in the velocity saturation region.

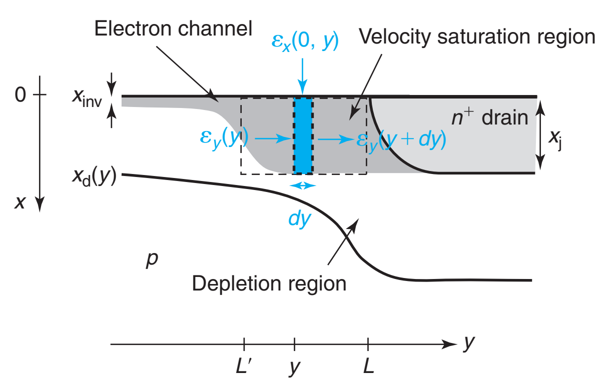


FIGURE 10.57
Geometry of the velocity saturation region at the drain end of a MOSFET biased in the saturation regime.

Dividing by dy and then taking the limit of small dy we can formulate this in the form of a differential equation:

$$x_j \frac{d\mathcal{E}_y}{dy} - \mathcal{E}_x(0, y) = -\frac{q(N_A + n)x_j}{\epsilon_s} \tag{10.122}$$

$\mathcal{E}_x(0, y)$ is the field entering the pill box from the top. Just as in the case of the electrostatics in the subthreshold regime [Eq. (10.115)], $\mathcal{E}_x(0, y)$ can be written as

$$\mathcal{E}_x(0, y) = \frac{\epsilon_{ox}}{\epsilon_s}\mathcal{E}_{ox}(y) = \frac{\epsilon_{ox}}{\epsilon_s}\frac{V_{GS} - V_{FB} - \phi_s(y)}{x_{ox}} \tag{10.123}$$

$y = L'$ represents the beginning of the velocity saturation region. Beyond this point, electron velocity saturation prevails and the GCA fails. If we can assume that the GCA still works at L', then using Eq. (10.123), we can write

$$\mathcal{E}_{ox}(L') = \frac{V_{GS} - V_{FB} - \phi_s(L')}{x_{ox}} \simeq \frac{q(N_A + n)x_j}{\epsilon_{ox}} \tag{10.124}$$

Substituting Eqs. (10.123) and (10.124) into Eq. (10.122), we get

$$x_j \frac{d\mathcal{E}_y}{dy} = -\frac{\epsilon_{ox}}{x_{ox}\epsilon_s}[\phi_s(y) - \phi_s(L')] \tag{10.125}$$

Using the definition of channel voltage [Eq. (9.7)], we can rewrite this as

$$\frac{d\mathcal{E}_y}{dy} = -\frac{\epsilon_{ox}}{x_{ox}x_j\epsilon_s}[V(y) - V(L')] \tag{10.126}$$

Since $V(L') = V_{DSsat}$ and expressing the electric field in terms of channel voltage, we get

$$\frac{d^2V}{dy^2} - \frac{V}{l^2} = -\frac{V_{DSsat}}{l^2} \tag{10.127}$$

where l is a characteristic length for this electrostatic problem given by

$$l = \sqrt{\frac{\epsilon_s}{\epsilon_{ox}}x_{ox}x_j} \tag{10.128}$$

This is the differential equation that governs the electrostatics of the velocity saturation region. It is solved in Section 10.4.1.

PROBLEMS

10.1 Consider a short n-channel MOSFET in the linear regime. Derive analytical expressions for $V(y)$, $\mathcal{E}_{ox}(y)$, $\mathcal{E}_y(x = 0, y)$, $v_e(y)$, and $Q_i(y)$ from the source to the drain that include the impact of velocity saturation. Sketch along y from the source to the drain.

10.2 Use the results of Problem 10.1 to study the lateral electrostatics of a short n-channel MOSFET biased at the boundary between the linear and saturation regimes, that is, $V_{DS} = V_{DSsat}$. Derive expressions for $V(y)$, $\mathcal{E}_{ox}(y)$, $\mathcal{E}_{y}(y)$, $v_e(y)$, and $Q_i(y)$ at $y = 0$ and $y = L$. Compare the obtained results against those from the mobility regime model.

10.3 Consider a state-of-the-art submicron nMOSFET. Indicate the impact of increasing the doping level of the body, N_A, on the parameters and figures of merit listed below. Nothing else is changed.

Circle one: *increase* $=\uparrow$, *decrease* $=\downarrow$, or *no effect*. For each item, give the reason for your choice.

threshold voltage of long device, $V_{T(long)}$:	$\uparrow$ $\downarrow$ *no effect*
maximum extent of the depletion region, x_{dmax}:	$\uparrow$ $\downarrow$ *no effect*
subthreshold swing, S:	$\uparrow$ $\downarrow$ *no effect*
for short device, $\lvert\Delta V_T\rvert$ at a given V_{DD}:	$\uparrow$ $\downarrow$ *no effect*
for short device, drain current for $V_{GS} = V_{DS} = V_{DD} > V_T$, I_{ON}:	$\uparrow$ $\downarrow$ *no effect*

10.4 This is about finding the limits of applicability of the gradual-channel approximation in the analysis of the drain current that includes velocity saturation that is performed in Section 10.1.2. To do this, proceed as in Advanced Topic AT9.1.3, that is, derive an expression for the constraint that the lateral electric field must fulfill for GCA to apply.

10.5 This homework is to illustrate some of the trade-offs of using two different gate materials in modern MOSFETs.

(a) Design the body doping of a n-channel MOSFET with a n^+-polySi gate and a 10-nm oxide thickness to have a "long-channel" threshold voltage of +1 V.
(b) Redesign the body using a gate metal that has a work function 0.55 eV larger than that of n^+-polySi. V_T should again be +1 V for a long-channel device.
(c) Compute the field at the Si/SiO_2 interface (on the semiconductor side) at threshold for both structures.
(d) Compute the oxide field at threshold for both structures.
(e) Compute the depletion region thickness at threshold for both structures.
(f) Estimate the minimum value of L_g that these two technologies can attain before short-channel effects become intolerable. Make judicious assumptions. Use $V_{DD} = 5$ V.
(g) Discuss the trade-offs of using one gate material versus the other.

10.6 At some point, the SIA (Semiconductor Industry Association) Road Map predicted that in the year 2007 CMOS IC's in production would have n-channel MOSFETs with a gate length of $L_g = 0.1$ μm, a gate oxide thickness of $x_{ox} = 3.4$ nm, a junction depth of $x_j = 14$ nm, and a threshold voltage of the order of $V_T = 0.2$ V. Let us examine some of the implications of these numbers. For this problem, use appropriate physical parameters and disregard any short-channel effects on V_T.

(a) What is the maximum value of V_{DD} that can be tolerated and still satisfy the reliability condition that the maximum oxide field not exceed 5 MV/cm?
(b) For $V_{DD} = 1$ V, calculate the inversion layer mobility.
(c) For $V_{DD} = 1$ V, what is the drive current per unit width that this transistor delivers?
(d) For $V_{DD} = 1$ V, what is the gate delay of this transistor?

10.7 Consider an optimized deep-submicron n-MOSFET technology. In a certain experiment, a device is fabricated with a W gate instead of the usual n^+-polysilicon gate. The well doping level is adjusted so as to obtain an identical threshold voltage to the standard devices. Nothing else is changed. Indicate the impact of the gate material substitution on the parameters and figures of merit listed below. $W_M(n^+ - \text{polySi}) = 4.04$ eV, $W_M(W) = 4.54$ eV.

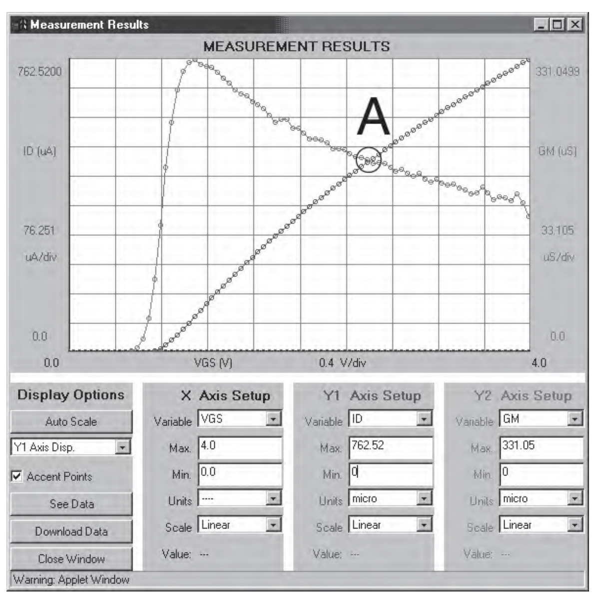
Circle one: *increase* $=\uparrow$, *decrease* $=\downarrow$, or *no effect*. For each item, give the reason for your choice.

how does N_A have to change to maintain V_T unaffected?	$\uparrow$ $\downarrow$ *no effect*
effective inversion layer mobility at a given V_{GS}, μ_{eff}:	$\uparrow$ $\downarrow$ *no effect*
subthreshold swing, S (in units of mV/dec):	$\uparrow$ $\downarrow$ *no effect*
electric field in the oxide on the source side at threshold, $\mathcal{E}_{ox}$:	$\uparrow$ $\downarrow$ *no effect*
minimum achievable gate length for a given *DIBL* criteria:	$\uparrow$ $\downarrow$ *no effect*

10.8 Consider a long-channel n-MOSFET with an n$^+$-polySi gate characterized by the following parameters: $L = 1$ μm, $W = 5$ μm, $x_{ox} = 20$ nm, $N_A = 10^{17}$ cm^{-3}. The device is biased with $V_{GS} = 3$ V, $V_{DS} = 4$ V, and $V_{BS} = 0$ V.

(a) In what regime is the MOSFET biased? Justify your answer with suitable calculations.
(b) Calculate the electron concentration per unit area at the source-end of the channel.
(c) Calculate the extent of the depletion region underneath the inversion layer at the source-end of the channel.
(d) Estimate the effective inversion layer mobility at the source-end of the channel.
(e) Estimate the value of f_T, the short-circuit current-gain cut-off frequency of this transistor at this bias point. Make whatever assumptions you need and state them.
(f) What is the value of V_{BS} that would drive this transistor to cutoff (V_{GS} and V_{DS} don't change).

10.9 The figure below graphs the transfer characteristics of a MOSFET measured through the MIT Microelectronics iLab. They are obtained in the linear regime with $V_{DS} = 0.1$ V.



This problem is about mapping the point labeled “A” in the above transfer characteristics onto the chart of the universal mobility curve of Figure 10.4 (top). That is, you must extract the values of μ_{eff} and $\mathcal{E}_{eff}$ from the experimental data in the graph above. Then plot the obtained data point on the μ_{eff} versus $\mathcal{E}_{eff}$ chart.

In case, you cannot clearly read the axis labels of the graph on the left, the horizontal scale plots V_{GS} at 0.4 V/div. The vertical scale on the left graphs I_D at 76.3 μA/div. The vertical scale on the right graphs g_m at 33.1 μS/div.

The structural parameters for this device are $L = 1.5$ μm, $W = 46.5$ μm, $x_{ox} = 20$ nm. This device has an n$^+$-polySi gate and an SiO_2 gate dielectric.

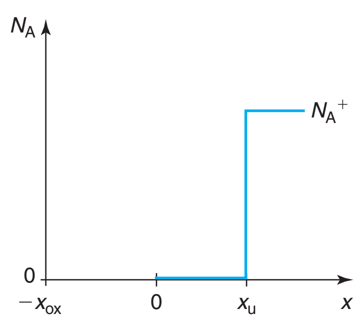
10.10 Consider a state-of-the-art deep-submicron n-channel MOSFET. Indicate the impact of applying a back bias, $V_{SB} > 0$, on the parameters and figures of merit listed below. Nothing else is changed.

Circle one: *increase* =↑, *decrease* =↓, or *no effect*. For each item, give the reason for your choice.

threshold voltage, V_T:	↑ ↓ *no effect*
subthreshold swing, S (in mV/dec):	↑ ↓ *no effect*
DIBL:	↑ ↓ *no effect*
effective inversion layer mobility at a given V_{GS}, μ_{eff}:	↑ ↓ *no effect*
drain current for $V_{GS} = V_{DS} = V_{DD} > V_T$, I_{ON}:	↑ ↓ *no effect*

10.11 A super-steep retrograde (SSR) body doping profile is claimed to have many advantages for deep-submicron MOSFETs. In this problem, we explore the essence of this claim.

Consider an n-channel MOSFET with a body doping profile distribution as sketched below:



At the limit, a super-steep retrograde body doping distribution can be thought of as a highly doped region with doping concentration N_A^+, buried under an undoped region of thickness x_u. The doping level in the highly doped region, N_A^+, is high enough so that it depletes negligibly. Also, in this region, the Fermi level is right at the valence band edge.

(a) Sketch the volume charge density, electric field, and electrostatic potential distribution across this MOS structure *at threshold*.

(b) Sketch the energy band diagram in the semiconductor at threshold. What would be a reasonable choice for the surface potential at threshold, ϕ_{sT}?

(c) Derive an expression for the threshold voltage of this structure in terms of V_{FB}, ϕ_{sT}, and the structural parameters.

(d) Let us put some numbers. Consider an n$^+$-polySi gate MOS structure with an oxide thickness of $x_{ox} = 4.5$ nm. Design x_u in order to obtain $V_T = 0.5$ V.

(e) Compare the scaling potential of this super-steep retrograde body doping distribution with a uniform body design of the same V_T. Which device exhibits better short-channel effects and why? Substantiate your answer quantitatively (note that the numbers for the uniform body design are the same as in Exercise 8.4).

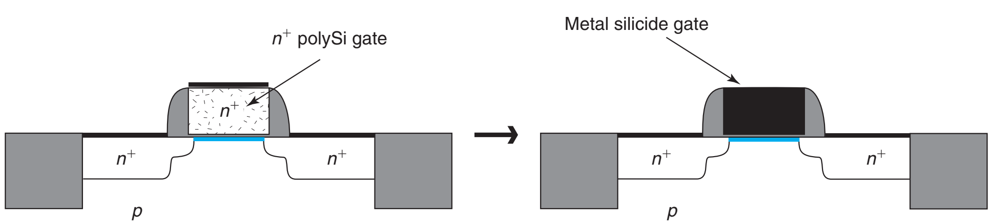
10.12 Consider an optimized deep-submicron n-MOSFET technology. In order to provide a higher voltage driving ability for the pads, a device is fabricated with a thicker gate oxide instead of the standard one. The well doping level is adjusted so as to obtain an identical threshold voltage to the standard device. Nothing else is changed.

Indicate the impact of the use of a thicker gate oxide on the parameters and figures of merit listed below. Circle one: *increase* $=\uparrow$, *decrease* $=\downarrow$, or *no effect*. For each item, give the reason for your choice.

how does N_A have to change to maintain V_T unaffected?	$\uparrow$ $\downarrow$ *no effect*
effective inversion layer mobility at a given V_{GS}, μ_{eff}:	$\uparrow$ $\downarrow$ *no effect*
subthreshold swing, S (in units of mV/dec):	$\uparrow$ $\downarrow$ *no effect*
electric field in the oxide at threshold, $\mathcal{E}_{ox}$:	$\uparrow$ $\downarrow$ *no effect*
minimum achievable gate length for a given *DIBL* criteria:	$\uparrow$ $\downarrow$ *no effect*

10.13 Consider a n-channel MOSFET process from the 0.18 μm-technology generation characterized by $x_{ox} = 4.5$ nm and $V_T = 0.3$ V. Estimate the sheet resistance of the inversion layer in the linear regime for $V_{GS} = 1.8$ V.

10.14 At the 2002 International Electron Devices Meeting, there were a number of papers that demonstrated a new technique to fully silicide the gate of MOSFETs. This is done by depositing a suitable metal on top of the polySi gate and then reacting it at high temperature until all the gate polySi is completely consumed. The remaining metal is then removed. This is sketched in the figure below:



This problem is about examining the changes that occur to the MOSFET device characteristics as a result of the complete silicidation of the gate. The key change that complete silicidation introduces in an n-channel MOSFET is an increase in the gate work function. To the first order, nothing else is changed.

Indicate the impact of the *increase* in the gate metal work function on the parameters and figures of merit listed below. Circle one: *increase* $=\uparrow$, *decrease* $=\downarrow$, or *no effect*. For each item, give the reason for your choice.

threshold voltage, V_T:	$\uparrow$ $\downarrow$ *no effect*
effective inversion layer mobility at a given V_{GS}, μ_{eff}:	$\uparrow$ $\downarrow$ *no effect*
subthreshold swing, S (in units of mV/dec):	$\uparrow$ $\downarrow$ *no effect*
off-state current, I_{off}:	$\uparrow$ $\downarrow$ *no effect*
minimum achievable gate length for a given *DIBL* criteria:	$\uparrow$ $\downarrow$ *no effect*

10.15 Consider a high-performance n-channel MOSFET from the 65 nm CMOS generation. The device is characterized by an $L_{g,eff} = 35$ nm, $x_{ox} = 1.2$ nm, and a $V_T = 0.37$ V at the nominal $V_{DS} = 1.2$ V.

We are also told that $DIBL = 130$ mV/V and the subthreshold swing is $S = 130$ mV/dec. The nominal voltage of operation for this technology is $V_{DD} = 1.2$ V.

(a) With the device biased in the linear regime with $V_{DS} = 0.05$ V, estimate the electron mobility in the channel at $V_{GS} = 1.2$ V.

(b) What is the sheet electron concentration in the inversion layer under the same conditions as in (a)?

(c) Estimate the drain current per unit width under the same conditions as in (a)?

(d) Now consider the device biased in the saturation regime with $V_{GS} = V_{DS} = V_{DD} = 1.2$ V. Estimate the drain current per unit width that flows under these conditions.

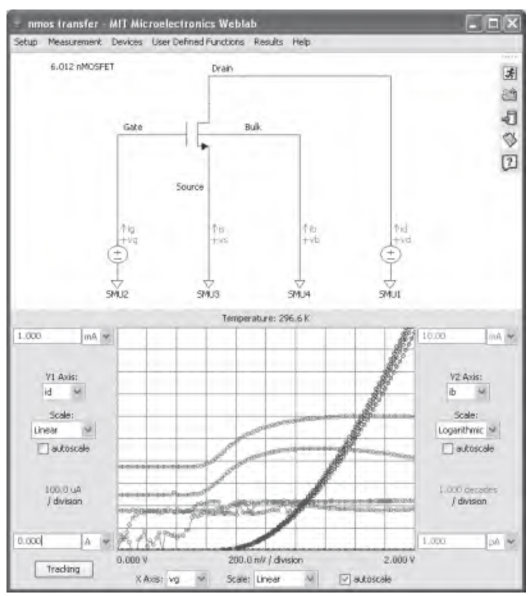
(e) Under the conditions of part (d), this device actually shows a drain current of about $I_D = 1.6$ mA/μm. The reason for this amazing current is the introduction of strain which boosts the electron velocity. Estimate the electron velocity in this device under these conditions ($V_{GS} = V_{DS} = V_{DD} = 1.2$ V).

(f) Estimate the OFF current of this device.

10.16 You asked a new undergraduate student in your lab to estimate the multiplication coefficient of impact ionization in an n-channel MOSFET at a certain bias condition. You told her to carry out a measurement of the transfer characteristics of the device while monitoring the body current.

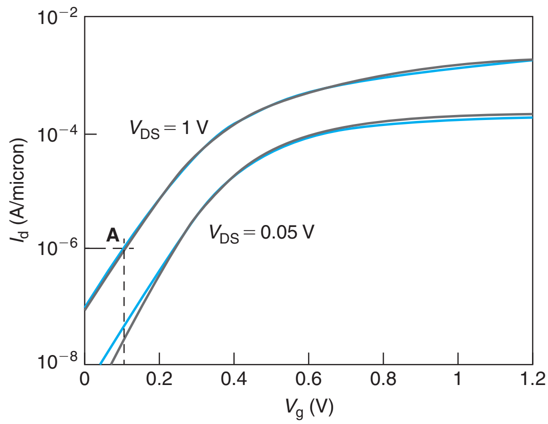
The student comes back with the poor quality black and white screenshot shown below. She tells you that the measurement conditions were $V_{DS} = 0$ to 4 V in steps of $\Delta V_{DS} = 1$ V and $V_{BS} = 0$ but she doesn't recall which line corresponds to what value of V_{DS} nor what axis. She doesn't know how to obtain the multiplication coefficient.

From the data in the graph below, estimate the multiplication coefficient at $V_{GS} = 1.8$ V and $V_{DS} = 4$ V.



Source: MIT Microelectronics iLab

10.17 Consider the room-temperature subthreshold characteristics of a 45-nm n-channel MOSFET with a high-K gate dielectric sketched below. As shown in the figure, for this transistor, a model has been developed that provides a reasonable fit to the measured characteristics. Some of the parameters of this model are $C_{ox} = 2.46\ \mu\text{F/cm}^2$, $V_T\ (@V_{DS} = 1\ \text{V}) = 0.38$ V, and $S = 95$ mV/dec.



Using these parameters, estimate the velocity of electrons at the source end of the channel when the device is biased at point **A** indicated in the figure above.

10.18 In high-k/metal-gate MOSFET technology, a thin SiO_2 interfacial layer is introduced between the high-k dielectric and the Si surface with the goal of obtaining a high-quality interface. The presence of the thin SiO_2 interfacial layer increases the equivalent-oxide thickness (EOT) of the gate stack and reduces the gate capacitance and the current drive that can be obtained from the transistor.

Consider a MOS gate stack with a metal gate and a composite gate dielectric that consists of a 2-nm high-k dielectric with a permittivity of 20 plus a 1-nm SiO_2 interfacial layer. Calculate the EOT of this gate stack.

10.19 A MOS gate stack utilizing an n^+-polysilicon gate with a doping level $N_p = 10^{20}$ cm^{-3} includes an oxide with an equivalent-oxide thickness (EOT) of 1.5 nm. At a $V_{GS} = 1$ V, the capacitance-equivalent thickness (CET) of this stack is 2.5 nm. At the same bias point, estimate the change in CET when the polysilicon gate is substituted by a metal gate.

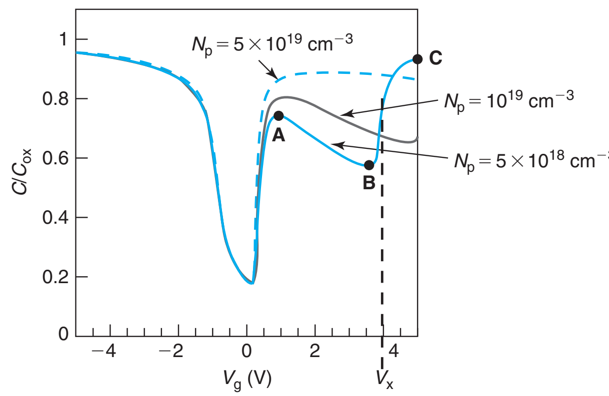
10.20 This is about the f_T of very short gate length MOSFETs in saturation.

(a) Derive an expression for the f_T of a very short MOSFET under large gate overdrive. State your assumptions. Comment on the gate length scaling of f_T.
(b) Estimate the f_T of a 90-nm nMOSFET with an effective gate length of 50 nm.
(c) Estimate the transit time of electrons through the channel of such a MOSFET.

Assume that the scattering length is much shorter than the gate length.

10.21 The figure below shows the low-frequency C–V characteristics of a MOS structure on a p-type substrate at room temperature.[13] The gate is made out of n^+-polysilicon doped at different donor concentrations. The figure reveals prominent "poly depletion", as is known, that is more pronounced the lower the doping level in the polysilicon gate is.

[13] R. Rios and N. D. Arora, Proc. IEEE International Electron Devices Meeting 1994, pp. 613–616.



In this problem, we focus on the trace that corresponds to a doping level in the polysilicon gate of $N_p = 5 \times 10^{18}$ cm^{-3}.

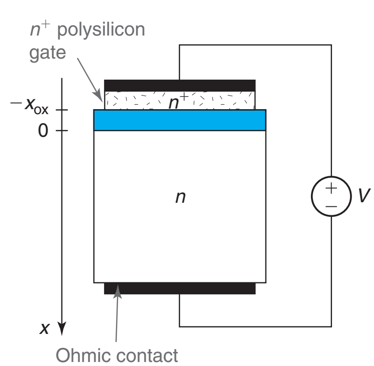
(a) First, consider point **A**. Carefully sketch an energy band diagram across the structure. In what regime (accumulation, depletion, or inversion) is the p-type body of the MOS structure? In what regime (accumulation, depletion, or inversion) is the n$^+$ polysilicon gate? Explain your answers.

(b) Now consider point **B**. Carefully sketch an energy band diagram across the structure. In what regime (accumulation, depletion, or inversion) is the p-type body of the MOS structure? In what regime (accumulation, depletion, or inversion) is the n$^+$ polysilicon gate? Why is the overall capacitance of the structure so much lower than at point **A**? Explain your answers.

(c) Derive an algebraic expression for the voltage V_x at which the capacitance abruptly rises right past point **B**. This should be in terms of the doping level in the substrate N_A, oxide thickness x_{ox}, and doping level in the polysilicon gate, N_p.

(d) Now consider point **C**. Carefully sketch an energy band diagram across the structure. In what regime (accumulation, depletion, or inversion) is the p-type body of the MOS structure? In what regime (accumulation, depletion, or inversion) is the n$^+$ polysilicon gate? Why is the overall capacitance of the structure so much higher than at point **B**? Explain your answers.

10.22 Polysilicon depletion is often experimentally studied on a semiconductor substrate of the same type as the polysilicon gate. Consider an ideal MOS structure with an n$^+$-polysilicon gate and an



n-type semiconductor, as sketched below. The doping in the polysilicon is $N_D^+ = 2.9 \times 10^{19}$ cm^{-3}. The doping in the n-type body is $N_D = 10^{17}$ cm^{-3}. The oxide thickness is $x_{ox} = 5$ nm.

Answer the questions below at room temperature. Assume nondegenerate statistics.

(a) Estimate the flatband voltage of this structure.
(b) Under a voltage $V = 5$ V and neglecting polysilicon depletion, estimate the charge per unit area in the n-type body of the structure.
(c) Under the same conditions as in (b) and now accounting for polysilicon depletion, estimate to the first order the extent of the depletion region into the polysilicon.
(d) Under the same conditions as in (b) and (c), estimate to the first order the charge per unit area in the n-type body now accounting for polysilicon depletion.
(e) Under the same conditions as in (b), sketch an energy band diagram including polysilicon depletion.

Chapter 11

The Bipolar Junction Transistor

The invention of the bipolar junction transistor (BJT) by William Shockley in 1948 and its subsequent demonstration in 1950 is widely considered as one of the most revolutionary electrical engineering events of the 20th century. A second invention involving bipolar transistors, that of the integrated circuit in 1958, unleashed the microelectronics revolution that went on to profoundly impact human society in the later part of the 20th century. The bipolar transistor enjoyed the limelight for about three decades until high-density CMOS integrated circuits emerged as a more suitable choice for many applications. Bipolar transistors chalked up many "firsts": very simple ICs based on the bipolar transistor were used in the computers that flew to the Moon with the Apollo astronauts, the first large mainframe computers were also based on bipolar ICs, supercomputers were implemented with bipolar technology all the way into the 1990s, and many others. Today, although the digital world has been taken over by CMOS, bipolar transistor technology is widely used in analog communications, IC test equipment, disk-drive, and many other applications that demand very fast and precise signal processing.

A schematic cross section of a modern integrated *npn* BJT, or bipolar transistor for short, is shown in Fig. 11.1(a). At its heart, the bipolar transistor consists of three distinct semiconductor regions that form two PN junctions back to back. The three regions are called *emitter*, *base*, and *collector*. The junctions are referred to as *base–emitter junction* and *base–collector junction*. The rest of the device is essentially overhead incurred in providing access from the outside world to these three regions as well as isolation to allow the integration of this device with others on the same wafer. Contact to the emitter is typically made through an n^+-poly-Si layer. Contact to the base is commonly made through p^+-poly-Si layers on both sides of the emitter. Contact to the collector is provided through an n^+-buried layer and an n^+-collector plug on the side of the device. The device is isolated through deep trenches that cut beyond the buried layer into the substrate. A "complementary" pnp BJT also exists. Its structure and operation is the dual of the npn BJT, although its performance is inferior.

Figure 11.1(b) shows a planar view of the same BJT. The emitter–base junction is nested inside the base–collector junction that is itself nested inside the collector–substrate junction. In this planar view, two important dimensions are to be noted. *Emitter-stripe length* and *emitter-stripe width* refer to the length and width of the emitter–base junction, respectively. These dimensions play a key role in several important figures of merit of the BJT.

The "intrinsic" region of a BJT where the bipolar action takes place represents a comparatively small fraction of the overall device. It is indicated in Fig. 11.1 by the dashed rectangle. In

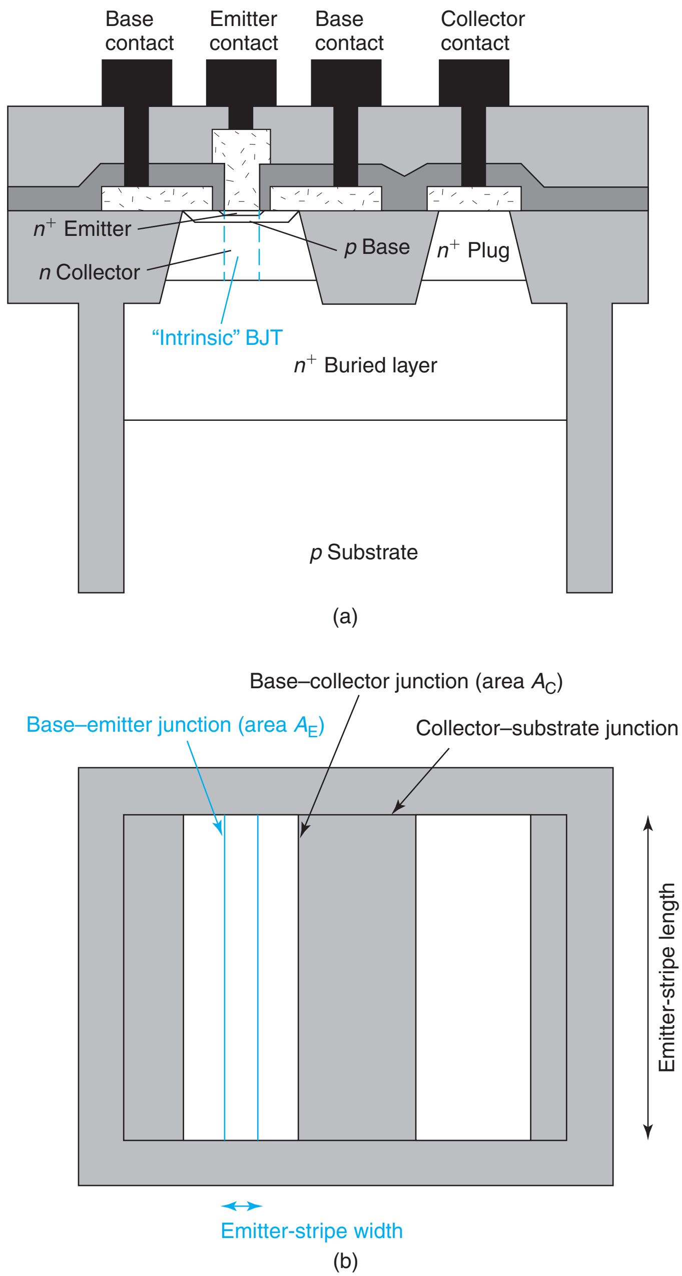


FIGURE 11.1
Simplified schematic diagram of modern bipolar junction transistor: cross section (a) and layout at the wafer surface (b).

the intrinsic region, there are essentially two PN junctions back to back. In the normal mode of operation of a BJT, called the *forward-active regime* (FAR), the base–emitter junction is forward biased and the base–collector junction is reverse biased. As a result, the emitter injects electrons into the base, while the collector extracts electrons from the base. If the base is thin enough, electrons injected from the emitter diffuse through the base and have a relatively high chance of getting extracted by the collector rather than recombining in the base. Once an electron enters the collector from the base, another electron pops out the collector contact to the outside circuit completing the current path. The main current flow in a BJT in the FAR therefore enters through the collector terminal and comes out of the emitter terminal.

In this basic mode of operation, the BJT features two key elements already encountered in the operation of the MOSFET: (1) a kind of remote-control action and (2) isolation between input and output. In a MOSFET biased in saturation, the drain current is controlled by the gate–source voltage. In a BJT, the collector current is controlled by the base–emitter voltage. This is because the higher the emitter–base voltage, the higher the number of electrons that are injected from the emitter into the base and the higher the number of them that are collected by the collector.

In a MOSFET in saturation, there is also isolation between input and output, that is, the drain current is independent of the drain voltage. Similarly, in the case of a BJT in the forward-active regime, the collector current is independent of the base–collector voltage. This results from electron transport through the base taking place through diffusion and once a small reverse bias is applied to the base–collector junction, the electron profile in the base is unaffected by any further bias increases. As a result, the collector current does not change.

This chapter deals with the operation, modeling, and design of integrated bipolar transistors. Section 11.1 defines the ideal bipolar transistor concept. This is a useful simplification of the real thing that allows us to focus on the essentials of BJT operation. Section 11.2 develops a set of models for the terminal currents of an ideal BJT in all regimes of operation. This section also presents a general-purpose equivalent-circuit model for the BJT. Section 11.3 discusses the charge–voltage and capacitance–voltage characteristics of the ideal bipolar transistor. Section 11.4 develops a small-signal equivalent circuit model for the ideal BJT in the FAR and discusses issues relevant to the high-frequency operation of bipolar transistors. Section 11.5 deals with significant nonideal effects and second-order effects of the BJT and their impact on device operation. Section 11.6 discusses the evolution of bipolar device design.

11.1 THE IDEAL BJT

In the study of the operation and design of the bipolar transistor, it is of great value to work with the concept of an ideal BJT. This fictional device, if properly defined, can effectively capture the essence of the BJT while removing obscuring details, second-order effects, and parasitics that complicate the picture. The abstraction of the ideal BJT provides great insight into the operation of the BJT and allows the fast derivation of useful first-order models.

A sketch of the ideal npn BJT is shown in Fig. 11.2. The ideal BJT consists of uniformly doped emitter, base, and collector regions. The emitter is nested inside the base so that the emitter–base junction area (or simply emitter area), A_E, is smaller than the collector–base junction area (or

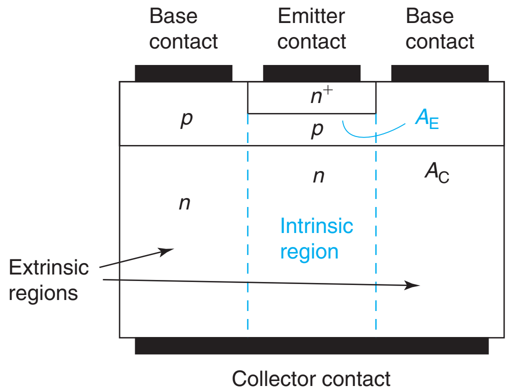


FIGURE 11.2
Sketch of the cross section of an ideal bipolar transistor. This conceptual device is a good basis for developing first-order understanding and useful models of BJT operation.

simply collector area), A_C. The area directly underneath the emitter is called the *intrinsic region*. The area outside the emitter is called the *extrinsic region*. We will therefore often distinguish between the *intrinsic base*, which is the region of the base directly underneath the emitter, and the *extrinsic base*, which lies on the sides. Similarly, we will differentiate between *intrinsic collector* and *extrinsic collector*. The emitter features an ohmic contact directly above. The base is contacted on the sides. In this ideal BJT, the collector is contacted directly underneath over the entire region.

In order to simplify its analysis, we are going to make a number of assumptions about this ideal BJT:

- All carrier flow is one dimensional.
- Doping levels are uniform throughout. The impact of non-uniform doping levels is addressed in Section 11.5.7.
- The base is thin enough so that minority carrier generation and recombination in it is negligible.
- The emitter and the collector are also assumed "short" from the minority carrier point of view, that is, all minority carrier recombination and generation takes place at their respective contacts.
- Low-level injection conditions prevail at all times in all regions.
- We assume nondegenerate statistics for electrons and holes.
- We assume the extent of the quasi-neutral regions in the emitter, base, and collector to be independent of V_{BE} and V_{BC} (but we *do* account for depletion capacitance). This is corrected in Section 11.5.1.
- We ignore any effect associated with the sidewall of the emitter–base junction.
- We ignore any resistance effects associated with the quasi-neutral regions or the ohmic contacts to these regions. The role of parasitic resistance is studied in Section 11.5.6.
- We ignore any effects associated with the substrate that surrounds the transistor.

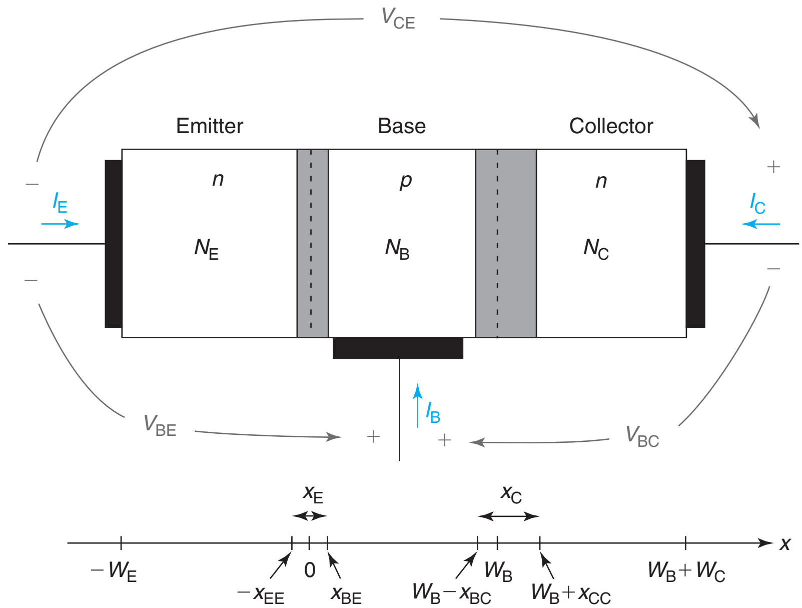


FIGURE 11.3
Sketch of ideal bipolar transistor indicating current, voltage, doping, and axis conventions. The gray-shaded areas are the space-charge regions associated with the PN junctions.

In the analysis of the ideal BJT it is useful to extract its intrinsic portion and place it on its side, as sketched in Fig. 11.3. This is the one-dimensional problem that we are going to study in detail. To make the contact to the base evident, it is often depicted on the side, as shown in the figure; this is even though it is actually not part of the intrinsic BJT.

Figure 11.3 also indicates the notation that will be used in this chapter. The doping levels in the emitter, base, and collector regions will be labeled N_E, N_B, and N_C, respectively. The emitter, base, and collector layers have thicknesses of W_E, W_B, and W_C, respectively. Around the two metallurgical PN junctions, there are two depletion regions. The base–emitter depletion region has an extent x_E. Of this, x_{EE} is on the emitter side and x_{BE} on the base side. Similarly, the base–collector depletion region has a total width x_C of which x_{BC} extends on the base side and x_{CC} extends on the collector side.

Figure 11.3 also shows the usual notation used to refer to the terminal voltages in an npn BJT. It additionally defines the terminal currents, which by convention are denoted as positive when they enter the device.

11.2 CURRENT–VOLTAGE CHARACTERISTICS OF THE IDEAL BJT

Depending on the relative signs of the base–emitter voltage, V_{BE}, and the base–collector voltage V_{BC}, there are four possible regimes of operation for the BJT. They are depicted together with

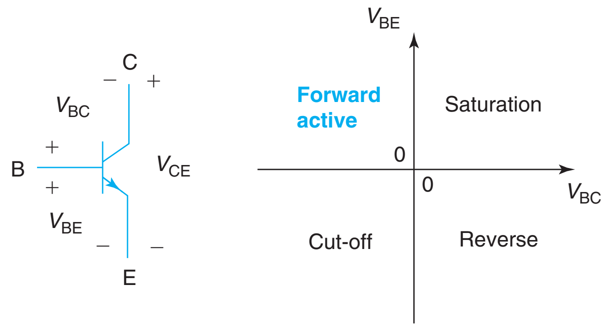


FIGURE 11.4
Symbol of npn BJT and illustration of the four regimes of operation of BJT.

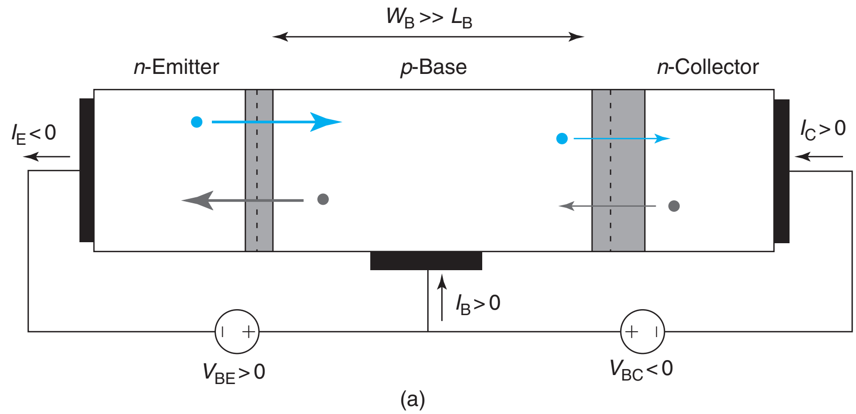
the symbol of an npn BJT in Fig. 11.4. The FAR is the most important regime of operation of a BJT. In this regime, the base–emitter junction is forward biased and the base–collector junction is reverse biased. In the FAR, the transistor exhibits good gain and isolation. This is the regime favored for most applications.

When both PN junctions are forward biased, the transistor is in *saturation*.[1] This is a much less useful regime of operation for a BJT. When the base–collector junction is forward biased and the base–emitter junction is reverse biased, the transistor is in the *reverse regime*. In this regime, the BJT works just like in FAR. However, since the transistor is optimized for operation in FAR, its performance in the reverse regime is typically very poor and is rarely used in this mode of operation. When both junctions are in reverse bias, the transistor is in *cut-off* and there is little current through it.

In a bipolar transistor, the base–emitter junction and the base–collector junction are close enough for them to interact with each other via the minority carriers (electrons in the base of an npn BJT), but not so close that the two junctions interact electrostatically. This is a deleterious condition named *punchthrough* that we will discuss later.

The basic operation of the BJT in the FAR is easy to understand with the help of Fig. 11.5. Figure 11.5(a) depicts two PN diodes back to back in an npn configuration. The middle p region is thick in the scale of the electron diffusion length in the base. The junction on the left is forward biased and as a consequence, electrons are injected from the n region on the left to the p region where they recombine. The p region also injects holes to the n region on the left where they recombine. There is then a current path through this structure that enters the p region and comes out of the n region on the left. This is a typical forward biased PN diode-like current with an $\exp \frac{qV_{BE}}{kT}$ dependence. The PN junction on the right is reverse biased. It therefore extracts minority carrier electrons from the p region and holes from the n region on the right. In an ideal PN junction, these are small generation currents that are independent of the reverse

[1]It is unfortunate that the term "saturation" has a very different meaning in a MOSFET and in a BJT. In a MOSFET in saturation, the drain current is independent of the drain voltage. This is the desirable regime of operation in many applications. In a BJT, the equivalent regime of operation is the forward-active bias regime, in which the collector current is independent of the collector voltage. A BJT in saturation, as we will see, exhibits a collector current that is a strong function of the collector voltage (akin to the linear regime in a MOSFET).



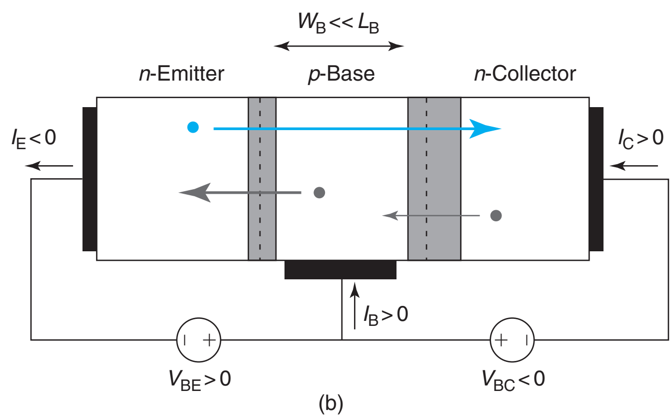


FIGURE 11.5
(a): Carrier flow in an npn back-to-back double-diode structure with the left junction in forward bias and the right junction in reverse bias. The middle p region is thick in the scale of the electron diffusion length. (b): Carrier flow in BJT in forward-active regime.

bias voltage. There is therefore a second current path in this device through the junction on the right that enters the n terminal on the right and comes out of the p terminal. The magnitude of this current is much smaller than the current flowing through the junction on the left. In this back-to-back two-diode structure, then

$$I_B \simeq -I_E \gg I_C$$

In contrast with the behavior of the double-diode structure, Fig. 11.5(b) depicts an npn BJT in the FAR. The base–emitter junction is forward biased and the base–collector junction is reverse biased. Just like in the diode back-to-back configuration, the base–emitter region injects minority carriers and the base–collector region collects minority carriers. What is significant here is that with the base region thin in the scale of the electron diffusion length, electrons injected from the emitter into the base diffuse through the base without recombining, reaching the edge of the base–collector space-charge region where they get collected. This constitutes a carrier flow path that did not exist in the back-to-back diode structure. This carrier flow path implies a current path that enters the collector and comes out of the emitter. This current is governed by the base–emitter voltage (because electron injection into the base is controlled by V_{BE}) but it is independent of the base–collector voltage. This is the main current path of a bipolar transistor.

In addition to this, there is a second significant current path that results from hole injection from the base into the emitter, just like the forward-biased PN junction. This current path flows through the base and emitter contacts. In a well-designed bipolar transistor, the magnitude of this current is much smaller than the current that flows between the emitter and the collector. All in all, in a BJT in FAR,

$$I_C \simeq -I_E \gg I_B \tag{11.1}$$

For completeness and as in the back-to-back two-diode structure, in a BJT in FAR there are additional current paths that result from carrier generation around the reverse-biased base–collector PN junction. These are negligible when compared with the currents mentioned above.

A complementary view of the situation can be obtained by looking at the carrier profiles across the transistor in equilibrium and in FAR, as qualitatively depicted in Fig. 11.6. In thermal equilibrium [11.6(a)] there are sharp transitions in the carrier concentration across the space-charge regions associated with the two metallurgical junctions.

Figure 11.6(b) sketches the carrier concentrations in the FAR. The base injects holes into the emitter where they diffuse toward the surface. The emitter ohmic contact brings the hole concentration at the surface to equilibrium. Hole injection into the emitter and recombination at its surface gives rise to the base current of the bipolar transistor in FAR.

In FAR, the emitter also injects electrons into the base. The electron concentration increases in the portion of the base closest to the emitter. However, since the base–collector junction is reverse biased, it brings the electron concentration down below equilibrium on the portion of the base closest to the collector. A steep gradient of electron concentration is therefore set across the base that drives electron diffusion from the emitter toward the collector that extracts the electrons away. Electron injection into the base, diffusion through it, and extraction by the collector constitute the collector current in a BJT in FAR.

There are some assumptions in this figure that are worth discussing briefly. First, the doping levels in the three regions follow a staircase distribution: the emitter doping is highest and the collector doping is lowest. As we will soon understand why, this is the hallmark of a well-designed bipolar transistor. Second, the emitter is assumed short in the scale of the hole diffusion length and injected holes reach the emitter surface. While a "short" emitter is in fact standard in modern transistors, having a "long" emitter would not change the operation of the transistor in any way. Third, the extent of the space-charge regions associated with both PN junctions is not affected by the application of voltage to the junctions. We know that this is not the case. We will study the impact of the modulation of the depletion region thicknesses in Section 11.5.1.

Table 11.1 summarizes what we have learned so far about the operation of the bipolar transistor in FAR and compares it with the operation of the ideal MOSFET in saturation. In a BJT,

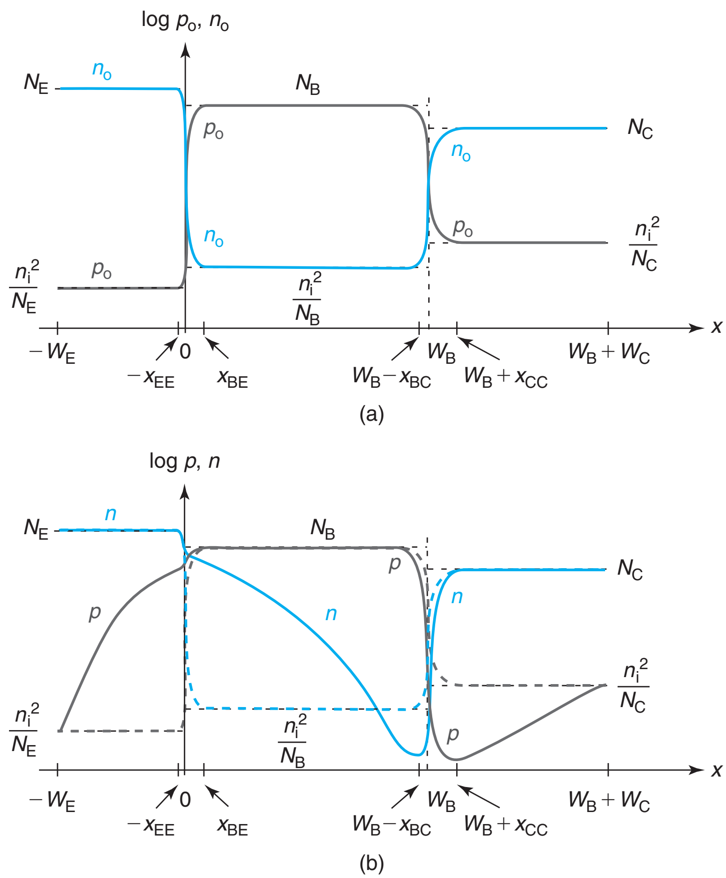


FIGURE 11.6
Sketch of equilibrium carrier concentration across intrinsic portion of bipolar transistor in equilibrium (a) and forward-active regime (b).

Table 11.1
Summary of key features of ideal BJT in FAR and comparison with ideal MOSFET in saturation.

Feature	Ideal BJT in FAR	Ideal MOSFET in saturation
Controlling terminal	Base	Gate
Controlled terminal	Collector	Drain
Common terminal	Emitter	Source
Functional dependence of controlled current	Exponential	Quadratic
DC current in controlling terminal	Exponential	0

the base is the controlling terminal, while the collector is the controlled terminal. This is parallel to the roles played by the gate and the drain in the MOSFET. In a BJT, the functional dependence of the collector current is exponential on the base–emitter voltage. This contrasts with the MOSFET in which the dependence of the drain current on the gate–source voltage is quadratic. Finally, while the base of the BJT draws DC current, that is not the case of the gate of the MOSFET due to the presence of the gate oxide.

We are now in a position to derive first-order models for the collector and base currents of an ideal BJT in FAR. This is carried out in the next subsection.

11.2.1 The forward-active regime

Based on our discussion in the previous paragraphs, in order to derive an expression for the collector current of a BJT, we must focus our attention onto the behavior of minority carrier electrons in the quasi-neutral base. Figure 11.7 depicts the electron concentration across the base. At the edge of the base with the base–emitter SCR, the boundary condition for electron concentration is determined by the base–emitter voltage,

$$n_B(x_{BE}) = n_{Bo} \exp \frac{qV_{BE}}{kT} \tag{11.2}$$

where $n_{Bo} = n_i^2/N_B$ is the equilibrium electron concentration in the quasi-neutral base. At the edge of the base with the base–collector SCR, the electron concentration boundary condition is determined by the base–collector voltage. Since this junction is reverse biased, then

$$n_B(W_B - x_{BC}) \simeq 0 \tag{11.3}$$

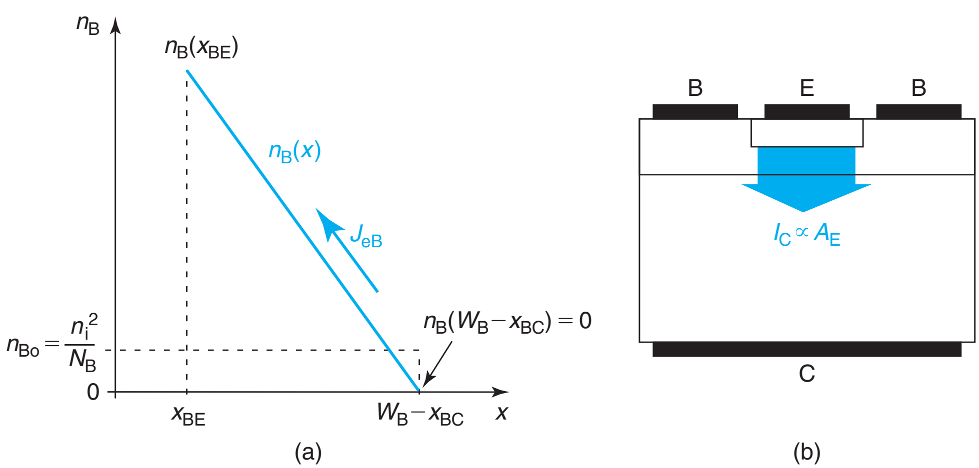


FIGURE 11.7
(a): Sketch of electron concentration across base of BJT in FAR. (b): Sketch of electron injection in BJT illustrating collector terminal current scaling with the area of the base–emitter junction.

If the base width W_B is thin enough in the scale of the electron diffusion length, negligible electron recombination takes place in the base and the electron concentration has a linear shape,

$$n_B(x) = n_B(x_{BE})\left(1 - \frac{x - x_{BE}}{W_B - x_{BE} - x_{BC}}\right) \tag{11.4}$$

The electron current density across the base is set by diffusion and given by

$$J_{eB} = qD_B\frac{dn_B}{dx} = -qD_B\frac{n_B(x_{BE})}{W_B - x_{BE} - x_{BC}} \tag{11.5}$$

which is independent of location since there is no recombination. The sign is negative because the current flows against the choice of axis made in Fig. 11.7.

As electrons diffuse through the base and reach the edge of the base–collector junction, they get extracted by the high electric field there giving rise to collector *terminal* current. In order to compute this current, we must first identify the area with which it scales. This is clarified on the right-hand side of Fig. 11.7. Electron injection from the emitter to the base takes place across the entire front of the base–emitter junction area, A_E. Since the base–collector junction area A_C is larger than A_E, increasing A_C does not change the amount of electrons that gets injected and hence those that contribute to the collector current. On the other hand, increasing A_E does increase the injection area and hence the total amount of electrons injected. In consequence, I_C scales with A_E. The collector terminal current is then given by

$$I_C = -J_{eB}A_E = qA_E\frac{n_i^2}{N_B}\frac{D_B}{W_B - x_{BE} - x_{BC}}\ \exp\frac{qV_{BE}}{kT} \tag{11.6}$$

where we have applied a minus sign to insure that the collector terminal current is positive, that is, entering into the transistor.

In general, the collector current is written as

$$I_C = I_S\ \exp\frac{qV_{BE}}{kT} \tag{11.7}$$

where I_S is referred to as the *collector saturation current*.

We now derive an expression for the base current. As discussed above, we must focus on the behavior of injected minority carriers in the emitter of the BJT. Figure 11.8 sketches the situation.

The boundary condition for the hole concentration at the base–emitter junction edge of the quasi-neutral emitter is set by the base–emitter voltage,

$$p_E(-x_{EE}) = p_{Eo}\ \exp\frac{qV_{BE}}{kT} \tag{11.8}$$

where p_{Eo} is the equilibrium hole concentration in the quasi-neutral emitter.

At the emitter surface, the hole concentration goes to equilibrium. Therefore,

$$p_E(-W_E) = p_{Eo} \tag{11.9}$$

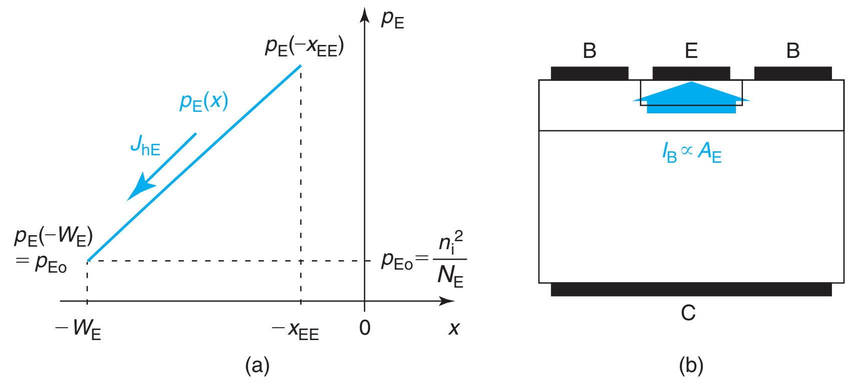


FIGURE 11.8
(a): Sketch of hole concentration across emitter of BJT in FAR. (b): Sketch of hole injection in BJT illustrating base terminal current scaling with the area of the base–emitter junction.

If the emitter is short, the hole profile follows a straight line distribution given by

$$p_E(x) = [p_E(-x_{EE}) - p_{Eo}]\left(1 + \frac{x + x_{EE}}{W_E - x_{EE}}\right) + p_{Eo} \tag{11.10}$$

The hole current density across the emitter is governed by diffusion and is given by

$$J_{hE} = -qD_E\frac{dp_E}{dx} = -qD_E\frac{p_E(-x_{EE}) - p_{Eo}}{W_E - x_{EE}} \tag{11.11}$$

which is independent of position since we have made the assumption of negligible recombination inside the emitter.

The hole injection current into the emitter becomes the base *terminal* current. To the first order, the base terminal current scales with the base–emitter area since this is the area of the injecting junction [Fig. 11.8(b)]. I_B is then given by

$$I_B = -J_{hE}A_E = qA_E\frac{n_i^2}{N_E}\frac{D_E}{W_E - x_{EE}}\left(\exp\frac{qV_{BE}}{kT} - 1\right) \tag{11.12}$$

As in I_C, the minus sign has been introduced to force the base terminal current to be positive, as mandated by common convention.

The pre-exponential factor in I_B in Eq. (11.12) is often expressed in terms of I_S as

$$I_B = \frac{I_S}{\beta_F}\left(\exp\frac{qV_{BE}}{kT} - 1\right) \tag{11.13}$$

We will discuss below the significance and physics of β_F. Sometimes the pre-exponential factor in I_B is referred to as the *emitter saturation current* and is given the symbol I_{Eo}. This notation reflects the fact that the base current is determined by the physics of minority carrier transport and recombination in the emitter.

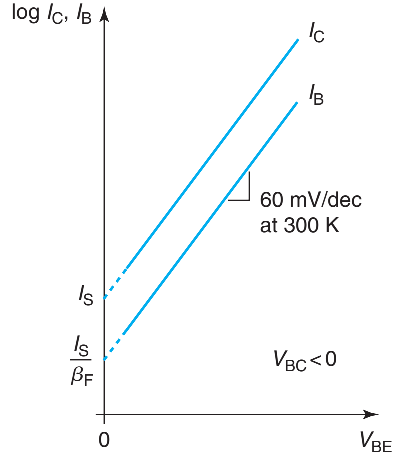


FIGURE 11.9
Gummel plot of an ideal BJT in FAR.

A typical way of plotting the characteristics of the bipolar transistor in FAR is in what is called a *Gummel plot*. In this plot, I_C and I_B are graphed versus V_{BE} in a semilogarithmic scale. A sketch for the Gummel plot of an ideal BJT in FAR is shown in Fig. 11.9. I_C and I_B appear as straight lines with an inverse slope characterized by an ideality factor of unity (the inverse slope of 60 mV/dec at 300 K). I_C and I_B extrapolate, respectively, to I_S and I_S/β_F for $V_{BE} = 0$.

If $V_{BE} \gg kT/q$, β_F as introduced in Eq. (11.13) represents the ratio of the collector current to the base current in FAR. β_F is called the *current gain* of the BJT and is given by

$$\beta_F \simeq \frac{N_E D_B W_E}{N_B D_E W_B} \simeq \frac{I_C}{I_B} \tag{11.14}$$

Here, we have neglected the extent of the depletion regions.

β_F indeed conveys well the concept of "gain" as it is the ratio of the controlled current (I_C) to the controlling current (I_B) in a BJT in FAR. The higher β_F, the smaller the amount of base current needed to control a certain amount of collector current. Several comments about β_F are appropriate at this time:

- In most circuit applications of bipolar transistors, a minimum value of β_F is required for proper circuit operation. Typically, this is around $\beta_{Fmin} \sim 50$.
- Looking at Eq. (11.14), there are in principle three device design strategies to get high values of β_F:
  - *Maximize N_E over N_B*. Since minority carrier injection into a quasi-neutral region is inversely proportional to the doping level, to get high β_F, we need to favor injection into the base relative to the emitter. Hence, the doping level in the emitter needs to be substantially higher than that of the base.
  - *Maximize D_B over D_E*. Since we wish to maximize I_C and minimize I_B, fast minority carriers should be used in the base and slow ones in the emitter. That calls for an *npn* design where electrons are minority carriers in the base and holes are minority carriers

in the emitter. All things being equal, a *pnp* design exhibits lower β_F than an *npn* design. For this and other reasons, pnp BJTs are much less favored and play a lesser role in IC design.

– *Maximize* W_E *over* W_B. A steeper gradient favors diffusion, hence W_B should be as thin as possible and W_E should be as thick as possible. From a technological point of view, it is not possible to make W_E too different from W_B. This is because the emitter is usually made in what is called a *subtractive* process, that is, the emitter is diffused into the base that has been prepared in advance. As a result, as W_E increases, W_B decreases. In a situation like this, proper process control demands that W_E not be too different from W_B. Hence, playing with this ratio does not offer a lot of room to maximize β_F.

- Equation (11.14) shows that β_F depends on several device design parameters such as emitter and base doping levels and thicknesses. In consequence, β_F is a parameter that is actually rather hard to control in a manufacturing environment. It is not uncommon for β_F to vary by as much as a factor of 2 even in a well-established process. A variety of circuit design strategies have been developed over the years to deal with this problem by rendering circuit operation insensitive to β_F variations provided that β_F is large enough, i.e., > 50. An important consequence of this is the recognition that there is little incentive in developing a bipolar technology that yields extraordinarily high current gains. As we will see later, there are actually some drawbacks to device operation that stem from unnecessarily high values of β_F.

The behavior of the bipolar transistor in the FAR can be easily captured in the simple equivalent circuit model shown in Fig. 11.10. In a BJT in FAR, the base current enters through the base terminal and flows out through the emitter terminal. This current is controlled by V_{BE}, the voltage across the same two terminals. Hence, the base current can be captured in a circuit representation by means of an ideal PN diode with a saturation current of I_S/β_F. The collector current flows through the collector and emitter terminals but it is controlled by V_{BE}. Hence, to represent it, a voltage-controlled current source is needed. The relationship between current and

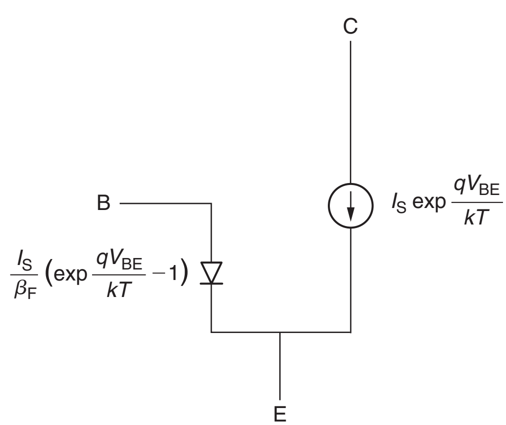


FIGURE 11.10
Equivalent circuit model of BJT in forward-active regime.

voltage is exponential. Both the base and collector currents flow through the emitter terminal. The emitter current is then

$$I_E = -I_C - I_B = -I_S \exp\frac{qV_{BE}}{kT} - \frac{I_S}{\beta_F}\left(\exp\frac{qV_{BE}}{kT} - 1\right) \tag{11.15}$$

The emitter current is negative because it comes out of the device.

To complete this discussion on the FAR of operation of the ideal BJT, Fig. 11.11 shows energy band diagrams in thermal equilibrium and in FAR. In thermal equilibrium, the energy band diagram exhibits a characteristic "hump" associated with the p-type base. The base region

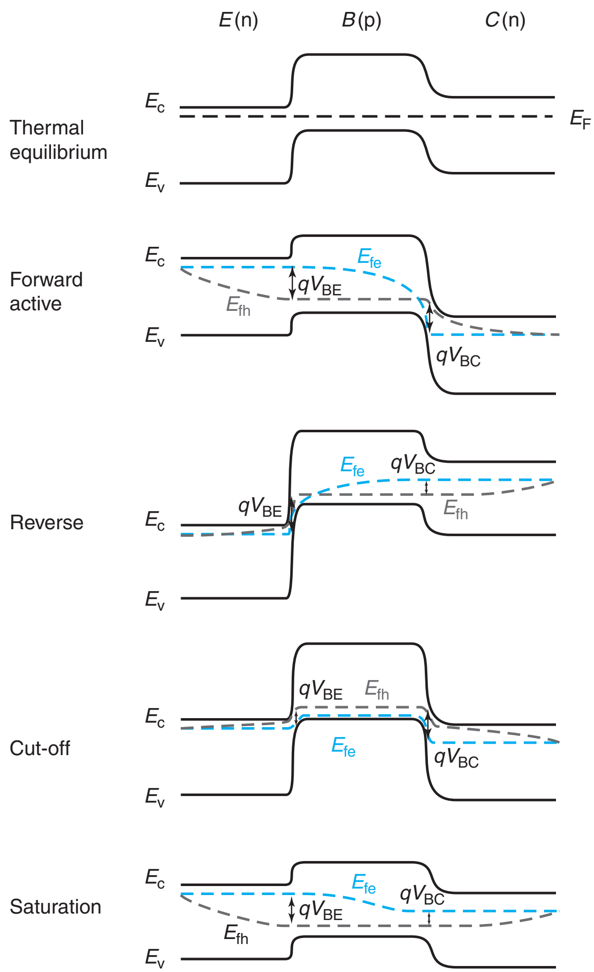


FIGURE 11.11
Schematic energy band diagrams of ideal BJT in (from top to bottom): thermal equilibrium, forward-active regime, reverse regime, saturation regime, and cut-off regime.

represents a barrier for electron flow between the emitter and the collector. In the FAR, the application of a forward voltage to the base–emitter junction reduces the barrier that electrons have to overcome to get from the emitter to the collector. As a result, electrons get injected from the emitter into the base. These electrons are collected by the base–collector junction. At this junction, the reverse bias increases the energy drop that these electrons experience as they are collected into the collector. The energy band diagram intuitively suggests that an increase in base–collector reverse bias does not result in an increase in the collector current, and it simply enhances the drop of the electrons as they get collected. I_C is "gated" by the energy barrier of the base–emitter junction.

EXERCISE 11.1

Consider a bipolar transistor with the following design: $N_E = 2 \times 10^{19}\ \text{cm}^{-3}$, $N_B = 5 \times 10^{17}\ \text{cm}^{-3}$, $W_E = W_B = 100$ nm, $A_E = 10\ \mu\text{m}^2$, *and* $A_C = 200\ \mu\text{m}^2$. *For a* $V_{BE} = 0.85$ V *and* $V_{BC} = -4$ V, *estimate: (i) the electron and hole concentrations at the edges of the E-B SCR, (ii) the collector and base currents, and (iii) the current gain.*

(i) For the indicated bias, the hole concentration at the emitter edge of the E-B SCR is obtained from Eq. (11.8),

$$p_E(-x_{EE}) = p_{Eo} \exp \frac{qV_{BE}}{kT} = \frac{n_i^2}{N_E} \exp \frac{qV_{BE}}{kT} = \frac{1.1 \times 10^{20}\ \text{cm}^{-6}}{2 \times 10^{19}\ \text{cm}^{-3}} \exp \frac{0.85\ \text{V}}{0.026\ \text{V}} = 8.7 \times 10^{14}\ \text{cm}^{-3}$$

The electron concentration at the base edge of the E-B SCR is obtained from Eq. (11.2),

$$n_B(x_{BE}) = n_{Bo} \exp \frac{qV_{BE}}{kT} = p_E(-x_{EE}) \frac{N_E}{N_B} = 8.7 \times 10^{14}\ \text{cm}^{-3} \frac{2 \times 10^{19}\ \text{cm}^{-3}}{5 \times 10^{17}\ \text{cm}^{-3}} = 3.5 \times 10^{16}\ \text{cm}^{-3}$$

(ii) The base current can be estimated by evaluating first the hole current density in the emitter using Eq. (11.11). For this, we need an estimate of the diffusion coefficient of holes in the emitter. For the given doping level, from Fig. 4.3, $\mu_h \simeq 140\ \text{cm}^2/\text{V}\cdot\text{s}$ and $D_E \simeq 3.5\ \text{cm}^2/\text{s}$. Then,

$$J_{hE} \simeq -qD_E \frac{p_E(-x_{EE})}{W_E} = -1.6 \times 10^{-19}\ \text{C} \times 3.5\ \text{cm}^2/\text{s}\ \frac{8.7 \times 10^{14}\ \text{cm}^{-3}}{100 \times 10^{-7}\ \text{cm}} = -49\ \text{A/cm}^2$$

where we have neglected p_{Eo} next to $p_E(-x_{EE})$ as well as the extent of the SCR into the emitter. The base current is then

$$I_B = -J_{hE} A_E = 49\ \text{A/cm}^2 \times 10 \times 10^{-8}\ \text{cm}^{-2} = 4.9\ \mu\text{A}$$

The collector current can be estimated by first evaluating the electron current density in the emitter. This can be obtained in a similar way from Eq. (11.5). For this, we also need an estimate of the diffusion coefficient for electrons in the base. Proceeding as above, we get $\mu_e \simeq 430\ \text{cm}^2/\text{V}\cdot\text{s}$ and $D_B \simeq 11\ \text{cm}^2/\text{s}$. Then,

$$J_{eB} \simeq -qD_B \frac{n_B(x_{BE})}{W_B} = -1.6 \times 10^{-19}\ \text{C} \times 11\ \text{cm}^2/\text{s} \frac{3.5 \times 10^{16}\ \text{cm}^{-3}}{100 \times 10^{-7}\ \text{cm}} = -6.2 \times 10^3\ \text{A/cm}^2$$

The collector current is then

$$I_C = -J_{eB}A_E = 6.2 \times 10^3 \text{ A/cm}^2 \times 10 \times 10^{-8} \text{ cm}^{-2} = 620\ \mu\text{A}$$

(iii) The current gain can be simply obtained by dividing I_C by I_B,

$$\beta_F = \frac{I_C}{I_B} = \frac{620\ \mu\text{A}}{4.9\ \mu\text{A}} = 127$$

We could also have used Eq. (11.14) to obtain the same value.

11.2.2 The reverse regime

The operation of a bipolar transistor in reverse is qualitatively identical to the FAR, except that the roles of the emitter and the collector are swapped. In reverse, the base–emitter junction is reverse biased and the base–collector junction is forward biased. In consequence, the collector injects electrons into the base; they diffuse through the base and they are collected by the base–emitter junction. As in FAR, base current results from hole injection and recombination into the collector. A sketch of the excess minority carrier concentrations across the transistor is shown in Fig. 11.12, together with those corresponding to the other three regimes of operation of the BJT.

The equations that describe the reverse regime can be obtained in a similar way as in FAR. Not surprisingly, they end up being identical to those of FAR, with the roles of the collector and the emitter reversed. Hence,

$$I_E = I_S \exp\frac{qV_{BC}}{kT} \tag{11.16}$$

$$I_B = \frac{I_S}{\beta_R}\left(\exp\frac{qV_{BC}}{kT} - 1\right) \tag{11.17}$$

$$I_C = -I_E - I_B = -I_S \exp\frac{qV_{BC}}{kT} - \frac{I_S}{\beta_R}\left(\exp\frac{qV_{BC}}{kT} - 1\right) \tag{11.18}$$

Two points are important to mention at this time. The prefactor of I_E in Eq. (11.16) is I_S, the same prefactor obtained in Eq. (11.7). This might seem like a surprise since I_S in Eq. (11.7) contains A_E and this should have been replaced by A_C. Actually, this is not the case, even in reverse, the main current path (now I_E) scales with A_E. This can be best understood by examining Fig. 11.13. The collector injects electrons into the base, over the whole base–collector junction. However, since the base is very thin, only those electrons injected into the intrinsic base are collected by the emitter. Hence, the emitter current in the reverse regime scales to the first order with A_E.

The second point regards the base current. In its expression in Eq. (11.17), a factor of β_R is introduced. This "reverse current gain" is necessarily different from β_F in FAR because the physics of the base current is now set by injection of holes and recombination in the collector. A first-order expression of β_R is not simply obtained by swapping "C" by "E" in Eq. (11.14). There are two reasons for this.

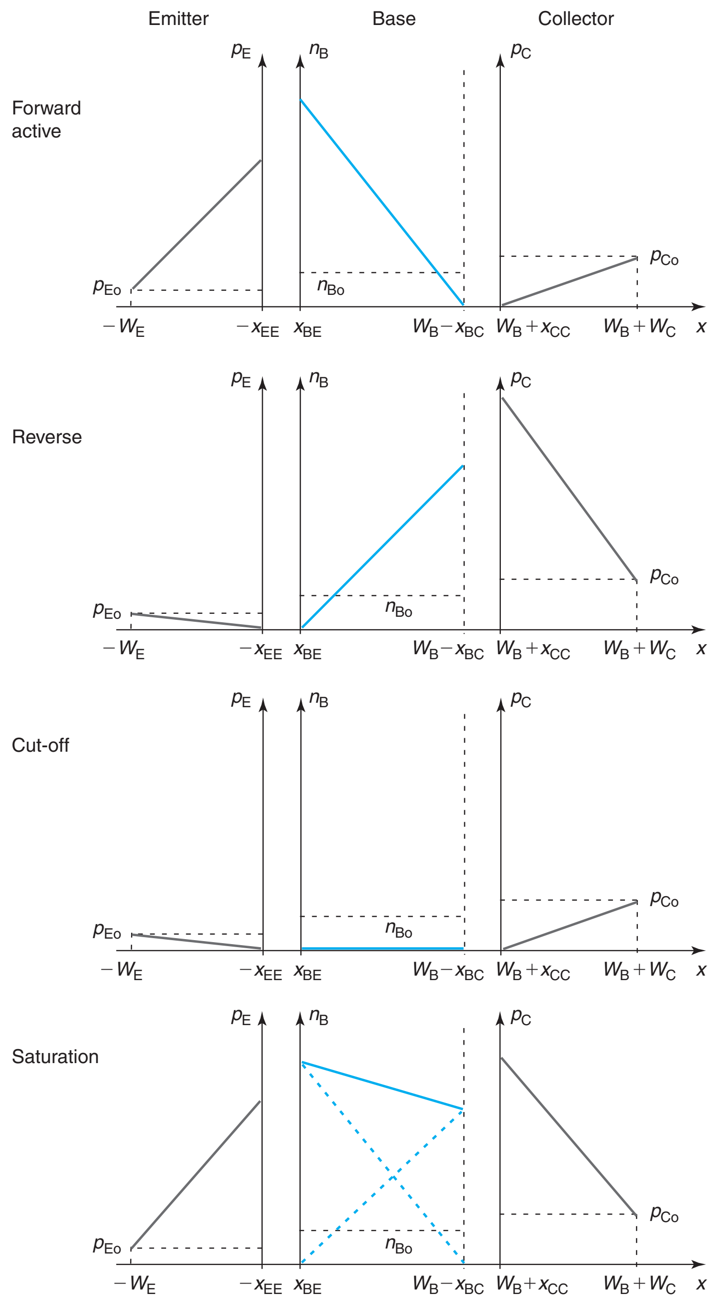


FIGURE 11.12
Schematic diagrams (not to scale) of excess minority carrier concentrations across ideal BJT in (from top to bottom): forward-active, reverse, cut-off, and saturation regimes. In saturation, the minority carrier profiles are given by the superposition of those corresponding to forward active and reverse.

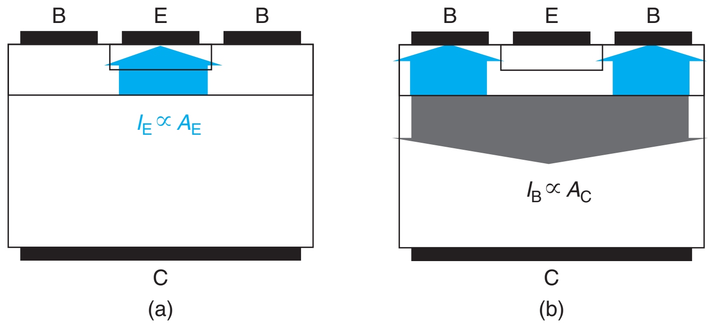


FIGURE 11.13
Schematic cross of BJT indicating main current paths for I_E (a) and I_B (b) in reverse regime of operation.

- First, in the reverse regime, though I_E scales with A_E, I_B scales roughly with A_C. As Fig. 11.13(b) illustrates, hole injection from the base into the collector takes place over the entire base–collector junction and it therefore scales with A_C. Since I_E scales with A_E, there is then an A_E/A_C factor in the expression of β_R that accounts for this geometrical issue.
- Second, in the reverse regime, there is an additional component of I_B that does not exist in FAR: electron injection from the collector into the *extrinsic* base. This is illustrated in Fig. 11.13(b). The electrons injected into the extrinsic base do not get collected by the emitter. Rather, they diffuse toward the surface where they recombine at the base ohmic contacts. This base current component scales with $A_C - A_E$.

We are not going to develop a detailed model here for β_R (see Problem 11.16). BJTs are rarely operated in this regime and are therefore optimized for operation in FAR. As a result, the reverse operation of the bipolar transistor is very poor. What is important to recognize here is that due to the asymmetry in the doping design of the BJT ($N_E > N_B > N_C$) and the fact that $A_C > A_E$, it then follows that $\beta_R \ll \beta_F$.

An equivalent circuit model description of the operation of the BJT in reverse is identical to that of FAR shown in Fig. 11.10, but with the roles of emitter and collector changed and the notation adjusted adequately. This is shown in Fig. 11.14.

The energy band diagram corresponding to the reverse regime of operation of a BJT is schematically drawn in Fig. 11.11. It shows an increased base–emitter energy barrier but a decreased base–collector barrier. This diagram leaves clear that from energy arguments alone, electrons will tend to flow from the collector to the emitter.

11.2.3 The cut-off regime

When both PN junctions are in reverse bias, the transistor is biased in the cut-off regime. A sketch of the excess minority carrier concentrations across the device is shown in Fig. 11.12. In this regime, small generation currents flow through the base–emitter and base–collector junctions. Holes generated in the emitter are extracted by the base–emitter junction and flow out as base current. Similarly, holes generated in the collector are extracted by the base–collector

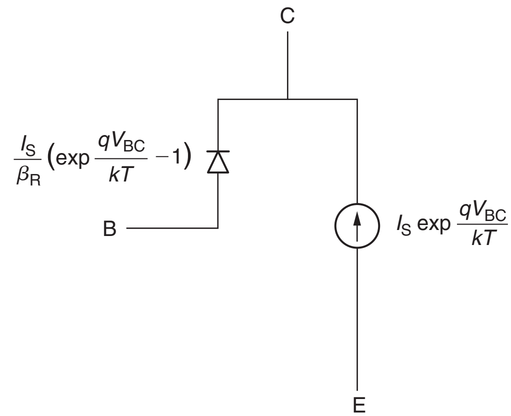


FIGURE 11.14
Equivalent circuit model for ideal BJT in reverse regime.

junction and flow out as base current. These are the dominant current paths. In our ideal BJT model, there is no recombination in the base of the transistor. As a result, when biased in cut-off, there is no generation in the base either.

The expression for the hole generation current in the emitter is easily obtained from the result obtained in Eq. (11.13). This equation gives the hole current injected in the emitter in the FAR. Since we used a general expression for the boundary condition at the edge of the emitter-base space-charge region, this equation remains correct for a negative value of V_{BE}. Hence, if V_{BE} is sufficiently negative, this current becomes $I_{B1} \simeq -\frac{I_S}{\beta_F}$.

Following a similar line of thought and working with Eq. (11.17), we conclude that the expression for the base current component associated with generation in the collector is given by $I_{B2} \simeq -\frac{I_S}{\beta_R}$. All together then,

$$I_E = \frac{I_S}{\beta_F} \tag{11.19}$$

$$I_B = -\frac{I_S}{\beta_F} - \frac{I_S}{\beta_R} \tag{11.20}$$

$$I_C = \frac{I_S}{\beta_R} \tag{11.21}$$

An equivalent circuit representation of the BJT in cut-off is shown in Fig. 11.15.

The energy band diagram corresponding to cut-off is schematically drawn in Fig. 11.11. When compared with the diagram in equilibrium shown at the top of this figure, in cut-off, the energy barrier that separates the emitter from the collector has grown, hence any carrier flow from the emitter to the collector or viceversa is suppressed. The diagram also shows that the quasi-Fermi level for holes is everywhere above the quasi-Fermi level for electrons. This means that net generation takes place in all regions of the transistor.

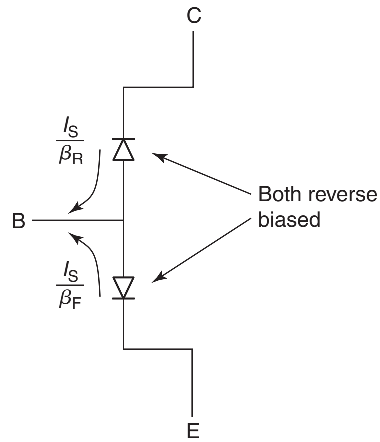


FIGURE 11.15
Equivalent circuit model for ideal BJT in cut-off.

11.2.4 The saturation regime

In saturation, both junctions of the BJT are forward biased. As Fig. 11.12 indicates, this results in excess carriers everywhere in the device. As this figure also suggests, depending on the relative values of V_{BE} and V_{BC}, the gradient of electron concentration in the base of the BJT can have either sign. In consequence, in saturation, the collector current can be either positive or negative.

Deriving expressions for the terminal currents of a BJT in saturation is straightforward since we can invoke the principle of superposition. The minority carrier profiles that are sketched in Fig. 11.12 corresponding to saturation are the superposition of those corresponding to the FAR and the reverse regime, as indicated in the figure. Hence, the terminal currents are simply the sum of Eqs. (11.7), (11.12), and (11.15) on the one hand, plus Eqs. (11.16)–(11.18), on the other. All together, we obtain:

$$I_C = I_S\left(\exp\frac{qV_{BE}}{kT} - \exp\frac{qV_{BC}}{kT}\right) - \frac{I_S}{\beta_R}\left(\exp\frac{qV_{BC}}{kT} - 1\right) \tag{11.22}$$

$$I_B = \frac{I_S}{\beta_F}\left(\exp\frac{qV_{BE}}{kT} - 1\right) + \frac{I_S}{\beta_R}\left(\exp\frac{qV_{BC}}{kT} - 1\right) \tag{11.23}$$

$$I_E = -\frac{I_S}{\beta_F}\left(\exp\frac{qV_{BE}}{kT} - 1\right) - I_S\left(\exp\frac{qV_{BE}}{kT} - \exp\frac{qV_{BC}}{kT}\right) \tag{11.24}$$

In the saturation regime, I_C and I_E can have either sign depending on the relative magnitude of V_{BE}, V_{BC}, β_F, and β_R. In saturation, I_B is always positive. Since this equation set has been built by superposition of FAR plus reverse, it summarizes all the regimes of operation of the BJT.

The equivalent circuit model for the BJT in saturation is also the superposition of those for FAR and reverse. It is shown in Fig. 11.16. This model captures the transistor operation in all four regimes. For historical reasons, this circuit model is often referred to as the *nonlinear hybrid-π model* of the BJT (more on the reason for the "π" later).

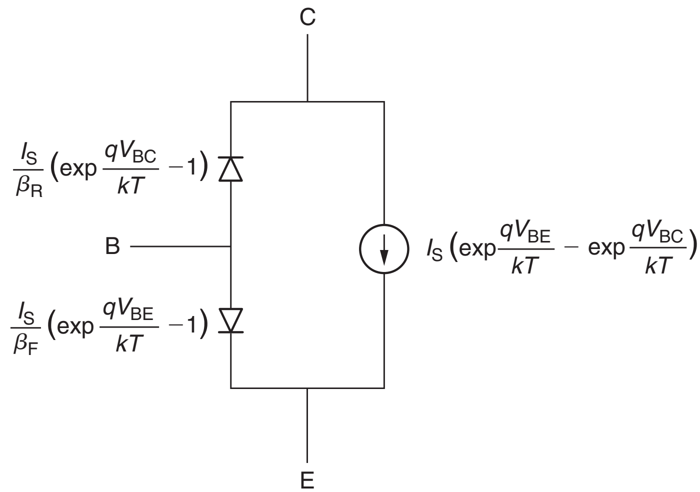


FIGURE 11.16
Equivalent circuit model for ideal BJT in saturation. Since this model is built by superposition of those corresponding to FAR (Fig. 11.10) and reverse (Fig. 11.14), this equivalent circuit model describes the BJT under all four possible regimes of operation. This equivalent circuit model is often called the *nonlinear hybrid-π model*.

The energy band diagram corresponding to saturation is sketched at the bottom of Fig. 11.11. It shows reduced emitter-base and base–collector energy barriers leading to injection of minority carriers everywhere. This is also manifested by the quasi-Fermi level for electrons being above the quasi-Fermi level for holes everywhere throughout the device. Net recombination takes place everywhere.

The saturation regime of operation of a BJT is generally avoided in most circuit applications. The main reason is that when a BJT enters saturation, it gets flooded with minority carriers everywhere in the intrinsic and extrinsic regions. Getting the transistor out of saturation takes a long time because we must either extract all the minority carriers or wait until they recombine. In either case, this is a slow process. Hence, since in most BJT applications, speed or bandwidth is of essence, circuit designers have developed techniques that avoid BJT saturation. The need for avoiding saturation is the reason why CMOS-like logic inverters using npn and pnp BJTs are not attractive. If such a circuit was built, one of the BJTs would alternatively be in saturation in either one of the logic states. Hence, switching would be very slow. To avoid this, ECL or emitter-coupled logic was developed. In this logic family, a pair of BJTs switch alternatively from FAR to cut-off. By avoiding saturation, this logic family is very fast. In fact, up to the mid-1990s all mainframes and supercomputers used ECL circuitry at their core. The problem with this logic family is that, unlike CMOS, it burns DC power in both logic states. Hence, even while idling, there is DC power consumption. This power dissipation puts a serious limit to the circuit density that can be attained. This makes ECL far less attractive than CMOS from a cost point of view.

11.2.5 Output *I–V* characteristics

A great deal of the DC behavior of a BJT is summarized in the output characteristics. These can take several forms. Figure 11.17(b) sketches a widely used set known as the *common-emitter output characteristics*. In this graph, the collector current is plotted as a function of the collector-emitter voltage with the base current as parameter. This section describes the salient features of this graph.

Before we describe the common-emitter output characteristics, it is best to examine Fig. 11(a). This figure graphs the collector current as a function of the collector-base voltage with the base current as parameter. While V_{CB} is positive (that is, the base–collector junction is reversed biased), the collector current is independent of V_{CB} and it depends linearly on I_B. This is the FAR. For V_{CB} negative, the base–collector junction becomes forward biased and the transistor goes into saturation. The collector current decreases quickly when this happens. For $I_B = 0$, the transistor is in cut-off and the collector current is negligible. The value of V_{BC} for which I_C starts going down is often denoted as V_{BCon}. For typical BJTs, V_{BCon} is of the order 0.3–0.5 V.

The common-emitter characteristics in Fig. 11.17(b) differ only from Fig. 11.7(a) in that the abscissa is V_{CE} instead of V_{CB}. Figure 11.7(b) can then be obtained by adding V_{BE} to the abscissa of Fig. 11.7(a). For the limited range of I_B values that can be graphed in these figures, V_{BE} cannot change too much (there is a logarithmic dependence of V_{BE} on I_B). Hence, Fig. 11.7(b) is basically Fig. 11.17(a) shifted by a nearly constant value of V_{BE} that we can denote as V_{BEon}. Hence, the collector current is independent of V_{CE} for values of V_{CE} higher than

$$V_{CEsat} = V_{BEon} - V_{BCon}$$

For typical BJTs, V_{BEon} is of the order 0.6–0.8 V, and V_{CEsat} is in the range 0.2–0.4 V.

There are several features in the common-emitter output characteristics of a BJT that are remarkably different from those of the common-source output characteristics of a MOSFET.

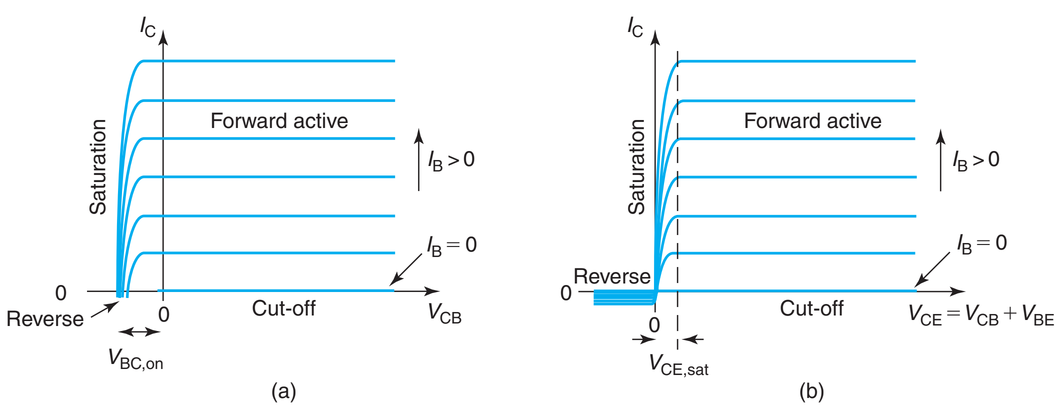
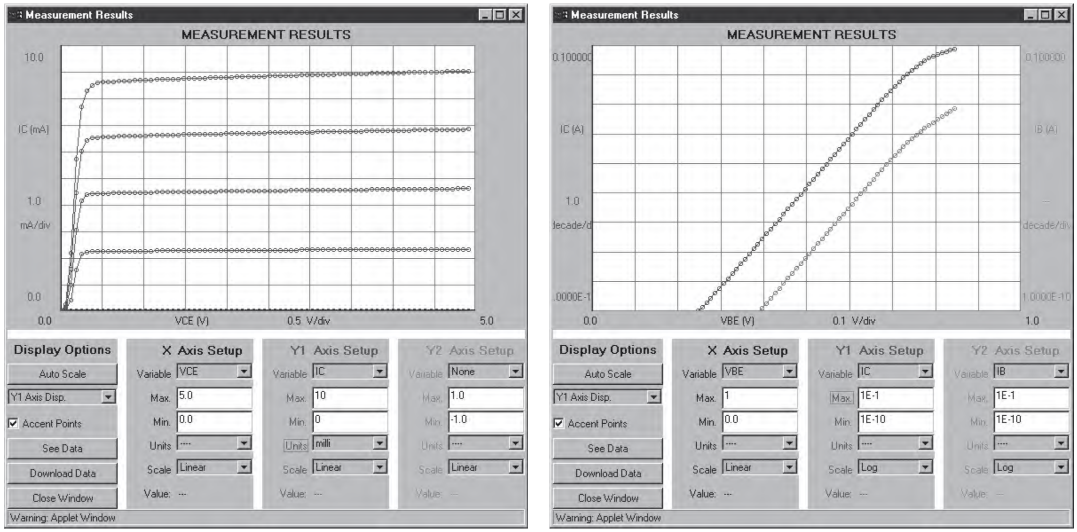


FIGURE 11.17
(a): I_C versus V_{CB}, with I_B as parameter. (b): Common-emitter output characteristics of ideal BJT (I_C vs. V_{CE}, with I_B as parameter).



(a) (b)

FIGURE 11.18
Measured I–V characteristics of a typical BJT. (a): Common-emitter output characteristics. (b): Gummel plot in the forward-active regime (Reprinted with permission from MIT Microelectronics iLab).

First, in an ideal BJT, the current characteristics in the saturation regime are equally spaced vertically reflecting the constant current gain. In the MOSFET, in contrast, the current characteristics become more widely spaced as the device is turned on reflecting the quadratic relationship between I_D and V_{GS}.

Second, in the ideal MOSFET, V_{DSsat} increases as the device is turned on. In the ideal BJT, V_{CEsat} is fairly independent of the current level. This is again because of the exponential relationship between the PN junction current and the voltage in forward bias.

Figure 11.18 shows the actual Gummel characteristics and the common-emitter output characteristics of a typical bipolar transistor. They are very much like those presented in this section though some nonidealities are also obvious. These will be discussed in the following section.

11.3 CHARGE–VOLTAGE CHARACTERISTICS OF IDEAL BJT

The I–V characteristics alone do not provide a complete description of a BJT. In order to capture its dynamic behavior, the charge–voltage or the capacitance–voltage characteristics must also be described. In this section, we develop simple models for the charge–voltage and the capacitance–voltage characteristics of the ideal BJT.

In a BJT, as in a PN junction diode, there are two types of stored charges: depletion charge in space-charge regions and minority carrier charge in quasi-neutral regions. These are schematically illustrated for the FAR in Fig. 11.19. Let us examine these two separately in the following two subsections.

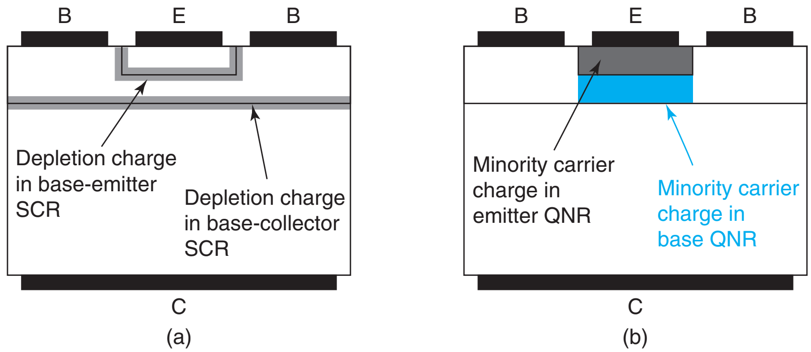


FIGURE 11.19
Schematic illustration of stored charge in ideal BJT in forward-active regime: depletion charge in base–emitter and base–collector space-charge regions (a) and minority carrier charge in emitter and base quasi-neutral regions (b).

11.3.1 Depletion charge

In an ideal BJT, there are depletion regions associated with the base–emitter and the base–collector junctions. The thickness of these depletion regions is modulated by V_{BE} and V_{BC}, respectively. The need to provide charge to the depletion regions when these voltages change has a major influence on the dynamic behavior of a BJT.

The physics of stored charge in a depletion region has been studied in the previous chapters. There is nothing special about the BJT in this regard. Using the model developed for the PN junction, the stored charge in the base–emitter and the base–collector junctions can, respectively, be written as:

$$Q_{\mathrm{jE}} = A_{\mathrm{E}}\sqrt{\frac{2\epsilon q N_{\mathrm{E}} N_{\mathrm{B}}(\phi_{\mathrm{biE}} - V_{\mathrm{BE}})}{N_{\mathrm{E}} + N_{\mathrm{B}}}} = Q_{\mathrm{jEo}}\sqrt{1 - \frac{V_{\mathrm{BE}}}{\phi_{\mathrm{biE}}}} \tag{11.25}$$

$$Q_{\mathrm{jC}} = A_{\mathrm{C}}\sqrt{\frac{2\epsilon q N_{\mathrm{B}} N_{\mathrm{C}}(\phi_{\mathrm{biC}} - V_{\mathrm{BC}})}{N_{\mathrm{B}} + N_{\mathrm{C}}}} = Q_{\mathrm{jCo}}\sqrt{1 - \frac{V_{\mathrm{BC}}}{\phi_{\mathrm{biC}}}} \tag{11.26}$$

where ϕ_{biE} and ϕ_{biC} are the built-in potentials of the emitter-base and base–collector junctions, respectively. Note that the depletion charge in the emitter-base and base–collector junctions scale with the respective junction areas.

Q_{jEo} and Q_{jCo} represent the depletion charge at the respective junctions in thermal equilibrium:

$$Q_{\mathrm{jEo}} = A_{\mathrm{E}}\sqrt{\frac{2\epsilon q N_{\mathrm{E}} N_{\mathrm{B}} \phi_{\mathrm{biE}}}{N_{\mathrm{E}} + N_{\mathrm{B}}}} \tag{11.27}$$

$$Q_{\mathrm{jCo}} = A_{\mathrm{C}}\sqrt{\frac{2\epsilon q N_{\mathrm{B}} N_{\mathrm{C}} \phi_{\mathrm{biC}}}{N_{\mathrm{B}} + N_{\mathrm{C}}}} \tag{11.28}$$

To the first order, since $N_E \gg N_B \gg N_C$, these expressions can be simplified to:

$$Q_{jEo} \simeq A_E\sqrt{2\epsilon q N_B \phi_{biE}} \tag{11.29}$$

$$Q_{jCo} \simeq A_C\sqrt{2\epsilon q N_C \phi_{biC}} \tag{11.30}$$

When the voltage across any one of these junctions changes, the stored charge also changes. This gives rise to a capacitive effect that is described by the junction capacitance. For each of the junctions, the junction capacitance is given by:

$$C_{je} = \frac{C_{jeo}}{\sqrt{1 - \frac{V_{BE}}{\phi_{biE}}}} \tag{11.31}$$

$$C_{jc} = \frac{C_{jco}}{\sqrt{1 - \frac{V_{BC}}{\phi_{biC}}}} \tag{11.32}$$

where C_{jeo} and C_{jco} are the junction capacitances that correspond to zero volts across each junction:

$$C_{jeo} \simeq A_E\sqrt{\frac{\epsilon q N_B}{2\phi_{biE}}} \tag{11.33}$$

$$C_{jco} \simeq A_C\sqrt{\frac{\epsilon q N_C}{2\phi_{biC}}} \tag{11.34}$$

C_{je} and C_{jc} display the familiar dependences of the space-charge region capacitance of a PN junction. They exhibit an inverse square-root dependence on the voltage across the junction as well as a square-root dependence on the doping level on the lowly doped side of the junction.

11.3.2 Minority carrier charge

In a BJT in the FAR, there are excess minority carriers in the emitter and base quasi-neutral regions. To maintain quasi-neutrality, an identical distribution of excess majority carriers is required. A capacitive effect ensues. If the voltage across the emitter-base junction is changed, the base and the emitter terminals must provide the required minority and majority carriers to continue to satisfy quasi-neutrality. In the PN diode we denoted this capacitive effect as diffusion capacitance.

Basic models for the minority carrier stored charge were developed for the PN junction in Section 6.4.2. In that section, we obtained the general result that the minority carrier stored charge in a quasi-neutral region can always be written as the product of the minority carrier current flowing through the region times the relevant time constant for the region in question. For the case of the "short" emitter of the ideal BJT being studied here, the minority carrier current is I_B, and the relevant time constant is the transit time through the emitter τ_{tE}. Hence, the minority carrier charge in the emitter, Q_E, can be written as

$$Q_E = \tau_{tE} I_B \tag{11.35}$$

τ_{tE} is the transit time of holes through the emitter and is given by

$$\tau_{tE} = \frac{W_E^2}{2D_E} \tag{11.36}$$

Here, we have neglected the extent of the depletion region into the emitter that tends to be small in the scale of W_E.

Similarly, the minority carrier stored charge in the base is given by the collector current times the transit time of electrons through the base,

$$Q_B = \tau_{tB} I_C \tag{11.37}$$

τ_{tB} is given by

$$\tau_{tB} = \frac{W_B^2}{2D_B} \tag{11.38}$$

Here, we have also neglected the extent of the depletion regions into the base.

Note that in Eqs. (11.35) and (11.37), the units of Q_E and Q_B are C. Also, note that both Q_E and Q_B scale with the area of the base–emitter junction, as sketched in Fig. 11.19 (both I_B and I_C in the FAR scale with A_E, as discussed above). As in the case of the PN diode, the most important result here is the linear dependence of the stored minority carrier charges, Q_E and Q_B, on the minority carrier currents, I_B and I_C, respectively.

The total minority carrier charge in the FAR is the sum of Q_E and Q_B,

$$Q_F = Q_E + Q_B = \left(\frac{\tau_{tE}}{\beta_F} + \tau_{tB}\right) I_C = \tau_F I_C \tag{11.39}$$

where we have used Eq. (11.14) and we have defined

$$\tau_F = \frac{\tau_{tE}}{\beta_F} + \tau_{tB} \tag{11.40}$$

τ_F is called the *intrinsic delay* and represents the overall time constant for minority carrier storage for the BJT in FAR. As seen in Eq. (11.40), the relative contribution of the emitter in τ_F is reduced by β_F because I_B is β_F times smaller than I_C.

The important result captured in Eq. (11.39) is that the total charge stored in a BJT in FAR is linear in I_C. In an alternate view, the total minority carrier stored charge scales as $\sim\exp\frac{qV_{BE}}{kT}$.

If the base–emitter voltage changes, the minority carrier charge in the emitter and base changes too. This is a capacitive effect that can be characterized by a diffusion capacitance. The capacitance is simply given by

$$C_F = \frac{dQ_F}{dV_{BE}} = \tau_F \frac{qI_C}{kT} \tag{11.41}$$

It is important to think about the location of this capacitance in the equivalent circuit model of the ideal BJT. The way to assess this is to identify the device terminals that supply the stored charge. For carrier storage in the emitter, the minority carrier holes are supplied by the base (hence, they flow through the base terminal) and the majority carrier electrons are provided by the emitter contact. For carrier storage in the base, the minority carrier electrons are injected

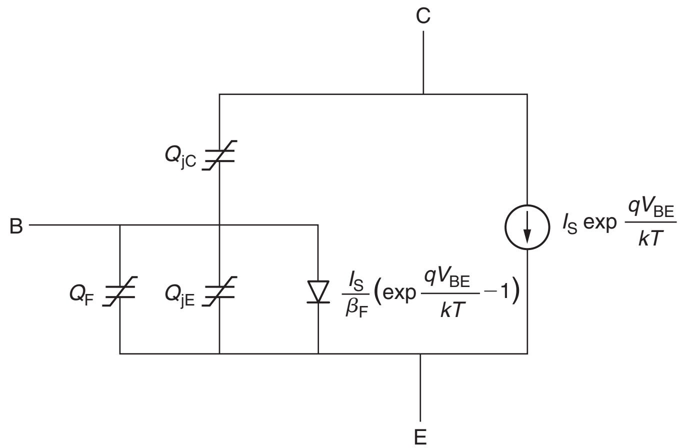


FIGURE 11.20
Equivalent circuit model for ideal BJT in forward-active regime including charge storage due to depletion charge and minority carrier charge.

from the emitter and come from the emitter contact, and the majority carrier holes are supplied by the base contact. Hence, the capacitance associated with both of these storage effects is located between the emitter and the base terminals. Figure 11.20 shows the equivalent circuit model of the ideal BJT in FAR incorporating all charge storage elements discussed here.

A similar analysis can be performed for the reverse regime of operation of the BJT. In reverse, there is minority carrier storage in the collector and in the base of the transistor, as sketched in Fig. 11.21. The situation is more complicated than in FAR because of the different dependencies associated with carrier storage in the intrinsic and extrinsic portions of the base. It is not very useful to take space here to provide a detailed analysis of this effect since a BJT is rarely operated outside FAR. Here, it suffices to say that, similarly to τ_F, there is a time constant τ_R that accounts

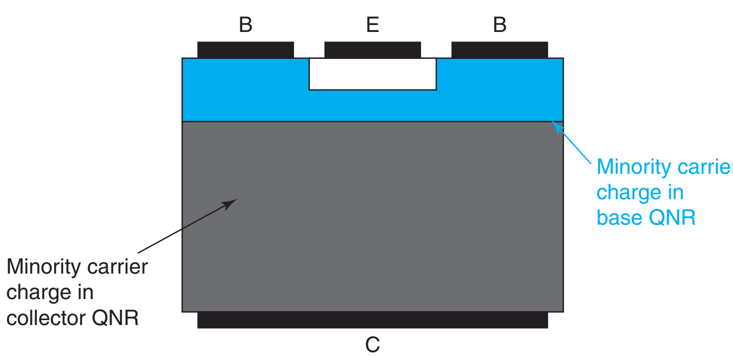


FIGURE 11.21
Schematic illustration of minority carrier stored charge in ideal BJT in reverse regime.

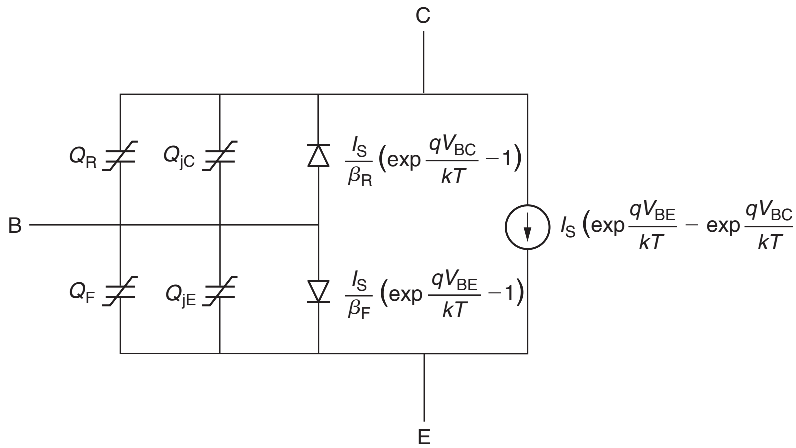


FIGURE 11.22
General equivalent circuit model for ideal BJT valid in all regimes of operation.

for all these details (see calculation of τ_R in Problem 11.16). Regardless, the total stored charge Q_R depends on τ_R and V_{BC} in a similar fashion as Q_F. Hence, we expect

$$Q_R = \tau_R I_E \tag{11.42}$$

The capacitance associated with charge storage is then given by

$$C_R = \frac{dQ_R}{dV_{BC}} = \tau_R \frac{qI_E}{kT} \tag{11.43}$$

Following similar arguments as above, this capacitance is located between the base and the collector terminals of the ideal BJT.

By superposition, in saturation, charge storage takes place everywhere in the transistor. This is then described by the combined effect of Q_F and Q_R. Figure 11.22 shows an equivalent circuit model for the ideal BJT that is valid in all regimes of operation. It includes all charge storage elements discussed here.

EXERCISE 11.2

Consider again the bipolar transistor of Exercise 11.1. For the same bias point of $V_{BE} = 0.85$ V and $V_{BC} = -4$ V, estimate: (i) the minority carrier transit times through the emitter and the base, (ii) the minority carrier charge in the emitter and the base, and (iii) the intrinsic delay.

(i) We use several results obtained in Exercise 11.1. The hole transit time through the emitter is given by Eq. (11.36),

$$\tau_{tE} = \frac{W_E^2}{2D_E} = \frac{(100 \times 10^{-7}\ \text{cm})^2}{2 \times 3.5\ \text{cm}^2/\text{s}} = 14\ \text{ps}$$

Similarly, the electron transit time through the base is obtained from Eq. (11.38),

$$\tau_{tB} = \frac{W_B^2}{2D_B} = \frac{(100 \times 10^{-7}\ \text{cm})^2}{2 \times 11\ \text{cm}^2/\text{s}} = 4.6\ \text{ps}$$

(ii) The hole charge in the emitter is obtained from Eq. (11.35),

$$Q_E = \tau_{tE} I_B = 14 \times 10^{-12}\ \text{s} \times 4.9 \times 10^{-6}\ \text{A} = 6.9 \times 10^{-17}\ \text{C}$$

For the electron charge in the base, we use Eq. (11.37),

$$Q_B = \tau_{tB} I_C = 4.6 \times 10^{-12}\ \text{s} \times 6.2 \times 10^{-4}\ \text{A} = 2.9 \times 10^{-15}\ \text{C}$$

(iii) The intrinsic delay is simply obtained using Eq. (11.40),

$$\tau_F = \frac{\tau_{tE}}{\beta_F} + \tau_{tB} = \frac{14\ \text{ps}}{127} + 4.6\ \text{ps} = 4.7\ \text{ps}$$

which is dominated by τ_{tB}.

11.4 SMALL-SIGNAL BEHAVIOR OF THE IDEAL BJT IN FORWARD-ACTIVE REGIME

In many analog and communication applications, a bipolar transistor is biased in the forward-active regime and a small signal is applied on top of this bias. In cases like this, there is an interest in understanding the response of the bipolar transistor to the small signal alone. As in previous chapters, a small-signal equivalent circuit model constitutes a very effective tool to deal with these kinds of situations.

In this section, we first derive a small-signal equivalent circuit model for the BJT operating in the FAR. We then use this model to introduce a very important figure of merit that characterizes the frequency response of a bipolar transistor.

11.4.1 Small-signal equivalent circuit model

Since we already have a general-purpose equivalent circuit model for the BJT (shown in Fig. 11.20 for FAR), deriving a small-signal equivalent circuit model requires only its linearization. In FAR, there are five elements in this circuit. Each, in turn, will yield a corresponding element in the small-signal equivalent circuit model. Let us derive them in order.

First, in Fig. 11.20 a voltage-controlled current source represents the collector current, I_C. This current source depends exponentially on V_{BE}. If a small-signal v_{be} is applied on top of the bias V_{BE}, the small-signal collector current, i_c, is easily obtained by the Taylor series expansion of the exponential,

$$I_C + i_c = I_S \exp\frac{q(V_{BE} + v_{be})}{kT} \simeq I_S \exp\frac{qV_{BE}}{kT}\left(1 + \frac{qv_{be}}{kT}\right) \tag{11.44}$$

Hence,

$$i_c = \frac{qI_C}{kT} v_{be} \tag{11.45}$$

The proportionality factor between i_c and v_{be} is called *transconductance* and as in the MOSFET, it is denoted by the symbol g_m. In a BJT, the transconductance is given by

$$g_m = \frac{qI_C}{kT} \tag{11.46}$$

The important result here is that in an ideal BJT, the transconductance depends only on the bias collector current and is independent of the layout of the device. Furthermore, the dependence of g_m with I_C is linear. This is unlike the case of the ideal MOSFET that we saw in Eq. (9.52), which exhibits a g_m dependence on the square root of not only the drain current but also the width-to-length ratio of the gate. The simple and in some way "universal" (it only depends on temperature) I_C dependence of g_m in a BJT is much appreciated by circuit designers.

The second element to linearize in the equivalent circuit model of Fig. 11.20 is the diode. Following a similar Taylor series expansion, the diode can be linearized to a conductance of value

$$g_\pi \simeq \frac{qI_B}{kT} = \frac{g_m}{\beta_F} \tag{11.47}$$

The unusual notation for this conductance (the "π" in g_π) stems from the topology of the small-signal equivalent circuit model (shown in Fig. 11.23). g_π also depends linearly on I_C but the dependence is not "universal" since β_F is also involved. Note that since in a well-designed transistor $\beta_F \gg 1$, then $g_\pi \ll g_m$.

Finally, there are three charge storage elements, Q_{je}, Q_{jc}, and Q_F, as described in the previous section. These get linearized to simple capacitances referred to as C_{je}, C_{jc}, and C_π, respectively. Their values were also given in the previous section. C_π is the usual notation for C_F in the small-signal equivalent circuit model. Going back to Eq. (11.41) and using Eq. (11.46), C_π can be written as

$$C_\pi = \tau_F g_m \tag{11.48}$$

which again only depends on I_C and is independent of the layout of the transistor.

Altogether, the small-signal equivalent circuit model for an ideal BJT in the FAR is shown in Fig. 11.23.

11.4.2 Common-emitter short-circuit current-gain cut-off frequency, f_T

Bipolar transistors are widely used in high-frequency analog and communications applications. In assessing the suitability of a given bipolar technology for a certain application, among other attributes, it is necessary to identify a figure of merit that gauges the ability of the transistor to operate at high frequencies. A widely used figure of merit is the so-called common-emitter short-circuit current-gain cut-off frequency, f_T. In this section, we derive an expression for f_T for the ideal BJT and we provide a physical interpretation for f_T. We also discuss the key dependences of f_T as well as the BJT design strategies to achieve a high f_T.

Let us first define f_T. When assessing f_T, the BJT is operated in the common-emitter configuration, as shown in Fig. 11.24. The base–emitter junction is biased with a current source I_B,

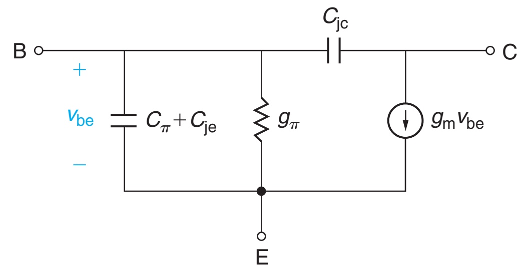


FIGURE 11.23
Small-signal equivalent circuit model of ideal BJT in forward-active regime.

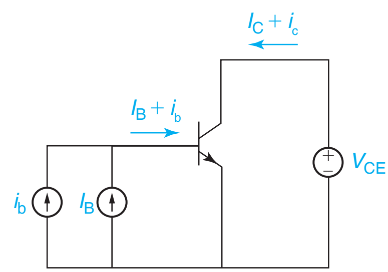


FIGURE 11.24
Bias and small-signal configuration used in defining h_{21} and f_T in a BJT.

and a voltage source is attached between the collector and the emitter. V_{CE} and I_B are selected so that the transistor operates in the FAR. On top of the bias current I_B, a small-signal sinusoidal steady-state base current i_b of frequency f (angular frequency ω) is applied. In response to the small signal, the transistor delivers a small-signal collector current, i_c. Notice that from the small-signal point of view, the output (the collector) is shorted out, that is, $v_{ce} = 0$.

In this configuration, one can define a small-signal current gain as the ratio of the small-signal collector current to the small-signal base current. This figure of merit is widely used in network theory and is often referred to as h_{21}. Then,

$$h_{21} = \left. \frac{i_c}{i_b} \right|_{v_{ce}=0} \tag{11.49}$$

It is clear that in an ideal BJT at low enough frequency, h_{21} should be equal to β_F, the DC current gain. However, as the frequency increases, h_{21} is expected to drop as the capacitors in the device begin drawing significant amounts of AC current when charging and discharging at high frequency. f_T is defined as the frequency at which the magnitude of h_{21} becomes unity, that is, the frequency for which there is no current gain left in the device. For most circuit applications, it is not practical to operate a bipolar transistor at frequencies beyond f_T. In fact, in many applications, one must operate the device with significant current gain left and therefore at a frequency that is a fraction of f_T.

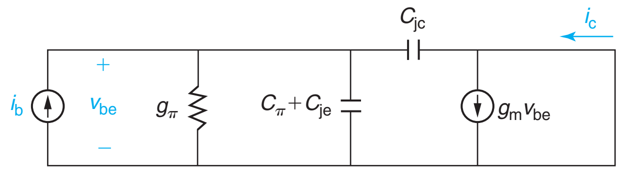


FIGURE 11.25
Small-signal equivalent circuit model for circuit configuration of Fig. 11.24.

Let us derive an expression for the f_T of an ideal BJT. The place to start is to draw a small-signal equivalent circuit model for the circuit of Fig. 11.24. This is shown in Fig. 11.25.

Obtaining h_{21} for this circuit is straightforward. The collector current is simply given by $i_c = (g_m - j\omega C_{jc})v_{be}$. The base current is given by $i_b = [g_\pi + j\omega(C_\pi + C_{je} + C_{jc})]v_{be}$. The ratio is therefore

$$h_{21} = \frac{g_m - j\omega C_{jc}}{g_\pi + j\omega(C_\pi + C_{je} + C_{jc})} \tag{11.50}$$

The magnitude of h_{21} is

$$|h_{21}| = \frac{\sqrt{g_m^2 + \omega^2 C_{jc}^2}}{\sqrt{g_\pi^2 + \omega^2(C_\pi + C_{je} + C_{jc})^2}} \tag{11.51}$$

$|h_{21}|$ is graphed in a Bode plot in Fig. 11.26. This represents $|h_{21}|$ as a function of angular frequency in a log–log plot. There are three clear regimes in $|h_{21}|$ separated by two corner frequencies ω_β and ω_c. For low enough frequencies, $\omega \ll \omega_\beta$, $|h_{21}|$ becomes

$$|h_{21}| \simeq \frac{g_m}{g_\pi} = \beta_F \tag{11.52}$$

as we expected.

For intermediate frequencies, $\omega_\beta \ll \omega \ll \omega_c$, $|h_{21}|$ is given by

$$|h_{21}| \simeq \frac{g_m}{\omega(C_\pi + C_{je} + C_{jc})} \tag{11.53}$$

and $|h_{21}|$ drops as $1/\omega$ as sketched in Fig. 11.26.

For high frequencies, $\omega \gg \omega_c$, $|h_{21}|$ becomes

$$|h_{21}| = \frac{C_{jc}}{C_\pi + C_{je} + C_{jc}} \tag{11.54}$$

The corner angular frequencies that separate the three frequency regimes are given by

$$\omega_\beta = \frac{g_\pi}{C_\pi + C_{je} + C_{jc}} = \frac{g_m}{\beta_F(C_\pi + C_{je} + C_{jc})} \tag{11.55}$$

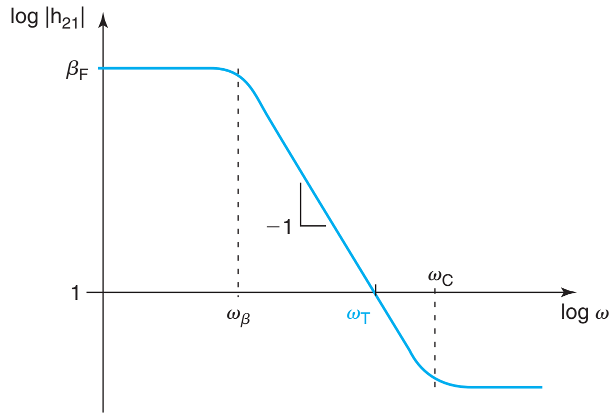


FIGURE 11.26
Frequency evolution of $|h_{21}|$ of ideal BJT.

and

$$\omega_c = \frac{g_m}{C_{jc}} \tag{11.56}$$

Since at high frequency, Eq. (11.54) suggests that $|h_{21}| < 1$, the angular frequency at which $|h_{21}|$ becomes unity can be derived from Eq. (11.53),

$$\omega_T = \frac{g_m}{C_\pi + C_{je} + C_{jc}} \tag{11.57}$$

The cut-off frequency, f_T, is then given by

$$f_T = \frac{g_m}{2\pi(C_\pi + C_{je} + C_{jc})} \tag{11.58}$$

f_T is found to be directly proportional to g_m and inversely proportional to the sum of all capacitors of the transistor. If we look at the small-signal equivalent circuit of Fig. 11.25, this is a result that makes sense. At high enough frequency, the capacitors short out the input. For a given i_b, the resulting v_{be} across the input of the device is reduced from the low-frequency value. Since it is v_{be} that "feeds" the transconductance generator, there is less collector current and an overall smaller current gain.

Physical meaning of f_T, delay time

As it turns out, f_T has a very rich physical meaning that provides great insight into the high-frequency operation of a transistor. This can be appreciated by realizing that $1/2\pi f_T$ has units of time and can be written as

$$\tau_d = \frac{1}{2\pi f_T} = \tau_{tB} + \frac{\tau_{tE}}{\beta_F} + \frac{C_{je}}{g_m} + \frac{C_{jc}}{g_m} \tag{11.59}$$

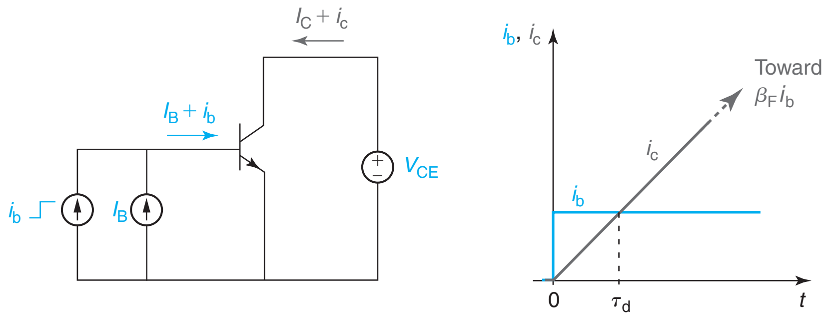


FIGURE 11.27
Response of ideal BJT to sudden rise of base current. $1/2\pi f_T$ is the time it takes for the collector current to rise by an amount equal to the step in the base current.

τ_d is called the *delay time* and it has four components. Each of these components has a physical significance of its own. This is best appreciated by considering for a moment a different dynamic situation depicted in Fig. 11.27.

Figure 11.27 shows a BJT biased in FAR for which at $t = 0$ the base current is suddenly increased from I_B to $I_B + i_b$. If we wait long enough, the base emitter voltage will rise from V_{BE} to $V_{BE} + v_{be}$ and the collector current will eventually rise from I_C to $I_C + i_c$, where $i_c = \beta_F i_b$. Suppose we want to estimate the time that it takes for the collector current to reach its final value. One way to do the calculation is to estimate the charge that must be provided by the base terminal and divide that by the rate at which the charge is being delivered, that is i_b.

When the transient is completely finished, charge has been delivered to four regions in the transistor:

- *Quasi-neutral emitter*. The extra base current means that the hole distribution in the emitter becomes steeper. There is then additional hole charge that must be delivered by the base terminal to build this profile. The extra charge, from Eq. (11.35), is given by

$$q_e = \tau_{tE} i_b \tag{11.60}$$

- *Quasi-neutral base*. When the collector current has fully risen to its final value, there is a steeper gradient of electrons in the base and extra stored charge. From Eq. (11.37), the additional charge stored in the base is given by

$$q_b = \tau_{tB} i_c \tag{11.61}$$

- *Emitter-base depletion region*. An increase in i_b also implies an increase in v_{be}. Therefore, the depletion region associated with the base–emitter junction shrinks slightly. The extra charge that is supplied, from Eq. (11.31), is given by

$$q_{je} = C_{je} v_{be} = \frac{C_{je}}{g_m} i_c \tag{11.62}$$

- *Base-collector depletion region*. In the circuit configuration selected for this experiment, the collector is shorted to the emitter. As a result, as v_{be} rises, v_{bc} increases by the same amount. Hence, the depletion region of the collector-base junction also shrinks slightly. The extra charge delivered to it, from Eq. (11.32), is given by

$$q_{jc} = C_{jc} v_{bc} = C_{jc} v_{be} = \frac{C_{jc}}{g_m} i_c \tag{11.63}$$

In Eqs. (11.62) and (11.63), the small-signal stored charge in the depletion regions decreases as g_m increases. This is because for a given value of i_b, a higher g_m implies a smaller change in v_{be}, which is what produces the small change in the depletion region thickness.

By the time the transient has finished, the base contact must have delivered the sum of these four charge components. The time it takes for this to happen is given by the total charge divided by the rate at which it is delivered, that is i_b. Hence, summing the above four terms, we get

$$\tau_\beta = \frac{q_e + q_b + q_{je} + q_{jc}}{i_b} = \tau_{tE} + \beta_F \left(\tau_{tB} + \frac{C_{je}}{g_m} + \frac{C_{jc}}{g_m} \right) \tag{11.64}$$

This is exactly $1/2\pi f_\beta$, where f_β is the corner frequency between the low-frequency and the high-frequency behavior of the transistor.

Now suppose that instead of asking how much it takes for the final value of $I_C + i_c$ to be reached, we inquire about the time it takes for the collector current to increase by an amount equal to the step in the base current, that is, for it to reach a value equal to $I_C + i_b$. To the first order, that takes β_F less time, since i_c is β_F larger than i_b. That delay time is then

$$\tau_d = \frac{\tau_\beta}{\beta_F} = \frac{\tau_{tE}}{\beta_F} + \tau_{tB} + \frac{C_{je}}{g_m} + \frac{C_{jc}}{g_m} \tag{11.65}$$

This is exactly $1/2\pi f_T$. Hence, τ_d represents the time that it takes for the collector current to build up by an amount equal to the increment in the base current. τ_d is a sort of delay time for the i_b perturbation to show up at the collector. Given more time, of course, the perturbation will be amplified by β_F, but that also takes β_F more time.

Coming back to f_T, it is now easy to understand what happens if instead of applying a step input, we apply a sinusoidal input, as in Fig. 11.24. For very low frequencies, the sinusoidal signal is fully amplified. However, for frequencies above the corner frequency f_β, the amplification factor gets reduced. For a frequency equal to f_T, the collector current cannot be amplified anymore because by the time i_c has responded to the rise in i_b, i_b has already started decreasing. At that frequency, the transistor displays a current gain of unity.

Another way to look at the situation is by examining the equivalent circuit model of Fig. 11.25. As discussed before, for high frequency as the frequency goes up, the capacitors increasingly short out g_π and v_{be} gets progressively smaller. The magnitude of i_c decreases too. f_T is the frequency at which $i_c = i_b$. At this frequency, only a $1/\beta_F$ fraction of i_b actually flows through g_π. The rest flows through the capacitors.

f_T dependencies

Let us now discuss the key dependencies of f_T. The most important one is that of the collector current that can be made explicit by expressing f_T in the following form:

$$f_T \simeq \frac{1}{2\pi\tau_F} \frac{1}{1 + \dfrac{kT}{q\tau_F}\dfrac{C_{je}+C_{jc}}{I_C}} \tag{11.66}$$

For small values of I_C, the delay terms associated with the capacitors dominate and f_T becomes

$$f_T \simeq \frac{q}{2\pi kT} \frac{I_C}{C_{je}+C_{jc}} \tag{11.67}$$

In this regime, f_T increases linearly with I_C. For low current levels, charging and discharging the junction capacitances constitute the bottleneck to the frequency response of the transistor. As I_C increases, these time delays shrink.

For high enough current levels, the delays associated with the capacitors become negligible and f_T reaches a plateau given by

$$f_T \simeq \frac{1}{2\pi\tau_F} \tag{11.68}$$

In this regime, the dominant delay in the transistor is the intrinsic delay that is associated with minority carrier storage in the transistor. Since both components of τ_F are current independent, at high enough collector current, f_T becomes independent of I_C. A sketch of the overall evolution of f_T with I_C is shown in Fig. 11.28.

Two alternative views to this picture are illustrated in Fig. 11.29. The individual delay terms of τ_d are sketched as a function of I_C in a log–log plot [Fig. 11.29(a)]. For low I_C, the delay terms associated with the capacitors dominate but they shrink as I_C increases. For sufficiently high current, τ_d is dominated by τ_F, the sum of the emitter and base delay times.

The chart in Fig. 11.29(b) plots the delay time τ_d versus $1/g_m$. When graphed in this form, one obtains a straight line with a slope given by $C_{je} + C_{jc}$. The line intersects the y-axis at τ_F.

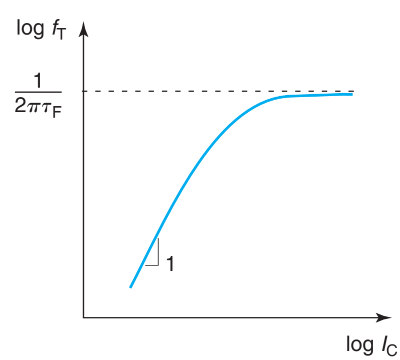
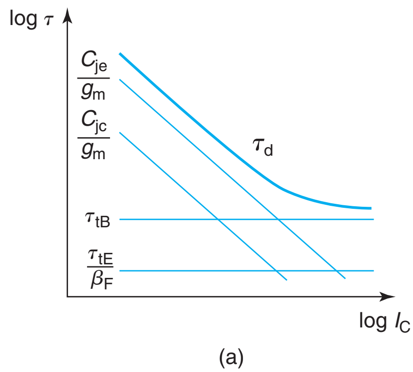


FIGURE 11.28
I_C dependence of f_T in a log–log plot.



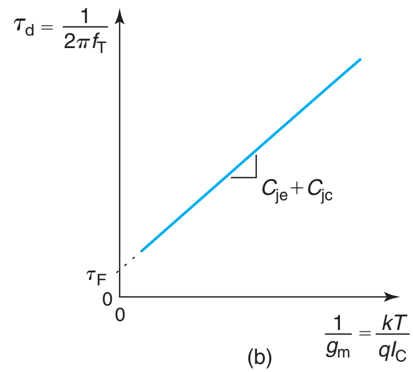


FIGURE 11.29
(a): I_C dependence of τ_d and its four delay components. (b): τ_d dependence on $1/g_m$ suggesting a procedure to determine τ_F and the sum of the junction capacitances.

This type of graph is widely used in practice to extract valuable parameter information from f_T measurements.

Figure 11.28 captures the most direct I_C dependence of f_T. There is, however, an additional indirect influence on I_C through C_{je} that one must be aware of. As I_C increases, V_{BE} increases too. This increases C_{je} to some extent and makes f_T not rise as fast with I_C. The dependence is not very strong because V_{BE} evolves as *log* I_C and C_{je} goes as $1/\sqrt{\phi_{biE} - V_{BE}}$. This effect manifests itself in the low I_C regime as a sublinear rise in f_T versus I_C, and τ_d versus $1/g_m$.

A second important dependence of f_T is on V_{CB}. In the ideal BJT, I_C itself does not depend on V_{CB}, but C_{jc} does. As V_{CB} increases, C_{jc} decreases. This affects f_T for low values of I_C (f_T goes up as V_{CB} increases), but it does not impact the high I_C regime. The dependence of f_T on V_{CB} can be graphically represented as in Fig. 11.30.

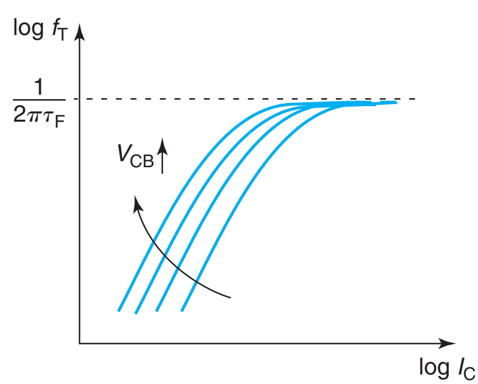


FIGURE 11.30
Impact of V_{CB} on I_C dependence of f_T.

A third set of dependencies of f_T worth noting at this time are those on the layout of the device, in particular, the emitter junction area A_E and the collector junction area A_C. In the limit of high collector current, f_T is dominated by the intrinsic delay of the transistor, which is entirely controlled by the vertical dimensions of the device. As a result, f_T is independent of A_E and A_C. For low currents, f_T depends on C_{je} and C_{jc}, and in consequence, on A_E and A_C. For constant I_C, if any of these areas increase, f_T decreases. There is a particularly important layout scaling mode that leaves f_T unaffected, even at low I_C. If an ideal BJT is biased at constant V_{BE} and V_{BC} and the emitter stripe length is increased, f_T does not change. This is because for constant-voltage bias, I_C, C_{je}, and C_{jc} all scale linearly with the emitter stripe length. This mode of scaling is very common since it is a scaling approach that keeps the collector current density, I_C/A_E, constant.

Design strategies for f_T optimization

We can now discuss device design strategies for increasing f_T. Looking again at Eq. (11.59) we see that there are four delay terms that contribute to f_T. Improving f_T consists of minimizing each one of these terms. Let us work on each of them at a time:

- τ_{tE}/β_F is the delay associated with minority carrier storage in the emitter. It is often called the *emitter charging time*. Its contribution to τ_d is usually rather small because of the presence of β_F in its denominator. Nevertheless, this delay component is to be minimized by making the emitter as shallow as possible and by having sufficient current gain in the device.
- τ_{tB} is the transit time of electrons through the base. This delay is associated with electron storage in the quasi-neutral base. This is an important component of the overall delay. Since $\tau_{tB} \sim W_B^2$, this delay is reduced by thinning W_B, the quasi-neutral base width.
- C_{je}/g_m is the delay term that results from charging and discharging the depletion capacitance associated with the base–emitter junction. This term is reduced by lowering N_B. Additionally, since C_{je} scales with A_E and g_m scales with I_C, then

$$\frac{C_{je}}{g_m} \propto \frac{A_E}{I_C} \tag{11.69}$$

  This delay component can then be minimized by operating the transistor at a higher *current density*, or in other words, by increasing V_{BE}.
- C_{jc}/g_m is the delay term that results from charging and discharging the depletion capacitance associated with the base–collector junction. This term is reduced by lowering N_C or by increasing the collector current density. Also, since C_{jc} scales with A_C, but I_C scales with A_E, then

$$\frac{C_{jc}}{g_m} \propto \frac{A_C}{I_C} = \frac{A_C}{A_E}\frac{A_E}{I_C} \tag{11.70}$$

  This delay component can then be minimized again by operating the transistor at high current density but also by making A_C as close to A_E as possible. That calls for a very tight device layout with minimum area overhead.

EXERCISE 11.3

Consider again the bipolar transistor of Exercises 11.1 and 11.2. At the same bias point of $V_{\mathrm{BE}} =$ 0.85 V *and* $V_{\mathrm{BC}} = -4$ V, *we are told that* $C_{\mathrm{je}} = 530$ fF *and* $C_{\mathrm{jc}} = 70$ *fF. At this bias point, estimate: (i) the delay time and each of its components and (ii)* f_{T}.

(i) The *emitter charging time* and the *base transit time* were already calculated in Exercise 11.2. They are, respectively,

$$\frac{\tau_{\mathrm{tE}}}{\beta_{\mathrm{F}}} = 0.1 \text{ ps}, \ \tau_{\mathrm{tB}} = 4.6 \text{ ps}$$

For the next two delay components, we need to compute g_{m}. From Exercise 11.1 we know that at this bias point, $I_{\mathrm{C}} = 620$ μA. Then, using Eq. (11.46), we have

$$g_{\mathrm{m}} = \frac{qI_{\mathrm{C}}}{kT} = \frac{6.2 \times 10^{-4} \text{ A}}{0.026 \text{ V}} = 2.4 \times 10^{-2} \text{ S}$$

The *emitter-base junction charging time* is obtained from

$$\frac{C_{\mathrm{je}}}{g_{\mathrm{m}}} = \frac{5.3 \times 10^{-13} \text{ F}}{2.4 \times 10^{-2} \text{ S}} = 22 \text{ ps}$$

The *base–collector junction charging time* is

$$\frac{C_{\mathrm{jc}}}{g_{\mathrm{m}}} = \frac{7 \times 10^{-14} \text{ F}}{2.4 \times 10^{-2} \text{ S}} = 2.9 \text{ ps}$$

The *delay time* is then

$$\tau_{\mathrm{d}} = \tau_{\mathrm{tB}} + \frac{\tau_{\mathrm{tE}}}{\beta_{\mathrm{F}}} + \frac{C_{\mathrm{je}}}{g_{\mathrm{m}}} + \frac{C_{\mathrm{jc}}}{g_{\mathrm{m}}} = 30 \text{ ps}$$

(ii) f_{T} can then easily be obtained from Eq. (11.59),

$$f_{\mathrm{T}} = \frac{1}{2\pi\tau_{\mathrm{d}}} = \frac{1}{2\pi 30 \text{ ps}} = 5.3 \text{ GHz}$$

11.5 NONIDEAL EFFECTS IN BJT

We have now completed a first pass through the basic physics of the ideal bipolar transistor. In this section, we will describe the most significant nonidealities encountered in “real” BJTs and we will discuss how they affect the device figures of merit that have been introduced in the previous sections. Many of these nonidealities have great significance to device operation and introduce important constraints to device design.

11.5.1 Base-width modulation

The thickness of the space-charge region of a PN junction depends on its voltage across. In a BJT, this impacts the capacitance of the junctions, an effect already considered earlier in this chapter. However, any expansion or contraction of a PN junction space-charge region comes at the expense of the extent of the neighboring quasi-neutral regions. SCR-width modulation then indirectly affects the minority carrier profiles and currents through the device. This effect was neglected in our treatment of the ideal BJT.

For a BJT in the FAR, we have to focus on the extent of the quasi-neutral emitter, $W_E - x_{EE}$ and the quasi-neutral base, $W_B - x_{BE} - x_{CE}$, which control I_B and I_C, respectively (refer to Fig. 11.3 for the detailed notation of our coordinate system). Since the emitter doping level is very high, emitter-width modulation is generally negligible. On the contrary, the base width can be affected by both V_{BE} and V_{BC} through the extent of their respective space-charge regions x_{BE} and x_{BC}. This is sketched in Fig. 11.31.

Specifically in FAR, $V_{BE} > 0$ and $V_{BC} < 0$. Let us examine the consequences of the application of these two voltages separately. With $V_{BC} < 0$, as $|V_{BC}|$ increases, x_{BC} increases too. This implies a reduction in the quasi-neutral base width. Since I_C is inversely proportional to this, I_C will increase. This is called the *Early effect*. Notice that while I_C increases, I_B does not change since the physics of I_B is dominated by minority carrier transport in the emitter.

Similarly, as V_{BE} increases, the emitter-base depletion region shrinks. In consequence, both the quasi-neutral emitter and the quasi-neutral base thicknesses increase. However, due to the large difference in doping levels, the increase in quasi-neutral emitter is negligible in comparison with the increase of the quasi-neutral base. The most important consequence is then that I_C increases more slowly with V_{BE} than calculated for the ideal BJT in Section 11.2.1. This is commonly referred to as the *reverse Early effect*. In the next two subsections, we discuss these two effects separately and independently from each other.

The Early effect

Figure 11.32 shows the impact of V_{BC} on the electron concentration across the quasi-neutral base of a BJT in FAR. For $V_{BC} < 0$, W_B decreases below its equilibrium value. In consequence,

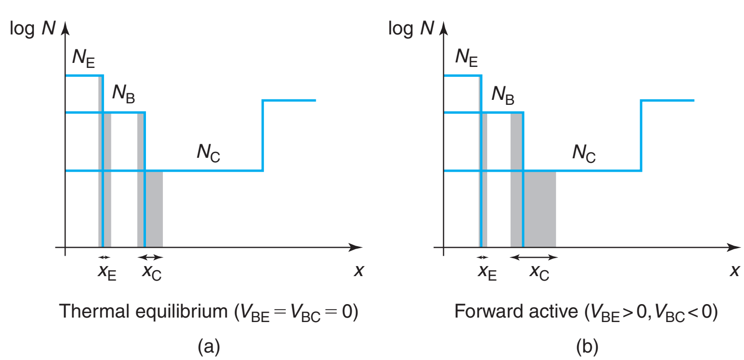


FIGURE 11.31
(a) Sketch depicting impact of V_{BE} and V_{BC} (b) in quasi-neutral regions of BJT in forward-active regime.

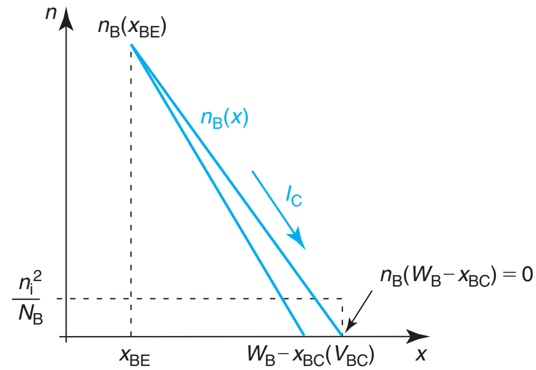


FIGURE 11.32
Sketch of electron concentration across quasi-neutral base of BJT in FAR as affected by application of base–collector voltage (the Early effect).

the slope in the electron concentration across the base increases and I_C goes up. Let us develop a first-order model for the dependence of I_C on V_{BC}.

In a well-designed bipolar transistor, base-width modulation effects usually represent a relatively small perturbation to the ideal characteristics of the device. For the Early effect, this means that the base width is only slightly perturbed by V_{BC}. As a result, a first-order model can be obtained by linearizing the dependence of W_B on V_{BC} in the following form:

$$(W_B - x_{BE} - x_{BC})|_{V_{BC}} \simeq W_{Bo} - \left.\frac{dx_{BC}}{dV_{BC}}\right|_{V_{BC}=0} V_{BC} \tag{11.71}$$

where

$$W_{Bo} = (W_B - x_{BE} - x_{BC})|_{V_{BE}=V_{BC}=0} \tag{11.72}$$

represents the quasi-neutral base-width in thermal equilibrium.

Note that $\frac{dx_{BC}}{dV_{BC}} < 0$ and the quasi-neutral base width shrinks as V_{BC} is made more negative. This derivative can easily be written in terms of the base–collector junction capacitance, since

$$C_{jco} = \left.\frac{dQ_{jC}}{dV_{BC}}\right|_{V_{BC}=0} = -A_C \left.\frac{d(qN_Bx_{BC})}{dV_{BC}}\right|_{V_{BC}=0} = -qA_CN_B \left.\frac{dx_{BC}}{dV_{BC}}\right|_{V_{BC}=0} \tag{11.73}$$

The units of C_{jco} are F.

Solving here for $\left.\frac{dx_{BC}}{dV_{BC}}\right|_{V_{BC}=0}$ and inserting this result into Eq. (11.71), we get

$$\begin{aligned}(W_B - x_{BE} - x_{BC})|_{V_{BC}} &\simeq W_{Bo}\left(1 + \frac{C_{jco}}{qA_CN_BW_{Bo}}V_{BC}\right) \\ &= W_{Bo}\left(1 + \frac{V_{BC}}{V_A}\right)\end{aligned} \tag{11.74}$$

V_A is called the *Early voltage* and it is the figure of merit that is used to characterize the Early effect. V_A is given by

$$V_A = \frac{qA_C N_B W_{Bo}}{C_{jco}} \tag{11.75}$$

It is clear from Eq. (11.74) that the higher the Early voltage, the weaker the Early effect is (i.e., the less the quasi-neutral base width is affected by V_{BC}). Equation (11.75) suggests that increasing the Early voltage requires higher doping level in the base, a thicker base width, and a smaller base–collector capacitance per unit area. All these dependences make good physical sense. In modern BJTs, V_A takes values in the range 20–50 V, depending on the details of the device design. In spite of what Eq. (11.75) seems to suggest, V_A is independent of A_C. This is because of the linear dependence between C_{jco} and A_C in the denominator.

There are several important consequences of the Early effect to the operation of the bipolar transistor that need to be understood. Let us first consider its impact on the DC characteristics. As mentioned above, as $|V_{BC}|$ increases, W_B decreases and I_C increases. A first-order model for this can be obtained by inserting Eq. (11.74) into Eq. (11.6),

$$\begin{aligned} I_C(V_{BC}) &= qA_E \frac{n_i^2}{N_B} \frac{D_B}{W_B - x_{BE} - x_{BC}} \exp \frac{qV_{BE}}{kT} \\ &\simeq qA_E \frac{n_i^2}{N_B} \frac{D_B}{W_{Bo}\left(1 + \frac{V_{BC}}{V_A}\right)} \exp \frac{qV_{BE}}{kT} \\ &= \frac{I_C(V_{BC} = 0)}{1 + \dfrac{V_{BC}}{V_A}} \end{aligned} \tag{11.76}$$

This equation exhibits the expected dependence of I_C on V_{BC}.

A common approximation to this equation relies on the fact that for a well-designed BJT, $V_A \gg V_{BC}$. In this case, one can approximate Eq. (11.76) by

$$I_C(V_{BC}) \simeq I_C(V_{BC} = 0)\left(1 - \frac{V_{BC}}{V_A}\right) \tag{11.77}$$

This is the preferred form used to capture the Early effect in CAD tools such as SPICE (Eq. (11.76) can lead to convergence problems if while searching for a solution, the simulator explores values of V_{BC} that are close to V_A).

The exponential dependence of I_C on V_{BE} is not affected by the Early effect. The Early effect only impacts the pre-exponential factor, which can therefore be written as

$$I_S = \frac{I_{So}}{1 + \frac{V_{BC}}{V_A}} \simeq I_{So}\left(1 - \frac{V_{BC}}{V_A}\right) \tag{11.78}$$

where I_{So} represents the expression of the saturation current of the ideal BJT derived earlier in this chapter.

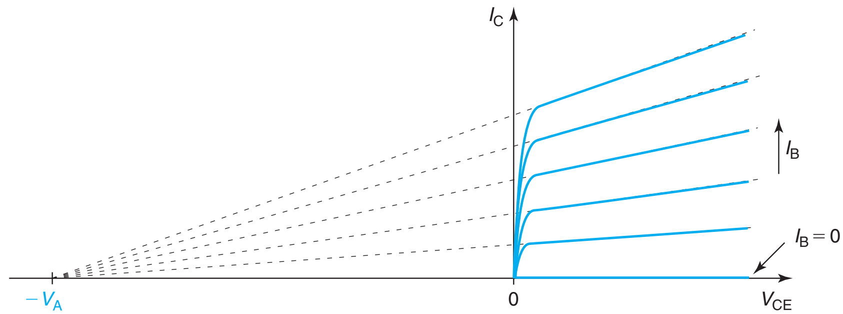


FIGURE 11.33
Sketch of common-emitter output characteristics of BJT in the presence of the Early effect.

As discussed above, the Early effect does not affect I_B, since I_B is set by the emitter physics. A consequence of this is that β_F inherits from I_C a dependence on V_{BC} of the following form:

$$\beta_F \simeq \frac{\beta_{Fo}}{1 + \dfrac{V_{BC}}{V_A}} \simeq \beta_{Fo}\left(1 - \frac{V_{BC}}{V_A}\right) \tag{11.79}$$

where β_{Fo} is the ideal value of β_F derived earlier in this chapter.

The clearest manifestation of the Early effect is in the output characteristics of the transistor. In an ideal BJT in FAR, I_C is independent of V_{CE} and the common-emitter output characteristics are flat. This is sketched in Fig. 11.17. The Early effect introduces a finite slope on the output characteristics in FAR. As Eq. (11.77) shows, the higher the value of $I_C(V_{BC} = 0)$, the higher the slope. The resulting common-emitter output characteristics look like those of Fig. 11.33.

An interesting feature of the Early effect is that if one extrapolates the I_C characteristics in FAR toward negative values of V_{CE}, one finds that they all cross the V_{CE} axis at around $-V_A$. Mathematically, this can be seen from Eq. (11.77), which can be rewritten as

$$I_C(V_{BC}) \simeq I_C(V_{BC} = 0)\left(1 + \frac{V_{CE} - V_{BEon}}{V_A}\right) \simeq I_C(V_{BC} = 0)\left(1 + \frac{V_{CE}}{V_A}\right) \tag{11.80}$$

since, in general, $V_{CE} \gg V_{BEon}$. This extrapolation technique provides for a rough way to determine the Early voltage from the output characteristics of a device.

The Early effect also impacts the AC behavior of the BJT in several ways. First of all, the fact that I_C now depends on V_{BC} means that there is a small-signal component in the collector current, i_c, in response to a small-signal base–collector voltage, v_{bc}. Following a similar procedure to that followed in Section 11.4.1 and using Eq. (11.77), we can write

$$I_C + i_c = I_C(V_{BC} = 0)\left(1 - \frac{V_{BC} + v_{bc}}{V_A}\right) = I_C(V_{BC} = 0)\left(1 - \frac{V_{BC}}{V_A}\right) - I_C(V_{BC} = 0)\frac{v_{bc}}{V_A} \tag{11.81}$$

Hence,

$$i_c \simeq -\frac{I_C}{V_A} v_{bc} = -g_o v_{bc} \tag{11.82}$$

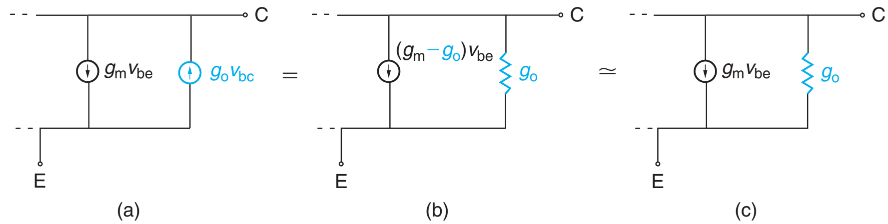


FIGURE 11.34
Modification to small-signal equivalent circuit model of BJT as a result of Early effect. Only the portion of the circuit affected by the Early effect is shown. (a): A new voltage-controlled current source is introduced between the collector and the emitter terminals. (b): Equivalent representation of new element. (c): Approximate representation of new element.

This new small-signal dependence can be captured in the small-signal equivalent circuit model by adding a new element. Since we are dealing with a component of the collector current, the new element is connected to the collector and emitter terminals. Equation (11.82) shows that this new component of i_c depends on v_{bc}. Hence, the element is a voltage-controlled current source, as sketched in Fig. 11.34(a).

This equivalent-circuit model representation of the Early effect can be simplified if we note that Eq. (11.82) can be written as

$$i_c = -g_o v_{be} + g_o v_{ce} \tag{11.83}$$

This suggests that an alternative representation is a voltage-controlled current source driven by v_{be} in parallel with a conductance of value g_o. This new current source can be integrated with the existing one on $g_m v_{be}$. This is represented in Fig. 11.34(b).

In a well-designed BJT, as we see below, $g_m \gg g_o$. As a result, it does not represent a significant error to neglect the new current source with value $g_o v_{be}$ when compared to the existing one with value $g_m v_{be}$. This results in the equivalent circuit model representation shown in Fig. 11.34(c), which is the most common one. In this representation, the dominant impact of the Early effect is the introduction of an *output conductance* of a value

$$g_o = \frac{I_C}{V_A} \tag{11.84}$$

It is easy to see that in general we should expect $g_m \gg g_o$. Looking at the definition of g_m in Eq. (11.46), it is clear that $g_m/g_o = qV_A/kT$, which easily exceeds a factor of 1000.

The second impact of the Early effect in the small-signal behavior of the transistor is in the base transit time. As we saw in Eq. (11.38), this is proportional to the square of the quasi-neutral base width. As $|V_{BC}|$ increases, W_B decreases and τ_{tB} decreases too. This improves the high-frequency performance of the BJT. To the first order, the dependence of τ_{tB} can be obtained by plugging Eq. (11.74) into Eq. (11.38) to get

$$\tau_{tB} = \tau_{tBo}\left(1 + \frac{V_{BC}}{V_A}\right)^2 \tag{11.85}$$

where τ_{tBo} is the base transit time of the ideal BJT.

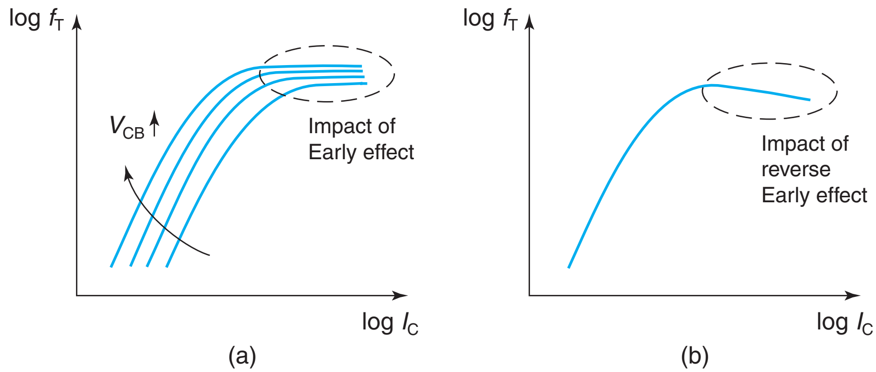


FIGURE 11.35
Manifestation of the Early effect (a) and reverse Early effect (b) in the I_C dependence of f_T for a BJT in FAR.

The Early effect manifests itself in the f_T-I_C characteristics by an improvement of f_T with increasing V_{CB} *at high current levels*. This is sketched in Fig. 11.35(a).

The reverse Early effect

Figure 11.36 shows the impact of the electron concentration across the quasi-neutral base of a BJT in FAR in the presence of the reverse Early effect. The electron concentration at the edge of the emitter-base space-charge region is set by V_{BE}. We have already discussed this. What we have not accounted for is the fact that the location of the edge of the SCR is also affected by V_{BE}. The consequence of this is that the extent of the quasi-neutral base width is also affected by V_{BE}.

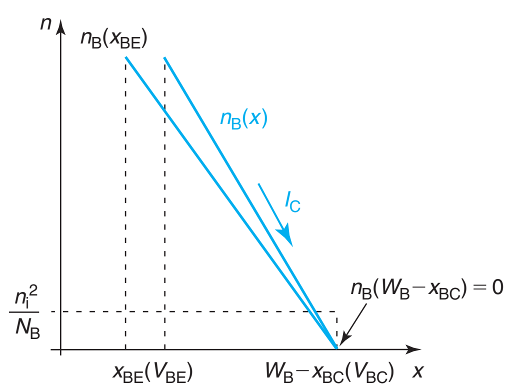


FIGURE 11.36
Sketch of electron concentration across quasi-neutral base of BJT in FAR as affected by base-width modulation due to application of base–emitter voltage (the reverse Early effect).

In particular, since in FAR $V_{BE} > 0$, the quasi-neutral base width W_B increases as V_{BE} increases. This implies that the ideal exponential dependence of I_C on V_{BE} is bound to be upset in the sense that I_C will not grow as fast as predicted by the ideal BJT theory. Developing a simple model for this is not difficult.

First, we derive a relationship between W_B and V_{BE}. If we proceed in a dual way to the approach followed for the Early effect above, we can easily find that

$$(W_B - x_{BE} - x_{BC})|_{V_{BE}} \simeq W_{Bo}\left(1 + \frac{V_{BE}}{V_B}\right) \tag{11.86}$$

where V_B is the so-called reverse Early voltage,

$$V_B = \frac{qA_E N_B W_{Bo}}{C_{jeo}} \tag{11.87}$$

The reverse Early voltage characterizes the reverse Early effect. For modern BJT technology, V_B is of the order 5–10 V. Increasing the reverse Early voltage requires an increase in the doping level or thickness of the base, or a decrease in the emitter-base junction capacitance per unit area.

The reverse Early effect impacts the ideal BJT characteristics in a variety of ways. In DC, the collector current is affected as already described. A simple model can be derived following an analogous approach to that of the Early effect,

$$\begin{aligned} I_C(V_{BE}) &= qA_E \frac{n_i^2}{N_B} \frac{D_B}{W_{Bo}\left(1 + \frac{V_{BE}}{V_B}\right)} \exp \frac{qV_{BE}}{kT} \\ &= \frac{I_{Ci}}{1 + \frac{V_{BE}}{V_B}} \\ &\simeq I_{Ci}\left(1 - \frac{V_{BE}}{V_B}\right) \end{aligned} \tag{11.88}$$

where I_{Ci} is the collector current in the ideal BJT. The last approximation in deriving Eq. (11.88) is warranted since typically $V_B \gg V_{BE}$. It is clear in this formulation that as V_{BE} increases, I_C deviates more and more from its ideal value. Similarly to the Early effect, we can capture the reverse Early effect as a perturbation of the ideal value of the collector saturation current I_{So},

$$I_S = \frac{I_{So}}{1 + \frac{V_{BE}}{V_B}} \simeq I_{So}\left(1 - \frac{V_{BE}}{V_B}\right) \tag{11.89}$$

The reverse Early effect does not affect the base current of the BJT in a significant way. This is because the emitter doping level is typically much higher than that of the base and in consequence, emitter-width modulation is much less significant than base-width modulation.

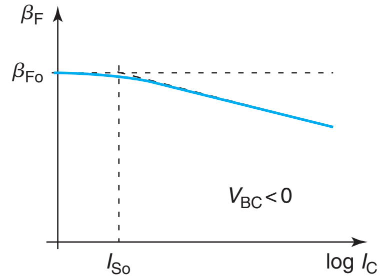


FIGURE 11.37
Sketch of β_F versus collector current in a semilog plot for a BJT in FAR. The reverse Early effect manifests itself as a linear decrease of β_F with $\log I_C$.

As a consequence, the reverse Early effect impacts the current gain of the transistor in a direct way,

$$\beta_F = \frac{\beta_{Fo}}{1 + \frac{V_{BE}}{V_B}} \simeq \beta_{Fo}\left(1 - \frac{V_{BE}}{V_B}\right) \tag{11.90}$$

This equation suggests that the Early effect brings down the current gain of the BJT for high values of I_C. This is best seen in a plot of β_F versus $\log I_C$, as sketched in Fig. 11.37.

Capturing the reverse Early effect in the small-signal equivalent circuit model does not require any changes to its topology. However, the AC characteristics of the ideal BJT are affected in two ways. First, since I_C does not rise as fast with V_{BE} as in the ideal model, the transconductance of the BJT will be smaller than that of the ideal BJT. Taking derivatives with V_{BE} in Eq. (11.88), we can easily see that

$$g_m \simeq \frac{qI_C}{kT}\left(1 - \frac{kT}{qV_B}\right) \tag{11.91}$$

The second impact of the reverse Early effect is in the base transit time. As in the Early effect, this is a natural consequence of base-width modulation. To first order, its impact is captured by an equation similar to Eq. (11.85),

$$\tau_{tB} = \tau_{tBo}\left(1 + \frac{V_{BE}}{V_B}\right)^2 \tag{11.92}$$

The expansion of the quasi-neutral base width with V_{BE} due to the reverse Early effect tends to increase τ_{tB} above its ideal value. Coupled with the decrease in g_m, this hampers the high-frequency behavior of the ideal BJT at high values of I_C. This is sketched in Fig. 11.35(b).

The Early effect and the reverse Early effect take place independently and simultaneously in a BJT. In well-designed devices, they are both relatively small effects. As a result, the combined effect of base-width modulation can be accounted for by adding their effects in a linear way, as follows:

$$I_S = \frac{I_{So}}{1 + \frac{V_{BC}}{V_A} + \frac{V_{BE}}{V_B}} \simeq I_{So}\left(1 - \frac{V_{BC}}{V_A} - \frac{V_{BE}}{V_B}\right) \tag{11.93}$$

$$\beta_F = \frac{\beta_{Fo}}{1 + \frac{V_{BC}}{V_A} + \frac{V_{BE}}{V_B}} \simeq \beta_{Fo}\left(1 - \frac{V_{BC}}{V_A} - \frac{V_{BE}}{V_B}\right) \tag{11.94}$$

$$\tau_{tB} = \tau_{tBo}\left(1 + \frac{V_{BC}}{V_A} + \frac{V_{BE}}{V_B}\right)^2 \tag{11.95}$$

It is of educational value to attempt to write the full current equations of the BJT in all four regimes in the presence of base-width modulation. This is left as an exercise to the reader.

11.5.2 Emitter-base space-charge region recombination

An important perturbation to the ideal behavior of a BJT in FAR arises from space-charge region recombination in the emitter-base junction. This is a phenomenon that we have already discussed in the context of the PN junction in Section 6.6.2. As a reminder, in a PN junction under forward bias, a certain amount of recombination takes place in the SCR. Conversely, under reverse bias, a fairly significant amount of generation can also take place inside the depletion region. Based on this understanding, it is therefore reasonable to expect that in a BJT in FAR, SCR recombination takes place in the base–emitter depletion region, while SCR generation occurs in the base–collector junction. The question is: are these mechanisms significant? And if so, how do they manifest themselves in the characteristics of the transistor?

SCR generation in the base–collector junction is not expected to be significant in a well-behaved transistor biased in FAR. This is because, as seen in Section 6.6.2, SCR generation would add a current component to I_B and I_C. For typical bias points, I_B and I_C are rather large and swamp any nonideal generation contributions.

SCR recombination in the emitter-base junction of a BJT in FAR can become noticeable if there is a significant concentration of traps inside the junction. As in the case of an isolated PN junction, the SCR recombination current represents a new current path that must be added to the ideal current flowing through the junction. In a BJT in FAR, emitter-base SCR recombination involves electrons coming from the emitter and holes coming from the base. In consequence, this new nonideal current component flows from the base to the emitter adding a contribution to the base current. As in the PN junction, we expect this current to exhibit an exponential behavior on the voltage across the junction with an ideality factor higher than unity. If we denote the new current component as $I_{B,SCR}$, we then have

$$I_{B,SCR} \simeq I_{Bo,SCR}\left(\exp\frac{qV_{BE}}{nkT} - 1\right) \tag{11.96}$$

where $I_{Bo,SCR}$ exhibits the same dependences as in the case of the isolated PN junction.

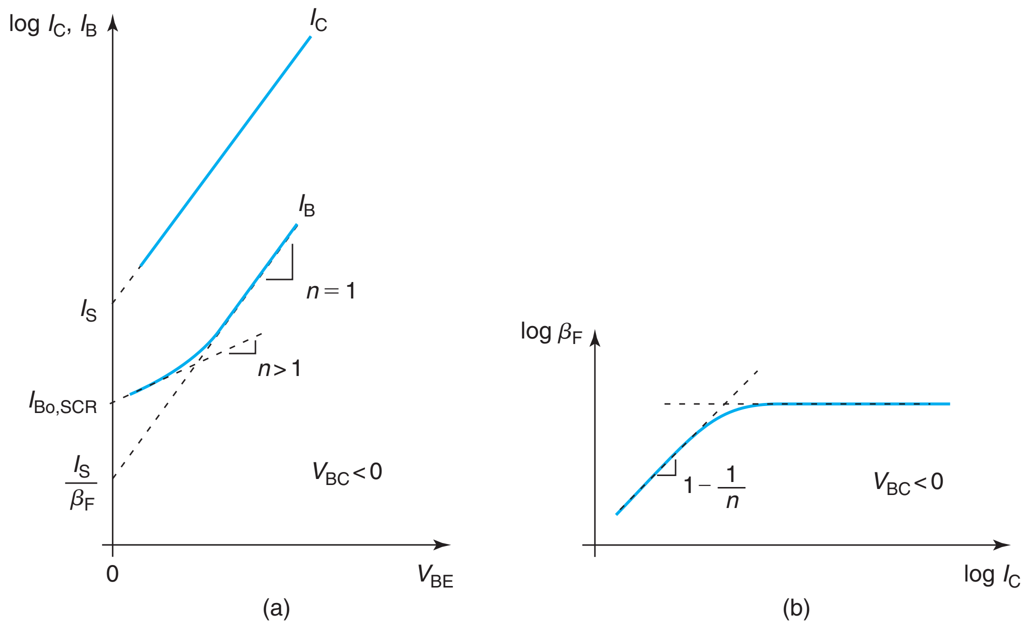


FIGURE 11.38
Impact of space-charge recombination in the base–emitter junction on the I–V characteristics of a BJT in FAR. (a): Gummel plot indicating new base current component. (b): Current gain versus collector current.

If SCR recombination in the base–emitter depletion region is significant in a BJT, it readily manifests itself in its Gummel plot. The presence of SCR recombination in the E–B junction adds a new current component to the ideal base current of the BJT but it does not modify the collector current. Hence, the line that depicts I_B in the ideal Gummel plot bends for low values of V_{BE} for which SCR recombination becomes comparatively significant. This is sketched in Fig. 11.38(a).

With I_B increasing over its ideal value but I_C unchanged, an important consequence of base–emitter junction SCR recombination is a drop of current gain for low values of I_C. This is sketched in Fig. 11.38(b). It is not difficult to show that in this log–log plot of β_F versus I_C, for low values of I_C, the dependence of $\log \beta_F$ on $\log I_C$ is linear with a slope of $1 - \frac{1}{n}$.

From an AC point of view, SCR recombination adds a component of conductance in parallel with g_π in the small-signal equivalent circuit model. This affects very little f_T and other high-frequency figures of merit that for a low value of I_C are dominated by the charging and discharging of the junction capacitances.

Emitter-base junction SCR recombination is very sensitive to contamination or improper processing of a BJT. In well-designed and fabricated BJTs, defect concentration is very small and this current component is negligible. This may not be true in new device structures or for a new process under development. Emitter-base SCR recombination is a good gauge of bipolar process cleanliness and maturity.

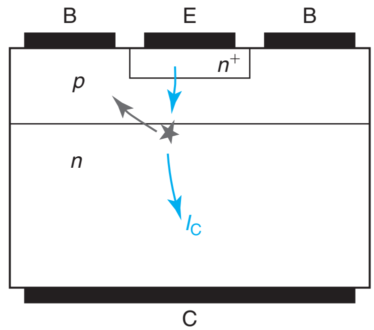


FIGURE 11.39
Sketch of impact ionization in a BJT biased in the forward-active regime with large V_{CB}.

11.5.3 Impact ionization

In a BJT in the FAR, as the reverse bias across the base–collector junction increases, its SCR widens and the electric field inside increases too. With a large value of V_{CB}, it becomes likely for impact ionization to take place inside the SCR. In this instance, the electrons that are generated in the process add to the collector current and the holes are swept toward the base where they become majority carriers. To maintain charge neutrality, once the holes reach the quasi-neutral base, an equal amount of holes escape through the base contact contributing *negative* base current. The process is illustrated in Fig. 11.39.

Impact ionization in a BJT in FAR adds new components to the base and the collector currents. This can be captured in an equivalent circuit model through a current generator that is placed between the collector and the base terminals, as shown in Fig. 11.40. The magnitude of this current source is $(M-1)I_{Ci}$, where I_{Ci} represents the collector current of the ideal BJT and M is the multiplication coefficient due to impact ionization. In this way, the total collector terminal current becomes

$$I_C = MI_{Ci} = MI_S \exp\frac{qV_{BE}}{kT} \tag{11.97}$$

The base current consists of the ideal hole current injected into the emitter *minus* the impact ionization current

$$I_B = \frac{I_S}{\beta_F}\left(\exp\frac{qV_{BE}}{kT} - 1\right) - (M-1)I_S \exp\frac{qV_{BE}}{kT} \tag{11.98}$$

For negligible impact ionization, $M = 1$ and the situation reverts to that of the ideal BJT.

Due to the fact that the base current in an ideal BJT is β_F smaller than the collector current, impact ionization becomes visible first in I_B and only for much higher values of M it eventually upsets I_C. This can be seen in Fig. 11.41. This graph sketches I_B and I_C for a BJT biased in FAR as a function of V_{CB} in a semilogarithmic scale. Also graphed here is the impact ionization current $(M-1)I_{Ci}$ with two values of M indicated at two key points along this line.

For low V_{CB}, both I_B and I_C take their ideal values. As V_{CB} increases, impact ionization events start taking place and impact ionization current starts flowing. As this current becomes comparable to the ideal value of I_B, the total base current I_B is initially reduced, then changes

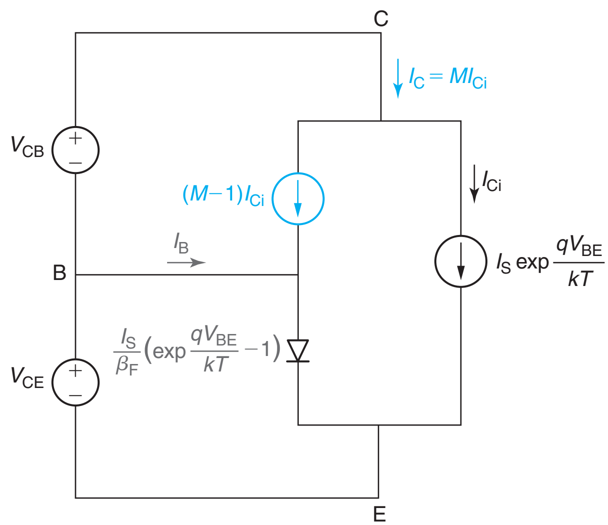


FIGURE 11.40
Equivalent circuit model for BJT biased in the forward-active regime with a current generator in the base–collector junction to capture impact ionization.

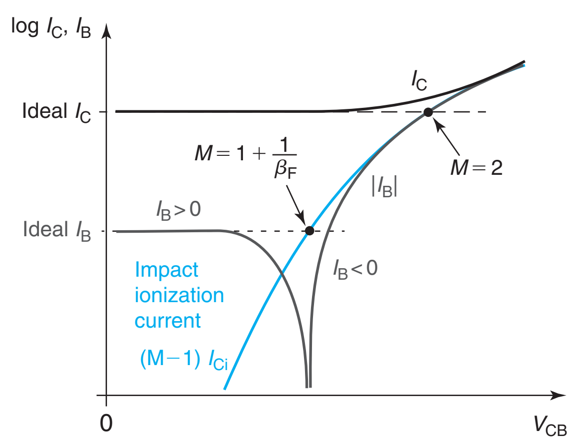


FIGURE 11.41
Semilogarithmic plot of I_C and I_B as a function of V_{CB} for a BJT biased in the forward-active regime. Also indicated is the impact ionization current in the BC-SCR that increases as V_{CB} increases. As the multiplication current increases, first I_B is reduced, then changes sign, and eventually increases again as the hole impact ionization current dominates. At a higher value of V_{CB}, I_C also gets to increase due to the impact-ionized electrons.

sign, and eventually increases along with the multiplication current. I_B goes to zero at a value of V_{CB} such that $M = 1 + \frac{1}{\beta_F}$.

I_C is only visibly affected at much higher values of V_{CB}. When $M = 2$, the multiplication current is identical to the ideal value of I_C and I_C has then doubled in magnitude. Further increases in V_{CB} produce noticeable increases in I_C.

Impact ionization is important because it can affect device operation in circuits. In advanced technologies with aggressive doping distributions and geometries, impact ionization can become quite high even under regular operating conditions and circuit models must account for it to be able to accurately predict device behavior. In addition, impact ionization leads to extra electrical noise that can be a problem in analog applications. Also, if unchecked, impact ionization might result in device breakdown, as discussed in the next section.

11.5.4 Breakdown voltage

In a bipolar transistor in the FAR, there is a limit to the maximum voltage that can be applied to the collector. This is called the *breakdown voltage*. Applying a voltage in excess of the breakdown voltage causes large currents to flow through the transistor, leading to degradation or even device destruction.

Figure 11.42 sketches the common-emitter output characteristics of a bipolar transistor in an expanded voltage scale. In an ideal BJT in FAR, the collector current is independent of V_{CE}. In real bipolar transistors, however, at a certain value of V_{CE}, I_C exhibits a sudden rise. This is referred to as breakdown, and the voltage at which this happens is called the *breakdown voltage*. The breakdown voltage is a key figure of merit of a transistor technology since it places an upper limit to the voltage-handling ability of the device.

There are two different physical phenomena that can lead to breakdown in BJTs. One is impact ionization and subsequent avalanche in the reverse-biased base–collector junction. The other is base punchthrough. The next two sections discuss the physics behind these two

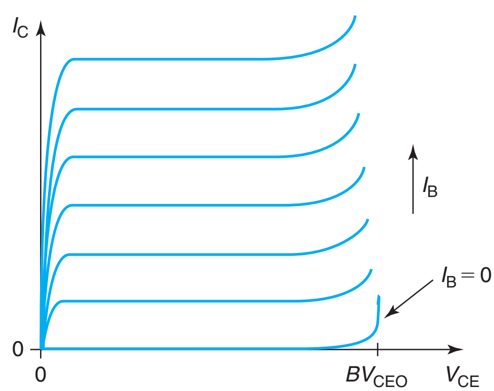


FIGURE 11.42
Common-emitter output characteristics of a bipolar transistor indicating the breakdown voltage. As V_{CE} approaches BV_{CEO}, the collector current increases without bounds.

breakdown modes as well as device design issues with relevance to the breakdown voltage. The third section that follows discusses how to identify which one of these two phenomena prevail in a given device.

Avalanche breakdown

In Section 11.5.3, we studied impact ionization in a BJT biased in the FAR. We learned that for large values of V_{CB}, electron–hole pair generation might take place inside the BC-SCR contributing new currents to the transistor. In Section **??**, we discussed current multiplication in detail and showed how this process can go "critical" and lead to avalanche and breakdown. This process, for example, limits the maximum reverse bias that can be applied to most PN diodes, as seen in Section 6.6.4.

In a completely analogous way, avalanche breakdown can take place in the base–collector junction of a BJT. A peculiarity of the BJT in FAR is that the voltage at which avalanche breakdown takes place depends on the way the transistor is connected. This is sketched in Fig. 11.43. In a common-emitter configuration, that is, with the base current fixed, the breakdown voltage is seen to be smaller than in the common-base configuration in which the emitter current is fixed. In Fig. 11.43, this is shown for $I_B = 0$ and $I_E = 0$, respectively.[2] However, the actual value of these currents matters less than the fact that their values are fixed.

Figure 11.43 also shows the common notation used to refer to these breakdown voltages. Typically, the breakdown voltage is given three subindices. The two that bear terminal letters indicate the two terminals across which the voltage is being measured. The third letter is associated with the third terminal and denotes its status. An "O" indicates that the third terminal is open. Hence, BV_{CEO} refers to the breakdown voltage between the collector and the emitter terminals with the base open and BV_{CBO} refers to the breakdown voltage between the collector

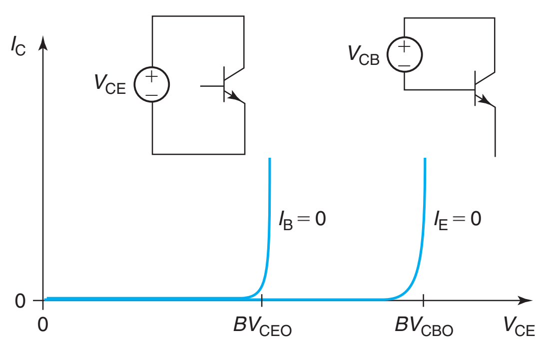


FIGURE 11.43
I_C versus V_{CE} for a BJT in the common-emitter configuration with $I_B = 0$ and in the common-base configuration with $I_E = 0$. The breakdown voltage under these two conditions is different and it is referred to as BV_{CEO} and BV_{CBO}, respectively.

[2]Under any of these conditions, the transistor is in cut-off. As suggested in Fig. 11.42, however, the breakdown voltage is mostly unchanged if the transistor is turned on by injecting a finite amount of I_B or I_E.

and the base terminals with the emitter open. As Fig. 11.43 suggests, it is a common observation in bipolar transistors that

$$BV_{CEO} < BV_{CBO} \tag{11.99}$$

Understanding the reason for this behavior and identifying the key device design criteria for breakdown are the goals of this section.

Insight into the physics of breakdown voltage in BJTs can be obtained from the sketches shown in Fig. 11.44 that depict a transistor in cut-off biased close to breakdown. In both common-emitter and common-base configurations, there is a trickle current that flows through the reverse-biased base–collector junction due to carrier generation in the quasi-neutral base and collector regions and in the base–collector space-charge region itself. With enough reverse voltage across this junction, as carriers flow through the SCR there is a chance that they produce impact ionization events in which more electrons and holes are generated.

In the common-base configuration, the electrons generated in the SCR are swept into the collector and contribute to the collector terminal current. Similarly, the generated holes are swept into the base and contribute to the base terminal current. These two currents are identical in magnitude since for every electron that it is generated, a hole is also generated. The situation is just like in an isolated PN junction, which we already studied in Chapter 6. We know that in these conditions, avalanche breakdown occurs when the multiplication coefficient, M, goes to infinity.

In the common-emitter configuration, the situation is rather different when it comes to the behavior of the holes. As holes are extracted by the base out of the base–collector SCR, they cannot escape through the base terminal because it is open. Hence, in order to preserve zero base current, they have to get injected into the emitter. This can easily be accommodated if the base–emitter junction becomes slightly forward biased. However, when this happens, not

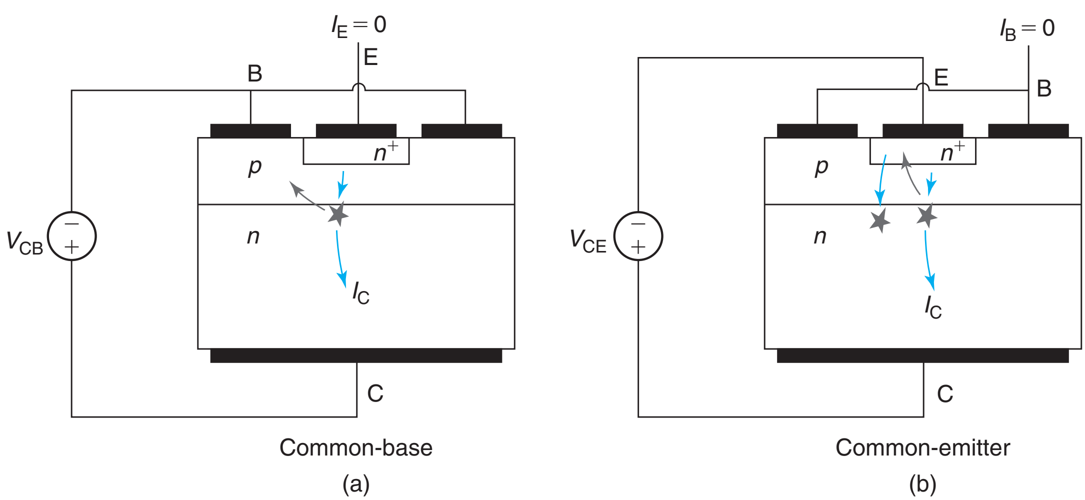


FIGURE 11.44
Sketch of major current paths at avalanche breakdown in common-base mode (a) and common-emitter mode (b).

only is there hole injection from the base into the emitter, as needed, but also electron injection from the emitter into the base. These injected electrons diffuse through the base, enter the base–collector SCR, and produce more impact ionization events in which more electrons and holes are generated.

It is clear that in the common-emitter configuration there is a positive feedback loop in action that is external to the carrier multiplication process. Since a well-designed bipolar transistor has high current gain, for every hole that is injected from the base into the emitter, there are many electrons that are injected from the emitter into the base. Hence, the external feedback mechanism that is in action in the common-emitter configuration fuels the overall carrier multiplication in a very effective way. The collector terminal current can therefore grow very quickly even with a moderate multiplication factor in the base–collector SCR. That is why the breakdown voltage in the common-emitter configuration can be substantially lower than in the common-base configuration.

It is not difficult to capture these arguments in an analytical formulation using the nonlinear hybrid-π model of the ideal BJT. Figure 11.45 shows an equivalent-circuit model description of an ideal BJT in the common-base configuration with $I_E = 0$. The base–collector junction is reverse biased and hence the terms in $\exp \frac{qV_{BC}}{kT}$ have been dropped. Added to the standard description of the ideal BJT is the current source that represents the impact ionization current, as discussed earlier.

To solve this circuit, we note that $I_E = 0$ implies that

$$\frac{I_S}{\beta_F}\left(\exp \frac{qV_{BE}}{kT} - 1\right) + I_S \exp \frac{qV_{BE}}{kT} = 0 \tag{11.100}$$

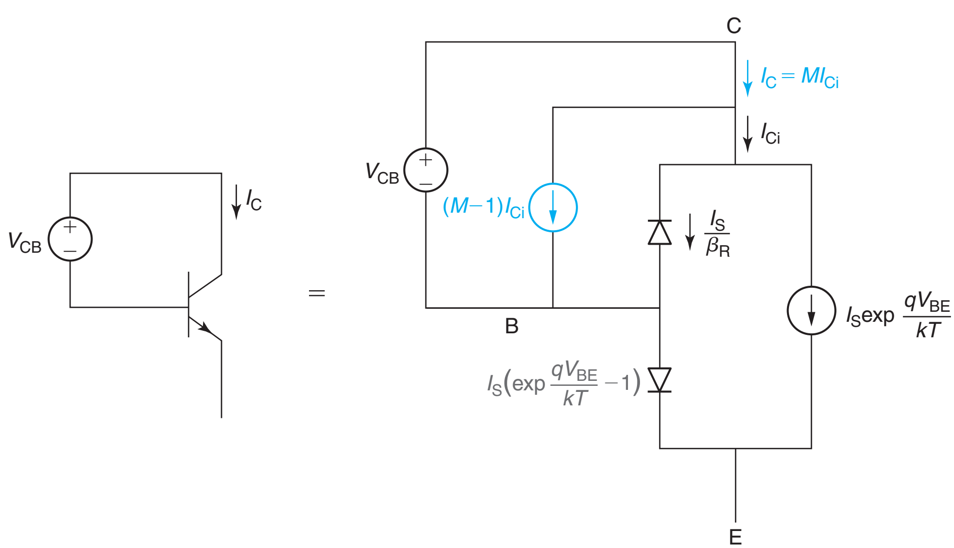


FIGURE 11.45
Equivalent-circuit model to compute BV_{CBO}.

Solving for V_{BE}, we find

$$V_{\mathrm{BE}} = -\frac{kT}{q}\ln(1+\beta_{\mathrm{F}}) \simeq -\frac{kT}{q}\ln\beta_{\mathrm{F}} \tag{11.101}$$

which suggests that the base–emitter junction is slightly reverse biased (about 100 mV or so).

The ideal collector current I_{Ci} can be written as

$$I_{\mathrm{Ci}} = I_{\mathrm{S}}\exp\frac{qV_{\mathrm{BE}}}{kT} + \frac{I_{\mathrm{S}}}{\beta_{\mathrm{R}}} \simeq \frac{I_{\mathrm{S}}}{\beta_{\mathrm{F}}} + \frac{I_{\mathrm{S}}}{\beta_{\mathrm{R}}} \simeq \frac{I_{\mathrm{S}}}{\beta_{\mathrm{R}}} \tag{11.102}$$

where we have inserted Eq. (11.101) and assumed that $\beta_{\mathrm{F}} \gg \beta_{\mathrm{R}}$.

I_{C} is then

$$I_{\mathrm{C}} \simeq M\frac{I_{\mathrm{S}}}{\beta_{\mathrm{R}}} \tag{11.103}$$

This suggests that breakdown occurs when $M \to \infty$. Hence,

$$BV_{\mathrm{CBO}} = V_{\mathrm{CB}}(M=\infty)|_{\mathrm{I_E=0}} \tag{11.104}$$

A similar algebraic exercise can be carried out for the common-emitter configuration. The equivalent circuit model is shown in Fig. 11.46. In this case, the $I_{\mathrm{B}} = 0$ condition implies that

$$(M-1)I_{\mathrm{Ci}} + \frac{I_{\mathrm{S}}}{\beta_{\mathrm{R}}} - \frac{I_{\mathrm{S}}}{\beta_{\mathrm{F}}}\left(\exp\frac{qV_{\mathrm{BE}}}{kT} - 1\right) = 0 \tag{11.105}$$

Additionally, the ideal collector current (before multiplication) is given by

$$I_{\mathrm{Ci}} = \frac{I_{\mathrm{S}}}{\beta_{\mathrm{R}}} + I_{\mathrm{S}}\exp\frac{qV_{\mathrm{BE}}}{kT} \tag{11.106}$$

This is a system of two equations with two unknowns for which we can solve for I_{Ci}. After some straightforward algebra, one finds that

$$I_{\mathrm{C}} \simeq M\frac{I_{\mathrm{S}}}{\beta_{\mathrm{R}}}\,\frac{\beta_{\mathrm{F}}}{1-\beta_{\mathrm{F}}(M-1)} \tag{11.107}$$

The presence of a minus sign in the denominator of this equation suggests that in this configuration, I_{C} grows boundless when $M = 1 + \frac{1}{\beta_{\mathrm{F}}}$. Hence,

$$BV_{\mathrm{CEO}} = V_{\mathrm{CE}}\left(M = 1 + \frac{1}{\beta_{\mathrm{F}}}\right)\Bigg|_{\mathrm{I_B=0}} \tag{11.108}$$

Since typically $\beta_{\mathrm{F}} \gg 1$, it only takes a relatively small amount of carrier multiplication in the base–collector SCR to drive the BJT into breakdown in the common-emitter configuration. When we compare this with the common-base configuration where we saw that M needs to go to ∞ for the breakdown condition to be reached, it is then clear that $BV_{\mathrm{CEO}} \ll BV_{\mathrm{CBO}}$.

This discussion reveals two key device design criteria that impact the avalanche breakdown voltage of BJTs. First is the collector design and most importantly, its doping level. The higher the doping level, the higher the electric field and the multiplication coefficient inside the

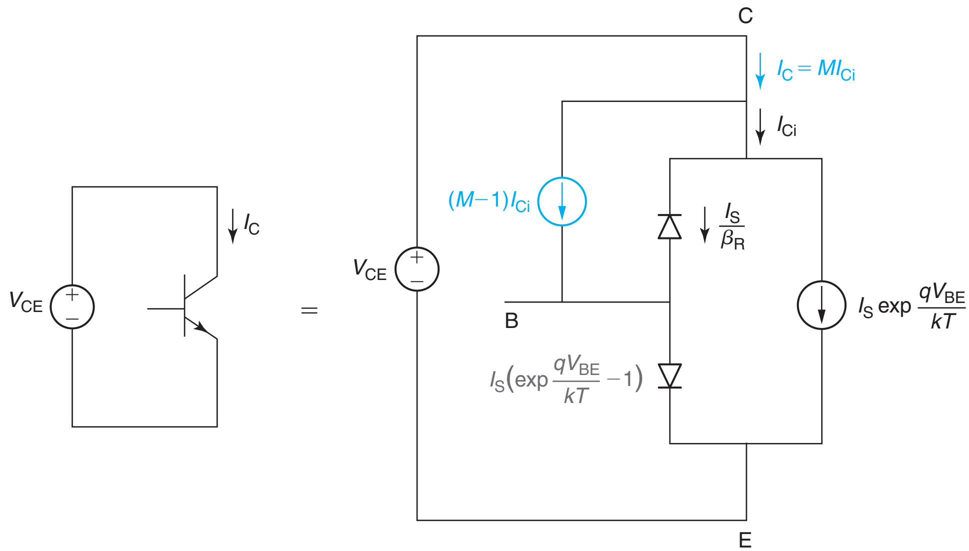


FIGURE 11.46
Equivalent-circuit model to compute BV_{CEO}.

base–collector SCR are for a given reverse voltage across. Hence, the lower the breakdown voltage becomes. The thickness of the collector also plays a role if it is so thin that the depletion region of the base–collector junction bumps into the buried collector before breakdown is reached. When this happens, the dependence of the peak field inside the SCR on V_{CB} increases and the breakdown condition is reached at a lower voltage than if the depletion region is allowed to fully expand.

The second device design issue of relevance to breakdown is the current gain of the transistor, which, as studied above, is set by the base and emitter design. All things being equal, the higher the current gain, the lower BV_{CEO} becomes (although BV_{CBO} remains unchanged). This of course makes good sense given our discussion so far. The higher the β_F is, the smaller the multiplication coefficient at the base–collector junction needs to be for I_C to go "critical" since the multiplicative factor contributed by the outside loop becomes more significant. An important consequence of this is that bipolar device designers engineer their transistors targeting a specific value of β_F as required by the circuit applications and do not attempt to maximize it. More β_F than needed buys little at the circuit level and detracts from the voltage-handling ability of the technology, which would jeopardize its use in many applications.

This discussion inevitably prompts the question: which breakdown voltage matters most, BV_{CEO} or BV_{CBO}? Actually, this is a circuit-level question. In most bipolar circuits, one does not typically find transistors in which either I_B or I_E are strictly fixed. In circuit jargon, the impedance of the base or the emitter node is rarely infinite. Hence, for most bipolar transistors in a circuit environment, an intermediate value of breakdown voltage between BV_{CEO} and BV_{CBO} most likely applies. Given the fact that when a bipolar technology is developed, it is intended to be used in a wide range of circuit applications, device designers have to design for the worst-case condition. This occurs when the base impedance is infinite. Hence, a bipolar technology is typically designed so that BV_{CEO} exceeds the supply voltage of the circuits that are going to be implemented. BV_{CEO} is then widely used as a key figure of merit that characterizes the voltage-handling ability of a bipolar technology. Together with f_T, BV_{CEO} is a figure of merit that is often used to *specify* a bipolar technology.

Punchthrough breakdown

Base punchthrough can also result in bipolar transistor breakdown. Base punchthrough can be thought of as an extreme case of base-width modulation. As the reverse bias across the base–collector junction increases, the depletion region associated with this junction widens too, both on the collector side as well as on the base side. The higher the V_{CB} is, the thinner the quasi-neutral base becomes. The modulation of the quasi-neutral base width results in an increase in the collector current. We denoted this as the Early effect that we studied in Section 11.5.1.

It is easy to see that for a large enough value of V_{CB}, it is possible for the quasi-neutral base to completely disappear. When this happens, a large uncontrolled rise in I_C will occur leading to a breakdown condition. The energy band diagram of Fig. 11.47 makes it clear how this happens. As the base–collector SCR widens, the quasi-neutral base width is reduced and eventually is completely wiped out. At this instance, the height of the energy barrier limiting electron injection from the emitter to the base is suddenly reduced and a large number of electrons "spill" into the collector. At punchthrough, the base loses its ability to control the collector current that increases very quickly with V_{CB}.

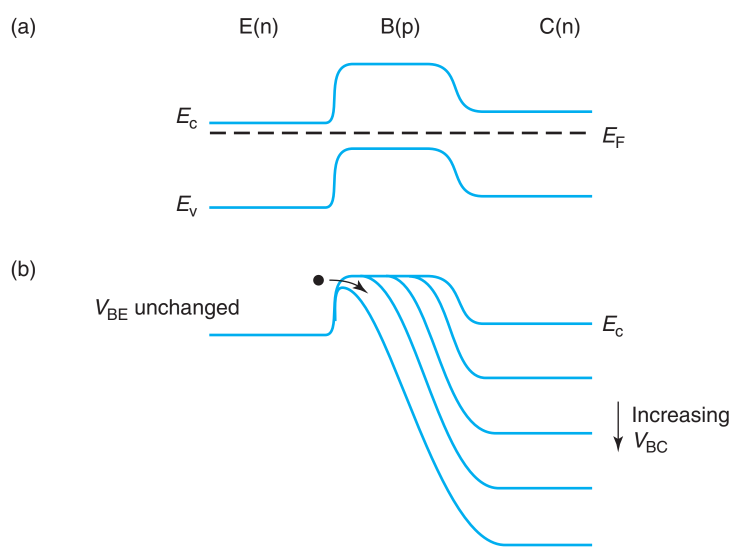


FIGURE 11.47
Energy band diagram of bipolar transistor in equilibrium (a), and with increasing values of V_{CB}, leading to base punchthrough (b).

It is fairly straightforward to compute the base–collector voltage that leads to base punchthrough. From Chapter 6, we know that the extent of the base–collector SCR into the base is given by

$$x_{BC} = \sqrt{\frac{2\epsilon N_C(\phi_{biC} - V_{BC})}{qN_B(N_B + N_C)}} \simeq \frac{1}{N_B}\sqrt{\frac{2\epsilon N_C V_{CB}}{q}} \tag{11.109}$$

where for simplicity, we have assumed that $N_C \gg N_B$ and that $V_{CB} \gg \phi_{biC}$.

At punchthrough, $x_{BC} \simeq W_B$. We can solve in Eq. (11.109) for the value of V_{CB} that leads to this condition,

$$V_{CBpunch} \simeq \frac{qW_B^2 N_B^2}{2\epsilon N_C} \tag{11.110}$$

This is the base–collector punchthrough voltage. How this result translates into the three-terminal breakdown voltages BV_{CBO} and BV_{CEO} requires some additional thinking. Figure 11.48 depicts the main current paths at punchthrough breakdown in these two configurations.

In the common-emitter configuration [Fig. 11.48(b)], BV_{CEO} should be the sum of $V_{CBpunch}$ plus the value of the base–emitter voltage. V_{BE} can be easily obtained from Eq. (11.105) by setting $M = 0$ and solving for V_{BE}. We find that

$$V_{BE} = \frac{kT}{q}\ln\left(1 + \frac{\beta_F}{\beta_R}\right) \tag{11.111}$$

This is actually rather small, of the order 0.1–0.2 V. Hence,

$$BV_{CEO} \simeq V_{CBpunch} \tag{11.112}$$

In the common-base configuration, the situation is more complicated. With the emitter terminal open, $I_E = 0$ and punchthrough, as described above, cannot take place since the emitter current has nowhere to go. What happens then when V_{CB} approaches $V_{CBpunch}$?

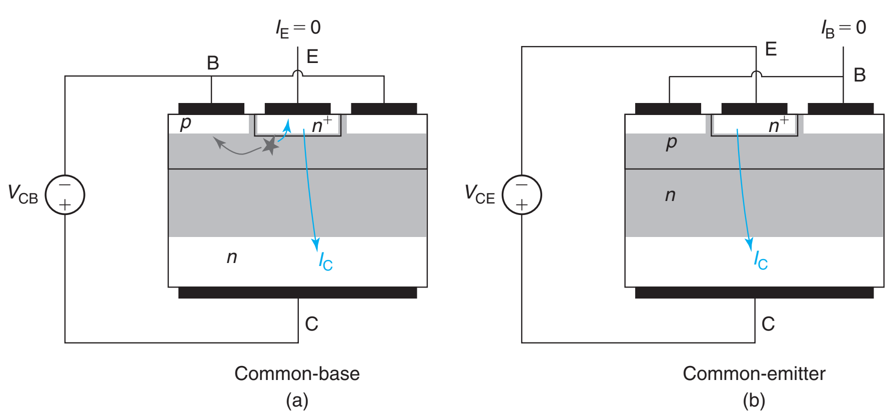


FIGURE 11.48
Sketch of major current paths at punchthrough breakdown in common-base mode (a) and common-emitter mode (b).

As discussed above, base punchthrough results in a reduction in the height of the emitter-base energy barrier that allows electron spillage into the collector. In the common-base configuration with $I_E = 0$, this cannot happen. The only possibility to prevent electron spillage is if a reverse bias appears across the base–emitter junction that restores back the energy barrier across it. Hence, in the common-base configuration, as V_{CB} approaches $V_{CBpunch}$, V_{BE} goes into reverse bias and I_E is kept at zero.

As V_{CB} gets closer to $V_{CBpunch}$, the reverse bias on the base–emitter junction can get so high that significant impact ionization starts taking place inside the base–emitter SCR! When this happens, the impact-ionized holes are extracted by the base and come out of the base terminal, and the generated electrons are extracted by the emitter. To keep $I_E = 0$, the same number of electrons are now allowed to spill into the collector becoming collector terminal current. Hence, there is now current flowing through the base and the collector terminals. These current paths are shown in Fig. 11.48(a). The closer V_{CB} gets to $V_{CBpunch}$, the closer the emitter-base junction is brought onto its own avalanche breakdown voltage. In the limit, then,

$$BV_{CBO} \simeq V_{CBpunch} + BV_{EBO} \tag{11.113}$$

where BV_{EBO} is the avalanche breakdown voltage of the isolated base–emitter junction. Interestingly, under punchthrough breakdown as under avalanche breakdown, $BV_{CBO} > BV_{CEO}$, although for very different reasons.

To prevent punchthrough breakdown, the design of the base is critical. This is clearly seen in Eq. (11.110). This equation suggests that the punchthrough breakdown depends on the square of the total quasi-neutral base integrated doping $W_B N_B$. Improving punchthrough breakdown therefore requires increasing the total amount of impurities in the base. Lightening the collector doping level also helps since it allows for the depletion region to expand more into it. For a given emitter design, the trade-off incurred in improving punchthrough breakdown is to degrade I_S and β_F, which are both inversely proportional to the $W_B N_B$ product.

Given that the engineering of punchthrough and avalanche breakdown is so different, it is essential to be able to accurately distinguish between these two breakdown mechanisms. This is discussed in the next section.

Distinguishing between punchthrough breakdown and avalanche breakdown

There are several ways to distinguish punchthrough from avalanche breakdown. The clearest one consists of measuring yet a fourth breakdown voltage, BV_{CES}. According to the notation outlined above, BV_{CES} is the collector to emitter breakdown voltage with the base *shorted* to the emitter, that is, with $V_{BE} = 0$.

Following similar procedures to those presented in the two previous sections, it is fairly straightforward to compute BV_{CES} for avalanche and punchthrough breakdown using the hybrid-π model of the BJT. When this is done, one finds that:

$$BV_{CES}|_{avalanche} = V_{CB}(M = \infty) = BV_{CBO}$$

$$BV_{CES}|_{punchthrough} = V_{CBpunch} \simeq BV_{CEO}$$

That is, while under avalanche breakdown $BV_{CES} \simeq BV_{CBO}$, under punchthrough breakdown $BV_{CES} \simeq BV_{CEO}$. This is understandable. In avalanche breakdown, shorting the

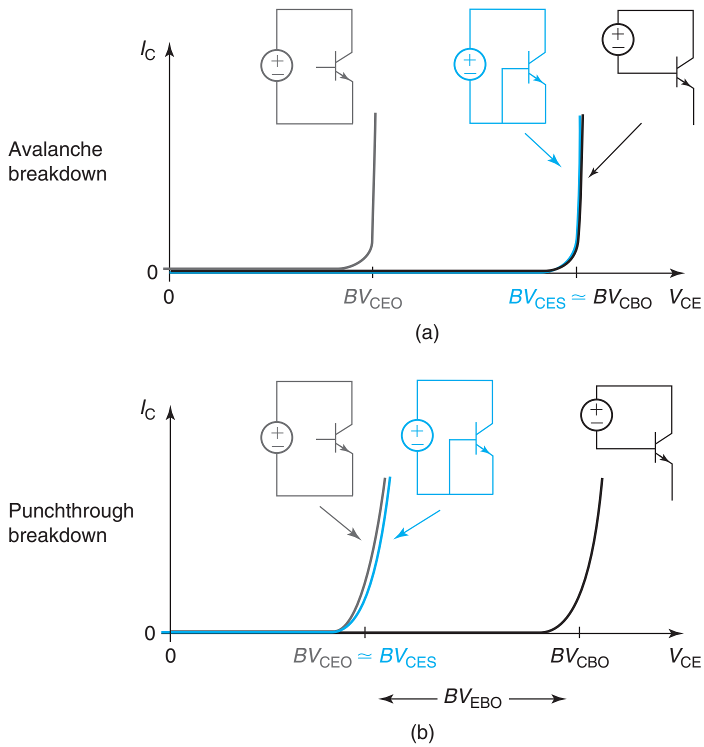


FIGURE 11.49
Sketch of breakdown characteristics for avalanche breakdown and punchthrough breakdown showing the relationship among BV_{CBO}, BV_{CEO}, BV_{CES}, and BV_{EBO}.

emitter-base junction prevents it from ever getting forward biased, effectively shutting off the feedback loop through the current gain of the BJT. Breakdown then happens when the multiplication coefficient in the base–collector junction blows up. In punchthrough breakdown, shorting the base–emitter junction prevents it from going into reverse bias and provides a path for the emitter current to flow. Hence, breakdown occurs simply when $V_{CB} = V_{CE} = V_{CBpunch}$, which is BV_{CEO}.

In a graphical way, Fig. 11.49 summarizes the three breakdown voltages that characterize each breakdown phenomena. The relationship between BV_{CES} and the other two breakdown voltages clearly reveals the nature of breakdown in a given device. In addition to this, in the case of punchthrough breakdown, the difference between BV_{CBO} and BV_{CEO} should be close to BV_{EBO}. Of course, observing this in a given device does not guarantee that the dominant breakdown mode is punchthrough.

A second, more subtle difference between punchthrough and avalanche breakdown is the shape of the collector current as it approaches breakdown. This is also sketched in Fig. 11.49. Avalanche breakdown is a multiplication process. Once impact ionization becomes significant, it grows very quickly as the reverse bias increases. Hence, the collector current rises very sharply at breakdown. In contrast, punchthrough breakdown consists of electrons spilling over an energy barrier of decreasing height. Hence, the rise of I_C with V_{CE} is softer.

The breakdown voltage of most modern npn bipolar transistors is typically limited by avalanche breakdown. In contrast, pnp bipolar transistors are often limited by punchthrough breakdown. This is because in pnp BJTs, in an effort to extract enough performance, device designers tend to be rather aggressive in the base design and engineer a relatively thin base so as to partially compensate for the reduced hole mobility. With a shallow base, pnp BJTs often suffer from base punchthrough before significant impact ionization occurs in the base–collector SCR.

The discussion of this subsection has one additional important consequence: at times, it might be beneficial for *circuit designers* to be able to distinguish between avalanche breakdown and punchthrough breakdown. In a circuit situation where the breakdown voltage might be marginal, a circuit designer can effectively enhance it by engineering the base impedance. But this only works if the breakdown voltage is limited by an avalanche process. If the underlying breakdown mechanism is base punchthrough, this is not possible. This is an excellent example of the benefits that having a solid grounding in device physics can bring to circuit designers.

11.5.5 High collector current effects

At high collector currents, bipolar transistors suffer from a number of significant anomalies. Figure 11.50 sketches β_F and f_T of a BJT biased in the FAR as a function of collector current. Beyond a certain collector current level, both β_F and f_T are seen to roll off. At high collector currents, the operation of the BJT is severely degraded. This obviously has important consequences. In order to maintain adequate performance, a limit has to be placed in the maximum current that a device can handle. This in turn constrains the minimum transistor size that can be used in a given circuit application. All this severely impacts the power dissipation, speed, and cost of bipolar ICs.

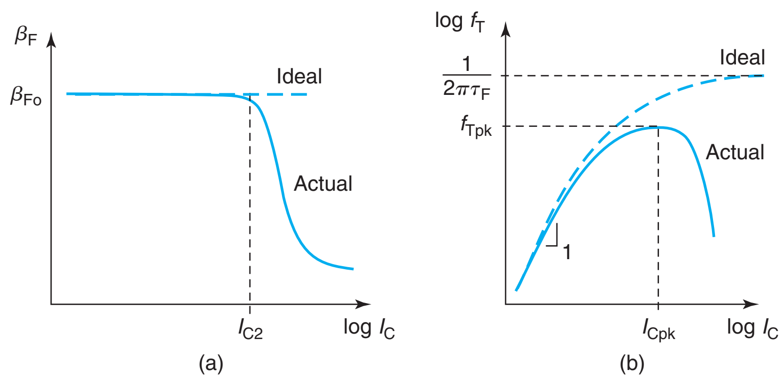


FIGURE 11.50
Current gain and f_T of a BJT in FAR as a function of collector current. At high collector current density, both β_F and f_T drop below their ideal values.

The physics behind the high collector current behavior of the BJT is rather complex. In most cases, several mechanisms are at play simultaneously. Their relative impact depends on the overall transistor design. It is for this reason that a variety of names are used in this context: *quasi-saturation*, *base widening*, *collector space-charge-limited transport*, *Kirk effect*, and so on. In this section, we will not attempt to present a comprehensive view of these phenomena. Our goal is, on the contrary, to describe the most important high collector current physics of relevance to contemporary microelectronics bipolar transistors so as to be able to develop simple analytical models and understand their implication in device design.

Underlying nearly all high collector current effects is electron velocity saturation in the base–collector junction and inside the quasi-neutral collector. This turns out to be important because as the maximum electron velocity is limited to the saturation velocity, high collector current requires a high electron concentration. In our analysis of the BJT so far, we have assumed negligible electron concentration inside the base–collector junction space-charge region. At a high collector current, this assumption fails. Due to velocity saturation, the electron concentration inside the B-C SCR builds up and upsets the electrostatics of the depletion region in a variety of interesting ways. They all conspire to degrade BJT operation.

There are two subsections in the rest of this presentation. We first study the changes in the electrostatics of the base–collector junction as a result of high collector current. We then study the impact of these changes on β_F and f_T.

Impact of high collector current on electrostatics of base–collector junction

Consider the geometry of Fig. 11.51. This figure shows the doping and carrier distribution in the base and collector of a bipolar transistor under various conditions. These sketches include the highly doped subcollector (also called buried collector) because, as we will see, it plays an important role at high I_C. To simplify the notation, the origin of the abscissa in this figure has been placed at the metallurgical base–collector junction. With this new x-axis, the locations of the edges of the base–collector space-charge region are at $-x_{BC}$ and x_{CC}, and the collector/subcollector metallurgical junction is located at W_{MC}. W_{MC} is the extent of the metallurgical collector.

All sketches in Fig. 11.51 correspond to a certain reverse bias of the base–collector junction $V_{BC} < 0$. The top diagram depicts a situation for $I_C = 0$ with a negligible concentration of electrons in the quasi-neutral base and the base–collector space-charge region. The second diagram shows a situation that corresponds to a small value of I_C flowing by. There is now a gradient in the electron concentration in the quasi-neutral base. The large electric field in the base–collector depletion region sweeps the electrons very fast and as a result their concentration inside the SCR is still negligible.

The third diagram depicts a situation for a substantially higher value of I_C. Since the electrons can only travel as fast as their saturation velocity, their concentration in the base–collector space-charge region is now appreciable when compared to the collector doping level (the base doping level is much higher so the impact on the base side of the junction remains negligible). Since electrons are negatively charged species and the ionized dopants on the collector side of the junction are donors, that is, positively charged species, the dipole of charge of the B-C SCR is upset. Reestablishing an overall charge neutral situation requires the edges of the SCR to

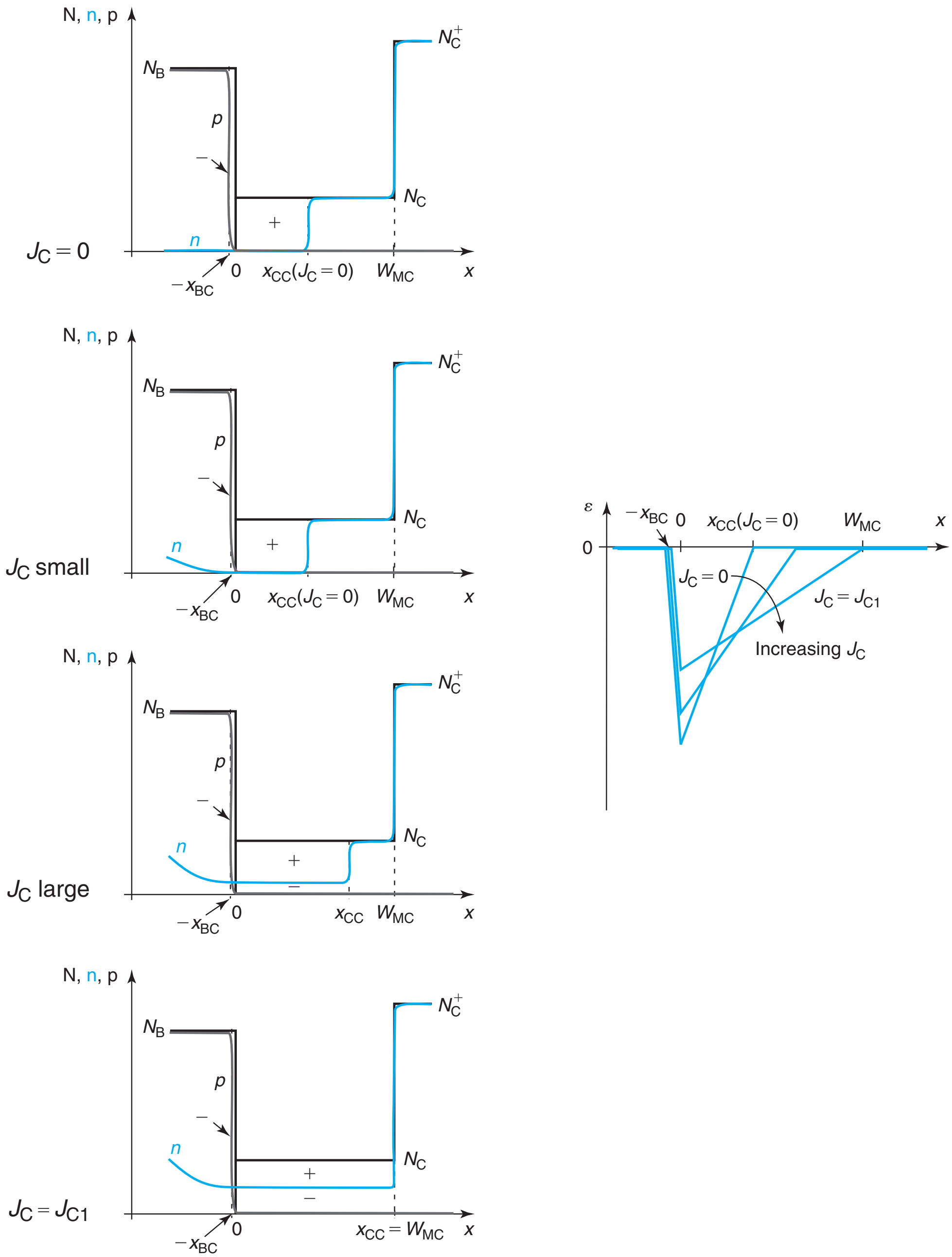


FIGURE 11.51
Sketches of doping levels and carrier concentrations in the base and collector of a bipolar transistor for different collector current levels. In all cases, $V_{BC} < 0$ remains unchanged.

move. The edge on the collector side will move further into the collector while the edge on the base side moves slightly in, closer to the metallurgical junction.

Deriving relationships for the movement of x_{BC} and x_{CC} with I_C is fairly straightforward. If we assume that the electric field in the space-charge region is high enough for electrons to travel at saturation velocity, then the electron concentration there is

$$n(x) = \frac{J_C}{qv_{sat}} \tag{11.114}$$

where J_C is defined as the collector current density computed as

$$J_C = \frac{I_C}{A_E} \tag{11.115}$$

since the cross-sectional area through which I_C flows is A_E.

Equation (11.114) suggests that the volume charge density on the collector side of the depletion region has gone down from qN_C, when I_C is small, to $qN_C - J_C/v_{sat}$. The effect of this on the location of the edges of the space-charge region can be easily assessed by going back to the formulation of the PN junction space-charge region of Chapter 6 and substituting N_C by $N_C - J_C/qv_{sat}$. Using the notation of this chapter, we easily get:

$$x_{CC} \simeq \sqrt{\frac{2\epsilon(\phi_{biC} - V_{BC})}{q\left(N_C - \frac{J_C}{qv_{sat}}\right)}} = \frac{x_{CC}(J_C = 0)}{\sqrt{1 - \frac{J_C}{J_{CK}}}} \tag{11.116}$$

$$x_{BC} \simeq \sqrt{\frac{2\epsilon\left(N_C - \frac{J_C}{qv_{sat}}\right)(\phi_{biC} - V_{BC})}{qN_B^2}} = x_{BC}(J_C = 0)\sqrt{1 - \frac{J_C}{J_{CK}}} \tag{11.117}$$

where we have also assumed that $N_B \gg N_C$. In these equations, we have defined

$$J_{CK} = qN_C v_{sat} \tag{11.118}$$

Equations (11.116), (11.117), and (11.118) show that as J_C increases, x_{CC} widens and x_{BC} shrinks.

It is clear that at some value of J_C, the space-charge region depletes the entire collector and bumps against the buried layer. This takes place when $x_{CC} = W_{MC}$. Using Eq. (11.116), this is found to happen at a collector current density given by

$$J_{C1} = J_{CK}\left[1 - \left(\frac{x_n(J_C = 0)}{W_{MC}}\right)^2\right] \tag{11.119}$$

Note that $J_{C1} < J_{CK}$.

The evolution of the electric field in the base–collector structure as J_C increases is depicted on the right-hand side of Fig. 11.51. For $J_C = 0$ and small values of J_C, we have the typical triangular shape of the electric field peaking at the metallurgical base–collector junction. As J_C increases, x_{BC} moves in, x_{CC} moves out, and the peak of the electric field drops. When $J_C = J_{C1}$, the electric field invades the entire lowly doped portion of the collector.

For $J_C > J_{C1}$, the electrostatics of the problem change. Figure 11.52 depicts what happens. As J_C increases beyond J_{C1}, the depletion region starts penetrating into the buried collector.

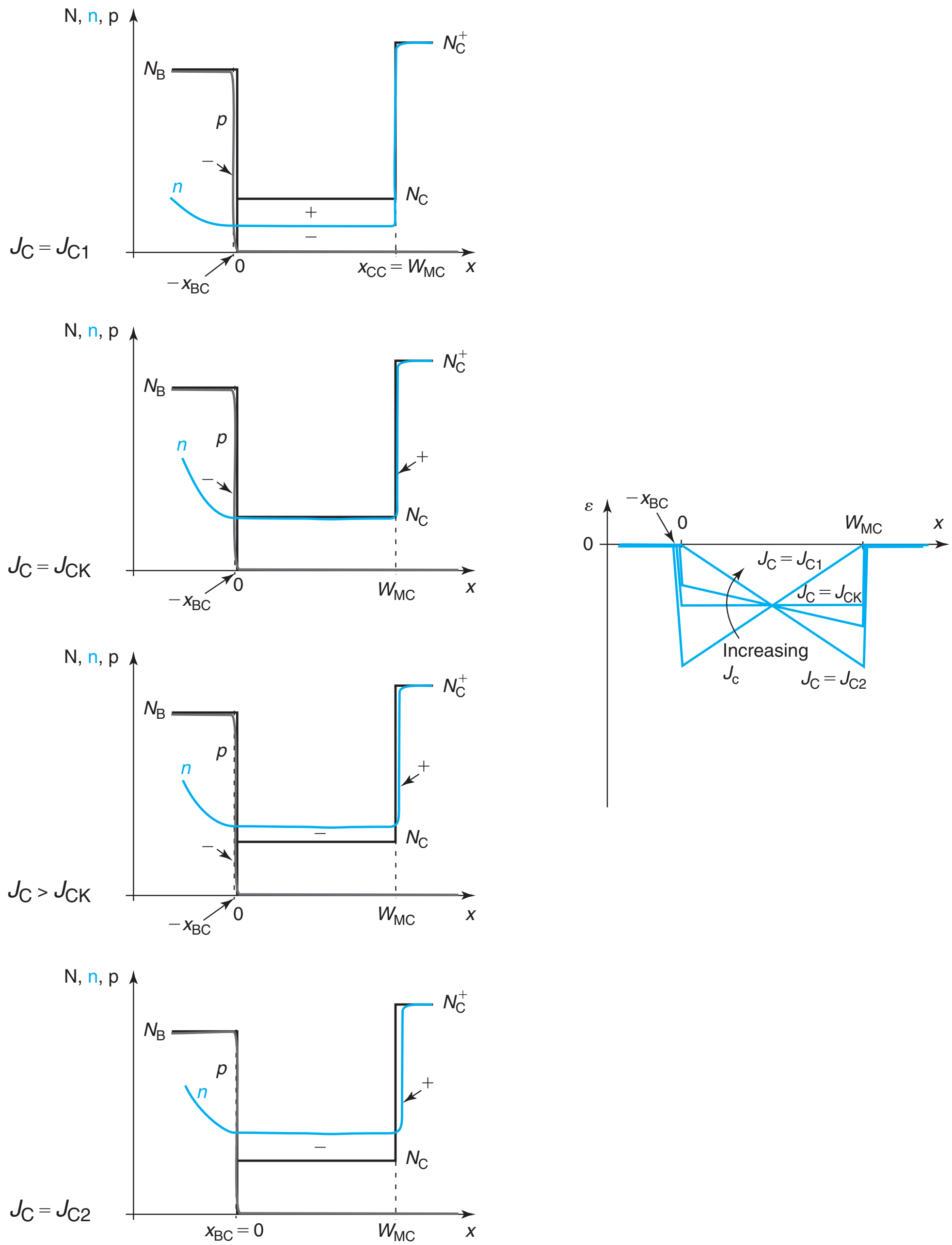


FIGURE 11.52
Sketches of doping levels and carrier concentrations in the base and collector of a bipolar transistor for different collector current levels in excess of J_{C1}. In all cases, $V_{BC} < 0$ remains unchanged.

Since it is heavily doped, the penetration can be assumed to be minimal and the overall field distribution now starts taking a trapezoidal shape.

At a current density $J_C = J_{CK}$, the electron concentration in the collector increases to a value equal to N_C. When this happens, the lowly doped portion of the collector becomes charge neutral and the electric field inside becomes uniform in space. The charge dipole that supports this electric field has its negative charge at the edge of the base and the positive charge at the edge of the buried collector.

For collector current densities beyond J_{CK}, the electron concentration inside the lowly doped portion of the collector exceeds the doping level. In this regime, the net volume charge in this region changes sign and becomes negative. Since the total potential build-up across the base–collector junction is fixed, the negative charge at the base space-charge region must be reduced in absolute terms. This can only happen by the depletion region shrinking on the base side. Consistent with this, the field at the metallurgical base–collector junction keeps getting smaller.

At a sufficient value of J_C, labeled here J_{C2}, the negative charge build-up in the collector increases to the point where the extent of the depletion region in the base disappears. When this takes place, the field at the metallurgical base–collector junction goes to zero. Estimating the collector current density at which this condition is reached is straight forward as this situation is now similar to a PN junction with a doping level on the n side equal to N_C^+, and a doping level on the p-side equal to $\frac{J_{C2}}{qv_{sat}} - N_C$. Assuming that this fictitious PN junction is asymmetric with the doping level on the n-side much higher than on the p side, the extent of the depletion region on the p side can be written as

$$x_p \simeq \sqrt{\frac{2\epsilon(\phi_{bi} - V_{BC})}{qN_A}} = \sqrt{\frac{2\epsilon(\phi_{bi} - V_{BC})}{q\left(\frac{J_{C2}}{qv_{sat}} - N_C\right)}} \tag{11.120}$$

At the critical current density $J_C = J_{C2}$, x_p becomes equal to W_{MC}. Plugging this into Eq. (11.120) and solving for J_{C2}, we find

$$J_{C2} = J_{CK} + \frac{2\epsilon v_{sat}(\phi_{bi} - V_{BC})}{W_{MC}^2} \tag{11.121}$$

The evolution of the electric field picture across the base–collector junction as J_C increases from J_{C1} to J_{C2} is shown on the right of Fig. 11.52. At $J_C = J_{C1}$, the field peaks at $x = 0$ and goes to zero at $x = W_{MC}$. At $J_C = J_{CK}$, the field is uniform throughout the lowly-doped collector. At $J_C = J_{C2}$, the field becomes zero at $x = 0$ and peaks at $x = W_{MC}$. Note that as the electric field distribution changes with J_C, its area underneath is constant reflecting the fact that V_{BC} is held unperturbed.

Is it possible to have a collector current density that exceeds J_{C2}? Indeed it is and the changes that occur in BJT operation in this high-current regime are fascinating. With the electrons already traversing through the lowly doped portion of the collector at saturation velocity, the only way to get a higher current density is to increase the electron concentration further in this region. At J_{C2}, the electron concentration in the collector is already of a value for which the electric field at $x = 0$ is zero. Since V_{CB} remains fixed, a further increase in the electron concentration must reduce the extent of the space-charge region in the lowly doped portion of the collector. How can this be accomplished? Easily, if holes spill over into the lowly doped collector right next to the base in a way that renders it charge neutral. This region is often called the *current-induced base* (or CIB), and the phenomenon is known as *base push-out* or the *Kirk effect*. The changes in the electrostatics that take place as J_C exceeds J_{C2} are shown in Fig. 11.53.

For $J_C > J_{C2}$, holes from the base spill over into the collector rendering a small region there quasi-neutral. In this CIB region, $p - n + N_C = 0$. The higher J_C, the higher n inside the collector needs to get, and the narrower the space region in the collector must become. In consequence, the CIB region has to widen.

The evolution of the electric field on the right-hand side of Fig. 11.53 shows a different perspective of the evolution of the electrostatics of the problem. At $J_C = J_{C2}$, the electric field at $x = 0$ is zero. As $J_C > J_{C2}$, the point of zero electric field has to move to the right so as to shrink the extent of the charge dipole. Hence, the peak electric field, which remains at $x = W_{MC}$, increases keeping the area under the triangle constant.

It is not difficult to derive a first-order expression for the extent of the CIB, W_{CIB}, as a function of J_C. In the high collector current regime pictorially described in Fig. 11.53, an equation similar to Eq. (11.120) (with a general J_C instead of J_{C2}) still applies and

$$W_{CIB} = W_{MC} - x_p = W_{MC} - \sqrt{\frac{2\epsilon(\phi_{bi} - V_{BC})}{q\left(\frac{J_C}{qv_{sat}} - N_C\right)}} \tag{11.122}$$

Using the definition of J_{C2} made in Eq. (11.121) and of J_{CK} made in Eq. (11.118), we can rewrite Eq. (11.122) in the following form:

$$W_{CIB} = W_{MC}\left(1 - \sqrt{\frac{J_{C2} - J_{CK}}{J_C - J_{CK}}}\right) \quad \text{for } J_C \geq J_{C2} \tag{11.123}$$

It is clear from this detailed discussion that major changes occur in the electrostatics of the base–collector junction as the collector current increases. What are the consequences of these changes to transistor operation, in particular β_F and f_T?

Impact of high collector current on β_F

The high-collector phenomena introduced in the previous subsection have a strong impact on the DC and the high-frequency characteristics of bipolar transistors. We discuss some of the most relevant issues by focusing first on the impact on β_F and then on f_T.

The current gain of a bipolar transistor in the FAR is upset at high collector currents because the quasi-neutral base width gets modified. In the range $J_C \leq J_{C1}$, the change in W_B is very small since the high doping of the base relative to the collector makes the movement of x_{BC} to be fairly minor. In the range $J_{C1} \leq J_C \leq J_{C2}$, the change in the base width is a bit more significant since essentially, the depletion region on the base side of the base–collector space-charge region gets wiped out as J_C approaches J_{C2}. However, since x_{BC} was small to begin with, to the first order this can still be ignored. Notice that in this regime, W_B tends to increase as J_C increases, and as a result, β_F tends to decrease slightly.

It is for $J_C \geq J_{C2}$ that β_F starts getting affected in a significant way as the CIB forms in the collector. As Fig. 11.53 reveals, the extent of the region over which electron diffusion takes place

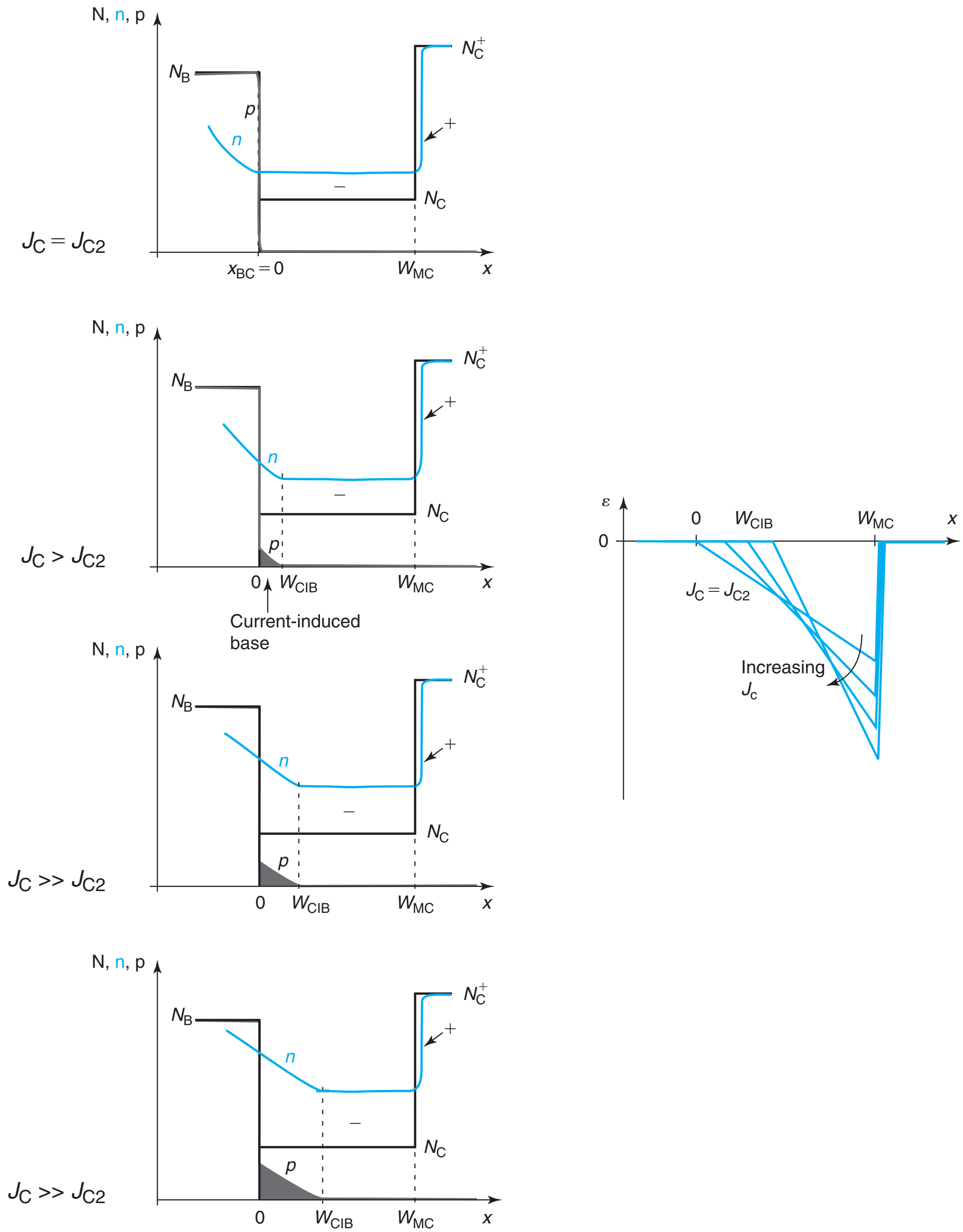


FIGURE 11.53
Sketches of doping levels and carrier concentrations in the base and collector of a bipolar transistor for different collector current levels in excess of J_{C2}. In all cases, $V_{BC} < 0$ remains unchanged.

on their way from the emitter to the collector is extended by an amount equal to W_{CIB}. Using Eq. (11.14), this suggests a simple first-order expression for the evolution of β_F with J_C,

$$\beta_F = \frac{N_E D_B W_E}{N_B D_E (W_B + W_{CIB})} = \frac{\beta_{Fo}}{1 + \frac{W_{CIB}}{W_B}} \quad \text{for } J_C \geq J_{C2} \tag{11.124}$$

with W_{CIB} given by Eq. (11.123).

There are some assumptions that have been made in deriving this simple result. First, transport through the CIB has been assumed to be purely diffusive. This is clearly not the case as the need to keep quasi-neutrality creates a gradient in the hole concentration in this region. This suggests the existence of a built-in electric field that affects electron transport. The second assumption is that the diffusion coefficient that characterizes diffusive transport through the CIB region is identical to the one that applies in the doped quasi-neutral base. This is also clearly not a good assumption since the doping level is rather different. In a way, the CIB is a highly injected region where transport occurs through what is termed "ambipolar diffusion," a complex mixture of diffusion and drift. It is beyond our scope here to attempt to construct a model of β_F degradation accounting for ambipolar diffusion.

The simple model that is embodied in Eq. (11.124) suggests an evolution in β_F with I_C for a bipolar transistor that looks like the sketch in Fig. 11.50(a). β_F drops significantly for $I_C > I_{C2}$. In this model, in the limit of high I_C, β_F saturates to a value equal to $\beta_{Fo}(W_B/W_{MC})$.

The output characteristics of a bipolar transistor are severely affected by the degradation in β_F just discussed. Figure 11.54 shows a sketch of the common-emitter output characteristics of a BJT indicating the locus of I_{C2} beyond which β_F degradation due to high-collector current effects takes place.

For $I_C > I_{C2}$, β_F is reduced as described by Eq. (11.124). In the common-emitter output characteristics, values of I_C versus V_{CE} corresponding to constant values of I_B are drawn. Hence, for

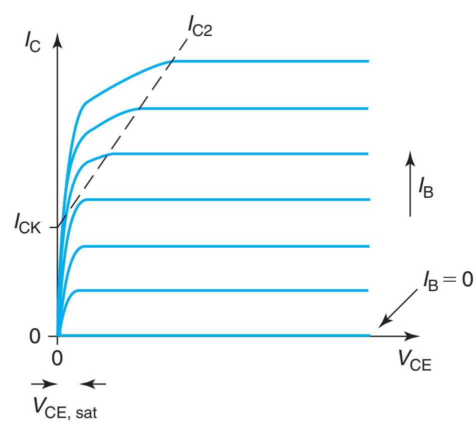


FIGURE 11.54
Sketch of common-emitter output characteristics of bipolar transistor indicating high-collector current effects.

$I_C > I_{C2}$, the amount of collector current that can be produced for a given I_B budget is smaller than in the low-current regime. This results in a severe deformation of the output characteristics in its low-V_{CE}/high-I_C region, as sketched in Fig. 11.54. This is often called *quasi-saturation*. Notice that the dependences of I_{C2} in Eq. (11.121) imply that the critical current beyond which degradation of the output characteristics occurs depends linearly on V_{CE} and it extrapolates to roughly I_{CK} for $V_{CE} = 0$, as sketched in Fig. 11.54. It is not possible to derive a simple explicit analytical equation for I_C versus V_{CE} in the high collector current regime. Nevertheless, Eqs. (11.121) and (11.124) describe in an implicit way the first-order model that results in the shape shown in Fig. 11.54.

Impact of high collector current on f_T

The change in the electrostatics of the base–collector junction at high collector currents also affects f_T in a drastic way. This occurs mainly through the impact of the collector current on the base-transit time, τ_{tB}. As discussed in Section 11.4.2, at high collector currents, τ_{tB} becomes the dominant term in f_T. τ_{tB} depends on the square of the quasi-neutral base width which, as we have seen, is affected by I_C.

In the moderate collector current regime, up to I_{C2}, the depletion region on the base side of the base–collector junction shrinks and eventually disappears. In consequence, the quasi-neutral base width expands. This should bring τ_{tB} up and f_T down. As argued before, this is a relatively weak effect because the depletion region is rather thin to begin with. Notice, however, that τ_{tB} goes as W_B^2, while β_F goes as W_B^{-1}. Hence, in the early stages of high collector current degradation, f_T is affected first and starts rolling down at lower currents than those that affect β_F.

The formation of a CIB for $I_C > I_{C2}$ has a dramatic impact on f_T. The CIB represents a new and significant storage site for electron and hole charge in a region of the collector that was previously depleted. In order to keep quasi-neutrality, when the electron concentration in the CIB region goes up, so does the hole concentration. Hence, this region contributes a new delay term to f_T. Since transport through this region is to first order due to diffusion, a first-order expression for the delay incurred in this new region is

$$\tau_{tCIB} \simeq \frac{W_{CIB}^2}{2D_B} \quad \text{for } J_C \geq J_{C2} \tag{11.125}$$

The quadratic dependence of τ_{tCIB} on W_{CIB}, combined with the square root-type dependence of W_{CIB} on I_C, makes this new delay term significant soon after I_C exceeds I_{C2}.

Figure 11.55 sketches the evolution of the various time delay components of f_T as a function of I_C. The most significant feature is the appearance of the new delay component at I_{C2} associated with the CIB. This new delay increases in a nearly linear way with I_C. There are other noticeable changes from the ideal BJT picture. τ_{tB} increases somehow in the range $I_{C1} \leq I_C \leq I_{C2}$ due to the modulation of the base width discussed above. τ_{tE} also increases beyond I_{C2} due to the degradation of β_F that takes place. The delay component associated with the base–collector junction capacitance C_{jc}/g_m broadly decreases as g_m increases. However, there is some fine structure associated with the initial widening of the base–collector region for values of I_C up to I_{C1} and the subsequent shrinking beyond I_{C2}. The delay component associated with the emitter-base junction capacitance C_{je}/g_m is roughly unaffected in the high-current regime.

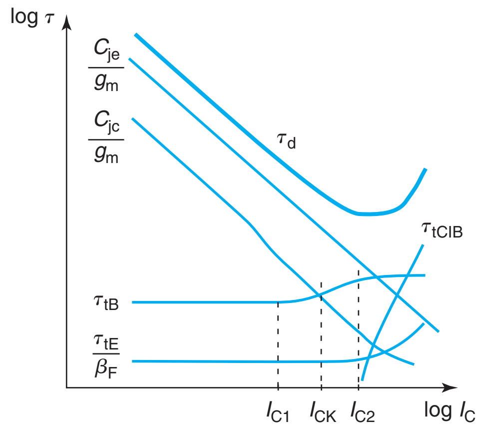


FIGURE 11.55
Sketch of evolution of the various time delay components of f_T with collector current. Also indicated is the sum of all of them, the delay time, $\tau_d = 1/2\pi f_T$.

The main consequence of all this is to bring f_T down. The detailed shape of the evolution of f_T with I_C depends on the overall transistor design. It is rather usual to see that as I_C increases, f_T increases too but never reaches the ultimate plateau determined by the intrinsic delay that is expected from simple theory. Instead f_T turns around and starts dropping. This is the situation sketched in Fig. 11.50(b), which shows f_T reaching a maximum value, f_{Tpk}, at a certain current level, I_{Cpk}. Our goal in the rest of this section is to be able to estimate I_{Cpk} and f_{Tpk} and to understand how to engineer them.

The sketch of the delay times of Fig. 11.55 suggests that f_T starts to seriously degrade at a collector current level around I_{C2}. To the first order, then

$$I_{Cpk} \simeq I_{C2} \tag{11.126}$$

f_{Tpk} can now be estimated using the ideal expression of f_T given in Eq. (11.66) at a current level equal to I_{C2}. This effectively assumes ideal transistor behavior up to I_{C2} but severe high collector current effects beyond. Under this approximation, f_{Tpk} then can be written as

$$f_{Tpk} \simeq \frac{1}{2\pi\tau_F} \frac{1}{1 + \frac{kT}{q\tau_F} \frac{C_{je}+C_{jc}}{I_{C2}}} \tag{11.127}$$

This simple expression provides us with valuable understanding as we attempt to identify how to maximize f_{Tpk}. Assuming that f_T does not reach its maximum ideal value given by $1/2\pi\tau_F$ (in which case f_{Tpk} is not affected by high current effects), then optimizing f_{Tpk} demands minimizing the ratio $(C_{je} + C_{jc})/I_{C2}$. Since both numerator and denominator in this ratio scale with the area of the transistor, then maximizing f_{Tpk} requires maximizing J_{C2}, the collector current density at the threshold of formation of the CIB.

This brings us to Eq. (11.121). There are basically three "knobs" that can be turned to maximize J_{C2}. Two of them can be used by the device designer, the third one is available to the circuit designer. The first and perhaps the most important transistor parameter with impact in J_{C2} is the collector doping level N_C. This enters directly into J_{CK}, the most important component in J_{C2}. Increasing the doping level obviously mitigates high-collector current effects because it allows more current to flow through the collector before velocity saturation effects become troublesome.

The second transistor parameter with impact in J_{C2} is W_{MC}, the extent of the lowly doped portion of the collector. W_{MC} should be minimized for J_{C2} to be maximized. W_{MC} enters the picture because the thinner the collector is, the higher is the current level that makes the integrated electron charge in the collector become comparable to the charge in the SCR portion of the base (see Fig. 11.52).

Finally, V_{BC} also affects J_{C2}. As the absolute value of V_{BC} increases, J_{C2} increases along with it. This is also to be expected since the higher $|V_{BC}|$ is, the higher the electric field in the base–collector junction is to begin with, and the higher the collector current becomes before the electrostatics of this junction are severely upset. This parameter, unlike the other two, is available for the circuit designer to engineer.

High collector current effects represent a fundamental limitation to the maximum current density that a transistor can support as well as the maximum frequency response that it can attain. Adequately managing high collector current effects represents a major goal in bipolar device design with special impact on the selection of the collector parameters.

11.5.6 Parasitic resistance

As with the MOSFET, the presence of parasitic resistance can severely affect the I–V characteristics of a BJT. Parasitic resistance is associated with the ohmic contacts to each of the regions of the BJT and the resistance of the quasi-neutral regions. To the first order, we can lump the parasitic resistance into three separate resistances located in series with the ideal BJT. This is sketched in Fig. 11.56.

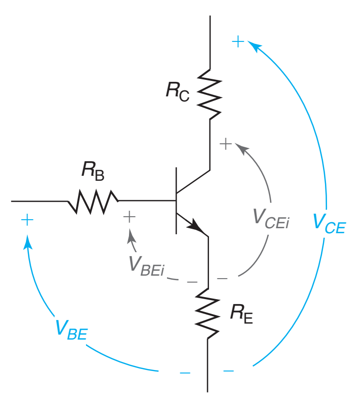


FIGURE 11.56
Schematic diagram showing parasitic emitter, base, and collector resistances in a BJT. The impact of these resistances is to reduce the internal voltages that are applied to the BJT.

The ohmic drop in the parasitic resistances of a BJT reduces the internal voltages that are applied to the device. This affects the I–V characteristics and the evolution of many figures of merit with bias. Here, we study some of the most important effects. The reader should be equipped to explore the rest.

If we denote the internal node voltages that drive the BJT as V_{CEi} and V_{BEi}, we can then write:

$$V_{BEi} = V_{BE} - I_B R_B - (I_B + I_C)R_E \simeq V_{BE} - I_C\left(R_E + \frac{R_B}{\beta_F}\right) \tag{11.128}$$

$$V_{CEi} = V_{CE} - I_C R_C - (I_B + I_C)R_E \simeq V_{CE} - I_C(R_C + R_E) \tag{11.129}$$

Here, we have assumed a well-behaved device with $\beta_F \gg 1$.

In the FAR, it is V_{BEi} that drives the base and the collector currents of the BJT. As a result of the presence of parasitic emitter and base resistance, V_{BEi} is smaller than the external voltage applied across the terminals. In this regard, R_E has much more relevance than R_B because $I_C + I_B$ flow through R_E, while only I_B flows through R_B. Looking at Eq. (11.128), the reduction in the internal voltage is more acute, the higher the I_C is. The clearest manifestation of this effect is in the Gummel plot. In effect, at high I_C, the V_{BE} axis is stretched from the ideal V_{BEi} by the ohmic drop indicated in Eq. (11.128). Both the I_B and I_C characteristics get deformed in the same way so that they run parallel. The result is shown in the sketch of Fig. 11.57. This behavior is commonly seen in BJTs. An interesting consequence of this is that the current gain of the bipolar transistor is not affected by this as both I_B and I_C are affected in a similar way at the same time.

The parasitic resistances of a BJT also appear in its small-signal equivalent circuit model. This is shown in Fig. 11.58 in the simplified form that is appropriate for the FAR. This is a lumped model with the parasitic resistances attached around the ideal equivalent circuit model. A more

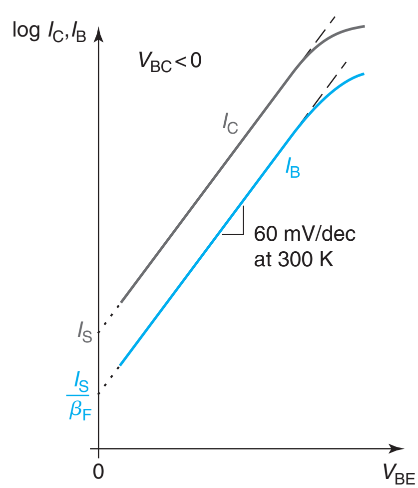


FIGURE 11.57
Schematic diagram showing the impact of parasitic resistance on the Gummel plot of a BJT in the forward-active regime.

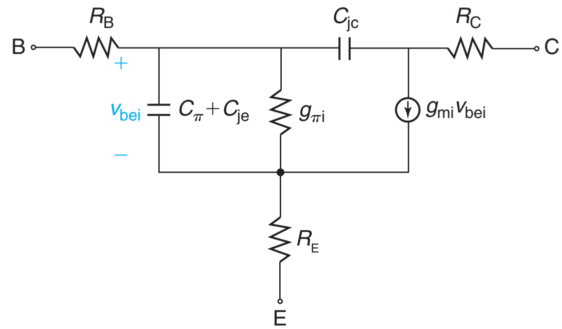


FIGURE 11.58
Small-signal equivalent circuit model of BJT in forward-active regime incorporating the parasitic terminal resistances. Note how the transconductance generator is driven by the internal small-signal base–emitter voltage v_{bei}.

accurate model, particularly appropriate for the base, would have an external resistance component and an internal component. Nevertheless, this simple model allows us to assess the impact of parasitic resistances on small-signal figures of merit.

Notice how in the circuit of Fig. 11.58 the transconductance generator that produces the collector current is driven by the internal voltage v_{bei}. As in the MOSFET, the external transconductance of the device (defined as $g_m = (\partial I_C/\partial V_{BE})|_{V_{CE}}$) is degraded from its intrinsic value ($g_{mi} = (\partial I_C/\partial V_{BEi})|_{V_{CEi}}$). Proceeding as in Section 9.7.5, it is easy to show that

$$g_m = \frac{g_{mi}}{1 + g_{mi}\left(R_E + \frac{R_B}{\beta_F}\right)} \tag{11.130}$$

The external observable transconductance of the device is degraded from its internal value by the emitter resistance and to a lesser extent by the base resistance. Similarly, one could derive an expression for g_π (external) versus $g_{\pi i}$ (internal). In contrast, it is interesting to show that f_T is not affected by the presence of parasitic resistances.

The parasitic resistances of a BJT are determined by the design of the device: the doping levels and the contact resistances. First-order expressions for R_E and R_C can be obtained following a similar approach to the PN diode in Section 6.7.2. The base resistance in a BJT is unique and we dedicate the next section to its analysis.

Base resistance

Of the three parasitic terminal resistances of a bipolar transistor, the base resistance is often the highest in value and also the one that affects circuit applications the most. Even though it does not directly impact f_T, the frequency response of BJTs in most analog and digital circuits is often seriously degraded by excessive R_B. In low-noise amplifiers based on BJTs, the noise contribution associated with the base resistance often dominates the amplifier's total noise at high frequencies. The reason why many bipolar transistors are designed with two base contacts is precisely the need to minimize the base resistance.

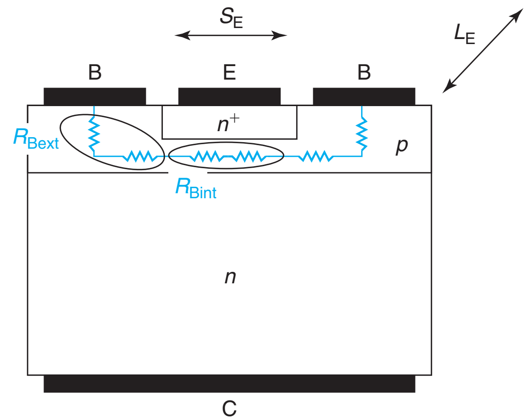


FIGURE 11.59
Components of base resistance in a bipolar transistor.

In a BJT, the base resistance tends to be relatively high because the intrinsic base has to be thin to enable high current gain and short base transit time. In addition to the intrinsic base resistance, there is also an extrinsic base resistance that is associated with the extrinsic base and the base ohmic contacts. In general, then,

$$R_B = R_{Bext} + R_{Bint} \tag{11.131}$$

Figure11.59 summarizes the contributions to base resistance in the ideal bipolar transistor.

Obviously, the intrinsic component of the base resistance scales with the emitter stripe width, S_E, and is inversely proportional to the emitter stripe length, L_E (these dimensions are also sketched in Fig. 11.59). It is not clear, however, how the intrinsic base resistance is to be computed, since in some way it is "distributed." What this means is that the entire base current does not flow laterally through the entire intrinsic base. The base current results from hole injection from the base into the emitter, which to first order is uniformly distributed across the base–emitter junction. Since the base current is fed from the edge of the intrinsic base, the lateral flow of base current increases from the center out.

Before we launch into a detailed study of this effect, let us define the *lateral resistance of the intrinsic base*, R_{blat}, as the geometrical resistance that it presents to lateral current flow across the entire base. That is,

$$R_{blat} = R_{shb}\frac{S_E}{L_E} \tag{11.132}$$

where R_{shb} is the sheet resistance of the base, given by

$$R_{shb} = \frac{1}{q\mu_B N_B W_B} \tag{11.133}$$

It is clear that $R_{Bint} < R_{blat}$ since the entire base current does not flow through the entire base. But how much smaller is R_{Bint} when compared to R_{blat}? We first answer this question in a bipolar transistor with a single base contact. The situation is shown in Fig. 11.60.

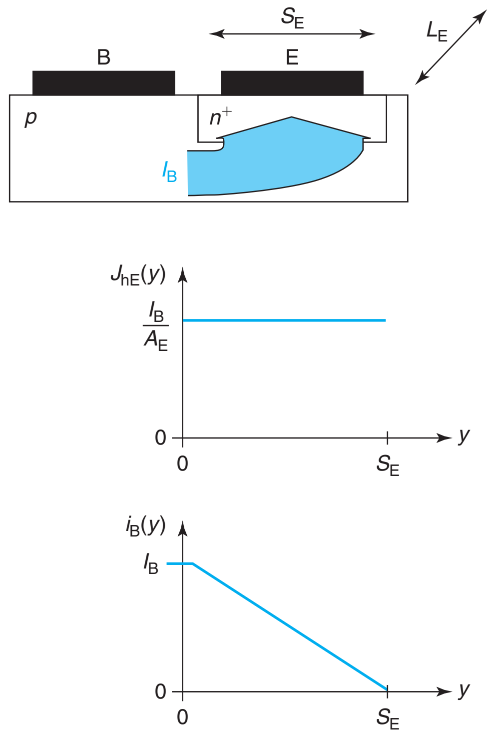


FIGURE 11.60
Sketch of base current flow through the intrinsic base of a bipolar transistor with a single base contact. $J_{hE(y)}$ and $i_B(y)$ represent, respectively, the hole current density injected into the emitter and the base current along the lateral coordinate y.

Assuming that the ohmic drop along the base is not very large, the hole current density injected into the emitter, J_{hE}, does not depend on the lateral dimension along the intrinsic base y. Hence, its value is simply $J_{hE} = I_B/A_E$ everywhere. This is sketched in Fig. 11.60.

A consequence of this is that the base current drops *linearly* along the length of the intrinsic base. Let us denote the lateral base current at any location y along the intrinsic base as $i_B(y)$. At its edge on the contact side, $i_B(y = 0) = I_B$, the full base current. At the other edge, $i_B(y = S_E) = 0$ since the transistor comes to an end there. The distribution of base current along the intrinsic base is then given by

$$i_B(y) = I_B\left(1 - \frac{y}{S_E}\right) \tag{11.134}$$

A simple way to derive the intrinsic base resistance associated with this situation is to compute the power dissipated in the intrinsic base, which should be equal to

$$p_{Bint} = I_B^2 R_{Bint} \tag{11.135}$$

The power dissipated across the intrinsic base in this distributed situation is easily computed by integrating the power dissipated in elemental lateral resistors of length dy, in the following way:

$$p_{\text{Bint}} = \int_{\text{intB}} i_{\text{B}}^2(y)dR = \int_0^{S_{\text{E}}} i_{\text{B}}^2(y)\frac{R_{\text{shb}}}{L_{\text{E}}}dy = I_{\text{B}}^2\frac{1}{3}R_{\text{shb}}\frac{S_{\text{E}}}{L_{\text{E}}} \tag{11.136}$$

where we have used Eq. (11.134).

Equating this result with Eq. (11.135) and using the definition made in Eq. (11.132), we finally obtain

$$R_{\text{Bint}} = \frac{1}{3}R_{\text{blat}} \tag{11.137}$$

As expected, R_{Bint} is smaller than R_{blat}. In fact, it is only one-third of R_{blat}.

The more common BJT design with two base contacts is shown in Fig. 11.61. In this case, half of the base current is fed through each end of the transistor. In consequence, the lateral base current flowing through the base goes to zero at the center of the base. With uniform hole

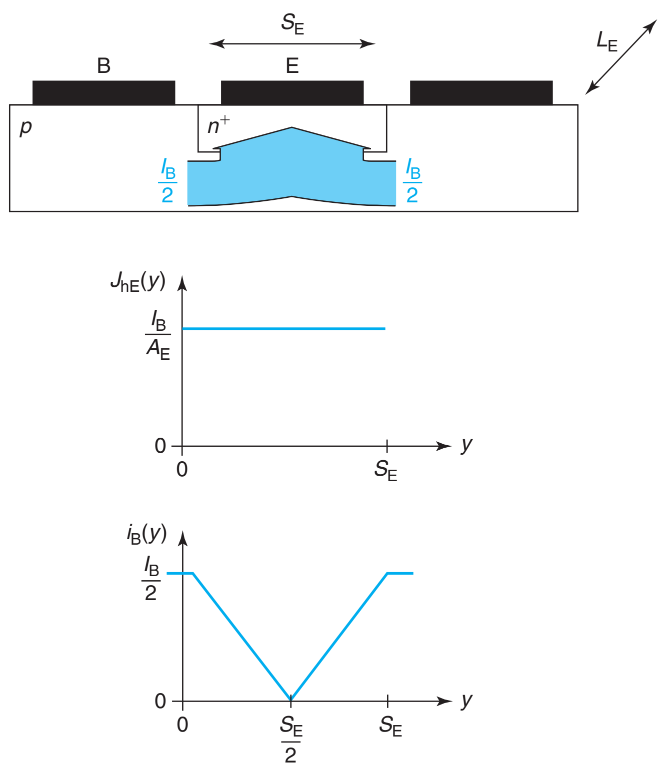


FIGURE 11.61
Sketch of base current flow through the intrinsic base of a bipolar transistor with two base contacts.

injection into the emitter across the emitter-base junction, this implies a "V"-shaped distribution for the base current. Analytically, we then have:

$$i_B(y) = \frac{I_B}{2}\left(1 - \frac{2y}{S_E}\right) \qquad \text{for } 0 \le y \le \frac{S_E}{2}$$

$$i_B(y) = \frac{I_B}{2}\left(\frac{2y}{S_E} - 1\right) \qquad \text{for } \frac{S_E}{2} \le y \le S_E$$

Following a procedure identical to that of a single base contact, we can quickly find that

$$R_{\text{Bint}} = \frac{1}{12} R_{\text{blat}} \tag{11.138}$$

Interestingly, this is a quarter of the resistance of the single base contact. This might seem surprising at first, but it makes good sense. In a two-base contact design, the base current is split into two, and each half flows half as far into the intrinsic base of the transistor as compared to the single base contact design, hence the factor of 4 reduction. In addition to this, the extrinsic base resistance of the BJT with the two base contacts is halved from that of the single contact design. This significant base resistance advantage over the single base contact design is the reason why the double base contact BJT is often the preferred choice for applications that are sensitive to base resistance. The trade-offs of using a two-base contact design are larger transistor size and increased base–collector junction capacitance.

In bipolar device design, minimizing the base resistance is often a high-priority goal. Besides introducing two base contacts, there are other device design knobs that can be turned to accomplish this. Let us look at the two components of R_B in turn. R_{Bext} is impacted by base contact resistance, the sheet resistance of the extrinsic base, and the overall distance between the base contacts and the edges of the emitter. Minimizing R_{Bext} requires high doping in the extrinsic base and close spacing between the base and the emitter contacts. As in the MOSFET, this is best accomplished by self-aligning the highly doped extrinsic base and the base ohmic contacts with the emitter. We will discuss how this is accomplished toward the end of this chapter.

Minimizing R_{Bint} can be accomplished in three ways. One is to increase the base width, W_B. However, the trade-off in frequency response through the base transit time is very severe. The second approach is to enhance the base doping level. The trade-offs of doing this are increased C_{je}, reduced V_{EBO} (which has some bearing on the reliability of the device), and smaller I_S and β_F. From a technology point of view, it is also hard to create a very shallow highly doped base buried underneath the emitter. Nevertheless, bipolar device design requires fine-tuning of the base doping level. The third approach to minimize R_{Bint} is to scale down the emitter stripe width S_E. Since S_E is set by the lithography and etching technology, it typically only changes from generation to generation, when the manufacturing tool set is updated. In bipolar technology, as in MOSFET technology, size downscaling is a long-term trend that most effectively improves performance, power efficiency, and circuit density.

11.5.7 Nonuniform doping levels

Up until this point in this chapter, we have been involved with the operation of bipolar transistors with uniform doping levels throughout. In reality, the doping distribution in a bipolar

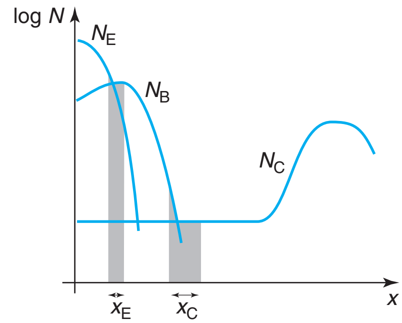


FIGURE 11.62
Sketch of doping distribution through the intrinsic region of a bipolar transistor.

transistor is usually highly nonuniform. If we take a cross section through the intrinsic portion of a modern bipolar transistor, we are likely to find a doping distribution that resembles that of Fig. 11.62. Doping nonuniformities are not only a result of the manufacturing technology, but also a desirable feature when it comes to optimizing certain important figures of merit, such as minimizing the transit time through the base or hole injection into the emitter.

An important part of the job of a bipolar device designer is to fine-tune the doping distributions throughout the entire device. This is typically done using two-dimensional computer-aided process and device simulation tools. Without the guidance that proper physical intuition provides, this can be arduous work. Hence, in this section, we discuss the most important consequences of nonuniform doping distributions in BJTs and we derive back-of-the-envelope relationships for some of the most important figures of merit.

This section heavily leverages the discussion about nonuniformly doped PN diodes in Section 6.6.5. As for PN diodes, the qualitative operation of a bipolar transistor does not change as a result of the doping distribution being nonuniform. However, the evaluation of important figures of merit is affected.

We start with the DC currents. As we discussed in Section 6.6.5, minority carrier injection into a nonuniformly doped quasi-neutral region is affected by any electric field that might exist there. This electric field can enhance or oppose carrier injection. In a bipolar transistor, this can be exploited to aid electron injection into the base and hinder hole injection into the emitter, hence maximizing the current gain. Following our discussion in Section 6.6.5, this requires a "downgoing" profile in the direction of electron flow in the base of the BJT, while an "upgoing" profile in the direction of hole flow in the emitter. This is exactly the doping distribution that one typically sees in a BJT and that is sketched in Fig. 11.62.

The computation of the collector and base currents is identical to that performed in Section 6.6.5. To keep the notation simple, we neglect at this point the extent of the space-charge region. We find, respectively:

$$I_C = \frac{qn_i^2 A_E}{\int_0^{W_B} \frac{N_B}{D_B} dx} \exp \frac{qV_{BE}}{kT} \simeq \frac{q < D_B > n_i^2 A_E}{\int_0^{W_B} N_B dx} \exp \frac{qV_{BE}}{kT} \tag{11.139}$$

$$I_B = \frac{qn_i^2 A_E}{\int_0^{-W_E} \frac{N_E}{D_E} dx} \left(\exp \frac{qV_{BE}}{kT} - 1 \right) \simeq \frac{q < D_E > n_i^2 A_E}{\int_0^{-W_E} N_E dx} \left(\exp \frac{qV_{BE}}{kT} - 1 \right) \tag{11.140}$$

where we have brought the minority carrier diffusion coefficients out of the integral since they are not very strong functions of doping.

The important piece of physical intuition that these results should provide is that to first order, the collector and base currents in a BJT are set by the integrated doping profile in the base and emitter regions, respectively. These integrals are often referred to as "integrated charge" or "Gummel number" of these two regions. An important reminder regarding these results is in order here. They apply only for "short" quasi-neutral regions with negligible bulk recombination. This of course is the case of the base of a well-designed BJT. It is also the case of the emitter in most BJT designs. For the emitter, furthermore, Eq. (11.140) demands that the minority-carrier boundary condition at its surface be characterized by an infinite surface recombination velocity. In modern designs with a poly-Si emitter, this is not quite the case and a modification of this formulation is required. This is usually done in a way similar to the approach that led to Eq. (6.118).

The current gain of the BJT is the ratio of Eqs. (11.139) and (11.140), and is then given by

$$\beta_F = \frac{< D_B > \int_0^{-W_E} N_E dx}{< D_E > \int_0^{W_B} N_B dx} \tag{11.141}$$

Not surprisingly, to first order, β_F is determined by the ratio of the integrated doping of the emitter to the base of the BJT.

The dynamics of the BJT are also affected by doping nonuniformities. The presence of an electric field inside the quasi-neutral region speeds up or slows down minority carrier flow, as studied in Section 6.6.5. In the BJT, this affects the transit time of electrons through the base and that of holes through the emitter. Analytical expressions for the respective transit times can be derived in a similar way as Eq. (6.123) was obtained. They are:

$$\tau_{tB} = \int_0^{W_B} \frac{1}{N_B} \left(\int_0^x \frac{N_B}{D_B} dx \right) dx \simeq \frac{1}{< D_B >} \int_0^{W_B} \frac{1}{N_B} \left(\int_0^x N_B dx \right) dx \tag{11.142}$$

$$\begin{aligned} \tau_{tE} &= \int_0^{-W_E} \frac{1}{N_E} \left(\int_0^x \frac{N_E}{D_E} dx \right) dx \\ &\simeq \frac{1}{< D_E >} \int_0^{-W_E} \frac{1}{N_E} \left(\int_0^x N_E dx \right) dx \end{aligned} \tag{11.143}$$

When compared with the expressions of I_C and I_B obtained above, it is important to realize (this was already discussed in Section 6.6.5) that the transit times do not depend on the integrated base dopant impurity concentration. On the contrary, they are affected by the details of the doping distribution. Different impurity profiles with the same integral can give rise to fairly different transit times. Problem 6.4 illustrated this point.

In the design of the doping profile of the base, current gain and transit time arguments favor a "downgoing" impurity profile in depth, as sketched in Fig. 11.62. This helps both to increase the collector current (and hence the current gain) and to decrease the base transit time. For the emitter, we have contradictory requirements. We wish to minimize hole injection, but at the same time, speed up the flow of holes through the quasi-neutral emitter. This places conflicting requirements on the shape of the impurity profile. Since the emitter transit time represents a relatively small contribution to the overall intrinsic delay of the BJT, hole injection suppression takes priority and the profile is designed as in Fig. 11.62 with the dopant concentration highest at the surface and going down into the wafer. This happens to be much easier to fabricate using standard technology and also yields smaller ohmic contact resistances.

The expressions of the Early voltages are also affected by the nonuniform distribution of dopants in the base. For V_A, the easiest place to start is from Eq. (11.77), which provides a simple definition for V_A

$$V_A \simeq -\frac{I_C(V_{BC}=0)}{\left.\frac{dI_C}{dV_{BC}}\right|_{V_{BE}}} \tag{11.144}$$

From Eq. (11.139), changing the upper limit of the integral to make explicit the depletion region thickness, we have

$$\left.\frac{dI_C}{dV_{BC}}\right|_{V_{BE}} \simeq \frac{-I_C}{\int_0^{W_B}\frac{N_B}{D_B}dx}\frac{d}{dV_{BC}}\left(\int_0^{W_B-x_{BC}}\frac{N_B}{D_B}dx\right) \simeq \frac{I_C(V_{BC}=0)}{\int_0^{W_B}\frac{N_B}{D_B}dx}\frac{N_B(W_B)}{D_B(W_B)}\frac{dx_{BC}}{dV_{BC}} \tag{11.145}$$

where we have used the Leibnitz' rule. We can now plug in the result of Eq. (11.73) and substitute the whole thing into Eq. (11.144) to get

$$V_A = \frac{qA_CD_B(W_B)}{C_{jco}}\int_0^{W_B}\frac{N_B}{D_B}dx \simeq \frac{qA_C}{C_{jco}}\int_0^{W_B}N_Bdx \tag{11.146}$$

In a similar way, we can derive an equivalent expression for the reverse Early voltage

$$V_B = \frac{qA_ED_B(0)}{C_{jeo}}\int_0^{W_B}\frac{N_B}{D_B}dx \simeq \frac{qA_E}{C_{jeo}}\int_0^{W_B}N_Bdx \tag{11.147}$$

Interestingly, we find again the result that both V_A and V_B depend only on the integrated base doping impurity concentration and not on the detailed dopant distribution itself.

To conclude the discussion on the impact of nonuniform doping distributions, we look at the intrinsic base resistance. The analysis of Section 11.5.6 expressed the intrinsic base resistance of the BJT in terms of the sheet resistance of the intrinsic base. Hence, the problem is reduced to suitably modifying the expression of R_{shb} in Eq. (11.132). This can easily be done using the definition made in Eq. (5.47), which, adapted to the notation at hand here, becomes

$$R_{shb} = \frac{1}{q\int_0^{W_B}\mu_BN_Bdx} \simeq \frac{1}{q<\mu_B>\int_0^{W_B}N_Bdx} \tag{11.148}$$

As other figures of merit, the intrinsic sheet resistance of the base of a BJT can be written in terms of the integral of the doping distribution in the quasi-neutral base.

This section not only provides with first-order expressions for important BJT figures of merit that account for the impact of nonuniform dopant profiles, but also reveals the significance of the integrated impurity concentration in the base of a BJT. The base Gummel number of a BJT sets to first order the collector current and the sheet resistance in the base. It also impacts in a big way the two Early voltages and the current gain. Selecting the integrated doping concentration in the base is one of the first and most important decisions that a bipolar designer must make.

11.6 EVOLUTION OF BJT DESIGN

This section provides an overview of some of the major milestones in the evolution of integrated bipolar transistor design. These devices have been used in multiple capacities over the years. Up to the early-1990s, BJT technology progress was largely fueled by its application in logic circuits, most notably in ECL, a fast logic family. This demanded aggressive transistor scaling, parasitic suppression, and voltage reduction. Eventually and due to power density considerations, ECL was displaced by CMOS. Nevertheless, BJT technology has remained vital for a broad range of applications in analog and mixed-signal electronics. In these, key concerns are footprint, supply voltage, frequency response, and breakdown voltage. In the 50 years of integrated BJT electronics, footprint has shrunk significantly and the frequency response has increased several orders of magnitude. Along the way, the breakdown voltage has also dropped substantially. This has been accommodated through a reduction in the supply voltage that has also been the key for the management of power consumption.

This review of bipolar transistor design skips the invention of the BJT, credited to William Shockley at Bell Labs in 1948 building on the first demonstration of the bipolar point contact transistor by John Bardeen and Walter Brattain also at Bell Labs in 1947. It also bypasses the demonstration of the first bipolar integrated circuit attributed to Jack Kilby at Texas Instruments in 1958. Instead, we start with the first demonstration of a *planar* bipolar transistor IC, credited to Robert Noyce of Fairchild in early 1959. It was the planar process that allowed high manufacturing yield and reliability and lead to the first commercial BJT ICs manufactured by Fairchild in the early-1960s.

A cross section of an early planar design is shown in Fig. 11.63. Many features in this design have lasted for several decades, some even to today. The process starts with the diffusion of the n^+ buried collector or subcollector into a p-type substrate. In an integrated BJT, the collector current flows first downward from the emitter-base junction through the intrinsic collector, then turns 90° and flows parallel to the surface through the subcollector, and finally, turns 90° one more time and flows upward through the collector plug toward the collector contact. To minimize the resistance of this long path, the subcollector and collector plugs are both heavily doped.

Once the subcollector is created, the intrinsic collector region is formed on top by epitaxy. Into this, the base and the emitter are diffused. Device isolation is achieved by means of a p^+ ring diffused from the surface that reaches into the p-type substrate. In this way, a PN junction

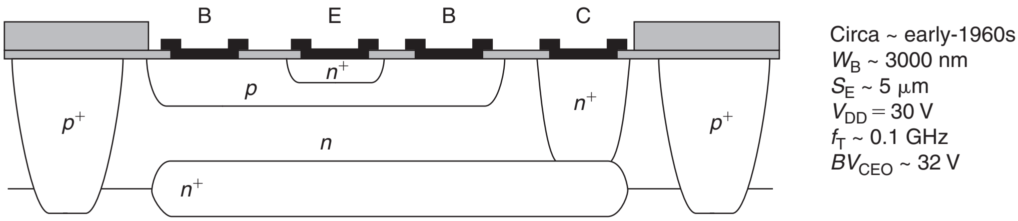


FIGURE 11.63
Sketch of an early planar junction-isolated bipolar transistor. Notable design aspects are junction isolation, metal contact to all regions, epitaxial collector, and buried subcollector.

is formed all around the transistor. This is referred to as "junction isolation." Electrical contacts to all regions are created directly with metal (Al in the early designs).

An interesting feature of the way this transistor is fabricated is that the base and the emitter are of comparable width (in the vertical direction). This is because the intrinsic base is formed through a subtractive process as the emitter is diffused into the base. Tight electrical base width control dictates that the emitter diffusion reach about halfway into the base diffusion, hence $W_E \simeq W_B$.

There are important consequences to this when it comes to device scaling. As with MOSFETs, BJT scaling has been driven by footprint reduction in an effort to boost transistor density. Size scaling has also been fueled by the desire to enhance frequency response and enable fast-switching logic circuits and high-frequency analog circuits. The scaling trajectory of BJTs goes through footprint shrinkage, a reduction in W_B and an increase in N_C to enhance frequency response. Consequences of this are the need to increase N_B to avoid excessive base resistance, a reduction in W_E to maintain tight control over the device characteristics, and a reduction in operating voltage as the breakdown voltage is reduced.

In the early junction isolated transistor design, emitter width reduction was problematic. With a metal contact made out of Al, junction spiking was a danger. In the contact annealing process, Al reacts with the Si, creating deep spikes that can penetrate through shallow junctions shorting them. This imposed a limit to the minimum emitter width and therefore to the minimum base width.

A solution to this problem was found in the mid-1970s by introducing an n^+-polysilicon layer between the emitter surface and the emitter contact. This is known as a "polysilicon emitter" and is illustrated in Fig. 11.64. The polysilicon layer is typically deposited through a chemical vapor deposition process and is then doped by ion implantation. The single crystal portion of the emitter is subsequently formed by diffusing the dopant species (As) at high temperature. The polysilicon layer effectively brings the contact away from the Si surface, preventing spiking and allowing further single-crystal emitter width and base width scaling.

A polysilicon emitter also resulted in improved current gain due to the removal of the metal contact further away from the junction and the presence of the poly/single-crystal interface that blocks holes. The extra current gain can be traded away against a higher doping level in the base which reduces the base resistance and speeds up circuits.

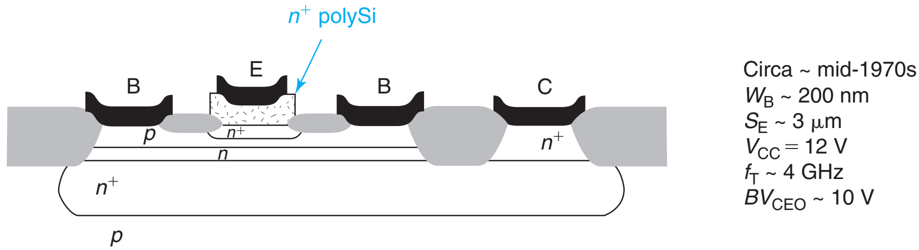


FIGURE 11.64
Sketch of a polysilicon-emitter bipolar transistor. Contact to the emitter is made through a heavily doped n$^+$-polysilicon region. Another notable feature is dielectric isolation.

An additional important feature introduced around the same time that is shown in Fig. 11.64 is *dielectric isolation*. This is the use of local oxidation (LOCOS) to isolate the devices from each other. This is far more compact than junction isolation and enabled footprint scaling, reduction in parasitics, and further performance enhancement.

Self-aligned BJT designs made their appearance in an effort to accelerate footprint scaling and parasitic reduction. With self-aligned designs, the separation between the device features is not governed by operator ability but by the construction of the process. This allows much tighter designs with lower parasitic resistance and capacitance and therefore higher performance.

Many BJT self-aligned processes were innovated in the 1980s and early-1990s. A particularly successful design that gained a lot of acceptance in various forms is the double-poly self-aligned BJT. A typical instance is sketched in Fig. 11.65. In this approach, two different polysilicon layers are used, an n$^+$ doped to form the emitter and a p$^+$ doped to form the extrinsic base. The n$^+$-polysilicon emitter contact is self-aligned to the opening in the p$^+$-polysilicon base contact. The single-crystal portion of the emitter is formed by outdiffusion of As from the n$^+$ emitter, while the extrinsic base doping is formed by outdiffusion of B from the p$^+$-poly. An additional feature of these designs is that the metal contact to the p$^+$-poly is placed on top of dielectric.

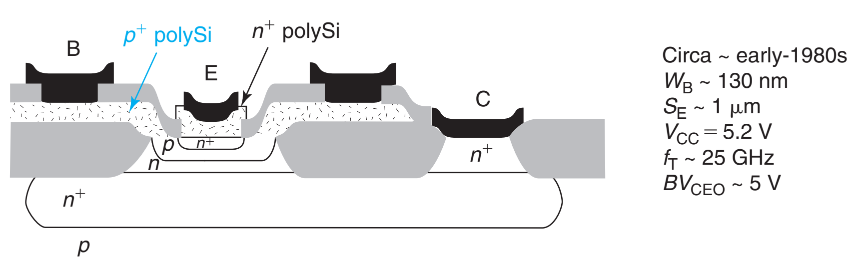


FIGURE 11.65
Sketch of a double-polysilicon self-aligned bipolar transistor. A notable feature of this design is that the base is contacted through a p$^+$-polysilicon layer and that the metal contact to the base is placed on top of isolation.

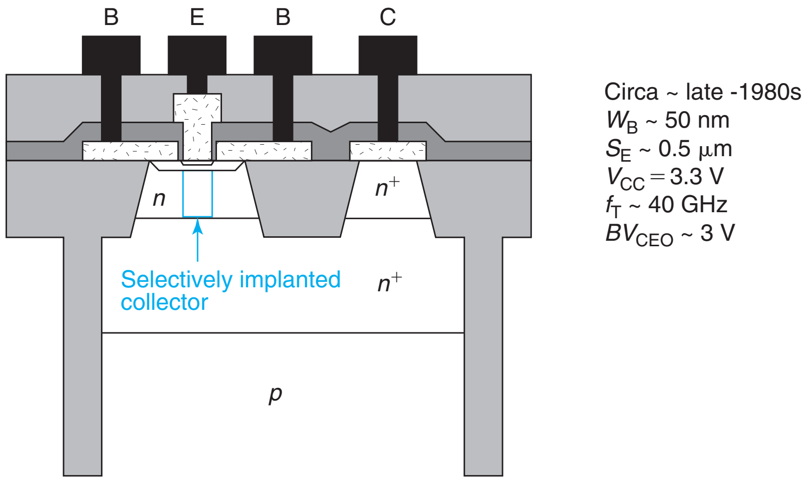


FIGURE 11.66
Sketch of a BJT with a selectively implanted collector and deep trench isolation.

This allows the minimization of the active base–emitter area and further contributes to parasitic capacitance reduction.

The next significant step in bipolar design evolution was the introduction of a selectively implanted collector (SIC), also known as collector pedestal implant. As sketched in Fig. 11.66, this design features a local higher doping level in the intrinsic portion of the collector (directly underneath the emitter). As we studied in Section 11.5.5, a higher collector doping level pushes the onset of the Kirk effect to a higher collector current resulting in an enhancement of the frequency response of the device (see Section 11.4.2). The problem with a higher donor concentration in the collector is that it leads to higher base–collector capacitance that is undesirable. With a SIC, the doping level is increased only where it is beneficial keeping the extrinsic collector with a lower doping level. A collector pedestal can be created in a self-aligned way through a deep implant at the point in the process when the emitter opening is created.

The design in Fig. 11.66 also features deep planar trench isolation that is much more compact than dielectric isolation and allows for further footprint reduction.

A revolutionary step in BJT design came with the use of epitaxy to fabricate the base and the emitter. This was introduced in the mid-1990s. Up to this point, the base of a BJT was fabricated by ion implantation of p-type dopants followed by thermal activation. Through the subtractive process described above, the intrinsic base is formed after the emitter is diffused in. This approach imposed a limit to how thin the base could be made that capped the base delay and the frequency response of the transistor.

Epitaxy is a layer-by-layer deposition technique that allows the growth of a semiconductor layer on top of a substrate with perfect atomic registry. A key advantage of epitaxy is that it affords very precise thickness and doping level control and therefore enables the use of a thinner emitter, base, and collector with precisely tailored doping profiles. This resulted in devices with superior electrical characteristics.

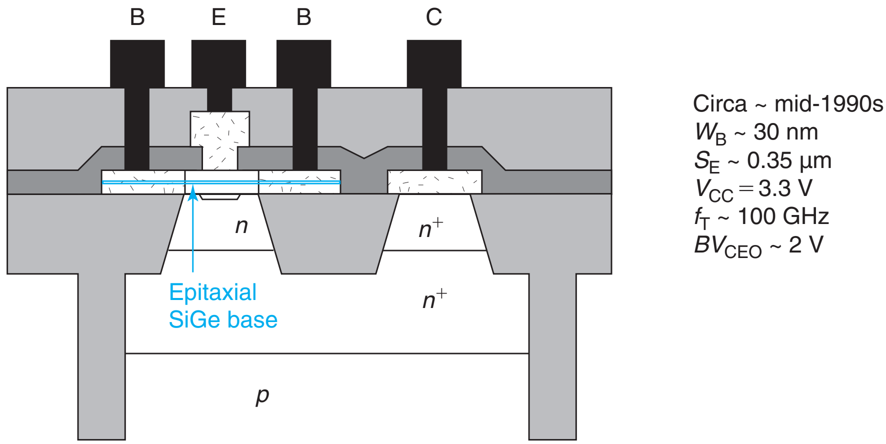


FIGURE 11.67
Sketch of a heterojunction bipolar transistor with a SiGe base.

But this was not all. The introduction of epitaxy brought along a very significant innovation: a SiGe base in what is known as a *heterojunction bipolar transistor* or *HBT*. In an HBT, during the epitaxial growth of the base it is possible to introduce a certain amount of Ge in the mix and in effect create a new "alloyed" semiconductor: SiGe. When properly prepared, SiGe is a semiconductor with a crystalline structure similar to that of Si but a larger lattice constant depending on the Ge composition. With increasing Ge composition, the bandgap of SiGe also narrows down. Engineering the Ge composition in the SiGe base of an HBT allows one to finely tune its band structure and transport characteristics. This is generally known as *bandgap engineering.* In particular, if the Ge composition is changed as the base is grown, it is possible to create a strong drift field even if the doping level is held constant. This can be used to speed up electron transport across the base and reduce the base transit time. Graded-base SiGe HBTs constitute the state of the art today and they have reached spectacular frequency response, with f_T exceeding 400 GHz. A typical cross section is shown in Fig. 11.67, with parameters characteristic of their time of introduction in the mid-1990s.

Bipolar transistors are used today in many different kinds of applications and there is a broad range of designs beyond those illustrated above. Complementary technology that combines npn and pnp BJTs has been developed for high-speed, low-power applications. Bipolar transistors, including HBTs and in some cases also including pnp BJTs, are routinely incorporated onto CMOS processes to enable mixed-signal (analog + digital) circuits. Bipolar transistor processes on SOI for very low parasitic substrate capacitance and small footprint are also commercial. Many other variations exist for specialized applications such as power management or microwave amplifiers.

While the dollar volume of bipolar-based ICs is dwarfed by CMOS, their importance in many communications, analog-signal processing, and mixed-signal systems cannot be overstated. Bipolar transistor technology remains most relevant. Understanding bipolar transistor physics and operation is essential for a semiconductor engineer.

11.7 SUMMARY

- The bipolar effect takes place in two PN junctions back to back when one of them is forward biased and the second one is reverse biased and the common region in the middle (the base) is thin in the scale of the minority carrier diffusion length. The emitter of the forward bias junction injects minority carriers into the base. These diffuse and are extracted by the collector. The collector current depends exponentially on the base–emitter voltage but is independent of the collector voltage.
- The current gain of a bipolar transistor operating in the FAR is maximized by maximizing the ratio of the emitter doping to the base doping. In general, the current gain is hard to control in a precise manner. Circuit designers have devised circuit design schemes that are tolerant to the specific value of the current gain provided that it is big enough ($\beta > 50$).
- A bipolar transistor in saturation is flooded with minority carriers everywhere. As a result, it takes time to get a bipolar transistor out of saturation.
- The high-frequency response of a bipolar transistor depends on the collector current. For low current level, the junction capacitances are dominant. For high current level, the transit time of minority carriers through the base becomes the dominant factor. This suggests that to enhance frequency response, the base should be made as thin as possible, a drift field should be introduced in the base, and the layout of the transistor should be tightened by making the emitter junction area as close as possible to the collector junction area.
- Emitter-base space-charge region recombination degrades the current gain at low collector current levels.
- The Early effect introduces a dependence of the collector current on the base–collector voltage. This introduces an output conductance that increases linearly with the collector current.
- In avalanche breakdown, the breakdown voltage of a bipolar transistor with the base open is smaller than with the emitter open. This is because of an internal gain mechanism that involves the current gain of the BJT. This implies that the current gain should not be any higher than necessary as it compromises the voltage-handling ability of the device. The key design feature for the breakdown voltage is the collector doping level. The higher the doping level, the lower the breakdown voltage.
- The doping level in the collector also limits the highest collector current that a BJT can handle. This is because of electron velocity saturation in the intrinsic collector. To enhance current density, the collector doping level needs to be increased.
- The historical scaling trajectory of bipolar transistors has been to increase the doping level of the base and collector, to thin down all layers, and to reduce the emitter stripe width and the overall footprint of the device. This results in higher current density, higher frequency response, and a reduced breakdown voltage. This is accommodated through a reduced operating voltage.

11.8 FURTHER READING

50th Anniversary of the Transistor! This Special Issue of Proceedings of the IEEE (Vol. 86, No. 1, pp. 1–308, January 1998) commemorates the invention of the bipolar junction transistor in 1948. This issue features articles from engineers involved in the transistor project at Bell Laboratories, reprints from original articles, and several accounts with technical as well as business angles of the semiconductor revolution that followed. This is a strongly recommended issue that includes easy-to-read accounts emphasizing the human side of the microelectronics revolution that started in 1948.

Crystal Fire – The Birth of the Information Age by M. Riordan and L. Hoddeson, W. W. Norton & Co. 1997 (ISBN 0-393-04124-7, TK7809.R56) is an easy-to-read book that tells the story of the invention of the bipolar transistor and the integrated circuit. The emphasis is on the personalities and their interactions, although interesting technical details also come through. This book makes it clear that the process of inventing the bipolar transistor was far from being a linear path. Along the way, fundamental discoveries about semiconductor physics had to be made and key technologies needed to be developed.

AT11.1 BIPOLAR ISSUES IN MOSFETS

One of the reasons to study the physics of bipolar transistors is to understand and appreciate the interesting bipolar effects that appear in MOSFETs and CMOS. Lurking inside a MOSFET is a bipolar transistor. For example, inside an n-channel MOSFET, there is an npn bipolar transistor that is formed by the n-type source, the p-type body, and the n-type drain. In a CMOS logic gate, there are several doped regions of different polarities in close proximity to each other. This potentially yields several bipolar transistors interconnected together.

Under normal circumstances, these parasitic bipolar transistors are all biased in their cut-off region and their presence is not felt. However, there are situations in which they can spring into action producing parasitic currents that can disrupt circuit operation or worse, resulting in device destruction.

In this advanced section, we study some of the most pervasive and important parasitic bipolar issues in MOSFETs and CMOS. Modern MOSFET design is not possible without detailed physical understanding of these effects.

AT11.1.1 Latch-up

Latch-up refers to a catastrophic situation that results when two bipolar transistors that are present in a CMOS logic gate turn on and introduce a shunt between the positive and the negative power supplies. This can lead to large and uncontrolled currents flowing through the transistors, excessive heat dissipation, and device destruction. Latch-up is a serious problem that can only be avoided through detailed modeling and careful device design. In this section, we discuss the physics of latch-up and the outline approaches to its mitigation.

In a CMOS logic gate, there are several PN junctions. A sketch was shown in Fig. 6.2. There we see an n-channel MOSFET side by side with a p-channel MOSFET. Each device has its own body contact. In that schematic, the n-MOSFET is junction isolated and the p-MOSFET is a bulk device. Since device density commands a huge premium in CMOS, the footprint of these transistors is very small and all these junctions are in close proximity to each other.

Lurking under these MOSFETs are several potential bipolar transistors. Two of them merit attention because of the way in which they are wired up. Figure 11.68 reproduces Fig. 6.2 but

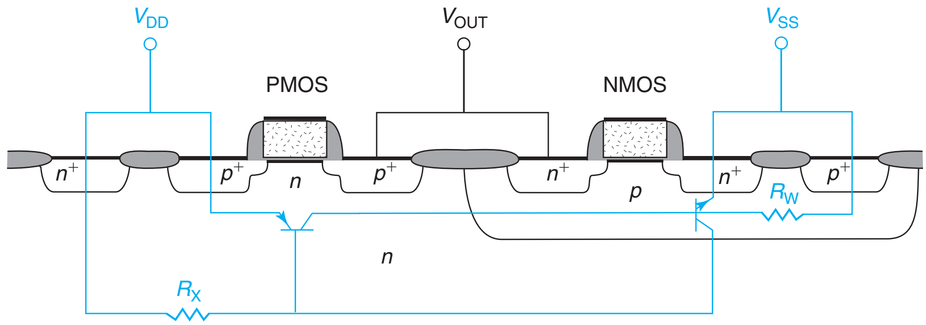


FIGURE 11.68
Sketch of CMOS logic gate exposing two parasitic bipolar transistors and two resistors that are responsible for latch-up.

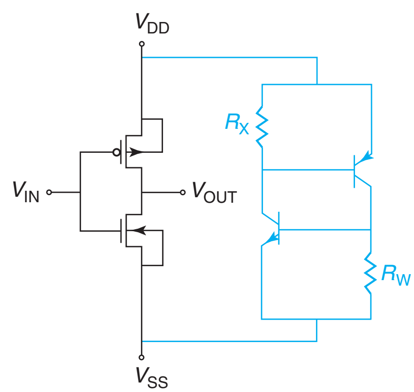


FIGURE 11.69
Circuit schematic of a CMOS logic gate with the two parasitic bipolar transistors and two resistors included.

now outlining the two BJTs and their connections (in blue). There is a vertical NPN formed by the n^+ source of the n-MOSFET, its p body, and the n substrate. There is also a lateral pnp formed by the p^+ source of the PMOS, its n body, and the p body of the NMOS. Relevant to our discussion here is also the presence of two lateral resistors formed by the body of the NMOS and the body of the PMOS, which is also the substrate.

This configuration is wired in a circuit that is redrawn in Fig. 11.69. In parallel with the familiar NMOS/PMOS pair, there is a parasitic network formed by the npn BJT and pnp BJT and the two resistors. This parasitic network connects from the positive power supply to the negative power supply.

Under normal circumstances, the body currents are negligible and the base–emitter voltage in these two bipolar transistors is zero. As a consequence, they are cut off and no current flows through the parasitic network. The situation changes if for any reason, even for a short instant, there is some current through resistor R_X. In this case, the pnp BJT turns on and current starts flowing through the right-hand branch. This produces an ohmic drop on R_W that can now turn on the npn BJT. This causes current to flow through the left-hand branch that increases the ohmic drop on R_X and turns the pnp further on, and so on. It is clear that there is a positive feedback loop going on here where if for any reason one of the BJTs turns on even momentarily, the second BJT will turn on too, and the current through the parasitic network will grow in an uncontrolled manner. At some point, the current becomes so high that the heat dissipation melts the semiconductor or destroys the contacts. A single instance in a single CMOS logic gate is enough to ruin an entire chip.

What could cause the initial disruption that triggers the latch-up process? Anything that produces a substrate current under the PMOS or an excessive body current in the NMOS could do it. An example is carrier generation due to ionizing radiation from residual radioactive contaminants in the package. Another example is impact ionization in one of the transistors as a result of a voltage spike. Latch-up is particularly problematic in input and output circuits that are in contact with the outside world and can be exposed to electrostatic discharge and voltage spikes.

Latch-up was a serious problem in the early days of CMOS. The quest for understanding latch-up spurred the development of two-dimensional and eventually three-dimensional device simulators. Once the roles of the two parasitic BJTs and the two parasitic resistors are recognized, options for the mitigation of this problem become evident. Essentially what needs to be done is to minimize the values of R_X and R_W and to reduce the current gains of the two BJTs.

An effective approach to minimize latch-up is to use a heavily doped substrate. This minimizes the value of R_X and it also degrades the gain of the lateral pnp since the substrate constitutes effectively its base. The high substrate doping cannot extend all the way to the surface or it would upset the threshold voltage of the MOSFETs and would make difficult obtaining the precise doping level required for the p-well. Therefore, above the heavily doped substrate an epitaxial layer of lower doping level is grown. The thinner this layer is, the more effective is the latch-up suppression.

A second approach is to increase the distance between the n^+-body contact of the pMOS and the p well. This also reduces the gain of the lateral pnp BJT. The trade-off here is the enlarged footprint of the transistors. This might be acceptable for input/output drivers of which there are relatively few but not for the high-density logic circuitry that implements the bulk of the system.

A third approach is to introduce guard rings. An example is shown in Fig. 11.70. In this scheme, p^+ regions are introduced between the nMOS and the pMOS and are biased to the most negative voltage available. These regions effectively behave as collectors for any holes that might be injected from the n^+ source of the nMOSFET. Effectively, this kills the gain of the lateral PNP BJT. The trade-off here is also evident, a larger footprint. Guard rings are therefore only used in particularly vulnerable locations such as the input/output circuitry.

AT11.1.2 Floating-body effects in SOI MOSFETs

In Section 10.5.6, we introduced the silicon-on-insulator (SOI) MOSFET. This was sketched in Fig. 10.51. SOI is a technology that allows for very compact and low-parasitic isolation of devices. SOI CMOS has established itself as a high-performance and low-power technology for specialized products.

An issue with SOI MOSFETs that was mentioned in Section 10.5.6 is its floating body. When the device is biased above threshold and at high V_{DS}, significant impact ionization can

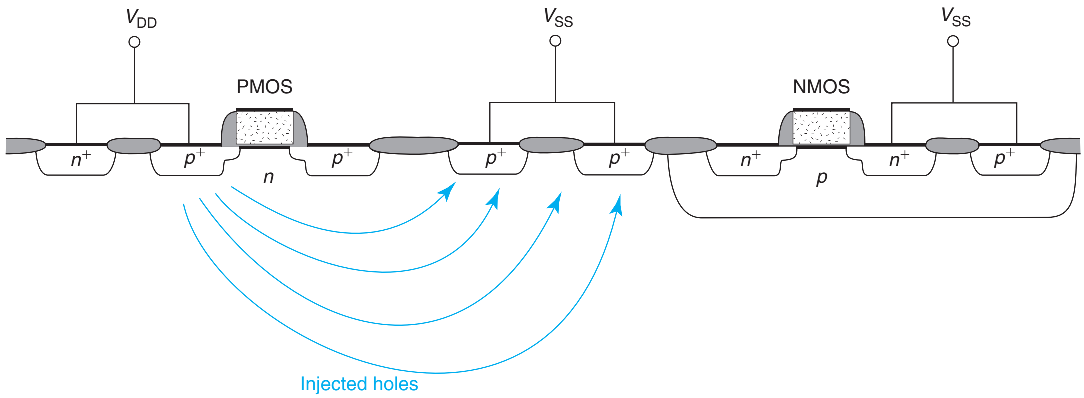


FIGURE 11.70
Schematic of protective guard ring to mitigate latch-up. The p^+ regions extract any holes that might be injected from the p^+ source of the p-MOSFET.

take place. In a bulk MOSFET, as we saw in Section 10.4.2, the body is firmly held at a certain voltage and the holes flow out through the body contact. An SOI MOSFET does not have a body contact and the holes cannot escape through this route. As a result, the body charges up and its potential rises. Eventually, this forward biases the body-source PN junction that allows the holes to be injected into the source.

There are several consequences to this. Depending on how much capacitance hangs from it, charging up of the body might take some time. This also depends on the impact ionization rate and the value of the saturation current of the source-body junction. As a result, the dynamics of the body can be relatively slow and are difficult to predict.

The positive voltage that the body takes also affects the threshold voltage of the MOSFET. As we studied in Section 9.7.2, when a positive voltage is applied to the body of an n-channel MOSFET with respect to its source, V_T shifts negative. For a given gate overdrive, this will cause an increase in the drain current. This is known as the “kink effect” and is illustrated in an actual device in Fig. 11.71(b).

Figure 11.71(a) shows the output characteristics of a 0.18-μm n-channel MOSFET. Superimposed are the body current characteristics on a semilog scale. The body current grows very fast with V_{DS} for values higher than about 1 V. This is consistent with what we studied in Section 10.4.2. On the right are the output characteristics of the same device with the body floating ($I_B = 0$). At about $V_{DS} = 1.3$ V, there is a noticeable step in the drain current. This is what is known as the “kink.” The extra drain current is caused by the negative shift of V_T which is itself the result of the rise of the body potential. At higher V_{DS}, the current flattens out again because once the body-source junction turns on, its I–V characteristics are also exponential and very little additional increase in body voltage is needed to accommodate the extra hole current produced by the increased level of impact ionization that comes with the higher V_{DS}.

In spite of the greater difficulty in modeling MOSFET dynamics, the kink effect is in a way beneficial for logic operation since the transistor delivers more current that allows faster charging and discharging of capacitances downstream.

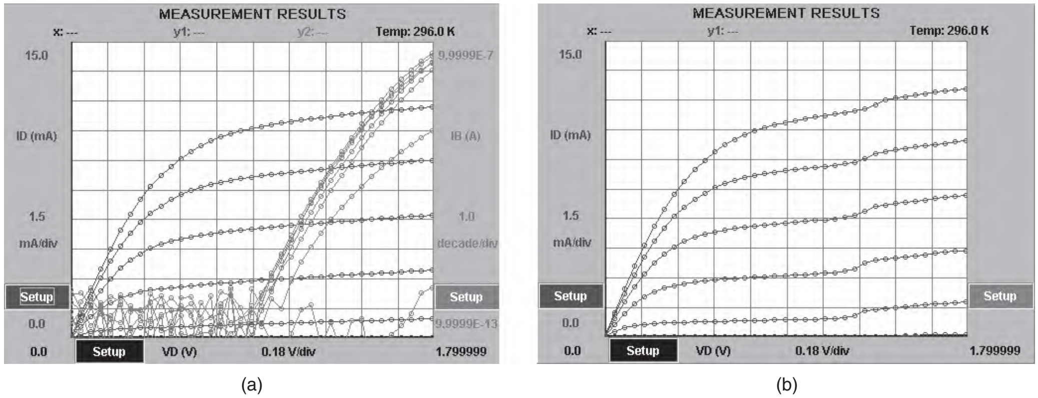


FIGURE 11.71
Output characteristics of 0.18-μm n-channel MOSFET with $V_{BS} = 0$ (a) and the body floating (b). In (a), the body current is also shown (Reprinted with permission from MIT Microelectronics iLab).

An equivalent circuit model description of the situation is shown in Fig. 11.72. Figure 11.72(a) shows the SOI-MOSFET model. Impact ionization is captured through a current generator that is controlled by the ideal drain current I_{Do}. The body source-diode is modeled through an ideal diode. With the body floating, the body voltage V_{BS} will take a value that makes the diode current match the impact ionization current. V_{BS} affects I_{Do} through the V_T dependence on V_{BS}.

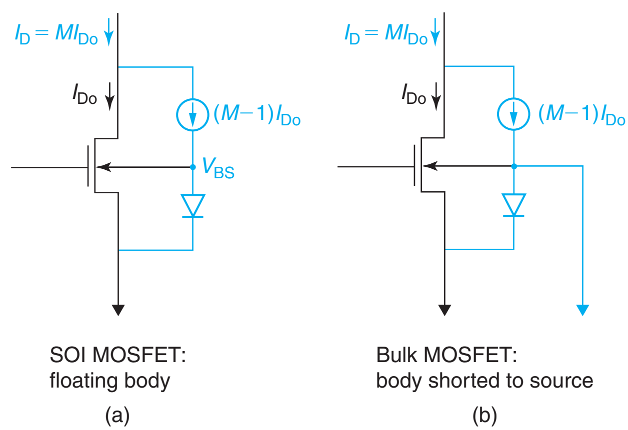


FIGURE 11.72
Equivalent circuit model for a MOSFET required to model the kink effect. (a) A SOI device with a current generator that describes impact ionization and the body-source PN diode. The impact ionization current forward biases the body-source diode. The body floats positive and this gives rise to the kink effect. (b) Bulk MOSFET with the body shorted to the source. In this case, the hole current flows to ground through the body contact.

Figure 11.72(b) shows the bulk MOSFET model with the body tied up to the source. In this case, the impact ionization current flows to ground and V_{BS} remains at zero.

AT11.1.3 MOSFET breakdown and snap-back

MOSFET breakdown in short-channel devices often occurs at a lower voltage than expected from a model based on the physics of impact ionization discussed in Section 10.4.2. In fact, breakdown often presents a complex behavior known as "snap-back," as sketched in Fig. 11.73(a).

If we trace the I–V characteristics for some value of $V_{GS} > V_T$, the expected behavior of the MOSFET is for the drain current to remain quite well saturated up to a certain value of V_{DS} at which the current suddenly rises. This is shown in the blue line in Fig. 11.73(a) and defines a breakdown voltage BV_{DSo}. This is generally the behavior of long-channel MOSFETs. In short-channel devices, it is often seen that at a certain voltage, the I–V characteristics "snap-back" and after reaching a maximum voltage BV_{DS}, the device goes into breakdown at a significantly lower breakdown voltage BV'_{DS}. This is an obvious problem because the device voltage-handling ability is limited by BV'_{DS} as opposed to BV_{DSo}.

The reason for snap-back in short-channel MOSFETs is the presence of two additional parasitic components that are lurking underneath. One is the npn bipolar transistor that is formed by the source, body, and drain regions. The second one is a body resistor associated with the finite resistance of the body and its external contact. An equivalent circuit schematic of the situation is shown in Fig. 11.73(b).

The best way to understand the snap-back behavior is to consider the MOSFET biased with a voltage source on the gate and a current source on the drain. We collect the I–V characteristics by ramping up the drain terminal current, while holding V_{GS} constant. Figure 11.73(a) traces the characteristics for a value of V_{GS} above threshold but a similar result is obtained below threshold.

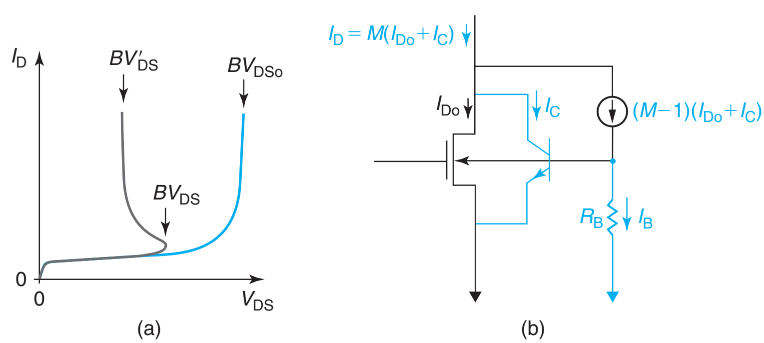


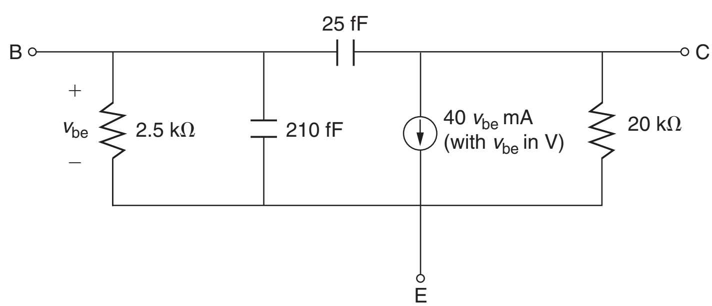
FIGURE 11.73
(a): Typical breakdown characteristics of a short-channel MOSFET (dark gray) versus a long-channel MOSFET (blue). The short-channel device exhibits the phenomenon known as *snap-back*. (b): The equivalent circuit model explains the essence of snap-back. The parasitic npn bipolar transistor and the body resistor create a positive feedback mechanism that makes the device "snap-back" to a lower breakdown voltage.

For V_{GS} above threshold, when the drain current is very small the transistor is biased in the linear regime and V_{DS} is small. As I_D increases, the device quickly saturates and V_{DS} increases rapidly. For a while, the underlying BJT is cut off as the body (base) is shorted to the source (emitter). For high enough V_{DS}, impact ionization starts taking place and a hole current flows through the body and out the body contact. This produces an ohmic drop on the body resistance that biases the body positive with respect to the source. There are two consequences to this. First, the V_T of the MOSFET shifts negatively giving the device extra current drive. In addition, the base–emitter junction of the BJT becomes forward biased and the source (emitter) injects electrons into the body (base) that are collected by the drain (collector). This also tends to increase the overall drain terminal current. Since the external terminal current is fixed, V_{DS} needs to drop so that the overall MOSFET+BJT current adds up correctly to the value that is imposed from the outside. As the terminal drain current increases further, V_{DS} shrinks more, producing the peculiar negative-resistance characteristics that are sketched in Fig. 11.73(a).

Snap-back is an excellent example of the importance of bipolar effects in MOSFETs. Understanding of BJT physics illuminates, for example, the gate length dependence of this effect. Long-channel MOSFETs tend not to suffer from snap-back and attain a higher breakdown voltage [blue line in Fig. 11.73(a)] because the current gain of the transistor is very small. As the gate length shortens, the effective thickness of the base of the BJT decreases and its current gain increases. Snap-back eventually becomes important.

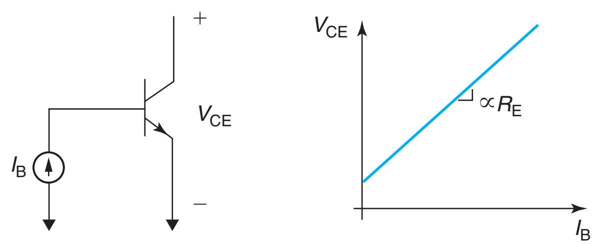
PROBLEMS

11.1 Consider the following equivalent circuit model of a bipolar transistor at a certain bias point (V_{BE}, V_{BC}) in the forward-active regime.



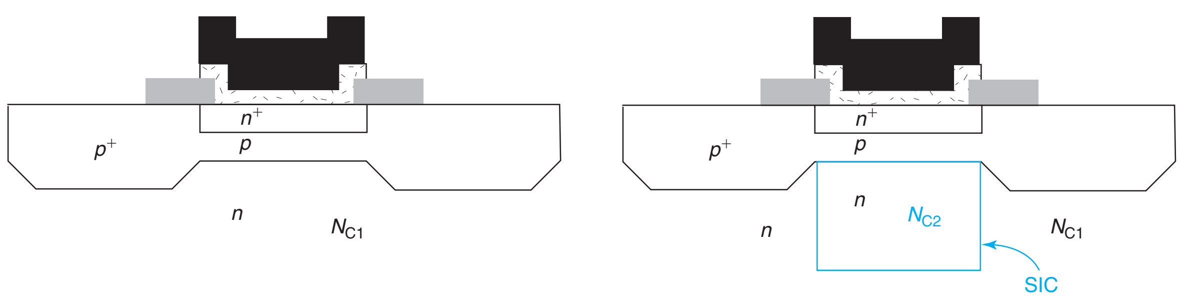
(a) Calculate I_C at which the device is biased.
(b) Calculate the current gain, β_F, of the transistor.
(c) Calculate the short-circuit current-gain cut-off frequency, f_T, of this device at this bias point.
(d) Calculate the Early voltage, V_A, of the transistor.
(e) Derive values for all small-signal elements of the equivalent circuit model of another transistor fabricated side-by-side to the one above but with an emitter stripe length that is twice as long. The device is biased at the same voltages (V_{BE}, V_{BC}) as the above device.

11.2 The emitter resistance is an undesired parasitic resistance that arises from the emitter contact and the quasi-neutral portion of the emitter itself. It hangs from the intrinsic emitter of the BJT. A technique to measure the emitter resistance in a bipolar transistor is shown below. Basically, with the BJT in the common-emitter configuration, a current source is applied to the base–emitter junction, while the collector is left open. If one plots the collector-emitter voltage versus the base current, a straight line results with a slope proportional to R_E. This problem is about showing how this result emerges in an ideal npn bipolar transistor.



(a) In what regime is the bipolar transistor biased in the course of this measurement? Explain.
(b) Using the nonlinear hybrid-π model, derive an expression that gives V_{BC} in terms of V_{BE}.
(c) Derive an expression for V_{CE} as a function of I_B. Derive expressions for the slope and intersection of the straight line that is obtained. Show how R_E can be experimentally obtained.

11.3 This problem is about estimating the improvement in the high-frequency response of a bipolar transistor technology that results from the introduction of a *selectively implanted collector (SIC)*. As the figure indicates below, the conventional transistor design features a uniformly doped collector with a doping level N_{C1}. The SIC design differs only in the introduction of an implantation step that increases the doping level of the collector directly underneath the emitter by a factor of 4, i.e., $N_{C2} = 4N_{C1}$. Nothing else is changed. The conventional design features are as follows: $C_{je} = 1$ pF, $C_{jc} = 0.4$ pF, $A_E = 2\ \mu\text{m}^2$, $A_C = 10\ \mu\text{m}^2$, and $N_{C1} = 2 \times 10^{16}\ \text{cm}^{-3}$. As high-frequency figure of merit, we have selected f_T.



(A) Conventional collector

(B) Selectively implanted collector

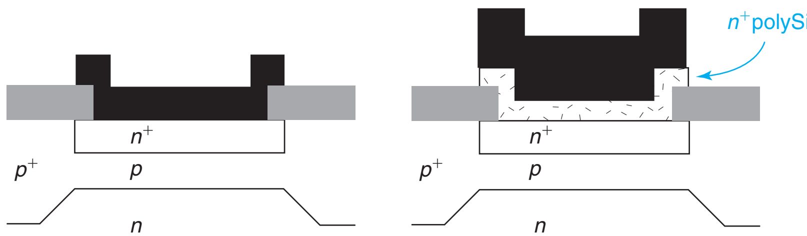
Assuming that τ_F (mainly the base delay) is much smaller than all other time delays of the transistor and that C_{je} and C_{jc} are weakly dependent on the current level, estimate:

(a) f_T of the conventional design at $I_C = 100\ \mu$A.
(b) *Peak* f_T, or highest attainable value of f_T in the conventional design.
(c) f_T of the SIC design at $I_C = 100\ \mu$A.

(d) *Peak* f_T, or highest attainable value of f_T in the SIC design.
(e) Sketch f_T versus I_C in a log–log plot for both transistor designs. Discuss the differences in this plot for both designs.

11.4 **Polysilicon-emitter bipolar transistors**. In the 1970s, a major problem in scaled npn bipolar transistors emerged. Device improvement requires scaling down of the base thickness which, in turn, demands scaling down of the emitter thickness (this is a fabrication issue). It was found that hole injection into the emitter (in the forward-active regime) increased as the emitter was thinned down. Therefore, the gain of scaled bipolar devices could not be improved. The root of the problem was hole recombination at the emitter ohmic contact.

A major breakthrough in Si BJTs took place with the insertion of a thin n^+-polysilicon layer between the emitter metal contact and the n^+-emitter (see figure below). This design was found to suppress the hole injection component of the emitter current and result in high gain.



(A) Metal-contacted emitter (B) PolySi-contacted emitter

Let us understand the nature of the problem and the physics of the solution. In order to focus the discussion, let us define the *emitter saturation current density* J_{hEo} as the pre-exponential term of the hole injection component of the base current density,

$$I_{B1} = A_E J_{hEo}\left(\exp\frac{qV_{BE}}{kT} - 1\right)$$

(a) Derive an expression for J_{hEo} for an n-type emitter with doping N_E and width W_E that is much *shorter* than the hole diffusion length L_E (Fig. A above). Assume an ohmic contact at the emitter surface. Qualitatively plot the excess hole concentration in forward bias across the emitter.
(b) Examine the scaling properties of J_{hEo} in this emitter design. What happens to J_{hEo} when W_E is scaled down?
(c) Now consider another emitter (see Fig. B above) that is identical to the previous one but includes a polysilicon layer of thickness W_p and doping $N_p \neq N_E$ located between the metal contact and the n^+ emitter. Derive the boundary condition for holes at the polysilicon–silicon interface.
(d) Derive an expression for J_{hEo} assuming that the diffusion length in the polysilicon layer, L_p, is much longer than the polysilicon width W_p. Compare the obtained J_{hEo} with the expression above. Which design has improved gain? What are the requirements on N_p with respect to N_E so as to maximize gain?
(e) Examine the scaling properties of J_{hEo} in the polysilicon emitter, as W_E is scaled down (not W_p). What can you conclude from all this?

11.5 In this exercise you have to design the base of an Si npn bipolar transistor. The collector has been designed to deliver a certain breakdown voltage. Its doping level is $N_C = 1 \times 10^{16}$ cm^{-3}. The emitter parameters will be designed to provide a certain gain and you do not have to worry about them now.

Your base design has to deliver the following figures of merit:

Early voltage: $V_A > 30$ V
Base transit time: $\tau_{tB} < 10$ ps
Base sheet resistance: $R_{shb} < 5000\ \Omega/\Box$

The technological constraints are:

Base doping: $N_B < 5 \times 10^{18}$ cm^{-3}
Base thickness: $W_B > 0.08$ μm

Assume that the doping level is flat across the base. Make any other assumptions that you might need. The end product of your design are values for N_B and W_B.

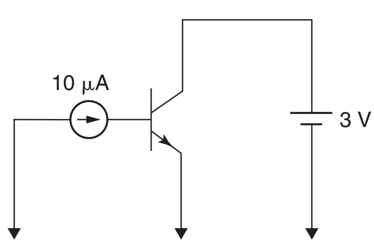
Proceed in order:

(a) What is the constraint to the base parameters imposed by the V_A requirement?
(b) What is the constraint to the base parameters imposed by the τ_{tB} requirement?
(c) What is the constraint to the base parameters imposed by R_{shb} requirement?
(d) Sketch a diagram of possible N_B and W_B values showing the parameter space where all constraints are satisfied.

11.6 A certain bipolar transistor biased at $I_C = 1$ mA and $V_{BC} = -2$ V has an intrinsic delay time $\tau_F = 10$ ps, an emitter–base junction capacitance $C_{je} = 450$ fF, and a collector–base junction capacitance $C_{jc} = 50$ fF. The built-in potentials for the emitter-base and base–collector junctions are $\phi_{biE} = 1$ V and $\phi_{biC} = 0.8$ V, respectively. We also know the nonlinear hybrid-π model parameter $I_S = 1.3 \times 10^{-17}$ A.

(a) Compute the f_T of the device at a bias given by $I_C = 1$ mA and $V_{BC} = -2$ V.
(b) Compute the f_T of the device at a bias given by $I_C = 1$ mA and $V_{BC} = -4$ V.
(c) Compute the f_T of the device at a bias given by $I_C = 0.5$ mA and $V_{BC} = -2$ V.
(d) Compute the f_T of an identical device of double size (double all areas) at a bias given by $I_C = 1$ mA and $V_{BC} = -2$ V.

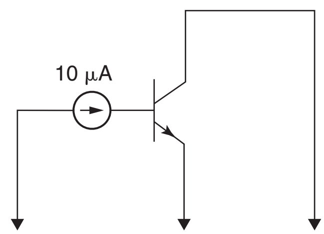
11.7 Consider a bipolar transistor with the following large-signal equivalent circuit model parameters $I_S = 10^{-16}$ A, $\beta_F = 100$, and $\beta_R = 1$. To answer some of the following questions, use the nonlinear hybrid-π model of the transistor presented in class.
Answer the following questions when the device is biased as sketched in the diagram below:



(a) In what regime is the device operating? Explain.
(b) Calculate the collector current, I_C.
(c) Calculate the base–emitter voltage, V_{BE}.

(d) If the doping level in the base is $N_B = 10^{17}$ cm^{-3} and the area of the base–emitter junction is $A_E = 40$ μm^2, calculate the electron velocity in the middle of the quasi-neutral base ($x = W_B/2$, assuming negligible SCRs).

Answer the following questions when the device is biased as sketched in the diagram below:



(e) In what regime is the device operating? Explain.
(f) Calculate the base–emitter voltage, V_{BE}.
(g) Calculate the collector current, I_C.

11.8 An npn bipolar transistor with $\beta_F = 50$ is biased in the forward-active regime with $I_C = 100$ μA at room temperature. The emitter-base junction area is $A_E = 1 \times 2$ μm^2 and the base–collector junction area is $A_C = 2 \times 6$ μm^2. In case you need them, the minority carrier diffusion coefficients in the respective regions are $D_E = 2$ cm^2/s and $D_B = 7$ cm^2/s.

(a) For the indicated bias condition, calculate g_m.
(b) For the indicated bias condition, calculate g_π.
(c) For the indicated bias condition, calculate the gradient of electrons in the base.
(d) For the indicated bias condition, calculate the flux of holes in the emitter.
(e) If without changing anything else structurally, the quasi-neutral base width of this device is *doubled*, how should V_{BE} change in order to maintain the same value of I_C?
(f) When comparing the new design in section (e) to the old design in the previous sections, both operating at the same I_C, how does the total excess minority carrier charge be stored in the base change?
(g) When comparing the new design to the old design operating at the same I_C, how does I_B change?

11.9 You have been asked to provide a design for the doping distribution in the base of an npn bipolar transistor that yields optimum performance. Your company uses an epitaxial technique to grow the base. The grower tells you that a base with a thickness as low as 50 nm can be reproducibly grown. You select this value for W_B. One of the goals for the technology is to provide an intrinsic base sheet resistance of $R_{shb} = 10$ kΩ/□. In your back-of-the-envelope calculation, you ignore the extension of the depletion regions into the base.

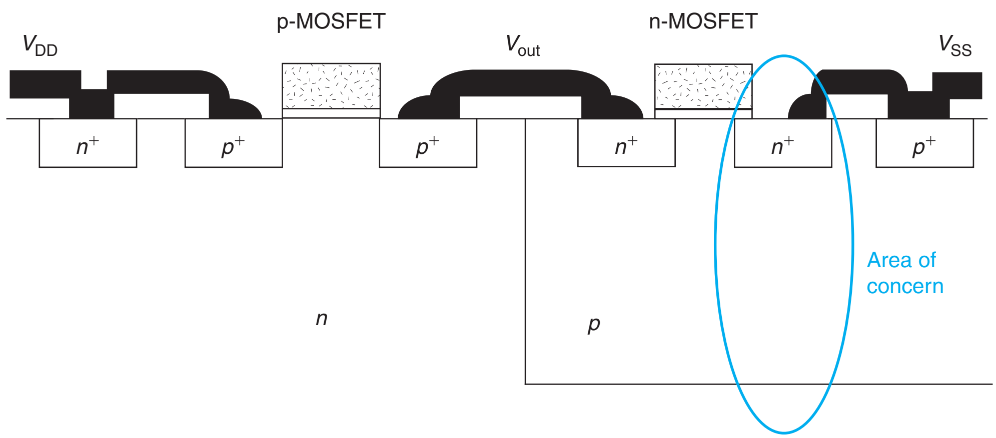
You first select a *uniform impurity profile.*

(a) Calculate the doping level in the base.
(b) Calculate the collector saturation current density (J_S) of this design.
(c) Calculate the base transit time (τ_{tB}) that this design yields.

Now you examine an *exponential impurity profile*. Thermal constraints in the process imply that the doping level cannot change by more than two orders of magnitude across the base. Make judicious choices for average values of carrier mobilities.

(d) Synthesize an exponential profile that meets all the design constraints and minimizes transit time, that is, provide an equation that completely describes the impurity distribution in the base. Take the location of the edge of the emitter-base SCR as $x = 0$. Sketch your selected profile.
(e) Compute the collector saturation current density of this design. Comment on how it compares with the result obtained in part (b) above.
(f) Compute the base transit time of this design. Comment on how it compares with the result obtained in part (c) above. Which impurity profile should you select for the BJT?

11.10 Estimate the forward current gain of the parasitic vertical npn bipolar transistor of the CMOS structure sketched below. The source doping is $N_{n+} = 10^{20}$ cm^{-3}, the p-well doping level is $N_{pw} = 3 \times 10^{17}$ cm^{-3}, and the n-substrate doping level is $N_{sub} = 5 \times 10^{16}$ cm^{-3}. The source junction depth is $x_j = 0.1$ μm, the well depth is $x_w = 2$ μm, and the substrate thickness is 400 μm.

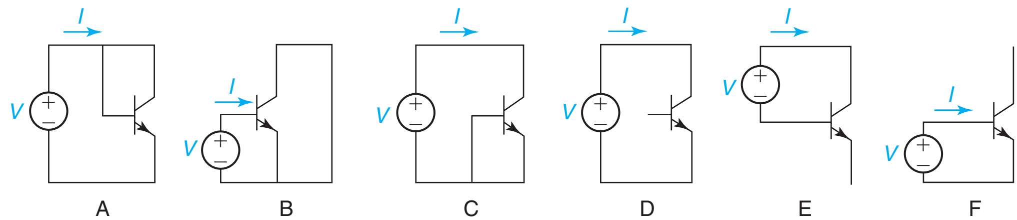


11.11 A certain well-balanced Si bipolar transistor technology features npn and pnp designs with identical structures, that is, $N_E(npn) = N_E(pnp)$, $W_E(npn) = W_E(pnp)$, and so on. For both transistors, the emitter is short and directly contacted with metal.

In the following scenarios, both transistors are biased in the forward-active regime. Answer the following questions:

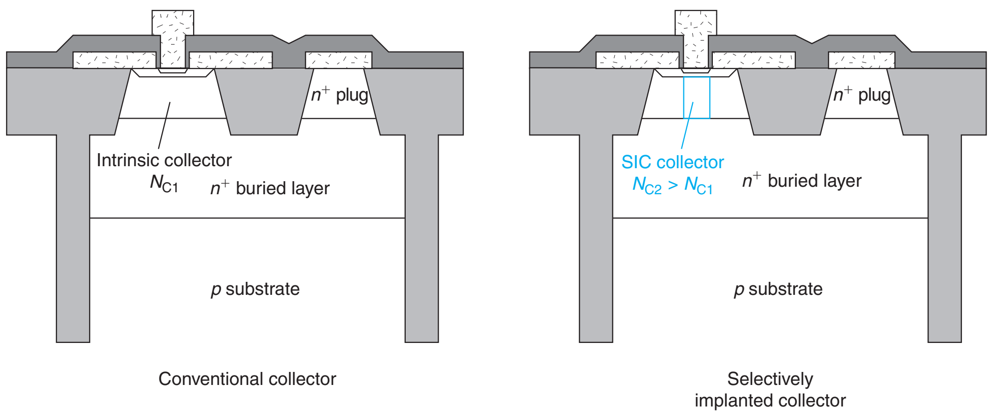
(a) Which device has the highest forward current gain β_F? Why?
(b) Which device has the highest value of output conductance, g_o, when biased at identical values of $|V_{BE}|$ and $|V_{CE}|$? Why?
(c) Which device has the highest excess minority carrier charge in the emitter, $|Q_E|$, when biased at identical $|I_C|$ and $|V_{CE}|$? Why?
(d) For a constant value of $|V_{CE}|$, which device has the highest collector current $|I_C|$ at the onset of high-level injection in the base?
(e) Which device has the highest value of C_π at identical values of $|I_C|$ and $|V_{CE}|$? Why?

11.12 The figure below shows six possible ways of connecting an npn bipolar transistor that may yield a diode-like behavior. Using the ideal *Non-linear Hybrid-π Model*, calculate the I–V characteristics of the two-terminal device in each configuration. Express your result as a function of I_S, β_F, and β_R.



Which of these configurations exhibit diode-like I–V characteristics?

11.13 Consider introducing a *selectively implanted collector* (SIC) to a well-designed bipolar transistor, as sketched below. The doping level of the SIC is higher than the existing collector doping concentration.



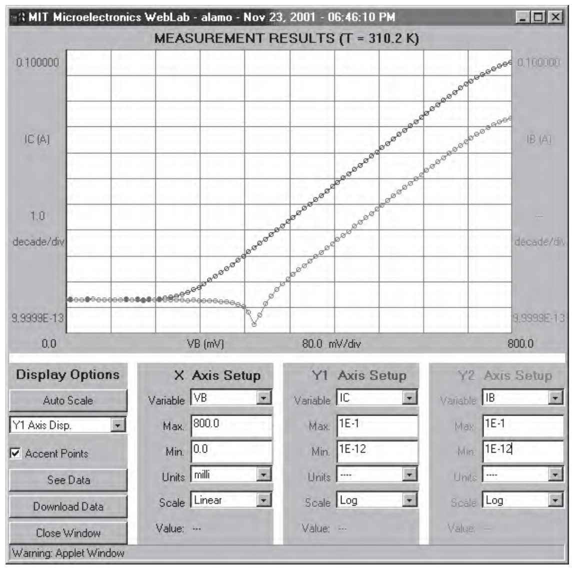
Indicate the impact of introducing the SIC on the following parameters and figures of merit in the forward-active regime at a certain V_{BC} (Select one: *increase* =↑, *decrease* =↓, or *no effect*). Write in a few words the reason for your choice.

Base-collector capacitance, C_{jc}:	↑ ↓ *no effect*
Maximum collector current, $I_{C,max}$:	↑ ↓ *no effect*
Intrinsic delay at low I_C, τ_F:	↑ ↓ *no effect*
f_T at low I_C:	↑ ↓ *no effect*
Maximum possible f_T, f_{Tpk}:	↑ ↓ *no effect*
BV_{CBO}:	↑ ↓ *no effect*
BV_{CEO}:	↑ ↓ *no effect*
Forward Early voltage, V_A:	↑ ↓ *no effect*
Reverse Early voltage, V_B:	↑ ↓ *no effect*
Forward current gain, β_F:	↑ ↓ *no effect*

11.14 This exercise is about reverse engineering a competitor's npn bipolar transistor. On the bench you measure $I_C = 4.2$ mA at $V_{BE} = 0.9$ V. You also extract a forward Early voltage $V_A = 23$ V and a reverse Early voltage $V_B = 8$ V. From the microscope you estimate $A_E = 10\ \mu m^2$ and $A_C = 80\ \mu m^2$. State any assumptions that you need to make.

(a) Estimate the collector doping level, N_C.
(b) Estimate the base doping level, N_B.
(c) Estimate the base width, W_B.
(d) Estimate the f_T of the device at $V_{BE} = 0.9$ V and $V_{BC} = -3$ V.

11.15 The figure below shows a Gummel plot in the forward-active regime obtained from an npn bipolar transistor measured through the MIT Microelectronics iLab. In this measurement, $V_{CE} = 3$ V.

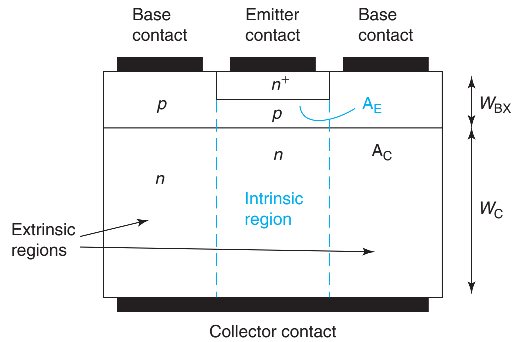


Source: Reprinted with permission from MIT Microelectronics iLab.

This plot reveals several interesting features. This problem is about discussing to what extent these features are captured by the ideal BJT model presented in this chapter. In answering the questions below, use the nonlinear hybrid-π model of the ideal BJT.

(a) For low V_{BE}, I_C is seen to saturate (it does not continue to decrease in an ideal fashion) to a value that we can denote as $I_C(V_{BE} = 0)$. Is this a feature of the ideal BJT? If so, can you derive an expression for $I_C(V_{BE} = 0)$ based on the nonlinear hybrid-π model?
(b) For low V_{BE}, I_B is seen to suddenly change sign and then saturate to a value that we can denote as $I_B(V_{BE} = 0)$. The absolute of this value seems close to $I_C(V_{BE} = 0)$. Is this a feature of the ideal BJT? If so, can you derive an expression for $I_B(V_{BE} = 0)$ based on the nonlinear hybrid-π model? Can you derive an expression for the value of V_{BE} at which I_B changes sign, $V_{BE}(I_B = 0)$?
(c) Now let us put numbers to this. This BJT is reasonably well described by the following parameters: $\beta_F = 160$, $I_S = 3.4 \times 10^{-15}$ A, $\beta_R = 1$, and $V_A = 135$ V. Using these parameters, estimate the expected value of $I_C(V_{BE} = 0)$, $I_B(V_{BE} = 0)$, and $V_{BE}(I_B = 0)$. Comment on agreement or discrepancies with the values that can be extracted from the figure above.

11.16 This problem is about deriving expression for β_R and τ_R for an ideal BJT. Consider the cross section of an ideal BJT where the thickness of the base in the extrinsic region is labeled as W_{BX} and the thickness of the collector is labeled as W_C. Use the usual symbols for all other parameters.



Consider this BJT ideal in all regards as defined in this chapter. This BJT is biased in the reverse regime.

(a) In the reverse regime, identify and derive expressions for all components of I_B in terms of structural parameters of the device and V_{BC}.
(b) Derive an expression for β_R in terms of the structural parameters of the device.
(c) Evaluate the order of β_R under the following assumptions: $N_B \simeq 10N_C$, $A_C \simeq 10A_E$, $W_C \simeq 10W_B$, $W_{BX} \simeq 2W_B$, and $D_B \simeq D_C$.
(d) Identify all components of Q_R, the minority carrier stored charge everywhere in the device in the reverse regime. Express your result in terms of V_{BC}, the structural parameters of the device, and the appropriate minority carrier transit times defined in the table below:

Region	**Minority carrier transit time**
Emitter	τ_{tE}
Base	τ_{tB}
Extrinsic base	τ_{tBX}
Collector	τ_{tC}

(e) Derive an expression for τ_R in terms of the appropriate minority carrier transit times and other structural parameters of the device.

11.17 At the 2002 IEDM, a world record SiGe HBT (heterojunction bipolar transistor) was presented. The device featured an $f_T = 350$ GHz, almost twice as high as the best result presented at the 2001 IEDM. This problem is about studying in some detail the reported results. To the first order, you can think of this transistor as a regular Si bipolar junction transistor.

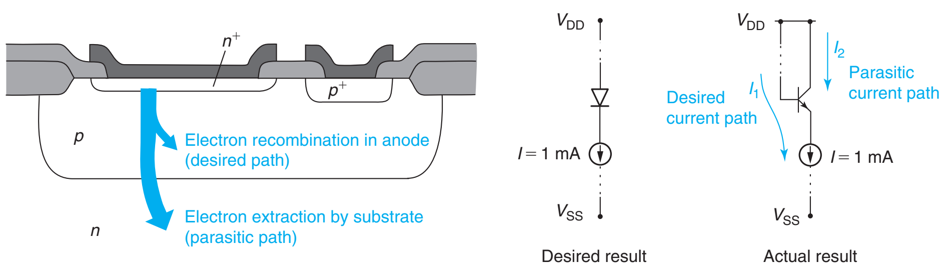
The following table gives the value of f_T at two different collector current levels, as reported in the paper ($V_{CB} = 1$ V):

I_C (mA)	f_T (GHz)
0.2	70
6	350

$I_{Cpk} = 6$ mA is the collector current at which f_T peaks. In the paper, the emitter-base junction area is listed as $A_E = 0.12 \times 2.5\ \mu m^2$. The current gain is listed as $\beta_F = 2300$.

(a) Estimate the collector doping level.
(b) Estimate the sum of the emitter-base and base–collector junction capacitances.
(c) Estimate the intrinsic delay.
(d) Estimate the quasi-neutral base width.
(e) At $I_C = I_{Cpk}$, estimate the total minority carrier charge in the device.

11.18 The figure below is a modified version of Fig. 6.44 in the notes. It describes the isolation problem when implementing PN diodes in a CMOS process. Depicted is a N^+P diode implemented using an n-MOSFET. The source/drain n^+ diffusion is used as the cathode. The p-type well is used as the anode. The p^+ body contact is used as the anode contact. Surrounding the p-type anode is the n-type substrate that is connected to the most positive voltage of this circuit, V_{DD}.



The problem with this "diode" is that it is actually a parasitic vertical npn bipolar transistor. Depending on the circuit environment, the actual current path can be very different from the desired current path.

An example is shown on the right where this "diode" is connected somewhere halfway between V_{DD} and V_{SS}, where V_{DD} is the positive rail of the circuit and V_{SS} is the negative rail. Immediately below the diode is a current source pulling down 1 mA. Ideally, this current should entirely flow from anode to cathode in the diode. Since this is in reality a bipolar transistor, some of this current can be diverted to the substrate through the bipolar effect and flow through a separate path, I_2. As a result, the current through the desired path, I_1, can be less than the intended value of 1 mA. The end result might be circuit misfunction. Computing these currents is the goal of this problem.

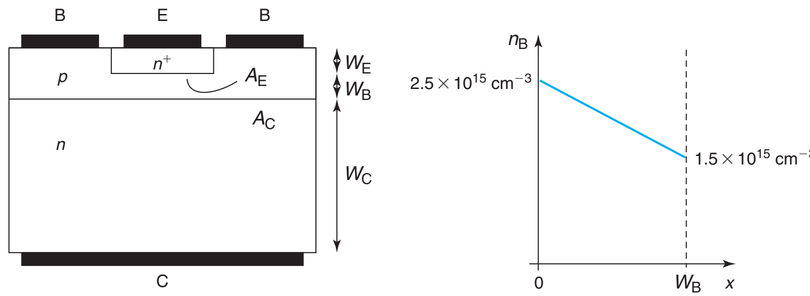
Here are specs for the three key regions:

Region	Type	Doping level (cm^{-3})	Thickness (μm)
Cathode	n^+	10^{19}	0.05
Anode	p	10^{17}	0.2
Substrate	n	10^{16}	400

The area of the N^+P junction is 100 μm^2.

(a) Under the conditions indicated above, identify the emitter, the base, and the collector of this bipolar transistor. In what regime is this bipolar transistor biased?
(b) Estimate the current gain of the bipolar transistor in the configuration indicated above.
(c) Estimate I_1 and I_2.
(d) Quantitatively sketch the excess minority carrier concentration across the base of the parasitic BJT.

11.19 Consider an ideal bipolar junction transistor, sketched below on the left, that is biased such that the electron carrier profile shown below on the right exists across the base ($x = 0$ is the base–emitter junction):

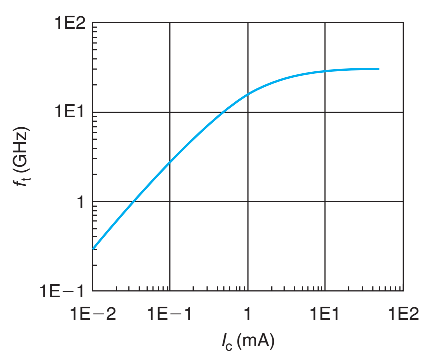


Values for the various device parameters are as follows: $N_E = 10^{19}$ cm^{-3}, $N_B = 10^{18}$ cm^{-3}, $N_C = 10^{17}$ cm^{-3}, $W_E = 0.1$ μm, $W_B = 0.1$ μm, $W_C = 0.5$ μm, $A_E = 5$ μm^2, and $A_C = 30$ μm^2.

For this problem, neglect the extent of the depletion regions across the junctions. That is, assume that the extent of the quasi-neutral regions in the emitter, base, and collector are, respectively, W_E, W_B, and W_C.

(a) Calculate V_{BE}.
(b) Calculate V_{BC}.
(c) In what regime is this device biased? Explain.
(d) Calculate the total minority carrier charge in the emitter Q_E.
(e) Calculate the total minority carrier charge in the base Q_B.
(f) Calculate the total minority carrier charge in the collector Q_C.

11.20 Consider an ideal bipolar transistor with the following $f_t - I_C$ characteristics at a certain V_{BC}:



(a) Estimate the sum of the emitter and collector junction capacitances, $C_{je} + C_{jc}$.
(b) Estimate the intrinsic delay, τ_F.
(c) For $I_C = 1$ mA, the transistor above has an $f_T \simeq 16$ GHz. At this bias point:

- Estimate the value of g_m.
- Estimate the value of C_π.
- Estimate the transit time of electrons through the base.

(d) For $I_C = 1$ mA, the transistor above has an $f_T \simeq 16$ GHz. What would the value of f_T be at the same value of I_C for a transistor with twice as long an emitter stripe length, L_E?
(e) For $I_C = 1$ mA, the transistor above has an $f_T \simeq 16$ GHz. What would the value of f_T be at the same value of V_{BE} for a transistor with twice as long an emitter stripe length, L_E?

11.21 This problem is about comparing basic figures of merit of npn and pnp bipolar transistors. Consider two ideal npn and pnp bipolar transistors with identical geometry and doping distributions. That is, the doping levels in the respective regions are identical and all the dimensions are identical. For example $W_B(npn) = W_B(pnp)$, $N_B(npn) = N_B(pnp)$, and so on. The only difference between these two transistors is that at any doping level, the mobilities of the respective carriers are such that

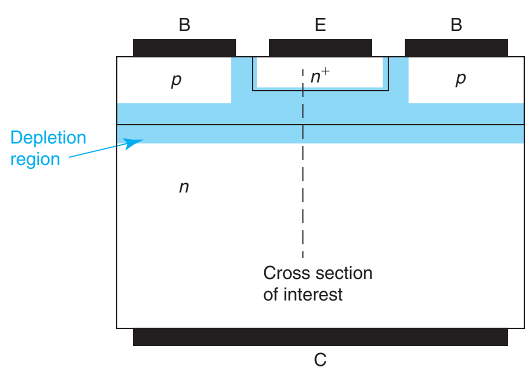
$$\left.\frac{\mu_e}{\mu_h}\right|_{N_A=N_D} = 2$$

This means, for example, that the mobility of electrons in the emitter in the npn is twice as high as the mobility of holes in the emitter of the pnp, or $\mu_{eE}(npn) = 2\mu_{hE}(pnp)$. It also means that the mobility of electrons in the emitter in the pnp is twice as high as the mobility of the holes in the emitter of the npn, or $\mu_{eE}(pnp) = 2\mu_{hE}(npn)$.

For this problem, assume complete symmetry in all other physical parameters, for example, $N_c = N_v$.

(a) What is the ratio of the base–collector junction capacitance in equilibrium in the npn with respect to the pnp?
(b) What is the ratio of the base transit time in the npn with respect to the pnp?
(c) What is the ratio of the Early voltage in the npn with respect to the pnp?
(d) What is the ratio of the current gain in the npn with respect to the pnp?
(e) What is the ratio of the collector saturation current in the npn with respect to the pnp?
(f) What is the ratio of f_T at a given low value of the collector current in the npn with respect to the pnp?
(g) What is the ratio of f_T at a given high value of the collector current in the npn with respect to the pnp?

11.22 An npn bipolar transistor was misfabricated with a base doping level much lower than the designed value. Instead of a doping sequence $N_E \gg N_B \gg N_C$, the transistor features $N_E \gg N_B = N_C$, where N_C is the design value. A consequence of this is that the base is in punchthrough in thermal equilibrium. This means that the two depletion regions associated with the base–emitter and base–collector junctions have merged in the base and there is no quasi-neutral base left. See the figure below.

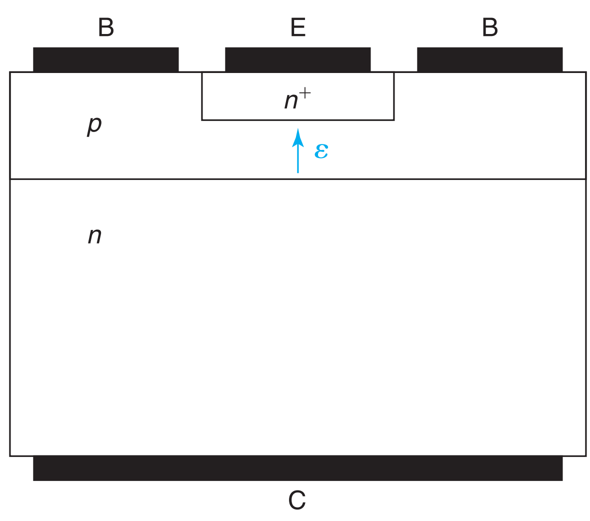


This problem is about solving the electrostatics of this situation in thermal equilibrium ($V_{BE} = V_{BC} = 0$) along the cross section indicated above. For this, the total thickness of the intrinsic base is W_B. The extent of the depletion region into the emitter and collector regions are, respectively, x_E and x_C.

(a) Sketch the volume charge density along the indicated cross section in space. Indicate key values.
(b) Sketch the electric field along the indicated cross section in space. Label key values.
(c) Sketch the electrostatic potential along the indicated cross section in space. Label key values.
(d) Derive an expression for the potential build up (built-in potential) across the structure from emitter to collector. Express in terms of fundamental constants, the doping levels and/or the thickness of the relevant regions, W_B, x_E, and x_C.
(e) Derive an expression for the height of the energy barrier that the electrons in the emitter face on the base side. Express in terms of fundamental constants, the doping levels and/or the thickness of the relevant regions, W_B, x_E, and x_C.
(f) Derive a system of equations that allows the calculation of the depletion region into the emitter and the collector, x_E and x_C.

11.23 This problem is to study the impact of introducing a strong electric field in the base of a bipolar transistor so as to enhance electron transport through the base and in this way boost transistor performance. A strong electric field can be created by means of a ramp of Ge composition. This is the essence of an SiGe heterojunction bipolar transistor.

Consider an ideal npn bipolar transistor in which a strong and constant electric field, $\mathcal{E}$, is introduced in the base with the sign indicated in the figure below.



Consider the device biased in the forward-active regime with $V_{BC} = 0$. In this regime, if you solve the problem, you will find that when the electric field is strong, the excess electron profile is given by

$$n'(x) = n'(0)\left[1 - e^{(x-W_B)/L}\right]$$

where $n'(0)$ is the excess electron concentration at the edge of the quasi-neutral base with the base–emitter junction and is set by the base–emitter voltage. The rest of the symbols have their usual meaning. L is a characteristic length for this problem, which is given by

$$L = \frac{kT}{q\mathcal{E}}$$

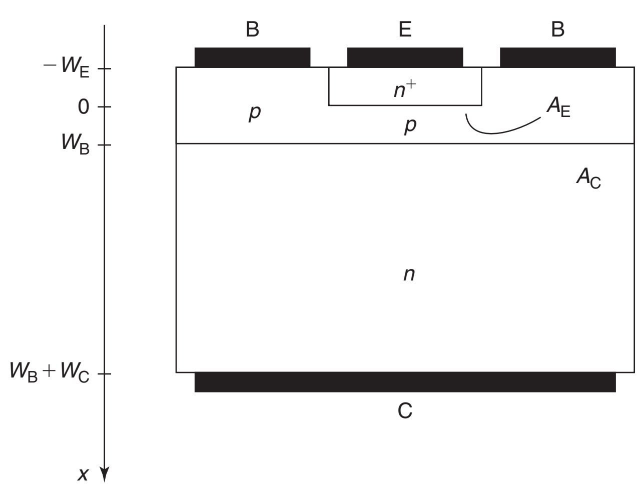
A strong electric field means that $W_B \gg L$.

(a) Sketch the profile of excess electron concentration across the quasi-neutral base.
(b) Derive an expression for the collector current of this bipolar transistor. Express your result in terms of $n'(0)$.
(c) Derive an expression for the stored electron charge in the base of this bipolar transistor. Express your result in terms of $n'(0)$.
(d) Derive an expression for the transit time of electrons across the base. Compare this transit time with that of a purely diffusive situation. Which one is shorter?
(e) Formulate a differential equation for electron transport in the base of this heterojunction bipolar transistor that leads to the expression of $n'(x)$ above.
(f) Solve this differential equation and obtain $n'(x)$. State and justify the assumptions that you need to make.

11.24 Assume a well-designed bipolar transistor. In a redesign, and without changing anything else, the base doping level is *increased*. Indicate in what way the following parameters change. Circle your choice and give reasons for it. The transistor is biased in the forward-active regime.

- Current gain, β_F: *increase*, *decrease*, *no change*
- f_T at low collector current: *increase*, *decrease*, *no change*
- BV_{CBO}: *increase*, *decrease*, *no change*
- BV_{CEO}: *increase*, *decrease*, *no change*
- Output conductance, g_o, at a certain I_C: *increase*, *decrease*, *no change*

11.25 Consider an ideal npn bipolar transistor (as defined in class) at room temperature. Relevant parameters for this BJT are as follows: $N_E = 10^{20}$ cm^{-3}, $N_B = 10^{18}$ cm^{-3}, and $N_C = 10^{17}$ cm^{-3}. The junction areas are $A_E = 10$ μm^2 and $A_C = 100$ μm^2.



This transistor is biased in the forward-active regime with $I_C = 1$ mA at $V_{BE} = 0.85$ V.

Estimate the magnitude of the electron velocity at the locations indicated below. Assume nondegenerate statistics. Make judicious approximations and state them.

(a) $x = -W_E/2$ (middle of the quasi-neutral emitter).
(b) $x = 0$ (metallurgical junction).
(c) $x = W_B/2$ (middle of the quasi-neutral base).
(d) $x = W_B + W_C$ (collector contact). Assume this is an ideal contact.

Appendix A

Table of Fundamental Physical Constants

Physical constant*	Symbol	Value SI units	Value "microelectronic" units
Boltzmann constant	k	1.38×10^{-23} J/K	8.62×10^{-5} eV/K
Electron charge	q	1.60×10^{-19} C	1.60×10^{-19} C = 1 e
Electron rest mass	m_o	9.11×10^{-31} kg	5.69×10^{-16} eV $\cdot$ s^2/cm^2
Planck's constant	h	6.63×10^{-34} J $\cdot$ s	4.14×10^{-15} eV $\cdot$ s
Speed of light in vacuum	c	3.00×10^{8} m/s	3.00×10^{10} cm/s
Permittivity of vacuum	ϵ_o	8.85×10^{-12} F/m	8.85×10^{-14} F/cm

*Values of fundamental constants taken from E. R. Cohen and B. N. Taylor, *Physics Today, BG9 (August 1994)*.

Conversion between both systems

$$1 \text{ eV} = 1.60 \times 10^{-19} \text{ J}$$

$$1 \text{ kg} = 6.24 \times 10^{14} \text{ eV} \cdot \text{s}^2/\text{cm}^2$$

Appendix B

Table of Important Material Parameters of Si, GaAs, and SiO_2 at 300 K

Physical parameter	Symbol	Si value	GaAs value	SiO_2 value	Units
Lattice constant	a	0.543	0.565	—	nm
Interatomic distance		0.235	0.245	—	nm
Atomic density	N_a	5.0×10^{22}	4.4×10^{22}	2.2×10^{22}	cm^{-3}
Density	d	2.33	5.32	2.0 – 2.3	$g \cdot cm^{-3}$
Linear thermal expansion coefficient	α	2.59×10^{-6}	5.73×10^{-6}	5.0×10^{-7}	K^{-1}
Thermal conductivity	κ	1.3	0.52	0.01	W/cm · (K)
Dielectric constant	ϵ	11.7	12.9	3.9	-
Refractive index	n	3.44	3.3	1.46	-
Electron affinity	χ	4.04	4.07	1.0	eV
Bandgap energy	E_g	1.124	1.422	9.0	eV
DOS electron effective mass	m^*_{de}	1.09 m_o	0.066 m_o		$eV \cdot s^2/cm^2$
DOS hole effective mass	m^*_{dh}	1.15 m_o	0.52 m_o		$eV \cdot s^2/cm^2$
Effective DOS of the conduction band	N_c	2.86×10^{19}	4.21×10^{17}		cm^{-3}
Effective DOS of the valence band	N_v	3.10×10^{19}	9.51×10^{18}		cm^{-3}
Intrinsic carrier concentration	n_i	1.07×10^{10}	2.25×10^{6}		cm^{-3}
Optical phonon energy	E_{opt}	0.063	0.035		eV
Conductivity electron effective mass	m^*_{ce}	0.28 m_o	0.070 m_o		$eV \cdot s^2/cm^2$
Conductivity hole effective mass	m^*_{ch}	0.41 m_o	0.44 m_o		$eV \cdot s^2/cm^2$
Phonon-limited electron mobility	μ_e	1430	8000		$cm^2/V \cdot s$
Phonon-limited hole mobility	μ_h	480	320		$cm^2/V \cdot s$
Electron saturation velocity	v_{esat}	1.0×10^{7}	$1 - 1.5 \times 10^{7}$		cm/s
Hole saturation velocity	v_{hsat}	6.0×10^{6}	5.0×10^{6}		cm/s
Optical G/R rate coefficient	k_{rad}	2.0×10^{-15}	7.2×10^{-10}		cm^3/s
Electron–electron Auger coefficient	k_{eeh}	1.8×10^{-31}	1.8×10^{-31}		cm^6/s
Hole–hole Auger coefficient	k_{ehh}	9.5×10^{-32}	4.0×10^{-30}		cm^6/s
Impact ionization threshold energy	E_{ii}	1.12	1.72		eV

Appendix C

Table of Acronyms

Acronym	Meaning
AC	Alternating current
BJT	Bipolar junction transistor
CAD	Computer-aided design
CCD	Charge-coupled device
CET	Capacitance equivalent thickness
CLM	Channel length modulation
CMOS	Complementary metal–oxide–semiconductor
DC	Direct current
DOS	Density of states
DRAM	Dynamic random access memory
ECL	Emitter-coupled logic
EEPROM	Electrically eraseable programmable read-only memory
EOT	Equivalent oxide thickness
ESD	Electrostatic discharge
FET	Field-effect transistor
HEMT	High-electron mobility transistor
IC	Integrated circuit
LED	Light-emitting diode
MESFET	Metal–semiconductor field-effect transistor
MOSFET	Metal–oxide–semiconductor field-effect transistor
ppm	Part per million
QNR	Quasi-neutral region
SCR	Space-charge region
SOI	Silicon on insulator
SRAM	Static random access memory
STI	Shallow trench isolation
TTL	Transistor–transistor logic
VLSI	Very large scale integration
XPS	X-ray photoelectron spectroscopy

Appendix D

Table of Taylor Series Expansions

Function	Expansion	Around
$\sin x$	$x - \frac{x^3}{6} + \frac{x^5}{120}$	$x = 0$
$\cos x$	$1 - \frac{x^2}{2} + \frac{x^4}{24}$	$x = 0$
$\tan x$	$x + \frac{x^3}{3} + \frac{2x^5}{15}$	$x = 0$
$\sinh x$	$x + \frac{x^3}{6} + \frac{x^5}{120}$	$x = 0$
$\cosh x$	$1 + \frac{x^2}{2} + \frac{x^4}{24}$	$x = 0$
$\tanh x$	$x - \frac{x^3}{3} + \frac{2x^5}{15}$	$x = 0$
$\coth x$	$\frac{1}{x} + \frac{x}{3} - \frac{x^3}{45}$	$x = 0$
e^x	$1 + x + \frac{x^2}{2}$	$x = 0$
$\frac{1}{1-x}$	$1 + x + x^2$	$x = 0$
$\frac{1}{1+x}$	$1 - x + x^2$	$x = 0$
$\sqrt{1+x}$	$1 + \frac{x}{2} - \frac{x^2}{8}$	$x = 0$
$\frac{1}{\sqrt{1+x}}$	$1 - \frac{x}{2} + \frac{3x^2}{8}$	$x = 0$
$\ln(1+x)$	$x - \frac{x^2}{2} + \frac{x^3}{3}$	$x = 0$
$(1+x)^p$	$1 + px + \frac{p(p-1)}{2}x^2$	$x = 0$

Appendix E

Analytical Expressions for Important Material Parameters of Si at 300 K

E.1 BANDGAP VERSUS TEMPERATURE

The temperature dependence of the bandgap of Si is graphed in Fig. 2.20. This is well represented by an equation of the form:

$$E_g(T) = E_g(0) - \frac{\alpha T^2}{T + \beta} \tag{E.1}$$

where $E_g(0)$ is the bandgap at 0 K and α and β are constants. For Si, $E_g(0) = 1.17$ eV, $\alpha = 3.83 \times 10^{-4}$ eV/K and $\beta = 453$ K.

E.2 BANDGAP NARROWING VERSUS DOPING LEVEL

The bandgap of Si narrows as the doping level increases. This is graphed in Fig. 2.23. Bandgap narrowing in Si has been found to depend only on the doping level and not on the doping type. A good working expression is

$$\Delta E_g = A \ln \frac{N}{B} \tag{E.2}$$

where $A = 22$ meV and $B = 7 \times 10^{17}$ cm^{-3}. In Si, to the first order, bandgap narrowing appears to be temperature independent.

In the Si literature, it is common to utilize an *effective* or *apparent* bandgap narrowing, ΔE_g^{app}, that lumps the impact of degenerate statistics together with the true bandgap shrinkage. It is common to use fits similar to Eq. (E.2) for this hybrid parameter. While with proper care this procedure works, it is not recommended. The main reason for this is that while ΔE_g is temperature independent, ΔE_g^{app} is not. This is because the impact of the statistics depends on temperature. Equation (2.81) makes this explicit and therefore one is less likely to forget this important dependence.

E.3 CARRIER LIFETIME VERSUS DOPING LEVEL

The carrier lifetime is a strong function of doping level. For Si, typical values at room temperature are shown in Fig. 3.17. For low and moderate doping levels, there is a wide scatter in the data as the carrier lifetime depends on process details, original material quality, and other factors.

Fits to the data are as follows. For n-type Si:

$$\frac{1}{\tau} = 7.8 \times 10^{-13} N_D + 1.8 \times 10^{-31} N_D^2 \tag{E.3}$$

For p-type Si:

$$\frac{1}{\tau} = 3.5 \times 10^{-13} N_A + 9.5 \times 10^{-32} N_A^2 \tag{E.4}$$

With N_A and N_D in cm^{-3} and τ in s.

E.4 MOBILITY VERSUS DOPING LEVEL

The carrier mobility is a strong function of doping level. For Si at room temperature, carrier mobilities are graphed in Fig. 4.3. For majority carriers, the carrier mobility dependence with doping level is quite tight. For minority carriers, it is less well known but it should also be very tight. The mobility depends to a small extent on the specific doping type, whether As, P, or Sb for n-type, for example. It is also affected by compensation.

A reasonable expression that does a relatively good job in describing the evolution of carrier mobility with doping level is the following:

$$\mu = \mu_1 + \frac{\mu_2}{1 + \left(\frac{N}{N_o}\right)^{\alpha}} \tag{E.5}$$

Values for μ_1, μ_2, N_o, and α for the four cases graphed in Fig. 4.3 are given in Table E.1.

Table E.1
Parameters for mobility expressions for Si at room temperature

Parameter	Value				Units
	n-type		p-type		
	μ_e	μ_h	μ_e	μ_h	
μ_1	80	130	232	60	$cm^2/V \cdot s$
μ_2	1360	349	1202	420	$cm^2/V \cdot s$
N_o	8×10^{16}	8×10^{17}	8×10^{16}	2×10^{17}	cm^{-3}
α	0.75	1.25	0.9	0.75	No units

E.5 IMPACT IONIZATION COEFFICIENT VERSUS ELECTRIC FIELD

The impact ionization coefficients for electrons and holes as a function of the electric field are graphed in Fig. 4.27. They follow an approximate expression of the form:

$$\alpha \simeq A \exp\left(-\frac{B}{|\mathcal{E}|}\right) \tag{E.6}$$

Appropriate values for A and B for electrons in Si at room temperature are $A = 7.03 \times 10^5$ cm^{-1} and $B = 1.23 \times 10^6$ V/cm. For holes, $A = 2.0 \times 10^6$ cm^{-1} and $B = 1.97 \times 10^6$ V/cm.

E.6 INVERSION LAYER MOBILITY VERSUS EFFECTIVE ELECTRIC FIELD

We saw in Section 10.1.1 that the inversion layer mobility in an MOS structure is a strong function of the sheet carrier concentration. We also discussed how for high enough sheet carrier concentrations, the evolution of the inversion layer mobility can be well described by a *universal mobility curve* that depends on an *effective vertical field*. Experimental data at 300 K and a fit to the data are shown in Fig. 10.4. The fit does a good job in describing the data. In this fit, the inversion mobility is given by

$$\mu_e = \frac{\mu_o}{1 + \left(\frac{|\mathcal{E}_{eff}|}{\mathcal{E}_o}\right)^{\theta}} \tag{E.7}$$

Table E.2 lists the values of the parameters in this equation for electrons and holes.

Table E.2
Parameters for universal mobility expressions for Si inversion layers at room temperature

Parameter	Value		Units
	μ_e	μ_h	
μ_o	880	260	cm^2/V · s
$\mathcal{E}_o$	4.2×10^5	2.35×10^5	cm/s
θ	1.05	0.94	No units

E.7 DRIFT VELOCITY VERSUS ELECTRIC FIELD

The drift velocity depends on the electric field. A simple phenomenological model was given in Eq. (4.13). A better description valid for electrons and holes drifting along the <111> direction in undoped Si is given by

$$\left|v^{drift}\right| = v_m \frac{\frac{\mathcal{E}}{\mathcal{E}_c}}{[1 + (\frac{\mathcal{E}}{\mathcal{E}_c})^{\beta}]^{1/\beta}} \tag{E.8}$$

Table E.3

Parameters for model of electron and hole velocity versus electric field in undoped Si at room temperature. The field is along the <111> direction

Parameter	Value v_e^{drift}	Value v_h^{drift}	Units
v_m	1.07×10^7	8.34×10^6	cm/s
$\mathcal{E}_c$	6.98×10^3	1.80×10^4	V/cm
β	1.11	1.21	No units

Values for the various parameters in this equation for electrons and holes are given in Table E.3.

Appendix F

Solutions to Selected Problems

Many of the problems in this book involve reading off values for relevant material parameters from charts distributed throughout the text. This will often introduce some ambiguities in the values that are used. Also, in some cases, there is some scatter in the available data and the reader must exercise appropriate judgment. Since several of the calculations are sensitive to the specific choice of parameter values, the results obtained might differ somewhat from those given below.

Chapter 1

1.1 (a) $E_K = 3.52 \times 10^{20}$ eV, $\lambda = 2.62 \times 10^{-32}$ cm. (b) $E_K = 3.60 \times 10^{24}$ eV, $\lambda = 1.67 \times 10^{-36}$ cm. (c) $E_K = 2.85 \times 10^{2}$ eV, $\lambda = 7.27 \times 10^{-9}$ cm. In all cases, the de Broglie wavelength is much smaller than the environment in which these objects are placed. Hence, a particle-type description is appropriate in the three examples.

1.2 $\lambda = 0.68$ nm, $\nu = 4.44 \times 10^{17}$ Hz.

1.3 For the situation on the left, λ_1 is absorbed in the GaAs wafer while λ_2 and λ_3 are absorbed in the Ge wafer. For the situation on the right, all three wavelengths are absorbed in the Ge wafer.

1.4 $T = 1.2 \times 10^{-9}$ K. $\lambda_{300\,K} = 3.0 \times 10^{-9}$ cm.

1.5 (a) $n_{ph} = 4.2 \times 10^{21}$ m$^{-2} \cdot$ s$^{-1} \cdot$ μm^{-1}. (b) $N_{ph} = 8.4 \times 10^{19}$ m$^{-2} \cdot$ s^{-1}. (c) $E_g = 2.1$ eV.

1.9 $\lambda = 0.51$ μm.

1.10 $E - E_F = -0.045$ eV.

1.11 $N_{ph} = 8.5 \times 10^{-11}$ cm^{-2}.

1.12 (a) From top to bottom, InGaP, InGaAs, and Ge. (b) InGaP absorbs light with $\lambda < 0.67$ μm. InGaAs absorbs light with $0.67 < \lambda < 0.89$ μm. Ge absorbs light with $0.89 < \lambda < 1.77$ μm.

Chapter 2

2.5 (a) $p_o = 1.0 \times 10^{17}$ cm^{-3}. (b) $n_o = 1.1 \times 10^{3}$ cm^{-3}. (c) $E_c - E_F = 0.98$ eV, $E_F - E_v = 0.15$ eV. (d) $f(E_c) = 3.7 \times 10^{-17}$. (e) $1 - f(E_v) = 3.1 \times 10^{-3}$.

2.6 $T = 784$ K. MB still holds at this temperature. MB fails at $T = 2850$ K but by then Si has melted.

2.7 $\lambda = 3.0 \times 10^{-7}$ cm.

2.8 (a) $N_c = 5.4 \times 10^{17}$ cm^{-3}, $N_v = 1.3 \times 10^{19}$ cm^{-3}, $n_i = 1.4 \times 10^7$ cm^{-3}, $E_i = 0.041$ eV above the middle of the gap. (b) $n_o = 2.5 \times 10^{14}$ cm^{-3}, $p_o = 0.8$ cm^{-3}, $n_o p_o = 2.0 \times 10^{14}$ cm^{-6}. This is precisely equal to n_i^2 calculated in part (a). (c) $n_o = 3.3 \times 10^{-7}$ cm^{-3}, $p_o = 7.7 \times 10^{19}$ cm^{-3}, $n_o p_o = 2.5 \times 10^{13}$ cm^{-6}. This is smaller than n_i^2 calculated in part (a) due to the degenerate nature of the holes.

2.12 (a) $N_D \ll 4.2 \times 10^{17}$ cm^{-3}. (b) $2.3 \times 10^6 \ll N_D \ll 4.2 \times 10^{17}$ cm^{-3}. (c) $2.3 \times 10^6 \ll N_D \ll 4.2 \times 10^{17}$ cm^{-3}.

2.16 $kT/2$ above the conduction band edge.

2.19 (a) $n_o = 0.1$ cm^{-3}. (b) $p_o = 1.8 \times 10^{20}$ cm^{-3}.

Chapter 3

3.1 $G_{rad} = 2.0 \times 10^5$ cm$^{-3} \cdot$ s^{-1}, $R_{rad} = 2.0 \times 10^{17}$ cm$^{-3} \cdot$ s^{-1}. $G_{eeh} = 1.8 \times 10^4$ cm$^{-3} \cdot$ s^{-1}, $R_{eeh} = 1.8 \times 10^{16}$ cm$^{-3} \cdot$ s^{-1}. $G_{ehh} = 9.5 \times 10^5$ cm$^{-3} \cdot$ s^{-1}, $R_{ehh} = 9.5 \times 10^{17}$ cm$^{-3} \cdot$ s^{-1}. $r_{ec} = 9.9 \times 10^{20}$ cm$^{-3} \cdot$ s^{-1}, $r_{ee} = 9.9 \times 10^{13}$ cm$^{-3} \cdot$ s^{-1}, $r_{hc} = 9.9 \times 10^{20}$ cm$^{-3} \cdot$ s^{-1}, $r_{he} = 9.9 \times 10^{15}$ cm$^{-3} \cdot$ s^{-1}.

3.2 (a) $n = 2.8 \times 10^{15}$ cm^{-3}, $p = 1.0 \times 10^{17}$ cm^{-3}, $U = 1.0 \times 10^{20}$ cm$^{-3} \cdot$ s^{-1}. (b) $n = 2.8 \times 10^{15}$ cm^{-3}, $p = 1.0 \times 10^{17}$ cm^{-3}, $U = 1.0 \times 10^{20}$ cm$^{-3} \cdot$ s^{-1}. (c) $n = 4.7 \times 10^{14}$ cm^{-3}, $p = 1.0 \times 10^{17}$ cm^{-3}, $U = 1.7 \times 10^{19}$ cm$^{-3} \cdot$ s^{-1}. (d) $n = 1.0 \times 10^3$ cm^{-3}, $p = 1.0 \times 10^{17}$ cm^{-3}, $U = 0$.

3.4 (a) $R_{rad} = 2.0 \times 10^5$ cm$^{-3} \cdot$ s^{-1}, $R_{eeh} = 1.8 \times 10^7$ cm$^{-3} \cdot$ s^{-1}, $R_{ehh} = 9.5 \times 10^{-10}$ cm$^{-3} \cdot$ s^{-1}, $r_{o,hc} = r_{o,he} = 8.2 \times 10^7$ cm$^{-3} \cdot$ s^{-1}. (b) $U_{rad} = 2.0 \times 10^{19}$ cm$^{-3} \cdot$ s^{-1}, $U_{Auger} = 1.8 \times 10^{21}$ cm$^{-3} \cdot$ s^{-1}, $U_{tr} = 8.1 \times 10^{21}$ cm$^{-3} \cdot$ s^{-1}.

3.9 $\tau = 0.46$ s.

3.10 (a) $P = 18$ mW. (b) $E = 1.8 \times 10^{-11}$ J. (c) $E = 4.2 \times 10^{-12}$ J. (d) $\tau \simeq 1$ μs.

3.12 $f = 6.1 \times 10^{-10}$.

3.14 $U_{rad} = 4.0 \times 10^{19}$ cm$^{-3} \cdot$ s^{-1}, $U_{eeh} = 7.2 \times 10^{20}$ cm$^{-3} \cdot$ s^{-1}, $U_{ehh} = 1.9 \times 10^{20}$ cm$^{-3} \cdot$ s^{-1}, $U_{tr} = 6.7 \times 10^{22}$ cm$^{-3} \cdot$ s^{-1}.

3.15 (a) $n = 1.0 \times 10^{18}$ cm^{-3}, $p = 1.0 \times 10^{13}$ cm^{-3}, $U = 1.0 \times 10^{19}$ cm$^{-3} \cdot$ s^{-1}. (b) $n = 1.0 \times 10^{18}$ cm^{-3}, $p = 9.0 \times 10^{12}$ cm^{-3}, $U = 9.0 \times 10^{18}$ cm$^{-3} \cdot$ s^{-1}.

3.16 $g_l = 2.8 \times 10^{29}$ cm$^{-3} \cdot$ s^{-1}, $W = 8.6 \times 10^5$ W/cm^2.

Chapter 4

4.1 (a) $l_{ce} = 11.2$ nm. (b) $\tau_t/\tau_{ce} = 24.6$.

4.2 (a) $\sigma = 3.4 \times 10^{-6}\ \Omega^{-1} \cdot \text{cm}^{-1}$. (b) $R_{sh} = 4.9\ \text{M}\Omega/\Box$. (c) Slightly p-type with $p_o = 1.9 \times 10^{10}\ \text{cm}^{-3}$. In this case: $\sigma = 2.9 \times 10^{-6}\ \Omega^{-1} \cdot \text{cm}^{-1}$.

4.15 (a) Thermal equilibrium, nonuniformly doped, p-type, non-degenerate, not applicable (thermal equilibrium), extrinsic. (b) $J_h = 0$. (c) $\mathcal{E} = -300$ V/cm. (d) $F_h = 9.1 \times 10^{21}\ \text{cm}^{-2} \cdot \text{s}^{-1}$.

Chapter 5

5.1 (a) $R_{sh} = 450\ \Omega/\Box$. (b) $N_D = 1 \times 10^{17}\ \text{cm}^{-3}$. (c) $x_j = 2$ μm.

5.4 (a) $J_e = -0.16$ A/cm^2. (b) $U_s = 10^{18}\ \text{cm}^{-2} \cdot \text{s}^{-1}$. (c) $dn'/dx = -5.5 \times 10^{16}\ \text{cm}^{-4}$. (d) $F_e = 10^{18}\ \text{cm}^{-2} \cdot \text{s}^{-1}$. (e) $v_e = 10^4$ cm/s. (f) $dE_{fe}/dx = -14.4$ eV/cm.

5.5 (a) Out of equilibrium, uniformly doped, p-type, low-level injection, extrinsic. (b) $J_e = 7.5 \times 10^{-3}$ A/cm^2. (c) $S = 8.2 \times 10^3$ cm/s. (d) $dn/dx = 4.4 \times 10^{15}\ \text{cm}^{-4}$.

5.6 (a) $t = 0.1$ ps. (b) $t = 8.5$ ns.

5.11 (a) Out of equilibrium, uniformly doped, p-type, low-level injection, extrinsic. (b) $J_{e,diff} = -143$ A/cm^2. (c) $J_{h,diff} = 63$ A/cm^2. (d) $J_{h,drift} = 80$ A/cm^2.

5.12 $v_h = 2.4 \times 10^3$ cm/s.

5.13 $\tau = 10\ \mu$s, $\mu_h = 306$ cm^2/V · s.

5.16 (a) Out of equilibrium, uniformly doped, p-type, low-level injection, extrinsic. (b) $J_e = 9.8 \times 10^{-11}$ A/cm^2. (c) $F_e = -6.1 \times 10^8\ \text{cm}^{-2} \cdot \text{s}^{-1}$. (d) $U_s = 6.1 \times 10^8\ \text{cm}^{-2} \cdot \text{s}^{-1}$. (e) $v_e = -3.6 \times 10^6$ cm/s. (f) $dn/dx = 5.9 \times 10^7\ \text{cm}^{-4}$. (g) $S = \infty$.

5.18 Worst case is with $S = \infty$ and for n-type material. In this case, the measured time constant is $\tau_1 \simeq 1$ ms which yields a 40% error.

5.21 (a) $W = 2$ μm, $L = 10$ μm. (b) $W = 20.8$ μm, $L = 104$ μm.

5.23 (a) $N_t = 7.7 \times 10^{15}\ \text{cm}^{-3}$. (b) $N_t = 1.8 \times 10^{17}\ \text{cm}^{-3}$. The minimum trap detection level is much higher when the doping level is higher. (c) $N_t = 1.0 \times 10^{10}\ \text{cm}^{-3}$.

5.25 (a) $\mathcal{E} = 2.0 \times 10^5$ V/cm. (b) $W = 3.0 \times 10^{10}$ W/cm^3. (c) $r = 3.0 \times 10^{30}\ \text{cm}^{-3} \cdot \text{s}^{-1}$.

5.26 (a) Out of equilibrium, uniformly doped, n-type, low-level injection, extrinsic. (b) $J_h = -3.3 \times 10^{-7}$ A/cm^2. (c) $U_s = 2.1 \times 10^{12}\ \text{cm}^{-2} \cdot \text{s}^{-1}$. (d) $S = 1.6 \times 10^6$ cm/s. (e) $J_{e,diff} = 3.3 \times 10^{-7}$ A/cm^2. (f) $J_{e,drift} \simeq 0$.

5.30 (a) Out of equilibrium, uniformly doped, p-type, non-degenerate, low-level injection, extrinsic. (b) $J_h = 940$ A/cm^2. (c) $J_e = 1.1$ A/cm^2. (d) $\mathcal{E} = 1.0 \times 10^3$ V/cm. (e) $U = 2.2 \times 10^{10}\ \text{cm}^{-2} \cdot \text{s}^{-1}$. (f) $I_t = 940$ μA.

Chapter 6

6.2 TT $= 24\ \mu$s.

6.6 (a) $N_D = 10^{17}\ \text{cm}^{-3}$. (b) $w_n = 11$ μm.

6.7 (a) $I_P/I_C = 2.0 \times 10^{-5}$. (b) $Q_P/Q_C = 0.2$. (c) Transit time: $\tau_C = 7.7$ ps. (d) Carrier lifetime: $\tau_P = 0.1$ μs.

6.8 (a) $I_p = 5.3 \times 10^{-8}$ A. (b) $I_n = 1.0 \times 10^{-8}$ A. (c) $C_{dn} = 9.2 \times 10^{-15}$ F. (d) $C_{dp} = 1.2 \times 10^{-15}$ F. (e) $\Delta V = 18$ mV. (f) Doubles. (g) $N_A/N_D = 10$. (h) Must decrease.

6.9 $x_d = 0.1$ μm.

6.10 (a) $J = 400$ A/cm^2. (b) $Q = 1.2 \times 10^{-8}$ C/cm^2. (c) $C_d = 4.6 \times 10^{-7}$ F/cm^2. (d) $\tau_t = 30$ ps.

6.11 (a) $\phi_{bi} = 0.95$ eV. (b) $N_L = 2.7 \times 10^{16}$ cm^{-3}. (b) $N_H = 2.7 \times 10^{19}$ cm^{-3}.

6.12 (a) The diode goes into breakdown. (b) $t = 37$ μs.

6.15 (a) $\phi_p = 0.071$ eV. (b) $n_o = 1.4 \times 10^4$ cm^{-3}, $p_o = 6.5 \times 10^{15}$ cm^{-3}. (c) $x = 85$ nm. (d) $Q_p = 1.2 \times 10^{-7}$ C/cm^2.

6.17 (a) $|J_e| = 520$ A/cm^2. (b) $U_s = 3.3 \times 10^{21}$ cm$^{-2} \cdot$ s^{-1}. (c) $S = 6.5 \times 10^5$ cm/s.

6.20 (a) $p' = 6.3 \times 10^{16}$ cm^{-3}. (b) $v_h = 1.0 \times 10^5$ cm/s. (c) $F_h = 6.3 \times 10^{21}$ cm$^{-2} \cdot$ s^{-1}. (d) $C_{dn} = 1.9 \times 10^{-12}$ F/cm^2.

6.21 (a) $I = 1.3$ mA. (b) $t_{rr} = 26$ ps. (c) $Q = 1.3 \times 10^{-11}$ C. (d) $C = 5 \times 10^{-10}$ F.

Chapter 7

7.3 $q\varphi_{Bn} = 0.81$ eV.

7.8 (i) $C = 3.5 \times 10^{-9}$ F/cm^2. (ii) $C = 1 \times 10^{-8}$ F/cm^2. (iii) $C = 1 \times 10^{-8}$ F/cm^2. (iv) $C = 8.7 \times 10^{-8}$ F/cm^2.

7.9 (a) $I = -9.3$ nA. (b) $V = 0.52$ V. (c) $C = 4.6$ pF.

7.11 (a) $n = 9.7 \times 10^{14}$ cm^{-3}. (b) $v_e = 1.0 \times 10^7$ cm/s. (c) $I = 0.16$ mA.

7.12 $f_{3dB} = 2.2$ GHz.

7.14 $N_D = 1.2 \times 10^{17}$ cm^{-3}, $q\varphi_{Bn} = 1.1$ eV.

7.16 (a) $C = 37$ fF. (b) $C = 48$ fF. (a) $C = 91$ fF. (a) $C = 149$ fF.

7.20 (a) Forward. (b) $l_{ce} = 10$ nm. (c) $V = 0.52$ V. (d) $J = 5.5 \times 10^4$ A/cm^2.

Chapter 8

8.1 $\Delta E_c = 3.0$ eV, $\Delta E_v = 4.8$ eV.

8.3 For $V = -3$ V: (a) $\phi_s = -0.14$ V. (b) $Q_s = 7.7 \times 10^{-7}$ C/cm^2. (c) $\mathcal{E}_{ox} = -2.2 \times 10^6$ V/cm. (d) $x_d = 0$. (e) $C_{LF} = 3.7 \times 10^{-7}$ F/cm^2. (f) $C_{HF} = 3.7 \times 10^{-7}$ F/cm^2.

For $V = 0$ V: (a) $\phi_s = 0.45$ V. (b) $Q_s = -2.1 \times 10^{-7}$ C/cm^2. (c) $\mathcal{E}_{ox} = 6.1 \times 10^5$ V/cm. (d) $x_d = 44$ nm. (e) $C_{LF} = 1.5 \times 10^{-7}$ F/cm^2. (f) $C_{HF} = 1.5 \times 10^{-7}$ F/cm^2.

For $V = 0.3$ V: (a) $\phi_s = 0.64$ V. (b) $Q_s = -2.5 \times 10^{-7}$ C/cm^2. (c) $\mathcal{E}_{ox} = 7.4 \times 10^5$ V/cm. (d) $x_d = 53$ nm. (e) $C_{LF} = 1.3 \times 10^{-7}$ F/cm^2. (f) $C_{HF} = 1.3 \times 10^{-7}$ F/cm^2.

For $V = 3$ V: (a) $\phi_s = 1.05$ V. (b) $Q_s = -1.2 \times 10^{-6}$ C/cm^2. (c) $\mathcal{E}_{ox} = 3.5 \times 10^6$ V/cm. (d) $x_d =$ 62 nm. (e) $C_{LF} = 3.8 \times 10^{-7}$ F/cm^2. (f) $C_{HF} = 1.2 \times 10^{-7}$ F/cm^2. (g) $V_T = 0.66$ V. (h) $Q_a =$ 1.4×10^{-6} C/cm^2. (i) $Q_i = -1.1 \times 10^{-6}$ C/cm^2.

8.7 (a) Accumulation: $Q_s = 4.0 \times 10^{-7}$ C/cm^2, $\mathcal{E}_{ox} = -1.1 \times 10^6$ V/cm. (b) Deep depletion: $Q_s = -3.6 \times 10^{-7}$ C/cm^2, $\mathcal{E}_{ox} = 1.0 \times 10^6$ V/cm. (c) Inversion: $Q_s = -1.2 \times 10^{-6}$ C/cm^2, $\mathcal{E}_{ox} =$ 3.5×10^6 V/cm.

8.8 (a) MOS will never reach inversion. (b) $t = 90$ ns. (c) $t = 9$ µs.

8.9 (a) At $V_{SB} = 3$ V, the entire p-type body is depleted. (b) $N_A = 1.2 \times 10^{17}$ cm^{-3}. (c) $x_{ox} =$ 9.6 nm.

8.10 (a) $V_{FB} = -0.97$ V. (b) $x_{dmax} = 0.1$ µm. (c) $\mathcal{E}_{ox} = 4.6 \times 10^5$ V/cm. (d) $Q_i =$ -1.9×10^{-7} C/cm^2. (e) $Q_a = 3.5 \times 10^{-7}$ C/cm^2.

8.13 $n = 1.5 \times 10^{12}$ cm^{-3}, $p = 6.5 \times 10^7$ cm^{-3}.

8.15 (a) $\mathcal{E}_{ox} = 6.7 \times 10^5$ V/cm. (b) $p = 6.7 \times 10^{13}$ cm^{-3}. (c) $\mathcal{E}_s = 6.4 \times 10^5$ V/cm. (d) $\phi_s = 3.5$ V. (e) $Q_g = 3.4 \times 10^{-7}$ C/cm^2. (f) $C_{HF} = 4.1 \times 10^{-8}$ F/cm^2.

8.18 (a) $\mathcal{E}_{ox} = -5.8 \times 10^5$ V/cm. (b) $C = 2.9 \times 10^{-7}$ F/cm^2.

8.19 (a) $\phi_s = 1$ V. (b) $p = 4$ cm^{-3}. (c) $n_s = 4.8 \times 10^{11}$ cm^{-2}. (d) $V = 0.6$ V.

8.22 (a) $\phi_s = 0.46$ V. (b) $Q_s = -3.0 \times 10^{-7}$ C/cm^2. (c) $\mathcal{E}_{ox} = 8.7 \times 10^5$ V/cm. (d) $V = -0.15$ V.

8.23 (a) $x_{ox} = 4.1$ nm. (b) $x_{dmax} = 17$ nm. (c) $N_A = 4.4 \times 10^{18}$ cm^{-3}. (d) $Q_i = -1.8 \times$ 10^{-6} C/cm^2.

8.25 (a) $N_A = 2.7 \times 10^{17}$ cm^{-3}. (b) $x_{ox} = 8.6$ nm. (c) $W_M = 4.0$ eV.

8.28 (a) $x_d = 0.24$ µm. (b) $N_s = 10^6$. (c) $x_d = 0.22$ µm.

Chapter 9

9.1 (a) $Q_i(y = 0) = -4.4 \times 10^{-7}$ C/cm^2. (b) $Q_d(y = 0) = -1.7 \times 10^{-7}$ C/cm^2. (c) $Q_i(y = L) =$ -1.3×10^{-7} C/cm^2. (d) $Q_d(y = L) = -2.2 \times 10^{-7}$ C/cm^2. (e) $Q_g(y = 0) = 6.1 \times 10^{-7}$ C/cm^2, $Q_g(y = L) = 3.5 \times 10^{-7}$ C/cm^2.

9.6 $I_{off} = 8.6$ µA/µm.

9.9 (a) $V_T = 0.4$ V. (b) $\mu_e = 400$ cm^2/V · s. (c) $g_m = 3.6$ mS. (d) $C_{gs} = 23$ fF. (e) $g_{mb} = 1.1$ mS. (f) I_D(with body effect)/I_D(ideal) = 0.77.

9.16 (a) $g'_m = 56.6$ mA/V, $r'_o = 1$ kΩ, $C'_{gs} = 150$ fF, $C'_{jD} = 100$ fF. (b) $g'_m = 40$ mA/V, $r'_o = 2$ kΩ, $C'_{gs} = 150$ fF, C'_{jD} cannot be calculated as we need ϕ_{bij} as well as the specific values of V_{DS} and

V'_{DS}. (c) $g'_{\mathrm{m}} = 80$ mA/V, $r'_{\mathrm{o}} = 1$ kΩ, $C'_{\mathrm{gs}} = 300$ fF, $C'_{\mathrm{jD}} = 200$ fF. (d) $g'_{\mathrm{m}} = 20$ mA/V, $r'_{\mathrm{o}} = 8$ kΩ, $C'_{\mathrm{gs}} = 300$ fF, $C'_{\mathrm{jD}} = 100$ fF.

Chapter 10

10.5 (a) $N_{\mathrm{A}} = 4.7 \times 10^{17}$ cm^{-3}. (b) $N_{\mathrm{A}} = 1.4 \times 10^{17}$ cm^{-3}. (c) $\mathcal{E}_{\mathrm{s}} = 3.6 \times 10^5$ V/cm (poly gate), $\mathcal{E}_{\mathrm{s}} = 1.9 \times 10^5$ V/cm (metal gate). (d) $\mathcal{E}_{\mathrm{ox}} = 1.1 \times 10^6$ V/cm (poly gate), $\mathcal{E}_{\mathrm{ox}} = 5.7 \times 10^5$ V/cm (metal gate). (e) $x_{\mathrm{dmax}} = 50$ nm (poly gate), $x_{\mathrm{dmax}} = 89$ nm (metal gate). (f) $L_{\mathrm{g,min}} = 201$ nm (poly gate), $L_{\mathrm{g,min}} = 240$ nm (metal gate). (g) Metal gate device does not scale as well due to the lower body doping. However, this will lead to higher drain current for the same overdrive.

10.6 (a) $V_{\mathrm{DD,max}} = 1.7$ V. (b) $\mu_{\mathrm{eff}} = 362$ cm^2/V · s. (c) $I_{\mathrm{D}} = 396$ μA/μm. (d) $\tau = 2.6$ ps.

10.8 (a) Saturation. (b) $n_{\mathrm{s}}(y = 0) = 2.3 \times 10^{12}$ cm^{-2}. (c) $x_{\mathrm{dmax}} = 0.1$ μm. (d) $\mu_{\mathrm{eff}} = 502$ cm^2/V · s. (e) $f_{\mathrm{t}} = 25$ GHz. (f) $V_{\mathrm{BS}} = -7.3$ V.

10.9 $\mathcal{E}_{\mathrm{eff}} = 2.8 \times 10^5$ V/cm, $\mu_{\mathrm{eff}} = 519$ cm^2/V · s.

10.13 $R_{\mathrm{sh}} = 2.9$ kΩ/□.

10.15 (a) $\mu_{\mathrm{eff}} = 122$ cm^2/V · s. (b) $n_{\mathrm{s}} = 1.2 \times 10^{13}$ cm^{-2}. (c) $I_{\mathrm{D}} = 333$ μA/μm. (d) $I_{\mathrm{D}} = 1.1$ mA/μm. (e) $v_{\mathrm{sat}} = 1.5 \times 10^7$ cm/s. (f) $I_{\mathrm{off}} = 9.2 \times 10^{-5}$ A/mm.

10.16 $M = 1.0013$.

10.18 $EOT = 1.4$ nm.

10.19 $\Delta CET = -0.5$ nm.

10.22 (a) $V_{\mathrm{FB}} = -0.15$ V. (b) $Q_{\mathrm{s}} = -3.6 \times 10^{-6}$ C/cm^2. (c) $x_{\mathrm{p}} = 7.7$ nm. (d) $Q_{\mathrm{s}} = -2.9 \times 10^{-6}$ C/cm^2.

Chapter 11

11.1 (a) $I_{\mathrm{C}} = 1$ mA. (b) $\beta_{\mathrm{F}} = 100$. (c) $f_{\mathrm{T}} = 27$ GHz. (d) $V_{\mathrm{A}} = 20$ V. (e) $g'_{\mathrm{m}} = 80$ mS, $r'_{\pi} = 1.25$ kΩ, $r'_{\mathrm{o}} = 10$ kΩ, $C'_{\pi} + C'_{\mathrm{je}} = 420$ fF, $C'_{\mathrm{jc}} = 50$ fF.

11.6 (a) $f_{\mathrm{T}} = 6.9$ GHz. (b) $f_{\mathrm{T}} = 7.0$ GHz. (c) $f_{\mathrm{T}} = 4.6$ GHz. (d) $f_{\mathrm{T}} = 4.6$ GHz.

11.7 (a) Forward active regime. (b) $I_{\mathrm{C}} = 1$ mA. (c) $V_{\mathrm{BE}} = 0.78$ V. (d) $v_{\mathrm{e}} = 3.0 \times 10^6$ cm/s. (e) Saturation. (f) $V_{\mathrm{BE}} = 0.66$ V. (g) $I_{\mathrm{C}} = -10$ μA.

11.8 (a) $g_{\mathrm{m}} = 3.9$ mS. (b) $g_{\pi} = 77$ μS. (c) $|dn_{\mathrm{B}}/dx| = 4.5 \times 10^{21}$ cm^{-4}. (d) $|F_{\mathrm{e}}| = 6.3 \times 10^{20}$ cm^{-2} · s^{-1}. (e) $\Delta V_{\mathrm{BE}} = 18$ mV. (f) $Q'_{\mathrm{B}} = 4Q_{\mathrm{B}}$. (g) $I'_{\mathrm{B}} = 2I_{\mathrm{B}}$.

11.10 $\beta_{\mathrm{F}} = 66$.

11.14 (a) $N_{\mathrm{C}} = 4.7 \times 10^{16}$ cm^{-3}. (b) $N_{\mathrm{B}} = 4.8 \times 10^{17}$ cm^{-3}. (c) $W_{\mathrm{B}} = 0.21$ μm. (d) $f_{\mathrm{T}} = 8.2$ GHz.

11.17 (a) $N_{\mathrm{C}} = 1.3 \times 10^{18}$ cm^{-3}. (b) $C_{\mathrm{je}} + C_{\mathrm{jc}} = 18$ fF. (c) $\tau_{\mathrm{F}} = 0.38$ ps. (d) $W_{\mathrm{B}} = 22$ nm. (e) $Q_{\mathrm{F}} = 2.3 \times 10^{-15}$ C.

11.19 (a) $V_{\mathrm{BE}} = 0.78$ V. (b) $V_{\mathrm{BC}} = 0.77$ V. (c) Saturation. (d) $Q_{\mathrm{E}} = 1.0 \times 10^{-17}$ C. (e) $Q_{\mathrm{B}} = 7.6 \times 10^{-16}$ C. (f) $Q_{\mathrm{C}} = 1.8 \times 10^{-14}$ C.

11.20 (a) $C_{je} + C_{jc} = 200$ fF. (b) $\tau_F = 5.3$ ps. (c) $g_m = 39$ mS, $C_\pi = 210$ fF, $\tau_{tB} = 5.3$ ps. (d) $f_T' = 10.2$ GHz. (e) $f_T' = 16$ GHz.

11.21 (a) $C_{jco}(npn)/C_{jco}(pnp) = 1$. (b) $\tau_B(npn)/\tau_B(pnp) = 0.5$. (c) $V_A(npn)/V_A(pnp) = 1$. (d) $\beta_F(npn)/\beta_F(pnp) = 4$. (e) $I_S(npn)/I_S(pnp) = 2$. (f) $f_F|_{\text{low } I_C}(npn)/f_F|_{\text{low } I_C}(pnp) = 1$. (g) $f_F|_{\text{high } I_C}(npn)/f_F|_{\text{high } I_C}(pnp) = 2$.

11.25 (a) $v_{e,E} = 625$ cm/s. (b) $v_e(0) = 687$ cm/s. (c) $v_e(W_B/2) = 3.7 \times 10^6$ cm/s. (d) $v_e(W_B + W_C) = 6.3 \times 10^4$ cm/s.

Index

M

N

O

P

T